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TELEPRINT OF THE SOURCE CODE -- GridAtlas

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WHAT THIS FILE IS

Everything the browser loaded to build the screen this was printed from, in full, plus a description of what was on that screen. It is meant to be attached to an AI chat and asked about directly.

HOW TO USE IT

Attach this file in ChatGPT, Claude or Gemini and describe what you saw. You do not need to know how to code, and you do not need GitHub.

WHAT IS NOT HERE

Map tiles and any other cross-origin response the browser will not let this page read. Every one of those is listed by URL under NOT READ, with the reason. Nothing has been guessed at or substituted.

CODE is here in full and is never shortened. DATA files (datasets, GeoJSON, CSV, JSON) are shown as their first 4000 characters and marked TRUNCATED, with their true size given, so this file stays small enough to attach to a chat on a phone.

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THE SCREEN THIS CAME FROM

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```
{
  "url": "https://ventusltd.github.io/gridatlas/atlas/",
  "title": "GlobalGrid2050 | UK Region v8 (Experimental)",
  "generation": "202609051510",
  "viewport": {
    "width": 440,
    "height": 796,
    "devicePixelRatio": 3,
    "orientation": "portrait"
  },
  "userAgent": "Mozilla/5.0 (iPhone; CPU iPhone OS 18_7 like Mac OS X)
AppleWebKit/605.1.15 (KHTML, like Gecko) Version/26.6.1 Mobile/15E148
Safari/604.1",
  "capturedAt": "2026-09-05T15:21:39.238Z",
  "selectedProject": null,
  "layersOn": [],
  "layersOff": [
    "400kv [OK]",
    "275kv [OK]",
    "220kv [OK]",
    "132kv [OK]",
    "66kv [OK]",
    "11kv (UKPN) [WAIT]",
    "Subs [OK]",
    "Nuclear [OK]",
    "Gas [OK]",
    "Industry [OK]",
    "Data Ctrs [OK]",
    "Airports [OK]",
    "Railways [OK]",
    "Tesco [WAIT]",
    "Sainsbury's [WAIT]",
    "Asda [WAIT]",
    "Morrisons [WAIT]",
    "Aldi [WAIT]",
    "Lidl [WAIT]"
  ]
}
```

```

"Waitrose [WAIT]",
"M&S Food [WAIT]",
"Co-op [WAIT]",
"Costco [WAIT]",
"Booths [WAIT]",
"Elizabeth Line [WAIT]",
"London Underground [WAIT]",
"DLR [WAIT]",
"UK Metro [WAIT]",
"Trams & Light Rail [WAIT]",
"UK Mainline Railways [WAIT]",
"HS2 [WAIT]",
"Eurostar (HS1) UK & Europe [WAIT]",
"Eurostar Stations UK & Europe [WAIT]",
"Major Stadiums [WAIT]",
"Subsea Data Cables (Fibre) (not all connections available yet) [WAIT]",
"Major Ports (Deep Water/Ferry) [WAIT]",
"Inland / Minor Harbours [WAIT]",
"Global Hydrocarbons (Oil, Gas, LNG) [WAIT]",
"Motorways [WAIT]",
"Major A-Roads (Trunk) [WAIT]",
"A-Roads (Primary) [WAIT]",
"Motorway Services [WAIT]",
"EV Rapid (100kW+) [WAIT]",
"Major Industrial Sites [WAIT]",
"Solar PV [WAIT]",
"Solar PV (Operational Only) [WAIT]",
"Solar Roof [WAIT]",
"Wind [WAIT]",
"Onshore Wind (Operational Only) [WAIT]",
"Offshore Wind (Operational Only) [WAIT]",
"Battery Storage [WAIT]",
"Battery Storage (Operational Only) [WAIT]",
"Biomass & EfW [WAIT]",
"Tidal & Wave [WAIT]",
"Hydrogen [WAIT]",
"Hydro & Pumped Storage [WAIT]",
"Flywheel [WAIT]",
"Advanced Conversion [WAIT]",
"Geothermal [WAIT]",
"Compressed Air Storage [WAIT]",
"Same technology [SELECT A PROJECT]",
"Wider fleet [SELECT A PROJECT]",
"All pipeline [SELECT A PROJECT]"
],
"attribution": ""
}

```

```

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CONTENTS -- 32 file(s), 1781659 characters
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```

1. <https://ventusltd.github.io/gridatlas/atlas/>
5221 chars · code · in full · found via the page itself
2. <https://cdn.jsdelivr.net/npm/maplibre-gl@3.6.2/dist/maplibre-gl.js>
762513 chars · code · in full · found via <script src>, loaded by the browser (script)
3. <https://cdn.jsdelivr.net/npm/maplibre-gl@3.6.2/dist/maplibre-gl.css>
63722 chars · code · in full · found via <link stylesheet>, loaded by the browser (link)
4. <https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/ventusv8.css>
14877 chars · code · in full · found via <link stylesheet>, loaded by the

browser (link)

5. <https://ventusltd.github.io/gridatlas/atlas/current.json>
46738 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
6. <https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/index.html>
21193 chars · code · in full · found via loaded by the browser (fetch)
7. <https://ventusltd.github.io/gridatlas/atlas/cartridges/202609041945-streaming-parquet-bridge-v9-5.js>
13316 chars · code · in full · found via loaded by the browser (fetch)
8. <https://ventusltd.github.io/gridatlas/atlas/cartridges/202609041945-place-global-search-v9-5.js>
34719 chars · code · in full · found via loaded by the browser (fetch)
9. <https://ventusltd.github.io/gridatlas/atlas/cartridges/202609051510-substation-intelligence-v9-63.js>
422014 chars · code · in full · found via loaded by the browser (fetch)
10. <https://ventusltd.github.io/gridatlas/atlas/cartridges/202609050354-sld-sandbox-v9-8.js>
359603 chars · code · in full · found via loaded by the browser (fetch)
11. <https://raw.githubusercontent.com/Ventusltd/data-grid-gb/1c9909d1138704b29235c27fd769436dda8a0b18/derived/connection-points.v3.json>
2896561 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
12. <https://basemaps.cartocdn.com/gl/dark-matter-gl-style/style.json>
70431 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
13. <https://tiles.basemaps.cartocdn.com/vector/carto.streets/v1/tiles.json>
44256 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
14. <https://tiles.basemaps.cartocdn.com/gl/dark-matter-gl-style/sprite@2x.json>
103 chars · data · in full · found via loaded by the browser (fetch)
15. <https://tiles.basemaps.cartocdn.com/gl/dark-matter-gl-style/sprite@2x.png>
378 chars · data · in full · found via loaded by the browser (fetch)
16. <https://tiles.basemaps.cartocdn.com/fonts/Montserrat%20Regular%20Italic%20Open%20Sans%20Italic%20Noto%20Sans%20Regular%20HanWangHeiLight%20Regular%20NanumBarunGothic%20Regular/0-255.pbf>
74101 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
17. https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/grid_substations.geojson
1192742 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
18. <https://tiles.basemaps.cartocdn.com/fonts/Montserrat%20Medium,Open%20Sans%20Bold,Noto%20Sans%20Regular,HanWangHeiLight%20Regular,NanumBarunGothic%20Regular/0-255.pbf>
77323 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
19. https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/grid_275kv.geojson
1022277 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
20. https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/grid_66kv.geojson
597382 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
21. https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/grid_220kv.geojson
62037 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
22. https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/grid_110kv.geojson

v9/data/grid_132kv.geojson
2835573 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
23. https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/power_plants.geojson
35625 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
24. https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/industrial_offtakers.geojson
1294980 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
25. <https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/datacentres.geojson>
42847 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
26. <https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/airports.geojson>
119564 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
27. <https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/railways.geojson>
508859 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
28. <https://cdn.jsdelivr.net/npm/@duckdb/duckdb-wasm@1.29.0/+esm>
26009 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (script)
29. <https://ventusltd.github.io/ventus-grid-engine/genome/engine-graph.json>
34334 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (fetch)
30. <https://cdn.jsdelivr.net/npm/apache-arrow@17.0.0/+esm>
200034 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (script)
31. <https://cdn.jsdelivr.net/npm/flatbuffers@24.3.25/+esm>
9269 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (script)
32. <https://cdn.jsdelivr.net/npm/tslib@2.6.3/+esm>
10708 chars · data · TRUNCATED to first 4000 · found via loaded by the browser (script)

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NOT READ -- 3 resource(s)

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- <https://ventusltd.github.io/npm/apache-arrow@17.0.0/+esm>
HTTP 404
- <https://ventusltd.github.io/npm/tslib@2.6.3/+esm>
HTTP 404
- <https://ventusltd.github.io/npm/flatbuffers@24.3.25/+esm>
HTTP 404

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THE LIVE PAGE AS IT STOOD (document.documentElement.outerHTML)

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```
<html lang="en" data-gridatlas-generation="202609051510" class="gridatlas-menu-hosted" style="--gridatlas-card-max: 722px; --gridatlas-menu-bar-clear: 42px;">
<head>
<base href="https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/">
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=1.0, user-scalable=no">
  <title>GlobalGrid2050 | UK Region v8 (Experimental)</title>
  <link href="https://cdn.jsdelivr.net/npm/maplibre-gl@3.6.2/dist/maplibre-
```

```

gl.css" rel="stylesheet">
  <link href="ventusv8.css" rel="stylesheet">
<style>.scada-wrapper[data-gridatlas-collapsed="1"]{max-
height:0;overflow:hidden;padding-top:0;padding-bottom:0;border:0;}#gridatlas-
dash-toggle{position:fixed;right:12px;bottom:12px;z-index:9999;min-
height:44px;min-width:44px;font:600 12px/1 ui-monospace,SFMono-
Regular,Menlo,monospace;letter-spacing:.08em;padding:8px 12px;border-
radius:6px;cursor:pointer;background:#0d1117;color:#7fe3d0;border:1px solid
#2b3a44;}#gridatlas-dash-toggle[hidden]{display:none !important;}#gridatlas-
dash-toggle:focus-visible{outline:2px solid #7fe3d0;outline-offset:2px;}</style>
<style id="gridatlas-neon-css">
.gridatlas-neon-block{margin-top:7px;padding-top:6px;border-top:1px solid
#123;font-family:monospace}
.gridatlas-neon-block .neon-hd{display:flex;align-items:center;gap:6px;font-
size:10px;letter-spacing:.06em;
  color:#5fbdc2;font-weight:bold;text-transform:uppercase}
.gridatlas-neon-block .neon-beta{font-size:8px;letter-spacing:.06em;padding:1px
4px;border-radius:2px;
  background:#3a2f12;color:#e0b050;border:1px solid #6a5320;text-
transform:uppercase}
.gridatlas-neon-block ol{list-style:none;margin:5px 0 0;padding:0}
.gridatlas-neon-block li{display:flex;align-items:baseline;gap:6px;padding:2px
0}
.gridatlas-neon-block .neon-km{color:#5fbdc2;font-weight:bold;font-variant-
numeric:tabular-nums;
  min-width:54px;text-shadow:0 0 6px rgba(95,189,194,.35)}
.gridatlas-neon-block .neon-name{color:#9fb3ba;overflow:hidden;text-
overflow:ellipsis;
  white-space:nowrap;flex:1;max-width:150px}
.gridatlas-neon-block .neon-kv{color:#ffae00;font-size:9px;white-
space:nowrap;cursor:help}
.gridatlas-neon-block .neon-kvnote{margin-top:6px;font-size:10px;line-
height:1.45}
.gridatlas-neon-block .neon-kvnote b{color:#ffae00;font-weight:normal}
.gridatlas-neon-block .neon-pin{display:block;width:100%;margin-
top:7px;padding:5px 6px;
  background:#0a1a1d;border:1px solid #2f6f75;border-radius:3px;color:#8b9aa1;
  font:inherit;font-size:10px;letter-spacing:.05em;cursor:pointer;text-
transform:uppercase}
.gridatlas-neon-block .neon-pin:hover{border-color:#5fbdc2;color:#bfe9ee}
.gridatlas-neon-block .neon-pin[aria-pressed="false"]{color:#5f7a80;border-
color:#1d3238}
.gridatlas-neon-block .neon-layout{display:block;width:100%;margin-
top:7px;padding:5px 6px;
  background:#0a1a1d;border:1px solid #2f6f75;border-radius:3px;color:#5fbdc2;
  font:inherit;font-size:10px;letter-spacing:.05em;cursor:pointer;text-
transform:uppercase}
.gridatlas-neon-block .neon-layout:hover{border-
color:#5fbdc2;color:#bfe9ee;background:#0d2429}
/* The measurement and its qualifiers, as one element. A hairline rule and
  six pixels of padding: nothing here colours or grades the number. */
.gridatlas-neon-block .neon-answer{margin:0 0 8px;padding:6px 0 8px;
  border-bottom:1px solid #123}
.gridatlas-neon-block .neon-answer > .neon-caveat:first-child{margin-top:0}
/* The card sits over the map and used to be immovable, with only MapLibre's
  own hairline close cross. It gets a bar: grab it to move the card out of the
  way, and two controls big enough to hit without aiming. */
/* Measured on the live map: the card was 563px tall inside a 319px map and
  hung 403px below it, so the caveat and the layout button could not be
  reached at all. The content is now bounded to the map and scrolls, and the
  bar stays put at the top of that scroll so the controls never leave. */
.maplibregl-popup-content{max-height:var(--gridatlas-card-max, 60vh) !important;

```

```

    overflow-y:auto !important;overflow-x:hidden;overscroll-behavior:contain}
.gridatlas-card-bar{position:sticky;top:-6px;z-index:2;flex:0 0 auto;
  display:flex;align-items:center;gap:6px;margin:-6px -6px 6px;
  padding:5px 6px;background:#0a1a1d;border-bottom:1px solid #1d3238;
  border-radius:3px 3px 0 0;cursor:grab;user-select:none;touch-action:none;font-
family:monospace}
.gridatlas-card-bar:active{cursor:grabbing}
.gridatlas-card-bar .grip{color:#3f6f75;letter-spacing:2px;font-size:11px}
.gridatlas-card-bar .label{color:#8b9aa1;font-size:10px;max-
width:190px;overflow:hidden;
  text-overflow:ellipsis;white-space:nowrap}
.maplibregl-popup.gridatlas-min .gridatlas-card-bar .label{color:#5fbdc2;font-
weight:bold;max-width:230px}
.gridatlas-card-bar .spacer{flex:1}
.gridatlas-card-bar button{background:#050a0d;border:1px solid
#2f6f75;color:#5fbdc2;
  font:inherit;font-size:14px;line-height:1;min-width:44px;height:44px;border-
radius:3px;
  cursor:pointer;padding:0 6px}
.gridatlas-card-bar button:hover{color:#bfe9ee;border-
color:#5fbdc2;background:#0d2429}
.gridatlas-card-bar button.close:hover{color:#ff8f8f;border-color:#ff5d5d}
.maplibregl-popup.gridatlas-free{position:fixed !important;transform:none
!important;
  left:var(--gx) !important;top:var(--gy) !important;z-index:12}
.maplibregl-popup.gridatlas-free .maplibregl-popup-tip{display:none !important}
.maplibregl-popup.gridatlas-min .maplibregl-popup-content > *:not(.gridatlas-
card-bar){display:none !important}
.maplibregl-popup.gridatlas-min .maplibregl-popup-content{padding:6px
!important;
  border:1px solid #2f6f75;border-radius:4px;box-shadow:0 0 14px
rgba(95,189,194,.25)}
.maplibregl-popup.gridatlas-min .gridatlas-card-bar{margin:0;border-bottom:0;
  border-radius:3px;background:#08171a}
.maplibregl-popup.gridatlas-min .gridatlas-card-bar button.min{border-
color:#5fbdc2;color:#bfe9ee}
.gridatlas-neon-block .neon-caveat{margin-top:6px;color:#68797f;font-
size:9px;line-height:1.5}
.gridatlas-neon-block .neon-caveat b{color:#8b9aa1;font-weight:bold}
/* The immutable shell predates phone-landscape use. These are composition
  overrides, not a mutation of the attested shell: the left control stack can
  scroll inside a short map, and search results cannot extend below it. */
@media (max-height:600px){
  .map-controls{max-height:min(70%,calc(100dvh - 100px));overflow-y:auto;
    overscroll-behavior:contain;scrollbar-width:thin}
  .search-results{max-height:calc(100dvh - 140px) !important}
}
@media (pointer:coarse){
  .map-ctrl-btn,.search-btn{min-height:44px}
  .search-input{min-height:44px;box-sizing:border-box}
  .gridatlas-neon-block .neon-pin,.gridatlas-neon-block .neon-layout{min-
height:44px}
}

```

/* — THE CARD IS A DOCKED SHEET ON A PHONE —

Measured on a verified iPhone-class device (393x852 at dpr 3, pointer:coarse, hover:none, 5 touch points, document.hidden false): the anchored card opened at y=426, 819px tall, so its bottom edge was 393px BELOW the screen and the end of it was unreachable at every scroll position. Its left edge sat at x=-89 at 390px wide. Anchoring is the wrong idea on a phone – a 340px box hung off a marker in a 393px viewport has nowhere to go – so on a coarse pointer or a narrow window the card docks to the bottom edge, full width, and nothing has to be dragged.

```

    No rule below reads technology: three buckets light no layer at all. */
html.gridatlas-sheet-open .maplibregl-popup.gridatlas-sheet{
    position:fixed !important;left:0 !important;right:0 !important;
    top:auto !important;bottom:0 !important;transform:none !important;
    width:100vw !important;max-width:100vw !important;
    margin:0 !important;padding:0 !important;z-index:400 !important}
.maplibregl-popup.gridatlas-sheet .maplibregl-popup-tip{display:none !important}
.maplibregl-popup.gridatlas-sheet .maplibregl-popup-content{
    max-height:var(--gridatlas-sheet-h,56dvh) !important;
    width:100% !important;max-width:100% !important;
    border-radius:12px 12px 0 0;box-sizing:border-box;
    border-top:1px solid #2f6f75;box-shadow:0 -8px 24px rgba(0,0,0,.55);
    padding-bottom:calc(8px + env(safe-area-inset-bottom,0px)) !important}
.maplibregl-popup.gridatlas-sheet.gridatlas-min .maplibregl-popup-content{
    max-height:none !important}
/* MapLibre's own hairline cross is 20x18 and now sits inside a full-width
   sheet whose bar already carries a 44px close. Two closes, one of them
   unhittable, is worse than one. */
.maplibregl-popup.gridatlas-sheet .maplibregl-popup-close-button{display:none
!important}
/* WHEN A CONTROL AND THE ANSWER WANT THE SAME PIXELS, THE ANSWER WINS.
   Four bars printed over the card's text on every load: the tray with the
   GB PRICES and VERSIONS bars (all inside .map-controls at y 698-822), HIDE
   LAYERS at y 796 on z-index 9999, and the credit strip at y 827. None is
   deleted and none goes UNDER the sheet, which would make it unreachable:
   they are lifted clear, still on the map and still 44px. The offset reads
   the same var the sheet is sized from, so the two cannot disagree. */
html.gridatlas-sheet-open .map-controls{
    bottom:calc(var(--gridatlas-sheet-h,56dvh) + 12px) !important;
    max-height:calc(100dvh - var(--gridatlas-sheet-h,56dvh) - 120px) !important;
    overflow-y:auto;overscroll-behavior:contain}
html.gridatlas-sheet-open #gridatlas-dash-toggle{
    bottom:calc(var(--gridatlas-sheet-h,56dvh) + 12px) !important}
/* Not an attribution: the OSM and CARTO credit is .custom-map-attrib at the
   top of the map and does not move. This is a shout-out that was printing
   across the card's sentences. Hidden only while a sheet is open. */
html.gridatlas-sheet-open .podcast-shoutout{display:none !important}</style>
<style id="gridatlas-gb-conditions-style">
#gridatlas-gb-conditions{margin-top:6px;font:11px/1.5 ui-monospace,SFMono-
Regular,Menlo,monospace}
#gridatlas-gb-conditions > button{display:block;width:100%;padding:6px
8px;background:#0a1a1d;
    border:1px solid #2f6f75;border-radius:3px;color:#8fb3b8;font:inherit;
    letter-spacing:.06em;text-transform:uppercase;cursor:pointer;text-align:left}
#gridatlas-gb-conditions > button:hover{border-color:#5fbdc2;color:#bfe9ee}
#gridatlas-gb-conditions .gb-body{display:none;margin-
top:5px;padding:8px;border:1px solid #1d3238;
    border-radius:3px;background:rgba(6,18,21,.94);max-width:min(88vw,260px);
    max-height:min(52vh,340px);overflow:auto;overscroll-behavior:contain}
#gridatlas-gb-conditions[data-open="1"] .gb-body{display:block}
#gridatlas-gb-conditions .gb-row{display:flex;justify-content:space-
between;gap:8px;
    padding:2px 0;border-bottom:1px solid #142226}
#gridatlas-gb-conditions .gb-k{color:#7d8f95}
#gridatlas-gb-conditions .gb-v{color:#bfe9ee;font-weight:bold}
#gridatlas-gb-conditions .gb-v em{color:#5f7a80;font-style:normal;font-
weight:normal;font-size:10px}
#gridatlas-gb-conditions .gb-note{margin:7px 0 0;color:#6f8288;font-
size:10px;line-height:1.45}
#gridatlas-gb-conditions .gb-note.gb-point{color:#9fb3ba;border-top:1px solid
#142226;padding-top:6px}
#gridatlas-gb-conditions .gb-note.gb-point b{color:#d8a76a}

```

```

#gridatlas-gb-conditions .gb-more{display:block;margin-
top:7px;color:#5fdbc2;font-size:10px;
  text-decoration:none;letter-spacing:.04em}
#gridatlas-gb-conditions .gb-more:hover{text-decoration:underline}</style><style
id="gridatlas-version-ledger-style">
#gridatlas-version-ledger{margin-top:6px;font:11px/1.5 ui-monospace,SFMono-
Regular,Menlo,monospace}
#gridatlas-version-ledger > button{display:block;width:100%;padding:6px
8px;background:#0a1a1d;
  border:1px solid #2f6f75;border-radius:3px;color:#8fb3b8;font:inherit;
  letter-spacing:.06em;text-transform:uppercase;cursor:pointer;text-align:left}
#gridatlas-version-ledger > button:hover{border-color:#5fdbc2;color:#bfe9ee}
#gridatlas-version-ledger .vl-body{display:none;margin-
top:5px;padding:8px;border:1px solid #1d3238;
  border-radius:3px;background:rgba(6,18,21,.94);max-width:min(88vw,300px);
  max-height:min(56vh,380px);overflow:auto;overscroll-behavior:contain}
#gridatlas-version-ledger[data-open="1"] .vl-body{display:block}
#gridatlas-version-ledger .vl-row{padding:4px 0;border-bottom:1px solid #142226}
#gridatlas-version-ledger .vl-head{display:flex;justify-content:space-
between;gap:8px}
#gridatlas-version-ledger .vl-ver{color:#bfe9ee;font-weight:bold}
#gridatlas-version-ledger .vl-status{margin-left:5px;color:#ff9b73;font-
size:9px;font-weight:bold}
#gridatlas-version-ledger .vl-when{color:#5f7a80;font-size:10px}
#gridatlas-version-ledger .vl-scope{color:#9fb3ba;font-size:10px;line-
height:1.4;margin-top:1px}
#gridatlas-version-ledger .vl-reason{color:#df9b83;font-size:9px;line-
height:1.35;margin-top:2px}
#gridatlas-version-ledger .vl-note{margin:7px 0 0;color:#6f8288;font-
size:10px;line-height:1.45}</style><style id="gridatlas-mobile-tray-style">
#gridatlas-mobile-tray{display:flex;gap:4px;font:11px/1.2 ui-monospace,SFMono-
Regular,Menlo,monospace}
#gridatlas-mobile-tray button{min-height:44px;padding:6px
10px;background:#0a1a1d;
  border:1px solid #2f6f75;border-radius:3px;color:#8fb3b8;font:inherit;
  letter-spacing:.06em;text-transform:uppercase;cursor:pointer}
#gridatlas-mobile-tray button[aria-pressed="true"]{border-
color:#5fdbc2;color:#bfe9ee;
  background:rgba(0,255,255,0.08)}
#gridatlas-mobile-tray button[disabled]{opacity:.45;cursor:default}
.map-controls.gm-tools-collapsed > .map-ctrl-btn{display:none}</style><style
id="gridatlas-sld-css">
/* Top RIGHT, below the search box. The Atlas keeps its own tool buttons down
the left edge -- EXPORT CSV, RADIUS SEARCH, ZONE DRAW, MEASURE -- and a
panel on that side covers them, and the search bar occupies 72-96px inside
the map container on the right, so the panel clears it at 112px. Both offsets
were measured on the live map: no headless test catches a collision with a
component the panel knows nothing about. */
#gridatlas-sld-panel{position:absolute;right:14px;top:112px;bottom:14px;z-
index:11;width:310px;
  max-width:calc(100% - 28px);box-sizing:border-box;overflow:auto;font:11px/1.5
'Courier New',monospace;
  color:#cfe9ee;background:rgba(2,8,11,.93);border:1px solid #0b5f63;border-
radius:5px;
  padding:11px 12px;box-shadow:0 0 22px rgba(0,255,255,.14);backdrop-
filter:blur(3px);display:none}
#gridatlas-sld-panel[data-open="true"]{display:block}
#gridatlas-sld-panel h4{margin:0 0 2px;font-size:10px;letter-
spacing:.09em;color:#5fdbc2;text-transform:uppercase;
  display:flex;align-items:center;gap:7px}
#gridatlas-sld-panel .sld-beta{font-size:8px;padding:1px 4px;border-
radius:2px;background:#3a2f12;

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    color:#e0b050;border:1px solid #6a5320}
#gridatlas-sld-panel .sld-site{color:#fff;font-size:12px;font-
weight:bold;margin:2px 0 8px;
    overflow:hidden;text-overflow:ellipsis;white-space:nowrap}
#gridatlas-sld-panel h4.sld-drag{cursor:grab;user-select:none;touch-action:none}
#gridatlas-sld-panel h4.sld-drag:active{cursor:grabbing}
#gridatlas-sld-panel .sld-min{margin-left:auto}
#gridatlas-sld-panel .sld-min,#gridatlas-sld-panel .sld-
close{cursor:pointer;background:#050a0d;
    border:1px solid #2f6f75;color:#5fbdc2;font:inherit;font-size:12px;line-
height:1;
    min-width:44px;height:44px;border-radius:3px;padding:0 5px}
#gridatlas-sld-panel .sld-min:hover{color:#bfe9ee;border-color:#5fbdc2}
#gridatlas-sld-panel .sld-close:hover{color:#ff8f8f;border-color:#ff5d5d}
#gridatlas-sld-panel[data-min="true"] > *:not(h4){display:none}
#gridatlas-sld-panel[data-min="true"]{width:auto;padding:7px 9px;
    box-shadow:0 0 14px rgba(95,189,194,.25)}
#gridatlas-sld-panel[data-min="true"] h4{margin:0}
#gridatlas-sld-panel[data-min="true"] .sld-min{border-
color:#5fbdc2;color:#bfe9ee}
#gridatlas-sld-panel .sld-to{color:#8b9aa1;font-size:9.5px;margin:-6px 0 8px}
#gridatlas-sld-panel .sld-target{margin:0 0 9px;padding:7px 8px;border:1px solid
#1d3238;
    border-radius:3px;background:#050a0d}
#gridatlas-sld-panel .sld-target-row{display:flex;justify-content:space-
between;align-items:baseline}
#gridatlas-sld-panel .sld-target-row b{color:#e0b050;font-variant-
numeric:tabular-nums}
#gridatlas-sld-panel .sld-basis{display:flex;align-items:center;gap:6px;margin-
top:5px}
#gridatlas-sld-panel .sld-basis span{color:#8b9aa1;font-size:10px;white-
space:nowrap}
#gridatlas-sld-panel .sld-basis select{flex:1}
#gridatlas-sld-panel .sld-danger{margin-top:6px;color:#ff5d5d;font-
size:9px;line-height:1.5;
    border-left:2px solid #ff5d5d;padding-left:6px}
#gridatlas-sld-panel .sld-fitted{margin-top:6px;color:#8b9aa1;font-
size:9px;line-height:1.5}
#gridatlas-sld-panel .sld-fitted b{color:#6fb582}
#gridatlas-sld-panel .sld-fitted b.sld-off{color:#ff5d5d}
#gridatlas-sld-panel .sld-ratio-note{margin-top:6px;color:#d9b45f;font-
size:9px;line-height:1.5;
    border-left:2px solid #8b6c28;padding-left:6px}
#gridatlas-sld-panel .sld-tabs{display:flex;gap:5px;margin-bottom:8px}
#gridatlas-sld-panel .sld-tabs button{flex:1;background:#050a0d;border:1px solid
#1d3238;color:#7f939a;
    font:inherit;font-size:9px;padding:4px;cursor:pointer;border-radius:3px;text-
transform:uppercase}
#gridatlas-sld-panel .sld-tabs button[data-on="true"]{color:#5fbdc2;border-
color:#5fbdc2}
#gridatlas-sld-panel .sld-grid{display:grid;grid-template-columns:1fr
62px;gap:3px 7px;align-items:center}
#gridatlas-sld-panel label{color:#8b9aa1;font-size:10px}
#gridatlas-sld-panel input,#gridatlas-sld-panel
select{width:100%;background:#050a0d;border:1px solid #1d3238;
    color:#d8dee6;font:inherit;font-size:10px;padding:2px 4px;border-radius:2px}
#gridatlas-sld-panel input:focus,#gridatlas-sld-panel select:focus{outline:1px
solid #5fbdc2}
#gridatlas-sld-panel .sld-out{margin-top:9px;padding-top:8px;border-top:1px
solid #10262b;
    display:grid;grid-template-columns:1fr auto;gap:2px 8px}
#gridatlas-sld-panel .sld-out b{color:#e0b050;font-variant-numeric:tabular-nums}

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#gridatlas-sld-panel .sld-out .lit{color:#5fbdc2}
#gridatlas-sld-panel .sld-warn{margin-top:7px;color:#d9963c;font-size:9px;line-height:1.45}
#gridatlas-sld-panel .sld-caveat{margin-top:7px;padding-top:7px;border-top:1px solid #10262b;color:#68797f;font-size:9px;line-height:1.5}
#gridatlas-sld-panel .sld-caveat b{color:#8b9aa1}
#gridatlas-sld-panel .sld-hint{margin-top:6px;color:#5f7a80;font-size:9px;line-height:1.45}
#gridatlas-sld-panel .sld-finance{margin-top:9px;border-top:1px solid #214047;padding-top:7px}
#gridatlas-sld-panel .sld-finance summary{min-height:32px;display:flex;align-items:center;cursor:pointer;color:#d9b45f;font-weight:bold;letter-spacing:.05em;user-select:none}
#gridatlas-sld-panel .sld-fin-grid{display:grid;grid-template-columns:1fr 76px;gap:3px 7px;align-items:center}
#gridatlas-sld-panel .sld-fin-section{grid-column:1/-1;margin-top:7px;padding-top:5px;border-top:1px solid #10262b;color:#5fbdc2;font-size:9px;text-transform:uppercase}
#gridatlas-sld-panel .sld-fin-grid input[type="checkbox"]{width:24px;justify-self:end}
#gridatlas-sld-panel .sld-fin-out{margin:8px 0;padding:7px;background:#050a0d;border:1px solid #1d3238;display:grid;grid-template-columns:1fr auto;gap:2px 8px}
#gridatlas-sld-panel .sld-fin-out b{color:#d9b45f;font-variant-numeric:tabular-nums;text-align:right}
#gridatlas-sld-panel .sld-fin-note{margin:6px 0;color:#8b9aa1;font-size:9px;line-height:1.5}
@media (max-width:700px){#gridatlas-sld-panel{width:auto;left:8px;right:8px;top:96px;bottom:8px}}
@media (pointer:coarse){
  #gridatlas-sld-panel .sld-tabs button,#gridatlas-sld-panel input,#gridatlas-sld-panel select,
  #gridatlas-sld-panel .sld-finance summary{min-height:44px}
}</style><style>.gridatlas-fs-layers{position:absolute !important;left:0;right:0;bottom:0;max-height:42vh;overflow:auto;z-index:9;background:rgba(2,8,11,.94);border-top:1px solid #0b5f63;backdrop-filter:blur(3px)}</style>
<style>.gridatlas-corridor-open{display:block;width:100%;min-height:44px;margin:8px 0 2px;font:600 12px/1.2 ui-monospace,SFMono-Regular,Menlo,monospace;letter-spacing:.06em;text-align:left;padding:12px 14px;border-radius:6px;cursor:pointer;background:#0d1117;color:#7fe3d0;border:1px solid #2b3a44;}.gridatlas-corridor-open:focus-visible{outline:2px solid #7fe3d0;outline-offset:2px;}#gridatlas-corridor-sheet{position:fixed;left:0;right:0;bottom:0;z-index:10000;max-height:min(70vh,560px);overflow:auto;transform:translateY(101%);transition:transform .18s ease-out;background:#070d11;color:#cfe6e8;border-top:1px solid #0b5f63;padding:14px 16px calc(16px + env(safe-area-inset-bottom));font:12px/1.5 ui-monospace,SFMono-Regular,Menlo,monospace;}#gridatlas-corridor-sheet[data-open="1"]{transform:translateY(0);}#gridatlas-corridor-sheet h4{margin:0 0 8px;font-size:13px;color:#7fe3d0;}#gridatlas-corridor-sheet .r{padding:8px 0;border-top:1px solid #14252b;}#gridatlas-corridor-sheet .c{color:#8fa6ab;}#gridatlas-corridor-sheet button{min-height:44px;min-width:44px;cursor:pointer;background:#0d1117;color:#7fe3d0;border:1px solid #2b3a44;border-radius:6px;padding:10px 14px;font:600 12px/1 ui-monospace,monospace;}@media (prefers-reduced-motion:reduce){#gridatlas-corridor-sheet{transition:none;}}</style><style id="gridatlas-menu-bar-css">#gridatlas-menu-bar{position:absolute;top:0;left:0;right:0;height:36px;z-index:10020;display:flex;align-items:stretch;gap:0;padding-left:env(safe-area-inset-left);padding-right:env(safe-area-inset-right);box-sizing:border-

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box;background:rgba(4,10,13,.95);border-bottom:1px solid
rgba(80,220,240,.3);font:11px/1 ui-monospace,SFMono-Regular,Menlo,monospace;-
webkit-backdrop-filter:blur(7px);backdrop-
filter:blur(7px);isolation:isolate;pointer-events:auto}#gridatlas-menu-bar .gm-
menu{position:relative;min-width:0}#gridatlas-menu-bar .gm-
title{appearance:none;border:0;background:transparent;color:#cfeef6;min-
height:36px;padding:0 11px;cursor:pointer;font:inherit;letter-
spacing:.05em;text-transform:uppercase;white-space:nowrap}#gridatlas-menu-bar
.gm-title:hover,#gridatlas-menu-bar .gm-title:focus-visible,#gridatlas-menu-bar
.gm-menu.gm-open>.gm-
title{background:rgba(80,220,240,.16);color:#fff}#gridatlas-menu-bar .gm-
title:focus-visible,#gridatlas-menu-bar .gm-panel :focus-visible{outline:2px
solid #6bebff;outline-offset:-2px}#gridatlas-menu-bar .gm-
side{display:flex;align-items:stretch;flex:1 1 0;min-width:0}#gridatlas-menu-bar
.gm-side-left{justify-content:flex-start}#gridatlas-menu-bar .gm-side-
right{justify-content:flex-end}#gridatlas-menu-bar .gm-brand-slot{flex:0 1
auto;min-width:0;max-width:64%;display:flex;align-items:center;justify-
content:center;overflow:hidden;padding:0 6px;text-align:center}#gridatlas-menu-
bar .gm-brand-slot .hud-
header{display:flex!important;position:static!important;width:auto!important;ali-
gn-items:center;justify-
content:center;gap:11px;margin:0!important;padding:0!important;background:none!i-
mportant;border:0!important}#gridatlas-menu-bar .gm-brand-slot .hud-
header>div{flex:0 0 auto;line-height:1.05}#gridatlas-menu-bar .gm-brand-slot
.hud-header small{font-size:6.5px;white-space:nowrap}#gridatlas-menu-bar .gm-
brand-slot .hud-header .hud-val{font-size:10.5px;text-shadow:none}#gridatlas-
menu-bar .gm-brand-slot .ventus-main{font-size:14px;font-weight:800;letter-
spacing:.2em;margin:0;color:#fff}#gridatlas-menu-bar .gm-brand-slot .ventus-
sub{font-size:5.5px;letter-spacing:.14em}#gridatlas-menu-bar .gm-
panel{position:absolute;top:100%;left:0;min-width:240px;max-
width:min(92vw,420px);max-height:min(72dvh,620px);overflow:auto;overscroll-
behavior:contain;padding:6px;background:rgba(4,10,13,.98);border:1px solid
rgba(80,220,240,.32);border-top:0;box-shadow:0 12px 34px rgba(0,0,0,.68);box-
sizing:border-box}#gridatlas-menu-bar .gm-panel[hidden]
{display:none!important}#gridatlas-menu-bar .gm-side-right .gm-
panel{left:auto;right:0}#gridatlas-menu-bar .gm-panel button,#gridatlas-menu-bar
.gm-panel [role="button"]{display:flex;align-items:center;width:100%;min-
height:44px;box-sizing:border-
box;position:static!important;inset:auto!important;transform:none!important;marg-
in:0 0 3px;padding:7px 10px;border:0;border-
radius:2px;background:transparent;color:#cfeef6;font:11px/1.35 ui-
monospace,SFMono-Regular,Menlo,monospace;text-align:left;letter-
spacing:.03em;text-transform:none;cursor:pointer}#gridatlas-menu-bar .gm-panel
button:hover,#gridatlas-menu-bar .gm-panel
[role="button"]:hover{background:rgba(80,220,240,.14);color:#fff}#gridatlas-
menu-bar .gm-panel a[data-gm-estate],#gridatlas-menu-bar .gm-panel a[data-gm-
engine],#gridatlas-menu-bar .gm-panel a[data-gm-study]{text-
decoration:none}#gridatlas-menu-bar .gm-panel a[data-gm-engine]{white-
space:nowrap;overflow:hidden;text-overflow:ellipsis;display:block;line-
height:30px;min-height:44px}#gridatlas-menu-bar .gm-panel .custom-map-
attrib{position:static!important;inset:auto!important;margin:4px 0
2px;padding:6px 8px;max-width:none;background:transparent;border:0;white-
space:normal;line-height:1.45;font:10px/1.45 ui-monospace,SFMono-
Regular,Menlo,monospace;color:#8fb6c0}#gridatlas-menu-bar .gm-layer-
group{margin:5px 0 2px;padding:6px 8px 3px;border-top:1px solid
#19343b;color:#6fa2ae;font-size:10px;letter-spacing:.08em;text-
transform:uppercase}#gridatlas-menu-bar .gm-
layer{position:relative;display:flex;align-items:center;gap:10px;min-
height:44px;box-sizing:border-box;padding:7px
9px;color:#cfeef6;cursor:pointer;line-height:1.35}#gridatlas-menu-bar .gm-
layer:hover{background:rgba(80,220,240,.12)}#gridatlas-menu-bar .gm-layer
input{position:absolute;inset:0;width:100%;height:100%;margin:0;opacity:0;cursor

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:pointer;z-index:1}#gridatlas-menu-bar .gm-layer-
box{width:20px;height:20px;flex:0 0 auto;border:1.5px solid #4a8b96;border-
radius:4px;position:relative;background:rgba(255,255,255,.04)}#gridatlas-menu-
bar .gm-layer input:checked~.gm-layer-box{background:#4fd7ee;border-
color:#4fd7ee}#gridatlas-menu-bar .gm-layer input:checked~.gm-layer-
box::after{content:"";position:absolute;left:6px;top:2px;width:5px;height:10px;b
order:solid #04141a;border-width:0 2px 2px 0;transform:rotate(38deg)}#gridatlas-
menu-bar .gm-layer input:focus-visible~.gm-layer-box{outline:2px solid
#6bebff;outline-offset:2px}#gridatlas-menu-bar .gm-layer-name{overflow-
wrap:anywhere}#gridatlas-menu-bar .gm-panel-
grid{padding:0;overflow:hidden;display:flex;flex-direction:column;min-
width:min(94vw,360px);max-width:min(96vw,900px)}#gridatlas-menu-bar .gm-panel-
head{flex:0 0 auto;padding:8px 8px 0}#gridatlas-menu-bar .gm-panel-head .scada-
brand{padding:0 0 6px;margin:0 0 6px}#gridatlas-menu-bar .gm-panel-head .status-
legend{padding:0 0 8px;margin:0 0;border:0}#gridatlas-menu-bar .gm-panel-
body{flex:1 1 auto;overflow:auto;overscroll-behavior:contain;padding:6px;min-
height:0}@media(min-width:560px){#gridatlas-menu-bar .gm-panel-body{column-
count:2;column-gap:14px}#gridatlas-menu-bar .gm-panel-body .gm-layer-
group{break-inside:avoid}#gridatlas-menu-bar .gm-panel-body .gm-layer{break-
inside:avoid}}@media(min-width:900px){#gridatlas-menu-bar .gm-panel-body{column-
count:3}}#gridatlas-menu-bar .search-bar-
wrapper{position:static!important;display:grid!important;grid-template-
columns:minmax(150px,1fr) auto;width:min(82vw,390px);gap:5px;margin:2px 0
6px}#gridatlas-menu-bar .search-bar-wrapper>div{position:relative}#gridatlas-
menu-bar .search-input{width:100%!important;min-height:44px;box-sizing:border-
box}#gridatlas-menu-bar .search-results{position:static!important;max-
height:42vh;overflow:auto}#gridatlas-menu-bar .disclaimer-box,#gridatlas-menu-
bar .podcast-shoutout{display:block!important;position:static!important;max-
width:380px;padding:8px;box-sizing:border-box;text-align:left;pointer-
events:auto}.gridatlas-menu-hosted .map-controls[data-gridatlas-menu-
emptied="1"]{display:none!important}.gridatlas-menu-hosted .scada-
wrapper{display:flex!important}#gridatlas-dash-toggle{display:inline-
flex!important}html.gridatlas-sheet-open #gridatlas-menu-bar{z-
index:10020!important;pointer-events:auto!important}html.gridatlas-sheet-open
#gridatlas-menu-bar .gm-panel{pointer-events:auto!important}body:not(.fs-active)
#gridatlas-menu-bar #btn-fullscreen-exit{display:none!important}body.fs-active
#gridatlas-menu-bar #btn-fullscreen-exit{display:flex!important}.gridatlas-menu-
hosted .custom-map-attrib{top:var(--gridatlas-menu-bar-clear,44px)!important;z-
index:10025!important}.gridatlas-menu-hosted #fs-
letterhead{display:none!important}@media(max-width:700px){#gridatlas-menu-
bar{height:34px}#gridatlas-menu-bar .gm-title{min-height:34px;padding:0
6px;font-size:9px;letter-spacing:.025em}#gridatlas-menu-bar .gm-brand-slot{max-
width:48%;padding:0 2px}#gridatlas-menu-bar .gm-brand-slot .hud-
header>div:first-child,#gridatlas-menu-bar .gm-brand-slot .hud-header>div:last-
child{display:none}#gridatlas-menu-bar .gm-brand-slot .ventus-main{font-
size:11px;letter-spacing:.14em}#gridatlas-menu-bar .gm-brand-slot .ventus-
sub{font-size:4.5px}#gridatlas-menu-bar .gm-
panel{position:fixed;top:34px;left:4px!important;right:4px!important;width:auto;
max-width:none;max-height:calc(100dvh - 40px);padding-bottom:calc(6px +
env(safe-area-inset-bottom))}#gridatlas-menu-bar .gm-panel-grid{max-width:none}}
</style></head>
<body>
<script>window.__GRIDATLAS_ATLAS__=
{"schema":"gridatlas.current.v2","generation":"202609051510","release_id":"20260
8300453-atlas-
v9","architecture":"IMMUTABLE_SHELL_PLUS_HASHED_CARTRIDGES","shell":
{"release_id":"202608300453-atlas-v9","index":"./releases/202608300453-atlas-
v9/index.html","base":"./releases/202608300453-atlas-v9/"},"cartridge_order":
["streaming-parquet-bridge","uk-gazetteer-flyto","substation-intelligence","sld-
sandbox"],"loaded_cartridges":[{"id":"streaming-parquet-
bridge","sha256":"4a6b594f7705bc4d7d62cbe69d2e40726d12f9a759896430cfe5eb00d830d2
3e"}, {"id":"uk-gazetteer-

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flyto","sha256":"21ab1507e98ee29961d03236e8b7853298104a33ecb1986d4d10ffc420a99c1c"}, {"id":"substation-intelligence","sha256":"2db05c263bc549ef5406a796c4f2c6ba278a92da04fd653f470f58564fdbb936"}, {"id":"sld-sandbox","sha256":"7c2df5dabb0082d4cef941b98bf31e92161d1e79a7d22bd6c84e81b171507022"}]};document.documentElement.dataset.gridatlasGeneration="202609051510";</script>
```

<div id="fatal-banner">CRITICAL ERROR: SYSTEM DEPENDENCIES FAILED TO LOAD</div>

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<div id="fs-curtain"><div id="fs-curtain-keys"><div class="key-group"><div class="key-title">Topology (GeoJSON)</div><label class="key-item"><input type="checkbox" data-layer-id="400"> <span id="fs-lbl-400" data-base-label="400kV" style="color: rgb(0, 84, 255); font-size: 11px; opacity: 1;">400kV [OK]</span></label><label class="key-item"><input type="checkbox" data-layer-id="275"> <span id="fs-lbl-275" data-base-label="275kV" style="color: rgb(255, 0, 0); font-size: 11px; opacity: 1;">275kV [OK]</span></label><label class="key-item"><input type="checkbox" data-layer-id="220"> <span id="fs-lbl-220" data-base-label="220kV" style="color: rgb(255, 153, 0); font-size: 11px; opacity: 1;">220kV [OK]</span></label><label class="key-item"><input type="checkbox" data-layer-id="132"> <span id="fs-lbl-132" data-base-label="132kV" style="color: rgb(0, 204, 0); font-size: 11px; opacity: 1;">132kV [OK]</span></label><label class="key-item"><input type="checkbox" data-layer-id="66"> <span id="fs-lbl-66" data-base-label="66kV" style="color: rgb(178, 0, 255); font-size: 11px; opacity: 1;">66kV [OK]</span></label><label class="key-item"><input type="checkbox" data-layer-id="11kv"> <span id="fs-lbl-11kv" data-base-label="11kV (UKPN)" style="color: rgb(255, 0, 255); font-size: 11px;">11kV (UKPN) [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="subs"> <span id="fs-lbl-subs" data-base-label="Subs" style="color: rgb(255, 255, 255); font-size: 11px; opacity: 1;">Subs [OK]</span></label></div><div class="key-group"><div class="key-title">Assets (GeoJSON)</div><label class="key-item"><input type="checkbox" data-layer-id="nuc"> <span id="fs-lbl-nuc" data-base-label="Nuclear" style="color: rgb(57, 255, 20); font-size: 11px; opacity: 1;">Nuclear [OK]</span></label><label class="key-item"><input type="checkbox" data-layer-id="gas"> <span id="fs-lbl-gas" data-base-label="Gas" style="color: rgb(255, 69, 0); font-size: 11px; opacity: 1;">Gas [OK]</span></label><label class="key-item"><input type="checkbox" data-layer-id="ind"> <span id="fs-lbl-ind" data-base-label="Industry" style="color: rgb(255, 102, 0); font-size: 11px; opacity: 1;">Industry [OK]</span></label><label class="key-item"><input type="checkbox" data-layer-id="dc"> <span id="fs-lbl-dc" data-base-label="Data Ctrs" style="color: rgb(0, 255, 255); font-size: 11px; opacity: 1;">Data Ctrs [OK]</span></label><label class="key-item"><input type="checkbox" data-layer-id="air"> <span id="fs-lbl-air" data-base-label="Airports" style="color: rgb(255, 0, 255); font-size: 11px; opacity: 1;">Airports [OK]</span></label><label class="key-item"><input type="checkbox" data-layer-id="rail"> <span id="fs-lbl-rail" data-base-label="Railways" style="color: rgb(255, 215, 0); font-size: 11px; opacity: 1;">Railways [OK]</span></label></div><div class="key-group"><div class="key-title">Supermarkets (GeoJSON)</div><label class="key-item"><input type="checkbox" data-layer-id="tesco"> <span id="fs-lbl-tesco" data-base-label="Tesco" style="color: rgb(238, 28, 46); font-size: 11px;">Tesco [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="sainsburys"> <span id="fs-lbl-sainsburys" data-base-label="Sainsbury's" style="color: rgb(255, 130, 0); font-size: 11px;">Sainsbury's [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="asda"> <span id="fs-lbl-asda" data-base-label="Asda" style="color: rgb(120, 190, 32); font-size: 11px;">Asda [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="morrisons"> <span id="fs-lbl-morrisons" data-base-label="Morrisons" style="color: rgb(255, 215, 0); font-size: 11px;">Morrisons [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="aldi"> <span id="fs-lbl-aldi" data-base-label="Aldi" style="color: rgb(0, 48, 135); font-size: 11px;">Aldi [WAIT]</span></label></div></div>
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</label><label class="key-item"><input type="checkbox" data-layer-id="lidl">
<span id="fs-lbl-lidl" data-base-label="Lidl" style="color: rgb(0, 80, 170);
font-size: 11px;">Lidl [WAIT]</span></label><label class="key-item"><input
type="checkbox" data-layer-id="waitrose"> <span id="fs-lbl-waitrose" data-base-
label="Waitrose" style="color: rgb(122, 184, 0); font-size: 11px;">Waitrose
[WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-
id="ms"> <span id="fs-lbl-ms" data-base-label="M&S Food" style="color:
rgb(0, 155, 119); font-size: 11px;">M&S Food [WAIT]</span></label><label
class="key-item"><input type="checkbox" data-layer-id="coop"> <span id="fs-lbl-
coop" data-base-label="Co-op" style="color: rgb(0, 177, 169); font-size:
11px;">Co-op [WAIT]</span></label><label class="key-item"><input type="checkbox"
data-layer-id="costco"> <span id="fs-lbl-costco" data-base-label="Costco"
style="color: rgb(0, 93, 170); font-size: 11px;">Costco [WAIT]</span></label>
<label class="key-item"><input type="checkbox" data-layer-id="booths"> <span
id="fs-lbl-booths" data-base-label="Booths" style="color: rgb(109, 32, 119);
font-size: 11px;">Booths [WAIT]</span></div><div class="key-group"><div
class="key-title">Transit (GeoJSON)</div><label class="key-item"><input
type="checkbox" data-layer-id="elizabeth"> <span id="fs-lbl-elizabeth" data-
base-label="Elizabeth Line" style="color: rgb(96, 57, 158); font-size:
11px;">Elizabeth Line [WAIT]</span></label><label class="key-item"><input
type="checkbox" data-layer-id="lu"> <span id="fs-lbl-lu" data-base-label="London
Underground" style="color: rgb(227, 32, 23); font-size: 11px;">London
Underground [WAIT]</span></label><label class="key-item"><input type="checkbox"
data-layer-id="dlr"> <span id="fs-lbl-dlr" data-base-label="DLR" style="color:
rgb(0, 175, 173); font-size: 11px;">DLR [WAIT]</span></label><label class="key-
item"><input type="checkbox" data-layer-id="metro"> <span id="fs-lbl-metro"
data-base-label="UK Metro" style="color: rgb(243, 167, 18); font-size: 11px;">UK
Metro [WAIT]</span></label><label class="key-item"><input type="checkbox" data-
layer-id="tram"> <span id="fs-lbl-tram" data-base-label="Trams & Light Rail"
style="color: rgb(132, 50, 155); font-size: 11px;">Trams & Light Rail [WAIT]
</span></label><label class="key-item"><input type="checkbox" data-layer-
id="mainline_rail"> <span id="fs-lbl-mainline_rail" data-base-label="UK Mainline
Railways" style="color: rgb(255, 136, 0); font-size: 11px;">UK Mainline Railways
[WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-
id="hs2"> <span id="fs-lbl-hs2" data-base-label="HS2" style="color: rgb(44, 111,
173); font-size: 11px;">HS2 [WAIT]</span></label><label class="key-item"><input
type="checkbox" data-layer-id="eurostar_route"> <span id="fs-lbl-eurostar_route"
data-base-label="Eurostar (HS1) UK & Europe" style="color: rgb(0, 20, 137);
font-size: 11px;">Eurostar (HS1) UK & Europe [WAIT]</span></label><label
class="key-item"><input type="checkbox" data-layer-id="eurostar_station"> <span
id="fs-lbl-eurostar_station" data-base-label="Eurostar Stations UK & Europe"
style="color: rgb(255, 215, 0); font-size: 11px;">Eurostar Stations UK &
Europe [WAIT]</span></div><div class="key-group"><div class="key-
title">Leisure & Sports (GeoJSON)</div><label class="key-item"><input
type="checkbox" data-layer-id="stadiums"> <span id="fs-lbl-stadiums" data-base-
label="Major Stadiums" style="color: rgb(229, 255, 0); font-size: 11px;">Major
Stadiums [WAIT]</span></div><div class="key-group"><div class="key-
title">Global Data Backbone</div><label class="key-item"><input type="checkbox"
data-layer-id="subsea_data_cables"> <span id="fs-lbl-subsea_data_cables" data-
base-label="Subsea Data Cables (Fibre) (not all connections available yet)"
style="color: rgb(0, 229, 255); font-size: 11px;">Subsea Data Cables (Fibre)
(not all connections available yet) [WAIT]</span></div><div class="key-
group"><div class="key-title">Global Supply Chain & Chokepoints</div><label
class="key-item"><input type="checkbox" data-layer-id="global_ports_major">
<span id="fs-lbl-global_ports_major" data-base-label="Major Ports (Deep
Water/Ferry)" style="color: rgb(0, 191, 255); font-size: 11px;">Major Ports
(Deep Water/Ferry) [WAIT]</span></label><label class="key-item"><input
type="checkbox" data-layer-id="global_ports_minor"> <span id="fs-lbl-
global_ports_minor" data-base-label="Inland / Minor Harbours" style="color:
rgb(0, 85, 170); font-size: 11px;">Inland / Minor Harbours [WAIT]</span></label>
<label class="key-item"><input type="checkbox" data-layer-
id="global_hydrocarbons"> <span id="fs-lbl-global_hydrocarbons" data-base-

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label="Global Hydrocarbons (Oil, Gas, LNG)" style="color: rgb(255, 51, 0); font-size: 11px;">Global Hydrocarbons (Oil, Gas, LNG) [WAIT]

Highway Infrastructure

☐ Motorways [WAIT]

☐ Major A-Roads (Trunk) [WAIT]

☐ A-Roads (Primary) [WAIT]

☐ Motorway Services [WAIT]

EV (GeoJSON)

☐ EV Rapid (100kW+) [WAIT]

Heavy Industry

☐ Major Industrial Sites [WAIT]

REPD (GeoJSON)

☐ Solar PV [WAIT]

☐ Solar PV (Operational Only) [WAIT]

☐ Solar Roof [WAIT]

☐ Wind [WAIT]

☐ Onshore Wind (Operational Only) [WAIT]

☐ Offshore Wind (Operational Only) [WAIT]

☐ Battery Storage [WAIT]

☐ Battery Storage (Operational Only) [WAIT]

☐ Biomass & EfW [WAIT]

☐ Tidal & Wave [WAIT]

☐ Hydrogen [WAIT]

☐ Hydro & Pumped Storage [WAIT]

☐

blob:https://ventusltd.github.io/184a2ca1-a0ae-42ea-b3ba-72ceefc889c4

05/09/2026, 16:22
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lbl-flywheel" data-base-label="Flywheel" style="color: rgb(255, 105, 180); font-size: 11px;">Flywheel [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="act"> <span id="fs-lbl-act" data-base-label="Advanced Conversion" style="color: rgb(255, 102, 0); font-size: 11px;">Advanced Conversion [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="geothermal"> <span id="fs-lbl-geothermal" data-base-label="Geothermal" style="color: rgb(255, 51, 0); font-size: 11px;">Geothermal [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="caes"> <span id="fs-lbl-caes" data-base-label="Compressed Air Storage" style="color: rgb(136, 170, 255); font-size: 11px;">Compressed Air Storage [WAIT]</span></label></div><div class="key-group"><div class="key-title">Basemap</div><label class="key-item"><input type="radio" name="bm-fs" value="dark" checked=""> Dark</label><label class="key-item"><input type="radio" name="bm-fs" value="sat"> Satellite</label></div><div class="key-group" data-pn-group="1"><div class="key-title">PIPELINE NEWS (REPD)</div><label class="key-item"><input type="checkbox" data-pn-layer="same"> <span data-pn-label="same" style="color: rgb(95, 189, 194);">Same technology [SELECT A PROJECT]</span></label><label class="key-item"><input type="checkbox" data-pn-layer="wider"> <span data-pn-label="wider" style="color: rgb(57, 255, 20);">Wider fleet [SELECT A PROJECT]</span></label><label class="key-item"><input type="checkbox" data-pn-layer="all"> <span data-pn-label="all" style="color: rgb(216, 182, 74);">All pipeline [SELECT A PROJECT]</span></label></div></div></div><div id="fs-letterhead"><div class="ventus-main">Ventus</div><div class="ventus-sub">Cables & Connectivity®</div></div><div class="dashboard">

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<div class="map-container" id="map-container"><nav id="gridatlas-menu-bar" aria-label="Atlas menu"><div class="gm-side gm-side-left"><div class="gm-menu gm-open"><button type="button" class="gm-title" id="gridatlas-menu-bar-title-0" aria-haspopup="menu" aria-expanded="true" aria-controls="gridatlas-menu-bar-panel-0">File</button><div class="gm-panel" id="gridatlas-menu-bar-panel-0" role="group" aria-labelledby="gridatlas-menu-bar-title-0"><div class="search-bar-wrapper"><div style="position: relative"><input class="search-input" id="search-input" type="text" placeholder="Search project, address, postcode or place..." autocomplete="off" aria-label="Search project, address, postcode or place"><div class="search-results" id="search-results"></div></div><button class="search-btn" id="search-btn">GO</button></div><button class="map-ctrl-btn" id="btn-export"><img alt="Download icon" data-bbox="695 650 715 665"/> Export CSV</button><div class="gm-layer-group">Export this view</div><button id="gridatlas-export-print" data-gm-export="print" type="button"><img alt="Print icon" data-bbox="755 680 775 695"/> Print · or save as PDF</button><button id="gridatlas-teleprint-pdf" type="button" data-gm-export="teleprint-pdf" data-teleprint="1"><img alt="Print icon" data-bbox="755 705 775 720"/> Print PDF · exactly this screen</button><button id="gridatlas-teleprint-source" type="button" data-gm-export="teleprint-source" data-teleprint="1"><img alt="Print icon" data-bbox="755 730 775 745"/> Print source code · for AI review</button><button id="gridatlas-export-pdf" data-gm-export="pdf" type="button">↓ Save this view as a PDF</button><button id="gridatlas-export-image" data-gm-export="image" type="button">↓ Save an image of this view</button><div class="gm-layer-group">Run the engine yourself · offline, no dependencies</div><button data-gm-engine="1" type="button" title="git clone https://github.com/VentusLtd/ventus-grid-engine & cd ventus-grid-engine & node verify.mjs"><img alt="Copy icon" data-bbox="755 830 775 845"/> Copy: clone the engine and run its 133 checks</button><div class="gm-layer-group">Engine · the maths this runs on · 11</div><a data-gm-engine="1" role="button"

```



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href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=deeplink%2Fcontract.js" target="_blank" rel="noopener"
title="buildDeepLink/parseDeepLink, LAYER_ID_FOR_BUCKET, the MAP button as one
testable thing">deeplink/contract.js</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=engine%2Fcorridor-estimate.js" target="_blank" rel="noopener"
title="PROMOTED this session: forCable(), calibrated straight-line-to-corridor
multiplier">engine/corridor-estimate.js</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=engine%2Felectrical-distance.js" target="_blank" rel="noopener"
title="PROMOTED this session: between/within, BFS hop-count over network-
topology.graph()">engine/electrical-distance.js</a><a data-gm-engine="1"
role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=engine%2Fgeo-area.js" target="_blank"
rel="noopener" title="polygonAreaKm2, polylinePerimeterKm,
circleCapAreaKm2">engine/geo-area.js</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=engine%2Fgeo-core.js" target="_blank" rel="noopener" title="the
one haversine + R_ATLAS/R_UK/R_MEAN, (lon,lat) order">engine/geo-core.js</a><a
data-gm-engine="1" role="button" href="https://ventusltd.github.io/ventus-grid-
engine/?graph=engine-graph&focus=engine%2Fgeo-geojson.js" target="_blank"
rel="noopener" title="circleFeatureCollection, GeoJSON shaping kept out of the
maths">engine/geo-geojson.js</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=engine%2Fgeo-shapes.js" target="_blank" rel="noopener"
title="destinationCirclePoints, the deduplicated circle generator">engine/geo-
shapes.js</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=engine%2Fnetwork-topology.js" target="_blank" rel="noopener"
title="PROMOTED this session: index/at/graph over one site's published
nodes/branches">engine/network-topology.js</a><a data-gm-engine="1"
role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=engine%2Frating-envelope.js" target="_blank"
rel="noopener" title="PROMOTED this session: at(), per-season lowest/highest
range, never summed">engine/rating-envelope.js</a><a data-gm-engine="1"
role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=engine%2Fv9-geodesy.js" target="_blank"
rel="noopener" title="distanceKm, destinationPoint, initialBearingDeg,
voltagesKv, representativePoint">engine/v9-geodesy.js</a><a data-gm-engine="1"
role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=engine%2Fv9-nearest-search.js" target="_blank"
rel="noopener" title="normalise + index, exhaustive scan, proven free of the
ring-search bug">engine/v9-nearest-search.js</a><div class="gm-layer-
group">Extracts · 1</div><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=sources%2Fv9-extracts%2Fsizing-arithmetic.mjs" target="_blank"
rel="noopener" title="staged verbatim, promotion DECLINED this session – impure
+ financial-domain, see population-plan.md §4">sources/v9-extracts/sizing-
arithmetic.mjs</a><div class="gm-layer-group">References · 1</div><a data-gm-
engine="1" role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=grid-distance-maths%2Fsrc%2Fgeodesy.mjs"
target="_blank" rel="noopener" title="the estate's canonical geodesy repository
– R_ATLAS/R_UK/R_MEAN identical to geo-core.js, by design">grid-distance-
maths/src/geodesy.mjs</a><div class="gm-layer-group">Copies elsewhere in the
estate · 31</div><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=atlas-pointer-deep-link.mjs%20buildAtlasV9DeepLink()"
target="_blank" rel="noopener" title="the LIVE emitter app.mjs actually imports;
hand-rolls its own URL construction">atlas-pointer-deep-link.mjs
buildAtlasV9DeepLink()</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=declared-connections.js%20nearestTransmission()" target="_blank"

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rel="noopener" title="hand-rolled running-min scan, correct by construction but
not importing the shared search">declared-connections.js nearestTransmission()
</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=grid-distance-column%2Fbuild_payload.py" target="_blank"
rel="noopener" title="the POSITIVE control: imports grid-distance-
maths/src/geodesy.py directly and refuses to run without it">grid-distance-
column/build_payload.py</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=grid-proximity%2Fbuild_payload.py%20haversine_km()"
target="_blank" rel="noopener" title="hand-rolled A WGS84/R ATLAS +
haversine_km, does not import grid-distance-maths despite a sibling cartridge
doing so">grid-proximity/build_payload.py haversine_km()</a><a data-gm-
engine="1" role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=grid-
proximity%2Fbuild_payload.py%20nearest_segment()%2Fnearest_substations()"
target="_blank" rel="noopener" title="the estate's REFERENCE ring-search: cell-
bucketed with a proven early-termination bound; grid-distance-maths credits it
as origin">grid-proximity/build_payload.py
nearest_segment()/nearest_substations()</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=gridatlas%20202609011950-geodesy.js" target="_blank"
rel="noopener" title="live, composed module; source that engine/v9-geodesy.js
was extracted from, byte-identical">gridatlas 202609011950-geodesy.js</a><a
data-gm-engine="1" role="button" href="https://ventusltd.github.io/ventus-grid-
engine/?graph=engine-graph&focus=gridatlas%20202609011950-substation-
lookup.js" target="_blank" rel="noopener" title="committed but DEAD – zero
manifests reference it; source engine/v9-nearest-search.js was extracted
from">gridatlas 202609011950-substation-lookup.js</a><a data-gm-engine="1"
role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=gridatlas%20202609012205-sizing-arithmetic.js"
target="_blank" rel="noopener" title="the CORRECTED port:
Math.min(inverter_ac_total, skid_ac_total) – the fix none of the five above
has">gridatlas 202609012205-sizing-arithmetic.js</a><a data-gm-engine="1"
role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=gridatlas%20202609012245-electrical-distance.js"
target="_blank" rel="noopener" title="live, composed module – the direct source
engine/electrical-distance.js was promoted from">gridatlas 202609012245-
electrical-distance.js</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=gridatlas%20202609012245-network-topology.js" target="_blank"
rel="noopener" title="live, composed module – the direct source engine/network-
topology.js was promoted from">gridatlas 202609012245-network-topology.js</a><a
data-gm-engine="1" role="button" href="https://ventusltd.github.io/ventus-grid-
engine/?graph=engine-graph&focus=gridatlas%20202609012250-rating-
envelope.js" target="_blank" rel="noopener" title="live, composed module – the
direct source engine/rating-envelope.js was promoted from">gridatlas
202609012250-rating-envelope.js</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=gridatlas%20202609030205-corridor-estimate.js" target="_blank"
rel="noopener" title="live, composed module – the direct source engine/corridor-
estimate.js was promoted from">gridatlas 202609030205-corridor-estimate.js</a><a
data-gm-engine="1" role="button" href="https://ventusltd.github.io/ventus-grid-
engine/?graph=engine-graph&focus=neon-substation-links-v9-
6.js%20R_ATLAS%2FdistanceKm" target="_blank" rel="noopener" title="hand-rolled
constant and distance function, own comment: "the house constant">neon-
substation-links-v9-6.js R_ATLAS/distanceKm</a><a data-gm-engine="1"
role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=place-global-search-arrival-
identity.js%20receiveExactRepdDeepLink()" target="_blank" rel="noopener"
title="the LIVE receiver-side parse – its identity regex is BROADER than the
emitter's, a live pattern drift">place-global-search-arrival-identity.js

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receiveExactRepdDeepLink()</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=sld-sandbox-technology-buckets.js%20LAYER_ID_FOR_BUCKET"
target="_blank" rel="noopener" title="the LIVE source table deeplink/contract.js
copied verbatim from; the v9.109 100%-failure fix">sld-sandbox-technology-
buckets.js LAYER_ID_FOR_BUCKET</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=solar-bess-topology-v5%2Fgis-sld-v5-calculations.js"
target="_blank" rel="noopener" title="ac_mw_direct double-count bug, line 147,
LIVE and unfixed">solar-bess-topology-v5/gis-sld-v5-calculations.js</a><a data-
gm-engine="1" role="button" href="https://ventusltd.github.io/ventus-grid-
engine/?graph=engine-graph&focus=solar-bess-topology-v6%2F...%2Fgis-sld-v5-
calculations.js" target="_blank" rel="noopener" title="the same bug, same line
number, byte-identical">solar-bess-topology-v6/.../gis-sld-v5-
calculations.js</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=solar-bess-topology-v7%2F...%2Fgis-sld-v5-calculations.js"
target="_blank" rel="noopener" title="the same bug, same line number, byte-
identical">solar-bess-topology-v7/.../gis-sld-v5-calculations.js</a><a data-gm-
engine="1" role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=solar-bess-topology-v8%2Fbess-gis-sld-financial-
sandbox%2Fgis-sld-v5-calculations.js" target="_blank" rel="noopener" title="the
same bug, same line number, byte-identical">solar-bess-topology-v8/bess-gis-sld-
financial-sandbox/gis-sld-v5-calculations.js</a><a data-gm-engine="1"
role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=solar-bess-topology-v8%2Fbess-pcs-standalone%2Fgis-
sld-v5-calculations.js" target="_blank" rel="noopener" title="the same bug, same
line number, byte-identical – fifth copy">solar-bess-topology-v8/bess-pcs-
standalone/gis-sld-v5-calculations.js</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=substation-intelligence%20cartridge%2C%20geodesy%20copy%20%231"
target="_blank" rel="noopener" title="inline legacy V8 haversine at line 57 of a
6,277-line composed cartridge">substation-intelligence cartridge, geodesy copy
#1</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=substation-intelligence%20cartridge%2C%20geodesy%20copy%20%232"
target="_blank" rel="noopener" title="a SECOND inline NS.geodesy IIFE at line
1544 of the SAME file as copy #1">substation-intelligence cartridge, geodesy
copy #2</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=substation-intelligence%20cartridge%2C%20inline%20nearest()"
target="_blank" rel="noopener" title="live duplicate, LESS defensive than the
dead module – can throw on a point with no voltages_kv">substation-intelligence
cartridge, inline nearest()</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=ventus-corev8engine-exact-repd-delegation.js" target="_blank"
rel="noopener" title="the engine's OWN former deep-link handler, now inert:
publishes DEFERRED_TO_EXACT_REPD_RECEIVER and returns">ventus-corev8engine-
exact-repd-delegation.js</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=ventus-corev8engine.js%20_zoneDrawCalcArea%20(Impl%20A)"
target="_blank" rel="noopener" title="spherical-excess polygon area, Zone Draw
tool">ventus-corev8engine.js _zoneDrawCalcArea (Impl A)</a><a data-gm-engine="1"
role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=ventus-corev8engine.js%20_zoneDrawCirclePoints"
target="_blank" rel="noopener" title="destination-point circle generator, Zone
Draw (n=24)">ventus-corev8engine.js _zoneDrawCirclePoints</a><a data-gm-
engine="1" role="button" href="https://ventusltd.github.io/ventus-grid-engine/?
graph=engine-graph&focus=ventus-corev8engine.js%20createGeoJSONCircle"
target="_blank" rel="noopener" title="byte-identical destination-point formula,
Radius tool (n=64/96/128), plus FeatureCollection wrapping">ventus-
corev8engine.js createGeoJSONCircle</a><a data-gm-engine="1" role="button"

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href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=ventus-corev8engine.js%20doRadiusAreaMeasure%20(Impl%20C)"
target="_blank" rel="noopener" title="spherical-cap area – a different question,
correctly kept separate">ventus-corev8engine.js doRadiusAreaMeasure (Impl C)</a>
<a data-gm-engine="1" role="button" href="https://ventusltd.github.io/ventus-
grid-engine/?graph=engine-graph&focus=ventus-corev8engine.js%20haversine()"
target="_blank" rel="noopener" title="the one distance primitive in the V8
monolith, correctly reused everywhere – not duplicated">ventus-corev8engine.js
haversine()</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=ventus-corev8engine.js%20updateMeasureDisplay%20(Impl%20B)"
target="_blank" rel="noopener" title="the identical shoelace-on-sphere formula,
copy-pasted into the Measure tool">ventus-corev8engine.js updateMeasureDisplay
(Impl B)</a><a data-gm-engine="1" role="button"
href="https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph&focus=wider-fleet.mjs%20atlasLink()" target="_blank" rel="noopener"
title="a SECOND, independent emitter for ~20 non-spine technology types; its own
URLSearchParams, own code path">wider-fleet.mjs atlasLink()</a><div
id="gridatlas-teleprint-status" role="status" aria-live="polite" style="padding:
6px 11px; font-style: normal; font-variant-caps: normal; font-weight: normal;
font-width: normal; font-size: 10px; line-height: 1.4; font-family: ui-
monospace, SFMono-Regular, Menlo, monospace; font-size-adjust: none; font-
kerning: auto; font-variant-alternates: normal; font-variant-ligatures: normal;
font-variant-numeric: normal; font-variant-east-asian: normal; font-variant-
position: normal; font-variant-emoji: normal; font-feature-settings: normal;
font-optical-sizing: auto; font-variation-settings: normal; color: rgb(159, 214,
228); opacity: 0.9; white-space: normal; max-width: 34ch;">Print PDF saves this
screen. Print source code saves the code behind it.</div></div><div><div
class="gm-menu"><button type="button" class="gm-title" id="gridatlas-menu-bar-
title-1" aria-haspopup="menu" aria-expanded="false" aria-controls="gridatlas-
menu-bar-panel-1">Edit</button><div class="gm-panel" id="gridatlas-menu-bar-
panel-1" hidden="" role="group" aria-labelledby="gridatlas-menu-bar-title-1">
<button class="map-ctrl-btn" id="btn-status">● Status Colours</button></div>
</div><div class="gm-menu"><button type="button" class="gm-title" id="gridatlas-
menu-bar-title-2" aria-haspopup="menu" aria-expanded="false" aria-
controls="gridatlas-menu-bar-panel-2">View</button><div class="gm-panel"
id="gridatlas-menu-bar-panel-2" hidden="" role="group" aria-
labelledby="gridatlas-menu-bar-title-2"><button id="btn-fullscreen"
onclick="enterFullscreen()" title="Fullscreen">[]</button><div id="gridatlas-gb-
conditions" data-open="0"><button type="button" aria-expanded="false">GB prices
· historic ▶</button><div class="gb-body"></div></div><button id="btn-
fullscreen-exit" onclick="exitFullscreen()" aria-label="Exit full screen">×
Exit</button><div class="gm-layer-group">Studies</div><a data-gm-study="1"
role="button"
href="https://globalgrid2050.com/data/grid_studies_public/great_britain_electric
ity_price_grid_constraint_trends_2016_2026.html" target="_blank"
rel="noopener">GB electricity price & grid constraint trends · 2016–2026</a>
</div></div></div><div class="gm-brand-slot"><div class="hud-header">
<div>
<small style="color:#888">SYSTEM TIME</small><br>
<span class="hud-val" id="clock">16:21:39</span><br>
<span id="date" style="font-size: 11px; color:
#888;">05/09/2026</span>
</div>
<div class="ventus-brand">
<div class="ventus-main">Ventus</div>
<div class="ventus-sub">Cables & Connectivity®</div>
</div>
<div style="text-align:right"><small style="color:#888">2050
TARGET</small><br><span class="hud-val" id="days" style="color:#ff9d00">8518
DAYS</span></div>
</div></div><div class="gm-side gm-side-right"><div class="gm-menu"><button

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type="button" class="gm-title" id="gridatlas-menu-bar-title-3" aria-
haspopup="menu" aria-expanded="false" aria-controls="gridatlas-menu-bar-panel-
3">Scope</button><div class="gm-panel" id="gridatlas-menu-bar-panel-3" hidden=""
role="group" aria-labelledby="gridatlas-menu-bar-title-3"><button class="map-
ctrl-btn" id="btn-radius">⊗ Radius Search</button><button class="map-ctrl-btn"
id="btn-radius-area">⊗ Radius Area</button><button class="map-ctrl-btn" id="btn-
zonedraw">⊙ Poly Zone</button><button class="map-ctrl-btn" id="btn-measure">
Measure</button><button type="button" aria-pressed="false">⊗ Scope</button>
<button type="button">× Clear</button><button class="map-ctrl-btn" id="btn-
gridpoint" type="button" aria-pressed="false" title="Click anywhere on the map
for the published network at the nearest mapped connection point">◆ Grid At
Point</button></div></div><div class="gm-menu"><button type="button" class="gm-
title" id="gridatlas-menu-bar-title-4" aria-haspopup="menu" aria-
expanded="false" aria-controls="gridatlas-menu-bar-panel-4">Grid</button><div
class="gm-panel gm-panel-grid" id="gridatlas-menu-bar-panel-4" hidden=""
role="group" aria-labelledby="gridatlas-menu-bar-title-4"><div class="gm-panel-
head"><div class="scada-brand">
  <div>
    <div class="scada-brand-main">Ventus</div>
    <div class="scada-brand-sub">Cables & ConnectivityⓈ</div>
  </div>
  <div class="scada-brand-tag">globalgrid2050.com | open
infrastructure intelligence</div>
</div><div class="status-legend">
  <div class="status-dot"><span style="background:#00ff88">
</span>Operational</div>
  <div class="status-dot"><span style="background:#ffcc00">
</span>Under Construction</div>
  <div class="status-dot"><span style="background:#ff8800">
</span>Consented</div>
  <div class="status-dot"><span style="background:#8888ff">
</span>Applied</div>
</div></div><div class="gm-panel-body"><div class="gm-layer-
group">Topology (GeoJSON)</div><label class="gm-layer" data-gridatlas-layer-
key="engine:400"><input type="checkbox" data-gridatlas-layer-proxy="engine:400"
aria-label="400kV [OK]"><span class="gm-layer-box" aria-hidden="true"></span>
<span class="gm-layer-name">400kV [OK]</span></label><label class="gm-layer"
data-gridatlas-layer-key="engine:275"><input type="checkbox" data-gridatlas-
layer-proxy="engine:275" aria-label="275kV [OK]"><span class="gm-layer-box"
aria-hidden="true"></span><span class="gm-layer-name">275kV [OK]</span></label>
<label class="gm-layer" data-gridatlas-layer-key="engine:220"><input
type="checkbox" data-gridatlas-layer-proxy="engine:220" aria-label="220kV [OK]">
<span class="gm-layer-box" aria-hidden="true"></span><span class="gm-layer-
name">220kV [OK]</span></label><label class="gm-layer" data-gridatlas-layer-
key="engine:132"><input type="checkbox" data-gridatlas-layer-proxy="engine:132"
aria-label="132kV [OK]"><span class="gm-layer-box" aria-hidden="true"></span>
<span class="gm-layer-name">132kV [OK]</span></label><label class="gm-layer"
data-gridatlas-layer-key="engine:66"><input type="checkbox" data-gridatlas-
layer-proxy="engine:66" aria-label="66kV [OK]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">66kV [OK]</span></label><label
class="gm-layer" data-gridatlas-layer-key="engine:11kv"><input type="checkbox"
data-gridatlas-layer-proxy="engine:11kv" aria-label="11kV (UKPN) [WAIT]"><span
class="gm-layer-box" aria-hidden="true"></span><span class="gm-layer-name">11kV
(UKPN) [WAIT]</span></label><label class="gm-layer" data-gridatlas-layer-
key="engine:subs"><input type="checkbox" data-gridatlas-layer-
proxy="engine:subs" aria-label="Subs [OK]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">Subs [OK]</span></label><div
class="gm-layer-group">Assets (GeoJSON)</div><label class="gm-layer" data-
gridatlas-layer-key="engine:nuc"><input type="checkbox" data-gridatlas-layer-
proxy="engine:nuc" aria-label="Nuclear [OK]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">Nuclear [OK]</span></label>

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<label class="gm-layer" data-gridatlas-layer-key="engine:gas"><input
type="checkbox" data-gridatlas-layer-proxy="engine:gas" aria-label="Gas [OK]">
<span class="gm-layer-box" aria-hidden="true"></span><span class="gm-layer-
name">Gas [OK]</span></label><label class="gm-layer" data-gridatlas-layer-
key="engine:ind"><input type="checkbox" data-gridatlas-layer-proxy="engine:ind"
aria-label="Industry [OK]"><span class="gm-layer-box" aria-hidden="true"></span>
<span class="gm-layer-name">Industry [OK]</span></label><label class="gm-layer"
data-gridatlas-layer-key="engine:dc"><input type="checkbox" data-gridatlas-
layer-proxy="engine:dc" aria-label="Data Ctrs [OK]"><span class="gm-layer-box"
aria-hidden="true"></span><span class="gm-layer-name">Data Ctrs [OK]</span>
</label><label class="gm-layer" data-gridatlas-layer-key="engine:air"><input
type="checkbox" data-gridatlas-layer-proxy="engine:air" aria-label="Airports
[OK]"><span class="gm-layer-box" aria-hidden="true"></span><span class="gm-
layer-name">Airports [OK]</span></label><label class="gm-layer" data-gridatlas-
layer-key="engine:rail"><input type="checkbox" data-gridatlas-layer-
proxy="engine:rail" aria-label="Railways [OK]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">Railways [OK]</span></label>
<div class="gm-layer-group">Supermarkets (GeoJSON)</div><label class="gm-layer"
data-gridatlas-layer-key="engine:tesco"><input type="checkbox" data-gridatlas-
layer-proxy="engine:tesco" aria-label="Tesco [WAIT]"><span class="gm-layer-box"
aria-hidden="true"></span><span class="gm-layer-name">Tesco [WAIT]</span>
</label><label class="gm-layer" data-gridatlas-layer-key="engine:sainsburys">
<input type="checkbox" data-gridatlas-layer-proxy="engine:sainsburys" aria-
label="Sainsbury's [WAIT]"><span class="gm-layer-box" aria-hidden="true"></span>
<span class="gm-layer-name">Sainsbury's [WAIT]</span></label><label class="gm-
layer" data-gridatlas-layer-key="engine:asda"><input type="checkbox" data-
gridatlas-layer-proxy="engine:asda" aria-label="Asda [WAIT]"><span class="gm-
layer-box" aria-hidden="true"></span><span class="gm-layer-name">Asda [WAIT]
</span></label><label class="gm-layer" data-gridatlas-layer-
key="engine:morrisons"><input type="checkbox" data-gridatlas-layer-
proxy="engine:morrisons" aria-label="Morrisons [WAIT]"><span class="gm-layer-
box" aria-hidden="true"></span><span class="gm-layer-name">Morrisons [WAIT]
</span></label><label class="gm-layer" data-gridatlas-layer-key="engine:aldi">
<input type="checkbox" data-gridatlas-layer-proxy="engine:aldi" aria-label="Aldi
[WAIT]"><span class="gm-layer-box" aria-hidden="true"></span><span class="gm-
layer-name">Aldi [WAIT]</span></label><label class="gm-layer" data-gridatlas-
layer-key="engine:lidl"><input type="checkbox" data-gridatlas-layer-
proxy="engine:lidl" aria-label="Lidl [WAIT]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">Lidl [WAIT]</span></label>
<label class="gm-layer" data-gridatlas-layer-key="engine:waitrose"><input
type="checkbox" data-gridatlas-layer-proxy="engine:waitrose" aria-
label="Waitrose [WAIT]"><span class="gm-layer-box" aria-hidden="true"></span>
<span class="gm-layer-name">Waitrose [WAIT]</span></label><label class="gm-
layer" data-gridatlas-layer-key="engine:ms"><input type="checkbox" data-
gridatlas-layer-proxy="engine:ms" aria-label="M&S Food [WAIT]"><span
class="gm-layer-box" aria-hidden="true"></span><span class="gm-layer-
name">M&S Food [WAIT]</span></label><label class="gm-layer" data-gridatlas-
layer-key="engine:coop"><input type="checkbox" data-gridatlas-layer-
proxy="engine:coop" aria-label="Co-op [WAIT]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">Co-op [WAIT]</span></label>
<label class="gm-layer" data-gridatlas-layer-key="engine:costco"><input
type="checkbox" data-gridatlas-layer-proxy="engine:costco" aria-label="Costco
[WAIT]"><span class="gm-layer-box" aria-hidden="true"></span><span class="gm-
layer-name">Costco [WAIT]</span></label><label class="gm-layer" data-gridatlas-
layer-key="engine:booths"><input type="checkbox" data-gridatlas-layer-
proxy="engine:booths" aria-label="Booths [WAIT]"><span class="gm-layer-box"
aria-hidden="true"></span><span class="gm-layer-name">Booths [WAIT]</span>
</label><div class="gm-layer-group">Transit (GeoJSON)</div><label class="gm-
layer" data-gridatlas-layer-key="engine:elizabeth"><input type="checkbox" data-
gridatlas-layer-proxy="engine:elizabeth" aria-label="Elizabeth Line [WAIT]">
<span class="gm-layer-box" aria-hidden="true"></span><span class="gm-layer-
name">Elizabeth Line [WAIT]</span></label><label class="gm-layer" data-

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gridatlas-layer-key="engine:lu"><input type="checkbox" data-gridatlas-layer-
proxy="engine:lu" aria-label="London Underground [WAIT]"><span class="gm-layer-
box" aria-hidden="true"></span><span class="gm-layer-name">London Underground
[WAIT]</span></label><label class="gm-layer" data-gridatlas-layer-
key="engine:dlr"><input type="checkbox" data-gridatlas-layer-proxy="engine:dlr"
aria-label="DLR [WAIT]"><span class="gm-layer-box" aria-hidden="true"></span>
<span class="gm-layer-name">DLR [WAIT]</span></label><label class="gm-layer"
data-gridatlas-layer-key="engine:metro"><input type="checkbox" data-gridatlas-
layer-proxy="engine:metro" aria-label="UK Metro [WAIT]"><span class="gm-layer-
box" aria-hidden="true"></span><span class="gm-layer-name">UK Metro [WAIT]
</span></label><label class="gm-layer" data-gridatlas-layer-key="engine:tram">
<input type="checkbox" data-gridatlas-layer-proxy="engine:tram" aria-
label="Trams & Light Rail [WAIT]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">Trams & Light Rail [WAIT]
</span></label><label class="gm-layer" data-gridatlas-layer-
key="engine:mainline_rail"><input type="checkbox" data-gridatlas-layer-
proxy="engine:mainline_rail" aria-label="UK Mainline Railways [WAIT]"><span
class="gm-layer-box" aria-hidden="true"></span><span class="gm-layer-name">UK
Mainline Railways [WAIT]</span></label><label class="gm-layer" data-gridatlas-
layer-key="engine:hs2"><input type="checkbox" data-gridatlas-layer-
proxy="engine:hs2" aria-label="HS2 [WAIT]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">HS2 [WAIT]</span></label><label
class="gm-layer" data-gridatlas-layer-key="engine:eurostar_route"><input
type="checkbox" data-gridatlas-layer-proxy="engine:eurostar_route" aria-
label="Eurostar (HS1) UK & Europe [WAIT]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">Eurostar (HS1) UK & Europe
[WAIT]</span></label><label class="gm-layer" data-gridatlas-layer-
key="engine:eurostar_station"><input type="checkbox" data-gridatlas-layer-
proxy="engine:eurostar_station" aria-label="Eurostar Stations UK & Europe
[WAIT]"><span class="gm-layer-box" aria-hidden="true"></span><span class="gm-
layer-name">Eurostar Stations UK & Europe [WAIT]</span></label><div
class="gm-layer-group">Leisure & Sports (GeoJSON)</div><label class="gm-
layer" data-gridatlas-layer-key="engine:stadiums"><input type="checkbox" data-
gridatlas-layer-proxy="engine:stadiums" aria-label="Major Stadiums [WAIT]"><span
class="gm-layer-box" aria-hidden="true"></span><span class="gm-layer-name">Major
Stadiums [WAIT]</span></label><div class="gm-layer-group">Global Data
Backbone</div><label class="gm-layer" data-gridatlas-layer-
key="engine:subsea_data_cables"><input type="checkbox" data-gridatlas-layer-
proxy="engine:subsea_data_cables" aria-label="Subsea Data Cables (Fibre) (not
all connections available yet) [WAIT]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">Subsea Data Cables (Fibre) (not
all connections available yet) [WAIT]</span></label><div class="gm-layer-
group">Global Supply Chain & Chokepoints</div><label class="gm-layer" data-
gridatlas-layer-key="engine:global_ports_major"><input type="checkbox" data-
gridatlas-layer-proxy="engine:global_ports_major" aria-label="Major Ports (Deep
Water/Ferry) [WAIT]"><span class="gm-layer-box" aria-hidden="true"></span><span
class="gm-layer-name">Major Ports (Deep Water/Ferry) [WAIT]</span></label><label
class="gm-layer" data-gridatlas-layer-key="engine:global_ports_minor"><input
type="checkbox" data-gridatlas-layer-proxy="engine:global_ports_minor" aria-
label="Inland / Minor Harbours [WAIT]"><span class="gm-layer-box" aria-
hidden="true"></span><span class="gm-layer-name">Inland / Minor Harbours [WAIT]
</span></label><label class="gm-layer" data-gridatlas-layer-
key="engine:global_hydrocarbons"><input type="checkbox" data-gridatlas-layer-
proxy="engine:global_hydrocarbons" aria-label="Global Hydrocarbons (Oil, Gas,
LNG) [WAIT]"><span class="gm-layer-box" aria-hidden="true"></span><span
class="gm-layer-name">Global Hydrocarbons (Oil, Gas, LNG) [WAIT]</span></label>
<div class="gm-layer-group">Highway Infrastructure</div><label class="gm-layer"
data-gridatlas-layer-key="engine:motorways"><input type="checkbox" data-
gridatlas-layer-proxy="engine:motorways" aria-label="Motorways [WAIT]"><span
class="gm-layer-box" aria-hidden="true"></span><span class="gm-layer-
name">Motorways [WAIT]</span></label><label class="gm-layer" data-gridatlas-
layer-key="engine:trunk_roads"><input type="checkbox" data-gridatlas-layer-

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proxy="engine:trunk_roads" aria-label="Major A-Roads (Trunk) [WAIT]">Major A-Roads (Trunk) [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:primary_roads"><input type="checkbox" data-gridatlas-layer-proxy="engine:primary_roads" aria-label="A-Roads (Primary) [WAIT]">A-Roads (Primary) [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:motorway_services"><input type="checkbox" data-gridatlas-layer-proxy="engine:motorway_services" aria-label="Motorway Services [WAIT]">Motorway Services [WAIT]</label><div class="gm-layer-group">EV (GeoJSON)</div><label class="gm-layer" data-gridatlas-layer-key="engine:ev"><input type="checkbox" data-gridatlas-layer-proxy="engine:ev" aria-label="EV Rapid (100kW+) [WAIT]">EV Rapid (100kW+) [WAIT]</label><div class="gm-layer-group">Heavy Industry</div><label class="gm-layer" data-gridatlas-layer-key="engine:naei_co2"><input type="checkbox" data-gridatlas-layer-proxy="engine:naei_co2" aria-label="Major Industrial Sites [WAIT]">Major Industrial Sites [WAIT]</label><div class="gm-layer-group">REPD (GeoJSON)</div><label class="gm-layer" data-gridatlas-layer-key="engine:solar"><input type="checkbox" data-gridatlas-layer-proxy="engine:solar" aria-label="Solar PV [WAIT]">Solar PV [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:solar_operational"><input type="checkbox" data-gridatlas-layer-proxy="engine:solar_operational" aria-label="Solar PV (Operational Only) [WAIT]">Solar PV (Operational Only) [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:solar_roof"><input type="checkbox" data-gridatlas-layer-proxy="engine:solar_roof" aria-label="Solar Roof [WAIT]">Solar Roof [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:wind"><input type="checkbox" data-gridatlas-layer-proxy="engine:wind" aria-label="Wind [WAIT]">Wind [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:wind_onshore_operational"><input type="checkbox" data-gridatlas-layer-proxy="engine:wind_onshore_operational" aria-label="Onshore Wind (Operational Only) [WAIT]">Onshore Wind (Operational Only) [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:wind_offshore_operational"><input type="checkbox" data-gridatlas-layer-proxy="engine:wind_offshore_operational" aria-label="Offshore Wind (Operational Only) [WAIT]">Offshore Wind (Operational Only) [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:bess"><input type="checkbox" data-gridatlas-layer-proxy="engine:bess" aria-label="Battery Storage [WAIT]">Battery Storage [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:bess_operational"><input type="checkbox" data-gridatlas-layer-proxy="engine:bess_operational" aria-label="Battery Storage (Operational Only) [WAIT]">Battery Storage (Operational Only) [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:biomass"><input type="checkbox" data-gridatlas-layer-proxy="engine:biomass" aria-label="Biomass & EfW [WAIT]">Biomass & EfW [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:tidal"><input type="checkbox" data-gridatlas-layer-proxy="engine:tidal" aria-label="Tidal & Wave [WAIT]">Tidal & Wave [WAIT]</label><label class="gm-layer" data-gridatlas-layer-key="engine:hydrogen"><input type="checkbox" data-gridatlas-layer-


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proxy="engine:hydrogen" aria-label="Hydrogen [WAIT]"><span class="gm-layer-box"
aria-hidden="true"></span><span class="gm-layer-name">Hydrogen [WAIT]</span>
</label><label class="gm-layer" data-gridatlas-layer-key="engine:hydro"><input
type="checkbox" data-gridatlas-layer-proxy="engine:hydro" aria-label="Hydro
& Pumped Storage [WAIT]"><span class="gm-layer-box" aria-hidden="true">
</span><span class="gm-layer-name">Hydro & Pumped Storage [WAIT]</span>
</label><label class="gm-layer" data-gridatlas-layer-key="engine:flywheel">
<input type="checkbox" data-gridatlas-layer-proxy="engine:flywheel" aria-
label="Flywheel [WAIT]"><span class="gm-layer-box" aria-hidden="true"></span>
<span class="gm-layer-name">Flywheel [WAIT]</span></label><label class="gm-
layer" data-gridatlas-layer-key="engine:act"><input type="checkbox" data-
gridatlas-layer-proxy="engine:act" aria-label="Advanced Conversion [WAIT]"><span
class="gm-layer-box" aria-hidden="true"></span><span class="gm-layer-
name">Advanced Conversion [WAIT]</span></label><label class="gm-layer" data-
gridatlas-layer-key="engine:geothermal"><input type="checkbox" data-gridatlas-
layer-proxy="engine:geothermal" aria-label="Geothermal [WAIT]"><span class="gm-
layer-box" aria-hidden="true"></span><span class="gm-layer-name">Geothermal
[WAIT]</span></label><label class="gm-layer" data-gridatlas-layer-
key="engine:caes"><input type="checkbox" data-gridatlas-layer-
proxy="engine:caes" aria-label="Compressed Air Storage [WAIT]"><span class="gm-
layer-box" aria-hidden="true"></span><span class="gm-layer-name">Compressed Air
Storage [WAIT]</span></label><div class="gm-layer-group">PIPELINE NEWS (REPD)
</div><label class="gm-layer" data-gridatlas-layer-key="pipeline:same"><input
type="checkbox" data-gridatlas-layer-proxy="pipeline:same" aria-label="Same
technology [SELECT A PROJECT]"><span class="gm-layer-box" aria-hidden="true">
</span><span class="gm-layer-name">Same technology [SELECT A PROJECT]</span>
</label><label class="gm-layer" data-gridatlas-layer-key="pipeline:wider"><input
type="checkbox" data-gridatlas-layer-proxy="pipeline:wider" aria-label="Wider
fleet [SELECT A PROJECT]"><span class="gm-layer-box" aria-hidden="true"></span>
<span class="gm-layer-name">Wider fleet [SELECT A PROJECT]</span></label><label
class="gm-layer" data-gridatlas-layer-key="pipeline:all"><input type="checkbox"
data-gridatlas-layer-proxy="pipeline:all" aria-label="All pipeline [SELECT A
PROJECT]"><span class="gm-layer-box" aria-hidden="true"></span><span class="gm-
layer-name">All pipeline [SELECT A PROJECT]</span></label><div class="gm-layer-
group">Basemap</div><label class="gm-layer"><input type="radio" name="gridatlas-
menu-basemap" value="dark"><span class="gm-layer-box" aria-hidden="true"></span>
<span class="gm-layer-name">Dark</span></label><label class="gm-layer"><input
type="radio" name="gridatlas-menu-basemap" value="sat"><span class="gm-layer-
box" aria-hidden="true"></span><span class="gm-layer-name">Satellite</span>
</label></div><div id="fs-curtain-tab" role="button" tabindex="0" aria-
label="Open the fullscreen layer curtain">⬇ Layers</div></div></div><div
class="gm-menu"><button type="button" class="gm-title" id="gridatlas-menu-bar-
title-5" aria-haspopup="menu" aria-expanded="false" aria-controls="gridatlas-
menu-bar-panel-5">About</button><div class="gm-panel" id="gridatlas-menu-bar-
panel-5" hidden="" role="group" aria-labelledby="gridatlas-menu-bar-title-5">
<div id="gridatlas-version-ledger" data-open="0"><button type="button" aria-
expanded="false">Versions · v9.130 ▶</button><div class="vl-body"><div
class="vl-row"><div class="vl-head"><span class="vl-ver">v9.130</span><span
class="vl-when">2026-09-05 15:10 UTC</span></div><div class="vl-scope">Print
source code keeps CODE in full and marks DATA truncated, so the teleprint is
small enough to attach to a chat on a phone.</div></div><div class="vl-row"><div
class="vl-head"><span class="vl-ver">v9.129</span><span class="vl-when">2026-09-
05 15:03 UTC</span></div><div class="vl-scope">Two separate Teleprinter commands
in the File menu: Print PDF (a picture of this screen, 1:1, our own writer) and
Print source code (the entire source and its browser dependencies, for AI
review).</div></div><div class="vl-row"><div class="vl-head"><span class="vl-
ver">v9.128</span><span class="vl-when">2026-09-05 14:55 UTC</span></div><div
class="vl-scope">File &gt; Print emits a screen grab of the reader's view:
nothing hidden, layout pinned to the viewport, the WebGL canvas substituted by
its own captured pixels in its own box, provenance strip moved below the view.
</div></div><div class="vl-row"><div class="vl-head"><span class="vl-
ver">v9.127</span><span class="vl-when">2026-09-05 13:40 UTC</span></div><div

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class="vl-scope">Print fills the sheet edge to edge with nothing on it but the map and its credit, and a new File control writes the PDF in the page so no print dialog or printer driver can lose it.</div></div><div class="vl-row"><div class="vl-head">v9.1212026-09-05 03:54 UTC</div><div class="vl-scope">File gains an Export group: print the view as a slide, or save it as an image. The print sizes to whatever page the reader chose, portrait or landscape, on a phone or a desktop. The image capture happens inside a render frame and is sampled for non-transparent pixels before it is offered. Both paths carry the attribution.</div></div><div class="vl-row"><div class="vl-head">v9.1202026-09-05 03:01 UTC</div><div class="vl-scope">View gains a Studies group carrying the GB electricity price and grid constraint series 2016-2026. No other panel, control or layer changed.</div></div><div class="vl-row"><div class="vl-head">v9.1192026-09-05 02:49 UTC</div><div class="vl-scope">The copyable engine command drops its install step, because the engine has no dependencies and its gate opens no socket. No other change.</div></div><div class="vl-row"><div class="vl-head">v9.1182026-09-05 02:44 UTC</div><div class="vl-scope">File: list every node the engine graph publishes, grouped canonical / extract / reference / fragment, not only the canonical eleven; add one copyable command that clones the engine and runs its own gate. No other panel, control or layer changed.</div></div><div class="vl-row"><div class="vl-head">v9.1172026-09-05 02:38 UTC</div><div class="vl-scope">About: the map attribution moves in, last and in small print, and an Estate group of three links sits above it. File: the engine's canonical modules are listed, fetched once from the engine's own published graph, each linking into it focused on itself. No other panel, control or layer changed.</div></div><div class="vl-row"><div class="vl-head">v9.1162026-09-04 21:23 UTC</div><div class="vl-scope">Bring the v8 layers panel back beneath the menu bar, open on a desktop and collapsed with its toggle visible on a phone, while keeping every dropdown - FILE, EDIT, VIEW, SCOPE, GRID and ABOUT - exactly as it is.</div></div><div class="vl-row"><div class="vl-head">v9.1152026-09-04 19:57 UTC</div><div class="vl-scope">Bring the arrival cartridge and the cartridge carrying the version ledger onto one generation, so every proof, the ledger's newest entry and the composition a reader is looking at all name the same thing.</div></div><div class="vl-row"><div class="vl-head">v9.1142026-09-04 19:56 UTC</div><div class="vl-scope">Bring the on-page version ledger up to the composition a reader is actually looking at, so its newest entry names this generation rather than an earlier one.</div></div><div class="vl-row"><div class="vl-head">v9.1122026-09-04 19:45 UTC</div><div class="vl-scope">Compose the written-but-never-composed iOS Safari visibility fix so a deep link opened in a background tab is not spent before anyone can see it; share one DuckDB runtime across the two cartridges that each built their own; and keep the GRID and SUBS chips on the map at phone widths instead of inside a menu.</div></div><div class="vl-row"><div class="vl-head">v9.1112026-09-04 13:30 UTC</div><div class="vl-scope">Hide the v8 fullscreen letterhead whenever the menu bar hosts the fused VENTUS masthead, so a phone arrival shows one wordmark in the bar instead of two and the SCOPE, GRID and ABOUT titles are unobscured.</div></div><div class="vl-row"><div class="vl-head">v9.1102026-09-04 12:50 UTC</div><div class="vl-scope">the v8 VENTUS masthead is fused into the six-menu bar's own centre so it is present at every width and cannot be torn into a closed panel and vanish after arrival; the SCADA layer panel is restored with the real, moved-not-cloned .scada-brand and .status-legend nodes and every layer control on a full-size >=44px hit target; every one of the six panels is anchored to its own side group and clamped into the viewport by measurement; and the OSM/CARTO/Open Charge Map credit now outranks every open panel by z-index, not only clearing the bar's own height</div></div><div

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class="vl-row"><div class="vl-head"><span class="vl-ver">v9.109</span><span
class="vl-when">2026-09-04 12:44 UTC</span></div><div class="vl-scope">Pipeline
News' three broken technology buckets (wind_onshore, wind_offshore, other)
resolve to the engine's real layer id through one table instead of a set-
membership test that read enabled while the layer sat off, on a third of the
register; substation-intelligence carried forward unchanged to keep the on-page
version ledger current</div></div><div class="vl-row"><div class="vl-head"><span
class="vl-ver">v9.108</span><span class="vl-when">2026-09-04 12:21 UTC</span>
</div><div class="vl-scope">the OSM/CARTO/Open Charge Map credit clears the menu
bar unconditionally, measured from the bar's own rendered height, at every width
-- not only while body.fs-active happens to be set</div></div><div class="vl-
row"><div class="vl-head"><span class="vl-ver">v9.107</span><span class="vl-
when">2026-09-04 04:03 UTC</span></div><div class="vl-scope">Move seven byte-
identical SLD stylesheet templates into the earlier substation cartridge, fail
closed if the module is absent, and restore bounded headroom without changing UI
behaviour.</div></div><div class="vl-row"><div class="vl-head"><span class="vl-
ver">v9.106</span><span class="vl-when">2026-09-04 03:37 UTC</span></div><div
class="vl-scope">Pipeline 0144 arrivals absent from the active register retain
their supplied identity and measurement with explicit provenance; real resolver
failures remain failures with a working single-epoch owner-to-measurement retry,
stale arrival ownership cannot reclaim the map, and the legacy receiver issues
no cross-domain Pipeline request.</div></div><div class="vl-row"><div class="vl-
head"><span class="vl-ver">v9.105</span><span class="vl-when">2026-09-04 02:19
UTC</span></div><div class="vl-scope">A successfully fetched transit source is
reported available only when its configured MapLibre layer can draw at least one
matching feature; zero eligible features become visibly EMPTY, unticked,
disabled and non-interactive while quarantined data and historical generations
remain unchanged.</div></div><div class="vl-row"><div class="vl-head"><span
class="vl-ver">v9.104</span><span class="vl-when">2026-09-04 01:34 UTC</span>
</div><div class="vl-scope">mobile fullscreen no longer creates an ancestor
cycle, and Grid keeps each layer tick plus its live V8 status visible</div>
</div><div class="vl-row"><div class="vl-head"><span class="vl-
ver">v9.103</span><span class="vl-when">2026-09-04 00:58 UTC</span></div><div
class="vl-scope">A late REPD identity can no longer resurrect an abandoned deep-
link arrival: every user selection and clear invalidates the pending token,
while intended first and corrected selections carry the current token through
asynchronous measurement.</div></div><div class="vl-row"><div class="vl-head">
<span class="vl-ver">v9.102<span class="vl-status">REJECTED_PRE_PROMOTION</span>
</span><span class="vl-when">2026-09-04 00:47 UTC</span></div><div class="vl-
scope">The production receiver uses its exported measure-first plan for all
valid Pipeline coordinates, records link provenance, reconciles resolved
identity atomically, accepts every canonical technology, and preserves mobile
layer hit targets plus executable transit hydration.</div><div class="vl-
reason">never live: a late identity could restore an arrival after a user
selection or clear</div></div><div class="vl-row"><div class="vl-head"><span
class="vl-ver">v9.101<span class="vl-status">REJECTED_PRE_PROMOTION</span>
</span><span class="vl-when">2026-09-04 00:46 UTC</span></div><div class="vl-
scope">Valid Pipeline coordinates measure before concurrent REPD verification;
every canonical technology reaches the exact receiver; the mobile menu remains
hittable above the docked project card; transit bridge hydration executes to a
populated source; pipeline layers move to the network cartridge to preserve SLD
headroom.</div><div class="vl-reason">never live: its generation proof reported
748/755</div></div><div class="vl-row"><div class="vl-head"><span class="vl-
ver">v9.100<span class="vl-status">REJECTED_PRE_PROMOTION</span></span><span
class="vl-when">2026-09-04 00:21 UTC</span></div><div class="vl-
scope">Consolidate the existing interface into exactly File, Edit, View, Scope,
Grid, and About menus; expose all 60 engine and 3 Pipeline News layer controls
through their original handlers; preserve nested Scope and Clear actions; remain
closed at rest and fail closed unless all 63 unique layer controls exist.</div>
<div class="vl-reason">never live: the mobile project card could cover the
engine layer panel</div></div><div class="vl-row"><div class="vl-head"><span
class="vl-ver">v9.99</span><span class="vl-when">2026-09-03 23:15 UTC</span>

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the ceiling warning light reports the limit that can actually fail the build

v9.982026-09-03 22:46 UTC

the arrival frames the project by the viewport, so a shared link is not wide and empty on a desktop

v9.972026-09-03 22:22 UTC

a deep link is not a search: the results list and the box it filled get out of the way

v9.962026-09-03 22:13 UTC

the layer panel opens closed, so the map is the first impression

v9.952026-09-03 20:41 UTC

withdraw the menu bar from the live Atlas on two testers' evidence

v9.942026-09-03 20:12 UTC

restamp both cartridges so every manifest names the generation it is part of

v9.932026-09-03 20:05 UTC

the menu bar fails soft where there is no timer, as its own comment claimed

v9.922026-09-03 20:01 UTC

collapse the chrome into File, Edit, View and About at the top, closed at rest

v9.912026-09-03 18:09 UTC

the register owns the coordinates on every repd_ref arrival, not only when the link is malformed, and which source won is published with the distance between them

v9.902026-09-03 17:51 UTC

the measurement arrives before the intelligence that qualifies it, and on a phone the card docks to the bottom edge instead of hanging off a marker: nothing is printed over it and nothing has to be dragged to reach it

v9.892026-09-03 13:16 UTC

the measurement stops reading technology: the one branch that gated it is removed, so offshore measures to the nearest mapped substation instead of withholding, and a module owns which sentences go under the distances rather than which technologies get one

v9.882026-09-03 02:34 UTC

a 44 pixel action opens the corridor figures in full, reachable by button, by right-click and by a long press that stands down for a pan, a pinch and any drag already under way

v9.872026-09-03 02:33 UTC

a straight line is not a route: beside every measured distance the card now prints an indicative highway-corridor estimate, with the calibration it came from and the questions it is not an answer to

v9.862026-09-03 02:00 UTC

the nearest 400 kV superlative carries the sample it was drawn from, counted at render time from the payload actually fetched rather than written into the sentence

v9.852026-09-03 01:56 UTC

the version ledger leaves the sandbox body for the cartridge that has room for it, because the sandbox stood 600 characters short of the guard its own proof asserts and every cut adds another row

v9.842026-09-03 01:51 UTC

the proof reads the published products through the pin the composition declares, by commit and by digest, and a product it cannot get fails those checks loudly instead of skipping them

v9.832026-09-03 01:37 UTC

the three products this Atlas reads at runtime are pinned to a commit and checked by their SHA-256, because a schema string defends shape and is blind to values

v9.82 2026-09-03 01:28 UTC

an unrecognised project technology abandoned the whole arrival - the card, the ring, the measurement and the substation layer - when the only thing that needs to know what a project generates is the one technology layer

v9.81 2026-09-03 01:19 UTC

a deep link carrying coordinates but no register identity moved no camera at all: both lanes that own the arrival stand down at the same identity test, and the zoom was then eased without a centre ever being set

v9.80 2026-09-03 01:16 UTC

HIDE LAYERS collapsed the element that contains the map, so on a phone the one control that gets past the layer panel blanked the application and the remembered choice blanked it again on reload

v9.79 2026-09-03 01:09 UTC

the site card counts transformers as machines rather than as winding connections: a site owns both ends of its own transformers, so every internal machine was published twice and Cowley's five read as ten

v9.78 2026-09-03 00:59 UTC

a PIPELINE NEWS (REPD) section in the layer dashboard that summons the rest of the pipeline around the selected project

v9.77 2026-09-02 00:18 UTC

the powerflow stops choosing an arbitrary withdrawal bus: the published 400 kV network has 238 connected components, a transfer across two of them does not exist, and the card was asking for one - it now uses a declared sink rule, refuses a cross-component transfer before the solver is asked, accepts an answer only on convergence AND a global residual AND Kirchhoff at every bus rather than at the injection bus alone, says plainly when no answer is available, and counts parallel circuits as the separate published rows they are rather than collapsing those that share a reactance

v9.76 2026-09-02 00:06 UTC

the computation moves to the cartridge that owns it: the five modules that read the operator's published network leave the sandbox for substation-intelligence, which is split into the two halves it has always been - the V8 engine carried verbatim and the intelligence itself - and gains the parts manifest it should always have had; the sandbox drops from 95% of its 400 kB boundary to about 79%, and the new owner-boundary module lands beside its siblings, naming which transmission owners the assets at a site belong to and where two of them meet on one circuit

v9.75 2026-09-01 23:45 UTC

the card reports what the operator has published as planned in its own sentence - by year, by status and by asset, with the published parameters carried - kept structurally apart from what exists today, because a row published for a future year is not a circuit now, not a commitment, not a consent and not a connection date

v9.74 2026-09-01 23:17 UTC

the grid computation is reachable from the map itself: a tool arms a point query, and a click on open map resolves the nearest published connection points and renders the published circuits, the seasonal ratings, the electrical distance in hops and the declared powerflow against the nearest one, saying every time that 384 of the 886 published points have no coordinates so the nearest MAPPED point may not be the nearest point; and the layers dash collapses and restores on any device without entering fullscreen, the choice remembered per browser

v9.73 2026-09-01 23:08 UTC

the Atlas solves a

declared DC injection response on the published node/branch model and reports which circuits would carry a project's stated capacity and what fraction each takes, with the equations, the 100 MVA base, the named slack and every assumption carried in the answer; validated to 1e-9 against networks whose solutions are exact by hand and checked for power conservation at every intermediate bus of the real 400 kV network; it never states a loading, because existing flows are published nowhere

v9.72	2026-09-01 22:49 UTC	the card reports every season the operator publishes rather than winter alone, each circuit keeping its own rating and its own season, scoped to the connection voltage; four circuits published at 9,999 MVA on spans of a kilometre or less are named as placeholders and excluded from the range while still being reported; and the module that produces all of this contains no code path that adds two ratings together, which its proof asserts structurally
v9.71	2026-09-01 22:43 UTC	the Atlas measures in the operator's own circuits as well as in kilometres: an electrical-distance module traverses the published node/branch model, a voltage changes only across a named transformer and a circuit that appears to change voltage is refused and recorded, planned changes are never walked as if they existed today, every hop carries its published rating and its R/X/B untouched, and the card names what lies two hops away
v9.70	2026-09-01 22:34 UTC	the composition manifest is proven against the bytes: substation-intelligence declares connection-points.v3 as it has required since v9.65 (the entry said v2 from v9.63 until this cut), sld-sandbox declares the transmission network and the price rollup it fetches, a data-contract parity proof holds every entry to its bytes in both directions, the source registry states what each fetching source requires in every state and registers the GB conditions loader, whose state the sandbox now publishes
v9.69	2026-09-01 22:11 UTC	sld-sandbox: the sizing arithmetic (physical inputs, the three named ratios, string and central nameplates, the finance port and the two-variable capacity fit) leaves the body for the sizing-arithmetic module, lifted mechanically expression for expression and proven value-for-value against the last inline copy; the body keeps one-line delegations so no caller changes
v9.68	2026-09-01 21:41 UTC	sld-sandbox: the declared-connections table moves out of the body into a module proven value-for-value against the last inline copy; the network-topology module (proven 47/47 at 202609012145, composed into nothing until now) is wired on demand - the 10 MB ETYS node/branch product is fetched on first click, never at load, and project and scope cards fill an ETYS topology block in place; the source registry reports the loader's true state instead of ready for a module on a shelf; and this stamp is read from the clock, which is why it sorts before the typed one it succeeds.
v9.67	2026-09-01 22:50 UTC	one geodesy for the whole cartridge, a click reports which cartridges answered it, and the zoom the deep link has always sent is finally read
v9.66	2026-09-01 21:55 UTC	the geodesy module measures on the estate's canonical haversine, so every version ever shipped returns the same distance to the last digit
v9.65	2026-09-01 21:30 UTC	the page's version ledger is written by the cut, so it cannot tell a reader it is running the generation before the one it is running
v9.64	2026-09-01 21:10 UTC	the Grid Finding Scope stops censusing a voltage that did not parse, and every assembled

cartridge is checked against the parts it was built from

Ver	When	Scope
v9.63	2026-09-01 20:45 UTC	fault current is quoted at the voltage the connection is made at, not across every busbar at the site
v9.62	2026-09-01 20:20 UTC	the Grid Finding Scope: a click on blank space says what grid is mapped there, in bands, and what that cannot tell you
v9.61	2026-09-01 19:15 UTC	the network sentence names its own scope: a site-wide envelope across the voltages present, never a value for the declared bus
v9.60	2026-09-01 18:45 UTC	the manifest states its own identity, and the Subs control is found by its attribute rather than by label text that changes
v9.59	2026-09-01 18:20 UTC	the substation cartridge returns through the engine slot, and the card gains what NESO publishes: circuits, ratings, fault current and planned changes
v9.58	2026-09-01 18:05 UTC	service restored: v9.57 claimed a shell script the shell never loads, so the composer refused it and the map went dark
v9.56	2026-09-01 17:18 UTC	pink for a point of connection not yet built, and a declared connection that is a circuit rather than a substation draws no line at all
v9.55	2026-09-01 16:15 UTC	the sales surface answers at once: a declared 400 kV connection and its citation are on the card before the payload arrives, the distance following it
v9.54	2026-09-01 16:12 UTC	the measurement no longer waits for the engine's layer controls: links draw at once, layers follow when they arrive
v9.53	2026-09-01 14:35 UTC	the consented works on the card: customer-substation and PoC interface quotes from the made Orders; the nearest-400 row gains its named companion
v9.52	2026-09-01 14:34 UTC	a recovered failure is not a failure: late-arrival entries move to their own ledger when the controls arrive, per the Codex supervision finding
v9.51	2026-09-01 14:33 UTC	the 400 kV public record: declared DCO connections drawn and carded, new customer substations named, nearest 400 kV measured for every project
v9.50	2026-09-01 12:51 UTC	the card precedes the lines: a register-absent arrival opened its card after drawing, and the lines-belong-to-the-card watcher rightly wiped them
v9.49	2026-09-01 12:44 UTC	supersedes v9.48, whose boundary shipped without its proof file; same composition, re-sealed whole
v9.48	2026-09-01 12:43 UTC	supersedes v9.47's boundary: the composition was sound, its proof shipped one stale check; re-sealed coherent
v9.47	2026-09-01 12:42 UTC	the arrival owns its card: 2,421 register-absent projects (873 solar) get a card from the link's own fields, yielded if the register's card lands

class="vl-ver">v9.462026-09-01 12:15 UTC
</div><div class="vl-scope">the distances survive the card: a keeper re-attaches
the measurement block when a late popup replaces the one it decorated</div>
</div><div class="vl-row"><div class="vl-head">v9.45
2026-09-01 12:05 UTC</div><div class="vl-
scope">arrival: fullscreen on touch, the identity wait runs to its end, and
every stage says what it is doing</div></div><div class="vl-row"><div class="vl-
head">v9.442026-09-01 11:41
UTC</div><div class="vl-scope">a repd_ref-only link computes the links:
identity resolved by the search lane is consumed, not re-required from the
URL</div></div><div class="vl-row"><div class="vl-head"><span class="vl-
ver">v9.432026-09-01 09:02 UTC</div><div
class="vl-scope">mobile: tools collapse behind one chip; grid and subs are one
tap</div></div><div class="vl-row"><div class="vl-head"><span class="vl-
ver">v9.422026-09-01 07:26 UTC</div><div
class="vl-scope">the price panel revalidates instead of pinning its first
sight</div></div><div class="vl-row"><div class="vl-head"><span class="vl-
ver">v9.412026-09-01 07:22 UTC</div><div
class="vl-scope">exact GB price evidence, beside the ledger</div></div><div
class="vl-row"><div class="vl-head">v9.40<span
class="vl-when">2026-09-01 02:04 UTC</div><div class="vl-scope">the
version ledger itself, on the page</div></div><div class="vl-row"><div
class="vl-head">v9.392026-09-
01 01:06 UTC</div><div class="vl-scope">remove duplicate BESS truth,
restore original central defaults and reject fractional topology counts</div>
</div><div class="vl-row"><div class="vl-head">v9.38
2026-09-01 00:58 UTC</div><div class="vl-
scope">restore topology-isolated physical inputs and the original mounting-to-
bifacial linkage</div></div><div class="vl-row"><div class="vl-head"><span
class="vl-ver">v9.372026-09-01 00:53 UTC
</div><div class="vl-scope">complete the original finance interaction contract
by linking development stage, cost and success</div></div><div class="vl-row">
<div class="vl-head">v9.36<span class="vl-
when">2026-09-01 00:40 UTC</div><div class="vl-scope">original financial-
model parity with explicit correction of the known central AC double-count</div>
</div><div class="vl-row"><div class="vl-head">v9.35
2026-09-01 00:21 UTC</div><div class="vl-
scope">phone pointer operation, viewport containment and named electrical
ratios</div></div><div class="vl-row"><div class="vl-head"><span class="vl-
ver">v9.342026-08-31 23:24 UTC</div><div
class="vl-scope">a missing source costs a drawing, never the session</div></div>
<div class="vl-row"><div class="vl-head">v9.33<span
class="vl-when">2026-08-31 23:21 UTC</div><div class="vl-scope">nothing
can rewrite the reference design, not even later</div></div><div class="vl-row">
<div class="vl-head">v9.32<span class="vl-
when">2026-08-31 23:17 UTC</div><div class="vl-scope">no substation can
display an impossible voltage</div></div><div class="vl-row"><div class="vl-
head">v9.312026-08-31 23:15
UTC</div><div class="vl-scope">Codex's LineAtlas cardinality gate
passes</div></div><div class="vl-row"><div class="vl-head"><span class="vl-
ver">v9.302026-08-31 23:13 UTC</div><div
class="vl-scope">the neon flow no longer exhausts the renderer</div></div><div
class="vl-row"><div class="vl-head">v9.29<span
class="vl-when">2026-08-31 23:06 UTC</div><div class="vl-scope">the
headline capacity actually moves the layout</div></div><div class="vl-row"><div
class="vl-head">v9.282026-08-
31 23:00 UTC</div><div class="vl-scope">voltage classes are explained,
and the whole dashboard is accepted</div></div><div class="vl-row"><div
class="vl-head">v9.272026-08-
31 22:57 UTC</div><div class="vl-scope">the MAP button works for every
technology in the register</div></div><div class="vl-row"><div class="vl-head">
v9.262026-08-31 22:44

UTC

Version	When	Content
v9.25	2026-08-31 22:38 UTC	late layer controls are used, and the repository is LF everywhere
v9.24	2026-08-31 22:27 UTC	one source of truth for GB prices: the data repository
v9.23	2026-08-31 22:22 UTC	the GB electricity tracker is connected to the map
v9.22	2026-08-31 22:08 UTC	card geometry resets on every selection
v9.21	2026-08-31 22:05 UTC	a symbol layer is added only once its text can be drawn
v9.20	2026-08-31 21:57 UTC	the MapLibre exception storm: symbol layers with no glyph atlas
v9.19	2026-08-31 21:54 UTC	the Atlas says what it is waiting for, sized for a phone
v9.18	2026-08-31 21:40 UTC	the grid maths installs even when the basemap never paints
v9.17	2026-08-31 21:33 UTC	the project marker is a ring, found by looking at it in Chrome
v9.16	2026-08-31 21:21 UTC	central AC sizing: the limiting nameplate, not a squared product

the project arriving from Pipeline News is visible: its own technology layer is enabled and a pin owned by this cartridge is dropped on it, with a toggle on the card

Each row is immutable audit evidence, not a promise that it was live. REJECTED_PRE_PROMOTION entries were never live and are not rollback targets. A deployed bad composition is never repaired in place; an earlier deployed one is composed again under a new timestamp. 16 earlier compositions predate the scope line. Extracted from the repository history at build time; nothing here is fetched or editable at runtime.

This interface is based on publicly available and open source infrastructure data, including public energy datasets such as REPD where applicable. It is presented for analytical and visualisation purposes only and does not include live operational or other non public operational data.

In support of The Future of Solar Photovoltaics podcast & all participants to date

<https://ventusltd.github.io/ventus-grid-engine/?graph=engine-graph>
 Estate
 Grid engine · the maths
https://ventusltd.github.io/data-federation-map-for-globalgrid2050-all-repos/dashboard/sandbox/spider_full_po_test.html
 Federation map
https://ventusltd.github.io/spiders/spider_printer_v1/
 Spider printer

Data © <https://www.openstreetmap.org/copyright>
 OpenStreetMap contributors | © CARTO | EV data © <https://openchargemap.org>
 Open Charge Map

Map

MapLibre GL Interactive

```
ctrl-top-left "></div><div class="maplibregl-ctrl-top-right "></div><div
class="maplibregl-ctrl-bottom-left "></div><div class="maplibregl-ctrl-bottom-
right "></div></div></div>
```

```
<div class="map-controls gm-tools-collapsed"><div id="gridatlas-mobile-
tray"><button type="button" aria-expanded="false" hidden="">Tools ▶</button>
<button type="button" aria-pressed="false">⚡ Grid</button><button type="button"
aria-pressed="false">● Subs</button></div>
```

```
</div>
```

```
<div class="radius-popup" id="radius-popup">
  <label>Radius from click</label>
  <div class="radius-input-row">
    <input type="number" id="radius-input" value="10" min="1"
max="160" step="1">
    <span class="radius-unit">km</span>
  </div>
  <div class="radius-hint">1 – 160 km (100 mi) • press Enter to
apply</div>
</div>
```

```
<div class="radius-popup" id="radius-area-popup">
  <label>Measure Area from click</label>
  <div class="radius-input-row">
    <input type="number" id="radius-area-input" value="1"
min="0.001" max="160" step="0.001">
    <span class="radius-unit">km</span>
  </div>
  <div class="radius-hint">Input radius: 0.001 (1m) to 160 km</div>
</div>
```

```
<div class="radius-popup" id="zonedraw-display">
  <label class="polyzone-label">◆ Poly Zone Tool</label>
  <div class="radius-input-row">
    <input type="number" id="zonedraw-radius-input" value="2"
min="0.1" max="160" step="0.1">
    <span class="radius-unit">km</span>
  </div>
  <div class="radius-hint">Set starting radius, then click map to
create zone</div>
  <button id="btn-zonedraw-undo" title="Undo last vertex"
style="margin-top:8px;background:#ff6600;color:#000;border:none;padding:6px
12px;border-radius:4px;cursor:pointer;font-weight:bold;font-size:12px;">↶ Undo
Point</button>
</div>
```

```
<div id="measure-display">
  <div class="m-row" id="m-line" style="display:none">— <span id="m-
km"></span> km | <span id="m-m"></span> m | <span id="m-mi"></span> mi</div>
  <div class="m-row" id="m-perim" style="display:none">Perimeter:
<span id="m-pkm"></span> km | <span id="m-pm"></span> m</div>
```

```

        <div class="m-row" id="m-area" style="display:none">Area: <span
id="m-km2"></span> km² | <span id="m-ha"></span> ha | <span id="m-ac"></span>
ac</div>

        <div class="m-hint" id="m-hint">Click to add points • Double-click
to close polygon</div>

        <button id="btn-measure-undo" title="Undo last point">↶ Undo
Point</button>
    </div>

    <div id="polyzone-display">
        <div class="m-row" id="pz-area" style="display:none">
            <span id="pz-km2"></span> km² • <span id="pz-ha"></span> ha<br>
            <span id="pz-ac"></span> ac • <span id="pz-mi2"></span> mi²<br>
            <span id="pz-m2"></span> m²<br>

            <span style="color:#ffae00">🎯 <span id="pz-pitch"></span>
pitches</span>
        </div>
        <div class="m-row" id="pz-perim" style="display:none">
            <span style="color:#555;font-size:9px">perimeter <span id="pz-
pkm"></span> km</span>
        </div>
        <div class="m-hint" id="pz-hint">Click map to drop circle</div>
        <button id="btn-polyzone-undo" title="Undo last vertex">↶ Undo
Vertex</button>
    </div>

</div>

<div class="scada-wrapper" data-gridatlas-collapsed="1">

    <div class="scada-keys" id="scada-ui-container"><div class="key-group">
<div class="key-title">Topology (GeoJSON)</div><label class="key-item"><input
type="checkbox" data-layer-id="400"> <span id="lbl-400" data-base-label="400kV"
style="color: rgb(0, 84, 255); opacity: 1;">400kV [OK]</span></label><label
class="key-item"><input type="checkbox" data-layer-id="275"> <span id="lbl-275"
data-base-label="275kV" style="color: rgb(255, 0, 0); opacity: 1;">275kV [OK]
</span></label><label class="key-item"><input type="checkbox" data-layer-
id="220"> <span id="lbl-220" data-base-label="220kV" style="color: rgb(255, 153,
0); opacity: 1;">220kV [OK]</span></label><label class="key-item"><input
type="checkbox" data-layer-id="132"> <span id="lbl-132" data-base-label="132kV"
style="color: rgb(0, 204, 0); opacity: 1;">132kV [OK]</span></label><label
class="key-item"><input type="checkbox" data-layer-id="66"> <span id="lbl-66"
data-base-label="66kV" style="color: rgb(178, 0, 255); opacity: 1;">66kV [OK]
</span></label><label class="key-item"><input type="checkbox" data-layer-
id="11kv"> <span id="lbl-11kv" data-base-label="11kV (UKPN)" style="color:
rgb(255, 0, 255);">11kV (UKPN) [WAIT]</span></label><label class="key-item">
<input type="checkbox" data-layer-id="subs"> <span id="lbl-subs" data-base-
label="Subs" style="color: rgb(255, 255, 255); opacity: 1;">Subs [OK]</span>
</label></div><div class="key-group"><div class="key-title">Assets (GeoJSON)
</div><label class="key-item"><input type="checkbox" data-layer-id="nuc"> <span
id="lbl-nuc" data-base-label="Nuclear" style="color: rgb(57, 255, 20);" opacity:
1;">Nuclear [OK]</span></label><label class="key-item"><input type="checkbox"
data-layer-id="gas"> <span id="lbl-gas" data-base-label="Gas" style="color:
rgb(255, 69, 0); opacity: 1;">Gas [OK]</span></label><label class="key-item">
<input type="checkbox" data-layer-id="ind"> <span id="lbl-ind" data-base-
label="Industry" style="color: rgb(255, 102, 0); opacity: 1;">Industry [OK]
</span></label><label class="key-item"><input type="checkbox" data-layer-
id="dc"> <span id="lbl-dc" data-base-label="Data Ctrs" style="color: rgb(0, 255,
255); opacity: 1;">Data Ctrs [OK]</span></label><label class="key-item"><input

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type="checkbox" data-layer-id="air"> <span id="lbl-air" data-base-label="Airports" style="color: rgb(255, 0, 255); opacity: 1;">Airports [OK]
</span></label><label class="key-item"><input type="checkbox" data-layer-id="rail"> <span id="lbl-rail" data-base-label="Railways" style="color: rgb(255, 215, 0); opacity: 1;">Railways [OK]</span></label></div><div class="key-group">
<div class="key-title">Supermarkets (GeoJSON)</div><label class="key-item">
<input type="checkbox" data-layer-id="tesco"> <span id="lbl-tesco" data-base-label="Tesco" style="color: rgb(238, 28, 46);">Tesco [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="sainsburys"> <span id="lbl-sainsburys" data-base-label="Sainsbury's" style="color: rgb(255, 130, 0);">Sainsbury's [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="asda"> <span id="lbl-asda" data-base-label="Asda" style="color: rgb(120, 190, 32);">Asda [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="morrisons"> <span id="lbl-morrisons" data-base-label="Morrisons" style="color: rgb(255, 215, 0);">Morrisons [WAIT]
</span></label><label class="key-item"><input type="checkbox" data-layer-id="aldi"> <span id="lbl-aldi" data-base-label="Aldi" style="color: rgb(0, 48, 135);">Aldi [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="lidl"> <span id="lbl-lidl" data-base-label="Lidl" style="color: rgb(0, 80, 170);">Lidl [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="waitrose"> <span id="lbl-waitrose" data-base-label="Waitrose" style="color: rgb(122, 184, 0);">Waitrose [WAIT]</span></label>
<label class="key-item"><input type="checkbox" data-layer-id="ms"> <span id="lbl-ms" data-base-label="M&S Food" style="color: rgb(0, 155, 119);">M&S Food [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="coop"> <span id="lbl-coop" data-base-label="Co-op" style="color: rgb(0, 177, 169);">Co-op [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="costco"> <span id="lbl-costco" data-base-label="Costco" style="color: rgb(0, 93, 170);">Costco [WAIT]
</span></label><label class="key-item"><input type="checkbox" data-layer-id="booths"> <span id="lbl-booths" data-base-label="Booths" style="color: rgb(109, 32, 119);">Booths [WAIT]</span></label></div><div class="key-group">
<div class="key-title">Transit (GeoJSON)</div><label class="key-item"><input type="checkbox" data-layer-id="elizabeth"> <span id="lbl-elizabeth" data-base-label="Elizabeth Line" style="color: rgb(96, 57, 158);">Elizabeth Line [WAIT]
</span></label><label class="key-item"><input type="checkbox" data-layer-id="lu"> <span id="lbl-lu" data-base-label="London Underground" style="color: rgb(227, 32, 23);">London Underground [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="dlr"> <span id="lbl-dlr" data-base-label="DLR" style="color: rgb(0, 175, 173);">DLR [WAIT]</span></label><label class="key-item"><input type="checkbox" data-layer-id="metro"> <span id="lbl-metro" data-base-label="UK Metro" style="color: rgb(243, 167, 18);">UK Metro [WAIT]
</span></label><label class="key-item"><input type="checkbox" data-layer-id="tram"> <span id="lbl-tram" data-base-label="Trams & Light Rail" style="color: rgb(132, 50, 155);">Trams & Light Rail [WAIT]</span></label>
<label class="key-item"><input type="checkbox" data-layer-id="mainline_rail"> <span id="lbl-mainline_rail" data-base-label="UK Mainline Railways" style="color: rgb(255, 136, 0);">UK Mainline Railways [WAIT]</span></label>
<label class="key-item"><input type="checkbox" data-layer-id="hs2"> <span id="lbl-hs2" data-base-label="HS2" style="color: rgb(44, 111, 173);">HS2 [WAIT]
</span></label><label class="key-item"><input type="checkbox" data-layer-id="eurostar_route"> <span id="lbl-eurostar_route" data-base-label="Eurostar (HS1) UK & Europe" style="color: rgb(0, 20, 137);">Eurostar (HS1) UK & Europe [WAIT]
</span></label><label class="key-item"><input type="checkbox" data-layer-id="eurostar_station"> <span id="lbl-eurostar_station" data-base-label="Eurostar Stations UK & Europe" style="color: rgb(255, 215, 0);">Eurostar Stations UK & Europe [WAIT]</span></label></div><div class="key-group"><div class="key-title">Leisure & Sports (GeoJSON)</div>
<label class="key-item"><input type="checkbox" data-layer-id="stadiums"> <span id="lbl-stadiums" data-base-label="Major Stadiums" style="color: rgb(229, 255, 0);">Major Stadiums [WAIT]</span></label></div><div class="key-group"><div class="key-title">Global Data Backbone</div><label class="key-item"><input

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type="checkbox" data-layer-id="subsea_data_cables"> <span id="lbl-
subsea_data_cables" data-base-label="Subsea Data Cables (Fibre) (not all
connections available yet)" style="color: rgb(0, 229, 255);">Subsea Data Cables
(Fibre) (not all connections available yet) [WAIT]</span></label></div><div
class="key-group"><div class="key-title">Global Supply Chain &
Chokepoints</div><label class="key-item"><input type="checkbox" data-layer-
id="global_ports_major"> <span id="lbl-global_ports_major" data-base-
label="Major Ports (Deep Water/Ferry)" style="color: rgb(0, 191, 255);">Major
Ports (Deep Water/Ferry) [WAIT]</span></label><label class="key-item"><input
type="checkbox" data-layer-id="global_ports_minor"> <span id="lbl-
global_ports_minor" data-base-label="Inland / Minor Harbours" style="color:
rgb(0, 85, 170);">Inland / Minor Harbours [WAIT]</span></label><label
class="key-item"><input type="checkbox" data-layer-id="global_hydrocarbons">
<span id="lbl-global_hydrocarbons" data-base-label="Global Hydrocarbons (Oil,
Gas, LNG)" style="color: rgb(255, 51, 0);">Global Hydrocarbons (Oil, Gas, LNG)
[WAIT]</span></label></div><div class="key-group"><div class="key-title">Highway
Infrastructure</div><label class="key-item"><input type="checkbox" data-layer-
id="motorways"> <span id="lbl-motorways" data-base-label="Motorways"
style="color: rgb(0, 84, 255);">Motorways [WAIT]</span></label><label
class="key-item"><input type="checkbox" data-layer-id="trunk_roads"> <span
id="lbl-trunk_roads" data-base-label="Major A-Roads (Trunk)" style="color:
rgb(0, 204, 0);">Major A-Roads (Trunk) [WAIT]</span></label><label class="key-
item"><input type="checkbox" data-layer-id="primary_roads"> <span id="lbl-
primary_roads" data-base-label="A-Roads (Primary)" style="color: rgb(255, 68,
0);">A-Roads (Primary) [WAIT]</span></label><label class="key-item"><input
type="checkbox" data-layer-id="motorway_services"> <span id="lbl-
motorway_services" data-base-label="Motorway Services" style="color: rgb(255, 0,
255);">Motorway Services [WAIT]</span></label></div><div class="key-group"><div
class="key-title">EV (GeoJSON)</div><label class="key-item"><input
type="checkbox" data-layer-id="ev"> <span id="lbl-ev" data-base-label="EV Rapid
(100kW+)" style="color: rgb(0, 255, 136);">EV Rapid (100kW+) [WAIT]</span>
</label></div><div class="key-group"><div class="key-title">Heavy Industry</div>
<label class="key-item"><input type="checkbox" data-layer-id="naei_co2"> <span
id="lbl-naei_co2" data-base-label="Major Industrial Sites" style="color:
rgb(255, 68, 0);">Major Industrial Sites [WAIT]</span></div><div
class="key-group"><div class="key-title">REPD (GeoJSON)</div><label class="key-
item"><input type="checkbox" data-layer-id="solar"> <span id="lbl-solar" data-
base-label="Solar PV" style="color: rgb(255, 255, 0);">Solar PV [WAIT]</span>
</label><label class="key-item"><input type="checkbox" data-layer-
id="solar_operational"> <span id="lbl-solar_operational" data-base-label="Solar
PV (Operational Only)" style="color: rgb(0, 255, 136);">Solar PV (Operational
Only) [WAIT]</span></label><label class="key-item"><input type="checkbox" data-
layer-id="solar_roof"> <span id="lbl-solar_roof" data-base-label="Solar Roof"
style="color: rgb(255, 204, 0);">Solar Roof [WAIT]</span></label><label
class="key-item"><input type="checkbox" data-layer-id="wind"> <span id="lbl-
wind" data-base-label="Wind" style="color: rgb(0, 255, 255);">Wind [WAIT]</span>
</label><label class="key-item"><input type="checkbox" data-layer-
id="wind_onshore_operational"> <span id="lbl-wind_onshore_operational" data-
base-label="Onshore Wind (Operational Only)" style="color: rgb(0, 255,
204);">Onshore Wind (Operational Only) [WAIT]</span></label><label class="key-
item"><input type="checkbox" data-layer-id="wind_offshore_operational"> <span
id="lbl-wind_offshore_operational" data-base-label="Offshore Wind (Operational
Only)" style="color: rgb(0, 102, 255);">Offshore Wind (Operational Only) [WAIT]
</span></label><label class="key-item"><input type="checkbox" data-layer-
id="bess"> <span id="lbl-bess" data-base-label="Battery Storage" style="color:
rgb(255, 174, 0);">Battery Storage [WAIT]</span></label><label class="key-item">
<input type="checkbox" data-layer-id="bess_operational"> <span id="lbl-
bess_operational" data-base-label="Battery Storage (Operational Only)"
style="color: rgb(255, 105, 180);">Battery Storage (Operational Only) [WAIT]
</span></label><label class="key-item"><input type="checkbox" data-layer-
id="biomass"> <span id="lbl-biomass" data-base-label="Biomass & EfW"
style="color: rgb(57, 255, 20);">Biomass & EfW [WAIT]</span></label><label

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class="key-item"><input type="checkbox" data-layer-id="tidal"> <span id="lbl-
tidal" data-base-label="Tidal & Wave" style="color: rgb(0, 191, 255);">Tidal
& Wave [WAIT]</span></label><label class="key-item"><input type="checkbox"
data-layer-id="hydrogen"> <span id="lbl-hydrogen" data-base-label="Hydrogen"
style="color: rgb(255, 255, 255);">Hydrogen [WAIT]</span></label><label
class="key-item"><input type="checkbox" data-layer-id="hydro"> <span id="lbl-
hydro" data-base-label="Hydro & Pumped Storage" style="color: rgb(0, 170,
255);">Hydro & Pumped Storage [WAIT]</span></label><label class="key-item">
<input type="checkbox" data-layer-id="flywheel"> <span id="lbl-flywheel" data-
base-label="Flywheel" style="color: rgb(255, 105, 180);">Flywheel [WAIT]</span>
</label><label class="key-item"><input type="checkbox" data-layer-id="act">
<span id="lbl-act" data-base-label="Advanced Conversion" style="color: rgb(255,
102, 0);">Advanced Conversion [WAIT]</span></label><label class="key-item">
<input type="checkbox" data-layer-id="geothermal"> <span id="lbl-geothermal"
data-base-label="Geothermal" style="color: rgb(255, 51, 0);">Geothermal [WAIT]
</span></label><label class="key-item"><input type="checkbox" data-layer-
id="caes"> <span id="lbl-caes" data-base-label="Compressed Air Storage"
style="color: rgb(136, 170, 255);">Compressed Air Storage [WAIT]</span></label>
</div><div class="key-group"><div class="key-title">Basemap</div><label
class="key-item"><input type="radio" name="bm" value="dark" checked="">
Dark</label><label class="key-item"><input type="radio" name="bm" value="sat">
Satellite</label></div><div class="key-group" data-pn-group="1"><div class="key-
title">PIPELINE NEWS (REPD)</div><label class="key-item"><input type="checkbox"
data-pn-layer="same"> <span data-pn-label="same" style="color: rgb(95, 189,
194);">Same technology [SELECT A PROJECT]</span></label><label class="key-item">
<input type="checkbox" data-pn-layer="wider"> <span data-pn-label="wider"
style="color: rgb(57, 255, 20);">Wider fleet [SELECT A PROJECT]</span></label>
<label class="key-item"><input type="checkbox" data-pn-layer="all"> <span data-
pn-label="all" style="color: rgb(216, 182, 74);">All pipeline [SELECT A PROJECT]
</span></label></div></div>

```

```

</div>
</div>

```

```

<script src="https://cdn.jsdelivr.net/npm/maplibre-gl@3.6.2/dist/maplibre-
gl.js"></script>

```

```

<script data-gridatlas-cartridge="streaming-parquet-bridge"
src="blob:https://ventusltd.github.io/09228574-19ac-406a-8b46-d0c215bfa7d8">
</script>

```

```

<script data-gridatlas-cartridge="uk-gazetteer-flyto"
src="blob:https://ventusltd.github.io/54fcb44e-529f-4698-b262-f1ec030c4c3d">
</script>

```

```

<script data-gridatlas-cartridge="substation-intelligence"
src="blob:https://ventusltd.github.io/b1b0ec0d-b93d-4e83-9246-2ea5aaf3f852">
</script>

```

```

<script data-gridatlas-cartridge="sld-sandbox"
src="blob:https://ventusltd.github.io/6c215f46-ab61-464e-ab64-1a81b477fc04">
</script><button id="gridatlas-dash-toggle" type="button" aria-pressed="true"
aria-label="Show the layers panel">▲ LAYERS</button>

```

```

<script>
  const ukConfig = [
    {
      group: "Topology (GeoJSON)",
      layers: [
        { id: "400", label: "400kV", color: "#0054ff", type: "line",
width: 2.5, url: "data/grid_400kv.geojson", snap: true, preload: true },
        { id: "275", label: "275kV", color: "#ff0000", type: "line",
width: 2.0, url: "data/grid_275kv.geojson", snap: true, preload: true },
        { id: "220", label: "220kV", color: "#ff9900", type: "line",

```

```

width: 1.8, url: "data/grid_220kv.geojson", snap: true, preload: true },
    { id: "132", label: "132kV", color: "#00cc00", type: "line",
width: 1.5, url: "data/grid_132kv.geojson", snap: true, preload: true },
    { id: "66", label: "66kV", color: "#b200ff", type: "line",
width: 1.2, url: "data/grid_66kv.geojson", snap: true, preload: true },
    { id: "11kv", label: "11kV (UKPN)", color: "#ff00ff", type:
"point", radius: ['interpolate', ['linear'], ['zoom'], 13.5, 4, 15, 8, 18, 18],
url: "data/grid_11kv_ukpn.geojson", minzoom: 13.5, preload: false },
    { id: "subs", label: "Subs", color: "#ffffff", type: "point",
radius: ['interpolate', ['linear'], ['zoom'], 5, 3, 10, 5, 14, 10, 18, 22], url:
"data/grid_substations.geojson", isSubs: true, preload: true }
    ]
  },
  {
    group: "Assets (GeoJSON)",
    layers: [
      { id: "nuc", label: "Nuclear", color: "#39ff14", type:
"point", radius: 4.5, url: "data/power_plants.geojson", filter: ['==', ['get',
'source']], 'nuclear'], preload: true },
      { id: "gas", label: "Gas", color: "#ff4500", type:
"point", radius: 4.5, url: "data/power_plants.geojson", filter: ['!=', ['get',
'source']], 'nuclear'], preload: true },
      { id: "ind", label: "Industry", color: "#ff6600", type:
"point", radius: 4.5, url: "data/industrial_offtakers.geojson", preload: true },
      { id: "dc", label: "Data Ctrs", color: "#00ffff", type:
"point", radius: 4.5, url: "data/datacentres.geojson", preload: true },
      { id: "air", label: "Airports", color: "#ff00ff", type:
"point", radius: 4.5, url: "data/airports.geojson", preload: true },
      { id: "rail", label: "Railways", color: "#ffd700", type:
"point", radius: 4.5, url: "data/railways.geojson", preload: true }
    ]
  },
  {
    group: "Supermarkets (GeoJSON)",
    layers: [
      { id: "tesco", label: "Tesco", color: "#eelc2e",
type: "point", radius: 4.5, url: "data/supermarkets_tesco.geojson", preload:
false },
      { id: "sainsburys", label: "Sainsbury's", color: "#ff8200",
type: "point", radius: 4.5, url: "data/supermarkets_sainsburys.geojson",
preload: false },
      { id: "asda", label: "Asda", color: "#78be20",
type: "point", radius: 4.5, url: "data/supermarkets_asda.geojson", preload:
false },
      { id: "morrison's", label: "Morrison's", color: "#ffd700",
type: "point", radius: 4.5, url: "data/supermarkets_morrison's.geojson", preload:
false },
      { id: "aldi", label: "Aldi", color: "#003087",
type: "point", radius: 4.5, url: "data/supermarkets_aldi.geojson", preload:
false },
      { id: "lidl", label: "Lidl", color: "#0050aa",
type: "point", radius: 4.5, url: "data/supermarkets_lidl.geojson", preload:
false },
      { id: "waitrose", label: "Waitrose", color: "#7ab800",
type: "point", radius: 4.5, url: "data/supermarkets_waitrose.geojson", preload:
false },
      { id: "ms", label: "M&S Food", color: "#009b77",
type: "point", radius: 4.5, url: "data/supermarkets_ms.geojson", preload: false
},
      { id: "coop", label: "Co-op", color: "#00b1a9",
type: "point", radius: 4.5, url: "data/supermarkets_coop.geojson", preload:
false },

```

```

        { id: "costco",      label: "Costco",      color: "#005daa",
type: "point", radius: 4.5, url: "data/supermarkets_costco.geojson", preload:
false },
        { id: "booths",     label: "Booths",     color: "#6d2077",
type: "point", radius: 4.5, url: "data/supermarkets_booths.geojson", preload:
false }
    ]
},
{
    group: "Transit (GeoJSON)",
    layers: [
        { id: "elizabeth", label: "Elizabeth Line", color: "#60399E",
type: "point", radius: 5, url: "data/elizabeth_line.geojson", preload: false },
        { id: "lu",        label: "London Underground", color: "#e32017",
type: "point", radius: 5, url: "data/london_underground.geojson", preload: false
},
        { id: "dlr",       label: "DLR", color: "#00afad", type: "point",
radius: 5, url: "/uk_metros_trams.geojson", preload: false,
filter: ['all', ['in', 'Docklands', ['get', 'operator']],
['!', ['in', 'London Underground', ['get', 'operator']]]] },
        { id: "metro", label: "UK Metro", color: "#f3a712", type:
"point", radius: 5, url: "/uk_metros_trams.geojson", preload: false,
filter: ['any', ['in', 'Tyne', ['get', 'operator']], ['in',
'Metrolink', ['get', 'operator']], ['in', 'Supertram', ['get', 'operator']],
['in', 'Nottingham', ['get', 'operator']], ['in', 'Edinburgh', ['get',
'operator']], ['in', 'Midland Metro', ['get', 'operator']]] },
        { id: "tram",      label: "Trams & Light Rail", color: "#84329b",
type: "point", radius: 5, url: "/uk_metros_trams.geojson", preload: false,
filter: ['all', ['==', ['get', 'type'], 'Tram / Light Rail'],
['!', ['in', 'London Underground', ['get', 'operator']]]] },
        { id: "mainline_rail", label: "UK Mainline Railways", color:
"#ff8800", type: "line", width: 1.5, url: "/uk_mainline_railways.geojson",
preload: false },
        { id: "hs2",       label: "HS2", color: "#2c6fad", type: "point",
radius: 4, url: "data/hs2.geojson", preload: false },
        { id: "eurostar_route", label: "Eurostar (HS1) UK & Europe",
color: "#001489", type: "line", width: 2.5, url: "data/eurostar.geojson",
filter: ['==', ['get', 'type'], 'route'], preload: false },
        { id: "eurostar_station", label: "Eurostar Stations UK &
Europe", color: "#ffd700", type: "point", radius: 5.5, url:
"data/eurostar.geojson", filter: ['==', ['get', 'type'], 'eurostar_station'],
preload: false }
    ]
},
{
    group: "Leisure & Sports (GeoJSON)",
    layers: [
        { id: "stadiums", label: "Major Stadiums", color: "#e5ff00",
type: "point", radius: 5.5, url: "data/stadiums.geojson", preload: false }
    ]
},
{
    group: "Global Data Backbone",
    layers: [
        { id: "subsea_data_cables", label: "Subsea Data Cables (Fibre)
(not all connections available yet)", color: "#00e5ff", type: "line", width:
1.5, url: "data/subsea_data_cables.geojson", preload: false }
    ]
},
{
    group: "Global Supply Chain & Chokepoints",
    layers: [

```



```

        { id: "global_ports_major", label: "Major Ports (Deep
Water/Ferry)", color: "#00bfff", type: "point", radius: ['interpolate',
['linear'], ['get', 'area_ha'], 0, 8, 80, 10, 500, 16], url:
"data/global_ports.geojson", filter: ['==', ['get', 'type'], 'Major
Cargo/Container Port'], preload: false },
        { id: "global_ports_minor", label: "Inland / Minor Harbours",
color: "#0055aa", type: "point", radius: ['interpolate', ['linear'], ['get',
'area_ha'], 0, 2.5, 50, 4.5, 200, 6], url: "data/global_ports.geojson", filter:
['==', ['get', 'type'], 'Minor/Local Harbour'], preload: false },
        { id: "global_hydrocarbons", label: "Global Hydrocarbons (Oil,
Gas, LNG)", color: "#ff3300", type: "point", radius: ['interpolate', ['linear'],
['get', 'area_ha'], 0, 4, 100, 8, 1000, 15], url:
"data/global_hydrocarbons.geojson", preload: false }
    ]
},
{
    group: "Highway Infrastructure",
    layers: [
        { id: "motorways", label: "Motorways", color: "#0054ff", type:
"line", width: 2.5, url: "/uk_motorways.geojson", preload: false },
        { id: "trunk_roads", label: "Major A-Roads (Trunk)", color:
"#00cc00", type: "line", width: 1.8, url: "/uk_trunk_roads.geojson", preload:
false },
        { id: "primary_roads", label: "A-Roads (Primary)", color:
"#ff4400", type: "line", width: 1.2, url: "/uk_primary_roads.geojson", preload:
false },
        { id: "motorway_services", label: "Motorway Services", color:
"#ff00ff", type: "point", radius: 5.5, url: "data/motorway_services.geojson",
preload: false }
    ]
},
{
    group: "EV (GeoJSON)",
    layers: [
        { id: "ev", label: "EV Rapid (100kW+)", color: "#00ff88", type:
"point", radius: 5, url: "data/ev_chargers.geojson", preload: false }
    ]
},
{
    group: "Heavy Industry",
    layers: [
        { id: "naei_co2", label: "Major Industrial Sites", color:
"#ff4400", type: "point", url: "/heavy_emitters_uk.json", preload: false }
    ]
},
{
    group: "REPD (GeoJSON)",
    layers: [
        { id: "solar", label: "Solar PV", color:
"#ffff00", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'solar'], preload: false },
        { id: "solar_operational", label: "Solar PV (Operational Only)",
color: "#00ff88", type: "point", radius: ['interpolate', ['linear'],
['coalesce', ['get', 'capacity'], 0], 0, 10, 10, 14, 50, 18, 100, 22, 200, 28,
350, 35, 500, 42], url: "/dist/repd_master.json", filter: ['all', ['==', ['get',
'tech'], 'solar'], ['==', ['get', 'status'], 'operational']], preload: false },
        { id: "solar_roof", label: "Solar Roof", color:
"#ffcc00", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'solar_roof'], preload: false },
        { id: "wind", label: "Wind", color:
"#00ffff", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'wind'], preload: false },

```

```

        { id: "wind_onshore_operational", label: "Onshore Wind
(Operational Only)", color: "#00ffcc", type: "point", radius: ['interpolate',
['linear'], ['coalesce', ['get', 'capacity'], 0], 0, 8, 10, 12, 50, 16, 100, 20,
200, 26, 350, 32, 500, 38], url: "/dist/repd_master.json", filter: ['all',
['==', ['get', 'raw_tech'], 'Wind Onshore'], ['==', ['get', 'status'],
'operational']], preload: false },
        { id: "wind_offshore_operational", label: "Offshore Wind
(Operational Only)", color: "#0066ff", type: "point", radius: ['interpolate',
['linear'], ['coalesce', ['get', 'capacity'], 0], 0, 8, 10, 12, 50, 16, 100, 20,
200, 26, 350, 32, 500, 38], url: "/dist/repd_master.json", filter: ['all',
['==', ['get', 'raw_tech'], 'Wind Offshore'], ['==', ['get', 'status'],
'operational']], preload: false },
        { id: "bess", label: "Battery Storage", color:
"#ffae00", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'bess'], preload: false },
        { id: "bess_operational", label: "Battery Storage (Operational
Only)", color: "#ff69b4", type: "point", radius: ['interpolate', ['linear'],
['coalesce', ['get', 'capacity'], 0], 0, 8, 10, 12, 50, 16, 100, 20, 200, 26,
350, 32, 500, 38], url: "/dist/repd_master.json", filter: ['all', ['==', ['get',
'tech'], 'bess'], ['==', ['get', 'status'], 'operational']], preload: false },
        { id: "biomass", label: "Biomass & EfW", color:
"#39ff14", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'biomass'], preload: false },
        { id: "tidal", label: "Tidal & Wave", color:
"#00bfff", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'tidal'], preload: false },
        { id: "hydrogen", label: "Hydrogen", color:
"#ffffff", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'hydrogen'], preload: false },
        { id: "hydro", label: "Hydro & Pumped Storage", color:
"#00aaff", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'hydro'], preload: false },
        { id: "flywheel", label: "Flywheel", color:
"#ff69b4", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'flywheel'], preload: false },
        { id: "act", label: "Advanced Conversion", color:
"#ff6600", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'act'], preload: false },
        { id: "geothermal", label: "Geothermal", color:
"#ff3300", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'geothermal'], preload: false },
        { id: "caes", label: "Compressed Air Storage", color:
"#88aaff", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'caes'], preload: false }
    ]
};

window.initVentusMap({
    config: ukConfig,
    center: [-3.5, 54.0],
    zoom: 4.2
});
</script>

```

```

<div id="gridatlas-corridor-sheet" role="dialog" aria-label="Route corridor
estimates" data-open="0" hidden=""></div></body></html>

```

```

=====
FILE: https://ventusltd.github.io/gridatlas/atlas/

```

5221 chars · code · shown in full · found via the page itself

```
=====
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width,initial-scale=1">
  <title>Grid Atlas</title>
  <style>
    html,body{height:100%;margin:0;background:#000;color:#00ffff;font:13px
monospace}
    #gridatlas-loader{display:grid;place-
items:center;height:100%;padding:24px;text-align:center}
    #gridatlas-loader strong{color:#ffae00}
  </style>
</head>
<body data-gridatlas-router="loading">
  <main id="gridatlas-loader"><div><strong>GRIDATLAS</strong><br>verifying
immutable shell and cartridges...</div></main>
  <script type="module">
    const SCRIPT_CLOSE = '</scr' + 'ipt>';
    const loader = document.getElementById('gridatlas-loader');

    function invariant(condition, message) {
      if (!condition) throw new Error(message);
    }

    function escapeAttribute(value) {
      return String(value).replace(/&/g, '&amp;').replace(/"/g,
'&quot;').replace(/</g, '&lt;');
    }

    async function sha256Hex(bytes) {
      const digest = await crypto.subtle.digest('SHA-256', bytes);
      return [...new Uint8Array(digest)].map(value =>
value.toString(16).padStart(2, '0')).join('');
    }

    function fail(error) {
      console.error('[GRIDATLAS COMPOSER]', error);
      document.body.dataset.gridatlasRouter = 'failed';
      loader.innerHTML = `<div><strong>GRIDATLAS COMPOSITION FAILED</strong>
<br>${String(error?.message || error).replace(/[<>]/g, '')}<br><small>The
immutable shell was not changed.</small></div>`;
    }

    try {
      const currentResponse = await fetch('./current.json', { cache: 'no-store'
});
      invariant(currentResponse.ok, `current.json HTTP
${currentResponse.status}`);
      const current = await currentResponse.json();
      invariant(current?.schema === 'gridatlas.current.v2', 'unsupported
current.json schema');
      invariant(current?.architecture ===
'IMMUTABLE_SHELL_PLUS_HASHED_CARTRIDGES', 'architecture contract mismatch');
      invariant(Array.isArray(current.cartridge_order) &&
Array.isArray(current.cartridges), 'cartridge registry malformed');

      const shellUrl = new URL(current.shell.index, window.location.href);
      const shellBase = new URL(current.shell.base, window.location.href).href;
      const shellResponse = await fetch(shellUrl, { cache: 'no-store' });
```

```

    invariant(shellResponse.ok, `immutable shell HTTP
${shellResponse.status}`);
    let html = await shellResponse.text();
    invariant(/<head\b/i.test(html) && /<body\b/i.test(html), 'immutable shell
is not HTML');
    html = html.replace(/<head([>]*)>/i, match => `${match}\n<base
href="${escapeAttribute(shellBase)}">`);

    const byId = new Map(current.cartridges.map(item => [item.id, item]));
    const loaded = [];
    const scriptTagPattern = new RegExp('<script\\b[>]*\\bsrc=([\"\\\'\\'])'
([^\\"\\\'\\']+)*\\b[>]*>\\s*</script>', 'gi');

    for (const id of current.cartridge_order) {
        const cartridge = byId.get(id);
        invariant(cartridge, `missing cartridge ${id}`);
        invariant(cartridge.type === 'script', `${id}: unsupported cartridge
type`);
        invariant(cartridge.slot === 'replace-script', `${id}: unsupported
cartridge slot`);
        invariant(typeof cartridge.replace_script === 'string' &&
cartridge.replace_script, `${id}: replace_script missing`);
        invariant(/^[a-f0-9]{64}$/.test(cartridge.sha256 || ''), `${id}: invalid
SHA-256`);

        const cartridgeUrl = new URL(cartridge.path, window.location.href);
        const response = await fetch(cartridgeUrl, { cache: 'no-store' });
        invariant(response.ok, `${id}: HTTP ${response.status}`);
        const bytes = await response.arrayBuffer();
        invariant(await sha256Hex(bytes) === cartridge.sha256, `${id}: SHA-256
mismatch`);
        const blobUrl = URL.createObjectURL(new Blob([bytes], { type:
'text/javascript' }));

        let replaced = false;
        html = html.replace(scriptTagPattern, (full, quote, src) => {
            if (replaced || String(src).split('/').pop() !==
cartridge.replace_script) return full;
            replaced = true;
            return `<script data-gridatlas-cartridge="${escapeAttribute(id)}"
src="${escapeAttribute(blobUrl)}">${SCRIPT_CLOSE}`;
        });
        invariant(replaced, `${id}: shell slot ${cartridge.replace_script} not
found`);
        loaded.push({ id, sha256: cartridge.sha256, blob_url: blobUrl });
    }

    const publicState = JSON.stringify({
        schema: current.schema,
        generation: current.generation,
        release_id: current.release_id,
        architecture: current.architecture,
        shell: current.shell,
        cartridge_order: current.cartridge_order,
        loaded_cartridges: loaded.map(item => ({ id: item.id, sha256:
item.sha256 })))
    }).replace(/</g, '\\u003c');
    const stateScript =
`<script>window.__GRIDATLAS_ATLAS__=${publicState};document.documentElement.data
set.gridatlasGeneration=${JSON.stringify(current.generation)};${SCRIPT_CLOSE}`;
    html = html.replace(/<body([>]*)>/i, match =>
`${match}\n${stateScript}`);

```

```

        document.open();
        document.write(html);
        document.close();
    } catch (error) {
        fail(error);
    }
</script>
</body>
</html>

```

```

=====
FILE: https://cdn.jsdelivr.net/npm/maplibre-gl@3.6.2/dist/maplibre-gl.js
762513 chars · code · shown in full · found via <script src>, loaded by the
browser (script)
=====

```

```

/* MapLibre GL JS is licensed under the 3-Clause BSD License. Full text of
license: https://github.com/maplibre/maplibre-gl-js/blob/v3.6.2/LICENSE.txt */
(function (global, factory) {
  typeof exports === 'object' && typeof module !== 'undefined' ? module.exports =
  factory() :
  typeof define === 'function' && define.amd ? define(factory) :
  (global = typeof globalThis !== 'undefined' ? globalThis : global || self,
  global.maplibregl = factory());
})(this, (function () { 'use strict';

/* eslint-disable */

var shared, worker, maplibregl;
// define gets called three times: one for each chunk. we rely on the order
// they're imported to know which is which
function define(_, chunk) {
  if (!shared) {
    shared = chunk;
  } else if (!worker) {
    worker = chunk;
  } else {
    var workerBundleString = 'var sharedChunk = {}; (' + shared + ')
(sharedChunk); (' + worker + ')(sharedChunk);'

    var sharedChunk = {};
    shared(sharedChunk);
    maplibregl = chunk(sharedChunk);
    if (typeof window !== 'undefined') {
      maplibregl.workerUrl = window.URL.createObjectURL(new
Blob([workerBundleString], { type: 'text/javascript' }));
    }
  }
}

```

```

define(["exports"], (function(t){ "use strict"; function e(t,e,r,n){return new(r||
(r=Promise))((function(i,a){function s(t){try{l(n.next(t));}catch(t)
{a(t);}}function o(t){try{l(n.throw(t));}catch(t){a(t);}}function l(t){var
e;t.done?i(t.value):(e=t.value,e instanceof r?e=new r((function(t)
{t(e);}))).then(s,o);l((n=n.apply(t,e|[])).next());}))}function r(t){return
t&&t.__esModule&&Object.prototype.hasOwnProperty.call(t,"default")?
t.default:t}"function"==typeof SuppressedError&&SuppressedError;var n=i;function
i(t,e){this.x=t,this.y=e;}i.prototype={clone:function(){return new
i(this.x,this.y),add:function(t){return this.clone()._add(t)},sub:function(t)
{return this.clone()._sub(t)},multByPoint:function(t){return

```

```

this.clone()._multByPoint(t)},divByPoint:function(t){return
this.clone()._divByPoint(t)},mult:function(t){return
this.clone()._mult(t)},div:function(t){return
this.clone()._div(t)},rotate:function(t){return
this.clone()._rotate(t)},rotateAround:function(t,e){return
this.clone()._rotateAround(t,e)},matMult:function(t){return
this.clone()._matMult(t)},unit:function(){return
this.clone()._unit()},perp:function(){return
this.clone()._perp()},round:function(){return
this.clone()._round()},mag:function(){return
Math.sqrt(this.x*this.x+this.y*this.y)},equals:function(t){return
this.x===t.x&&this.y===t.y},dist:function(t){return
Math.sqrt(this.distSqr(t))},distSqr:function(t){var e=t.x-this.x,r=t.y-
this.y;return e*e+r*r},angle:function(){return
Math.atan2(this.y,this.x)},angleTo:function(t){return Math.atan2(this.y-
t.y,this.x-t.x)},angleWith:function(t){return
this.angleWithSep(t.x,t.y)},angleWithSep:function(t,e){return
Math.atan2(this.x*e-this.y*t,this.x*t+this.y*e)},_matMult:function(t){var
e=t[2]*this.x+t[3]*this.y;return
this.x=t[0]*this.x+t[1]*this.y,this.y=e,this},_add:function(t){return
this.x+=t.x,this.y+=t.y,this},_sub:function(t){return this.x-=t.x,this.y-
=t.y,this},_mult:function(t){return this.x*=t,this.y*=t,this},_div:function(t)
{return this.x/=t,this.y/=t,this},_multByPoint:function(t){return
this.x*=t.x,this.y*=t.y,this},_divByPoint:function(t){return
this.x/=t.x,this.y/=t.y,this},_unit:function(){return
this._div(this.mag()),this},_perp:function(){var t=this.y;return
this.y=this.x,this.x=-t,this},_rotate:function(t){var
e=Math.cos(t),r=Math.sin(t),n=r*this.x+e*this.y;return this.x=e*this.x-
r*this.y,this.y=n,this},_rotateAround:function(t,e){var
r=Math.cos(t),n=Math.sin(t),i=e.y+n*(this.x-e.x)+r*(this.y-e.y);return
this.x=e.x+r*(this.x-e.x)-n*(this.y-e.y),this.y=i,this},_round:function(){return
this.x=Math.round(this.x),this.y=Math.round(this.y),this},i.convert=function(t)
{return t instanceof i?t:Array.isArray(t)?new i(t[0],t[1]):t;var
a=r(n),s=o;function o(t,e,r,n){this.cx=3*t,this.bx=3*(r-t)-this.cx,this.ax=1-
this.cx-this.bx,this.cy=3*e,this.by=3*(n-e)-this.cy,this.ay=1-this.cy-
this.by,this.plx=t,this.ply=e,this.p2x=r,this.p2y=n;o.prototype=
{sampleCurveX:function(t){return
((this.ax*t+this.bx)*t+this.cx)*t},sampleCurveY:function(t){return
((this.ay*t+this.by)*t+this.cy)*t},sampleCurveDerivativeX:function(t){return
(3*this.ax*t+2*this.bx)*t+this.cx},solveCurveX:function(t,e){if(void 0===e&&
(e=1e-6),t<0)return 0;if(t>1)return 1;for(var r=t,n=0;n<8;n++){var
i=this.sampleCurveX(r)-t;if(Math.abs(i)<e)return r;var
a=this.sampleCurveDerivativeX(r);if(Math.abs(a)<1e-6)break;r-=i/a;}var
s=0,o=1;for(r=t,n=0;n<20&&(i=this.sampleCurveX(r),!(Math.abs(i-t)<e));n++)t>i?
s=r:o=r,r=.5*(o-s)+s;return r},solve:function(t,e){return
this.sampleCurveY(this.solveCurveX(t,e))};var l=r(s);let u,c;function h()
{return null==u&&(u="undefined"!==typeof OffscreenCanvas&&new
OffscreenCanvas(1,1).getContext("2d")&&"function"===typeof
createImageBitmap),u}function p(){if(null==c&&(c=!1,h())){const t=5,e=new
OffscreenCanvas(t,t).getContext("2d",{willReadFrequently:!0});if(e){for(let
r=0;r<t*t;r++){const
n=4*r;e.fillStyle=`rgb(${n},${n+1},${n+2})`,e.fillRect(r*t,Math.floor(r/t),1,1);
}const r=e.getImageData(0,0,t,t).data;for(let
e=0;e<t*t*4;e++)if(e%4!=3&&r[e]!==e){c=!0;break}}return c||!1}function
f(t,e,r,n){const i=new l(t,e,r,n);return function(t){return i.solve(t)}}const
d=f(.25,.1,.25,1);function y(t,e,r){return Math.min(r,Math.max(e,t))}function
m(t,e,r){const n=r-e,i=((t-e)%n+n)%n+e;return i===e?r:i}function g(t,...e)
{for(const r of e)for(const e in r)t[e]=r[e];return t}let x=1;function v(t,e,r)
{const n={};for(const i in t)n[i]=e.call(r||this,t[i],i,t);return n}function
b(t,e,r){const n={};for(const i in t)e.call(r||this,t[i],i,t)&&
(n[i]=t[i]);return n}function w(t){return Array.isArray(t)?
t.map(w):"object"===typeof t&&t?v(t,w):t}const _={};function A(t){_[t]||

```

```

("undefined"! =typeof console&&console.warn(t),_[t]=!0);}function S(t,e,r){return
(r.y-t.y)*(e.x-t.x)>(e.y-t.y)*(r.x-t.x)}function k(t){let e=0;for(let
r,n,i=0,a=t.length,s=a-1;i<a;s=i++)r=t[i],n=t[s],e+=(n.x-r.x)*(r.y+n.y);return
e}function I(){return "undefined"! =typeof WorkerGlobalScope&&"undefined"! =typeof
self&&self instanceof WorkerGlobalScope}let z=null;function M(t){return
"undefined"! =typeof ImageBitmap&&t instanceof ImageBitmap}const
C="
YV2NgAAIAAAUAArVyFEAAAAASUVORK5CYII=";function B(t,r,n,i,a){return e(this,void
0,void 0,(function*(){if("undefined"==typeof VideoFrame)throw new
Error("VideoFrame not supported");const e=new VideoFrame(t,
{timestamp:0});try{const s=null==e?void
0:e.format;if(!s||!s.startsWith("BGR")&&!s.startsWith("RGB"))throw new
Error("Unrecognized format "+s)};const o=s.startsWith("BGR"),l=new
Uint8ClampedArray(i*a*4);if(yield e.copyTo(l,function(t,e,r,n,i){const
a=4*Math.max(-e,0),s=(Math.max(0,r)-
r)*n*4+a,o=4*n,l=Math.max(0,e),u=Math.max(0,r);return {rect:
{x:l,y:u,width:Math.min(t.width,e+n)-l,height:Math.min(t.height,r+i)-u},layout:
[{offset:s,stride:o}]}(t,r,n,i,a),o)for(let t=0;t<l.length;t+=4){const
e=l[t];l[t]=l[t+2],l[t+2]=e;}return l}finally{e.close();}))}let P,V,E,F;const
T={now:"undefined"! =typeof performance&&performance&&performance.now?
performance.now.bind(performance):Date.now.bind(Date),frame(t){const
e=requestAnimationFrame(t);return {cancel:
()=>cancelAnimationFrame(e)}},getImageData(t,e=0){return
this.getImageCanvasContext(t).getImageData(-e,-
e,t.width+2*e,t.height+2*e)},getImageCanvasContext(t){const
e=window.document.createElement("canvas"),r=e.getContext("2d",
{willReadFrequently:!0});if(!r)throw new Error("failed to create canvas 2d
context");return
e.width=t.width,e.height=t.height,r.drawImage(t,0,0,t.width,t.height),r},resolve
URL:t=>(E||
(E=document.createElement("a")),E.href=t,E.href),hardwareConcurrency:"undefined"
!=typeof navigator&&navigator.hardwareConcurrency||4,get prefersReducedMotion()
{return !!matchMedia&&(null==F&&(F=matchMedia("(prefers-reduced-motion:
reduce)")),F.matches)}},$=
{MAX_PARALLEL_IMAGE_REQUESTS:16,MAX_PARALLEL_IMAGE_REQUESTS_PER_FRAME:8,MAX_TILE
_CACHE_ZOOM_LEVELS:5,REGISTERED_PROTOCOLS:{},WORKER_URL:""};class L extends
Error{constructor(t,e,r,n){super(`AJAXError: ${e} (${t})`);
this.status=t,this.statusText=e,this.url=r,this.body=n;}}const D=I()?
()=>self.worker&&self.worker.referrer:()=>("blob:"===window.location.protocol?
window.parent:window).location.href,O=t=>$.REGISTERED_PROTOCOLS[t.substring(0,t.
indexOf(":/"))];function U(t,e){const r=new AbortController,n=new
Request(t.url,
{method:t.method||"GET",body:t.body,credentials:t.credentials,headers:t.headers,
cache:t.cache,referrer:D(),signal:r.signal});let
i=!1,a=!1;"json"===t.type&&n.headers.set("Accept","application/json");return
a||fetch(n).then((r=>r.ok?(r=>{"arrayBuffer"===t.type||"image"===t.type?
r.arrayBuffer():"json"===t.type?r.json():r.text()).then((t=>{a||
(i=!0,e(null,t,r.headers.get("Cache-
Control"),r.headers.get("Expires")));})).catch((t=>{a||e(new
Error(t.message));}));}))(r:r.blob().then((n=>e(new
L(r.status,r.statusText,t.url,n))))).catch((t=>{20!==(t.code&&e(new
Error(t.message));})),{cancel:()=>{a=!0,i||r.abort();}})}const R=function(t,e)
{if(/:\//.test(t.url)&&!/^https?:|^file:/.test(t.url))
{if(I())&&self.worker&&self.worker.actor)return
self.worker.actor.send("getResource",t,e);if(!I())return (O(t.url)||U)(t,e)}if(!
(/^file:/.test(r=t.url)||/^file:/.test(D())&&!/^w+/.test(r)))
{if(fetch&&Request&&AbortController&&Object.prototype.hasOwnProperty.call(Reques
t.prototype,"signal"))return
U(t,e);if(I())&&self.worker&&self.worker.actor)return
self.worker.actor.send("getResource",t,e,void 0,!0)}var r;return function(t,e)
{const r=new
XMLHttpRequest;r.open(t.method||"GET",t.url,!0),"arrayBuffer"! =t.type&&"image"!

```

```

==t.type||(r.responseType="arraybuffer");for(const e in
t.headers)r.setRequestHeader(e,t.headers[e]);return "json"===t.type&&
(r.responseType="text",r.setRequestHeader("Accept","application/json")),r.withCr
edentials="include"===t.credentials,r.onerror={()=>{e(new
Error(r.statusText));},r.onload={()=>
{if((r.status>=200&&r.status<300||0===r.status)&&null!==r.response){let
n=r.response;if("json"===t.type)try{n=JSON.parse(r.response);}catch(t){return
e(t)}e(null,n,r.getResponseHeader("Cache-
Control"),r.getResponseHeader("Expires"))};else {const n=new Blob([r.response],
{type:r.getResponseHeader("Content-Type")});e(new
L(r.status,r.statusText,t.url,n));}},r.send(t.body),{cancel:()=>r.abort()}}
(t,e)},q=function(t,e){return R(g(t,{type:"arrayBuffer"}),e)};function j(t)
{if(!t||t.indexOf("://")
<=0||0===t.indexOf("data:image/")||0===t.indexOf("blob:"))return !0;const e=new
URL(t),r=window.location;return
e.protocol===r.protocol&&e.host===r.host}function N(t,e,r)
{r[t]&&-1!==r[t].indexOf(e)||r[t]=r[t]||[],r[t].push(e)};function Z(t,e,r)
{if(r&&r[t]){const n=r[t].indexOf(e);-1!==n&&r[t].splice(n,1)}}class
K{constructor(t,e={}){g(this,e),this.type=t}}class G extends K{constructor(t,e=
{}){super("error",g({error:t},e))}}class J{on(t,e){return
this._listeners=this._listeners||[],N(t,e,this._listeners),this}off(t,e){return
Z(t,e,this._listeners),Z(t,e,this._oneTimeListeners),this}once(t,e){return e?
(this._oneTimeListeners=this._oneTimeListeners||
[],N(t,e,this._oneTimeListeners),this):new
Promise((e=>this.once(t,e))){fire(t,e){"string"===typeof t&&(t=new K(t,e||
{}))};const r=t.type;if(this._listeners(r)){t.target=this;const
e=this._listeners&&this._listeners[r]?this._listeners[r].slice():[];for(const r
of e)r.call(this,t);const n=this._oneTimeListeners&&this._oneTimeListeners[r]?
this._oneTimeListeners[r].slice():[];for(const e of
n)Z(r,e,this._oneTimeListeners),e.call(this,t);const i=this._eventedParent;i&&
(g(t,"function"===typeof this._eventedParentData?
this._eventedParentData():this._eventedParentData),i.fire(t))};else t instanceof
G&&console.error(t.error);return this}listens(t){return
this._listeners&&this._listeners[t]&&this._listeners[t].length>0||this._oneTimeL
isteners&&this._oneTimeListeners[t]&&this._oneTimeListeners[t].length>0||this._e
ventedParent&&this._eventedParent.listens(t)}setEventedParent(t,e){return
this._eventedParent=t,this._eventedParentData=e,this}}var X={$version:8,$root:
{version:{required:!0,type:"enum",values:[8]},name:{type:"string"},metadata:
{type:"*"},center:{type:"array",value:"number"},zoom:{type:"number"},bearing:
{type:"number",default:0,period:360,units:"degrees"},pitch:
{type:"number",default:0,units:"degrees"},light:{type:"light"},terrain:
{type:"terrain"},sources:{required:!0,type:"sources"},sprite:
{type:"sprite"},glyphs:{type:"string"},transition:{type:"transition"},layers:
{required:!0,type:"array",value:"layer"},sources:{type:"source"},source:
["source_vector","source_raster","source_raster_dem","source_geojson","source_vi
deo","source_image"],source_vector:{type:{required:!0,type:"enum",values:
{vector:{}}},url:{type:"string"},tiles:{type:"array",value:"string"},bounds:
{type:"array",value:"number",length:4,default:
[-180,-85.051129,180,85.051129]},scheme:{type:"enum",values:{xyz:{},tms:
{}},default:"xyz"},minzoom:{type:"number",default:0},maxzoom:
{type:"number",default:22},attribution:{type:"string"},promoteId:
{type:"promoteId"},volatile:{type:"boolean",default:!1},"*":
{type:"*"}},source_raster:{type:{required:!0,type:"enum",values:{raster:
{}},url:{type:"string"},tiles:{type:"array",value:"string"},bounds:
{type:"array",value:"number",length:4,default:
[-180,-85.051129,180,85.051129]},minzoom:{type:"number",default:0},maxzoom:
{type:"number",default:22},tileSize:
{type:"number",default:512,units:"pixels"},scheme:{type:"enum",values:{xyz:
{}},tms:{},default:"xyz"},attribution:{type:"string"},volatile:
{type:"boolean",default:!1},"*":{type:"*"}},source_raster_dem:{type:
{required:!0,type:"enum",values:{raster-dem:{}}},url:{type:"string"},tiles:
{type:"array",value:"string"},bounds:

```



```

{type:"array",value:"number",length:4,default:
[-180,-85.051129,180,85.051129]},minzoom:{type:"number",default:0},maxzoom:
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{type:"number",default:512,units:"pixels"},attribution:{type:"string"},encoding:
{type:"enum",values:{terrarium:{},mapbox:{},custom:
{}},default:"mapbox"},redFactor:{type:"number",default:1},blueFactor:
{type:"number",default:1},greenFactor:{type:"number",default:1},baseShift:
{type:"number",default:0},volatile:{type:"boolean",default:!1},"*":
{type:"*"}},source_geojson:{type:{required:!0,type:"enum",values:{geojson:
{}}},data:{required:!0,type:"*"},maxzoom:{type:"number",default:18},attribution:
{type:"string"},buffer:{type:"number",default:128,maximum:512,minimum:0},filter:
{type:"*"},tolerance:{type:"number",default:.375},cluster:
{type:"boolean",default:!1},clusterRadius:
{type:"number",default:50,minimum:0},clusterMaxZoom:
{type:"number"},clusterMinPoints:{type:"number"},clusterProperties:
{type:"*"},lineMetrics:{type:"boolean",default:!1},generateId:
{type:"boolean",default:!1},promoteId:{type:"promoteId"}},source_video:{type:
{required:!0,type:"enum",values:{video:{}}},urls:
{required:!0,type:"array",value:"string"},coordinates:
{required:!0,type:"array",length:4,value:
{type:"array",length:2,value:"number"}}},source_image:{type:
{required:!0,type:"enum",values:{image:{}}},url:
{required:!0,type:"string"},coordinates:
{required:!0,type:"array",length:4,value:
{type:"array",length:2,value:"number"}}},layer:{id:
{type:"string",required:!0},type:{type:"enum",values:{fill:{},line:{},symbol:
{}},circle:{},heatmap:{},fill-extrusion:{},raster:{},hillshade:{},background:
{}},required:!0},metadata:{type:"*"},source:{type:"string"},"source-layer":
{type:"string"},minzoom:{type:"number",minimum:0,maximum:24},maxzoom:
{type:"number",minimum:0,maximum:24},filter:{type:"filter"},layout:
{type:"layout"},paint:{type:"paint"}},layout:
["layout_fill","layout_line","layout_circle","layout_heatmap","layout_fill-
extrusion","layout_symbol","layout_raster","layout_hillshade","layout_background
"],layout_background:{visibility:{type:"enum",values:{visible:{},none:
{}},default:"visible"},"property-type":"constant"}},layout_fill:{fill-sort-key":
{type:"number",expression:{interpolated:!1,parameters:
["zoom","feature"]},"property-type":"data-driven"},visibility:
{type:"enum",values:{visible:{},none:{},default:"visible"},"property-
type":"constant"}},layout_circle:{circle-sort-key":{type:"number",expression:
{interpolated:!1,parameters:["zoom","feature"]},"property-type":"data-
driven"},visibility:{type:"enum",values:{visible:{},none:
{}},default:"visible"},"property-type":"constant"}},layout_heatmap:{visibility:
{type:"enum",values:{visible:{},none:{},default:"visible"},"property-
type":"constant"}},layout_fill-extrusion:{visibility:{type:"enum",values:
{visible:{},none:{},default:"visible"},"property-type":"constant"}},layout_line:
{"line-cap":{type:"enum",values:{butt:{},round:{},square:
{}},default:"butt",expression:{interpolated:!1,parameters:["zoom"]},"property-
type":"data-constant"},"line-join":{type:"enum",values:{bevel:{},round:{},miter:
{}},default:"miter",expression:{interpolated:!1,parameters:
["zoom","feature"]},"property-type":"data-driven"},"line-miter-limit":
{type:"number",default:2,requires:[{"line-join":"miter"}],expression:
{interpolated:!0,parameters:["zoom"]},"property-type":"data-constant"},"line-
round-limit":{type:"number",default:1.05,requires:[{"line-
join":"round"}],expression:{interpolated:!0,parameters:["zoom"]},"property-
type":"data-constant"},"line-sort-key":{type:"number",expression:
{interpolated:!1,parameters:["zoom","feature"]},"property-type":"data-
driven"},visibility:{type:"enum",values:{visible:{},none:
{}},default:"visible"},"property-type":"constant"}},layout_symbol:{symbol-
placement":{type:"enum",values:{point:{},line:{},line-center":
{}},default:"point",expression:{interpolated:!1,parameters:["zoom"]},"property-
type":"data-constant"},"symbol-spacing":
{type:"number",default:250,minimum:1,units:"pixels",requires:[{"symbol-

```

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placement": "line"]], expression: {interpolated: !0, parameters: ["zoom"]}, "property-
type": "data-constant"}, "symbol-avoid-edges":
{type: "boolean", default: !1, expression: {interpolated: !1, parameters:
["zoom"]}, "property-type": "data-constant"}, "symbol-sort-key":
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["zoom", "feature"]}, "property-type": "data-driven"}, "symbol-z-order":
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{}}, default: "auto", expression: {interpolated: !1, parameters: ["zoom"]}, "property-
type": "data-constant"}, "icon-allow-overlap": {type: "boolean", default: !1, requires:
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["zoom"]}, "property-type": "data-constant"}, "icon-overlap": {type: "enum", values:
{never: {}, always: {}, cooperative: {}}, requires: ["icon-image"], expression:
{interpolated: !1, parameters: ["zoom"]}, "property-type": "data-constant"}, "icon-
ignore-placement": {type: "boolean", default: !1, requires: ["icon-image"], expression:
{interpolated: !1, parameters: ["zoom"]}, "property-type": "data-constant"}, "icon-
optional": {type: "boolean", default: !1, requires: ["icon-image", "text-
field"], expression: {interpolated: !1, parameters: ["zoom"]}, "property-type": "data-
constant"}, "icon-rotation-alignment": {type: "enum", values: {map: {}, viewport:
{}}, auto: {}}, default: "auto", requires: ["icon-image"], expression:
{interpolated: !1, parameters: ["zoom"]}, "property-type": "data-constant"}, "icon-
size": {type: "number", default: 1, minimum: 0, units: "factor of the original icon
size", requires: ["icon-image"], expression: {interpolated: !0, parameters:
["zoom", "feature"]}, "property-type": "data-driven"}, "icon-text-fit":
{type: "enum", values: {none: {}, width: {}, height: {}, both:
{}}, default: "none", requires: ["icon-image", "text-field"], expression:
{interpolated: !1, parameters: ["zoom"]}, "property-type": "data-constant"}, "icon-
text-fit-padding": {type: "array", value: "number", length: 4, default:
[0, 0, 0, 0], units: "pixels", requires: ["icon-image", "text-field", {"icon-text-fit":
["both", "width", "height"]}], expression: {interpolated: !0, parameters:
["zoom"]}, "property-type": "data-constant"}, "icon-image":
{type: "resolvedImage", tokens: !0, expression: {interpolated: !1, parameters:
["zoom", "feature"]}, "property-type": "data-driven"}, "icon-rotate":
{type: "number", default: 0, period: 360, units: "degrees", requires: ["icon-
image"], expression: {interpolated: !0, parameters: ["zoom", "feature"]}, "property-
type": "data-driven"}, "icon-padding": {type: "padding", default:
[2], units: "pixels", requires: ["icon-image"], expression:
{interpolated: !0, parameters: ["zoom", "feature"]}, "property-type": "data-
driven"}, "icon-keep-upright": {type: "boolean", default: !1, requires: ["icon-image",
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["type","source","source-layer","minzoom","maxzoom","filter","layout"];function
H(t,e){const r={};for(const e in t)"ref"!==e&&(r[e]=t[e]);return Y.forEach((t=>

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{t in e&&(r[t]=e[t]);}),r}function W(t,e){if(Array.isArray(t))
{if(!Array.isArray(e)||t.length!==e.length)return !1;for(let
r=0;r<t.length;r++)if(!W(t[r],e[r]))return !1;return !0}if("object"===typeof
t&&null!==t&&null!==e){if("object"!==typeof e)return
!1;if(Object.keys(t).length!==Object.keys(e).length)return !1;for(const r in
t)if(!W(t[r],e[r]))return !1;return !0}return t===e}const Q=
{setStyle:"setStyle",addLayer:"addLayer",removeLayer:"removeLayer",setPaintPrope
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oJSONSourceData",setLayerZoomRange:"setLayerZoomRange",setLayerProperty:"setLaye
rProperty",setCenter:"setCenter",setZoom:"setZoom",setBearing:"setBearing",setPi
tch:"setPitch",setSprite:"setSprite",setGlyphs:"setGlyphs",setTransition:"setTra
nsition",setLight:"setLight"};function tt(t,e,r)
{r.push({command:Q.addSource,args:[t,e[t]]});}function et(t,e,r)
{e.push({command:Q.removeSource,args:[t]}),r[t]=!0;}function rt(t,e,r,n)
{et(t,r,n),tt(t,e,r);}function nt(t,e,r){let n;for(n in
t[r])if(Object.prototype.hasOwnProperty.call(t[r],n)&&"data"!==n&&!W(t[r]
[n],e[r][n]))return !1;for(n in
e[r])if(Object.prototype.hasOwnProperty.call(e[r],n)&&"data"!==n&&!W(t[r]
[n],e[r][n]))return !1;return !0}function it(t,e,r,n,i,a){let s;for(s in e=e||
{}),t=t||{}Object.prototype.hasOwnProperty.call(t,s)&&
(W(t[s],e[s])||r.push({command:a,args:[n,s,e[s],i]}));for(s in
e)Object.prototype.hasOwnProperty.call(e,s)&&!Object.prototype.hasOwnProperty.ca
ll(t,s)&&(W(t[s],e[s])||r.push({command:a,args:[n,s,e[s],i]}));}function at(t)
{return t.id}function st(t,e){return t[e.id]=e,t}class ot{constructor(t,e,r,n)
{this.message=(t?`${t}: `:"")+r,n&&(this.identifier=n),null!=e&&e.__line__&&
(this.line=e.__line__);}function lt(t,...e){for(const r of e)for(const e in
r)t[e]=r[e];return t}class ut extends Error{constructor(t,e)
{super(e),this.message=e,this.key=t;}class ct{constructor(t,e=[] )
{this.parent=t,this.bindings={};for(const[t,r]of e)this.bindings[t]=r;}concat(t)
{return new ct(this,t)}get(t){if(this.bindings[t])return
this.bindings[t];if(this.parent)return this.parent.get(t);throw new Error(`${t}
not found in scope.`)}has(t){return
!!this.bindings[t]||!this.parent&&this.parent.has(t)}}const ht=
{kind:"null"},pt={kind:"number"},ft={kind:"string"},dt={kind:"boolean"},yt=
{kind:"color"},mt={kind:"object"},gt={kind:"value"},xt={kind:"collator"},vt=
{kind:"formatted"},bt={kind:"padding"},wt={kind:"resolvedImage"},_t=
{kind:"variableAnchorOffsetCollection"};function At(t,e){return
{kind:"array",itemType:t,N:e}}function St(t){if("array"===t.kind){const
e=St(t.itemType);return "number"===typeof t.N?`array<${e},
${t.N}>`:`value`===t.itemType.kind?`array<${e}>`:t.kind}const kt=
[ht,pt,ft,dt,yt,vt,mt,At(gt),bt,wt,_t];function It(t,e)
{if("error"===e.kind)return null;if("array"===t.kind){if("array"===e.kind&&
(0===e.N&&"value"===e.itemType.kind)||!It(t.itemType,e.itemType))&&
("number"!==typeof t.N||t.N===e.N))return null}else {if(t.kind===e.kind)return
null;if("value"===t.kind)for(const t of kt)if(!It(t,e))return null}return
`Expected ${St(t)} but found ${St(e)} instead.`}function zt(t,e){return
e.some((e=>e.kind===t.kind))}function Mt(t,e){return e.some((e=>"null"===e?
null===t:"array"===e?Array.isArray(t):"object"===e?
t&&!Array.isArray(t)&&"object"===typeof t:e===typeof t))}function Ct(t,e){return
"array"===t.kind&&"array"===e.kind?
t.itemType.kind===e.itemType.kind&&"number"===typeof t.N:t.kind===e.kind}const
Bt=.96422,Pt=.82521,Vt=4/29,Et=6/29,Ft=3*Et*Et,Tt=Et*Et*Et,$t=Math.PI/180,Lt=180
/Math.PI;function Dt(t){return (t%360)<0&&(t+=360),t}function Ot([t,e,r,n]){let
i,a;const s=Rt((.2225045*(t=Ut(t))+.7168786*(e=Ut(e))+.0606169*
(r=Ut(r)))/1);t===e&&e===r?i=a:s:
(i=Rt((.4360747*t+.3850649*e+.1430804*r)/Bt),a=Rt((.0139322*t+.0971045*e+.714173
3*r)/Pt));const o=116*s-16;return [o<0?0:o,500*(i-s),200*(s-a),n]}function Ut(t)
{return t<=.04045?t/12.92:Math.pow((t+.055)/1.055,2.4)}function Rt(t){return
t>Tt?Math.pow(t,1/3):t/Ft+Vt}function qt([t,e,r,n]){let i=(t+16)/116,a=isNaN(e)?
i:i+e/500,s=isNaN(r)?i:i-r/200;return i=1*Nt(i),a=Bt*Nt(a),s=Pt*Nt(s),
[jt(3.1338561*a-

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1.6168667*i-.4906146*s),jt(-.9787684*a+1.9161415*i+.033454*s),jt(.0719453*a-.228
9914*i+1.4052427*s),n]}function jt(t){return (t=t<=.00304?
12.92*t:1.055*Math.pow(t,1/2.4)-.055)<0?0:t>1?1:t}function Nt(t){return t>Et?
t*t*t:Ft*(t-Vt)}function Zt(t){return parseInt(t.padEnd(2,t),16)/255}function
Kt(t,e){return Gt(e?t/100:t,0,1)}function Gt(t,e,r){return
Math.min(Math.max(e,t),r)}function Jt(t){return !t.some(Number.isNaN)}const Xt=
{aliceblue:[240,248,255],antiquewhite:[250,235,215],aqua:[0,255,255],aquamarine:
[127,255,212],azure:[240,255,255],beige:[245,245,220],bisque:
[255,228,196],black:[0,0,0],blanchedalmond:[255,235,205],blue:
[0,0,255],blueviolet:[138,43,226],brown:[165,42,42],burlywood:
[222,184,135],cadetblue:[95,158,160],chartreuse:[127,255,0],chocolate:
[210,105,30],coral:[255,127,80],cornflowerblue:[100,149,237],cornsilk:
[255,248,220],crimson:[220,20,60],cyan:[0,255,255],darkblue:[0,0,139],darkcyan:
[0,139,139],darkgoldenrod:[184,134,11],darkgray:[169,169,169],darkgreen:
[0,100,0],darkgrey:[169,169,169],darkkhaki:[189,183,107],darkmagenta:
[139,0,139],darkolivegreen:[85,107,47],darkorange:[255,140,0],darkorchid:
[153,50,204],darkred:[139,0,0],darksalmon:[233,150,122],darkseagreen:
[143,188,143],darkslateblue:[72,61,139],darkslategray:[47,79,79],darkslategrey:
[47,79,79],darkturquoise:[0,206,209],darkviolet:[148,0,211],deeppink:
[255,20,147],deepskyblue:[0,191,255],dimgray:[105,105,105],dimgrey:
[105,105,105],dodgerblue:[30,144,255],firebrick:[178,34,34],floralwhite:
[255,250,240],forestgreen:[34,139,34],fuchsia:[255,0,255],gainsboro:
[220,220,220],ghostwhite:[248,248,255],gold:[255,215,0],goldenrod:
[218,165,32],gray:[128,128,128],green:[0,128,0],greenyellow:[173,255,47],grey:
[128,128,128],honeydew:[240,255,240],hotpink:[255,105,180],indianred:
[205,92,92],indigo:[75,0,130],ivory:[255,255,240],khaki:[240,230,140],lavender:
[230,230,250],lavenderblush:[255,240,245],lawngreen:[124,252,0],lemonchiffon:
[255,250,205],lightblue:[173,216,230],lightcoral:[240,128,128],lightcyan:
[224,255,255],lightgoldenrodyellow:[250,250,210],lightgray:
[211,211,211],lightgreen:[144,238,144],lightgrey:[211,211,211],lightpink:
[255,182,193],lightsalmon:[255,160,122],lightseagreen:[32,178,170],lightskyblue:
[135,206,250],lightslategray:[119,136,153],lightslategrey:
[119,136,153],lightsteelblue:[176,196,222],lightyellow:[255,255,224],lime:
[0,255,0],limegreen:[50,205,50],linen:[250,240,230],magenta:[255,0,255],maroon:
[128,0,0],mediumaquamarine:[102,205,170],mediumblue:[0,0,205],mediumorchid:
[186,85,211],mediumpurple:[147,112,219],mediumseagreen:
[60,179,113],mediumslateblue:[123,104,238],mediumspringgreen:
[0,250,154],mediumturquoise:[72,209,204],mediumvioletred:
[199,21,133],midnightblue:[25,25,112],mintcream:[245,255,250],mistyrose:
[255,228,225],moccasin:[255,228,181],navajowhite:[255,222,173],navy:
[0,0,128],oldlace:[253,245,230],olive:[128,128,0],olivedrab:[107,142,35],orange:
[255,165,0],orangered:[255,69,0],orchid:[218,112,214],palegoldenrod:
[238,232,170],palegreen:[152,251,152],paleturquoise:[175,238,238],palevioletred:
[219,112,147],papayawhip:[255,239,213],peachpuff:[255,218,185],peru:
[205,133,63],pink:[255,192,203],plum:[221,160,221],powderblue:
[176,224,230],purple:[128,0,128],rebeccapurple:[102,51,153],red:
[255,0,0],rosybrown:[188,143,143],royalblue:[65,105,225],saddlebrown:
[139,69,19],salmon:[250,128,114],sandybrown:[244,164,96],seagreen:
[46,139,87],seashell:[255,245,238],sienna:[160,82,45],silver:
[192,192,192],skyblue:[135,206,235],slateblue:[106,90,205],slategray:
[112,128,144],slategrey:[112,128,144],snow:[255,250,250],springgreen:
[0,255,127],steelblue:[70,130,180],tan:[210,180,140],teal:[0,128,128],thistle:
[216,191,216],tomato:[255,99,71],turquoise:[64,224,208],violet:
[238,130,238],wheat:[245,222,179],white:[255,255,255],whitesmoke:
[245,245,245],yellow:[255,255,0],yellowgreen:[154,205,50]};class
Yt{constructor(t,e,r,n=1,i=!0){this.r=t,this.g=e,this.b=r,this.a=n,i||
(this.r*=n,this.g*=n,this.b*=n,n||this.overwriteGetter("rgb",[t,e,r,n]));}static
parse(t){if(t instanceof Yt)return t;if("string"!==typeof t)return;const
e=function(t){if("transparent"===t)t=t.toLowerCase().trim();return
[0,0,0,0];const e=Xt[t];if(e){const[t,r,n]=e;return
[t/255,r/255,n/255,1]}if(t.startsWith("#")&&/^#(?:[0-9a-f]{3,4}|[0-9a-f]{6}|[0-
9a-f]{8})$/).test(t)){const e=t.length<6?1:2;let r=1;return

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[Zt(t.slice(r,r+=e)),Zt(t.slice(r,r+=e)),Zt(t.slice(r,r+=e)),Zt(t.slice(r,r+=e))||
"ff"]}]if(t.startsWith("rgb")){const e=t.match(/^rgba?\s*([\de.+-]+)(%)?
(?:\s+|\s*(,)\s*)([\de.+-]+)(%)?(?:\s+|\s*(,)\s*)([\de.+-]+)(%)?(?:\s*([\s\/])\s*
([\de.+-]+)(%)?)?\s*\$)/);if(e){const[t,r,n,i,a,s,o,l,u,c,h,p]=e,f=[i||" ",o||"
",c].join("");if(" "===f||"/"===f||","===f||",,"===f){const t=
[n,s,u].join(""),e="%%%"===t?100:""===t?255:0;if(e){const t=
[Gt(+r/e,0,1),Gt(+a/e,0,1),Gt(+l/e,0,1),h?Kt(+h,p):1];if(Jt(t))return
t}}return}}const r=t.match(/^hsla?\s*([\de.+-]+)(?:deg)?(?:\s+|\s*(,)\s*)
([\de.+-]+)(%)?(?:\s+|\s*(,)\s*)([\de.+-]+)(%)?(?:\s*([\s\/])\s*([\de.+-]+)(%)?)?
\s*\$)/);if(r){const[t,e,n,i,a,s,o,l,u]=r,c=[n||" ",a||" ",o].join("");if("
"===c||"/"===c||","===c||",,"===c){const t=[+e,Gt(+i,0,100),Gt(+s,0,100),1?
Kt(+l,u):1];if(Jt(t))return function([t,e,r,n]){function i(n){const i=
(n+t/30)%12,a=e*Math.min(r,1-r);return r-a*Math.max(-1,Math.min(i-3,9-
i,1))}return t=Dt(t),e/=100,r/=100,[i(0),i(8),i(4),n]}(t)};return e?new
Yt(...e,!1):void 0}get rgb(){const{r:t,g:e,b:r,a:n}=this,i=n|1/0;return
this.overwriteGetter("rgb",[t/i,e/i,r/i,n])}get hcl(){return
this.overwriteGetter("hcl",function(t)
{const[e,r,n,i]=Ot(t),a=Math.sqrt(r*r+n*n);return [Math.round(1e4*a)?
Dt(Math.atan2(n,r)*Lt):NaN,a,e,i]}(this.rgb))}get lab(){return
this.overwriteGetter("lab",Ot(this.rgb))}overwriteGetter(t,e){return
Object.defineProperty(this,t,{value:e}),e}toString()
{const[t,e,r,n]=this.rgb;return
`rgba(${[t,e,r].map((t=>Math.round(255*t))).join(",")},${n})`}Yt.black=new
Yt(0,0,0,1),Yt.white=new Yt(1,1,1,1),Yt.transparent=new Yt(0,0,0,0),Yt.red=new
Yt(1,0,0,1);class Ht{constructor(t,e,r){this.sensitivity=t?
e?"variant":"case":e?"accent":"base",this.locale=r,this.collator=new
Intl.Collator(this.locale?this.locale:[],
{sensitivity:this.sensitivity,usage:"search"});}compare(t,e){return
this.collator.compare(t,e)}resolvedLocale(){return new
Intl.Collator(this.locale?this.locale:[]).resolvedOptions().locale}}class
Wt{constructor(t,e,r,n,i)
{this.text=t,this.image=e,this.scale=r,this.fontStack=n,this.textColor=i;}}class
Qt{constructor(t){this.sections=t;}static fromString(t){return new Qt([new
Wt(t,null,null,null,null)])}isEmpty(){return
0===this.sections.length||!this.sections.some((t=>0!==(t.text.length||t.image&&0!
==t.image.name.length))}static factory(t){return t instanceof Qt?
t:Qt.fromString(t)}toString(){return
0===this.sections.length?"":this.sections.map((t=>t.text)).join("")}}class
te{constructor(t){this.values=t.slice();}static parse(t){if(t instanceof
te)return t;if("number"===typeof t)return new
te([t,t,t,t]);if(Array.isArray(t)&&!(t.length<1||t.length>4)){for(const e of
t)if("number"!==typeof e)return;switch(t.length){case 1:t=
[t[0],t[0],t[0],t[0]];break;case 2:t=[t[0],t[1],t[0],t[1]];break;case 3:t=
[t[0],t[1],t[1],t[2],t[1]];return new te(t)}toString(){return
JSON.stringify(this.values)}}const ee=new
Set(["center","left","right","top","bottom","top-left","top-right","bottom-
left","bottom-right"]);class re{constructor(t){this.values=t.slice();}static
parse(t){if(t instanceof re)return t;if(Array.isArray(t)&&!(
t.length<1)&&t.length%2==0){for(let e=0;e<t.length;e+=2){const
r=t[e],n=t[e+1];if("string"!==typeof
r||!ee.has(r))return;if(!Array.isArray(n)||2!==n.length||"number"!==typeof
n[0]||"number"!==typeof n[1])return}return new re(t)}toString(){return
JSON.stringify(this.values)}}class ne{constructor(t)
{this.name=t.name,this.available=t.available;}toString(){return this.name}static
fromString(t){return t?new ne({name:t,available:!1}):null}function ie(t,e,r,n)
{return "number"===typeof t&&t>=0&&t<=255&&"number"===typeof
e&&e>=0&&e<=255&&"number"===typeof r&&r>=0&&r<=255?void 0===n||"number"===typeof
n&&n>=0&&n<=1?null:Invalid rgba value [`${[t,e,r,n].join(", ")}`]: 'a' must be
between 0 and 1.`:Invalid rgba value [`${("number"===typeof n?[t,e,r,n]:
[t,e,r]).join(", ")}`]: 'r', 'g', and 'b' must be between 0 and 255.`}function
ae(t){if(null===t||"string"===typeof t||"boolean"===typeof t||"number"===typeof
t||t instanceof Yt||t instanceof Ht||t instanceof Qt||t instanceof te||t

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instanceof re||t instanceof ne)return !0;if(Array.isArray(t)){for(const e of
t)if(!ae(e))return !1;return !0}if("object"===typeof t){for(const e in
t)if(!ae(t[e]))return !1;return !0}return !1}function se(t){if(null===t)return
ht;if("string"===typeof t)return ft;if("boolean"===typeof t)return
dt;if("number"===typeof t)return pt;if(t instanceof Yt)return yt;if(t instanceof
Ht)return xt;if(t instanceof Qt)return vt;if(t instanceof te)return bt;if(t
instanceof re)return _t;if(t instanceof ne)return wt;if(Array.isArray(t)){const
e=t.length;let r;for(const e of t){const t=se(e);if(r)
{if(r===t)continue;r=gt;break}r=t;}return At(r||gt,e)}return mt}function oe(t)
{const e=typeof t;return null===t?"":"string"===e||"number"===e||"boolean"===e?
String(t):t instanceof Yt||t instanceof Qt||t instanceof te||t instanceof re||t
instanceof ne?t.toString():JSON.stringify(t)}class le{constructor(t,e)
{this.type=t,this.value=e;}static parse(t,e){if(2!==t.length)return
e.error(`literal` expression requires exactly one argument, but found
${t.length-1} instead.`);if(!ae(t[1]))return e.error("invalid value");const
r=t[1];let n=se(r);const i=e.expectedType;return
"array"!==n.kind||0!==n.N||!i||"array"!==i.kind||"number"===typeof i.N&&0!==i.N||
(n=i),new le(n,r)}evaluate(){return this.value}eachChild(){return outputDefined()}
{return !0}}class ue{constructor(t)
{this.name="ExpressionEvaluationError",this.message=t;}toJSON(){return
this.message}}const ce={string:ft,number:pt,boolean:dt,object:mt};class
he{constructor(t,e){this.type=t,this.args=e;}static parse(t,e)
{if(t.length<2)return e.error("Expected at least one argument.");let r,n=1;const
i=t[0];if("array"===i){let i,a;if(t.length>2){const r=t[1];if("string"!==typeof
r||!(r in ce)||"object"===r)return e.error('The item type argument of "array"
must be one of string, number, boolean',1);i=ce[r],n++;}else i=gt;if(t.length>3)
{if(null!==t[2]&&("number"!==typeof t[2]||t[2]<0||t[2]!==Math.floor(t[2])))return
e.error('The length argument to "array" must be a positive integer
literal',2);a=t[2],n++;}r=At(i,a);}else {if(!ce[i])throw new Error(`Types
doesn't contain name = ${i}`);r=ce[i];const a=[];for(;n<t.length;n++){const
r=e.parse(t[n],n,gt);if(!r)return null;a.push(r);}return new he(r,a)}evaluate(t)
{for(let e=0;e<this.args.length;e++){const
r=this.args[e].evaluate(t);if(!It(this.type,se(r)))return
r;if(e===this.args.length-1)throw new ue(`Expected value to be of type
${St(this.type)}, but found ${St(se(r))} instead.`)}throw new Error}eachChild(t)
{this.args.forEach(t);}outputDefined(){return
this.args.every((t=>t.outputDefined()))}const pe={"to-boolean":dt,"to-
color":yt,"to-number":pt,"to-string":ft};class fe{constructor(t,e)
{this.type=t,this.args=e;}static parse(t,e){if(t.length<2)return
e.error("Expected at least one argument.");const r=t[0];if(!pe[r])throw new
Error(`Can't parse ${r} as it is not part of the known types`);if(("to-
boolean"===r||"to-string"===r)&&2!==t.length)return e.error("Expected one
argument.");const n=pe[r],i=[];for(let r=1;r<t.length;r++){const
n=e.parse(t[r],r,gt);if(!n)return null;i.push(n);}return new fe(n,i)}evaluate(t)
{switch(this.type.kind){case"boolean":return
Boolean(this.args[0].evaluate(t));case"color":{let e,r;for(const n of this.args)
{if(e=n.evaluate(t),r=null,e instanceof Yt)return e;if("string"===typeof e){const
r=t.parseColor(e);if(r)return r}else if(Array.isArray(e)&&
(r=e.length<3||e.length>4?`Invalid rgba value ${JSON.stringify(e)}: expected an
array containing either three or four numeric
values.`:ie(e[0],e[1],e[2],e[3]),!r))return new
Yt(e[0]/255,e[1]/255,e[2]/255,e[3])}throw new ue(r||`Could not parse color from
value '${JSON.stringify(e)}'`)}case"padding":{let
e;for(const r of this.args){e=r.evaluate(t);const n=re.parse(e);if(n)return
n}throw new ue(`Could not parse padding from value '${JSON.stringify(e)}'`)}case"variableAnchorOffsetCollection":{let e;for(const r
of this.args){e=r.evaluate(t);const n=re.parse(e);if(n)return n}throw new
ue(`Could not parse variableAnchorOffsetCollection from value
'${JSON.stringify(e)}'`)}case"number":{let e=null;for(const
r of this.args){if(e=r.evaluate(t),null===e)return 0;const
n=Number(e);if(!isNaN(n))return n}throw new ue(`Could not convert
${JSON.stringify(e)} to number.`)}case"formatted":return

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Qt.fromString(oe(this.args[0].evaluate(t)));case"resolvedImage":return
ne.fromString(oe(this.args[0].evaluate(t)));default:return
oe(this.args[0].evaluate(t))}eachChild(t){this.args.forEach(t);}outputDefined()
{return this.args.every((t=>t.outputDefined()))}const de=
["Unknown","Point","LineString","Polygon"];class ye{constructor()
{this.globals=null,this.feature=null,this.featureState=null,this.formattedSectio
n=null,this._parseColorCache=
{}},this.availableImages=null,this.canonical=null;}id(){return
this.feature&&"id"in this.feature?this.feature.id:null}geometryType(){return
this.feature?"number"==typeof this.feature.type?
de[this.feature.type]:this.feature.type:null}geometry(){return
this.feature&&"geometry"in this.feature?this.feature.geometry:null}canonicalID()
{return this.canonical}properties(){return
this.feature&&this.feature.properties||{}}parseColor(t){let
e=this._parseColorCache[t];return e||
(e=this._parseColorCache[t]=Yt.parse(t),e)}class me{constructor(t,e,r=
[],n,i=new ct,a=[])
{this.registry=t,this.path=r,this.key=r.map((t=>`${t}`).join(" "),this.scope=i
,this.errors=a,this.expectedType=n,this._isConstant=e;}parse(t,e,r,n,i={})
{return e?this.concat(e,r,n)._parse(t,i):this._parse(t,i)}_parse(t,e){function
r(t,e,r){return "assert"===r?new he(e,[t]):"coerce"===r?new fe(e,
[t]):t}if(null!==t&&"string"!==typeof t&&"boolean"!==typeof t&&"number"!==typeof
t||t=["literal",t],Array.isArray(t)){if(0===t.length)return
this.error('Expected an array with at least one element. If you wanted a literal
array, use ["literal", []].');const n=t[0];if("string"!==typeof n)return
this.error(`Expression name must be a string, but found ${typeof n} instead. If
you wanted a literal array, use ["literal", [...]].`,0),null;const
i=this.registry[n];if(i){let n=i.parse(t,this);if(!n)return
null;if(this.expectedType){const
t=this.expectedType,i=n.type;if("string"!==t.kind&&"number"!==t.kind&&"boolean"!
==t.kind&&"object"!==t.kind&&"array"!==t.kind||"value"!==i.kind)if("color"!==t.k
ind&&"formatted"!==t.kind&&"resolvedImage"!==t.kind||"value"!==i.kind&&"string"!
==i.kind)if("padding"!==t.kind||"value"!==i.kind&&"number"!==i.kind&&"array"!==i
.kind)if("variableAnchorOffsetCollection"!==t.kind||"value"!==i.kind&&"array"!==
i.kind){if(this.checkSubtype(t,i))return null}else
n=r(n,t,e.typeAnnotation||"coerce");else
n=r(n,t,e.typeAnnotation||"coerce");else
n=r(n,t,e.typeAnnotation||"coerce");else
n=r(n,t,e.typeAnnotation||"assert");}if(!(n instanceof
le)&&"resolvedImage"!==n.type.kind&&this._isConstant(n)){const t=new
ye;try{n=new le(n.type,n.evaluate(t));}catch(t){return
this.error(t.message,null)}return n}return this.error(`Unknown expression
"${n}". If you wanted a literal array, use ["literal", [...]].`,0)}return
this.error(void 0===t?"'undefined' value invalid. Use null
instead.":"object"===typeof t?'Bare objects invalid. Use ["literal", {...}]
instead.':`Expected an array, but found ${typeof t} instead.`)}concat(t,e,r)
{const n="number"===typeof t?this.path.concat(t):this.path,i=r?
this.scope.concat(r):this.scope;return new
me(this.registry,this._isConstant,n,e||null,i,this.errors)}error(t,...e){const
r=`${this.key}${e.map((t=>`${t}`).join(" "))}.join(" ")`;this.errors.push(new
ut(r,t));}checkSubtype(t,e){const r=It(t,e);return r&&this.error(r,r)}class
ge{constructor(t,e,r)
{this.type=xt,this.locale=r,this.caseSensitive=t,this.diacriticSensitive=e;}stat
ic parse(t,e){if(2!==t.length)return e.error("Expected one argument.");const
r=t[1];if("object"!==typeof r||Array.isArray(r))return e.error("Collator options
argument must be an object.");const n=e.parse(void 0!==r["case-
sensitive"]&&r["case-sensitive"],1,dt);if(!n)return null;const i=e.parse(void
0!==r["diacritic-sensitive"]&&r["diacritic-sensitive"],1,dt);if(!i)return
null;let a=null;return r.locale&&(a=e.parse(r.locale,1,ft),!a)?null:new
ge(n,i,a)}evaluate(t){return new
Ht(this.caseSensitive.evaluate(t),this.diacriticSensitive.evaluate(t),this.local
e?this.locale.evaluate(t):null)}eachChild(t)

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{t(this.caseSensitive),t(this.diacriticSensitive),this.locale&&t(this.locale);}o
utputDefined(){return !1}const xe=8192;function ve(t,e)
{t[0]=Math.min(t[0],e[0]),t[1]=Math.min(t[1],e[1]),t[2]=Math.max(t[2],e[0]),t[3]
=Math.max(t[3],e[1]);}function be(t,e){return !(t[0]<=e[0]||t[2]>=e[2]||t[1]
<=e[1]||t[3]>=e[3])}function we(t,e){const r=(180+t[0])/360,n=(180-
180/Math.PI*Math.log(Math.tan(Math.PI/4+t[1]*Math.PI/360)))/360,i=Math.pow(2,e.z
);return [Math.round(r*i*xe),Math.round(n*i*xe)]}function _e(t,e,r){const
n=t[0]-e[0],i=t[1]-e[1],a=t[0]-r[0],s=t[1]-r[1];return n*s-
a*i==0&&n*a<=0&&i*s<=0}function Ae(t,e){let r=!1;for(let s=0,o=e.length;s<o;s++)
{const o=e[s];for(let e=0,s=o.length;e<s-1;e++){if(_e(t,o[e],o[e+1]))return !1;
(i=o[e])[1]>(n=t)[1]!=(a=o[e+1])[1]>n[1]&&n[0]<(a[0]-i[0])*(n[1]-i[1])/(a[1]-
i[1])+i[0]&&(r=!r);}}var n,i,a;return r}function Se(t,e){for(let
r=0;r<e.length;r++)if(Ae(t,e[r]))return !0;return !1}function ke(t,e,r,n){const
i=n[0]-r[0],a=n[1]-r[1],s=(t[0]-r[0])*a-i*(t[1]-r[1]),o=(e[0]-r[0])*a-i*(e[1]-
r[1]);return s>0&&o<0||s<0&&o>0}function Ie(t,e,r){for(const u of r)for(let
r=0;r<u.length-1;++r)if(0!=(o=[(s=u[r+1])[0]-(a=u[r])[0],s[1]-a[1]])[0]*(l=
[(i=e)[0]-(n=t)[0],i[1]-n[1]])[1]-o[1]*l[0]&&ke(n,i,a,s)&&ke(a,s,n,i))return
!0;var n,i,a,s,o,l;return !1}function ze(t,e){for(let
r=0;r<t.length;++r)if(!Ae(t[r],e))return !1;for(let r=0;r<t.length-
1;++r)if(Ie(t[r],t[r+1],e))return !1;return !0}function Me(t,e){for(let
r=0;r<e.length;r++)if(ze(t,e[r]))return !0;return !1}function Ce(t,e,r){const n=
[];for(let i=0;i<t.length;i++){const a=[];for(let n=0;n<t[i].length;n++){const
s=we(t[i][n],r);ve(e,s),a.push(s);}n.push(a);}return n}function Be(t,e,r){const
n=[];for(let i=0;i<t.length;i++){const a=Ce(t[i],e,r);n.push(a);}return
n}function Pe(t,e,r,n){if(t[0]<r[0]||t[0]>r[2]){const e=.5*n;let i=t[0]-r[0]>e?-
n:r[0]-t[0]>e?n:0;0===i&&(i=t[0]-r[2]>e?-n:r[2]-t[0]>e?
n:0),t[0]+=i;}ve(e,t);}function Ve(t,e,r,n){const i=Math.pow(2,n.z)*xe,a=
[n.x*xe,n.y*xe],s=[];for(const n of t)for(const t of n){const n=
[t.x+a[0],t.y+a[1]];Pe(n,e,r,i),s.push(n);}return s}function Ee(t,e,r,n){const
i=Math.pow(2,n.z)*xe,a=[n.x*xe,n.y*xe],s=[];for(const r of t){const t=
[];for(const n of r){const r=
[n.x+a[0],n.y+a[1]];ve(e,r),t.push(r);}s.push(t);}if(e[2]-e[0]<=i/2){(o=e)
[0]=o[1]=1/0,o[2]=o[3]=-1/0;for(const t of s)for(const n of t)Pe(n,e,r,i);}var
o;return s}class Fe{constructor(t,e)
{this.type=dt,this.geojson=t,this.geometries=e;}static parse(t,e)
{if(2!==t.length)return e.error(`'within' expression requires exactly one
argument, but found ${t.length-1} instead.`);if(ae(t[1])){const
e=t[1];if("FeatureCollection"===e.type)for(let t=0;t<e.features.length;++t)
{const r=e.features[t].geometry.type;if("Polygon"===r||"MultiPolygon"===r)return
new Fe(e,e.features[t].geometry)}else if("Feature"===e.type){const
t=e.geometry.type;if("Polygon"===t||"MultiPolygon"===t)return new
Fe(e,e.geometry)}else if("Polygon"===e.type||"MultiPolygon"===e.type)return new
Fe(e,e)}return e.error("'within' expression requires valid geojson object that
contains polygon geometry type.")}evaluate(t)
{if(null!=t.geometry()&&null!=t.canonicalID())
{if("Point"===t.geometryType())return function(t,e){const r=
[1/0,1/0,-1/0,-1/0],n=
[1/0,1/0,-1/0,-1/0],i=t.canonicalID();if("Polygon"===e.type){const
a=Ce(e.coordinates,n,i),s=Ve(t.geometry(),r,n,i);if(!be(r,n))return !1;for(const
t of s)if(!Ae(t,a))return !1}if("MultiPolygon"===e.type){const
a=Be(e.coordinates,n,i),s=Ve(t.geometry(),r,n,i);if(!be(r,n))return !1;for(const
t of s)if(!Se(t,a))return !1}return !0}
(t,this.geometries);if("LineString"===t.geometryType())return function(t,e)
{const r=[1/0,1/0,-1/0,-1/0],n=
[1/0,1/0,-1/0,-1/0],i=t.canonicalID();if("Polygon"===e.type){const
a=Ce(e.coordinates,n,i),s=Ee(t.geometry(),r,n,i);if(!be(r,n))return !1;for(const
t of s)if(!ze(t,a))return !1}if("MultiPolygon"===e.type){const
a=Be(e.coordinates,n,i),s=Ee(t.geometry(),r,n,i);if(!be(r,n))return !1;for(const
t of s)if(!Me(t,a))return !1}return !0}(t,this.geometries)}return !1}eachChild()
{}outputDefined(){return !0}}class Te{constructor(t,e)
{this.type=e.type,this.name=t,this.boundExpression=e;}static parse(t,e)
{if(2!==t.length||"string"!==typeof t[1])return e.error("'var' expression

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requires exactly one string literal argument.");const r=t[1];return
e.scope.has(r)?new Te(r,e.scope.get(r)):e.error(`Unknown variable "${r}". Make
sure "${r}" has been bound in an enclosing "let" expression before using
it.`),1)}evaluate(t){return this.boundExpression.evaluate(t)}eachChild()
{}outputDefined(){return !1}}class $e{constructor(t,e,r,n)
{this.name=t,this.type=e,this._evaluate=r,this.args=n;}evaluate(t){return
this._evaluate(t,this.args)}eachChild(t){this.args.forEach(t);}outputDefined()
{return !1}static parse(t,e){const r=t[0],n=$e.definitions[r];if(!n)return
e.error(`Unknown expression "${r}". If you wanted a literal array, use
["literal", [...]].`,0);const i=Array.isArray(n)?n[0]:n.type,a=Array.isArray(n)?
[[n[1],n[2]]]:n.overloads,s=a.filter(((e)=>!Array.isArray(e)||e.length===t.len
gth-1));let o=null;for(const[n,a]of s){o=new
me(e.registry,Le,e.path,null,e.scope);const s=[];let l=!1;for(let
e=1;e<t.length;e++){const r=t[e],i=Array.isArray(n)?n[e-
1]:n.type,a=o.parse(r,l+s.length,i);if(!a)
{l=!0;break}s.push(a);}if(!l)if(Array.isArray(n)&&n.length!==s.length)o.error(`E
xpected ${n.length} arguments, but found ${s.length} instead.`);else {for(let
t=0;t<s.length;t++){const e=Array.isArray(n)?
n[t]:n.type,r=s[t];o.concat(t+1).checkSubtype(e,r.type);}if(0===o.errors.length)
return new $e(r,i,a,s)}if(l===s.length)e.errors.push(...o.errors);else {const
r=(s.length?s:a).map(((t))=>{return e=t,Array.isArray(e)?`(${e.map(St).join(",
")})`:`${St(e.type)}...`;var e;})).join(" | "),n=[];for(let
r=1;r<t.length;r++){const i=e.parse(t[r],l+n.length);if(!i)return
null;n.push(St(i.type));}e.error(`Expected arguments of type ${r}, but found
`${n.join(", ")} instead.`);}return null}static register(t,e)
{$e.definitions=e;for(const r in e)t[r]=$e;}function Le(t){if(t instanceof
Te)return Le(t.boundExpression);if(t instanceof $e&&"error"===t.name)return
!1;if(t instanceof ge)return !1;if(t instanceof Fe)return !1;const e=t
instanceof fe||t instanceof he;let r=!0;return t.eachChild((t=>{r=e?
r&&Le(t):r&&t instanceof le;})),!!r&&De(t)&&Ue(t,["zoom","heatmap-
density","line-progress","accumulated","is-supported-script"])}function De(t)
{if(t instanceof $e){if("get"===t.name&&l===t.args.length)return !1;if("feature-
state"===t.name)return !1;if("has"===t.name&&l===t.args.length)return
!1;if("properties"===t.name||"geometry-type"===t.name||"id"===t.name)return
!1;if(/^filter-/i.test(t.name))return !1;if(t instanceof Fe)return !1;let
e=!0;return t.eachChild((t=>{e&&!De(t)&&(e=!1);})),e}function Oe(t){if(t
instanceof $e&&"feature-state"===t.name)return !1;let e=!0;return
t.eachChild((t=>{e&&!Oe(t)&&(e=!1);})),e}function Ue(t,e){if(t instanceof
$e&&e.indexOf(t.name)>=0)return !1;let r=!0;return t.eachChild((t=>
{r&&!Ue(t,e)&&(r=!1);})),r}function Re(t,e){const r=t.length-1;let
n,i,a=0,s=r,o=0;for(;a<=s;)if(o=Math.floor((a+s)/2),n=t[o],i=t[o+1],n<=e)
{if(o===r||e<i)return o;a=o+1;}else {if(!(n>e))throw new ue("Input is not a
number.");s=o-1;}return 0}class ge{constructor(t,e,r)
{this.type=t,this.input=e,this.labels=[],this.outputs=[];for(const[t,e]of
r)this.labels.push(t),this.outputs.push(e);}static parse(t,e){if(t.length-
1<4)return e.error(`Expected at least 4 arguments, but found only ${t.length-
1}`);if((t.length-1)%2!==0)return e.error("Expected an even number of
arguments.");const r=e.parse(t[1],1,pt);if(!r)return null;const n=[];let
i=null;e.expectedType&&"value"!==e.expectedType.kind&&(i=e.expectedType);for(let
r=1;r<t.length;r+=2){const a=l===r?
-1/0:t[r],s=t[r+1],o=r,l=r+1;if("number"!==typeof a)return e.error('Input/output
pairs for "step" expressions must be defined using literal numeric values (not
computed expressions) for the input values.',o);if(n.length&&n[n.length-1]
[0]>=a)return e.error('Input/output pairs for "step" expressions must be
arranged with input values in strictly ascending order.',o);const
u=e.parse(s,l,i);if(!u)return null;i=i||u.type,n.push([a,u]);}return new
ge(i,r,n)}evaluate(t){const e=this.labels,r=this.outputs;if(l===e.length)return
r[0].evaluate(t);const n=this.input.evaluate(t);if(n<=e[0])return
r[0].evaluate(t);const i=e.length;return n>=e[i-1]?r[i-
1].evaluate(t):r[Re(e,n)].evaluate(t)}eachChild(t){t(this.input);for(const e of
this.outputs)t(e);}outputDefined(){return
this.outputs.every((t=>t.outputDefined()))}}function je(t,e,r){return t+r*(e-

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t)}function Ne(t,e,r){return t.map(((t,n)=>je(t,e[n],r)))}const Ze=
{number:je,color:function(t,e,r,n="rgb"){switch(n){case"rgb":
{const[n,i,a,s]=Ne(t.rgb,e.rgb,r);return new Yt(n,i,a,s,!1)}case"hcl":
{const[n,i,a,s]=t.hcl,[o,l,u,c]=e.hcl;let h,p;if(isNaN(n)||isNaN(o))isNaN(n)?
isNaN(o)?h=NaN:(h=o,l!=a&&0!=a||(p=1)):(h=n,l!=u&&0!=u||(p=i));else {let
t=o-n;o>n&&t>180?t-=360:o<n&&n-o>180&&
(t+=360),h=n+r*t;}const[f,d,y,m]=function([t,e,r,n]){return t=isNaN(t)?
0:t*$t,qt([r,Math.cos(t)*e,Math.sin(t)*e,n])}([h,null!=p?
p:je(i,l,r),je(a,u,r),je(s,c,r)]);return new Yt(f,d,y,m,!1)}case"lab":
{const[n,i,a,s]=qt(Ne(t.lab,e.lab,r));return new
Yt(n,i,a,s,!1)}}},array:Ne,padding:function(t,e,r){return new
te(Ne(t.values,e.values,r)),variableAnchorOffsetCollection:function(t,e,r)
{const n=t.values,i=e.values;if(n.length!=i.length)throw new ue(`Cannot
interpolate values of different length. from: ${t.toString()}, to:
${e.toString()}`);const a=[];for(let t=0;t<n.length;t+=2){if(n[t]!=i[t])throw
new ue(`Cannot interpolate values containing mismatched anchors. from[${t}]:
${n[t]}, to[${t}]: ${i[t]}`);a.push(n[t]);const[e,s]=n[t+1],
[o,l]=i[t+1];a.push([je(e,o,r),je(s,l,r)]);}return new re(a)}};class
Ke{constructor(t,e,r,n,i)
{this.type=t,this.operator=e,this.interpolation=r,this.input=n,this.labels=
[],this.outputs=[];for(const[t,e]of
i)this.labels.push(t),this.outputs.push(e);}static interpolationFactor(t,e,r,n)
{let i=0;if("exponential"===t.name)i=Ge(e,t.base,r,n);else
if("linear"===t.name)i=Ge(e,1,r,n);else if("cubic-bezier"===t.name){const
a=t.controlPoints;i=new l(a[0],a[1],a[2],a[3]).solve(Ge(e,1,r,n));}return
i}static parse(t,e){let[r,n,i,...a]=t;if(!Array.isArray(n)||0===n.length)return
e.error("Expected an interpolation type expression.",1);if("linear"===n[0])n=
{name:"linear"};else if("exponential"===n[0]){const t=n[1];if("number"!==typeof
t)return e.error("Exponential interpolation requires a numeric base.",1,1);n=
{name:"exponential",base:t};}else {if("cubic-bezier"!==n[0])return
e.error(`Unknown interpolation type ${String(n[0])}`,1,0);}const
t=n.slice(1);if(4!==t.length||t.some((t=>"number"!==typeof t||t<0||t>1)))return
e.error("Cubic bezier interpolation requires four numeric arguments with values
between 0 and 1.",1);n={name:"cubic-bezier",controlPoints:t};}if(t.length-
1<4)return e.error(`Expected at least 4 arguments, but found only ${t.length-
1}.`);if((t.length-1)%2!=0)return e.error("Expected an even number of
arguments.");if(i=e.parse(i,2,pt),!i)return null;const s=[];let
o=null;"interpolate-hcl"===r||"interpolate-lab"===r?
o=yt:e.expectedType&&"value"!==e.expectedType.kind&&(o=e.expectedType);for(let
t=0;t<a.length;t+=2){const r=a[t],n=a[t+1],i=t+3,l=t+4;if("number"!==typeof
r)return e.error('Input/output pairs for "interpolate" expressions must be
defined using literal numeric values (not computed expressions) for the input
values.',i);if(s.length&&s[s.length-1][0]>=r)return e.error('Input/output pairs
for "interpolate" expressions must be arranged with input values in strictly
ascending order.',i);const u=e.parse(n,l,o);if(!u)return
null;o=o||u.type,s.push([r,u]);}return
Ct(o,pt)||Ct(o,yt)||Ct(o,bt)||Ct(o,_t)||Ct(o,At(pt))?new
Ke(o,r,n,i,s):e.error(`Type ${St(o)} is not interpolatable.`)}evaluate(t){const
e=this.labels,r=this.outputs;if(1===e.length)return r[0].evaluate(t);const
n=this.input.evaluate(t);if(n<=e[0])return r[0].evaluate(t);const
i=e.length;if(n>=e[i-1])return r[i-1].evaluate(t);const
a=Re(e,n),s=Ke.interpolationFactor(this.interpolation,n,e[a],e[a+1]),o=r[a].eval
uate(t),l=r[a+1].evaluate(t);switch(this.operator){case"interpolate":return
Ze[this.type.kind](o,l,s);case"interpolate-hcl":return
Ze.color(o,l,s,"hcl");case"interpolate-lab":return
Ze.color(o,l,s,"lab")}}eachChild(t){t(this.input);for(const e of
this.outputs)t(e);}outputDefined(){return
this.outputs.every((t=>t.outputDefined())}}function Ge(t,e,r,n){const i=n-
r,a=t-r;return 0===i?0:1===e?a/i:(Math.pow(e,a)-1)/(Math.pow(e,i)-1)}class
Je{constructor(t,e){this.type=t,this.args=e;}static parse(t,e)
{if(t.length<2)return e.error("Expected at least one argument.");let
r=null;const n=e.expectedType;n&&"value"!==n.kind&&(r=n);const i=[];for(const n

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of t.slice(1)){const t=e.parse(n,1+i.length,r,void 0,
{typeAnnotation:"omit"});if(!t)return null;r=r||t.type,i.push(t);}if(!r)throw
new Error("No output type");const a=n&&i.some((t=>It(n,t.type)));return new
Je(a?gt:r,i)}evaluate(t){let e,r=null,n=0;for(const i of
this.args)if(n++,r=i.evaluate(t),r&&r instanceof ne&&!r.available&&(e||
(e=r.name),r=null,n===this.args.length&&(r=e)),null!==r)break;return
r}eachChild(t){this.args.forEach(t);}outputDefined(){return
this.args.every((t=>t.outputDefined()))}class Xe{constructor(t,e)
{this.type=e.type,this.bindings=[].concat(t),this.result=e;evaluate(t){return
this.result.evaluate(t)}eachChild(t){for(const e of
this.bindings)t(e[1]);t(this.result);}static parse(t,e){if(t.length<4)return
e.error(`Expected at least 3 arguments, but found ${t.length-1} instead.`);const
r=[];for(let n=1;n<t.length-1;n+=2){const i=t[n];if("string"!==typeof i)return
e.error(`Expected string, but found ${typeof i} instead.`);if(/^a-zA-Z0-
9_]/.test(i))return e.error("Variable names must contain only alphanumeric
characters or '_'");const a=e.parse(t[n+1],n+1);if(!a)return
null;r.push([i,a]);}const n=e.parse(t[t.length-1],t.length-
1,e.expectedType,r);return n?new Xe(r,n):null}outputDefined(){return
this.result.outputDefined()}}class Ye{constructor(t,e,r)
{this.type=t,this.index=e,this.input=r;}static parse(t,e){if(3!==t.length)return
e.error(`Expected 2 arguments, but found ${t.length-1} instead.`);const
r=e.parse(t[1],1,pt),n=e.parse(t[2],2,At(e.expectedType||gt));return r&&n?new
Ye(n.type.itemType,r,n):null}evaluate(t){const
e=this.index.evaluate(t),r=this.input.evaluate(t);if(e<0)throw new ue(`Array
index out of bounds: ${e} < 0.`);if(e>r.length)throw new ue(`Array index out of
bounds: ${e} > ${r.length-1}.`);if(e!==Math.floor(e))throw new ue(`Array index
must be an integer, but found ${e} instead.`);return r[e]}eachChild(t)
{t(this.index),t(this.input);}outputDefined(){return !1}}class
He{constructor(t,e){this.type=dt,this.needle=t,this.haystack=e;}static
parse(t,e){if(3!==t.length)return e.error(`Expected 2 arguments, but found
${t.length-1} instead.`);const r=e.parse(t[1],1,gt),n=e.parse(t[2],2,gt);return
r&&n?zt(r.type,[dt,ft,pt,ht,gt])?new He(r,n):e.error(`Expected first argument to
be of type boolean, string, number or null, but found ${St(r.type)}
instead`):null}evaluate(t){const
e=this.needle.evaluate(t),r=this.haystack.evaluate(t);if(!r)return !1;if(!Mt(e,
["boolean","string","number","null"]))throw new ue(`Expected first argument to
be of type boolean, string, number or null, but found ${St(se(e))}
instead.`);if(!Mt(r,["string","array"]))throw new ue(`Expected second argument
to be of type array or string, but found ${St(se(r))} instead.`);return
r.indexOf(e)>=0}eachChild(t){t(this.needle),t(this.haystack);}outputDefined()
{return !0}}class We{constructor(t,e,r)
{this.type=pt,this.needle=t,this.haystack=e,this.fromIndex=r;}static parse(t,e)
{if(t.length<=2||t.length>=5)return e.error(`Expected 3 or 4 arguments, but
found ${t.length-1} instead.`);const
r=e.parse(t[1],1,gt),n=e.parse(t[2],2,gt);if(!r||!n)return null;if(!zt(r.type,
[dt,ft,pt,ht,gt]))return e.error(`Expected first argument to be of type boolean,
string, number or null, but found ${St(r.type)} instead`);if(4===t.length){const
i=e.parse(t[3],3,pt);return i?new We(r,n,i):null}return new We(r,n)}evaluate(t)
{const e=this.needle.evaluate(t),r=this.haystack.evaluate(t);if(!Mt(e,
["boolean","string","number","null"]))throw new ue(`Expected first argument to
be of type boolean, string, number or null, but found ${St(se(e))}
instead.`);if(!Mt(r,["string","array"]))throw new ue(`Expected second argument
to be of type array or string, but found ${St(se(r))}
instead.`);if(this.fromIndex){const n=this.fromIndex.evaluate(t);return
r.indexOf(e,n)}return r.indexOf(e)}eachChild(t)
{t(this.needle),t(this.haystack),this.fromIndex&&t(this.fromIndex);}outputDefine
d(){return !1}}class Qe{constructor(t,e,r,n,i,a)
{this.inputType=t,this.type=e,this.input=r,this.cases=n,this.outputs=i,this.othe
rwise=a;}static parse(t,e){if(t.length<5)return e.error(`Expected at least 4
arguments, but found only ${t.length-1}.`);if(t.length%2!==1)return
e.error("Expected an even number of arguments.");let
r,n;e.expectedType&&"value"!==e.expectedType.kind&&(n=e.expectedType);const i=

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    {},a=[];for(let s=2;s<t.length-1;s+=2){let o=t[s];const
    l=t[s+1];Array.isArray(o)|| (o=[o]);const u=e.concat(s);if(0===o.length)return
    u.error("Expected at least one branch label.");for(const t of o)
    {if("number"!==typeof t&&"string"!==typeof t)return u.error("Branch labels must be
    numbers or strings.");if("number"===typeof
    t&&Math.abs(t)>Number.MAX_SAFE_INTEGER)return u.error(`Branch labels must be
    integers no larger than ${Number.MAX_SAFE_INTEGER}.`);if("number"===typeof
    t&&Math.floor(t)!==t)return u.error("Numeric branch labels must be integer
    values.");if(r){if(u.checkSubtype(r,se(t)))return null}else r=se(t);if(void
    0!==i[String(t)])return u.error("Branch labels must be
    unique.");i[String(t)]=a.length;}const c=e.parse(l,s,n);if(!c)return
    null;n=n||c.type,a.push(c);}const s=e.parse(t[1],1,gt);if(!s)return null;const
    o=e.parse(t[t.length-1],t.length-1,n);return
    o?"value"!==s.type.kind&&e.concat(1).checkSubtype(r,s.type)?null:new
    Qe(r,n,s,i,a,o):null}evaluate(t){const e=this.input.evaluate(t);return
    (se(e)===this.inputType&&this.outputs[this.cases[e]]||this.otherwise).evaluate(t
    )}eachChild(t)
    {t(this.input),this.outputs.forEach(t),t(this.otherwise);}outputDefined(){return
    this.outputs.every((t=>t.outputDefined()))&&this.otherwise.outputDefined()}class
    s tr{constructor(t,e,r){this.type=t,this.branches=e,this.otherwise=r;}static
    parse(t,e){if(t.length<4)return e.error(`Expected at least 3 arguments, but
    found only ${t.length-1}.`);if(t.length%2!==0)return e.error("Expected an odd
    number of arguments.");let r;e.expectedType&&"value"!==e.expectedType.kind&&
    (r=e.expectedType);const n=[];for(let i=1;i<t.length-1;i+=2){const
    a=e.parse(t[i],i,dt);if(!a)return null;const
    s=e.parse(t[i+1],i+1,r);if(!s)return null;n.push([a,s]),r=r||s.type;}const
    i=e.parse(t[t.length-1],t.length-1,r);if(!i)return null;if(!r)throw new
    Error("Can't infer output type");return new tr(r,n,i)}evaluate(t)
    {for(const[e,r]of this.branches)if(e.evaluate(t))return r.evaluate(t);return
    this.otherwise.evaluate(t)}eachChild(t){for(const[e,r]of
    this.branches)t(e),t(r);t(this.otherwise);}outputDefined(){return
    this.branches.every((([t,e])=>e.outputDefined()))&&this.otherwise.outputDefined(
    )}class er{constructor(t,e,r,n)
    {this.type=t,this.input=e,this.beginIndex=r,this.endIndex=n;}static parse(t,e)
    {if(t.length<=2||t.length>=5)return e.error(`Expected 3 or 4 arguments, but
    found ${t.length-1} instead.`);const
    r=e.parse(t[1],1,gt),n=e.parse(t[2],2,pt);if(!r||!n)return null;if(!zt(r.type,
    [At(gt),ft,gt]))return e.error(`Expected first argument to be of type array or
    string, but found ${St(r.type)} instead`);if(4===t.length){const
    i=e.parse(t[3],3,pt);return i?new er(r.type,r,n,i):null}return new
    er(r.type,r,n)}evaluate(t){const
    e=this.input.evaluate(t),r=this.beginIndex.evaluate(t);if(!Mt(e,
    ["string","array"]))throw new ue(`Expected first argument to be of type array or
    string, but found ${St(se(e))} instead.`);if(this.endIndex){const
    n=this.endIndex.evaluate(t);return e.slice(r,n)}return e.slice(r)}eachChild(t)
    {t(this.input),t(this.beginIndex),this.endIndex&&t(this.endIndex);}outputDefined
    (){return !1}}function rr(t,e){return
    "=="===t||"!="===t?"boolean"===e.kind||"string"===e.kind||"number"===e.kind||"nu
    ll"===e.kind||"value"===e.kind:"string"===e.kind||"number"===e.kind||"value"===e
    .kind}function nr(t,e,r,n){return 0===n.compare(e,r)}function ir(t,e,r){const
    n="=="!==t&&"!="!==t;return class i{constructor(t,e,r)
    {this.type=dt,this.lhs=t,this.rhs=e,this.collator=r,this.hasUntypedArgument="val
    ue"===t.type.kind||"value"===e.type.kind;}static parse(t,e)
    {if(3!==t.length&&4!==t.length)return e.error("Expected two or three
    arguments.");const r=t[0];let a=e.parse(t[1],1,gt);if(!a)return
    null;if(!rr(r,a.type))return e.concat(1).error(`"${r}" comparisons are not
    supported for type '${St(a.type)}'.`);let s=e.parse(t[2],2,gt);if(!s)return
    null;if(!rr(r,s.type))return e.concat(2).error(`"${r}" comparisons are not
    supported for type
    '${St(s.type)}'.`);if(a.type.kind!==s.type.kind&&"value"!==a.type.kind&&"value"!
    ==s.type.kind)return e.error(`Cannot compare types '${St(a.type)}' and
    '${St(s.type)}'.`);n&&("value"===a.type.kind&&"value"!==s.type.kind?a=new

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he(s.type,[a]):"value"!==a.type.kind&&"value"===s.type.kind&&(s=new he(a.type,
[s]));let o=null;if(4===t.length)
{if("string"!==a.type.kind&&"string"!==s.type.kind&&"value"!==a.type.kind&&"valu
e"!==s.type.kind)return e.error("Cannot use collator to compare non-string
types.");if(o=e.parse(t[3],3,xt),!o)return null}return new i(a,s,o)}evaluate(i)
{const
a=this.lhs.evaluate(i),s=this.rhs.evaluate(i);if(n&&this.hasUntypedArgument)
{const
e=se(a),r=se(s);if(e.kind!==r.kind||"string"!==e.kind&&"number"!==e.kind)throw
new ue(`Expected arguments for "${t}" to be (string, string) or (number,
number), but found (${e.kind}, ${r.kind})
instead.`)}if(this.collator&&!n&&this.hasUntypedArgument){const
t=se(a),r=se(s);if("string"!==t.kind||"string"!==r.kind)return e(i,a,s)}return
this.collator?r(i,a,s,this.collator.evaluate(i)):e(i,a,s)}eachChild(t)
{t(this.lhs),t(this.rhs),this.collator&&t(this.collator);}outputDefined(){return
!0}}const ar=ir("==",(function(t,e,r){return e===r}),nr),sr=ir("!=",(
function(t,e,r){return e!==r}),(function(t,e,r,n){return
!nr(0,e,r,n)}),or=ir("<",(function(t,e,r){return e<r}),(function(t,e,r,n)
{return n.compare(e,r)<0})),lr=ir(">",(function(t,e,r){return e>r})),
(function(t,e,r,n){return n.compare(e,r)>0})),ur=ir("<=",(function(t,e,r){return
e<=r}),(function(t,e,r,n){return n.compare(e,r)<=0})),cr=ir(">=",(
function(t,e,r){return e>=r}),(function(t,e,r,n){return
n.compare(e,r)>=0}));class hr{constructor(t,e,r,n,i)
{this.type=ft,this.number=t,this.locale=e,this.currency=r,this.minFractionDigits
=n,this.maxFractionDigits=i;}static parse(t,e){if(3!==t.length)return
e.error("Expected two arguments.");const r=e.parse(t[1],1,pt);if(!r)return
null;const n=t[2];if("object"!==typeof n||Array.isArray(n))return
e.error("NumberFormat options argument must be an object.");let
i=null;if(n.locale&&(i=e.parse(n.locale,1,ft),!i))return null;let
a=null;if(n.currency&&(a=e.parse(n.currency,1,ft),!a))return null;let
s=null;if(n["min-fraction-digits"]&&(s=e.parse(n["min-fraction-
digits"],1,pt),!s))return null;let o=null;return n["max-fraction-digits"]&&
(o=e.parse(n["max-fraction-digits"],1,pt),!o)?null:new hr(r,i,a,s,o)}evaluate(t)
{return new Intl.NumberFormat(this.locale?this.locale.evaluate(t):[],
{style:this.currency?"currency":"decimal",currency:this.currency?
this.currency.evaluate(t):void 0,minimumFractionDigits:this.minFractionDigits?
this.minFractionDigits.evaluate(t):void
0,maximumFractionDigits:this.maxFractionDigits?
this.maxFractionDigits.evaluate(t):void
0}).format(this.number.evaluate(t))}eachChild(t)
{t(this.number),this.locale&&t(this.locale),this.currency&&t(this.currency),this
.minFractionDigits&&t(this.minFractionDigits),this.maxFractionDigits&&t(this.max
FractionDigits);}outputDefined(){return !1}}class pr{constructor(t)
{this.type=vt,this.sections=t;}static parse(t,e){if(t.length<2)return
e.error("Expected at least one argument.");const
r=t[1];if(!Array.isArray(r)&&"object"===typeof r)return e.error("First argument
must be an image or text section.");const n=[];let i=!1;for(let r=1;r<=t.length-
1;++r){const a=t[r];if(i&&"object"===typeof a&&!Array.isArray(a)){i=!1;let
t=null;if(a["font-scale"]&&(t=e.parse(a["font-scale"],1,pt),!t))return null;let
r=null;if(a["text-font"]&&(r=e.parse(a["text-font"],1,At(ft),!r))return
null;let s=null;if(a["text-color"]&&(s=e.parse(a["text-color"],1,yt),!s))return
null;const o=n[n.length-1];o.scale=t,o.font=r,o.textColor=s;}else {const
a=e.parse(t[r],1,gt);if(!a)return null;const
s=a.type.kind;if("string"!==s&&"value"!==s&&"null"!==s&&"resolvedImage"!==s)retu
rn e.error("Formatted text type must be 'string', 'value', 'image' or
'null'.");i=!0,n.push({content:a,scale:null,font:null,textColor:null});}}return
new pr(n)}evaluate(t){return new Qt(this.sections.map((e=>{const
r=e.content.evaluate(t);return se(r)===wt?new Wt("",r,null,null,null):new
Wt(oe(r),null,e.scale?e.scale.evaluate(t):null,e.font?
e.font.evaluate(t).join(","):null,e.textColor?
e.textColor.evaluate(t):null)}))}eachChild(t){for(const e of
this.sections)t(e.content),e.scale&&t(e.scale),e.font&&t(e.font),e.textColor&&t(

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e.textColor);}outputDefined(){return !1}}class fr{constructor(t)
{this.type=wt,this.input=t;}static parse(t,e){if(2!==t.length)return
e.error("Expected two arguments.");const r=e.parse(t[1],1,ft);return r?new
fr(r):e.error("No image name provided.")}evaluate(t){const
e=this.input.evaluate(t),r=ne.fromString(e);return r&&t.availableImages&&
(r.available=t.availableImages.indexOf(e)>-1),r}eachChild(t)
{t(this.input);}outputDefined(){return !1}}class dr{constructor(t)
{this.type=pt,this.input=t;}static parse(t,e){if(2!==t.length)return
e.error(`Expected 1 argument, but found ${t.length-1} instead.`);const
r=e.parse(t[1],1);return
r?"array"!==r.type.kind&&"string"!==r.type.kind&&"value"!==r.type.kind?
e.error(`Expected argument of type string or array, but found ${St(r.type)}
instead.`):new dr(r):null}evaluate(t){const
e=this.input.evaluate(t);if("string"===typeof e)return
e.length;if(Array.isArray(e))return e.length;throw new ue(`Expected value to be
of type string or array, but found ${St(se(e))} instead.`)}eachChild(t)
{t(this.input);}outputDefined(){return !1}}const yr={"===":ar,"!=":sr,">":lr,"
<":or,">=":cr,"
<=":ur,array:he,at:Ye,boolean:he,case:tr,coalesce:Je,collator:ge,format:pr,image
:fr,in:He,"index-of":We,interpolate:Ke,"interpolate-hcl":Ke,"interpolate-
lab":Ke,length:dr,let:Xe,literal:le,match:Qe,number:he,"number-
format":hr,object:he,slice:er,step:qe,string:he,"to-boolean":fe,"to-
color":fe,"to-number":fe,"to-string":fe,var:Te,within:Fe};function mr(t,
[e,r,n,i]){e=e.evaluate(t),r=r.evaluate(t),n=n.evaluate(t);const a=i?
i.evaluate(t):1,s=ie(e,r,n,a);if(s)throw new ue(s);return new
Yt(e/255,r/255,n/255,a,!1)}function gr(t,e){return t in e}function xr(t,e){const
r=e[t];return void 0===r?null:r}function vr(t){return {type:t}}function br(t)
{return {result:"success",value:t}}function wr(t){return
{result:"error",value:t}}function _r(t){return "data-driven"===t["property-
type"]||"cross-faded-data-driven"===t["property-type"]}function Ar(t){return
!!t.expression&&t.expression.parameters.indexOf("zoom")>-1}function Sr(t){return
!!t.expression&&t.expression.interpolated}function kr(t){return t instanceof
Number?"number":t instanceof String?"string":t instanceof
Boolean?"boolean":Array.isArray(t)?"array":null===t?"null":typeof t}function
Ir(t){return "object"===typeof t&&null!==t&&!Array.isArray(t)}function zr(t)
{return t}function Mr(t,e){const r="color"===e.type,n=t.stops&&"object"===typeof
t.stops[0][0],i=n||!(n||void 0!==t.property),a=t.type||
(Sr(e)?"exponential":"interval");if(r||"padding"===e.type){const n=r?
Yt.parse:te.parse;(t=lt({},t)).stops&&(t.stops=t.stops.map((t=>
[t[0],n(t[1])])),t.default=n(t.default?
t.default:e.default));if(t.colorSpace&&"rgb"!==
(s=t.colorSpace)&&"hcl"!==s&&"lab"!==s)throw new Error(`Unknown color space:
"${t.colorSpace}"`);var s;let o,l,u;if("exponential"===a)o=Vr;else
if("interval"===a)o=Pr;else if("categorical"===a)
{o=Br,l=Object.create(null);for(const e of t.stops)l[e[0]]=e[1];u=typeof
t.stops[0][0];}else {if("identity"===a)throw new Error(`Unknown function type
"${a}"`);o=Er;}if(n){const r={},n=[];for(let e=0;e<t.stops.length;e++){const
i=t.stops[e],a=i[0].zoom;void 0===r[a]&&(r[a]=
{zoom:a,type:t.type,property:t.property,default:t.default,stops:
[],n.push(a)),r[a].stops.push([i[0].value,i[1]]);}const i=[];for(const t of
n)i.push([r[t].zoom,Mr(r[t],e)]);const a={name:"linear"};return
{kind:"composite",interpolationType:a,interpolationFactor:Ke.interpolationFactor
.bind(void 0,a),zoomStops:i.map((t=>t[0])),evaluate:
({zoom:r},n)=>Vr({stops:i,base:t.base},e,r).evaluate(r,n)}}if(i){const
r="exponential"===a?{name:"exponential",base:void 0!==t.base?
t.base:1}:null;return
{kind:"camera",interpolationType:r,interpolationFactor:Ke.interpolationFactor.bi
nd(void 0,r),zoomStops:t.stops.map((t=>t[0])),evaluate:
({zoom:r})=>o(t,e,r,l,u)}}return {kind:"source",evaluate(r,n){const
i=n&&n.properties?n.properties[t.property]:void 0;return void 0===i?
Cr(t.default,e.default):o(t,e,i,l,u)}}}function Cr(t,e,r){return void 0!==t?
t:void 0!==e?e:void 0!==r?r:void 0}function Br(t,e,r,n,i){return Cr(typeof

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r===i?n[r]:void 0,t.default,e.default)}function Pr(t,e,r)
{if("number"!==kr(r))return Cr(t.default,e.default);const
n=t.stops.length;if(1===n)return t.stops[0][1];if(r<=t.stops[0][0])return
t.stops[0][1];if(r>=t.stops[n-1][0])return t.stops[n-1][1];const
i=Re(t.stops.map((t=>t[0])),r);return t.stops[i][1]}function Vr(t,e,r){const
n=void 0!==t.base?t.base:1;if("number"!==kr(r))return
Cr(t.default,e.default);const i=t.stops.length;if(1===i)return t.stops[0]
[1];if(r<=t.stops[0][0])return t.stops[0][1];if(r>=t.stops[i-1][0])return
t.stops[i-1][1];const a=Re(t.stops.map((t=>t[0])),r),s=function(t,e,r,n){const
i=n-r,a=t-r;return 0===i?0:1===e?a/i:(Math.pow(e,a)-1)/(Math.pow(e,i)-1)}
(r,n,t.stops[a][0],t.stops[a+1][0]),o=t.stops[a][1],l=t.stops[a+1]
[1],u=Ze[e.type]||zr;return "function"===typeof o.evaluate?{evaluate(...e){const
r=o.evaluate.apply(void 0,e),n=l.evaluate.apply(void 0,e);if(void 0!==r&&void
0!==n)return u(r,n,s,t.colorSpace)}}:u(o,l,s,t.colorSpace)}function Er(t,e,r)
{switch(e.type)
{case"color":r=Yt.parse(r);break;case"formatted":r=Qt.fromString(r.toString());b
reak;case"resolvedImage":r=ne.fromString(r.toString());break;case"padding":r=te.
parse(r);break;default:kr(r)===e.type||"enum"===e.type&&e.values[r]||(r=void
0);}return Cr(r,t.default,e.default)}$e.register(yr,{error:[{kind:"error"},[ft],
(t,[e])=>{throw new ue(e.evaluate(t))}],typeof:[ft,[gt],(t,
[e])=>St(se(e.evaluate(t))),,"to-rgba":[At(pt,4),[yt],(t,[e])=>
{const[r,n,i,a]=e.evaluate(t).rgb;return [255*r,255*n,255*i,a]}],rgb:[yt,
[pt,pt,pt],mr],rgba:[yt,[pt,pt,pt,pt],mr],has:{type:dt,overloads:[[[ft],(t,
[e])=>gr(e.evaluate(t),t.properties())],[[ft,mt],(t,
[e,r])=>gr(e.evaluate(t),r.evaluate(t))]}],get:{type:gt,overloads:[[[ft],(t,
[e])=>xr(e.evaluate(t),t.properties())],[[ft,mt],(t,
[e,r])=>xr(e.evaluate(t),r.evaluate(t))]}],,"feature-state":[gt,[ft],(t,
[e])=>xr(e.evaluate(t),t.featureState|{}]},properties:[mt,
[],t=>t.properties()],,"geometry-type":[ft,[],t=>t.geometryType()],id:[gt,
[],t=>t.id()],zoom:[pt,[],t=>t.globals.zoom],,"heatmap-density":[pt,
[],t=>t.globals.heatmapDensity|0],,"line-progress":[pt,
[],t=>t.globals.lineProgress|0],accumulated:[gt,[],t=>void
0===t.globals.accumulated?null:t.globals.accumulated],"+":[pt,vr(pt),(t,e)>{let
r=0;for(const n of e)r+=n.evaluate(t);return r}],"*":[pt,vr(pt),(t,e)>{let
r=1;for(const n of e)r*=n.evaluate(t);return r}],"-":{type:pt,overloads:
[[[pt,pt],(t,[e,r])=>e.evaluate(t)-r.evaluate(t)],[[pt],(t,[e])=>-
e.evaluate(t)]}],"/":[pt,[pt,pt],(t,[e,r])=>e.evaluate(t)/r.evaluate(t)],"%":
[pt,[pt,pt],(t,[e,r])=>e.evaluate(t)%r.evaluate(t)],ln2:[pt,[],()=>Math.LN2],pi:
[pt,[],()=>Math.PI],e:[pt,[],()=>Math.E],"^":[pt,[pt,pt],(t,
[e,r])=>Math.pow(e.evaluate(t),r.evaluate(t))],sqrt:[pt,[pt],(t,
[e])=>Math.sqrt(e.evaluate(t))],log10:[pt,[pt],(t,
[e])=>Math.log(e.evaluate(t))/Math.LN10],ln:[pt,[pt],(t,
[e])=>Math.log(e.evaluate(t))],log2:[pt,[pt],(t,
[e])=>Math.log(e.evaluate(t))/Math.LN2],sin:[pt,[pt],(t,
[e])=>Math.sin(e.evaluate(t))],cos:[pt,[pt],(t,
[e])=>Math.cos(e.evaluate(t))],tan:[pt,[pt],(t,
[e])=>Math.tan(e.evaluate(t))],asin:[pt,[pt],(t,
[e])=>Math.asin(e.evaluate(t))],acos:[pt,[pt],(t,
[e])=>Math.acos(e.evaluate(t))],atan:[pt,[pt],(t,
[e])=>Math.atan(e.evaluate(t))],min:[pt,vr(pt),
(t,e)>Math.min(...e.map((e=>e.evaluate(t))))],max:[pt,vr(pt),
(t,e)>Math.max(...e.map((e=>e.evaluate(t))))],abs:[pt,[pt],(t,
[e])=>Math.abs(e.evaluate(t))],round:[pt,[pt],(t,[e])=>{const
r=e.evaluate(t);return r<0?-Math.round(-r):Math.round(r)}],floor:[pt,[pt],(t,
[e])=>Math.floor(e.evaluate(t))],ceil:[pt,[pt],(t,
[e])=>Math.ceil(e.evaluate(t))],,"filter-==":[dt,[ft,gt],(t,
[e,r])=>t.properties()[e.value]===r.value],,"filter-id-==":[dt,[gt],(t,
[e,r])=>t.id()===e.value],,"filter-type-==":[dt,[ft],(t,
[e])=>t.geometryType()===e.value],,"filter-<":[dt,[ft,gt],(t,[e,r])=>{const
n=t.properties()[e.value],i=r.value;return typeof n===typeof i&&n<i}],,"filter-id-
<":[dt,[gt],(t,[e])=>{const r=t.id(),n=e.value;return typeof r===typeof
n&&r<n}],,"filter->":[dt,[ft,gt],(t,[e,r])=>{const n=t.properties()

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[e.value],i=r.value;return typeof n==typeof i&&n>i}], "filter-id->":[dt,[gt],(t,
[e])=>{const r=t.id(),n=e.value;return typeof r==typeof n&&r>n}], "filter-<=":
[dt,[ft,gt],(t,[e,r])=>{const n=t.properties()[e.value],i=r.value;return typeof
n==typeof i&&n<=i}], "filter-id-<=":[dt,[gt],(t,[e])=>{const
r=t.id(),n=e.value;return typeof r==typeof n&&r<=n}], "filter->=":[dt,[ft,gt],(t,
[e,r])=>{const n=t.properties()[e.value],i=r.value;return typeof n==typeof
i&&n>=i}], "filter-id->=":[dt,[gt],(t,[e])=>{const r=t.id(),n=e.value;return
typeof r==typeof n&&r>=n}], "filter-has":[dt,[gt],(t,[e])=>e.value in
t.properties()], "filter-has-id":[dt,[],t=>null!=t.id()&&void
0!=t.id()], "filter-type-in":[dt,[At(ft)],(t,
[e])=>e.value.indexOf(t.geometryType())>=0], "filter-id-in":[dt,[At(gt)],(t,
[e])=>e.value.indexOf(t.id())>=0], "filter-in-small":[dt,[ft,At(gt)],(t,
[e,r])=>r.value.indexOf(t.properties()[e.value])>=0], "filter-in-large":[dt,
[ft,At(gt)],(t,[e,r])=>function(t,e,r,n){for(;r<=n;){const
i=r+n>>1;if(e[i]==t)return !0;e[i]>t?n=i-1:r=i+1;}return !1}(t.properties()
[e.value],r.value,0,r.value.length-1)], all:{type:dt,overloads:[[dt,dt],(t,
[e,r])=>e.evaluate(t)&&r.evaluate(t)], [vr(dt),(t,e)=>{for(const r of
e)if(!r.evaluate(t))return !1;return !0}]]], any:{type:dt,overloads:[[dt,dt],(t,
[e,r])=>e.evaluate(t)||r.evaluate(t)], [vr(dt),(t,e)=>{for(const r of
e)if(r.evaluate(t))return !0;return !1}]]], "!":[dt,[dt],(t,
[e])=>!e.evaluate(t)], "is-supported-script":[dt,[ft],(t,[e])=>{const
r=t.globals&&t.globals.isSupportedScript;return !r||r(e.evaluate(t))}], upcase:
[ft,[ft],(t,[e])=>e.evaluate(t).toUpperCase()], downcase:[ft,[ft],(t,
[e])=>e.evaluate(t).toLowerCase()], concat:[ft,vt(gt),
(t,e)=>e.map((e=>oe(e.evaluate(t))).join(""))], "resolved-locale":[ft,[xt],(t,
[e])=>e.evaluate(t).resolvedLocale()}}];class Fr{constructor(t,e){var
r;this.expression=t,this._warningHistory={},this._evaluator=new
ye,this._defaultValue=e?"color"===r.type?Yt(r.default)?new
Yt(0,0,0,0):"color"===r.type?Yt.parse(r.default)||null:"padding"===r.type?
te.parse(r.default)||null:"variableAnchorOffsetCollection"===r.type?
re.parse(r.default)||null:void 0===r.default?
null:r.default:null,this._enumValues=e&&"enum"===e.type?
e.values:null;}evaluateWithoutErrorHandling(t,e,r,n,i,a){return
this._evaluator.globals=t,this._evaluator.feature=e,this._evaluator.featureState
=r,this._evaluator.canonical=n,this._evaluator.availableImages=i||null,this._eva
luator.formattedSection=a,this.expression.evaluate(this._evaluator)}evaluate(t,e
,r,n,i,a)
{this._evaluator.globals=t,this._evaluator.feature=e||null,this._evaluator.featu
reState=r||null,this._evaluator.canonical=n,this._evaluator.availableImages=i||n
ull,this._evaluator.formattedSection=a||null;try{const
t=this.expression.evaluate(this._evaluator);if(null==t||"number"===typeof
t&&t!=t)return this._defaultValue;if(this._enumValues&&!(t in
this._enumValues))throw new ue(`Expected value to be one of
${Object.keys(this._enumValues).map((t=>JSON.stringify(t)).join(", ")), but
found ${JSON.stringify(t)} instead.`);return t}catch(t){return
this._warningHistory[t.message]||
(this._warningHistory[t.message]=!0,"undefined"!=typeof
console&&console.warn(t.message)),this._defaultValue}}function Tr(t){return
Array.isArray(t)&&t.length>0&&"string"===typeof t[0]&&t[0]in yr}function $r(t,e)
{const r=new me(yr,Le,[],e?function(t){const e=
{color:yt,string:ft,number:pt,enum:ft,boolean:dt,formatted:vt,padding:bt,resolve
dImage:wt,variableAnchorOffsetCollection:_t};return "array"===t.type?
At(e[t.value]||gt,t.length):e[t.type]}(e):void 0),n=r.parse(t,void 0,void 0,void
0,e&&"string"===e.type?{typeAnnotation:"coerce"}:void 0);return n?br(new
Fr(n,e)):wr(r.errors)}class Lr{constructor(t,e)
{this.kind=t,this._styleExpression=e,this.isStateDependent="constant"!=t&&!Oe(e
.expression);}evaluateWithoutErrorHandling(t,e,r,n,i,a){return
this._styleExpression.evaluateWithoutErrorHandling(t,e,r,n,i,a)}evaluate(t,e,r,n
,i,a){return this._styleExpression.evaluate(t,e,r,n,i,a)}}class
Dr{constructor(t,e,r,n)
{this.kind=t,this.zoomStops=r,this._styleExpression=e,this.isStateDependent="cam
era"!=t&&!Oe(e.expression),this.interpolationType=n;}evaluateWithoutErrorHandli

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ng(t,e,r,n,i,a){return
this._styleExpression.evaluateWithoutErrorHandling(t,e,r,n,i,a)}evaluate(t,e,r,n,i,a){return
this._styleExpression.evaluate(t,e,r,n,i,a)}interpolationFactor(t,e,r){return
this.interpolationType?
Ke.interpolationFactor(this.interpolationType,t,e,r):0}}function Or(t,e){const
r=$r(t,e);if("error"===r.result)return r;const
n=r.value.expression,i=De(n);if(!i&&!_r(e))return wr([new ut("", "data
expressions not supported")]);const a=Ue(n,["zoom"]);if(!a&&!Ar(e))return
wr([new ut("", "zoom expressions not supported")]);const s=Rr(n);return s||a?s
instanceof ut?wr([s]):s instanceof Ke&&!Sr(e)?wr([new ut("", "'interpolate"
expressions cannot be used with this property')]):br(s?new
Dr(i?"camera":"composite",r.value,s.labels,s instanceof Ke?s.interpolation:void
0):new Lr(i?"constant":"source",r.value)):wr([new ut("", "'zoom" expression may
only be used as input to a top-level "step" or "interpolate"
expression.')])}class Ur{constructor(t,e)
{this._parameters=t,this._specification=e,lt(this,Mr(this._parameters,this._spec
ification));}static deserialize(t){return new
Ur(t._parameters,t._specification)}static serialize(t){return
{_parameters:t._parameters,_specification:t._specification}}}function Rr(t){let
e=null;if(t instanceof Xe)e=Rr(t.result);else if(t instanceof Je){for(const r of
t.args)if(e=Rr(r),e)break}else (t instanceof qe||t instanceof Ke)&&t.input
instanceof $e&&"zoom"===t.input.name&&(e=t);return e instanceof
ut||t.eachChild((t=>{const r=Rr(t);r instanceof ut?e=r:!e&&r?e=new ut("", "'zoom"
expression may only be used as input to a top-level "step" or "interpolate"
expression.'):e&&r&&e!==r&&(e=new ut("", 'Only one zoom-based "step" or
"interpolate" subexpression may be used in an expression.'))});}),e}function
qr(t){if(!0===t||!1===t)return !0;if(!Array.isArray(t)||0===t.length)return
!1;switch(t[0]){case"has":return
t.length>=2&&"$id"!==t[1]&&"$type"!==t[1];case"in":return t.length>=3&&
("string"!==typeof
t[1]||Array.isArray(t[2]));case"!in":case"!has":case"none":return
!1;case"===":case"!=":case">":case">=":case"<":case"<=":return
3!==t.length||Array.isArray(t[1])||Array.isArray(t[2]);case"any":case"all":for(c
onst e of t.slice(1))if(!qr(e)&&"boolean"!==typeof e)return !1;return
!0;default:return !0}}const jr=
{type:"boolean",default:!1,transition:!1,"property-type":"data-
driven",expression:{interpolated:!1,parameters:["zoom","feature"]}};function
Nr(t){if(null===t)return {filter:()=>!0,needGeometry:!1};qr(t)||t=Gr(t);const
e=$r(t,jr);if("error"===e.result)throw new Error(e.value.map((t=>`${t.key}:
${t.message}`)).join(", "));return {filter:(t,r,n)=>e.value.evaluate(t,r,
{n},n),needGeometry:Kr(t)}}function Zr(t,e){return t<e?-1:t>e?1:0}function Kr(t)
{if(!Array.isArray(t))return !1;if("within"===t[0])return !0;for(let
e=1;e<t.length;e++)if(Kr(t[e]))return !0;return !1}function Gr(t){if(!t)return
!0;const e=t[0];return t.length<=1?"any"!==e:"==="===e?
Jr(t[1],t[2], "=="):"!="===e?Hr(Jr(t[1],t[2], "==")):"<===e||">===e||"
<===e||">===e?Jr(t[1],t[2],e):"any"===e?(r=t.slice(1),
["any"].concat(r.map(Gr)):"all"===e?
["all"].concat(t.slice(1).map(Gr)):"none"===e?
["all"].concat(t.slice(1).map(Gr).map(Hr)):"in"===e?
Xr(t[1],t.slice(2)):"!in"===e?Hr(Xr(t[1],t.slice(2)):"has"===e?
Yr(t[1]):"!has"===e?Hr(Yr(t[1])):"within"!==e||t;var r;}function Jr(t,e,r)
{switch(t){case"$type":return [`filter-type-${r}`,e];case"$id":return [`filter-
id-${r}`,e];default:return [`filter-${r}`,t,e]}}function Xr(t,e)
{if(0===e.length)return !1;switch(t){case"$type":return ["filter-type-in",
["literal",e]];case"$id":return ["filter-id-in",["literal",e]];default:return
e.length>200&&!e.some((t=>typeof t!==typeof e[0]))?["filter-in-large",t,
["literal",e.sort(Zr)]]:["filter-in-small",t,["literal",e]}}function Yr(t)
{switch(t){case"$type":return !0;case"$id":return ["filter-has-
id"];default:return ["filter-has",t]}}function Hr(t){return ["!",t]}function
Wr(t){const e=typeof
t;if("number"===e||"boolean"===e||"string"===e||null===t)return

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JSON.stringify(t);if(Array.isArray(t)){let e="[";for(const r of
t)e+=`${Wr(r)},`;return `${e]}`const r=Object.keys(t).sort();let n="{";for(let
e=0;e<r.length;e++)n+=`${JSON.stringify(r[e])}:${Wr(t[r[e]])},`;return
`${n}}`function Qr(t){let e="";for(const r of Y)e+=`${Wr(t[r])}`;return
e}function tn(t){const e=t.value;return e?[new ot(t.key,e,"constants have been
deprecated as of v8")]:[]}function en(t){return t instanceof Number||t
instanceof String||t instanceof Boolean?t.valueOf():t}function rn(t)
{if(Array.isArray(t))return t.map(rn);if(t instanceof Object&&!(t instanceof
Number||t instanceof String||t instanceof Boolean)){const e={};for(const r in
t)e[r]=rn(t[r]);return e}return en(t)}function nn(t){const
e=t.key,r=t.value,n=t.valueSpec||{},i=t.objectElementValidators||
{},a=t.style,s=t.styleSpec,o=t.validateSpec;let l=[];const
u=kr(r);if("object"!==u)return [new ot(e,r,"object expected, ${u}
found`)]};for(const t in r){const u=t.split(".")[0],c=n[u]||n["*"];let
h;if(i[u])h=i[u];else if(n[u])h=o;else if(i["*"])h=i["*"];else {if(!n["*"])
{l.push(new ot(e,r[t],"unknown property
`${t}``));continue}h=o;}l=l.concat(h({key:(e?
`${e}.`+t,value:r[t],valueSpec:c,style:a,styleSpec:s,object:r,objectKey:t,valid
ateSpec:o},r));}for(const t in n)i[t]||n[t].required&&void
0===n[t].default&&void 0===r[t]&&l.push(new ot(e,r,"missing required property
`${t}``));return l}function an(t){const
e=t.value,r=t.valueSpec,n=t.style,i=t.styleSpec,a=t.key,s=t.arrayElementValidato
r||t.validateSpec;if("array"!==kr(e))return [new ot(a,e,"array expected,
${kr(e)} found`)]};if(r.length&&e.length!==r.length)return [new ot(a,e,"array
length ${r.length} expected, length ${e.length} found`)]};if(r["min-
length"]&&e.length<r["min-length"])return [new ot(a,e,"array length at least
${r["min-length"]} expected, length ${e.length} found`)]};let o=
{type:r.value,values:r.values};i.$version<7&&
(o.function=r.function),"object"===kr(r.value)&&(o=r.value);let l=[];for(let
r=0;r<e.length;r++)l=l.concat(s({array:e,arrayIndex:r,value:e[r],valueSpec:o,valid
ateSpec:t.validateSpec,style:n,styleSpec:i,key:`${a}[${r}]`)}));return
l}function sn(t){const e=t.key,r=t.value,n=t.valueSpec;let i=kr(r);return
"number"===i&&r!=""&&(i="NaN"),"number"!==i?[new ot(e,r,"number expected, ${i}
found`)]:"minimum" in n&&r<n.minimum?[new ot(e,r,"${r} is less than the minimum
value ${n.minimum}`)]:"maximum" in n&&r>n.maximum?[new ot(e,r,"${r} is greater
than the maximum value ${n.maximum}`)]:[]}function on(t){const
e=t.valueSpec,r=en(t.value.type);let n,i,a,s={};const o="categorical"!==r&&void
0===t.value.property,l=!o,u="array"===kr(t.value.stops)&&"array"===kr(t.value.st
ops[0])&&"object"===kr(t.value.stops[0]
[0]),c=nn({key:t.key,value:t.value,valueSpec:t.styleSpec,function,validateSpec:t
.validateSpec,style:t.style,styleSpec:t.styleSpec,objectElementValidators:
{stops:function(t){if("identity"===r)return [new ot(t.key,t.value,"identity
function may not have a \"stops\" property')]};let e=[];const n=t.value;return
e.concat(an({key:t.key,value:n,valueSpec:t.valueSpec,validateSpec:t.validateSp
ec,style:t.style,styleSpec:t.styleSpec,arrayElementValidator:h})),,"array"===kr(n
)&&0===n.length&&e.push(new ot(t.key,n,"array must have at least one
stop")),e,default:function(t){return
t.validateSpec({key:t.key,value:t.value,valueSpec:e,validateSpec:t.validateSpec,
style:t.style,styleSpec:t.styleSpec}})});return "identity"===r&&o&&c.push(new
ot(t.key,t.value,"missing required property
\"property\"")),,"identity"===r||t.value.stops||c.push(new
ot(t.key,t.value,"missing required property
\"stops\"")),,"exponential"===r&&t.valueSpec.expression&&!Sr(t.valueSpec)&&c.push(n
ew ot(t.key,t.value,"exponential functions not
supported")),t.styleSpec.$version>=8&&(l&&!_r(t.valueSpec)?c.push(new
ot(t.key,t.value,"property functions not
supported")):o&&!Ar(t.valueSpec)&&c.push(new ot(t.key,t.value,"zoom functions
not supported"))),"categorical"!==r&&!u||void 0!==t.value.property||c.push(new
ot(t.key,t.value,"property property is required")),c;function h(t){let r=
[];const n=t.value,o=t.key;if("array"!==kr(n))return [new ot(o,n,"array
expected, ${kr(n)} found`)]};if(2!==n.length)return [new ot(o,n,"array length 2
expected, length ${n.length} found`)]};if(u){if("object"!==kr(n[0]))return [new

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ot(o,n,`object expected, ${kr(n[0])} found`);if(void 0===n[0].zoom)return [new
ot(o,n,"object stop key must have zoom");if(void 0===n[0].value)return [new
ot(o,n,"object stop key must have value");if(a&&a>en(n[0].zoom))return [new
ot(o,n[0].zoom,"stop zoom values must appear in ascending
order")];en(n[0].zoom)!=a&&(a=en(n[0].zoom),i=void 0,s=
{}),r=r.concat(nn({key:`${o}[0]`,value:n[0],valueSpec:{zoom:
{}},validateSpec:t.validateSpec,style:t.style,styleSpec:t.styleSpec,objectElemen
tValidators:{zoom:sn,value:p}}));}else r=r.concat(p({key:`${o}
[0]`,value:n[0],valueSpec:
{}},validateSpec:t.validateSpec,style:t.style,styleSpec:t.styleSpec},n));return
Tr(rn(n[1]))?r.concat([new ot(`${o}[1]`,n[1],"expressions are not allowed in
function stops.")]:r.concat(t.validateSpec({key:`${o}
[1]`,value:n[1],valueSpec:e,validateSpec:t.validateSpec,style:t.style,styleSpec:
t.styleSpec}))}function p(t,a){const
o=kr(t.value),l=en(t.value),u=null!==(t.value?>t.value:a;if(n){if(o!==(n)return
[new ot(t.key,u,`${o} stop domain type must match previous stop domain type
${n}`)]}else n=o;if("number"!=o&&"string"!=o&&"boolean"!=o)return [new
ot(t.key,u,"stop domain value must be a number, string, or
boolean");if("number"!=o&&"categorical"!=r){let n=`number expected, ${o}
found`;return _r(e)&&void 0===r&&(n+=`\nIf you intended to use a categorical
function, specify `type`: "categorical"`.'),[new ot(t.key,u,n)]}return
"categorical"!=r||"number"!=o||isFinite(l)&&Math.floor(l)===l?"categorical"!=
r&&"number"===o&&void 0!=i&&l<i?[new ot(t.key,u,"stop domain values must appear
in ascending order")]:(i=l,"categorical"===r&&l in s?[new ot(t.key,u,"stop
domain values must be unique")]:(s[l]=!0,[]):[new ot(t.key,u,`integer expected,
found ${l}`)]}function ln(t){const e=("property"===t.expressionContext?Or:$r)
(rn(t.value),t.valueSpec);if("error"===e.result)return e.value.map((e=>new
ot(`${t.key}${e.key}`,t.value,e.message)));const
r=e.value.expression|e.value._styleExpression.expression;if("property"===t.expr
essionContext&&"text-font"===t.propertyKey&&!r.outputDefined())return [new
ot(t.key,t.value,`Invalid data expression for "${t.propertyKey}". Output values
must be contained as literals within the
expression.`)];if("property"===t.expressionContext&&"layout"===t.propertyType&&!
Oe(r))return [new ot(t.key,t.value,"feature-state" data expressions are not
supported with layout
properties.`)];if("filter"===t.expressionContext&&!Oe(r))return [new
ot(t.key,t.value,"feature-state" data expressions are not supported with
filters.`)];if(t.expressionContext&&0===t.expressionContext.indexOf("cluster"))
{if(!Ue(r,["zoom","feature-state"]))return [new ot(t.key,t.value,"zoom" and
"feature-state" expressions are not supported with cluster
properties.`)];if("cluster-initial"===t.expressionContext&&!De(r))return [new
ot(t.key,t.value,"Feature data expressions are not supported with initial
expression part of cluster properties.`)]}return []}function un(t){const
e=t.key,r=t.value,n=t.valueSpec,i=[];return Array.isArray(n.values)?
-1===n.values.indexOf(en(r))&&i.push(new ot(e,r,`expected one of
[${n.values.join(", ")}], ${JSON.stringify(r)}
found`)): -1===Object.keys(n.values).indexOf(en(r))&&i.push(new ot(e,r,`expected
one of [${Object.keys(n.values).join(", ")}], ${JSON.stringify(r)}
found`)),i}function cn(t){return qr(rn(t.value))?ln(lt({},t,
{expressionContext:"filter",valueSpec:{value:"boolean"}})):hn(t)}function hn(t)
{const e=t.value,r=t.key;if("array"!=kr(e))return [new ot(r,e,`array expected,
${kr(e)} found`)];const n=t.styleSpec;let i,a=[];if(e.length<1)return [new
ot(r,e,"filter array must have at least 1
element")];switch(a=a.concat(un({key:`${r}
[0]`,value:e[0],valueSpec:n.filter_operator,style:t.style,styleSpec:t.styleSpec}
)),en(e[0])){case"<":case"
<=":case">":case">=":e.length>=2&&"$type"===en(e[1])&&a.push(new ot(r,e,`"$type"
cannot be use with operator
"${e[0]}"`));case"===":case"!=":3!==(e.length&&a.push(new ot(r,e,`filter array for
operator "${e[0]}" must have 3 elements`));case"in":case"!in":e.length>=2&&
(i=kr(e[1]),"string"!=i&&a.push(new ot(`${r}[1]`,e[1],`string expected, ${i}
found`)));for(let s=2;s<e.length;s++)i=kr(e[s]),"$type"===en(e[1])?

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a=a.concat(un({key:`${r}
[${s}]`,value:e[s],valueSpec:n.geometry_type,style:t.style,styleSpec:t.styleSpec
})): "string"!==i&&"number"!==i&&"boolean"!==i&&a.push(new ot(`${r}
[${s}]`,e[s],`string, number, or boolean expected, ${i}
found`));break;case"any":case"all":case"none":for(let
n=1;n<e.length;n++)a=a.concat(hn({key:`${r}
[${n}]`,value:e[n],style:t.style,styleSpec:t.styleSpec}));break;case"has":case"!
has":i=kr(e[1]),2!==e.length?a.push(new ot(r,e,`filter array for "${e[0]}"
operator must have 2 elements`)): "string"!==i&&a.push(new ot(`${r}
[1]`,e[1],`string expected, ${i}
found`));break;case"within":i=kr(e[1]),2!==e.length?a.push(new ot(r,e,`filter
array for "${e[0]}" operator must have 2 elements`)): "object"!==i&&a.push(new
ot(`${r}[1]`,e[1],`object expected, ${i} found`));}return a}function pn(t,e)
{const
r=t.key,n=t.validateSpec,i=t.style,a=t.styleSpec,s=t.value,o=t.objectKey,l=a[`${
e}_${t.layerType}`];if(!l)return [];const u=o.match(/^(.*)-
transition$/);if("paint"===e&&u&&l[u[1]]&&l[u[1]].transition)return
n({key:r,value:s,valueSpec:a.transition,style:i,styleSpec:a});const
c=t.valueSpec||l[o];if(!c)return [new ot(r,s,`unknown property "${o}"`)];let
h;if("string"===kr(s)&&_r(c)&&!c.tokens&&(h=/^{\[\^}\]\+})$/).exec(s))return [new
ot(r,s,`${o}" does not support interpolation syntax\nUse an identity property
function instead: \`${"type": "identity", "property": ${JSON.stringify(h[1])}
}\``)];const p=[];return "symbol"===t.layerType&&("text-
field"===o&&i&&i.glyphs&&p.push(new ot(r,s,`use of "text-field" requires a
style "glyphs" property`)), "text-
font"===o&&lr(rn(s))&&"identity"===en(s.type)&&p.push(new ot(r,s,`"text-font"
does not support identity
functions`))),p.concat(n({key:t.key,value:s,valueSpec:c,style:i,styleSpec:a,expr
essionContext:"property",propertyType:e,propertyKey:o}))}function fn(t){return
pn(t,"paint")}function dn(t){return pn(t,"layout")}function yn(t){let e=[];const
r=t.value,n=t.key,i=t.style,a=t.styleSpec;r.type||r.ref||e.push(new
ot(n,r,`either "type" or "ref" is required`));let s=en(r.type);const
o=en(r.ref);if(r.id){const a=en(r.id);for(let s=0;s<t.arrayIndex;s++){const
t=i.layers[s];en(t.id)===a&&e.push(new ot(n,r.id,`duplicate layer id "${r.id}",
previously used at line ${t.id.__line__}`));}}if("ref" in r){let t;
["type","source","source-layer","filter","layout"].forEach((t=>{t in
r&&e.push(new ot(n,r[t],`${t}" is prohibited for ref
layers`));})),i.layers.forEach((e=>{en(e.id)===o&&(t=e);})),t?t.ref?e.push(new
ot(n,r.ref,`ref cannot reference another ref layer`)):s=en(t.type):e.push(new
ot(n,r.ref,`ref layer "${o}" not found`));}else if("background"!==s)if(r.source)
{const
t=i.sources&&i.sources[r.source],a=t&&en(t.type);t?"vector"===a&&"raster"===s?
e.push(new ot(n,r.source,`layer "${r.id}" requires a raster source`)): "raster-
dem"===a&&"hillshade"===s?e.push(new ot(n,r.source,`layer "${r.id}" requires a
raster-dem source`)): "raster"===a&&"raster"!==s?e.push(new ot(n,r.source,`layer
"${r.id}" requires a vector source`)): "vector"!==a||r["source-layer"]?"raster-
dem"===a&&"hillshade"!==s?e.push(new ot(n,r.source,`raster-dem source can only
be used with layer type 'hillshade'`)): "line"!==s||!r.paint||!r.paint["line-
gradient"]||"geojson"===a&&t.lineMetrics||e.push(new ot(n,r,`layer "${r.id}"
specifies a line-gradient, which requires a GeoJSON source with \`lineMetrics\`
enabled`)):e.push(new ot(n,r,`layer "${r.id}" must specify a "source-
layer`)):e.push(new ot(n,r.source,`source "${r.source}" not found`));}else
e.push(new ot(n,r,`missing required property "source"`));return
e=e.concat(nn({key:n,value:r,valueSpec:a.layer,style:t.style,styleSpec:t.styleSp
ec,validateSpec:t.validateSpec,objectElementValidators:{"*":()=>[],type:
()=>t.validateSpec({key:`${n}.type`,value:r.type,valueSpec:a.layer.type,style:t.
style,styleSpec:t.styleSpec,validateSpec:t.validateSpec,object:r,objectKey:"type
"},filter:cn,layout:t=>nn({layer:r,key:t.key,value:t.value,style:t.style,styleS
pec:t.styleSpec,validateSpec:t.validateSpec,objectElementValidators:
{"*":t=>dn(lt({layerType:s},t))}),paint:t=>nn({layer:r,key:t.key,value:t.value,
style:t.style,styleSpec:t.styleSpec,validateSpec:t.validateSpec,objectElementVal
idators:{"*":t=>fn(lt({layerType:s},t))}})})),e}function mn(t){const

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e=t.value,r=t.key,n=kr(e);return "string"!==n?[new ot(r,e,`string expected, ${n}
found`)]:[{}]}const gn={promoteId:function({key:t,value:e})
{if("string"===kr(e))return mn({key:t,value:e});{const r=[];for(const n in
e)r.push(...mn({key:`${t}.${n}`,value:e[n]}));return r}}};function xn(t){const
e=t.value,r=t.key,n=t.styleSpec,i=t.style,a=t.validateSpec;if(!e.type)return
[new ot(r,e,`"type" is required`)];const s=en(e.type);let o;switch(s)
{case"vector":case"raster":return
o=nn({key:r,value:e,valueSpec:n[`${source}_${s.replace("-","_")}`],style:t.style,styleSpec:n,objectElementValidators:gn,validateSpec:a}),
o;case"raster-dem":return o=function(t){var e;const r=null!==(
e=t.sourceName)&&void 0!==(e=
e:""),n=t.value,i=t.styleSpec,a=i.source_raster_dem,s=t.style;let o=[];const
l=kr(n);if(void 0===n)return o;if("object"!==l)return o.push(new
ot("source_raster_dem",n,`object expected, ${l} found`)),o;const
u="custom"===en(n.encoding),c=
["redFactor","greenFactor","blueFactor","baseShift"],h=t.value.encoding?
`${t.value.encoding}`:"Default";for(const e in n)!u&&c.includes(e)?o.push(new
ot(e,n[e],`In "${r}": "${e}" is only valid when "encoding" is set to "custom".
${h} encoding found`)):a[e]?
o=o.concat(t.validateSpec({key:e,value:n[e],valueSpec:a[e],validateSpec:t.valida
teSpec,style:s,styleSpec:i})):o.push(new ot(e,n[e],`unknown property
"${e}"`));return o}
({sourceName:r,value:e,style:t.style,styleSpec:n,validateSpec:a}),o;case"geojson
":if(o=nn({key:r,value:e,valueSpec:n.source_geojson,style:i,styleSpec:n,validate
Spec:a,objectElementValidators:gn}),e.cluster)for(const t in
e.clusterProperties){const[n,i]=e.clusterProperties[t],s="string"===typeof n?[n,
["accumulated"]],
["get",t]]:n;o.push(...ln({key:`${r}.${t}.map`,value:i,validateSpec:a,expression
Context:"cluster-
map"})),o.push(...ln({key:`${r}.${t}.reduce`,value:s,validateSpec:a,expressionCo
ntext:"cluster-reduce"}))};return o;case"video":return
nn({key:r,value:e,valueSpec:n.source_video,style:i,validateSpec:a,styleSpec:n});
case"image":return
nn({key:r,value:e,valueSpec:n.source_image,style:i,validateSpec:a,styleSpec:n});
case"canvas":return [new ot(r,null,"Please use runtime APIs to add canvas
sources, rather than including them in
stylesheets.",`source.canvas`)];default:return
un({key:`${r}.type`,value:e.type,valueSpec:{values:["vector","raster","raster-
dem","geojson","video","image"],style:i,validateSpec:a,styleSpec:n}})}function
vn(t){const e=t.value,r=t.styleSpec,n=r.light,i=t.style;let a=[];const
s=kr(e);if(void 0===e)return a;if("object"!==s)return a=a.concat([new
ot("light",e,`object expected, ${s} found`)]),a;for(const s in e){const
o=s.match(/^(.*)-transition$/);a=a.concat(o&&n[o[1]]&&n[o[1]].transition?
t.validateSpec({key:s,value:e[s],valueSpec:r.transition,validateSpec:t.validates
pec,style:i,styleSpec:r}):n[s]?
t.validateSpec({key:s,value:e[s],valueSpec:n[s],validateSpec:t.validateSpec,styl
e:i,styleSpec:r}):[new ot(s,e[s],`unknown property "${s}"`)]};return a}function
bn(t){const e=t.value,r=t.styleSpec,n=r.terrain,i=t.style;let a=[];const
s=kr(e);if(void 0===e)return a;if("object"!==s)return a=a.concat([new
ot("terrain",e,`object expected, ${s} found`)]),a;for(const s in
e)a=a.concat(n[s]?
t.validateSpec({key:s,value:e[s],valueSpec:n[s],validateSpec:t.validateSpec,styl
e:i,styleSpec:r}):[new ot(s,e[s],`unknown property "${s}"`)]};return a}function
wn(t){let e=[];const r=t.value,n=t.key;if(Array.isArray(r)){const i=[],a=
[];for(const s in r)r[s].id&&i.includes(r[s].id)&&e.push(new ot(n,r,`all the
sprites' ids must be unique, but ${r[s].id} is
duplicated`)),i.push(r[s].id),r[s].url&&a.includes(r[s].url)&&e.push(new
ot(n,r,`all the sprites' URLs must be unique, but ${r[s].url} is
duplicated`)),a.push(r[s].url),e=e.concat(nn({key:`${n}
[${s}]`,value:r[s],valueSpec:{id:{type:"string",required:!0},url:
{type:"string",required:!0}},validateSpec:t.validateSpec)}));return e}return
mn({key:n,value:r})}const _n={"*":()=>=>[],array:an,boolean:function(t){const

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e=t.value,r=t.key,n=kr(e);return "boolean"!==n?[new ot(r,e,`boolean expected,
${n} found`)]:[],number:sn,color:function(t){const
e=t.key,r=t.value,n=kr(r);return "string"!==n?[new ot(e,r,`color expected, ${n}
found`)]:Yt.parse(String(r))?[[]:[new ot(e,r,`color expected, "${r}"
found`)]],constants:tn,enum:un,filter:cn,function:on,layer:yn,object:nn,source:x
n,light:vn,terrain:bn,string:mn,formatted:function(t){return 0===mn(t).length?
[]:ln(t)},resolvedImage:function(t){return 0===mn(t).length?
[]:ln(t)},padding:function(t){const e=t.key,r=t.value;if("array"===kr(r))
{if(r.length<1||r.length>4)return [new ot(e,r,`padding requires 1 to 4 values;
${r.length} values found`)]};const n={type:"number"};let i=[];for(let
a=0;a<r.length;a++)i=i.concat(t.validateSpec({key:`${e}
[${a}]`,value:r[a],validateSpec:t.validateSpec,valueSpec:n}));return i}return
sn({key:e,value:r,valueSpec:{}})},variableAnchorOffsetCollection:function(t)
{const
e=t.key,r=t.value,n=kr(r),i=t.styleSpec;if("array"!==n||r.length<1||r.length%2!=
0)return [new ot(e,r,"variableAnchorOffsetCollection requires a non-empty array
of even length")];let a=[];for(let n=0;n<r.length;n+=2)a=a.concat(un({key:`${e}
[${n}]`,value:r[n],valueSpec:i.layout_symbol["text-
anchor"]}))},a=a.concat(an({key:`${e}[${n+1}]`,value:r[n+1],valueSpec:
{length:2,value:"number"},validateSpec:t.validateSpec,style:t.style,styleSpec:i}
));return a},sprite:wn};function An(t){const
e=t.value,r=t.valueSpec,n=t.styleSpec;return
t.validateSpec=An,r.expression&&lr(en(e))?on(t):r.expression&&Tr(rn(e))?
ln(t):r.type&&n[r.type]?_n[r.type](t):nn(lt({},t,{valueSpec:r.type?
n[r.type]:r}))}function Sn(t){const e=t.value,r=t.key,n=mn(t);return n.length||
(-1===e.indexOf("{fontstack}")&&n.push(new ot(r,e,"glyphs" url must include a "
{fontstack}" token')),(-1===e.indexOf("{range}")&&n.push(new ot(r,e,"glyphs" url
must include a "{range}" token'))),n}function kn(t,e=X){let r=[];return
r=r.concat(An({key:" ",value:t,valueSpec:e.$root,styleSpec:e,style:t,validateSpec
:An,objectElementValidators:{glyphs:Sn,"*":()=>[]}))},t.constants&&
(r=r.concat(tn({key:"constants",value:t.constants,style:t,styleSpec:e,validateSp
ec:An}))),zn(r)}function In(t){return function(e){return
t({...e,validateSpec:An})}}function zn(t){return
[].concat(t).sort((t,e)=>t.line-e.line)}function Mn(t){return function(...e)
{return
zn(t.apply(this,e))}}kn.source=Mn(In(xn)),kn.sprite=Mn(In(wn)),kn.glyphs=Mn(In(S
n)),kn.light=Mn(In(vn)),kn.terrain=Mn(In(bn)),kn.layer=Mn(In(yn)),kn.filter=Mn(In
(cn)),kn.paintProperty=Mn(In(fn)),kn.layoutProperty=Mn(In(dn));const
Cn=kn,Bn=Cn.light,Pn=Cn.paintProperty,Vn=Cn.layoutProperty;function En(t,e){let
r=!1;if(e&&e.length)for(const n of e)t.fire(new G(new
Error(n.message))),r=!0;return r}class Fn{constructor(t,e,r){const n=this.cells=
[];if(t instanceof ArrayBuffer){this.arrayBuffer=t;const i=new
Int32Array(this.arrayBuffer);t=i[0],this.d=(e=i[1])+2*(r=i[2]);for(let
t=0;t<this.d*this.d;t++){const e=i[3+t],r=i[3+t+1];n.push(e===r?
null:i.subarray(e,r));}const
a=i[3+n.length+1];this.keys=i.subarray(i[3+n.length],a),this.bboxes=i.subarray(a
),this.insert=this._insertReadonly;}else {this.d=e+2*r;for(let
t=0;t<this.d*this.d;t++)n.push([]);this.keys=[],this.bboxes=
[];}this.n=e,this.extent=t,this.padding=r,this.scale=e/t,this.uid=0;const
i=r/e*t,this.min=-i,this.max=t+i;insert(t,e,r,n,i)
{this._forEachCell(e,r,n,i,this._insertCell,this.uid++,void 0,void
0),this.keys.push(t),this.bboxes.push(e),this.bboxes.push(r),this.bboxes.push(n)
,this.bboxes.push(i);}_insertReadonly(){throw new Error("Cannot insert into a
GridIndex created from an ArrayBuffer.")}_insertCell(t,e,r,n,i,a)
{this.cells[i].push(a);}query(t,e,r,n,i){const
a=this.min,s=this.max;if(t<=a&&e<=a&&s<=r&&s<=n&&!i)return
Array.prototype.slice.call(this.keys);{const a=[];return
this._forEachCell(t,e,r,n,this._queryCell,a,{},i,a)}_queryCell(t,e,r,n,i,a,s,o)
{const l=this.cells[i];if(null!==l){const i=this.keys,u=this.bboxes;for(let
c=0;c<l.length;c++){const h=l[c];if(void 0===s[h]){const l=4*h;(o?
o(u[l+0],u[l+1],u[l+2],u[l+3]):t<=u[l+2]&&e<=u[l+3]&&r>=u[l+0]&&n>=u[l+1]))?
(s[h]=!0,a.push(i[h])):s[h]=!1;}}}_forEachCell(t,e,r,n,i,a,s,o){const

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l=this._convertToCellCoord(t),u=this._convertToCellCoord(e),c=this._convertToCellCoord(r),h=this._convertToCellCoord(n);for(let p=1;p<=c;p++)for(let l=u;l<=h;l++){const u=this.d*l+p;if(!o||o(this._convertFromCellCoord(p),this._convertFromCellCoord(l),this._convertFromCellCoord(p+1),this._convertFromCellCoord(l+1)))&&i.call(this,s,t,e,r,n,u,a,s,o))return}}_convertFromCellCoord(t){return (t-this.padding)/this.scale}_convertToCellCoord(t){return Math.max(0,Math.min(this.d-1,Math.floor(t*this.scale)+this.padding))}toArrayBuffer(){if(this.arrayBuffer)return this.arrayBuffer;const t=this.cells,e=3+this.cells.length+1+1;let r=0;for(let t=0;t<this.cells.length;t++)r+=this.cells[t].length;const n=new Int32Array(e+r+this.keys.length+this.bboxes.length);n[0]=this.extent,n[1]=this.n,n[2]=this.padding;let i=e;for(let e=0;e<t.length;e++){const r=t[e];n[3+e]=i,n.set(r,i),i+=r.length;return n[3+t.length]=i,n.set(this.keys,i),i+=this.keys.length,n[3+t.length+1]=i,n.set(this.bboxes,i),i+=this.bboxes.length,n.buffer}static serialize(t,e){const r=t.toArrayBuffer();return e&&e.push(r,{buffer:r})static deserialize(t){return new Fn(t.buffer)}}const Tn={};function $n(t,e,r={}){if(Tn[t])throw new Error(`${t} is already registered.`);Object.defineProperty(e,"_classRegistryKey",{value:t,writeable:!1}),Tn[t]={klass:e,omit:r.omit||[],shallow:r.shallow||[]};$n("Object",Object),$n("TransferableGridIndex",Fn),$n("Color",Yt),$n("Error",Error),$n("AJAXError",L),$n("ResolvedImage",ne),$n("StylePropertyFunction",Ur),$n("StyleExpression",Fr,{omit:["_evaluator"]}),$n("ZoomDependentExpression",Dr),$n("ZoomConstantExpression",Lr),$n("CompoundExpression",Se,{omit:["_evaluate"]});for(const t in yr)yr[t]._classRegistryKey|$n(`Expression_${t}`,yr[t]);function Ln(t){return t&&"undefined"!=typeof ArrayBuffer&&(t instanceof ArrayBuffer||t.constructor&&"ArrayBuffer"===t.constructor.name)}function Dn(t,e){if(null==t||"boolean"===typeof t||"number"===typeof t||"string"===typeof t||t instanceof Boolean||t instanceof Number||t instanceof String||t instanceof Date||t instanceof RegExp||t instanceof Blob)return t;if(Ln(t))return e&&e.push(t),t;if(M(t))return e&&e.push(t),t;if(ArrayBuffer.isView(t)){const r=t;return e&&e.push(r.buffer),r}if(t instanceof ImageData)return e&&e.push(t.data.buffer),t;if(Array.isArray(t)){const r=[];for(const n of t)r.push(Dn(n,e));return r}if("object"===typeof t){const r=t.constructor,n=r._classRegistryKey;if(!n)throw new Error("can't serialize object of unregistered class");if(!Tn[n])throw new Error(`${n} is not registered.`);const i=r.serialize?r.serialize(t,e):{};if(r.serialize){if(e&&i===e[e.length-1])throw new Error("statically serialized object won't survive transfer of $name property")}else {for(const r in t){if(!t.hasOwnProperty(r))continue;if(Tn[n].omit.indexOf(r)>=0)continue;const a=t[r];i[r]=Tn[n].shallow.indexOf(r)>=0?a:Dn(a,e);}t instanceof Error&&(i.message=t.message);}if(i.$name)throw new Error("$name property is reserved for worker serialization logic.");return "Object"!==n&&(i.$name=n),i}throw new Error("can't serialize object of type "+typeof t)}function On(t){if(null==t||"boolean"===typeof t||"number"===typeof t||"string"===typeof t||t instanceof Boolean||t instanceof Number||t instanceof String||t instanceof Date||t instanceof RegExp||t instanceof Blob||Ln(t)||M(t)||ArrayBuffer.isView(t)||t instanceof ImageData)return t;if(Array.isArray(t))return t.map(On);if("object"===typeof t){const e=t.$name||"Object";if(!Tn[e])throw new Error(`can't deserialize unregistered class ${e}`);const {klass:r}=Tn[e];if(!r)throw new Error(`can't deserialize unregistered class ${e}`);if(r.deserialize)return r.deserialize(t);const n=Object.create(r.prototype);for(const r of Object.keys(t)){if("$name"===r)continue;const i=t[r];n[r]=Tn[e].shallow.indexOf(r)>=0?i:On(i);}return n}throw new Error("can't deserialize object of type "+typeof t)}class Un{constructor(){this.first=!0;update(t,e){const r=Math.floor(t);return this.first?(this.first=!1,this.lastIntegerZoom=r,this.lastIntegerZoomTime=0,this.lastZoom=t,this.lastFloorZoom=r,!0):(this.lastFloorZoom>r?

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(this.lastIntegerZoom=r+1,this.lastIntegerZoomTime=e):this.lastFloorZoom<r&&
(this.lastIntegerZoom=r,this.lastIntegerZoomTime=e),t!==this.lastZoom&&
(this.lastZoom=t,this.lastFloorZoom=r,!0))}}const Rn={"Latin-1
Supplement":t=>t>=128&&t<=255,Arabic:t=>t>=1536&&t<=1791,"Arabic
Supplement":t=>t>=1872&&t<=1919,"Arabic Extended-A":t=>t>=2208&&t<=2303,"Hangul
Jamo":t=>t>=4352&&t<=4607,"Unified Canadian Aboriginal
Syllabics":t=>t>=5120&&t<=5759,Khmer:t=>t>=6016&&t<=6143,"Unified Canadian
Aboriginal Syllabics Extended":t=>t>=6320&&t<=6399,"General
Punctuation":t=>t>=8192&&t<=8303,"Letterlike
Symbols":t=>t>=8448&&t<=8527,"Number Forms":t=>t>=8528&&t<=8591,"Miscellaneous
Technical":t=>t>=8960&&t<=9215,"Control Pictures":t=>t>=9216&&t<=9279,"Optical
Character Recognition":t=>t>=9280&&t<=9311,"Enclosed
Alphanumerics":t=>t>=9312&&t<=9471,"Geometric
Shapes":t=>t>=9632&&t<=9727,"Miscellaneous
Symbols":t=>t>=9728&&t<=9983,"Miscellaneous Symbols and
Arrows":t=>t>=11008&&t<=11263,"CJK Radicals
Supplement":t=>t>=11904&&t<=12031,"Kangxi
Radicals":t=>t>=12032&&t<=12255,"Ideographic Description
Characters":t=>t>=12272&&t<=12287,"CJK Symbols and
Punctuation":t=>t>=12288&&t<=12351,Hiragana:t=>t>=12352&&t<=12447,Katakana:t=>t>
=12448&&t<=12543,Bopomofo:t=>t>=12544&&t<=12591,"Hangul Compatibility
Jamo":t=>t>=12592&&t<=12687,Kanbun:t=>t>=12688&&t<=12703,"Bopomofo
Extended":t=>t>=12704&&t<=12735,"CJK Strokes":t=>t>=12736&&t<=12783,"Katakana
Phonetic Extensions":t=>t>=12784&&t<=12799,"Enclosed CJK Letters and
Months":t=>t>=12800&&t<=13055,"CJK Compatibility":t=>t>=13056&&t<=13311,"CJK
Unified Ideographs Extension A":t=>t>=13312&&t<=19903,"Yijing Hexagram
Symbols":t=>t>=19904&&t<=19967,"CJK Unified
Ideographs":t=>t>=19968&&t<=40959,"Yi Syllables":t=>t>=40960&&t<=42127,"Yi
Radicals":t=>t>=42128&&t<=42191,"Hangul Jamo Extended-
A":t=>t>=43360&&t<=43391,"Hangul Syllables":t=>t>=44032&&t<=55215,"Hangul Jamo
Extended-B":t=>t>=55216&&t<=55295,"Private Use Area":t=>t>=57344&&t<=63743,"CJK
Compatibility Ideographs":t=>t>=63744&&t<=64255,"Arabic Presentation Forms-
A":t=>t>=64336&&t<=65023,"Vertical Forms":t=>t>=65040&&t<=65055,"CJK
Compatibility Forms":t=>t>=65072&&t<=65103,"Small Form
Variants":t=>t>=65104&&t<=65135,"Arabic Presentation Forms-
B":t=>t>=65136&&t<=65279,"Halfwidth and Fullwidth
Forms":t=>t>=65280&&t<=65519};function qn(t){for(const e of
t)if(Zn(e.charCodeAt(0)))return !0;return !1}function jn(t){for(const e of
t)if(!Nn(e.charCodeAt(0)))return !1;return !0}function Nn(t){return !
(Rn.Arabic(t)||Rn["Arabic Supplement"](t)||Rn["Arabic Extended-A"]
(t)||Rn["Arabic Presentation Forms-A"](t)||Rn["Arabic Presentation Forms-B"]
(t))}function Zn(t){return !(746!==t&&747!==t&&(t<4352||!(Rn["Bopomofo
Extended"](t)||Rn.Bopomofo(t)||Rn["CJK Compatibility Forms"](t)&&
(t>=65097&&t<=65103)||Rn["CJK Compatibility Ideographs"](t)||Rn["CJK
Compatibility"](t)||Rn["CJK Radicals Supplement"](t)||Rn["CJK Strokes"](t)||
(!Rn["CJK Symbols and Punctuation"]
(t)||t>=12296&&t<=12305||t>=12308&&t<=12319||12336===t)||Rn["CJK Unified
Ideographs Extension A"](t)||Rn["CJK Unified Ideographs"](t)||Rn["Enclosed CJK
Letters and Months"](t)||Rn["Hangul Compatibility Jamo"](t)||Rn["Hangul Jamo
Extended-A"](t)||Rn["Hangul Jamo Extended-B"](t)||Rn["Hangul Jamo"]
(t)||Rn["Hangul Syllables"](t)||Rn.Hiragana(t)||Rn["Ideographic Description
Characters"](t)||Rn.Kanbun(t)||Rn["Kangxi Radicals"](t)||Rn["Katakana Phonetic
Extensions"](t)||Rn.Katakana(t)&&12540!==t||!(Rn["Halfwidth and Fullwidth
Forms"]
(t)||65288===t||65289===t||65293===t||t>=65306&&t<=65310||65339===t||65341===t||
65343===t||t>=65371&&t<=65503||65507===t||t>=65512&&t<=65519)||!(Rn["Small Form
Variants"](t)||t>=65112&&t<=65118||t>=65123&&t<=65126)||Rn["Unified Canadian
Aboriginal Syllabics"](t)||Rn["Unified Canadian Aboriginal Syllabics Extended"]
(t)||Rn["Vertical Forms"](t)||Rn["Yijing Hexagram Symbols"](t)||Rn["Yi
Syllables"](t)||Rn["Yi Radicals"](t))))}function Kn(t){return !
(Zn(t)||function(t){return !(Rn["Latin-1 Supplement"](t)&&
(167===t||169===t||174===t||177===t||188===t||189===t||190===t||215===t||247===t

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)|Rn["General Punctuation"])(t)&&
(8214===t|8224===t|8225===t|8240===t|8241===t|8251===t|8252===t|8258===t|
|8263===t|8264===t|8265===t|8273===t)|Rn["Letterlike Symbols"]
(t)|Rn["Number Forms"])(t)|Rn["Miscellaneous Technical"])(t)&&
(t>=8960&&t<=8967|t>=8972&&t<=8991|t>=8996&&t<=9e3|9003===t|t>=9085&&t<=9114
|t>=9150&&t<=9165|9167===t|t>=9169&&t<=9179|t>=9186&&t<=9215)|Rn["Control
Pictures"])(t)&&9251!===t|Rn["Optical Character Recognition"])(t)|Rn["Enclosed
Alphanumerics"])(t)|Rn["Geometric Shapes"])(t)|Rn["Miscellaneous Symbols"])(t)&&!
(t>=9754&&t<=9759)|Rn["Miscellaneous Symbols and Arrows"])(t)&&
(t>=11026&&t<=11055|t>=11088&&t<=11097|t>=11192&&t<=11243)|Rn["CJK Symbols
and Punctuation"])(t)|Rn.Katakana(t)|Rn["Private Use Area"])(t)|Rn["CJK
Compatibility Forms"])(t)|Rn["Small Form Variants"])(t)|Rn["Halfwidth and
Fullwidth Forms"]
(t)|8734===t|8756===t|8757===t|t>=9984&&t<=10087|t>=10102&&t<=10131|65532=
==t|65533===t)}(t))function Gn(t){return t>=1424&&t<=2303|Rn["Arabic
Presentation Forms-A"])(t)|Rn["Arabic Presentation Forms-B"])(t)}function Jn(t,e)
{return !(e&&Gn(t)|t>=2304&&t<=3583|t>=3840&&t<=4255|Rn.Khmer(t))}function
Xn(t){for(const e of t)if(Gn(e.charCodeAt(0)))return !0;return !1}const
Yn="deferred",Hn="loading",Wn="loaded";let
Qn=null,ti="unavailable",ei=null;const ri=function(t){t&&"string"===typeof
t&&t.indexOf("NetworkError")>-1&&(ti="error"),Qn&&Qn(t);};function ni()
{ii.fire(new K("pluginStateChange",{pluginStatus:ti,pluginURL:ei}));}const
ii=new J,ai=function(){return ti},si=function(){if(ti!==Yn|!ei)throw new
Error("rtl-text-plugin cannot be downloaded unless a pluginURL is
specified");ti=Hn,ni(),ei&&q({url:ei},{t=>{t?ri(t):(ti=Wn,ni());}});},oi=
{applyArabicShaping:null,processBidirectionalText:null,processStyledBidirectiona
lText:null,isLoaded:()=>ti===Wn|null!=oi.applyArabicShaping,isLoading:
()=>ti===Hn,setState(t){if(!I())throw new Error("Cannot set the state of the
rtl-text-plugin when not in the web-worker
context");ti=t.pluginStatus,ei=t.pluginURL;},isParsed(){if(!I())throw new
Error("rtl-text-plugin is only parsed on the worker-threads");return
null!=oi.applyArabicShaping&&null!=oi.processBidirectionalText&&null!=oi.process
StyledBidirectionalText},getPluginURL(){if(!I())throw new Error("rtl-text-plugin
url can only be queried from the worker threads");return ei}};class
li{constructor(t,e){this.zoom=t,e?
(this.now=e.now,this.fadeDuration=e.fadeDuration,this.zoomHistory=e.zoomHistory,
this.transition=e.transition):
(this.now=0,this.fadeDuration=0,this.zoomHistory=new Un,this.transition=
{});}isSupportedScript(t){return function(t,e){for(const r of
t)if(!Jn(r.charCodeAt(0),e))return !1;return !0}
(t,oi.isLoaded())}crossFadingFactor(){return 0===this.fadeDuration?
1:Math.min((this.now-
this.zoomHistory.lastIntegerZoomTime)/this.fadeDuration,1)}getCrossfadeParameter
s(){const t=this.zoom,e=t-Math.floor(t),r=this.crossFadingFactor();return
t>this.zoomHistory.lastIntegerZoom?{fromScale:2,toScale:1,t:e+(1-e)*r}:
{fromScale:.5,toScale:1,t:1-(1-r)*e}}class ui{constructor(t,e)
{this.property=t,this.value=e,this.expression=function(t,e){if(Ir(t))return new
Ur(t,e);if(Tr(t)){const r=Or(t,e);if("error"===r.result)throw new
Error(r.value.map((t=>`${t.key}: ${t.message}`)).join(", "));return r.value}{let
r=t;return "color"===e.type&&"string"===typeof t?
r=Yt.parse(t):"padding"!==e.type||"number"!==typeof
t&&!Array.isArray(t)?"variableAnchorOffsetCollection"===e.type&&Array.isArray(t)
&&(r=re.parse(t)):r=te.parse(t),{kind:"constant",evaluate:()=>r}}}(void 0===e?
t.specification.default:e,t.specification);}isDataDriven(){return
"source"===this.expression.kind||"composite"===this.expression.kind}possiblyEval
uate(t,e,r){return this.property.possiblyEvaluate(this,t,e,r)}}class
ci{constructor(t){this.property=t,this.value=new ui(t,void 0);}transitioned(t,e)
{return new
pi(this.property,this.value,e,g({},t.transition,this.transition),t.now)}untransi
tioned(){return new pi(this.property,this.value,null,{},0)}}class
hi{constructor(t)
{this._properties=t,this._values=Object.create(t.defaultTransitionablePropertyVa

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lues);}getValue(t){return w(this._values[t].value.value)}setValue(t,e)
{Object.prototype.hasOwnProperty.call(this._values,t)|| (this._values[t]=new
ci(this._values[t].property)),this._values[t].value=new
ui(this._values[t].property,null===e?void 0:w(e));}getTransition(t){return
w(this._values[t].transition)}setTransition(t,e)
{Object.prototype.hasOwnProperty.call(this._values,t)|| (this._values[t]=new
ci(this._values[t].property)),this._values[t].transition=w(e)||void
0;}serialize(){const t={};for(const e of Object.keys(this._values)){const
r=this.getValue(e);void 0!==r&&(t[e]=r);const n=this.getTransition(e);void
0!==n&&(t[`$${e}-transition`]=n);}return t}transitioned(t,e){const r=new
fi(this._properties);for(const n of
Object.keys(this._values))r._values[n]=this._values[n].transitioned(t,e._values[
n]);return r}untransitioned(){const t=new fi(this._properties);for(const e of
Object.keys(this._values))t._values[e]=this._values[e].untransitioned();return
t}}class pi{constructor(t,e,r,n,i)
{this.property=t,this.value=e,this.begin=i+n.delay||0,this.end=this.begin+n.dura
tion||0,t.specification.transition&&(n.delay||n.duration)&&
(this.prior=r);}possiblyEvaluate(t,e,r){const
n=t.now||0,i=this.value.possiblyEvaluate(t,e,r),a=this.prior;if(a)
{if(n>this.end)return this.prior=null,i;if(this.value.isDataDriven())return
this.prior=null,i;if(n<this.begin)return a.possiblyEvaluate(t,e,r);}const s=(n-
this.begin)/(this.end-this.begin);return
this.property.interpolate(a.possiblyEvaluate(t,e,r),i,function(t){if(t<=0)return
0;if(t>=1)return 1;const e=t*t,r=e*t;return 4*(t<.5?r:3*(t-e)+r-.75)}
(s))}}return i}}class fi{constructor(t)
{this._properties=t,this._values=Object.create(t.defaultTransitioningPropertyVal
ues);}possiblyEvaluate(t,e,r){const n=new mi(this._properties);for(const i of
Object.keys(this._values))n._values[i]=this._values[i].possiblyEvaluate(t,e,r);r
eturn n}hasTransition(){for(const t of
Object.keys(this._values))if(this._values[t].prior)return !0;return !1}}class
di{constructor(t)
{this._properties=t,this._values=Object.create(t.defaultPropertyValues);}hasValu
e(t){return void 0!==this._values[t].value}getValue(t){return
w(this._values[t].value)}setValue(t,e){this._values[t]=new
ui(this._values[t].property,null===e?void 0:w(e));}serialize(){const t=
{};for(const e of Object.keys(this._values)){const r=this.getValue(e);void
0!==r&&(t[e]=r);}return t}possiblyEvaluate(t,e,r){const n=new
mi(this._properties);for(const i of
Object.keys(this._values))n._values[i]=this._values[i].possiblyEvaluate(t,e,r);r
eturn n}}class yi{constructor(t,e,r)
{this.property=t,this.value=e,this.parameters=r;}isConstant(){return
"constant"===this.value.kind}constantOr(t){return "constant"===this.value.kind?
this.value.value:t}evaluate(t,e,r,n){return
this.property.evaluate(this.value,this.parameters,t,e,r,n)}}class
mi{constructor(t)
{this._properties=t,this._values=Object.create(t.defaultPossiblyEvaluatedValues)
;}get(t){return this._values[t]}}class gi{constructor(t)
{this.specification=t;}possiblyEvaluate(t,e){if(t.isDataDriven())throw new
Error("Value should not be data driven");return
t.expression.evaluate(e)}interpolate(t,e,r){const
n=Ze[this.specification.type];return n?n(t,e,r):t}}class xi{constructor(t,e)
{this.specification=t,this.overrides=e;}possiblyEvaluate(t,e,r,n){return new
yi(this,"constant"===t.expression.kind||"camera"===t.expression.kind?
{kind:"constant",value:t.expression.evaluate(e,null,
{}),r,n}:t.expression,e)}interpolate(t,e,r)
{if("constant"!==t.value.kind||"constant"!==e.value.kind)return t;if(void
0===t.value.value||void 0===e.value.value)return new yi(this,
{kind:"constant",value:void 0},t.parameters);const
n=Ze[this.specification.type];if(n){const
i=n(t.value.value,e.value.value,r);return new yi(this,
{kind:"constant",value:i},t.parameters)}return t}evaluate(t,e,r,n,i,a){return
"constant"===t.kind?t.value:t.evaluate(e,r,n,i,a)}}class vi extends

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xi{possiblyEvaluate(t,e,r,n){if(void 0===t.value)return new yi(this,
{kind:"constant",value:void 0},e);if("constant"===t.expression.kind){const
i=t.expression.evaluate(e,null,
{},r,n),a="resolvedImage"===t.property.specification.type&&"string"!==typeof i?
i.name:i,s=this._calculate(a,a,a,e);return new yi(this,
{kind:"constant",value:s},e)}if("camera"===t.expression.kind){const
r=this._calculate(t.expression.evaluate({zoom:e.zoom-
1}),t.expression.evaluate({zoom:e.zoom}),t.expression.evaluate({zoom:e.zoom+1}),
e);return new yi(this,{kind:"constant",value:r},e)}return new
yi(this,t.expression,e)}evaluate(t,e,r,n,i,a){if("source"===t.kind){const
s=t.evaluate(e,r,n,i,a);return this._calculate(s,s,s,e)}return
"composite"===t.kind?
this._calculate(t.evaluate({zoom:Math.floor(e.zoom)-1},r,n),t.evaluate({zoom:Mat
h.floor(e.zoom)},r,n),t.evaluate({zoom:Math.floor(e.zoom)+1},r,n),e):t.value}_ca
lculate(t,e,r,n){return n.zoom>n.zoomHistory.lastIntegerZoom?{from:t,to:e}:
{from:r,to:e}}interpolate(t){return t}}class bi{constructor(t)
{this.specification=t;}possiblyEvaluate(t,e,r,n){if(void 0!==t.value)
{if("constant"===t.expression.kind){const i=t.expression.evaluate(e,null,
{},r,n);return this._calculate(i,i,i,e)}return
this._calculate(t.expression.evaluate(new li(Math.floor(e.zoom-
1),e)),t.expression.evaluate(new
li(Math.floor(e.zoom),e)),t.expression.evaluate(new
li(Math.floor(e.zoom+1),e)),e)}_calculate(t,e,r,n){return
n.zoom>n.zoomHistory.lastIntegerZoom?{from:t,to:e}:{from:r,to:e}}interpolate(t)
{return t}}class wi{constructor(t)
{this.specification=t;}possiblyEvaluate(t,e,r,n){return
!!t.expression.evaluate(e,null,{},r,n)}interpolate(){return !1}}class
_i{constructor(t){this.properties=t,this.defaultPropertyValues=
{},this.defaultTransitionablePropertyValues=
{},this.defaultTransitioningPropertyValues=
{},this.defaultPossiblyEvaluatedValues={},this.overridableProperties=
[];for(const e in t){const
r=t[e];r.specification.overridable&&this.overridableProperties.push(e);const
n=this.defaultPropertyValues[e]=new ui(r,void
0),i=this.defaultTransitionablePropertyValues[e]=new
ci(r);this.defaultTransitioningPropertyValues[e]=i.untransitioned(),this.default
PossiblyEvaluatedValues[e]=n.possiblyEvaluate({});}}$n("DataDrivenProperty",xi)
,$n("DataConstantProperty",gi),$n("CrossFadedDataDrivenProperty",vi),$n("CrossFa
dedProperty",bi),$n("ColorRampProperty",wi);const Ai="-transition";class Si
extends J{constructor(t,e)
{if(super(),this.id=t.id,this.type=t.type,this._featureFilter={filter:
()}=>!0,needGeometry:!1),"custom"!==t.type&&
(this.metadata=t.metadata,this.minzoom=t.minzoom,this.maxzoom=t.maxzoom,"backgro
und"!==t.type&&(this.source=t.source,this.sourceLayer=t["source-
layer"],this.filter=t.filter),e.layout&&(this._unevaluatedLayout=new
di(e.layout),e.paint)){this._transitionablePaint=new hi(e.paint);for(const e in
t.paint)this.setPaintProperty(e,t.paint[e],{validate:!1});for(const e in
t.layout)this.setLayoutProperty(e,t.layout[e],
{validate:!1});this._transitioningPaint=this._transitionablePaint.untransitioned
(),this.paint=new mi(e.paint);}}getCrossfadeParameters(){return
this._crossfadeParameters}getLayoutProperty(t){return "visibility"===t?
this.visibility:this._unevaluatedLayout.getValue(t)}setLayoutProperty(t,e,r={})
{null!=e&&this._validate(Vn,`layers.${this.id}.layout.${t}`,t,e,r)||
("visibility"!==t?
this._unevaluatedLayout.setValue(t,e):this.visibility=e);}getPaintProperty(t)
{return t.endsWith(Ai)?
this._transitionablePaint.getTransition(t.slice(0,-11)):this._transitionablePain
t.getValue(t)}setPaintProperty(t,e,r={})
{if(null!=e&&this._validate(Pn,`layers.${this.id}.paint.${t}`,t,e,r))return
!1;if(t.endsWith(Ai))return
this._transitionablePaint.setTransition(t.slice(0,-11),e||void 0),!1;const
r=this._transitionablePaint._values[t],n="cross-faded-data-

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driven"===r.property.specification["property-
type"],i=r.value.isDataDriven(),a=r.value;this._transitionablePaint.setValue(t,e
),this._handleSpecialPaintPropertyUpdate(t);const
s=this._transitionablePaint._values[t].value;return
s.isDataDriven()||i||n||this._handleOverridablePaintPropertyUpdate(t,a,s)}_hand
leSpecialPaintPropertyUpdate(t){}_handleOverridablePaintPropertyUpdate(t,e,r)
{return !1}isHidden(t){return !!(this.minzoom&&t<this.minzoom)||!!
(this.maxzoom&&t>=this.maxzoom)||"none"===this.visibility}updateTransitions(t)
{this._transitioningPaint=this._transitionablePaint.transitioned(t,this._transit
ioningPaint);}hasTransition(){return
this._transitioningPaint.hasTransition()}recalculate(t,e)
{t.getCrossfadeParameters&&
(this._crossfadeParameters=t.getCrossfadeParameters()),this._unevaluatedLayout&&
(this.layout=this._unevaluatedLayout.possiblyEvaluate(t,void
0,e)),this.paint=this._transitioningPaint.possiblyEvaluate(t,void
0,e);}serialize(){const t={id:this.id,type:this.type,source:this.source,"source-
layer":this.sourceLayer,metadata:this.metadata,minzoom:this.minzoom,maxzoom:this
.maxzoom,filter:this.filter,layout:this._unevaluatedLayout&&this._unevaluatedLay
out.serialize(),paint:this._transitionablePaint&&this._transitionablePaint.seria
lize()};return this.visibility&&(t.layout=t.layout||
{}),t.layout.visibility=this.visibility),b(t,((t,e)=>!(void
0===t||"layout"===e&&!Object.keys(t).length||"paint"===e&&!Object.keys(t).length
)))}_validate(t,e,r,n,i={}){return (!i||!1!==i.validate)&&En(this,t.call(Cn,
{key:e,layerType:this.type,objectKey:r,value:n,styleSpec:X,style:
{glyphs:!0,sprite:!0}}))}is3D(){return !1}isTileClipped(){return
!1}hasOffscreenPass(){return !1}resize(){_isStateDependent(){for(const t in
this.paint._values){const e=this.paint.get(t);if(e instanceof
yi&&_r(e.property.specification)&&
("source"===e.value.kind||"composite"===e.value.kind)&&e.value.isStateDependent)
return !0}return !1}}const ki=
{Int8:Int8Array,Uint8:Uint8Array,Int16:Int16Array,Uint16:Uint16Array,Int32:Int32
Array,Uint32:Uint32Array,Float32:Float32Array};class Ii{constructor(t,e)
{this._structArray=t,this._pos1=e*this.size,this._pos2=this._pos1/2,this._pos4=t
his._pos1/4,this._pos8=this._pos1/8;}}class zi{constructor()
{this.isTransferred=!1,this.capacity=-1,this.resize(0);}static serialize(t,e)
{return t._trim(),e&&(t.isTransferred=!0,e.push(t.arrayBuffer)),
{length:t.length,arrayBuffer:t.arrayBuffer}}static deserialize(t){const
e=Object.create(this.prototype);return
e.arrayBuffer=t.arrayBuffer,e.length=t.length,e.capacity=t.arrayBuffer.byteLengt
h/e.bytesPerElement,e._refreshViews(),e._trim(){this.length!==this.capacity&&
(this.capacity=this.length,this.arrayBuffer=this.arrayBuffer.slice(0,this.length
*this.bytesPerElement),this._refreshViews());}clear(){this.length=0;}resize(t)
{this.reserve(t),this.length=t;}reserve(t){if(t>this.capacity)
{this.capacity=Math.max(t,Math.floor(5*this.capacity)),this.arrayBuffer=new
ArrayBuffer(this.capacity*this.bytesPerElement);const
e=this.uint8;this._refreshViews(),e&&this.uint8.set(e);}_refreshViews(){throw
new Error("_refreshViews() must be implemented by each concrete StructArray
layout")}}function Mi(t,e=1){let r=0,n=0;return {members:t.map((t=>{const
i=ki[t.type].BYTES_PER_ELEMENT,a=r=Ci(r,Math.max(e,i)),s=t.components||1;return
n=Math.max(n,i),r+=i*s,
{name:t.name,type:t.type,components:s,offset:a}))),size:Ci(r,Math.max(n,e)),alig
nment:e}}function Ci(t,e){return Math.ceil(t/e)*e}class Bi extends
zi{_refreshViews(){this.uint8=new Uint8Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer);}emplaceBack(t,e){const r=this.length;return
this.resize(r+1),this.emplace(r,t,e)}emplace(t,e,r){const n=2*t;return
this.int16[n+0]=e,this.int16[n+1]=r,t}}Bi.prototype.bytesPerElement=4,$n("Struct
ArrayLayout2i4",Bi);class Pi extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer);}emplaceBack(t,e,r){const n=this.length;return
this.resize(n+1),this.emplace(n,t,e,r)}emplace(t,e,r,n){const i=3*t;return
this.int16[i+0]=e,this.int16[i+1]=r,this.int16[i+2]=n,t}}Pi.prototype.bytesPerEl
ement=6,$n("StructArrayLayout3i6",Pi);class Vi extends zi{_refreshViews()

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{this.uint8=new Uint8Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer);}emplaceBack(t,e,r,n){const i=this.length;return
this.resize(i+1),this.emplace(i,t,e,r,n)}emplace(t,e,r,n,i){const a=4*t;return
this.int16[a+0]=e,this.int16[a+1]=r,this.int16[a+2]=n,this.int16[a+3]=i,t}}Vi.pr
ototype.bytesPerElement=8,$n("StructArrayLayout4i8",Vi);class Ei extends
zi{_refreshViews(){this.uint8=new Uint8Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer);}emplaceBack(t,e,r,n,i,a){const
s=this.length;return
this.resize(s+1),this.emplace(s,t,e,r,n,i,a)}emplace(t,e,r,n,i,a,s){const
o=6*t;return
this.int16[o+0]=e,this.int16[o+1]=r,this.int16[o+2]=n,this.int16[o+3]=i,this.int
16[o+4]=a,this.int16[o+5]=s,t}}Ei.prototype.bytesPerElement=12,$n("StructArrayLa
yout2i4i12",Ei);class Fi extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer);}emplaceBack(t,e,r,n,i,a){const
s=this.length;return
this.resize(s+1),this.emplace(s,t,e,r,n,i,a)}emplace(t,e,r,n,i,a,s){const
o=4*t,l=8*t;return
this.int16[o+0]=e,this.int16[o+1]=r,this.uint8[l+4]=n,this.uint8[l+5]=i,this.uin
t8[l+6]=a,this.uint8[l+7]=s,t}}Fi.prototype.bytesPerElement=8,$n("StructArrayLay
out2i4ub8",Fi);class Ti extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.float32=new
Float32Array(this.arrayBuffer);}emplaceBack(t,e){const r=this.length;return
this.resize(r+1),this.emplace(r,t,e)}emplace(t,e,r){const n=2*t;return
this.float32[n+0]=e,this.float32[n+1]=r,t}}Ti.prototype.bytesPerElement=8,$n("St
ructArrayLayout2f8",Ti);class $i extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.uint16=new
Uint16Array(this.arrayBuffer);}emplaceBack(t,e,r,n,i,a,s,o,l,u){const
c=this.length;return
this.resize(c+1),this.emplace(c,t,e,r,n,i,a,s,o,l,u)}emplace(t,e,r,n,i,a,s,o,l,u
,c){const h=10*t;return
this.uint16[h+0]=e,this.uint16[h+1]=r,this.uint16[h+2]=n,this.uint16[h+3]=i,this
.uint16[h+4]=a,this.uint16[h+5]=s,this.uint16[h+6]=o,this.uint16[h+7]=l,this.uin
t16[h+8]=u,this.uint16[h+9]=c,t}}$i.prototype.bytesPerElement=20,$n("StructArray
Layout10ui20",$i);class Li extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer),this.uint16=new
Uint16Array(this.arrayBuffer);}emplaceBack(t,e,r,n,i,a,s,o,l,u,c,h){const
p=this.length;return
this.resize(p+1),this.emplace(p,t,e,r,n,i,a,s,o,l,u,c,h)}emplace(t,e,r,n,i,a,s,o
,l,u,c,h,p){const f=12*t;return
this.int16[f+0]=e,this.int16[f+1]=r,this.int16[f+2]=n,this.int16[f+3]=i,this.uin
t16[f+4]=a,this.uint16[f+5]=s,this.uint16[f+6]=o,this.uint16[f+7]=l,this.int16[f
+8]=u,this.int16[f+9]=c,this.int16[f+10]=h,this.int16[f+11]=p,t}}Li.prototype.by
tesPerElement=24,$n("StructArrayLayout4i4ui4i24",Li);class Di extends
zi{_refreshViews(){this.uint8=new Uint8Array(this.arrayBuffer),this.float32=new
Float32Array(this.arrayBuffer);}emplaceBack(t,e,r){const n=this.length;return
this.resize(n+1),this.emplace(n,t,e,r)}emplace(t,e,r,n){const i=3*t;return
this.float32[i+0]=e,this.float32[i+1]=r,this.float32[i+2]=n,t}}Di.prototype.byte
sPerElement=12,$n("StructArrayLayout3f12",Di);class Oi extends
zi{_refreshViews(){this.uint8=new Uint8Array(this.arrayBuffer),this.uint32=new
Uint32Array(this.arrayBuffer);}emplaceBack(t){const e=this.length;return
this.resize(e+1),this.emplace(e,t)}emplace(t,e){return
this.uint32[1*t+0]=e,t}}Oi.prototype.bytesPerElement=4,$n("StructArrayLayout1ul4
",Oi);class Ui extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer),this.uint32=new
Uint32Array(this.arrayBuffer),this.uint16=new
Uint16Array(this.arrayBuffer);}emplaceBack(t,e,r,n,i,a,s,o,l){const
u=this.length;return
this.resize(u+1),this.emplace(u,t,e,r,n,i,a,s,o,l)}emplace(t,e,r,n,i,a,s,o,l,u)
{const c=10*t,h=5*t;return

```

```

this.int16[c+0]=e,this.int16[c+1]=r,this.int16[c+2]=n,this.int16[c+3]=i,this.int
16[c+4]=a,this.int16[c+5]=s,this.uint32[h+3]=o,this.uint16[c+8]=l,this.uint16[c+
9]=u,t}}Ui.prototype.bytesPerElement=20,$n("StructArrayLayout6ilul2ui20",Ui);cla
ss Ri extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer);}emplaceBack(t,e,r,n,i,a){const
s=this.length;return
this.resize(s+1),this.emplace(s,t,e,r,n,i,a)}emplace(t,e,r,n,i,a,s){const
o=6*t;return
this.int16[o+0]=e,this.int16[o+1]=r,this.int16[o+2]=n,this.int16[o+3]=i,this.int
16[o+4]=a,this.int16[o+5]=s,t}}Ri.prototype.bytesPerElement=12,$n("StructArrayLa
yout2i2i2i12",Ri);class qi extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.float32=new
Float32Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer);}emplaceBack(t,e,r,n,i){const a=this.length;return
this.resize(a+1),this.emplace(a,t,e,r,n,i)}emplace(t,e,r,n,i,a){const
s=4*t,o=8*t;return
this.float32[s+0]=e,this.float32[s+1]=r,this.float32[s+2]=n,this.int16[o+6]=i,th
is.int16[o+7]=a,t}}qi.prototype.bytesPerElement=16,$n("StructArrayLayout2flf2i16
",qi);class ji extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.float32=new
Float32Array(this.arrayBuffer);}emplaceBack(t,e,r,n){const i=this.length;return
this.resize(i+1),this.emplace(i,t,e,r,n)}emplace(t,e,r,n,i){const
a=12*t,s=3*t;return
this.uint8[a+0]=e,this.uint8[a+1]=r,this.float32[s+1]=n,this.float32[s+2]=i,t}}j
i.prototype.bytesPerElement=12,$n("StructArrayLayout2ub2f12",ji);class Ni
extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.uint16=new
Uint16Array(this.arrayBuffer);}emplaceBack(t,e,r){const n=this.length;return
this.resize(n+1),this.emplace(n,t,e,r)}emplace(t,e,r,n){const i=3*t;return
this.uint16[i+0]=e,this.uint16[i+1]=r,this.uint16[i+2]=n,t}}Ni.prototype.bytesPe
rElement=6,$n("StructArrayLayout3ui6",Ni);class Zi extends zi{_refreshViews()
{this.uint8=new Uint8Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer),this.uint16=new
Uint16Array(this.arrayBuffer),this.uint32=new
Uint32Array(this.arrayBuffer),this.float32=new
Float32Array(this.arrayBuffer);}emplaceBack(t,e,r,n,i,a,s,o,l,u,c,h,p,f,d,y,m)
{const g=this.length;return
this.resize(g+1),this.emplace(g,t,e,r,n,i,a,s,o,l,u,c,h,p,f,d,y,m)}emplace(t,e,r
,n,i,a,s,o,l,u,c,h,p,f,d,y,m,g){const x=24*t,v=12*t,b=48*t;return
this.int16[x+0]=e,this.int16[x+1]=r,this.uint16[x+2]=n,this.uint16[x+3]=i,this.u
int32[v+2]=a,this.uint32[v+3]=s,this.uint32[v+4]=o,this.uint16[x+10]=l,this.uint
16[x+11]=u,this.uint16[x+12]=c,this.float32[v+7]=h,this.float32[v+8]=p,this.uint
8[b+36]=f,this.uint8[b+37]=d,this.uint8[b+38]=y,this.uint32[v+10]=m,this.int16[x
+22]=g,t}}Zi.prototype.bytesPerElement=48,$n("StructArrayLayout2i2ui3ul3ui2f3ub1
ulli48",Zi);class Ki extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.int16=new
Int16Array(this.arrayBuffer),this.uint16=new
Uint16Array(this.arrayBuffer),this.uint32=new
Uint32Array(this.arrayBuffer),this.float32=new
Float32Array(this.arrayBuffer);}emplaceBack(t,e,r,n,i,a,s,o,l,u,c,h,p,f,d,y,m,g,
x,v,b,w,_A,S,k,I,z){const M=this.length;return
this.resize(M+1),this.emplace(M,t,e,r,n,i,a,s,o,l,u,c,h,p,f,d,y,m,g,x,v,b,w,_A,
S,k,I,z)}emplace(t,e,r,n,i,a,s,o,l,u,c,h,p,f,d,y,m,g,x,v,b,w,_A,S,k,I,z,M)
{const C=32*t,B=16*t;return
this.int16[C+0]=e,this.int16[C+1]=r,this.int16[C+2]=n,this.int16[C+3]=i,this.int
16[C+4]=a,this.int16[C+5]=s,this.int16[C+6]=o,this.int16[C+7]=l,this.uint16[C+8]
=u,this.uint16[C+9]=c,this.uint16[C+10]=h,this.uint16[C+11]=p,this.uint16[C+12]=
f,this.uint16[C+13]=d,this.uint16[C+14]=y,this.uint16[C+15]=m,this.uint16[C+16]=
g,this.uint16[C+17]=x,this.uint16[C+18]=v,this.uint16[C+19]=b,this.uint16[C+20]=
w,this.uint16[C+21]=_,this.uint16[C+22]=A,this.uint32[B+12]=S,this.float32[B+13]
=k,this.float32[B+14]=I,this.uint16[C+30]=z,this.uint16[C+31]=M,t}}Ki.prototype.

```

```

bytesPerElement=64,$n("StructArrayLayout8i15ui1ul2f2ui64",Ki);class Gi extends
zi{_refreshViews(){this.uint8=new Uint8Array(this.arrayBuffer),this.float32=new
Float32Array(this.arrayBuffer);}emplaceBack(t){const e=this.length;return
this.resize(e+1),this.emplace(e,t)}emplace(t,e){return
this.float32[1*t+0]=e,t}}Gi.prototype.bytesPerElement=4,$n("StructArrayLayout1f4
",Gi);class Ji extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.uint16=new
Uint16Array(this.arrayBuffer),this.float32=new
Float32Array(this.arrayBuffer);}emplaceBack(t,e,r){const n=this.length;return
this.resize(n+1),this.emplace(n,t,e,r)}emplace(t,e,r,n){const i=3*t;return
this.uint16[6*t+0]=e,this.float32[i+1]=r,this.float32[i+2]=n,t}}Ji.prototype.by
tesPerElement=12,$n("StructArrayLayout1ui2f12",Ji);class Xi extends
zi{_refreshViews(){this.uint8=new Uint8Array(this.arrayBuffer),this.uint32=new
Uint32Array(this.arrayBuffer),this.uint16=new
Uint16Array(this.arrayBuffer);}emplaceBack(t,e,r){const n=this.length;return
this.resize(n+1),this.emplace(n,t,e,r)}emplace(t,e,r,n){const i=4*t;return
this.uint32[2*t+0]=e,this.uint16[i+2]=r,this.uint16[i+3]=n,t}}Xi.prototype.bytes
PerElement=8,$n("StructArrayLayout1ul2ui8",Xi);class Yi extends
zi{_refreshViews(){this.uint8=new Uint8Array(this.arrayBuffer),this.uint16=new
Uint16Array(this.arrayBuffer);}emplaceBack(t,e){const r=this.length;return
this.resize(r+1),this.emplace(r,t,e)}emplace(t,e,r){const n=2*t;return
this.uint16[n+0]=e,this.uint16[n+1]=r,t}}Yi.prototype.bytesPerElement=4,$n("Stru
ctArrayLayout2ui4",Yi);class Hi extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.uint16=new
Uint16Array(this.arrayBuffer);}emplaceBack(t){const e=this.length;return
this.resize(e+1),this.emplace(e,t)}emplace(t,e){return
this.uint16[1*t+0]=e,t}}Hi.prototype.bytesPerElement=2,$n("StructArrayLayout1ui2
",Hi);class Wi extends zi{_refreshViews(){this.uint8=new
Uint8Array(this.arrayBuffer),this.float32=new
Float32Array(this.arrayBuffer);}emplaceBack(t,e,r,n){const i=this.length;return
this.resize(i+1),this.emplace(i,t,e,r,n)}emplace(t,e,r,n,i){const a=4*t;return
this.float32[a+0]=e,this.float32[a+1]=r,this.float32[a+2]=n,this.float32[a+3]=i,
t}}Wi.prototype.bytesPerElement=16,$n("StructArrayLayout4f16",Wi);class Qi
extends Ii{get anchorPointX(){return this._structArray.int16[this._pos2+0]}get
anchorPointY(){return this._structArray.int16[this._pos2+1]}get x1(){return
this._structArray.int16[this._pos2+2]}get y1(){return
this._structArray.int16[this._pos2+3]}get x2(){return
this._structArray.int16[this._pos2+4]}get y2(){return
this._structArray.int16[this._pos2+5]}get featureIndex(){return
this._structArray.uint32[this._pos4+3]}get sourceLayerIndex(){return
this._structArray.uint16[this._pos2+8]}get bucketIndex(){return
this._structArray.uint16[this._pos2+9]}get anchorPoint(){return new
a(this.anchorPointX,this.anchorPointY)}Qi.prototype.size=20;class ta extends
Ui{get(t){return new Qi(this,t)}}$n("CollisionBoxArray",ta);class ea extends
Ii{get anchorX(){return this._structArray.int16[this._pos2+0]}get anchorY()
{return this._structArray.int16[this._pos2+1]}get glyphStartIndex(){return
this._structArray.uint16[this._pos2+2]}get numGlyphs(){return
this._structArray.uint16[this._pos2+3]}get vertexStartIndex(){return
this._structArray.uint32[this._pos4+2]}get lineStartIndex(){return
this._structArray.uint32[this._pos4+3]}get lineLength(){return
this._structArray.uint32[this._pos4+4]}get segment(){return
this._structArray.uint16[this._pos2+10]}get lowerSize(){return
this._structArray.uint16[this._pos2+11]}get upperSize(){return
this._structArray.uint16[this._pos2+12]}get lineOffsetX(){return
this._structArray.float32[this._pos4+7]}get lineOffsetY(){return
this._structArray.float32[this._pos4+8]}get writingMode(){return
this._structArray.uint8[this._pos1+36]}get placedOrientation(){return
this._structArray.uint8[this._pos1+37]}set placedOrientation(t)
{this._structArray.uint8[this._pos1+37]=t;}get hidden(){return
this._structArray.uint8[this._pos1+38]}set hidden(t)
{this._structArray.uint8[this._pos1+38]=t;}get crossTileID(){return
this._structArray.uint32[this._pos4+10]}set crossTileID(t)

```

```

{this._structArray.uint32[this._pos4+10]=t;}get associatedIconIndex(){return
this._structArray.int16[this._pos2+22]}ea.prototype.size=48;class ra extends
Zi{get(t){return new ea(this,t)}}$n("PlacedSymbolArray",ra);class na extends
Ii{get anchorX(){return this._structArray.int16[this._pos2+0]}get anchorY()
{return this._structArray.int16[this._pos2+1]}get
rightJustifiedTextSymbolIndex(){return this._structArray.int16[this._pos2+2]}get
centerJustifiedTextSymbolIndex(){return
this._structArray.int16[this._pos2+3]}get leftJustifiedTextSymbolIndex(){return
this._structArray.int16[this._pos2+4]}get verticalPlacedTextSymbolIndex(){return
this._structArray.int16[this._pos2+5]}get placedIconSymbolIndex(){return
this._structArray.int16[this._pos2+6]}get verticalPlacedIconSymbolIndex(){return
this._structArray.int16[this._pos2+7]}get key(){return
this._structArray.uint16[this._pos2+8]}get textBoxStartIndex(){return
this._structArray.uint16[this._pos2+9]}get textBoxEndIndex(){return
this._structArray.uint16[this._pos2+10]}get verticalTextBoxStartIndex(){return
this._structArray.uint16[this._pos2+11]}get verticalTextBoxEndIndex(){return
this._structArray.uint16[this._pos2+12]}get iconBoxStartIndex(){return
this._structArray.uint16[this._pos2+13]}get iconBoxEndIndex(){return
this._structArray.uint16[this._pos2+14]}get verticalIconBoxStartIndex(){return
this._structArray.uint16[this._pos2+15]}get verticalIconBoxEndIndex(){return
this._structArray.uint16[this._pos2+16]}get featureIndex(){return
this._structArray.uint16[this._pos2+17]}get numHorizontalGlyphVertices(){return
this._structArray.uint16[this._pos2+18]}get numVerticalGlyphVertices(){return
this._structArray.uint16[this._pos2+19]}get numIconVertices(){return
this._structArray.uint16[this._pos2+20]}get numVerticalIconVertices(){return
this._structArray.uint16[this._pos2+21]}get useRuntimeCollisionCircles(){return
this._structArray.uint16[this._pos2+22]}get crossTileID(){return
this._structArray.uint32[this._pos4+12]}set crossTileID(t)
{this._structArray.uint32[this._pos4+12]=t;}get textBoxScale(){return
this._structArray.float32[this._pos4+13]}get collisionCircleDiameter(){return
this._structArray.float32[this._pos4+14]}get textAnchorOffsetStartIndex(){return
this._structArray.uint16[this._pos2+30]}get textAnchorOffsetEndIndex(){return
this._structArray.uint16[this._pos2+31]}na.prototype.size=64;class ia extends
Ki{get(t){return new na(this,t)}}$n("SymbolInstanceArray",ia);class aa extends
Gi{getoffsetX(t){return this.float32[1*t+0]}}$n("GlyphOffsetArray",aa);class sa
extends Pi{getx(t){return this.int16[3*t+0]}gety(t){return
this.int16[3*t+1]}gettileUnitDistanceFromAnchor(t){return
this.int16[3*t+2]}}$n("SymbolLineVertexArray",sa);class oa extends Ii{get
textAnchor(){return this._structArray.uint16[this._pos2+0]}get textOffset0()
{return this._structArray.float32[this._pos4+1]}get textOffset1(){return
this._structArray.float32[this._pos4+2]}oa.prototype.size=12;class la extends
Ji{get(t){return new oa(this,t)}}$n("TextAnchorOffsetArray",la);class ua extends
Ii{get featureIndex(){return this._structArray.uint32[this._pos4+0]}get
sourceLayerIndex(){return this._structArray.uint16[this._pos2+2]}get
bucketIndex(){return
this._structArray.uint16[this._pos2+3]}ua.prototype.size=8;class ca extends
Xi{get(t){return new ua(this,t)}}$n("FeatureIndexArray",ca);class ha extends
Bi{}class pa extends Bi{}class fa extends Bi{}class da extends Ei{}class ya
extends Fi{}class ma extends Ti{}class ga extends Si{}class xa extends Li{}class
va extends Di{}class ba extends Oi{}class wa extends Ri{}class _a extends
ji{}class Aa extends Ni{}class Sa extends Yi{}const
ka=Mt([{"name":"a_pos",components:2,type:"Int16"}],4),{members:Ia}=ka;class
za{constructor(t=[]){this.segments=t;}prepareSegment(t,e,r,n){let
i=this.segments[this.segments.length-1];return
t>za.MAX_VERTEX_ARRAY_LENGTH&&A(`Max vertices per segment is
${za.MAX_VERTEX_ARRAY_LENGTH}: bucket requested ${t}`),
(!i||i.vertexLength+t>za.MAX_VERTEX_ARRAY_LENGTH||i.sortKey!==n)&&(i=
{vertexOffset:e.length,primitiveOffset:r.length,vertexLength:0,primitiveLength:0
},void 0!==n&&(i.sortKey=n),this.segments.push(i)),i}get(){return
this.segments}destroy(){for(const t of this.segments)for(const e in
t.vaos)t.vaos[e].destroy();}static simpleSegment(t,e,r,n){return new
za([{"vertexOffset:t,primitiveOffset:e,vertexLength:r,primitiveLength:n,vaos:

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```

    {} ,sortKey:0}})function Ma(t,e){return 256*
    (t=y(Math.floor(t),0,255))+y(Math.floor(e),0,255)}za.MAX_VERTEX_ARRAY_LENGTH=Math
    h.pow(2,16)-1,$n("SegmentVector",za);const
    Ca=Mi([ {name:"a_pattern_from",components:4,type:"Uint16"},
    {name:"a_pattern_to",components:4,type:"Uint16"},
    {name:"a_pixel_ratio_from",components:1,type:"Uint16"},
    {name:"a_pixel_ratio_to",components:1,type:"Uint16"}]);var Ba={exports:{}},Pa=
    {exports:{}},Pa.exports=function(t,e){var r,n,i,a,s,o,l,u;for(n=t.length-
    (r=3&t.length),i=e,s=3432918353,o=461845907,u=0;u<n;l=255&t.charCodeAt(u)|
    (255&t.charCodeAt(++u))<<8|(255&t.charCodeAt(++u))<<16|(255&t.charCodeAt(++u))
    <<24,++u,i=27492+(65535&(a=5*(65535&(i=(i^=1=(65535&(l=(l=(65535&l)*s+
    ((l>>>16)*s&65535)<<16)&4294967295)<<15|l>>>17))*o+((l>>>16)*o&65535)
    <<16)&4294967295)<<13|i>>>19))+((5*(i>>>16)&65535)<<16)&4294967295))+((58964+
    (a>>>16)&65535)<<16);switch(l=0,r){case 3:l^=(255&t.charCodeAt(u+2))<<16;case
    2:l^=(255&t.charCodeAt(u+1))<<8;case 1:i^=l=(65535&(l=(l=(65535&
    (l^=255&t.charCodeAt(u))*s+((l>>>16)*s&65535)<<16)&4294967295)<<15|l>>>17))*o+
    ((l>>>16)*o&65535)<<16)&4294967295;}return i^=t.length,i=2246822507*(65535&
    (i^=i>>>16))+((2246822507*(i>>>16)&65535)<<16)&4294967295,i=3266489909*(65535&
    (i^=i>>>13))+((3266489909*(i>>>16)&65535)<<16)&4294967295,(i^=i>>>16)>>>0};var
    Va=Pa.exports,Ea={exports:{}},Ea.exports=function(t,e){for(var
    r,n=t.length,i=e^n,a=0;n>=4;)r=1540483477*(65535&(r=255&t.charCodeAt(a)|
    (255&t.charCodeAt(++a))<<8|(255&t.charCodeAt(++a))<<16|(255&t.charCodeAt(++a))
    <<24))+((1540483477*(r>>>16)&65535)<<16),i=1540483477*(65535&i)+((1540483477*
    (i>>>16)&65535)<<16)^(r=1540483477*(65535&(r^=r>>>24))+((1540483477*
    (r>>>16)&65535)<<16)),n-=4,++a;switch(n){case 3:i^=(255&t.charCodeAt(a+2))
    <<16;case 2:i^=(255&t.charCodeAt(a+1))<<8;case 1:i=1540483477*(65535&
    (i^=255&t.charCodeAt(a)))+(1540483477*(i>>>16)&65535)<<16);}return
    i=1540483477*(65535&(i^=i>>>13))+((1540483477*(i>>>16)&65535)<<16),
    (i^=i>>>15)>>>0};var
    Fa=Va,Ta=Ea.exports;Ba.exports=Fa,Ba.exports.murmur3=Fa,Ba.exports.murmur2=Ta;var
    $a=r(Ba.exports);class La{constructor(){this.ids=[],this.positions=
    [],this.indexed=!1;}add(t,e,r,n)
    {this.ids.push(Da(t)),this.positions.push(e,r,n);}getPositions(t)
    {if(!this.indexed)throw new Error("Trying to get index, but feature positions
    are not indexed");const e=Da(t);let r=0,n=this.ids.length-1;for(;r<n;){const
    t=r+n>>1;this.ids[t]>=e?n=t:r=t+1;}const i=
    [];for(;this.ids[r]==e;)i.push({index:this.positions[3*r],start:this.positions[
    3*r+1],end:this.positions[3*r+2]}),r++;return i}static serialize(t,e){const
    r=new Float64Array(t.ids),n=new Uint32Array(t.positions);return
    Oa(r,n,0,r.length-1),e&&e.push(r.buffer,n.buffer),{ids:r,positions:n}}static
    deserialize(t){const e=new La;return
    e.ids=t.ids,e.positions=t.positions,e.indexed=!0,e}}function Da(t){const
    e=+t;return !isNaN(e)&&e<=Number.MAX_SAFE_INTEGER?e:$a(String(t))}function
    Oa(t,e,r,n){for(;r<n;){const i=t[r+n>>1];let a=r-1,s=n+1;for(;;)
    {do{a++;}while(t[a]<i);do{s--
    ;}while(t[s]>i);if(a==s)break;Ua(t,a,s),Ua(e,3*a,3*s),Ua(e,3*a+1,3*s+1),Ua(e,3*a
    +2,3*s+2);}s-r<n-s?(Oa(t,e,r,s),r=s+1):(Oa(t,e,s+1,n),n=s);}function Ua(t,e,r)
    {const n=t[e];t[e]=t[r],t[r]=n;$n("FeaturePositionMap",La);class
    Ra{constructor(t,e){this.gl=t.gl,this.location=e;}}class qa extends
    Ra{constructor(t,e){super(t,e),this.current=0;}set(t){this.current!==t&&
    (this.current=t,this.gl.uniform1f(this.location,t));}}class ja extends
    Ra{constructor(t,e){super(t,e),this.current=[0,0,0,0];}set(t)
    {t[0]==this.current[0]&&t[1]==this.current[1]&&t[2]==this.current[2]&&t[3]==
    this.current[3]||
    (this.current=t,this.gl.uniform4f(this.location,t[0],t[1],t[2],t[3]));}class Na
    extends Ra{constructor(t,e){super(t,e),this.current=Yt.transparent;}set(t)
    {t.r===this.current.r&&t.g===this.current.g&&t.b===this.current.b&&t.a===this.cu
    rrent.a||
    (this.current=t,this.gl.uniform4f(this.location,t.r,t.g,t.b,t.a));}}const Za=new
    Float32Array(16);function Ka(t){return
    [Ma(255*t.r,255*t.g),Ma(255*t.b,255*t.a)]}class Ga{constructor(t,e,r)
    {this.value=t,this.uniformNames=e.map((t=>`u_${t}`)),this.type=r;}setUniform(t,e

```

```

,r){t.set(r.constantOr(this.value));}getBinding(t,e,r){return
"color"===this.type?new Na(t,e):new qa(t,e)}}class Ja{constructor(t,e)
{this.uniformNames=e.map((t=>`u_${t}`)),this.patternFrom=null,this.patternTo=null,
this.pixelRatioFrom=1,this.pixelRatioTo=1;}setConstantPatternPositions(t,e)
{this.pixelRatioFrom=e.pixelRatio,this.pixelRatioTo=t.pixelRatio,this.patternFrom=e.tlbr,
this.patternTo=t.tlbr;}setUniform(t,e,r,n){const i="u_pattern_to"===n?
this.patternTo:"u_pattern_from"===n?this.patternFrom:"u_pixel_ratio_to"===n?
this.pixelRatioTo:"u_pixel_ratio_from"===n?
this.pixelRatioFrom:null;i&&t.set(i);}getBinding(t,e,r){return
"u_pattern"===r.substr(0,9)?new ja(t,e):new qa(t,e)}}class
Xa{constructor(t,e,r,n)
{this.expression=t,this.type=r,this.maxValue=0,this.paintVertexAttributes=e.map(
(t=>({name:`a_${t}`,type:"Float32",components:"color"===r?
2:1,offset:0}))),this.paintVertexArray=new n;}populatePaintArray(t,e,r,n,i)
{const a=this.paintVertexArray.length,s=this.expression.evaluate(new li(0),e,
{}),n,
[],i);this.paintVertexArray.resize(t),this._setPaintValue(a,t,s);}updatePaintArray(t,e,r,n){const
i=this.expression.evaluate({zoom:0},r,n);this._setPaintValue(t,e,i);}setPaintValue(t,e,r){if("color"===this.type){const n=Ka(r);for(let
r=t;r<e;r++)this.paintVertexArray.emplace(r,n[0],n[1]);}else {for(let
n=t;n<e;n++)this.paintVertexArray.emplace(n,r);this.maxValue=Math.max(this.maxValue,Math.abs(r));}}upload(t)
{this.paintVertexArray&&this.paintVertexArray.arrayBuffer&&
(this.paintVertexBuffer&&this.paintVertexBuffer.buffer?
this.paintVertexBuffer.updateData(this.paintVertexArray):this.paintVertexBuffer=
t.createVertexBuffer(this.paintVertexArray,this.paintVertexAttributes,this.expression.isStateDependent));}destroy()
{this.paintVertexBuffer&&this.paintVertexBuffer.destroy();}}class
Ya{constructor(t,e,r,n,i,a)
{this.expression=t,this.uniformNames=e.map((t=>`u_${t}_t`)),this.type=r,this.useIntegerZoom=n,
this.zoom=i,this.maxValue=0,this.paintVertexAttributes=e.map((t=>
({name:`a_${t}`,type:"Float32",components:"color"===r?
4:2,offset:0}))),this.paintVertexArray=new a;}populatePaintArray(t,e,r,n,i)
{const a=this.expression.evaluate(new li(this.zoom),e,{}),n,
[],i),s=this.expression.evaluate(new li(this.zoom+1),e,{}),n,
[],i),o=this.paintVertexArray.length;this.paintVertexArray.resize(t),this._setPaintValue(o,t,a,s);}updatePaintArray(t,e,r,n){const
i=this.expression.evaluate({zoom:this.zoom},r,n),a=this.expression.evaluate({zoom:this.zoom+1},r,n);this._setPaintValue(t,e,i,a);}setPaintValue(t,e,r,n)
{if("color"===this.type){const i=Ka(r),a=Ka(n);for(let
r=t;r<e;r++)this.paintVertexArray.emplace(r,i[0],i[1],a[0],a[1]);}else {for(let
i=t;i<e;i++)this.paintVertexArray.emplace(i,r,n);this.maxValue=Math.max(this.maxValue,Math.abs(r),Math.abs(n));}}upload(t)
{this.paintVertexArray&&this.paintVertexArray.arrayBuffer&&
(this.paintVertexBuffer&&this.paintVertexBuffer.buffer?
this.paintVertexBuffer.updateData(this.paintVertexArray):this.paintVertexBuffer=
t.createVertexBuffer(this.paintVertexArray,this.paintVertexAttributes,this.expression.isStateDependent));}destroy()
{this.paintVertexBuffer&&this.paintVertexBuffer.destroy();}setUniform(t,e){const
r=this.useIntegerZoom?
Math.floor(e.zoom):e.zoom,n=y(this.expression.interpolationFactor(r,this.zoom,this.zoom+1),0,1);t.set(n);}getBinding(t,e,r){return new qa(t,e)}}class
Ha{constructor(t,e,r,n,i,a)
{this.expression=t,this.type=e,this.useIntegerZoom=r,this.zoom=n,this.layerId=a,
this.zoomInPaintVertexArray=new i,this.zoomOutPaintVertexArray=new
i;}populatePaintArray(t,e,r){const
n=this.zoomInPaintVertexArray.length;this.zoomInPaintVertexArray.resize(t),this.zoomOutPaintVertexArray.resize(t),this._setPaintValues(n,t,e.patterns&&e.patterns[this.layerId],r);}updatePaintArray(t,e,r,n,i)
{this._setPaintValues(t,e,r.patterns&&r.patterns[this.layerId],i);}setPaintValues(t,e,r,n)

```

```

{if(!n||!r)return;const{min:i,mid:a,max:s}=r,o=n[i],l=n[a],u=n[s];if(o&l&u)for
(let
r=t;r<e;r++)this.zoomInPaintVertexArray.emplace(r,l.tl[0],l.tl[1],l.br[0],l.br[1
],o.tl[0],o.tl[1],o.br[0],o.br[1],l.pixelRatio,o.pixelRatio),this.zoomOutPaintVe
rtexArray.emplace(r,l.tl[0],l.tl[1],l.br[0],l.br[1],u.tl[0],u.tl[1],u.br[0],u.br
[1],l.pixelRatio,u.pixelRatio);}upload(t)
{this.zoomInPaintVertexArray&&this.zoomInPaintVertexArray.arrayBuffer&&this.zoom
OutPaintVertexArray&&this.zoomOutPaintVertexArray.arrayBuffer&&
(this.zoomInPaintVertexBuffer=t.createVertexBuffer(this.zoomInPaintVertexArray,C
a.members,this.expression.isStateDependent),this.zoomOutPaintVertexBuffer=t.crea
teVertexBuffer(this.zoomOutPaintVertexArray,Ca.members,this.expression.isStateDe
pendent));}destroy()
{this.zoomOutPaintVertexBuffer&&this.zoomOutPaintVertexBuffer.destroy(),this.zoo
mInPaintVertexBuffer&&this.zoomInPaintVertexBuffer.destroy();}class
Wa{constructor(t,e,r){this.binders={},this._buffers=[];const n=[];for(const i in
t.paint._values){if(!r(i))continue;const a=t.paint.get(i);if(!(a instanceof
yi&&_r(a.property.specification)))continue;const
s=ts(i,t.type),o=a.value,l=a.property.specification.type,u=a.property.useInteger
Zoom,c=a.property.specification["property-type"],h="cross-faded"===c||"cross-
faded-data-driven"===c;if("constant"===o.kind)this.binders[i]=h?new
Ja(o.value,s):new Ga(o.value,s,l),n.push(`/u_${i}`);else
if("source"===o.kind|h){const r=es(i,l,"source");this.binders[i]=h?new
Ha(o,l,u,e,r,t.id):new Xa(o,s,l,r),n.push(`/a_${i}`);}else {const
t=es(i,l,"composite");this.binders[i]=new
Ya(o,s,l,u,e,t),n.push(`/z_${i}`);}this.cacheKey=n.sort().join("");}getMaxValue
(t){const e=this.binders[t];return e instanceof Xa||e instanceof Ya?
e.maxValue:0}populatePaintArrays(t,e,r,n,i){for(const a in this.binders){const
s=this.binders[a];(s instanceof Xa||s instanceof Ya||s instanceof
Ha)&&s.populatePaintArray(t,e,r,n,i);}setConstantPatternPositions(t,e)
{for(const r in this.binders){const n=this.binders[r];n instanceof
Ja&&n.setConstantPatternPositions(t,e);}updatePaintArrays(t,e,r,n,i){let
a=!1;for(const s in t){const o=e.getPositions(s);for(const e of o){const
o=r.feature(e.index);for(const r in this.binders){const l=this.binders[r];if((l
instanceof Xa||l instanceof Ya||l instanceof
Ha)&&!0===l.expression.isStateDependent){const
u=n.paint.get(r);l.expression=u.value,l.updatePaintArray(e.start,e.end,o,t[s],i
,a=!0;}}}}return a}defines(){const t=[];for(const e in this.binders){const
r=this.binders[e];(r instanceof Ga||r instanceof
Ja)&&t.push(...r.uniformNames.map((t=>`#define HAS_UNIFORM_${t}`)));}return
t}getBinderAttributes(){const t=[];for(const e in this.binders){const
r=this.binders[e];if(r instanceof Xa||r instanceof Ya)for(let
e=0;e<r.paintVertexAttributes.length;e++)t.push(r.paintVertexAttributes[e].name)
;else if(r instanceof Ha)for(let
e=0;e<Ca.members.length;e++)t.push(Ca.members[e].name);}return
t}getBinderUniforms(){const t=[];for(const e in this.binders){const
r=this.binders[e];if(r instanceof Ga||r instanceof Ja||r instanceof Ya)for(const
e of r.uniformNames)t.push(e);}return t}getPaintVertexBuffers(){return
this._buffers}getUniforms(t,e){const r=[];for(const n in this.binders){const
i=this.binders[n];if(i instanceof Ga||i instanceof Ja||i instanceof Ya)for(const
a of i.uniformNames)if(e[a]){const
s=i.getBinding(t,e[a],a);r.push({name:a,property:n,binding:s});}}return
r}setUniforms(t,e,r,n){for(const{name:t,property:i,binding:a}of
e)this.binders[i].setUniform(a,n,r.get(i),t);}updatePaintBuffers(t)
{this._buffers=[];for(const e in this.binders){const r=this.binders[e];if(t&&r
instanceof Ha){const e=2===t.fromScale?
r.zoomInPaintVertexBuffer:r.zoomOutPaintVertexBuffer;e&&this._buffers.push(e);}e
lse (r instanceof Xa||r instanceof
Ya)&&r.paintVertexBuffer&&this._buffers.push(r.paintVertexBuffer);}upload(t)
{for(const e in this.binders){const r=this.binders[e];(r instanceof Xa||r
instanceof Ya||r instanceof
Ha)&&r.upload(t);}this.updatePaintBuffers();}destroy(){for(const t in
this.binders){const e=this.binders[t];(e instanceof Xa||e instanceof Ya||e

```



```

instanceof Ha)&&e.destroy();}}class Qa{constructor(t,e,r=((>!0))
{this.programConfigurations={};for(const n of
t)this.programConfigurations[n.id]=new
Wa(n,e,r);this.needsUpload=!1,this._featureMap=new
La,this._bufferOffset=0;}populatePaintArrays(t,e,r,n,i,a){for(const r in
this.programConfigurations)this.programConfigurations[r].populatePaintArrays(t,e
,n,i,a);void
0!:=e.id&&this._featureMap.add(e.id,r,this._bufferOffset,t),this._bufferOffset=t
,this.needsUpload=!0;}updatePaintArrays(t,e,r,n){for(const i of
r)this.needsUpload=this.programConfigurations[i.id].updatePaintArrays(t,this._fe
atureMap,e,i,n)||this.needsUpload;}get(t){return
this.programConfigurations[t].upload(t){if(this.needsUpload){for(const e in
this.programConfigurations)this.programConfigurations[e].upload(t);this.needsUplo
ad=!1;}}destroy(){for(const t in
this.programConfigurations)this.programConfigurations[t].destroy();}}function
ts(t,e){return {"text-opacity":["opacity"],"icon-opacity":["opacity"],"text-
color":["fill_color"],"icon-color":["fill_color"],"text-halo-color":
["halo_color"],"icon-halo-color":["halo_color"],"text-halo-blur":
["halo_blur"],"icon-halo-blur":["halo_blur"],"text-halo-width":
["halo_width"],"icon-halo-width":["halo_width"],"line-gap-width":
["gapwidth"],"line-pattern":
["pattern_to","pattern_from","pixel_ratio_to","pixel_ratio_from"],"fill-
pattern":
["pattern_to","pattern_from","pixel_ratio_to","pixel_ratio_from"],"fill-
extrusion-pattern":
["pattern_to","pattern_from","pixel_ratio_to","pixel_ratio_from"]}[t]||
[t.replace(`${e}-`,`").replace(/-/g,`_`)]}function es(t,e,r){const n={color:
{source:Ti,composite:Wi},number:{source:Gi,composite:Ti}},i=function(t){return
{"line-pattern":{source:ga,composite:ga},"fill-pattern":
{source:ga,composite:ga},"fill-extrusion-pattern":{source:ga,composite:ga}}[t]}
(t);return i&&i[r]||n[e]
[r]}$n("ConstantBinder",Ga),$n("CrossFadedConstantBinder",Ja),$n("SourceExpressi
onBinder",Xa),$n("CrossFadedCompositeBinder",Ha),$n("CompositeExpressionBinder",
Ya),$n("ProgramConfiguration",Wa,{omit:
["_buffers"]}),$n("ProgramConfigurationSet",Qa);const
rs=8192,ns=Math.pow(2,14)-1,is=-ns-1;function as(t){const
e=rs/t.extent,r=t.loadGeometry();for(let t=0;t<r.length;t++){const
n=r[t];for(let t=0;t<n.length;t++){const
r=n[t],i=Math.round(r.x*e),a=Math.round(r.y*e);r.x=y(i,is,ns),r.y=y(a,is,ns),
(i<r.x||i>r.x+1||a<r.y||a>r.y+1)&&A("Geometry exceeds allowed extent, reduce
your vector tile buffer size");}}return r}function ss(t,e){return
{type:t.type,id:t.id,properties:t.properties,geometry:e?as(t):[]}}function
os(t,e,r,n,i){t.emplaceBack(2*e+(n+1)/2,2*r+(i+1)/2);}class ls{constructor(t)
{this.zoom=t.zoom,this.overscaling=t.overscaling,this.layers=t.layers,this.layer
Ids=this.layers.map((t=>t.id)),this.index=t.index,this.hasPattern=!1,this.layout
VertexArray=new pa,this.indexArray=new Aa,this.segments=new
za,this.programConfigurations=new
Qa(t.layers,t.zoom),this.stateDependentLayerIds=this.layers.filter((t=>t.isState
Dependent())).map((t=>t.id));}populate(t,e,r){const n=this.layers[0],i=[];let
a=null,s=!1;"circle"===n.type&&(a=n.layout.get("circle-sort-
key"),s=!a.isConstant());for(const{feature:e,id:n,index:o,sourceLayerIndex:l}of
t){const
t=this.layers[0]._featureFilter.needGeometry,u=ss(e,t);if(!this.layers[0]._featu
reFilter.filter(new li(this.zoom),u,r))continue;const c=s?a.evaluate(u,
{}),r):void 0,h=
{id:n,properties:e.properties,type:e.type,sourceLayerIndex:l,index:o,geometry:t?
u.geometry:as(e),patterns:{},sortKey:c};i.push(h);}s&&i.sort((t,e)=>t.sortKey-
e.sortKey));for(const n of i)
{const{geometry:i,index:a,sourceLayerIndex:s}=n,o=t[a].feature;this.addFeature(n
,i,a,r),e.featureIndex.insert(o,i,a,s,this.index);}update(t,e,r)
{this.stateDependentLayers.length&&this.programConfigurations.updatePaintArrays(
t,e,this.stateDependentLayers,r);}isEmpty(){return

```

```

0===this.layoutVertexArray.length}uploadPending(){return
!this.uploaded||this.programConfigurations.needsUpload}upload(t){this.uploaded||
(this.layoutVertexBuffer=t.createVertexBuffer(this.layoutVertexArray,Ia),this.in
dexBuffer=t.createIndexBuffer(this.indexArray)),this.programConfigurations.uploa
d(t),this.uploaded=!0;}destroy(){this.layoutVertexBuffer&&
(this.layoutVertexBuffer.destroy(),this.indexBuffer.destroy(),this.programConfig
urations.destroy(),this.segments.destroy());}addFeature(t,e,r,n){for(const r of
e)for(const e of r){const r=e.x,n=e.y;if(r<0||r>=rs||n<0||n>=rs)continue;const
i=this.segments.prepareSegment(4,this.layoutVertexArray,this.indexArray,t.sortKe
y),a=i.vertexLength;os(this.layoutVertexArray,r,n,-1,-1),os(this.layoutVertexArr
ay,r,n,1,-1),os(this.layoutVertexArray,r,n,1,1),os(this.layoutVertexArray,r,n,-1
,1),this.indexArray.emplaceBack(a,a+1,a+2),this.indexArray.emplaceBack(a,a+3,a+2
),i.vertexLength+=4,i.primitiveLength+=2;}this.programConfigurations.populatePai
ntArrays(this.layoutVertexArray.length,t,r,{},n);}}function us(t,e){for(let
r=0;r<t.length;r++)if(xs(e,t[r]))return !0;for(let
r=0;r<t.length;r++)if(xs(t,e[r]))return !0;return !fs(t,e)}function cs(t,e,r)
{return !xs(t,e)||!ys(e,t,r)}function hs(t,e){if(1===t.length)return
gs(e,t[0]);for(let r=0;r<t.length;r++){const n=e[r];for(let
e=0;e<n.length;e++)if(xs(t,n[e]))return !0}for(let
r=0;r<t.length;r++)if(gs(e,t[r]))return !0;for(let
r=0;r<t.length;r++)if(fs(t,e[r]))return !0;return !1}function ps(t,e,r)
{if(t.length>1){if(fs(t,e))return !0;for(let
n=0;n<t.length;n++)if(ys(e[n],t,r))return !0}for(let
n=0;n<t.length;n++)if(ys(t[n],e,r))return !0;return !1}function fs(t,e)
{if(0===t.length||0===e.length)return !1;for(let r=0;r<t.length-1;r++){const
n=t[r],i=t[r+1];for(let t=0;t<e.length-1;t++)if(ds(n,i,e[t],e[t+1]))return
!0}return !1}function ds(t,e,r,n){return
S(t,r,n)!=S(e,r,n)&&S(t,e,r)!=S(t,e,n)}function ys(t,e,r){const
n=r*r;if(1===e.length)return t.distSqr(e[0])<n;for(let
r=1;r<e.length;r++)if(ms(t,e[r-1],e[r])<n)return !0;return !1}function ms(t,e,r)
{const n=e.distSqr(r);if(0===n)return t.distSqr(e);const i=((t.x-e.x)*(r.x-e.x)+
(t.y-e.y)*(r.y-e.y))/n;return t.distSqr(i<0?e:i>1?
r:r.sub(e)._mult(i)._add(e))}function gs(t,e){let r,n,i,a=!1;for(let
s=0;s<t.length;s++){r=t[s];for(let t=0,t=s.length-
1;t<r.length;s=t++)n=r[t],i=r[s],n.y>e.y!=i.y>e.y&&e.x<(i.x-n.x)*(e.y-n.y)/(i.y-
n.y)+n.x&&(a=!a);}return a}function xs(t,e){let r=!1;for(let n=0,i=t.length-
1;n<t.length;i=n++){const a=t[n],s=t[i];a.y>e.y!=s.y>e.y&&e.x<(s.x-a.x)*(e.y-
a.y)/(s.y-a.y)+a.x&&(r=!r);}return r}function vs(t,e,r){const
n=r[0],i=r[2];if(t.x<n.x&&e.x<n.x||t.x>i.x&&e.x>i.x||t.y<n.y&&e.y<n.y||t.y>i.y&&
e.y>i.y)return !1;const a=S(t,e,r[0]);return
a!=S(t,e,r[1])||a!=S(t,e,r[2])||a!=S(t,e,r[3])}function bs(t,e,r){const
n=e.paint.get(t).value;return "constant"===n.kind?
n.value:r.programConfigurations.get(e.id).getMaxValue(t)}function ws(t){return
Math.sqrt(t[0]*t[0]+t[1]*t[1])}function _s(t,e,r,n,i){if(!e[0]&&!e[1])return
t;const s=a.convert(e)._mult(i);"viewport"===r&&s._rotate(-n);const o=[];for(let
e=0;e<t.length;e++)o.push(t[e].sub(s));return o}let As,Ss,$n("CircleBucket",ls,
{omit:["layers"]});var ks={get paint(){return Ss=Ss||new _i({"circle-radius":new
xi(X.paint_circle["circle-radius"]), "circle-color":new
xi(X.paint_circle["circle-color"]), "circle-blur":new xi(X.paint_circle["circle-
blur"]), "circle-opacity":new xi(X.paint_circle["circle-opacity"]), "circle-
translate":new gi(X.paint_circle["circle-translate"]), "circle-translate-
anchor":new gi(X.paint_circle["circle-translate-anchor"]), "circle-pitch-
scale":new gi(X.paint_circle["circle-pitch-scale"]), "circle-pitch-alignment":new
gi(X.paint_circle["circle-pitch-alignment"]), "circle-stroke-width":new
xi(X.paint_circle["circle-stroke-width"]), "circle-stroke-color":new
xi(X.paint_circle["circle-stroke-color"]), "circle-stroke-opacity":new
xi(X.paint_circle["circle-stroke-opacity"])}),get layout(){return As=As||new
_i({"circle-sort-key":new xi(X.layout_circle["circle-sort-key"])}),Is=le-
6,zs="undefined"!=typeof Float32Array?Float32Array:Array;function Ms(t){return
t[0]=1,t[1]=0,t[2]=0,t[3]=0,t[4]=0,t[5]=1,t[6]=0,t[7]=0,t[8]=0,t[9]=0,t[10]=1,t[
11]=0,t[12]=0,t[13]=0,t[14]=0,t[15]=1,t}function Cs(t,e,r){var
n=e[0],i=e[1],a=e[2],s=e[3],o=e[4],l=e[5],u=e[6],c=e[7],h=e[8],p=e[9],f=e[10],d=

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e[11],y=e[12],m=e[13],g=e[14],x=e[15],v=r[0],b=r[1],w=r[2],_r[3];return
t[0]=v*n+b*o+w*h+_y,t[1]=v*i+b*l+w*p+_m,t[2]=v*a+b*u+w*f+_g,t[3]=v*s+b*c+w*d+_x,t[4]=(v=r[4])*n+(b=r[5])*o+(w=r[6])*h+
(_r[7])*y,t[5]=v*i+b*l+w*p+_m,t[6]=v*a+b*u+w*f+_g,t[7]=v*s+b*c+w*d+_x,t[8]=
(v=r[8])*n+(b=r[9])*o+(w=r[10])*h+
(_r[11])*y,t[9]=v*i+b*l+w*p+_m,t[10]=v*a+b*u+w*f+_g,t[11]=v*s+b*c+w*d+_x,t[12]=(v=r[12])*n+(b=r[13])*o+(w=r[14])*h+
(_r[15])*y,t[13]=v*i+b*l+w*p+_m,t[14]=v*a+b*u+w*f+_g,t[15]=v*s+b*c+w*d+_x,t
Math.hypot||(Math.hypot=function(){for(var t=0,e=arguments.length;e--
;)t+=arguments[e]*arguments[e];return Math.sqrt(t)});var Bs,Ps=Cs;function
Vs(t,e,r){var n=e[0],i=e[1],a=e[2],s=e[3];return
t[0]=r[0]*n+r[4]*i+r[8]*a+r[12]*s,t[1]=r[1]*n+r[5]*i+r[9]*a+r[13]*s,t[2]=r[2]*n+
r[6]*i+r[10]*a+r[14]*s,t[3]=r[3]*n+r[7]*i+r[11]*a+r[15]*s,t}Bs=new
zs(4),zs!=Float32Array&&(Bs[0]=0,Bs[1]=0,Bs[2]=0,Bs[3]=0);class Es extends
Si{constructor(t){super(t,ks);}createBucket(t){return new ls(t)}queryRadius(t)
{const e=t;return bs("circle-radius",this,e)+bs("circle-stroke-
width",this,e)+ws(this.paint.get("circle-
translate"))}queryIntersectsFeature(t,e,r,n,i,a,s,o){const
l=_s(t,this.paint.get("circle-translate"),this.paint.get("circle-translate-
anchor"),a.angle,s),u=this.paint.get("circle-
radius").evaluate(e,r)+this.paint.get("circle-stroke-
width").evaluate(e,r),c="map"===this.paint.get("circle-pitch-alignment"),h=c?
l:function(t,e){return t.map((t=>Fs(t,e)))}(l,o),p=c?u*s:u;for(const t of
n)for(const e of t){const t=c?e:Fs(e,o);let r=p;const n=Vs([],
[e.x,e.y,0,1],o);if("viewport"===this.paint.get("circle-pitch-
scale")&&"map"===this.paint.get("circle-pitch-alignment"))?
r*=n[3]/a.cameraToCenterDistance:"map"===this.paint.get("circle-pitch-
scale")&&"viewport"===this.paint.get("circle-pitch-alignment")&&
(r*=a.cameraToCenterDistance/n[3]),cs(h,t,r))return !0}return !1}}function
Fs(t,e){const r=Vs([],[t.x,t.y,0,1],e);return new a(r[0]/r[3],r[1]/r[3])}class
Ts extends ls{let $s;$n("HeatmapBucket",Ts,{omit:["layers"]});var Ls={get
paint(){return $s=$s||new _i({"heatmap-radius":new xi(X.paint_heatmap["heatmap-
radius"]), "heatmap-weight":new xi(X.paint_heatmap["heatmap-weight"]), "heatmap-
intensity":new gi(X.paint_heatmap["heatmap-intensity"]), "heatmap-color":new
wi(X.paint_heatmap["heatmap-color"]), "heatmap-opacity":new
gi(X.paint_heatmap["heatmap-opacity"])});};function Ds(t,{width:e,height:r},n,i)
{if(i){if(i instanceof Uint8ClampedArray)i=new Uint8Array(i.buffer);else
if(i.length!==e*r*n)throw new RangeError(`mismatched image size. expected:
${i.length} but got: ${e*r*n}`)}else i=new Uint8Array(e*r*n);return
t.width=e,t.height=r,t.data=i,t}function Os(t,{width:e,height:r},n)
{if(e===t.width&&r===t.height)return;const i=Ds({}, {width:e,height:r},n);Us(t,i,
{x:0,y:0},{x:0,y:0},
{width:Math.min(t.width,e),height:Math.min(t.height,r)},n),t.width=e,t.height=r,
t.data=i.data;}function Us(t,e,r,n,i,a){if(0===i.width||0===i.height)return
e;if(i.width>t.width||i.height>t.height||r.x>t.width-i.width||r.y>t.height-
i.height)throw new RangeError("out of range source coordinates for image
copy");if(i.width>e.width||i.height>e.height||n.x>e.width-i.width||n.y>e.height-
i.height)throw new RangeError("out of range destination coordinates for image
copy");const s=t.data,o=e.data;if(s===o)throw new Error("srcData equals dstData,
so image is already copied");for(let l=0;l<i.height;l++){const u=
((r.y+l)*t.width+r.x)*a,c=((n.y+l)*e.width+n.x)*a;for(let
t=0;t<i.width*a;t++)o[c+t]=s[u+t];}return e}class Rs{constructor(t,e)
{Ds(this,t,1,e);}resize(t){Os(this,t,1);}clone(){return new
Rs({width:this.width,height:this.height},new Uint8Array(this.data))}static
copy(t,e,r,n,i){Us(t,e,r,n,i,1);}class qs{constructor(t,e)
{Ds(this,t,4,e);}resize(t){Os(this,t,4);}replace(t,e){e?
this.data.set(t):this.data=t instanceof Uint8ClampedArray?new
Uint8Array(t.buffer):t;}clone(){return new
qs({width:this.width,height:this.height},new Uint8Array(this.data))}static
copy(t,e,r,n,i){Us(t,e,r,n,i,4);}function js(t){const e=
{},r=t.resolution||256,n=t.clips?t.clips.length:1,i=t.image||new
qs({width:r,height:n});if(Math.log(r)/Math.LN2%1!=0)throw new Error(`width is

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not a power of 2 - ${r}`);const a=(r,n,a)=>{e[t.evaluationKey]=a;const
s=t.expression.evaluate(e);i.data[r+n+0]=Math.floor(255*s.r/s.a),i.data[r+n+1]=M
ath.floor(255*s.g/s.a),i.data[r+n+2]=Math.floor(255*s.b/s.a),i.data[r+n+3]=Math.
floor(255*s.a);};if(t.clips)for(let e=0,i=0;e<n;++e,i+=4*r)for(let
n=0,s=0;n<r;n++,s+=4){const o=n/(r-1),{start:l,end:u}=t.clips[e];a(i,s,l*(1-
o)+u*o);}else for(let t=0,e=0;t<r;t++,e+=4)a(0,e,t/(r-1));return
i}$n("AlphaImage",Rs),$n("RGBAImage",qs);class Ns extends Si{createBucket(t)
{return new Ts(t)}constructor(t)
{super(t,Ls),this._updateColorRamp();}_handleSpecialPaintPropertyUpdate(t)
{"heatmap-color"===t&&this._updateColorRamp();}_updateColorRamp()
{this.colorRamp=js({expression:this._transitionablePaint._values["heatmap-
color"].value.expression,evaluationKey:"heatmapDensity",image:this.colorRamp}),t
his.colorRampTexture=null;resize(){this.heatmapFbo&&
(this.heatmapFbo.destroy(),this.heatmapFbo=null);}queryRadius(){return
0}queryIntersectsFeature(){return !1}hasOffscreenPass(){return
0!==(this.paint.get("heatmap-opacity")&&"none"!==this.visibility)}}let Zs;var Ks=
{get paint(){return Zs=Zs||new _i({"hillshade-illumination-direction":new
gi(X.paint_hillshade["hillshade-illumination-direction"]),"hillshade-
illumination-anchor":new gi(X.paint_hillshade["hillshade-illumination-
anchor"]),"hillshade-exaggeration":new gi(X.paint_hillshade["hillshade-
exaggeration"]),"hillshade-shadow-color":new gi(X.paint_hillshade["hillshade-
shadow-color"]),"hillshade-highlight-color":new gi(X.paint_hillshade["hillshade-
highlight-color"]),"hillshade-accent-color":new gi(X.paint_hillshade["hillshade-
accent-color"])});}class Gs extends Si{constructor(t)
{super(t,Ks);}hasOffscreenPass(){return 0!==(this.paint.get("hillshade-
exaggeration")&&"none"!==this.visibility)}const
Js=M([name:"a_pos",components:2,type:"Int16"],4),{members:Xs}=Js;var Ys=
{exports:{}};function Hs(t,e,r){r=r||2;var n,i,a,s,o,l,u,c=e&&e.length,h=c?
e[0]*r:t.length,p=Ws(t,0,h,r,!0),f=[];if(!p||p.next===p.prev)return f;if(c&&
(p=function(t,e,r,n){var i,a,s,o=[];for(i=0,a=e.length;i<a;i++)
(s=Ws(t,e[i]*n,i<a-1?e[i+1]*n:t.length,n,!1))===s.next&&
(s.steiner=!0),o.push(uo(s));for(o.sort(ao),i=0;i<o.length;i++)r=so(o[i],r);retu
rn r}(t,e,p,r)),t.length>80*r){n=a=t[0],i=s=t[1];for(var d=r;d<h;d+=r)(o=t[d])
<n&&(n=o),(l=t[d+1])<i&&(i=l),o>a&&(a=o),l>s&&(s=l);u=0!==(u=Math.max(a-n,s-i))?
32767/u:0;}return to(p,f,r,n,i,u,0),f}function Ws(t,e,r,n,i){var
a,s;if(i===Ao(t,e,r,n)>0)for(a=e;a<r;a+=n)s=bo(a,t[a],t[a+1],s);else for(a=r-
n;a>=e;a-=n)s=bo(a,t[a],t[a+1],s);return s&&fo(s,s.next)&&
(wo(s),s=s.next),s}function Qs(t,e){if(!t)return t;e||(e=t);var
r,n=t;do{if(r=!l,n.steiner||!fo(n,n.next)&&0!==(po(n.prev,n,n.next))n=n.next;else
{if(wo(n),(n=e=n.prev)===n.next)break;r=!0;}}while(r||n!==(e));return e}function
to(t,e,r,n,i,a,s){if(t){!s&&a&&function(t,e,r,n){var i=t;do{0===i.z&&
(i.z=lo(i.x,i.y,e,r,n)),i.prevZ=i.prev,i.nextZ=i.next,i=i.next;}while(i!==(t));i.p
revZ.nextZ=null,i.prevZ=null,function(t){var
e,r,n,i,a,s,o,l,u=1;do{for(r=t,t=null,a=null,s=0;r;){for(s++,n=r,o=0,e=0;e<u&&
(o++,n=n.nextZ;e++);for(l=u;o>0||l>0&&n;)0!==(o&&(0===l||!n||r.z<=n.z)?
(i=r,r=r.nextZ,o--):(i=n,n=n.nextZ,l--),a?
a.nextZ=i:t=i,i.prevZ=a,a=i;r=n;}a.nextZ=null,u*=2;}while(s>1)}(i);}
(t,n,i,a);for(var o,l,u=t;t.prev!==t.next;)if(o=t.prev,l=t.next,a?
ro(t,n,i,a):eo(t))e.push(o.i/r|0),e.push(t.i/r|0),e.push(l.i/r|0),wo(t),t=l.next
,u=l.next;else if((t=l)===u){s!==(s?
to(t=no(Qs(t),e,r),e,r,n,i,a,2):2===s&&io(t,e,r,n,i,a):to(Qs(t),e,r,n,i,a,1);bre
ak)}}function eo(t){var e=t.prev,r=t,n=t.next;if(po(e,r,n)>=0)return !1;for(var
i=e.x,a=r.x,s=n.x,o=e.y,l=r.y,u=n.y,c=i<a?i<s?i:s:a<s?a:s,h=o<l?o<u?o:u:l<u?
l:u,p=i>a?i>s?i:s:a>s?a:s,f=o>l?o>u?o:u:l>u?l:u,d=n.next;d!==(e);}
{if(d.x>=c&&d.x<=p&&d.y>=h&&d.y<=f&&co(i,o,a,l,s,u,d.x,d.y)&&po(d.prev,d,d.next)
>=0)return !1;d=d.next;}return !0}function ro(t,e,r,n){var
i=t.prev,a=t,s=t.next;if(po(i,a,s)>=0)return !1;for(var
o=i.x,l=a.x,u=s.x,c=i.y,h=a.y,p=s.y,f=o<l?o<u?o:u:l<u?l:u,d=c<h?c<p?c:p:h<p?
h:p,y=o>l?o>u?o:u:l>u?l:u,m=c>h?c>p?c:p:h>p?
h:p,g=lo(f,d,e,r,n),x=lo(y,m,e,r,n),v=t.prevZ,b=t.nextZ;v&&v.z>=g&&b&&b.z<=x;
){if(v.x>=f&&v.x<=y&&v.y>=d&&v.y<=m&&v!==(i&&v!==(s&&co(o,c,l,h,u,p,v.x,v.y)&&po(v.
prev,v,v.next)>=0)return

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!l;if(v=v.prevZ,b.x>=f&&b.x<=y&&b.y>=d&&b.y<=m&&b!==(i&&b!==(s&&co(o,c,l,h,u,p,b.x,b.y)&&po(b.prev,b,b.next)>=0))return !l;b=b.nextZ;}for(;v&&v.z>=g;){if(v.x>=f&&v.x<=y&&v.y>=d&&v.y<=m&&v!==(i&&v!==(s&&co(o,c,l,h,u,p,v.x,v.y)&&po(v.prev,v,v.next)>=0))return !l;v=v.prevZ;}for(;b&&b.z<=x;){if(b.x>=f&&b.x<=y&&b.y>=d&&b.y<=m&&b!==(i&&b!==(s&&co(o,c,l,h,u,p,b.x,b.y)&&po(b.prev,b,b.next)>=0))return !l;b=b.nextZ;}return !0}function no(t,e,r){var n=t;do{var i=n.prev,a=n.next.next;!fo(i,a)&&yo(i,n,n.next,a)&&xo(i,a)&&xo(a,i)&&(e.push(i.i/r|0),e.push(n.i/r|0),e.push(a.i/r|0),wo(n),wo(n.next),n=t=a),n=n.next;t;}while(n!==(t));return Qs(n)}function io(t,e,r,n,i,a){var s=t;do{for(var o=s.next.next;o!==(s.prev);}if(s.i!==(o.i&&ho(s,o)){var l=vo(s,o);return s=Qs(s,s.next),l=Qs(l,l.next),to(s,e,r,n,i,a,0),void to(l,e,r,n,i,a,0)}o=o.next;s=s.next;}while(s!==(t))}function ao(t,e){return t.x-e.x}function so(t,e){var r=function(t,e){var r,n=e,i=t.x,a=t.y,s=-1/0;do{if(a<n.y&&a>n.next.y&&n.next.y!==(n.y){var o=n.x+(a-n.y)*(n.next.x-n.x)/(n.next.y-n.y);if(o<=i&&o>s&&(s=o,r=n.x<n.next.x?n:n.next,o===i))return r}n=n.next;}while(n!==(e));if(!r)return null;var l,u=r,c=r.x,h=r.y,p=1/0;n=r;do{if(i>n.x&&n.x>=c&&i!==(n.x&&co(a<h?i:s,a,c,h,a<h?s:i,a,n.x,n.y)&&(l=Math.abs(a-n.y)/(i-n.x),xo(n,t)&&(l<p||l===p&&(n.x>r.x|n.x===r.x&&oo(r,n))&&(r=n,p=1)),n=n.next);}while(n!==(u));return r}(t,e);if(!r)return e;var n=vo(r,t);return Qs(n,n.next),Qs(r,r.next)}function oo(t,e){return po(t.prev,t,e.prev)<0&&po(e.next,t,t.next)<0}function lo(t,e,r,n,i){return (t=1431655765&((t=858993459&((t=252645135&((t=16711935&((t=(t-r)*i|0)|t<<8))|t<<4))|t<<2))|t<<1))|(e=1431655765&((e=858993459&((e=252645135&((e=16711935&((e=(e-n)*i|0)|e<<8))|e<<4))|e<<2))|e<<1))<<1)}function uo(t){var e=t,r=t;do{(e.x<r.x|e.x===r.x&&e.y<r.y)&&(r=e),e=e.next;}while(e!==(t));return r}function co(t,e,r,n,i,a,s,o){return (i-s)*(e-o)>=(t-s)*(a-o)&&(t-s)*(n-o)>=(r-s)*(e-o)&&(r-s)*(a-o)>=(i-s)*(n-o)}function ho(t,e){return t.next.i!==(e.i&&t.prev.i!==(e.i&&!function(t,e){var r=t;do{if(r.i!==(t.i&&r.next.i!==(t.i&&r.next.i!==(e.i&&r.next.i!==(e.i&&yo(r,r.next,t,e))return !0;r=r.next;}while(r!==(t));return !l}(t,e)&&(xo(t,e)&&xo(e,t)&&function(t,e){var r=t,n=!l,i=(t.x+e.x)/2,a=(t.y+e.y)/2;do{r.y>a!==(r.next.y>a&&r.next.y!==(r.y&&i<(r.next.x-r.x)*(a-r.y)/(r.next.y-r.y)+r.x&&(n=!n),r=r.next);}while(r!==(t));return n}(t,e)&&(po(t.prev,t,e.prev)|po(t,e.prev,e))|fo(t,e)&&po(t.prev,t,t.next)>0&&po(e.prev,e,e.next)>0)}function po(t,e,r){return (e.y-t.y)*(r.x-e.x)-(e.x-t.x)*(r.y-e.y)}function fo(t,e){return t.x===e.x&&t.y===e.y}function yo(t,e,r,n){var i=go(po(t,e,r)),a=go(po(t,e,n)),s=go(po(r,n,t)),o=go(po(r,n,e));return i!==(a&&s!==(o||!(0!==(i||!mo(t,r,e))||!(0!==(a||!mo(t,n,e))||!(0!==(s||!mo(r,t,n))||!(0!==(o||!mo(r,e,n))}function mo(t,e,r){return e.x<=Math.max(t.x,r.x)&&e.x>=Math.min(t.x,r.x)&&e.y<=Math.max(t.y,r.y)&&e.y>=Math.min(t.y,r.y)}function go(t){return t>0?1:t<0?-1:0}function xo(t,e){return po(t.prev,t,t.next)<0?po(t,e,t.next)>0&&po(t,t.prev,e)>0:po(t,e,t.prev)<0|po(t,t.next,e)<0}function vo(t,e){var r=new _o(t.i,t.x,t.y),n=new _o(e.i,e.x,e.y),i=t.next,a=e.prev;return t.next=e,e.prev=t,r.next=i,i.prev=r,n.next=r,r.prev=n,a.next=n,n.prev=a,n}function bo(t,e,r,n){var i=new _o(t,e,r);return n?(i.next=n.next,i.prev=n,n.next.prev=i,n.next=i):(i.prev=i,i.next=i),i}function wo(t){t.next.prev=t.prev,t.prev.next=t.next,t.prevZ&&(t.prevZ.nextZ=t.nextZ),t.nextZ&&(t.nextZ.prevZ=t.prevZ);}function _o(t,e,r){this.i=t,this.x=e,this.y=r,this.prev=null,this.next=null,this.z=0,this.prevZ=null,this.nextZ=null,this.steiner=!l;}function Ao(t,e,r,n){for(var i=0,a=e,s=r-n;a<r;a+=n)i+=(t[s]-t[a])*(t[a+1]+t[s+1]),s=a;return i}Ys.exports=Hs,Ys.exports.default=Hs,Hs.deviation=function(t,e,r,n){var i=e&&e.length,a=Math.abs(Ao(t,0,i?e[0]*r:t.length,r));if(i)for(var s=0,o=e.length;s<o;s++)a-=Math.abs(Ao(t,e[s]*r,s<o-1?e[s+1]*r:t.length,r));var l=0;for(s=0;s<n.length;s+=3){var u=n[s]*r,c=n[s+1]*r,h=n[s+2]*r;l+=Math.abs((t[u]-t[h])*(t[c+1]-t[u+1])-(t[u]-t[c])*(t[h+1]-t[u+1]));}return 0===a&&0===l?0:Math.abs((l-a)/a)},Hs.flatten=function(t){for(var e=t[0][0].length,r={vertices:[],holes:[],dimensions:e},n=0,i=0;i<t.length;i++){for(var a=0;a<t[i].length;a++)for(var s=0;s<e;s++)r.vertices.push(t[i][a][s]);i>0&&r.holes.push(n+=t[i-1].length);}return r};var So=r(Ys.exports);function ko(t,e,r,n,i)

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{Io(t,e,r||0,n||t.length-1,i||Mo);}function Io(t,e,r,n,i){for(;n>r;){if(n-r>600)
{var a=n-r+1,s=e-r+1,o=Math.log(a),l=.5*Math.exp(2*o/3),u=.5*Math.sqrt(o*l*(a-
l)/a)*(s-a/2<0?-1:1);Io(t,e,Math.max(r,Math.floor(e-
s*l/a+u)),Math.min(n,Math.floor(e+(a-s)*l/a+u)),i);}var
c=t[e],h=r,p=n;for(zo(t,r,e),i(t[n],c)>0&&zo(t,r,n);h<p;){for(zo(t,h,p),h++,p--
;i(t[h],c)<0;)h++;for(;i(t[p],c)>0;)p--;}0===i(t[r],c)?
zo(t,r,p):zo(t,++p,n),p<=e&&(r=p+1),e<=p&&(n=p-1);}}function zo(t,e,r){var
n=t[e];t[e]=t[r],t[r]=n;}function Mo(t,e){return t<e?-1:t>e?1:0}function Co(t,e)
{const r=t.length;if(r<=1)return [t];const n=[];let i,a;for(let e=0;e<r;e++)
{const r=k(t[e]);0!==(r&&(t[e].area=Math.abs(r),void 0===a&&(a=r<0),a===r<0?
(i&&n.push(i),i=[t[e]]):i.push(t[e])));if(i&&n.push(i),e>1)for(let
t=0;t<n.length;t++)n[t].length<=e||(ko(n[t],e,1,n[t].length-
1,Bo),n[t]=n[t].slice(0,e));return n}function Bo(t,e){return e.area-
t.area}function Po(t,e,r){const n=r.patternDependencies;let i=!1;for(const r of
e){const e=r.paint.get(`$${t}-pattern`);e.isConstant()||(i=!0);const
a=e.constantOr(null);a&&(i=!0,n[a.to]=!0,n[a.from]=!0);}return i}function
Vo(t,e,r,n,i){const a=i.patternDependencies;for(const s of e){const
e=s.paint.get(`$${t}-pattern`).value;if("constant"!==e.kind){let
t=e.evaluate({zoom:n-1},r,{},i.availableImages),o=e.evaluate({zoom:n},r,
{},i.availableImages),l=e.evaluate({zoom:n+1},r,
{},i.availableImages);t=t&&t.name?t.name:t,o=o&&o.name?o.name:o,l=l&&l.name?
l.name:l,a[t]=!0,a[o]=!0,a[l]=!0,r.patterns[s.id]={min:t,mid:o,max:l};}}return
r}class Eo{constructor(t)
{this.zoom=t.zoom,this.overscaling=t.overscaling,this.layers=t.layers,this.layer
Ids=this.layers.map((t=>t.id)),this.index=t.index,this.hasPattern=!1,this.patter
nFeatures=[],this.layoutVertexArray=new fa,this.indexArray=new
Aa,this.indexArray2=new Sa,this.programConfigurations=new
Qa(t.layers,t.zoom),this.segments=new za,this.segments2=new
za,this.stateDependentLayerIds=this.layers.filter((t=>t.isStateDependent()))
.map((t=>t.id));}populate(t,e,r){this.hasPattern=Po("fill",this.layers,e);const
n=this.layers[0].layout.get("fill-sort-key"),i=!n.isConstant(),a=
[];for(const{feature:s,id:o,index:l,sourceLayerIndex:u}of t){const
t=this.layers[0]._featureFilter.needGeometry,c=ss(s,t);if(!this.layers[0]._featu
reFilter.filter(new li(this.zoom),c,r))continue;const h=i?n.evaluate(c,
{},r,e.availableImages):void 0,p=
{id:o,properties:s.properties,type:s.type,sourceLayerIndex:u,index:l,geometry:t?
c.geometry:as(s),patterns:{},sortKey:h};a.push(p);i&&a.sort((t,e)=>t.sortKey-
e.sortKey));for(const n of a)
{const{geometry:i,index:a,sourceLayerIndex:s}=n;if(this.hasPattern){const
t=Vo("fill",this.layers,n,this.zoom,e);this.patternFeatures.push(t);}else
this.addFeature(n,i,a,r,
{});e.featureIndex.insert(t[a].feature,i,a,s,this.index);}}update(t,e,r)
{this.stateDependentLayers.length&&this.programConfigurations.updatePaintArrays(
t,e,this.stateDependentLayers,r);}addFeatures(t,e,r){for(const t of
this.patternFeatures)this.addFeature(t,t.geometry,t.index,e,r);}isEmpty(){return
0===this.layoutVertexArray.length}uploadPending(){return
!this.uploaded||this.programConfigurations.needsUpload}upload(t){this.uploaded||
(this.layoutVertexBuffer=t.createVertexBuffer(this.layoutVertexArray,Xs),this.in
dexBuffer=t.createIndexBuffer(this.indexArray),this.indexBuffer2=t.createIndexBu
ffer(this.indexArray2)),this.programConfigurations.upload(t),this.uploaded=!0;}d
estroy(){this.layoutVertexBuffer&&
(this.layoutVertexBuffer.destroy(),this.indexBuffer.destroy(),this.indexBuffer2.
destroy(),this.programConfigurations.destroy(),this.segments.destroy(),this.segm
ents2.destroy());}addFeature(t,e,r,n,i){for(const t of Co(e,500)){let
e=0;for(const r of t)e+=r.length;const
r=this.segments.prepareSegment(e,this.layoutVertexArray,this.indexArray),n=r.ver
texLength,i=[],a=[];for(const e of t)
{if(0===e.length)continue;e!==t[0]&&a.push(i.length/2);const
r=this.segments2.prepareSegment(e.length,this.layoutVertexArray,this.indexArray2
),n=r.vertexLength;this.layoutVertexArray.emplaceBack(e[0].x,e[0].y),this.indexA
rray2.emplaceBack(n+e.length-1,n),i.push(e[0].x),i.push(e[0].y);for(let
t=1;t<e.length;t++)this.layoutVertexArray.emplaceBack(e[t].x,e[t].y),this.indexA

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rray2.emplaceBack(n+t-
1,n+t),i.push(e[t].x),i.push(e[t].y);r.vertexLength+=e.length,r.primitiveLength+=
=e.length;}const s=So(i,a);for(let
t=0;t<s.length;t+=3)this.indexArray.emplaceBack(n+s[t],n+s[t+1],n+s[t+2]);r.vert
exLength+=e,r.primitiveLength+=s.length/3;}this.programConfigurations.populatePa
intArrays(this.layoutVertexArray.length,t,r,i,n);}let Fo,To;$n("FillBucket",Eo,
{omit:["layers","patternFeatures"]});var $o={get paint(){return To=To|new
_i({"fill-antialias":new gi(X.paint_fill["fill-antialias"]), "fill-opacity":new
xi(X.paint_fill["fill-opacity"]), "fill-color":new xi(X.paint_fill["fill-
color"]), "fill-outline-color":new xi(X.paint_fill["fill-outline-color"]), "fill-
translate":new gi(X.paint_fill["fill-translate"]), "fill-translate-anchor":new
gi(X.paint_fill["fill-translate-anchor"]), "fill-pattern":new
vi(X.paint_fill["fill-pattern"])}),get layout(){return Fo=Fo|new _i({"fill-
sort-key":new xi(X.layout_fill["fill-sort-key"])}))};class Lo extends
Si{constructor(t){super(t,$o);}recalculate(t,e){super.recalculate(t,e);const
r=this.paint._values["fill-outline-color"]; "constant"===r.value.kind&&void
0===r.value.value&&(this.paint._values["fill-outline-
color"]=this.paint._values["fill-color"]);}createBucket(t){return new
Eo(t)}queryRadius(){return ws(this.paint.get("fill-
translate"))}queryIntersectsFeature(t,e,r,n,i,a,s){return
hs(_s(t,this.paint.get("fill-translate"),this.paint.get("fill-translate-
anchor"),a.angle,s),n)}isTileClipped(){return !0}}const
Do=Mi([ {name:"a_pos",components:2,type:"Int16"},
{name:"a_normal_ed",components:4,type:"Int16"} ],4),Oo=Mi([ {name:"a_centroid",com
ponents:2,type:"Int16"} ],4),{members:Uo}=Do;var Ro={},qo=n,jo=No;function
No(t,e,r,n,i){this.properties=
{},this.extent=r,this.type=0,this._pbf=t,this._geometry=-1,this._keys=n,this._va
lues=i,t.readFields(Zo,this,e);}function Zo(t,e,r){l==t?
e.id=r.readVarint():2==t?function(t,e){for(var r=t.readVarint()+t.pos;t.pos<r;)
{var n=e._keys[t.readVarint()],i=e._values[t.readVarint()];e.properties[n]=i;}}
(r,e):3==t?e.type=r.readVarint():4==t&&(e._geometry=r.pos);}function Ko(t)
{for(var e,r,n=0,i=0,a=t.length,s=a-1;i<a;s=i++)n+=(r=t[s]).x-(e=t[i]).x)*
(e.y+r.y);return n}No.types=
["Unknown","Point","LineString","Polygon"],No.prototype.loadGeometry=function()
{var t=this._pbf;t.pos=this._geometry;for(var
e,r=t.readVarint()+t.pos,n=1,i=0,a=0,s=0,o=[];t.pos<r;){if(i<=0){var
l=t.readVarint();n=7&l,i=1>>3;}if(i-
-,l===n|2===n)a+=t.readSVarint(),s+=t.readSVarint(),l===n&&(e&&o.push(e),e=
[]),e.push(new qo(a,s));else {if(7!==n)throw new Error("unknown command
"+n);e&&e.push(e[0].clone());}}return
e&&o.push(e),o},No.prototype.bbox=function(){var
t=this._pbf;t.pos=this._geometry;for(var
e=t.readVarint()+t.pos,r=1,n=0,i=0,a=0,s=1/0,o=-1/0,l=1/0,u=-1/0;t.pos<e;)
{if(n<=0){var c=t.readVarint();r=7&c,n=c>>3;}if(n--,l===r|2===r)
(i+=t.readSVarint())<s&&(s=i),i>o&&(o=i),(a+=t.readSVarint())<l&&(l=a),a>u&&
(u=a);else if(7!==r)throw new Error("unknown command "+r)}return
[s,l,o,u]},No.prototype.toGeoJSON=function(t,e,r){var
n,i,a=this.extent*Math.pow(2,r),s=this.extent*t,o=this.extent*e,l=this.loadGeome
try(),u=No.types[this.type];function c(t){for(var e=0;e<t.length;e++){var
r=t[e];t[e]=[360*(r.x+s)/a-180,360/Math.PI*Math.atan(Math.exp((180-360*
(r.y+o)/a)*Math.PI/180))-90];}}switch(this.type){case 1:var h=
[];for(n=0;n<l.length;n++)h[n]=l[n][0];c(l=h);break;case
2:for(n=0;n<l.length;n++)c(l[n]);break;case 3:for(l=function(t){var
e=t.length;if(e<=1)return [t];for(var r,n,i=[],a=0;a<e;a++){var
s=Ko(t[a]);0!==s&&(void 0===n&&(n=s<0),n===s<0?(r&&i.push(r),r=
[t[a]]):r.push(t[a]));}return r&&i.push(r),i)
(l),n=0;n<l.length;n++)for(i=0;i<l[n].length;i++)c(l[n][i]);}l===l.length?
l=l[0]:u="Multi"+u;var p={type:"Feature",geometry:
{type:u,coordinates:l},properties:this.properties};return "id"in this&&
(p.id=this.id),p;var Go=jo,Jo=Xo;function Xo(t,e)
{this.version=1,this.name=null,this.extent=4096,this.length=0,this._pbf=t,this._
keys=[],this._values=[],this._features=

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[ ],t.readFields(Yo,this,e),this.length=this._features.length;}function Yo(t,e,r)
{15===t?e.version=r.readVarint():1===t?e.name=r.readString():5===t?
e.extent=r.readVarint():2===t?e._features.push(r.pos):3===t?
e._keys.push(r.readString()):4===t&&e._values.push(function(t){for(var
e=null,r=t.readVarint()+t.pos;t.pos<r){var n=t.readVarint()>>3;e=1===n?
t.readString():2===n?t.readFloat():3===n?t.readDouble():4===n?
t.readVarint64():5===n?t.readVarint():6===n?t.readSVarint():7===n?
t.readBoolean():null;}return e})(r));}Xo.prototype.feature=function(t)
{if(t<0||t>=this._features.length)throw new Error("feature index out of
bounds");this._pbf.pos=this._features[t];var
e=this._pbf.readVarint()+this._pbf.pos;return new
Go(this._pbf,e,this.extent,this._keys,this._values)};var Ho=Jo;function
Wo(t,e,r){if(3===t){var n=new Ho(r,r.readVarint()+r.pos);n.length&&
(e[n.name]=n);}Ro.VectorTile=function(t,e){this.layers=t.readFields(Wo,
{}),e};Ro.VectorTileFeature=jo,Ro.VectorTileLayer=Jo;const
Qo=Ro.VectorTileFeature.types,tl=Math.pow(2,13);function el(t,e,r,n,i,a,s,o)
{t.emplaceBack(e,r,2*Math.floor(n*tl)+s,i*tl*2,a*tl*2,Math.round(o));}class
rl{constructor(t)
{this.zoom=t.zoom,this.overscaling=t.overscaling,this.layers=t.layers,this.layer
Ids=this.layers.map((t=>t.id)),this.index=t.index,this.hasPattern=!1,this.layout
VertexArray=new da,this.centroidVertexArray=new ha,this.indexArray=new
Aa,this.programConfigurations=new Qa(t.layers,t.zoom),this.segments=new
za,this.stateDependentLayerIds=this.layers.filter((t=>t.isStateDependent()))
.map((t=>t.id));}populate(t,e,r){this.features=[],this.hasPattern=Po("fill-
extrusion",this.layers,e);for(const{feature:n,id:i,index:a,sourceLayerIndex:s}of
t){const
t=this.layers[0]._featureFilter.needGeometry,o=ss(n,t);if(!this.layers[0]._featu
reFilter.filter(new li(this.zoom),o,r))continue;const l=
{id:i,sourceLayerIndex:s,index:a,geometry:t?
o.geometry:as(n),properties:n.properties,type:n.type,patterns:
{}};this.hasPattern?this.features.push(Vo("fill-
extrusion",this.layers,l,this.zoom,e)):this.addFeature(l,l.geometry,a,r,
{}),e.featureIndex.insert(n,l.geometry,a,s,this.index,!0);}addFeatures(t,e,r)
{for(const t of this.features)
{const{geometry:n}=t;this.addFeature(t,n,t.index,e,r);}update(t,e,r)
{this.stateDependentLayers.length&&this.programConfigurations.updatePaintArrays(
t,e,this.stateDependentLayers,r);}isEmpty(){return
0===this.layoutVertexArray.length&&0===this.centroidVertexArray.length}uploadPen
ding(){return !this.uploaded||this.programConfigurations.needsUpload}upload(t)
{this.uploaded||
(this.layoutVertexBuffer=t.createVertexBuffer(this.layoutVertexArray,Uo),this.ce
ntroidVertexBuffer=t.createVertexBuffer(this.centroidVertexArray,Oo.members,!0),
this.indexBuffer=t.createIndexBuffer(this.indexArray)),this.programConfiguratio
n.s.upload(t),this.uploaded=!0;}destroy(){this.layoutVertexBuffer&&
(this.layoutVertexBuffer.destroy(),this.indexBuffer.destroy(),this.programConfig
urations.destroy(),this.segments.destroy(),this.centroidVertexBuffer.destroy());
}addFeature(t,e,r,n,i){const a={x:0,y:0,vertexCount:0};for(const r of Co(e,500))
{let e=0;for(const t of r)e+=t.length;let
n=this.segments.prepareSegment(4,this.layoutVertexArray,this.indexArray);for(con
st t of r){if(0===t.length)continue;if(il(t))continue;let e=0;for(let
r=0;r<t.length;r++){const i=t[r];if(r>=1){const s=t[r-1];if(!nl(i,s))
{n.vertexLength+4>za.MAX_VERTEX_ARRAY_LENGTH&&
(n=this.segments.prepareSegment(4,this.layoutVertexArray,this.indexArray));const
t=i.sub(s)._perp()._unit(),r=s.dist(i);e+r>32768&&
(e=0),el(this.layoutVertexArray,i.x,i.y,t.x,t.y,0,0,e),el(this.layoutVertexArray
,i.x,i.y,t.x,t.y,0,1,e),a.x+=2*i.x,a.y+=2*i.y,a.vertexCount+=2,e+=r,el(this.layo
utVertexArray,s.x,s.y,t.x,t.y,0,0,e),el(this.layoutVertexArray,s.x,s.y,t.x,t.y,0
,1,e),a.x+=2*s.x,a.y+=2*s.y,a.vertexCount+=2;const
o=n.vertexLength;this.indexArray.emplaceBack(o,o+2,o+1),this.indexArray.emplaceB
ack(o+1,o+2,o+3),n.vertexLength+=4,n.primitiveLength+=2;}}}}if(n.vertexLength+e>
za.MAX_VERTEX_ARRAY_LENGTH&&
(n=this.segments.prepareSegment(e,this.layoutVertexArray,this.indexArray)),"Poly

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gon"!==Qo[t.type])continue;const i=[],s=[],o=n.vertexLength;for(const t of
r)if(0!==t.length){t!==(r[0]&&s.push(i.length/2));for(let e=0;e<t.length;e++)
{const
r=t[e];el(this.layoutVertexArray,r.x,r.y,0,0,1,1,0),a.x+=r.x,a.y+=r.y,a.vertexCo
unt+=1,i.push(r.x),i.push(r.y);}}const l=So(i,s);for(let
t=0;t<l.length;t+=3)this.indexArray.emplaceBack(o+l[t],o+l[t+2],o+l[t+1]);n.prim
itiveLength+=l.length/3,n.vertexLength+=e;}for(let
t=0;t<a.vertexCount;t++)this.centroidVertexArray.emplaceBack(Math.floor(a.x/a.ve
rtexCount),Math.floor(a.y/a.vertexCount));this.programConfigurations.populatePai
ntArrays(this.layoutVertexArray.length,t,r,i,n);}}function nl(t,e){return
t.x===e.x&&(t.x<0||t.x>rs)||t.y===e.y&&(t.y<0||t.y>rs)}function il(t){return
t.every((t=>t.x<0))||t.every((t=>t.x>rs))||t.every((t=>t.y<0))||t.every((t=>t.y>
rs))}let al;$n("FillExtrusionBucket",rl,{omit:["layers","features"]});var sl=
{get paint(){return al=al||new _i({"fill-extrusion-opacity":new
gi(X["paint_fill-extrusion"]["fill-extrusion-opacity"]),"fill-extrusion-
color":new xi(X["paint_fill-extrusion"]["fill-extrusion-color"]),"fill-
extrusion-translate":new gi(X["paint_fill-extrusion"]["fill-extrusion-
translate"]),"fill-extrusion-translate-anchor":new gi(X["paint_fill-extrusion"]
["fill-extrusion-translate-anchor"]),"fill-extrusion-pattern":new
vi(X["paint_fill-extrusion"]["fill-extrusion-pattern"]),"fill-extrusion-
height":new xi(X["paint_fill-extrusion"]["fill-extrusion-height"]),"fill-
extrusion-base":new xi(X["paint_fill-extrusion"]["fill-extrusion-base"]),"fill-
extrusion-vertical-gradient":new gi(X["paint_fill-extrusion"]["fill-extrusion-
vertical-gradient"])});}};class ol extends Si{constructor(t)
{super(t,sl);}createBucket(t){return new rl(t)}queryRadius(){return
ws(this.paint.get("fill-extrusion-translate"))}is3D(){return
!0}queryIntersectsFeature(t,e,r,n,i,s,o,l){const u=_s(t,this.paint.get("fill-
extrusion-translate"),this.paint.get("fill-extrusion-translate-
anchor"),s.angle,o),c=this.paint.get("fill-extrusion-
height").evaluate(e,r),h=this.paint.get("fill-extrusion-
base").evaluate(e,r),p=function(t,e,r,n){const i=[];for(const r of t){const t=
[r.x,r.y,0,1];Vs(t,t,e),i.push(new a(t[0]/t[3],t[1]/t[3]));}return i}
(u,l),f=function(t,e,r,n){const i=[],s=
[],o=n[8]*e,l=n[9]*e,u=n[10]*e,c=n[11]*e,h=n[8]*r,p=n[9]*r,f=n[10]*r,d=n[11]*r;f
or(const e of t){const t=[],r=[];for(const i of e){const
e=i.x,s=i.y,y=n[0]*e+n[4]*s+n[12],m=n[1]*e+n[5]*s+n[13],g=n[2]*e+n[6]*s+n[14],x=
n[3]*e+n[7]*s+n[15],v=g+u,b=x+c,w=y+h,_=m+p,A=g+f,S=x+d,k=new a((y+o)/b,
(m+l)/b);k.z=v/b,t.push(k);const I=new
a(w/S,_/S);I.z=A/S,r.push(I);i.push(t),s.push(r);}return [i,s]}(n,h,c,l);return
function(t,e,r){let n=1/0;hs(r,e)&&(n=ul(r,e[0]));for(let i=0;i<e.length;i++)
{const a=e[i],s=t[i];for(let t=0;t<a.length-1;t++){const e=a[t],i=
[e,a[t+1],s[t+1],s[t],e];us(r,i)&&(n=Math.min(n,ul(r,i)));}}return n!==1/0&&n}
(f[0],f[1],p)}function ll(t,e){return t.x*e.x+t.y*e.y}function ul(t,e)
{if(1===t.length){let r=0;const n=e[r++];let
i;for(;;!i||n.equals(i);)if(i=e[r++],!i)return 1/0;for(;r<e.length;r++){const
a=e[r],s=t[0],o=i.sub(n),l=a.sub(n),u=s.sub(n),c=ll(o,o),h=ll(o,l),p=ll(l,l),f=l
l(u,o),d=ll(u,l),y=c*p-h*h,m=(p*f-h*d)/y,g=(c*d-h*f)/y,x=n.z*(1-m-
g)+i.z*m+a.z*g;if(isFinite(x))return x}return 1/0}{let t=1/0;for(const r of
e)t=Math.min(t,r.z);return t}}const
cl=Mi([ {name:"a_pos_normal",components:2,type:"Int16"},
{name:"a_data",components:4,type:"Uint8"}],4),
{members:hl}=cl,pl=Mi([ {name:"a_uv_x",components:1,type:"Float32"},
{name:"a_split_index",components:1,type:"Float32"}]),
{members:fl}=pl,dl=Ro.VectorTileFeature.types,yl=Math.cos(Math.PI/180*37.5),ml=M
ath.pow(2,14)/.5;class gl{constructor(t)
{this.zoom=t.zoom,this.overscaling=t.overscaling,this.layers=t.layers,this.layer
Ids=this.layers.map((t=>t.id)),this.index=t.index,this.hasPattern=!1,this.patter
nFeatures=[],this.lineClipsArray=[],this.gradients={},this.layers.forEach((t=>
{this.gradients[t.id]={};})),this.layoutVertexArray=new
ya,this.layoutVertexArray2=new ma,this.indexArray=new
Aa,this.programConfigurations=new Qa(t.layers,t.zoom),this.segments=new
za,this.maxLineLength=0,this.stateDependentLayerIds=this.layers.filter((t=>t.isS

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tateDependent()).map((t=>t.id));}populate(t,e,r)
{this.hasPattern=Po("line",this.layers,e);const
n=this.layers[0].layout.get("line-sort-key"),i=!n.isConstant(),a=
[];for(const{feature:e,id:s,index:o,sourceLayerIndex:l}of t){const
t=this.layers[0]._featureFilter.needGeometry,u=ss(e,t);if(!this.layers[0]._featu
reFilter.filter(new li(this.zoom),u,r))continue;const c=i?n.evaluate(u,
{}),r):void 0,h=
{id:s,properties:e.properties,type:e.type,sourceLayerIndex:l,index:o,geometry:t?
u.geometry:as(e),patterns:{},sortKey:c};a.push(h);i&&a.sort((t,e)=>t.sortKey-
e.sortKey));for(const n of a)
{const{geometry:i,index:a,sourceLayerIndex:s}=n;if(this.hasPattern){const
t=Vo("line",this.layers,n,this.zoom,e);this.patternFeatures.push(t);}else
this.addFeature(n,i,a,r,
{});e.featureIndex.insert(t[a].feature,i,a,s,this.index);}update(t,e,r)
{this.stateDependentLayers.length&&this.programConfigurations.updatePaintArrays(
t,e,this.stateDependentLayers,r);}addFeatures(t,e,r){for(const t of
this.patternFeatures)this.addFeature(t,t.geometry,t.index,e,r);}isEmpty(){return
0===this.layoutVertexArray.length}uploadPending(){return
!this.uploaded||this.programConfigurations.needsUpload}upload(t){this.uploaded||
(0!==this.layoutVertexArray2.length&&
(this.layoutVertexBuffer2=t.createVertexBuffer(this.layoutVertexArray2,fl)),this
.layoutVertexBuffer=t.createVertexBuffer(this.layoutVertexArray,h1),this.indexBu
ffer=t.createIndexBuffer(this.indexArray)),this.programConfigurations.upload(t),
this.uploaded=!0;}destroy(){this.layoutVertexBuffer&&
(this.layoutVertexBuffer.destroy(),this.indexBuffer.destroy(),this.programConfig
urations.destroy(),this.segments.destroy());}lineFeatureClips(t)
{if(t.properties&&Object.prototype.hasOwnProperty.call(t.properties,"mapbox_clip
_start")&&Object.prototype.hasOwnProperty.call(t.properties,"mapbox_clip_end"))r
eturn
{start:+t.properties.mapbox_clip_start,end:+t.properties.mapbox_clip_end}}addFea
ture(t,e,r,n,i){const a=this.layers[0].layout,s=a.get("line-join").evaluate(t,
{}),o=a.get("line-cap"),l=a.get("line-miter-limit"),u=a.get("line-round-
limit");this.lineClips=this.lineFeatureClips(t);for(const r of
e)this.addLine(r,t,s,o,l,u);this.programConfigurations.populatePaintArrays(this.
layoutVertexArray.length,t,r,i,n);}addLine(t,e,r,n,i,a)
{if(this.distance=0,this.scaledDistance=0,this.totalDistance=0,this.lineClips)
{this.lineClipsArray.push(this.lineClips);for(let e=0;e<t.length-
1;e++)this.totalDistance+=t[e].dist(t[e+1]);this.updateScaledDistance(),this.max
LineLength=Math.max(this.maxLineLength,this.totalDistance);}const
s="Polygon"===dl[e.type];let o=t.length;for(;o>=2&&t[o-1].equals(t[o-2]));o--
;let l=0;for(;l<o-1&&t[l].equals(t[l+1]));l++;if(o<(s?3:2))return;"bevel"===r&&
(i=1.05);const u=this.overscaling<=16?
15*rs/(512*this.overscaling):0,c=this.segments.prepareSegment(10*o,this.layoutVe
rtexArray,this.indexArray);let h,p,f,d,y;this.e1=this.e2=-1,s&&(h=t[o-
2],y=t[l].sub(h)._unit()._perp());for(let e=1;e<o;e++){if(f=e===o-1?s?
t[l+1]:void 0:t[e+1],f&&t[e].equals(f))continue;y&&(d=y),h&&(p=h),h=t[e],y=f?
f.sub(h)._unit()._perp():d,d=d||y;let
m=d.add(y);0===m.x&&0===m.y||m._unit();const
g=d.x*y.x+d.y*y.y,x=m.x*y.x+m.y*y.y,v=0!==(x?1/x:1/0),b=2*Math.sqrt(2-
2*x),w=x<y1&&p&&f,_=d.x*y.y-d.y*y.x>0;if(w&&e>1){const t=h.dist(p);if(t>2*u)
{const
e=h.sub(h.sub(p)._mult(u/t)._round());this.updateDistance(p,e),this.addCurrentVe
rtex(e,d,0,0,c),p=e;}const A=p&&f;let S=A?r:s?"butt":n;if(A&&"round"===S&&(v<a?
S="miter":v<=2&&(S="fakaround")), "miter"===S&&v>i&&(S="bevel"), "bevel"===S&&
(v>2&&(S="flipbevel"),v<i&&
(S="miter")),p&&this.updateDistance(p,h), "miter"===S)m._mult(v),this.addCurrentV
ertex(h,m,0,0,c);else if("flipbevel"===S){if(v>100)m=y.mult(-1);else {const
t=v*d.add(y).mag()/d.sub(y).mag();m._perp()._mult(t*(?
-1:1));}this.addCurrentVertex(h,m,0,0,c),this.addCurrentVertex(h,m.mult(-1),0,0,
c);}else if("bevel"===S||"fakaround"===S){const t=-Math.sqrt(v*v-1),e=?t:0,r=?
0:t;if(p&&this.addCurrentVertex(h,d,e,r,c), "fakaround"===S){const
t=Math.round(180*b/Math.PI/20);for(let e=1;e<t;e++){let r=e/t;if(.5!==(r){const

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t=r-.5;r+=r*t*(r-1)*((1.0904+g*(g*(3.55645-1.43519*g)-3.2452))*t*t+.848013+g*
(.215638*g-1.06021));}const n=y.sub(d)._mult(r)._add(d)._unit()._mult(_?
-1:1);this.addHalfVertex(h,n.x,n.y,!l,_0,c);}f&&this.addCurrentVertex(h,y,-e,-
r,c);}else if("butt"===S)this.addCurrentVertex(h,m,0,0,c);else if("square"===S)
{const t=p?1:-1;this.addCurrentVertex(h,m,t,t,c);}else "round"===S&&(p&&
(this.addCurrentVertex(h,d,0,0,c),this.addCurrentVertex(h,d,1,1,c,!0)),f&&
(this.addCurrentVertex(h,y,-1,-1,c,!0),this.addCurrentVertex(h,y,0,0,c)));if(w&&
e<o-1){const t=h.dist(f);if(t>2*u){const
e=h.add(f.sub(h)._mult(u/t)._round());this.updateDistance(h,e),this.addCurrentVe
rtex(e,y,0,0,c),h=e;}}}}addCurrentVertex(t,e,r,n,i,a=!1){const s=e.y*n-e.x,o=-
e.y-e.x*n;this.addHalfVertex(t,e.x+e.y*r,e.y-
e.x*r,a,!1,r,i),this.addHalfVertex(t,s,o,a,!0,-
n,i),this.distance>ml/2&&0===this.totalDistance&&
(this.distance=0,this.updateScaledDistance(),this.addCurrentVertex(t,e,r,n,i,a))
;addHalfVertex({x:t,y:e},r,n,i,a,s,o){const l=.5*(this.lineClips?
this.scaledDistance*(ml-
1):this.scaledDistance);this.layoutVertexArray.emplaceBack((t<<1)+(i?1:0),
(e<<1)+(a?1:0),Math.round(63*r)+128,Math.round(63*n)+128,1+(0===s?s<0?-1:1)|
(63&l)
<<2,l>>6),this.lineClips&&this.layoutVertexArray2.emplaceBack((this.scaledDistan
ce-this.lineClips.start)/(this.lineClips.end-
this.lineClips.start),this.lineClipsArray.length);const
u=o.vertexLength++;this.e1>=0&&this.e2>=0&&
(this.indexArray.emplaceBack(this.e1,this.e2,u),o.primitiveLength++),a?
this.e2=u:this.e1=u;}updateScaledDistance(){this.scaledDistance=this.lineClips?
this.lineClips.start+(this.lineClips.end-
this.lineClips.start)*this.distance/this.totalDistance:this.distance;}updateDist
ance(t,e){this.distance+=t.dist(e),this.updateScaledDistance();}let
xl,vl;$n("LineBucket",gl,{omit:["layers","patternFeatures"]});var bl={get
paint(){return vl=vl||new _i({{"line-opacity":new xi(X.paint_line["line-
opacity"]),{"line-color":new xi(X.paint_line["line-color"]),{"line-translate":new
gi(X.paint_line["line-translate"]),{"line-translate-anchor":new
gi(X.paint_line["line-translate-anchor"]),{"line-width":new
xi(X.paint_line["line-width"]),{"line-gap-width":new xi(X.paint_line["line-gap-
width"]),{"line-offset":new xi(X.paint_line["line-offset"]),{"line-blur":new
xi(X.paint_line["line-blur"]),{"line-dasharray":new bi(X.paint_line["line-
dasharray"]),{"line-pattern":new vi(X.paint_line["line-pattern"]),{"line-
gradient":new wi(X.paint_line["line-gradient"])}))},get layout(){return
xl=xl||new _i({{"line-cap":new gi(X.layout_line["line-cap"]),{"line-join":new
xi(X.layout_line["line-join"]),{"line-miter-limit":new gi(X.layout_line["line-
miter-limit"]),{"line-round-limit":new gi(X.layout_line["line-round-
limit"]),{"line-sort-key":new xi(X.layout_line["line-sort-key"])}))}};class wl
extends xi{possiblyEvaluate(t,e){return e=new li(Math.floor(e.zoom),
{now:e.now,fadeDuration:e.fadeDuration,zoomHistory:e.zoomHistory,transition:e.tr
ansition}),super.possiblyEvaluate(t,e)}evaluate(t,e,r,n){return e=g({},e,
{zoom:Math.floor(e.zoom)}),super.evaluate(t,e,r,n)}let _l;class Al extends
Si{constructor(t){super(t,bl),this.gradientVersion=0,_l||(_l=new
wl(bl.paint.properties["line-
width"].specification),_l.useIntegerZoom=!0);}_handleSpecialPaintPropertyUpdate(
t){if("line-gradient"===t){const
t=this.gradientExpression();this.stepInterpolant=!!function(t){return void
0!==t._styleExpression}(t)&t._styleExpression.expression instanceof
qe,this.gradientVersion=
(this.gradientVersion+1)%Number.MAX_SAFE_INTEGER;}}gradientExpression(){return
this._transitionablePaint._values["line-
gradient"].value.expression}recalculate(t,e)
{super.recalculate(t,e),this.paint._values["line-
floorwidth"]=_l.possiblyEvaluate(this._transitioningPaint._values["line-
width"].value,t);}createBucket(t){return new gl(t)}queryRadius(t){const
e=t,r=S1(bs("line-width",this,e),bs("line-gap-width",this,e)),n=bs("line-
offset",this,e);return r/2+Math.abs(n)+ws(this.paint.get("line-
translate"))}queryIntersectsFeature(t,e,r,n,i,s,o){const

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l=_s(t,this.paint.get("line-translate"),this.paint.get("line-translate-
anchor"),s.angle,o),u=o/2*S1(this.paint.get("line-
width").evaluate(e,r),this.paint.get("line-gap-
width").evaluate(e,r)),c=this.paint.get("line-offset").evaluate(e,r);return c&&
(n=function(t,e){const r=[];for(let n=0;n<t.length;n++){const i=t[n],s=
[];for(let t=0;t<i.length;t++){const r=i[t-1],n=i[t],o=i[t+1],l=0===t?new
a(0,0):n.sub(r)._unit()._perp(),u=t===i.length-1?new
a(0,0):o.sub(n)._unit()._perp(),c=l._add(u)._unit(),h=c.x*u.x+c.y*u.y;0!==h&&c._
mult(1/h),s.push(c._mult(e)._add(n));}r.push(s);}return r}
(n,c*o)),function(t,e,r){for(let n=0;n<e.length;n++){const
i=e[n];if(t.length>=3)for(let e=0;e<i.length;e++)if(xs(t,i[e]))return
!0;if(ps(t,i,r))return !0}return !1}(l,n,u)}isTileClipped(){return !0}}function
S1(t,e){return e>0?e+2*t:t}const
kl=Mi([ {name:"a_pos_offset",components:4,type:"Int16"},
{name:"a_data",components:4,type:"UInt16"},
{name:"a_pixeloffset",components:4,type:"Int16"}],4),Il=Mi([ {name:"a_projected_p
os",components:3,type:"Float32"}],4);Mi([ {name:"a_fade_opacity",components:1,typ
e:"UInt32"}],4);const zl=Mi([ {name:"a_placed",components:2,type:"UInt8"},
{name:"a_shift",components:2,type:"Float32"}]);Mi([ {type:"Int16",name:"anchorPoi
ntX"}, {type:"Int16",name:"anchorPointY"}, {type:"Int16",name:"x1"},
{type:"Int16",name:"y1"}, {type:"Int16",name:"x2"}, {type:"Int16",name:"y2"},
{type:"UInt32",name:"featureIndex"}, {type:"UInt16",name:"sourceLayerIndex"},
{type:"UInt16",name:"bucketIndex"}]);const
Ml=Mi([ {name:"a_pos",components:2,type:"Int16"},
{name:"a_anchor_pos",components:2,type:"Int16"},
{name:"a_extrude",components:2,type:"Int16"}],4),Cl=Mi([ {name:"a_pos",components
:2,type:"Float32"}, {name:"a_radius",components:1,type:"Float32"},
{name:"a_flags",components:2,type:"Int16"}],4);function Bl(t,e,r){return
t.sections.forEach((t=>{t.text=function(t,e,r){const n=e.layout.get("text-
transform").evaluate(r,{});return "uppercase"===n?
t=t.toLocaleUpperCase():"lowercase"===n&&
(t=t.toLocaleLowerCase()),oi.applyArabicShaping&&(t=oi.applyArabicShaping(t)),t}
(t.text,e,r);})),t}Mi([ {name:"triangle",components:3,type:"UInt16"}]),Mi([ {type:
"Int16",name:"anchorX"}, {type:"Int16",name:"anchorY"},
{type:"UInt16",name:"glyphStartIndex"}, {type:"UInt16",name:"numGlyphs"},
{type:"UInt32",name:"vertexStartIndex"}, {type:"UInt32",name:"lineStartIndex"},
{type:"UInt32",name:"lineLength"}, {type:"UInt16",name:"segment"},
{type:"UInt16",name:"lowerSize"}, {type:"UInt16",name:"upperSize"},
{type:"Float32",name:"lineOffsetX"}, {type:"Float32",name:"lineOffsetY"},
{type:"UInt8",name:"writingMode"}, {type:"UInt8",name:"placedOrientation"},
{type:"UInt8",name:"hidden"}, {type:"UInt32",name:"crossTileID"},
{type:"Int16",name:"associatedIconIndex"}]),Mi([ {type:"Int16",name:"anchorX"},
{type:"Int16",name:"anchorY"},
{type:"Int16",name:"rightJustifiedTextSymbolIndex"},
{type:"Int16",name:"centerJustifiedTextSymbolIndex"},
{type:"Int16",name:"leftJustifiedTextSymbolIndex"},
{type:"Int16",name:"verticalPlacedTextSymbolIndex"},
{type:"Int16",name:"placedIconSymbolIndex"},
{type:"Int16",name:"verticalPlacedIconSymbolIndex"}, {type:"UInt16",name:"key"},
{type:"UInt16",name:"textBoxStartIndex"}, {type:"UInt16",name:"textBoxEndIndex"},
{type:"UInt16",name:"verticalTextBoxStartIndex"},
{type:"UInt16",name:"verticalTextBoxEndIndex"},
{type:"UInt16",name:"iconBoxStartIndex"}, {type:"UInt16",name:"iconBoxEndIndex"},
{type:"UInt16",name:"verticalIconBoxStartIndex"},
{type:"UInt16",name:"verticalIconBoxEndIndex"},
{type:"UInt16",name:"featureIndex"},
{type:"UInt16",name:"numHorizontalGlyphVertices"},
{type:"UInt16",name:"numVerticalGlyphVertices"},
{type:"UInt16",name:"numIconVertices"},
{type:"UInt16",name:"numVerticalIconVertices"},
{type:"UInt16",name:"useRuntimeCollisionCircles"},
{type:"UInt32",name:"crossTileID"}, {type:"Float32",name:"textBoxScale"},

```

[illegible]

```

this.readFields(t,e,this.readVarint()+this.pos)},readFixed32:function(){var
t=Wl(this.buf,this.pos);return this.pos+=4,t},readSFixed32:function(){var
t=tu(this.buf,this.pos);return this.pos+=4,t},readFixed64:function(){var
t=Wl(this.buf,this.pos)+Wl(this.buf,this.pos+4)*Ll;return
this.pos+=8,t},readSFixed64:function(){var
t=Wl(this.buf,this.pos)+tu(this.buf,this.pos+4)*Ll;return
this.pos+=8,t},readFloat:function(){var t=Fl(this.buf,this.pos,!0,23,4);return
this.pos+=4,t},readDouble:function(){var t=Fl(this.buf,this.pos,!0,52,8);return
this.pos+=8,t},readVarint:function(t){var e,r,n=this.buf;return e=127&
(r=n[this.pos++]),r<128?e:(e|=(127&(r=n[this.pos++]))<<7,r<128?e:(e|=(127&
(r=n[this.pos++]))<<14,r<128?e:(e|=(127&(r=n[this.pos++]))<<21,r<128?
e:function(t,e,r){var n,i,a=r.buf;if(n=(112&(i=a[r.pos++]))>>4,i<128)return
Rl(t,n,e);if(n|=(127&(i=a[r.pos++]))<<3,i<128)return Rl(t,n,e);if(n|=(127&
(i=a[r.pos++]))<<10,i<128)return Rl(t,n,e);if(n|=(127&(i=a[r.pos++]))<<17,i<128)return Rl(t,n,e);if(n|=(127&(i=a[r.pos++]))<<24,i<128)return
Rl(t,n,e);if(n|=(1&(i=a[r.pos++]))<<31,i<128)return Rl(t,n,e);throw new
Error("Expected varint not more than 10 bytes")}(e|=(15&(r=n[this.pos]))
<<28,t,this))),readVarint64:function(){return
this.readVarint(!0)},readSVarint:function(){var t=this.readVarint();return
t%2==1?(t+1)/-2:t/2},readBoolean:function(){return
Boolean(this.readVarint())},readString:function(){var
t=this.readVarint()+this.pos,e=this.pos;return this.pos=t,t-e>=12&&0l?
function(t,e,r){return Ol.decode(t.subarray(e,r))}(this.buf,e,t):function(t,e,r)
{for(var n="",i=e;i<r;){var a,s,o,l=t[i],u=null,c=l>239?4:l>223?3:l>191?
2:1;if(i+c>r)break;l===c?l<128&&(u=l):2===c?128==(192&(a=t[i+1]))&&(u=(31&l)
<<6|63&a)<=127&&(u=null):3===c?(s=t[i+2],128==(192&(a=t[i+1]))&&128==(192&s)&&
((u=(15&l)<<12|(63&a)<<6|63&s)<=2047||u>=55296&&u<=57343)&&(u=null)):4===c&&
(s=t[i+2],o=t[i+3],128==(192&(a=t[i+1]))&&128==(192&s)&&128==(192&o)&&((u=(15&l)
<<18|(63&a)<<12|(63&s)<<6|63&o)<=65535||u>=1114112)&&(u=null)),null===u?
(u=65533,c=1):u>65535&&(u=
=65536,n+=String.fromCharCode(u>>>10&1023|55296),u=56320|1023&u),n+=String.fromC
harCode(u),i+=c;return n}(this.buf,e,t)},readBytes:function(){var
t=this.readVarint()+this.pos,e=this.buf.subarray(this.pos,t);return
this.pos=t,e},readPackedVarint:function(t,e){if(this.type!=$l.Bytes)return
t.push(this.readVarint(e));var r=Ul(this);for(t=t||
[];this.pos<r;)t.push(this.readVarint(e));return
t},readPackedSVarint:function(t){if(this.type!=$l.Bytes)return
t.push(this.readSVarint());var e=Ul(this);for(t=t||
[];this.pos<e;)t.push(this.readSVarint());return
t},readPackedBoolean:function(t){if(this.type!=$l.Bytes)return
t.push(this.readBoolean());var e=Ul(this);for(t=t||
[];this.pos<e;)t.push(this.readBoolean());return t},readPackedFloat:function(t)
{if(this.type!=$l.Bytes)return t.push(this.readFloat());var
e=Ul(this);for(t=t||[];this.pos<e;)t.push(this.readFloat());return
t},readPackedDouble:function(t){if(this.type!=$l.Bytes)return
t.push(this.readDouble());var e=Ul(this);for(t=t||
[];this.pos<e;)t.push(this.readDouble());return t},readPackedFixed32:function(t)
{if(this.type!=$l.Bytes)return t.push(this.readFixed32());var
e=Ul(this);for(t=t||[];this.pos<e;)t.push(this.readFixed32());return
t},readPackedSFixed32:function(t){if(this.type!=$l.Bytes)return
t.push(this.readSFixed32());var e=Ul(this);for(t=t||
[];this.pos<e;)t.push(this.readSFixed32());return
t},readPackedFixed64:function(t){if(this.type!=$l.Bytes)return
t.push(this.readFixed64());var e=Ul(this);for(t=t||
[];this.pos<e;)t.push(this.readFixed64());return
t},readPackedSFixed64:function(t){if(this.type!=$l.Bytes)return
t.push(this.readSFixed64());var e=Ul(this);for(t=t||
[];this.pos<e;)t.push(this.readSFixed64());return t},skip:function(t){var
e=7&t;if(e=== $l.Varint)for(;this.buf[this.pos++]>127;);else
if(e=== $l.Bytes)this.pos=this.readVarint()+this.pos;else
if(e=== $l.Fixed32)this.pos+=4;else {if(e!=$l.Fixed64)throw new
Error("Unimplemented type: "+e);this.pos+=8;}},writeTag:function(t,e)

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{this.writeVarint(t<<3|e);},realloc:function(t){for(var
e=this.length||16;e<this.pos+t;)e*=2;if(e!==this.length){var r=new
Uint8Array(e);r.set(this.buf),this.buf=r,this.length=e;}},finish:function()
{return
this.length=this.pos,this.pos=0,this.buf.subarray(0,this.length)},writeFixed32:f
unction(t)
{this.realloc(4),Ql(this.buf,t,this.pos),this.pos+=4;},writeSFixed32:function(t)
{this.realloc(4),Ql(this.buf,t,this.pos),this.pos+=4;},writeFixed64:function(t)
{this.realloc(8),Ql(this.buf,-1&t,this.pos),Ql(this.buf,Math.floor(t*Dl),this.po
s+4),this.pos+=8;},writeSFixed64:function(t)
{this.realloc(8),Ql(this.buf,-1&t,this.pos),Ql(this.buf,Math.floor(t*Dl),this.po
s+4),this.pos+=8;},writeVarint:function(t){(t+=t|0)>268435455|t<0?
function(t,e){var r,n;if(t>=0?(r=t%4294967296|0,n=t/4294967296|0):(n=~(-
t/4294967296),4294967295^(r=~(-t%4294967296)))?r=r+1|0:
(r=0,n=n+1|0)),t>=0x10000000000000000|t<-0x10000000000000000)throw new
Error("Given varint doesn't fit into 10 bytes");e.realloc(10),function(t,e,r)
{r.buf[r.pos++]=127&t|128,t>>=7,r.buf[r.pos++]=127&t|128,t>>=7,r.buf[r.pos++]=
127&t|128,t>>=7,r.buf[r.pos++]=127&t|128,r.buf[r.pos]=127&(t>>=7);}
(r,0,e),function(t,e){var r=(7&t)<<4;e.buf[e.pos++]|=r|((t>>=3)?128:0),t&&
(e.buf[e.pos++]=127&t|((t>>=7)?128:0),t&&(e.buf[e.pos++]=127&t|((t>>=7)?
128:0),t&&(e.buf[e.pos++]=127&t|((t>>=7)?128:0),t&&(e.buf[e.pos++]=127&t|
((t>>=7)?128:0),t&&(e.buf[e.pos++]=127&t)))));(n,e);}(t,this):
(this.realloc(4),this.buf[this.pos++]=127&t|(t>127?128:0),t<=127|
(this.buf[this.pos++]=127&(t>>=7)|(t>127?128:0),t<=127|
(this.buf[this.pos++]=127&(t>>=7)|(t>127?128:0),t<=127|
(this.buf[this.pos++]=t>>7&127)))));},writeSVarint:function(t)
{this.writeVarint(t<0?2*-t-1:2*t);},writeBoolean:function(t)
{this.writeVarint(Boolean(t));},writeString:function(t)
{t=String(t),this.realloc(4*t.length),this.pos++;var
e=this.pos;this.pos=function(t,e,r){for(var n,i,a=0;a<e.length;a++)
{if((n=e.charCodeAt(a))>55295&&n<57344){if(!i){n>56319||a+1===e.length?
(t[r++]=239,t[r++]=191,t[r++]=189):i=n;continue}if(n<56320)
{t[r++]=239,t[r++]=191,t[r++]=189,i=n;continue}n=i-55296<<10|n-
56320|65536,i=null;}else i&&(t[r++]=239,t[r++]=191,t[r++]=189,i=null);n<128?
t[r++]=n:(n<2048?t[r++]=n>>6|192:(n<65536?t[r++]=n>>12|224:
(t[r++]=n>>18|240,t[r++]=n>>12&63|128),t[r++]=n>>6&63|128),t[r++]=63&n|128);}ret
urn r}(this.buf,t,this.pos);var r=this.pos-e;r>=128&&q1(e,r,this),this.pos=e-
1,this.writeVarint(r),this.pos+=r;},writeFloat:function(t)
{this.realloc(4),Tl(this.buf,t,this.pos,!0,23,4),this.pos+=4;},writeDouble:functi
on(t)
{this.realloc(8),Tl(this.buf,t,this.pos,!0,52,8),this.pos+=8;},writeBytes:functi
on(t){var e=t.length;this.writeVarint(e),this.realloc(e);for(var
r=0;r<e;r++)this.buf[this.pos++]=t[r];},writeRawMessage:function(t,e)
{this.pos++;var r=this.pos;t(e,this);var n=this.pos-
r;n>=128&&q1(r,n,this),this.pos=r-
1,this.writeVarint(n),this.pos+=n;},writeMessage:function(t,e,r)
{this.writeTag(t,$l.Bytes),this.writeRawMessage(e,r);},writePackedVarint:functio
n(t,e){e.length&&this.writeMessage(t,jl,e);},writePackedSVarint:function(t,e)
{e.length&&this.writeMessage(t,Nl,e);},writePackedBoolean:function(t,e)
{e.length&&this.writeMessage(t,Gl,e);},writePackedFloat:function(t,e)
{e.length&&this.writeMessage(t,Zl,e);},writePackedDouble:function(t,e)
{e.length&&this.writeMessage(t,Kl,e);},writePackedFixed32:function(t,e)
{e.length&&this.writeMessage(t,Jl,e);},writePackedSFixed32:function(t,e)
{e.length&&this.writeMessage(t,Xl,e);},writePackedFixed64:function(t,e)
{e.length&&this.writeMessage(t,Yl,e);},writePackedSFixed64:function(t,e)
{e.length&&this.writeMessage(t,Hl,e);},writeBytesField:function(t,e)
{this.writeTag(t,$l.Bytes),this.writeBytes(e);},writeFixed32Field:function(t,e)
{this.writeTag(t,$l.Fixed32),this.writeFixed32(e);},writeSFixed32Field:function(
t,e)
{this.writeTag(t,$l.Fixed32),this.writeSFixed32(e);},writeFixed64Field:function(
t,e)
{this.writeTag(t,$l.Fixed64),this.writeFixed64(e);},writeSFixed64Field:function(

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t,e)
{this.writeTag(t,$l.Fixed64),this.writeSFixed64(e);},writeVarintField:function(t
,e)
{this.writeTag(t,$l.Varint),this.writeVarint(e);},writeSVarintField:function(t,e
)
{this.writeTag(t,$l.Varint),this.writeSVarint(e);},writeStringField:function(t,e
){this.writeTag(t,$l.Bytes),this.writeString(e);},writeFloatField:function(t,e)
{this.writeTag(t,$l.Fixed32),this.writeFloat(e);},writeDoubleField:function(t,e)
{this.writeTag(t,$l.Fixed64),this.writeDouble(e);},writeBooleanField:function(t,
e){this.writeVarintField(t,Boolean(e));};var eu=r(E1);const ru=3;function
nu(t,e,r){l===t&&r.readMessage(iu,e);}function iu(t,e,r){if(3===t)
{const{id:t,bitmap:n,width:i,height:a,left:s,top:o,advance:l}=r.readMessage(au,
{});e.push({id:t,bitmap:new Rs({width:i+2*ru,height:a+2*ru},n),metrics:
{width:i,height:a,left:s,top:o,advance:l}});}}function au(t,e,r){l===t?
e.id=r.readVarint():2===t?e.bitmap=r.readBytes():3===t?
e.width=r.readVarint():4===t?e.height=r.readVarint():5===t?
e.left=r.readSVarint():6===t?e.top=r.readSVarint():7===t&&
(e.advance=r.readVarint());}const su=ru;function ou(t){let e=0,r=0;for(const n
of t)e+=n.w*n.h,r=Math.max(r,n.w);t.sort((t,e)=>e.h-t.h);const n=
[{x:0,y:0,w:Math.max(Math.ceil(Math.sqrt(e/.95)),r),h:1/0}];let
i=0,a=0;for(const e of t)for(let t=n.length-1;t>=0;t--){const r=n[t];if(!
(e.w>r.w||e.h>r.h))
{if(e.x=r.x,e.y=r.y,a=Math.max(a,e.y+e.h),i=Math.max(i,e.x+e.w),e.w===r.w&&e.h==
=r.h){const e=n.pop();t<n.length&&(n[t]=e);}else e.h===r.h?(r.x+=e.w,r.w-
=e.w):e.w===r.w?(r.y+=e.h,r.h-=e.h):(n.push({x:r.x+e.w,y:r.y,w:r.w-
e.w,h:e.h}),r.y+=e.h,r.h-=e.h);break}}return {w:i,h:a,fill:e/(i*a)||0}}const
lu=1;class uu{constructor(t,
{pixelRatio:e,version:r,stretchX:n,stretchY:i,content:a})
{this.paddedRect=t,this.pixelRatio=e,this.stretchX=n,this.stretchY=i,this.conten
t=a,this.version=r;}get tl(){return
[this.paddedRect.x+lu,this.paddedRect.y+lu]}get br(){return
[this.paddedRect.x+this.paddedRect.w-lu,this.paddedRect.y+this.paddedRect.h-
lu]}get tlbr(){return this.tl.concat(this.br)}get displaySize(){return
[(this.paddedRect.w-2*lu)/this.pixelRatio,(this.paddedRect.h-
2*lu)/this.pixelRatio]}class cu{constructor(t,e){const r={},n=
{};this.haveRenderCallbacks=[];const i=
[];this.addImages(t,r,i),this.addImages(e,n,i);const{w:a,h:s}=ou(i),o=new
qs({width:a||1,height:s||1});for(const e in t){const
n=t[e],i=r[e].paddedRect;qs.copy(n.data,o,{x:0,y:0},
{x:i.x+lu,y:i.y+lu},n.data);}for(const t in e){const
r=e[t],i=n[t].paddedRect,a=i.x+lu,s=i.y+lu,l=r.data.width,u=r.data.height;qs.cop
y(r.data,o,{x:0,y:0},{x:a,y:s},r.data),qs.copy(r.data,o,{x:0,y:u-1},{x:a,y:s-1},
{width:l,height:1}),qs.copy(r.data,o,{x:0,y:0},{x:a,y:s+u},
{width:l,height:1}),qs.copy(r.data,o,{x:l-1,y:0},{x:a-1,y:s},
{width:1,height:u}),qs.copy(r.data,o,{x:0,y:0},{x:a+1,y:s},
{width:1,height:u});}this.image=o,this.iconPositions=r,this.patternPositions=n;}
addImages(t,e,r){for(const n in t){const i=t[n],a=
{x:0,y:0,w:i.data.width+2*lu,h:i.data.height+2*lu};r.push(a),e[n]=new
uu(a,i),i.hasRenderCallback&&this.haveRenderCallbacks.push(n);}}patchUpdatedImag
es(t,e){t.dispatchRenderCallbacks(this.haveRenderCallbacks);for(const r in
t.updatedImages)this.patchUpdatedImage(this.iconPositions[r],t.getImage(r),e),th
is.patchUpdatedImage(this.patternPositions[r],t.getImage(r),e);}patchUpdatedImag
e(t,e,r)
{if(!t||!e)return;if(t.version===e.version)return;t.version=e.version;const[n,i]
=t.tl;r.update(e.data,void 0,{x:n,y:i});}var
hu;$n("ImagePosition",uu),$n("ImageAtlas",cu),t.ai=void 0,(hu=t.ai||(t.ai={}))
[hu.none=0]="none",hu[hu.horizontal=1]="horizontal",hu[hu.vertical=2]="vertical"
,hu[hu.horizontalOnly=3]="horizontalOnly";const pu=-17;class fu{constructor()
{this.scale=1,this.fontStack="",this.imageName=null;}static forText(t,e){const
r=new fu;return r.scale=t||1,r.fontStack=e,r}static forImage(t){const e=new
fu;return e.imageName=t,e}}class du{constructor()
{this.text="",this.sectionIndex=[],this.sections=

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[],this.imageSectionID=null;};static fromFeature(t,e){const r=new du;for(let
n=0;n<t.sections.length;n++){const i=t.sections[n];i.image?
r.addImageSection(i):r.addTextSection(i,e);}return r}length(){return
this.text.length}getSection(t){return
this.sections[this.sectionIndex[t]]}getSectionIndex(t){return
this.sectionIndex[t]}getCharCode(t){return
this.text.charCodeAt(t)}verticalizePunctuation(){this.text=function(t){let
e="";for(let r=0;r<t.length;r++){const
n=t.charCodeAt(r+1)||null,i=t.charCodeAt(r-
1)||null;e+=n&&Kn(n)&&!Pl[t[r+1]]||i&&Kn(i)&&!Pl[t[r-1]]||!Pl[t[r]]?
t[r]:Pl[t[r]];}return e}(this.text);}trim(){let t=0;for(let
e=0;e<this.text.length&&mu[this.text.charCodeAt(e)];e++)t++;let
e=this.text.length;for(let r=this.text.length-
1;r>=0&&r>=t&&mu[this.text.charCodeAt(r)];r--)e--
;this.text=this.text.substring(t,e);this.sectionIndex=this.sectionIndex.slice(t,
e);}substring(t,e){const r=new du;return
r.text=this.text.substring(t,e),r.sectionIndex=this.sectionIndex.slice(t,e),r.se
ctions=this.sections,r}toString(){return this.text}getMaxScale(){return
this.sectionIndex.reduce((t,e)=>Math.max(t,this.sections[e].scale)),0)}addTextS
ection(t,e)
{this.text+=t.text,this.sections.push(fu.forText(t.scale,t.fontStack||e));const
r=this.sections.length-1;for(let
e=0;e<t.text.length;++e)this.sectionIndex.push(r);}addImageSection(t){const
e=t.image?t.image.name:"";if(0===e.length)return void A("Can't add
FormattedSection with an empty image.");const
r=this.getNextImageSectionCharCode();r?
(this.text+=String.fromCharCode(r),this.sections.push(fu.forImage(e)),this.secti
onIndex.push(this.sections.length-1)):A("Reached maximum number of images
6401");}getNextImageSectionCharCode(){return this.imageSectionID?
this.imageSectionID>=63743?null:++this.imageSectionID:
(this.imageSectionID=57344,this.imageSectionID)}}function
yu(e,r,n,i,a,s,o,l,u,c,h,p,f,d,y,m){const g=du.fromFeature(e,a);let
x;p===t.ai.vertical&&g.verticalizePunctuation();const{processBidirectionalText:v
,processStyledBidirectionalText:b}=oi;if(v&&l===g.sections.length){x=[];const
t=v(g.toString(),Au(g,c,s,r,i,d,y));for(const e of t){const t=new
du;t.text=e,t.sections=g.sections;for(let
r=0;r<e.length;r++)t.sectionIndex.push(0);x.push(t);}else if(b){x=[];const
t=b(g.text,g.sectionIndex,Au(g,c,s,r,i,d,y));for(const e of t){const t=new
du;t.text=e[0],t.sectionIndex=e[1],t.sections=g.sections,x.push(t);}else
x=function(t,e){const r=[],n=t.text;let i=0;for(const n of
e)r.push(t.substring(i,n)),i=n;return
i<n.length&&r.push(t.substring(i,n.length)),r}(g,Au(g,c,s,r,i,d,y));const w=
[],_=
{positionedLines:w,text:g.toString(),top:h[1],bottom:h[1],left:h[0],right:h[0],w
ritingMode:p,iconsInText:!1,verticalizable:!1};return
function(e,r,n,i,a,s,o,l,u,c,h,p){let f=0,d=pu,y=0,m=0;const g="right"===l?
l:"left"===l?0:.5;let x=0;for(const o of a){o.trim();const a=o.getMaxScale(),l=
(a-1)*Vl,b={positionedGlyphs:[],lineOffset:0};e.positionedLines[x]=b;const
w=b.positionedGlyphs;let _=0;if(!o.length()){d+=s,++x;continue}for(let
s=0;s<o.length();s++){const
y=o.getSection(s),m=o.getSectionIndex(s),g=o.getCharCode(s);let
x=0,b=null,A=null,S=null,k=Vl;const I=!(u===t.ai.horizontal||!h&&!Zn(g)||h&&
(mu[g]||(v=g,Rn.Arabic(v)||Rn["Arabic Supplement"])(v)||Rn["Arabic Extended-A"]
(v)||Rn["Arabic Presentation Forms-A"])(v)||Rn["Arabic Presentation Forms-B"]
(v)))));if(y.imageName){const
t=i[y.imageName];if(!t)continue;S=y.imageName,e.iconsInText=e.iconsInText||!0,A=
t.paddedRect;const r=t.displaySize;y.scale=y.scale*Vl/p,b=
{width:r[0],height:r[1],left:lu,top:-su,advance:I?r[1]:r[0]},x=l+(Vl-
r[1]*y.scale),k=b.advance;const n=I?r[0]*y.scale-Vl*a:r[1]*y.scale-
Vl*a;n>0&&n>_&&(_=n);}else {const
t=n[y.fontStack],e=t&&t[g];if(e&&e.rect)A=e.rect,b=e.metrics;else {const
t=r[y.fontStack],e=t&&t[g];if(!e)continue;b=e.metrics;}x=(a-y.scale)*Vl;}I?

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(e.verticalizable=!0,w.push({glyph:g,imageName:S,x:f,y:d+x,vertical:I,scale:y.scale,fontStack:y.fontStack,sectionIndex:m,metrics:b,rect:A}),f+=k*y.scale+c):
(w.push({glyph:g,imageName:S,x:f,y:d+x,vertical:I,scale:y.scale,fontStack:y.fontStack,sectionIndex:m,metrics:b,rect:A}),f+=b.advance*y.scale+c);}0!==(w.length&&
(y=Math.max(f-c,y),ku(w,0,w.length-1,g,_)),f=0;const
A=s*a+_b.lineOffset=Math.max(_,1),d+=A,m=Math.max(A,m),++x;}var v;const b=d-pu,
{horizontalAlign:w,verticalAlign:_}=Su(o);((function(t,e,r,n,i,a,s,o,l){const u=
(e-r)*i;let c=0;c=a!==(s?-o*n-pu:(-n*1+.5)*s;for(const e of t)for(const t of
e.positionedGlyphs)t.x+=u,t.y+=c;}))
(e.positionedLines,g,w,_,y,m,s,b,a.length),e.top+=-
_*b,e.bottom=e.top+b,e.left+=-w*y,e.right=e.left+y;}
(_,r,n,i,x,o,l,u,p,c,f,m),!function(t){for(const e of
t)if(0!==(e.positionedGlyphs.length)return !1;return !0}(w)&&_)const mu=
{9:!0,10:!0,11:!0,12:!0,13:!0,32:!0},gu=
{10:!0,32:!0,38:!0,40:!0,41:!0,43:!0,45:!0,47:!0,173:!0,183:!0,8203:!0,8208:!0,8
211:!0,8231:!0};function xu(t,e,r,n,i,a){if(e.imageName){const
t=n[e.imageName];return t?t.displaySize[0]*e.scale*v1/a+i:0}{const
n=r[e.fontStack],a=n&&n[t];return a?a.metrics.advance*e.scale+i:0}}function
vu(t,e,r,n){const i=Math.pow(t-e,2);return n?t<e?
i/2*2*i:i+Math.abs(r)*r}function bu(t,e,r){let n=0;return 10===t&&(n-=1e4),r&&
(n+=150),40!==(t&&65288!==(t|(n+=50),41!==(e&&65289!==(e|(n+=50),n)}function
wu(t,e,r,n,i,a){let s=null,o=vu(e,r,i,a);for(const t of n){const n=vu(e-
t.x,r,i,a)+t.badness;n<=o&&(s=t,o=n);}return
{index:t,x:e,priorBreak:s,badness:o}}function _u(t){return t?
_u(t.priorBreak).concat(t.index):[]}function Au(t,e,r,n,i,a,s)
{if("point"!==a)return [];if(!t)return [];const o=[],l=function(t,e,r,n,i,a){let
s=0;for(let r=0;r<t.length();r++){const
o=t.getSection(r);s+=xu(t.getCharCode(r),o,n,i,e,a);}return
s/Math.max(1,Math.ceil(s/r))}(t,e,r,n,i,s),u=t.text.indexOf("")>=0;let
c=0;for(let r=0;r<t.length();r++){const
a=t.getSection(r),p=t.getCharCode(r);if(mu[p]||
(c+=xu(p,a,n,i,e,s)),r<t.length()-1){const e=!((h=p)<11904||!(Rn["Bopomofo
Extended"])(h)||Rn["Bopomofo"])(h)||Rn["CJK Compatibility Forms"])(h)||Rn["CJK
Compatibility Ideographs"])(h)||Rn["CJK Compatibility"])(h)||Rn["CJK Radicals
Supplement"])(h)||Rn["CJK Strokes"])(h)||Rn["CJK Symbols and Punctuation"]
)(h)||Rn["CJK Unified Ideographs Extension A"])(h)||Rn["CJK Unified Ideographs"]
)(h)||Rn["Enclosed CJK Letters and Months"])(h)||Rn["Halfwidth and Fullwidth
Forms"])(h)||Rn["Hiragana"])(h)||Rn["Ideographic Description Characters"]
)(h)||Rn["Kangxi Radicals"])(h)||Rn["Katakana Phonetic Extensions"]
)(h)||Rn["Katakana"])(h)||Rn["Vertical Forms"])(h)||Rn["Yi Radicals"])(h)||Rn["Yi
Syllables"])(h))};
(gu[p]||e||a.imageName)&&o.push(wu(r+1,c,l,o,bu(p,t.getCharCode(r+1),e&&u,!1));
}var h;return _u(wu(t.length(),c,l,o,!0))}function Su(t){let
e=.5,r=.5;switch(t){case"right":case"top-right":case"bottom-
right":e=1;break;case"left":case"top-left":case"bottom-left":e=0;}switch(t)
{case"bottom":case"bottom-right":case"bottom-left":r=1;break;case"top":case"top-
right":case"top-left":r=0;}return {horizontalAlign:e,verticalAlign:r}}function
ku(t,e,r,n,i){if(!n&&!i)return;const a=t[r],s=
(t[r].x+a.metrics.advance*a.scale)*n;for(let n=e;n<=r;n++)t[n].x-
=s,t[n].y+=i;}function Iu(t,e,r)
{const{horizontalAlign:n,verticalAlign:i}=Su(r),a=e[0]-
t.displaySize[0]*n,s=e[1]-t.displaySize[1]*i;return
{image:t,top:s,bottom:s+t.displaySize[1],left:a,right:a+t.displaySize[0]}}functi
on zu(t,e,r,n,i,a){const s=t.image;let o;if(s.content){const
t=s.content,e=s.pixelRatio|1;o=[t[0]/e,t[1]/e,s.displaySize[0]-
t[2]/e,s.displaySize[1]-t[3]/e];}const l=e.left*a,u=e.right*a;let
c,h,p,f;"width"===r||"both"===r?(f=i[0]+1-n[3],h=i[0]+u+n[1]):(f=i[0]+(1+u-
s.displaySize[0])/2,h=f+s.displaySize[0]);const d=e.top*a,y=e.bottom*a;return
"height"===r||"both"===r?(c=i[1]+d-n[0],p=i[1]+y+n[2]):(c=i[1]+(d+y-
s.displaySize[1])/2,p=c+s.displaySize[1]),
{image:s,top:c,right:h,bottom:p,left:f,collisionPadding:o}}const
Mu=255,Cu=128,Bu=Mu*Cu;function Pu(t,e)

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{const{expression:r}=e;if("constant"===r.kind)return
{kind:"constant",layoutSize:r.evaluate(new li(t+1))};if("source"===r.kind)return
{kind:"source"};const{zoomStops:e,interpolationType:n}=r;let
i=0;for(;i<e.length&&e[i]<=t;)i++;i=Math.max(0,i-1);let
a=i;for(;a<e.length&&e[a]<t+1;)a++;a=Math.min(e.length-1,a);const
s=e[i],o=e[a];return "composite"===r.kind?
{kind:"composite",minZoom:s,maxZoom:o,interpolationType:n}:
{kind:"camera",minZoom:s,maxZoom:o,minSize:r.evaluate(new
li(s)),maxSize:r.evaluate(new li(o)),interpolationType:n}}function Vu(t,e,r)
{let n="never";const i=t.get(e);return i?n=i:t.get(r)&&(n="always"),n}const
Eu=Ro.VectorTileFeature.types,Fu=
[ {name:"a_fade_opacity",components:1,type:"Uint8",offset:0} ];function
Tu(t,e,r,n,i,a,s,o,l,u,c,h,p){const f=o?Math.min(Bu,Math.round(o[0])):0,d=o?
Math.min(Bu,Math.round(o[1])):0;t.emplaceBack(e,r,Math.round(32*n),Math.round(32
*i),a,s,(f<1)+(l?1:0),d,16*u,16*c,256*h,256*p);}function $u(t,e,r)
{t.emplaceBack(e.x,e.y,r),t.emplaceBack(e.x,e.y,r),t.emplaceBack(e.x,e.y,r),t.em
placeBack(e.x,e.y,r);}function Lu(t){for(const e of
t.sections)if(Xn(e.text))return !0;return !1}class Du{constructor(t)
{this.layoutVertexArray=new xa,this.indexArray=new
Aa,this.programConfigurations=t,this.segments=new
za,this.dynamicLayoutVertexArray=new va,this.opacityVertexArray=new
ba,this.hasVisibleVertices=!1,this.placedSymbolArray=new ra;}isEmpty(){return
0===this.layoutVertexArray.length&&0===this.indexArray.length&&0===this.dynamicL
ayoutVertexArray.length&&0===this.opacityVertexArray.length}upload(t,e,r,n)
{this.isEmpty()||(r&&
(this.layoutVertexBuffer=t.createVertexBuffer(this.layoutVertexArray,kl.members)
,this.indexBuffer=t.createIndexBuffer(this.indexArray,e),this.dynamicLayoutVerte
xBuffer=t.createVertexBuffer(this.dynamicLayoutVertexArray,il.members,!0),this.o
pacityVertexBuffer=t.createVertexBuffer(this.opacityVertexArray,Fu,!0),this.opac
ityVertexBuffer.itemSize=1),
(r|n)&&this.programConfigurations.upload(t));}destroy()
{this.layoutVertexBuffer&&
(this.layoutVertexBuffer.destroy(),this.indexBuffer.destroy(),this.programConfig
urations.destroy(),this.segments.destroy(),this.dynamicLayoutVertexBuffer.destro
y(),this.opacityVertexBuffer.destroy());}$n("SymbolBuffers",Du);class
Ou{constructor(t,e,r){this.layoutVertexArray=new
t,this.layoutAttributes=e,this.indexArray=new r,this.segments=new
za,this.collisionVertexArray=new _a;}upload(t)
{this.layoutVertexBuffer=t.createVertexBuffer(this.layoutVertexArray,this.layout
Attributes),this.indexBuffer=t.createIndexBuffer(this.indexArray),this.collision
VertexBuffer=t.createVertexBuffer(this.collisionVertexArray,zl.members,!0);}dest
roy(){this.layoutVertexBuffer&&
(this.layoutVertexBuffer.destroy(),this.indexBuffer.destroy(),this.segments.dest
roy(),this.collisionVertexBuffer.destroy());}$n("CollisionBuffers",Ou);class
Uu{constructor(e)
{this.collisionBoxArray=e.collisionBoxArray,this.zoom=e.zoom,this.overscaling=e.
overscaling,this.layers=e.layers,this.layerIds=this.layers.map((t=>t.id)),this.i
ndex=e.index,this.pixelRatio=e.pixelRatio,this.sourceLayerIndex=e.sourceLayerInd
ex,this.hasPattern=!1,this.hasRTLText=!1,this.sortKeyRanges=
[],this.collisionCircleArray=
[],this.placementInvProjMatrix=Ms([]),this.placementViewportMatrix=Ms([]);const
r=this.layers[0]._unevaluatedLayout._values;this.textSizeData=Pu(this.zoom,r["te
xt-size"]),this.iconSizeData=Pu(this.zoom,r["icon-size"]);const
n=this.layers[0].layout,i=n.get("symbol-sort-key"),a=n.get("symbol-z-
order");this.canOverlap="never"!==Vu(n,"text-overlap","text-allow-
overlap")||"never"!==Vu(n,"icon-overlap","icon-allow-overlap")||n.get("text-
ignore-placement")||n.get("icon-ignore-
placement"),this.sortFeaturesByKey="viewport-
y"!==a&&!i.isConstant(),this.sortFeaturesByY=("viewport-
y"===a||"auto"===a&&!this.sortFeaturesByKey)&&this.canOverlap,"point"===n.get("s
ymbol-placement")&&(this.writingModes=n.get("text-writing-
mode")).map((e=>t.ai[e])),this.stateDependentLayerIds=this.layers.filter((t=>t.i

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sStateDependent()).map((t=>t.id)),this.sourceID=e.sourceID;}createArrays()
{this.text=new Du(new Qa(this.layers,this.zoom,
(t=>/^text/.test(t))),this.icon=new Du(new Qa(this.layers,this.zoom,
(t=>/^icon/.test(t))),this.glyphOffsetArray=new aa,this.lineVertexArray=new
sa,this.symbolInstances=new ia,this.textAnchorOffsets=new
la;}calculateGlyphDependencies(t,e,r,n,i){for(let
a=0;a<t.length;a++)if(e[t.charCodeAt(a)]!=0,(r|n)&&i){const
r=Pl[t.charAt(a)];r&&(e[r.charCodeAt(0)]!=0);}populate(e,r,n){const
i=this.layers[0],a=i.layout,s=a.get("text-font"),o=a.get("text-
field"),l=a.get("icon-image"),u=("constant"!=o.value.kind||o.value.value
instanceof Qt&&o.value.value.isEmpty()||o.value.value.toString().length>0)&&
("constant"!=s.value.kind||s.value.value.length>0),c="constant"!=l.value.kind|
|!l.value.value|Object.keys(l.parameters).length>0,h=a.get("symbol-sort-
key");if(this.features=[],!u&&!c)return;const
p=r.iconDependencies,f=r.glyphDependencies,d=r.availableImages,y=new
li(this.zoom);for(const{feature:r,id:o,index:l,sourceLayerIndex:m}of e){const
e=i._featureFilter.needGeometry,g=ss(r,e);if(!i._featureFilter.filter(y,g,n))con
tinue;let x,v;if(e||(g.geometry=as(r)),u){const
t=i.getValueAndResolveTokens("text-field",g,n,d),e=Qt.factory(t);Lu(e)&&
(this.hasRTLText!=0),
(!this.hasRTLText||"unavailable"===ai()||this.hasRTLText&&oi.isParsed())&&
(x=Bl(e,i,g));}if(c){const t=i.getValueAndResolveTokens("icon-image",g,n,d);v=t
instanceof ne?t.ne.fromString(t);}if(!x&&!v)continue;const
b=this.sortFeaturesByKey?h.evaluate(g,{},n):void
0;if(this.features.push({id:o,text:x,icon:v,index:l,sourceLayerIndex:m,geometry:
g.geometry,properties:r.properties,type:Eu[r.type],sortKey:b}),v&&
(p[v.name]!=0),x){const e=s.evaluate(g,
{ },n).join(", "),r="viewport"!=a.get("text-rotation-
alignment")&&"point"!=a.get("symbol-
placement");this.allowVerticalPlacement=this.writingModes&&this.writingModes.ind
exOf(t.ai.vertical)>=0;for(const t of
x.sections)if(t.image)p[t.image.name]!=0;else {const
n=qn(x.toString()),i=t.fontStack||e,a=f[i]=f[i]||
{};this.calculateGlyphDependencies(t.text,a,r,this.allowVerticalPlacement,n);}
"line"===a.get("symbol-placement")&&(this.features=function(t){const e={},r=
{ },n=[];let i=0;function a(e){n.push(t[e]),i++;}function s(t,e,i){const
a=r[t];return delete
r[t],r[e]=a,n[a].geometry[0].pop(),n[a].geometry[0]=n[a].geometry[0].concat(i[0]
),a}function o(t,r,i){const a=e[r];return delete
e[r],e[t]=a,n[a].geometry[0].shift(),n[a].geometry[0]=i[0].concat(n[a].geometry[
0]),a}function l(t,e,r){const n=r?e[0][e[0].length-1]:e[0][0];return
`$${t}.$${n.x}.$${n.y}`}for(let u=0;u<t.length;u++){const
c=t[u],h=c.geometry,p=c.text?c.text.toString():null;if(!p){a(u);continue}const
f=l(p,h),d=l(p,h,!0);if(f in r&&d in e&&r[f]!=e[d]){const
t=o(f,d,h),i=s(f,d,n[t].geometry);delete e[f],delete
r[d],r[l(p,n[i].geometry,!0)]=i,n[t].geometry=null;}else f in r?s(f,d,h):d in e?
o(f,d,h):(a(u),e[f]=i-1,r[d]=i-1);}return n.filter((t=>t.geometry))}
(this.features)),this.sortFeaturesByKey&&this.features.sort((t,e)=>t.sortKey-
e.sortKey));}update(t,e,r){this.stateDependentLayers.length&&
(this.text.programConfigurations.updatePaintArrays(t,e,this.layers,r),this.icon.
programConfigurations.updatePaintArrays(t,e,this.layers,r));}isEmpty(){return
0===this.symbolInstances.length&&!this.hasRTLText}uploadPending(){return
!this.uploaded||this.text.programConfigurations.needsUpload||this.icon.programCo
nfigurations.needsUpload}upload(t){!this.uploaded&&this.hasDebugData()&&
(this.text.CollisionBox.upload(t),this.icon.CollisionBox.upload(t)),this.text.uploa
d(t,this.sortFeaturesByY,!this.uploaded,this.text.programConfigurations.needsUp
load),this.icon.upload(t,this.sortFeaturesByY,!this.uploaded,this.icon.programCo
nfigurations.needsUpload),this.uploaded=!0;}destroyDebugData()
{this.text.CollisionBox.destroy(),this.icon.CollisionBox.destroy();}destroy()
{this.text.destroy(),this.icon.destroy(),this.hasDebugData()&&this.destroyDebugD
ata();}addToLineVertexArray(t,e){const r=this.lineVertexArray.length;if(void
0!=t.segment){let r=t.dist(e[t.segment+1]),n=t.dist(e[t.segment]);const i=

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{};for(let n=t.segment+1;n<e.length;n++)i[n]=
{x:e[n].x,y:e[n].y,tileUnitDistanceFromAnchor:r},n<e.length-1&&
(r+=e[n+1].dist(e[n]));for(let r=t.segment||0;r>=0;r--)i[r]=
{x:e[r].x,y:e[r].y,tileUnitDistanceFromAnchor:n},r>0&&(n+=e[r-
1].dist(e[r]));for(let t=0;t<e.length;t++){const
e=i[t];this.lineVertexArray.emplaceBack(e.x,e.y,e.tileUnitDistanceFromAnchor);}}
return {lineStartIndex:r,lineLength:this.lineVertexArray.length-
r}}addSymbols(e,r,n,i,a,s,o,l,u,c,h,p){const
f=e.indexArray,d=e.layoutVertexArray,y=e.segments.prepareSegment(4*r.length,d,f,
this.canOverlap?s.sortKey:void
0),m=this.glyphOffsetArray.length,g=y.vertexLength,x=this.allowVerticalPlacement
&&o==t.ai.vertical?Math.PI/2:0,v=s.text&&s.text.sections;for(let
t=0;t<r.length;t++){const{tl:i,tr:a,bl:o,br:u,tex:c,pixelOffsetTL:h,pixelOffsetBR:m,minFontScaleX:g
,minFontScaleY:b,glyphOffset:w,isSDF:_,sectionIndex:A}=r[t],S=y.vertexLength,k=w
[1];Tu(d,l.x,l.y,i.x,k+i.y,c.x,c.y,n,_,h.x,h.y,g,b),Tu(d,l.x,l.y,a.x,k+a.y,c.x+c
.w,c.y,n,_,m.x,h.y,g,b),Tu(d,l.x,l.y,o.x,k+o.y,c.x,c.y+c.h,n,_,h.x,m.y,g,b),Tu(d
,l.x,l.y,u.x,k+u.y,c.x+c.w,c.y+c.h,n,_,m.x,m.y,g,b),$(u(e.dynamicLayoutVertexArra
y,l,x),f.emplaceBack(S,S+1,S+2),f.emplaceBack(S+1,S+2,S+3),y.vertexLength+=4,y.p
rimitiveLength+=2,this.glyphOffsetArray.emplaceBack(w[0]),t!=r.length-
1&&A==r[t+1].sectionIndex|e.programConfigurations.populatePaintArrays(d.length
,s,s.index,
{}),p,v&&v[A]);}e.placedSymbolArray.emplaceBack(l.x,l.y,m,this.glyphOffsetArray.l
ength-m,g,u,c,l.segment,n?n[0]:0,n?
n[1]:0,i[0],i[1],o,0,!1,0,h);}_addCollisionDebugVertex(t,e,r,n,i,a){return
e.emplaceBack(0,0),t.emplaceBack(r.x,r.y,n,i,Math.round(a.x),Math.round(a.y))}ad
dCollisionDebugVertices(t,e,r,n,i,s,o){const
l=i.segments.prepareSegment(4,i.layoutVertexArray,i.indexArray),u=l.vertexLength
,c=i.layoutVertexArray,h=i.collisionVertexArray,p=o.anchorX,f=o.anchorY;this._ad
dCollisionDebugVertex(c,h,s,p,f,new
a(t,e)),this._addCollisionDebugVertex(c,h,s,p,f,new
a(r,e)),this._addCollisionDebugVertex(c,h,s,p,f,new
a(r,n)),this._addCollisionDebugVertex(c,h,s,p,f,new
a(t,n)),l.vertexLength+=4;const
d=i.indexArray;d.emplaceBack(u,u+1),d.emplaceBack(u+1,u+2),d.emplaceBack(u+2,u+3
),d.emplaceBack(u+3,u),l.primitiveLength+=4;}addDebugCollisionBoxes(t,e,r,n)
{for(let i=t;i<e;i++){const
t=this.collisionBoxArray.get(i);this.addCollisionDebugVertices(t.x1,t.y1,t.x2,t.
y2,n?
this.textCollisionBox:this.iconCollisionBox,t.anchorPoint,r);}}generateCollision
DebugBuffers()
{this.hasDebugData()&&this.destroyDebugData(),this.textCollisionBox=new
Ou(wa,Ml.members,Sa),this.iconCollisionBox=new Ou(wa,Ml.members,Sa);for(let
t=0;t<this.symbolInstances.length;t++){const
e=this.symbolInstances.get(t);this.addDebugCollisionBoxes(e.textBoxStartIndex,e.
textBoxEndIndex,e,!0),this.addDebugCollisionBoxes(e.verticalTextBoxStartIndex,e.
verticalTextBoxEndIndex,e,!0),this.addDebugCollisionBoxes(e.iconBoxStartIndex,e.
iconBoxEndIndex,e,!1),this.addDebugCollisionBoxes(e.verticalIconBoxStartIndex,e.
verticalIconBoxEndIndex,e,!1);}_deserializeCollisionBoxesForSymbol(t,e,r,n,i,a,
s,o,l){const u={};for(let n=e;n<r;n++){const e=t.get(n);u.textBox=
{x1:e.x1,y1:e.y1,x2:e.x2,y2:e.y2,anchorPointX:e.anchorPointX,anchorPointY:e.anch
orPointY},u.textFeatureIndex=e.featureIndex;break}for(let e=n;e<i;e++){const
r=t.get(e);u.verticalTextBox=
{x1:r.x1,y1:r.y1,x2:r.x2,y2:r.y2,anchorPointX:r.anchorPointX,anchorPointY:r.anch
orPointY},u.verticalTextFeatureIndex=r.featureIndex;break}for(let e=a;e<s;e++)
{const r=t.get(e);u.iconBox=
{x1:r.x1,y1:r.y1,x2:r.x2,y2:r.y2,anchorPointX:r.anchorPointX,anchorPointY:r.anch
orPointY},u.iconFeatureIndex=r.featureIndex;break}for(let e=o;e<l;e++){const
r=t.get(e);u.verticalIconBox=
{x1:r.x1,y1:r.y1,x2:r.x2,y2:r.y2,anchorPointX:r.anchorPointX,anchorPointY:r.anch
orPointY},u.verticalIconFeatureIndex=r.featureIndex;break}return
u}deserializeCollisionBoxes(t){this.collisionArrays=[];for(let

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e=0;e<this.symbolInstances.length;e++){const
r=this.symbolInstances.get(e);this.collisionArrays.push(this._deserializeCollisionBoxesForSymbol(t,r.textBoxStartIndex,r.textBoxEndIndex,r.verticalTextBoxStartIndex,r.verticalTextBoxEndIndex,r.iconBoxStartIndex,r.iconBoxEndIndex,r.verticalIconBoxStartIndex,r.verticalIconBoxEndIndex));}}hasTextData(){return
this.text.segments.get().length>0}hasIconData(){return
this.icon.segments.get().length>0}hasDebugData(){return
this.textCollisionBox&&this.iconCollisionBox}hasTextCollisionBoxData(){return
this.hasDebugData()&&this.textCollisionBox.segments.get().length>0}hasIconCollisionBoxData(){return
this.hasDebugData()&&this.iconCollisionBox.segments.get().length>0}addIndicesForPlacedSymbol(t,e){const
r=t.placedSymbolArray.get(e),n=r.vertexStartIndex+4*r.numGlyphs;for(let
e=r.vertexStartIndex;e<n;e+=4)t.indexArray.emplaceBack(e,e+1,e+2),t.indexArray.emplaceBack(e+1,e+2,e+3);}getSortedSymbolIndexes(t){if(this.sortedAngle===t&&void
0!==this.symbolInstanceIndexes)return this.symbolInstanceIndexes;const
e=Math.sin(t),r=Math.cos(t),n=[],i=[],a=[];for(let
t=0;t<this.symbolInstances.length;++t){a.push(t);const
s=this.symbolInstances.get(t);n.push(0|Math.round(e*s.anchorX+r*s.anchorY)),i.push(s.featureIndex);}return a.sort(((t,e)=>n[t]-n[e]||i[e]-i[t])),a}addToSortKeyRanges(t,e){const
r=this.sortKeyRanges[this.sortKeyRanges.length-1];r&&r.sortKey===e?
r.symbolInstanceEnd=t+1:this.sortKeyRanges.push({sortKey:e,symbolInstanceStart:t,symbolInstanceEnd:t+1});}sortFeatures(t)
{if(this.sortFeaturesByY&&this.sortedAngle!==t&&!(this.text.segments.get().length>1||this.icon.segments.get().length>1))
{this.symbolInstanceIndexes=this.getSortedSymbolIndexes(t),this.sortedAngle=t,this.text.indexArray.clear(),this.icon.indexArray.clear(),this.featureSortOrder=[];for(const t of this.symbolInstanceIndexes){const
e=this.symbolInstances.get(t);this.featureSortOrder.push(e.featureIndex),[e.rightJustifiedTextSymbolIndex,e.centerJustifiedTextSymbolIndex,e.leftJustifiedTextSymbolIndex].forEach(((t,e,r)=>
{t>=0&&r.indexOf(t)===e&&this.addIndicesForPlacedSymbol(this.text,t);})),e.verticalPlacedTextSymbolIndex>=0&&this.addIndicesForPlacedSymbol(this.text,e.verticalPlacedTextSymbolIndex),e.placedIconSymbolIndex>=0&&this.addIndicesForPlacedSymbol(this.icon,e.placedIconSymbolIndex),e.verticalPlacedIconSymbolIndex>=0&&this.addIndicesForPlacedSymbol(this.icon,e.verticalPlacedIconSymbolIndex);}this.text.indexBuffer&&this.text.indexBuffer.updateData(this.text.indexArray),this.icon.indexBuffer&&this.icon.indexBuffer.updateData(this.icon.indexArray);}}let
Ru,qu;$n("SymbolBucket",Uu,{omit:
["layers","collisionBoxArray","features","compareText"]}),Uu.MAX_GLYPHS=65535,Uu.addDynamicAttributes=$u;var ju={get paint(){return qu=qu||new _i({"icon-opacity":new xi(X.paint_symbol["icon-opacity"]),"icon-color":new xi(X.paint_symbol["icon-color"]),"icon-halo-color":new xi(X.paint_symbol["icon-halo-color"]),"icon-halo-width":new xi(X.paint_symbol["icon-halo-width"]),"icon-halo-blur":new xi(X.paint_symbol["icon-halo-blur"]),"icon-translate":new gi(X.paint_symbol["icon-translate"]),"icon-translate-anchor":new gi(X.paint_symbol["icon-translate-anchor"]),"text-opacity":new xi(X.paint_symbol["text-opacity"]),"text-color":new xi(X.paint_symbol["text-color"])}),{runtimeType:yt,getOverride:t=>t.textColor,hasOverride:t=>!!t.textColor},"text-halo-color":new xi(X.paint_symbol["text-halo-color"]),"text-halo-width":new xi(X.paint_symbol["text-halo-width"]),"text-halo-blur":new xi(X.paint_symbol["text-halo-blur"]),"text-translate":new gi(X.paint_symbol["text-translate"]),"text-translate-anchor":new gi(X.paint_symbol["text-translate-anchor"])}},get layout(){return Ru=Ru||new _i({"symbol-placement":new gi(X.layout_symbol["symbol-placement"]),"symbol-spacing":new gi(X.layout_symbol["symbol-spacing"]),"symbol-avoid-edges":new gi(X.layout_symbol["symbol-avoid-edges"]),"symbol-sort-key":new xi(X.layout_symbol["symbol-sort-key"]),"symbol-z-order":new gi(X.layout_symbol["symbol-z-order"]),"icon-allow-overlap":new gi(X.layout_symbol["icon-allow-overlap"]),"icon-overlap":new

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gi(X.layout_symbol["icon-overlap"]), "icon-ignore-placement":new
gi(X.layout_symbol["icon-ignore-placement"]), "icon-optional":new
gi(X.layout_symbol["icon-optional"]), "icon-rotation-alignment":new
gi(X.layout_symbol["icon-rotation-alignment"]), "icon-size":new
xi(X.layout_symbol["icon-size"]), "icon-text-fit":new gi(X.layout_symbol["icon-
text-fit"]), "icon-text-fit-padding":new gi(X.layout_symbol["icon-text-fit-
padding"]), "icon-image":new xi(X.layout_symbol["icon-image"]), "icon-rotate":new
xi(X.layout_symbol["icon-rotate"]), "icon-padding":new xi(X.layout_symbol["icon-
padding"]), "icon-keep-upright":new gi(X.layout_symbol["icon-keep-
upright"]), "icon-offset":new xi(X.layout_symbol["icon-offset"]), "icon-
anchor":new xi(X.layout_symbol["icon-anchor"]), "icon-pitch-alignment":new
gi(X.layout_symbol["icon-pitch-alignment"]), "text-pitch-alignment":new
gi(X.layout_symbol["text-pitch-alignment"]), "text-rotation-alignment":new
gi(X.layout_symbol["text-rotation-alignment"]), "text-field":new
xi(X.layout_symbol["text-field"]), "text-font":new xi(X.layout_symbol["text-
font"]), "text-size":new xi(X.layout_symbol["text-size"]), "text-max-width":new
xi(X.layout_symbol["text-max-width"]), "text-line-height":new
gi(X.layout_symbol["text-line-height"]), "text-letter-spacing":new
xi(X.layout_symbol["text-letter-spacing"]), "text-justify":new
xi(X.layout_symbol["text-justify"]), "text-radial-offset":new
xi(X.layout_symbol["text-radial-offset"]), "text-variable-anchor":new
gi(X.layout_symbol["text-variable-anchor"]), "text-variable-anchor-offset":new
xi(X.layout_symbol["text-variable-anchor-offset"]), "text-anchor":new
xi(X.layout_symbol["text-anchor"]), "text-max-angle":new
gi(X.layout_symbol["text-max-angle"]), "text-writing-mode":new
gi(X.layout_symbol["text-writing-mode"]), "text-rotate":new
xi(X.layout_symbol["text-rotate"]), "text-padding":new gi(X.layout_symbol["text-
padding"]), "text-keep-upright":new gi(X.layout_symbol["text-keep-
upright"]), "text-transform":new xi(X.layout_symbol["text-transform"]), "text-
offset":new xi(X.layout_symbol["text-offset"]), "text-allow-overlap":new
gi(X.layout_symbol["text-allow-overlap"]), "text-overlap":new
gi(X.layout_symbol["text-overlap"]), "text-ignore-placement":new
gi(X.layout_symbol["text-ignore-placement"]), "text-optional":new
gi(X.layout_symbol["text-optional"])});class Nu{constructor(t){if(void
0===t.property.overrides)throw new Error("overrides must be provided to
instantiate FormatSectionOverride class");this.type=t.property.overrides?
t.property.overrides.runtimeType:ht,this.defaultValue=t;}evaluate(t)
{if(t.formattedSection){const
e=this.defaultValue.property.overrides;if(e&&e.hasOverride(t.formattedSection))r
eturn e.getOverride(t.formattedSection)}return t.feature&&t.featureState?
this.defaultValue.evaluate(t.feature,t.featureState):this.defaultValue.property.
specification.default}eachChild(t)
{this.defaultValue.isConstant()||t(this.defaultValue.value._styleExpression.expr
ession);}outputDefined(){return !1}serialize(){return
null}}$n("FormatSectionOverride",Nu,{omit:["defaultValue"]});class Zu extends
Si{constructor(t){super(t,ju);}recalculate(t,e)
{if(super.recalculate(t,e),"auto"===this.layout.get("icon-rotation-alignment")&&
(this.layout._values["icon-rotation-
alignment"]="point"!==this.layout.get("symbol-
placement")?"map":"viewport"),"auto"===this.layout.get("text-rotation-
alignment")&&(this.layout._values["text-rotation-
alignment"]="point"!==this.layout.get("symbol-
placement")?"map":"viewport"),"auto"===this.layout.get("text-pitch-alignment")&&
(this.layout._values["text-pitch-alignment"]="map"===this.layout.get("text-
rotation-alignment")?"map":"viewport"),"auto"===this.layout.get("icon-pitch-
alignment")&&(this.layout._values["icon-pitch-alignment"]=this.layout.get("icon-
rotation-alignment")),"point"===this.layout.get("symbol-placement")){const
t=this.layout.get("text-writing-mode");if(t){const e=[];for(const r of
t)e.indexOf(r)<0&&e.push(r);this.layout._values["text-writing-mode"]=e;}else
this.layout._values["text-writing-mode"]=
["horizontal"]};this._setPaintOverrides();}getValueAndResolveTokens(t,e,r,n)
{const i=this.layout.get(t).evaluate(e,

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    {} , r , n ) , a = this . _unevaluatedLayout . _values [ t ] ; return
    a . isDataDriven () || Tr ( a . value ) || ! i ? i : function ( t , e ) { return
    e . replace ( / ( [ ^ { } ] + ) / g , ( ( e , r ) => t & r in t ? String ( t [ r ] ) : "" ) ) }
    ( e . properties , i ) } createBucket ( t ) { return new Uu ( t ) } queryRadius () { return
    0 } queryIntersectsFeature () { throw new Error ( "Should take a different path in
    FeatureIndex" ) } _setPaintOverrides () { for ( const t of
    ju . paint . overridableProperties )
    { if ( ! Zu . hasPaintOverride ( this . layout , t ) ) continue ; const e = this . paint . get ( t ) , r = new
    Nu ( e ) , n = new Fr ( r , e . property . specification ) ; let
    i = null ; i = "constant" === e . value . kind || "source" === e . value . kind ? new
    Lr ( "source" , n ) : new Dr ( "composite" , n , e . value . zoomStops ) , this . paint . _values [ t ] = new
    yi ( e . property , i , e . parameters ) ; } } _handleOverridablePaintPropertyUpdate ( t , e , r )
    { return !
    ( ! this . layout || e . isDataDriven () || r . isDataDriven () ) & Zu . hasPaintOverride ( this . lay
    out , t ) } static hasPaintOverride ( t , e ) { const r = t . get ( "text-
    field" ) , n = ju . paint . properties [ e ] ; let i = ! 1 ; const a = t => { for ( const e of
    t ) if ( n . overrides & n . overrides . hasOverride ( e ) ) return
    void ( i = ! 0 ) ; if ( "constant" === r . value . kind & r . value . value instanceof
    Qt ) a ( r . value . value . sections ) ; else if ( "source" === r . value . kind ) { const t = e => { i || ( e
    instanceof le & se ( e . value ) === vt ? a ( e . value . sections ) : e instanceof pr ?
    a ( e . sections ) : e . eachChild ( t ) ) ; } , e = r . value ; e . _styleExpression & t ( e . _styleExpressi
    on . expression ) ; } return i } } let Ku ; var Gu = { get paint () { return Ku = Ku || new
    _i ( { "background-color" : new gi ( X . paint _background [ "background-
    color" ] ) , "background-pattern" : new bi ( X . paint _background [ "background-
    pattern" ] ) , "background-opacity" : new gi ( X . paint _background [ "background-
    opacity" ] ) } ) } ; class Ju extends Si { constructor ( t ) { super ( t , Gu ) ; } } let Xu ; var Yu =
    { get paint () { return Xu = Xu || new _i ( { "raster-opacity" : new
    gi ( X . paint _raster [ "raster-opacity" ] ) , "raster-hue-rotate" : new
    gi ( X . paint _raster [ "raster-hue-rotate" ] ) , "raster-brightness-min" : new
    gi ( X . paint _raster [ "raster-brightness-min" ] ) , "raster-brightness-max" : new
    gi ( X . paint _raster [ "raster-brightness-max" ] ) , "raster-saturation" : new
    gi ( X . paint _raster [ "raster-saturation" ] ) , "raster-contrast" : new
    gi ( X . paint _raster [ "raster-contrast" ] ) , "raster-resampling" : new
    gi ( X . paint _raster [ "raster-resampling" ] ) , "raster-fade-duration" : new
    gi ( X . paint _raster [ "raster-fade-duration" ] ) } ) } ; class Hu extends
    Si { constructor ( t ) { super ( t , Yu ) ; } } class Wu extends Si { constructor ( t ) { super ( t ,
    {} ) , this . onAdd = t =>
    { this . implementation . onAdd & this . implementation . onAdd ( t , t . painter . context . gl ) ; } ,
    this . onRemove = t =>
    { this . implementation . onRemove & this . implementation . onRemove ( t , t . painter . context .
    gl ) ; } , this . implementation = t ; } is3D () { return
    "3d" === this . implementation . renderingMode } hasOffscreenPass () { return void
    0 ! = this . implementation . prerender () recalculate () { } updateTransitions ()
    { } hasTransition () { return ! 1 } serialize () { throw new Error ( "Custom layers cannot be
    serialized" ) } } class Qu { constructor ( t )
    { this . _callback = t , this . _triggered = ! 1 , "undefined" != typeof MessageChannel &
    ( this . _channel = new MessageChannel , this . _channel . port2 . onmessage = ( ) =>
    { this . _triggered = ! 1 , this . _callback () ; } ) ; } trigger () { this . _triggered ||
    ( this . _triggered = ! 0 , this . _channel ?
    this . _channel . port1 . postMessage ( ! 0 ) : setTimeout ( ( ) =>
    { this . _triggered = ! 1 , this . _callback () ; } , 0 ) ) ; } remove () { delete
    this . _channel , this . _callback = ( ) => { } ; } } const tc = 6371008.8 ; class
    ec { constructor ( t , e ) { if ( isNaN ( t ) || isNaN ( e ) ) throw new Error ( `Invalid LngLat
    object: (${t},
    ${e})` ) ; if ( this . lng = + t , this . lat = + e , this . lat > 90 || this . lat < - 90 ) throw new
    Error ( "Invalid LngLat latitude value: must be between -90 and 90" ) } wrap () { return
    new ec ( m ( this . lng , - 180 , 180 ) , this . lat ) } toArray () { return
    [ this . lng , this . lat ] } toString () { return `LngLat(${this.lng},
    ${this.lat})` } distanceTo ( t ) { const
    e = Math . PI / 180 , r = this . lat * e , n = t . lat * e , i = Math . sin ( r ) * Math . sin ( n ) + Math . cos ( r ) * Math .
    cos ( n ) * Math . cos ( ( t . lng - this . lng ) * e ) ; return tc * Math . acos ( Math . min ( i , 1 ) ) } static
    convert ( t ) { if ( t instanceof ec ) return t ; if ( Array . isArray ( t ) &

```



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(2===t.length||3===t.length))return new
ec(Number(t[0]),Number(t[1]));if(!Array.isArray(t)&&"object"===typeof
t&&null!==t)return new ec(Number("lng"in t?t.lng:t.lon),Number(t.lat));throw new
Error("`LngLatLike` argument must be specified as a LngLat instance, an object
{lng: <lng>, lat: <lat>}, an object {lon: <lng>, lat: <lat>}, or an array of
[<lng>, <lat>]")}const rc=2*Math.PI*t;function nc(t){return
rc*Math.cos(t*Math.PI/180)}function ic(t){return (180+t)/360}function ac(t)
{return (180-
180/Math.PI*Math.log(Math.tan(Math.PI/4+t*Math.PI/360)))/360}function sc(t,e)
{return t/nc(e)}function oc(t){return 360*t-180}function lc(t){return
360/Math.PI*Math.atan(Math.exp((180-360*t)*Math.PI/180))-90}class
uc{constructor(t,e,r=0){this.x+=t,this.y+=e,this.z+=r;}static fromLngLat(t,e=0)
{const r=ec.convert(t);return new uc(ic(r.lng),ac(r.lat),sc(e,r.lat))}toLngLat()
{return new ec(oc(this.x),lc(this.y))}toAltitude(){return
this.z*nc(lc(this.y))}meterInMercatorCoordinateUnits(){return 1/rc*
(t=lc(this.y),1/Math.cos(t*Math.PI/180));var t;}}function cc(t,e,r){var
n=2*Math.PI*6378137/256/Math.pow(2,r);return [t*n-2*Math.PI*6378137/2,e*n-
2*Math.PI*6378137/2]}class hc{constructor(t,e,r)
{if(t<0||t>25||r<0||r>=Math.pow(2,t)||e<0||e>=Math.pow(2,t))throw new
Error(`x=${e}, y=${r}, z=${t} outside of bounds. 0<=x<${Math.pow(2,t)},
0<=y<${Math.pow(2,t)} 0<=z<=25
`);this.z=t,this.x=e,this.y=r,this.key=dc(0,t,t,e,r);}equals(t){return
this.z===t.z&&this.x===t.x&&this.y===t.y}url(t,e,r){const n=
(a=this.y,s=this.z,o=cc(256*(i=this.x),256*(a=Math.pow(2,s)-a-1),s),l=cc(256*
(i+1),256*(a+1),s),o[0]+","+o[1]+","+l[0]+","+l[1]);var i,a,s,o,l;const
u=function(t,e,r){let n,i="";for(let a=t;a>0;a--)n=1<a-1,i+=(e&n?1:0)+(r&n?
2:0);return i}(this.z,this.x,this.y);return
t[(this.x+this.y)%t.length].replace(/ {prefix}/g,(this.x%16).toString(16)+
(this.y%16).toString(16)).replace(/ {z}/g,String(this.z)).replace(/ {x}/g,String(t
his.x)).replace(/ {y}/g,String("tms"===r?Math.pow(2,this.z)-this.y-
1:this.y)).replace(/ {ratio}/g,e>1?"@2x":"").replace(/ {quadkey}/g,u).replace(/ {bb
ox-epsg-3857}/g,n)}isChildOf(t){const e=this.z-t.z;return
e>0&&t.x===this.x>>e&&t.y===this.y>>e}getTilePoint(t){const
e=Math.pow(2,this.z);return new a((t.x*e-this.x)*rs,(t.y*e-
this.y)*rs)}toString(){return `${this.z}/${this.x}/${this.y}`}}class
pc{constructor(t,e)
{this.wrap=t,this.canonical=e,this.key=dc(t,e.z,e.z,e.x,e.y);}class
fc{constructor(t,e,r,n,i){if(t<r)throw new Error(`overscaledZ should be >= z;
overscaledZ = ${t}; z = ${r}`);this.overscaledZ=t,this.wrap=e,this.canonical=new
hc(r,+n,+i),this.key=dc(e,t,r,n,i);}clone(){return new
fc(this.overscaledZ,this.wrap,this.canonical.z,this.canonical.x,this.canonical.y
)}equals(t){return
this.overscaledZ===t.overscaledZ&&this.wrap===t.wrap&&this.canonical.equals(t.ca
nonical)}scaledTo(t){if(t>this.overscaledZ)throw new Error(`targetZ >
this.overscaledZ; targetZ = ${t}; overscaledZ = ${this.overscaledZ}`);const
e=this.canonical.z-t;return t>this.canonical.z?new
fc(t,this.wrap,this.canonical.z,this.canonical.x,this.canonical.y):new
fc(t,this.wrap,t,this.canonical.x>>e,this.canonical.y>>e)}calculateScaledKey(t,e
){if(t>this.overscaledZ)throw new Error(`targetZ > this.overscaledZ; targetZ =
${t}; overscaledZ = ${this.overscaledZ}`);const r=this.canonical.z-t;return
t>this.canonical.z?
dc(this.wrap+e,t,this.canonical.z,this.canonical.x,this.canonical.y):dc(this.wr
ap+e,t,t,this.canonical.x>>r,this.canonical.y>>r)}isChildOf(t)
{if(t.wrap!==this.wrap)return !1;const e=this.canonical.z-t.canonical.z;return
0===t.overscaledZ||t.overscaledZ<this.overscaledZ&&t.canonical.x===this.canonica
l.x>>e&&t.canonical.y===this.canonical.y>>e}children(t)
{if(this.overscaledZ>=t)return [new
fc(this.overscaledZ+1,this.wrap,this.canonical.z,this.canonical.x,this.canonical
.y)];const e=this.canonical.z+1,r=2*this.canonical.x,n=2*this.canonical.y;return
[new fc(e,this.wrap,e,r,n),new fc(e,this.wrap,e,r+1,n),new
fc(e,this.wrap,e,r,n+1),new fc(e,this.wrap,e,r+1,n+1)]}isLessThan(t){return
this.wrap<t.wrap||!(this.wrap>t.wrap)&&(this.overscaledZ<t.overscaledZ||!

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(this.overscaledZ>t.overscaledZ)&&(this.canonical.x<t.canonical.x||!
(this.canonical.x>t.canonical.x)&&this.canonical.y<t.canonical.y))}wrapped()
{return new
fc(this.overscaledZ,0,this.canonical.z,this.canonical.x,this.canonical.y)}unwrap
To(t){return new
fc(this.overscaledZ,t,this.canonical.z,this.canonical.x,this.canonical.y)}oversc
aleFactor(){return Math.pow(2,this.overscaledZ-this.canonical.z)}toUnwrapped()
{return new pc(this.wrap,this.canonical)}toString(){return
`$${this.overscaledZ}/${this.canonical.x}/${this.canonical.y}`}getTilePoint(t)
{return this.canonical.getTilePoint(new uc(t.x-this.wrap,t.y))}}function
dc(t,e,r,n,i){(t*=2)<0&&(t=-1*t-1);const a=1<<r;return
(a*a*t+a*i+n).toString(36)+r.toString(36)+e.toString(36)}$n("CanonicalTileID",hc
),$n("OverscaledTileID",fc,{omit:["posMatrix"]});class
yc{constructor(t,e,r,n=1,i=1,a=1,s=0){if(this.uid=t,e.height!==e.width)throw new
RangeError("DEM tiles must be square");if(r&&!
["mapbox","terrarium","custom"].includes(r))return void A(`$${r}` is not a valid
encoding type. Valid types include "mapbox", "terrarium" and
"custom".`);this.stride=e.height;const o=this.dim=e.height-
2;switch(this.data=new Uint32Array(e.data.buffer),r)
{case"terrarium":this.redFactor=256,this.greenFactor=1,this.blueFactor=1/256,thi
s.baseShift=32768;break;case"custom":this.redFactor=n,this.greenFactor=i,this.bl
ueFactor=a,this.baseShift=s;break;default:this.redFactor=6553.6,this.greenFactor
=25.6,this.blueFactor=.1,this.baseShift=1e4;}for(let
t=0;t<o;t++)this.data[this._idx(-1,t)]=this.data[this._idx(0,t)],this.data[this.
_idx(o,t)]=this.data[this._idx(o-
1,t)],this.data[this._idx(t,-1)]=this.data[this._idx(t,0)],this.data[this._idx(t
,o)]=this.data[this._idx(t,o-
1)];this.data[this._idx(-1,-1)]=this.data[this._idx(0,0)],this.data[this._idx(o,
-1)]=this.data[this._idx(o-
1,0)],this.data[this._idx(-1,o)]=this.data[this._idx(0,o-
1)],this.data[this._idx(o,o)]=this.data[this._idx(o-1,o-
1)],this.min=Number.MAX_SAFE_INTEGER,this.max=Number.MIN_SAFE_INTEGER;for(let
t=0;t<o;t++)for(let e=0;e<o;e++){const r=this.get(t,e);r>this.max&&
(this.max=r),r<this.min&&(this.min=r);}}get(t,e){const r=new
Uint8Array(this.data.buffer),n=4*this._idx(t,e);return
this.unpack(r[n],r[n+1],r[n+2])}getUnpackVector(){return
[this.redFactor,this.greenFactor,this.blueFactor,this.baseShift]}_idx(t,e)
{if(t<-1||t>=this.dim+1||e<-1||e>=this.dim+1)throw new RangeError("out of range
source coordinates for DEM data");return (e+1)*this.stride+(t+1)}unpack(t,e,r)
{return t*this.redFactor+e*this.greenFactor+r*this.blueFactor-
this.baseShift}getPixels(){return new
qs({width:this.stride,height:this.stride},new
Uint8Array(this.data.buffer))}backfillBorder(t,e,r){if(this.dim!==t.dim)throw
new Error("dem dimension mismatch");let
n=e*this.dim,i=e*this.dim+this.dim,a=r*this.dim,s=r*this.dim+this.dim;switch(e)
{case-1:n=i-1;break;case 1:i=n+1;}switch(r){case-1:a=s-1;break;case
1:s=a+1;}const o=-e*this.dim,l=-r*this.dim;for(let e=a;e<s;e++)for(let
r=n;r<i;r++)this.data[this._idx(r,e)]=t.data[this._idx(r+o,e+1)];}$n("DEMData",
yc);class mc{constructor(t){this._stringToNumber={},this._numberToString=
[];for(let e=0;e<t.length;e++){const
r=t[e];this._stringToNumber[r]=e,this._numberToString[e]=r;}}encode(t){return
this._stringToNumber[t]}decode(t){if(t>=this._numberToString.length)throw new
Error(`Out of bounds. Index requested n=${t} can't be >=
this._numberToString.length $${this._numberToString.length}`);return
this._numberToString[t]}class gc{constructor(t,e,r,n,i)
{this.type="Feature",this._vectorTileFeature=t,t._z=e,t._x=r,t._y=n,this.propert
ies=t.properties,this.id=i;}get geometry(){return void 0===this._geometry&&
(this._geometry=this._vectorTileFeature.toGeoJSON(this._vectorTileFeature._x,thi
s._vectorTileFeature._y,this._vectorTileFeature._z).geometry),this._geometry}set
geometry(t){this._geometry=t;}toJSON(){const t=
{geometry:this._geometry};for(const e in
this)"_geometry"!==e&&"_vectorTileFeature"!==e&&(t[e]=this[e]);return t}}class

```

```

xc{constructor(t,e)
{this.tileID=t,this.x=t.canonical.x,this.y=t.canonical.y,this.z=t.canonical.z,this.grid=new Fn(rs,16,0),this.grid3D=new Fn(rs,16,0),this.featureIndexArray=new ca,this.promoteId=e;}insert(t,e,r,n,i,a){const s=this.featureIndexArray.length;this.featureIndexArray.emplaceBack(r,n,i);const o=a?this.grid3D:this.grid;for(let t=0;t<e.length;t++){const r=e[t],n=[1/0,1/0,-1/0,-1/0];for(let t=0;t<r.length;t++){const e=r[t];n[0]=Math.min(n[0],e.x),n[1]=Math.min(n[1],e.y),n[2]=Math.max(n[2],e.x),n[3]=Math.max(n[3],e.y);}n[0]<rs&&n[1]<rs&&n[2]>=0&&n[3]>=0&&o.insert(s,n[0],n[1],n[2],n[3]);}}loadVTLayers(){return this.vtLayers||(this.vtLayers=new Ro.VectorTile(new eu(this.rawTileData)).layers,this.sourceLayerCoder=new mc(this.vtLayers?Object.keys(this.vtLayers).sort():["_geojsonTileLayer"])),this.vtLayers}query(t,e,r,n){this.loadVTLayers();const i=t.params||{},s=rs/t.tileSize/t.scale,o=Nr(i.filter),l=t.queryGeometry,u=t.queryPadding*s,c=bc(l),h=this.grid.query(c.minX-u,c.minY-u,c.maxX+u,c.maxY+u),p=bc(t.cameraQueryGeometry),f=this.grid3D.query(p.minX-u,p.minY-u,p.maxX+u,p.maxY+u,((e,r,n,i)=>function(t,e,r,n,i){for(const a of t)if(e<=a.x&&r<=a.y&&n>=a.x&&i>=a.y)return !0;const s=[new a(e,r),new a(e,i),new a(n,i),new a(n,r)];if(t.length>2)for(const e of s)if(xs(t,e))return !0;for(let e=0;e<t.length-1;e++)if(vs(t[e],t[e+1],s))return !0;return !1}(t.cameraQueryGeometry,e-u,r-u,n+u,i+u)));for(const t of f)h.push(t);h.sort(wc);const d={};let y;for(let a=0;a<h.length;a++){const u=h[a];if(u===y)continue;y=u;const c=this.featureIndexArray.get(u);let p=null;this.loadMatchingFeature(d,c.bucketIndex,c.sourceLayerIndex,c.featureIndex,o,i.layers,i.availableImages,e,r,n,((e,r,n)=>(p||(p=as(e)),r.queryIntersectsFeature(l,e,n,p,this.z,t.transform,s,t.pixelPosMatrix)))));}return d}loadMatchingFeature(t,e,r,n,i,a,s,o,l,u,c){const h=this.bucketLayerIDs[e];if(a&&!function(t,e){for(let r=0;r<t.length;r++)if(e.indexOf(t[r])>=0)return !0;return !1}(a,h))return;const p=this.sourceLayerCoder.decode(r),f=this.vtLayers[p].feature(n);if(i.needGeometry){const t=ss(f,!0);if(!i.filter(new li(this.tileID.overscaledZ),t,this.tileID.canonical))return}else if(!i.filter(new li(this.tileID.overscaledZ),f))return;const d=this.getId(f,p);for(let e=0;e<h.length;e++){const r=h[e];if(a&&a.indexOf(r)<0)continue;const i=o[r];if(!i)continue;let p={};d&&u&&(p=u.getState(i.sourceLayer||"_geojsonTileLayer",d));const y=g({},l[r]);y.paint=vc(y.paint,i.paint,f,p,s),y.layout=vc(y.layout,i.layout,f,p,s);const m=!c||c(f,i,p);if(!m)continue;const x=new gc(f,this.z,this.x,this.y,d);x.layer=y;let v=t[r];void 0===v&&(v=t[r]=[]),v.push({featureIndex:n,feature:x,intersectionZ:m});}lookupSymbolFeatures(t,e,r,n,i,a,s,o){const l={};this.loadVTLayers();const u=Nr(i);for(const i of t)this.loadMatchingFeature(l,r,n,i,u,a,s,o,e);return l}hasLayer(t){for(const e of this.bucketLayerIDs)for(const r of e)if(t===r)return !0;return !1}getId(t,e){let r=t.id;return this.promoteId&&(r=t.properties["string"]==typeof this.promoteId?this.promoteId:this.promoteId[e],"boolean")==typeof r&&(r=Number(r)),r}}function vc(t,e,r,n,i){return v(t,((t,a)=>{const s=e instanceof mi?e.get(a):null;return s&&s.evaluate?s.evaluate(r,n,i):s}))}function bc(t){let e=1/0,r=1/0,n=-1/0,i=-1/0;for(const a of t)e=Math.min(e,a.x),r=Math.min(r,a.y),n=Math.max(n,a.x),i=Math.max(i,a.y);return {minX:e,minY:r,maxX:n,maxY:i}}function wc(t,e){return e-t}function _c(t,e,r,n,i){const s=[];for(let o=0;o<t.length;o++){const l=t[o];let u;for(let t=0;t<l.length-1;t++){let o=l[t],c=l[t+1];o.x<e&&c.x<e||(o.x<e?o=new a(e,o.y+(e-o.x)/(c.x-o.x)*(c.y-o.y))._round():c.x<e&&(c=new a(e,o.y+(e-o.x)/(c.x-o.x)*(c.y-o.y))._round()),o.y<r&&c.y<r||(o.y<r?o=new a(o.x+(r-o.y)/(c.y-o.y)*(c.x-o.x),r)._round():c.y<r&&(c=new a(o.x+(r-o.y)/(c.y-o.y)*(c.x-o.x),r)._round()),o.x>=n&&c.x>=n||(o.x>=n?o=new a(n,o.y+(n-o.x)/(c.x-o.x)*(c.y-o.y))._round():c.x>=n&&(c=new a(n,o.y+(n-o.x)/(c.x-o.x)*(c.y-o.y))._round()),o.y>=i&&c.y>=i||(o.y>=i?o=new a(o.x+(i-o.y)/(c.y-o.y)*(c.x-o.x),i)._round():c.y>=i&&(c=new a(o.x+(i-o.y)/(c.y-o.y)*(c.x-o.x),i)._round()),u&&o.equals(u[u.length-1])||(u=

```

```

[o],s.push(u)),u.push(c))));}}return s}$n("FeatureIndex",xc,{omit:
["rawTileData","sourceLayerCoder"]});class Ac extends a{constructor(t,e,r,n)
{super(t,e),this.angle=r,void 0!==(this.segment=n)}clone(){return new
Ac(this.x,this.y,this.angle,this.segment)}function Sc(t,e,r,n,i){if(void
0===e.segment||0===r)return !0;let a=e,s=e.segment+1,o=0;for(;o>-r/2;){if(s-
-,s<0)return !1;o-=t[s].dist(a),a=t[s];}o+=t[s].dist(t[s+1]),s++;const l=[];let
u=0;for(;o<r/2;){const e=t[s],r=t[s+1];if(!r)return !1;let a=t[s-1].angleTo(e)-
e.angleTo(r);for(a=Math.abs((a+3*Math.PI)%(2*Math.PI))-
Math.PI),l.push({distance:o,angleDelta:a}),u+=a;o-=l[0].distance>n;)u-
=l.shift().angleDelta;if(u>i)return !1;s++,o+=e.dist(r);}return !0}function
kc(t){let e=0;for(let r=0;r<t.length-1;r++)e+=t[r].dist(t[r+1]);return
e}function Ic(t,e,r){return t?.6*e*r:0}function zc(t,e){return Math.max(t?
t.right-t.left:0,e?e.right-e.left:0)}function Mc(t,e,r,n,i,a){const
s=Ic(r,i,a),o=zc(r,n)*a;let l=0;const u=kc(t)/2;for(let r=0;r<t.length-1;r++)
{const n=t[r],i=t[r+1],a=n.dist(i);if(l+a>u){const c=(u-
l)/a,h=Ze.number(n.x,i.x,c),p=Ze.number(n.y,i.y,c),f=new
Ac(h,p,i.angleTo(n),r);return f._round(),!s||Sc(t,f,o,s,e)?f:void
0}l+=a;}}function Cc(t,e,r,n,i,a,s,o,l){const
u=Ic(n,a,s),c=zc(n,i),h=c*s,p=0===t[0].x||t[0].x===1||0===t[0].y||t[0].y===1;ret
urn e-h<e/4&&(e=h+e/4),Bc(t,p?e/2*o%e:(c/2+2*a)*s*o%e,e,u,r,h,p,!1,l)}function
Bc(t,e,r,n,i,a,s,o,l){const u=a/2,c=kc(t);let h=0,p=e-r,f=[];for(let
e=0;e<t.length-1;e++){const
s=t[e],o=t[e+1],d=s.dist(o),y=o.angleTo(s);for(;p+r<h+d;){p+=r;const m=(p-
h)/d,g=Ze.number(s.x,o.x,m),x=Ze.number(s.y,o.y,m);if(g>=0&&g<l&&x>=0&&x<l&&p-
u>=0&&p+u<=c){const r=new
Ac(g,x,y,e);r._round(),n&&!Sc(t,r,a,n,i)||f.push(r);}h+=d;}return
o||f.length||s||(f=Bc(t,h/2,r,n,i,a,s,!0,l)),f}$n("Anchor",Ac);const
Pc=pc,function Vc(t,e,r,n){const i=[],s=t.image,o=s.pixelRatio,l=s.paddedRect.w-
2*Pc,u=s.paddedRect.h-2*Pc,c=t.right-t.left,h=t.bottom-t.top,p=s.stretchX||
[[0,l]],f=s.stretchY||[[0,u]],d=(t,e)>t+e[1]-
e[0],y=p.reduce(d,0),m=f.reduce(d,0),g=l-y,x=u-m;let
v=0,b=y,w=0,_=m,A=0,S=g,k=0,I=x;if(s.content&&n){const
t=s.content;v=Ec(p,0,t[0]),w=Ec(f,0,t[1]),b=Ec(p,t[0],t[2]),_=Ec(f,t[1],t[3]),A=
t[0]-v,k=t[1]-w,S=t[2]-t[0]-b,I=t[3]-t[1]-_;}const z=(n,i,l,u)>=>{const
p=Tc(n.stretch-v,b,c,t.left),f=$c(n.fixed-A,S,n.stretch,y),d=Tc(i.stretch-
w,_,h,t.top),g=$c(i.fixed-k,I,i.stretch,m),x=Tc(l.stretch-
v,b,c,t.left),z=$c(l.fixed-A,S,l.stretch,y),M=Tc(u.stretch-
w,_,h,t.top),C=$c(u.fixed-k,I,u.stretch,m),B=new a(p,d),P=new a(x,d),V=new
a(x,M),E=new a(p,M),F=new a(f/o,g/o),T=new a(z/o,C/o),S=e*Math.PI/180;if($){
const t=Math.sin($),e=Math.cos($),r=[e,-
t,t,e];B._matMult(r),P._matMult(r),E._matMult(r),V._matMult(r);}const
L=n.stretch+n.fixed,D=i.stretch+i.fixed;return {tl:B,tr:P,bl:E,br:V,tex:
{x:s.paddedRect.x+Pc+L,y:s.paddedRect.y+Pc+D,w:l.stretch+l.fixed-
L,h:u.stretch+u.fixed-D},writingMode:void 0,glyphOffset:
[0,0],sectionIndex:0,pixelOffsetTL:F,pixelOffsetBR:T,minFontScaleX:S/o/c,minFont
ScaleY:I/o/h,isSDF:r}};if(n&&(s.stretchX||s.stretchY)){const
t=Fc(p,g,y),e=Fc(f,x,m);for(let r=0;r<t.length-1;r++){const
n=t[r],a=t[r+1];for(let t=0;t<e.length-1;t++)i.push(z(n,e[t],a,e[t+1]));}else
i.push(z({fixed:0,stretch:-1},{fixed:0,stretch:-1},{fixed:0,stretch:l+1},
{fixed:0,stretch:u+1}));return i}function Ec(t,e,r){let n=0;for(const i of
t)n+=Math.max(e,Math.min(r,i[l]))-Math.max(e,Math.min(r,i[0]));return n}function
Fc(t,e,r){const n=[{fixed:-Pc,stretch:0}];for(const[e,r]of t){const
t=n[n.length-1];n.push({fixed:e-t.stretch,stretch:t.stretch}),n.push({fixed:e-
t.stretch,stretch:t.stretch+(r-e)});}return
n.push({fixed:e+Pc,stretch:r}),n}function Tc(t,e,r,n){return t/e*r+n}function
Sc(t,e,r,n){return t-e*r/n}class Lc{constructor(t,e,r,n,i,s,o,l,u,c)
{if(this.boxStartIndex=t.length,u){let t=s.top,e=s.bottom;const
r=s.collisioPadding;r&&(t-=r[1],e+=r[3]);let n=e-t;n>0&&
(n=Math.max(10,n),this.circleDiameter=n);}else {let u=s.top*o-
l[0],h=s.bottom*o+l[2],p=s.left*o-l[3],f=s.right*o+l[1];const
d=s.collisioPadding;if(d&&(p-=d[0]*o,u-=d[1]*o,f+=d[2]*o,h+=d[3]*o),c){const
t=new a(p,u),e=new a(f,u),r=new a(p,h),n=new

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a(f,h),i=c*Math.PI/180;t._rotate(i),e._rotate(i),r._rotate(i),n._rotate(i),p=Math
h.min(t.x,e.x,r.x,n.x),f=Math.max(t.x,e.x,r.x,n.x),u=Math.min(t.y,e.y,r.y,n.y),h
=Math.max(t.y,e.y,r.y,n.y);}t.emplaceBack(e.x,e.y,p,u,f,h,r,n,i);}this.boxEndInd
ex=t.length;}}class Dc{constructor(t=[],e=0c)
{if(this.data=t,this.length=this.data.length,this.compare=e,this.length>0)for(1e
t t=(this.length>>1)-1;t>=0;t--)this._down(t);}push(t)
{this.data.push(t),this.length++,this._up(this.length-1);}pop()
{if(0===this.length)return;const t=this.data[0],e=this.data.pop();return
this.length--,this.length>0&&(this.data[0]=e,this._down(0)),t}peek(){return
this.data[0]}_up(t){const{data:e,compare:r}=this,n=e[t];for(;t>0;){const i=t-
1>>1,a=e[i];if(r(n,a)>=0)break;e[t]=a,t=i;}e[t]=n;}_down(t)
{const{data:e,compare:r}=this,n=this.length>>1,i=e[t];for(;t<n;){let n=1+
(t<<1),a=e[n];const s=n+1;if(s<this.length&&r(e[s],a)<0&&
(n=s,a=e[s]),r(a,i)>=0)break;e[t]=a,t=n;}e[t]=i;}function Oc(t,e){return t<e?
-1:t>e?1:0}function Uc(t,e=1,r=!1){let n=1/0,i=1/0,s=-1/0,o=-1/0;const
l=t[0];for(let t=0;t<l.length;t++){const e=l[t];(!t||e.x<n)&&(n=e.x),
(!t||e.y<i)&&(i=e.y),(!t||e.x>s)&&(s=e.x),(!t||e.y>o)&&(o=e.y);}const
u=Math.min(s-n,o-i);let c=u/2;const h=new Dc([],Rc);if(0===u)return new
a(n,i);for(let e=n;e<s;e+=u)for(let r=i;r<o;r+=u)h.push(new qc(e+c,r+c,c,t));let
p=function(t){let e=0,r=0,n=0;const i=t[0];for(let t=0,a=i.length,s=a-
1;t<a;s=t++){const a=i[t],o=i[s],l=a.x*o.y-o.x*a.y;r+=(a.x+o.x)*1,n+=
(a.y+o.y)*1,e+=3*1;}return new qc(r/e,n/e,0,t)}(t),f=h.length;for(;h.length;){
const n=h.pop();(n.d>p.d||!p.d)&&(p=n,r&&console.log("found best %d after %d
probes",Math.round(1e4*n.d)/1e4,f)),n.max-p.d<=e||((c=n.h/2,h.push(new qc(n.p.x-
c,n.p.y-c,c,t)),h.push(new qc(n.p.x+c,n.p.y-c,c,t)),h.push(new qc(n.p.x-
c,n.p.y+c,c,t)),h.push(new qc(n.p.x+c,n.p.y+c,c,t)),f+=4);}return r&&
(console.log(`num probes: ${f}`),console.log(`best distance:
${p.d}`)),p.p}function Rc(t,e){return e.max-t.max}function qc(t,e,r,n)
{this.p=new a(t,e),this.h=r,this.d=function(t,e){let r=!1,n=1/0;for(let
i=0;i<e.length;i++){const a=e[i];for(let e=0,i=a.length,s=i-1;e<i;s=e++){const
i=a[e],o=a[s];i.y>t.y!=o.y>t.y&&t.x<(o.x-i.x)*(t.y-i.y)/(o.y-i.y)+i.x&&
(r=!r),n=Math.min(n,ms(t,i,o));}}return (r?1:-1)*Math.sqrt(n)}
(this.p,n),this.max=this.d+this.h*Math.SQRT2;}var jc;t.aq=void 0,(jc=t.aq||
(t.aq={}))
[jc.center=1]="center",jc[jc.left=2]="left",jc[jc.right=3]="right",jc[jc.top=4]=
"top",jc[jc.bottom=5]="bottom",jc[jc["top-left"]=6]="top-left",jc[jc["top-
right"]=7]="top-right",jc[jc["bottom-left"]=8]="bottom-left",jc[jc["bottom-
right"]=9]="bottom-right";const Nc=7,Zc=Number.POSITIVE_INFINITY;function
Kc(t,e){return e[1]!==Zc?function(t,e,r){let
n=0,i=0;switch(e=Math.abs(e),r=Math.abs(r),t){case"top-right":case"top-
left":case"top":i=r-Nc;break;case"bottom-right":case"bottom-
left":case"bottom":i=-r+Nc;}switch(t){case"top-right":case"bottom-
right":case"right":n=-e;break;case"top-left":case"bottom-
left":case"left":n=e;}return [n,i]}(t,e[0],e[1]):function(t,e){let r=0,n=0;e<0&&
(e=0);const i=e/Math.SQRT2;switch(t){case"top-right":case"top-left":n=i-
Nc;break;case"bottom-right":case"bottom-left":n=-i+Nc;break;case"bottom":n=-
e+Nc;break;case"top":n=e-Nc;}switch(t){case"top-right":case"bottom-right":r=-
i;break;case"top-left":case"bottom-
left":r=i;break;case"left":r=e;break;case"right":r=-e;}return [r,n]}
(t,e[0])}function Gc(t,e,r){var n;const i=t.layout,a=null===(n=i.get("text-
variable-anchor-offset"))||void 0===n?void 0:n.evaluate(e,{},r);if(a){const
t=a.values,e=[];for(let r=0;r<t.length;r+=2){const
n=e[r]=t[r],i=t[r+1].map((t=>t*Vl));n.startsWith("top")?i[1]-
=Nc:n.startsWith("bottom")&&(i[1]+=Nc),e[r+1]=i;}return new re(e)}const
s=i.get("text-variable-anchor");if(s){let n;n=void
0!==t._unevaluatedLayout.getValue("text-radial-offset")?[i.get("text-radial-
offset").evaluate(e,{},r)*Vl,Zc]:i.get("text-offset").evaluate(e,
{},{},r).map((t=>t*Vl));const a=[];for(const t of s)a.push(t,Kc(t,n));return new
re(a)}return null}function Jc(t){switch(t){case"right":case"top-
right":case"bottom-right":return "right";case"left":case"top-left":case"bottom-
left":return "left"}return "center"}function Xc(e,r,n,i,a,s,o,l,u,c,h){let
p=s.textMaxSize.evaluate(r,{});void 0===p&&(p=0);const

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f=e.layers[0].layout,d=f.get("icon-offset").evaluate(r,
{},{h},y=Hc(n.horizontal),m=o/24,g=e.tilePixelRatio*m,x=e.tilePixelRatio*p/24,v=e
.tilePixelRatio*l,b=e.tilePixelRatio*f.get("symbol-spacing"),w=f.get("text-
padding")*e.tilePixelRatio,_=function(t,e,r,n=1){const i=t.get("icon-
padding").evaluate(e,{},{r}),a=i&&i.values;return [a[0]*n,a[1]*n,a[2]*n,a[3]*n]}
(f,r,h,e.tilePixelRatio),S=f.get("text-max-
angle")/180*Math.PI,k="viewport"!==f.get("text-rotation-
alignment")&&"point"!==f.get("symbol-placement"),I="map"===f.get("icon-rotation-
alignment")&&"point"!==f.get("symbol-placement"),z=f.get("symbol-
placement"),M=b/2,C=f.get("icon-text-fit");let B;i&&"none"!==C&&
(e.allowVerticalPlacement&&n.vertical&&(B=zu(i,n.vertical,C,f.get("icon-text-
fit-padding"),d,m)),y&&(i=zu(i,y,C,f.get("icon-text-fit-padding"),d,m)));const
P=(l,p)=>
{p.x<0||p.x>rs||p.y<0||p.y>rs||function(e,r,n,i,a,s,o,l,u,c,h,p,f,d,y,m,g,x,v,
b,w,_S,k,I){const z=e.addToLineVertexArray(r,n);let
M,C,B,P,V=0,E=0,F=0,T=0,$=-1,L=-1;const D={};let
O=$a("");if(e.allowVerticalPlacement&&i.vertical){const t=l.layout.get("text-
rotate").evaluate(w,{},{k})+90;B=new Lc(u,r,c,h,p,i.vertical,f,d,y,t),o&&(P=new
Lc(u,r,c,h,p,o,g,x,y,t));}if(a){const n=l.layout.get("icon-rotate").evaluate(w,
{},{i="none"!==l.layout.get("icon-text-fit"),s=Vc(a,n,S,i),f=o?Vc(o,n,S,i):void
0;C=new Lc(u,r,c,h,p,a,g,x,!l,n),V=4*s.length;const d=e.iconSizeData;let
y=null;"source"===d.kind?(y=[Cu*l.layout.get("icon-size").evaluate(w,
{},{})],y[0]>Bu&&A(`$${e.layerIds[0]}: Value for "icon-size" is >= ${Mu}. Reduce
your "icon-size".`)):"composite"===d.kind&&(y=
[Cu*_compositeIconSizes[0].evaluate(w,
{},{k}),Cu*_compositeIconSizes[1].evaluate(w,{},{k})],
(y[0]>Bu||y[1]>Bu)&&A(`$${e.layerIds[0]}: Value for "icon-size" is >= ${Mu}.
Reduce your "icon-
size".`)),e.addSymbols(e.icon,s,y,b,v,w,t.ai.none,r,z.lineStartIndex,z.lineLengt
h,-1,k),$=e.icon.placedSymbolArray.length-1,f&&
(E=4*f.length,e.addSymbols(e.icon,f,y,b,v,w,t.ai.vertical,r,z.lineStartIndex,z.l
ineLength,-1,k),L=e.icon.placedSymbolArray.length-1);}const
U=Object.keys(i.horizontal);for(const n of U){const a=i.horizontal[n];if(!M)
{O=$a(a.text);const t=l.layout.get("text-rotate").evaluate(w,{},{k});M=new
Lc(u,r,c,h,p,a,f,d,y,t);}const
o=l===a.positionedLines.length;if(F+=Yc(e,r,a,s,l,y,w,m,z,i.vertical?
t.ai.horizontal:t.ai.horizontalOnly,o?U:[n],D,$,_k),o)break}i.vertical&&
(T+=Yc(e,r,i.vertical,s,l,y,w,m,z,t.ai.vertical,["vertical"],D,L,_k));const
R=M?M.boxStartIndex:e.collisionBoxArray.length,q=M?
M.boxEndIndex:e.collisionBoxArray.length,j=B?
B.boxStartIndex:e.collisionBoxArray.length,N=B?
B.boxEndIndex:e.collisionBoxArray.length,Z=C?
C.boxStartIndex:e.collisionBoxArray.length,K=C?
C.boxEndIndex:e.collisionBoxArray.length,G=P?
P.boxStartIndex:e.collisionBoxArray.length,J=P?
P.boxEndIndex:e.collisionBoxArray.length;let X=-1;const Y=
(t,e)=>t&&t.circleDiameter?
Math.max(t.circleDiameter,e):e;X=Y(M,X),X=Y(B,X),X=Y(C,X),X=Y(P,X);const H=X>-1?
1:0;H&&(X*=I/Vl),e.glyphOffsetArray.length>=Uu.MAX_GLYPHS&&A("Too many glyphs
being rendered in a tile. See https://github.com/mapbox/mapbox-gl-
js/issues/2907"),void
0!==(w.sortKey&&e.addToSortKeyRanges(e.symbolInstances.length,w.sortKey);const
W=Gc(l,w,k),[Q,tt]=function(e,r){const n=e.length,i=null==r?void
0:r.values;if((null==i?void 0:i.length)>0)for(let r=0;r<i.length;r+=2){const
n=i[r+1];e.emplaceBack(t.aq[i[r]],n[0],n[1]);}return [n,e.length]}
(e.textAnchorOffsets,W);e.symbolInstances.emplaceBack(r.x,r.y,D.right>=0?
D.right:-1,D.center>=0?D.center:-1,D.left>=0?
D.left:-1,D.vertical|=-1,$,L,O,R,q,j,N,Z,K,G,J,c,F,T,V,E,H,0,f,X,Q,tt);}
(e,p,l,n,i,a,B,e.layers[0],e.collisionBoxArray,r.index,r.sourceLayerIndex,e.inde
x,g,[w,w,w,w],k,u,v,_I,d,r,s,c,h,o));if("line"===z)for(const t of
_c(r.geometry,0,0,rs,rs)){const
r=Cc(t,b,S,n.vertical||y,i,24,x,e.overscaling,rs);for(const n of

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r)y&&Wc(e,y.text,M,n)||P(t,n);}else if("line-center"===z){for(const t of
r.geometry)if(t.length>1){const e=Mc(t,S,n.vertical||y,i,24,x);e&&P(t,e);}else
if("Polygon"===r.type)for(const t of Co(r.geometry,0)){const
e=Uc(t,16);P(t[0],new Ac(e.x,e.y,0));}else if("LineString"===r.type)for(const t
of r.geometry)P(t,new Ac(t[0].x,t[0].y,0));else if("Point"===r.type)for(const t
of r.geometry)for(const e of t)P([e],new Ac(e.x,e.y,0));}function
Yc(t,e,r,n,i,s,o,l,u,c,h,p,f,d,y){const m=function(t,e,r,n,i,s,o,l){const
u=n.layout.get("text-rotate").evaluate(s,{})*Math.PI/180,c=[];for(const t of
e.positionedLines)for(const n of t.positionedGlyphs){if(!n.rect)continue;const
s=n.rect||{};let h=su+1,p=!0,f=1,d=0;const y=
(i||l)&&n.vertical,m=n.metrics.advance*n.scale/2;if(l&&e.verticalizable&&
(d=t.lineOffset/2-(n.imageName?-(Vl-n.metrics.width*n.scale)/2:(n.scale-
1)*Vl)),n.imageName){const t=o[n.imageName];p=t.sdf,f=t.pixelRatio,h=lu/f;}const
g=i?[n.x+m,n.y]:[0,0];let x=i?[0,0]:[n.x+m+r[0],n.y+r[1]-d],v=[0,0];y&&(v=x,x=
[0,0]);const b=n.metrics.isDoubleResolution?2:1,w=(n.metrics.left-h)*n.scale-
m+x[0],_=(-n.metrics.top-
h)*n.scale+x[1],A=w+s.w/b*n.scale/f,S=+s.h/b*n.scale/f,k=new a(w,_),I=new
a(A,_),z=new a(w,S),M=new a(A,S);if(y){const t=new a(-m,m-pu),e=-
Math.PI/2,r=Vl/2-m,i=new a(5-pu-r,-(n.imageName?r:0)),s=new
a(...v);k._rotateAround(e,t)._add(i)._add(s),I._rotateAround(e,t)._add(i)._add(s),
z._rotateAround(e,t)._add(i)._add(s),M._rotateAround(e,t)._add(i)._add(s);}if(
u){const t=Math.sin(u),e=Math.cos(u),r=[e,-
t,t,e];k._matMult(r),I._matMult(r),z._matMult(r),M._matMult(r);}const C=new
a(0,0),B=new
a(0,0);c.push({tl:k,tr:I,bl:z,br:M,tex:s,writingMode:e.writingMode,glyphOffset:g
,sectionIndex:n.sectionIndex,isSDF:p,pixelOffsetTL:C,pixelOffsetBR:B,minFontScal
eX:0,minFontScaleY:0});}return c}
(0,r,l,i,s,o,n,t.allowVerticalPlacement),g=t.textSizeData;let
x=null;"source"===g.kind?(x=[Cu*i.layout.get("text-size").evaluate(o,
{})],x[0]>Bu&&A(`$t.layerIds[0]`: Value for "text-size" is >= ${Mu}. Reduce
your "text-size".`)): "composite"===g.kind&&(x=
[Cu*d.compositeTextSizes[0].evaluate(o,
{}),y),Cu*d.compositeTextSizes[1].evaluate(o,{},y)],
(x[0]>Bu||x[1]>Bu)&&A(`$t.layerIds[0]`: Value for "text-size" is >= ${Mu}.
Reduce your "text-
size".`)),t.addSymbols(t.text,m,x,l,s,o,c,e,u.lineStartIndex,u.lineLength,f,y);f
or(const e of h)p[e]=t.text.placedSymbolArray.length-1;return
4*m.length}function Hc(t){for(const e in t)return t[e];return null}function
Wc(t,e,r,n){const i=t.compareText;if(e in i){const t=i[e];for(let e=t.length-
1;e>=0;e--)if(n.dist(t[e])<r)return !0}else i[e]=[];return i[e].push(n),!1}const
Qc=
[Int8Array,Uint8Array,Uint8ClampedArray,Int16Array,Uint16Array,Int32Array,Uint32
Array,Float32Array,Float64Array];class th{static from(t){if(!(t instanceof
ArrayBuffer))throw new Error("Data must be an instance of
ArrayBuffer.");const[e,r]=new Uint8Array(t,0,2);if(219!==e)throw new Error("Data
does not appear to be in a KDBush format.");const n=r>>4;if(1!==n)throw new
Error(`Got v${n} data when expected vl.`);const i=Qc[15&r];if(!i)throw new
Error("Unrecognized array type.");const[a]=new Uint16Array(t,2,1),[s]=new
Uint32Array(t,4,1);return new th(s,a,i,t)}constructor(t,e=64,r=Float64Array,n)
{if(isNaN(t)||t<0)throw new Error(`Unexpected numItems value:
${t}.`);this.numItems=+t,this.nodeSize=Math.min(Math.max(+e,2),65535),this.Array
Type=r,this.IndexArrayType=t<65536?Uint16Array:Uint32Array;const
i=Qc.indexOf(this.ArrayType),a=2*t*this.ArrayType.BYTES_PER_ELEMENT,s=t*this.Ind
exArrayType.BYTES_PER_ELEMENT,o=(8-s%8)%8;if(i<0)throw new Error(`Unexpected
typed array class: ${r}.`);n&&n instanceof ArrayBuffer?(this.data=n,this.ids=new
this.IndexArrayType(this.data,8,t),this.coords=new
this.ArrayType(this.data,8+s+o,2*t),this._pos=2*t,this._finished=!0):
(this.data=new ArrayBuffer(8+a+s+o),this.ids=new
this.IndexArrayType(this.data,8,t),this.coords=new
this.ArrayType(this.data,8+s+o,2*t),this._pos=0,this._finished=!1,new
Uint8Array(this.data,0,2).set([219,16+i]),new Uint16Array(this.data,2,1)
[0]=e,new Uint32Array(this.data,4,1)[0]=t);}add(t,e){const r=this._pos>>1;return

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this.ids[r]=r,this.coords[this._pos++]=t,this.coords[this._pos++]=e,r}finish()
{const t=this._pos>>1;if(t!==this.numItems)throw new Error(`Added ${t} items
when expected ${this.numItems}.`);return
eh(this.ids,this.coords,this.nodeSize,0,this.numItems-
1,0),this._finished=!0,this}range(t,e,r,n){if(!this._finished)throw new
Error("Data not yet indexed - call
index.finish().");const{ids:i,coords:a,nodeSize:s}=this,o=[0,i.length-1,0],l=
[];for(;o.length;){const u=o.pop()||0,c=o.pop()||0,h=o.pop()||0;if(c-h<=s)
{for(let s=h;s<=c;s++){const
o=a[2*s],u=a[2*s+1];o>=t&&o<=r&&u>=e&&u<=n&&l.push(i[s]);}continue}const
p=h+c>>1,f=a[2*p],d=a[2*p+1];f>=t&&f<=r&&d>=e&&d<=n&&l.push(i[p]),(0===u?
t<=f:e<=d)&&(o.push(h),o.push(p-1),o.push(1-u)),(0===u?r>=f:n>=d)&&
(o.push(p+1),o.push(c),o.push(1-u));}return l}within(t,e,r)
{if(!this._finished)throw new Error("Data not yet indexed - call
index.finish().");const{ids:n,coords:i,nodeSize:a}=this,s=[0,n.length-1,0],o=
[],l=r*r;for(;s.length;){const u=s.pop()||0,c=s.pop()||0,h=s.pop()||0;if(c-h<=a)
{for(let r=h;r<=c;r++)ah(i[2*r],i[2*r+1],t,e)<=l&&o.push(n[r]);continue}const
p=h+c>>1,f=i[2*p],d=i[2*p+1];ah(f,d,t,e)<=l&&o.push(n[p]),(0===u?t-r<=f:e-
r<=d)&&(s.push(h),s.push(p-1),s.push(1-u)),(0===u?t+r>=f:e+r>=d)&&
(s.push(p+1),s.push(c),s.push(1-u));}return o}}function eh(t,e,r,n,i,a){if(i-
n<=r)return;const s=n+i>>1,rh(t,e,s,n,i,a),eh(t,e,r,n,s-1,1-a),eh(t,e,r,s+1,i,1-
a);}function rh(t,e,r,n,i,a){for(;i>n;){if(i-n>600){const s=i-n+1,o=r-
n+1,l=Math.log(s),u=.5*Math.exp(2*l/3),c=.5*Math.sqrt(1*u*(s-u)/s)*(o-s/2<0?
-1:1);rh(t,e,r,Math.max(n,Math.floor(r-o*u/s+c)),Math.min(i,Math.floor(r+(s-
o)*u/s+c)),a);}const s=e[2*r+a];let
o=n,l=i;for(nh(t,e,n,r),e[2*i+a]>s&&nh(t,e,n,i);o<1;){for(nh(t,e,o,l),o++,l--
;e[2*o+a]<s;)o++;for(;e[2*l+a]>s;)l--;e[2*n+a]===s?nh(t,e,n,l):
(l++,nh(t,e,l,i)),l<=r&&(n=l+1),r<=l&&(i=l-1);}}function nh(t,e,r,n)
{ih(t,r,n),ih(e,2*r,2*n),ih(e,2*r+1,2*n+1);}function ih(t,e,r){const
n=t[e];t[e]=t[r],t[r]=n;}function ah(t,e,r,n){const i=t-r,a=e-n;return
i*i+a*a}var sh;t.bh=void 0,(sh=t.bh||(t.bh=
{})).create="create",sh.load="load",sh.fullLoad="fullLoad";let oh=null,lh=
[];const uh=1e3/60,ch="loadTime",hh="fullLoadTime",ph={mark(t)
{performance.mark(t);},frame(t){const e=t;null!=oh&&lh.push(e-
oh),oh=e;},clearMetrics(){oh=null,lh=
[]},performance.clearMeasures(ch),performance.clearMeasures(hh);for(const e in
t.bh)performance.clearMarks(t.bh[e]);},getPerformanceMetrics()
{performance.measure(ch,t.bh.create,t.bh.load),performance.measure(hh,t.bh.creat
e,t.bh.fullLoad);const e=performance.getEntriesByName(ch)
[0].duration,r=performance.getEntriesByName(hh)
[0].duration,n=lh.length,i=1/(lh.reduce((t,e)=>t+e,0)/n/1e3),a=lh.filter((t=>t
>uh)).reduce((t,e)=>t+(e-uh)/uh,0);return
{loadTime:e,fullLoadTime:r,fps:i,percentDroppedFrames:a/(n+a)*100,totalFrames:n}
};t.$=function(t,e,r){var n,i,a,s,o,l,u,c,h,p,f,d,y=r[0],m=r[1],g=r[2];return
e===t?
(t[12]=e[0]*y+e[4]*m+e[8]*g+e[12],t[13]=e[1]*y+e[5]*m+e[9]*g+e[13],t[14]=e[2]*y+
e[6]*m+e[10]*g+e[14],t[15]=e[3]*y+e[7]*m+e[11]*g+e[15]):
(i=e[1],a=e[2],s=e[3],o=e[4],l=e[5],u=e[6],c=e[7],h=e[8],p=e[9],f=e[10],d=e[11],
t[0]=n=e[0],t[1]=i,t[2]=a,t[3]=s,t[4]=o,t[5]=l,t[6]=u,t[7]=c,t[8]=h,t[9]=p,t[10]
=f,t[11]=d,t[12]=n*y+o*m+h*g+e[12],t[13]=i*y+l*m+p*g+e[13],t[14]=a*y+u*m+f*g+e[1
4],t[15]=s*y+c*m+d*g+e[15]),t},t.A=zs,t.B=Ze,t.C=class{constructor(t,e,r)
{this.receive=t=>{const e=t.data,r=e.id;if(r&&
(!e.targetMapId||this.mapId===e.targetMapId))if("<cancel>"===e.type){delete
this.tasks[r];const t=this.cancelCallbacks[r];delete
this.cancelCallbacks[r],t&&t();}else I()||e.mustQueue?
(this.tasks[r]=e,this.taskQueue.push(r),this.invoker.trigger()):this.processTask
(r,e);},this.process=()=>{if(!this.taskQueue.length)return;const
t=this.taskQueue.shift(),e=this.tasks[t];delete
this.tasks[t],this.taskQueue.length&&this.invoker.trigger(),e&&this.processTask(
t,e);},this.target=t,this.parent=e,this.mapId=r,this.callbacks={},this.tasks=
{},this.taskQueue=[],this.cancelCallbacks={},this.invoker=new
Qu(this.process),this.target.addEventListener("message",this.receive,!1),this.gl

```



```

obalScope=I()?t:window;}send(t,e,r,n,i=!1){const
a=Math.round(1e18*Math.random()).toString(36).substring(0,10);r&&
(this.callbacks[a]=r);const s=[],o=
{id:a,type:t,hasCallback:!!r,targetMapId:n,mustQueue:i,sourceMapId:this.mapId,da
ta:Dn(e,s)};return this.target.postMessage(o,{transfer:s}},{cancel:()=>
{r&&delete this.callbacks[a],this.target.postMessage({id:a,type:"
<cancel>",targetMapId:n,sourceMapId:this.mapId});}}processTask(t,e){if("
<response>"===e.type){const r=this.callbacks[t];delete this.callbacks[t],r&&
(e.error?r(On(e.error)):r(null,On(e.data)));}else {let r=!1;const n=
[],i=e.hasCallback?(e,i)=>{r=!0,delete this.cancelCallbacks[t];const a=
{id:t,type:"<response>",sourceMapId:this.mapId,error:e?
Dn(e):null,data:Dn(i,n)};this.target.postMessage(a,{transfer:n});}:t=>
{r=!0};let a=null;const
s=On(e.data);if(this.parent[e.type])a=this.parent[e.type]
(e.sourceMapId,s,i);else if("getWorkerSource"in this.parent){const
t=e.type.split(".");a=this.parent.getWorkerSource(e.sourceMapId,t[0],s.source)
[t[1]](s,i);}else i(new Error(`Could not find function
${e.type}`));!r&&a&&a.cancel&&(this.cancelCallbacks[t]=a.cancel);}remove()
{this.invoker.remove(),this.target.removeEventListener("message",this.receive,!1
)}},t.D=gi,t.E=J,t.F=function(t,e){const r={};for(let n=0;n<e.length;n++){const
i=e[n];i in t&&(r[i]=t[i]);}return
r},t.G=ic,t.H=ac,t.I=uu,t.J=h,t.K=p,t.L=ec,t.M=B,t.N=rs,t.O=fc,t.P=a,t.Q=Mi,t.R=
qs,t.S=za,t.T=hi,t.U=uc,t.V=class extends Vi{},t.W=hc,t.X=function(t,e){const
r=window.document.createElement("video");r.muted=!0,r.onloadstart=function()
{e(null,r)};for(let e=0;e<t.length;e++){const
n=window.document.createElement("source");j(t[e])||
(r.crossOrigin="Anonymous"),n.src=t[e],r.appendChild(n);}return {cancel:()=>>
{}}},t.Y=ot,t.Z=function(){var t=new zs(16);return zs!=Float32Array&&
(t[1]=0,t[2]=0,t[3]=0,t[4]=0,t[6]=0,t[7]=0,t[8]=0,t[9]=0,t[11]=0,t[12]=0,t[13]=0
,t[14]=0),t[0]=1,t[5]=1,t[10]=1,t[15]=1,t},t._=e,t.a=M,t.a$=class extends
Hi{},t.a0=function(t,e,r){var n=r[0],i=r[1],a=r[2];return
t[0]=e[0]*n,t[1]=e[1]*n,t[2]=e[2]*n,t[3]=e[3]*n,t[4]=e[4]*i,t[5]=e[5]*i,t[6]=e[6
]*i,t[7]=e[7]*i,t[8]=e[8]*a,t[9]=e[9]*a,t[10]=e[10]*a,t[11]=e[11]*a,t[12]=e[12],
t[13]=e[13],t[14]=e[14],t[15]=e[15],t},t.a1=Cs,t.a2=function(){return
x++},t.a3=ta,t.a4=Uu,t.a5=function()
{oi.isLoading()||oi.isLoaded()||"deferred"!==ai()||si();},t.a6=Nr,t.a7=ss,t.a8=1
i,t.a9=gc,t.aA=ri,t.aB=function(t){t=t.slice();const
e=Object.create(null);for(let r=0;r<t.length;r++)e[t[r].id]=t[r];for(let
r=0;r<t.length;r++)"ref"in t[r]&&(t[r]=H(t[r],e[t[r].ref]));return
t},t.aC=function(t){if("custom"===t.type)return new Wu(t);switch(t.type)
{case"background":return new Ju(t);case"circle":return new
Es(t);case"fill":return new Lo(t);case"fill-extrusion":return new
ol(t);case"heatmap":return new Ns(t);case"hillshade":return new
Gs(t);case"line":return new Al(t);case"raster":return new
Hu(t);case"symbol":return new Zu(t)}},t.aD=w,t.aE=function(t,e){if(!t)return
[command:Q.setStyle,args:[e]];let r=[];try{if(!W(t.version,e.version))return
[command:Q.setStyle,args:
[e]];W(t.center,e.center)||r.push({command:Q.setCenter,args:
[e.center]}),W(t.zoom,e.zoom)||r.push({command:Q.setZoom,args:
[e.zoom]}),W(t.bearing,e.bearing)||r.push({command:Q.setBearing,args:
[e.bearing]}),W(t.pitch,e.pitch)||r.push({command:Q.setPitch,args:
[e.pitch]}),W(t.sprite,e.sprite)||r.push({command:Q.setSprite,args:
[e.sprite]}),W(t.glyphs,e.glyphs)||r.push({command:Q.setGlyphs,args:
[e.glyphs]}),W(t.transition,e.transition)||r.push({command:Q.setTransition,args:
[e.transition]}),W(t.light,e.light)||r.push({command:Q.setLight,args:
[e.light]});const n={},i=[];!function(t,e,r,n){let i;for(i in e=e||{},t=t||
{})Object.prototype.hasOwnProperty.call(t,i)&&
(Object.prototype.hasOwnProperty.call(e,i)||et(i,r,n));for(i in
e)Object.prototype.hasOwnProperty.call(e,i)&&
(Object.prototype.hasOwnProperty.call(t,i)?W(t[i],e[i])||
("geojson"===t[i].type&&"geojson"===e[i].type&&nt(t,e,i)?
r.push({command:Q.setGeoJSONSourceData,args:

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```

[i,e[i].data]}}:rt(i,e,r,n)):tt(i,e,r));}(t.sources,e.sources,i,n);const a=
[];t.layers&&t.layers.forEach((t=>{n[t.source]?
r.push({command:Q.removeLayer,args:
[t.id]}):a.push(t)})),r=r.concat(i),function(t,e,r){e=e||[];const n=(t=t||
[]).map(at),i=e.map(at),a=t.reduce(st,{}),s=e.reduce(st,
{}),o=n.slice(),l=Object.create(null);let
u,c,h,p,f,d,y;for(u=0,c=0;u<n.length;u++)h=n[u],Object.prototype.hasOwnProperty.
call(s,h)?c++:(r.push({command:Q.removeLayer,args:
[h]}),o.splice(o.indexOf(h,c),1));for(u=0,c=0;u<i.length;u++)h=i[i.length-1-
u],o[o.length-1-u]!==h&&(Object.prototype.hasOwnProperty.call(a,h)?
(r.push({command:Q.removeLayer,args:[h]}),o.splice(o.lastIndexOf(h,o.length-
c),1)):c++,d=o[o.length-u],r.push({command:Q.addLayer,args:
[s[h],d]}),o.splice(o.length-
u,0,h),l[h]=!0);for(u=0;u<i.length;u++)if(h=i[u],p=a[h],f=s[h],!l[h]&&!W(p,f))if
(W(p.source,f.source)&&W(p["source-layer"],f["source-layer"])&&W(p.type,f.type))
for(y in
it(p.layout,f.layout,r,h,null,Q.setLayoutProperty),it(p.paint,f.paint,r,h,null,Q
.setPaintProperty),W(p.filter,f.filter)||r.push({command:Q.setFilter,args:
[h,f.filter]}),W(p.minzoom,f.minzoom)&&W(p.maxzoom,f.maxzoom)||r.push({command:Q
.setLayerZoomRange,args:
[h,f.minzoom,f.maxzoom]}),p)Object.prototype.hasOwnProperty.call(p,y)&&"layout"!
==y&&"paint"!==y&&"filter"!==y&&"metadata"!==y&&"minzoom"!==y&&"maxzoom"!==y&&
(0===y.indexOf("paint."))?
it(p[y],f[y],r,h,y.slice(6),Q.setPaintProperty):W(p[y],f[y])||r.push({command:Q.
setLayerProperty,args:[h,y,f[y]]});for(y in
f)Object.prototype.hasOwnProperty.call(f,y)&&!Object.prototype.hasOwnProperty.ca
ll(p,y)&&"layout"!==y&&"paint"!==y&&"filter"!==y&&"metadata"!==y&&"minzoom"!==y&&
"maxzoom"!==y&&(0===y.indexOf("paint."))?
it(p[y],f[y],r,h,y.slice(6),Q.setPaintProperty):W(p[y],f[y])||r.push({command:Q.
setLayerProperty,args:[h,y,f[y]]});}else r.push({command:Q.removeLayer,args:
[h]}),d=o[o.lastIndexOf(h)+1],r.push({command:Q.addLayer,args:[f,d]});}
(a,e.layers,r);}catch(t){console.warn("Unable to compute style diff:",t),r=
[{command:Q.setStyle,args:[e]}];}return r},t.aF=function(t){const e=
[],r=t.id;return void 0===r&&e.push({message:`layers.${r}: missing required
property "id"`}),void 0===t.render&&e.push({message:`layers.${r}: missing
required method
"render"`}),t.renderingMode&&"2d"!==t.renderingMode&&"3d"!==t.renderingMode&&e.p
ush({message:`layers.${r}: property "renderingMode" must be either "2d" or
"3d"`}),e},t.aG=function t(e,r){if(Array.isArray(e))
{if(!Array.isArray(r)||e.length!==r.length)return !1;for(let
n=0;n<e.length;n++)if(!t(e[n],r[n]))return !1;return !0}if("object"===typeof
e&&null!==e&&null!==r){if("object"!==typeof r)return
!1;if(Object.keys(e).length===Object.keys(r).length)return !1;for(const n in
e)if(!t(e[n],r[n]))return !1;return !0}return
e===r},t.aH=v,t.aI=b,t.aJ=ii,t.aK=function(t){return
t({pluginStatus:ti,pluginURL:ei}),ii.on("pluginStateChange",t),t},t.aL=class
extends Ra{constructor(t,e){super(t,e),this.current=0;}set(t){this.current!==t&&
(this.current=t,this.gl.uniform1i(this.location,t));}},t.aM=qa,t.aN=class
extends Ra{constructor(t,e){super(t,e),this.current=Za;}set(t)
{if(t[12]!==this.current[12]||t[0]!==this.current[0])return this.current=t,void
this.gl.uniformMatrix4fv(this.location,!1,t);for(let
e=1;e<16;e++)if(t[e]!==this.current[e])
{this.current=t,this.gl.uniformMatrix4fv(this.location,!1,t);break}}},t.aO=ja,t.
aP=class extends Ra{constructor(t,e){super(t,e),this.current=[0,0,0];}set(t)
{t[0]===this.current[0]&&t[1]===this.current[1]&&t[2]===this.current[2]||
(this.current=t,this.gl.uniform3f(this.location,t[0],t[1],t[2]));}},t.aQ=class
extends Ra{constructor(t,e){super(t,e),this.current=[0,0];}set(t)
{t[0]===this.current[0]&&t[1]===this.current[1]||
(this.current=t,this.gl.uniform2f(this.location,t[0],t[1]));}},t.aR=Na,t.aS=func
tion(t,e,r,n,i,a,s){var o=1/(e-r),l=1/(n-i),u=1/(a-s);return
t[0]=-2*o,t[1]=0,t[2]=0,t[3]=0,t[4]=0,t[5]=-2*l,t[6]=0,t[7]=0,t[8]=0,t[9]=0,t[10
]=2*u,t[11]=0,t[12]=(e+r)*o,t[13]=(i+n)*l,t[14]=

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(s+a)*u,t[15]=1,t},t.aT=Yt,t.aU=Ps,t.aV=class extends qi{},t.aW=C1,t.aX=class
extends Ni{},t.aY=function(t){return t<=1?
1:Math.pow(2,Math.ceil(Math.log(t)/Math.LN2))},t.aZ=js,t.a_ =ha,t.aa=function(t)
{const e={};if(t.replace(/(?:^|(?!\s*\s))(\^\x00-\x20\\(<>@\\,;\:\\"\/\[\]\?
\\=\{\}\x7F)+)(?:\=(?:(^\x00-\x20\\(<>@\\,;\:\\"\/\[\]\?\\=\{\}\x7F)+)|(?:\"(?:
[^\\"\\]|\\.)*\\\")))/g,((t,r,n,i)=>{const a=n||i;return
e[r]=!a||a.toLowerCase(),""})),e["max-age"]){const t=parseInt(e["max-
age"],10);isNaN(t)?delete e["max-age"]:e["max-age"]=t;}return
e},t.ab=function(t,e){const r=[];for(const n in t)n in e||r.push(n);return
r},t.ac=function(t){if(null==z){const e=t.navigator?
t.navigator.userAgent:null;z=!t.safari||!(e||
(/b(iPad|iPhone|iPod)\b/.test(e)||e.match("Safari")&&!e.match("Chrome")));}retu
rn z},t.ad=y,t.ae=function(t,e,r){var
n=Math.sin(r),i=Math.cos(r),a=e[0],s=e[1],o=e[2],l=e[3],u=e[4],c=e[5],h=e[6],p=e
[7];return e!=t&&
(t[8]=e[8],t[9]=e[9],t[10]=e[10],t[11]=e[11],t[12]=e[12],t[13]=e[13],t[14]=e[14]
,t[15]=e[15]),t[0]=a*i+u*n,t[1]=s*i+c*n,t[2]=o*i+h*n,t[3]=l*i+p*n,t[4]=u*i-
a*n,t[5]=c*i-s*n,t[6]=h*i-o*n,t[7]=p*i-l*n,t},t.af=function(t){var e=new
zs(16);return
e[0]=t[0],e[1]=t[1],e[2]=t[2],e[3]=t[3],e[4]=t[4],e[5]=t[5],e[6]=t[6],e[7]=t[7],
e[8]=t[8],e[9]=t[9],e[10]=t[10],e[11]=t[11],e[12]=t[12],e[13]=t[13],e[14]=t[14],
e[15]=t[15],e},t.ag=Vs,t.ah=function(t,e){let
r=0,n=0;if("constant"===t.kind)n=t.layoutSize;else if("source"!==t.kind)
{const{interpolationType:i,minZoom:a,maxZoom:s}=t,o=i?
y(Ke.interpolationFactor(i,e,a,s),0,1):0;"camera"===t.kind?
n=Ze.number(t.minSize,t.maxSize,o):r=o;}return
{uSizeT:r,uSize:n}},t.aj=function(t,{uSize:e,uSizeT:r},
{lowerSize:n,upperSize:i}){return "source"===t.kind?n/Cu:"composite"===t.kind?
Ze.number(n/Cu,i/Cu,r):e},t.ak=$u,t.al=function(t,e,r,n){const i=e.y-t.y,s=e.x-
t.x,o=n.y-r.y,l=n.x-r.x,u=o*s-l*i;if(0===u)return null;const c=(l*(t.y-r.y)-o*
(t.x-r.x))/u;return new
a(t.x+c*s,t.y+c*i)},t.am=_c,t.an=us,t.ao=Ms,t.ap=Vl,t.ar=Vu,t.as=function(t,e)
{var
r=e[0],n=e[1],i=e[2],a=e[3],s=e[4],o=e[5],l=e[6],u=e[7],c=e[8],h=e[9],p=e[10],f=
e[11],d=e[12],y=e[13],m=e[14],g=e[15],x=r*o-n*s,v=r*l-i*s,b=r*u-a*s,w=n*l-
i*o,_=n*u-a*o,A=i*u-a*l,S=c*y-h*d,k=c*m-p*d,I=c*g-f*d,z=h*m-p*y,M=h*g-f*y,C=p*g-
f*m,B=x*C-v*M+b*z+w*I-_k+A*S;return B?(t[0]=(o*C-l*M+u*z)*(B=1/B),t[1]=(i*M-
n*C-a*z)*B,t[2]=(y*A-m*_+g*w)*B,t[3]=(p*_h*A-f*w)*B,t[4]=(l*I-s*C-u*k)*B,t[5]=
(r*C-i*I+a*k)*B,t[6]=(m*b-d*A-g*v)*B,t[7]=(c*A-p*b+f*v)*B,t[8]=(s*M-
o*I+u*S)*B,t[9]=(n*I-r*M-a*S)*B,t[10]=(d*_y*b+g*x)*B,t[11]=(h*b-c*_-
f*x)*B,t[12]=(o*k-s*z-l*S)*B,t[13]=(r*z-n*k+i*S)*B,t[14]=(y*v-d*w-m*x)*B,t[15]=
(c*w-h*v+p*x)*B,t):null},t.at=Jc,t.au=Su,t.av=th,t.aw=function(){const t=
{e=X.$version;for(const r in X.$root){const n=X.$root[r];if(n.required){let
i=null;i="version"===r?e:"array"===n.type?[]:{},null!=i&&(t[r]=i);}}return
t},t.ax=Q,t.ay=Un,t.az=D,t.b=function(t,e){const r=new Blob([new Uint8Array(t)],
{type:"image/png"});createImageBitmap(r).then((t=>{e(null,t);})).catch((t=>
{e(new Error(`Could not load image because of ${t.message}. Please make sure to
use a supported image type such as PNG or JPEG. Note that SVGs are not
supported.`))});});t.b0=Aa,t.b1=function(t,e){var
r=t[0],n=t[1],i=t[2],a=t[3],s=t[4],o=t[5],l=t[6],u=t[7],c=t[8],h=t[9],p=t[10],f=
t[11],d=t[12],y=t[13],m=t[14],g=t[15],x=e[0],v=e[1],b=e[2],w=e[3],_=e[4],A=e[5],
S=e[6],k=e[7],I=e[8],z=e[9],M=e[10],C=e[11],B=e[12],P=e[13],V=e[14],E=e[15];retu
rn Math.abs(r-x)<=Is*Math.max(1,Math.abs(r),Math.abs(x))&&Math.abs(n-v)
<=Is*Math.max(1,Math.abs(n),Math.abs(v))&&Math.abs(i-b)
<=Is*Math.max(1,Math.abs(i),Math.abs(b))&&Math.abs(a-w)
<=Is*Math.max(1,Math.abs(a),Math.abs(w))&&Math.abs(s-_ )
<=Is*Math.max(1,Math.abs(s),Math.abs(_))&&Math.abs(o-A)
<=Is*Math.max(1,Math.abs(o),Math.abs(A))&&Math.abs(l-S)
<=Is*Math.max(1,Math.abs(l),Math.abs(S))&&Math.abs(u-k)
<=Is*Math.max(1,Math.abs(u),Math.abs(k))&&Math.abs(c-I)
<=Is*Math.max(1,Math.abs(c),Math.abs(I))&&Math.abs(h-z)
<=Is*Math.max(1,Math.abs(h),Math.abs(z))&&Math.abs(p-M)

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<=Is*Math.max(1,Math.abs(p),Math.abs(M))&&Math.abs(f-C)
<=Is*Math.max(1,Math.abs(f),Math.abs(C))&&Math.abs(d-B)
<=Is*Math.max(1,Math.abs(d),Math.abs(B))&&Math.abs(y-P)
<=Is*Math.max(1,Math.abs(y),Math.abs(P))&&Math.abs(m-V)
<=Is*Math.max(1,Math.abs(m),Math.abs(V))&&Math.abs(g-E)
<=Is*Math.max(1,Math.abs(g),Math.abs(E))},t.b2=function(t,e){return
t[0]=e[0],t[1]=e[1],t[2]=e[2],t[3]=e[3],t[4]=e[4],t[5]=e[5],t[6]=e[6],t[7]=e[7],
t[8]=e[8],t[9]=e[9],t[10]=e[10],t[11]=e[11],t[12]=e[12],t[13]=e[13],t[14]=e[14],
t[15]=e[15],t},t.b3=function(t,e,r){return
t[0]=e[0]*r[0],t[1]=e[1]*r[1],t[2]=e[2]*r[2],t[3]=e[3]*r[3],t},t.b4=function(t,e
){return
t[0]*e[0]+t[1]*e[1]+t[2]*e[2]+t[3]*e[3]},t.b5=m,t.b6=pc,t.b7=sc,t.b8=function(t,
e,r,n,i){var a,s=1/Math.tan(e/2);return
t[0]=s/r,t[1]=0,t[2]=0,t[3]=0,t[4]=0,t[5]=s,t[6]=0,t[7]=0,t[8]=0,t[9]=0,t[11]=-1
,t[12]=0,t[13]=0,t[15]=0,null!=i&&i!=1/0?(t[10]=(i+n)*(a=1/(n-
i)),t[14]=2*i*n*a):(t[10]=-1,t[14]=-2*n),t},t.b9=function(t,e,r){var
n=Math.sin(r),i=Math.cos(r),a=e[4],s=e[5],o=e[6],l=e[7],u=e[8],c=e[9],h=e[10],p=
e[11];return e!=t&&
(t[0]=e[0],t[1]=e[1],t[2]=e[2],t[3]=e[3],t[12]=e[12],t[13]=e[13],t[14]=e[14],t[1
5]=e[15]),t[4]=a*i+u*n,t[5]=s*i+c*n,t[6]=o*i+h*n,t[7]=l*i+p*n,t[8]=u*i-
a*n,t[9]=c*i-s*n,t[10]=h*i-o*n,t[11]=p*i-
l*n,t},t.bA=n,t.bB=El,t.bC=$r,t.bD=oi,t.ba=f,t.bb=d,t.bc=function(t,e){return
t[0]=e[0],t[1]=0,t[2]=0,t[3]=0,t[4]=0,t[5]=e[1],t[6]=0,t[7]=0,t[8]=0,t[9]=0,t[10
]=e[2],t[11]=0,t[12]=0,t[13]=0,t[14]=0,t[15]=1,t},t.bd=class extends
Pi{},t.be=tc,t.bf=oc,t.bg=ph,t.bi=L,t.bj=function(t,e,r=!1)
{if(ti===Yn||ti===Hn||ti===Wn)throw new Error("setRTLTextPlugin cannot be called
multiple
times.");ei=T.resolveURL(t,ti=Yn,Qn=e,ni(),r||si());},t.bk=ai,t.bl=function(t,e)
{const r={};for(let n=0;n<t.length;n++){const i=e&&e[t[n].id]||Qr(t[n]);e&&
(e[t[n].id]=i);let a=r[i];a||(a=r[i]=[]),a.push(t[n]);}const n=[];for(const t in
r)n.push(r[t]);return n},t.bm=$n,t.bn=mc,t.bo=xc,t.bp=cu,t.bq=function(e)
{e.bucket.createArrays(),e.bucket.tilePixelRatio=rs/(512*e.bucket.overscaling),e
.bucket.compareText={},e.bucket.iconsNeedLinear=!1;const
r=e.bucket.layers[0],n=r.layout,i=r.unevaluatedLayout._values,a=
{layoutIconSize:i["icon-size"].possiblyEvaluate(new
li(e.bucket.zoom+1),e.canonical),layoutTextSize:i["text-
size"].possiblyEvaluate(new
li(e.bucket.zoom+1),e.canonical),textMaxSize:i["text-size"].possiblyEvaluate(new
li(18))};if("composite"===e.bucket.textSizeData.kind)
{const{minZoom:t,maxZoom:r}=e.bucket.textSizeData;a.compositeTextSizes=[i["text-
size"].possiblyEvaluate(new li(t),e.canonical),i["text-
size"].possiblyEvaluate(new
li(r),e.canonical)];}if("composite"===e.bucket.iconSizeData.kind)
{const{minZoom:t,maxZoom:r}=e.bucket.iconSizeData;a.compositeIconSizes=[i["icon-
size"].possiblyEvaluate(new li(t),e.canonical),i["icon-
size"].possiblyEvaluate(new li(r),e.canonical)];}const s=n.get("text-line-
height")*Vl,o="viewport"!==n.get("text-rotation-
alignment")&&"point"!==n.get("symbol-placement"),l=n.get("text-keep-
upright"),u=n.get("text-size");for(const i of e.bucket.features){const
c=n.get("text-font").evaluate(i,{},e.canonical).join(","),h=u.evaluate(i,
{},e.canonical),p=a.layoutTextSize.evaluate(i,
{},e.canonical),f=a.layoutIconSize.evaluate(i,{},e.canonical),d={horizontal:
{},vertical:void 0},y=i.text;let m,g=[0,0];if(y){const
a=y.toString(),u=n.get("text-letter-spacing").evaluate(i,
{},e.canonical)*Vl,f=jn(a)?u:0,m=n.get("text-anchor").evaluate(i,
{},e.canonical),x=Gc(r,i,e.canonical);if(!x){const t=n.get("text-radial-
offset").evaluate(i,{},e.canonical);g=t?Kc(m,[t*Vl,Zc]):n.get("text-
offset").evaluate(i,{},e.canonical).map((t=>t*Vl));}let
v=o?"center":n.get("text-justify").evaluate(i,{},e.canonical);const
b=n.get("symbol-placement"),w="point"===b?n.get("text-max-width").evaluate(i,
{},e.canonical)*Vl:0,_=()=>{e.bucket.allowVerticalPlacement&&qn(a)&&
(d.vertical=yu(y,e.glyphMap,e.glyphPositions,e.imagePositions,c,w,s,m,"left",f,g

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,t.ai.vertical,!0,b,p,h));};if(!o&&x){const r=new Set;if("auto"===v)for(let
t=0;t<x.values.length;t+=2)r.add(Jc(x.values[t]));else r.add(v);let
n=!1;for(const i of
r)if(!d.horizontal[i])if(n)d.horizontal[i]=d.horizontal[0];else {const
r=yu(y,e.glyphMap,e.glyphPositions,e.imagePositions,c,w,s,"center",i,f,g,t.ai.ho
rizontal,!1,b,p,h);r&&
(d.horizontal[i]=r,n=!1===r.positionedLines.length);}_();}else {"auto"===v&&
(v=Jc(m));const
r=yu(y,e.glyphMap,e.glyphPositions,e.imagePositions,c,w,s,m,v,f,g,t.ai.horizonta
l,!1,b,p,h);r&&(d.horizontal[v]=r),_(),qn(a)&&o&&l&&
(d.vertical=yu(y,e.glyphMap,e.glyphPositions,e.imagePositions,c,w,s,m,v,f,g,t.ai
.vertical,!1,b,p,h));}}let x=!1;if(i.icon&&i.icon.name){const
t=e.imageMap[i.icon.name];t&&(m=Iu(e.imagePositions[i.icon.name],n.get("icon-
offset").evaluate(i,{},e.canonical),n.get("icon-anchor").evaluate(i,
{},{},e.canonical)),x=!t.sdf,void 0===e.bucket.sdfIcons?
e.bucket.sdfIcons=x:e.bucket.sdfIcons!==x&&A("Style sheet warning: Cannot mix
SDF and non-SDF icons in one buffer"),
(t.pixelRatio!==e.bucket.pixelRatio||0!==n.get("icon-rotate").constantOr(1))&&
(e.bucket.iconsNeedLinear=!0));}const
v=Hc(d.horizontal)||d.vertical;e.bucket.iconsInText=!!v&&v.iconsInText,
(v||m)&&Xc(e.bucket,i,d,m,e.imageMap,a,p,f,g,x,e.canonical);}e.showCollisionBoxe
s&&e.bucket.generateCollisionDebugBuffers();},t.br=gl,t.bs=Eo,t.bt=rl,t.bu=class
{constructor(t){this._marks={start:[t.url,"start"].join("#"),end:
[t.url,"end"].join("#"),measure:t.url.toString()},performance.mark(this._marks.s
tart);}finish(){performance.mark(this._marks.end);}let
t=performance.getEntriesByName(this._marks.measure);return 0===t.length&&
(performance.measure(this._marks.measure,this._marks.start,this._marks.end),t=pe
rformance.getEntriesByName(this._marks.measure),performance.clearMarks(this._mar
ks.start),performance.clearMarks(this._marks.end),performance.clearMeasures(this
._marks.measure)),t}},t.bv=eu,t.bw=Ro,t.bx=function(t,r,n,i,a){return
e(this,void 0,void 0,(function*(){if(p())try{return yield B(t,r,n,i,a)}catch(t)
{return function(t,e,r,n,i){const a=t.width,s=t.height;P&&V||(P=new
OffscreenCanvas(a,s),V=P.getContext("2d",
{willReadFrequently:!0})),P.width=a,P.height=s,V.drawImage(t,0,0,a,s);const
o=V.getImageData(e,r,n,i);return V.clearRect(0,0,a,s),o.data}
(t,r,n,i,a))})),t.by=yc,t.bz=r,t.c=$,t.d=function(t,e){const r=new
Image;r.onload=(()=>>
{e(null,r),URL.revokeObjectURL(r.src),r.onload=null>window.requestAnimationFrame
((()=>>{r.src=C;}));},r.onerror=(()=>>e(new Error("Could not load image. Please
make sure to use a supported image type such as PNG or JPEG. Note that SVGs are
not supported."));const n=new Blob([new Uint8Array(t)],
{type:"image/png"});r.src=t.byteLength?
URL.createObjectURL(n):C;},t.e=g,t.f=function(t,e){return R(g(t,
{type:"json"}),e)},t.g=O,t.h=T,t.i=I,t.j=G,t.k=K,t.l=q,t.m=R,t.n=function(t)
{return new eu(t).readFields(nu,[])},t.o=function(t,e,r){if(!t.length)return
r(null,[]);let n=t.length;const i=new Array(t.length);let
a=null;t.forEach((t,s)=>{e(t,((t,e)=>{t&&(a=t),i[s]=e,0===--
n&&r(a,i);}));});},t.p=ou,t.q=Rs,t.r=_i,t.s=j,t.t=Bn,t.u=Rn,t.v=X,t.w=A,t.x=En,
t.y=Cn,t.z=function([t,e,r]){return e+=90,e*=Math.PI/180,r*=Math.PI/180,
{x:t*Math.cos(e)*Math.sin(r),y:t*Math.sin(e)*Math.sin(r),z:t*Math.cos(r)}};});
define(["./shared"],(function(e){"use strict";class t{constructor(e)
{this.keyCache={},e&&this.replace(e);}replace(e){this._layerConfigs=
{},this._layers={},this.update(e,[]);}update(t,i){for(const i of t)
{this._layerConfigs[i.id]=i;const
t=this._layers[i.id]=e.aC(i);t._featureFilter=e.a6(t.filter),this.keyCache[i.id]
&&delete this.keyCache[i.id];}for(const e of i)delete this.keyCache[e],delete
this._layerConfigs[e],delete this._layers[e];this.familiesBySource={};const
o=e.bl(Object.values(this._layerConfigs),this.keyCache);for(const e of o){const
t=e.map((e=>this._layers[e.id])),i=t[0];if("none"===i.visibility)continue;const
o=i.source||"";let r=this.familiesBySource[o];r||(r=this.familiesBySource[o]=
{});const s=i.sourceLayer||"_geojsonTileLayer";let n=r[s];n||(n=r[s]=

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[]),n.push(t);}}class i{constructor(t){const i={},o=[];for(const e in t){const
r=t[e],s=i[e]={};for(const e in r){const
t=r[+e];if(!t||0===t.bitmap.width||0===t.bitmap.height)continue;const i=
{x:0,y:0,w:t.bitmap.width+2,h:t.bitmap.height+2};o.push(i),s[e]=
{rect:i,metrics:t.metrics}};}}const{w:r,h:s}=e.p(o),n=new
e.q({width:r||1,height:s||1});for(const o in t){const r=t[o];for(const t in r)
{const s=r[+t];if(!s||0===s.bitmap.width||0===s.bitmap.height)continue;const
a=i[o][t].rect,e.q.copy(s.bitmap,n,{x:0,y:0},
{x:a.x+1,y:a.y+1},s.bitmap)}}this.image=n,this.positions=i;}}e.bm("GlyphAtlas",
i);class o{constructor(t){this.tileID=new
e.O(t.tileID.overscaledZ,t.tileID.wrap,t.tileID.canonical.z,t.tileID.canonical.x
,t.tileID.canonical.y),this.uid=t.uid,this.zoom=t.zoom,this.pixelRatio=t.pixelRa
tio,this.tileSize=t.tileSize,this.source=t.source,this.overscaling=this.tileID.o
verscaleFactor(),this.showCollisionBoxes=t.showCollisionBoxes,this.collectResour
ceTiming=!t.collectResourceTiming,this.returnDependencies=!t.returnDependencies,
this.promoteId=t.promoteId,this.inFlightDependencies=
[],this.dependencySentinel=-1;}}parse(t,o,s,n,a)
{this.status="parsing",this.data=t,this.collisionBoxArray=new e.a3;const l=new
e.bn(Object.keys(t.layers).sort()),c=new
e.bo(this.tileID,this.promoteId);c.bucketLayerIDs=[];const h={},u=
{featureIndex:c,iconDependencies:{},patternDependencies:{},glyphDependencies:
{},availableImages:s},d=o.familiesBySource[this.source];for(const i in d){const
o=t.layers[i];if(!o)continue;l===o.version&&e.w(`Vector tile source
"${this.source}" layer "${i}" does not use vector tile spec v2 and therefore may
have some rendering errors.`);const n=l.encode(i),a=[];for(let
e=0;e<o.length;e++){const
t=o.feature(e),r=c.getId(t,i);a.push({feature:t,id:r,index:e,sourceLayerIndex:n}
);}for(const t of d[i]){const i=t[0];i.source!==this.source&&e.w(`layer.source =
${i.source} does not equal this.source =
${this.source}`),i.minzoom&&this.zoom<Math.floor(i.minzoom)||i.maxzoom&&this.zoo
m>=i.maxzoom||"none"!==i.visibility&&(r(t,this.zoom,s),
(h[i.id]=i.createBucket({index:c.bucketLayerIDs.length,layers:t,zoom:this.zoom,p
ixelRatio:this.pixelRatio,overscaling:this.overscaling,collisionBoxArray:this.co
llisionBoxArray,sourceLayerIndex:n,sourceID:this.source})).populate(a,u,this.til
eID.canonical),c.bucketLayerIDs.push(t.map((e=>e.id))))};let p,f,g,m;const
y=e.aH(u.glyphDependencies,
(e=>Object.keys(e).map(Number)));this.inFlightDependencies.forEach((e=>null==e?
void 0:e.cancel()));this.inFlightDependencies=[];const
v=++this.dependencySentinel;Object.keys(y).length?
this.inFlightDependencies.push(n.send("getGlyphs",
{uid:this.uid,stacks:y,source:this.source,tileID:this.tileID,type:"glyphs"},
((e,t)=>{v===this.dependencySentinel&&(p||(p=e,f=t,b.call(this))))):f=
{};const w=Object.keys(u.iconDependencies);w.length?
this.inFlightDependencies.push(n.send("getImages",
{icons:w,source:this.source,tileID:this.tileID,type:"icons"},((e,t)=>
{v===this.dependencySentinel&&(p||(p=e,g=t,b.call(this))))):g={};const
x=Object.keys(u.patternDependencies);function b(){if(p)return a(p);if(f&&g&&m)
{const t=new i(f),o=new e.bp(g,m);for(const i in h){const n=h[i];n instanceof
e.a4?
(r(n.layers,this.zoom,s),e.bq({bucket:n,glyphMap:f,glyphPositions:t.positions,im
ageMap:g,imagePositions:o.iconPositions,showCollisionBoxes:this.showCollisionBox
es,canonical:this.tileID.canonical})):n.hasPattern&&(n instanceof e.br||n
instanceof e.bs||n instanceof e.bt)&&
(r(n.layers,this.zoom,s),n.addFeatures(u,this.tileID.canonical,o.patternPosition
s));}}this.status="done",a(null,
{buckets:Object.values(h).filter((e=>!e.isEmpty()))},featureIndex:c,collisionBoxA
rray:this.collisionBoxArray,glyphAtlasImage:t.image,imageAtlas:o,glyphMap:this.r
eturnDependencies?f:null,iconMap:this.returnDependencies?
g:null,glyphPositions:this.returnDependencies?t.positions:null));}}x.length?
this.inFlightDependencies.push(n.send("getImages",
{icons:x,source:this.source,tileID:this.tileID,type:"patterns"},((e,t)=>
{v===this.dependencySentinel&&(p||(p=e,m=t,b.call(this))))):m=

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    },b.call(this));}}function r(t,i,o){const r=new e.a8(i);for(const e of
t)e.recalculate(r,o);}function s(t,i){const o=e.l(t.request,((o,r,s,n)=>
{if(o)i(o);else if(r)try{const t=new e.bw.VectorTile(new e.bv(r));i(null,
{vectorTile:t,rawData:r,cacheControl:s,expires:n});}catch(e){const o=new
Uint8Array(r);let s=`Unable to parse the tile at ${t.request.url},
`;s+=31===o[0]&&139===o[1]?"please make sure the data is not gzipped and that
you have configured the relevant header in the server":`got error:
${e.messge}`;i(new Error(s));}}));return ()=>{o.cancel(),i();}}class
n{constructor(e,t,i,o)
{this.actor=e,this.layerIndex=t,this.availableImages=i,this.loadVectorData=o||s,
this.fetching={},this.loading={},this.loaded={};}loadTile(t,i){const
r=t.uid;this.loading||(this.loading={});const s=!!
(t&&t.request&&t.request.collectResourceTiming)&&new
e.bu(t.request,n=this.loading[r]=new o(t);n.abort=this.loadVectorData(t,
((t,o)=>{if(delete this.loading[r],t||!o)return
n.status="done",this.loaded[r]=n,i(t);const a=o.rawData,l={};o.expires&&
(l.expires=o.expires),o.cacheControl&&(l.cacheControl=o.cacheControl);const c=
{};if(s){const e=s.finish();e&&
(c.resourceTiming=JSON.parse(JSON.stringify(e)));}n.vectorTile=o.vectorTile,n.pa
rse(o.vectorTile,this.layerIndex,this.availableImages,this.actor,((t,o)=>
{if(delete this.fetching[r],t||!o)return
i(t);i(null,e.e({rawTileData:a.slice(0)},o,l,c));})),this.loaded=this.loaded||
{};this.loaded[r]=n,this.fetching[r]=
{rawTileData:a,cacheControl:l,resourceTiming:c});}reloadTile(t,i){const
o=this.loaded,r=t.uid;if(o&&o[r]){const
s=o[r];s.showCollisionBoxes=t.showCollisionBoxes,"parsing"===s.status?
s.parse(s.vectorTile,this.layerIndex,this.availableImages,this.actor,((t,o)=>
{if(t||!o)return i(t,o);let s;if(this.fetching[r])
{const{rawTileData:t,cacheControl:i,resourceTiming:n}=this.fetching[r];delete
this.fetching[r],s=e.e({rawTileData:t.slice(0)},o,i,n);}else
s=o;i(null,s);}):"done"===s.status&&(s.vectorTile?
s.parse(s.vectorTile,this.layerIndex,this.availableImages,this.actor,i):i());}a
bortTile(e,t){const i=this.loading,o=e.uid;i&&i[o]&&i[o].abort&&
(i[o].abort(),delete i[o]),t();}removeTile(e,t){const
i=this.loaded,o=e.uid;i&&i[o]&&delete i[o],t();}}class a{constructor()
{this.loaded={};}loadTile(t,i){return e._(this,void 0,void 0,(function*()
{const{uid:o,encoding:r,rawImageData:s,redFactor:n,greenFactor:a,blueFactor:l,ba
seShift:c}=t,h=s.width+2,u=s.height+2,d=e.a(s)?new e.R({width:h,height:u},yield
e.bx(s,-1,-1,h,u)):s,p=new e.by(o,d,r,n,a,l,c);this.loaded=this.loaded||
{};this.loaded[o]=p,i(null,p);}))}removeTile(e){const
t=this.loaded,i=e.uid;t&&t[i]&&delete t[i];}}function l(e,t){if(0!==e.length)
{c(e[0],t);for(var i=1;i<e.length;i++)c(e[i],!t);}}function c(e,t){for(var
i=0,o=0,r=0,s=e.length,n=s-1;r<s;n=r++){var a=(e[r][0]-e[n][0])*(e[n][1]+e[r]
[1]),l=i+a;o+=Math.abs(i)>Math.abs(a)?i-l+a:a-
l+i,i=1;i+=o>=0?!t&&e.reverse();}var h=e.bz((function e(t,i){var
o,r=t&&t.type;if("FeatureCollection"===r)for(o=0;o<t.features.length;o++)e(t.fea
tures[o],i);else
if("GeometryCollection"===r)for(o=0;o<t.geometries.length;o++)e(t.geometries[o],
i);else if("Feature"===r)e(t.geometry,i);else
if("Polygon"===r)l(t.coordinates,i);else
if("MultiPolygon"===r)for(o=0;o<t.coordinates.length;o++)l(t.coordinates[o],i);r
eturn t}));const u=e.bw.VectorTileFeature.prototype.toGeoJSON;var d={exports:
{}},p=e.bA,f=e.bw.VectorTileFeature,g=m;function m(e,t){this.options=t||
{};this.features=e,this.length=e.length;}function y(e,t)
{this.id="number"===typeof e.id?e.id:void
0,this.type=e.type,this.rawGeometry=l===e.type?
[e.geometry]:e.geometry,this.properties=e.tags,this.extent=t||4096;}m.prototype.
feature=function(e){return new
y(this.features[e],this.options.extent)},y.prototype.loadGeometry=function(){var
e=this.rawGeometry;this.geometry=[];for(var t=0;t<e.length;t++){for(var
i=e[t],o=[],r=0;r<i.length;r++)o.push(new p(i[r][0],i[r]
[1]));this.geometry.push(o);}return this.geometry},y.prototype.bbox=function()

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{this.geometry||this.loadGeometry();for(var
e=this.geometry,t=1/0,i=-1/0,o=1/0,r=-1/0,s=0;s<e.length;s++)for(var
n=e[s],a=0;a<n.length;a++){var
l=n[a];t=Math.min(t,l.x),i=Math.max(i,l.x),o=Math.min(o,l.y),r=Math.max(r,l.y);}
return [t,o,i,r]},y.prototype.toGeoJSON=f.prototype.toGeoJSON;var
v=e.bb,w=g;function x(e){var t=new v;return function(e,t){for(var i in
e.layers)t.writeMessage(3,b,e.layers[i]);}(e,t),t.finish()}function b(e,t){var
i;t.writeVarintField(15,e.version||1),t.writeStringField(1,e.name||""),t.writeVa
rintField(5,e.extent||4096);var o={keys:[],values:[],keycache:{},valuecache:
{}};for(i=0;i<e.length;i++)o.feature=e.feature(i),t.writeMessage(2,S,o);var
r=o.keys;for(i=0;i<r.length;i++)t.writeStringField(3,r[i]);var
s=o.values;for(i=0;i<s.length;i++)t.writeMessage(4,_,s[i]);}function S(e,t){var
i=e.feature;void
0!==i.id&&t.writeVarintField(1,i.id),t.writeMessage(2,M,e),t.writeVarintField(3,
i.type),t.writeMessage(4,T,i);}function M(e,t){var
i=e.feature,o=e.keys,r=e.values,s=e.keycache,n=e.valuecache;for(var a in
i.properties){var l=i.properties[a],c=s[a];if(null!==l){void 0===c&&
(o.push(a),s[a]=c,o.length-1),t.writeVarint(c);var h=typeof
l;"string"!==h&&"boolean"!==h&&"number"!==h&&(l=JSON.stringify(l));var
u=h+": "+l,d=n[u];void 0===d&&(r.push(l),n[u]=d=r.length-
1),t.writeVarint(d);}}}function I(e,t){return (t<<3)+(7&e)}function P(e){return
e<<1^e>>31}function T(e,t){for(var
i=e.loadGeometry(),o=e.type,r=0,s=0,n=i.length,a=0;a<n;a++){var
l=i[a],c=1;l===o&&(c=l.length),t.writeVarint(I(1,c));for(var h=3===o?l.length-
1:l.length,u=0;u<h;u++){1===u&&l===o&&t.writeVarint(I(2,h-1));var d=l[u].x-
r,p=l[u].y-
s;t.writeVarint(P(d)),t.writeVarint(P(p)),r+=d,s+=p;}3===o&&t.writeVarint(I(7,1)
);}}function _(e,t){var i=typeof e;"string"===i?
t.writeStringField(1,e):"boolean"===i?t.writeBooleanField(7,e):"number"===i&&
(e%1!=0?t.writeDoubleField(3,e):e<0?
t.writeSVarintField(6,e):t.writeVarintField(5,e));}d.exports=x,d.exports.fromVec
torTileJs=x,d.exports.fromGeojsonVt=function(e,t){t=t||{};var i={};for(var o in
e)i[o]=new
w(e[o].features,t),i[o].name=o,i[o].version=t.version,i[o].extent=t.extent;retur
n x({layers:i}),d.exports.GeoJSONWrapper=w;var k=e.bz(d.exports);const D=
{minZoom:0,maxZoom:16,minPoints:2,radius:40,extent:512,nodeSize:64,log:!1,genera
teId:!1,reduce:null,map:e=>e},O=Math.fround||(C=new Float32Array(1),e=>
(C[0]=+e,C[0]));var C;const L=3,F=5,z=6;class N{constructor(e)
{this.options=Object.assign(Object.create(D),e),this.trees=new
Array(this.options.maxZoom+1),this.stride=this.options.reduce?
7:6,this.clusterProps=[];}load(e)
{const{log:t,minZoom:i,maxZoom:o}=this.options;t&&console.time("total
time");const r=`prepare ${e.length}
points`;t&&console.time(r),this.points=e;const s=[];for(let t=0;t<e.length;t++)
{const
i=e[t];if(!i.geometry)continue;const[o,r]=i.geometry.coordinates,n=0(Z(o)),a=0(G
(r));s.push(n,a,1/0,t,-1,1),this.options.reduce&&s.push(0);}let
n=this.trees[o+1]=this._createTree(s);t&&console.timeEnd(r);for(let e=0;e>=i;e--
){const
i+=Date.now();n=this.trees[e]=this._createTree(this._cluster(n,e)),t&&console.lo
g("z%d: %d clusters in %dms",e,n.numItems,+Date.now()-i);}return
t&&console.timeEnd("total time"),this}getClusters(e,t){let i=
((e[0]+180)%360+360)%360-180;const o=Math.max(-90,Math.min(90,e[1]));let
r=180===e[2]?180:((e[2]+180)%360+360)%360-180;const
s=Math.max(-90,Math.min(90,e[3]));if(e[2]-e[0]>=360)i=-180,r=180;else if(i>r)
{const
e=this.getClusters([i,o,180,s],t),n=this.getClusters([-180,o,r,s],t);return
e.concat(n)}const
n=this.trees[this._limitZoom(t)],a=n.range(Z(i),G(s),Z(r),G(o)),l=n.data,c=
[];for(const e of a){const t=this.stride*e;c.push(l[t+F]>1?
E(1,t,this.clusterProps):this.points[l[t+L]);}return c}getChildren(e){const
t=this._getOriginId(e),i=this._getOriginZoom(e),o="No cluster with the specified

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id.",r=this.trees[i];if(!r)throw new Error(o);const
s=r.data;if(t*this.stride>=s.length)throw new Error(o);const
n=this.options.radius/(this.options.extent*Math.pow(2,i-
1)),a=r.within(s[t*this.stride],s[t*this.stride+1],n),l=[];for(const t of a)
{const i=t*this.stride;s[i+4]==e&&l.push(s[i+F]>1?
E(s,i,this.clusterProps):this.points[s[i+L]]);}if(0===l.length)throw new
Error(o);return l}getLeaves(e,t,i){const o=[];return
this._appendLeaves(o,e,t=t||10,i=i||0,0,o)}getTile(e,t,i){const
o=this.trees[this._limitZoom(e)],r=Math.pow(2,e),
{extent:s,radius:n}=this.options,a=n/s,l=(i-a)/r,c=(i+1+a)/r,h={features:
[]};return this._addTileFeatures(o.range((t-a)/r,l,
(t+1+a)/r,c),o.data,t,i,r,h),0===t&&this._addTileFeatures(o.range(1-
a/r,l,l,c),o.data,r,i,r,h),t===r-
1&&this._addTileFeatures(o.range(0,l,a/r,c),o.data,-1,i,r,h),h.features.length?
h:null}getClusterExpansionZoom(e){let
t=this._getOriginZoom(e)-1;for(;t<=this.options.maxZoom;){const
i=this.getChildren(e);if(t++,1!==i.length)break;e=i[0].properties.cluster_id;ret
urn t}_appendLeaves(e,t,i,o,r){const s=this.getChildren(t);for(const t of s)
{const s=t.properties;if(s&&s.cluster?r+s.point_count<=o?
r+=s.point_count:r=this._appendLeaves(e,s.cluster_id,i,o,r):r<o?
r++:e.push(t),e.length===i)break}return r}_createTree(t){const i=new
e.av(t.length/this.stride|0,this.options.nodeSize,Float32Array);for(let
e=0;e<t.length;e+=this.stride)i.add(t[e],t[e+1]);return
i.finish(),i.data=t,i}_addTileFeatures(e,t,i,o,r,s){for(const n of e){const
e=n*this.stride,a=t[e+F]>1;let
l,c,h;if(a)l=j(t,e,this.clusterProps),c=t[e],h=t[e+1];else {const
i=this.points[t[e+L]];l=i.properties;const[o,r]=i.geometry.coordinates;c=Z(o),h=
G(r);}const u={type:l,geometry:[Math.round(this.options.extent*(c*r-
i)),Math.round(this.options.extent*(h*r-o))],tags:l};let
d;d=a||this.options.generateId?t[e+L]:this.points[t[e+L]].id,void 0!==d&&
(u.id=d),s.features.push(u);}_limitZoom(e){return
Math.max(this.options.minZoom,Math.min(Math.floor(+e),this.options.maxZoom+1))}_
cluster(e,t)
{const{radius:i,extent:o,reduce:r,minPoints:s}=this.options,n=i/(o*Math.pow(2,t)
),a=e.data,l=[],c=this.stride;for(let i=0;i<a.length;i+=c){if(a[i+2]
<=t)continue;a[i+2]=t;const
o=a[i],h=a[i+1],u=e.within(a[i],a[i+1],n),d=a[i+F];let p=d;for(const e of u)
{const i=e*c;a[i+2]>t&&(p+=a[i+F]);}if(p>d&&p>=s){let e,s=o*d,n=h*d,f=-1;const
g=((i/c|0)<<5)+(t+1)+this.points.length;for(const o of u){const l=o*c;if(a[l+2]
<=t)continue;a[l+2]=t;const h=a[l+F];s+=a[l]*h,n+=a[l+1]*h,a[l+4]=g,r&&(e||
(e=this._map(a,i,!0),f=this.clusterProps.length,this.clusterProps.push(e)),r(e,t
his._map(a,l)));}a[i+4]=g,l.push(s/p,n/p,1/0,g,-1,p),r&&l.push(f);}else {for(let
e=0;e<c;e++)l.push(a[i+e]);if(p>1)for(const e of u){const i=e*c;if(!(a[i+2]<=t))
{a[i+2]=t;for(let e=0;e<c;e++)l.push(a[i+e]);}}}}return l}_getOriginId(e){return
e-this.points.length>>5}_getOriginZoom(e){return (e-
this.points.length)%32}_map(e,t,i){if(e[t+F]>1){const
o=this.clusterProps[e[t+z]];return i?Object.assign({},o):o}const
o=this.points[e[t+L]].properties,r=this.options.map(o);return i&&r===o?
Object.assign({},r):r}}function E(e,t,i){return
{type:"Feature",id:e[t+L],properties:j(e,t,i),geometry:
{type:"Point",coordinates:[(o=e[t],360*(o-.5)),J(e[t+1])]};var o;}function
j(e,t,i){const o=e[t+F],r=o>=1e4?`$${Math.round(o/1e3)}k`:o>=1e3?
Math.round(o/100)/10+"k":o,s=e[t+z],n=-1===s?{}:Object.assign({},i[s]);return
Object.assign(n,
{cluster:!0,cluster_id:e[t+L],point_count:o,point_count_abbreviated:r})}function
Z(e){return e/360+.5}function G(e){const
t=Math.sin(e*Math.PI/180),i=.5-.25*Math.log((1+t)/(1-t))/Math.PI;return i<0?
0:i>1?1:i}function J(e){const t=(180-360*e)*Math.PI/180;return
360*Math.atan(Math.exp(t))/Math.PI-90}function Y(e,t,i,o){for(var r,s=o,n=i-
t>>1,a=i-t,l=e[t],c=e[t+1],h=e[i],u=e[i+1],d=t+3;d<i;d+=3){var
p=A(e[d],e[d+1],l,c,h,u);if(p>s)r=d,s=p;else if(p===s){var f=Math.abs(d-n);f<a&&
(r=d,a=f);}}s>o&&(r-t>3&&Y(e,t,r,o),e[r+2]=s,i-r>3&&Y(e,r,i,o));}function

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A(e,t,i,o,r,s){var n=r-i,a=s-o;if(0!==n||0!==a){var l=((e-i)*n+(t-
o)*a)/(n*n+a*a);l>1?(i=r,o=s):l>0&&(i+=n*l,o+=a*l);}return (n=e-i)*n+(a=t-
o)*a}function V(e,t,i,o){var r={id:void 0===e?
null:e,type:t,geometry:i,tags:o,minX:1/0,minY:1/0,maxX:-1/0,maxY:-1/0};return
function(e){var
t=e.geometry,i=e.type;if("Point"===i||"MultiPoint"===i||"LineString"===i)B(e,t);
else if("Polygon"===i||"MultiLineString"===i)for(var
o=0;o<t.length;o++)B(e,t[o]);else
if("MultiPolygon"===i)for(o=0;o<t.length;o++)for(var
r=0;r<t[o].length;r++)B(e,t[o][r]);}(r),r}function B(e,t){for(var
i=0;i<t.length;i+=3)e.minX=Math.min(e.minX,t[i]),e.minY=Math.min(e.minY,t[i+1]),
e.maxX=Math.max(e.maxX,t[i]),e.maxY=Math.max(e.maxY,t[i+1]);}function X(e,t,i,o)
{if(t.geometry){var
r=t.geometry.coordinates,s=t.geometry.type,n=Math.pow(i.tolerance/((1<<i.maxZoom
)*i.extent),2),a=[],l=t.id;if(i.promoteId?
l=t.properties[i.promoteId]:i.generateId&&(l=o||0),"Point"===s)q(r,a);else
if("MultiPoint"===s)for(var c=0;c<r.length;c++)q(r[c],a);else
if("LineString"===s)R(r,a,n,!1);else if("MultiLineString"===s){if(i.lineMetrics)
{for(c=0;c<r.length;c++)R(r[c],a=
[],n,!1),e.push(V(l,"LineString",a,t.properties));return}W(r,a,n,!1);}else
if("Polygon"===s)W(r,a,n,!0);else {if("MultiPolygon"!==s)
{if("GeometryCollection"===s){for(c=0;c<t.geometry.geometries.length;c++)X(e,
{id:l,geometry:t.geometry.geometries[c],properties:t.properties},i,o);return}thr
ow new Error("Input data is not a valid GeoJSON
object.")}for(c=0;c<r.length;c++){var h=
[];W(r[c],h,n,!0),a.push(h);}}e.push(V(l,s,a,t.properties));}}function q(e,t)
{t.push($(e[0])),t.push(U(e[1])),t.push(0);}function R(e,t,i,o){for(var
r,s,n=0,a=0;a<e.length;a++){var l=$(e[a][0]),c=U(e[a]
[1]);t.push(l),t.push(c),t.push(0),a>0&&(n+=o?(r*c-l*s)/2:Math.sqrt(Math.pow(l-
r,2)+Math.pow(c-s,2))),r=l,s=c;}var h=t.length-
3;t[2]=1,Y(t,0,h,i),t[h+2]=1,t.size=Math.abs(n),t.start=0,t.end=t.size;}function
W(e,t,i,o){for(var r=0;r<e.length;r++){var s=
[];R(e[r],s,i,o),t.push(s);}function $(e){return e/360+.5}function U(e){var
t=Math.sin(e*Math.PI/180),i=.5-.25*Math.log((1+t)/(1-t))/Math.PI;return i<0?
0:i>1?1:i}function H(e,t,i,o,r,s,n,a){if(o/=t,s>=(i/=t)&&n<o)return
e;if(n<i||s>=o)return null;for(var l=[],c=0;c<e.length;c++){var
h=e[c],u=h.geometry,d=h.type,p=0===r?h.minX:h.minY,f=0===r?
h.maxX:h.maxY;if(p>=i&&f<o)l.push(h);else if(!(f<i||p>=o)){var g=
[];if("Point"===d||"MultiPoint"===d)K(u,g,i,o,r);else
if("LineString"===d)Q(u,g,i,o,r,!1,a.lineMetrics);else
if("MultiLineString"===d)te(u,g,i,o,r,!1);else
if("Polygon"===d)te(u,g,i,o,r,!0);else if("MultiPolygon"===d)for(var
m=0;m<u.length;m++){var y=
[];te(u[m],y,i,o,r,!0),y.length&&g.push(y);}if(g.length)
{if(a.lineMetrics&&"LineString"===d)
{for(m=0;m<g.length;m++)l.push(V(h.id,d,g[m],h.tags));continue}"LineString"!==d&
&"MultiLineString"!==d||l===g.length?
(d="LineString",g=g[0]):(d="MultiLineString"),("Point"!==d&&"MultiPoint"!==d||
(d=3===g.length?"Point":"MultiPoint")),l.push(V(h.id,d,g,h.tags));}}return
l.length?l:null}function K(e,t,i,o,r){for(var s=0;s<e.length;s+=3){var
n=e[s+r];n>=i&&n<=o&&(t.push(e[s]),t.push(e[s+1]),t.push(e[s+2]));}}function
Q(e,t,i,o,r,s,n){for(var a,l,c=ee(e),h=0===r?oe:re,u=e.start,d=0;d<e.length-
3;d+=3){var p=e[d],f=e[d+1],g=e[d+2],m=e[d+3],y=e[d+4],v=0===r?p:f,w=0===r?
m:y,x=!1;n&&(a=Math.sqrt(Math.pow(p-m,2)+Math.pow(f-y,2))),v<i?w>i&&
(l=h(c,p,f,m,y,i),n&&(c.start=u+a*l)):v>o?w<o&&(l=h(c,p,f,m,y,o),n&&
(c.start=u+a*l)):ie(c,p,f,g),w<i&&v>=i&&(l=h(c,p,f,m,y,i),x=!0),w>o&&v<=o&&
(l=h(c,p,f,m,y,o),x=!0),!s&&x&&(n&&(c.end=u+a*l),t.push(c),c=ee(e)),n&&
(u+=a);}var b=e.length-3;p=e[b],f=e[b+1],g=e[b+2],(v=0===r?
p:f)>=i&&v<=o&&ie(c,p,f,g),b<c.length-3,s&&b>=3&&
(c[b]!==c[0]||c[b+1]!==c[1])&&ie(c,c[0],c[1],c[2]),c.length&&t.push(c);}function
ee(e){var t=[];return t.size=e.size,t.start=e.start,t.end=e.end,t}function
te(e,t,i,o,r,s){for(var n=0;n<e.length;n++)Q(e[n],t,i,o,r,s,!1);}function

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ie(e,t,i,o){e.push(t),e.push(i),e.push(o);}function oe(e,t,i,o,r,s){var n=(s-
t)/(o-t);return e.push(s),e.push(i+(r-i)*n),e.push(1),n}function re(e,t,i,o,r,s)
{var n=(s-i)/(r-i);return e.push(t+(o-t)*n),e.push(s),e.push(1),n}function
se(e,t){for(var i=[],o=0;o<e.length;o++){var
r,s=e[o],n=s.type;if("Point"===n||"MultiPoint"===n||"LineString"===n)r=ne(s.geom
etry,t);else if("MultiLineString"===n||"Polygon"===n){r=[];for(var
a=0;a<s.geometry.length;a++)r.push(ne(s.geometry[a],t));}else
if("MultiPolygon"===n)for(r=[],a=0;a<s.geometry.length;a++){for(var l=
[],c=0;c<s.geometry[a].length;c++)l.push(ne(s.geometry[a]
[c],t));r.push(l);}i.push(V(s.id,n,r,s.tags));}return i}function ne(e,t){var i=
[];i.size=e.size,void 0!==e.start&&(i.start=e.start,i.end=e.end);for(var
o=0;o<e.length;o+=3)i.push(e[o]+t,e[o+1],e[o+2]);return i}function ae(e,t)
{if(e.transformed)return e;var
i,o,r,s=1<=e.z,n=e.x,a=e.y;for(i=0;i<e.features.length;i++){var
l=e.features[i],c=l.geometry,h=l.type;if(l.geometry=
[],l===h)for(o=0;o<c.length;o+=2)l.geometry.push(le(c[o],c[o+1],t,s,n,a));else
for(o=0;o<c.length;o++){var u=[];for(r=0;r<c[o].length;r+=2)u.push(le(c[o]
[r],c[o][r+1],t,s,n,a));l.geometry.push(u);}return e.transformed=!0,e}function
le(e,t,i,o,r,s){return [Math.round(i*(e*o-r)),Math.round(i*(t*o-s))]}function
ce(e,t,i,o,r){for(var s=t===r.maxZoom?0:r.tolerance/((1<<t)*r.extent),n=
{features:
[],numPoints:0,numSimplified:0,numFeatures:0,source:null,x:i,y:o,z:t,transformed
:!1,minX:2,minY:1,maxX:-1,maxY:0},a=0;a<e.length;a++)
{n.numFeatures++,he(n,e[a],s,r);var
l=e[a].minX,c=e[a].minY,h=e[a].maxX,u=e[a].maxY;l<n.minX&&(n.minX=l),c<n.minY&&
(n.minY=c),h>n.maxX&&(n.maxX=h),u>n.maxY&&(n.maxY=u);}return n}function
he(e,t,i,o){var r=t.geometry,s=t.type,n=
[];if("Point"===s||"MultiPoint"===s)for(var
a=0;a<r.length;a+=3)n.push(r[a]),n.push(r[a+1]),e.numPoints++,e.numSimplified++;
else if("LineString"===s)ue(n,r,e,i,!1,!1);else
if("MultiLineString"===s||"Polygon"===s)for(a=0;a<r.length;a++)ue(n,r[a],e,i,"Po
lygon"===s,0===a);else if("MultiPolygon"===s)for(var l=0;l<r.length;l++){var
c=r[l];for(a=0;a<c.length;a++)ue(n,c[a],e,i,!0,0===a);}if(n.length){var
h=t.tags||null;if("LineString"===s&&o.lineMetrics){for(var u in h=
{},{t.tags}h[u]=t.tags[u];h.mapbox_clip_start=r.start/r.size,h.mapbox_clip_end=r.
end/r.size;}var d={geometry:n,type:"Polygon"===s||"MultiPolygon"===s?
3:"LineString"===s||"MultiLineString"===s?2:1,tags:h};null!==t.id&&
(d.id=t.id),e.features.push(d);}function ue(e,t,i,o,r,s){var
n=o*o;if(o>0&&t.size<(r?n:o))i.numPoints+=t.length/3;else {for(var a=
[],l=0;l<t.length;l+=3)(0===o||t[l+2]>n)&&
(i.numSimplified++,a.push(t[l]),a.push(t[l+1])),i.numPoints++;r&&function(e,t)
{for(var i=0,o=0,r=e.length,s=r-2;o<r;s=o,o+=2)i+=(e[o]-e[s])*
(e[o+1]+e[s+1]);if(i>0===t)for(o=0,r=e.length;o<r/2;o+=2){var
n=e[o],a=e[o+1];e[o]=e[r-2-o],e[o+1]=e[r-1-o],e[r-2-o]=n,e[r-1-o]=a;}}
(a,s),e.push(a);}}function de(e,t){var i=(t=this.options=function(e,t){for(var i
in t)e[i]=t[i];return e}
(Object.create(this.options),t)).debug;if(i&&console.time("preprocess
data"),t.maxZoom<0||t.maxZoom>24)throw new Error("maxZoom should be in the 0-24
range");if(t.promoteId&&t.generateId)throw new Error("promoteId and generateId
cannot be used together.");var o=function(e,t){var i=
[];if("FeatureCollection"===e.type)for(var
o=0;o<e.features.length;o++)X(i,e.features[o],t,o);else X(i,"Feature"===e.type?
e:{geometry:e},t);return i}(e,t);this.tiles={},this.tileCoords=[],i&&
(console.timeEnd("preprocess data"),console.log("index: maxZoom: %d, maxPoints:
%d",t.indexMaxZoom,t.indexMaxPoints),console.time("generate tiles"),this.stats=
{},{this.total=0},o=function(e,t){var i=t.buffer/t.extent,o=e,r=H(e,1,-1-
i,i,0,-1,2,t),s=H(e,1,1-i,2+i,0,-1,2,t);return (r||s)&&(o=H(e,1,-
i,1+i,0,-1,2,t)||[],r&&(o=se(r,1).concat(o)),s&&(o=o.concat(se(s,-1))))},o)
(o,t),o.length&&this.splitTile(o,0,0,0),i&&(o.length&&console.log("features: %d,
points:
%d",this.tiles[0].numFeatures,this.tiles[0].numPoints),console.timeEnd("generate
tiles"),console.log("tiles

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generated:",this.total,JSON.stringify(this.stats));}function pe(e,t,i){return
32*((1<<e)*i+t)+e}function fe(e,t){return t?e.properties[t]:e.id}function
ge(e,t){if(null==e)return !0;if("Feature"===e.type)return
null!=fe(e,t);if("FeatureCollection"===e.type){const i=new Set;for(const o of
e.features){const e=fe(o,t);if(null==e)return !1;if(i.has(e))return
!1;i.add(e);}return !0}return !1}function me(e,t){const i=new
Map;if(null==e);else if("Feature"===e.type)i.set(fe(e,t),e);else for(const o of
e.features)i.set(fe(o,t),o);return i}de.prototype.options=
{maxZoom:14,indexMaxZoom:5,indexMaxPoints:1e5,tolerance:3,extent:4096,buffer:64,
lineMetrics:!1,promoteId:null,generateId:!1,debug:0},de.prototype.splitTile=func
tion(e,t,i,o,r,s,n){for(var a=[e,t,i,o],l=this.options,c=l.debug;a.length;){
o=a.pop(),i=a.pop(),t=a.pop(),e=a.pop();var
h=1<<t,u=pe(t,i,o),d=this.tiles[u];if(!d&&
(c>1&&console.time("creation"),d=this.tiles[u]=ce(e,t,i,o,l),this.tileCoords.pus
h({z:t,x:i,y:o}),c))){c>1&&(console.log("tile z%d-%d-%d (features: %d, points:
%d, simplified:
%d)",t,i,o,d.numFeatures,d.numPoints,d.numSimplified),console.timeEnd("creation"
));var p="z"+t;this.stats[p]=(this.stats[p]||0)+1,this.total++;}if(d.source=e,r)
{if(t===l.maxZoom||t===r)continue;var f=1<<r-
t;if(i!==Math.floor(s/f)||o!==Math.floor(n/f))continue}else
if(t===l.indexMaxZoom||d.numPoints<=l.indexMaxPoints)continue;if(d.source=null,0
!=e.length){c>1&&console.time("clipping");var
g,m,y,v,w,x,b=.5*l.buffer/l.extent,S=.5-b,M=.5+b,I=1+b;g=m=y=v=null,w=H(e,h,i-
b,i+M,0,d.minX,d.maxX,l),x=H(e,h,i+S,i+I,0,d.minX,d.maxX,l),e=null,w&&
(g=H(w,h,o-
b,o+M,1,d.minY,d.maxY,l),m=H(w,h,o+S,o+I,1,d.minY,d.maxY,l),w=null),x&&
(y=H(x,h,o-
b,o+M,1,d.minY,d.maxY,l),v=H(x,h,o+S,o+I,1,d.minY,d.maxY,l),x=null),c>1&&console
.timeEnd("clipping"),a.push(g||[],t+1,2*i,2*o),a.push(m||
[],t+1,2*i,2*o+1),a.push(y||[],t+1,2*i+1,2*o),a.push(v||
[],t+1,2*i+1,2*o+1);}}},de.prototype.getTile=function(e,t,i){var
o=this.options,r=o.extent,s=o.debug;if(e<0||e>24)return null;var
n=1<<e,a=pe(e,t=(t%n+n)%n,i);if(this.tiles[a])return
ae(this.tiles[a],r);s>1&&console.log("drilling down to z%d-%d-%d",e,t,i);for(var
l,c=e,h=t,u=i;!l&&c>0;){c-
-,h=Math.floor(h/2),u=Math.floor(u/2),l=this.tiles[pe(c,h,u)];return
l&&l.source?(s>1&&console.log("found parent tile z%d-%d-
%d",c,h,u),s>1&&console.time("drilling
down"),this.splitTile(l.source,c,h,u,e,t,i),s>1&&console.timeEnd("drilling
down"),this.tiles[a]?ae(this.tiles[a],r):null):null};class ye extends
n{constructor(t,i,o,r){super(t,i,o),this._dataUpdateable=new
Map,this.loadGeoJSON=(t,i)=>{const{promoteId:o}=t;if(t.request)return
e.f(t.request,((e,t,r,s)=>{this._dataUpdateable=ge(t,o)?me(t,o):void
0,i(e,t,r,s)}));if("string"===typeof t.data)try{const
e=JSON.parse(t.data);this._dataUpdateable=ge(e,o)?me(e,o):void
0,i(null,e);}catch(e){i(new Error(`Input data given to '${t.source}' is not a
valid GeoJSON object.`));}else t.dataDiff?this._dataUpdateable?(function(e,t,i)
{var o,r,s,n;if(t.removeAll&&e.clear(),t.remove)for(const i of
t.remove)e.delete(i);if(t.add)for(const o of t.add){const
t=fe(o,i);null!=t&&e.set(t,o);}if(t.update)for(const i of t.update){let
t=e.get(i.id);if(null==t)continue;const a=!i.removeAllProperties&&((null===
(o=i.removeProperties)||void 0===o?void 0:o.length)>0)||((null===
(r=i.addOrUpdateProperties)||void 0===r?void
0:r.length)>0);if((i.newGeometry||i.removeAllProperties||a)&&
(t=Object.assign({},t),e.set(i.id,t),a&&
(t.properties=Object.assign({},t.properties))),i.newGeometry&&
(t.geometry=i.newGeometry),i.removeAllProperties)t.properties={};else
if((null===(s=i.removeProperties)||void 0===s?void 0:s.length)>0)for(const e of
i.removeProperties)Object.prototype.hasOwnProperty.call(t.properties,e)&&delete
t.properties[e];if((null===(n=i.addOrUpdateProperties)||void 0===n?void
0:n.length)>0)for(const{key:e,value:o}of
i.addOrUpdateProperties)t.properties[e]=o;}}

```

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(this._dataUpdateable,t.dataDiff,o),i(null,
{type:"FeatureCollection",features:Array.from(this._dataUpdateable.values())}):i(
i(new Error(`Cannot update existing geojson data in ${t.source}`)):i(new
Error(`Input data given to '${t.source}' is not a valid GeoJSON
object.`));return {cancel:()=>{}}},this.loadVectorData=this.loadGeoJSONTile,r&&
(this.loadGeoJSON=r);}loadGeoJSONTile(t,i){const
o=t.tileID.canonical;if(!this._geoJSONIndex)return i(null,null);const
r=this._geoJSONIndex.getTile(o.z,o.x,o.y);if(!r)return i(null,null);const s=new
class{constructor(t){this.layers=
{_geojsonTileLayer:this},this.name="_geojsonTileLayer",this.extent=e.N,this.leng
th=t.length,this._features=t;}feature(t){return new class{constructor(t)
{this._feature=t,this.extent=e.N,this.type=t.type,this.properties=t.tags,"id" in
t&&!isNaN(t.id)&&(this.id=parseInt(t.id,10));}loadGeometry()
{if(1===this._feature.type){const t=[];for(const i of
this._feature.geometry)t.push([new e.P(i[0],i[1])]);return t}{const t=
[];for(const i of this._feature.geometry){const o=[];for(const t of i)o.push(new
e.P(t[0],t[1]));t.push(o);}return t}}toGeoJSON(e,t,i){return
u.call(this,e,t,i)}(this._features[t])}(r.features);let
n=k(s);0===n.byteOffset&&n.byteLength===n.buffer.byteLength||(n=new
Uint8Array(n)),i(null,{vectorTile:s,rawData:n.buffer});}loadData(t,i){var
o;null===(o=this._pendingRequest)||void
0===o||o.cancel(),this._pendingCallback&&this._pendingCallback(null,
{abandoned:!0});const r=!(t&&t.request&&t.request.collectResourceTiming)&&new
e.bu(t.request);this._pendingCallback=i,this._pendingRequest=this.loadGeoJSON(t,
((o,s)=>{if(delete this._pendingCallback,delete
this._pendingRequest,o||!s)return i(o);if("object"!==typeof s)return i(new
Error(`Input data given to '${t.source}' is not a valid GeoJSON object.`));
{h(s,!0);try{if(t.filter){const i=e.bC(t.filter,{type:"boolean","property-
type":"data-driven",overridable:!1,transition:!1});if("error"===i.result)throw
new Error(i.value.map((e=>`${e.key}: ${e.message}`)).join(", "));const
o=s.features.filter((e=>i.value.evaluate({zoom:0},e)));s=
{type:"FeatureCollection",features:o};}this._geoJSONIndex=t.cluster?new
N(function({superclusterOptions:t,clusterProperties:i}){if(!i||!t)return t;const
o={},r={},s={accumulated:null,zoom:0},n=
{properties:null},a=Object.keys(i);for(const t of a)
{const[s,n]=i[t],a=e.bC(n),l=e.bC("string"===typeof s?[s,["accumulated"],
["get",t]]:s);o[t]=a.value,r[t]=l.value;}return t.map=e=>{n.properties=e;const
t={};for(const e of a)t[e]=o[e].evaluate(s,n);return t},t.reduce=(e,t)=>
{n.properties=t;for(const t of a)s.accumulated=e[t],e[t]=r[t].evaluate(s,n);},t)
(t)).load(s.features):function(e,t){return new de(e,t)}
(s,t.geojsonVtOptions);}catch(o){return i(o)}this.loaded={};const n={};if(r)
{const e=r.finish();e&&(n.resourceTiming=
{,n.resourceTiming[t.source]=JSON.stringify(e));}i(null,n);}}));}re
loadTile(e,t){const i=this.loaded;return i&&i[e.uid]?
super.reloadTile(e,t):this.loadTile(e,t)}removeSource(e,t)
{this._pendingCallback&&this._pendingCallback(null,
{abandoned:!0}),t();}getClusterExpansionZoom(e,t)
{try{t(null,this._geoJSONIndex.getClusterExpansionZoom(e.clusterId));}catch(e)
{t(e);}}getClusterChildren(e,t)
{try{t(null,this._geoJSONIndex.getChildren(e.clusterId));}catch(e)
{t(e);}}getClusterLeaves(e,t)
{try{t(null,this._geoJSONIndex.getLeaves(e.clusterId,e.limit,e.offset));}catch(e)
{t(e);}}}class ve{constructor(t){this.self=t,this.actor=new
e.C(t,this),this.layerIndexes={},this.availableImages={},this.workerSourceTypes=
{vector:n,geojson:ye},this.workerSources={},this.demWorkerSources=
{},this.self.registerWorkerSource=(e,t)=>{if(this.workerSourceTypes[e])throw new
Error(`Worker source with name "${e}" already
registered.`);this.workerSourceTypes[e]=t;},this.self.registerRTLTextPlugin=t=>
{if(e.bd.isParsed())throw new Error("RTL text plugin already
registered.");e.bd.applyArabicShaping=t.applyArabicShaping,e.bd.processBidirecti
onalText=t.processBidirectionalText,e.bd.processStyledBidirectionalText=t.proces
sStyledBidirectionalText;}}setReferrer(e,t){this.referrer=t;}setImages(e,t,i)

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```

{this.availableImages[e]=t;for(const i in this.workerSources[e]){const
o=this.workerSources[e][i];for(const e in
o)o[e].availableImages=t;i();}setLayers(e,t,i)
{this.getLayerIndex(e).replace(t,i());}updateLayers(e,t,i)
{this.getLayerIndex(e).update(t.layers,t.removedIds,i());}loadTile(e,t,i)
{this.getWorkerSource(e,t.type,t.source).loadTile(t,i);}loadDEMTile(e,t,i)
{this.getDEMWorkerSource(e,t.source).loadTile(t,i);}reloadTile(e,t,i)
{this.getWorkerSource(e,t.type,t.source).reloadTile(t,i);}abortTile(e,t,i)
{this.getWorkerSource(e,t.type,t.source).abortTile(t,i);}removeTile(e,t,i)
{this.getWorkerSource(e,t.type,t.source).removeTile(t,i);}removeDEMTile(e,t)
{this.getDEMWorkerSource(e,t.source).removeTile(t);}removeSource(e,t,i)
{if(!this.workerSources[e]||!this.workerSources[e]
[t.type]||!this.workerSources[e][t.type][t.source])return;const
o=this.workerSources[e][t.type][t.source];delete this.workerSources[e][t.type]
[t.source],void 0!==o.removeSource?
o.removeSource(t,i):i();}loadWorkerSource(e,t,i)
{try{this.self.importScripts(t.url),i();}catch(e)
{i(e.toString());}}syncRTLPluginState(t,i,o){try{e.bD.setState(i);const
t=e.bD.getPluginURL();if(e.bD.isLoaded()&&!e.bD.isParsed()&&null!=t)
{this.self.importScripts(t);const i=e.bD.isParsed();o(i?void 0:new Error(`RTL
Text Plugin failed to import scripts from ${t}`),i);}}catch(e)
{o(e.toString());}}getAvailableImages(e){let t=this.availableImages[e];return
t||(t=[]),t}getLayerIndex(e){let i=this.layerIndexes[e];return i||
(i=this.layerIndexes[e]=new t),i}getWorkerSource(e,t,i){return
this.workerSources[e]||(this.workerSources[e]={}),this.workerSources[e][t]||
(this.workerSources[e][t]={}),this.workerSources[e][t][i]||
(this.workerSources[e][t][i]=new this.workerSourceTypes[t]({send:(t,i,o)=>
{this.actor.send(t,i,o,e)};}),this.getLayerIndex(e),this.getAvailableImages(e))),
this.workerSources[e][t][i]}getDEMWorkerSource(e,t,i){return
this.demWorkerSources[e]||(this.demWorkerSources[e]={}),this.demWorkerSources[e]
[t]||(this.demWorkerSources[e][t]=new a),this.demWorkerSources[e][t]}return
e.i()&&(self.worker=new ve(self),ve));

```

```

define(["./shared"],(function(t){
"use strict";var e="3.6.2";class i{static
testProp(t){if(!i.docStyle)return t[0];for(let e=0;e<t.length;e++)if(t[e]in
i.docStyle)return t[e];return t[0]}static create(t,e,i){const
s=window.document.createElement(t);return void 0!==e&&
(s.className=e),i&&i.appendChild(s),s}static createNS(t,e){return
window.document.createElementNS(t,e)}static disableDrag()
{i.docStyle&&i.selectProp&&
(i.userSelect=i.docStyle[i.selectProp],i.docStyle[i.selectProp]="none");}static
enableDrag(){i.docStyle&&i.selectProp&&
(i.docStyle[i.selectProp]=i.userSelect);}static setTransform(t,e)
{t.style[i.transformProp]=e;}static addEventListener(t,e,i,s={})
{t.addEventListener(e,i,"passive"in s?s:s.capture);}static
removeEventListener(t,e,i,s={}){t.removeEventListener(e,i,"passive"in s?
s:s.capture);}static suppressClickInternal(t)
{t.preventDefault(),t.stopPropagation(),window.removeEventListener("click",i.sup
pressClickInternal,!0);}static suppressClick()
{window.addEventListener("click",i.suppressClickInternal,!0),window.setTimeout((
()=>
{window.removeEventListener("click",i.suppressClickInternal,!0)}),0);}static
mousePos(e,i){const s=e.getBoundingClientRect();return new t.P(i.clientX-s.left-
e.clientLeft,i.clientY-s.top-e.clientTop)}static touchPos(e,i){const
s=e.getBoundingClientRect(),a=[];for(let o=0;o<i.length;o++)a.push(new
t.P(i[o].clientX-s.left-e.clientLeft,i[o].clientY-s.top-e.clientTop));return
a}static mouseButton(t){return t.button}static remove(t)
{t.parentNode&&t.parentNode.removeChild(t);}i.docStyle="undefined"!==typeof
window&&window.document&&window.document.documentElement.style,i.selectProp=i.te
stProp(["userSelect","MozUserSelect","WebkitUserSelect","msUserSelect"]),i.trans
formProp=i.testProp(["transform","WebkitTransform"]);const s=
{supported:!1,testSupport:function(t){!r&&o&&(n?l(t):a=t)};};let

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```

a,o,r=!1,n=!1;function l(t){const
e=t.createTexture();t.bindTexture(t.TEXTURE_2D,e);try{if(t.texImage2D(t.TEXTURE_
2D,0,t.RGBA,t.RGBA,t.UNSIGNED_BYTE,o),t.isContextLost())return;s.supported=!0;}c
atch(t){t.deleteTexture(e),r=!0;}var h,c;"undefined"!==typeof document&&
(o=document.createElement("img"),o.onload=function()
{a&&l(a),a=null,n=!0;},o.onerror=function()
{r=!0,a=null;},o.src="
Q//73v/+BiOh/AAA="),function(e){let i,a,o,r;e.resetRequestQueue=()=>{i=
[],a=0,o=0,r={}};e.addThrottleControl=t=>{const e=o++;return
r[e]=t,e},e.removeThrottleControl=t=>{delete r[t],h();},e.getImage=(t,e,o=!0)=>
{s.supported&&(t.headers||t.headers=
{}),t.headers.accept="image/webp,*/*");const r=
{requestParameters:t,supportImageRefresh:o,callback:e,cancelled:!1,completed:!1,
cancel:()=>{r.completed||r.cancelled||(r.cancelled=!0,r.innerRequest&&
(r.innerRequest.cancel(),a--),h());}};return i.push(r),h(),r};const n=e=>
{const{requestParameters:i,supportImageRefresh:s,callback:a}=e;return t.e(i,
{type:"image"}),
(!1!==s||t.i()||t.g(i.url)||i.headers&&!Object.keys(i.headers).reduce(((t,e)=>t&
&"accept"===e),!0)?t.m:c)(i,((t,i,s,o)=>{l(e,a,t,i,s,o);})))},l=(e,i,s,o,r,n)=>
{s?i(s):o instanceof HTMLImageElement||t.a(o)?i(null,o):o&&((e,i)=>
{"function"===typeof createImageBitmap?t.b(e,i):t.d(e,i)})(o,((t,e)=>{null!=t?
i(t):null!=e&&i(null,e,{cacheControl:r,expires:n});})),e.cancelled||
(e.completed=!0,a--,h());},h=()=>{const e=()=>{const t=Object.keys(r);let
e=!1;if(t.length>0)for(const i of t)if(e=r[i](),e)break;return e}}()
t.c.MAX_PARALLEL_IMAGE_REQUESTS_PER_FRAME:t.c.MAX_PARALLEL_IMAGE_REQUESTS;for(le
t t=a;t<e&&i.length>0;t++){const e=i.shift();if(e.cancelled){t--;continue}const
s=n(e),a++,e.innerRequest=s;},c=(e,i)=>{const s=new Image,a=e.url;let
o=!1;const r=e.credentials;return r&&"include"===r?s.crossOrigin="use-
credentials":(r&&"same-origin"===r||!t.s(a))&&
(s.crossOrigin="anonymous"),s.fetchPriority="high",s.onload=()=>
{i(null,s),s.onerror=s.onload=null;},s.onerror=()=>{o||i(new Error("Could not
load image. Please make sure to use a supported image type such as PNG or JPEG.
Note that SVGs are not supported.")),s.onerror=s.onload=null;},s.src=a,{cancel:
()=>{o=!0,s.src="";}};}(h||{h={}}),h.resetRequestQueue(),function(t)
{t.Glyphs="Glyphs",t.Image="Image",t.Source="Source",t.SpriteImage="SpriteImage",
t.SpriteJSON="SpriteJSON",t.Style="Style",t.Tile="Tile",t.Unknown="Unknown";}
(c||{c={}}));class u{constructor(t)
{this._transformRequestFn=t;}transformRequest(t,e){return
this._transformRequestFn&&this._transformRequestFn(t,e)||
{url:t}}normalizeSpriteURL(t,e,i){const s=function(t){const
e=t.match(d);if(!e)throw new Error(`Unable to parse URL "${t}"`);return
{protocol:e[1],authority:e[2],path:e[3]||"/",params:e[4]?e[4].split("&"):[]}}
(t);return s.path+`${e}${i}`,function(t){const e=t.params.length?`
${t.params.join("&")}`:"";return`${t.protocol}://${t.authority}${t.path}${e}`}
(s)}setTransformRequest(t){this._transformRequestFn=t;}const
d=/^(\\w+):\\/\\/([\\^/?]*)\\/[\\^?]+)?\\??(\\+)?\\/;function_(e){var i=new t.A(3);return
i[0]=e[0],i[1]=e[1],i[2]=e[2],i}var p,m=function(t,e,i){return t[0]=e[0]-
i[0],t[1]=e[1]-i[1],t[2]=e[2]-i[2],t};p=new t.A(3),t.A!=Float32Array&&
(p[0]=0,p[1]=0,p[2]=0);var f=function(t){var e=t[0],i=t[1];return
e*e+i*i};function g(t){const e=[];if("string"===typeof
t)e.push({id:"default",url:t});else if(t&&t.length>0){const i=
[];for(const{id:s,url:a}of t){const t=`${s}${a}`;-1===i.indexOf(t)&&
(i.push(t),e.push({id:s,url:a}))};}return e}function v(e,i,s,a,o){if(a)return
void
e(a);if(o!==Object.values(i).length||o!==Object.values(s).length)return;const r=
{};for(const e in i){r[e]={};const
a=t.h.getImageCanvasContext(s[e]),o=i[e];for(const t in o)
{const{width:i,height:s,x:n,y:l,sdf:h,pixelRatio:c,stretchX:u,stretchY:d,content
:}=o[t];r[e][t]=
{data:null,pixelRatio:c,sdf:h,stretchX:u,stretchY:d,content:_,spriteData:
{width:i,height:s,x:n,y:l,context:a}};}}e(null,r);!function(){var e=new
t.A(2);t.A!=Float32Array&&(e[0]=0,e[1]=0);}();class x{constructor(t,e,i,s)

```

```

{this.context=t,this.format=i,this.texture=t.gl.createTexture(),this.update(e,s)
;}update(e,i,s){const{width:a,height:o}=e,r=!
(this.size&&this.size[0]===a&&this.size[1]===o|s),{context:n}=this,
{gl:l}=n;if(this.useMipmap=Boolean(i&&i.useMipmap),l.bindTexture(l.TEXTURE_2D,th
is.texture),n.pixelStoreUnpackFlipY.set(!1),n.pixelStoreUnpack.set(1),n.pixelSto
reUnpackPremultiplyAlpha.set(this.format===l.RGBA&&
(!i||!1!==i.premultiply)),r)this.size=[a,o],e instanceof HTMLElement||e
instanceof HTMLCanvasElement||e instanceof HTMLVideoElement||e instanceof
ImageData||t.a(e)?
l.texImage2D(l.TEXTURE_2D,0,this.format,this.format,l.UNSIGNED_BYTE,e):l.texImag
e2D(l.TEXTURE_2D,0,this.format,a,o,0,this.format,l.UNSIGNED_BYTE,e.data);else
{const{x:i,y:r}=s||{x:0,y:0};e instanceof HTMLElement||e instanceof
HTMLCanvasElement||e instanceof HTMLVideoElement||e instanceof
ImageData||t.a(e)?
l.texSubImage2D(l.TEXTURE_2D,0,i,r,l.RGBA,l.UNSIGNED_BYTE,e):l.texSubImage2D(l.T
EXTURE_2D,0,i,r,a,o,l.RGBA,l.UNSIGNED_BYTE,e.data);}this.useMipmap&&this.isSizeP
owerOfTwo()&&l.generateMipmap(l.TEXTURE_2D);}bind(t,e,i){const{context:s}=this,
{gl:a}=s;a.bindTexture(a.TEXTURE_2D,this.texture),i!==a.LINEAR_MIPMAP_NEAREST||t
his.isSizePowerOfTwo()||(i=a.LINEAR),t!==this.filter&&
(a.texParameteri(a.TEXTURE_2D,a.TEXTURE_MAG_FILTER,t),a.texParameteri(a.TEXTURE_
2D,a.TEXTURE_MIN_FILTER,i|t),this.filter=t),e!==this.wrap&&
(a.texParameteri(a.TEXTURE_2D,a.TEXTURE_WRAP_S,e),a.texParameteri(a.TEXTURE_2D,a
.TEXTURE_WRAP_T,e),this.wrap=e);}isSizePowerOfTwo(){return
this.size[0]===this.size[1]&&Math.log(this.size[0])/Math.LN2%1===0}destroy()
{const{gl:t}=this.context;t.deleteTexture(this.texture),this.texture=null;}func
tion y(t){const{userImage:e}=t;return !(e&&e.render&&e.render())&&
(t.data.replace(new Uint8Array(e.data.buffer)),!0)}class b extends
t.E{constructor(){super(),this.images={},this.updatedImages=
{},this.callbackDispatchedThisFrame={},this.loaded=!1,this.requestors=
[],this.patterns={},this.atlasImage=new
t.R({width:1,height:1}),this.dirty=!0;}isLoading(){return
this.loaded}setLoaded(t){if(this.loaded!==t&&(this.loaded=t,t))
{for(const{ids:t,callback:e}of
this.requestors)this._notify(t,e);this.requestors=[];}}getImage(e){const
i=this.images[e];if(i&&!i.data&&i.spriteData){const e=i.spriteData;i.data=new
t.R({width:e.width,height:e.height},e.context.getImageData(e.x,e.y,e.width,e.hei
ght).data),i.spriteData=null;}return i}addImage(t,e){if(this.images[t])throw new
Error(`Image id ${t} already exist, use updateImage
instead`);this._validate(t,e)&&(this.images[t]=e);}_validate(e,i){let s=!0;const
a=i.data||i.spriteData;return this._validateStretch(i.stretchX,a&&a.width)||
(this.fire(new t.j(new Error(`Image "${e}" has invalid "stretchX"
value`))),s=!1),this._validateStretch(i.stretchY,a&&a.height)||
(this.fire(new t.j(new Error(`Image "${e}" has invalid "stretchY"
value`))),s=!1),this._validateContent(i.content,i)||
(this.fire(new t.j(new Error(`Image "${e}" has invalid "content"
value`))),s=!1),s}_validateStretch(t,e){if(!t)return !0;let i=0;for(const s of
t){if(s[0]<i||s[1]<s[0]||e<s[1])return !1;i=s[1];}return
!0}_validateContent(t,e){if(!t)return !0;if(4!==t.length)return !1;const
i=e.spriteData,s=i&&i.width||e.data.width,a=i&&i.height||e.data.height;return !
(t[0]<0||s<t[0]||t[1]<0||a<t[1]||t[2]<0||s<t[2]||t[3]<0||a<t[3]||t[2]<t[0]||t[3]
<t[1])}updateImage(t,e,i=!0){const s=this.getImage(t);if(i&&
(s.data.width!==e.data.width||s.data.height!==e.data.height))throw new
Error(`size mismatch between old image (${s.data.width}x${s.data.height}) and
new image
(${e.data.width}x${e.data.height}).`);e.version=s.version+1,this.images[t]=e,thi
s.updatedImages[t]=!0;}removeImage(t){const e=this.images[t];delete
this.images[t],delete
this.patterns[t],e.userImage&&e.userImage.onRemove&&e.userImage.onRemove();}list
Images(){return Object.keys(this.images)}getImages(t,e){let
i=!0;if(!this.isLoading())for(const e of t)this.images[e]||
(i=!1);this.isLoading()||i?
this._notify(t,e):this.requestors.push({ids:t,callback:e});}_notify(e,i){const

```



```

s={};for(const i of e){let e=this.getImage(i);e||(this.fire(new
t.k("styleimagemissing",{id:i})),e=this.getImage(i)),e?s[i]=
{data:e.data.clone(),pixelRatio:e.pixelRatio,sdf:e.sdf,version:e.version,stretch
X:e.stretchX,stretchY:e.stretchY,content:e.content,hasRenderCallback:Boolean(e.u
serImage&&e.userImage.render)}:t.w(`Image "${i}" could not be loaded. Please
make sure you have added the image with map.addImage() or a "sprite" property in
your style. You can provide missing images by listening for the
"styleimagemissing" map event.`);}i(null,s);}getPixelSize()
{const{width:t,height:e}=this.atlasImage;return {width:t,height:e}}getPattern(e)
{const i=this.patterns[e],s=this.getImage(e);if(!s)return
null;if(i&&i.position.version===s.version)return
i.position;if(i).position.version=s.version;else {const i=
{w:s.data.width+2,h:s.data.height+2,x:0,y:0},a=new t.I(i,s);this.patterns[e]=
{bin:i,position:a};}return
this._updatePatternAtlas(),this.patterns[e].position}bind(t){const
e=t.gl;this.atlasTexture?this.dirty&&
(this.atlasTexture.update(this.atlasImage),this.dirty=!1):this.atlasTexture=new
x(t,this.atlasImage,e.RGBA),this.atlasTexture.bind(e.LINEAR,e.CLAMP_TO_EDGE);}_u
pdatePatternAtlas(){const e=[];for(const t in
this.patterns)e.push(this.patterns[t].bin);const{w:i,h:s}=t.p(e),a=this.atlasIma
ge;a.resize({width:i||1,height:s||1});for(const e in this.patterns)
{const{bin:i}=this.patterns[e],s=i.x+1,o=i.y+1,r=this.getImage(e).data,n=r.width
,l=r.height;t.R.copy(r,a,{x:0,y:0},{x:s,y:o},{width:n,height:l}),t.R.copy(r,a,
{x:0,y:l-1},{x:s,y:o-1},{width:n,height:l}),t.R.copy(r,a,{x:0,y:0},{x:s,y:o+1},
{width:n,height:l}),t.R.copy(r,a,{x:n-1,y:0},{x:s-1,y:o},
{width:1,height:l}),t.R.copy(r,a,{x:0,y:0},{x:s+n,y:o},
{width:1,height:l});}this.dirty=!0;}beginFrame()
{this.callbackDispatchedThisFrame={};}dispatchRenderCallbacks(e){for(const i of
e)
{if(this.callbackDispatchedThisFrame[i])continue;this.callbackDispatchedThisFram
e[i]=!0;const e=this.getImage(i);e||t.w(`Image with ID: "${i}" was not
found`),y(e)&&this.updateImage(i,e);}}const w=1e20;function
T(t,e,i,s,a,o,r,n,l){for(let h=e;h<e+s;h++)I(t,i*o+h,o,a,r,n,l);for(let
h=i;h<i+a;h++)I(t,h*o+e,l,s,r,n,l);}function I(t,e,i,s,a,o,r){o[0]=0,r[0]=--
w,r[1]=w,a[0]=t[e];for(let n=1,l=0,h=0;n<s;n++){a[n]=t[e+n*i];const
s=n*n;do{const t=o[l];h=(a[n]-a[t]+s-t*t)/(n-t)/2;}while(h<=r[l]&&--
l>-1);l++,o[l]=n,r[l]=h,r[l+1]=w;}for(let n=0,l=0;n<s;n++){for(;r[l+1]
<n;)l++;const s=o[l],h=n-s;t[e+n*i]=a[s]+h*h;}}class E{constructor(t,e)
{this.requestManager=t,this.localIdeographFontFamily=e,this.entries=
{};}}setURL(t){this.url=t;}getGlyphs(e,i){const s=[];for(const t in e)for(const i
of e[t])s.push({stack:t,id:i});t.o(s,(({stack:t,id:e},i)=>{let
s=this.entries[t];s||(s=this.entries[t]={glyphs:{},requests:{},ranges:{}});let
a=s.glyphs[e];if(void 0!==a)return void i(null,
{stack:t,id:e,glyph:a});if(a=this._tinySDF(s,t,e),a)return s.glyphs[e]=a,void
i(null,{stack:t,id:e,glyph:a});const o=Math.floor(e/256);if(256*o>65535)return
void i(new Error("glyphs > 65535 not supported"));if(s.ranges[o])return void
i(null,{stack:t,id:e,glyph:a});if(!this.url)return void i(new Error("glyphsUrl
is not set"));let r=s.requests[o];r||(r=s.requests[o]=
[],E.loadGlyphRange(t,o,this.url,this.requestManager,((t,e)=>{if(e){for(const t
in e)this._doesCharSupportLocalGlyph(+t)||
(s.glyphs[+t]=e[+t]);s.ranges[o]=!0;}for(const i of r)i(t,e);delete
s.requests[o];})),r.push(((s,a)=>{s?i(s):a&&i(null,
{stack:t,id:e,glyph:a[e]||null});})),((t,e)=>{if(t)i(t);else if(e){const t=
{};for(const{stack:i,id:s,glyph:a}of e)(t[i]||(t[i]={}))[s]=a&&
{id:a.id,bitmap:a.bitmap.clone(),metrics:a.metrics};i(null,t);}));}_doesCharSup
portLocalGlyph(e){return !!this.localIdeographFontFamily&&(t.u["CJK Unified
Ideographs"])(e)||t.u["Hangul Syllables"]
(e)||t.u.Hiragana(e)||t.u.Katakana(e))}_tinySDF(e,i,s){const
a=this.localIdeographFontFamily;if(!a)return;if(!this._doesCharSupportLocalGlyph
(s))return;let o=e.tinySDF;if(!o){let t="400";/bold/i.test(i)?
t="900":/medium/i.test(i)?t="500":/light/i.test(i)&&(t="200"),o=e.tinySDF=new
E.TinySDF({fontSize:48,buffer:6,radius:16,cutoff:.25,fontFamily:a,fontWeight:t})}

```

```

; }const r=o.draw(String.fromCharCode(s));return {id:s,bitmap:new
t.q({width:r.width||60,height:r.height||60},r.data),metrics:
{width:r.glyphWidth/2||24,height:r.glyphHeight/2||24,left:r.glyphLeft/2+.5||0,to
p:r.glyphTop/2-
27.5||-8,advance:r.glyphAdvance/2||24,isDoubleResolution:!0}}}}E.loadGlyphRange=
function(e,i,s,a,o){const r=256*i,n=r+255,l=a.transformRequest(s.replace("
{fontstack}",e).replace("{range}",`${r}-${n}`),c.Glyphs);t.l(1,((e,i)=>
{if(e)o(e);else if(i){const e={};for(const s of
t.n(i))e[s.id]=s;o(null,e);}}));E.TinySDF=class{constructor({fontSize:t=24,buf
fer:e=3,radius:i=8,cutoff:s=.25,fontFamily:a="sans-
serif",fontWeight:o="normal",fontStyle:r="normal"}={})
{this.buffer=e,this.cutoff=s,this.radius=i;const
n=this.size=t+4*e,l=this._createCanvas(n),h=this.ctx=l.getContext("2d",
{willReadFrequently:!0});h.font=`$ {r} $ {o} $ {t}px
$ {a}`;h.textBaseline="alphabetic",h.textAlign="left",h.fillStyle="black",this.gr
idOuter=new Float64Array(n*n),this.gridInner=new Float64Array(n*n),this.f=new
Float64Array(n),this.z=new Float64Array(n+1),this.v=new
Uint16Array(n);}_createCanvas(t){const e=document.createElement("canvas");return
e.width=e.height=t,e.draw(t)
{const{width:e,actualBoundingBoxAscent:i,actualBoundingBoxDescent:s,actualBoundi
ngBoxLeft:a,actualBoundingBoxRight:o}=this.ctx.measureText(t),r=Math.ceil(i),n=M
ath.max(0,Math.min(this.size-this.buffer,Math.ceil(o-a))),l=Math.min(this.size-
this.buffer,r+Math.ceil(s)),h=n+2*this.buffer,c=l+2*this.buffer,u=Math.max(h*c,0
),d=new Uint8ClampedArray(u),_=
{data:d,width:h,height:c,glyphWidth:n,glyphHeight:l,glyphTop:r,glyphLeft:0,glyph
Advance:e};if(0===n||0===l)return
_;const{ctx:p,buffer:m,gridInner:f,gridOuter:g}=this;p.clearRect(m,m,n,l),p.fill
Text(t,m,m+r);const
v=p.getImageData(m,m,n,l);g.fill(w,0,u),f.fill(0,0,u);for(let
t=0;t<l;t++)for(let e=0;e<n;e++){const i=v.data[4*
(t*n+e)+3]/255;if(0===i)continue;const s=(t+m)*h+e+m;if(1===i)g[s]=0,f[s]=w;else
{const t=.5-i;g[s]=t>0?t*t:0,f[s]=t<0?
t*t:0;}}T(g,0,0,h,c,h,this.f,this.v,this.z),T(f,m,m,n,l,h,this.f,this.v,this.z);
for(let t=0;t<u;t++){const e=Math.sqrt(g[t])-
Math.sqrt(f[t]);d[t]=Math.round(255-255*(e/this.radius+this.cutoff));}return
_};class S{constructor()
{this.specification=t.v.light.position;}possiblyEvaluate(e,i){return
t.z(e.expression.evaluate(i))}interpolate(e,i,s){return
{x:t.B.number(e.x,i.x,s),y:t.B.number(e.y,i.y,s),z:t.B.number(e.z,i.z,s)}}let
C;class P extends t.E{constructor(e){super(),C=C||new t.r({anchor:new
t.D(t.v.light.anchor),position:new S,color:new
t.D(t.v.light.color),intensity:new
t.D(t.v.light.intensity)}),this._transitionable=new
t.T(C),this.setLight(e),this._transitioning=this._transitionable.untransitioned(
);}getLight(){return this._transitionable.serialize()}setLight(e,i={})
{if(!this._validate(t.t,e,i))for(const t in e){const i=e[t];t.endsWith("-
transition")?
this._transitionable.setTransition(t.slice(0,-11),i):this._transitionable.setVal
ue(t,i);}updateTransitions(t)
{this._transitioning=this._transitionable.transitioned(t,this._transitioning);}h
asTransition(){return this._transitioning.hasTransition()}recalculate(t)
{this.properties=this._transitioning.possiblyEvaluate(t);}_validate(e,i,s)
{return (!s||!1!==s.validate)&&t.x(this,e.call(t.y,t.e({value:i,style:
{glyphs:!0,sprite:!0},styleSpec:t.v})))}}class D{constructor(t,e)
{this.width=t,this.height=e,this.nextRow=0,this.data=new
Uint8Array(this.width*this.height),this.dashEntry={};getDash(t,e){const
i=t.join(",")+String(e);return this.dashEntry[i]||
(this.dashEntry[i]=this.addDash(t,e)),this.dashEntry[i]}getDashRanges(t,e,i)
{const s=[];let a=t.length%2==1?-t[t.length-
1]*i:0,o=t[0]*i,r=!0;s.push({left:a,right:o,isDash:r,zeroLength:0===t[0]});let
n=t[0];for(let e=1;e<t.length;e++){r=!r;const
l=t[e];a=n*i,n+=l,o=n*i,s.push({left:a,right:o,isDash:r,zeroLength:0===l});}retu

```

```

rn s}addRoundDash(t,e,i){const s=e/2;for(let e=-i;e<=i;e++){const a=this.width*
(this.nextRow+i+e);let o=0,r=t[o];for(let n=0;n<this.width;n++){n/r.right>1&&
(r=t[++o]);const l=Math.abs(n-r.left),h=Math.abs(n-r.right),c=Math.min(l,h);let
u;const d=e/i*(s+1);if(r.isDash){const t=s-
Math.abs(d);u=Math.sqrt(c*c+t*t);}else u=s-
Math.sqrt(c*c+d*d);this.data[a+n]=Math.max(0,Math.min(255,u+128));}}addRegularD
ash(t){for(let e=t.length-1;e>=0;--e){const i=t[e],s=t[e+1];i.zeroLength?
t.splice(e,1):s&&s.isDash===i.isDash&&(s.left=i.left,t.splice(e,1));}const
e=t[0],i=t[t.length-1];e.isDash===i.isDash&&(e.left=i.left-
this.width,i.right=e.right+this.width);const s=this.width*this.nextRow;let
a=0,o=t[a];for(let e=0;e<this.width;e++){e/o.right>1&&(o=t[++a]);const
i=Math.abs(e-o.left),r=Math.abs(e-
o.right),n=Math.min(i,r);this.data[s+e]=Math.max(0,Math.min(255,(o.isDash?n:-
n)+128));}}addDash(e,i){const s=i?
7:0,a=2*s+1;if(this.nextRow+a>this.height)return t.w("LineAtlas out of
space"),null;let o=0;for(let t=0;t<e.length;t++)o+=e[t];if(0!==o){const
t=this.width/o,a=this.getDashRanges(e,this.width,t);i?
this.addRoundDash(a,t,s):this.addRegularDash(a);}const r={y:
(this.nextRow+s+.5)/this.height,height:2*s/this.height,width:o};return
this.nextRow+=a,this.dirty=!0,r}bind(t){const e=t.gl;this.texture?
(e.bindTexture(e.TEXTURE_2D,this.texture),this.dirty&&
(this.dirty=!1,e.texSubImage2D(e.TEXTURE_2D,0,0,0,this.width,this.height,e.ALPHA
,e.UNSIGNED_BYTE,this.data))):
(this.texture=e.createTexture(),e.bindTexture(e.TEXTURE_2D,this.texture),e.texPa
rameteri(e.TEXTURE_2D,e.TEXTURE_WRAP_S,e.REPEAT),e.texParameteri(e.TEXTURE_2D,e.
TEXTURE_WRAP_T,e.REPEAT),e.texParameteri(e.TEXTURE_2D,e.TEXTURE_MIN_FILTER,e.LIN
EAR),e.texParameteri(e.TEXTURE_2D,e.TEXTURE_MAG_FILTER,e.LINEAR),e.texImage2D(e.
TEXTURE_2D,0,e.ALPHA,this.width,this.height,0,e.ALPHA,e.UNSIGNED_BYTE,this.data)
);}class M{constructor(e,i,s){this.workerPool=e,this.actors=
[],this.currentActor=0,this.id=s;const a=this.workerPool.acquire(s);for(let
e=0;e<a.length;e++){const o=new t.C(a[e],i,s);o.name=`Worker
${e}`,this.actors.push(o);}if(!this.actors.length)throw new Error("No actors
found")}broadcast(e,i,s){t.o(this.actors,((t,s)=>
{t.send(e,i,s)}),s=s||function(){});}getActor(){return this.currentActor=
(this.currentActor+1)%this.actors.length,this.actors[this.currentActor]}remove(
t){this.actors.forEach((t=>{t.remove()})),this.actors=
[],t&&this.workerPool.release(this.id);}function z(e,i,s){const a=function(i,a)
{if(i)return s(i);if(a){const i=t.F(t.e(a,e),
["tiles","minzoom","maxzoom","attribution","bounds","scheme","tileSize","encodin
g"]);a.vector_layers&&
(i.vectorLayers=a.vector_layers,i.vectorLayerIds=i.vectorLayers.map((t=>t.id))),
s(null,i)};}}return e.url?
t.f(i.transformRequest(e.url,c.Source),a):t.h.frame((())=>a(null,e))}class
L{constructor(t,e){t&&(e?this.setSouthWest(t).setNorthEast(e):Array.isArray(t)&&
(4===t.length?
this.setSouthWest([t[0],t[1]]).setNorthEast([t[2],t[3]]):this.setSouthWest(t[0])
.setNorthEast(t[1])));}setNorthEast(e){return this._ne=e instanceof t.L?new
t.L(e.lng,e.lat):t.L.convert(e,this)}setSouthWest(e){return this._sw=e
instanceof t.L?new t.L(e.lng,e.lat):t.L.convert(e,this)}extend(e){const
i=this._sw,s=this._ne;let a,o;if(e instanceof t.L)a=e,o=e;else if(!(e
instanceof L))return Array.isArray(e)?4===e.length||e.every(Array.isArray)?
this.extend(L.convert(e)):this.extend(t.L.convert(e)):e&&("lng"in e||"lon"in
e)&&"lat"in e?this.extend(t.L.convert(e)):this;if(a=e._sw,o=e._ne,!a||!o)return
this}return i||s?
(i.lng=Math.min(a.lng,i.lng),i.lat=Math.min(a.lat,i.lat),s.lng=Math.max(o.lng,s.
lng),s.lat=Math.max(o.lat,s.lat)):this._sw=new t.L(a.lng,a.lat),this._ne=new
t.L(o.lng,o.lat),this}getCenter(){return new t.L((this._sw.lng+this._ne.lng)/2,
(this._sw.lat+this._ne.lat)/2)}getSouthWest(){return this._sw}getNorthEast(){
return this._ne}getNorthWest(){return new
t.L(this.getWest(),this.getNorth())}getSouthEast(){return new
t.L(this.getEast(),this.getSouth())}getWest(){return this._sw.lng}getSouth(){
return this._sw.lat}getEast(){return this._ne.lng}getNorth(){return

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this._ne.lat}toArray(){return [this._sw.toArray(),this._ne.toArray()]}toString()
{return `LngLatBounds(${this._sw.toString()}, ${this._ne.toString()})`}isEmpty()
{return !(this._sw&&this._ne)}contains(e){const{lng:i,lat:s}=t.L.convert(e);let
a=this._sw.lng<=i&&i<=this._ne.lng;return this._sw.lng>this._ne.lng&&
(a=this._sw.lng>=i&&i>=this._ne.lng),this._sw.lat<=s&&s<=this._ne.lat&&a}static
convert(t){return t instanceof L?t:t?new L(t):t}static fromLngLat(e,i=0){const
s=360*i/40075017,a=s/Math.cos(Math.PI/180*e.lat);return new L(new t.L(e.lng-
a,e.lat-s),new t.L(e.lng+a,e.lat+s))}class A{constructor(t,e,i)
{this.bounds=L.convert(this.validateBounds(t)),this.minzoom=e|0,this.maxzoom=i|
24;}validateBounds(t){return Array.isArray(t)&&4===t.length?
[Math.max(-180,t[0]),Math.max(-90,t[1]),Math.min(180,t[2]),Math.min(90,t[3])]:
[-180,-90,180,90]}contains(e){const
i=Math.pow(2,e.z),s=Math.floor(t.G(this.bounds.getWest())*i),a=Math.floor(t.H(th
is.bounds.getNorth())*i),o=Math.ceil(t.G(this.bounds.getEast())*i),r=Math.ceil(t
.H(this.bounds.getSouth())*i);return e.x>=s&&e.x<o&&e.y>=a&&e.y<r}}class R
extends t.E{constructor(e,i,s,a){if(super(),this.load()==>
{this._loaded=!1,this.fire(new t.k("dataloading",
{dataType:"source"})),this._tileJSONRequest=z(this._options,this.map._requestMan
ager,((e,i)=>
{this._tileJSONRequest=null,this._loaded=!0,this.map.style.sourceCaches[this.id]
.clearTiles(),e?this.fire(new t.j(e)):i&&(t.e(this,i),i.bounds&&
(this.tileBounds=new A(i.bounds,this.minzoom,this.maxzoom)),this.fire(new
t.k("data",{dataType:"source",sourceDataType:"metadata"})),this.fire(new
t.k("data",{dataType:"source",sourceDataType:"content"}))));});this.serialize=
()==>t.e({},this._options),this.id=e,this.dispatcher=s,this.type="vector",this.mi
nzoom=0,this.maxzoom=22,this.scheme="xyz",this.tileSize=512,this.reparseOverscal
ed=!0,this.isTileClipped=!0,this._loaded=!1,t.e(this,t.F(i,
["url","scheme","tileSize","promoteId"])),this._options=t.e({type:"vector"},i),t
his._collectResourceTiming=i.collectResourceTiming,512!==this.tileSize)throw new
Error("vector tile sources must have a tileSize of
512");this.setEventedParent(a);}loaded(){return this._loaded}hasTile(t){return
!this.tileBounds||this.tileBounds.contains(t.canonical)}onAdd(t)
{this.map=t,this.load();}setSourceProperty(t)
{this._tileJSONRequest&&this._tileJSONRequest.cancel(),t(),this.load();}setTiles
(t){return this.setSourceProperty((()=>{this._options.tiles=t;})),this}setUrl(t)
{return this.setSourceProperty((()=>
{this.url=t,this._options.url=t;})),this}onRemove(){this._tileJSONRequest&&
(this._tileJSONRequest.cancel(),this._tileJSONRequest=null);}loadTile(t,e){const
i=t.tileID.canonical.url(this.tiles,this.map.getPixelRatio(),this.scheme),s=
{request:this.map._requestManager.transformRequest(i,c.Tile),uid:t.uid,tileID:t.
tileID,zoom:t.tileID.overscaledZ,tileSize:this.tileSize*t.tileID.overscaleFactor
(),type:this.type,source:this.id,pixelRatio:this.map.getPixelRatio(),showCollisi
onBoxes:this.map.showCollisionBoxes,promoteId:this.promoteId};function a(i,s)
{return delete t.request,t.aborted?e(null):i&&404!==i.status?e(i):
(s&&s.resourceTiming&&
(t.resourceTiming=s.resourceTiming),this.map._refreshExpiredTiles&&s&&t.setExpir
yData(s),t.loadVectorData(s,this.map.painter),e(null),void(t.reloadCallback&&
(this.loadTile(t,t.reloadCallback),t.reloadCallback=null)))}s.request.collectRes
ourceTiming=this._collectResourceTiming,t.actor&&"expired"!==t.state?"loading"==
=t.state?t.reloadCallback=e:t.request=t.actor.send("reloadTile",s,a.bind(this)):
(t.actor=this.dispatcher.getActor(),t.request=t.actor.send("loadTile",s,a.bind(t
his)));}abortTile(t){t.request&&(t.request.cancel(),delete
t.request),t.actor&&t.actor.send("abortTile",
{uid:t.uid,type:this.type,source:this.id},void 0);}unloadTile(t)
{t.unloadVectorData(),t.actor&&t.actor.send("removeTile",
{uid:t.uid,type:this.type,source:this.id},void 0);}hasTransition(){return
!1}}class k extends t.E{constructor(e,i,s,a)
{super(),this.id=e,this.dispatcher=s,this.setEventedParent(a),this.type="raster"
,this.minzoom=0,this.maxzoom=22,this.roundZoom=!0,this.scheme="xyz",this.tileSiz
e=512,this._loaded=!1,this._options=t.e({type:"raster"},i),t.e(this,t.F(i,
["url","scheme","tileSize"]));}load(){this._loaded=!1,this.fire(new
t.k("dataloading",

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{dataType:"source"})),this._tileJSONRequest=z(this._options,this.map._requestManager,((e,i)=>{this._tileJSONRequest=null,this._loaded=!0,e?this.fire(new t.j(e)):i&&(t.e(this,i),i.bounds&&(this.tileBounds=new A(i.bounds,this.minzoom,this.maxzoom)),this.fire(new t.k("data",{dataType:"source",sourceDataType:"metadata"})),this.fire(new t.k("data",{dataType:"source",sourceDataType:"content"}))));});loaded(){return this._loaded}onAdd(t){this.map=t,this.load();}onRemove(){this._tileJSONRequest&&(this._tileJSONRequest.cancel(),this._tileJSONRequest=null);}setSourceProperty(t){this._tileJSONRequest&&this._tileJSONRequest.cancel(),t(),this.load();}setTiles(t){return this.setSourceProperty((()=>{this._options.tiles=t;})),this.serialize(){return t.e({},this._options)}hasTile(t){return !this.tileBounds||this.tileBounds.contains(t.canonical)}loadTile(t,e){const i=t.tileID.canonical.url(this.tiles,this.map.getPixelRatio(),this.scheme);t.request=h.getImage(this.map._requestManager.transformRequest(i,c.Tile),((i,s,a)=>{if(delete t.request,t.aborted)t.state="unloaded",e(null);else if(i)t.state="errored",e(i);else if(s){this.map._refreshExpiredTiles&&t.setExpiryData(a);const i=this.map.painter.context,o=i.gl;t.texture=this.map.painter.getTileTexture(s.width),t.texture?t.texture.update(s,{useMipmap:!0}):(t.texture=new x(i,s,o.RGBA,{useMipmap:!0}),t.texture.bind(o.LINEAR,o.CLAMP_TO_EDGE,o.LINEAR_MIPMAP_NEAREST),i.extTextureFilterAnisotropic&o.texParameterf(o.TEXTURE_2D,i.extTextureFilterAnisotropic.TEXTURE_MAX_ANISOTROPY_EXT,i.extTextureFilterAnisotropicMax)),t.state="loaded",e(null);})),this.map._refreshExpiredTiles);}abortTile(t,e){t.request&&(t.request.cancel(),delete t.request),e();}unloadTile(t,e){t.texture&&this.map.painter.saveTileTexture(t.texture),e();}hasTransition(){return !1}}class F extends k{constructor(e,i,s,a){super(e,i,s,a),this.type="raster-dem",this.maxzoom=22,this._options=t.e({type:"raster-dem"},i),this.encoding=i.encoding||"mapbox",this.redFactor=i.redFactor,this.greenFactor=i.greenFactor,this.blueFactor=i.blueFactor,this.baseShift=i.baseShift;}loadTile(e,i){const s=e.tileID.canonical.url(this.tiles,this.map.getPixelRatio(),this.scheme),a=this.map._requestManager.transformRequest(s,c.Tile);function o(t,s){t&&(e.state="errored",i(t)),s&&(e.dem=s,e.needsHillshadePrepare=!0,e.needsTerrainPrepare=!0,e.state="loaded",i(null));}e.neighboringTiles=this._getNeighboringTiles(e.tileID),e.request=h.getImage(a,((s,a,r)=>t._(this,void 0,void 0,(function*(){if(delete e.request,e.aborted)e.state="unloaded",i(null);else if(s)e.state="errored",i(s);else if(a){this.map._refreshExpiredTiles&&e.setExpiryData(r);const i=t.a(a)&&t.J()?a:yield function(e){return t._(this,void 0,void 0,(function*(){if("undefined"!==typeof VideoFrame&t.K()){const i=e.width+2,s=e.height+2;try{return new t.R({width:i,height:s},yield t.M(e,-1,-1,i,s))}catch(t){})}return t.h.getImageData(e,1)})))(a),s={uid:e.uid,coord:e.tileID,source:this.id,rawImageData:i,encoding:this.encoding,redFactor:this.redFactor,greenFactor:this.greenFactor,blueFactor:this.blueFactor,baseShift:this.baseShift};e.actor&&"expired"!==e.state||(e.actor=this.dispatcher.getActor(),e.actor.send("loadDEMtile",s,o));}))),this.map._refreshExpiredTiles);}_getNeighboringTiles(e){const i=e.canonical,s=Math.pow(2,i.z),a=(i.x-1+s)%s,o=0===i.x?e.wrap-1:e.wrap,r=(i.x+1+s)%s,n=i.x+1===s?e.wrap+1:e.wrap,l={};return l[new t.O(e.overscaledZ,o,i.z,a,i.y).key]={backfilled:!1},l[new t.O(e.overscaledZ,n,i.z,r,i.y).key]={backfilled:!1},i.y>0&&(l[new t.O(e.overscaledZ,o,i.z,a,i.y-1).key]={backfilled:!1},l[new t.O(e.overscaledZ,e.wrap,i.z,i.x,i.y-1).key]={backfilled:!1},l[new t.O(e.overscaledZ,n,i.z,r,i.y-1).key]={backfilled:!1}),i.y+1<s&&(l[new t.O(e.overscaledZ,o,i.z,a,i.y+1).key]={backfilled:!1},l[new t.O(e.overscaledZ,e.wrap,i.z,i.x,i.y+1).key]={backfilled:!1},l[new t.O(e.overscaledZ,n,i.z,r,i.y+1).key]={backfilled:!1}),l}unloadTile(t){t.demTexture&&this.map.painter.saveTileTexture(t.demTexture),t.fbo&&

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(t.fbo.destroy(),delete t.fbo),t.dem&&delete t.dem,delete
t.neighboringTiles,t.state="unloaded",t.actor&&t.actor.send("removeDEMTile",
{uid:t.uid,source:this.id});}}class B extends t.E{constructor(e,i,s,a)
{super(),this.load(=>{this._updateWorkerData();},this.serialize=
(=>t.e({},this._options,
{type:this.type,data:this._data})),this.id=e,this.type="geojson",this.minzoom=0,t
his.maxzoom=18,this.tileSize=512,this.isTileClipped=!0,this.reparseOverscaled=!0
,this._removed=!1,this._pendingLoads=0,this.actor=s.getActor(),this.setEventedPa
rent(a),this._data=i.data,this._options=t.e({},i),this._collectResourceTiming=i.
collectResourceTiming,void 0!==i.maxzoom&&(this.maxzoom=i.maxzoom),i.type&&
(this.type=i.type),i.attribution&&
(this.attribution=i.attribution),this.promoteId=i.promoteId;const
o=t.N/this.tileSize;this.workerOptions=t.e({source:this.id,cluster:i.cluster||!1
,geojsonVtOptions:{buffer:(void 0!==i.buffer?i.buffer:128)*o,tolerance:(void
0!==i.tolerance?
i.tolerance:.375)*o,extent:t.N,maxZoom:this.maxzoom,lineMetrics:i.lineMetrics||!
1,generateId:i.generateId||!1},superclusterOptions:{maxZoom:void
0!==i.clusterMaxZoom?i.clusterMaxZoom:this.maxzoom-
1,minPoints:Math.max(2,i.clusterMinPoints||2),extent:t.N,radius:
(i.clusterRadius||50)*o,log:!1,generateId:i.generateId||!1},clusterProperties:i.
clusterProperties,filter:i.filter},i.workerOptions),"string"===typeof
this.promoteId&&(this.workerOptions.promoteId=this.promoteId);}onAdd(t)
{this.map=t,this.load();}setData(t){return
this._data=t,this._updateWorkerData(),this}updateData(t){return
this._updateWorkerData(t),this}setClusterOptions(t){return
this.workerOptions.cluster=t.cluster,t&&(void 0!==t.clusterRadius&&
(this.workerOptions.superclusterOptions.radius=t.clusterRadius),void
0!==t.clusterMaxZoom&&
(this.workerOptions.superclusterOptions.maxZoom=t.clusterMaxZoom)),this._updateW
orkerData(),this}getClusterExpansionZoom(t,e){return
this.actor.send("geojson.getClusterExpansionZoom",
{clusterId:t,source:this.id},e),this}getClusterChildren(t,e){return
this.actor.send("geojson.getClusterChildren",
{clusterId:t,source:this.id},e),this}getClusterLeaves(t,e,i,s){return
this.actor.send("geojson.getClusterLeaves",
{source:this.id,clusterId:t,limit:e,offset:i},s),this}_updateWorkerData(e){const
i=t.e({},this.workerOptions);e?i.dataDiff=e:"string"===typeof this._data?
(i.request=this.map._requestManager.transformRequest(t.h.resolveURL(this._data),
c.Source),i.request.collectResourceTiming=this._collectResourceTiming):i.data=JS
ON.stringify(this._data),this._pendingLoads++,this.fire(new t.k("dataloading",
{dataType:"source"})),this.actor.send(`${this.type}.loadData`,i,((e,i)=>
{if(this._pendingLoads--,this._removed||i&&i.abandoned)return void this.fire(new
t.k("dataabort",{dataType:"source"}));let
s=null;if(i&&i.resourceTiming&&i.resourceTiming[this.id]&&
(s=i.resourceTiming[this.id].slice(0)),e)return void this.fire(new t.j(e));const
a={dataType:"source"};this._collectResourceTiming&&s&&s.length>0&&t.e(a,
{resourceTiming:s}),this.fire(new t.k("data",Object.assign(Object.assign({},a),
{sourceDataType:"metadata"}))),this.fire(new
t.k("data",Object.assign(Object.assign({},a),
{sourceDataType:"content"})))));}loaded(){return
0===this._pendingLoads}loadTile(t,e){const
i=t.actor?"reloadTile":"loadTile";t.actor=this.actor;const s=
{type:this.type,uid:t.uid,tileID:t.tileID,zoom:t.tileID.overscaledZ,maxZoom:this
.maxzoom,tileSize:this.tileSize,source:this.id,pixelRatio:this.map.getPixelRatio
(),showCollisionBoxes:this.map.showCollisionBoxes,promoteId:this.promoteId};t.re
quest=this.actor.send(i,s,((s,a)=>(delete
t.request,t.unloadVectorData(),t.aborted?e(null):s?e(s):
(t.loadVectorData(a,this.map.painter,"reloadTile"===i),e(null)))));}abortTile(t)
{t.request&&(t.request.cancel(),delete t.request),t.aborted=!0;}unloadTile(t)
{t.unloadVectorData(),this.actor.send("removeTile",
{uid:t.uid,type:this.type,source:this.id});}onRemove()
{this._removed=!0,this.actor.send("removeSource",

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{type:this.type,source:this.id});}hasTransition(){return !1}}var
O=t.Q([{"name":"a_pos",type:"Int16",components:2},
{"name":"a_texture_pos",type:"Int16",components:2}]);class N extends
t.E{constructor(e,i,s,a){super(),this.load=(e,i)=>{this._loaded=!1,this.fire(new
t.k("dataloading",
{dataType:"source"})),this.url=this.options.url,this._request=h.getImage(this.ma
p._requestManager.transformRequest(this.url,c.Image),((s,a)=>
{this._request=null,this._loaded=!0,s?this.fire(new t.j(s)):a&&(this.image=a,e&&
(this.coordinates=e),i&&i(),this._finishLoading());}));},this.prepare=()=>>
{if(0===Object.keys(this.tiles).length||!this.image)return;const
e=this.map.painter.context,i=e.gl;this.boundsBuffer||
(this.boundsBuffer=e.createVertexBuffer(this._boundsArray,O.members)),this.bound
sSegments||(this.boundsSegments=t.S.simpleSegment(0,0,4,2)),this.texture||
(this.texture=new
x(e,this.image,i.RGBA),this.texture.bind(i.LINEAR,i.CLAMP_TO_EDGE));let
s=!1;for(const t in this.tiles){const e=this.tiles[t];"loaded"!==e.state&&
(e.state="loaded",e.texture=this.texture,s=!0);}s&&this.fire(new t.k("data",
{dataType:"source",sourceDataType:"idle",sourceId:this.id}));},this.serialize=
()=>>
({type:"image",url:this.options.url,coordinates:this.coordinates}),this.id=e,thi
s.dispatcher=s,this.coordinates=i.coordinates,this.type="image",this.minzoom=0,t
his.maxzoom=22,this.tileSize=512,this.tiles=
{}},this._loaded=!1,this.setEventedParent(a),this.options=i;}loaded(){return
this._loaded}updateImage(t){return t.url?(this._request&&
(this._request.cancel(),this._request=null),this.options.url=t.url,this.load(t.c
oordinates,(()=>>{this.texture=null;})),this):this._finishLoading(){this.map&&
(this.setCoordinates(this.coordinates),this.fire(new t.k("data",
{dataType:"source",sourceDataType:"metadata"}))});}onAdd(t)
{this.map=t,this.load();}onRemove(){this._request&&
(this._request.cancel(),this._request=null);}setCoordinates(e)
{this.coordinates=e;const i=e.map(t.U.fromLngLat);this.tileID=function(e){let
i=1/0,s=1/0,a=-1/0,o=-1/0;for(const t of
e)i=Math.min(i,t.x),s=Math.min(s,t.y),a=Math.max(a,t.x),o=Math.max(o,t.y);const
r=Math.max(a-i,o-s),n=Math.max(0,Math.floor(-
Math.log(r)/Math.LN2)),l=Math.pow(2,n);return new
t.W(n,Math.floor((i+a)/2*1),Math.floor((s+o)/2*1))}
(i),this.minzoom=this.maxzoom=this.tileID.z;const
s=i.map((t=>this.tileID.getTilePoint(t)._round()));return this._boundsArray=new
t.V,this._boundsArray.emplaceBack(s[0].x,s[0].y,0,0),this._boundsArray.emplaceBa
ck(s[1].x,s[1].y,t.N,0),this._boundsArray.emplaceBack(s[3].x,s[3].y,0,t.N),this.
_boundsArray.emplaceBack(s[2].x,s[2].y,t.N,t.N),this.boundsBuffer&&
(this.boundsBuffer.destroy(),delete this.boundsBuffer),this.fire(new t.k("data",
{dataType:"source",sourceDataType:"content"})),this.loadTile(t,e)
{this.tileID&&this.tileID.equals(t.tileID.canonical)?
(this.tiles[String(t.tileID.wrap)]=t,t.buckets={},e(null)):
(t.state="errored",e(null));}hasTransition(){return !1}}class U extends
N{constructor(e,i,s,a){super(e,i,s,a),this.load=()=>>{this._loaded=!1;const
e=this.options;this.urls=[];for(const t of
e.urls)this.urls.push(this.map._requestManager.transformRequest(t,c.Source).url)
;t.X(this.urls,((e,i)=>{this._loaded=!0,e?this.fire(new t.j(e)):i&&
(this.video=i,this.video.loop=!0,this.video.addEventListener("playing",(()=>>
{this.map.triggerRepaint();})),this.map&&this.video.play(),this._finishLoading()
);}));},this.prepare=()=>>
{if(0===Object.keys(this.tiles).length||this.video.readyState<2)return;const
e=this.map.painter.context,i=e.gl;this.boundsBuffer||
(this.boundsBuffer=e.createVertexBuffer(this._boundsArray,O.members)),this.bound
sSegments||(this.boundsSegments=t.S.simpleSegment(0,0,4,2)),this.texture?
this.video.paused||
(this.texture.bind(i.LINEAR,i.CLAMP_TO_EDGE),i.texSubImage2D(i.TEXTURE_2D,0,0,0,
i.RGBA,i.UNSIGNED_BYTE,this.video)):this.texture=new
x(e,this.video,i.RGBA),this.texture.bind(i.LINEAR,i.CLAMP_TO_EDGE));let
s=!1;for(const t in this.tiles){const e=this.tiles[t];"loaded"!==e.state&&

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(e.state="loaded",e.texture=this.texture,s=!0);}s&&this.fire(new t.k("data",
{dataType:"source",sourceDataType:"idle",sourceId:this.id}));},this.serialize=
()=>
({type:"video",urls:this.urls,coordinates:this.coordinates}),this.roundZoom=!0,t
his.type="video",this.options=i;}pause(){this.video&&this.video.pause();}play()
{this.video&&this.video.play();}seek(e){if(this.video){const
i=this.video.seekable;e<i.start(0)||e>i.end(0)?this.fire(new t.j(new
t.Y(`sources.${this.id}`,null,`Playback for this video can be set only between
the ${i.start(0)} and ${i.end(0)}-second
mark.`)):this.video.currentTime=e;}}getVideo(){return this.video}onAdd(t)
{this.map||(this.map=t,this.load(),this.video&&
(this.video.play(),this.setCoordinates(this.coordinates)));}hasTransition()
{return this.video&&!this.video.paused}class Z extends N{constructor(e,i,s,a)
{super(e,i,s,a),this.load()=>{this._loaded=!0,this.canvas||
(this.canvas=this.options.canvas instanceof HTMLCanvasElement?
this.options.canvas:document.getElementById(this.options.canvas)),this.width=thi
s.canvas.width,this.height=this.canvas.height,this._hasInvalidDimensions()?
this.fire(new t.j(new Error("Canvas dimensions cannot be less than or equal to
zero."))):(this.play=function()
{this._playing=!0,this.map.triggerRepaint();},this.pause=function()
{this._playing&&
(this.prepare(),this._playing=!1);},this._finishLoading());},this.prepare=()=>
{let e=!1;if(this.canvas.width!==this.width&&
(this.width=this.canvas.width,e=!0),this.canvas.height!==this.height&&
(this.height=this.canvas.height,e=!0),this._hasInvalidDimensions())return;if(0==
=Object.keys(this.tiles).length)return;const
i=this.map.painter.context,s=i.gl;this.boundsBuffer||
(this.boundsBuffer=i.createVertexBuffer(this._boundsArray,0.members)),this.bound
sSegments||(this.boundsSegments=t.S.simpleSegment(0,0,4,2)),this.texture?
(e||this._playing&&this.texture.update(this.canvas,
{premultiply:!0}):this.texture=new x(i,this.canvas,s.RGBA,{premultiply:!0});let
a=!1;for(const t in this.tiles){const e=this.tiles[t];"loaded"!==e.state&&
(e.state="loaded",e.texture=this.texture,a=!0);}a&&this.fire(new t.k("data",
{dataType:"source",sourceDataType:"idle",sourceId:this.id}));},this.serialize=
()=>({type:"canvas",coordinates:this.coordinates}),i.coordinates?
Array.isArray(i.coordinates)&&4===i.coordinates.length&&!i.coordinates.some((t=>
!Array.isArray(t)||2!==t.length||t.some((t=>"number"!==typeof
t))))||this.fire(new t.j(new t.Y(`sources.${e}`,null,'"coordinates" property
must be an array of 4 longitude/latitude array pairs'))):this.fire(new t.j(new
t.Y(`sources.${e}`,null,'missing required property
"coordinates"'))),i.animate&&"boolean"!==typeof i.animate&&this.fire(new t.j(new
t.Y(`sources.${e}`,null,'optional "animate" property must be a boolean
value'))),i.canvas?"string"===typeof i.canvas||i.canvas instanceof
HTMLCanvasElement||this.fire(new t.j(new t.Y(`sources.${e}`,null,'"canvas" must
be either a string representing the ID of the canvas element from which to read,
or an HTMLCanvasElement instance'))):this.fire(new t.j(new
t.Y(`sources.${e}`,null,'missing required property
"canvas"'))),this.options=i,this.animate=void
0===i.animate||i.animate;}getCanvas(){return this.canvas}onAdd(t)
{this.map=t,this.load(),this.canvas&&this.animate&&this.play();}onRemove()
{this.pause();}hasTransition(){return this._playing}_hasInvalidDimensions()
{for(const t of [this.canvas.width,this.canvas.height])if(isNaN(t)||t<=0)return
!0;return !1}const G={},j=t=>{switch(t){case"geojson":return
B;case"image":return N;case"raster":return k;case"raster-dem":return
F;case"vector":return R;case"video":return U;case"canvas":return Z}return
G[t]};function V(e,i){const s=t.Z();return t.$(s,s,[1,1,0]),t.a0(s,s,
[.5*e.width,.5*e.height,1]),t.a1(s,s,e.calculatePosMatrix(i.toUnwrapped()))}func
tion q(t,e,i,s,a,o){const r=function(t,e,i){if(t)for(const s of t){const
t=e[s];if(t&&t.source===i&&"fill-extrusion"===t.type)return !0}else for(const t
in e){const s=e[t];if(s.source===i&&"fill-extrusion"===s.type)return !0}return
!1}
(a&&a.layers,e,t.id),n=o.maxPitchScaleFactor(),l=t.tilesIn(s,n,r);l.sort($);cons

```



```

t h=[];for(const s of
l)h.push({wrappedTileID:s.tileID.wrapped().key,queryResults:s.tile.queryRendered
Features(e,i,t.state,s.queryGeometry,s.cameraQueryGeometry,s.scale,a,o,n,V(t.tr
ansform,s.tileID)}));const c=function(t){const e={},i={};for(const s of t){const
t=s.queryResults,a=s.wrappedTileID,o=i[a]=i[a]||{};for(const i in t){const
s=t[i],a=o[i]=o[i]||{};r=e[i]=e[i]||[];for(const t of s)a[t.featureIndex]||
(a[t.featureIndex]=!0,r.push(t));}}return e}(h);for(const e in
c)c[e].forEach((e=>{const i=e.feature,s=t.getFeatureState(i.layer["source-
layer"],i.id);i.source=i.layer.source,i.layer["source-layer"]&&
(i.sourceLayer=i.layer["source-layer"],i.state=s;}));return c}function $(t,e)
{const i=t.tileID,s=e.tileID;return i.overscaledZ-s.overscaledZ||i.canonical.y-
s.canonical.y||i.wrap-s.wrap||i.canonical.x-s.canonical.x}class
W{constructor(e,i)
{this.timeAdded=0,this.fadeEndTime=0,this.tileID=e,this.uid=t.a2(),this.uses=0,t
his.tileSize=i,this.buckets=
{},this.expirationTime=null,this.queryPadding=0,this.hasSymbolBuckets=!1,this.ha
sRTLText=!1,this.dependencies={},this.rtt=[],this.rttCoords=
{},this.expiredRequestCount=0,this.state="loading";}registerFadeDuration(t)
{const e=t+this.timeAdded;e<this.fadeEndTime||
(this.fadeEndTime=e);}wasRequested(){return
"errored"===this.state||"loaded"===this.state||"reloading"===this.state}clearTex
tures(t)
{this.demTexture&t.saveTileTexture(this.demTexture),this.demTexture=null;}loadV
ectorData(e,i,s)
{if(this.hasData()&&this.unloadVectorData(),this.state="loaded",e)
{e.featureIndex&&(this.latestFeatureIndex=e.featureIndex,e.rawTileData?
(this.latestRawTileData=e.rawTileData,this.latestFeatureIndex.rawTileData=e.rawT
ileData):this.latestRawTileData&&
(this.latestFeatureIndex.rawTileData=this.latestRawTileData)),this.collisionBoxA
rray=e.collisionBoxArray,this.buckets=function(t,e){const i={};if(!e)return
i;for(const s of t){const
t=s.layerIds.map((t=>e.getLayer(t))).filter(Boolean);if(0!==t.length)
{s.layers=t,s.stateDependentLayerIds&&
(s.stateDependentLayers=s.stateDependentLayerIds.map((e=>t.filter((t=>t.id===e)
[0])));for(const e of t)i[e.id]=s;}}return i}
(e.buckets,i.style),this.hasSymbolBuckets=!1;for(const e in this.buckets){const
i=this.buckets[e];if(i instanceof t.a4)
{if(this.hasSymbolBuckets=!0,!s)break;i.justReloaded=!0;}}if(this.hasRTLText=!1,
this.hasSymbolBuckets)for(const e in this.buckets){const i=this.buckets[e];if(i
instanceof t.a4&&i.hasRTLText)
{this.hasRTLText=!0,t.a5();break}}this.queryPadding=0;for(const t in
this.buckets){const
e=this.buckets[t];this.queryPadding=Math.max(this.queryPadding,i.style.getLayer(
t).queryRadius(e));e.imageAtlas&&
(this.imageAtlas=e.imageAtlas),e.glyphAtlasImage&&
(this.glyphAtlasImage=e.glyphAtlasImage);}else this.collisionBoxArray=new
t.a3;}unloadVectorData(){for(const t in
this.buckets)this.buckets[t].destroy();this.buckets=
{},this.imageAtlasTexture&&this.imageAtlasTexture.destroy(),this.imageAtlas&&
(this.imageAtlas=null),this.glyphAtlasTexture&&this.glyphAtlasTexture.destroy(),
this.latestFeatureIndex=null,this.state="unloaded";}getBucket(t){return
this.buckets[t.id]}upload(t){for(const e in this.buckets){const
i=this.buckets[e];i.uploadPending()&&i.upload(t);}const
e=t.gl;this.imageAtlas&&!this.imageAtlas.uploaded&&(this.imageAtlasTexture=new
x(t,this.imageAtlas.image,e.RGBA),this.imageAtlas.uploaded=!0),this.glyphAtlasIm
age&&(this.glyphAtlasTexture=new
x(t,this.glyphAtlasImage,e.ALPHA),this.glyphAtlasImage=null);}prepare(t)
{this.imageAtlas&&this.imageAtlas.patchUpdatedImages(t,this.imageAtlasTexture);}
queryRenderedFeatures(t,e,i,s,a,o,r,n,l,h){return
this.latestFeatureIndex&&this.latestFeatureIndex.rawTileData?
this.latestFeatureIndex.query({queryGeometry:s,cameraQueryGeometry:a,scale:o,til
eSize:this.tileSize,pixelPosMatrix:h,transform:n,params:r,queryPadding:this.quer

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yPadding*1},t,e,i):{}}querySourceFeatures(e,i){const
s=this.latestFeatureIndex;if(!s||!s.rawTileData)return;const
a=s.loadVTLayers(),o=i&&i.sourceLayer?
i.sourceLayer:"",r=a._geojsonTileLayer||a[0];if(!r)return;const
n=t.a6(i&&i.filter),{z:l,x:h,y:c}=this.tileID.canonical,u={z:l,x:h,y:c};for(let
i=0;i<r.length;i++){const a=r.feature(i);if(n.needGeometry){const
e=t.a7(a,!0);if(!n.filter(new
t.a8(this.tileID.overscaledZ),e,this.tileID.canonical))continue}else
if(!n.filter(new t.a8(this.tileID.overscaledZ),a))continue;const
d=s.getId(a,o),_=new t.a9(a,l,h,c,d);_.tile=u,e.push(_);}}hasData(){return
"loaded"===this.state||"reloading"===this.state||"expired"===this.state}patterns
Loaded(){return
this.imageAtlas&&!!Object.keys(this.imageAtlas.patternPositions).length}setExpir
yData(e){const i=this.expirationTime;if(e.cacheControl){const
i=t.aa(e.cacheControl);i["max-age"]&&(this.expirationTime=Date.now()+1e3*i["max-
age"]);}else e.expires&&(this.expirationTime=new
Date(e.expires).getTime());if(this.expirationTime){const t=Date.now();let
e=!1;if(this.expirationTime>t)e=!1;else if(i)if(this.expirationTime<i)e=!0;else
{const s=this.expirationTime-i;s?
this.expirationTime=t+Math.max(s,3e4):e=!0;}else e=!0;e?
(this.expiredRequestCount++,this.state="expired"):this.expiredRequestCount=0;}}g
etExpiryTimeout(){if(this.expirationTime)return this.expiredRequestCount?1e3*
(1<<Math.min(this.expiredRequestCount-1,31)):Math.min(this.expirationTime-(new
Date).getTime(),Math.pow(2,31)-1)}setFeatureState(t,e)
{if(!this.latestFeatureIndex||!this.latestFeatureIndex.rawTileData||0===Object.k
eys(t).length)return;const i=this.latestFeatureIndex.loadVTLayers();for(const s
in this.buckets){if(!e.style.hasLayer(s))continue;const
a=this.buckets[s],o=a.layers[0].sourceLayer||"_geojsonTileLayer",r=i[o],n=t[o];i
f(!r||!n||0===Object.keys(n).length)continue;a.update(n,r,this.imageAtlas&&this.
imageAtlas.patternPositions||{});const l=e&&e.style&&e.style.getLayer(s);l&&
(this.queryPadding=Math.max(this.queryPadding,l.queryRadius(a)));}}holdingForFad
e(){return void 0!==(this.symbolFadeHoldUntil?symbolFadeFinished():return
!this.symbolFadeHoldUntil||this.symbolFadeHoldUntil<t.h.now())clearFadeHold()
{this.symbolFadeHoldUntil=void 0;}setHoldDuration(e)
{this.symbolFadeHoldUntil=t.h.now()+e;}setDependencies(t,e){const i={};for(const
t of e)i[t]=!0;this.dependencies[t]=i;}hasDependency(t,e){for(const i of t)
{const t=this.dependencies[i];if(t)for(const i of e)if(t[i])return !0}return
!1}}class H{constructor(t,e){this.max=t,this.onRemove=e,this.reset();}reset()
{for(const t in this.data)for(const e of
this.data[t])e.timeout&&clearTimeout(e.timeout),this.onRemove(e.value);return
this.data={},this.order=[],this.add(t,e,i){const s=t.wrapped().key;void
0===this.data[s]&&(this.data[s]=[]);const a={value:e,timeout:void 0};if(void
0!==(i&&a.timeout=setTimeout(()=>>
{this.remove(t,a);}),i)),this.data[s].push(a),this.order.push(s),this.order.leng
th>this.max){const
t=this._getAndRemoveByKey(this.order[0]);t&&this.onRemove(t);}return this}has(t)
{return t.wrapped().key in this.data}getAndRemove(t){return this.has(t)?
this._getAndRemoveByKey(t.wrapped().key):null}_getAndRemoveByKey(t){const
e=this.data[t].shift();return
e.timeout&&clearTimeout(e.timeout),0===this.data[t].length&&delete
this.data[t],this.order.splice(this.order.indexOf(t),1),e.value}getByKey(t)
{const e=this.data[t];return e?e[0].value:null}get(t){return this.has(t)?
this.data[t.wrapped().key][0].value:null}remove(t,e){if(!this.has(t))return
this;const i=t.wrapped().key,s=void 0===e?
0:this.data[i].indexOf(e),a=this.data[i][s];return
this.data[i].splice(s,1),a.timeout&&clearTimeout(a.timeout),0===this.data[i].len
gth&&delete
this.data[i],this.onRemove(a.value),this.order.splice(this.order.indexOf(i),1),t
his.setMaxSize(t){for(this.max=t;this.order.length>this.max;){const
t=this._getAndRemoveByKey(this.order[0]);t&&this.onRemove(t);}return
this}filter(t){const e=[];for(const i in this.data)for(const s of
this.data[i])t(s.value)||e.push(s);for(const t of

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e)this.remove(t.value.tileID,t);}}class X{constructor(){this.state=
{},this.stateChanges={},this.deletedStates={};}updateState(e,i,s){const
a=String(i);if(this.stateChanges[e]=this.stateChanges[e]||
{},this.stateChanges[e][a]=this.stateChanges[e][a]||{},t.e(this.stateChanges[e]
[a],s),null===this.deletedStates[e]){this.deletedStates[e]={};for(const t in
this.state[e])t!=a&&(this.deletedStates[e][t]=null);}else
if(this.deletedStates[e]&&null===this.deletedStates[e][a]){this.deletedStates[e]
[a]={};for(const t in this.state[e][a])s[t]||(this.deletedStates[e][a]
[t]=null);}else for(const t in s)this.deletedStates[e]&&this.deletedStates[e]
[a]&&null===this.deletedStates[e][a][t]&&delete this.deletedStates[e][a]
[t];}removeFeatureState(t,e,i){if(null===this.deletedStates[t])return;const
s=String(e);if(this.deletedStates[t]=this.deletedStates[t]||{},i&&void
0!=e)null!=this.deletedStates[t][s]&&(this.deletedStates[t]
[s]=this.deletedStates[t][s]||{},this.deletedStates[t][s][i]=null);else if(void
0!=e)if(this.stateChanges[t]&&this.stateChanges[t][s])for(i in
this.deletedStates[t][s]={},this.stateChanges[t][s])this.deletedStates[t][s]
[i]=null;else this.deletedStates[t][s]=null;else
this.deletedStates[t]=null;}getState(e,i){const s=String(i),a=t.e({},
(this.state[e]||{})[s],(this.stateChanges[e]||{})[
s]);if(null===this.deletedStates[e])return {};if(this.deletedStates[e]){const
t=this.deletedStates[e][i];if(null===t)return {};for(const e in t)delete
a[e];}return a}initializeTileState(t,e)
{t.setFeatureState(this.state,e);}coalesceChanges(e,i){const s={};for(const e in
this.stateChanges){this.state[e]=this.state[e]||{};const i={};for(const s in
this.stateChanges[e])this.state[e][s]||(this.state[e][s]={}),t.e(this.state[e]
[s],this.stateChanges[e][s]),i[s]=this.state[e][s];s[e]=i;}for(const e in
this.deletedStates){this.state[e]=this.state[e]||{};const i=
{};if(null===this.deletedStates[e])for(const t in this.state[e])i[t]=
{};this.state[e][t]={};else for(const t in this.deletedStates[e])
{if(null===this.deletedStates[e][t])this.state[e][t]={};else for(const i of
Object.keys(this.deletedStates[e][t]))delete this.state[e][t]
[i];i[t]=this.state[e][t];}s[e]=s[e]||{},t.e(s[e],i);}if(this.stateChanges=
{},this.deletedStates={},0!=Object.keys(s).length)for(const t in
e)e[t].setFeatureState(s,i);}}class K extends t.E{constructor(t,e,i)
{super(),this.id=t,this.dispatcher=i,this.on("data",(t=>
{"source"===t.dataType&&"metadata"===t.sourceDataType&&
(this._sourceLoaded=!0),this._sourceLoaded&&!this._paused&&"source"===t.dataType
&&"content"===t.sourceDataType&&
(this.reload(),this.transform&&this.update(this.transform,this.terrain),this._di
dEmitContent=!0);})),this.on("dataloading",(()=>>
{this._sourceErrored=!1;})),this.on("error",(()=>>
{this._sourceErrored=this._source.loaded();})),this._source=((t,e,i,s)=>{const
a=new(j(e.type))(t,e,i,s);if(a.id!==t)throw new Error(`Expected Source id to be
${t} instead of ${a.id}`);return a})(t,e,i,this),this._tiles={},this._cache=new
H(0,this._unloadTile.bind(this)),this._timers={},this._cacheTimers=
{},this._maxTileCacheSize=null,this._maxTileCacheZoomLevels=null,this._loadedPar
entTiles={},this._coveredTiles={},this._state=new
X,this._didEmitContent=!1,this._updated=!1;}onAdd(t)
{this.map=t,this._maxTileCacheSize=t?
t._maxTileCacheSize:null,this._maxTileCacheZoomLevels=t?
t._maxTileCacheZoomLevels:null,this._source&&this._source.onAdd&&this._source.on
Add(t);}onRemove(t)
{this.clearTiles(),this._source&&this._source.onRemove&&this._source.onRemove(t)
;}loaded(){if(this._sourceErrored)return !0;if(!this._sourceLoaded)return
!1;if(!this._source.loaded())return !1;if(!(void 0===this.used&&void
0===this.usedForTerrain||this.used||this.usedForTerrain))return
!0;if(!this._updated)return !1;for(const t in this._tiles){const
e=this._tiles[t];if("loaded"!==e.state&&"errored"!==e.state)return !1}return
!0}getSource(){return this._source}pause(){this._paused=!0;}resume()
{if(!this._paused)return;const
t=this._shouldReloadOnResume;this._paused=!1,this._shouldReloadOnResume=!1,t&&th
is.reload(),this.transform&&this.update(this.transform,this.terrain);}loadTile(

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t,e){return this._source.loadTile(t,e)}_unloadTile(t)
{if(this._source.unloadTile)return this._source.unloadTile(t,(()=>
{}))}_abortTile(e){this._source.abortTile&&this._source.abortTile(e,(()=>
{})),this._source.fire(new t.k("dataabort",
{tile:e,coord:e.tileID,dataType:"source"}));}serialize(){return
this._source.serialize()}prepare(t)
{this._source.prepare&&this._source.prepare(),this._state.coalesceChanges(this._
tiles,this.map?this.map.painter:null);for(const e in this._tiles){const
i=this._tiles[e];i.upload(t),i.prepare(this.map.style.imageManager);}}getIds()
{return
Object.values(this._tiles).map((t=>t.tileID)).sort(Q).map((t=>t.key))}getRendera
bleIds(e){const i=[];for(const t in
this._tiles)this._isIdRenderable(t,e)&&i.push(this._tiles[t]);return e?
i.sort((e,i)=>{const s=e.tileID,a=i.tileID,o=new
t.P(s.canonical.x,s.canonical.y)._rotate(this.transform.angle),r=new
t.P(a.canonical.x,a.canonical.y)._rotate(this.transform.angle);return
s.overscaledZ-a.overscaledZ||r.y-o.y||r.x-
o.x}))}.map((t=>t.tileID.key)):i.map((t=>t.tileID)).sort(Q).map((t=>t.key))}hasRe
nderableParent(t){const e=this.findLoadedParent(t,0);return
!!e&&this._isIdRenderable(e.tileID.key)}_isIdRenderable(t,e){return
this._tiles[t]&&this._tiles[t].hasData()&&!this._coveredTiles[t]&&
(e||!this._tiles[t].holdingForFade())}reload()
{if(this._paused)this._shouldReloadOnResume=!0;else
{this._cache.reset();for(const t in
this._tiles)"errored"!==this._tiles[t].state&&this._reloadTile(t,"reloading");}}
_reloadTile(t,e){const i=this._tiles[t];i&&("loading"!==i.state&&
(i.state=e),this._loadTile(i,this._tileLoaded.bind(this,i,t,e)));}_tileLoaded(e,
i,s,a){if(a)return e.state="errored",void(404!==a.status?this._source.fire(new
t.j(a,
{tile:e})):this.update(this.transform,this.terrain));e.timeAdded=t.h.now(),"expi
red"===s&&(e.refreshedUponExpiration=!0),this._setTileReloadTimer(i,e,"raster-
dem"===this.getSource().type&&e.dem&&this._backfillDEM(e),this._state.initialize
TileState(e,this.map?this.map.painter:null),e.aborted||this._source.fire(new
t.k("data",{dataType:"source",tile:e,coord:e.tileID}));}_backfillDEM(t){const
e=this.getRenderableIds();for(let s=0;s<e.length;s++){const
a=e[s];if(t.neighboringTiles&&t.neighboringTiles[a]){const
e=this.getTileByID(a);i(t,e),i(e,t);}}function i(t,e)
{t.needsHillshadePrepare=!0,t.needsTerrainPrepare=!0;let i=e.tileID.canonical.x-
t.tileID.canonical.x;const s=e.tileID.canonical.y-
t.tileID.canonical.y,a=Math.pow(2,t.tileID.canonical.z),o=e.tileID.key;0===i&&0=
==s||Math.abs(s)>1||Math.abs(i)>1&&(1===Math.abs(i+a)?i+=a:1===Math.abs(i-a)&&
(i-=a)),e.dem&&t.dem&&
(t.dem.backfillBorder(e.dem,i,s),t.neighboringTiles&&t.neighboringTiles[o]&&
(t.neighboringTiles[o].backfilled=!0));}}getTile(t){return
this.getTileByID(t.key)}getTileByID(t){return
this._tiles[t]}_retainLoadedChildren(t,e,i,s){for(const a in this._tiles){let
o=this._tiles[a];if(s[a]||!o.hasData()||o.tileID.overscaledZ<=e||o.tileID.oversc
aledZ>i)continue;let r=o.tileID;for(;o&&o.tileID.overscaledZ>e+1;){const
t=o.tileID.scaledTo(o.tileID.overscaledZ-
1);o=this._tiles[t.key],o&&o.hasData()&&(r=t);}let
n=r;for(;n.overscaledZ>e;){if(n=n.scaledTo(n.overscaledZ-1),t[n.key])
{s[r.key]=r;break}}findLoadedParent(t,e){if(t.key in this._loadedParentTiles)
{const i=this._loadedParentTiles[t.key];return i&&i.tileID.overscaledZ>=e?
i:null}for(let i=t.overscaledZ-1;i>=e;i--){const
e=t.scaledTo(i),s=this._getLoadedTile(e);if(s)return s}}_getLoadedTile(t){const
e=this._tiles[t.key];return e&&e.hasData()?
e:this._cache.getByKey(t.wrapped().key)}updateCacheSize(e){const
i=Math.ceil(e.width/this._source.tileSize)+1,s=Math.ceil(e.height/this._source.t
ileSize)+1,a=Math.floor(i*s*(null===this._maxTileCacheZoomLevels?
t.c.MAX_TILE_CACHE_ZOOM_LEVELS:this._maxTileCacheZoomLevels)),o="number"===typeof
this._maxTileCacheSize?
Math.min(this._maxTileCacheSize,a):a;this._cache.setMaxSize(o);}handleWrapJump(t

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){const e=Math.round((t-(void 0===this._prevLng?
t:this._prevLng))/360);if(this._prevLng=t,e){const t={};for(const i in
this._tiles){const
s=this._tiles[i];s.tileID=s.tileID.unwrapTo(s.tileID.wrap+e),t[s.tileID.key]=s;}
this._tiles=t;for(const t in this._timers)clearTimeout(this._timers[t]),delete
this._timers[t];for(const t in
this._tiles)this._setTileReloadTimer(t,this._tiles[t]);}update(e,i)
{if(this.transform=e,this.terrain=i,!this._sourceLoaded||this._paused)return;let
s;this.updateCacheSize(e),this.handleWrapJump(this.transform.center.lng),this._c
overedTiles={},this.used||this.usedForTerrain?this._source.tileID?
s=e.getVisibleUnwrappedCoordinates(this._source.tileID).map((e=>new
t.O(e.canonical.z,e.wrap,e.canonical.z,e.canonical.x,e.canonical.y))):
(s=e.coveringTiles({tileSize:this.usedForTerrain?
this.tileSize:this._source.tileSize,minzoom:this._source.minzoom,maxzoom:this._s
ource.maxzoom,roundZoom:!this.usedForTerrain&&this._source.roundZoom,reparseOver
scaled:this._source.reparseOverscaled,terrain:i})),this._source.hasTile&&
(s=s.filter((t=>this._source.hasTile(t))));s=[];const
a=e.coveringZoomLevel(this._source),o=Math.max(a-
K.maxOverzooming,this._source.minzoom),r=Math.max(a+K.maxUnderzooming,this._sour
ce.minzoom);if(this.usedForTerrain){const t={};for(const e of
s)if(e.canonical.z>this._source.minzoom){const i=e.scaledTo(e.canonical.z-
1);t[i.key]=i;const
s=e.scaledTo(Math.max(this._source.minzoom,Math.min(e.canonical.z,5)));t[s.key]=
s;}s=s.concat(Object.values(t));}const
n=0===s.length&&!this._updated&&this._didEmitContent;this._updated=!0,n&&this.fi
re(new t.k("data",
{sourceDataType:"idle",dataType:"source",sourceId:this.id}));const
l=this._updateRetainedTiles(s,a);if(Y(this._source.type)){const e={},n=
{},h=Object.keys(l),c=t.h.now();for(const t of h){const
i=l[t],s=this._tiles[t];if(!s||0!==s.fadeEndTime&&s.fadeEndTime<=c)continue;cons
t a=this.findLoadedParent(i,o);a&&
(this._addTile(a.tileID),e[a.tileID.key]=a.tileID,n[t]=i);}this._retainLoadedChi
ldren(n,a,r,l);for(const t in e)l[t]||(this._coveredTiles[t]=!0,l[t]=e[t]);if(i)
{const t={},e={};for(const i of s)this._tiles[i.key].hasData()?
t[i.key]=i:e[i.key]=i;for(const i in e){const
s=e[i].children(this._source.maxzoom);this._tiles[s[0].key]&&this._tiles[s[1].ke
y]&&this._tiles[s[2].key]&&this._tiles[s[3].key]&&
(t[s[0].key]=l[s[0].key]=s[0],t[s[1].key]=l[s[1].key]=s[1],t[s[2].key]=l[s[2].ke
y]=s[2],t[s[3].key]=l[s[3].key]=s[3],delete e[i]);}for(const i in e){const
s=this.findLoadedParent(e[i],this._source.minzoom);if(s)
{t[s.tileID.key]=l[s.tileID.key]=s.tileID;for(const e in
t)t[e].isChildOf(s.tileID)&&delete t[e];}}for(const e in this._tiles)t[e]||
(this._coveredTiles[e]=!0);}}for(const t in
l)this._tiles[t].clearFadeHold();const h=t.ab(this._tiles,l);for(const t of h)
{const e=this._tiles[t];e.hasSymbolBuckets&&!e.holdingForFade()?
e.setHoldDuration(this.map._fadeDuration):e.hasSymbolBuckets&&!e.symbolFadeFinis
hed()||this._removeTile(t);}this._updateLoadedParentTileCache();}releaseSymbolFa
deTiles(){for(const t in
this._tiles)this._tiles[t].holdingForFade()&&this._removeTile(t);}updateRetaine
dTiles(t,e){const i={},s={},a=Math.max(e-
K.maxOverzooming,this._source.minzoom),o=Math.max(e+K.maxUnderzooming,this._sour
ce.minzoom),r={};for(const s of t){const
t=this._addTile(s);i[s.key]=s,t.hasData()||e<this._source.maxzoom&&
(r[s.key]=s);}this._retainLoadedChildren(r,e,o,i);for(const o of t){let
t=this._tiles[o.key];if(t.hasData())continue;if(e+1>this._source.maxzoom){const
t=o.children(this._source.maxzoom)[0],e=this._tiles[t];if(e&&e.hasData())
{i[t.key]=t;continue}}else {const
t=o.children(this._source.maxzoom);if(i[t[0].key]&&i[t[1].key]&&i[t[2].key]&&i[t
[3].key])continue}let r=t.wasRequested();for(let e=o.overscaledZ-1;e>=a;--e)
{const
a=o.scaledTo(e);if(s[a.key])break;if(s[a.key]=!0,t=this._tiles[a],!t&&r&&
(t=this._addTile(a),t){const e=t.hasData();if((r||e)&&

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(i[a.key]=a),r=t.wasRequested(),e)break}}return
i}_updateLoadedParentTileCache(){this._loadedParentTiles={};for(const t in
this._tiles){const e=[];let i,s=this._tiles[t].tileID;for(;s.overscaledZ>0;)
{if(s.key in this._loadedParentTiles)
{i=this._loadedParentTiles[s.key];break}e.push(s.key);const
t=s.scaledTo(s.overscaledZ-1);if(i=this._getLoadedTile(t),i)break;s=t;}for(const
t of e)this._loadedParentTiles[t]=i;}}_addTile(e){let
i=this._tiles[e.key];if(i)return i;i=this._cache.getAndRemove(e),i&&
(this._setTileReloadTimer(e.key,i),i.tileID=e,this._state.initializeTileState(i,
this.map?this.map.painter:null),this._cacheTimers[e.key]&&
(clearTimeout(this._cacheTimers[e.key]),delete
this._cacheTimers[e.key],this._setTileReloadTimer(e.key,i)));const s=i;return
i||i=new
W(e,this._source.tileSize*e.overscaleFactor()),this._loadTile(i,this._tileLoaded
.bind(this,i,e.key,i.state))),i.uses++,this._tiles[e.key]=i,s|this._source.fire
(new t.k("dataloading",
{tile:i,coord:i.tileID,dataType:"source"})),i}_setTileReloadTimer(t,e){t in
this._timers&&(clearTimeout(this._timers[t]),delete this._timers[t]);const
i=e.getExpiryTimeout();i&&(this._timers[t]=setTimeout(()=>>
{this._reloadTile(t,"expired"),delete this._timers[t];}),i));}_removeTile(t)
{const e=this._tiles[t];e&&(e.uses--,delete this._tiles[t],this._timers[t]&&
(clearTimeout(this._timers[t]),delete this._timers[t]),e.uses>0|
(e.hasData()&&"reloading"!==e.state?
this._cache.add(e.tileID,e,e.getExpiryTimeout()):
(e.aborted=!0,this._abortTile(e),this._unloadTile(e)));}clearTiles()
{this._shouldReloadOnResume=!1,this._paused=!1;for(const t in
this._tiles)this._removeTile(t);this._cache.reset();}tilesIn(e,i,s){const a=
[],o=this.transform;if(!o)return a;const r=s?
o.getCameraQueryGeometry(e):e,n=e.map((t=>o.pointCoordinate(t,this.terrain))),l=
r.map((t=>o.pointCoordinate(t,this.terrain))),h=this.getIds();let
c=1/0,u=1/0,d=-1/0,_=-1/0;for(const t of
l)c=Math.min(c,t.x),u=Math.min(u,t.y),d=Math.max(d,t.x),_=Math.max(_,t.y);for(le
t e=0;e<h.length;e++){const
s=this._tiles[h[e]];if(s.holdingForFade())continue;const
r=s.tileID,p=Math.pow(2,o.zoom-
s.tileID.overscaledZ),m=i*s.queryPadding*t.N/s.tileSize/p,f=[r.getTilePoint(new
t.U(c,u)),r.getTilePoint(new t.U(d,_))];if(f[0].x-m<t.N&&f[0].y-
m<t.N&&f[1].x+m>=0&&f[1].y+m>=0){const
t=n.map((t=>r.getTilePoint(t))),e=l.map((t=>r.getTilePoint(t)));a.push({tile:s,t
ileID:r,queryGeometry:t,cameraQueryGeometry:e,scale:p});}return
a}getVisibleCoordinates(t){const
e=this._getRenderableIds(t).map((t=>this._tiles[t].tileID));for(const t of
e)t.posMatrix=this.transform.calculatePosMatrix(t.toUnwrapped());return
e}hasTransition(){if(this._source.hasTransition())return
!0;if(Y(this._source.type)){const e=t.h.now();for(const t in
this._tiles)if(this._tiles[t].fadeEndTime>=e)return !0}return
!1}setFeatureState(t,e,i)
{this._state.updateState(t=t|"_geojsonTileLayer",e,i);}removeFeatureState(t,e,i)
{this._state.removeFeatureState(t=t|"_geojsonTileLayer",e,i);}getFeatureState(t
,e){return
this._state.getState(t=t|"_geojsonTileLayer",e)}setDependencies(t,e,i){const
s=this._tiles[t];s&&s.setDependencies(e,i);}reloadTilesForDependencies(t,e)
{for(const i in
this._tiles)this._tiles[i].hasDependency(t,e)&&this._reloadTile(i,"reloading");t
his._cache.filter((i=>!i.hasDependency(t,e)));}function Q(t,e){const
i=Math.abs(2*t.wrap)-+(t.wrap<0),s=Math.abs(2*e.wrap)-+(e.wrap<0);return
t.overscaledZ-e.overscaledZ|s-i|e.canonical.y-t.canonical.y|e.canonical.x-
t.canonical.x}function Y(t){return
"raster"===t||"image"===t||"video"===t?K.maxOverzooming=10,K.maxUnderzooming=3;c
onst J="mapboxgl_preloaded_worker_pool";class tt{constructor(){this.active=
{};}}acquire(e){if(!this.workers)for(this.workers=

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[];this.workers.length<tt.workerCount;)this.workers.push(new
Worker(t.c.WORKER_URL));return this.active[e]!=0,this.workers.slice()}release(t)
{delete this.active[t],0===this.numActive()&&(this.workers.forEach((t=>
{t.terminate()})),this.workers=null);}isPreloaded(){return
!!this.active[J]}numActive(){return Object.keys(this.active).length}}const
et=Math.floor(t.h.hardwareConcurrency/2);let it;function st(){return it||(it=new
tt),it}tt.workerCount=t.ac(globalThis)?Math.max(Math.min(et,3),1):1;class
at{constructor(t,e){this.reset(t,e);}reset(t,e){this.points=t|
[],this._distances=[0];for(let
t=1;t<this.points.length;t++)this._distances[t]=this._distances[t-
1]+this.points[t].dist(this.points[t-
1]);this.length=this._distances[this._distances.length-
1],this.padding=Math.min(e||0,.5*this.length),this.paddedLength=this.length-
2*this.padding;let lp(e){if(1===this.points.length)return
this.points[0];e=t.ad(e,0,1);let i=1,s=this._distances[i];const
a=e*this.paddedLength+this.padding;for(;s<a&&i<this._distances.length;)s=this._d
istances[++i];const o=i-1,r=this._distances[o],n=s-r,l=n>0?(a-r)/n:0;return
this.points[o].mult(1-l).add(this.points[i].mult(l))}function ot(t,e){let
i=!0;return "always"===t||"never"!==t&&"never"!==e||(i=!1),i}class
rt{constructor(t,e,i){const s=this.boxCells=[],a=this.circleCells=
[];this.xCellCount=Math.ceil(t/i),this.yCellCount=Math.ceil(e/i);for(let
t=0;t<this.xCellCount*this.yCellCount;t++)s.push([]),a.push([]);this.circleKeys=
[],this.boxKeys=[],this.bboxes=[],this.circles=
[],this.width=t,this.height=e,this.xScale=this.xCellCount/t,this.yScale=this.yCe
llCount/e,this.boxUid=0,this.circleUid=0;}keysLength(){return
this.boxKeys.length+this.circleKeys.length}insert(t,e,i,s,a)
{this._forEachCell(e,i,s,a,this._insertBoxCell,this.boxUid++),this.boxKeys.push(
t),this.bboxes.push(e),this.bboxes.push(i),this.bboxes.push(s),this.bboxes.push(
a);}insertCircle(t,e,i,s){this._forEachCell(e-s,i-
s,e+s,i+s,this._insertCircleCell,this.circleUid++),this.circleKeys.push(t),this.
circles.push(e),this.circles.push(i),this.circles.push(s);}_insertBoxCell(t,e,i,
s,a,o){this.boxCells[a].push(o);}_insertCircleCell(t,e,i,s,a,o)
{this.circleCells[a].push(o);}_query(t,e,i,s,a,o,r)
{if(i<0||t>this.width||s<0||e>this.height)return [];const n=
[];if(t<=0&&e<=0&&this.width<=i&&this.height<=s){if(a)return
[{key:null,x1:t,y1:e,x2:i,y2:s}];for(let
t=0;t<this.boxKeys.length;t++)n.push({key:this.boxKeys[t],x1:this.bboxes[4*t],y1
:this.bboxes[4*t+1],x2:this.bboxes[4*t+2],y2:this.bboxes[4*t+3]});for(let
t=0;t<this.circleKeys.length;t++){const
e=this.circles[3*t],i=this.circles[3*t+1],s=this.circles[3*t+2];n.push({key:this
.circleKeys[t],x1:e-s,y1:i-s,x2:e+s,y2:i+s});}else
this._forEachCell(t,e,i,s,this._queryCell,n,{hitTest:a,overlapMode:o,seenUids:
{box:{},circle:{}}},r);return n}query(t,e,i,s){return
this._query(t,e,i,s,!1,null)}hitTest(t,e,i,s,a,o){return
this._query(t,e,i,s,!0,a,o).length>0}hitTestCircle(t,e,i,s,a){const o=t-
i,r=t+i,n=e-i,l=e+i;if(r<0||o>this.width||l<0||n>this.height)return !1;const h=
[];return this._forEachCell(o,n,r,l,this._queryCellCircle,h,
{hitTest:!0,overlapMode:s,circle:{x:t,y:e,radius:i},seenUids:{box:{},circle:
{}}},a),h.length>0}_queryCell(t,e,i,s,a,o,r,n)
{const{seenUids:l,hitTest:h,overlapMode:c}=r,u=this.boxCells[a];if(null!==u)
{const a=this.bboxes;for(const r of u)if(!l.box[r]){l.box[r]=!0;const
u=4*r,d=this.boxKeys[r];if(t<=a[u+2]&&e<=a[u+3]&&i>=a[u+0]&&s>=a[u+1]&&
(!n||n(d))&&(!h||!ot(c,d.overlapMode))&&
(o.push({key:d,x1:a[u],y1:a[u+1],x2:a[u+2],y2:a[u+3]}),h))return !0}}const
d=this.circleCells[a];if(null!==d){const a=this.circles;for(const r of
d)if(!l.circle[r]){l.circle[r]=!0;const
u=3*r,d=this.circleKeys[r];if(this._circleAndRectCollide(a[u],a[u+1],a[u+2],t,e,
i,s)&&(!n||n(d))&&(!h||!ot(c,d.overlapMode)))}{const
t=a[u],e=a[u+1],i=a[u+2];if(o.push({key:d,x1:t-i,y1:e-i,x2:t+i,y2:e+i}),h)return
!0}}return !1}_queryCellCircle(t,e,i,s,a,o,r,n)
{const{circle:l,seenUids:h,overlapMode:c}=r,u=this.boxCells[a];if(null!==u)
{const t=this.bboxes;for(const e of u)if(!h.box[e]){h.box[e]=!0;const

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i=4*e,s=this.boxKeys[e];if(this._circleAndRectCollide(l.x,l.y,l.radius,t[i+0],t[
i+1],t[i+2],t[i+3])&&(!n|n(s))&&!ot(c,s.overlapMode))return
o.push(!0,!0)}const d=this.circleCells[a];if(null!==d){const
t=this.circles;for(const e of d){if(!h.circle[e]){h.circle[e]=!0;const
i=3*e,s=this.circleKeys[e];if(this._circlesCollide(t[i],t[i+1],t[i+2],l.x,l.y,l.
radius)&&(!n|n(s))&&!ot(c,s.overlapMode))return
o.push(!0,!0)}}_forEachCell(t,e,i,s,a,o,r,n){const
l=this._convertToXCellCoord(t),h=this._convertToYCellCoord(e),c=this._convertToX
CellCoord(i),u=this._convertToYCellCoord(s);for(let d=l;d<=c;d++)for(let
l=h;l<=u;l++)if(a.call(this,t,e,i,s,this.xCellCount*l+d,o,r,n))return}_convertTo
XCellCoord(t){return Math.max(0,Math.min(this.xCellCount-
1,Math.floor(t*this.xScale)))}_convertToYCellCoord(t){return
Math.max(0,Math.min(this.yCellCount-
1,Math.floor(t*this.yScale)))}_circlesCollide(t,e,i,s,a,o){const r=s-t,n=a-
e,l=i+o;return l*l>r*r+n*n}_circleAndRectCollide(t,e,i,s,a,o,r){const n=(o-
s)/2,l=Math.abs(t-(s+n));if(l>n+i)return !1;const h=(r-a)/2,c=Math.abs(e-
(a+h));if(c>h+i)return !1;if(l<=n|c<=h)return !0;const u=l-n,d=c-h;return
u*u+d*d<=i*i}}function nt(e,i,s,a,o){const r=t.Z();return i?(t.a0(r,r,
[1/o,1/o,1]),s||t.ae(r,r,a.angle)):t.a1(r,a.labelPlaneMatrix,e),r}function
lt(e,i,s,a,o){if(i){const i=t.af(e);return t.a0(i,i,[o,o,1]),s||t.ae(i,i,-
a.angle),i}return a.glCoordMatrix}function ht(e,i,s){let a;s?(a=
[e.x,e.y,s(e.x,e.y),1],t.ag(a,a,i)):a=[e.x,e.y,0,1],Tt(a,a,i);const
o=a[3];return {point:new t.P(a[0]/o,a[1]/o),signedDistanceFromCamera:o}}function
ct(t,e){return .5+t/e*.5}function ut(t,e){const i=t[0]/t[3],s=t[1]/t[3];return
i>=-e[0]&&i<=e[0]&&s>=-e[1]&&s<=e[1]}function dt(e,i,s,a,o,r,n,l,h,c){const u=a?
e.textSizeData:e.iconSizeData,d=t.ah(u,s.transform.zoom),_=
[256/s.width*2+1,256/s.height*2+1],p=a?
e.text.dynamicLayoutVertexArray:e.icon.dynamicLayoutVertexArray;p.clear();const
m=e.lineVertexArray,f=a?
e.text.placedSymbolArray:e.icon.placedSymbolArray,g=s.transform.width/s.transfor
m.height;let v=!1;for(let a=0;a<f.length;a++){const
x=f.get(a);if(x.hidden||x.writingMode===t.ai.vertical&&!v)
{wt(x.numGlyphs,p);continue}let y;if(v=!1,c?(y=
[x.anchorX,x.anchorY,c(x.anchorX,x.anchorY),1],t.ag(y,y,i)):y=
[x.anchorX,x.anchorY,0,1],Tt(y,y,i),!ut(y,_)){wt(x.numGlyphs,p);continue}const
b=ct(s.transform.cameraToCenterDistance,y[3]),w=t.aj(u,d,x),T=n?w/b:w*b,I=new
t.P(x.anchorX,x.anchorY),E=ht(I,o,c).point,S={projections:{},offsets:
{}},C=mt(x,T,!1,l,i,o,r,e.glyphOffsetArray,m,p,E,I,S,g,h,c);v=C.useVertical,
(C.notEnoughRoom||v||C.needsFlipping&&mt(x,T,!0,l,i,o,r,e.glyphOffsetArray,m,p,E
,I,S,g,h,c).notEnoughRoom)&&wt(x.numGlyphs,p);}a?
e.text.dynamicLayoutVertexBuffer.updateData(p):e.icon.dynamicLayoutVertexBuffer.
updateData(p);}function _t(t,e,i,s,a,o,r,n,l,h,c,u,d){const
_=n.glyphStartIndex+n.numGlyphs,p=n.lineStartIndex,m=n.lineStartIndex+n.lineLeng
th,f=e.getOffsetX(n.glyphStartIndex),g=e.getOffsetX(_-1),v=yt(t*f,i,s,a,o,r,n.se
gment,p,m,l,h,c,u,d);if(!v)return null;const
x=yt(t*g,i,s,a,o,r,n.segment,p,m,l,h,c,u,d);return x?
{first:v,last:x}:null}function pt(e,i,s,a){return
e===t.ai.horizontal&&Math.abs(s.y-i.y)>Math.abs(s.x-i.x)*a?{useVertical:!0}:
(e===t.ai.vertical?i.y<s.y:i.x>s.x)?{needsFlipping:!0}:null}function
mt(e,i,s,a,o,r,n,l,h,c,u,d,_p,m,f){const
g=i/24,v=e.lineOffsetX*g,x=e.lineOffsetY*g;let y;if(e.numGlyphs>1){const
t=e.glyphStartIndex+e.numGlyphs,i=e.lineStartIndex,o=e.lineStartIndex+e.lineLeng
th,c=_t(g,l,v,x,s,u,d,e,h,r,_m,f);if(!c)return {notEnoughRoom:!0};const
b=ht(c.first.point,n,f).point,w=ht(c.last.point,n,f).point;if(a&&!s){const
t=pt(e.writingMode,b,w,p);if(t)return t}y=[c.first];for(let
a=e.glyphStartIndex+1;a<t-
1;a++)y.push(yt(g*l.getOffsetX(a),v,x,s,u,d,e.segment,i,o,h,r,_m,f));y.push(c.l
ast);}else {if(a&&!s){const
i=ht(d,o,f).point,s=e.lineStartIndex+e.segment+1,a=new
t.P(h.getx(s),h.gety(s)),r=ht(a,o,f),n=r.signedDistanceFromCamera>0?
r.point:ft(d,a,i,l,o,f),l=pt(e.writingMode,i,n,p);if(l)return l}const
i=yt(g*l.getOffsetX(e.glyphStartIndex),v,x,s,u,d,e.segment,e.lineStartIndex,e.li

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neStartIndex+e.lineLength,h,r,_,m,f);if(!i)return {notEnoughRoom:!0};y=
[i];}for(const e of y)t.ak(c,e.point,e.angle);return {}}function ft(t,e,i,s,a,o)
{const r=ht(t.add(t.sub(e)._unit()),a,o).point,n=i.sub(r);return
i.add(n._mult(s/n.mag()))}function gt(e,i)
{const{projectionCache:s,lineVertexArray:a,labelPlaneMatrix:o,tileAnchorPoint:r,
distanceFromAnchor:n,getElevation:l,previousVertex:h,direction:c,absOffsetX:u}=i
;if(s.projections[e])return s.projections[e];const d=new
t.P(a.getx(e),a.gety(e)),_ht(d,o,l);if(_.signedDistanceFromCamera>0)return
s.projections[e]=_.point,_.point;const p=e-c;return ft(0===n?r:new
t.P(a.getx(p),a.gety(p)),d,h,u-n+1,o,l)}function vt(t,e,i){return
t._unit()._perp()._mult(e*i)}function xt(e,i,s,a,o,r,n,l)
{const{projectionCache:h,direction:c}=l;if(h.offsets[e])return
h.offsets[e];const u=s.add(i);if(e+c<a||e+c>o)return h.offsets[e]=u,u;const
d=gt(e+c,l),_=vt(d.sub(s),n,c),p=s.add(_),m=d.add(_);return
h.offsets[e]=t.al(r,u,p,m)||u,h.offsets[e]}function
yt(t,e,i,s,a,o,r,n,l,h,c,u,d,_){const p=s?t-e:t+e;let m=p>0?1:-1,f=0;s&&
(m*=-1,f=Math.PI),m<0&&(f+=Math.PI);let g,v,x=m>0?
n+r:n+r+1,y=a,b=a,w=0,T=0;const I=Math.abs(p),E=[];let S;for(;w+T<=I;)
{if(x+m,x<n||x>=l)return null;w+=T,b=y,v=g;const t=
{projectionCache:u,lineVertexArray:h,labelPlaneMatrix:c,tileAnchorPoint:o,distan
ceFromAnchor:w,getElevation:_,previousVertex:b,direction:m,absOffsetX:I};if(y=gt
(x,t),0===i)E.push(b),S=y.sub(b);else {let e;const s=y.sub(b);e=0===s.mag()?
vt(gt(x+m,t).sub(y),i,m):vt(s,i,m),v||
(v=b.add(e)),g=xt(x,e,y,n,l,v,i,t),E.push(v),S=g.sub(v);}T=S.mag();}const
C=S._mult((I-w)/T)._add(v||b),P=f+Math.atan2(y.y-b.y,y.x-b.x);return E.push(C),
{point:C,angle:d?P:0,path:E}}const bt=new
Float32Array([-1/0,-1/0,0,-1/0,-1/0,0,-1/0,-1/0,0,-1/0,-1/0,0]);function wt(t,e)
{for(let i=0;i<t;i++){const
t=e.length;e.resize(t+4),e.float32.set(bt,3*t);}}function Tt(t,e,i){const
s=e[0],a=e[1];return
t[0]=i[0]*s+i[4]*a+i[12],t[1]=i[1]*s+i[5]*a+i[13],t[3]=i[3]*s+i[7]*a+i[15],t}con
st It=100;class Et{constructor(t,e=new rt(t.width+200,t.height+200,25),i=new
rt(t.width+200,t.height+200,25))
{this.transform=t,this.grid=e,this.ignoredGrid=i,this.pitchfactor=Math.cos(t._pi
tch)*t.cameraToCenterDistance,this.screenRightBoundary=t.width+It,this.screenBot
tomBoundary=t.height+It,this.gridRightBoundary=t.width+200,this.gridBottomBounda
ry=t.height+200,this.perspectiveRatioCutoff=.6;}placeCollisionBox(t,e,i,s,a,o)
{const
r=this.projectAndGetPerspectiveRatio(s,t.anchorPointX,t.anchorPointY,o),n=i*r.pe
rspectiveRatio,l=t.x1*n+r.point.x,h=t.y1*n+r.point.y,c=t.x2*n+r.point.x,u=t.y2*n
+r.point.y;return
!this.isInsideGrid(l,h,c,u)||"always"!==e&&this.grid.hitTest(l,h,c,u,e,a)||r.per
spectiveRatio<this.perspectiveRatioCutoff?{box:[],offscreen:!1}:{box:
[l,h,c,u],offscreen:this.isOffscreen(l,h,c,u)}}placeCollisionCircles(e,i,s,a,o,r
,n,l,h,c,u,d,_,p){const m=[],f=new
t.P(i.anchorX,i.anchorY),g=ht(f,r,p),v=ct(this.transform.cameraToCenterDistance,
g.signedDistanceFromCamera),x=(c?
o/v:o*v)/t.ap,y=ht(f,n,p).point,b=_t(x,a,i.lineOffsetX*x,i.lineOffsetY*x,!1,y,f,
i,s,n,{projections:{},offsets:{}},!1,p);let w=!1,T=!1,I=!0;if(b){const
i=.5*d*v+_s=new t.P(-100,-100),a=new
t.P(this.screenRightBoundary,this.screenBottomBoundary),o=new
at,r=b.first,n=b.last;let c=[];for(let t=r.path.length-1;t>=1;t--
)c.push(r.path[t]);for(let t=1;t<n.path.length;t++)c.push(n.path[t]);const
f=2.5*i;if(l){const
t=c.map((t=>ht(t,l,p)));c=t.some((t=>t.signedDistanceFromCamera<=0))?
[]:t.map((t=>t.point));}let g=[];if(c.length>0){const
e=c[0].clone(),i=c[0].clone();for(let
t=1;t<c.length;t++)e.x=Math.min(e.x,c[t].x),e.y=Math.min(e.y,c[t].y),i.x=Math.ma
x(i.x,c[t].x),i.y=Math.max(i.y,c[t].y);g=e.x>=s.x&&i.x<=a.x&&e.y>=s.y&&i.y<=a.y?
[c]:i.x<s.x||e.x>a.x||i.y<s.y||e.y>a.y?[]:t.am([c],s.x,s.y,a.x,a.y);}for(const t
of g){o.reset(t,.25*i);let s=0;s=o.length<=.5*i?
1:Math.ceil(o.paddedLength/f)+1;for(let t=0;t<s;t++){const a=t/Math.max(s-

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1,l),r=o.lerp(a),n=r.x+It,l=r.y+It;m.push(n,l,i,0);const c=n-i,d=l-
i,_=n+i,p=l+i;if(I=I&&this.isOffscreen(c,d,_,p),T=T||this.isInsideGrid(c,d,_,p),
"always"!==e&&this.grid.hitTestCircle(n,l,i,e,u)&&(w!=0,!h))return {circles:
[],offscreen:!l,collisionDetected:w}}}}return
{circles:!h&&w||!T||v<this.perspectiveRatioCutoff?
[]:m,offscreen:I,collisionDetected:w}}queryRenderedSymbols(e)
{if(0===e.length||0===this.grid.keysLength()&&0===this.ignoredGrid.keysLength())
return {};const i=[];let s=1/0,a=1/0,o=-1/0,r=-1/0;for(const n of e){const e=new
t.P(n.x+It,n.y+It);s=Math.min(s,e.x),a=Math.min(a,e.y),o=Math.max(o,e.x),r=Math.
max(r,e.y),i.push(e);}const
n=this.grid.query(s,a,o,r).concat(this.ignoredGrid.query(s,a,o,r)),l={},h=
{};for(const e of n){const s=e.key;if(void 0===l[s.bucketInstanceId]&&
(l[s.bucketInstanceId]={}),l[s.bucketInstanceId][s.featureIndex])continue;const
a=[new t.P(e.x1,e.y1),new t.P(e.x2,e.y1),new t.P(e.x2,e.y2),new
t.P(e.x1,e.y2)];t.an(i,a)&&(l[s.bucketInstanceId][s.featureIndex]!=0,void
0===h[s.bucketInstanceId]&&(h[s.bucketInstanceId]=
[]),h[s.bucketInstanceId].push(s.featureIndex));}return
h}insertCollisionBox(t,e,i,s,a,o){(i?
this.ignoredGrid:this.grid).insert({bucketInstanceId:s,featureIndex:a,collisionG
roupID:o,overlapMode:e},t[0],t[1],t[2],t[3]);}insertCollisionCircles(t,e,i,s,a,o
){const r=i?this.ignoredGrid:this.grid,n=
{bucketInstanceId:s,featureIndex:a,collisionGroupID:o,overlapMode:e};for(let
e=0;e<t.length;e+=4)r.insertCircle(n,t[e],t[e+1],t[e+2]);}projectAndGetPerspecti
veRatio(e,i,s,a){let o;return a?(o=[i,s,a(i,s),l],t.ag(o,o,e)):o=
[i,s,0,l],Tt(o,o,e),{point:new t.P((o[0]/o[3]+1)/2*this.transform.width+It,(-
o[1]/o[3]+1)/2*this.transform.height+It),perspectiveRatio:.5+this.transform.came
raToCenterDistance/o[3]*.5}}isOffscreen(t,e,i,s){return
i<It||t>this.screenRightBoundary||s<It||e>this.screenBottomBoundary}isInsideGri
d(t,e,i,s){return
i>=0&&t<this.gridRightBoundary&&s>=0&&e<this.gridBottomBoundary}getViewPortMatri
x(){const e=t.ao([]);return t.$(e,e,[-100,-100,0],e)}function St(e,i,s){return
i*(t.N/(e.tileSize*Math.pow(2,s-e.tileID.overscaledZ)))}class
Ct{constructor(t,e,i,s){this.opacity=t?Math.max(0,Math.min(1,t.opacity+
(t.placed?e:-e))):s&&i!0,this.placed=i;}isHidden(){return
0===this.opacity&&!this.placed}}class Pt{constructor(t,e,i,s,a){this.text=new
Ct(t?t.text:null,e,i,a),this.icon=new Ct(t?t.icon:null,e,s,a);}isHidden(){return
this.text.isHidden()&&this.icon.isHidden()}}class Dt{constructor(t,e,i)
{this.text=t,this.icon=e,this.skipFade=i;}}class Mt{constructor()
{this.invProjMatrix=t.Z(),this.viewportMatrix=t.Z(),this.circles=[];}}class
zt{constructor(t,e,i,s,a)
{this.bucketInstanceId=t,this.featureIndex=e,this.sourceLayerIndex=i,this.bucket
Index=s,this.tileID=a;}}class Lt{constructor(t)
{this.crossSourceCollisions=t,this.maxGroupID=0,this.collisionGroups={};}get(t)
{if(this.crossSourceCollisions)return
{ID:0,predicate:null};if(!this.collisionGroups[t]){const
e=++this.maxGroupID;this.collisionGroups[t]=
{ID:e,predicate:t=>t.collisionGroupID===e};}return
this.collisionGroups[t]}function At(e,i,s,a,o)
{const{horizontalAlign:r,verticalAlign:n}=t.au(e);return new t.P(-
(r-.5)*i+a[0]*o,-(n-.5)*s+a[1]*o)}function Rt(e,i,s,a,o,r)
{const{x1:n,x2:l,y1:h,y2:c,anchorPointX:u,anchorPointY:d}=e,_=new
t.P(i,s);return a&&_.rotate(o?r:-r),
{x1:n+_x,y1:h+_y,x2:l+_x,y2:c+_y,anchorPointX:u,anchorPointY:d}}class
kt{constructor(t,e,i,s,a)
{this.transform=t.clone(),this.terrain=e,this.collisionIndex=new
Et(this.transform),this.placements={},this.opacities={},this.variableOffsets=
{},this.stale=!l,this.commitTime=0,this.fadeDuration=i,this.retainedQueryData=
{},this.collisionGroups=new Lt(s),this.collisionCircleArrays=
{},this.prevPlacement=a,a&&(a.prevPlacement=void 0),this.placedOrientations=
{};}}getBucketParts(e,i,s,a){const
o=s.getBucket(i),r=s.latestFeatureIndex;if(!o||!r||i.id!==o.layerIds[0])return;c
onst

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n=s.collisionBoxArray,l=o.layers[0].layout,h=Math.pow(2,this.transform.zoom-
s.tileID.overscaledZ),c=s.tileSize/t.N,u=this.transform.calculatePosMatrix(s.til
eID.toUnwrapped()),d="map"===l.get("text-pitch-
alignment"),_="map"===l.get("text-rotation-
alignment"),p=St(s,l,this.transform.zoom),m=nt(u,d,_,this.transform,p);let
f=null;if(d){const
e=lt(u,d,_,this.transform,p);f=t.al([],this.transform.labelPlaneMatrix,e);}this.
retainedQueryData[o.bucketInstanceId]=new
zt(o.bucketInstanceId,r,o.sourceLayerIndex,o.index,s.tileID);const g=
{bucket:o,layout:l,posMatrix:u,textLabelPlaneMatrix:m,labelToScreenMatrix:f,sca
le:h,textPixelRatio:c,holdingForFade:s.holdingForFade(),collisionBoxArray:n,parti
allyEvaluatedTextSize:t.ah(o.textSizeData,this.transform.zoom),collisionGroup:th
is.collisionGroups.get(o.sourceID)};if(a)for(const t of o.sortKeyRanges)
{const{sortKey:i,symbolInstanceStart:s,symbolInstanceEnd:a}=t;e.push({sortKey:i,
symbolInstanceStart:s,symbolInstanceEnd:a,parameters:g});}else
e.push({symbolInstanceStart:0,symbolInstanceEnd:o.symbolInstances.length,paramet
ers:g});}attemptAnchorPlacement(e,i,s,a,o,r,n,l,h,c,u,d,_,p,m,f){const
g=t.aq[e.textAnchor],v=
[e.textOffset0,e.textOffset1],x=At(g,s,a,v,o),y=this.collisionIndex.placeCollisi
onBox(Rt(i,x.x,x.y,r,n,this.transform.angle),u,l,h,c.predicate,f);if((!m||0!==th
is.collisionIndex.placeCollisionBox(Rt(m,x.x,x.y,r,n,this.transform.angle),u,l,h
,c.predicate,f).box.length)&&y.box.length>0){let
t;if(this.prevPlacement&&this.prevPlacement.variableOffsets[d.crossTileID]&&this
.prevPlacement.placements[d.crossTileID]&&this.prevPlacement.placements[d.crossT
ileID].text&&
(t=this.prevPlacement.variableOffsets[d.crossTileID].anchor),0===d.crossTileID)t
hrow new Error("symbolInstance.crossTileID can't be 0");return
this.variableOffsets[d.crossTileID]=
{textOffset:v,width:s,height:a,anchor:g,textBoxScale:o,prevAnchor:t},this.markUs
edJustification(_,g,d,p),_.allowVerticalPlacement&&
(this.markUsedOrientation(_,p,d),this.placedOrientations[d.crossTileID]=p),
{shift:x,placedGlyphBoxes:y}}}placeLayerBucketPart(e,i,s)
{const{bucket:a,layout:o,posMatrix:r,textLabelPlaneMatrix:n,labelToScreenMatrix:
l,textPixelRatio:h,holdingForFade:c,collisionBoxArray:u,partiallyEvaluatedTextSi
ze:d,collisionGroup:_}=e.parameters,p=o.get("text-optional"),m=o.get("icon-
optional"),f=t.ar(o,"text-overlap","text-allow-
overlap"),g="always"===f,v=t.ar(o,"icon-overlap","icon-allow-
overlap"),x="always"===v,y="map"===o.get("text-rotation-
alignment"),b="map"===o.get("text-pitch-alignment"),w="none"!==o.get("icon-text-
fit"),T="viewport-y"===o.get("symbol-z-order"),I=g&&
(x||!a.hasIconData())||m,E=x&&
(g||!a.hasTextData())||p;!a.collisionArrays&&u&&a.deserializeCollisionBoxes(u);c
onst S=this.retainedQueryData[a.bucketInstanceId].tileID,C=this.terrain?
(t,e)=>this.terrain.getElevation(S,t,e):null,P=(e,u)=>{var
x,T;if(i[e.crossTileID])return;if(c)return
void(this.placements[e.crossTileID]=new Dt(!1,!1,!1));let
S=!1,P=!1,D=!0,M=null,z={box:null,offscreen:null},L=
{box:null,offscreen:null},A=null,R=null,k=null,F=0,B=0,O=0;u.textFeatureIndex?
F=u.textFeatureIndex:e.useRuntimeCollisionCircles&&
(F=e.featureIndex),u.verticalTextFeatureIndex&&
(B=u.verticalTextFeatureIndex);const N=u.textBox;if(N){const i=i=>{let
s=t.ai.horizontal;if(a.allowVerticalPlacement&&!i&&this.prevPlacement){const
t=this.prevPlacement.placedOrientations[e.crossTileID];t&&
(this.placedOrientations[e.crossTileID]=t,s=t,this.markUsedOrientation(a,s,e));}
return s},s=(i,s)=>
{if(a.allowVerticalPlacement&&e.numVerticalGlyphVertices>0&&u.verticalTextBox)
{for(const e of a.writingModes)if(e===t.ai.vertical?
(z=s(),L=z):z=i(),z&&z.box&&z.box.length)break}else
z=i();},o=e.textAnchorOffsetStartIndex,n=e.textAnchorOffsetEndIndex;if(n===o)
{const o=(t,i)=>{const
s=this.collisionIndex.placeCollisionBox(t,f,h,r,_.predicate,C);return
s&&s.box&&s.box.length&&

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(this.markUsedOrientation(a,i,e),this.placedOrientations[e.crossTileID]=i),s};s(
(())=>o(N,t.ai.horizontal)),(())=>{const i=u.verticalTextBox;return
a.allowVerticalPlacement&&e.numVerticalGlyphVertices>0&&i?o(i,t.ai.vertical):
{box:null,offscreen:null}})),i(z&&z.box&&z.box.length);}else {let l=t.aq[null===
(T=null===(x=this.prevPlacement))||void 0===x?void
0:x.variableOffsets[e.crossTileID]]||void 0===T?void 0:T.anchor];const c=
(t,i,s)=>{const c=t.x2-t.x1,u=t.y2-t.y1,d=e.textBoxScale,p=w&&"never"===v?
i:null;let m={box:[],offscreen:!1},g="never"===f?1:2,x="never";l&&g++;for(let
i=0;i<g;i++){for(let i=0;i<n;i++){const
o=a.textAnchorOffsets.get(i);if(l&&o.textAnchor!==l)continue;const
n=this.attemptAnchorPlacement(o,t,c,u,d,y,b,h,r,_,x,e,a,s,p,C);if(n&&
(m=n.placedGlyphBoxes,m&&m.box&&m.box.length))return S=!0,M=n.shift,m}l?
l=null:x=f;}return m};s(())=>c(N,u.iconBox,t.ai.horizontal)),(())=>{const
i=u.verticalTextBox;return a.allowVerticalPlacement&&
(z&&z.box&&z.box.length)&&e.numVerticalGlyphVertices>0&&i?
c(i,u.verticalIconBox,t.ai.vertical):{box:null,offscreen:null}})),z&&
(S=z.box,D=z.offscreen);const d=i(z&&z.box);if(!S&&this.prevPlacement){const
t=this.prevPlacement.variableOffsets[e.crossTileID];t&&
(this.variableOffsets[e.crossTileID]=t,this.markUsedJustification(a,t.anchor,e,d
));}}if(A=z,S=A&&A.box&&A.box.length>0,D=A&&A.offscreen,e.useRuntimeCollisionCi
rcles){const
i=a.text.placedSymbolArray.get(e.centerJustifiedTextSymbolIndex),h=t.aj(a.textSi
zeData,d,i),c=o.get("text-
padding");R=this.collisionIndex.placeCollisionCircles(f,i,a.lineVertexArray,a.gl
yphOffsetArray,h,r,n,l,s,b,_.predicate,e.collisionCircleDiameter,c,C),R.circles.
length&&R.collisionDetected&&!s&&t.w("Collisions detected, but collision boxes
are not
shown"),S=g||R.circles.length>0&&!R.collisionDetected,D=D&&R.offscreen;}if(u.ico
nFeatureIndex&&(O=u.iconFeatureIndex),u.iconBox){const t=t=>{const e=w&&M?
Rt(t,M.x,M.y,y,b,this.transform.angle):t;return
this.collisionIndex.placeCollisionBox(e,v,h,r,_.predicate,C)};L&&L.box&&L.box.le
ngth&&u.verticalIconBox?(k=t(u.verticalIconBox),P=k.box.length>0):
(k=t(u.iconBox),P=k.box.length>0),D=D&&k.offscreen;}const
U=p||0===e.numHorizontalGlyphVertices&&0===e.numVerticalGlyphVertices,Z=m||0===e
.numIconVertices;if(U||Z?Z?U||
(P=P&&S):S=P&&S:P=S&&P&&S,S&&A&&A.box&&this.collisionIndex.insertCollisionBox(A.b
ox,f,o.get("text-ignore-placement"),a.bucketInstanceId,L&&L.box&&B?
B:F,_.ID),P&&k&&this.collisionIndex.insertCollisionBox(k.box,v,o.get("icon-
ignore-placement"),a.bucketInstanceId,O,_.ID),R&&
(S&&this.collisionIndex.insertCollisionCircles(R.circles,f,o.get("text-ignore-
placement"),a.bucketInstanceId,F,_.ID),s)){const t=a.bucketInstanceId;let
e=this.collisionCircleArrays[t];void 0===e&&(e=this.collisionCircleArrays[t]=new
Mt);for(let
t=0;t<R.circles.length;t+=4)e.circles.push(R.circles[t+0]),e.circles.push(R.circ
les[t+1]),e.circles.push(R.circles[t+2]),e.circles.push(R.collisionDetected?
1:0);}if(0===e.crossTileID)throw new Error("symbolInstance.crossTileID can't be
0");if(0===a.bucketInstanceId)throw new Error("bucket.bucketInstanceId can't be
0");this.placements[e.crossTileID]=new
Dt(S||I,P||E,D||a.justReloaded),i[e.crossTileID]=!0;};if(T)
{if(0!==e.symbolInstanceStart)throw new Error("bucket.bucketInstanceId should be
0");const t=a.getSortedSymbolIndexes(this.transform.angle);for(let e=t.length-
1;e>=0;--e){const i=t[e];P(a.symbolInstances.get(i),a.collisionArrays[i]);}}else
for(let
t=e.symbolInstanceStart;t<e.symbolInstanceEnd;t++)P(a.symbolInstances.get(t),a.c
ollisionArrays[t]);if(s&&a.bucketInstanceId in this.collisionCircleArrays){const
e=this.collisionCircleArrays[a.bucketInstanceId];t.as(e.invProjMatrix,r),e.viewp
ortMatrix=this.collisionIndex.getViewportMatrix();}a.justReloaded=!1;}markUsedJu
stification(e,i,s,a){let o;o===t.ai.vertical?s.verticalPlacedTextSymbolIndex:
{left:s.leftJustifiedTextSymbolIndex,center:s.centerJustifiedTextSymbolIndex,rig
ht:s.rightJustifiedTextSymbolIndex}[t.at(i)];const r=
[s.leftJustifiedTextSymbolIndex,s.centerJustifiedTextSymbolIndex,s.rightJustifie
dTextSymbolIndex,s.verticalPlacedTextSymbolIndex];for(const t of r)t>=0&&

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(e.text.placedSymbolArray.get(t).crossTileID=o>=0&&t!=o?
0:s.crossTileID);}markUsedOrientation(e,i,s){const
a=i===t.ai.horizontal||i===t.ai.horizontalOnly?i:0,o=i===t.ai.vertical?i:0,r=
[s.leftJustifiedTextSymbolIndex,s.centerJustifiedTextSymbolIndex,s.rightJustifiedTextSymbolIndex];for(const t of
r)e.text.placedSymbolArray.get(t).placedOrientation=a;s.verticalPlacedTextSymbolIndex&&
(e.text.placedSymbolArray.get(s.verticalPlacedTextSymbolIndex).placedOrientation=o);}commit(t)
{this.commitTime=t,this.zoomAtLastRecencyCheck=this.transform.zoom;const
e=this.prevPlacement;let i=!1;this.prevZoomAdjustment=e?
e.zoomAdjustment(this.transform.zoom):0;const s=e?e.symbolFadeChange(t):1,a=e?
e.opacities:{},o=e?e.variableOffsets:{},r=e?e.placedOrientations:{};for(const t
in this.placements){const e=this.placements[t],o=a[t];o?(this.opacities[t]=new
Pt(o,s,e.text,e.icon),i=i||e.text!=o.text.placed||e.icon!=o.icon.placed):
(this.opacities[t]=new
Pt(null,s,e.text,e.icon,e.skipFade),i=i||e.text||e.icon);}for(const t in a)
{const e=a[t];if(!this.opacities[t]){const a=new Pt(e,s,!1,!1);a.isHidden()||
(this.opacities[t]=a,i=i||e.text.placed||e.icon.placed);}for(const t in
o)this.variableOffsets[t]||!this.opacities[t]||this.opacities[t].isHidden()||
(this.variableOffsets[t]=o[t]);for(const t in
r)this.placedOrientations[t]||!this.opacities[t]||this.opacities[t].isHidden()||
(this.placedOrientations[t]=r[t]);if(e&&void 0===e.lastPlacementChangeTime)throw
new Error("Last placement time for previous placement is not defined");i?
this.lastPlacementChangeTime=t:"number"!==typeof this.lastPlacementChangeTime&&
(this.lastPlacementChangeTime=e?
e.lastPlacementChangeTime:t);}updateLayerOpacities(t,e){const i={};for(const s
of e){const
e=s.getBucket(t);e&&s.latestFeatureIndex&&t.id===e.layerIds[0]&&this.updateBucketOpacities(e,i,s.collisio
nBoxArray);}updateBucketOpacities(e,i,s)
{e.hasTextData()&&
(e.text.opacityVertexArray.clear(),e.text.hasVisibleVertices=!1),e.hasIconData()
&&
(e.icon.opacityVertexArray.clear(),e.icon.hasVisibleVertices=!1),e.hasIconCollisionBoxData()&&e.iconCollisionBox.collisionVertexArray.clear(),e.hasTextCollisionBoxData()&&e.textCollisionBox.collisionVertexArray.clear();const
a=e.layers[0],o=a.layout,r=new Pt(null,0,!1,!1,!0),n=o.get("text-allow-overlap"),l=o.get("icon-allow-overlap"),h=a._unevaluatedLayout.hasValue("text-variable-anchor")||a._unevaluatedLayout.hasValue("text-variable-anchor-offset"),c="map"===o.get("text-rotation-alignment"),u="map"===o.get("text-pitch-alignment"),d="none"===o.get("icon-text-fit"),_new Pt(null,0,n&&
(l||!e.hasIconData()||o.get("icon-optional")),l&&
(n||!e.hasTextData()||o.get("text-optional")),!0);!e.collisionArrays&&s&&
(e.hasIconCollisionBoxData()||e.hasTextCollisionBoxData())&&e.deserializeCollisionBoxes(s);const p=(t,e,i)=>{for(let
s=0;s<e/4;s++)t.opacityVertexArray.emplaceBack(i);t.hasVisibleVertices=t.hasVisibleVertices||i!=qt;};for(let s=0;s<e.symbolInstances.length;s++){const
a=e.symbolInstances.get(s),
{numHorizontalGlyphVertices:o,numVerticalGlyphVertices:n,crossTileID:l}=a;let
m=this.opacities[l];i[l]?m=r.m||(m=_this.opacities[l]=m),i[l]=!0;const
f=a.numIconVertices>0,g=this.placedOrientations[a.crossTileID],v=g===t.ai.vertical,x=g===t.ai.horizontal||g===t.ai.horizontalOnly;if(o>0||n>0){const
t=Vt(m.text);p(e.text,o,v?qt:t),p(e.text,n,x?qt:t);const i=m.text.isHidden();
[a.rightJustifiedTextSymbolIndex,a.centerJustifiedTextSymbolIndex,a.leftJustifiedTextSymbolIndex].forEach((t=>{t>=0&&
(e.text.placedSymbolArray.get(t).hidden=i||v?
1:0;}))a.verticalPlacedTextSymbolIndex>=0&&
(e.text.placedSymbolArray.get(a.verticalPlacedTextSymbolIndex).hidden=i||x?
1:0);const
s=this.variableOffsets[a.crossTileID];s&&this.markUsedJustification(e,s.anchor,a,g);const r=this.placedOrientations[a.crossTileID];r&&
(this.markUsedJustification(e,"left",a,r),this.markUsedOrientation(e,r,a));}if(f

```

```

){const t=Vt(m.icon),i=!
(d&&a.verticalPlacedIconSymbolIndex&&v);a.placedIconSymbolIndex>=0&&
(p(e.icon,a.numIconVertices,i?
t:qt),e.icon.placedSymbolArray.get(a.placedIconSymbolIndex).hidden=m.icon.isHidd
en()),a.verticalPlacedIconSymbolIndex>=0&&(p(e.icon,a.numVerticalIconVertices,i?
qt:t),e.icon.placedSymbolArray.get(a.verticalPlacedIconSymbolIndex).hidden=m.ico
n.isHidden());}if(e.hasIconCollisionBoxData()||e.hasTextCollisionBoxData())
{const i=e.collisionArrays[s];if(i){let s=new
t.P(0,0);if(i.textBox||i.verticalTextBox){let t=!0;if(h){const
e=this.variableOffsets[l];e?
(s=At(e.anchor,e.width,e.height,e.textOffset,e.textBoxScale),c&&s._rotate(u?
this.transform.angle):-
this.transform.angle)):t=!1;}i.textBox&&Ft(e.textCollisionBox.collisionVertexArr
ay,m.text.placed,!t||v,s.x,s.y),i.verticalTextBox&&Ft(e.textCollisionBox.collisi
onVertexArray,m.text.placed,!t||x,s.x,s.y);}const
a=Boolean(!x&&i.verticalIconBox);i.iconBox&&Ft(e.iconCollisionBox.collisionVerte
xArray,m.icon.placed,a,d?s.x:0,d?
s.y:0),i.verticalIconBox&&Ft(e.iconCollisionBox.collisionVertexArray,m.icon.plac
ed,!a,d?s.x:0,d?
s.y:0);}}if(e.sortFeatures(this.transform.angle),this.retainedQueryData[e.bucke
tInstanceId]&&
(this.retainedQueryData[e.bucketInstanceId].featureSortOrder=e.featureSortOrder)
,e.hasTextData()&&e.text.opacityVertexBuffer&&e.text.opacityVertexBuffer.updated
ata(e.text.opacityVertexArray),e.hasIconData()&&e.icon.opacityVertexBuffer&&e.ic
on.opacityVertexBuffer.updateData(e.icon.opacityVertexArray),e.hasIconCollisionB
oxData()&&e.iconCollisionBox.collisionVertexBuffer&&e.iconCollisionBox.collision
VertexBuffer.updateData(e.iconCollisionBox.collisionVertexArray),e.hasTextCollis
ionBoxData()&&e.textCollisionBox.collisionVertexBuffer&&e.textCollisionBox.colli
sionVertexBuffer.updateData(e.textCollisionBox.collisionVertexArray),e.text.opac
ityVertexArray.length!==e.text.layoutVertexArray.length/4)throw new
Error(`bucket.text.opacityVertexArray.length (=
${e.text.opacityVertexArray.length}) !== bucket.text.layoutVertexArray.length (=
${e.text.layoutVertexArray.length}) /
4`);if(e.icon.opacityVertexArray.length!==e.icon.layoutVertexArray.length/4)thro
w new Error(`bucket.icon.opacityVertexArray.length (=
${e.icon.opacityVertexArray.length}) !== bucket.icon.layoutVertexArray.length (=
${e.icon.layoutVertexArray.length}) / 4`);if(e.bucketInstanceId in
this.collisionCircleArrays){const
t=this.collisionCircleArrays[e.bucketInstanceId];e.placementInvProjMatrix=t.invP
rojMatrix,e.placementViewportMatrix=t.viewportMatrix,e.collisionCircleArray=t.ci
rcles,delete
this.collisionCircleArrays[e.bucketInstanceId];}symbolFadeChange(t){return
0===this.fadeDuration?1:(t-
this.commitTime)/this.fadeDuration+this.prevZoomAdjustment}zoomAdjustment(t)
{return Math.max(0,(this.transform.zoom-t)/1.5)}hasTransitions(t){return
this.stale||t-this.lastPlacementChangeTime<this.fadeDuration}stillRecent(t,e)
{const i=this.zoomAtLastRecencyCheck===e?1-this.zoomAdjustment(e):1;return
this.zoomAtLastRecencyCheck=e,this.commitTime+this.fadeDuration*i>t}setStale()
{this.stale=!0;}function Ft(t,e,i,s,a){t.emplaceBack(e?1:0,i?
1:0,s||0,a||0),t.emplaceBack(e?1:0,i?1:0,s||0,a||0),t.emplaceBack(e?1:0,i?
1:0,s||0,a||0),t.emplaceBack(e?1:0,i?1:0,s||0,a||0);}const
Bt=Math.pow(2,25),Ot=Math.pow(2,24),Nt=Math.pow(2,17),Ut=Math.pow(2,16),Zt=Math.
pow(2,9),Gt=Math.pow(2,8),jt=Math.pow(2,1);function Vt(t)
{if(0===t.opacity&&!t.placed)return 0;if(1===t.opacity&&t.placed)return
4294967295;const e=t.placed?1:0,i=Math.floor(127*t.opacity);return
i*Bt+e*Ot+i*Nt+e*Ut+i*Zt+e*Gt+i*jt+e}const qt=0;class $t{constructor(t)
{this._sortAcrossTiles="viewport-y"!==t.layout.get("symbol-z-
order")&&t.layout.get("symbol-sort-
key").isConstant(),this._currentTileIndex=0,this._currentPartIndex=0,this._seenC
rossTileIDs={},this._bucketParts=[];}continuePlacement(t,e,i,s,a){const
o=this._bucketParts;for(;this._currentTileIndex<t.length;)if(e.getBucketParts(o,
s,t[this._currentTileIndex],this._sortAcrossTiles),this._currentTileIndex++,a())

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return !0;for(this._sortAcrossTiles&&
(this._sortAcrossTiles=!1,o.sort(((t,e)=>t.sortKey-
e.sortKey)));this._currentPartIndex<o.length;)if(e.placeLayerBucketPart(o[this._
currentPartIndex],this._seenCrossTileIDs,i),this._currentPartIndex++,a())return
!0;return !1}}class Wt{constructor(t,e,i,s,a,o,r,n){this.placement=new
kt(t,e,o,r,n),this._currentPlacementIndex=i.length-
1,this._forceFullPlacement=s,this._showCollisionBoxes=a,this._done=!1;}isDone()
{return this._done}continuePlacement(e,i,s){const a=t.h.now(),o=
()=>!this._forceFullPlacement&&t.h.now()-
a>2;for(;this._currentPlacementIndex>=0;){const
t=i[e[this._currentPlacementIndex]],a=this.placement.collisionIndex.transform.zo
om;if("symbol"===t.type&&(!t.minzoom||t.minzoom<=a)&&(!t.maxzoom||t.maxzoom>a))
{if(this._inProgressLayer||(this._inProgressLayer=new
$t(t)),this._inProgressLayer.continuePlacement(s[t.source],this.placement,this._
showCollisionBoxes,t,o))return;delete
this._inProgressLayer;}this._currentPlacementIndex--;}this._done=!0;}commit(t)
{return this.placement.commit(t),this.placement}}const Ht=512/t.N/2;class
Xt{constructor(e,i,s){this.tileID=e,this.bucketInstanceId=s,this._symbolsByKey=
{};const a=new Map;for(let t=0;t<i.length;t++){const
e=i.get(t),s=e.key,o=a.get(s);o?o.push(e):a.set(s,[e]);}for(const[e,i]of a)
{const s={positions:i.map((t=>
({x:Math.floor(t.anchorX*Ht),y:Math.floor(t.anchorY*Ht)})),crossTileIDs:i.map((
t=>t.crossTileID)));if(s.positions.length>128){const e=new
t.av(s.positions.length,16,Uint16Array);for(const{x:t,y:i}of
s.positions)e.add(t,i);e.finish(),delete
s.positions,s.index=e;}this._symbolsByKey[e]=s;}}getScaledCoordinates(e,i)
{const{x:s,y:a,z:o}=this.tileID.canonical,
{x:r,y:n,z:l}=i.canonical,h=Ht/Math.pow(2,l-o),c=
(n*t.N+e.anchorY)*h,u=a*t.N*Ht;return {x:Math.floor((r*t.N+e.anchorX)*h-
s*t.N*Ht),y:Math.floor(c-u)}}findMatches(t,e,i){const
s=this.tileID.canonical.z<e.canonical.z?1:Math.pow(2,this.tileID.canonical.z-
e.canonical.z);for(let a=0;a<t.length;a++){const
o=t.get(a);if(o.crossTileID)continue;const
r=this._symbolsByKey[o.key];if(!r)continue;const
n=this.getScaledCoordinates(o,e);if(r.index){const t=r.index.range(n.x-s,n.y-
s,n.x+s,n.y+s).sort();for(const e of t){const t=r.crossTileIDs[e];if(!i[t])
{i[t]=!0,o.crossTileID=t;break}}}else if(r.positions)for(let
t=0;t<r.positions.length;t++){const
e=r.positions[t],a=r.crossTileIDs[t];if(Math.abs(e.x-n.x)<=s&&Math.abs(e.y-n.y)
<=s&&!i[a]){i[a]=!0,o.crossTileID=a;break}}}}getCrossTileIDsLists(){return
Object.values(this._symbolsByKey).map((({crossTileIDs:t})=>t))}class
Kt{constructor(){this.maxCrossTileID=0;}generate(){return
++this.maxCrossTileID}}class Qt{constructor(){this.indexes=
{};this.usedCrossTileIDs={},this.lng=0;}handleWrapJump(t){const e=Math.round((t-
this.lng)/360);if(0!==e)for(const t in this.indexes){const i=this.indexes[t],s=
{};for(const t in i){const
a=i[t];a.tileID=a.tileID.unwrapTo(a.tileID.wrap+e),s[a.tileID.key]=a;}this.index
es[t]=s;}this.lng=t;}addBucket(t,e,i)
{if(this.indexes[t.overscaledZ]&&this.indexes[t.overscaledZ][t.key])
{if(this.indexes[t.overscaledZ]
[t.key].bucketInstanceId===e.bucketInstanceId)return
!1;this.removeBucketCrossTileIDs(t.overscaledZ,this.indexes[t.overscaledZ]
[t.key]);}for(let
t=0;t<e.symbolInstances.length;t++)e.symbolInstances.get(t).crossTileID=0;this.u
sedCrossTileIDs[t.overscaledZ]|=(this.usedCrossTileIDs[t.overscaledZ]={});const
s=this.usedCrossTileIDs[t.overscaledZ];for(const i in this.indexes){const
a=this.indexes[i];if(Number(i)>t.overscaledZ)for(const i in a){const
o=a[i];o.tileID.isChildOf(t)&&o.findMatches(e.symbolInstances,t,s);}else {const
o=a[t.scaledTo(Number(i)).key];o&o.findMatches(e.symbolInstances,t,s);}}for(let
t=0;t<e.symbolInstances.length;t++){const
a=e.symbolInstances.get(t);a.crossTileID|
(a.crossTileID=i.generate(),s[a.crossTileID]=!0);}return void

```

```

0===this.indexes[t.overscaledZ]&&(this.indexes[t.overscaledZ]=
{}),this.indexes[t.overscaledZ][t.key]=new
Xt(t,e.symbolInstances,e.bucketInstanceId,!0)removeBucketCrossTileIDs(t,e)
{for(const i of e.getCrossTileIDsLists())for(const e of i)delete
this.usedCrossTileIDs[t][e];}removeStaleBuckets(t){let e=!1;for(const i in
this.indexes){const s=this.indexes[i];for(const a in
s)t[s[a].bucketInstanceId]||(this.removeBucketCrossTileIDs(i,s[a]),delete
s[a],e=!0);}return e}class Yt{constructor(){this.layerIndexes=
{}},this.crossTileIDs=new
Kt,t,this.maxBucketInstanceId=0,this.bucketsInCurrentPlacement={};}addLayer(t,e,i)
{let s=this.layerIndexes[t.id];void 0===s&&(s=this.layerIndexes[t.id]=new
Qt);let a=!1;const o={};s.handleWrapJump(i);for(const i of e){const
e=i.getBucket(t);e&&t.id===e.layerIds[0]&&(e.bucketInstanceId|
(e.bucketInstanceId===this.maxBucketInstanceId),s.addBucket(i.tileID,e,this.cros
sTileIDs)&&(a=!0),o[e.bucketInstanceId]=!0);}return s.removeStaleBuckets(o)&&
(a=!0),a}pruneUnusedLayers(t){const e={};t.forEach((t=>{e[t]=!0;}));for(const t
in this.layerIndexes)e[t]||delete this.layerIndexes[t];}const Jt=
(e,i)=>t.x(e,i&&i.filter((t=>"source.canvas"!==t.identifier))),te=t.F(t.ax,
["addLayer","removeLayer","setPaintProperty","setLayoutProperty","setFilter","ad
dSource","removeSource","setLayerZoomRange","setLight","setTransition","setGeoJS
ONSourceData","setGlyphs","setSprite"]),ee=t.F(t.ax,
["setCenter","setZoom","setBearing","setPitch"]),ie=t.aw();class se extends
t.E{constructor(e,i={}){super(),this.map=e,this.dispatcher=new
M(st(),this,e._getMapId()),this.imageManager=new
b,this.imageManager.setEventedParent(this),this.glyphManager=new
E(e._requestManager,i.localIdeographFontFamily),this.lineAtlas=new
D(256,512),this.crossTileSymbolIndex=new Yt,this._spritesImagesIds=
{}},this._layers={},this._order=[],this.sourceCaches={},this.zoomHistory=new
t.ay,this._loaded=!1,this._availableImages=
[],this._resetUpdates(),this.dispatcher.broadcast("setReferrer",t.az());const
s=this;this._rtlTextPluginCallback=se.registerForPluginStateChange((e=>
{s.dispatcher.broadcast("syncRTLPluginState",
{pluginStatus:e.pluginStatus,pluginURL:e.pluginURL}),((e,i)=>
{if(t.aA(e),i&&i.every((t=>t)))for(const t in s.sourceCaches){const
e=s.sourceCaches[t].getSource().type;"vector"!==e&&"geojson"!==e||s.sourceCaches
[t].reload();}})})),this.on("data",(t=>
{if("source"!==t.dataType||"metadata"!==t.sourceDataType)return;const
e=this.sourceCaches[t.sourceId];if(!e)return;const
i=e.getSource();if(i&&i.vectorLayerIds)for(const t in this._layers){const
e=this._layers[t];e.source===i.id&&this._validateLayer(e);}));}loadURL(e,i=
{s},s){this.fire(new t.k("dataloading",
{dataType:"style"})),i.validate="boolean"!==typeof i.validate||i.validate;const
a=this.map._requestManager.transformRequest(e,c.Style);this._request=t.f(a,
((e,a)=>{this._request=null,e?this.fire(new
t.j(e)):a&&this._load(a,i,s)}));}loadJSON(e,i={s},s){this.fire(new
t.k("dataloading",{dataType:"style"})),this._request=t.h.frame((()=>>
{this._request=null,i.validate=!1===i.validate,this._load(e,i,s)}));}loadEmpty(
){this.fire(new t.k("dataloading",{dataType:"style"})),this._load(ie,
{validate:!1});}_load(e,i,s){var a;const o=i.transformStyle?
i.transformStyle(s,e):e;if(!i.validate||!Jt(this,t.y(o)))
{this._loaded=!0,this.stylesheet=o;for(const t in
o.sources)this.addSource(t,o.sources[t],{validate:!1});o.sprite?
this._loadSprite(o.sprite):this.imageManager.setLoaded(!0),this.glyphManager.set
URL(o.glyphs),this._createLayers(),this.light=new
P(this.stylesheet.light),this.map.setTerrain(null!==(
a=this.stylesheet.terrain)&&void 0!==a?a:null),this.fire(new t.k("data",
{dataType:"style"})),this.fire(new t.k("style.load"));}_createLayers(){const
e=t.aB(this.stylesheet.layers);this.dispatcher.broadcast("setLayers",e),this._or
der=e.map((t=>t.id),this._layers={},this._serializedLayers=null;for(const i of
e){const e=t.aC(i);e.setEventedParent(this,{layer:
{id:i.id}}),this._layers[i.id]=e;}_loadSprite(e,i=!1,s=void 0)
{this.imageManager.setLoaded(!1),this._spriteRequest=function(e,i,s,a){const

```



```

o=g(e),r=o.length,n=s>1?"@2x":"" ,l={},u={},d={};for(const{id:e,url:s}of o){const
o=i.transformRequest(i.normalizeSpriteURL(s,n,".json"),c.SpriteJSON),_=`${e}_${o
.url}`;l[_]=t.f(o,((t,i)=>{delete l[_],u[e]=i,v(a,u,d,t,r);}));const
p=i.transformRequest(i.normalizeSpriteURL(s,n,".png"),c.SpriteImage),m=`${e}_${p
.url}`;l[m]=h.getImage(p,((t,i)=>{delete l[m],d[e]=i,v(a,u,d,t,r);}));}return
{cancel(){for(const t of Object.values(l))t.cancel();}}
(e,this.map._requestManager,this.map.getPixelRatio(),((e,a)=>
{if(this._spriteRequest=null,e)this.fire(new t.j(e));else if(a)for(const t in a)
{this._spritesImagesIds[t]=[];const e=this._spritesImagesIds[t]?
this._spritesImagesIds[t].filter((t=>!(t in a))):[];for(const t of
e)this.imageManager.removeImage(t),this._changedImages[t]=!0;for(const e in
a[t]){const s="default"===t?e:`${t}:${e}`;this._spritesImagesIds[t].push(s),s in
this.imageManager.images?this.imageManager.updateImage(s,a[t]
[e],!1):this.imageManager.addImage(s,a[t][e]),i&&
(this._changedImages[s]=!0);}}this.imageManager.setLoaded(!0),this._availableIma
ges=this.imageManager.listImages(),i&&
(this._changed=!0),this.dispatcher.broadcast("setImages",this._availableImages),
this.fire(new t.k("data",{dataType:"style"})),s&&s(e);}});_unloadSprite()
{for(const t of
Object.values(this._spritesImagesIds).flat())this.imageManager.removeImage(t),th
is._changedImages[t]=!0;this._spritesImagesIds=
{},this._availableImages=this.imageManager.listImages(),this._changed=!0,this.di
spatcher.broadcast("setImages",this._availableImages),this.fire(new t.k("data",
{dataType:"style"}));_validateLayer(e){const
i=this.sourceCaches[e.source];if(!i)return;const
s=e.sourceLayer;if(!s)return;const a=i.getSource();
("geojson"===a.type||a.vectorLayerIds&&-1===a.vectorLayerIds.indexOf(s))&&this.f
ire(new t.j(new Error(`Source layer "${s}" does not exist on source "${a.id}" as
specified by style layer "${e.id}"`)));}loaded(){if(!this._loaded)return
!1;if(Object.keys(this._updatedSources).length)return !1;for(const t in
this.sourceCaches)if(!this.sourceCaches[t].loaded())return !1;return
!!this.imageManager.isLoaded())_serializeByIds(t){const
e=this._serializedAllLayers();if(!t||0===t.length)return Object.values(e);const
i=[];for(const s of t)e[s]&&i.push(e[s]);return i}_serializedAllLayers(){let
t=this._serializedLayers;if(t)return t;t=this._serializedLayers={};const
e=Object.keys(this._layers);for(const i of e){const
e=this._layers[i];"custom"!==e.type&&(t[i]=e.serialize());}return
t}hasTransitions(){if(this.light&&this.light.hasTransition())return !0;for(const
t in this.sourceCaches)if(this.sourceCaches[t].hasTransition())return
!0;for(const t in this._layers)if(this._layers[t].hasTransition())return
!0;return !1}_checkLoaded(){if(!this._loaded)throw new Error("Style is not done
loading.")}update(e){if(!this._loaded)return;const
i=this._changed;if(this._changed){const
t=Object.keys(this._updatedLayers),i=Object.keys(this._removedLayers);
(t.length||i.length)&&this._updateWorkerLayers(t,i);for(const t in
this._updatedSources){const
e=this._updatedSources[t];if("reload"===e)this._reloadSource(t);else
{if("clear"===e)throw new Error(`Invalid action
${e}`);this._clearSource(t);}}this._updateTilesForChangedImages(),this._updateTi
lesForChangedGlyphs();for(const t in
this._updatedPaintProps)this._layers[t].updateTransitions(e);this.light.updateTr
ansitions(e),this._resetUpdates();}const s={};for(const t in this.sourceCaches)
{const e=this.sourceCaches[t];s[t]=e.used,e.used=!1;}for(const t of this._order)
{const
i=this._layers[t];i.recalculate(e,this._availableImages,!i.isHidden(e.zoom)&&i.
source&&(this.sourceCaches[i.source].used=!0);}for(const e in s){const
i=this.sourceCaches[e];s[e]!==i.used&&i.fire(new t.k("data",
{sourceDataType:"visibility",dataType:"source",sourceId:e}));}this.light.recalcu
late(e),this.z=e.zoom,i&&this.fire(new t.k("data",
{dataType:"style"}));}_updateTilesForChangedImages(){const
t=Object.keys(this._changedImages);if(t.length){for(const e in
this.sourceCaches)this.sourceCaches[e].reloadTilesForDependencies(["icons","patt

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erns"],t);this._changedImages={};}}_updateTilesForChangedGlyphs()
{if(this._glyphsDidChange){for(const t in
this.sourceCaches)this.sourceCaches[t].reloadTilesForDependencies(["glyphs"],
[""]);this._glyphsDidChange=!1;}}_updateWorkerLayers(t,e)
{this.dispatcher.broadcast("updateLayers",
{layers:this._serializeByIds(t),removedIds:e});}_resetUpdates()
{this._changed=!1,this._updatedLayers={},this._removedLayers=
{},this._updatedSources={},this._updatedPaintProps={},this._changedImages=
{},this._glyphsDidChange=!1;}}setState(e,i={}){this._checkLoaded();const
s=this.serialize();if(e=i.transformStyle?
i.transformStyle(s,e):e,Jt(this,t.y(e)))return !1;
(e=t.aD(e)).layers=t.aB(e.layers);const a=t.aE(s,e).filter((t=>!(t.command in
ee)));if(0===a.length)return !1;const o=a.filter((t=>!(t.command in
te)));if(o.length>0)throw new Error(`Unimplemented:
${o.map((t=>t.command)).join(", ")}.`);for(const t of
a)"setTransition"!==t.command&&this[t.command].apply(this,t.args);return
this.stylesheet=e,this._serializedLayers=null,!0}addImage(e,i)
{if(this.getImage(e))return this.fire(new t.j(new Error(`An image named "${e}"
already
exists.`)));this.imageManager.addImage(e,i),this._afterImageUpdated(e);}updateIm
age(t,e){this.imageManager.updateImage(t,e);}getImage(t){return
this.imageManager.getImage(t)}removeImage(e){if(!this.getImage(e))return
this.fire(new t.j(new Error(`An image named "${e}" does not
exist.`)));this.imageManager.removeImage(e),this._afterImageUpdated(e);}_afterIm
ageUpdated(e)
{this._availableImages=this.imageManager.listImages(),this._changedImages[e]=!0,
this._changed=!0,this.dispatcher.broadcast("setImages",this._availableImages),th
is.fire(new t.k("data",{dataType:"style"}));}listImages(){return
this._checkLoaded(),this.imageManager.listImages()}addSource(e,i,s={})
{if(this._checkLoaded(),void 0!==this.sourceCaches[e])throw new Error(`Source
"${e}" already exists.`);if(!i.type)throw new Error(`The type property must be
defined, but only the following properties were given: ${Object.keys(i).join(",
")}.`);if(["vector","raster","geojson","video","image"].indexOf(i.type)>=0&&this
._validate(t.y.source,`sources.${e}` ,i,null,s))return;this.map&&this.map._collec
tResourceTiming&&(i.collectResourceTiming=!0);const a=this.sourceCaches[e]=new
K(e,i,this.dispatcher);a.style=this,a.setEventedParent(this,(()=>>
({isSourceLoaded:a.loaded(),source:a.serialize(),sourceId:e}))),a.onAdd(this.map
),this._changed=!0;}removeSource(e){if(this._checkLoaded(),void
0===this.sourceCaches[e])throw new Error("There is no source with this
ID");for(const i in this._layers)if(this._layers[i].source===e)return
this.fire(new t.j(new Error(`Source "${e}" cannot be removed while layer "${i}"
is using it.`)));const i=this.sourceCaches[e];delete this.sourceCaches[e],delete
this._updatedSources[e],i.fire(new t.k("data",
{sourceDataType:"metadata",dataType:"source",sourceId:e})),i.setEventedParent(nu
ll),i.onRemove(this.map),this._changed=!0;}setGeoJSONSourceData(t,e)
{if(this._checkLoaded(),void 0===this.sourceCaches[t])throw new Error(`There is
no source with this ID=${t}`);const
i=this.sourceCaches[t].getSource();if("geojson"!==i.type)throw new
Error(`geojsonSource.type is ${i.type}, which is !==
'geojson`);i.setData(e),this._changed=!0;}getSource(t){return
this.sourceCaches[t]&&this.sourceCaches[t].getSource()}addLayer(e,i,s={})
{this._checkLoaded();const a=e.id;if(this.getLayer(a))return void this.fire(new
t.j(new Error(`Layer "${a}" already exists on this map.`)));let
o;if("custom"===e.type){if(Jt(this,t.aF(e)))return;o=t.aC(e);}else
{if("source" in e&&"object"===typeof e.source&&
(this.addSource(a,e.source),e=t.aD(e),e=t.e(e,
{source:a})),this._validate(t.y.layer,`layers.${a}` ,e,
{arrayIndex:-1},s))return;o=t.aC(e),this._validateLayer(o),o.setEventedParent(th
is,{layer:{id:a}});}const r=i?
this._order.indexOf(i):this._order.length;if(i&&-1===r)this.fire(new t.j(new
Error(`Cannot add layer "${a}" before non-existing layer "${i}"`));else
{if(this._order.splice(r,0,a),this._layerOrderChanged=!0,this._layers[a]=o,this.

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_removedLayers[a]&&o.source&&"custom"!==o.type){const
t=this._removedLayers[a];delete this._removedLayers[a],t.type!==o.type?
this._updatedSources[o.source]="clear":
(this._updatedSources[o.source]="reload",this.sourceCaches[o.source].pause());}t
his._updateLayer(o,o.onAdd&&o.onAdd(this.map));}moveLayer(e,i)
{if(this._checkLoaded(),this._changed=!0,!this._layers[e])return void
this.fire(new t.j(new Error(`The layer '${e}' does not exist in the map's style
and cannot be moved.`)));if(e===i)return;const
s=this._order.indexOf(e);this._order.splice(s,1);const a=i?
this._order.indexOf(i):this._order.length;i&&-1===a?this.fire(new t.j(new
Error(`Cannot move layer "${e}" before non-existing layer "${i}".`))):
(this._order.splice(a,0,e),this._layerOrderChanged=!0);}removeLayer(e)
{this._checkLoaded();const i=this._layers[e];if(!i)return void this.fire(new
t.j(new Error(`Cannot remove non-existing layer
"${e}".`)));i.setEventedParent(null);const
s=this._order.indexOf(e);this._order.splice(s,1),this._layerOrderChanged=!0,this
._changed=!0,this._removedLayers[e]=i,delete
this._layers[e],this._serializedLayers&&delete this._serializedLayers[e],delete
this._updatedLayers[e],delete
this._updatedPaintProps[e],i.onRemove&&i.onRemove(this.map);}getLayer(t){return
this._layers[t]}getLayersOrder(){return [...this._order]}hasLayer(t){return t in
this._layers}setLayerZoomRange(e,i,s){this._checkLoaded();const
a=this.getLayer(e);a?a.minzoom===i&&a.maxzoom===s||!(null!=i&&
(a.minzoom=i),null!=s&&(a.maxzoom=s)),this._updateLayer(a):this.fire(new t.j(new
Error(`Cannot set the zoom range of non-existing layer
"${e}".`)));}setFilter(e,i,s){this._checkLoaded();const
a=this.getLayer(e);if(a){if(!t.aG(a.filter,i))return null==i?(a.filter=void
0,void
this._updateLayer(a)):void(this._validate(t.y.filter,`layers.${a.id}.filter`,i,n
ull,s)||!(a.filter=t.aD(i),this._updateLayer(a)))}else this.fire(new t.j(new
Error(`Cannot filter non-existing layer "${e}".`)));}getFilter(e){return
t.aD(this.getLayer(e).filter)}setLayoutProperty(e,i,s,a){
{this._checkLoaded();const o=this.getLayer(e);o?t.aG(o.getLayoutProperty(i),s)||
(o.setLayoutProperty(i,s,a),this._updateLayer(o)):this.fire(new t.j(new
Error(`Cannot style non-existing layer "${e}".`)));}getLayoutProperty(e,i){const
s=this.getLayer(e);if(s)return s.getLayoutProperty(i);this.fire(new t.j(new
Error(`Cannot get style of non-existing layer
"${e}".`)));}setPaintProperty(e,i,s,a){this._checkLoaded();const
o=this.getLayer(e);o?t.aG(o.getPaintProperty(i),s)||
(o.setPaintProperty(i,s,a)&&this._updateLayer(o),this._changed=!0,this._updatedP
aintProps[e]=!0):this.fire(new t.j(new Error(`Cannot style non-existing layer
"${e}".`)));}getPaintProperty(t,e){return
this.getLayer(t).getPaintProperty(e)}setFeatureState(e,i)
{this._checkLoaded();const
s=e.source,a=e.sourceLayer,o=this.sourceCaches[s];if(void 0===o)return void
this.fire(new t.j(new Error(`The source '${s}' does not exist in the map's
style.`)));const r=o.getSource().type;"geojson"===r&&a?this.fire(new t.j(new
Error("GeoJSON sources cannot have a sourceLayer parameter."))):"vector"!==r||a?
(void 0===e.id&&this.fire(new t.j(new Error("The feature id parameter must be
provided."))),o.setFeatureState(a,e.id,i):this.fire(new t.j(new Error("The
sourceLayer parameter must be provided for vector source
types.")));}removeFeatureState(e,i){this._checkLoaded();const
s=e.source,a=this.sourceCaches[s];if(void 0===a)return void this.fire(new
t.j(new Error(`The source '${s}' does not exist in the map's style.`)));const
o=a.getSource().type,r="vector"===o?e.sourceLayer:void 0;"vector"!==o||r?
i&&"string"!==typeof e.id&&"number"!==typeof e.id?this.fire(new t.j(new Error("A
feature id is required to remove its specific state
property."))):a.removeFeatureState(r,e.id,i):this.fire(new t.j(new Error("The
sourceLayer parameter must be provided for vector source
types.")));}getFeatureState(e){this._checkLoaded();const
i=e.source,s=e.sourceLayer,a=this.sourceCaches[i];if(void 0!==a)return
"vector"!==a.getSource().type||s?(void 0===e.id&&this.fire(new t.j(new

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Error("The feature id parameter must be
provided.")),a.getFeatureState(s,e.id)):void this.fire(new t.j(new Error("The
sourceLayer parameter must be provided for vector source
types.")));this.fire(new t.j(new Error(`The source '${i}' does not exist in the
map's style.`)));getTransition(){return
t.e({duration:300,delay:0},this.stylesheet&&this.stylesheet.transition)}serializ
e(){if(!this._loaded)return;const e=t.aH(this.sourceCaches,
(t=>t.serialize()),i=this._serializeByIds(this._order),s=this.map.getTerrain()|
|void 0,a=this.stylesheet;return
t.aI({version:a.version,name:a.name,metadata:a.metadata,light:a.light,center:a.c
enter,zoom:a.zoom,bearing:a.bearing,pitch:a.pitch,sprite:a.sprite,glyphs:a.glyph
s,transition:a.transition,sources:e,layers:i,terrain:s},(t=>void
0!==(t))}_updateLayer(t)
{this._updatedLayers[t.id]=!0,t.source&&!this._updatedSources[t.source]&&"raster
"!==(this.sourceCaches[t.source].getSource().type&&
(this._updatedSources[t.source]="reload",this.sourceCaches[t.source].pause()),th
is._serializedLayers=null,this._changed=!0;}_flattenAndSortRenderedFeatures(t)
{const e=t=>"fill-extrusion"===this._layers[t].type,i={},s=[];for(let
a=this._order.length-1;a>=0;a--){const o=this._order[a];if(e(o))
{i[o]=a;for(const e of t){const t=e[o];if(t)for(const e of
t)s.push(e);}}s.sort(((t,e)=>e.intersectionZ-t.intersectionZ));const a=
[];for(let o=this._order.length-1;o>=0;o--){const
r=this._order[o];if(e(r))for(let t=s.length-1;t>=0;t--){const
e=s[t].feature;if(i[e.layer.id]<o)break;a.push(e),s.pop();}else for(const e of
t){const t=e[r];if(t)for(const e of t)a.push(e.feature);}}return
a}queryRenderedFeatures(e,i,s)
{i&&i.filter&&this._validate(t.y.filter,"queryRenderedFeatures.filter",i.filter,
null,i);const a={};if(i&&i.layers){if(!Array.isArray(i.layers))return
this.fire(new t.j(new Error("parameters.layers must be an Array."))),
[];for(const e of i.layers){const i=this._layers[e];if(!i)return this.fire(new
t.j(new Error(`The layer '${e}' does not exist in the map's style and cannot be
queried for features.`))),[];a[i.source]=!0;}}const o=
[];i.availableImages=this._availableImages;const
r=this._serializedAllLayers();for(const t in
this.sourceCaches)i.layers&&!a[t]||o.push(q(this.sourceCaches[t],this._layers,r,
e,i,s));return this.placement&&o.push(function(t,e,i,s,a,o,r){const n=
{},l=o.queryRenderedSymbols(s),h=[];for(const t of
Object.keys(l).map(Number))h.push(r[t]);h.sort($);for(const i of h){const
s=i.featureIndex.lookupSymbolFeatures(l[i.bucketInstanceId],e,i.bucketIndex,i.so
urceLayerIndex,a.filter,a.layers,a.availableImages,t);for(const t in s){const
e=n[t]=n[t]||[],a=s[t];a.sort(((t,e)=>{const s=i.featureSortOrder;if(s){const
i=s.indexOf(t.featureIndex);return s.indexOf(e.featureIndex)-i}return
e.featureIndex-t.featureIndex));for(const t of a)e.push(t);}}for(const e in
n)n[e].forEach((s=>{const
a=s.feature,o=i[t[e].source].getFeatureState(a.layer["source-
layer"],a.id);a.source=a.layer.source,a.layer["source-layer"]&&
(a.sourceLayer=a.layer["source-layer"]),a.state=o;});return n}
(this._layers,r,this.sourceCaches,e,i,this.placement.collisionIndex,this.placeme
nt.retainedQueryData)),this._flattenAndSortRenderedFeatures(o)}querySourceFeatur
es(e,i)
{i&&i.filter&&this._validate(t.y.filter,"querySourceFeatures.filter",i.filter,nu
ll,i);const s=this.sourceCaches[e];return s?function(t,e){const
i=t.getRenderableIds().map((e=>t.getTileByID(e))),s=[],a={};for(let
t=0;t<i.length;t++){const o=i[t],r=o.tileID.canonical.key;a[r]||
(a[r]=!0,o.querySourceFeatures(s,e));}return s}(s,i):[]}_addSourceType(t,e,i)
{return j(t)?i(new Error(`A source type called "${t}" already exists.`)):
((t,e)=>{G[t]=e;})(t,e),e.workerSourceURL?void
this.dispatche.broadcast("loadWorkerSource",
{name:t,url:e.workerSourceURL},i):i(null,null))}getLight(){return
this.light.getLight()}setLight(e,i={}){this._checkLoaded();const
s=this.light.getLight();let a=!1;for(const i in e)if(!t.aG(e[i],s[i]))
{a=!0;break}if(!a)return;const o=

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{now:t.h.now(),transition:t.e({duration:300,delay:0},this.stylesheet.transition)
};this.light.setLight(e,i),this.light.updateTransitions(o);}_validate(e,i,s,a,o=
{})){return
(!o||!l==o.validate)&&Jt(this,e.call(t.y,t.e({key:i,style:this.serialize(),valu
e:s,styleSpec:t.v},a)))}_remove(e=!0){this._request&&
(this._request.cancel(),this._request=null),this._spriteRequest&&
(this._spriteRequest.cancel(),this._spriteRequest=null),t.aJ.off("pluginStateCha
nge",this._rtlTextPluginCallback);for(const t in
this._layers)this._layers[t].setEventedParent(null);for(const t in
this.sourceCaches){const
e=this.sourceCaches[t];e.setEventedParent(null),e.onRemove(this.map);}this.image
Manager.setEventedParent(null),this.setEventedParent(null),this.dispatcher.remov
e(e);}_clearSource(t){this.sourceCaches[t].clearTiles();}_reloadSource(t)
{this.sourceCaches[t].resume(),this.sourceCaches[t].reload();}_updateSources(t)
{for(const e in
this.sourceCaches)this.sourceCaches[e].update(t,this.map.terrain);}_generateColl
isionBoxes(){for(const t in
this.sourceCaches)this._reloadSource(t);}_updatePlacement(e,i,s,a,o=!1){let
r=!1,n=!1;const l={};for(const t of this._order){const
i=this._layers[t];if("symbol"!==i.type)continue;if(!l[i.source]){const
t=this.sourceCaches[i.source];l[i.source]=t.getRenderableIds(!0).map((e=>t.getTi
leByID(e))).sort(((t,e)>=>e.tileID.overscaledZ-t.tileID.overscaledZ||
(t.tileID.isLessThan(e.tileID)?-1:1)));}const
s=this.crossTileSymbolIndex.addLayer(i,l[i.source],e.center.lng);r=r||s;if(this
.crossTileSymbolIndex.pruneUnusedLayers(this._order),
((o=o||this._layerOrderChanged||0===s)||!this.pauseablePlacement||this.pauseable
Placement.isDone())&&!this.placement.stillRecent(t.h.now(),e.zoom))&&
(this.pauseablePlacement=new
Wt(e,this.map.terrain,this._order,o,i,s,a,this.placement),this._layerOrderChange
d=!1),this.pauseablePlacement.isDone()?this.placement.setStale():
(this.pauseablePlacement.continuePlacement(this._order,this._layers,l),this.paus
eablePlacement.isDone())&&
(this.placement=this.pauseablePlacement.commit(t.h.now()),n=!0),r&&this.pauseabl
ePlacement.placement.setStale()),n||r)for(const t of this._order){const
e=this._layers[t];"symbol"===e.type&&this.placement.updateLayerOpacities(e,l[e.s
ource]);}return
!this.pauseablePlacement.isDone()||this.placement.hasTransitions(t.h.now())}_rel
easeSymbolFadeTiles(){for(const t in
this.sourceCaches)this.sourceCaches[t].releaseSymbolFadeTiles();}getImages(t,e,i
){
this.imageManager.getImages(e.icons,i),this._updateTilesForChangedImages();cons
t
s=this.sourceCaches[e.source];s&&s.setDependencies(e.tileID.key,e.type,e.icons);
}getGlyphs(t,e,i){this.glyphManager.getGlyphs(e.stacks,i);const
s=this.sourceCaches[e.source];s&&s.setDependencies(e.tileID.key,e.type,
[""]);}getResource(e,i,s){return t.m(i,s)}getGlyphsUrl(){return
this.stylesheet.glyphs||null}setGlyphs(e,i={})
{this._checkLoaded(),e&&this._validate(t.y.glyphs,"glyphs",e,null,i)||
(this._glyphsDidChange=!0,this.stylesheet.glyphs=e,this.glyphManager.entries=
{},this.glyphManager.setURL(e));}addSprite(e,i,s={},a){this._checkLoaded();const
o=[{id:e,url:i}],r=
[...g(this.stylesheet.sprite,...o);this._validate(t.y.sprite,"sprite",r,null,s)
||this.stylesheet.sprite=r,this._loadSprite(o,!0,a)];}removeSprite(e)
{this._checkLoaded();const i=g(this.stylesheet.sprite);if(i.find((t=>t.id===e)))
{if(this._spritesImagesIds[e])for(const t of
this._spritesImagesIds[e])this.imageManager.removeImage(t),this._changedImages[t
]=!0;i.splice(i.findIndex((t=>t.id===e)),1),this.stylesheet.sprite=i.length>0?
i:void 0,delete
this._spritesImagesIds[e],this._availableImages=this.imageManager.listImages(),t
his._changed=!0,this.dispatcher.broadcast("setImages",this._availableImages),thi
s.fire(new t.k("data",{dataType:"style"}));}else this.fire(new t.j(new
Error(`Sprite "${e}" doesn't exists on this map.`)));}getSprite(){return

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g(this.stylesheet.sprite)}setSprite(e,i={},s)
{this._checkLoaded(),e&&this._validate(t.y.sprite,"sprite",e,null,i)||
(this.stylesheet.sprite=e,e?this._loadSprite(e,!0,s):
(this._unloadSprite(),s&&s(null)));}se.registerForPluginStateChange=t.aK;var
ae=t.Q([{name:"a_pos",type:"Int16",components:2}],oe="attribute vec3
a_pos3d;uniform mat4 u_matrix;uniform float u_ele_delta;varying vec2
v_texture_pos;varying float v_depth;void main() {float extent=8192.0;float
ele_delta=a_pos3d.z==1.0 ? u_ele_delta :
0.0;v_texture_pos=a_pos3d.xy/extent;gl_Position=u_matrix*vec4(a_pos3d.xy,get_ele
vation(a_pos3d.xy)-ele_delta,1.0);v_depth=gl_Position.z/gl_Position.w;}";const
re={prelude:ne("#ifdef GL_ES\nprecision mediump float;\n#else\n#if
!defined(lowp)\n#define lowp\n#endif\n#if !defined(mediump)\n#define
mediump\n#endif\n#if !defined(highp)\n#define highp\n#endif\n#endif\n",
"#ifdef GL_ES\nprecision highp float;\n#else\n#if !defined(lowp)\n#define
lowp\n#endif\n#if !defined(mediump)\n#define mediump\n#endif\n#if
!defined(highp)\n#define highp\n#endif\n#endif\nvec2 unpack_float(const float
packedValue) {int packedIntValue=int(packedValue);int
v0=packedIntValue/256;return vec2(v0,packedIntValue-v0*256);}vec2
unpack_opacity(const float packedOpacity) {int
intOpacity=int(packedOpacity)/2;return
vec2(float(intOpacity)/127.0,mod(packedOpacity,2.0));}vec4 decode_color(const
vec2 encodedColor) {return
vec4(unpack_float(encodedColor[0])/255.0,unpack_float(encodedColor[1])/255.0\n);
}float unpack_mix_vec2(const vec2 packedValue,const float t) {return
mix(packedValue[0],packedValue[1],t);}vec4 unpack_mix_color(const vec4
packedColors,const float t) {vec4
minColor=decode_color(vec2(packedColors[0],packedColors[1]));vec4
maxColor=decode_color(vec2(packedColors[2],packedColors[3]));return
mix(minColor,maxColor,t);}vec2 get_pattern_pos(const vec2
pixel_coord_upper,const vec2 pixel_coord_lower,const vec2 pattern_size,const
float tile_units_to_pixels,const vec2 pos) {vec2
offset=mod(mod(mod(pixel_coord_upper,pattern_size)*256.0,pattern_size)*256.0+pix
el_coord_lower,pattern_size);return
(tile_units_to_pixels*pos+offset)/pattern_size;}#ifdef TERRAIN3D\nuniform
sampler2D u_terrain;uniform float u_terrain_dim;uniform mat4
u_terrain_matrix;uniform vec4 u_terrain_unpack;uniform float
u_terrain_exaggeration;uniform highp sampler2D u_depth;\n#endif\nconst highp
vec4 bitSh=vec4(256.*256.*256.,256.*256.,256.,1.);const highp vec4
bitShifts=vec4(1.)/bitSh;highp float unpack(highp vec4 color) {return
dot(color,bitShifts);}highp float depthOpacity(vec3 frag) {\n#ifdef
TERRAIN3D\nhighp float d=unpack(texture2D(u_depth,frag.xy*0.5+0.5))+0.0001-
frag.z;return 1.0-max(0.0,min(1.0,-d*500.0));\n#else\nreturn
1.0;\n#endif\n}float calculate_visibility(vec4 pos) {\n#ifdef TERRAIN3D\nvec3
frag=pos.xyz/pos.w;highp float d=depthOpacity(frag);if (d > 0.95) return
1.0;return (d+depthOpacity(frag+vec3(0.0,0.01,0.0)))/2.0;\n#else\nreturn
1.0;\n#endif\n}float ele(vec2 pos) {\n#ifdef TERRAIN3D\nvec4 rgb=
(texture2D(u_terrain,pos)*255.0)*u_terrain_unpack;return rgb.r+rgb.g+rgb.b-
u_terrain_unpack.a;\n#else\nreturn 0.0;\n#endif\n}float get_elevation(vec2 pos)
{\n#ifdef TERRAIN3D\nvec2 coord=
(u_terrain_matrix*vec4(pos,0.0,1.0)).xy*u_terrain_dim+1.0;vec2
f=fract(coord);vec2 c=(floor(coord)+0.5)/(u_terrain_dim+2.0);float
d=1.0/(u_terrain_dim+2.0);float tl=ele(c);float tr=ele(c+vec2(d,0.0));float
bl=ele(c+vec2(0.0,d));float br=ele(c+vec2(d,d));float
elevation=mix(mix(tl,tr,f.x),mix(bl,br,f.x),f.y);return
elevation*u_terrain_exaggeration;\n#else\nreturn
0.0;\n#endif\n}"),background:ne("uniform vec4 u_color;uniform float
u_opacity;void main() {gl_FragColor=u_color*u_opacity;\n#ifdef
OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n}","attribute vec2
a_pos;uniform mat4 u_matrix;void main()
{gl_Position=u_matrix*vec4(a_pos,0,1);}"),backgroundPattern:ne("uniform vec2
u_pattern_tl_a;uniform vec2 u_pattern_br_a;uniform vec2 u_pattern_tl_b;uniform
vec2 u_pattern_br_b;uniform vec2 u_texsize;uniform float u_mix;uniform float

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u_opacity;uniform sampler2D u_image;varying vec2 v_pos_a;varying vec2
v_pos_b;void main() {vec2 imagecoord=mod(v_pos_a,1.0);vec2
pos=mix(u_pattern_tl_a/u_texsize,u_pattern_br_a/u_texsize,imagecoord);vec4
color1=texture2D(u_image,pos);vec2 imagecoord_b=mod(v_pos_b,1.0);vec2
pos2=mix(u_pattern_tl_b/u_texsize,u_pattern_br_b/u_texsize,imagecoord_b);vec4
color2=texture2D(u_image,pos2);gl_FragColor=mix(color1,color2,u_mix)*u_opacity;\
n#ifdef OVERDRAW_INSPECTOR\negl_FragColor=vec4(1.0);\n#endif\n"},"uniform mat4
u_matrix;uniform vec2 u_pattern_size_a;uniform vec2 u_pattern_size_b;uniform
vec2 u_pixel_coord_upper;uniform vec2 u_pixel_coord_lower;uniform float
u_scale_a;uniform float u_scale_b;uniform float u_tile_units_to_pixels;attribute
vec2 a_pos;varying vec2 v_pos_a;varying vec2 v_pos_b;void main()
{gl_Position=u_matrix*vec4(a_pos,0,1);v_pos_a=get_pattern_pos(u_pixel_coord_upper,u_pixel_coord_lower,u_scale_a*u_pattern_size_a,u_tile_units_to_pixels,a_pos);v
_pos_b=get_pattern_pos(u_pixel_coord_upper,u_pixel_coord_lower,u_scale_b*u_pattern_size_b,u_tile_units_to_pixels,a_pos);})",circle:ne("varying vec3
v_data;varying float v_visibility;\n#pragma mapbox: define highp vec4
color\n#pragma mapbox: define mediump float radius\n#pragma mapbox: define lowp
float blur\n#pragma mapbox: define lowp float opacity\n#pragma mapbox: define
highp vec4 stroke_color\n#pragma mapbox: define mediump float
stroke_width\n#pragma mapbox: define lowp float stroke_opacity\nvoid main()
{\n#pragma mapbox: initialize highp vec4 color\n#pragma mapbox: initialize
mediump float radius\n#pragma mapbox: initialize lowp float blur\n#pragma
mapbox: initialize lowp float opacity\n#pragma mapbox: initialize highp vec4
stroke_color\n#pragma mapbox: initialize mediump float stroke_width\n#pragma
mapbox: initialize lowp float stroke_opacity\nvec2 extrude=v_data.xy;float
extrude_length=length(extrude);lowp float antialiasblur=v_data.z;float
antialiased_blur=-max(blur,antialiasblur);float
opacity_t=smoothstep(0.0,antialiased_blur,extrude_length-1.0);float
color_t=stroke_width < 0.01 ? 0.0 :
smoothstep(antialiased_blur,0.0,extrude_length-
radius/(radius+stroke_width));gl_FragColor=v_visibility*opacity_t*mix(color*opacity,stroke_color*stroke_opacity,color_t);\n#ifdef
OVERDRAW_INSPECTOR\negl_FragColor=vec4(1.0);\n#endif\n"},"uniform mat4
u_matrix;uniform bool u_scale_with_map;uniform bool u_pitch_with_map;uniform
vec2 u_extrude_scale;uniform lowp float u_device_pixel_ratio;uniform highp float
u_camera_to_center_distance;attribute vec2 a_pos;varying vec3 v_data;varying
float v_visibility;\n#pragma mapbox: define highp vec4 color\n#pragma mapbox:
define mediump float radius\n#pragma mapbox: define lowp float blur\n#pragma
mapbox: define lowp float opacity\n#pragma mapbox: define highp vec4
stroke_color\n#pragma mapbox: define mediump float stroke_width\n#pragma mapbox:
define lowp float stroke_opacity\nvoid main(void) {\n#pragma mapbox: initialize
highp vec4 color\n#pragma mapbox: initialize mediump float radius\n#pragma
mapbox: initialize lowp float blur\n#pragma mapbox: initialize lowp float
opacity\n#pragma mapbox: initialize highp vec4 stroke_color\n#pragma mapbox:
initialize mediump float stroke_width\n#pragma mapbox: initialize lowp float
stroke_opacity\nvec2 extrude=vec2(mod(a_pos,2.0)*2.0-1.0);vec2
circle_center=floor(a_pos*0.5);float
ele=get_elevation(circle_center);v_visibility=calculate_visibility(u_matrix*vec4
(circle_center,ele,1.0));if (u_pitch_with_map) {vec2
corner_position=circle_center;if (u_scale_with_map) {corner_position+=extrude*
(radius+stroke_width)*u_extrude_scale;} else {vec4
projected_center=u_matrix*vec4(circle_center,0,1);corner_position+=extrude*
(radius+stroke_width)*u_extrude_scale*
(projected_center.w/u_camera_to_center_distance);}gl_Position=u_matrix*vec4(corn
er_position,ele,1);} else {gl_Position=u_matrix*vec4(circle_center,ele,1);if
(u_scale_with_map) {gl_Position.xy+=extrude*
(radius+stroke_width)*u_extrude_scale*u_camera_to_center_distance;} else
{gl_Position.xy+=extrude*
(radius+stroke_width)*u_extrude_scale*gl_Position.w;}}lowp float
antialiasblur=1.0/u_device_pixel_ratio/(radius+stroke_width);v_data=vec3(extrude
.x,extrude.y,antialiasblur);})",clippingMask:ne("void main()
{gl_FragColor=vec4(1.0);},"attribute vec2 a_pos;uniform mat4 u_matrix;void

```

```

main() {gl_Position=u_matrix*vec4(a_pos,0,1);}),heatmap:ne("uniform highp float
u_intensity;varying vec2 v_extrude;\n#pragma mapbox: define highp float
weight\n#define GAUSS_COEF 0.3989422804014327\nvoid main() {\n#pragma mapbox:
initialize highp float weight\nfloat
d=-0.5*3.0*3.0*dot(v_extrude,v_extrude);float
val=weight*u_intensity*GAUSS_COEF*exp(d);gl_FragColor=vec4(val,1.0,1.0,1.0);\n#i
fdef OVERDRAW_INSPECTOR\negl_FragColor=vec4(1.0);\n#endif\n}", "uniform mat4
u_matrix;uniform float u_extrude_scale;uniform float u_opacity;uniform float
u_intensity;attribute vec2 a_pos;varying vec2 v_extrude;\n#pragma mapbox: define
highp float weight\n#pragma mapbox: define mediump float radius\nconst highp
float ZERO=1.0/255.0/16.0;\n#define GAUSS_COEF 0.3989422804014327\nvoid
main(void) {\n#pragma mapbox: initialize highp float weight\n#pragma mapbox:
initialize mediump float radius\nvec2 unscaled_extrude=vec2(mod(a_pos,2.0)*2.0-
1.0);float
S=sqrt(-2.0*log(ZERO/weight/u_intensity/GAUSS_COEF))/3.0;v_extrude=S*unscaled_ex
trude;vec2 extrude=v_extrude*radius*u_extrude_scale;vec4
pos=vec4(floor(a_pos*0.5)+extrude,0,1);gl_Position=u_matrix*pos;}),heatmapTextu
re:ne("uniform sampler2D u_image;uniform sampler2D u_color_ramp;uniform float
u_opacity;varying vec2 v_pos;void main() {float
t=texture2D(u_image,v_pos).r;vec4
color=texture2D(u_color_ramp,vec2(t,0.5));gl_FragColor=color*u_opacity;\n#i
fdef OVERDRAW_INSPECTOR\negl_FragColor=vec4(0.0);\n#endif\n}", "uniform mat4
u_matrix;uniform vec2 u_world;attribute vec2 a_pos;varying vec2 v_pos;void
main()
{gl_Position=u_matrix*vec4(a_pos*u_world,0,1);v_pos.x=a_pos.x;v_pos.y=1.0-
a_pos.y;}),collisionBox:ne("varying float v_placed;varying float v_notUsed;void
main() {float alpha=0.5;gl_FragColor=vec4(1.0,0.0,0.0,1.0)*alpha;if (v_placed >
0.5) {gl_FragColor=vec4(0.0,0.0,1.0,0.5)*alpha;}\nif (v_notUsed > 0.5)
{gl_FragColor*=.1;}\n}", "attribute vec2 a_pos;attribute vec2
a_anchor_pos;attribute vec2 a_extrude;attribute vec2 a_placed;attribute vec2
a_shift;uniform mat4 u_matrix;uniform vec2 u_extrude_scale;uniform float
u_camera_to_center_distance;varying float v_placed;varying float v_notUsed;void
main() {vec4 projectedPoint=u_matrix*vec4(a_anchor_pos,0,1);highp float
camera_to_anchor_distance=projectedPoint.w;highp float
collision_perspective_ratio=clamp(0.5+0.5*
(u_camera_to_center_distance/camera_to_anchor_distance),0.0,4.0);gl_Position=u_m
atrix*vec4(a_pos,get_elevation(a_pos),1.0);gl_Position.xy+=
(a_extrude+a_shift)*u_extrude_scale*gl_Position.w*collision_perspective_ratio;v_
placed=a_placed.x;v_notUsed=a_placed.y;}),collisionCircle:ne("varying float
v_radius;varying vec2 v_extrude;varying float v_perspective_ratio;varying float
v_collision;void main() {float alpha=0.5*min(v_perspective_ratio,1.0);float
stroke_radius=0.9*max(v_perspective_ratio,1.0);float
distance_to_center=length(v_extrude);float
distance_to_edge=abs(distance_to_center-v_radius);float opacity_t=smoothstep(-
stroke_radius,0.0,-distance_to_edge);vec4
color=mix(vec4(0.0,0.0,1.0,0.5),vec4(1.0,0.0,0.0,1.0),v_collision);gl_FragColor=
color*alpha*opacity_t;}\n", "attribute vec2 a_pos;attribute float
a_radius;attribute vec2 a_flags;uniform mat4 u_matrix;uniform mat4
u_inv_matrix;uniform vec2 u_viewport_size;uniform float
u_camera_to_center_distance;varying float v_radius;varying vec2
v_extrude;varying float v_perspective_ratio;varying float v_collision;vec3
toTilePosition(vec2 screenPos) {vec4
rayStart=u_inv_matrix*vec4(screenPos,-1.0,1.0);vec4 rayEnd
=u_inv_matrix*vec4(screenPos, 1.0,1.0);rayStart.xyz/=rayStart.w;rayEnd.xyz
/=rayEnd.w;highp float t=(0.0-rayStart.z)/(rayEnd.z-rayStart.z);return
mix(rayStart.xyz,rayEnd.xyz,t);}\nvoid main() {vec2 quadCenterPos=a_pos;float
radius=a_radius;float collision=a_flags.x;float vertexIdx=a_flags.y;vec2
quadVertexOffset=vec2(mix(-1.0,1.0,float(vertexIdx
>=2.0)),mix(-1.0,1.0,float(vertexIdx >=1.0 && vertexIdx <=2.0)));vec2
quadVertexExtent=quadVertexOffset*radius;vec3
tilePos=toTilePosition(quadCenterPos);vec4
clipPos=u_matrix*vec4(tilePos,1.0);highp float

```



```

camera_to_anchor_distance=clipPos.w;highp float
collision_perspective_ratio=clamp(0.5+0.5*
(u_camera_to_center_distance/camera_to_anchor_distance),0.0,4.0);float
padding_factor=1.2;v_radius=radius;v_extrude=quadVertexExtent*padding_factor;v_p
erspective_ratio=collision_perspective_ratio;v_collision=collision;gl_Position=v
ec4(clipPos.xyz/clipPos.w,1.0)+vec4(quadVertexExtent*padding_factor/u_viewport_s
ize*2.0,0.0,0.0);}"),debug:ne("uniform highp vec4 u_color;uniform sampler2D
u_overlay;varying vec2 v_uv;void main() {vec4
overlay_color=texture2D(u_overlay,v_uv);gl_FragColor=mix(u_color,overlay_color,o
verlay_color.a);}","attribute vec2 a_pos;varying vec2 v_uv;uniform mat4
u_matrix;uniform float u_overlay_scale;void main()
{v_uv=a_pos/8192.0;gl_Position=u_matrix*vec4(a_pos*u_overlay_scale,get_elevation
(a_pos),1);}"),fill:ne("#pragma mapbox: define highp vec4 color\n#pragma mapbox:
define lowp float opacity\nvoid main() {\n#pragma mapbox: initialize highp vec4
color\n#pragma mapbox: initialize lowp float
opacity\nogl_FragColor=color*opacity;\n#ifdef
OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n}","attribute vec2
a_pos;uniform mat4 u_matrix;\n#pragma mapbox: define highp vec4 color\n#pragma
mapbox: define lowp float opacity\nvoid main() {\n#pragma mapbox: initialize
highp vec4 color\n#pragma mapbox: initialize lowp float
opacity\nogl_Position=u_matrix*vec4(a_pos,0,1);}"),fillOutline:ne("varying vec2
v_pos;\n#pragma mapbox: define highp vec4 outline_color\n#pragma mapbox: define
lowp float opacity\nvoid main() {\n#pragma mapbox: initialize highp vec4
outline_color\n#pragma mapbox: initialize lowp float opacity\nfloat
dist=length(v_pos-gl_FragCoord.xy);float alpha=1.0-
smoothstep(0.0,1.0,dist);gl_FragColor=outline_color*(alpha*opacity);\n#ifdef
OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n}","attribute vec2
a_pos;uniform mat4 u_matrix;uniform vec2 u_world;varying vec2 v_pos;\n#pragma
mapbox: define highp vec4 outline_color\n#pragma mapbox: define lowp float
opacity\nvoid main() {\n#pragma mapbox: initialize highp vec4
outline_color\n#pragma mapbox: initialize lowp float
opacity\nogl_Position=u_matrix*vec4(a_pos,0,1);v_pos=
(gl_Position.xy/gl_Position.w+1.0)/2.0*u_world;}"),fillOutlinePattern:ne("unifor
m vec2 u_textsize;uniform sampler2D u_image;uniform float u_fade;varying vec2
v_pos_a;varying vec2 v_pos_b;varying vec2 v_pos;\n#pragma mapbox: define lowp
float opacity\n#pragma mapbox: define lowp vec4 pattern_from\n#pragma mapbox:
define lowp vec4 pattern_to\nvoid main() {\n#pragma mapbox: initialize lowp
float opacity\n#pragma mapbox: initialize mediump vec4 pattern_from\n#pragma
mapbox: initialize mediump vec4 pattern_to\nvec2
pattern_tl_a=pattern_from.xy;vec2 pattern_br_a=pattern_from.zw;vec2
pattern_tl_b=pattern_to.xy;vec2 pattern_br_b=pattern_to.zw;vec2
imagecoord=mod(v_pos_a,1.0);vec2
pos=mix(pattern_tl_a/u_textsize,pattern_br_a/u_textsize,imagecoord);vec4
color1=texture2D(u_image,pos);vec2 imagecoord_b=mod(v_pos_b,1.0);vec2
pos2=mix(pattern_tl_b/u_textsize,pattern_br_b/u_textsize,imagecoord_b);vec4
color2=texture2D(u_image,pos2);float dist=length(v_pos-gl_FragCoord.xy);float
alpha=1.0-
smoothstep(0.0,1.0,dist);gl_FragColor=mix(color1,color2,u_fade)*alpha*opacity;\n
#ifdef OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n}","uniform mat4
u_matrix;uniform vec2 u_world;uniform vec2 u_pixel_coord_upper;uniform vec2
u_pixel_coord_lower;uniform vec3 u_scale;attribute vec2 a_pos;varying vec2
v_pos_a;varying vec2 v_pos_b;varying vec2 v_pos;\n#pragma mapbox: define lowp
float opacity\n#pragma mapbox: define lowp vec4 pattern_from\n#pragma mapbox:
define lowp vec4 pattern_to\n#pragma mapbox: define lowp float
pixel_ratio_from\n#pragma mapbox: define lowp float pixel_ratio_to\nvoid main()
{\n#pragma mapbox: initialize lowp float opacity\n#pragma mapbox: initialize
mediump vec4 pattern_from\n#pragma mapbox: initialize mediump vec4
pattern_to\n#pragma mapbox: initialize lowp float pixel_ratio_from\n#pragma
mapbox: initialize lowp float pixel_ratio_to\nvec2
pattern_tl_a=pattern_from.xy;vec2 pattern_br_a=pattern_from.zw;vec2
pattern_tl_b=pattern_to.xy;vec2 pattern_br_b=pattern_to.zw;float
tileRatio=u_scale.x;float fromScale=u_scale.y;float

```

```

toScale=u_scale.z;gl_Position=u_matrix*vec4(a_pos,0,1);vec2 display_size_a=
(pattern_br_a-pattern_tl_a)/pixel_ratio_from;vec2 display_size_b=(pattern_br_b-
pattern_tl_b)/pixel_ratio_to;v_pos_a=get_pattern_pos(u_pixel_coord_upper,u_pixel
_coord_lower,fromScale*display_size_a,tileRatio,a_pos);v_pos_b=get_pattern_pos(u
_pixel_coord_upper,u_pixel_coord_lower,toScale*display_size_b,tileRatio,a_pos);v
_pos=(gl_Position.xy/gl_Position.w+1.0)/2.0*u_world;})",fillPattern:ne("#ifdef
GL_ES\nprecision highp float;\n#endif\nuniform vec2 u_textsize;uniform float
u_fade;uniform sampler2D u_image;varying vec2 v_pos_a;varying vec2
v_pos_b;\n#pragma mapbox: define lowp float opacity\n#pragma mapbox: define lowp
vec4 pattern_from\n#pragma mapbox: define lowp vec4 pattern_to\nvoid main()
{\n#pragma mapbox: initialize lowp float opacity\n#pragma mapbox: initialize
mediump vec4 pattern_from\n#pragma mapbox: initialize mediump vec4
pattern_to\nvec2 pattern_tl_a=pattern_from.xy;vec2
pattern_br_a=pattern_from.zw;vec2 pattern_tl_b=pattern_to.xy;vec2
pattern_br_b=pattern_to.zw;vec2 imagecoord=mod(v_pos_a,1.0);vec2
pos=mix(pattern_tl_a/u_textsize,pattern_br_a/u_textsize,imagecoord);vec4
color1=texture2D(u_image,pos);vec2 imagecoord_b=mod(v_pos_b,1.0);vec2
pos2=mix(pattern_tl_b/u_textsize,pattern_br_b/u_textsize,imagecoord_b);vec4
color2=texture2D(u_image,pos2);gl_FragColor=mix(color1,color2,u_fade)*opacity;\n
#ifdef OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n}", "uniform mat4
u_matrix;uniform vec2 u_pixel_coord_upper;uniform vec2
u_pixel_coord_lower;uniform vec3 u_scale;attribute vec2 a_pos;varying vec2
v_pos_a;varying vec2 v_pos_b;\n#pragma mapbox: define lowp float
opacity\n#pragma mapbox: define lowp vec4 pattern_from\n#pragma mapbox: define
lowp vec4 pattern_to\n#pragma mapbox: define lowp float
pixel_ratio_from\n#pragma mapbox: define lowp float pixel_ratio_to\nvoid main()
{\n#pragma mapbox: initialize lowp float opacity\n#pragma mapbox: initialize
mediump vec4 pattern_from\n#pragma mapbox: initialize mediump vec4
pattern_to\n#pragma mapbox: initialize lowp float pixel_ratio_from\n#pragma
mapbox: initialize lowp float pixel_ratio_to\nvec2
pattern_tl_a=pattern_from.xy;vec2 pattern_br_a=pattern_from.zw;vec2
pattern_tl_b=pattern_to.xy;vec2 pattern_br_b=pattern_to.zw;float
tileZoomRatio=u_scale.x;float fromScale=u_scale.y;float toScale=u_scale.z;vec2
display_size_a=(pattern_br_a-pattern_tl_a)/pixel_ratio_from;vec2 display_size_b=
(pattern_br_b-
pattern_tl_b)/pixel_ratio_to;gl_Position=u_matrix*vec4(a_pos,0,1);v_pos_a=get_pa
ttern_pos(u_pixel_coord_upper,u_pixel_coord_lower,fromScale*display_size_a,tileZ
oomRatio,a_pos);v_pos_b=get_pattern_pos(u_pixel_coord_upper,u_pixel_coord_lower,
toScale*display_size_b,tileZoomRatio,a_pos);})",fillExtrusion:ne("varying vec4
v_color;void main() {gl_FragColor=v_color;\n#ifdef
OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n}", "uniform mat4
u_matrix;uniform vec3 u_lightcolor;uniform lowp vec3 u_lightpos;uniform lowp
float u_lightintensity;uniform float u_vertical_gradient;uniform lowp float
u_opacity;attribute vec2 a_pos;attribute vec4 a_normal_ed;\n#ifdef
TERRAIN3D\nattribute vec2 a_centroid;\n#endif\nvarying vec4 v_color;\n#pragma
mapbox: define highp float base\n#pragma mapbox: define highp float
height\n#pragma mapbox: define highp vec4 color\nvoid main() {\n#pragma mapbox:
initialize highp float base\n#pragma mapbox: initialize highp float
height\n#pragma mapbox: initialize highp vec4 color\nvec3
normal=a_normal_ed.xyz;\n#ifdef TERRAIN3D\nfloat
height_terrain3d_offset=get_elevation(a_centroid);float
base_terrain3d_offset=height_terrain3d_offset-(base > 0.0 ? 0.0 :
10.0);\n#else\nfloat height_terrain3d_offset=0.0;float
base_terrain3d_offset=0.0;\n#endif\nbase=max(0.0,base)+base_terrain3d_offset;hei
ght=max(0.0,height)+height_terrain3d_offset;float
t=mod(normal.x,2.0);gl_Position=u_matrix*vec4(a_pos,t > 0.0 ? height :
base,1);float
colorvalue=color.r*0.2126+color.g*0.7152+color.b*0.0722;v_color=vec4(0.0,0.0,0.0
,1.0);vec4 ambientlight=vec4(0.03,0.03,0.03,1.0);color+=ambientlight;float
directional=clamp(dot(normal/16384.0,u_lightpos),0.0,1.0);directional=mix((1.0-
u_lightintensity),max((1.0-colorvalue+u_lightintensity),1.0),directional);if
(normal.y !=0.0) {directional*=((1.0-u_vertical_gradient)+

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```

(u_vertical_gradient*clamp((t+base)*pow(height/150.0,0.5),mix(0.7,0.98,1.0-
u_lightintensity),1.0)));v_color.r+=clamp(color.r*directional*u_lightcolor.r,mix(0.0,0.3,1.0-
u_lightcolor.r),1.0);v_color.g+=clamp(color.g*directional*u_lightcolor.g,mix(0.0
,0.3,1.0-
u_lightcolor.g),1.0);v_color.b+=clamp(color.b*directional*u_lightcolor.b,mix(0.0
,0.3,1.0-
u_lightcolor.b),1.0);v_color*=u_opacity;})",fillExtrusionPattern:ne("uniform
vec2 u_texsize;uniform float u_fade;uniform sampler2D u_image;varying vec2
v_pos_a;varying vec2 v_pos_b;varying vec4 v_lighting;\n#pragma mapbox: define
lowp float base\n#pragma mapbox: define lowp float height\n#pragma mapbox:
define lowp vec4 pattern_from\n#pragma mapbox: define lowp vec4
pattern_to\n#pragma mapbox: define lowp float pixel_ratio_from\n#pragma mapbox:
define lowp float pixel_ratio_to\n\nvoid main() {\n#pragma mapbox: initialize lowp
float base\n#pragma mapbox: initialize lowp float height\n#pragma mapbox:
initialize mediump vec4 pattern_from\n#pragma mapbox: initialize mediump vec4
pattern_to\n#pragma mapbox: initialize lowp float pixel_ratio_from\n#pragma
mapbox: initialize lowp float pixel_ratio_to\n\nvec2
pattern_tl_a=pattern_from.xy;vec2 pattern_br_a=pattern_from.zw;vec2
pattern_tl_b=pattern_to.xy;vec2 pattern_br_b=pattern_to.zw;vec2
imagecoord=mod(v_pos_a,1.0);vec2
pos=mix(pattern_tl_a/u_texsize,pattern_br_a/u_texsize,imagecoord);vec4
color1=texture2D(u_image,pos);vec2 imagecoord_b=mod(v_pos_b,1.0);vec2
pos2=mix(pattern_tl_b/u_texsize,pattern_br_b/u_texsize,imagecoord_b);vec4
color2=texture2D(u_image,pos2);vec4
mixedColor=mix(color1,color2,u_fade);gl_FragColor=mixedColor*v_lighting;\n#ifdef
OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n}", "uniform mat4
u_matrix;uniform vec2 u_pixel_coord_upper;uniform vec2
u_pixel_coord_lower;uniform float u_height_factor;uniform vec3 u_scale;uniform
float u_vertical_gradient;uniform lowp float u_opacity;uniform vec3
u_lightcolor;uniform lowp vec3 u_lightpos;uniform lowp float
u_lightintensity;attribute vec2 a_pos;attribute vec4 a_normal_ed;\n#ifdef
TERRAIN3D\nattribute vec2 a_centroid;\n#endif\n\nvarying vec2 v_pos_a;varying vec2
v_pos_b;varying vec4 v_lighting;\n#pragma mapbox: define lowp float
base\n#pragma mapbox: define lowp float height\n#pragma mapbox: define lowp vec4
pattern_from\n#pragma mapbox: define lowp vec4 pattern_to\n#pragma mapbox:
define lowp float pixel_ratio_from\n#pragma mapbox: define lowp float
pixel_ratio_to\n\nvoid main() {\n#pragma mapbox: initialize lowp float
base\n#pragma mapbox: initialize lowp float height\n#pragma mapbox: initialize
mediump vec4 pattern_from\n#pragma mapbox: initialize mediump vec4
pattern_to\n#pragma mapbox: initialize lowp float pixel_ratio_from\n#pragma
mapbox: initialize lowp float pixel_ratio_to\n\nvec2
pattern_tl_a=pattern_from.xy;vec2 pattern_br_a=pattern_from.zw;vec2
pattern_tl_b=pattern_to.xy;vec2 pattern_br_b=pattern_to.zw;float
tileRatio=u_scale.x;float fromScale=u_scale.y;float toScale=u_scale.z;vec3
normal=a_normal_ed.xyz;float edgedistance=a_normal_ed.w;vec2 display_size_a=
(pattern_br_a-pattern_tl_a)/pixel_ratio_from;vec2 display_size_b=(pattern_br_b-
pattern_tl_b)/pixel_ratio_to;\n#ifdef TERRAIN3D\nfloat
height_terrain3d_offset=get_elevation(a_centroid);float
base_terrain3d_offset=height_terrain3d_offset-(base > 0.0 ? 0.0 :
10.0);\n#else\nfloat height_terrain3d_offset=0.0;float
base_terrain3d_offset=0.0;\n#endif\n\nbase=max(0.0,base)+base_terrain3d_offset;hei
ght=max(0.0,height)+height_terrain3d_offset;float t=mod(normal.x,2.0);float z=t
> 0.0 ? height : base;gl_Position=u_matrix*vec4(a_pos,z,1);vec2
pos=normal.x==1.0 && normal.y==0.0 && normal.z==16384.0\n? a_pos\n:
vec2(edgedistance,z*u_height_factor);v_pos_a=get_pattern_pos(u_pixel_coord_upper
,u_pixel_coord_lower,fromScale*display_size_a,tileRatio,pos);v_pos_b=get_pattern
_pos(u_pixel_coord_upper,u_pixel_coord_lower,toScale*display_size_b,tileRatio,po
s);v_lighting=vec4(0.0,0.0,0.0,1.0);float
directional=clamp(dot(normal/16383.0,u_lightpos),0.0,1.0);directional=mix((1.0-
u_lightintensity),max((0.5+u_lightintensity),1.0),directional);if (normal.y
!=0.0) {directional*=((1.0-u_vertical_gradient)+

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(u_vertical_gradient*clamp((t+base)*pow(height/150.0,0.5),mix(0.7,0.98,1.0-
u_lightintensity),1.0)));}v_lighting.rgb+=clamp(directional*u_lightcolor,mix(vec
3(0.0),vec3(0.3),1.0-
u_lightcolor),vec3(1.0));v_lighting*=u_opacity;})",hillshadePrepare:ne("#ifdef
GL_ES\nprecision highp float;\n#endif\nuniform sampler2D u_image;varying vec2
v_pos;uniform vec2 u_dimension;uniform float u_zoom;uniform vec4 u_unpack;float
getElevation(vec2 coord,float bias) {vec4
data=texture2D(u_image,coord)*255.0;data.a=-1.0;return
dot(data,u_unpack)/4.0;}void main() {vec2 epsilon=1.0/u_dimension;float
a=getElevation(v_pos+vec2(-epsilon.x,-epsilon.y),0.0);float
b=getElevation(v_pos+vec2(0,-epsilon.y),0.0);float
c=getElevation(v_pos+vec2(epsilon.x,-epsilon.y),0.0);float
d=getElevation(v_pos+vec2(-epsilon.x,0),0.0);float
e=getElevation(v_pos,0.0);float
f=getElevation(v_pos+vec2(epsilon.x,0),0.0);float g=getElevation(v_pos+vec2(-
epsilon.x,epsilon.y),0.0);float
h=getElevation(v_pos+vec2(0,epsilon.y),0.0);float
i=getElevation(v_pos+vec2(epsilon.x,epsilon.y),0.0);float
exaggerationFactor=u_zoom < 2.0 ? 0.4 : u_zoom < 4.5 ? 0.35 : 0.3;float
exaggeration=u_zoom < 15.0 ? (u_zoom-15.0)*exaggerationFactor : 0.0;vec2
deriv=vec2((c+f+f+i)-(a+d+d+g),(g+h+h+i)-(a+b+b+c))/pow(2.0,exaggeration+
(19.2562-
u_zoom));gl_FragColor=clamp(vec4(deriv.x/2.0+0.5,deriv.y/2.0+0.5,1.0,1.0),0.0,1.
0);\n#ifdef OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n", "uniform
mat4 u_matrix;uniform vec2 u_dimension;attribute vec2 a_pos;attribute vec2
a_texture_pos;varying vec2 v_pos;void main()
{gl_Position=u_matrix*vec4(a_pos,0,1);highp vec2 epsilon=1.0/u_dimension;float
scale=(u_dimension.x-2.0)/u_dimension.x;v_pos=
(a_texture_pos/8192.0)*scale+epsilon;}",hillshade:ne("uniform sampler2D
u_image;varying vec2 v_pos;uniform vec2 u_latrange;uniform vec2 u_light;uniform
vec4 u_shadow;uniform vec4 u_highlight;uniform vec4 u_accent;\n#define PI
3.141592653589793\nvoid main() {vec4 pixel=texture2D(u_image,v_pos);vec2 deriv=
((pixel.rg*2.0)-1.0);float scaleFactor=cos(radians((u_latrange[0]-
u_latrange[1])*(1.0-v_pos.y)+u_latrange[1]));float
slope=atan(1.25*length(deriv)/scaleFactor);float aspect=deriv.x !=0.0 ?
atan(deriv.y,-deriv.x) : PI/2.0*(deriv.y > 0.0 ? 1.0 :-1.0);float
intensity=u_light.x;float azimuth=u_light.y+PI;float base=1.875-
intensity*1.75;float maxValue=0.5*PI;float scaledSlope=intensity !=0.5 ?
((pow(base,slope)-1.0)/(pow(base,maxValue)-1.0))*maxValue : slope;float
accent=cos(scaledSlope);vec4 accent_color=(1.0-
accent)*u_accent*clamp(intensity*2.0,0.0,1.0);float
shade=abs(mod((aspect+azimuth)/PI+0.5,2.0)-1.0);vec4
shade_color=mix(u_shadow,u_highlight,shade)*sin(scaledSlope)*clamp(intensity*2.0
,0.0,1.0);gl_FragColor=accent_color*(1.0-shade_color.a)+shade_color;\n#ifdef
OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n", "uniform mat4
u_matrix;attribute vec2 a_pos;attribute vec2 a_texture_pos;varying vec2
v_pos;void main()
{gl_Position=u_matrix*vec4(a_pos,0,1);v_pos=a_texture_pos/8192.0;}",line:ne("un
iform lowp float u_device_pixel_ratio;varying vec2 v_width2;varying vec2
v_normal;varying float v_gamma_scale;\n#pragma mapbox: define highp vec4
color\n#pragma mapbox: define lowp float blur\n#pragma mapbox: define lowp float
opacity\nvoid main() {\n#pragma mapbox: initialize highp vec4 color\n#pragma
mapbox: initialize lowp float blur\n#pragma mapbox: initialize lowp float
opacity\nfloat dist=length(v_normal)*v_width2.s;float blur2=
(blur+1.0/u_device_pixel_ratio)*v_gamma_scale;float alpha=clamp(min(dist-
(v_width2.t-blur2),v_width2.s-dist)/blur2,0.0,1.0);gl_FragColor=color*
(alpha*opacity);\n#ifdef
OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n", "\n#define scale
0.015873016\nattribute vec2 a_pos_normal;attribute vec4 a_data;uniform mat4
u_matrix;uniform mediump float u_ratio;uniform vec2 u_units_to_pixels;uniform
lowp float u_device_pixel_ratio;varying vec2 v_normal;varying vec2
v_width2;varying float v_gamma_scale;varying highp float v_linesofar;\n#pragma

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mapbox: define highp vec4 color\n#pragma mapbox: define lowp float blur\n#pragma
mapbox: define lowp float opacity\n#pragma mapbox: define mediump float
gapwidth\n#pragma mapbox: define lowp float offset\n#pragma mapbox: define
mediump float width\nvoid main() {\n#pragma mapbox: initialize highp vec4
color\n#pragma mapbox: initialize lowp float blur\n#pragma mapbox: initialize
lowp float opacity\n#pragma mapbox: initialize mediump float gapwidth\n#pragma
mapbox: initialize lowp float offset\n#pragma mapbox: initialize mediump float
width\nfloat ANTIALIASING=1.0/u_device_pixel_ratio/2.0;vec2 a_extrude=a_data.xy-
128.0;float a_direction=mod(a_data.z,4.0)-1.0;v_linesofar=
(floor(a_data.z/4.0)+a_data.w*64.0)*2.0;vec2 pos=floor(a_pos_normal*0.5);mediump
vec2 normal=a_pos_normal-2.0*pos;normal.y=normal.y*2.0-
1.0;v_normal=normal;gapwidth=gapwidth/2.0;float
halfwidth=width/2.0;offset=-1.0*offset;float inset=gapwidth+(gapwidth > 0.0 ?
ANTIALIASING : 0.0);float outset=gapwidth+halfwidth*(gapwidth > 0.0 ? 2.0 :
1.0)+(halfwidth==0.0 ? 0.0 : ANTIALIASING);mediump vec2
dist=outset*a_extrude*scale;mediump float u=0.5*a_direction;mediump float t=1.0-
abs(u);mediump vec2 offset2=offset*a_extrude*scale*normal.y*mat2(t,-u,u,t);vec4
projected_extrude=u_matrix*vec4(dist/u_ratio,0.0,0.0);gl_Position=u_matrix*vec4(
pos+offset2/u_ratio,0.0,1.0)+projected_extrude;\n#ifdef
TERRAIN3D\nv_gamma_scale=1.0;\n#else\nfloat
extrude_length_without_perspective=length(dist);float
extrude_length_with_perspective=length(projected_extrude.xy/gl_Position.w*u_unit
s_to_pixels);v_gamma_scale=extrude_length_without_perspective/extrude_length_wit
h_perspective;\n#endif\nv_width2=vec2(outset,inset);}"),lineGradient:ne("uniform
lowp float u_device_pixel_ratio;uniform sampler2D u_image;varying vec2
v_width2;varying vec2 v_normal;varying float v_gamma_scale;varying highp vec2
v_uv;\n#pragma mapbox: define lowp float blur\n#pragma mapbox: define lowp float
opacity\nvoid main() {\n#pragma mapbox: initialize lowp float blur\n#pragma
mapbox: initialize lowp float opacity\nfloat
dist=length(v_normal)*v_width2.s;float blur2=
(blur+1.0/u_device_pixel_ratio)*v_gamma_scale;float alpha=clamp(min(dist-
(v_width2.t-blur2),v_width2.s-dist)/blur2,0.0,1.0);vec4
color=texture2D(u_image,v_uv);gl_FragColor=color*(alpha*opacity);\n#ifdef
OVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n}", "\n#define scale
0.015873016\nattribute vec2 a_pos_normal;attribute vec4 a_data;attribute float
a_uv_x;attribute float a_split_index;uniform mat4 u_matrix;uniform mediump float
u_ratio;uniform lowp float u_device_pixel_ratio;uniform vec2
u_units_to_pixels;uniform float u_image_height;varying vec2 v_normal;varying
vec2 v_width2;varying float v_gamma_scale;varying highp vec2 v_uv;\n#pragma
mapbox: define lowp float blur\n#pragma mapbox: define lowp float
opacity\n#pragma mapbox: define mediump float gapwidth\n#pragma mapbox: define
lowp float offset\n#pragma mapbox: define mediump float width\nvoid main()
{\n#pragma mapbox: initialize lowp float blur\n#pragma mapbox: initialize lowp
float opacity\n#pragma mapbox: initialize mediump float gapwidth\n#pragma
mapbox: initialize lowp float offset\n#pragma mapbox: initialize mediump float
width\nfloat ANTIALIASING=1.0/u_device_pixel_ratio/2.0;vec2 a_extrude=a_data.xy-
128.0;float a_direction=mod(a_data.z,4.0)-1.0;highp float
texel_height=1.0/u_image_height;highp float
half_texel_height=0.5*texel_height;v_uv=vec2(a_uv_x,a_split_index*texel_height-
half_texel_height);vec2 pos=floor(a_pos_normal*0.5);mediump vec2
normal=a_pos_normal-2.0*pos;normal.y=normal.y*2.0-
1.0;v_normal=normal;gapwidth=gapwidth/2.0;float
halfwidth=width/2.0;offset=-1.0*offset;float inset=gapwidth+(gapwidth > 0.0 ?
ANTIALIASING : 0.0);float outset=gapwidth+halfwidth*(gapwidth > 0.0 ? 2.0 :
1.0)+(halfwidth==0.0 ? 0.0 : ANTIALIASING);mediump vec2
dist=outset*a_extrude*scale;mediump float u=0.5*a_direction;mediump float t=1.0-
abs(u);mediump vec2 offset2=offset*a_extrude*scale*normal.y*mat2(t,-u,u,t);vec4
projected_extrude=u_matrix*vec4(dist/u_ratio,0.0,0.0);gl_Position=u_matrix*vec4(
pos+offset2/u_ratio,0.0,1.0)+projected_extrude;\n#ifdef
TERRAIN3D\nv_gamma_scale=1.0;\n#else\nfloat
extrude_length_without_perspective=length(dist);float
extrude_length_with_perspective=length(projected_extrude.xy/gl_Position.w*u_unit

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s_to_pixels);v_gamma_scale=extrude_length_without_perspective/extrude_length_with_perspective;\n#endif\nv_width2=vec2(outset,inset);}"),linePattern:ne("#ifdef GL_ES\nprecision highp float;\n#endif\nuniform lowp float u_device_pixel_ratio;uniform vec2 u_texsize;uniform float u_fade;uniform mediump vec3 u_scale;uniform sampler2D u_image;varying vec2 v_normal;varying vec2 v_width2;varying float v_linesofar;varying float v_gamma_scale;varying float v_width;\n#pragma mapbox: define lowp vec4 pattern_from\n#pragma mapbox: define lowp vec4 pattern_to\n#pragma mapbox: define lowp float pixel_ratio_from\n#pragma mapbox: define lowp float pixel_ratio_to\n#pragma mapbox: define lowp float blur\n#pragma mapbox: define lowp float opacity\nvoid main() {\n#pragma mapbox: initialize mediump vec4 pattern_from\n#pragma mapbox: initialize mediump vec4 pattern_to\n#pragma mapbox: initialize lowp float pixel_ratio_from\n#pragma mapbox: initialize lowp float pixel_ratio_to\n#pragma mapbox: initialize lowp float blur\n#pragma mapbox: initialize lowp float opacity\nvec2 pattern_tl_a=pattern_from.xy;vec2 pattern_br_a=pattern_from.zw;vec2 pattern_tl_b=pattern_to.xy;vec2 pattern_br_b=pattern_to.zw;float tileZoomRatio=u_scale.x;float fromScale=u_scale.y;float toScale=u_scale.z;vec2 display_size_a=(pattern_br_a-pattern_tl_a)/pixel_ratio_from;vec2 display_size_b=(pattern_br_b-pattern_tl_b)/pixel_ratio_to;vec2 pattern_size_a=vec2(display_size_a.x*fromScale/tileZoomRatio,display_size_a.y);vec2 pattern_size_b=vec2(display_size_b.x*toScale/tileZoomRatio,display_size_b.y);float aspect_a=display_size_a.y/v_width;float aspect_b=display_size_b.y/v_width;float dist=length(v_normal)*v_width2.s;float blur2=(blur+1.0/u_device_pixel_ratio)*v_gamma_scale;float alpha=clamp(min(dist-(v_width2.t-blur2),v_width2.s-dist)/blur2,0.0,1.0);float x_a=mod(v_linesofar/pattern_size_a.x*aspect_a,1.0);float x_b=mod(v_linesofar/pattern_size_b.x*aspect_b,1.0);float y=0.5*v_normal.y+0.5;vec2 texel_size=1.0/u_texsize;vec2 pos_a=mix(pattern_tl_a*texel_size-texel_size,pattern_br_a*texel_size+texel_size,vec2(x_a,y));vec2 pos_b=mix(pattern_tl_b*texel_size-texel_size,pattern_br_b*texel_size+texel_size,vec2(x_b,y));vec4 color=mix(texture2D(u_image,pos_a),texture2D(u_image,pos_b),u_fade);gl_FragColor=color*alpha*opacity;\n#endif\nOVERDRAW_INSPECTOR\nogl_FragColor=vec4(1.0);\n#endif\n}\n", "\n#define scale 0.015873016\n#define LINE_DISTANCE_SCALE 2.0\nattribute vec2 a_pos_normal;attribute vec4 a_data;uniform mat4 u_matrix;uniform vec2 u_units_to_pixels;uniform mediump float u_ratio;uniform lowp float u_device_pixel_ratio;varying vec2 v_normal;varying vec2 v_width2;varying float v_linesofar;varying float v_gamma_scale;varying float v_width;\n#pragma mapbox: define lowp float blur\n#pragma mapbox: define lowp float opacity\n#pragma mapbox: define lowp float offset\n#pragma mapbox: define mediump float gapwidth\n#pragma mapbox: define mediump float width\n#pragma mapbox: define lowp float floorwidth\n#pragma mapbox: define lowp vec4 pattern_from\n#pragma mapbox: define lowp vec4 pattern_to\n#pragma mapbox: define lowp float pixel_ratio_from\n#pragma mapbox: define lowp float pixel_ratio_to\nvoid main() {\n#pragma mapbox: initialize lowp float blur\n#pragma mapbox: initialize lowp float opacity\n#pragma mapbox: initialize lowp float offset\n#pragma mapbox: initialize mediump float gapwidth\n#pragma mapbox: initialize mediump float width\n#pragma mapbox: initialize lowp float floorwidth\n#pragma mapbox: initialize mediump vec4 pattern_from\n#pragma mapbox: initialize mediump vec4 pattern_to\n#pragma mapbox: initialize lowp float pixel_ratio_from\n#pragma mapbox: initialize lowp float pixel_ratio_to\nfloat ANTIALIASING=1.0/u_device_pixel_ratio/2.0;vec2 a_extrude=a_data.xy-128.0;float a_direction=mod(a_data.z,4.0)-1.0;float a_linesofar=(floor(a_data.z/4.0)+a_data.w*64.0)*LINE_DISTANCE_SCALE;vec2 pos=floor(a_pos_normal*0.5);mediump vec2 normal=a_pos_normal-2.0*pos;normal.y=normal.y*2.0-1.0;v_normal=normal;gapwidth=gapwidth/2.0;float halfwidth=width/2.0;offset=-1.0*offset;float inset=gapwidth+(gapwidth > 0.0 ? ANTIALIASING : 0.0);float outset=gapwidth+halfwidth*(gapwidth > 0.0 ? 2.0 :

```

```

1.0)+(halfwidth==0.0 ? 0.0 : ANTIALIASING);mediump vec2
dist=outset*a_extrude*scale;mediump float u=0.5*a_direction;mediump float t=1.0-
abs(u);mediump vec2 offset2=offset*a_extrude*scale*normal.y*mat2(t,-u,u,t);vec4
projected_extrude=u_matrix*vec4(dist/u_ratio,0.0,0.0);gl_Position=u_matrix*vec4(
pos+offset2/u_ratio,0.0,1.0)+projected_extrude;\n#elseif
TERRAIN3D\nv_gamma_scale=1.0;\n#else\nfloat
extrude_length_without_perspective=length(dist);float
extrude_length_with_perspective=length(projected_extrude.xy/gl_Position.w*u_unit
s_to_pixels);v_gamma_scale=extrude_length_without_perspective/extrude_length_wit
h_perspective;\n#endif\nv_linesofar=a_linesofar;v_width2=vec2(outset,inset);v_wi
dth=floorwidth;})",lineSDF:ne("uniform lowp float u_device_pixel_ratio;uniform
sampler2D u_image;uniform float u_sdfgamma;uniform float u_mix;varying vec2
v_normal;varying vec2 v_width2;varying vec2 v_tex_a;varying vec2 v_tex_b;varying
float v_gamma_scale;\n#pragma mapbox: define highp vec4 color\n#pragma mapbox:
define lowp float blur\n#pragma mapbox: define lowp float opacity\n#pragma
mapbox: define mediump float width\n#pragma mapbox: define lowp float
floorwidth\n\nvoid main() {\n#pragma mapbox: initialize highp vec4 color\n#pragma
mapbox: initialize lowp float blur\n#pragma mapbox: initialize lowp float
opacity\n#pragma mapbox: initialize mediump float width\n#pragma mapbox:
initialize lowp float floorwidth\n\nfloat dist=length(v_normal)*v_width2.s;float
blur2=(blur+1.0/u_device_pixel_ratio)*v_gamma_scale;float alpha=clamp(min(dist-
(v_width2.t-blur2),v_width2.s-dist)/blur2,0.0,1.0);float
sdfdist_a=texture2D(u_image,v_tex_a).a;float
sdfdist_b=texture2D(u_image,v_tex_b).a;float
sdfdist=mix(sdfdist_a,sdfdist_b,u_mix);alpha*=smoothstep(0.5-
u_sdfgamma/floorwidth,0.5+u_sdfgamma/floorwidth,sdfdist);gl_FragColor=color*
(alpha*opacity);\n#endif
OVERDRAW_INSPECTOR\negl_FragColor=vec4(1.0);\n#endif\n}", "\n#define scale
0.015873016\n#define LINE_DISTANCE_SCALE 2.0\n\nattribute vec2
a_pos_normal;attribute vec4 a_data;uniform mat4 u_matrix;uniform mediump float
u_ratio;uniform lowp float u_device_pixel_ratio;uniform vec2
u_patternscale_a;uniform float u_tex_y_a;uniform vec2 u_patternscale_b;uniform
float u_tex_y_b;uniform vec2 u_units_to_pixels;varying vec2 v_normal;varying
vec2 v_width2;varying vec2 v_tex_a;varying vec2 v_tex_b;varying float
v_gamma_scale;\n#pragma mapbox: define highp vec4 color\n#pragma mapbox: define
lowp float blur\n#pragma mapbox: define lowp float opacity\n#pragma mapbox:
define mediump float gapwidth\n#pragma mapbox: define lowp float offset\n#pragma
mapbox: define mediump float width\n#pragma mapbox: define lowp float
floorwidth\n\nvoid main() {\n#pragma mapbox: initialize highp vec4 color\n#pragma
mapbox: initialize lowp float blur\n#pragma mapbox: initialize lowp float
opacity\n#pragma mapbox: initialize mediump float gapwidth\n#pragma mapbox:
initialize lowp float offset\n#pragma mapbox: initialize mediump float
width\n\n#pragma mapbox: initialize lowp float floorwidth\n\nfloat
ANTIALIASING=1.0/u_device_pixel_ratio/2.0;vec2 a_extrude=a_data.xy-128.0;float
a_direction=mod(a_data.z,4.0)-1.0;float a_linesofar=
(floor(a_data.z/4.0)+a_data.w*64.0)*LINE_DISTANCE_SCALE;vec2
pos=floor(a_pos_normal*0.5);mediump vec2 normal=a_pos_normal-
2.0*pos;normal.y=normal.y*2.0-1.0;v_normal=normal;gapwidth=gapwidth/2.0;float
halfwidth=width/2.0;offset=-1.0*offset;float inset=gapwidth+(gapwidth > 0.0 ?
ANTIALIASING : 0.0);float outset=gapwidth+halfwidth*(gapwidth > 0.0 ? 2.0 :
1.0)+(halfwidth==0.0 ? 0.0 : ANTIALIASING);mediump vec2
dist=outset*a_extrude*scale;mediump float u=0.5*a_direction;mediump float t=1.0-
abs(u);mediump vec2 offset2=offset*a_extrude*scale*normal.y*mat2(t,-u,u,t);vec4
projected_extrude=u_matrix*vec4(dist/u_ratio,0.0,0.0);gl_Position=u_matrix*vec4(
pos+offset2/u_ratio,0.0,1.0)+projected_extrude;\n#elseif
TERRAIN3D\nv_gamma_scale=1.0;\n#else\nfloat
extrude_length_without_perspective=length(dist);float
extrude_length_with_perspective=length(projected_extrude.xy/gl_Position.w*u_unit
s_to_pixels);v_gamma_scale=extrude_length_without_perspective/extrude_length_wit
h_perspective;\n#endif\nv_tex_a=vec2(a_linesofar*u_patternscale_a.x/floorwidth,n
ormal.y*u_patternscale_a.y+u_tex_y_a);v_tex_b=vec2(a_linesofar*u_patternscale_b.
x/floorwidth,normal.y*u_patternscale_b.y+u_tex_y_b);v_width2=vec2(outset,inset);

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}"),raster:ne("uniform float u_fade_t;uniform float u_opacity;uniform sampler2D
u_image0;uniform sampler2D u_image1;varying vec2 v_pos0;varying vec2
v_pos1;uniform float u_brightness_low;uniform float u_brightness_high;uniform
float u_saturation_factor;uniform float u_contrast_factor;uniform vec3
u_spin_weights;void main() {vec4 color0=texture2D(u_image0,v_pos0);vec4
color1=texture2D(u_image1,v_pos1);if (color0.a > 0.0)
{color0.rgb=color0.rgb/color0.a;}if (color1.a > 0.0)
{color1.rgb=color1.rgb/color1.a;}vec4
color=mix(color0,color1,u_fade_t);color.a*=u_opacity;vec3
rgb=color.rgb;rgb=vec3(dot(rgb,u_spin_weights.xyz),dot(rgb,u_spin_weights.zxy),d
ot(rgb,u_spin_weights.yzx));float average=(color.r+color.g+color.b)/3.0;rgb+=
(average-rgb)*u_saturation_factor;rgb=(rgb-0.5)*u_contrast_factor+0.5;vec3
u_high_vec=vec3(u_brightness_low,u_brightness_low,u_brightness_low);vec3
u_low_vec=vec3(u_brightness_high,u_brightness_high,u_brightness_high);gl_FragCol
or=vec4(mix(u_high_vec,u_low_vec,rgb)*color.a,color.a);\n#ifdef
OVERDRAW_INSPECTOR\negl_FragColor=vec4(1.0);\n#endif\n","uniform mat4
u_matrix;uniform vec2 u_tl_parent;uniform float u_scale_parent;uniform float
u_buffer_scale;attribute vec2 a_pos;attribute vec2 a_texture_pos;varying vec2
v_pos0;varying vec2 v_pos1;void main()
{gl_Position=u_matrix*vec4(a_pos,0,1);v_pos0=
(((a_texture_pos/8192.0)-0.5)/u_buffer_scale )+0.5;v_pos1=
(v_pos0*u_scale_parent)+u_tl_parent;}),symbolIcon:ne("uniform sampler2D
u_texture;varying vec2 v_tex;varying float v_fade_opacity;\n#pragma mapbox:
define lowp float opacity\nvoid main() {\n#pragma mapbox: initialize lowp float
opacity\nlowp float
alpha=opacity*v_fade_opacity;gl_FragColor=texture2D(u_texture,v_tex)*alpha;\n#if
def OVERDRAW_INSPECTOR\negl_FragColor=vec4(1.0);\n#endif\n}","const float
PI=3.141592653589793;attribute vec4 a_pos_offset;attribute vec4 a_data;attribute
vec4 a_pixeloffset;attribute vec3 a_projected_pos;attribute float
a_fade_opacity;uniform bool u_is_size_zoom_constant;uniform bool
u_is_size_feature_constant;uniform highp float u_size_t;uniform highp float
u_size;uniform highp float u_camera_to_center_distance;uniform highp float
u_pitch;uniform bool u_rotate_symbol;uniform highp float u_aspect_ratio;uniform
float u_fade_change;uniform mat4 u_matrix;uniform mat4
u_label_plane_matrix;uniform mat4 u_coord_matrix;uniform bool u_is_text;uniform
bool u_pitch_with_map;uniform vec2 u_textsize;varying vec2 v_tex;varying float
v_fade_opacity;\n#pragma mapbox: define lowp float opacity\nvoid main()
{\n#pragma mapbox: initialize lowp float opacity\nvec2
a_pos=a_pos_offset.xy;vec2 a_offset=a_pos_offset.zw;vec2 a_tex=a_data.xy;vec2
a_size=a_data.zw;float a_size_min=floor(a_size[0]*0.5);vec2
a_pxoffset=a_pixeloffset.xy;vec2 a_minFontScale=a_pixeloffset.zw/256.0;float
ele=get_elevation(a_pos);highp float segment_angle=-a_projected_pos[2];float
size;if (!u_is_size_zoom_constant && !u_is_size_feature_constant)
{size=mix(a_size_min,a_size[1],u_size_t)/128.0;} else if
(u_is_size_zoom_constant && !u_is_size_feature_constant)
{size=a_size_min/128.0;} else {size=u_size;}vec4
projectedPoint=u_matrix*vec4(a_pos,ele,1);highp float
camera_to_anchor_distance=projectedPoint.w;highp float
distance_ratio=u_pitch_with_map ?
\ncamera_to_anchor_distance/u_camera_to_center_distance
:\nu_camera_to_center_distance/camera_to_anchor_distance;highp float
perspective_ratio=clamp(0.5+0.5*distance_ratio,0.0,4.0);size*=perspective_ratio;
float fontScale=u_is_text ? size/24.0 : size;highp float symbol_rotation=0.0;if
(u_rotate_symbol) {vec4
offsetProjectedPoint=u_matrix*vec4(a_pos+vec2(1,0),ele,1);vec2
a=projectedPoint.xy/projectedPoint.w;vec2
b=offsetProjectedPoint.xy/offsetProjectedPoint.w;symbol_rotation=atan((b.y-
a.y)/u_aspect_ratio,b.x-a.x);}highp float
angle_sin=sin(segment_angle+symbol_rotation);highp float
angle_cos=cos(segment_angle+symbol_rotation);mat2
rotation_matrix=mat2(angle_cos,-1.0*angle_sin,angle_sin,angle_cos);vec4
projected_pos=u_label_plane_matrix*vec4(a_projected_pos.xy,ele,1.0);float

```



```

z=float(u_pitch_with_map)*projected_pos.z/projected_pos.w;gl_Position=u_coord_matrix*vec4(projected_pos.xy/projected_pos.w+rotation_matrix*(a_offset/32.0*max(a_minFontScale,fontScale)+a_poffset/16.0),z,1.0);v_tex=a_tex/u_texsize;vec2 fade_opacity=unpack_opacity(a_fade_opacity);float fade_change=fade_opacity[1] > 0.5 ? u_fade_change : -u_fade_change;float visibility=calculate_visibility(projectedPoint);v_fade_opacity=max(0.0,min(visibility,fade_opacity[0]+fade_change));}),symbolSDF:ne("#define SDF_PX 8.0\nuniform bool u_is_halo;uniform sampler2D u_texture;uniform highp float u_gamma_scale;uniform lowp float u_device_pixel_ratio;uniform bool u_is_text;varying vec2 v_data0;varying vec3 v_data1;\n#pragma mapbox: define highp vec4 fill_color\n#pragma mapbox: define highp vec4 halo_color\n#pragma mapbox: define lowp float opacity\n#pragma mapbox: define lowp float halo_width\n#pragma mapbox: define lowp float halo_blur\n\nvoid main() {\n#pragma mapbox: initialize highp vec4 fill_color\n#pragma mapbox: initialize highp vec4 halo_color\n#pragma mapbox: initialize lowp float opacity\n#pragma mapbox: initialize lowp float halo_width\n#pragma mapbox: initialize lowp float halo_blur\n\nfloat EDGE_GAMMA=0.105/u_device_pixel_ratio;vec2 tex=v_data0.xy;float gamma_scale=v_data1.x;float size=v_data1.y;float fade_opacity=v_data1[2];float fontScale=u_is_text ? size/24.0 : size;lowp vec4 color=fill_color;highp float gamma=EDGE_GAMMA/(fontScale*u_gamma_scale);lowp float inner_edge=(256.0-64.0)/256.0;if (u_is_halo) {color=halo_color;gamma=(halo_blur*1.19/SDF_PX+EDGE_GAMMA)/(fontScale*u_gamma_scale);inner_edge=inner_edge+gamma*gamma_scale;}lowp float dist=texture2D(u_texture,tex).a;highp float gamma_scaled=gamma*gamma_scale;highp float alpha=smoothstep(inner_edge-gamma_scaled,inner_edge+gamma_scaled,dist);if (u_is_halo) {lowp float halo_edge=(6.0-halo_width/fontScale)/SDF_PX;alpha=min(smoothstep(halo_edge-gamma_scaled,halo_edge+gamma_scaled,dist),1.0-alpha);}gl_FragColor=color*(alpha*opacity*fade_opacity);\n#ifdef OVERDRAW_INSPECTOR\n    gl_FragColor=vec4(1.0);\n#endif\n}","const float PI=3.141592653589793;attribute vec4 a_pos_offset;attribute vec4 a_data;attribute vec4 a_pixeloffset;attribute vec3 a_projected_pos;attribute float a_fade_opacity;uniform bool u_is_size_zoom_constant;uniform bool u_is_size_feature_constant;uniform highp float u_size_t;uniform highp float u_size;uniform mat4 u_matrix;uniform mat4 u_label_plane_matrix;uniform mat4 u_coord_matrix;uniform bool u_is_text;uniform bool u_pitch_with_map;uniform highp float u_pitch;uniform bool u_rotate_symbol;uniform highp float u_aspect_ratio;uniform highp float u_camera_to_center_distance;uniform float u_fade_change;uniform vec2 u_texsize;varying vec2 v_data0;varying vec3 v_data1;\n#pragma mapbox: define highp vec4 fill_color\n#pragma mapbox: define highp vec4 halo_color\n#pragma mapbox: define lowp float opacity\n#pragma mapbox: define lowp float halo_width\n#pragma mapbox: define lowp float halo_blur\n\nvoid main() {\n#pragma mapbox: initialize highp vec4 fill_color\n#pragma mapbox: initialize highp vec4 halo_color\n#pragma mapbox: initialize lowp float opacity\n#pragma mapbox: initialize lowp float halo_width\n#pragma mapbox: initialize lowp float halo_blur\n\nvec2 a_pos=a_pos_offset.xy;vec2 a_offset=a_pos_offset.zw;vec2 a_tex=a_data.xy;vec2 a_size=a_data.zw;float a_size_min=floor(a_size[0]*0.5);vec2 a_poffset=a_pixeloffset.xy;float ele=get_elevation(a_pos);highp float segment_angle=-a_projected_pos[2];float size;if (!u_is_size_zoom_constant && !u_is_size_feature_constant) {size=mix(a_size_min,a_size[1],u_size_t)/128.0;} else if (u_is_size_zoom_constant && !u_is_size_feature_constant) {size=a_size_min/128.0;} else {size=u_size;}vec4 projectedPoint=u_matrix*vec4(a_pos,ele,1);highp float camera_to_anchor_distance=projectedPoint.w;highp float distance_ratio=u_pitch_with_map ? \n    camera_to_anchor_distance/u_camera_to_center_distance : \n    camera_to_center_distance/camera_to_anchor_distance;highp float perspective_ratio=clamp(0.5+0.5*distance_ratio,0.0,4.0);size*=perspective_ratio;float fontScale=u_is_text ? size/24.0 : size;highp float symbol_rotation=0.0;if (u_rotate_symbol) {vec4 offsetProjectedPoint=u_matrix*vec4(a_pos+vec2(1,0),ele,1);vec2 a=projectedPoint.xy/projectedPoint.w;vec2

```

```

b=offsetProjectedPoint.xy/offsetProjectedPoint.w;symbol_rotation=atan((b.y-
a.y)/u_aspect_ratio,b.x-a.x);}highp float
angle_sin=sin(segment_angle+symbol_rotation);highp float
angle_cos=cos(segment_angle+symbol_rotation);mat2
rotation_matrix=mat2(angle_cos,-1.0*angle_sin,angle_sin,angle_cos);vec4
projected_pos=u_label_plane_matrix*vec4(a_projected_pos.xy,ele,1.0);float
z=float(u_pitch_with_map)*projected_pos.z/projected_pos.w;gl_Position=u_coord_ma
trix*vec4(projected_pos.xy/projected_pos.w+rotation_matrix*
(a_offset/32.0*fontScale+a_poffset),z,1.0);float gamma_scale=gl_Position.w;vec2
fade_opacity=unpack_opacity(a_fade_opacity);float
visibility=calculate_visibility(projectedPoint);float
fade_change=fade_opacity[1] > 0.5 ? u_fade_change :-u_fade_change;float
interpolated_fade_opacity=max(0.0,min(visibility,fade_opacity[0]+fade_change));v
_data0=a_tex/u_textsize;v_data1=vec3(gamma_scale,size,interpolated_fade_opacity);
}),symbolTextAndIcon:ne("#define SDF_PX 8.0\n#define SDF 1.0\n#define ICON
0.0\nuniform bool u_is_halo;uniform sampler2D u_texture;uniform sampler2D
u_texture_icon;uniform highp float u_gamma_scale;uniform lowp float
u_device_pixel_ratio;varying vec4 v_data0;varying vec4 v_data1;\n#pragma mapbox:
define highp vec4 fill_color\n#pragma mapbox: define highp vec4
halo_color\n#pragma mapbox: define lowp float opacity\n#pragma mapbox: define
lowp float halo_width\n#pragma mapbox: define lowp float halo_blur\nvoid main()
{\n#pragma mapbox: initialize highp vec4 fill_color\n#pragma mapbox: initialize
highp vec4 halo_color\n#pragma mapbox: initialize lowp float opacity\n#pragma
mapbox: initialize lowp float halo_width\n#pragma mapbox: initialize lowp float
halo_blur\nfloat fade_opacity=v_data1[2];if (v_data1.w==ICON) {vec2
tex_icon=v_data0.zw;lowp float
alpha=opacity*fade_opacity;gl_FragColor=texture2D(u_texture_icon,tex_icon)*alpha
;\n#ifdef OVERDRAW_INSPECTOR\negl_FragColor=vec4(1.0);\n#endif\nreturn;}vec2
tex=v_data0.xy;float EDGE_GAMMA=0.105/u_device_pixel_ratio;float
gamma_scale=v_data1.x;float size=v_data1.y;float fontScale=size/24.0;lowp vec4
color=fill_color;highp float gamma=EDGE_GAMMA/(fontScale*u_gamma_scale);lowp
float buff=(256.0-64.0)/256.0;if (u_is_halo) {color=halo_color;gamma=
(halo_blur*1.19/SDF_PX+EDGE_GAMMA)/(fontScale*u_gamma_scale);buff=(6.0-
halo_width/fontScale)/SDF_PX;}lowp float dist=texture2D(u_texture,tex).a;highp
float gamma_scaled=gamma*gamma_scale;highp float alpha=smoothstep(buff-
gamma_scaled,buff+gamma_scaled,dist);gl_FragColor=color*
(alpha*opacity*fade_opacity);\n#endif
OVERDRAW_INSPECTOR\negl_FragColor=vec4(1.0);\n#endif\n}","const float
PI=3.141592653589793;attribute vec4 a_pos_offset;attribute vec4 a_data;attribute
vec3 a_projected_pos;attribute float a_fade_opacity;uniform bool
u_is_size_zoom_constant;uniform bool u_is_size_feature_constant;uniform highp
float u_size_t;uniform highp float u_size;uniform mat4 u_matrix;uniform mat4
u_label_plane_matrix;uniform mat4 u_coord_matrix;uniform bool u_is_text;uniform
bool u_pitch_with_map;uniform highp float u_pitch;uniform bool
u_rotate_symbol;uniform highp float u_aspect_ratio;uniform highp float
u_camera_to_center_distance;uniform float u_fade_change;uniform vec2
u_textsize;uniform vec2 u_textsize_icon;varying vec4 v_data0;varying vec4
v_data1;\n#pragma mapbox: define highp vec4 fill_color\n#pragma mapbox: define
highp vec4 halo_color\n#pragma mapbox: define lowp float opacity\n#pragma
mapbox: define lowp float halo_width\n#pragma mapbox: define lowp float
halo_blur\nvoid main() {\n#pragma mapbox: initialize highp vec4
fill_color\n#pragma mapbox: initialize highp vec4 halo_color\n#pragma mapbox:
initialize lowp float opacity\n#pragma mapbox: initialize lowp float
halo_width\n#pragma mapbox: initialize lowp float halo_blur\nvec2
a_pos=a_pos_offset.xy;vec2 a_offset=a_pos_offset.zw;vec2 a_tex=a_data.xy;vec2
a_size=a_data.zw;float a_size_min=floor(a_size[0]*0.5);float
is_sdf=a_size[0]-2.0*a_size_min;float ele=get_elevation(a_pos);highp float
segment_angle=-a_projected_pos[2];float size;if (!u_is_size_zoom_constant &&
!u_is_size_feature_constant) {size=mix(a_size_min,a_size[1],u_size_t)/128.0;}
else if (u_is_size_zoom_constant && !u_is_size_feature_constant)
{size=a_size_min/128.0;} else {size=u_size;}vec4
projectedPoint=u_matrix*vec4(a_pos,ele,1);highp float

```

```

camera_to_anchor_distance=projectedPoint.w;highp float
distance_ratio=u_pitch_with_map ?
\ncamera_to_anchor_distance/u_camera_to_center_distance
:\nu_camera_to_center_distance/camera_to_anchor_distance;highp float
perspective_ratio=clamp(0.5+0.5*distance_ratio,0.0,4.0);size*=perspective_ratio;
float fontScale=size/24.0;highp float symbol_rotation=0.0;if (u_rotate_symbol)
{vec4 offsetProjectedPoint=u_matrix*vec4(a_pos+vec2(1,0),ele,1);vec2
a=projectedPoint.xy/projectedPoint.w;vec2
b=offsetProjectedPoint.xy/offsetProjectedPoint.w;symbol_rotation=atan((b.y-
a.y)/u_aspect_ratio,b.x-a.x);}highp float
angle_sin=sin(segment_angle+symbol_rotation);highp float
angle_cos=cos(segment_angle+symbol_rotation);mat2
rotation_matrix=mat2(angle_cos,-1.0*angle_sin,angle_sin,angle_cos);vec4
projected_pos=u_label_plane_matrix*vec4(a_projected_pos.xy,ele,1.0);float
z=float(u_pitch_with_map)*projected_pos.z/projected_pos.w;gl_Position=u_coord_ma
trix*vec4(projected_pos.xy/projected_pos.w+rotation_matrix*
(a_offset/32.0*fontScale),z,1.0);float gamma_scale=gl_Position.w;vec2
fade_opacity=unpack_opacity(a_fade_opacity);float
visibility=calculate_visibility(projectedPoint);float
fade_change=fade_opacity[1] > 0.5 ? u_fade_change :-u_fade_change;float
interpolated_fade_opacity=max(0.0,min(visibility,fade_opacity[0]+fade_change));v
_data0.xy=a_tex/u_texsize;v_data0.zw=a_tex/u_texsize_icon;v_data1=vec4(gamma_sca
le,size,interpolated_fade_opacity,is_sdf);}"),terrain:ne("uniform sampler2D
u_texture;varying vec2 v_texture_pos;void main()
{gl_FragColor=texture2D(u_texture,v_texture_pos);}"),oe),terrainDepth:ne("varying
float v_depth;const highp vec4
bitSh=vec4(256.*256.*256.,256.*256.,256.,1.);const highp vec4
bitMsk=vec4(0.,vec3(1./256.0));highp vec4 pack(highp float value) {highp vec4
comp=fract(value*bitSh);comp-=comp.xxyz*bitMsk;return comp;}void main()
{gl_FragColor=pack(v_depth);}"),oe),terrainCoords:ne("precision mediump
float;uniform sampler2D u_texture;uniform float u_terrain_coords_id;varying vec2
v_texture_pos;void main() {vec4
rgba=texture2D(u_texture,v_texture_pos);gl_FragColor=vec4(rgba.r,rgba.g,rgba.b,u
_terrain_coords_id);}"),oe});function ne(t,e){const i=/#pragma mapbox: ([\w]+)
([\w]+) ([\w]+) ([\w]+)/g,s=e.match(/attribute ([\w]+)
([\w]+)/g),a=t.match(/uniform ([\w]+) ([\w]+)([\s]*)
([\w]*)/g),o=e.match(/uniform ([\w]+) ([\w]+)([\s]*)([\w]*)/g),r=o?
o.concat(a):a,n={};return {fragmentSource:t=t.replace(i,((t,e,i,s,a)=>
(n[a]!=""?"define"===e?"\n#ifndef HAS_UNIFORM_u_${a}\nvarying ${i} ${s}
${a};\n#else\nuniform ${i} ${s} u_${a};\n#endif\n`:\n#ifdef
HAS_UNIFORM_u_${a}\n    ${i} ${s} ${a} =
u_${a};\n#endif\n`)),vertexSource:e=e.replace(i,((t,e,i,s,a)=>{const
o="float"===s?"vec4":"vec4",r=a.match(/color/)?"color":o;return
n[a]?"define"===e?"\n#ifndef HAS_UNIFORM_u_${a}\nuniform lowp float
u_${a}_t;\nattribute ${i} ${o} a_${a};\nvarying ${i} ${s} ${a};\n#else\nuniform
${i} ${s} u_${a};\n#endif\n`:"vec4"===r?"\n#ifndef HAS_UNIFORM_u_${a}\n    ${a}
= a_${a};\n#else\n    ${i} ${s} ${a} = u_${a};\n#endif\n`:\n#ifndef
HAS_UNIFORM_u_${a}\n    ${a} = unpack_mix_${r}(a_${a}, u_${a}_t);\n#else\n
${i} ${s} ${a} = u_${a};\n#endif\n`:"define"===e?"\n#ifndef
HAS_UNIFORM_u_${a}\nuniform lowp float u_${a}_t;\nattribute ${i} ${o}
a_${a};\n#else\nuniform ${i} ${s} u_${a};\n#endif\n`:"vec4"===r?"\n#ifndef
HAS_UNIFORM_u_${a}\n    ${i} ${s} ${a} = a_${a};\n#else\n    ${i} ${s} ${a} =
u_${a};\n#endif\n`:\n#ifndef HAS_UNIFORM_u_${a}\n    ${i} ${s} ${a} =
unpack_mix_${r}(a_${a}, u_${a}_t);\n#else\n    ${i} ${s} ${a} =
u_${a};\n#endif\n`))},staticAttributes:s,staticUniforms:r}}class
le{constructor()
{this.boundProgram=null,this.boundLayoutVertexBuffer=null,this.boundPaintVertexB
uffers=
[,this.boundIndexBuffer=null,this.boundVertexOffset=null,this.boundDynamicVerte
xBuffer=null,this.vao=null;}bind(t,e,i,s,a,o,r,n,l){this.context=t;let
h=this.boundPaintVertexBuffers.length!==s.length;for(let
t=0;!h&&t<s.length;t++)this.boundPaintVertexBuffers[t]!==s[t]&&

```

```

(h=!0);!this.vao||this.bindProgram!==e||this.bindLayoutVertexBuffer!==i||h||this.bindIndexBuffer!==a||this.bindVertexOffset!==o||this.bindDynamicVertexBuffer!==r||this.bindDynamicVertexBuffer2!==n||this.bindDynamicVertexBuffer3!==l?this.freshBind(e,i,s,a,o,r,n,l):(t.bindVertexArray.set(this.vao),r&&r.bind(),a&&a.dynamicDraw&&a.bind(),n&&n.bind(),l&&l.bind());}freshBind(t,e,i,s,a,o,r,n){const l=t.numAttributes,h=this.context,c=h.gl;this.vao&&this.destroy(),this.vao=h.createVertexArray(),h.bindVertexArray.set(this.vao),this.bindProgram=t,this.bindLayoutVertexBuffer=e,this.bindPaintVertexBuffers=i,this.bindIndexBuffer=s,this.bindVertexOffset=a,this.bindDynamicVertexBuffer=o,this.bindDynamicVertexBuffer2=r,this.bindDynamicVertexBuffer3=n,e.enableAttributes(c,t);for(const e of i)e.enableAttributes(c,t);o&&o.enableAttributes(c,t),r&&r.enableAttributes(c,t),n&&n.enableAttributes(c,t),e.bind(),e.setVertexAttribPointers(c,t,a);for(const e of i)e.bind(),e.setVertexAttribPointers(c,t,a);o&&(o.bind(),o.setVertexAttribPointers(c,t,a)),s&&s.bind(),r&&(r.bind(),r.setVertexAttribPointers(c,t,a)),n&&(n.bind(),n.setVertexAttribPointers(c,t,a)),h.currentNumAttributes=1;}destroy(){this.vao&&(this.context.deleteVertexArray(this.vao),this.vao=null);}function he(t){const e=[];for(let i=0;i<t.length;i++){if(null===t[i])continue;const s=t[i].split(" ");e.push(s.pop());}return e}class ce{constructor(e,i,s,a,o,r){const n=e.gl;this.program=n.createProgram();const l=he(i.staticAttributes),h=s?s.getBinderAttributes():[],c=l.concat(h),u=re.prelude.staticUniforms?he(re.prelude.staticUniforms):[],d=i.staticUniforms?he(i.staticUniforms):[],_s?s.getBinderUniforms():[],p=u.concat(d).concat(_),m=[];for(const t of p)m.indexOf(t)<0&&m.push(t);const f=s?s.defines():[];o&&f.push("#define OVERDRAW_INSPECTOR;"),r&&f.push("#define TERRAIN3D;");const g=f.concat(re.prelude.fragmentSource,i.fragmentSource).join("\n"),v=f.concat(re.prelude.vertexSource,i.vertexSource).join("\n"),x=n.createShader(n.FRAGMENT_SHADER);if(n.isContextLost())return void(this.failedToCreate=!0);if(n.shaderSource(x,g),n.compileShader(x),!n.getShaderParameter(x,n.COMPILE_STATUS))throw new Error(`Could not compile fragment shader: ${n.getShaderInfoLog(x)}`);n.attachShader(this.program,x);const y=n.createShader(n.VERTEX_SHADER);if(n.isContextLost())return void(this.failedToCreate=!0);if(n.shaderSource(y,v),n.compileShader(y),!n.getShaderParameter(y,n.COMPILE_STATUS))throw new Error(`Could not compile vertex shader: ${n.getShaderInfoLog(y)}`);n.attachShader(this.program,y),this.attributes={};const b={};this.numAttributes=c.length;for(let t=0;t<this.numAttributes;t++)c[t]&&(n.bindAttribLocation(this.program,t,c[t]),this.attributes[c[t]]=t);if(n.linkProgram(this.program),!n.getProgramParameter(this.program,n.LINK_STATUS))throw new Error(`Program failed to link: ${n.getProgramInfoLog(this.program)}`);n.deleteShader(y),n.deleteShader(x);for(let t=0;t<m.length;t++){const e=m[t];if(e&&!b[e]){const t=n.getUniformLocation(this.program,e);t&&(b[e]=t);}}this.fixedUniforms=a(e,b),this.terrainUniforms=((e,i)=>({u_depth:new t.aL(e,i.u_depth),u_terrain:new t.aL(e,i.u_terrain),u_terrain_dim:new t.aM(e,i.u_terrain_dim),u_terrain_matrix:new t.aN(e,i.u_terrain_matrix),u_terrain_unpack:new t.aO(e,i.u_terrain_unpack),u_terrain_exaggeration:new t.aM(e,i.u_terrain_exaggeration)}))(e,b),this.binderUniforms=s?s.getUniforms(e,b):[];draw(t,e,i,s,a,o,r,n,l,h,c,u,d,_p,m,f,g){const v=t.gl;if(this.failedToCreate)return;if(t.program.set(this.program),t.setDepthMode(i),t.setStencilMode(s),t.setColorMode(a),t.setCullFace(o),n){t.activeTexture.set(v.TEXTURE2),v.bindTexture(v.TEXTURE_2D,n.depthTexture),t.activeTexture.set(v.TEXTURE3),v.bindTexture(v.TEXTURE_2D,n.texture);for(const t in this.terrainUniforms)this.terrainUniforms[t].set(n[t]);}for(const t in this.fixedUniforms)this.fixedUniforms[t].set(r[t]);p&&p.setUniforms(t,this.binderUniforms,d,{zoom:_});let x=0;switch(e){case v.LINES:x=2;break;case v.TRIANGLES:x=3;break;case v.LINE_STRIP:x=1;}for(const i of u.get()){const s=i.vaos||(i.vaos={});(s[l]||(s[l]=new le)).bind(t,this,h,p)p.getPaintVertexBuffers():

```

```

[,c,i.vertexOffset,m,f,g),v.drawElements(e,i.primitiveLength*x,v.UNSIGNED_SHORT
,i.primitiveOffset*x*2);}}function ue(t,e,i){const
s=1/St(i,l,e.transform.tileZoom),a=Math.pow(2,i.tileID.overscaledZ),o=i.tileSize
*Math.pow(2,e.transform.tileZoom)/a,r=o*
(i.tileID.canonical.x+i.tileID.wrap*a),n=o*i.tileID.canonical.y;return
{u_image:0,u_textsize:i.imageAtlasTexture.size,u_scale:
[s,t.fromScale,t.toScale],u_fade:t.t,u_pixel_coord_upper:
[r>>16,n>>16],u_pixel_coord_lower:[65535&r,65535&n]}}const de=(e,i,s,a)=>{const
o=i.style.light,r=o.properties.get("position"),n=[r.x,r.y,r.z],l=function(){var
e=new t.A(9);return t.A!=Float32Array&&
(e[1]=0,e[2]=0,e[3]=0,e[5]=0,e[6]=0,e[7]=0),e[0]=1,e[4]=1,e[8]=1,e
();"viewport"===o.properties.get("anchor")&&function(t,e){var
i=Math.sin(e),s=Math.cos(e);t[0]=s,t[1]=i,t[2]=0,t[3]=-
i,t[4]=s,t[5]=0,t[6]=0,t[7]=0,t[8]=1;}(1,-i.transform.angle),function(t,e,i){var
s=e[0],a=e[1],o=e[2];t[0]=s*i[0]+a*i[3]+o*i[6],t[1]=s*i[1]+a*i[4]+o*i[7],t[2]=s*
i[2]+a*i[5]+o*i[8];}(n,n,l);const h=o.properties.get("color");return
{u_matrix:e,u_lightpos:n,u_lightintensity:o.properties.get("intensity"),u_lightc
olor:[h.r,h.g,h.b],u_vertical_gradient:+s,u_opacity:a}},_e=
(e,i,s,a,o,r,n)=>t.e(de(e,i,s,a),ue(r,i,n),{u_height_factor:-
Math.pow(2,o.overscaledZ)/n.tileSize/8}),pe=t=>({u_matrix:t}),me=
(e,i,s,a)=>t.e(pe(e),ue(s,i,a)),fe=(t,e)=>({u_matrix:t,u_world:e}),ge=
(e,i,s,a,o)=>t.e(me(e,i,s,a),{u_world:o}),ve=(t,e,i,s)=>{const a=t.transform;let
o,r;if("map"===s.paint.get("circle-pitch-alignment")){const
t=St(i,l,a.zoom);o=!0,r=[t,t];}else o=!1,r=a.pixelsToGLUnits;return
{u_camera_to_center_distance:a.cameraToCenterDistance,u_scale_with_map:+
("map"===s.paint.get("circle-pitch-
scale")),u_matrix:t.translatePosMatrix(e.posMatrix,i,s.paint.get("circle-
translate")),s.paint.get("circle-translate-
anchor")),u_pitch_with_map:+o,u_device_pixel_ratio:t.pixelRatio,u_extrude_scale:
r}},xe=(t,e,i)=>{const s=St(i,l,e.zoom),a=Math.pow(2,e.zoom-
i.tileID.overscaledZ),o=i.tileID.overscaleFactor();return
{u_matrix:t,u_camera_to_center_distance:e.cameraToCenterDistance,u_pixels_to_til
e_units:s,u_extrude_scale:
[e.pixelsToGLUnits[0]/(s*a),e.pixelsToGLUnits[1]/(s*a)],u_overscale_factor:o}},y
e=(t,e,i,l)=>({u_matrix:t,u_color:e,u_overlay:0,u_overlay_scale:i}),be=t=>
({u_matrix:t}),we=(t,e,i,s)=>
({u_matrix:t,u_extrude_scale:St(e,l,i),u_intensity:s});function Te(e,i){const
s=Math.pow(2,i.canonical.z),a=i.canonical.y;return [new
t.U(0,a/s).toLngLat().lat,new t.U(0,(a+1)/s).toLngLat().lat]}const Ie=
(t,e,i,s)=>{const a=t.transform;return
{u_matrix:De(t,e,i,s),u_ratio:1/St(e,l,a.zoom),u_device_pixel_ratio:t.pixelRatio
,u_units_to_pixels:[1/a.pixelsToGLUnits[0],1/a.pixelsToGLUnits[1]]}},Ee=
(e,i,s,a,o)=>t.e(Ie(e,i,s,o),{u_image:0,u_image_height:a}),Se=(t,e,i,s,a)=>
{const o=t.transform,r=Pe(e,o);return
{u_matrix:De(t,e,i,a),u_textsize:e.imageAtlasTexture.size,u_ratio:1/St(e,l,o.zoom
),u_device_pixel_ratio:t.pixelRatio,u_image:0,u_scale:
[r,s.fromScale,s.toScale],u_fade:s.t,u_units_to_pixels:
[1/o.pixelsToGLUnits[0],1/o.pixelsToGLUnits[1]]}},Ce=(e,i,s,a,o,r)=>{const
n=e.lineAtlas,l=Pe(i,e.transform),h="round"===s.layout.get("line-
cap"),c=n.getDash(a.from,h),u=n.getDash(a.to,h),d=c.width*o.fromScale,_=u.width*
o.toScale;return t.e(Ie(e,i,s,r),{u_patternscale_a:[l/d,-
c.height/2],u_patternscale_b:[l/_,-
u.height/2],u_sdfgamma:n.width/(256*Math.min(d,_)*e.pixelRatio)/2,u_image:0,u_te
x_y_a:c.y,u_tex_y_b:u.y,u_mix:o.t}});function Pe(t,e){return
1/St(t,l,e.tileZoom)}function De(t,e,i,s){return t.translatePosMatrix(s?
s.posMatrix:e.tileID.posMatrix,e,i.paint.get("line-
translate"),i.paint.get("line-translate-anchor"))}const Me=(t,e,i,s,a)=>{return
{u_matrix:t,u_tl_parent:e,u_scale_parent:i,u_buffer_scale:1,u_fade_t:s.mix,u_opa
city:s.opacity*a.paint.get("raster-
opacity"),u_image0:0,u_image1:1,u_brightness_low:a.paint.get("raster-brightness-
min"),u_brightness_high:a.paint.get("raster-brightness-
max"),u_saturation_factor:(r=a.paint.get("raster-saturation"),r>0?1-1/(1.001-

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r):-r),u_contrast_factor:(o=a.paint.get("raster-contrast"),o>0?1/(1-
o):1+o),u_spin_weights:ze(a.paint.get("raster-hue-rotate"));var o,r;};function
ze(t){t*=Math.PI/180;const e=Math.sin(t),i=Math.cos(t);return [(2*i+1)/3,(-
Math.sqrt(3)*e-i+1)/3,(Math.sqrt(3)*e-i+1)/3]}const Le=(t,e,i,s,a,o,r,n,l,h)=>
{const c=a.transform;return {u_is_size_zoom_constant:+
("constant"===t||"source"===t),u_is_size_feature_constant:+
("constant"===t||"camera"===t),u_size_t:e?e.uSizeT:0,u_size:e?
e.uSize:0,u_camera_to_center_distance:c.cameraToCenterDistance,u_pitch:c.pitch/3
60*2*Math.PI,u_rotate_symbol:+i,u_aspect_ratio:c.width/c.height,u_fade_change:a.
options.fadeDuration?
a.symbolFadeChange:1,u_matrix:o,u_label_plane_matrix:r,u_coord_matrix:n,u_is_tex
t:+1,u_pitch_with_map:+s,u_textsize:h,u_texture:0}},Ae=(e,i,s,a,o,r,n,l,h,c,u)=>
{const d=o.transform;return t.e(Le(e,i,s,a,o,r,n,l,h,c),{u_gamma_scale:a?
Math.cos(d._pitch)*d.cameraToCenterDistance:1,u_device_pixel_ratio:o.pixelRatio,
u_is_halo:+u}),Re=(e,i,s,a,o,r,n,l,h,c)=>t.e(Ae(e,i,s,a,o,r,n,l,!0,h,!0),
{u_textsize:icon:c,u_texture_icon:1}),ke=(t,e,i)=>
({u_matrix:t,u_opacity:e,u_color:i}),Fe=(e,i,s,a,o,r)=>t.e(function(t,e,i,s)
{const
a=i.imageManager.getPattern(t.from.toString()),o=i.imageManager.getPattern(t.to.
toString()),
{width:r,height:n}=i.imageManager.getPixelSize(),l=Math.pow(2,s.tileID.overscale
dZ),h=s.tileSize*Math.pow(2,i.transform.tileZoom)/l,c=h*
(s.tileID.canonical.x+s.tileID.wrap*1),u=h*s.tileID.canonical.y;return
{u_image:0,u_pattern_tl_a:a.tl,u_pattern_br_a:a.br,u_pattern_tl_b:o.tl,u_pattern
_br_b:o.br,u_textsize:
[r,n],u_mix:e.t,u_pattern_size_a:a.displaySize,u_pattern_size_b:o.displaySize,u
scale_a:e.fromScale,u_scale_b:e.toScale,u_tile_units_to_pixels:1/St(s,l,i.transf
orm.tileZoom),u_pixel_coord_upper:[c>>16,u>>16],u_pixel_coord_lower:
[65535&c,65535&u]})(a,r,s,o),{u_matrix:e,u_opacity:i}),Be={fillExtrusion:(e,i)=>
({u_matrix:new t.aN(e,i.u_matrix),u_lightpos:new
t.aP(e,i.u_lightpos),u_lightintensity:new
t.aM(e,i.u_lightintensity),u_lightcolor:new
t.aP(e,i.u_lightcolor),u_vertical_gradient:new
t.aM(e,i.u_vertical_gradient),u_opacity:new
t.aM(e,i.u_opacity)}),fillExtrusionPattern:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_lightpos:new t.aP(e,i.u_lightpos),u_lightintensity:new
t.aM(e,i.u_lightintensity),u_lightcolor:new
t.aP(e,i.u_lightcolor),u_vertical_gradient:new
t.aM(e,i.u_vertical_gradient),u_height_factor:new
t.aM(e,i.u_height_factor),u_image:new t.aL(e,i.u_image),u_textsize:new
t.aQ(e,i.u_textsize),u_pixel_coord_upper:new
t.aQ(e,i.u_pixel_coord_upper),u_pixel_coord_lower:new
t.aQ(e,i.u_pixel_coord_lower),u_scale:new t.aP(e,i.u_scale),u_fade:new
t.aM(e,i.u_fade),u_opacity:new t.aM(e,i.u_opacity)}),fill:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix)}),fillPattern:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_image:new t.aL(e,i.u_image),u_textsize:new
t.aQ(e,i.u_textsize),u_pixel_coord_upper:new
t.aQ(e,i.u_pixel_coord_upper),u_pixel_coord_lower:new
t.aQ(e,i.u_pixel_coord_lower),u_scale:new t.aP(e,i.u_scale),u_fade:new
t.aM(e,i.u_fade)}),fillOutline:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_world:new t.aQ(e,i.u_world)}),fillOutlinePattern:(e,i)=>
({u_matrix:new t.aN(e,i.u_matrix),u_world:new t.aQ(e,i.u_world),u_image:new
t.aL(e,i.u_image),u_textsize:new t.aQ(e,i.u_textsize),u_pixel_coord_upper:new
t.aQ(e,i.u_pixel_coord_upper),u_pixel_coord_lower:new
t.aQ(e,i.u_pixel_coord_lower),u_scale:new t.aP(e,i.u_scale),u_fade:new
t.aM(e,i.u_fade)}),circle:(e,i)=>({u_camera_to_center_distance:new
t.aM(e,i.u_camera_to_center_distance),u_scale_with_map:new
t.aL(e,i.u_scale_with_map),u_pitch_with_map:new
t.aL(e,i.u_pitch_with_map),u_extrude_scale:new
t.aQ(e,i.u_extrude_scale),u_device_pixel_ratio:new
t.aM(e,i.u_device_pixel_ratio),u_matrix:new t.aN(e,i.u_matrix)}),collisionBox:
(e,i)=>({u_matrix:new t.aN(e,i.u_matrix),u_camera_to_center_distance:new

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t.aM(e,i.u_camera_to_center_distance),u_pixels_to_tile_units:new
t.aM(e,i.u_pixels_to_tile_units),u_extrude_scale:new
t.aQ(e,i.u_extrude_scale),u_overscale_factor:new
t.aM(e,i.u_overscale_factor)),collisionCircle:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_inv_matrix:new
t.aN(e,i.u_inv_matrix),u_camera_to_center_distance:new
t.aM(e,i.u_camera_to_center_distance),u_viewport_size:new
t.aQ(e,i.u_viewport_size)),debug:(e,i)=>({u_color:new
t.aR(e,i.u_color),u_matrix:new t.aN(e,i.u_matrix),u_overlay:new
t.aL(e,i.u_overlay),u_overlay_scale:new
t.aM(e,i.u_overlay_scale)),clippingMask:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix)),heatmap:(e,i)=>({u_extrude_scale:new
t.aM(e,i.u_extrude_scale),u_intensity:new t.aM(e,i.u_intensity),u_matrix:new
t.aN(e,i.u_matrix)),heatmapTexture:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_world:new t.aQ(e,i.u_world),u_image:new
t.aL(e,i.u_image),u_color_ramp:new t.aL(e,i.u_color_ramp),u_opacity:new
t.aM(e,i.u_opacity)),hillshade:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_image:new t.aL(e,i.u_image),u_latrange:new
t.aQ(e,i.u_latrange),u_light:new t.aQ(e,i.u_light),u_shadow:new
t.aR(e,i.u_shadow),u_highlight:new t.aR(e,i.u_highlight),u_accent:new
t.aR(e,i.u_accent)),hillshadePrepare:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_image:new t.aL(e,i.u_image),u_dimension:new
t.aQ(e,i.u_dimension),u_zoom:new t.aM(e,i.u_zoom),u_unpack:new
t.aO(e,i.u_unpack)),line:(e,i)=>({u_matrix:new t.aN(e,i.u_matrix),u_ratio:new
t.aM(e,i.u_ratio),u_device_pixel_ratio:new
t.aM(e,i.u_device_pixel_ratio),u_units_to_pixels:new
t.aQ(e,i.u_units_to_pixels)),lineGradient:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_ratio:new t.aM(e,i.u_ratio),u_device_pixel_ratio:new
t.aM(e,i.u_device_pixel_ratio),u_units_to_pixels:new
t.aQ(e,i.u_units_to_pixels),u_image:new t.aL(e,i.u_image),u_image_height:new
t.aM(e,i.u_image_height)),linePattern:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_textsize:new t.aQ(e,i.u_textsize),u_ratio:new
t.aM(e,i.u_ratio),u_device_pixel_ratio:new
t.aM(e,i.u_device_pixel_ratio),u_image:new
t.aL(e,i.u_image),u_units_to_pixels:new t.aQ(e,i.u_units_to_pixels),u_scale:new
t.aP(e,i.u_scale),u_fade:new t.aM(e,i.u_fade)),lineSDF:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_ratio:new t.aM(e,i.u_ratio),u_device_pixel_ratio:new
t.aM(e,i.u_device_pixel_ratio),u_units_to_pixels:new
t.aQ(e,i.u_units_to_pixels),u_patternscale_a:new
t.aQ(e,i.u_patternscale_a),u_patternscale_b:new
t.aQ(e,i.u_patternscale_b),u_sdfgamma:new t.aM(e,i.u_sdfgamma),u_image:new
t.aL(e,i.u_image),u_tex_y_a:new t.aM(e,i.u_tex_y_a),u_tex_y_b:new
t.aM(e,i.u_tex_y_b),u_mix:new t.aM(e,i.u_mix)),raster:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_tl_parent:new t.aQ(e,i.u_tl_parent),u_scale_parent:new
t.aM(e,i.u_scale_parent),u_buffer_scale:new
t.aM(e,i.u_buffer_scale),u_fade_t:new t.aM(e,i.u_fade_t),u_opacity:new
t.aM(e,i.u_opacity),u_image0:new t.aL(e,i.u_image0),u_image1:new
t.aL(e,i.u_image1),u_brightness_low:new
t.aM(e,i.u_brightness_low),u_brightness_high:new
t.aM(e,i.u_brightness_high),u_saturation_factor:new
t.aM(e,i.u_saturation_factor),u_contrast_factor:new
t.aM(e,i.u_contrast_factor),u_spin_weights:new
t.aP(e,i.u_spin_weights)),symbolIcon:(e,i)=>({u_is_size_zoom_constant:new
t.aL(e,i.u_is_size_zoom_constant),u_is_size_feature_constant:new
t.aL(e,i.u_is_size_feature_constant),u_size_t:new t.aM(e,i.u_size_t),u_size:new
t.aM(e,i.u_size),u_camera_to_center_distance:new
t.aM(e,i.u_camera_to_center_distance),u_pitch:new
t.aM(e,i.u_pitch),u_rotate_symbol:new
t.aL(e,i.u_rotate_symbol),u_aspect_ratio:new
t.aM(e,i.u_aspect_ratio),u_fade_change:new t.aM(e,i.u_fade_change),u_matrix:new
t.aN(e,i.u_matrix),u_label_plane_matrix:new
t.aN(e,i.u_label_plane_matrix),u_coord_matrix:new

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t.aN(e,i.u_coord_matrix),u_is_text:new t.aL(e,i.u_is_text),u_pitch_with_map:new
t.aL(e,i.u_pitch_with_map),u_textsize:new t.aQ(e,i.u_textsize),u_texture:new
t.aL(e,i.u_texture)}),symbolSDF:(e,i)=>({u_is_size_zoom_constant:new
t.aL(e,i.u_is_size_zoom_constant),u_is_size_feature_constant:new
t.aL(e,i.u_is_size_feature_constant),u_size_t:new t.aM(e,i.u_size_t),u_size:new
t.aM(e,i.u_size),u_camera_to_center_distance:new
t.aM(e,i.u_camera_to_center_distance),u_pitch:new
t.aM(e,i.u_pitch),u_rotate_symbol:new
t.aL(e,i.u_rotate_symbol),u_aspect_ratio:new
t.aM(e,i.u_aspect_ratio),u_fade_change:new t.aM(e,i.u_fade_change),u_matrix:new
t.aN(e,i.u_matrix),u_label_plane_matrix:new
t.aN(e,i.u_label_plane_matrix),u_coord_matrix:new
t.aN(e,i.u_coord_matrix),u_is_text:new t.aL(e,i.u_is_text),u_pitch_with_map:new
t.aL(e,i.u_pitch_with_map),u_textsize:new t.aQ(e,i.u_textsize),u_texture:new
t.aL(e,i.u_texture),u_gamma_scale:new
t.aM(e,i.u_gamma_scale),u_device_pixel_ratio:new
t.aM(e,i.u_device_pixel_ratio),u_is_halo:new
t.aL(e,i.u_is_halo)}),symbolTextAndIcon:(e,i)=>({u_is_size_zoom_constant:new
t.aL(e,i.u_is_size_zoom_constant),u_is_size_feature_constant:new
t.aL(e,i.u_is_size_feature_constant),u_size_t:new t.aM(e,i.u_size_t),u_size:new
t.aM(e,i.u_size),u_camera_to_center_distance:new
t.aM(e,i.u_camera_to_center_distance),u_pitch:new
t.aM(e,i.u_pitch),u_rotate_symbol:new
t.aL(e,i.u_rotate_symbol),u_aspect_ratio:new
t.aM(e,i.u_aspect_ratio),u_fade_change:new t.aM(e,i.u_fade_change),u_matrix:new
t.aN(e,i.u_matrix),u_label_plane_matrix:new
t.aN(e,i.u_label_plane_matrix),u_coord_matrix:new
t.aN(e,i.u_coord_matrix),u_is_text:new t.aL(e,i.u_is_text),u_pitch_with_map:new
t.aL(e,i.u_pitch_with_map),u_textsize:new t.aQ(e,i.u_textsize),u_textsize_icon:new
t.aQ(e,i.u_textsize_icon),u_texture:new t.aL(e,i.u_texture),u_texture_icon:new
t.aL(e,i.u_texture_icon),u_gamma_scale:new
t.aM(e,i.u_gamma_scale),u_device_pixel_ratio:new
t.aM(e,i.u_device_pixel_ratio),u_is_halo:new t.aL(e,i.u_is_halo)}),background:
(e,i)=>({u_matrix:new t.aN(e,i.u_matrix),u_opacity:new
t.aM(e,i.u_opacity),u_color:new t.aR(e,i.u_color)}),backgroundPattern:(e,i)=>
({u_matrix:new t.aN(e,i.u_matrix),u_opacity:new t.aM(e,i.u_opacity),u_image:new
t.aL(e,i.u_image),u_pattern_tl_a:new t.aQ(e,i.u_pattern_tl_a),u_pattern_br_a:new
t.aQ(e,i.u_pattern_br_a),u_pattern_tl_b:new
t.aQ(e,i.u_pattern_tl_b),u_pattern_br_b:new
t.aQ(e,i.u_pattern_br_b),u_textsize:new t.aQ(e,i.u_textsize),u_mix:new
t.aM(e,i.u_mix),u_pattern_size_a:new
t.aQ(e,i.u_pattern_size_a),u_pattern_size_b:new
t.aQ(e,i.u_pattern_size_b),u_scale_a:new t.aM(e,i.u_scale_a),u_scale_b:new
t.aM(e,i.u_scale_b),u_pixel_coord_upper:new
t.aQ(e,i.u_pixel_coord_upper),u_pixel_coord_lower:new
t.aQ(e,i.u_pixel_coord_lower),u_tile_units_to_pixels:new
t.aM(e,i.u_tile_units_to_pixels)}),terrain:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_texture:new t.aL(e,i.u_texture),u_ele_delta:new
t.aM(e,i.u_ele_delta)}),terrainDepth:(e,i)=>({u_matrix:new
t.aN(e,i.u_matrix),u_ele_delta:new t.aM(e,i.u_ele_delta)}),terrainCoords:(e,i)=>
({u_matrix:new t.aN(e,i.u_matrix),u_texture:new
t.aL(e,i.u_texture),u_terrain_coords_id:new
t.aM(e,i.u_terrain_coords_id),u_ele_delta:new t.aM(e,i.u_ele_delta)}});class
Oe{constructor(t,e,i){this.context=t;const
s=t.gl;this.buffer=s.createBuffer(),this.dynamicDraw=Boolean(i),this.context.unb
indVAO(),t.bindElementBuffer.set(this.buffer),s.bufferData(s.ELEMENT_ARRAY_BUF
R,e.arrayBuffer,this.dynamicDraw?
s.DYNAMIC_DRAW:s.STATIC_DRAW),this.dynamicDraw||delete e.arrayBuffer;}bind()
{this.context.bindElementBuffer.set(this.buffer);}updateData(t){const
e=this.context.gl;if(!this.dynamicDraw)throw new Error("Attempted to update data
while not in dynamic
mode.");this.context.unbindVAO(),this.bind(),e.bufferSubData(e.ELEMENT_ARRAY_BUF

```



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FER,0,t.arrayBuffer);}destroy(){this.buffer&&
(this.context.gl.deleteBuffer(this.buffer),delete this.buffer);}const Ne=
{Int8:"BYTE",Uint8:"UNSIGNED_BYTE",Int16:"SHORT",Uint16:"UNSIGNED_SHORT",Int32:"
INT",Uint32:"UNSIGNED_INT",Float32:"FLOAT"};class Ue{constructor(t,e,i,s)
{this.length=e.length,this.attributes=i,this.itemSize=e.bytesPerElement,this.dyn
amicDraw=s,this.context=t;const
a=t.gl;this.buffer=a.createBuffer(),t.bindVertexBuffer.set(this.buffer),a.buffer
Data(a.ARRAY_BUFFER,e.arrayBuffer,this.dynamicDraw?
a.DYNAMIC_DRAW:a.STATIC_DRAW),this.dynamicDraw||delete e.arrayBuffer;}bind()
{this.context.bindVertexBuffer.set(this.buffer);}updateData(t)
{if(t.length!==this.length)throw new Error(`Length of new data is ${t.length},
which doesn't match current length of ${this.length}`);const
e=this.context.gl;this.bind(),e.bufferSubData(e.ARRAY_BUFFER,0,t.arrayBuffer);}e
nableAttributes(t,e){for(let i=0;i<this.attributes.length;i++){const
s=e.attributes[this.attributes[i].name];void
0!==s&&t.enableVertexAttribArray(s);}setVertexAttribPointers(t,e,i){for(let
s=0;s<this.attributes.length;s++){const
a=this.attributes[s],o=e.attributes[a.name];void
0!==o&&t.vertexAttribPointer(o,a.components,t[Ne[a.type]],!1,this.itemSize,a.off
set+this.itemSize*(i|0));}}destroy(){this.buffer&&
(this.context.gl.deleteBuffer(this.buffer),delete this.buffer);}const Ze=new
WeakMap;function Ge(t){var e;if(Ze.has(t))return Ze.get(t);{const i=null===
(e=t.getParameter(t.VERSION))||void 0===e?void 0:e.startsWith("WebGL
2.0");return Ze.set(t,i),i}}class je{constructor(t)
{this.gl=t.gl,this.default=this.getDefault(),this.current=this.default,this.dirt
y=!1;}get(){return this.current}set(t){getDefaultValue(){return
this.default}setDefault(){this.set(this.default);}}class Ve extends
je{getDefaultValue(){return t.aT.transparent}set(t){const e=this.current;
(t.r!==e.r||t.g!==e.g||t.b!==e.b||t.a!==e.a||this.dirty)&&
(this.gl.clearColor(t.r,t.g,t.b,t.a),this.current=t,this.dirty=!1);}}class qe
extends je{getDefaultValue(){return 1}set(t){(t!==this.current||this.dirty)&&
(this.gl.clearDepth(t),this.current=t,this.dirty=!1);}}class $e extends
je{getDefaultValue(){return 0}set(t){(t!==this.current||this.dirty)&&
(this.gl.clearStencil(t),this.current=t,this.dirty=!1);}}class We extends
je{getDefaultValue(){return [!0,!0,!0,!0]}set(t){const e=this.current;
(t[0]!==e[0]||t[1]!==e[1]||t[2]!==e[2]||t[3]!==e[3]||this.dirty)&&
(this.gl.colorMask(t[0],t[1],t[2],t[3]),this.current=t,this.dirty=!1);}}class He
extends je{getDefaultValue(){return !0}set(t){(t!==this.current||this.dirty)&&
(this.gl.depthMask(t),this.current=t,this.dirty=!1);}}class Xe extends
je{getDefaultValue(){return 255}set(t){(t!==this.current||this.dirty)&&
(this.gl.stencilMask(t),this.current=t,this.dirty=!1);}}class Ke extends
je{getDefaultValue(){return {func:this.gl.ALWAYS,ref:0,mask:255}}set(t){const
e=this.current;(t.func!==e.func||t.ref!==e.ref||t.mask!==e.mask||this.dirty)&&
(this.gl.stencilFunc(t.func,t.ref,t.mask),this.current=t,this.dirty=!1);}}class
Qe extends je{getDefaultValue(){const t=this.gl;return [t.KEEP,t.KEEP,t.KEEP]}set(t)
{const e=this.current;(t[0]!==e[0]||t[1]!==e[1]||t[2]!==e[2]||this.dirty)&&
(this.gl.stencilOp(t[0],t[1],t[2]),this.current=t,this.dirty=!1);}}class Ye
extends je{getDefaultValue(){return !1}set(t)
{if(t===this.current&&!this.dirty)return;const e=this.gl;t?
e.enable(e.STENCIL_TEST):e.disable(e.STENCIL_TEST),this.current=t,this.dirty=!1;
}}class Je extends je{getDefaultValue(){return [0,1]}set(t){const e=this.current;
(t[0]!==e[0]||t[1]!==e[1]||this.dirty)&&
(this.gl.depthRange(t[0],t[1]),this.current=t,this.dirty=!1);}}class ti extends
je{getDefaultValue(){return !1}set(t){if(t===this.current&&!this.dirty)return;const
e=this.gl;t?
e.enable(e.DEPTH_TEST):e.disable(e.DEPTH_TEST),this.current=t,this.dirty=!1;}}cl
ass ei extends je{getDefaultValue(){return this.gl.LESS}set(t)
{(t!==this.current||this.dirty)&&
(this.gl.depthFunc(t),this.current=t,this.dirty=!1);}}class ii extends
je{getDefaultValue(){return !1}set(t){if(t===this.current&&!this.dirty)return;const
e=this.gl;t?
e.enable(e.BLEND):e.disable(e.BLEND),this.current=t,this.dirty=!1;}}class si

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extends je{getDefault(){const t=this.gl;return [t.ONE,t.ZERO]}set(t){const
e=this.current;(t[0]!==e[0]||t[1]!==e[1]||this.dirty)&&
(this.gl.blendFunc(t[0],t[1]),this.current=t,this.dirty=!1);}}class ai extends
je{getDefault(){return t.aT.transparent}set(t){const e=this.current;
(t.r!==e.r||t.g!==e.g||t.b!==e.b||t.a!==e.a||this.dirty)&&
(this.gl.blendColor(t.r,t.g,t.b,t.a),this.current=t,this.dirty=!1);}}class oi
extends je{getDefault(){return this.gl.FUNC_ADD}set(t)
{(t!==this.current||this.dirty)&&
(this.gl.blendEquation(t),this.current=t,this.dirty=!1);}}class ri extends
je{getDefault(){return !1}set(t){if(t===this.current&&!this.dirty)return;const
e=this.gl;t?
e.enable(e.CULL_FACE):e.disable(e.CULL_FACE),this.current=t,this.dirty=!1;}}clas
s ni extends je{getDefault(){return this.gl.BACK}set(t)
{(t!==this.current||this.dirty)&&
(this.gl.cullFace(t),this.current=t,this.dirty=!1);}}class li extends
je{getDefault(){return this.gl.CCW}set(t){(t!==this.current||this.dirty)&&
(this.gl.frontFace(t),this.current=t,this.dirty=!1);}}class hi extends
je{getDefault(){return null}set(t){(t!==this.current||this.dirty)&&
(this.gl.useProgram(t),this.current=t,this.dirty=!1);}}class ci extends
je{getDefault(){return this.gl.TEXTURE0}set(t){(t!==this.current||this.dirty)&&
(this.gl.activeTexture(t),this.current=t,this.dirty=!1);}}class ui extends
je{getDefault(){const t=this.gl;return
[0,0,t.drawingBufferWidth,t.drawingBufferHeight]}set(t){const e=this.current;
(t[0]!==e[0]||t[1]!==e[1]||t[2]!==e[2]||t[3]!==e[3]||this.dirty)&&
(this.gl.viewport(t[0],t[1],t[2],t[3]),this.current=t,this.dirty=!1);}}class di
extends je{getDefault(){return null}set(t)
{if(t===this.current&&!this.dirty)return;const
e=this.gl;e.bindFramebuffer(e.FRAMEBUFFER,t),this.current=t,this.dirty=!1;}}clas
s _i extends je{getDefault(){return null}set(t)
{if(t===this.current&&!this.dirty)return;const
e=this.gl;e.bindRenderbuffer(e.RENDERBUFFER,t),this.current=t,this.dirty=!1;}}cl
ass pi extends je{getDefault(){return null}set(t)
{if(t===this.current&&!this.dirty)return;const
e=this.gl;e.bindTexture(e.TEXTURE_2D,t),this.current=t,this.dirty=!1;}}class mi
extends je{getDefault(){return null}set(t)
{if(t===this.current&&!this.dirty)return;const
e=this.gl;e.bindBuffer(e.ARRAY_BUFFER,t),this.current=t,this.dirty=!1;}}class fi
extends je{getDefault(){return null}set(t){const
e=this.gl;e.bindBuffer(e.ELEMENT_ARRAY_BUFFER,t),this.current=t,this.dirty=!1;}}
class gi extends je{getDefault(){return null}set(t){var
e;if(t===this.current&&!this.dirty)return;const i=this.gl;Ge(i)?
i.bindVertexArray(t):null==(e=i.getExtension("OES_vertex_array_object"))||void
0===e||e.bindVertexArrayOES(t),this.current=t,this.dirty=!1;}}class vi extends
je{getDefault(){return 4}set(t){if(t===this.current&&!this.dirty)return;const
e=this.gl;e.pixelStorei(e.UNPACK_ALIGNMENT,t),this.current=t,this.dirty=!1;}}cla
ss xi extends je{getDefault(){return !1}set(t)
{if(t===this.current&&!this.dirty)return;const
e=this.gl;e.pixelStorei(e.UNPACK_PREMULTIPLY_ALPHA_WEBGL,t),this.current=t,this.
dirty=!1;}}class yi extends je{getDefault(){return !1}set(t)
{if(t===this.current&&!this.dirty)return;const
e=this.gl;e.pixelStorei(e.UNPACK_FLIP_Y_WEBGL,t),this.current=t,this.dirty=!1;}}
class bi extends je{constructor(t,e)
{super(t),this.context=t,this.parent=e;}getDefault(){return null}}class wi
extends bi{setDirty(){this.dirty=!0;}set(t)
{if(t===this.current&&!this.dirty)return;this.context.bindFramebuffer.set(this.p
arent);const
e=this.gl;e.framebufferTexture2D(e.FRAMEBUFFER,e.COLOR_ATTACHMENT0,e.TEXTURE_2D,
t,0),this.current=t,this.dirty=!1;}}class Ti extends bi{set(t)
{if(t===this.current&&!this.dirty)return;this.context.bindFramebuffer.set(this.p
arent);const
e=this.gl;e.framebufferRenderbuffer(e.FRAMEBUFFER,e.DEPTH_ATTACHMENT,e.RENDERBU
FER,t),this.current=t,this.dirty=!1;}}class Ii extends bi{set(t)

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{if(t===this.current&&!this.dirty)return;this.context.bindFramebuffer.set(this.p
arent);const
e=this.gl;e.framebufferRenderbuffer(e.FRAMEBUFFER,e.DEPTH_STENCIL_ATTACHMENT,e.R
ENDERBUFFER,t),this.current=t,this.dirty=!1;}}class Ei{constructor(t,e,i,s,a)
{this.context=t,this.width=e,this.height=i;const
o=t.gl,r=this.framebuffer=o.createFramebuffer();if(this.colorAttachment=new
wi(t,r),s)this.depthAttachment=a?new Ii(t,r):new Ti(t,r);else if(a)throw new
Error("Stencil cannot be setted without
depth");if(o.checkFramebufferStatus(o.FRAMEBUFFER)!=o.FRAMEBUFFER_COMPLETE)thro
w new Error("Framebuffer is not complete")}destroy(){const
t=this.context.gl,e=this.colorAttachment.get();if(e&&t.deleteTexture(e),this.dep
thAttachment){const
e=this.depthAttachment.get();e&&t.deleteRenderbuffer(e);}t.deleteFramebuffer(thi
s.framebuffer);}}class Si{constructor(t,e,i)
{this.blendFunction=t,this.blendColor=e,this.mask=i;}}Si.Replace=
[1,0],Si.disabled=new Si(Si.Replace,t.aT.transparent,
[!1,!1,!1,!1]),Si.unblended=new Si(Si.Replace,t.aT.transparent,
[!0,!0,!0,!0]),Si.alphaBlended=new Si([1,771],t.aT.transparent,
[!0,!0,!0,!0]);class Ci{constructor(t){var e,i;if(this.gl=t,this.clearColor=new
Ve(this),this.clearDepth=new qe(this),this.clearStencil=new
$e(this),this.colorMask=new We(this),this.depthMask=new
He(this),this.stencilMask=new Xe(this),this.stencilFunc=new
Ke(this),this.stencilOp=new Qe(this),this.stencilTest=new
Ye(this),this.depthRange=new Je(this),this.depthTest=new
ti(this),this.depthFunc=new ei(this),this.blend=new ii(this),this.blendFunc=new
si(this),this.blendColor=new ai(this),this.blendEquation=new
oi(this),this.cullFace=new ri(this),this.cullFaceSide=new
ni(this),this.frontFace=new li(this),this.program=new
hi(this),this.activeTexture=new ci(this),this.viewport=new
ui(this),this.bindFramebuffer=new di(this),this.bindRenderbuffer=new
_i(this),this.bindTexture=new pi(this),this.bindVertexBuffer=new
mi(this),this.bindElementBuffer=new fi(this),this.bindVertexArray=new
gi(this),this.pixelStoreUnpack=new
vi(this),this.pixelStoreUnpackPremultiplyAlpha=new
xi(this),this.pixelStoreUnpackFlipY=new
yi(this),this.extTextureFilterAnisotropic=t.getExtension("EXT_texture_filter_ani
sotropic")||t.getExtension("MOZ_EXT_texture_filter_anisotropic")||t.getExtension
("WEBKIT_EXT_texture_filter_anisotropic"),this.extTextureFilterAnisotropic&&
(this.extTextureFilterAnisotropicMax=t.getParameter(this.extTextureFilterAnisotr
opic.MAX_TEXTURE_MAX_ANISOTROPY_EXT)),this.maxTextureSize=t.getParameter(t.MAX_T
EXTURE_SIZE),Ge(t)){this.HALF_FLOAT=t.HALF_FLOAT;const
s=t.getExtension("EXT_color_buffer_half_float");this.RGBA16F=null!==(
e=t.RGBA16F)&&void 0!==(e?e:null==s?void 0:s.RGBA16F_EXT,this.RGB16F=null!==(
i=t.RGB16F)&&void 0!==(i?i:null==s?void
0:s.RGB16F_EXT,t.getExtension("EXT_color_buffer_float"));}else
{t.getExtension("EXT_color_buffer_half_float"),t.getExtension("OES_texture_half_
float_linear");const
e=t.getExtension("OES_texture_half_float");this.HALF_FLOAT=null==e?void
0:e.HALF_FLOAT_OES;}}setDefault()
{this.unbindVAO(),this.clearColor.setDefault(),this.clearDepth.setDefault(),this
.clearStencil.setDefault(),this.colorMask.setDefault(),this.depthMask.setDefault
(),this.stencilMask.setDefault(),this.stencilFunc.setDefault(),this.stencilOp.se
tDefault(),this.stencilTest.setDefault(),this.depthRange.setDefault(),this.depth
Test.setDefault(),this.depthFunc.setDefault(),this.blend.setDefault(),this.blend
Func.setDefault(),this.blendColor.setDefault(),this.blendEquation.setDefault(),t
his.cullFace.setDefault(),this.cullFaceSide.setDefault(),this.frontFace.setDefau
lt(),this.program.setDefault(),this.activeTexture.setDefault(),this.bindFramebuf
fer.setDefault(),this.pixelStoreUnpack.setDefault(),this.pixelStoreUnpackPremult
iplyAlpha.setDefault(),this.pixelStoreUnpackFlipY.setDefault();}setDirty()
{this.clearColor.dirty=!0,this.clearDepth.dirty=!0,this.clearStencil.dirty=!0,th
is.colorMask.dirty=!0,this.depthMask.dirty=!0,this.stencilMask.dirty=!0,this.ste
ncilFunc.dirty=!0,this.stencilOp.dirty=!0,this.stencilTest.dirty=!0,this.depthRa

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nge.dirty=!0,this.depthTest.dirty=!0,this.depthFunc.dirty=!0,this.blend.dirty=!0
,this.blendFunc.dirty=!0,this.blendColor.dirty=!0,this.blendEquation.dirty=!0,th
is.cullFace.dirty=!0,this.cullFaceSide.dirty=!0,this.frontFace.dirty=!0,this.pro
gram.dirty=!0,this.activeTexture.dirty=!0,this.viewport.dirty=!0,this.bindFrameb
uffer.dirty=!0,this.bindRenderbuffer.dirty=!0,this.bindTexture.dirty=!0,this.bin
dVertexBuffer.dirty=!0,this.bindElementBuffer.dirty=!0,this.bindVertexArray.dirt
y=!0,this.pixelStoreUnpack.dirty=!0,this.pixelStoreUnpackPremultiplyAlpha.dirty=
!0,this.pixelStoreUnpackFlipY.dirty=!0;}createIndexBuffer(t,e){return new
Oe(this,t,e)}createVertexBuffer(t,e,i){return new
Ue(this,t,e,i)}createRenderbuffer(t,e,i){const
s=this.gl,a=s.createRenderbuffer();return
this.bindRenderbuffer.set(a),s.renderbufferStorage(s.RENDERBUFFER,t,e,i),this.bi
ndRenderbuffer.set(null),a}createFramebuffer(t,e,i,s){return new
Ei(this,t,e,i,s)}clear({color:t,depth:e,stencil:i}){const s=this.gl;let a=0;t&&
(a|=s.COLOR_BUFFER_BIT,this.clearColor.set(t),this.colorMask.set([!0,!0,!0,!0]))
,void 0!==e&&
(a|=s.DEPTH_BUFFER_BIT,this.depthRange.set([0,1]),this.clearDepth.set(e),this.de
pthMask.set(!0)),void 0!==i&&
(a|=s.STENCIL_BUFFER_BIT,this.clearStencil.set(i),this.stencilMask.set(255)),s.c
lear(a);}setCullFace(t){!1===t.enable?this.cullFace.set(!1):
(this.cullFace.set(!0),this.cullFaceSide.set(t.mode),this.frontFace.set(t.frontF
ace));}setDepthMode(t){t.func!==this.gl.ALWAYS||t.mask?
(this.depthTest.set(!0),this.depthFunc.set(t.func),this.depthMask.set(t.mask),th
is.depthRange.set(t.range)):this.depthTest.set(!1);}setStencilMode(t)
{t.test.func!==this.gl.ALWAYS||t.mask?
(this.stencilTest.set(!0),this.stencilMask.set(t.mask),this.stencilOp.set([t.fai
l,t.depthFail,t.pass]),this.stencilFunc.set({func:t.test.func,ref:t.ref,mask:t.t
est.mask})):this.stencilTest.set(!1);}setColorMode(e)
{t.aG(e.blendFunction,Si.Replace)?this.blend.set(!1):
(this.blend.set(!0),this.blendFunc.set(e.blendFunction),this.blendColor.set(e.bl
endColor)),this.colorMask.set(e.mask);}createVertexArray(){var t;return
Ge(this.gl)?this.gl.createVertexArray():null===
(t=this.gl.getExtension("OES_vertex_array_object"))||void 0===t?void
0:t.createVertexArrayOES()}deleteVertexArray(t){var e;return Ge(this.gl)?
this.gl.deleteVertexArray(t):null===
(e=this.gl.getExtension("OES_vertex_array_object"))||void 0===e?void
0:e.deleteVertexArrayOES(t)}unbindVAO(){this.bindVertexArray.set(null);}class
Pi{constructor(t,e,i)
{this.func=t,this.mask=e,this.range=i;}}Pi.ReadOnly=!1,Pi.ReadWrite=!0,Pi.disabl
ed=new Pi(519,Pi.ReadOnly,[0,1]);const Di=7680;class Mi{constructor(t,e,i,s,a,o)
{this.test=t,this.ref=e,this.mask=i,this.fail=s,this.depthFail=a,this.pass=o;}}M
i.disabled=new Mi({func:519,mask:0},0,0,Di,Di,Di);class zi{constructor(t,e,i)
{this.enable=t,this.mode=e,this.frontFace=i;}}let Li;function Ai(e,i,s,a,o,r,n)
{const l=e.context,h=l.gl,c=e.useProgram("collisionBox"),u=[];let
d=0,_=0;for(let p=0;p<a.length;p++){const
m=a[p],f=i.getTile(m),g=f.getBucket(s);if(!g)continue;let
v=m.posMatrix;0===o[0]&&0===o[1]||
(v=e.translatePosMatrix(m.posMatrix,f,o,r));const x=n?
g.textCollisionBox:g.iconCollisionBox,y=g.collisionCircleArray;if(y.length>0)
{const
i=t.Z(),s=v;t.aU(i,g.placementInvProjMatrix,e.transform.glCoordMatrix),t.aU(i,i,
g.placementViewportMatrix),u.push({circleArray:y,circleOffset:_,transform:s,invT
ransform:i,coord:m}),d+=y.length/4,_=d;}x&&c.draw(l,h.LINES,Pi.disabled,Mi.disab
led,e.colorModeForRenderPass(),zi.disabled,xe(v,e.transform,f),e.style.map.terra
in&&e.style.map.terrain.getTerrainData(m),s.id,x.layoutVertexBuffer,x.indexBuffe
r,x.segments,null,e.transform.zoom,null,null,x.collisionVertexBuffer);}if(!n||!u
.length)return;const p=e.useProgram("collisionCircle"),m=new
t.aV;m.resize(4*d),m._trim();let f=0;for(const t of u)for(let
e=0;e<t.circleArray.length/4;e++){const
i=4*e,s=t.circleArray[i+0],a=t.circleArray[i+1],o=t.circleArray[i+2],r=t.circleA
rray[i+3];m.emplace(f++,s,a,o,r,0),m.emplace(f++,s,a,o,r,1),m.emplace(f++,s,a,o,
r,2),m.emplace(f++,s,a,o,r,3);}(!Li||Li.length<2*d)&&(Li=function(e){const

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i=2*e,s=new t.aX;s.resize(i),s._trim();for(let t=0;t<i;t++){const
e=6*t;s.uint16[e+0]=4*t+0,s.uint16[e+1]=4*t+1,s.uint16[e+2]=4*t+2,s.uint16[e+3]=
4*t+2,s.uint16[e+4]=4*t+3,s.uint16[e+5]=4*t+0;}return s}(d));const
g=l.createIndexBuffer(Li,!0),v=l.createVertexBuffer(m,t.aW.members,!0);for(const
i of u){const a=
{u_matrix:i.transform,u_inv_matrix:i.invTransform,u_camera_to_center_distance:
(x=e.transform).cameraToCenterDistance,u_viewport_size:
[x.width,x.height]};p.draw(1,h.TRIANGLES,Pi.disabled,Mi.disabled,e.colorModeForR
enderPass(),zi.disabled,a,e.style.map.terrain&&e.style.map.terrain.getTerrainDat
a(i.coord),s.id,v,g,t.S.simpleSegment(0,2*i.circleOffset,i.circleArray.length,i.
circleArray.length/2),null,e.transform.zoom,null,null,null);}var
x;v.destroy(),g.destroy();}zi.disabled=new zi(!1,1029,2305),zi.backCCW=new
zi(!0,1029,2305);const Ri=t.ao(new Float32Array(16));function ki(e,i,s,a,o,r)
{const{horizontalAlign:n,verticalAlign:l}=t.au(e);return new t.P((-
(n-.5)*i/o+a[0])*r,(-(l-.5)*s/o+a[1])*r)}function Fi(e,i,s,a,o,r,n,l,h,c,u)
{const
d=e.text.placedSymbolArray,_=e.text.dynamicLayoutVertexArray,p=e.icon.dynamicLay
outVertexArray,m={};_.clear();for(let p=0;p<d.length;p++){const
f=d.get(p),g=f.hidden||!f.crossTileID||e.allowVerticalPlacement&&!f.placedOrient
ation?null:a[f.crossTileID];if(g){const a=new t.P(f.anchorX,f.anchorY),d=ht(a,s?
n:r,u),p=ct(o.cameraToCenterDistance,d.signedDistanceFromCamera);let
v=t.aj(e.textSizeData,h,f)*p/t.ap;s&&
(v*=e.tilePixelRatio/l);const{width:x,height:y,anchor:b,textOffset:w,textBoxScal
e:T}=g,I=ki(b,x,y,w,T,v),E=s?ht(a.add(I),r,u).point:d.point.add(i?I.rotate(-
o.angle):I),S=e.allowVerticalPlacement&&f.placedOrientation===t.ai.vertical?
Math.PI/2:0;for(let
e=0;e<f.numGlyphs;e++)t.ak(_,E,S);c&&f.associatedIconIndex>=0&&
(m[f.associatedIconIndex]={shiftedAnchor:E,angle:S});}else
wt(f.numGlyphs,_);}if(c){p.clear();const i=e.icon.placedSymbolArray;for(let
e=0;e<i.length;e++){const s=i.get(e);if(s.hidden)wt(s.numGlyphs,p);else {const
i=m[e];if(i)for(let e=0;e<s.numGlyphs;e++)t.ak(p,i.shiftedAnchor,i.angle);else
wt(s.numGlyphs,p);}}e.icon.dynamicLayoutVertexBuffer.updateData(p);}e.text.dynam
icLayoutVertexBuffer.updateData(_);}function Bi(t,e,i){return
i.iconsInText&&e?"symbolTextAndIcon":t?"symbolSDF":"symbolIcon"}function
Oi(e,i,s,a,o,r,n,l,h,c,u,d){const
_=e.context,p=_.gl,m=e.transform,f="map"===l,g="map"===h,v="viewport"!==l&&"poin
t"!==s.layout.get("symbol-placement"),x=f&&!g&&!v,y=!s.layout.get("symbol-sort-
key").isConstant();let b=!1;const
w=e.depthModeForSublayer(0,Pi.ReadOnly),T=s._unevaluatedLayout.hasValue("text-
variable-anchor")||s._unevaluatedLayout.hasValue("text-variable-anchor-
offset"),I=[];for(const l of a){const
a=i.getTile(l),h=a.getBucket(s);if(!h)continue;const u=o?
h.text:h.icon;if(!u||!u.segments.get().length||!u.hasVisibleVertices)continue;co
nst d=u.programConfigurations.get(s.id),_=o||h.sdfIcons,w=o?
h.textSizeData:h.iconSizeData,E=g||0!==m.pitch,S=e.useProgram(Bi(_,o,h),d),C=t.a
h(w,m.zoom),P=e.style.map.terrain&&e.style.map.terrain.getTerrainData(l);let
D,M,z,L,A=
[0,0],R=null;if(o)M=a.glyphAtlasTexture,z=p.LINEAR,D=a.glyphAtlasTexture.size,h.
iconsInText&&
(A=a.imageAtlasTexture.size,R=a.imageAtlasTexture,L=E||e.options.rotating||e.opt
ions.zooming||"composite"===w.kind||"camera"===w.kind?p.LINEAR:p.NEAREST);else
{const t=1!==s.layout.get("icon-
size").constantOr(0)||h.iconsNeedLinear;M=a.imageAtlasTexture,z=_||e.options.rot
ating||e.options.zooming||t||E?
p.LINEAR:p.NEAREST,D=a.imageAtlasTexture.size;}const
k=St(a,l,e.transform.zoom),F=nt(l.posMatrix,g,f,e.transform,k),B=lt(l.posMatrix,
g,f,e.transform,k),O=T&&h.hasTextData(),N="none"!==s.layout.get("icon-text-
fit")&&O&&h.hasIconData();if(v){const t=e.style.map.terrain?
(t,i)>=e.style.map.terrain.getElevation(l,t,i):null,i="map"===s.layout.get("text
-rotation-alignment");dt(h,l.posMatrix,e,o,F,B,g,c,i,t);}const
U=e.translatePosMatrix(l.posMatrix,a,r,n),Z=v||O&&T||N?
Ri:F,G=e.translatePosMatrix(B,a,r,n,!0),j=_&&O!==s.paint.get(o?"text-halo-

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width": "icon-halo-width").constantOr(1);let V;V=_?h.iconsInText?
Re(w.kind,C,x,g,e,U,Z,G,D,A):Ae(w.kind,C,x,g,e,U,Z,G,o,D,!0):Le(w.kind,C,x,g,e,U
,Z,G,o,D);const q=
{program:S,buffers:u,uniformValues:V,atlasTexture:M,atlasTextureIcon:R,atlasInte
rpolation:z,atlasInterpolationIcon:L,isSDF:_,hasHalo:j};if(y&&h.canOverlap)
{b=!0;const e=u.segments.get();for(const i of e)I.push({segments:new
t.S([i]),sortKey:i.sortKey,state:q,terrainData:P});}else
I.push({segments:u.segments,sortKey:0,state:q,terrainData:P});}b&&I.sort(((t,e)=
>t.sortKey-e.sortKey));for(const t of I){const
i=t.state;if(_.activeTexture.set(p.TEXTURE0),i.atlasTexture.bind(i.atlasInterpol
ation,p.CLAMP_TO_EDGE),i.atlasTextureIcon&&
(_.activeTexture.set(p.TEXTURE1),i.atlasTextureIcon&&i.atlasTextureIcon.bind(i.a
tlasInterpolationIcon,p.CLAMP_TO_EDGE)),i.isSDF){const
a=i.uniformValues;i.hasHalo&&
(a.u_is_halo=1,Ni(i.buffers,t.segments,s,e,i.program,w,u,d,a,t.terrainData)),a.u
_is_halo=0;}Ni(i.buffers,t.segments,s,e,i.program,w,u,d,i.uniformValues,t.terrai
nData);}}function Ni(t,e,i,s,a,o,r,n,l,h){const
c=s.context;a.draw(c,c.gl.TRIANGLES,o,r,n,zi.disabled,l,h,i.id,t.layoutVertexBuf
fer,t.indexBuffer,e,i.paint,s.transform.zoom,t.programConfigurations.get(i.id),t
.dynamicLayoutVertexBuffer,t.opacityVertexBuffer);}function Ui(t,e,i,s,a)
{if(!i||!s||!s.imageAtlas)return;const o=s.imageAtlas.patternPositions;let
r=o[i.to.toString()],n=o[i.from.toString()];if(!r&&n&&(r=n),!n&&r&&(n=r),!r||!n)
{const
t=a.getPaintProperty(e);r=o[t],n=o[t];}r&&n&&t.setConstantPatternPositions(r,n);
}function Zi(t,e,i,s,a,o,r){const n=t.context.gl,l="fill-
pattern",h=i.paint.get(l),c=h&&h.constantOr(1),u=i.getCrossfadeParameters();let
d,_p,m,f;r?(_=c&&!i.getPaintProperty("fill-outline-
color")?"fillOutlinePattern":"fillOutline",d=n.LINES):
(_=c?"fillPattern":"fill",d=n.TRIANGLES);const g=h.constantOr(null);for(const h
of s){const s=e.getTile(h);if(c&&!s.patternsLoaded())continue;const
v=s.getBucket(i);if(!v)continue;const
x=v.programConfigurations.get(i.id),y=t.useProgram(_,x),b=t.style.map.terrain&&t
.style.map.terrain.getTerrainData(h);c&&
(t.context.activeTexture.set(n.TEXTURE0),s.imageAtlasTexture.bind(n.LINEAR,n.CLA
MP_TO_EDGE),x.updatePaintBuffers(u)),Ui(x,l,g,s,i);const w=b?
h:null,T=t.translatePosMatrix(w?w.posMatrix:h.posMatrix,s,i.paint.get("fill-
translate"),i.paint.get("fill-translate-anchor"));if(r)
{m=v.indexBuffer2,f=v.segments2;const e=
[n.drawingBufferWidth,n.drawingBufferHeight];p="fillOutlinePattern"===_&&c?
ge(T,t,u,s,e):fe(T,e);}else m=v.indexBuffer,f=v.segments,p=c?
me(T,t,u,s):pe(T);y.draw(t.context,d,a,t.stencilModeForClipping(h),o,zi.disabled
,p,b,i.id,v.layoutVertexBuffer,m,f,i.paint,t.transform.zoom,x);}function
Gi(t,e,i,s,a,o,r){const n=t.context,l=n.gl,h="fill-extrusion-
pattern",c=i.paint.get(h),u=c.constantOr(1),d=i.getCrossfadeParameters(),_=i.pai
nt.get("fill-extrusion-opacity"),p=c.constantOr(null);for(const c of s){const
s=e.getTile(c),m=s.getBucket(i);if(!m)continue;const
f=t.style.map.terrain&&t.style.map.terrain.getTerrainData(c),g=m.programConfigur
ations.get(i.id),v=t.useProgram(u?"fillExtrusionPattern":"fillExtrusion",g);u&&
(t.context.activeTexture.set(l.TEXTURE0),s.imageAtlasTexture.bind(l.LINEAR,l.CLA
MP_TO_EDGE),g.updatePaintBuffers(d)),Ui(g,h,p,s,i);const
x=t.translatePosMatrix(c.posMatrix,s,i.paint.get("fill-extrusion-
translate"),i.paint.get("fill-extrusion-translate-anchor")),y=i.paint.get("fill-
extrusion-vertical-gradient"),b=u?
_e(x,t,y,_,c,d,s):de(x,t,y,_)v.draw(n,n.gl.TRIANGLES,a,o,r,zi.backCCW,b,f,i.id,
m.layoutVertexBuffer,m.indexBuffer,m.segments,i.paint,t.transform.zoom,g,t.style
.map.terrain&&m.centroidVertexBuffer);}function ji(t,e,i,s,a,o,r){const
n=t.context,l=n.gl,h=i.fbo;if(!h)return;const
c=t.useProgram("hillshade"),u=t.style.map.terrain&&t.style.map.terrain.getTerrai
nData(e);n.activeTexture.set(l.TEXTURE0),l.bindTexture(l.TEXTURE_2D,h.colorAttac
hment.get()),c.draw(n,l.TRIANGLES,a,o,r,zi.disabled,((t,e,i,s)=>{const
a=i.paint.get("hillshade-shadow-color"),o=i.paint.get("hillshade-highlight-
color"),r=i.paint.get("hillshade-accent-color");let n=i.paint.get("hillshade-

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illumination-direction" )*(Math.PI/180); "viewport"===i.paint.get("hillshade-
illumination-anchor" )&&(n-=t.transform.angle);const l=!t.options.moving;return
{u_matrix:s?
s.posMatrix:t.transform.calculatePosMatrix(e.tileID.toUnwrapped(),l),u_image:0,u
_latrange:Te(0,e.tileID),u_light:[i.paint.get("hillshade-
exaggeration"),n],u_shadow:a,u_highlight:o,u_accent:r}}(t,i,s,u?
e:null),u,s.id,t.rasterBoundsBuffer,t.quadTriangleIndexBuffer,t.rasterBoundsSegm
ents);}function Vi(e,i,s,a,o,r){const n=e.context,l=n.gl,h=i.dem;if(h&&h.data)
{const
c=h.dim,u=h.stride,d=h.getPixels();if(n.activeTexture.set(l.TEXTURE1),n.pixelSto
reUnpackPremultiplyAlpha.set(!1),i.demTexture=i.demTexture|e.getTileTexture(u),
i.demTexture){const t=i.demTexture;t.update(d,
{premultiply:!1}),t.bind(l.NEAREST,l.CLAMP_TO_EDGE);}else i.demTexture=new
x(n,d,l.RGBA,
{premultiply:!1}),i.demTexture.bind(l.NEAREST,l.CLAMP_TO_EDGE);n.activeTexture.s
et(l.TEXTURE0);let _=i.fbo;if(!_){const t=new x(n,
{width:c,height:c,data:null},l.RGBA);t.bind(l.LINEAR,l.CLAMP_TO_EDGE),_=i.fbo=n.
createFramebuffer(c,c,!0,!1),_.colorAttachment.set(t.texture);}n.bindFramebuffer
.set(_.framebuffer),n.viewport.set([0,0,c,c]),e.useProgram("hillshadePrepare").d
raw(n,l.TRIANGLES,a,o,r,zi.disabled,((e,i)=>{const s=i.stride,a=t.Z();return
t.aS(a,0,t.N,-t.N,0,0,1),t.$(a,a,[0,-t.N,0]),{u_matrix:a,u_image:1,u_dimension:
[s,s],u_zoom:e.overscaledZ,u_unpack:i.getUnpackVector()}})
(i.tileID,h),null,s.id,e.rasterBoundsBuffer,e.quadTriangleIndexBuffer,e.rasterBo
undsSegments),i.needsHillshadePrepare=!1;}}function qi(e,i,s,a,o,r){const
n=a.paint.get("raster-fade-duration");if(!r&&n>0){const a=t.h.now(),r=(a-
e.timeAdded)/n,l=i?(a-
i.timeAdded)/n:-1,h=s.getSource(),c=o.coveringZoomLevel({tileSize:h.tileSize,rou
ndZoom:h.roundZoom}),u=!i|Math.abs(i.tileID.overscaledZ-
c)>Math.abs(e.tileID.overscaledZ-c),d=u&&e.refreshedUponExpiration?1:t.ad(u?r:l-
1,0,1);return e.refreshedUponExpiration&&r>=1&&(e.refreshedUponExpiration=!1),i?
{opacity:1,mix:l-d}:{opacity:d,mix:0}}return {opacity:1,mix:0}}const $i=new
t.aT(1,0,0,1),Wi=new t.aT(0,1,0,1),Hi=new t.aT(0,0,1,1),Xi=new
t.aT(1,0,1,1),Ki=new t.aT(0,1,1,1);function Qi(t,e,i,s)
{Ji(t,0,e+i/2,t.transform.width,i,s);}function Yi(t,e,i,s){Ji(t,e-
i/2,0,i,t.transform.height,s);}function Ji(t,e,i,s,a,o){const
r=t.context,n=r.gl;n.enable(n.SCISSOR_TEST),n.scissor(e*t.pixelRatio,i*t.pixelRa
tio,s*t.pixelRatio,a*t.pixelRatio),r.clear({color:o}),n.disable(n.SCISSOR_TEST);
}function ts(e,i,s){const
a=e.context,o=a.gl,r=s.posMatrix,n=e.useProgram("debug"),l=Pi.disabled,h=Mi.disa
bled,c=e.colorModeForRenderPass(),u="$debug",d=e.style.map.terrain&&e.style.map.
terrain.getTerrainData(s);a.activeTexture.set(o.TEXTURE0);const
_=i.getTileByID(s.key).latestRawTileData,p=Math.floor((_&&_.byteLength||0)/1024)
,m=i.getTile(s).tileSize,f=512/Math.min(m,512)*
(s.overscaledZ/e.transform.zoom)*.5;let
g=s.canonical.toString();s.overscaledZ!==s.canonical.z&&(g+=`=>
${s.overscaledZ}`),function(t,e){t.initDebugOverlayCanvas();const
i=t.debugOverlayCanvas,s=t.context.gl,a=t.debugOverlayCanvas.getContext("2d");a.
clearRect(0,0,i.width,i.height),a.shadowColor="white",a.shadowBlur=2,a.lineWidth
=1.5,a.strokeStyle="white",a.textBaseline="top",a.font="bold 36px Open Sans,
sans-
serif",a.fillText(e,5,5),a.strokeText(e,5,5),t.debugOverlayTexture.update(i),t.d
ebugOverlayTexture.bind(s.LINEAR,s.CLAMP_TO_EDGE);} (e,`${g}
${p}kB`),n.draw(a,o.TRIANGLES,l,h,Si.alphaBlended,zi.disabled, ye(r,t.aT.transpar
ent,f),null,u,e.debugBuffer,e.quadTriangleIndexBuffer,e.debugSegments),n.draw(a,
o.LINE_STRIP,l,h,c,zi.disabled, ye(r,t.aT.red),d,u,e.debugBuffer,e.tileBorderInde
xBuffer,e.debugSegments);}function es(t,e,i){const
s=t.context,a=s.gl,o=t.colorModeForRenderPass(),r=new
Pi(a.LEQUAL,Pi.ReadWrite,t.depthRangeFor3D),n=t.useProgram("terrain"),l=e.getTer
rainMesh();s.bindFramebuffer.set(null),s.viewport.set([0,0,t.width,t.height]);fo
r(const h of i){const
i=t.renderToTexture.getTexture(h),c=e.getTerrainData(h.tileID);s.activeTexture.s
et(a.TEXTURE0),a.bindTexture(a.TEXTURE_2D,i.texture);const u=

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{u_matrix:t.transform.calculatePosMatrix(h.tileID.toUnwrapped()),u_texture:0,u_e
le_delta:e.getMeshFrameDelta(t.transform.zoom)};n.draw(s,a.TRIANGLES,r,Mi.disabl
ed,o,zi.backCCW,u,c,"terrain",l.vertexBuffer,l.indexBuffer,l.segments);}class
is{constructor(e,i){this.context=new Ci(e),this.transform=i,this._tileTextures=
{}},this.terrainFacilitator=
{dirty:!0,matrix:t.Z(),renderTime:0},this.setup(),this.numSublayers=K.maxUnderzo
oming+K.maxOverzooming+1,this.depthEpsilon=1/Math.pow(2,16),this.crossTileSymbol
Index=new Yt;}resize(t,e,i)
{if(this.width=Math.floor(t*i),this.height=Math.floor(e*i),this.pixelRatio=i,thi
s.context.viewport.set([0,0,this.width,this.height]),this.style)for(const t of
this.style._order)this.style._layers[t].resize();}setup(){const
e=this.context,i=new
t.a_;i.emplaceBack(0,0),i.emplaceBack(t.N,0),i.emplaceBack(0,t.N),i.emplaceBack(
t.N,t.N),this.tileExtentBuffer=e.createVertexBuffer(i,ae.members),this.tileExten
tSegments=t.S.simpleSegment(0,0,4,2);const s=new
t.a_;s.emplaceBack(0,0),s.emplaceBack(t.N,0),s.emplaceBack(0,t.N),s.emplaceBack(
t.N,t.N),this.debugBuffer=e.createVertexBuffer(s,ae.members),this.debugSegments=
t.S.simpleSegment(0,0,4,5);const a=new
t.V;a.emplaceBack(0,0,0,0),a.emplaceBack(t.N,0,t.N,0),a.emplaceBack(0,t.N,0,t.N)
,a.emplaceBack(t.N,t.N,t.N,t.N),this.rasterBoundsBuffer=e.createVertexBuffer(a,O
.members),this.rasterBoundsSegments=t.S.simpleSegment(0,0,4,2);const o=new
t.a_;o.emplaceBack(0,0),o.emplaceBack(1,0),o.emplaceBack(0,1),o.emplaceBack(1,1)
,this.viewportBuffer=e.createVertexBuffer(o,ae.members),this.viewportSegments=t.
S.simpleSegment(0,0,4,2);const r=new
t.a$;r.emplaceBack(0),r.emplaceBack(1),r.emplaceBack(3),r.emplaceBack(2),r.empla
ceBack(0),this.tileBorderIndexBuffer=e.createIndexBuffer(r);const n=new
t.b0;n.emplaceBack(0,1,2),n.emplaceBack(2,1,3),this.quadTriangleIndexBuffer=e.cr
eateIndexBuffer(n);const l=this.context.gl;this.stencilClearMode=new
Mi({func:l.ALWAYS,mask:0},0,255,l.ZERO,l.ZERO,l.ZERO);}clearStencil(){const
e=this.context,i=e.gl;this.nextStencilID=1,this.currentStencilSource=void
0;const s=t.Z();t.aS(s,0,this.width,this.height,0,0,1),t.a0(s,s,
[i.drawingBufferWidth,i.drawingBufferHeight,0]),this.useProgram("clippingMask").
draw(e,i.TRIANGLES,Pi.disabled,this.stencilClearMode,Si.disabled,zi.disabled,be(
s),null,"$clipping",this.viewportBuffer,this.quadTriangleIndexBuffer,this.viewpo
rtSegments);}_renderTileClippingMasks(t,e)
{if(this.currentStencilSource===t.source||!t.isTileClipped()||!e||!e.length)retu
rn;this.currentStencilSource=t.source;const
i=this.context,s=i.gl;this.nextStencilID+e.length>256&&this.clearStencil(),i.set
ColorMode(Si.disabled),i.setDepthMode(Pi.disabled);const
a=this.useProgram("clippingMask");this._tileClippingMaskIDs={};for(const t of e)
{const
e=this._tileClippingMaskIDs[t.key]=this.nextStencilID++,o=this.style.map.terrain
&&this.style.map.terrain.getTerrainData(t);a.draw(i,s.TRIANGLES,Pi.disabled,new
Mi({func:s.ALWAYS,mask:0},e,255,s.KEEP,s.KEEP,s.REPLACE),Si.disabled,zi.disabled
,be(t.posMatrix),o,"$clipping",this.tileExtentBuffer,this.quadTriangleIndexBuffe
r,this.tileExtentSegments);}stencilModeFor3D(){this.currentStencilSource=void
0,this.nextStencilID+1>256&&this.clearStencil();const
t=this.nextStencilID++,e=this.context.gl;return new
Mi({func:e.NOTEQUAL,mask:255},t,255,e.KEEP,e.KEEP,e.REPLACE)}stencilModeForClipp
ing(t){const e=this.context.gl;return new
Mi({func:e.EQUAL,mask:255},this._tileClippingMaskIDs[t.key],0,e.KEEP,e.KEEP,e.RE
PLACE)}stencilConfigForOverlap(t){const
e=this.context.gl,i=t.sort((t,e)=>e.overscaledZ-t.overscaledZ),s=i[i.length-
1].overscaledZ,a=i[0].overscaledZ-s+1;if(a>1){this.currentStencilSource=void
0,this.nextStencilID+a>256&&this.clearStencil();const t={};for(let
i=0;i<a;i++)t[i+s]=new
Mi({func:e.GEQUAL,mask:255},i+this.nextStencilID,255,e.KEEP,e.KEEP,e.REPLACE);re
turn this.nextStencilID+=a,[t,i]}return
[{{s}:Mi.disabled},i]}colorModeForRenderPass(){const
e=this.context.gl;if(this._showOverdrawInspector){const i=1/8;return new
Si([e.CONSTANT_COLOR,e.ONE],new t.aT(i,i,i,0),[!0,!0,!0,!0])}return
"opaque"===this.renderPass?

```



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Si.unblended:Si.alphaBlended}depthModeForSublayer(t,e,i)
{if(!this.opaquePassEnabledForLayer())return Pi.disabled;const s=1-
((1+this.currentLayer)*this.numSublayers+t)*this.depthEpsilon;return new
Pi(i||this.context.gl.LEQUAL,e,[s,s])}opaquePassEnabledForLayer(){return
this.currentLayer<this.opaquePassCutoff}render(e,i)
{this.style=e,this.options=i,this.lineAtlas=e.lineAtlas,this.imageManager=e.imag
eManager,this.glyphManager=e.glyphManager,this.symbolFadeChange=e.placement.symb
olFadeChange(t.h.now()),this.imageManager.beginFrame();const
s=this.style._order,a=this.style.sourceCaches,o={},r={},n={};for(const t in a)
{const
e=a[t];e.used&&e.prepare(this.context),o[t]=e.getVisibleCoordinates(),r[t]=o[t].
slice().reverse(),n[t]=e.getVisibleCoordinates(!0).reverse();}this.opaquePassCut
off=1/0;for(let t=0;t<s.length;t++)if(this.style._layers[s[t]].is3D())
{this.opaquePassCutoff=t;break}if(this.renderToTexture)
{this.renderToTexture.prepareForRender(this.style,this.transform.zoom),this.opaq
uePassCutoff=0;const
e=this.style.map.terrain.sourceCache.tilesAfterTime(this.terrainFacilitator.rend
erTime);
(this.terrainFacilitator.dirty||!t.bl(this.terrainFacilitator.matrix,this.transf
orm.projMatrix)||e.length)&&
(t.b2(this.terrainFacilitator.matrix,this.transform.projMatrix),this.terrainFaci
litator.renderTime=Date.now(),this.terrainFacilitator.dirty=!1,function(e,i)
{const s=e.context,a=s.gl,o=Si.unblended,r=new Pi(a.LEQUAL,Pi.ReadWrite,
[0,1]),n=i.getTerrainMesh(),l=i.sourceCache.getRenderableTiles(),h=e.useProgram(
"terrainDepth");s.bindFramebuffer.set(i.getFramebuffer("depth").framebuffer),s.v
iewport.set([0,0,e.width/devicePixelRatio,e.height/devicePixelRatio]),s.clear({c
olor:t.aT.transparent,depth:1});for(const t of l){const
l=i.getTerrainData(t.tileID),c=
{u_matrix:e.transform.calculatePosMatrix(t.tileID.toUnwrapped()),u_ele_delta:i.g
etMeshFrameDelta(e.transform.zoom)};h.draw(s,a.TRIANGLES,r,Mi.disabled,o,zi.back
CCW,c,l,"terrain",n.vertexBuffer,n.indexBuffer,n.segments);s.bindFramebuffer.se
t(null),s.viewport.set([0,0,e.width,e.height]);}
(this,this.style.map.terrain),function(e,i){const
s=e.context,a=s.gl,o=Si.unblended,r=new Pi(a.LEQUAL,Pi.ReadWrite,
[0,1]),n=i.getTerrainMesh(),l=i.getCoordsTexture(),h=i.sourceCache.getRenderable
Tiles(),c=e.useProgram("terrainCoords");s.bindFramebuffer.set(i.getFramebuffer("
coords").framebuffer),s.viewport.set([0,0,e.width/devicePixelRatio,e.height/devi
cePixelRatio]),s.clear({color:t.aT.transparent,depth:1}),i.coordsIndex=
[];for(const t of h){const
h=i.getTerrainData(t.tileID);s.activeTexture.set(a.TEXTURE0),a.bindTexture(a.TEX
TURE_2D,l.texture);const u=
{u_matrix:e.transform.calculatePosMatrix(t.tileID.toUnwrapped()),u_terrain_coord
s_id:(255-
i.coordsIndex.length)/255,u_texture:0,u_ele_delta:i.getMeshFrameDelta(e.transfor
m.zoom)};c.draw(s,a.TRIANGLES,r,Mi.disabled,o,zi.backCCW,u,h,"terrain",n.vertexB
uffer,n.indexBuffer,n.segments),i.coordsIndex.push(t.tileID.key);s.bindFramebuf
fer.set(null),s.viewport.set([0,0,e.width,e.height]);}
(this,this.style.map.terrain));}this.renderPass="offscreen";for(const t of s)
{const
e=this.style._layers[t];if(!e.hasOffscreenPass()||e.isHidden(this.transform.zoom
))continue;const i=r[e.source];
("custom"===e.type||i.length)&&this.renderLayer(this,a[e.source],e,i);}if(this.c
ontext.bindFramebuffer.set(null),this.context.clear({color:i.showOverdrawInspect
or?
t.aT.black:t.aT.transparent,depth:1}),this.clearStencil(),this._showOverdrawInsp
ector=i.showOverdrawInspector,this.depthRangeFor3D=[0,1-
(e._order.length+2)*this.numSublayers*this.depthEpsilon],!this.renderToTexture)f
or(this.renderPass="opaque",this.currentLayer=s.length-
1;this.currentLayer>=0;this.currentLayer--){const
t=this.style._layers[s[this.currentLayer]],e=a[t.source],i=o[t.source];this._ren
derTileClippingMasks(t,i),this.renderLayer(this,e,t,i);}for(this.renderPass="tra
nslucent",this.currentLayer=0;this.currentLayer<s.length;this.currentLayer++)

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{const
t=this.style._layers[s[this.currentLayer]],e=a[t.source];if(this.renderToTexture
&&this.renderToTexture.renderLayer(t))continue;const i=("symbol"===t.type?n:r)
[t.source];this._renderTileClippingMasks(t,o[t.source]),this.renderLayer(this,e,
t,i);}if(this.options.showTileBoundaries){const t=function(t,e){let i=null;const
s=Object.values(t._layers).flatMap((i=>i.source&&!i.isHidden(e)?
[t.sourceCaches[i.source]]:
[])),a=s.filter((t=>"vector"===t.getSource().type)),o=s.filter((t=>"vector"!==t.
getSource().type)),r=t=>{(!i||i.getSource().maxzoom<t.getSource().maxzoom)&&
(i=t)};return a.forEach((t=>r(t))),i||o.forEach((t=>r(t))),i}
(this.style,this.transform.zoom);t&&function(t,e,i){for(let
s=0;s<i.length;s++)ts(t,e,i[s]);}
(this,t,t.getVisibleCoordinates());}this.options.showPadding&&function(t){const
e=t.transform.padding;Qi(t,t.transform.height-
(e.top||0),3,$i),Qi(t,e.bottom||0,3,Wi),Yi(t,e.left||0,3,Hi),Yi(t,t.transform.wi
dth-(e.right||0),3,Xi);const i=t.transform.centerPoint;!function(t,e,i,s)
{Ji(t,e-1,i-10,2,20,s),Ji(t,e-10,i-1,20,2,s);} (t,i.x,t.transform.height-
i.y,Ki);} (this),this.context.setDefault();}renderLayer(e,i,s,a)
{if(!s.isHidden(this.transform.zoom)&&
("background"===s.type||"custom"===s.type||a||
[]).length))switch(this.id=s.id,s.type){case"symbol":!function(e,i,s,a,o)
{if("translucent"!==e.renderPass)return;const
r=Mi.disabled,n=e.colorModeForRenderPass();(s._unevaluatedLayout.hasValue("text-
variable-anchor")||s._unevaluatedLayout.hasValue("text-variable-anchor-
offset"))&&function(e,i,s,a,o,r,n){const
l=i.transform,h="map"===o,c="map"===r;for(const o of e){const
e=a.getTile(o),r=e.getBucket(s);if(!r||!r.text||!r.text.segments.get().length)co
ntinue;const
u=t.ah(r.textSizeData,l.zoom),d=St(e,l,i.transform.zoom),_=nt(o.posMatrix,c,h,i.
transform,d),p="none"!==s.layout.get("icon-text-fit")&&r.hasIconData();if(u)
{const t=Math.pow(2,l.zoom-
e.tileID.overscaledZ);Fi(r,h,c,n,l,_,o.posMatrix,t,u,p,i.style.map.terrain?
(t,e)=>i.style.map.terrain.getElevation(o,t,e):null);}}}
(a,e,s,i,s.layout.get("text-rotation-alignment"),s.layout.get("text-pitch-
alignment"),o),0!==s.paint.get("icon-
opacity").constantOr(1)&&Oi(e,i,s,a,!l,s.paint.get("icon-
translate"),s.paint.get("icon-translate-anchor"),s.layout.get("icon-rotation-
alignment"),s.layout.get("icon-pitch-alignment"),s.layout.get("icon-keep-
upright"),r,n),0!==s.paint.get("text-
opacity").constantOr(1)&&Oi(e,i,s,a,!0,s.paint.get("text-
translate"),s.paint.get("text-translate-anchor"),s.layout.get("text-rotation-
alignment"),s.layout.get("text-pitch-alignment"),s.layout.get("text-keep-
upright"),r,n),i.map.showCollisionBoxes&&(Ai(e,i,s,a,s.paint.get("text-
translate"),s.paint.get("text-translate-
anchor"),!0),Ai(e,i,s,a,s.paint.get("icon-translate"),s.paint.get("icon-
translate-anchor"),!1));}
(e,i,s,a,this.style.placement.variableOffsets);break;case"circle":!function(e,i,
s,a){if("translucent"!==e.renderPass)return;const o=s.paint.get("circle-
opacity"),r=s.paint.get("circle-stroke-width"),n=s.paint.get("circle-stroke-
opacity"),l=!s.layout.get("circle-sort-
key").isConstant();if(0===o.constantOr(1)&&
(0===r.constantOr(1)||0===n.constantOr(1)))return;const
h=e.context,c=h.gl,u=e.depthModeForSublayer(0,Pi.ReadOnly),d=Mi.disabled,_=e.col
orModeForRenderPass(),p=[];for(let o=0;o<a.length;o++){const
r=a[o],n=i.getTile(r),h=n.getBucket(s);if(!h)continue;const
c=h.programConfigurations.get(s.id),u=e.useProgram("circle",c),d=h.layoutVertexB
uffer,_=h.indexBuffer,m=e.style.map.terrain&&e.style.map.terrain.getTerrainData(
r),f=
{programConfiguration:c,program:u,layoutVertexBuffer:d,indexBuffer:_,uniformValu
es:ve(e,r,n,s),terrainData:m};if(l){const e=h.segments.get();for(const i of
e)p.push({segments:new t.S([i]),sortKey:i.sortKey,state:f});}else
p.push({segments:h.segments,sortKey:0,state:f});}l&&p.sort(((t,e)=>t.sortKey-

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e.sortKey));for(const t of p)
{const{programConfiguration:i,program:a,layoutVertexBuffer:o,indexBuffer:r,uniformValues:n,terrainData:l}=t.state;a.draw(h,c.TRIANGLES,u,d,_,zi.disabled,n,l,s.id,o,r,t.segments,s.paint,e.transform.zoom,i);}}
(e,i,s,a);break;case"heatmap":!function(e,i,s,a){if(0!==s.paint.get("heatmap-opacity"))if("offscreen"===e.renderPass){const
o=e.context,r=o.gl,n=Mi.disabled,l=new Si([r.ONE,r.ONE],t.aT.transparent,[!0,!0,!0,!0]);!function(t,e,i){const
s=t.gl;t.activeTexture.set(s.TEXTURE1),t.viewport.set([0,0,e.width/4,e.height/4]);let
a=i.heatmapFbo;if(a)s.bindTexture(s.TEXTURE_2D,a.colorAttachment.get()),t.bindFramebuffer.set(a.framebuffer);else {const
o=s.createTexture();s.bindTexture(s.TEXTURE_2D,o),s.texParameteri(s.TEXTURE_2D,s.TEXTURE_WRAP_S,s.CLAMP_TO_EDGE),s.texParameteri(s.TEXTURE_2D,s.TEXTURE_WRAP_T,s.CLAMP_TO_EDGE),s.texParameteri(s.TEXTURE_2D,s.TEXTURE_MIN_FILTER,s.LINEAR),s.texParameteri(s.TEXTURE_2D,s.TEXTURE_MAG_FILTER,s.LINEAR),a=i.heatmapFbo=t.createFramebuffer(e.width/4,e.height/4,!1,!1),function(t,e,i,s){var a,o;const
r=t.gl,n=null!==(a=t.HALF_FLOAT)&&void 0!==a?a:r.UNSIGNED_BYTE,l=null!==(o=t.RGBA16F)&&void 0!==o?
o:r.RGBA;r.texImage2D(r.TEXTURE_2D,0,l,e.width/4,e.height/4,0,r.RGBA,n,null),s.colorAttachment.set(i);}(t,e,o,a);}}
(o,e,s),o.clear({color:t.aT.transparent});for(let t=0;t<a.length;t++){const
h=a[t];if(i.hasRenderableParent(h))continue;const
c=i.getTile(h),u=c.getBucket(s);if(!u)continue;const
d=u.programConfigurations.get(s.id),_=e.useProgram("heatmap",d),
{zoom:p}=e.transform;_.draw(o,r.TRIANGLES,Pi.disabled,n,l,zi.disabled,we(h.posMatrix,c,p,s.paint.get("heatmap-intensity")),null,s.id,u.layoutVertexBuffer,u.indexBuffer,u.segments,s.paint,e.transform.zoom,d);o.viewport.set([0,0,e.width,e.height]);}else
"translucent"===e.renderPass&&
(e.context.setColorMode(e.colorModeForRenderPass()),function(e,i){const
s=e.context,a=s.gl,o=i.heatmapFbo;if(!o)return;s.activeTexture.set(a.TEXTURE0),a.bindTexture(a.TEXTURE_2D,o.colorAttachment.get()),s.activeTexture.set(a.TEXTURE1);let r=i.colorRampTexture;r||(r=i.colorRampTexture=new
x(s,i.colorRamp,a.RGBA)),r.bind(a.LINEAR,a.CLAMP_TO_EDGE),e.useProgram("heatmapTexture").draw(s,a.TRIANGLES,Pi.disabled,Mi.disabled,e.colorModeForRenderPass(),zi.disabled,((e,i,s,a)=>{const o=t.Z();t.aS(o,0,e.width,e.height,0,0,1);const
r=e.context.gl;return {u_matrix:o,u_world:
[r.drawingBufferWidth,r.drawingBufferHeight],u_image:0,u_color_ramp:1,u_opacity:
i.paint.get("heatmap-opacity")}})
(e,i),null,i.id,e.viewportBuffer,e.quadTriangleIndexBuffer,e.viewportSegments,i.paint,e.transform.zoom);}(e,s));}(e,i,s,a);break;case"line":!function(e,i,s,a)
{if("translucent"!==e.renderPass)return;const o=s.paint.get("line-opacity"),r=s.paint.get("line-width");if(0===o.constantOr(1)||0===r.constantOr(1))return;const
n=e.depthModeForSublayer(0,Pi.ReadOnly),l=e.colorModeForRenderPass(),h=s.paint.get("line-dasharray"),c=s.paint.get("line-pattern"),u=c.constantOr(1),d=s.paint.get("line-gradient"),_=s.getCrossfadeParameters(),p=u?"linePattern":h?"lineSDF":d?"lineGradient":"line",m=e.context,f=m.gl;let g=!0;for(const o of a){const
a=i.getTile(o);if(u&&!a.patternsLoaded())continue;const
r=a.getBucket(s);if(!r)continue;const
v=r.programConfigurations.get(s.id),y=e.context.program.get(),b=e.useProgram(p,v),w=g|b.program!==y,T=e.style.map.terrain&&e.style.map.terrain.getTerrainData(o),I=c.constantOr(null);if(I&&a.imageAtlas){const
t=a.imageAtlas,e=t.patternPositions[I.toString()],i=t.patternPositions[I.toString()];e&&i&&v.setConstantPatternPositions(e,i);}const E=T?o:null,S=u?
Se(e,a,s,_,E):h?Ce(e,a,s,h,_,E):d?
Ee(e,a,s,r.lineClipsArray.length,E):Ie(e,a,s,E);if(u)m.activeTexture.set(f.TEXTURE0),a.imageAtlasTexture.bind(f.LINEAR,f.CLAMP_TO_EDGE),v.updatePaintBuffers(_);
else if(h&&
(w|e.lineAtlas.dirty))m.activeTexture.set(f.TEXTURE0),e.lineAtlas.bind(m);else

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if(d){const a=r.gradients[s.id];let
n=a.texture;if(s.gradientVersion!==a.version){let l=256;if(s.stepInterpolant)
{const s=i.getSource().maxzoom,a=o.canonical.z===s?
Math.ceil(1<e.transform.maxZoom-
o.canonical.z):1;l=t.ad(t.aY(r.maxLineLength/t.N*1024*a),256,m.maxTextureSize);}
a.gradient=t.aZ({expression:s.gradientExpression(),evaluationKey:"lineProgress",
resolution:l,image:a.gradient||void 0,clips:r.lineClipsArray}),a.texture?
a.texture.update(a.gradient):a.texture=new
x(m,a.gradient,f.RGBA),a.version=s.gradientVersion,n=a.texture;}m.activeTexture.
set(f.TEXTURE0),n.bind(s.stepInterpolant?
f.NEAREST:f.LINEAR,f.CLAMP_TO_EDGE);}b.draw(m,f.TRIANGLES,n,e.stencilModeForClip
ping(o),l,zi.disabled,S,T,s.id,r.layoutVertexBuffer,r.indexBuffer,r.segments,s.p
aint,e.transform.zoom,v,r.layoutVertexBuffer2),g=!1;}}
(e,i,s,a);break;case"fill":!function(e,i,s,a){const o=s.paint.get("fill-
color"),r=s.paint.get("fill-opacity");if(0===r.constantOr(1))return;const
n=e.colorModeForRenderPass(),l=s.paint.get("fill-
pattern"),h=e.opaquePassEnabledForLayer()&&!l.constantOr(1)&&l===o.constantOr(t.
aT.transparent).a&&l===r.constantOr(0)?"opaque":"translucent";if(e.renderPass===
h){const t=e.depthModeForSublayer(1,"opaque"===e.renderPass?
Pi.ReadWrite:Pi.ReadOnly);Zi(e,i,s,a,t,n,!1);}if("translucent"===e.renderPass&&s
.paint.get("fill-antialias")){const
t=e.depthModeForSublayer(s.getPaintProperty("fill-outline-color")?
2:0,Pi.ReadOnly);Zi(e,i,s,a,t,n,!0);} }(e,i,s,a);break;case"fill-
extrusion":!function(t,e,i,s){const a=i.paint.get("fill-extrusion-
opacity");if(0!==a&&"translucent"===t.renderPass){const o=new
Pi(t.context.gl.LEQUAL,Pi.ReadWrite,t.depthRangeFor3D);if(1!==a||i.paint.get("fi
ll-extrusion-
pattern").constantOr(1))Gi(t,e,i,s,o,Mi.disabled,Si.disabled),Gi(t,e,i,s,o,t.ste
ncilModeFor3D(),t.colorModeForRenderPass());else {const
a=t.colorModeForRenderPass();Gi(t,e,i,s,o,Mi.disabled,a);}}}
(e,i,s,a);break;case"hillshade":!function(t,e,i,s)
{if("offscreen"!==t.renderPass&&"translucent"!==t.renderPass)return;const
a=t.context,o=t.depthModeForSublayer(0,Pi.ReadOnly),r=t.colorModeForRenderPass()
,[n,l]="translucent"===t.renderPass?t.stencilConfigForOverlap(s):
[{},s];for(const s of l){const a=e.getTile(s);void
0!==a.needsHillshadePrepare&&a.needsHillshadePrepare&&"offscreen"===t.renderPass
?
Vi(t,a,i,o,Mi.disabled,r):"translucent"===t.renderPass&&ji(t,s,a,i,o,n[s.oversca
ledZ],r);}a.viewport.set([0,0,t.width,t.height]);}
(e,i,s,a);break;case"raster":!function(t,e,i,s)
{if("translucent"!==t.renderPass)return;if(0===i.paint.get("raster-
opacity"))return;if(!s.length)return;const
a=t.context,o=a.gl,r=e.getSource(),n=t.useProgram("raster"),l=t.colorModeForRend
erPass(),[h,c]=r instanceof N?[{},s]:t.stencilConfigForOverlap(s),u=c[c.length-
1].overscaledZ,d=!t.options.moving;for(const s of c){const
c=t.depthModeForSublayer(s.overscaledZ-u,l===i.paint.get("raster-opacity")?
Pi.ReadWrite:Pi.ReadOnly,o.LESS),_=e.getTile(s);_.registerFadeDuration(i.paint.g
et("raster-fade-duration"));const
p=e.findLoadedParent(s,0),m=qi(_,p,e,i,t.transform,t.style.map.terrain);let
f,g;const v="nearest"===i.paint.get("raster-resampling")?
o.NEAREST:o.LINEAR;a.activeTexture.set(o.TEXTURE0),_.texture.bind(v,o.CLAMP_TO_E
DGE,o.LINEAR_MIPMAP_NEAREST),a.activeTexture.set(o.TEXTURE1),p?
(p.texture.bind(v,o.CLAMP_TO_EDGE,o.LINEAR_MIPMAP_NEAREST),f=Math.pow(2,p.tileID
.overscaledZ-_.tileID.overscaledZ),g=
[_tileID.canonical.x*f%1,_tileID.canonical.y*f%1]):_.texture.bind(v,o.CLAMP_TO
_EDGE,o.LINEAR_MIPMAP_NEAREST);const
x=t.style.map.terrain&&t.style.map.terrain.getTerrainData(s),y=x?s:null,b=y?
y.posMatrix:t.transform.calculatePosMatrix(s.toUnwrapped(),d),w=Me(b,g||
[0,0],f||1,m,i);r instanceof N?
n.draw(a,o.TRIANGLES,c,Mi.disabled,l,zi.disabled,w,x,i.id,r.boundsBuffer,t.quadT
riangleIndexBuffer,r.boundsSegments):n.draw(a,o.TRIANGLES,c,h[s.overscaledZ],l,z
i.disabled,w,x,i.id,t.rasterBoundsBuffer,t.quadTriangleIndexBuffer,t.rasterBound

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sSegments);}}(e,i,s,a);break;case"background":!function(t,e,i,s){const
a=i.paint.get("background-color"),o=i.paint.get("background-
opacity");if(0===o)return;const
r=t.context,n=r.gl,l=t.transform,h=l.tileSize,c=i.paint.get("background-
pattern");if(t.isPatternMissing(c))return;const
u=!c&&l===a.a&&l===o&&t.opaquePassEnabledForLayer()?"opaque":"translucent";if(t.
renderPass!==u)return;const
d=Mi.disabled,_=t.depthModeForSublayer(0,"opaque"===u?
Pi.ReadWrite:Pi.ReadOnly),p=t.colorModeForRenderPass(),m=t.useProgram(c?"backgro
undPattern":"background"),f=s||l.coveringTiles({tileSize:h,terrain:t.style.map.t
errain});c&&
(r.activeTexture.set(n.TEXTURE0),t.imageManager.bind(t.context));const
g=i.getCrossfadeParameters();for(const e of f){const l=s?
e.posMatrix:t.transform.calculatePosMatrix(e.toUnwrapped()),u=c?Fe(l,o,t,c,
{tileID:e,tileSize:h},g):ke(l,o,a),f=t.style.map.terrain&&t.style.map.terrain.ge
tTerrainData(e);m.draw(r,n.TRIANGLES,_,d,p,zi.disabled,u,f,i.id,t.tileExtentBuff
er,t.quadTriangleIndexBuffer,t.tileExtentSegments);}}
(e,0,s,a);break;case"custom":!function(t,e,i){const
s=t.context,a=i.implementation;if("offscreen"===t.renderPass){const
e=a.prerender;e&&
(t.setCustomLayerDefaults(),s.setColorMode(t.colorModeForRenderPass()),e.call(a,
s.gl,t.transform.customLayerMatrix()),s.setDirty(),t.setBaseState());}else
if("translucent"===t.renderPass)
{t.setCustomLayerDefaults(),s.setColorMode(t.colorModeForRenderPass()),s.setSten
cilMode(Mi.disabled);const e="3d"===a.renderingMode?new
Pi(t.context.gl.LEQUAL,Pi.ReadWrite,t.depthRangeFor3D):t.depthModeForSublayer(0,
Pi.ReadOnly);s.setDepthMode(e),a.render(s.gl,t.transform.customLayerMatrix()),s.
setDirty(),t.setBaseState(),s.bindFramebuffer.set(null);}}
(e,0,s);}}translatePosMatrix(e,i,s,a,o){if(!s[0]&&!s[1])return e;const
r=o?"map"===a?this.transform.angle:0:"viewport"===a?-
this.transform.angle:0;if(r){const t=Math.sin(r),e=Math.cos(r);s=[s[0]*e-
s[1]*t,s[0]*t+s[1]*e];}const n=[o?s[0]:St(i,s[0],this.transform.zoom),o?
s[1]:St(i,s[1],this.transform.zoom),0],l=new Float32Array(16);return
t.$(l,e,n),l}saveTileTexture(t){const e=this._tileTextures[t.size[0]];e?
e.push(t):this._tileTextures[t.size[0]]=[t];}getTileTexture(t){const
e=this._tileTextures[t];return e&&e.length>0?e.pop():null}isPatternMissing(t)
{if(!t)return !1;if(!t.from||!t.to)return !0;const
e=this.imageManager.getPattern(t.from.toString()),i=this.imageManager.getPattern
(t.to.toString());return !e||!i}useProgram(t,e){this.cache=this.cache||{};const
i=t+(e?e.cacheKey:"")+(this._showOverdrawInspector?"overdraw:"")+
(this.style.map.terrain?"terrain:"");return this.cache[i]||(this.cache[i]=new
ce(this.context,re[t],e,Be[t],this._showOverdrawInspector,this.style.map.terrain
)),this.cache[i]}setCustomLayerDefaults()
{this.context.unbindVAO(),this.context.cullFace.setDefault(),this.context.active
Texture.setDefault(),this.context.pixelStoreUnpack.setDefault(),this.context.pix
elStoreUnpackPremultiplyAlpha.setDefault(),this.context.pixelStoreUnpackFlipY.se
tDefault();}setBaseState(){const
t=this.context.gl;this.context.cullFace.set(!1),this.context.viewport.set([0,0,t
his.width,this.height]),this.context.blendEquation.set(t.FUNC_ADD);}initDebugOve
rlayCanvas(){null===this.debugOverlayCanvas&&
(this.debugOverlayCanvas=document.createElement("canvas"),this.debugOverlayCanva
s.width=512,this.debugOverlayCanvas.height=512,this.debugOverlayTexture=new
x(this.context,this.debugOverlayCanvas,this.context.gl.RGBA));}destroy()
{this.debugOverlayTexture&&this.debugOverlayTexture.destroy();}overLimit()
{const{drawingBufferWidth:t,drawingBufferHeight:e}=this.context.gl;return
this.width!==t||this.height!==e}}class ss{constructor(t,e)
{this.points=t,this.planes=e;}static fromInvProjectionMatrix(e,i,s){const
a=Math.pow(2,s),o=[[-1,1,-1,1],[1,1,-1,1],[1,-1,-1,1],[-1,-1,-1,1],[-1,1,1,1],
[1,1,1,1],[1,-1,1,1],[-1,-1,1,1]].map((s=>{const o=1/(s=t.ag([],s,e))
[3]/i*a;return t.b3(s,s,[o,o,1/s[3],o]))),r=[[0,1,2],[6,5,4],[0,3,7],[2,1,5],
[3,2,6],[0,4,5]].map((t=>{const e=function(t,e){var
i=e[0],s=e[1],a=e[2],o=i*i+s*s+a*a;return o>0&&

```

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(o=1/Math.sqrt(o)),t[0]=e[0]*o,t[1]=e[1]*o,t[2]=e[2]*o,t}([],function(t,e,i){var
s=e[0],a=e[1],o=e[2],r=i[0],n=i[1],l=i[2];return t[0]=a*1-o*n,t[1]=o*r-
s*1,t[2]=s*n-a*r,t}([],m([],o[t[0]],o[t[1]]),m([],o[t[2]],o[t[1]]))),i=-((s=e)
[0]*(a=o[t[1]])[0]+s[1]*a[1]+s[2]*a[2]);var s,a;return e.concat(i)}));return new
ss(o,r)}}class as{constructor(t,e)
{this.min=t,this.max=e,this.center=function(t,e,i){return
t[0]=.5*e[0],t[1]=.5*e[1],t[2]=.5*e[2],t}([],function(t,e,i){return
t[0]=e[0]+i[0],t[1]=e[1]+i[1],t[2]=e[2]+i[2],t}
([],this.min,this.max));}quadrant(t){const e=
[t%2==0,t<2],i=(this.min),s=(this.max);for(let t=0;t<e.length;t++)i[t]=e[t]?
this.min[t]:this.center[t],s[t]=e[t]?this.center[t]:this.max[t];return
s[2]=this.max[2],new as(i,s)}distanceX(t){return
Math.max(Math.min(this.max[0],t[0]),this.min[0])-t[0]}distanceY(t){return
Math.max(Math.min(this.max[1],t[1]),this.min[1])-t[1]}intersects(e){const i=
[[this.min[0],this.min[1],this.min[2],1],
[this.max[0],this.min[1],this.min[2],1],[this.max[0],this.max[1],this.min[2],1],
[this.min[0],this.max[1],this.min[2],1],[this.min[0],this.min[1],this.max[2],1],
[this.max[0],this.min[1],this.max[2],1],[this.max[0],this.max[1],this.max[2],1],
[this.min[0],this.max[1],this.max[2],1]];let s=!0;for(let
a=0;a<e.planes.length;a++){const o=e.planes[a];let r=0;for(let
e=0;e<i.length;e++)t.b4(o,i[e])>=0&&r++;if(0===r)return 0;r!==i.length&&
(s=!1);}if(s)return 2;for(let t=0;t<3;t++){let i=Number.MAX_VALUE,s=-
Number.MAX_VALUE;for(let a=0;a<e.points.length;a++){const o=e.points[a][t]-
this.min[t];i=Math.min(i,o),s=Math.max(s,o);}if(s<0||i>this.max[t]-
this.min[t])return 0}return 1}}class os{constructor(t=0,e=0,i=0,s=0)
{if(isNaN(t)||t<0||isNaN(e)||e<0||isNaN(i)||i<0||isNaN(s)||s<0)throw new
Error("Invalid value for edge-insets, top, bottom, left and right must all be
numbers");this.top=t,this.bottom=e,this.left=i,this.right=s;}interpolate(e,i,s)
{return null!=i.top&&null!=e.top&&
(this.top=t.B.number(e.top,i.top,s)),null!=i.bottom&&null!=e.bottom&&
(this.bottom=t.B.number(e.bottom,i.bottom,s)),null!=i.left&&null!=e.left&&
(this.left=t.B.number(e.left,i.left,s)),null!=i.right&&null!=e.right&&
(this.right=t.B.number(e.right,i.right,s)),this}getCenter(e,i){const
s=t.ad((this.left+e-this.right)/2,0,e),a=t.ad((this.top+i-
this.bottom)/2,0,i);return new t.P(s,a)}equals(t){return
this.top===t.top&&this.bottom===t.bottom&&this.left===t.left&&this.right===t.rig
ht}clone(){return new os(this.top,this.bottom,this.left,this.right)}toJSON()
{return {top:this.top,bottom:this.bottom,left:this.left,right:this.right}}}class
rs{constructor(e,i,s,a,o)
{this.tileSize=512,this.maxValidLatitude=85.051129,this._renderWorldCopies=void
0===o||!o,this._minZoom=e||0,this._maxZoom=i||22,this._minPitch=null===s?
0:s,this._maxPitch=null==a?
60:a,this.setMaxBounds(),this.width=0,this.height=0,this._center=new
t.L(0,0),this._elevation=0,this.zoom=0,this.angle=0,this._fov=.6435011087932844,
this._pitch=0,this._unmodified=!0,this._edgeInsets=new os,this._posMatrixCache=
{},this._alignedPosMatrixCache={},this._minElevationForCurrentTile=0;}clone()
{const t=new
rs(this._minZoom,this._maxZoom,this._minPitch,this.maxPitch,this._renderWorldCop
ies);return t.apply(this),t}apply(t)
{this.tileSize=t.tileSize,this.latRange=t.latRange,this.width=t.width,this.heigh
t=t.height,this._center=t._center,this._elevation=t._elevation,this._minElevati
onForCurrentTile=t._minElevationForCurrentTile,this.zoom=t.zoom,this.angle=t.an
gle,this._fov=t._fov,this._pitch=t._pitch,this._unmodified=t._unmodified,this._e
dgeInsets=t._edgeInsets.clone(),this._calcMatrices();}get minZoom(){return
this._minZoom}set minZoom(t){this._minZoom!==t&&
(this._minZoom=t,this.zoom=Math.max(this.zoom,t));}get maxZoom(){return
this._maxZoom}set maxZoom(t){this._maxZoom!==t&&
(this._maxZoom=t,this.zoom=Math.min(this.zoom,t));}get minPitch(){return
this._minPitch}set minPitch(t){this._minPitch!==t&&
(this._minPitch=t,this.pitch=Math.max(this.pitch,t));}get maxPitch(){return
this._maxPitch}set maxPitch(t){this._maxPitch!==t&&
(this._maxPitch=t,this.pitch=Math.min(this.pitch,t));}get renderWorldCopies()

```

```

{return this._renderWorldCopies}set renderWorldCopies(t){void 0===t?
t=!0:null===t&&(t=!1),this._renderWorldCopies=t;}get worldSize(){return
this.tileSize*this.scale}get centerOffset(){return
this.centerPoint._sub(this.size._div(2))}get size(){return new
t.P(this.width,this.height)}get bearing(){return -this.angle/Math.PI*180}set
bearing(e){const i=-t.b5(e,-180,180)*Math.PI/180;this.angle!==i&&
(this._unmodified=!1,this.angle=i,this._calcMatrices(),this.rotationMatrix=funct
ion()){var e=new t.A(4);return t.A!=Float32Array&&
(e[1]=0,e[2]=0),e[0]=1,e[3]=1,e[4]=1,function(t,e,i){var
s=e[0],a=e[1],o=e[2],r=e[3],n=Math.sin(i),l=Math.cos(i);t[0]=s*l+o*n,t[1]=a*l+r*
n,t[2]=s*-n+o*l,t[3]=a*-n+r*l;}
(this.rotationMatrix,this.rotationMatrix,this.angle));}get pitch(){return
this._pitch/Math.PI*180}set pitch(e){const
i=t.ad(e,this.minPitch,this.maxPitch)/180*Math.PI;this._pitch!==i&&
(this._unmodified=!1,this._pitch=i,this._calcMatrices());}get fov(){return
this._fov/Math.PI*180}set fov(t){t=Math.max(.01,Math.min(60,t)),this._fov!==t&&
(this._unmodified=!1,this._fov=t/180*Math.PI,this._calcMatrices());}get zoom()
{return this._zoom}set zoom(t){const
e=Math.min(Math.max(t,this.minZoom),this.maxZoom);this._zoom!==e&&
(this._unmodified=!1,this._zoom=e,this.tileZoom=Math.max(0,Math.floor(e)),this.s
cale=this.zoomScale(e),this._constrain(),this._calcMatrices());}get center()
{return this._center}set center(t)
{t.lat===this._center.lat&&t.lng===this._center.lng||
(this._unmodified=!1,this._center=t,this._constrain(),this._calcMatrices());}get
elevation(){return this._elevation}set elevation(t){t!==this._elevation&&
(this._elevation=t,this._constrain(),this._calcMatrices());}get padding(){return
this._edgeInsets.toJSON()}set padding(t){this._edgeInsets.equals(t)||
(this._unmodified=!1,this._edgeInsets.interpolate(this._edgeInsets,t,1),this._ca
lcMatrices());}get centerPoint(){return
this._edgeInsets.getCenter(this.width,this.height)}isPaddingEqual(t){return
this._edgeInsets.equals(t)}interpolatePadding(t,e,i)
{this._unmodified=!1,this._edgeInsets.interpolate(t,e,i),this._constrain(),this.
_calcMatrices();}coveringZoomLevel(t){const e=(t.roundZoom?
Math.round:Math.floor)
(this.zoom+this.scaleZoom(this.tileSize/t.tileSize));return
Math.max(0,e)}getVisibleUnwrappedCoordinates(e){const i=[new
t.b6(0,e)];if(this._renderWorldCopies){const s=this.pointCoordinate(new
t.P(0,0)),a=this.pointCoordinate(new
t.P(this.width,0)),o=this.pointCoordinate(new
t.P(this.width,this.height)),r=this.pointCoordinate(new
t.P(0,this.height)),n=Math.floor(Math.min(s.x,a.x,o.x,r.x)),l=Math.floor(Math.ma
x(s.x,a.x,o.x,r.x)),h=1;for(let s=n-h;s<=l+h;s++)0!==s&&i.push(new
t.b6(s,e));}return i}coveringTiles(e){var i,s;let
a=this.coveringZoomLevel(e);const o=a;if(void 0!==e.minzoom&&a<e.minzoom)return
[];void 0!==e.maxzoom&&a>e.maxzoom&&(a=e.maxzoom);const
r=this.pointCoordinate(this.getCameraPoint()),n=t.U.fromLngLat(this.center),l=Ma
th.pow(2,a),h=[l*r.x,l*r.y,0],c=
[l*n.x,l*n.y,0],u=ss.fromInvProjectionMatrix(this.invProjMatrix,this.worldSize,a
);let d=e.minzoom||0;!e.terrain&&this.pitch<=60&&this._edgeInsets.top<.1&&
(d=a);const _=e.terrain?
2/Math.min(this.tileSize,e.tileSize)*this.tileSize:3,p=t=>({aabb:new
as([t*1,0,0],[t+1*1,1,0]),zoom:0,x:0,y:0,wrap:t,fullyVisible:!1}),m=[],g=
[],v=a,x=e.reparseOverscaled?o:a;if(this._renderWorldCopies)for(let
t=1;t<=3;t++)m.push(p(-t)),m.push(p(t));for(m.push(p(0));m.length>0){const
a=m.pop(),o=a.x,r=a.y;let n=a.fullyVisible;if(!n){const
t=a.aabb.intersects(u);if(0===t)continue;n=2===t;}const l=e.terrain?
h:c,p=a.aabb.distanceX(1),y=a.aabb.distanceY(1),b=Math.max(Math.abs(p),Math.abs(
y));if(a.zoom===v||b>_+(1<<v-a.zoom)-2&&a.zoom>=d){const e=v-a.zoom,i=h[0]-.5-
(o<<e),s=h[1]-.5-(r<<e);g.push({tileID:new t.O(a.zoom===v?
x:a.zoom,a.wrap,a.zoom,o,r),distanceSq:f([c[0]-.5-o,c[1]-.5-
r]),tileDistanceToCamera:Math.sqrt(i*i+s*s)});}else for(let l=0;l<4;l++){const
h=(o<<1)+l%2,c=(r<<1)+(l>>1),u=a.zoom+1;let d=a.aabb.quadrant(l);if(e.terrain)

```

```

{const o=new t.O(u,a.wrap,u,h,c),r=e.terrain.getMinMaxElevation(o),n=null!==(i=r.minElevation)&&void 0!==(i?i:this.elevation,l=null!==(s=r.maxElevation)&&void 0!==(s?s:this.elevation;d=new as([d.min[0],d.min[1],n],[d.max[0],d.max[1],l]));m.push({aabb:d,zoom:u,x:h,y:c,wrap:a.wrap,fullyVisible:n});}}return g.sort((t,e)=>t.distanceSq-e.distanceSq)).map((t=>t.tileID))resize(t,e)
{this.width=t,this.height=e,this.pixelsToGLUnits=[2/t,-2/e],this._constrain(),this._calcMatrices();}get unmodified(){return this._unmodified}zoomScale(t){return Math.pow(2,t)}scaleZoom(t){return Math.log(t)/Math.LN2}project(e){const i=t.ad(e.lat,-this.maxValidLatitude,this.maxValidLatitude);return new t.P(t.G(e.lng)*this.worldSize,t.H(i)*this.worldSize)}unproject(e){return new t.U(e.x/this.worldSize,e.y/this.worldSize).toLngLat()}get point(){return this.project(this.center)}getCameraPosition(){return {lngLat:this.pointLocation(this.getCameraPoint()),altitude:Math.cos(this._pitch)*this.cameraToCenterDistance/this._pixelPerMeter+this.elevation}}recalculateZoom(e){const i=this.pointLocation(this.centerPoint,e),s=e.getElevationForLngLatZoom(i,this.tileZoom);if(!(this.elevation-s))return;const a=this.getCameraPosition(),o=t.U.fromLngLat(a.lngLat,a.altitude),r=t.U.fromLngLat(i,s),n=o.x-r.x,l=o.y-r.y,h=o.z-r.z,c=Math.sqrt(n*n+l*l+h*h),u=this.scaleZoom(this.cameraToCenterDistance/c/this.tileSize);this._elevation=s,this._center=i,this.zoom=u;}setLocationAtPoint(e,i){const s=this.pointCoordinate(i),a=this.pointCoordinate(this.centerPoint),o=this.locationCoordinate(e),r=new t.U(o.x-(s.x-a.x),o.y-(s.y-a.y));this.center=this.coordinateLocation(r),this._renderWorldCopies&&(this.center=this.center.wrap());}locationPoint(t,e){return e?this.coordinatePoint(this.locationCoordinate(t),e.getElevationForLngLatZoom(t,this.tileZoom),this.pixelMatrix3D):this.coordinatePoint(this.locationCoordinate(t))}pointLocation(t,e){return this.coordinateLocation(this.pointCoordinate(t,e))}locationCoordinate(e){return t.U.fromLngLat(e)}coordinateLocation(t){return t&&t.toLngLat()}pointCoordinate(e,i){if(i){const t=i.pointCoordinate(e);if(null!=t)return t}const s=[e.x,e.y,0,1],a=[e.x,e.y,1,1];t.ag(s,s,this.pixelMatrixInverse),t.ag(a,a,this.pixelMatrixInverse);const o=s[3],r=a[3],n=s[1]/o,l=a[1]/r,h=s[2]/o,c=a[2]/r,u=h===c?0:(0-h)/(c-h);return new t.U(t.B.number(s[0]/o,a[0]/r,u)/this.worldSize,t.B.number(n,l,u)/this.worldSize)}coordinatePoint(e,i=0,s=this.pixelMatrix){const a=[e.x*this.worldSize,e.y*this.worldSize,i,1];return t.ag(a,a,s),new t.P(a[0]/a[3],a[1]/a[3])}getBounds(){const e=Math.max(0,this.height/2-this.getHorizon());return (new L).extend(this.pointLocation(new t.P(0,e))).extend(this.pointLocation(new t.P(this.width,e))).extend(this.pointLocation(new t.P(this.width,this.height))).extend(this.pointLocation(new t.P(0,this.height)))}getMaxBounds(){return this.latRange&&2===this.latRange.length&&this.lngRange&&2===this.lngRange.length?new L([this.lngRange[0],this.latRange[0]],[this.lngRange[1],this.latRange[1]]):null}getHorizon(){return Math.tan(Math.PI/2-this._pitch)*this.cameraToCenterDistance*.85}setMaxBounds(t){t?(this.lngRange=[t.getWest(),t.getEast()],this.latRange=[t.getSouth(),t.getNorth()],this._constrain()):this.lngRange=null,this.latRange=[-this.maxValidLatitude,this.maxValidLatitude];}calculatePosMatrix(e,i=!1){const s=e.key,a=i?this._alignedPosMatrixCache:this._posMatrixCache;if(a[s])return a[s];const o=e.canonical,r=this.worldSize/this.zoomScale(o.z),n=o.x+Math.pow(2,o.z)*e.wrap,l=t.a0(new Float64Array(16));return t.$(1,l,[n*r,o.y*r,0]),t.a0(1,l,[r/t.N,r/t.N,1]),t.a1(1,i?this.alignedProjMatrix:this.projMatrix,l),a[s]=new Float32Array(1),a[s].customLayerMatrix(){return this.mercatorMatrix.slice()}_constrain()}

```



```

{if(!this.center||!this.width||!this.height||this._constraining)return;this._con
straining=!0;let e,i,s,a,o=-90,r=90,n=-180,l=180;const
h=this.size,c=this._unmodified;if(this.latRange){const
i=this.latRange;o=t.H(i[1])*this.worldSize,r=t.H(i[0])*this.worldSize,e=r-o<h.y?
h.y/(r-o):0;}if(this.lngRange){const
e=this.lngRange;n=t.b5(t.G(e[0])*this.worldSize,0,this.worldSize),l=t.b5(t.G(e[1
])*this.worldSize,0,this.worldSize),l<n&&(l+=this.worldSize),i=l-n<h.x?h.x/(l-
n):0;}const u=this.point,d=Math.max(i||0,e||0);if(d)return
this.center=this.unproject(new t.P(i?(l+n)/2:u.x,e?
(r+o)/2:u.y)),this.zoom+=this.scaleZoom(d),this._unmodified=c,void(this._constra
ining=!1);if(this.latRange){const t=u.y,e=h.y/2;t-e<o&&(a=o+e),t+e>r&&(a=r-
e);}if(this.lngRange){const e=(n+l)/2,i=t.b5(u.x,e-
this.worldSize/2,e+this.worldSize/2),a=h.x/2;i-a<n&&(s=n+a),i+a>l&&(s=l-a);}void
0===s&&void 0===a||(this.center=this.unproject(new t.P(void 0!==s?s:u.x,void
0!==a?a:u.y)).wrap()),this._unmodified=c,this._constraining=!1;}_calcMatrices()
{if(!this.height)return;const
e=this.centerOffset,i=this.point.x,s=this.point.y;this.cameraToCenterDistance=.5
/Math.tan(this._fov/2)*this.height,this._pixelPerMeter=t.b7(1,this.center.lat)*t
his.worldSize;let a=t.a0(new Float64Array(16));t.a0(a,a,[this.width/2,-
this.height/2,1]),t.$(a,a,[1,-1,0]),this.labelPlaneMatrix=a,a=t.a0(new
Float64Array(16)),t.a0(a,a,[1,-1,1]),t.$(a,a,[-1,-1,0]),t.a0(a,a,
[2/this.width,2/this.height,1]),this.glCoordMatrix=a;const
o=this.cameraToCenterDistance+this._elevation*this._pixelPerMeter/Math.cos(this.
_pitch),r=Math.min(this.elevation,this._minElevationForCurrentTile),n=o-
r*this._pixelPerMeter/Math.cos(this._pitch),l=r<0?
n:o,h=Math.PI/2+this._pitch,c=this._fov*
(.5+e.y/this.height),u=Math.sin(c)*l/Math.sin(t.ad(Math.PI-h-
c,.01,Math.PI-.01)),d=this.getHorizon(),_=2*Math.atan(d/this.cameraToCenterDista
nce)*(0.5+e.y/(2*d)),p=Math.sin(_)*l/Math.sin(t.ad(Math.PI-h-
_,.01,Math.PI-.01)),m=Math.min(u,p),f=1.01*(Math.cos(Math.PI/2-
this._pitch)*m+1),g=this.height/50;a=new
Float64Array(16),t.b8(a,this._fov,this.width/this.height,g,f),a[8]=2*-
e.x/this.width,a[9]=2*e.y/this.height,t.a0(a,a,[1,-1,1]),t.$(a,a,[0,0,-
this.cameraToCenterDistance]),t.b9(a,a,this._pitch),t.ae(a,a,this.angle),t.$(a,a,
[-i,-s,0]),this.mercatorMatrix=t.a0([],a,
[this.worldSize,this.worldSize,this.worldSize]),t.a0(a,a,
[1,1,this._pixelPerMeter]),this.pixelMatrix=t.a1(new
Float64Array(16),this.labelPlaneMatrix,a),t.$(a,a,[0,0,-
this.elevation]),this.projMatrix=a,this.invProjMatrix=t.as([],a),this.pixelMatri
x3D=t.a1(new Float64Array(16),this.labelPlaneMatrix,a);const
v=this.width%2/2,x=this.height%2/2,y=Math.cos(this.angle),b=Math.sin(this.angle)
,w=i-Math.round(i)+y*v+b*x,T=s-Math.round(s)+y*x+b*v,I=new
Float64Array(a);if(t.$(I,I,[w>.5?w-1:w,T>.5?T-
1:T,0]),this.alignedProjMatrix=I,a=t.as(new
Float64Array(16),this.pixelMatrix,!a))throw new Error("failed to invert
matrix");this.pixelMatrixInverse=a,this._posMatrixCache=
{},this._alignedPosMatrixCache={};}maxPitchScaleFactor()
{if(!this.pixelMatrixInverse)return 1;const e=this.pointCoordinate(new
t.P(0,0)),i=[e.x*this.worldSize,e.y*this.worldSize,0,1];return
t.ag(i,i,this.pixelMatrix)[3]/this.cameraToCenterDistance}getCameraPoint(){const
e=Math.tan(this._pitch)*(this.cameraToCenterDistance||1);return
this.centerPoint.add(new t.P(0,e))}getCameraQueryGeometry(e){const
i=this.getCameraPoint();if(1===e.length)return [e[0],i];let
s=i.x,a=i.y,o=i.x,r=i.y;for(const t of
e)s=Math.min(s,t.x),a=Math.min(a,t.y),o=Math.max(o,t.x),r=Math.max(r,t.y);return
[new t.P(s,a),new t.P(o,a),new t.P(o,r),new t.P(s,r),new t.P(s,a)]}}function
ns(t,e){let i,s=!1,a=null,o=null;const r={()=>{a=null,s&&
(t.apply(o,i),a=setTimeout(r,e),s=!1);};return (...t)=>
(s?!0,o=this,i=t,a|!r(),a)}class ls{constructor(t){this._getCurrentHash={()=>
{const t=window.location.hash.replace("#","");if(this._hashName){let e;return
t.split("&").map((t=>t.split("="))).forEach((t=>{t[0]===this._hashName&&
(e=t);})),(e&&e[1]||"").split("/")}}return t.split("/"),this._onHashChange={()=>

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```

{const t=this._getCurrentHash();if(t.length>=3&&!t.some((t=>isNaN(t)))){const
e=this._map.dragRotate.isEnabled()&&this._map.touchZoomRotate.isEnabled()?+
(t[3]||0):this._map.getBearing();return this._map.jumpTo({center:
[+t[2],+t[1]],zoom:+t[0],bearing:e,pitch:+(t[4]||0)}),!0}return
!1},this._updateHashUnthrottled=()=>{const
t=window.location.href.replace(/(#+)?$/ ,this._getHashString());try{window.history.replaceState(window.history.state,nu
ll,t);}catch(t)
{}},this._updateHash=ns(this._updateHashUnthrottled,300),this._hashName=t&&encod
eURIComponent(t);}addTo(t){return
this._map=t,addEventListener("hashchange",this._onHashChange,!1),this._map.on("m
oveend",this._updateHash),this.remove(){return
removeEventListener("hashchange",this._onHashChange,!1),this._map.off("moveend",
this._updateHash),clearTimeout(this._updateHash()),delete
this._map,this._getHashString(t){const
e=this._map.getCenter(),i=Math.round(100*this._map.getZoom())/100,s=Math.ceil((i
*Math.LN2+Math.log(512/360/.5))/Math.LN10),a=Math.pow(10,s),o=Math.round(e.lng*a
)/a,r=Math.round(e.lat*a)/a,n=this._map.getBearing(),l=this._map.getPitch();let
h="";if(h+=t?`/${o}/${r}/${i}`:`/${i}/${r}/${o}`,n||l)&&
(h+=`/${Math.round(10*n)/10},l&&(h+=`/${Math.round(l)} `,this._hashName){const
t=this._hashName;let e=!1;const
i=window.location.hash.slice(1).split("&").map((i=>{const s=i.split("=")
[0];return s===t?(e=!0,`${s}=${h}`):i})).filter((t=>t));return
e||i.push(`${t}=${h}`),`#${i.join("&")}`}return `${h}`}const hs=
{linearity:.3,easing:t.ba(0,0,.3,1)},cs=t.e({deceleration:2500,maxSpeed:1400},hs
),us=t.e({deceleration:20,maxSpeed:1400},hs),ds=t.e({deceleration:1e3,maxSpeed:3
60},hs),_s=t.e({deceleration:1e3,maxSpeed:90},hs);class ps{constructor(t)
{this._map=t,this.clear();}clear(){this._inertiaBuffer=[];}record(e)
{this._drainInertiaBuffer(),this._inertiaBuffer.push({time:t.h.now(),settings:e)
});_drainInertiaBuffer(){const
e=this._inertiaBuffer,i=t.h.now();for(;e.length>0&&i-
e[0].time>160;)e.shift();}_onMoveEnd(e)
{if(this._drainInertiaBuffer(),this._inertiaBuffer.length<2)return;const i=
{zoom:0,bearing:0,pitch:0,pan:new t.P(0,0),pinchAround:void 0,around:void
0};for(const{settings:t}of
this._inertiaBuffer)i.zoom+=t.zoomDelta||0,i.bearing+=t.bearingDelta||0,i.pitch+
=t.pitchDelta||0,t.panDelta&&i.pan._add(t.panDelta),t.around&&
(i.around=t.around),t.pinchAround&&(i.pinchAround=t.pinchAround);const
s=this._inertiaBuffer[this._inertiaBuffer.length-1].time-
this._inertiaBuffer[0].time,a={};if(i.pan.mag()){const
o=fs(i.pan.mag(),s,t.e({},cs,e||
{}));a.offset=i.pan.mult(o.amount/i.pan.mag()),a.center=this._map.transform.cent
er,ms(a,o);}if(i.zoom){const
t=fs(i.zoom,s,us);a.zoom=this._map.transform.zoom+t.amount,ms(a,t);}if(i.bearing
){const
e=fs(i.bearing,s,ds);a.bearing=this._map.transform.bearing+t.ad(e.amount,-179,17
9),ms(a,e);}if(i.pitch){const
t=fs(i.pitch,s,_s);a.pitch=this._map.transform.pitch+t.amount,ms(a,t);}if(a.zoom
||a.bearing){const t=void 0===i.pinchAround?i.around:i.pinchAround;a.around=t?
this._map.unproject(t):this._map.getCenter();}return this.clear(),t.e(a,
{noMoveStart:!0})}}function ms(t,e){(!t.duration||t.duration<e.duration)&&
(t.duration=e.duration,t.easing=e.easing);}function fs(e,i,s)
{const{maxSpeed:a,linearity:o,deceleration:r}=s,n=t.ad(e*o/(i/1e3),-
a,a),l=Math.abs(n)/(r*o);return {easing:s.easing,duration:1e3*l,amount:n*
(1/2)}}class gs extends t.k{preventDefault(){this._defaultPrevented=!0;}get
defaultPrevented(){return this._defaultPrevented}constructor(e,s,a,o={}){const
r=i.mousePos(s.getCanvasContainer(),a),n=s.unproject(r);super(e,t.e({point:r,lng
Lat:n,originalEvent:a,o}),this._defaultPrevented=!1,this.target=s);}class vs
extends t.k{preventDefault(){this._defaultPrevented=!0;}get defaultPrevented()
{return this._defaultPrevented}constructor(e,s,a){const o="touchend"===e?
a.changedTouches:a.touches,r=i.touchPos(s.getCanvasContainer(),o),n=r.map((t=>s.
unproject(t))),l=r.reduce((t,e,i,s)=>t.add(e.div(s.length))),new

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t.P(0,0));super(e,
{points:r,point:l,lngLats:n,lngLat:s.unproject(l),originalEvent:a}),this._default
tPrevented=!1;}}class xs extends t.k{preventDefault()
{this._defaultPrevented=!0;}get defaultPrevented(){return
this._defaultPrevented}constructor(t,e,i){super(t,
{originalEvent:i}),this._defaultPrevented=!1;}}class ys{constructor(t,e)
{this._map=t,this._clickTolerance=e.clickTolerance;}reset(){delete
this._mousedownPos;}wheel(t){return this._firePreventable(new
xs(t.type,this._map,t))}mousedown(t,e){return
this._mousedownPos=e,this._firePreventable(new
gs(t.type,this._map,t))}mouseup(t){this._map.fire(new
gs(t.type,this._map,t));}click(t,e)
{this._mousedownPos&&this._mousedownPos.dist(e)>=this._clickTolerance||this._map
.fire(new gs(t.type,this._map,t));}dblclick(t){return this._firePreventable(new
gs(t.type,this._map,t))}mouseover(t){this._map.fire(new
gs(t.type,this._map,t));}mouseout(t){this._map.fire(new
gs(t.type,this._map,t));}touchstart(t){return this._firePreventable(new
vs(t.type,this._map,t))}touchmove(t){this._map.fire(new
vs(t.type,this._map,t));}touchend(t){this._map.fire(new
vs(t.type,this._map,t));}touchcancel(t){this._map.fire(new
vs(t.type,this._map,t));}_firePreventable(t)
{if(this._map.fire(t),t.defaultPrevented)return {}}}isEnabled(){return
!0}isActive(){return !1}enable(){disable()}class bs{constructor(t)
{this._map=t;}reset()
{this._delayContextMenu=!1,this._ignoreContextMenu=!0,delete
this._contextMenuEvent;}mousemove(t){this._map.fire(new
gs(t.type,this._map,t));}mousedown()
{this._delayContextMenu=!0,this._ignoreContextMenu=!1;}mouseup()
{this._delayContextMenu=!1,this._contextMenuEvent&&(this._map.fire(new
gs("contextmenu",this._map,this._contextMenuEvent)),delete
this._contextMenuEvent);}contextmenu(t){this._delayContextMenu?
this._contextMenuEvent=t:this._ignoreContextMenu||this._map.fire(new
gs(t.type,this._map,t)),this._map.listens("contextmenu")&&t.preventDefault();}is
Enabled(){return !0}isActive(){return !1}enable(){disable()}class
ws{constructor(t){this._map=t;}get transform(){return
this._map._requestedCameraState||this._map.transform}get center(){return
{lng:this.transform.center.lng,lat:this.transform.center.lat}}get zoom(){return
this.transform.zoom}get pitch(){return this.transform.pitch}get bearing(){return
this.transform.bearing}unproject(e){return
this.transform.pointLocation(t.P.convert(e),this._map.terrain)}}class
Ts{constructor(t,e){this._map=t,this._tr=new
ws(t),this._el=t.getCanvasContainer(),this._container=t.getContainer(),this._cli
ckTolerance=e.clickTolerance||1;}isEnabled(){return !!this._enabled}isActive()
{return !!this._active}enable(){this.isEnabled()||(this._enabled=!0);}disable()
{this.isEnabled()&&(this._enabled=!1);}mousedown(t,e)
{this.isEnabled()&&t.shiftKey&&0===t.button&&
(i.disableDrag(),this._startPos=this._lastPos=e,this._active=!0);}mousemoveWindo
w(t,e){if(!this._active)return;const
s=e;if(this._lastPos.equals(s)||!this._box&&s.dist(this._startPos)
<this._clickTolerance)return;const a=this._startPos,this._lastPos=s,this._box||
(this._box=i.create("div","maplibregl-
boxzoom",this._container),this._container.classList.add("maplibregl-
crosshair"),this._fireEvent("boxzoomstart",t));const
o=Math.min(a.x,s.x),r=Math.max(a.x,s.x),n=Math.min(a.y,s.y),l=Math.max(a.y,s.y);
i.setTransform(this._box,`translate(${o}px,${n}px)`),this._box.style.width=r-
o+"px",this._box.style.height=l-n+"px";}mouseupWindow(e,s)
{if(!this._active)return;if(0!==e.button)return;const
a=this._startPos,o=s;if(this._reset(),i.suppressClick(),a.x!==o.x||a.y!==o.y)retu
rn this._map.fire(new t.k("boxzoomend",{originalEvent:e})),
{cameraAnimation:t=>t.fitScreenCoordinates(a,o,this._tr.bearing,
{linear:!0})};this._fireEvent("boxzoomcancel",e);}keydown(t)
{this._active&&27===t.keyCode&&

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(this.reset(),this._fireEvent("boxzoomcancel",t));}reset()
{this._active=!1,this._container.classList.remove("maplibregl-
crosshair"),this._box&&
(i.remove(this._box),this._box=null),i.enableDrag(),delete this._startPos,delete
this._lastPos;}_fireEvent(e,i){return this._map.fire(new t.k(e,
{originalEvent:i}))}}function Is(t,e){if(t.length!==e.length)throw new
Error(`The number of touches and points are not equal - touches ${t.length},
points ${e.length}`);const i={};for(let
s=0;s<t.length;s++)i[t[s].identifier]=e[s];return i}class Es{constructor(t)
{this.reset(),this.numTouches=t.numTouches;}reset(){delete this.centroid,delete
this.startTime,delete this.touches,this.aborted=!1;}touchstart(e,i,s)
{((this.centroid||s.length>this.numTouches)&&(this.aborted=!0),this.aborted||
(void 0===this.startTime&&
(this.startTime=e.timeStamp),s.length===this.numTouches&&
(this.centroid=function(e){const i=new t.P(0,0);for(const t of
e)i._add(t);return i.div(e.length)}(i),this.touches=Is(s,i))));}touchmove(t,e,i)
{if(this.aborted||!this.centroid)return;const s=Is(i,e);for(const t in
this.touches){const e=s[t];(!e||e.dist(this.touches[t])>30)&&
(this.aborted=!0);}touchend(t,e,i){if((!this.centroid||t.timeStamp-
this.startTime>500)&&(this.aborted=!0),0===i.length){const
t=!this.aborted&&this.centroid;if(this.reset(),t)return t}}class
Ss{constructor(t){this.singleTap=new
Es(t),this.numTaps=t.numTaps,this.reset();}reset(){this.lastTime=1/0,delete
this.lastTap,this.count=0,this.singleTap.reset();}touchstart(t,e,i)
{this.singleTap.touchstart(t,e,i);}touchmove(t,e,i)
{this.singleTap.touchmove(t,e,i);}touchend(t,e,i){const
s=this.singleTap.touchend(t,e,i);if(s){const e=t.timeStamp-
this.lastTime<500,i=!this.lastTap||this.lastTap.dist(s)
<30;if(e&&i||this.reset(),this.count++,this.lastTime=t.timeStamp,this.lastTap=s,
this.count===this.numTaps)return this.reset(),s}}class Cs{constructor(t)
{this._tr=new ws(t),this._zoomIn=new
Ss({numTouches:1,numTaps:2}),this._zoomOut=new
Ss({numTouches:2,numTaps:1}),this.reset();}reset()
{this._active=!1,this._zoomIn.reset(),this._zoomOut.reset();}touchstart(t,e,i)
{this._zoomIn.touchstart(t,e,i),this._zoomOut.touchstart(t,e,i);}touchmove(t,e,i)
{this._zoomIn.touchmove(t,e,i),this._zoomOut.touchmove(t,e,i);}touchend(t,e,i)
{const
s=this._zoomIn.touchend(t,e,i),a=this._zoomOut.touchend(t,e,i),o=this._tr;return
s?(this._active=!0,t.preventDefault(),setTimeout((()=>this.reset()),0),
{cameraAnimation:e=>e.easeTo({duration:300,zoom:o.zoom+1,around:o.unproject(s)},
{originalEvent:t}}):a?
(this._active=!0,t.preventDefault(),setTimeout((()=>this.reset()),0),
{cameraAnimation:e=>e.easeTo({duration:300,zoom:o.zoom-1,around:o.unproject(a)},
{originalEvent:t}}):void 0}touchcancel(){this.reset();}enable()
{this._enabled=!0;}disable(){this._enabled=!1,this.reset();}isEnabled(){return
this._enabled}isActive(){return this._active}}class Ps{constructor(t)
{this._enabled=!!t.enable,this._moveStateManager=t.moveStateManager,this._clickT
olerance=t.clickTolerance||1,this._moveFunction=t.move,this._activateOnStart=!!t
.activateOnStart,t.assignEvents(this),this.reset();}reset(t)
{this._active=!1,this._moved=!1,delete
this._lastPoint,this._moveStateManager.endMove(t);}_move(...t){const
e=this._moveFunction(...t);if(e.bearingDelta||e.pitchDelta||e.around||e.panDelta)
return this._active=!0,e}dragStart(t,e)
{this.isEnabled()&&!this._lastPoint&&this._moveStateManager.isValidStartEvent(t)
&&(this._moveStateManager.startMove(t),this._lastPoint=e.length?
e[0]:e,this._activateOnStart&&this._lastPoint&&(this._active=!0));}dragMove(t,e)
{if(!this.isEnabled())return;const
i=this._lastPoint;if(!i)return;if(t.preventDefault(),!this._moveStateManager.isV
alidMoveEvent(t))return void this.reset(t);const s=e.length?e[0]:e;return
!this._moved&&s.dist(i)<this._clickTolerance?void 0:
(this._moved=!0,this._lastPoint=s,this._move(i,s))}dragEnd(t)
{this.isEnabled()&&this._lastPoint&&this._moveStateManager.isValidEndEvent(t)&&

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(this._moved&&i.suppressClick(),this.reset());}enable()
{this._enabled=!0;}disable(){this._enabled=!1,this.reset();}isEnabled(){return
this._enabled}isActive(){return this._active}getClickTolerance(){return
this._clickTolerance}}const Ds={0:1,2:2};class Ms{constructor(t)
{this._correctEvent=t.checkCorrectEvent;}startMove(t){const
e=i.mouseButton(t);this._eventButton=e;}endMove(t){delete
this._eventButton;}isValidStartEvent(t){return
this._correctEvent(t)}isValidMoveEvent(t){return !function(t,e){const
i=Ds[e];return void 0===t.buttons||!(t.buttons&i)!=i}
(t,this._eventButton)}isValidEndEvent(t){return
i.mouseButton(t)===this._eventButton}}class zs{constructor()
{this._firstTouch=void 0;}_isOneFingerTouch(t){return
1===t.targetTouches.length}_isSameTouchEvent(t){return
t.targetTouches[0].identifier===this._firstTouch}startMove(t)
{this._firstTouch=t.targetTouches[0].identifier;}endMove(t){delete
this._firstTouch;}isValidStartEvent(t){return
this._isOneFingerTouch(t)}isValidMoveEvent(t){return
this._isOneFingerTouch(t)&&this._isSameTouchEvent(t)}isValidEndEvent(t){return
this._isOneFingerTouch(t)&&this._isSameTouchEvent(t)}}const Ls=t=>
{t.mousedown=t.dragStart,t.mousemoveWindow=t.dragMove,t.mouseup=t.dragEnd,t.cont
extmenu=function(t){t.preventDefault();};},As=
({enable:t,clickTolerance:e,bearingDegreesPerPixelMoved:s=.8})=>{const a=new
Ms({checkCorrectEvent:t=>0===i.mouseButton(t)&&t.ctrlKey||2===i.mouseButton(t)})
;return new Ps({clickTolerance:e,move:(t,e)=>({bearingDelta:(e.x-
t.x)*s}),moveStateManager:a,enable:t,assignEvents:Ls}),Rs=
({enable:t,clickTolerance:e,pitchDegreesPerPixelMoved:s=-.5})=>{const a=new
Ms({checkCorrectEvent:t=>0===i.mouseButton(t)&&t.ctrlKey||2===i.mouseButton(t)})
;return new Ps({clickTolerance:e,move:(t,e)=>({pitchDelta:(e.y-
t.y)*s}),moveStateManager:a,enable:t,assignEvents:Ls})};class
ks{constructor(t,e){this._minTouches=t.cooperativeGestures?
2:1,this._clickTolerance=t.clickTolerance||1,this._map=e,this.reset();}reset()
{this._active=!1,this._touches={},this._sum=new t.P(0,0),setTimeout(((()=>
{this._cancelCooperativeMessage=!1;}),200));}touchstart(t,e,i){return
this._calculateTransform(t,e,i)}touchmove(t,e,i)
{if(this._map._cooperativeGestures&&
(2===this._minTouches&&i.length<2&&!this._cancelCooperativeMessage?
this._map._onCooperativeGesture(t,!1,i.length):this._cancelCooperativeMessage||
(this._cancelCooperativeMessage=!0)),this._active&&!
(i.length<this._minTouches))return
t.preventDefault(),this._calculateTransform(t,e,i)}touchend(t,e,i)
{this._calculateTransform(t,e,i),this._active&&i.length<this._minTouches&&this.r
eset();}touchcancel(){this.reset();}_calculateTransform(e,i,s){s.length>0&&
(this._active=!0);const a=Is(s,i),o=new t.P(0,0),r=new t.P(0,0);let
n=0;for(const t in a){const e=a[t],i=this._touches[t];i&&
(o._add(e),r._add(e.sub(i)),n++,a[t]=e);}if(this._touches=a,n<this._minTouches||
!r.mag())return;const l=r.div(n);return this._sum._add(l),this._sum.mag()
<this._clickTolerance?void 0:{around:o.div(n),panDelta:l}}enable()
{this._enabled=!0;}disable(){this._enabled=!1,this.reset();}isEnabled(){return
this._enabled}isActive(){return this._active}}class Fs{constructor()
{this.reset();}reset(){this._active=!1,delete
this._firstTwoTouches;}touchstart(t,e,i){this._firstTwoTouches||i.length<2||
(this._firstTwoTouches=
[i[0].identifier,i[1].identifier],this._start([e[0],e[1]]));}touchmove(t,e,i)
{if(!this._firstTwoTouches)return;t.preventDefault();const[s,a]=this._firstTwoTo
uches,o=Bs(i,e,s),r=Bs(i,e,a);if(!o||!r)return;const n=this._aroundCenter?
null:o.add(r).div(2);return this._move([o,r],n,t)}touchend(t,e,s)
{if(!this._firstTwoTouches)return;const[a,o]=this._firstTwoTouches,r=Bs(s,e,a),n
=Bs(s,e,o);r&&n||!(this._active&&i.suppressClick(),this.reset());}touchcancel()
{this.reset();}enable(t)
{this._enabled=!0,this._aroundCenter=!t&&"center"===t.around;}disable()
{this._enabled=!1,this.reset();}isEnabled(){return this._enabled}isActive()
{return this._active}}function Bs(t,e,i){for(let

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s=0;s<t.length;s++)if(t[s].identifier===i)return e[s]}function Os(t,e){return
Math.log(t/e)/Math.LN2}class Ns extends Fs{reset(){super.reset(),delete
this._distance,delete this._startDistance;}_start(t)
{this._startDistance=this._distance=t[0].dist(t[1]);}_move(t,e){const
i=this._distance;if(this._distance=t[0].dist(t[1]),this._active||!
(Math.abs(Os(this._distance,this._startDistance))<.1))return this._active=!0,
{zoomDelta:Os(this._distance,i),pinchAround:e}}function Us(t,e){return
180*t.angleWith(e)/Math.PI}class Zs extends Fs{reset(){super.reset(),delete
this._minDiameter,delete this._startVector,delete this._vector;}_start(t)
{this._startVector=this._vector=t[0].sub(t[1]),this._minDiameter=t[0].dist(t[1])
;}_move(t,e){const
i=this._vector;if(this._vector=t[0].sub(t[1]),this._active||!this._isBelowThresh
old(this._vector))return this._active=!0,
{bearingDelta:Us(this._vector,i),pinchAround:e}}_isBelowThreshold(t)
{this._minDiameter=Math.min(this._minDiameter,t.mag());const
e=25/(Math.PI*this._minDiameter)*360,i=Us(t,this._startVector);return
Math.abs(i)<e}}function Gs(t){return Math.abs(t.y)>Math.abs(t.x)}class js
extends Fs{constructor(t){super(),this._map=t;}reset()
{super.reset(),this._valid=void 0,delete this._firstMove,delete
this._lastPoints;}touchstart(t,e,i)
{super.touchstart(t,e,i),this._currentTouchCount=i.length;}_start(t)
{this._lastPoints=t,Gs(t[0].sub(t[1]))&&(this._valid=!1);}_move(t,e,i)
{if(this._map._cooperativeGestures&&this._currentTouchCount<3)return;const
s=t[0].sub(this._lastPoints[0]),a=t[1].sub(this._lastPoints[1]);return
this._valid=this.gestureBeginsVertically(s,a,i.timeStamp),this._valid?
(this._lastPoints=t,this._active=!0,{pitchDelta:(s.y+a.y)/2*-.5}):void
0}gestureBeginsVertically(t,e,i){if(void 0!==this._valid)return
this._valid;const s=t.mag()>=2,a=e.mag()>=2;if(!s&&!a)return;if(!s||!a)return
void 0===this._firstMove&&(this._firstMove=i),i-this._firstMove<100&&void
0;const o=t.y>0===e.y>0;return Gs(t)&&Gs(e)&&o}}const Vs=
{panStep:100,bearingStep:15,pitchStep:10};class qs{constructor(t){this._tr=new
ws(t);const
e=Vs;this._panStep=e.panStep,this._bearingStep=e.bearingStep,this._pitchStep=e.p
itchStep,this._rotationDisabled=!1;}reset(){this._active=!1;}keydown(t)
{if(t.altKey||t.ctrlKey||t.metaKey)return;let
e=0,i=0,s=0,a=0,o=0;switch(t.keyCode){case 61:case 107:case 171:case
187:e=1;break;case 189:case 109:case 173:e=-1;break;case 37:t.shiftKey?i=-1:
(t.preventDefault(),a=-1);break;case 39:t.shiftKey?i=1:
(t.preventDefault(),a=1);break;case 38:t.shiftKey?s=1:
(t.preventDefault(),o=-1);break;case 40:t.shiftKey?s=-1:
(t.preventDefault(),o=1);break;default:return}return this._rotationDisabled&&
(i=0,s=0),{cameraAnimation:r=>{const
n=this._tr;r.easeTo({duration:300,easeId:"keyboardHandler",easing:$s,zoom:e?
Math.round(n.zoom)+e*(t.shiftKey?
2:1):n.zoom,bearing:n.bearing+i*this._bearingStep,pitch:n.pitch+s*this._pitchSte
p,offset:[-a*this._panStep,-o*this._panStep],center:n.center},
{originalEvent:t}})}enable(){this._enabled=!0;}disable()
{this._enabled=!1,this.reset();}isEnabled(){return this._enabled}isActive()
{return this._active}disableRotation()
{this._rotationDisabled=!0;}enableRotation()
{this._rotationDisabled=!1;}}function $s(t){return t*(2-t)}const
Ws=4.000244140625;class Hs{constructor(t,e){this._onTimeout=t=>
{this._type="wheel",this._delta=
this._lastValue,this._active||this._start(t);},this._map=t,this._tr=new
ws(t),this._el=t.getCanvasContainer(),this._triggerRenderFrame=e,this._delta=0,t
his._defaultZoomRate=.01,this._wheelZoomRate=.0022222222222222222;}setZoomRate(t)
{this._defaultZoomRate=t;}setWheelZoomRate(t)
{this._wheelZoomRate=t;}isEnabled(){return !!this._enabled}isActive(){return
!!this._active||void 0!==this._finishTimeout}isZooming(){return
!!this._zooming}enable(t){this.isEnabled()||
(this._enabled=!0,this._aroundCenter=!!t&&"center"===t.around);}disable()
{this.isEnabled()&&(this._enabled=!1);}wheel(e)

```

```

{if(!this.isEnabled())return;if(this._map._cooperativeGestures)
{if(!e[this._map._metaKey])return;e.preventDefault();}let
i=e.deltaMode===WheelEvent.DOM_DELTA_LINE?40*e.deltaY:e.deltaY;const
s=t.h.now(),a=s-
(this._lastWheelEventTime||0);this._lastWheelEventTime=s,0!==i&&i%Ws==0?
this._type="wheel":0!==i&&Math.abs(i)<4?this._type="trackpad":a>400?
(this._type=null,this._lastValue=i,this._timeout=setTimeout(this._onTimeout,40,e
)):this._type||(this._type=Math.abs(a*i)<200?"trackpad":"wheel",this._timeout&&
(clearTimeout(this._timeout),this._timeout=null,i+=this._lastValue)),e.shiftKey&
&i&&(i/=4),this._type&&(this._lastWheelEvent=e,this._delta-
=i,this._active||this._start(e),e.preventDefault());}_start(e)
{if(!this._delta)return;this._frameId&&
(this._frameId=null),this._active=!0,this.isZooming()||
(this._zooming=!0),this._finishTimeout&&
(clearTimeout(this._finishTimeout),delete this._finishTimeout);const
s=i.mousePos(this._el,e),a=this._tr;this._around=t.L.convert(this._aroundCenter?
a.center:a.unproject(s)),this._aroundPoint=a.transform.locationPoint(this._around
d),this._frameId||(this._frameId=!0,this._triggerRenderFrame());}renderFrame()
{if(!this._frameId)return;if(this._frameId=null,!this.isActive())return;const
e=this._tr.transform;if(0!==this._delta){const
t="wheel"===this._type&&Math.abs(this._delta)>Ws?
this._wheelZoomRate:this._defaultZoomRate;let i=2/(1+Math.exp(-
Math.abs(this._delta*t)));this._delta<0&&0!==i&&(i=1/i);const s="number"===typeof
this._targetZoom?
e.zoomScale(this._targetZoom):e.scale;this._targetZoom=Math.min(e.maxZoom,Math.m
ax(e.minZoom,e.scaleZoom(s*i))),"wheel"===this._type&&
(this._startZoom=e.zoom,this._easing=this._smoothOutEasing(200)),this._delta=0;}
const i="number"===typeof this._targetZoom?
this._targetZoom:e.zoom,s=this._startZoom,a=this._easing;let
o,r=!1;if("wheel"===this._type&&s&&a){const e=Math.min((t.h.now()-
this._lastWheelEventTime)/200,1),n=a(e);o=t.B.number(s,i,n),e<1?this._frameId||
(this._frameId=!0):r=!0;}else o=i,r=!0;return this._active=!0,r&&
(this._active=!1,this._finishTimeout=setTimeout((()=>>
{this._zooming=!1,this._triggerRenderFrame(),delete this._targetZoom,delete
this._finishTimeout;}),200)),{noInertia:!0,needsRenderFrame:!r,zoomDelta:o-
e.zoom,around:this._aroundPoint,originalEvent:this._lastWheelEvent}}_smoothOutEa
sing(e){let i=t.bb;if(this._prevEase){const e=this._prevEase,s=(t.h.now()-
e.start)/e.duration,a=e.easing(s+.01)-e.easing(s),o=.27/Math.sqrt(a*a+1e-
4)*.01,r=Math.sqrt(.0729-o*o);i=t.ba(o,r,.25,1);}return this._prevEase=
{start:t.h.now(),duration:e,easing:i},i}reset()
{this._active=!1,this._zooming=!1,delete this._targetZoom,this._finishTimeout&&
(clearTimeout(this._finishTimeout),delete this._finishTimeout);}class
Xs{constructor(t,e){this._clickZoom=t,this._tapZoom=e;}enable()
{this._clickZoom.enable(),this._tapZoom.enable();}disable()
{this._clickZoom.disable(),this._tapZoom.disable();}isEnabled(){return
this._clickZoom.isEnabled()&&this._tapZoom.isEnabled()}isActive(){return
this._clickZoom.isActive()||this._tapZoom.isActive()}class Ks{constructor(t)
{this._tr=new ws(t),this.reset();}reset(){this._active=!1;}dblclick(t,e){return
t.preventDefault(),{cameraAnimation:i=>
{i.easeTo({duration:300,zoom:this._tr.zoom+(t.shiftKey?
-1:1),around:this._tr.unproject(e)},{originalEvent:t});}}enable()
{this._enabled=!0;}disable(){this._enabled=!1,this.reset();}isEnabled(){return
this._enabled}isActive(){return this._active}}class Qs{constructor()
{this._tap=new Ss({numTouches:1,numTaps:1}),this.reset();}reset()
{this._active=!1,delete this._swipePoint,delete this._swipeTouch,delete
this._tapTime,delete this._tapPoint,this._tap.reset();}touchstart(t,e,i)
{if(!this._swipePoint)if(this._tapTime){const s=e[0],a=t.timeStamp-
this._tapTime<500,o=this._tapPoint.dist(s)<30;a&&o&&i.length>0&&
(this._swipePoint=s,this._swipeTouch=i[0].identifier):this.reset();}else
this._tap.touchstart(t,e,i);}touchmove(t,e,i){if(this._tapTime)
{if(this._swipePoint){if(i[0].identifier!==this._swipeTouch)return;const
s=e[0],a=s.y-this._swipePoint.y;return

```

```

this._swipePoint=s,t.preventDefault(),this._active=!0,{zoomDelta:a/128}}}else
this._tap.touchmove(t,e,i);}touchend(t,e,i)
{if(this._tapTime)this._swipePoint&&0===i.length&&this.reset();else {const
s=this._tap.touchend(t,e,i);s&&
(this._tapTime=t.timeStamp,this._tapPoint=s);}touchcancel()
{this.reset();}enable(){this._enabled=!0;}disable()
{this._enabled=!1,this.reset();}isEnabled(){return this._enabled}isActive()
{return this._active}}class Ys{constructor(t,e,i)
{this._el=t,this._mousePan=e,this._touchPan=i;}enable(t)
{this._inertiaOptions=t||
{},this._mousePan.enable(),this._touchPan.enable(),this._el.classList.add("mapli
bregl-touch-drag-pan");}disable()
{this._mousePan.disable(),this._touchPan.disable(),this._el.classList.remove("ma
plibregl-touch-drag-pan");}isEnabled(){return
this._mousePan.isEnabled()}&&this._touchPan.isEnabled()}isActive(){return
this._mousePan.isActive()}||this._touchPan.isActive()}class
Js{constructor(t,e,i)
{this._pitchWithRotate=t.pitchWithRotate,this._mouseRotate=e,this._mousePitch=i;
}enable()
{this._mouseRotate.enable(),this._pitchWithRotate&&this._mousePitch.enable();}di
sable(){this._mouseRotate.disable(),this._mousePitch.disable();}isEnabled()
{return this._mouseRotate.isEnabled()}&&
(!this._pitchWithRotate||this._mousePitch.isEnabled())}isActive(){return
this._mouseRotate.isActive()}||this._mousePitch.isActive()}class
ta{constructor(t,e,i,s)
{this._el=t,this._touchZoom=e,this._touchRotate=i,this._tapDragZoom=s,this._rota
tionDisabled=!1,this._enabled=!0;}enable(t)
{this._touchZoom.enable(t),this._rotationDisabled||this._touchRotate.enable(t),t
his._tapDragZoom.enable(),this._el.classList.add("maplibregl-touch-zoom-
rotate");}disable()
{this._touchZoom.disable(),this._touchRotate.disable(),this._tapDragZoom.disable
(),this._el.classList.remove("maplibregl-touch-zoom-rotate");}isEnabled(){return
this._touchZoom.isEnabled()}&&
(this._rotationDisabled||this._touchRotate.isEnabled())&&this._tapDragZoom.isEna
bled()}isActive(){return
this._touchZoom.isActive()}||this._touchRotate.isActive()}||this._tapDragZoom.isAc
tive()}disableRotation()
{this._rotationDisabled=!0,this._touchRotate.disable();}enableRotation()
{this._rotationDisabled=!1,this._touchZoom.isEnabled()}&&this._touchRotate.enable
();}const ea=t=>t.zoom|t.drag|t.pitch|t.rotate;class ia extends
t.k{}function sa(t){return
t.panDelta&&t.panDelta.mag()||t.zoomDelta|t.bearingDelta|t.pitchDelta}class
aa{constructor(t,e){this.handleWindowEvent=t=>
{this.handleEvent(t,`${t.type}Window`);},this.handleEvent=(t,e)=>
{if("blur"===t.type)return void this.stop(!0);this._updatingCamera=!0;const
s="renderFrame"===t.type?void 0:t,a={needsRenderFrame:!1},o={},r=
{},n=t.touches,l=n?this._getMapTouches(n):void 0,h=l?
i.touchPos(this._el,l):i.mousePos(this._el,t);for(const{handlerName:i,handler:n,
allowed:c}of this._handlers){if(!n.isEnabled())continue;let
u;this._blockedByActive(r,c,i)?n.reset():n[e|t.type]&&(u=n[e|t.type]
(t,h,l),this.mergeHandlerResult(a,o,u,i,s),u&&u.needsRenderFrame&&this._triggerR
enderFrame()),(u|n.isActive())&&(r[i]=n);}const c={};for(const t in
this._previousActiveHandlers)r[t]||(c[t]=s);this._previousActiveHandlers=r,
(Object.keys(c).length|sa(a))&&
(this._changes.push([a,o,c]),this._triggerRenderFrame()),
(Object.keys(r).length|sa(a))&&this._map._stop(!0),this._updatingCamera=!1;cons
t{cameraAnimation:u}=a;u&&(this._inertia.clear(),this._fireEvents({},
{},!0),this._changes=
[],u(this._map));},this._map=t,this._el=this._map.getCanvasContainer(),this._han
dlers=[],this._handlersById={},this._changes=[],this._inertia=new
ps(t),this._bearingSnap=e.bearingSnap,this._previousActiveHandlers=
{},this._eventsInProgress={},this._addDefaultHandlers(e);const

```



```

s=this._el;this._listeners=[s,"touchstart",{passive:!0}],[s,"touchmove",
{passive:!1}],[s,"touchend",void 0],[s,"touchcancel",void 0],[s,"mousedown",void
0],[s,"mousemove",void 0],[s,"mouseup",void 0],[document,"mousemove",
{capture:!0}],[document,"mouseup",void 0],[s,"mouseover",void 0],
[s,"mouseout",void 0],[s,"dblclick",void 0],[s,"click",void 0],[s,"keydown",
{capture:!1}],[s,"keyup",void 0],[s,"wheel",{passive:!1}],[s,"contextmenu",void
0],[window,"blur",void 0]];for(const[t,e,s]of
this._listeners)i.addEventListener(t,e,t===document?
this.handleWindowEvent:this.handleEvent,s);}destroy(){for(const[t,e,s]of
this._listeners)i.removeEventListener(t,e,t===document?
this.handleWindowEvent:this.handleEvent,s);}_addDefaultHandlers(t){const
e=this._map,s=e.getCanvasContainer();this._add("mapEvent",new ys(e,t));const
a=e.boxZoom=new
Ts(e,t);this._add("boxZoom",a),t.interactive&&t.boxZoom&&a.enable();const o=new
Cs(e),r=new Ks(e);e.doubleClickZoom=new
Xs(r,o),this._add("tapZoom",o),this._add("clickZoom",r),t.interactive&&t.doubleC
lickZoom&&e.doubleClickZoom.enable();const n=new
Qs;this._add("tapDragZoom",n);const l=e.touchPitch=new
js(e);this._add("touchPitch",l),t.interactive&&t.touchPitch&&e.touchPitch.enable
(t.touchPitch);const h=As(t),c=Rs(t);e.dragRotate=new
Js(t,h,c),this._add("mouseRotate",h,["mousePitch"]),this._add("mousePitch",c,
["mouseRotate"]),t.interactive&&t.dragRotate&&e.dragRotate.enable();const u=
(({enable:t,clickTolerance:e})=>{const s=new
Ms({checkCorrectEvent:t>0===i.mouseButton(t)&&!t.ctrlKey});return new
Ps({clickTolerance:e,move:(t,e)>=>
({around:e,panDelta:e.sub(t)}),activateOnStart:!0,moveStateManager:s,enable:t,as
signEvents:Ls})))(t),d=new ks(t,e);e.dragPan=new
Ys(s,u,d),this._add("mousePan",u),this._add("touchPan",d,
["touchZoom","touchRotate"]),t.interactive&&t.dragPan&&e.dragPan.enable(t.dragPa
n);const _=new Zs,p=new Ns;e.touchZoomRotate=new
ta(s,p,_),this._add("touchRotate",_,
["touchPan","touchZoom"]),this._add("touchZoom",p,
["touchPan","touchRotate"]),t.interactive&&t.touchZoomRotate&&e.touchZoomRotate.
enable(t.touchZoomRotate);const m=e.scrollZoom=new Hs(e,
(())=>this._triggerRenderFrame());this._add("scrollZoom",m,
["mousePan"]),t.interactive&&t.scrollZoom&&e.scrollZoom.enable(t.scrollZoom);con
st f=e.keyboard=new
qs(e);this._add("keyboard",f),t.interactive&&t.keyboard&&e.keyboard.enable(),thi
s._add("blockableMapEvent",new bs(e));}_add(t,e,i)
{this._handlers.push({handlerName:t,handler:e,allowed:i}),this._handlersById[t]=
e;}stop(t){if(!this._updatingCamera){for(const{handler:t}of
this._handlers)t.reset();this._inertia.clear(),this._fireEvents({},
{}),t),this._changes=[];}}isActive(){for(const{handler:t}of
this._handlers)if(t.isActive())return !0;return !1}isZooming(){return
!!this._eventsInProgress.zoom||this._map.scrollZoom.isZooming()}isRotating()
{return !!this._eventsInProgress.rotate}isMoving(){return
Boolean(ea(this._eventsInProgress))||this.isZooming()}_blockedByActive(t,e,i)
{for(const s in t)if(s!==i&&(!e||e.indexOf(s)<0))return !0;return
!1}_getMapTouches(t){const e=[];for(const i of
t)this._el.contains(i.target)&&e.push(i);return e}mergeHandlerResult(e,i,s,a,o)
{if(!s)return;t.e(e,s);const r=
{handlerName:a,originalEvent:s.originalEvent||o;void 0!==s.zoomDelta&&
(i.zoom=r),void 0!==s.panDelta&&(i.drag=r),void 0!==s.pitchDelta&&
(i.pitch=r),void 0!==s.bearingDelta&&(i.rotate=r);}_applyChanges(){const e={},i=
{},s={};for(const[a,o,r]of this._changes)a.panDelta&&(e.panDelta=
(e.panDelta||new t.P(0,0))._add(a.panDelta)),a.zoomDelta&&(e.zoomDelta=
(e.zoomDelta||0)+a.zoomDelta),a.bearingDelta&&(e.bearingDelta=
(e.bearingDelta||0)+a.bearingDelta),a.pitchDelta&&(e.pitchDelta=
(e.pitchDelta||0)+a.pitchDelta),void 0!==a.around&&(e.around=a.around),void
0!==a.pinchAround&&(e.pinchAround=a.pinchAround),a.noInertia&&
(e.noInertia=a.noInertia),t.e(i,o),t.e(s,r);this._updateMapTransform(e,i,s),this
._changes=[];}_updateMapTransform(t,e,i){const

```

```

s=this._map,a=s._getTransformForUpdate(),o=s.terrain;if(!
(sa(t)||o&&this._terrainMovement))return
this._fireEvents(e,i,!0);let{panDelta:r,zoomDelta:n,bearingDelta:l,pitchDelta:h,
around:c,pinchAround:u}=t;void 0!==u&&
(c=u),s._stop(!0),c=c||s.transform.centerPoint;const d=a.pointLocation(r?
c.sub(r):c);l&&(a.bearing+=l),h&&(a.pitch+=h),n&&(a.zoom+=n),o?
this._terrainMovement||!e.drag&&!e.zoom?e.drag&&this._terrainMovement?
a.center=a.pointLocation(a.centerPoint.sub(r)):a.setLocationAtPoint(d,c):
(this._terrainMovement=!0,this._map._elevationFreeze=!0,a.setLocationAtPoint(d,c
),this._map.once("moveend",(()=>>
{this._map._elevationFreeze=!1,this._terrainMovement=!1,a.recalculateZoom(s.terr
ain);})):a.setLocationAtPoint(d,c),s._applyUpdatedTransform(a),this._map._updat
e(),t.noInertia||this._inertia.record(t),this._fireEvents(e,i,!0);}
_fireEvents(e
,i,s){const a=ea(this._eventsInProgress),o=ea(e),r={};for(const t in e)
{const{originalEvent:i}=e[t];this._eventsInProgress[t]||
(r[`$${t}start`]=i),this._eventsInProgress[t]=e[t];}!a&&o&&this._fireEvent("moves
tart",o.originalEvent);for(const t in
r)this._fireEvent(t,r[t]);o&&this._fireEvent("move",o.originalEvent);for(const t
in e){const{originalEvent:i}=e[t];this._fireEvent(t,i);}const n={};let
l;for(const t in this._eventsInProgress)
{const{handlerName:e,originalEvent:s}=this._eventsInProgress[t];this._handlersBy
Id[e].isActive()||(delete
this._eventsInProgress[t],l=i[e]||s,n[`$${t}end`]=l);}for(const t in
n)this._fireEvent(t,n[t]);const h=ea(this._eventsInProgress);if(s&&(a||o)&&!h)
{this._updatingCamera=!0;const
e=this._inertia._onMoveEnd(this._map.dragPan._inertiaOptions),i=t=>0!==t&&-
this._bearingSnap<t&&t<this._bearingSnap;!e||!e.essential&&t.h.prefersReducedMot
ion?(this._map.fire(new t.k("moveend",
{originalEvent:l})),i(this._map.getBearing())&&this._map.resetNorth()):
(i(e.bearing)||this._map.getBearing())&&
(e.bearing=0),e.freezeElevation=!0,this._map.easeTo(e,
{originalEvent:l})),this._updatingCamera=!1;}_fireEvent(e,i){this._map.fire(new
t.k(e,i?{originalEvent:i}:{}));}_requestFrame(){return
this._map.triggerRepaint(),this._map._renderTaskQueue.add((t=>{delete
this._frameId,this._handleEvent(new ia("renderFrame",
{timestamp:t})),this._applyChanges();}));}_triggerRenderFrame(){void
0===this._frameId&&(this._frameId=this._requestFrame());}class oa extends
t.E{constructor(e,i){super(),this._renderFrameCallback=()=>{const
e=Math.min((t.h.now()-
this._easeStart)/this._easeOptions.duration,1);this._onEaseFrame(this._easeOptio
ns.easing(e)),e<1&&this._easeFrameId?
this._easeFrameId=this._requestRenderFrame(this._renderFrameCallback):this.stop(
);},this._moving=!1,this._zooming=!1,this.transform=e,this._bearingSnap=i.bearin
gSnap,this.on("moveend",(()=>>{delete this._requestedCameraState;}));}getCenter()
{return new
t.L(this.transform.center.lng,this.transform.center.lat)}setCenter(t,e){return
this.jumpTo({center:t},e)}panBy(e,i,s){return
e=t.P.convert(e).mult(-1),this.panTo(this.transform.center,t.e({offset:e},i),s)}
panTo(e,i,s){return this.easeTo(t.e({center:e},i),s)}getZoom(){return
this.transform.zoom}setZoom(t,e){return
this.jumpTo({zoom:t},e),this.zoomTo(e,i,s)}return
this.easeTo(t.e({zoom:e},i),s)}zoomIn(t,e){return
this.zoomTo(this.getZoom()+1,t,e),this.zoomOut(t,e)}return
this.zoomTo(this.getZoom()-1,t,e),this.getBearing(){return
this.transform.bearing}setBearing(t,e){return
this.jumpTo({bearing:t},e),this.getPadding(){return
this.transform.padding}setPadding(t,e){return
this.jumpTo({padding:t},e),this.rotateTo(e,i,s)}return
this.easeTo(t.e({bearing:e},i),s)}resetNorth(e,i){return
this.rotateTo(0,t.e({duration:1e3},e),i),this.resetNorthPitch(e,i)}return
this.easeTo(t.e({bearing:0,pitch:0,duration:1e3},e),i),this.snapToNorth(t,e)
{return Math.abs(this.getBearing())<this._bearingSnap?

```

```

this.resetNorth(t,e):this}getPitch(){return this.transform.pitch}setPitch(t,e)
{return this.jumpTo({pitch:t},e),this}cameraForBounds(t,e){t=L.convert(t);const
i=e&&e.bearing||0;return
this._cameraForBoxAndBearing(t.getNorthWest(),t.getSouthEast(),i,e)}_cameraForBo
xAndBearing(e,i,s,a){const o=
{top:0,bottom:0,right:0,left:0};if("number"===typeof(a=t.e({padding:o,offset:
[0,0],maxZoom:this.transform.maxZoom},a)).padding){const t=a.padding;a.padding=
{top:t,bottom:t,right:t,left:t};}a.padding=t.e(o,a.padding);const
r=this.transform,n=r.padding,l=r.project(t.L.convert(e)),h=r.project(t.L.convert
(i)),c=l.rotate(-s*Math.PI/180),u=h.rotate(-s*Math.PI/180),d=new
t.P(Math.max(c.x,u.x),Math.max(c.y,u.y)),_=new
t.P(Math.min(c.x,u.x),Math.min(c.y,u.y)),p=d.sub(_),m=(r.width-
(n.left+n.right+a.padding.left+a.padding.right))/p.x,f=(r.height-
(n.top+n.bottom+a.padding.top+a.padding.bottom))/p.y;if(f<0||m<0)return void
t.w("Map cannot fit within canvas with the given bounds, padding, and/or
offset.");const
g=Math.min(r.scaleZoom(r.scale*Math.min(m,f)),a.maxZoom),v=t.P.convert(a.offset)
,x=new t.P((a.padding.left-a.padding.right)/2,(a.padding.top-
a.padding.bottom)/2).rotate(s*Math.PI/180),y=v.add(x).mult(r.scale/r.zoomScale(g
));return
{center:r.unproject(l.add(h).div(2).sub(y)),zoom:g,bearing:s}}fitBounds(t,e,i)
{return
this._fitInternal(this.cameraForBounds(t,e),e,i)}fitScreenCoordinates(e,i,s,a,o)
{return
this._fitInternal(this._cameraForBoxAndBearing(this.transform.pointLocation(t.P.
convert(e)),this.transform.pointLocation(t.P.convert(i)),s,a),a,o)}_fitInternal(
e,i,s){return e?(delete(i=t.e(e,i)).padding,i.linear?
this.easeTo(i,s):this.flyTo(i,s)):this.jumpTo(e,i){this.stop();const
s=this._getTransformForUpdate();let a=!1,o=!1,r=!1;return "zoom"in
e&&s.zoom!=="e.zoom"&&(a=!0,s.zoom+=e.zoom),void 0!=="e.center"&&
(s.center=t.L.convert(e.center)),"bearing"in e&&s.bearing!=="e.bearing"&&
(o=!0,s.bearing+=e.bearing),"pitch"in e&&s.pitch!=="e.pitch"&&
(r=!0,s.pitch+=e.pitch),null=="e.padding"||s.isPaddingEqual(e.padding)||
(s.padding=e.padding),this._applyUpdatedTransform(s),this.fire(new
t.k("movestart",i)).fire(new t.k("move",i)),a&&this.fire(new
t.k("zoomstart",i)).fire(new t.k("zoom",i)).fire(new
t.k("zoomend",i)),o&&this.fire(new t.k("rotatestart",i)).fire(new
t.k("rotate",i)).fire(new t.k("rotateend",i)),r&&this.fire(new
t.k("pitchstart",i)).fire(new t.k("pitch",i)).fire(new
t.k("pitchend",i)),this.fire(new
t.k("moveend",i))}calculateCameraOptionsFromTo(e,i,s,a=0){const
o=t.U.fromLngLat(e,i),r=t.U.fromLngLat(s,a),n=r.x-o.x,l=r.y-o.y,h=r.z-
o.z,c=Math.hypot(n,l,h);if(0===c)throw new Error("Can't calculate camera options
with same From and To");const
u=Math.hypot(n,l),d=this.transform.scaleZoom(this.transform.cameraToCenterDistan
ce/c/this.transform.tileSize),_=180*Math.atan2(n,-l)/Math.PI;let
p=180*Math.acos(u/c)/Math.PI;return p=h<0?90-p:90+p,
{center:r.toLngLat(),zoom:d,pitch:p,bearing:_}}easeTo(e,i)
{this._stop(!1,e.easeId),(!!1===e=t.e({offset:
[0,0],duration:500,easing:t.bb},e)).animate||!e.essential&&t.h.prefersReducedMot
ion)&&(e.duration=0);const
s=this._getTransformForUpdate(),a=this.getZoom(),o=this.getBearing(),r=this.getP
itch(),n=this.getPadding(),l="zoom"in e?+e.zoom:a,h="bearing"in e?
this._normalizeBearing(e.bearing,o):o,c="pitch"in e?+e.pitch:r,u="padding"in e?
e.padding:s.padding,d=t.P.convert(e.offset);let _=s.centerPoint.add(d);const
p=s.pointLocation(_),m=t.L.convert(e.center|p);this._normalizeCenter(m);const
f=s.project(p),g=s.project(m).sub(f),v=s.zoomScale(1-a);let x,y;e.around&&
(x=t.L.convert(e.around),y=s.locationPoint(x));const b=
{moving:this._moving,zooming:this._zooming,rotating:this._rotating,pitching:this
._pitching};return
this._zooming=this._zooming||!1===a,this._rotating=this._rotating||o!=="h",this._pi
tching=this._pitching||c!=="r",this._padding=!s.isPaddingEqual(u),this._easeId=e.e

```

```

aseId,this._prepareEase(i,e.noMoveStart,b),this.terrain&&this._prepareElevation(
m),this._ease((p=>{if(this._zooming&&(s.zoom=t.B.number(a,l,p)),this._rotating&&
(s.bearing=t.B.number(o,h,p)),this._pitching&&
(s.pitch=t.B.number(r,c,p)),this._padding&&
(s.interpolatePadding(n,u,p),_=s.centerPoint.add(d)),this.terrain&&!e.freezeElev
ation&&this._updateElevation(p),x)s.setLocationAtPoint(x,y);else {const
t=s.zoomScale(s.zoom-a),e=1>a?Math.min(2,v):Math.max(.5,v),i=Math.pow(e,1-
p),o=s.unproject(f.add(g.mult(p*i)).mult(t));s.setLocationAtPoint(s.renderWorldC
opies?o.wrap():o,_);}this._applyUpdatedTransform(s),this._fireMoveEvents(i);}),
(t=>
{this.terrain&&this._finalizeElevation(),this._afterEase(i,t);}),e),this}_prepar
eEase(e,i,s={}){this._moving=!0,i||s.moving||this.fire(new
t.k("movestart",e)),this._zooming&&!s.zooming&&this.fire(new
t.k("zoomstart",e)),this._rotating&&!s.rotating&&this.fire(new
t.k("rotatestart",e)),this._pitching&&!s.pitching&&this.fire(new
t.k("pitchstart",e));}_prepareElevation(t)
{this._elevationCenter=t,this._elevationStart=this.transform.elevation,this._ele
vationTarget=this.terrain.getElevationForLngLatZoom(t,this.transform.tileZoom),t
his._elevationFreeze=!0;}_updateElevation(e)
{this.transform._minElevationForCurrentTile=this.terrain.getMinTileElevationFor
LngLatZoom(this._elevationCenter,this.transform.tileZoom);const
i=this.terrain.getElevationForLngLatZoom(this._elevationCenter,this.transform.ti
leZoom);if(e<1&&i!==this._elevationTarget){const t=this._elevationTarget-
this._elevationStart;this._elevationStart+=e*(t-(i-
(t*e+this._elevationStart)))/(1-
e)),this._elevationTarget=i;}this.transform.elevation=t.B.number(this._elevation
Start,this._elevationTarget,e);}_finalizeElevation()
{this._elevationFreeze=!1,this.transform.recalculateZoom(this.terrain);}_getTran
sformForUpdate(){return this.transformCameraUpdate?(this._requestedCameraState||
(this._requestedCameraState=this.transform.clone()),this._requestedCameraState):
this.transform}_applyUpdatedTransform(t)
{if(!this.transformCameraUpdate)return;const e=t.clone(),
{center:i,zoom:s,pitch:a,bearing:o,elevation:r}=this.transformCameraUpdate(e);i&
&(e.center=i),void 0!==s&&(e.zoom=s),void 0!==a&&(e.pitch=a),void 0!==o&&
(e.bearing=o),void 0!==r&&
(e.elevation=r),this.transform.apply(e);}_fireMoveEvents(e){this.fire(new
t.k("move",e)),this._zooming&&this.fire(new
t.k("zoom",e)),this._rotating&&this.fire(new
t.k("rotate",e)),this._pitching&&this.fire(new t.k("pitch",e));}_afterEase(e,i)
{if(this._easeId&&i&&this._easeId===i)return;delete this._easeId;const
s=this._zooming,a=this._rotating,o=this._pitching;this._moving=!1,this._zooming=
!1,this._rotating=!1,this._pitching=!1,this._padding=!1,s&&this.fire(new
t.k("zoomend",e)),a&&this.fire(new t.k("rotateend",e)),o&&this.fire(new
t.k("pitchend",e)),this.fire(new t.k("moveend",e));}_flyTo(e,i)
{if(!e.essential&&t.h.prefersReducedMotion){const s=t.F(e,
["center","zoom","bearing","pitch","around"]);return
this.jumpTo(s,i)}this.stop(),e=t.e({offset:
[0,0],speed:1.2,curve:1.42,easing:t.bb},e);const
s=this._getTransformForUpdate(),a=this.getZoom(),o=this.getBearing(),r=this.getP
itch(),n=this.getPadding(),l="zoom"in e?
t.ad(+e.zoom,s.minZoom,s.maxZoom):a,h="bearing"in e?
this._normalizeBearing(e.bearing,o):o,c="pitch"in e?+e.pitch:r,u="padding"in e?
e.padding:s.padding,d=s.zoomScale(1-a),_=t.P.convert(e.offset);let
p=s.centerPoint.add(_);const
m=s.pointLocation(p),f=t.L.convert(e.center||m);this._normalizeCenter(f);const
g=s.project(m),v=s.project(f).sub(g);let x=e.curve;const
y=Math.max(s.width,s.height),b=y/d,w=v.mag();if("minZoom"in e){const
i=t.ad(Math.min(e.minZoom,a,l),s.minZoom,s.maxZoom),o=y/s.zoomScale(i-
a);x=Math.sqrt(o/w*2);}const T=x*x;function I(t){const e=(b*b-y*y+(t?
-1:1)*T*T*w*w)/(2*(t?b:y)*T*w);return Math.log(Math.sqrt(e*e+1)-e)}function E(t)
{return (Math.exp(t)-Math.exp(-t))/2}function S(t){return
(Math.exp(t)+Math.exp(-t))/2}const C=I(!1);let P=function(t){return

```

```

S(C)/S(C+x*t)},D=function(t){return y*((S(C)*(E(e=C+x*t)/S(e))-E(C))/T)/w;var
e;},M=(I(!0)-C)/x;if(Math.abs(w)<1e-6||!isFinite(M)){if(Math.abs(y-b)<1e-
6)return this.easeTo(e,i);const t=b<y?
-1:1;M=Math.abs(Math.log(b/y))/x,D=function(){return 0},P=function(e){return
Math.exp(t*x*e)};return e.duration="duration"in e?
+e.duration:1e3*M/("screenSpeed"in e?
+e.screenSpeed/x:+e.speed),e.maxDuration&&e.duration>e.maxDuration&&
(e.duration=0),this._zooming=!0,this._rotating=o!==(h,this._pitching=c!==(r,this._
padding=!s.isPaddingEqual(u),this._prepareEase(i,!1),this.terrain&&this._prepare
Elevation(f),this._ease((d=>{const m=d*M,x=1/P(m);s.zoom=1===d?
l:a+s.scaleZoom(x),this._rotating&&
(s.bearing=t.B.number(o,h,d)),this._pitching&&
(s.pitch=t.B.number(r,c,d)),this._padding&&
(s.interpolatePadding(n,u,d),p=s.centerPoint.add(_),this.terrain&&!e.freezeElev
ation&&this._updateElevation(d);const y=1===d?
f:s.unproject(g.add(v.mult(D(m))).mult(x));s.setLocationAtPoint(s.renderWorldCop
ies?y.wrap():y,p),this._applyUpdatedTransform(s),this._fireMoveEvents(i);})),
(())=>
{this.terrain&&this._finalizeElevation(),this._afterEase(i);}),e,this}isEasing(
){return !!this._easeFrameId}stop(){return this._stop()}_stop(t,e)
{if(this._easeFrameId&&(this._cancelRenderFrame(this._easeFrameId),delete
this._easeFrameId,delete this._onEaseFrame),this._onEaseEnd){const
t=this._onEaseEnd;delete this._onEaseEnd,t.call(this,e);}if(!t){const
t=this.handlers;t&&t.stop(!1);}return this}_ease(e,i,s)
{!1===s.animate||0===s.duration?(e(1),i()):
(this._easeStart=t.h.now(),this._easeOptions=s,this._onEaseFrame=e,this._onEaseE
nd=i,this._easeFrameId=this._requestRenderFrame(this._renderFrameCallback));}_no
rmalizeBearing(e,i){e=t.b5(e,-180,180);const s=Math.abs(e-i);return Math.abs(e-
360-i)<s&&(e=-360),Math.abs(e+360-i)<s&&(e+=360),e}_normalizeCenter(t){const
e=this.transform;if(!e.renderWorldCopies||e.lngRange)return;const i=t.lng-
e.center.lng;t.lng+=i>180?-360:i<-180?360:0;}queryTerrainElevation(e){return
this.terrain?
this.terrain.getElevationForLngLatZoom(t.L.convert(e),this.transform.tileZoom)-
this.transform.elevation:null}}class ra{constructor(t={})
{this._toggleAttribution=()=>{this._container.classList.contains("maplibregl-
compact")&&(this._container.classList.contains("maplibregl-compact-show")?
(this._container.setAttribute("open",""),this._container.classList.remove("mapli
bregl-compact-show")):(this._container.classList.add("maplibregl-compact-
show"),this._container.removeAttribute("open"))};},this._updateData=t=>
{!t||"metadata"!==t.sourceDataType&&"visibility"!==t.sourceDataType&&"style"!==t
.dataType&&"terrain"!==t.type||this._updateAttributions();},this._updateCompact=
()=>>
{this._map.getCanvasContainer().offsetWidth<=640||this._compact?!1===this._compa
ct?
this._container.setAttribute("open",""):this._container.classList.contains("mapl
ibregl-compact")||this._container.classList.contains("maplibregl-attrib-
empty")||
(this._container.setAttribute("open",""),this._container.classList.add("maplibre
gl-compact","maplibregl-compact-show")):
(this._container.setAttribute("open",""),this._container.classList.contains("map
libregl-compact")&&this._container.classList.remove("maplibregl-
compact","maplibregl-compact-show"))};},this._updateCompactMinimize=()=>>
{this._container.classList.contains("maplibregl-
compact")&&this._container.classList.contains("maplibregl-compact-
show")&&this._container.classList.remove("maplibregl-compact-
show");},this.options=t;}getDefaultPosition(){return "bottom-right"}onAdd(t)
{return
this._map=t,this._compact=this.options&&this.options.compact,this._container=i.c
reate("details","maplibregl-ctrl maplibregl-ctrl-
attrib"),this._compactButton=i.create("summary","maplibregl-ctrl-attrib-
button",this._container),this._compactButton.addEventListener("click",this._togg
leAttribution),this._setElementTitle(this._compactButton,"ToggleAttribution"),th

```

```

is._innerContainer=i.create("div","maplibregl-ctrl-attri-
inner",this._container),this._updateAttributions(),this._updateCompact(),this._m
ap.on("styledata",this._updateData),this._map.on("sourcedata",this._updateData),
this._map.on("terrain",this._updateData),this._map.on("resize",this._updateCompa
ct),this._map.on("drag",this._updateCompactMinimize),this._container}onRemove()
{i.remove(this._container),this._map.off("styledata",this._updateData),this._map
.off("sourcedata",this._updateData),this._map.off("terrain",this._updateData),th
is._map.off("resize",this._updateCompact),this._map.off("drag",this._updateCompa
ctMinimize),this._map=void 0,this._compact=void 0,this._attribHTML=void
0;}_setElementTitle(t,e){const
i=this._map._getUIString(`AttributionControl.${e}`);t.title=i,t.setAttribute("ar
ia-label",i);}_updateAttributions(){if(!this._map.style)return;let t=
[];if(this.options.customAttribution&&
(Array.isArray(this.options.customAttribution)?
t=t.concat(this.options.customAttribution.map((t=>"string"!==typeof
t?"":t)):"string"!==typeof
this.options.customAttribution&&t.push(this.options.customAttribution)),this._ma
p.style.stylesheet){const
t=this._map.style.stylesheet;this.styleOwner=t.owner,this.styleId=t.id;}const
e=this._map.style.sourceCaches;for(const i in e){const
s=e[i];if(s.used||s.usedForTerrain){const
e=s.getSource();e.attribution&&t.indexOf(e.attribution)
<0&&t.push(e.attribution);}}t=t.filter((t=>String(t).trim()),t.sort(((t,e)=>t.l
ength-e.length)),t=t.filter(((e,i)=>{for(let
s=i+1;s<t.length;s++)if(t[s].indexOf(e)>=0)return !1;return !0})));const
i=t.join(" | ");i!==this._attribHTML&&(this._attribHTML=i,t.length?
(this._innerContainer.innerHTML=i,this._container.classList.remove("maplibregl-
attrib-empty")):this._container.classList.add("maplibregl-attri-
empty"),this._updateCompact(),this._editLink=null);}}class na{constructor(t={})
{this._updateCompact=()=>{const t=this._container.children;if(t.length){const
e=t[0];this._map.getCanvasContainer().offsetWidth<=640||this._compact?!1!==this.
_compact&&e.classList.add("maplibregl-compact"):e.classList.remove("maplibregl-
compact");}},this.options=t;}getDefaultPosition(){return "bottom-left"}onAdd(t)
{this._map=t,this._compact=this.options&&this.options.compact,this._container=i.
create("div","maplibregl-ctrl");const e=i.create("a","maplibregl-ctrl-
logo");return e.target="_blank",e.rel="noopener
nofollow",e.href="https://maplibre.org/",e.setAttribute("aria-
label",this._map._getUIString("LogoControl.Title")),e.setAttribute("rel","noopen
er
nofollow"),this._container.appendChild(e),this._container.style.display="block",
this._map.on("resize",this._updateCompact),this._updateCompact(),this._container
}onRemove()
{i.remove(this._container),this._map.off("resize",this._updateCompact),this._map
=void 0,this._compact=void 0;}}class la{constructor(){this._queue=
[],this._id=0,this._cleared=!1,this._currentlyRunning=!1;}add(t){const
e=++this._id;return this._queue.push({callback:t,id:e,cancelled:!1}),e}remove(t)
{const e=this._currentlyRunning,i=e?this._queue.concat(e):this._queue;for(const
e of i)if(e.id===t)return void(e.cancelled=!0)}run(t=0)
{if(this._currentlyRunning)throw new Error("Attempting to run(), but is already
running.");const e=this._currentlyRunning=this._queue;this._queue=[];for(const i
of e)if(!i.cancelled&&
(i.callback(t),this._cleared))break;this._cleared=!1,this._currentlyRunning=!1;}
clear(){this._currentlyRunning&&(this._cleared=!0),this._queue=[];}const ha=
{"AttributionControl.ToggleAttribution":"Toggle
attribution","AttributionControl.MapFeedback":"Map
feedback","FullscreenControl.Enter":"Enter
fullscreen","FullscreenControl.Exit":"Exit
fullscreen","GeolocateControl.FindMyLocation":"Find my
location","GeolocateControl.LocationNotAvailable":"Location not
available","LogoControl.Title":"Mapbox
logo","NavigationControl.ResetBearing":"Reset bearing to
north","NavigationControl.ZoomIn":"Zoom in","NavigationControl.ZoomOut":"Zoom

```



```

is._emptyDemMatrix=t.ao([]);}const
i=this.sourceCache.getSourceTile(e,!0);if(i&&i.dem&&
(!i.demTexture||i.needsTerrainPrepare)){const
t=this.painter.context;i.demTexture=this.painter.getTileTexture(i.dem.stride),i.
demTexture?i.demTexture.update(i.dem.getPixels(),
{premultiply:!1}):i.demTexture=new x(t,i.dem.getPixels(),t.gl.RGBA,
{premultiply:!1}),i.demTexture.bind(t.gl.NEAREST,t.gl.CLAMP_TO_EDGE),i.needsTerr
ainPrepare=!1;}const s=i&&i.tileID.key+e.key;if(s&&!this._demMatrixCache[s])
{const s=this.sourceCache.sourceCache._source.maxzoom;let a=e.canonical.z-
i.tileID.canonical.z;e.overscaledZ>e.canonical.z&&(e.canonical.z>=s?
a=e.canonical.z-s:t.w("cannot calculate elevation if elevation maxzoom >
source.maxzoom"));const o=e.canonical.x-(e.canonical.x>>a<<a),r=e.canonical.y-
(e.canonical.y>>a<<a),n=t.bc(new Float64Array(16),
[1/(t.N<<a),1/(t.N<<a),0]);t.$(n,n,[o*t.N,r*t.N,0]),this._demMatrixCache[e.key]=
{matrix:n,coord:e};}return
{u_depth:2,u_terrain:3,u_terrain_dim:i&&i.dem&&i.dem.dim||1,u_terrain_matrix:s?
this._demMatrixCache[e.key].matrix:this._emptyDemMatrix,u_terrain_unpack:i&&i.de
m&&i.dem.getUnpackVector()||this._emptyDemUnpack,u_terrain_exaggeration:this.exa
ggeration,texture:(i&&i.demTexture||this._emptyDemTexture).texture,depthTexture:
(this._fboDepthTexture||this._emptyDepthTexture).texture,tile:i}}getFramebuffer(
t){const
e=this.painter,i=e.width/devicePixelRatio,s=e.height/devicePixelRatio;return
!this._fbo||this._fbo.width===i&&this._fbo.height===s||
(this._fbo.destroy(),this._fboCoordsTexture.destroy(),this._fboDepthTexture.dest
roy(),delete this._fbo,delete this._fboDepthTexture,delete
this._fboCoordsTexture),this._fboCoordsTexture||(this._fboCoordsTexture=new
x(e.context,{width:i,height:s,data:null},e.context.gl.RGBA,
{premultiply:!1}),this._fboCoordsTexture.bind(e.context.gl.NEAREST,e.context.gl.
CLAMP_TO_EDGE)),this._fboDepthTexture||(this._fboDepthTexture=new x(e.context,
{width:i,height:s,data:null},e.context.gl.RGBA,
{premultiply:!1}),this._fboDepthTexture.bind(e.context.gl.NEAREST,e.context.gl.C
LAMP_TO_EDGE)),this._fbo||
(this._fbo=e.context.createFramebuffer(i,s,!0,!1),this._fbo.depthAttachment.set(
e.context.createRenderbuffer(e.context.gl.DEPTH_COMPONENT16,i,s)),this._fbo.col
orAttachment.set("coords"===t?
this._fboCoordsTexture.texture:this._fboDepthTexture.texture),this._fbo)}getCoord
sTexture(){const e=this.painter.context;if(this._coordsTexture)return
this._coordsTexture;const i=new
Uint8Array(this._coordsTextureSize*this._coordsTextureSize*4);for(let
t=0,e=0;t<this._coordsTextureSize;t++)for(let
s=0;s<this._coordsTextureSize;s++,e+=4)i[e+0]=255&s,i[e+1]=255&t,i[e+2]=s>>8<<4|
t>>8,i[e+3]=0;const s=new
t.R({width:this._coordsTextureSize,height:this._coordsTextureSize},new
Uint8Array(i.buffer)),a=new x(e,s,e.gl.RGBA,{premultiply:!1});return
a.bind(e.gl.NEAREST,e.gl.CLAMP_TO_EDGE),this._coordsTexture=a,a.pointCoordinate(
e){const i=new
Uint8Array(4),s=this.painter.context,a=s.gl;s.bindFramebuffer.set(this.getFrameb
uffer("coords").framebuffer),a.readPixels(e.x,this.painter.height/devicePixelRat
io-e.y-1,1,1,a.RGBA,a.UNSIGNED_BYTE,i),s.bindFramebuffer.set(null);const o=i[0]+
(i[2]>>4<<8),r=i[1]+((15&i[2])<<8),n=this.coordsIndex[255-
i[3]],l=n&&this.sourceCache.getTileByID(n);if(!l)return null;const
h=this._coordsTextureSize,c=(1<<l.tileID.canonical.z)*h;return new
t.U(this._allowMercatorOverflow(e,(l.tileID.canonical.x*h+o)/c),
(l.tileID.canonical.y*h+r)/c,this.getElevation(l.tileID,o,r,h))}getTerrainMesh()
{if(this._mesh)return this._mesh;const e=this.painter.context,i=new t.bd,s=new
t.b0,a=this.meshSize,o=t.N/a,r=a*a;for(let t=0;t<=a;t++)for(let
e=0;e<=a;e++)i.emplaceBack(e*o,t*o,0);for(let t=0;t<r;t+=a+1)for(let
e=0;e<a;e++)s.emplaceBack(e+t,a+e+t+1,a+e+t+2),s.emplaceBack(e+t,a+e+t+2,e+t+1);
const n=i.length,l=n+2*(a+1);for(const e of [0,1])for(let s=0;s<=a;s++)for(const
a of [0,1])i.emplaceBack(s*o,e*t.N,a);for(let
t=0;t<2*a;t+=2)s.emplaceBack(1+t,1+t+1,1+t+3),s.emplaceBack(1+t,1+t+3,1+t+2),s.e
mplaceBack(n+t,n+t+3,n+t+1),s.emplaceBack(n+t,n+t+2,n+t+3);const

```



```

h=i.length,c=h+2*(a+1);for(const e of [0,1])for(let s=0;s<=a;s++)for(const a of
[0,1])i.emplaceBack(e*t.N,s*o,a);for(let
t=0;t<2*a;t+=2)s.emplaceBack(h+t,h+t+1,h+t+3),s.emplaceBack(h+t,h+t+3,h+t+2),s.e
mplaceBack(c+t,c+t+3,c+t+1),s.emplaceBack(c+t,c+t+2,c+t+3);return this._mesh=
{indexBuffer:e.createIndexBuffer(s),vertexBuffer:e.createVertexBuffer(i,ca.membe
rs),segments:t.S.simpleSegment(0,0,i.length,s.length)},this._mesh}getMeshFrameDe
lta(e){return
2*Math.PI*t.be/Math.pow(2,e)/5}getMinTileElevationForLngLatZoom(t,e){var
i;const{tileID:s}=this._getOverscaledTileIDFromLngLatZoom(t,e);return null!==(
i=this.getMinMaxElevation(s).minElevation)&&void 0!==(i?
i:0)}getMinMaxElevation(t){const e=this.getTerrainData(t).tile,i=
{minElevation:null,maxElevation:null};return e&&e.dem&&
(i.minElevation=e.dem.min*this.exaggeration,i.maxElevation=e.dem.max*this.exagge
ration),i}_getOverscaledTileIDFromLngLatZoom(e,i){const
s=t.U.fromLngLat(e.wrap()),a=
(1<<i)*t.N,o=s.x*a,r=s.y*a,n=Math.floor(o/t.N),l=Math.floor(r/t.N);return
{tileID:new t.O(i,0,i,n,l),mercatorX:o,mercatorY:r}}_allowMercatorOverflow(e,i)
{const s=e.x<this.painter.width/2;let a=t.bf(i);const
o=this.painter.transform.center.lng;return s&&Math.sign(a)>0&&Math.sign(o)
<0||!s&&Math.sign(a)<0&&Math.sign(o)>0?(a=360*Math.sign(o)+a,t.G(a)):i}}class
_a{constructor(t,e,i)
{this._context=t,this._size=e,this._tileSize=i,this._objects=
[],this._recentlyUsed=[],this._stamp=0;}destruct(){for(const t of
this._objects)t.texture.destroy(),t.fbo.destroy();}_createObject(t){const
e=this._context.createFramebuffer(this._tileSize,this._tileSize,!0,!0),i=new
x(this._context,
{width:this._tileSize,height:this._tileSize,data:null},this._context.gl.RGBA);re
turn
i.bind(this._context.gl.LINEAR,this._context.gl.CLAMP_TO_EDGE),e.depthAttachment
.set(this._context.createRenderbuffer(this._context.gl.DEPTH_STENCIL,this._tileS
ize,this._tileSize)),e.colorAttachment.set(i.texture),
{id:t,fbo:e,texture:i,stamp:-1,inUse:!1}}getObjectForId(t){return
this._objects[t]}useObject(t)
{t.inUse=!0,this._recentlyUsed=this._recentlyUsed.filter((e=>t.id!==e)),this._re
centlyUsed.push(t.id);}stampObject(t)
{t.stamp++this._stamp;}getOrCreateFreeObject(){for(const t of
this._recentlyUsed)if(!this._objects[t].inUse)return
this._objects[t];if(this._objects.length>=this._size)throw new Error("No free
RenderPool available, call freeAllObjects() required!");const
t=this._createObject(this._objects.length);return
this._objects.push(t),t}freeObject(t){t.inUse=!1;}freeAllObjects(){for(const t
of this._objects)this.freeObject(t);}isFull(){return !
(this._objects.length<this._size)&&!1===this._objects.some((t=>!t.inUse))}const
pa={background:!0,fill:!0,line:!0,raster:!0,hillshade:!0};class
ma{constructor(t,e){this.painter=t,this.terrain=e,this.pool=new
_a(t.context,30,e.sourceCache.tileSize*e.qualityFactor);}destruct()
{this.pool.destruct();}getTexture(t){return
this.pool.getObjectForId(t.rtt[this._stacks.length-
1].id).texture}prepareForRender(t,e){this._stacks=
[],this._prevType=null,this._rttTiles=
[],this._renderableTiles=this.terrain.sourceCache.getRenderableTiles(),this._ren
derableLayerIds=t._order.filter((i=>!t._layers[i].isHidden(e))),this._coordsDesc
endingInv={};for(const e in t.sourceCaches){this._coordsDescendingInv[e]=
{};const i=t.sourceCaches[e].getVisibleCoordinates();for(const t of i){const
i=this.terrain.sourceCache.getTerrainCoords(t);for(const t in
i)this._coordsDescendingInv[e][t]||(this._coordsDescendingInv[e][t]=
[]),this._coordsDescendingInv[e][t].push(i[t]);}}this._coordsDescendingInvStr=
{};for(const e of t._order){const
i=t._layers[e],s=i.source;if(pa[i.type]&&!this._coordsDescendingInvStr[s])
{this._coordsDescendingInvStr[s]={};for(const t in
this._coordsDescendingInv[s])this._coordsDescendingInvStr[s]
[t]=this._coordsDescendingInv[s][t].map((t=>t.key)).sort().join();}}for(const t

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of this._renderableTiles)for(const e in this._coordsDescendingInvStr){const
i=this._coordsDescendingInvStr[e][t.tileID.key];i&&i!==t.rttCoords[e]&&(t.rtt=
[]);}}renderLayer(e){if(e.isHidden(this.painter.transform.zoom))return !1;const
i=e.type,s=this.painter,a=this._renderableLayerIds[this._renderableLayerIds.leng
th-1]===e.id;if(pa[i]&&
(this._prevType&&pa[this._prevType]||this._stacks.push([]),this._prevType=i,this
._stacks[this._stacks.length-1].push(e.id),!a))return
!0;if(pa[this._prevType]||pa[i]&&a){this._prevType=i;const
e=this._stacks.length-1,a=this._stacks[e][0];for(const i of
this._renderableTiles){if(this.pool.isFull())&&
(es(this.painter,this.terrain,this._rttTiles),this._rttTiles=
[],this.pool.freeAllObjects()),this._rttTiles.push(i),i.rtt[e]){const
t=this.pool.getObjectForId(i.rtt[e].id);if(t.stamp===i.rtt[e].stamp)
{this.pool.useObject(t);continue}}const
o=this.pool.getOrCreateFreeObject();this.pool.useObject(o),this.pool.stampObject
(o),i.rtt[e]=
{id:o.id,stamp:o.stamp},s.context.bindFramebuffer.set(o.fbo.framebuffer),s.conte
xt.clear({color:t.aT.transparent,stencil:0}),s.currentStencilSource=void
0;for(let t=0;t<a.length;t++){const e=s.style._layers[a[t]],r=e.source?
this._coordsDescendingInv[e.source][i.tileID.key]:
[i.tileID];s.context.viewport.set([0,0,o.fbo.width,o.fbo.height]),s._renderTileC
lippingMasks(e,r),s.renderLayer(s,s.style.sourceCaches[e.source],e,r),e.source&&
(i.rttCoords[e.source]=this._coordsDescendingInvStr[e.source]
[i.tileID.key]);}}return
es(this.painter,this.terrain,this._rttTiles),this._rttTiles=
[],this.pool.freeAllObjects(),pa[i]}return !1}}const fa=e,ga={center:
[0,0],zoom:0,bearing:0,pitch:0,minZoom:-2,maxZoom:22,minPitch:0,maxPitch:60,inte
ractive:!0,scrollZoom:!0,boxZoom:!0,dragRotate:!0,dragPan:!0,keyboard:!0,doubleC
lickZoom:!0,touchZoomRotate:!0,touchPitch:!0,cooperativeGestures:void
0,bearingSnap:7,clickTolerance:3,pitchWithRotate:!0,hash:!1,attributionControl:!
0,maplibreLogo:!1,failIfMajorPerformanceCaveat:!1,preserveDrawingBuffer:!1,track
Resize:!0,renderWorldCopies:!0,refreshExpiredTiles:!0,maxTileCacheSize:null,maxT
ileCacheZoomLevels:t.c.MAX_TILE_CACHE_ZOOM_LEVELS,localIdeographFontFamily:"sans
-
serif",transformRequest:null,transformCameraUpdate:null,fadeDuration:300,crossSo
urceCollisions:!0,validateStyle:!0,maxCanvasSize:[4096,4096]},va=t=>
{t.touchstart=t.dragStart,t.touchmoveWindow=t.dragMove,t.touchend=t.dragEnd;},xa
={showCompass:!0,showZoom:!0,visualizePitch:!1};class ya{constructor(e,s,a=!1)
{this.mousedown=e=>{this.startMouse(t.e({},e,{ctrlKey:!0,preventDefault:
()=>e.preventDefault()}),i.mousePos(this.element,e)),i.addEventListener(window,"
mousemove",this.mousemove),i.addEventListener(window,"mouseup",this.mouseup);},t
his.mousemove=t=>
{this.moveMouse(t,i.mousePos(this.element,t));},this.mouseup=t=>
{this.mouseRotate.dragEnd(t),this.mousePitch&&this.mousePitch.dragEnd(t),this.of
fTemp();},this.touchstart=t=>{1!==t.targetTouches.length?this.reset():
(this._startPos=this._lastPos=i.touchPos(this.element,t.targetTouches)
[0],this.startTouch(t,this._startPos),i.addEventListener(window,"touchmove",this
.touchmove,
{passive:!1}),i.addEventListener(window,"touchend",this.touchend));},this.touchm
ove=t=>{1!==t.targetTouches.length?this.reset():
(this._lastPos=i.touchPos(this.element,t.targetTouches)
[0],this.moveTouch(t,this._lastPos));},this.touchend=t=>
{0===t.targetTouches.length&&this._startPos&&this._lastPos&&this._startPos.dist(
this._lastPos)<this._clickTolerance&&this.element.click(),delete
this._startPos,delete this._lastPos,this.offTemp();},this.reset=()=>>
{this.mouseRotate.reset(),this.mousePitch&&this.mousePitch.reset(),this.touchRot
ate.reset(),this.touchPitch&&this.touchPitch.reset(),delete
this._startPos,delete
this._lastPos,this.offTemp();},this._clickTolerance=10;const
o=e.dragRotate._mouseRotate.getClickTolerance(),r=e.dragRotate._mousePitch.getCl
ickTolerance();this.element=s,this.mouseRotate=As({clickTolerance:o,enable:!0}),
this.touchRotate=

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(({enable:t,clickTolerance:e,bearingDegreesPerPixelMoved:i=.8})=>{const s=new
zs;return new Ps({clickTolerance:e,move:(t,e)=>({bearingDelta:(e.x-
t.x)*i}),moveStateManager:s,enable:t,assignEvents:va}))
({clickTolerance:o,enable:!0}),this.map=e,a&&
(this.mousePitch=Rs({clickTolerance:r,enable:!0}),this.touchPitch=
(({enable:t,clickTolerance:e,pitchDegreesPerPixelMoved:i=-.5})=>{const s=new
zs;return new Ps({clickTolerance:e,move:(t,e)=>({pitchDelta:(e.y-
t.y)*i}),moveStateManager:s,enable:t,assignEvents:va}))
({clickTolerance:r,enable:!0})),i.addEventListener(s,"mousedown",this.mousedown)
,i.addEventListener(s,"touchstart",this.touchstart,
{passive:!1}),i.addEventListener(s,"touchcancel",this.reset);}startMouse(t,e)
{this.mouseRotate.dragStart(t,e),this.mousePitch&&this.mousePitch.dragStart(t,e)
,i.disableDrag();}startTouch(t,e)
{this.touchRotate.dragStart(t,e),this.touchPitch&&this.touchPitch.dragStart(t,e)
,i.disableDrag();}moveMouse(t,e){const i=this.map,
{bearingDelta:s}=this.mouseRotate.dragMove(t,e)||
{};if(s&&i.setBearing(i.getBearing()+s),this.mousePitch)
{const{pitchDelta:s}=this.mousePitch.dragMove(t,e)||
{};s&&i.setPitch(i.getPitch()+s);}moveTouch(t,e){const i=this.map,
{bearingDelta:s}=this.touchRotate.dragMove(t,e)||
{};if(s&&i.setBearing(i.getBearing()+s),this.touchPitch)
{const{pitchDelta:s}=this.touchPitch.dragMove(t,e)||
{};s&&i.setPitch(i.getPitch()+s);}off(){const
t=this.element;i.removeEventListener(t,"mousedown",this.mousedown),i.removeEvent
Listener(t,"touchstart",this.touchstart,
{passive:!1}),i.removeEventListener(window,"touchmove",this.touchmove,
{passive:!1}),i.removeEventListener(window,"touchend",this.touchend),i.removeEve
ntListener(t,"touchcancel",this.reset),this.offTemp();}offTemp()
{i.enableDrag(),i.removeEventListener(window,"mousemove",this.mousemove),i.remov
eEventListener(window,"mouseup",this.mouseup),i.removeEventListener(window,"touc
hmove",this.touchmove,
{passive:!1}),i.removeEventListener(window,"touchend",this.touchend);}let
ba;function wa(e,i,s){if(e=new t.L(e.lng,e.lat),i){const a=new t.L(e.lng-
360,e.lat),o=new
t.L(e.lng+360,e.lat),r=s.locationPoint(e).distSqr(i);s.locationPoint(a).distSqr(
i)<r?e=a:s.locationPoint(o).distSqr(i)<r&&(e=o);}for(;Math.abs(e.lng-
s.center.lng)>180;){const
t=s.locationPoint(e);if(t.x>=0&&t.y>=0&&t.x<=s.width&&t.y<=s.height)break;e.lng>
s.center.lng?e.lng-=360:e.lng+=360;}return e}const Ta=
{center:"translate(-50%,-50%)",top:"translate(-50%,0)","top-
left":"translate(0,0)","top-
right":"translate(-100%,0)","bottom":"translate(-50%,-100%)","bottom-
left":"translate(0,-100%)","bottom-
right":"translate(-100%,-100%)",left:"translate(0,-50%)",right:"translate(-100%,
-50%)"};function Ia(t,e,i){const s=t.classList;for(const t in
Ta)s.remove(`maplibregl-${i}-anchor-${t}`);s.add(`maplibregl-${i}-anchor-
${e}`);}class Ea extends t.E{constructor(e){if(super(),this._onKeyPress=t=>
{const
e=t.code,i=t.charCode||t.keyCode;"Space"!==e&&"Enter"!==e&&32!==i&&13!==i||this.
togglePopup();},this._onMapClick=t=>{const
e=t.originalEvent.target,i=this._element;this._popup&&
(e===i||i.contains(e))&&this.togglePopup();},this._update=t=>
{if(!this._map)return;const e=this._map.loaded()&&!this._map.isMoving();
("terrain"===t?(null===t?void 0:t.type)||"render"===t?(null===t?void
0:t.type)&&!e)&&this._map.once("render",this._update),this._map.transform.render
WorldCopies&&
(this._lngLat=wa(this._lngLat,this._pos,this._map.transform)),this._pos=this._ma
p.project(this._lngLat)._add(this._offset);let
s="";"viewport"===this._rotationAlignment||"auto"===this._rotationAlignment?
s=`rotateZ(${this._rotation}deg)`:"map"===this._rotationAlignment&&
(s=`rotateZ(${this._rotation-this._map.getBearing()}deg)`);let
a="";"viewport"===this._pitchAlignment||"auto"===this._pitchAlignment?

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a="rotateX(0deg)": "map"===this._pitchAlignment&&
(a=`rotateX(${this._map.getPitch()}deg)`),t&&"moveend"!==t.type||
(this._pos=this._pos.round()),i.setTransform(this._element,`${Ta[this._anchor]}
translate(${this._pos.x}px, ${this._pos.y}px) ${a}
${s}`),this._map.terrain&&!this._opacityTimeout&&
(this._opacityTimeout=setTimeout((()=>{const
t=this._map.unproject(this._pos),e=40075016.686*Math.abs(Math.cos(this._lngLat.l
at*Math.PI/180))/Math.pow(2,this._map.transform.tileZoom+8);this._element.style.
opacity=t.distanceTo(this._lngLat)>20*e?"0.2":"1.0",this._opacityTimeout=null;})
,100));},this._onMove=e=>{if(!this._isDragging){const
t=this._clickTolerance||this._map._clickTolerance;this._isDragging=e.point.dist(
this._pointerdownPos)>=t;}}this._isDragging&&
(this._pos=e.point.sub(this._positionDelta),this._lngLat=this._map.unproject(thi
s._pos),this.setLngLat(this._lngLat),this._element.style.pointerEvents="none","p
ending"===this._state&&(this._state="active",this.fire(new
t.k("dragstart"))),this.fire(new t.k("drag")));},this._onUp=(()=>>
{this._element.style.pointerEvents="auto",this._positionDelta=null,this._pointer
downPos=null,this._isDragging=!1,this._map.off("mousemove",this._onMove),this._m
ap.off("touchmove",this._onMove),"active"===this._state&&this.fire(new
t.k("dragend")),this._state="inactive";},this._addDragHandler=t=>
{this._element.contains(t.originalEvent.target)&&
(t.preventDefault(),this._positionDelta=t.point.sub(this._pos).add(this._offset)
,this._pointerdownPos=t.point,this._state="pending",this._map.on("mousemove",thi
s._onMove),this._map.on("touchmove",this._onMove),this._map.once("mouseup",this.
_onUp),this._map.once("touchend",this._onUp));},this._anchor=e&&e.anchor||"cente
r",this._color=e&&e.color||"#3FB1CE",this._scale=e&&e.scale||1,this._draggable=e
&&e.draggable||!1,this._clickTolerance=e&&e.clickTolerance||0,this._isDragging=!
1,this._state="inactive",this._rotation=e&&e.rotation||0,this._rotationAlignment
=e&&e.rotationAlignment||"auto",this._pitchAlignment=e&&e.pitchAlignment&&"auto"
!==e.pitchAlignment?
e.pitchAlignment:this._rotationAlignment,e&&e.element)this._element=e.element,th
is._offset=t.P.convert(e&&e.offset||[0,0]);else
{this._defaultMarker=!0,this._element=i.create("div"),this._element.setAttribute
("aria-label","Map marker");const
s=i.createNS("http://www.w3.org/2000/svg","svg"),a=41,o=27;s.setAttributeNS(null
,"display","block"),s.setAttributeNS(null,"height",`${a}px`),s.setAttributeNS(nu
ll,"width",`${o}px`),s.setAttributeNS(null,"viewBox",`0 0 ${o} ${a}`);const
r=i.createNS("http://www.w3.org/2000/svg","g");r.setAttributeNS(null,"stroke","n
one"),r.setAttributeNS(null,"stroke-
width","1"),r.setAttributeNS(null,"fill","none"),r.setAttributeNS(null,"fill-
rule","evenodd");const
n=i.createNS("http://www.w3.org/2000/svg","g");n.setAttributeNS(null,"fill-
rule","nonzero");const
l=i.createNS("http://www.w3.org/2000/svg","g");l.setAttributeNS(null,"transform"
,"translate(3.0, 29.0)"),l.setAttributeNS(null,"fill","#000000");const h=
[{rx:"10.5",ry:"5.25002273"},{rx:"10.5",ry:"5.25002273"},
{rx:"9.5",ry:"4.77275007"},{rx:"8.5",ry:"4.29549936"},
{rx:"7.5",ry:"3.81822308"},{rx:"6.5",ry:"3.34094679"},
{rx:"5.5",ry:"2.86367051"},{rx:"4.5",ry:"2.38636864"}];for(const t of h){const
e=i.createNS("http://www.w3.org/2000/svg","ellipse");e.setAttributeNS(null,"opac
ity","0.04"),e.setAttributeNS(null,"cx","10.5"),e.setAttributeNS(null,"cy","5.80
029008"),e.setAttributeNS(null,"rx",t.rx),e.setAttributeNS(null,"ry",t.ry),l.app
endChild(e);}const
c=i.createNS("http://www.w3.org/2000/svg","g");c.setAttributeNS(null,"fill",this
._color);const
u=i.createNS("http://www.w3.org/2000/svg","path");u.setAttributeNS(null,"d","M27
13.5 C27,19.074644 20.250001,27.000002 14.75,34.500002 C14.016665,35.500004
12.983335,35.500004 12.25,34.500002 C6.7499993,27.000002 0,19.222562 0,13.5
C0,6.0441559 6.0441559,0 13.5,0 C20.955844,0 27,6.0441559 27,13.5
Z"),c.appendChild(u);const
d=i.createNS("http://www.w3.org/2000/svg","g");d.setAttributeNS(null,"opacity","
0.25"),d.setAttributeNS(null,"fill","#000000");const

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```

    _=i.createNS("http://www.w3.org/2000/svg","path");_.setAttributeNS(null,"d","M13
    .5,0 C6.0441559,0 0,6.0441559 0,13.5 C0,19.222562 6.7499993,27 12.25,34.5
    C13,35.522727 14.016664,35.500004 14.75,34.5 C20.250001,27 27,19.074644 27,13.5
    C27,6.0441559 20.955844,0 13.5,0 Z M13.5,1 C20.415404,1 26,6.584596 26,13.5
    C26,15.898657 24.495584,19.181431 22.220703,22.738281 C19.945823,26.295132
    16.705119,30.142167 13.943359,33.908203 C13.743445,34.180814 13.612715,34.322738
    13.5,34.441406 C13.387285,34.322738 13.256555,34.180814 13.056641,33.908203
    C10.284481,30.127985 7.4148684,26.314159 5.015625,22.773438 C2.6163816,19.232715
    1,15.953538 1,13.5 C1,6.584596 6.584596,1 13.5,1 Z");d.appendChild(_);const
    p=i.createNS("http://www.w3.org/2000/svg","g");p.setAttributeNS(null,"transform"
    ,"translate(6.0, 7.0)");p.setAttributeNS(null,"fill","#FFFFFF");const
    m=i.createNS("http://www.w3.org/2000/svg","g");m.setAttributeNS(null,"transform"
    ,"translate(8.0, 8.0)");const
    f=i.createNS("http://www.w3.org/2000/svg","circle");f.setAttributeNS(null,"fill"
    ,"#000000");f.setAttributeNS(null,"opacity","0.25");f.setAttributeNS(null,"cx",
    "5.5");f.setAttributeNS(null,"cy","5.5");f.setAttributeNS(null,"r","5.4999962");c
    onst
    g=i.createNS("http://www.w3.org/2000/svg","circle");g.setAttributeNS(null,"fill"
    ,"#FFFFFF");g.setAttributeNS(null,"cx","5.5");g.setAttributeNS(null,"cy","5.5");
    g.setAttributeNS(null,"r","5.4999962");m.appendChild(f),m.appendChild(g),n.appen
    dChild(l),n.appendChild(c),n.appendChild(d),n.appendChild(p),n.appendChild(m),s.
    appendChild(n),s.setAttributeNS(null,"height",a*this._scale+"px"),s.setAttribute
    NS(null,"width",o*this._scale+"px"),this._element.appendChild(s),this._offset=t.
    P.convert(e&&e.offset||[0,-14]);}if(this._element.classList.add("maplibregl-
    marker"),this._element.addEventListener("dragstart",(t=>
    {t.preventDefault();})),this._element.addEventListener("mousedown",(t=>
    {t.preventDefault();})),Ia(this._element,this._anchor,"marker"),e&&e.className)f
    or(const t of e.className.split("
    "))this._element.classList.add(t);this._popup=null;addTo(t){return
    this.remove(),this._map=t,t.getCanvasContainer().appendChild(this._element),t.on
    ("move",this._update),t.on("moveend",this._update),t.on("terrain",this._update),
    this.setDraggable(this._draggable),this._update(),this._map.on("click",this._onM
    apClick),this.remove(){return this._opacityTimeout&&
    (clearTimeout(this._opacityTimeout),delete this._opacityTimeout),this._map&&
    (this._map.off("click",this._onMapClick),this._map.off("move",this._update),this
    ._map.off("moveend",this._update),this._map.off("mousedown",this._addDragHandler
    ),this._map.off("touchstart",this._addDragHandler),this._map.off("mouseup",this.
    _onUp),this._map.off("touchend",this._onUp),this._map.off("mousemove",this._onMo
    ve),this._map.off("touchmove",this._onMove),delete
    this._map),i.remove(this._element),this._popup&&this._popup.remove(),this}getLng
    Lat(){return this._lngLat}setLngLat(e){return
    this._lngLat=t.L.convert(e),this._pos=null,this._popup&&this._popup.setLngLat(th
    is._lngLat),this._update(),this}getElement(){return this._element}setPopup(t)
    {if(this._popup&&
    (this._popup.remove(),this._popup=null,this._element.removeEventListener("keypre
    ss",this._onKeyPress),this._originalTabIndex|this._element.removeAttribute("tab
    index")),t){if(!("offset"in t.options)){const
    e=38.1,i=13.5,s=Math.abs(i)/Math.SQRT2;t.options.offset=this._defaultMarker?
    {top:[0,0],"top-left":[0,0],"top-right":[0,0],bottom:[0,-e],"bottom-left":[s,-1*
    (e-i+s)],"bottom-right":[-s,-1*(e-i+s)],left:[i,-1*(e-i)],right:[-i,-1*(e-
    i)]}:this._offset;}this._popup=t,this._lngLat&&this._popup.setLngLat(this._lngLa
    t),this._originalTabIndex=this._element.getAttribute("tabindex"),this._originalT
    abIndex|this._element.setAttribute("tabindex","0"),this._element.addEventListener
    ("keypress",this._onKeyPress);}return this}getPopup(){return
    this._popup}togglePopup(){const t=this._popup;return t?(t.isOpen())?
    t.remove():t.addTo(this._map),this}:this}getOffset(){return
    this._offset}setOffset(e){return
    this._offset=t.P.convert(e),this._update(),this}addClassName(t)
    {this._element.classList.add(t);}removeClassName(t)
    {this._element.classList.remove(t);}toggleClassName(t){return
    this._element.classList.toggle(t)}setDraggable(t){return
    this._draggable=!!t,this._map&&(t?

```

```

(this._map.on("mousedown",this._addDragHandler),this._map.on("touchstart",this._
addDragHandler)):
(this._map.off("mousedown",this._addDragHandler),this._map.off("touchstart",this
._addDragHandler))),this.isDraggable(){return this._draggable}setRotation(t)
{return this._rotation=t||0,this._update(),this}getRotation(){return
this._rotation}setRotationAlignment(t){return
this._rotationAlignment=t||"auto",this._update(),this}getRotationAlignment()
{return this._rotationAlignment}setPitchAlignment(t){return
this._pitchAlignment=t&"auto"!==t?
t:this._rotationAlignment,this._update(),this}getPitchAlignment(){return
this._pitchAlignment}}const Sa={positionOptions:
{enableHighAccuracy:!1,maximumAge:0,timeout:6e3},fitBoundsOptions:
{maxZoom:15},trackUserLocation:!1,showAccuracyCircle:!0,showUserLocation:!0};let
Ca=0,Pa=!1;const Da={maxWidth:100,unit:"metric"};function Ma(t,e,i){const
s=i&&i.maxWidth||100,a=t._container.clientHeight/2,o=t.unproject([0,a]),r=t.unpr
oject([s,a]),n=o.distanceTo(r);if(i&&"imperial"===i.unit){const
i=3.2808*n;i>5280?
za(e,s,i/5280,t._getUIString("ScaleControl.Miles")):za(e,s,i,t._getUIString("Sca
leControl.Feet"));}else i&&"nautical"===i.unit?
za(e,s,n/1852,t._getUIString("ScaleControl.NauticalMiles")):n>=1e3?
za(e,s,n/1e3,t._getUIString("ScaleControl.Kilometers")):za(e,s,n,t._getUIString(
"ScaleControl.Meters"));}function za(t,e,i,s){const a=function(t){const
e=Math.pow(10,Math.floor(t)).length-1;let i=t/e;return i=i>=10?10:i>=5?
5:i>=3?3:i>=2?2:i>=1?1:function(t){const e=Math.pow(10,Math.ceil(-
Math.log(t)/Math.LN10));return Math.round(t*e)/e}(i),e*i}(i);t.style.width=e*
(a/i)+"px",t.innerHTML=`${a}&nbsp;${s}`;const La=
{closeButton:!0,closeOnClick:!0,focusAfterOpen:!0,className:"",maxWidth:"240px"}
,Aa=["a[href]","[tabindex]:not([tabindex='-1'])","[
contenteditable]:not([contenteditable='false'])","button:not([disabled])","inpu
t:not([disabled])","select:not([disabled])","textarea:not([disabled])"].join(",
");function Ra(e){if(e){if("number"===typeof e){const
i=Math.round(Math.abs(e)/Math.SQRT2);return {center:new t.P(0,0),top:new
t.P(0,e),"top-left":new t.P(i,i),"top-right":new t.P(-i,i),bottom:new t.P(0,-
e),"bottom-left":new t.P(i,-i),"bottom-right":new t.P(-i,-i),left:new
t.P(e,0),right:new t.P(-e,0)}}if(e instanceof t.P||Array.isArray(e)){const
i=t.P.convert(e);return {center:i,top:i,"top-left":i,"top-
right":i,bottom:i,"bottom-left":i,"bottom-right":i,left:i,right:i}}return
{center:t.P.convert(e.center||[0,0]),top:t.P.convert(e.top||[0,0]),"top-
left":t.P.convert(e["top-left"]||[0,0]),"top-right":t.P.convert(e["top-right"]||
[0,0]),bottom:t.P.convert(e.bottom||[0,0]),"bottom-left":t.P.convert(e["bottom-
left"]||[0,0]),"bottom-right":t.P.convert(e["bottom-right"]||
[0,0]),left:t.P.convert(e.left||[0,0]),right:t.P.convert(e.right||[0,0])}}return
Ra(new t.P(0,0))}const ka={extend:(e,...i)=>t.e(e,...i),run(t)
{t(),logToElement(t,e=!1,i="log")}{const
s=window.document.getElementById(i);s&&(e&&
(s.innerHTML=""),s.innerHTML+=`<br>${t}`)}};Fa=e;class Ba{static get version()
{return Fa}static get workerCount(){return tt.workerCount}static set
workerCount(t){tt.workerCount=t;}static get maxParallelImageRequests(){return
t.c.MAX_PARALLEL_IMAGE_REQUESTS}static set maxParallelImageRequests(e)
{t.c.MAX_PARALLEL_IMAGE_REQUESTS=e;}static get workerUrl(){return
t.c.WORKER_URL}static set workerUrl(e){t.c.WORKER_URL=e;}static addProtocol(e,i)
{t.c.REGISTERED_PROTOCOLS[e]=i;}static removeProtocol(e){delete
t.c.REGISTERED_PROTOCOLS[e];}return Ba.Map=class extends oa{constructor(e)
{if(t.bg.mark(t.bh.create),null!=
(e=t.e({},ga,e)).minZoom&&null!=e.maxZoom&&e.minZoom>e.maxZoom)throw new
Error("maxZoom must be greater than or equal to
minZoom");if(null!=e.minPitch&&null!=e.maxPitch&&e.minPitch>e.maxPitch)throw new
Error("maxPitch must be greater than or equal to
minPitch");if(null!=e.minPitch&&e.minPitch<0)throw new Error("minPitch must be
greater than or equal to 0");if(null!=e.maxPitch&&e.maxPitch>85)throw new
Error("maxPitch must be less than or equal to 85");if(super(new
rs(e.minZoom,e.maxZoom,e.minPitch,e.maxPitch,e.renderWorldCopies),

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{bearingSnap:e.bearingSnap}),this._cooperativeGesturesOnWheel=t=>
{this._onCooperativeGesture(t,t[this._metaKey],1);},this._contextLost=e=>
{e.preventDefault(),this._frame&&
(this._frame.cancel(),this._frame=null),this.fire(new t.k("webglcontextlost",
{originalEvent:e}));},this._contextRestored=e=>
{this._setupPainter(),this.resize(),this._update(),this.fire(new
t.k("webglcontextrestored",{originalEvent:e}));},this._onMapScroll=t=>
{if(t.target===this._container)return
this._container.scrollTop=0,this._container.scrollLeft=0,!1},this._onWindowOnlin
e=()=>
{this._update();},this._interactive=e.interactive,this._cooperativeGestures=e.co
operativeGestures,this._metaKey=0===navigator.platform.indexOf("Mac")?"metaKey":
"ctrlKey",this._maxTileCacheSize=e.maxTileCacheSize,this._maxTileCacheZoomLevels
=e.maxTileCacheZoomLevels,this._failIfMajorPerformanceCaveat=e.failIfMajorPerfor
manceCaveat,this._preserveDrawingBuffer=e.preserveDrawingBuffer,this._antialias=
e.antialias,this._trackResize=e.trackResize,this._bearingSnap=e.bearingSnap,this
._refreshExpiredTiles=e.refreshExpiredTiles,this._fadeDuration=e.fadeDuration,th
is._crossSourceCollisions=e.crossSourceCollisions,this._crossFadingFactor=1,this
._collectResourceTiming=e.collectResourceTiming,this._renderTaskQueue=new
la,this._controls=
[],this._mapId=t.a2(),this._locale=t.e({},ha,e.locale),this._clickTolerance=e.cl
ickTolerance,this._overridePixelRatio=e.pixelRatio,this._maxCanvasSize=e.maxCanv
asSize,this.transformCameraUpdate=e.transformCameraUpdate,this._imageQueueHandle
=h.addThrottleControl((()=>this.isMoving()),this._requestManager=new
u(e.transformRequest),"string"===typeof e.container)
{if(this._container=document.getElementById(e.container),!this._container)throw
new Error(`Container '${e.container}' not found.`)}else {if(!(e.container
instanceof HTMLElement))throw new Error("Invalid type: 'container' must be a
String or
HTMLElement.");this._container=e.container;}if(e.maxBounds&&this.setMaxBounds(e.
maxBounds),this._setupContainer(),this._setupPainter(),this.on("move",
((()=>this._update(!1))),this.on("moveend",
((()=>this._update(!1))),this.on("zoom",
((()=>this._update(!0))),this.on("terrain",((()=>
{this.painter.terrainFacilitator.dirty=!0,this._update(!0);})),this.once("idle",
((()=>{this._idleTriggered=!0;})),this.on("undefined"!=typeof window)
{addEventListener("online",this._onWindowOnline,!1);let t=!1;const e=ns((t=>
{this._trackResize&&!this._removed&&this.resize(t)._update();}),50);this._resize
Observer=new ResizeObserver((i=>{t?
e(i):t=!0;})),this._resizeObserver.observe(this._container);}this.handlers=new
aa(this,e),this._cooperativeGestures&&this._setupCooperativeGestures(),this._has
h=e.hash&&new ls("string"===typeof e.hash&&e.hash||void
0).addTo(this),this._hash&&this._hash._onHashChange()||
(this.jumpTo({center:e.center,zoom:e.zoom,bearing:e.bearing,pitch:e.pitch}),e.bo
unds&&(this.resize(),this.fitBounds(e.bounds,t.e({},e.fitBoundsOptions,
{duration:0}))),this.resize(),this._localIdeographFontFamily=e.localIdeographFo
ntFamily,this._validateStyle=e.validateStyle,e.style&&this.setStyle(e.style,
{localIdeographFontFamily:e.localIdeographFontFamily}),e.attributionControl&&thi
s.addControl(new
ra({customAttribution:e.customAttribution})),e.maplibreLogo&&this.addControl(new
na,e.logoPosition),this.on("style.load",((()=>
{this.transform.unmodified&&this.jumpTo(this.style.stylesheet);})),this.on("data
",e=>{this._update("style"===e.dataType),this.fire(new
t.k(`${e.dataType}data`,e)})),this.on("dataloading",e=>{this.fire(new
t.k(`${e.dataType}dataloading`,e)})),this.on("dataabort",e=>{this.fire(new
t.k("sourcedataabort",e)}));}_getMapId(){return this._mapId}addControl(e,i)
{if(void 0===i&&(i=e.getDefaultPosition?e.getDefaultPosition():"top-
right"),!e||!e.onAdd)return this.fire(new t.j(new Error("Invalid argument to
map.addControl(). Argument must be a control with onAdd and onRemove
methods.")));const s=e.onAdd(this);this._controls.push(e);const
a=this._controlPositions[i];return -1!==i.indexOf("bottom"?
a.insertBefore(s,a.firstChild):a.appendChild(s),this}removeControl(e)

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    if(!e||!e.onRemove)return this.fire(new t.j(new Error("Invalid argument to
    map.removeControl(). Argument must be a control with onAdd and onRemove
    methods.")));const i=this._controls.indexOf(e);return
    i>-1&&this._controls.splice(i,1),e.onRemove(this),this}hasControl(t){return
    this._controls.indexOf(t)>-1}calculateCameraOptionsFromTo(t,e,i,s){return
    null==s&&this.terrain&&
    (s=this.terrain.getElevationForLngLatZoom(i,this.transform.tileZoom)),super.calc
    ulateCameraOptionsFromTo(t,e,i,s)}resize(e){var i;const
    s=this._containerDimensions(),a=s[0],o=s[1],r=this._getClampedPixelRatio(a,o);if
    (this._resizeCanvas(a,o,r),this.painter.resize(a,o,r),this.painter.overLimit())
    {const t=this.painter.context.gl;this._maxCanvasSize=
    [t.drawingBufferWidth,t.drawingBufferHeight];const
    e=this._getClampedPixelRatio(a,o);this._resizeCanvas(a,o,e),this.painter.resize(
    a,o,e);}this.transform.resize(a,o),null===(i=this._requestedCameraState)||void
    0===i||i.resize(a,o);const n=!this._moving;return n&&(this.stop(),this.fire(new
    t.k("movestart",e)).fire(new t.k("move",e))),this.fire(new
    t.k("resize",e)),n&&this.fire(new
    t.k("moveend",e)),this}_getClampedPixelRatio(t,e)
    {const{0:i,1:s}=this._maxCanvasSize,a=this.getPixelRatio(),o=t*a,r=e*a;return
    Math.min(o>i?i/o:1,r>s?s/r:1)*a}getPixelRatio(){var t;return null!==
    (t=this._overridePixelRatio)&&void 0!==t?t:devicePixelRatio}setPixelRatio(t)
    {this._overridePixelRatio=t,this.resize();}getBounds(){return
    this.transform.getBounds()}getMaxBounds(){return
    this.transform.getMaxBounds()}setMaxBounds(t){return
    this.transform.setMaxBounds(L.convert(t)),this._update()}setMinZoom(t)
    {if((t=null==t?-2:t)>=-2&&t<=this.transform.maxZoom)return
    this.transform.minZoom=t,this._update(),this.getZoom()
    <t&&this.setZoom(t),this;throw new Error("minZoom must be between -2 and the
    current maxZoom, inclusive")}getMinZoom(){return
    this.transform.minZoom}setMaxZoom(t){if((t=null==t?
    22:t)>=this.transform.minZoom)return
    this.transform.maxZoom=t,this._update(),this.getZoom()>t&&this.setZoom(t),this;
    throw new Error("maxZoom must be greater than the current minZoom")}getMaxZoom()
    {return this.transform.maxZoom}setMinPitch(t){if((t=null==t?0:t)<0)throw new
    Error("minPitch must be greater than or equal to
    0");if(t>=0&&t<=this.transform.maxPitch)return
    this.transform.minPitch=t,this._update(),this.getPitch()
    <t&&this.setPitch(t),this;throw new Error("minPitch must be between 0 and the
    current maxPitch, inclusive")}getMinPitch(){return
    this.transform.minPitch}setMaxPitch(t){if((t=null==t?60:t)>85)throw new
    Error("maxPitch must be less than or equal to
    85");if(t>=this.transform.minPitch)return
    this.transform.maxPitch=t,this._update(),this.getPitch()>t&&this.setPitch(t),thi
    s;throw new Error("maxPitch must be greater than the current
    minPitch")}getMaxPitch(){return this.transform.maxPitch}getRenderWorldCopies()
    {return this.transform.renderWorldCopies}setRenderWorldCopies(t){return
    this.transform.renderWorldCopies=t,this._update()}getCooperativeGestures()
    {return this._cooperativeGestures}setCooperativeGestures(t){return
    this._cooperativeGestures=t,this._cooperativeGestures?
    this._setupCooperativeGestures():this._destroyCooperativeGestures(),this}project
    (e){return
    this.transform.locationPoint(t.L.convert(e),this.style&&this.terrain)}unproject(
    e){return this.transform.pointLocation(t.P.convert(e),this.terrain)}isMoving()
    {var t;return this._moving||(null===(t=this.handlers)||void 0===t?void
    0:t.isMoving())}isZooming(){var t;return this._zooming||(null===
    (t=this.handlers)||void 0===t?void 0:t.isZooming())}isRotating(){var t;return
    this._rotating||(null===(t=this.handlers)||void 0===t?void
    0:t.isRotating())}createDelegatedListener(t,e,i)
    {if("mouseenter"===t||"mouseover"===t){let s=!1;const a=a=>{const
    o=this.getLayer(e)?this.queryRenderedFeatures(a.point,{layers:[e]}):[];o.length?
    s||(s=!0,i.call(this,new gs(t,this,a.originalEvent,{features:o})):s=!1};return
    {layer:e,listener:i,delegates:{mousemove:a,mouseout:()=>

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{s=!1;}}}}if("mouseleave"===t||"mouseout"===t){let s=!1;const a=a=>
{(this.getLayer(e)?this.queryRenderedFeatures(a.point,{layers:[e]}):[]).length?
s=!0:s&&(s=!1,i.call(this,new gs(t,this,a.originalEvent)));},o=e=>{s&&
(s=!1,i.call(this,new gs(t,this,e.originalEvent)));};return
{layer:e,listener:i,delegates:{mousemove:a,mouseout:o}}}{const s=t=>{const
s=this.getLayer(e)?this.queryRenderedFeatures(t.point,{layers:[e]}):
[];s.length&&(t.features=s,i.call(this,t),delete t.features);};return
{layer:e,listener:i,delegates:[t]:s}}}}on(t,e,i){if(void 0===i)return
super.on(t,e);const
s=this._createDelegatedListener(t,e,i);this._delegatedListeners=this._delegatedL
isteners||{},this._delegatedListeners[t]=this._delegatedListeners[t]||
[],this._delegatedListeners[t].push(s);for(const t in
s.delegates)this.on(t,s.delegates[t]);return this}once(t,e,i){if(void
0===i)return super.once(t,e);const
s=this._createDelegatedListener(t,e,i);for(const t in
s.delegates)this.once(t,s.delegates[t]);return this}off(t,e,i){return void
0===i?super.off(t,e):(this._delegatedListeners&&this._delegatedListeners[t]&&
(s=>{const a=this._delegatedListeners[t];for(let t=0;t<a.length;t++){const
s=a[t];if(s.layer===e&&s.listener===i){for(const t in
s.delegates)this.off(t,s.delegates[t]);return a.splice(t,1),this}}})
(),this)}queryRenderedFeatures(e,i){if(!this.style)return [];let s;const a=e
instanceof t.P||Array.isArray(e),o=a?e:[0,0],
[this.transform.width,this.transform.height];if(i=i||(a?{}:e)||{},o instanceof
t.P||"number"===typeof o[0])s=[t.P.convert(o)];else {const
e=t.P.convert(o[0]),i=t.P.convert(o[1]);s=[e,new t.P(i.x,e.y),i,new
t.P(e.x,i.y),e];}return
this.style.queryRenderedFeatures(s,i,this.transform)}querySourceFeatures(t,e)
{return this.style.querySourceFeatures(t,e)}setStyle(e,i){return !!(i=t.e({,
{localIdeographFontFamily:this._localIdeographFontFamily,validate:this._validate
Style},i)).diff&&i.localIdeographFontFamily===this._localIdeographFontFamily&&th
is.style&&e?(this._diffStyle(e,i),this):
(this._localIdeographFontFamily=i.localIdeographFontFamily,this._updateStyle(e,i
))}setTransformRequest(t){return
this._requestManager.setTransformRequest(t),this}_getUIString(t){const
e=this._locale[t];if(null==e)throw new Error(`Missing UI string '${t}'`);return
e}_updateStyle(t,e){if(e.transformStyle&&this.style&&!this.style._loaded)return
void this.style.once("style.load",(()=>this._updateStyle(t,e)));const
i=this.style&&e.transformStyle?this.style.serialize():void 0;return this.style&&
(this.style.setEventedParent(null),this.style._remove(!t)),t?(this.style=new
se(this,e||{}),this.style.setEventedParent(this,
{style:this.style}),"string"===typeof t?
this.style.loadURL(t,e,i):this.style.loadJSON(t,e,i),this):(delete
this.style,this)._lazyInitEmptyStyle(){this.style||(this.style=new se(this,
{}),this.style.setEventedParent(this,
{style:this.style})),this.style.loadEmpty();}_diffStyle(e,i){if("string"===typeof
e){const s=this._requestManager.transformRequest(e,c.Style);t.f(s,((e,s)=>{e?
this.fire(new t.j(e)):s&&this._updateDiff(s,i)}));}else "object"===typeof
e&&this._updateDiff(e,i);_updateDiff(e,i)
{try{this.style.setState(e,i)&&this._update(!0);}catch(s){t.w(`Unable to perform
style diff: ${s.message}|${s.error}|${s}. Rebuilding the style from
scratch.`),this._updateStyle(e,i);}getStyle(){if(this.style)return
this.style.serialize()}isStyleLoaded(){return this.style?
this.style.loaded():t.w("There is no style added to the map.")}addSource(t,e)
{return
this._lazyInitEmptyStyle(),this.style.addSource(t,e),this._update(!0)}isSourceLo
aded(e){const i=this.style&&this.style.sourceCaches[e];if(void 0!==i)return
i.loaded();this.fire(new t.j(new Error(`There is no source with ID
'${e}'`)));}setTerrain(e)
{if(this.style._checkLoaded(),this._terrainDataCallback&&this.style.off("data",t
his._terrainDataCallback),e){const
i=this.style.sourceCaches[e.source];if(!i)throw new Error(`cannot load terrain,
because there exists no source with ID: ${e.source}`);for(const i in

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this.style._layers){const
s=this.style._layers[i];"hillshade"===s.type&&s.source===e.source&&t.w("You are
using the same source for a hillshade layer and for 3D terrain. Please consider
using two separate sources to improve rendering quality.");}this.terrain=new
da(this.painter,i,e),this.painter.renderToTexture=new
ma(this.painter,this.terrain),this.transform._minElevationForCurrentTile=this.t
errain.getMinTileElevationForLngLatZoom(this.transform.center,this.transform.til
eZoom),this.transform.elevation=this.terrain.getElevationForLngLatZoom(this.tran
sform.center,this.transform.tileZoom),this._terrainDataCallback=t=>
{"style"===t.dataType?
this.terrain.sourceCache.freeRtt():"source"===t.dataType&&t.tile&&
(t.sourceId!==e.source||this._elevationFreeze||
(this.transform._minElevationForCurrentTile=this.terrain.getMinTileElevationFor
LngLatZoom(this.transform.center,this.transform.tileZoom),this.transform.elevati
on=this.terrain.getElevationForLngLatZoom(this.transform.center,this.transform.t
ileZoom)),this.terrain.sourceCache.freeRtt(t.tile.tileID));},this.style.on("data
",this._terrainDataCallback);}else
this.terrain&&this.terrain.sourceCache.destruct(),this.terrain=null,this.painter
.renderToTexture&&this.painter.renderToTexture.destruct(),this.painter.renderToT
exture=null,this.transform._minElevationForCurrentTile=0,this.transform.elevati
on=0;return this.fire(new t.k("terrain",{terrain:e})),this}getTerrain(){var
t,e;return null!==(e=null===(t=this.terrain)||void 0===t?void 0:t.options)&&void
0!==e?e:null}areTilesLoaded(){const
t=this.style&&this.style.sourceCaches;for(const e in t){const
i=t[e]._tiles;for(const t in i){const
e=i[t];if("loaded"!==e.state&&"errored"!==e.state)return !1}}return
!0}addSourceType(t,e,i){return
this._lazyInitEmptyStyle(),this.style.addSourceType(t,e,i)}removeSource(t)
{return this.style.removeSource(t),this._update(!0)}getSource(t){return
this.style.getSource(t)}addImage(e,i,s={})
{const{pixelRatio:a=1,sdf:o=!1,stretchX:r,stretchY:n,content:l}=s;if(this._lazyI
nitEmptyStyle(),!(i instanceof HTMLImageElement||t.a(i)))if(void
0===i.width||void 0===i.height)return this.fire(new t.j(new Error("Invalid
arguments to map.addImage(). The second argument must be an `HTMLImageElement`,
`ImageData`, `ImageBitmap`, or object with `width`, `height`, and `data`
properties with the same format as `ImageData`")));
{const{width:s,height:h,data:c}=i,u=i;return this.style.addImage(e,{data:new
t.R({width:s,height:h},new
Uint8Array(c)),pixelRatio:a,stretchX:r,stretchY:n,content:l,sdf:o,version:0,user
Image:u}),u.onAdd&&u.onAdd(this,e),this}
{const{width:s,height:h,data:c}=t.h.getImageData(i);this.style.addImage(e,
{data:new
t.R({width:s,height:h},c),pixelRatio:a,stretchX:r,stretchY:n,content:l,sdf:o,ver
sion:0});}}updateImage(e,i){const s=this.style.getImage(e);if(!s)return
this.fire(new t.j(new Error("The map has no image with that id. If you are
adding a new image use `map.addImage(...)` instead.")));const a=i instanceof
HTMLImageElement||t.a(i)?t.h.getImageData(i):i,
{width:o,height:r,data:n}=a;if(void 0===o||void 0===r)return this.fire(new
t.j(new Error("Invalid arguments to map.updateImage(). The second argument must
be an `HTMLImageElement`, `ImageData`, `ImageBitmap`, or object with `width`,
`height`, and `data` properties with the same format as
`ImageData`")));if(o!==s.data.width||r!==s.data.height)return this.fire(new
t.j(new Error("The width and height of the updated image must be that same as
the previous version of the image")));const l=(i instanceof
HTMLImageElement||t.a(i));return
s.data.replace(n,l),this.style.updateImage(e,s),this}getImage(t){return
this.style.getImage(t)}hasImage(e){return e?!!this.style.getImage(e):
(this.fire(new t.j(new Error("Missing required image id"))),!1)}removeImage(t)
{this.style.removeImage(t);}loadImage(t,e)
{h.getImage(this._requestManager.transformRequest(t,c.Image),e);}listImages()
{return this.style.listImages()}addLayer(t,e){return
this._lazyInitEmptyStyle(),this.style.addLayer(t,e),this._update(!0)}moveLayer(t

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,e){return this.style.moveLayer(t,e),this._update(!0)}removeLayer(t){return
this.style.removeLayer(t),this._update(!0)}getLayer(t){return
this.style.getLayer(t)}getLayersOrder(){return
this.style.getLayersOrder()}setLayerZoomRange(t,e,i){return
this.style.setLayerZoomRange(t,e,i),this._update(!0)}setFilter(t,e,i={}){return
this.style.setFilter(t,e,i),this._update(!0)}getFilter(t){return
this.style.getFilter(t)}setPaintProperty(t,e,i,s={}){return
this.style.setPaintProperty(t,e,i,s),this._update(!0)}getPaintProperty(t,e)
{return this.style.getPaintProperty(t,e)}setLayoutProperty(t,e,i,s={}){return
this.style.setLayoutProperty(t,e,i,s),this._update(!0)}getLayoutProperty(t,e)
{return this.style.getLayoutProperty(t,e)}setGlyphs(t,e={}){return
this._lazyInitEmptyStyle(),this.style.setGlyphs(t,e),this._update(!0)}getGlyphs(
){return this.style.getGlyphsUrl()}addSprite(t,e,i={}){return
this._lazyInitEmptyStyle(),this.style.addSprite(t,e,i,(t=>
{t||this._update(!0);})),this.removeSprite(t)}return
this._lazyInitEmptyStyle(),this.style.removeSprite(t),this._update(!0)}getSprite(
){return this.style.getSprite()}setSprite(t,e={}){return
this._lazyInitEmptyStyle(),this.style.setSprite(t,e,(t=>
{t||this._update(!0);})),this.setLight(t,e={}){return
this._lazyInitEmptyStyle(),this.style.setLight(t,e),this._update(!0)}getLight()
{return this.style.getLight()}setFeatureState(t,e){return
this.style.setFeatureState(t,e),this._update(!0)}removeFeatureState(t,e){return
this.style.removeFeatureState(t,e),this._update(!0)}getFeatureState(t){return
this.style.getFeatureState(t)}getContainer(){return
this._container}getCanvasContainer(){return this._canvasContainer}getCanvas()
{return this._canvas}_containerDimensions(){let t=0,e=0;return this._container&&
(t=this._container.clientWidth||400,e=this._container.clientHeight||300),
[t,e]}_setupContainer(){const t=this._container;t.classList.add("maplibregl-
map");const e=this._canvasContainer=i.create("div","maplibregl-canvas-
container",t);this._interactive&&e.classList.add("maplibregl-
interactive"),this._canvas=i.create("canvas","maplibregl-
canvas",e),this._canvas.addEventListener("webglcontextlost",this._contextLost,!1
),this._canvas.addEventListener("webglcontextrestored",this._contextRestored,!1
),this._canvas.setAttribute("tabindex","0"),this._canvas.setAttribute("aria-
label","Map"),this._canvas.setAttribute("role","region");const
s=this._containerDimensions(),a=this._getClampedPixelRatio(s[0],s[1]);this._resi
zeCanvas(s[0],s[1],a);const o=this._controlContainer=i.create("div","maplibregl-
control-container",t),r=this._controlPositions={};["top-left","top-
right","bottom-left","bottom-right"].forEach((t=>
{r[t]=i.create("div","maplibregl-ctrl-${t}
`,o);})),this._container.addEventListener("scroll",this._onMapScroll,!1);}_setup
CooperativeGestures()
{this._cooperativeGesturesScreen=i.create("div","maplibregl-cooperative-gesture-
screen",this._container);let t="boolean"!=typeof
this._cooperativeGestures&&this._cooperativeGestures.windowsHelpText?
this._cooperativeGestures.windowsHelpText:"Use Ctrl + scroll to zoom the
map";0===navigator.platform.indexOf("Mac")&&(t="boolean"!=typeof
this._cooperativeGestures&&this._cooperativeGestures.macHelpText?
this._cooperativeGestures.macHelpText:"Use ⌘ + scroll to zoom the
map"),this._cooperativeGesturesScreen.innerHTML=`<div
class="maplibregl-desktop-message">${t}</div><div
class="maplibregl-mobile-message">${"boolean"!=typeof
this._cooperativeGestures&&this._cooperativeGestures.mobileHelpText?
this._cooperativeGestures.mobileHelpText:"Use two fingers to move the map"}
</div>`<div></div>`,this._cooperativeGesturesScreen.setAttribute("aria-
hidden","true"),this._canvasContainer.addEventListener("wheel",this._cooperative
GesturesOnWheel,!1),this._canvasContainer.classList.add("maplibregl-cooperative-
gestures");}_destroyCooperativeGestures()
{i.remove(this._cooperativeGesturesScreen),this._canvasContainer.removeEventList
ener("wheel",this._cooperativeGesturesOnWheel,!1),this._canvasContainer.classLis
t.remove("maplibregl-cooperative-gestures");}_resizeCanvas(t,e,i)
{this._canvas.width=Math.floor(i*t),this._canvas.height=Math.floor(i*e),this._ca

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nvas.style.width=`${t}px`,this._canvas.style.height=`${e}px`;}_setupPainter()
{const t=
{alpha:!0,stencil:!0,depth:!0,failIfMajorPerformanceCaveat:this._failIfMajorPerf
ormanceCaveat,preserveDrawingBuffer:this._preserveDrawingBuffer,antialias:this._
antialias||!1};let
e=null;this._canvas.addEventListener("webglcontextcreationerror",(i=>{e=
{requestedAttributes:t},i&&(e.statusMessage=i.statusMessage,e.type=i.type)})),
{once:!0});const
i=this._canvas.getContext("webgl2",t)||this._canvas.getContext("webgl",t);if(!i)
{const t="Failed to initialize WebGL";throw e?(e.message=t,new
Error(JSON.stringify(e))):new Error(t)}this.painter=new
is(i,this.transform),s.testSupport(i);}_onCooperativeGesture(t,e,i){return
!e&&i<2&&(this._cooperativeGesturesScreen.classList.add("maplibregl-
show"),setTimeout((()=>>
{this._cooperativeGesturesScreen.classList.remove("maplibregl-
show");}),100)),!1}loaded(){return
!this._styleDirty&&!this._sourcesDirty&&!this.style&&this.style.loaded()}_updat
e(t){return this.style&&this.style.loaded?
(this._styleDirty=this._styleDirty||t,this._sourcesDirty=!0,this.triggerRepaint(
),this):this}_requestRenderFrame(t){return
this._update(),this._renderTaskQueue.add(t)}_cancelRenderFrame(t)
{this._renderTaskQueue.remove(t);}_render(e){const i=this._idleTriggered?
this._fadeDuration:0;if(this.painter.context.setDirty(),this.painter.setBaseStat
e(),this._renderTaskQueue.run(e),this._removed)return;let
s=!1;if(this.style&&this._styleDirty){this._styleDirty=!1;const
e=this.transform.zoom,a=t.h.now();this.style.zoomHistory.update(e,a);const o=new
t.a8(e,
{now:a,fadeDuration:i,zoomHistory:this.style.zoomHistory,transition:this.style.g
etTransition()}),r=o.crossFadingFactor();l===r&&r===this._crossFadingFactor||
(s=!0,this._crossFadingFactor=r),this.style.update(o);}this.style&&this._sources
Dirty&&
(this._sourcesDirty=!1,this.style._updateSources(this.transform),this.terrain?
(this.terrain.sourceCache.update(this.transform,this.terrain),this.transform._mi
nElevationForCurrentTile=this.terrain.getMinTileElevationForLngLatZoom(this.tran
sform.center,this.transform.tileZoom),this._elevationFreeze||
(this.transform.elevation=this.terrain.getElevationForLngLatZoom(this.transform.
center,this.transform.tileZoom))) :
(this.transform._minElevationForCurrentTile=0,this.transform.elevation=0),this.
_placementDirty=this.style&&this.style._updatePlacement(this.painter.transform,t
his.showCollisionBoxes,i,this._crossSourceCollisions),this.painter.render(this.s
tyle,
{showTileBoundaries:this.showTileBoundaries,showOverdrawInspector:this._showOver
drawInspector,rotating:this.isRotating(),zooming:this.isZooming(),moving:this.is
Moving(),fadeDuration:i,showPadding:this.showPadding}),this.fire(new
t.k("render")),this.loaded()&&!this._loaded&&
(this._loaded=!0,t.bg.mark(t.bh.load),this.fire(new t.k("load"))),this.style&&
(this.style.hasTransitions()||s)&&
(this._styleDirty=!0),this.style&&!this._placementDirty&&this.style._releaseSymb
olFadeTiles();const
a=this._sourcesDirty||this._styleDirty||this._placementDirty;return
a||this._repaint?
this.triggerRepaint():!this.isMoving()&&this.loaded()&&this.fire(new
t.k("idle")),!this._loaded||this._fullyLoaded||a||
(this._fullyLoaded=!0,t.bg.mark(t.bh.fullLoad)),this}redraw(){return
this.style&&(this._frame&&
(this._frame.cancel(),this._frame=null),this._render(0)),this}remove(){var
e;this._hash&&this._hash.remove();for(const t of
this._controls)t.onRemove(this);this._controls=[],this._frame&&
(this._frame.cancel(),this._frame=null),this._renderTaskQueue.clear(),this.paint
er.destroy(),this.handlers.destroy(),delete
this.handlers,this.setStyle(null),"undefined"!=typeof
window&&removeEventListener("online",this._onWindowOnline,!1),h.removeThrottleCo

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ntrol(this._imageQueueHandle),null===(e=this._resizeObserver)||void
0===e||e.disconnect();const
s=this.painter.context.gl.getExtension("WEBGL_lose_context");s&&s.loseContext(),
this._canvas.removeEventListener("webglcontextrestored",this._contextRestored,!1
),this._canvas.removeEventListener("webglcontextlost",this._contextLost,!1),i.re
move(this._canvasContainer),i.remove(this._controlContainer),this._cooperativeGe
stures&&this._destroyCooperativeGestures(),this._container.classList.remove("map
libregl-map"),t.bg.clearMetrics(),this._removed=!0,this.fire(new
t.k("remove"));};triggerRepaint(){this.style&&!this._frame&&
(this._frame=t.h.frame(e=>
{t.bg.frame(e),this._frame=null,this._render(e);}));};get showTileBoundaries()
{return !!this._showTileBoundaries}set showTileBoundaries(t)
{this._showTileBoundaries!==t&&(this._showTileBoundaries=t,this._update());}get
showPadding(){return !!this._showPadding}set showPadding(t)
{this._showPadding!==t&&(this._showPadding=t,this._update());}get
showCollisionBoxes(){return !!this._showCollisionBoxes}set showCollisionBoxes(t)
{this._showCollisionBoxes!==t&&(this._showCollisionBoxes=t,t?
this.style._generateCollisionBoxes():this._update());}get
showOverdrawInspector(){return !!this._showOverdrawInspector}set
showOverdrawInspector(t){this._showOverdrawInspector!==t&&
(this._showOverdrawInspector=t,this._update());}get repaint(){return
!!this._repaint}set repaint(t){this._repaint!==t&&
(this._repaint=t,this.triggerRepaint());}get vertices(){return
!!this._vertices}set vertices(t){this._vertices=t,this._update();}get version()
{return fa}getCameraTargetElevation(){return
this.transform.elevation}},Ba.NavigationControl=class{constructor(e)
{this._updateZoomButtons=()=>{const
t=this._map.getZoom(),e=t===this._map.getMaxZoom(),i=t===this._map.getMinZoom();
this._zoomInButton.disabled=e,this._zoomOutButton.disabled=i,this._zoomInButton.
setAttribute("aria-
disabled",e.toString()),this._zoomOutButton.setAttribute("aria-
disabled",i.toString());},this._rotateCompassArrow=()=>{const
t=this.options.visualizePitch?
`scale(${1/Math.pow(Math.cos(this._map.transform.pitch*(Math.PI/180))),.5)})
rotateX(${this._map.transform.pitch}deg) rotateZ(${this._map.transform.angle*
(180/Math.PI)}deg)`:`rotate(${this._map.transform.angle*
(180/Math.PI)}deg)`;this._compassIcon.style.transform=t;},this._setButtonTitle=
(t,e)=>{const
i=this._map._getUIString(`NavigationControl.${e}`);t.title=i,t.setAttribute("ari
a-
label",i);},this.options=t.e({},xa,e),this._container=i.create("div","maplibregl
-ctrl maplibregl-ctrl-group"),this._container.addEventListener("contextmenu",
(t=>t.preventDefault()),this.options.showZoom&&
(this._zoomInButton=this._createButton("maplibregl-ctrl-zoom-in",
(t=>this._map.zoomIn({},{originalEvent:t}))),i.create("span","maplibregl-ctrl-
icon",this._zoomInButton).setAttribute("aria-
hidden","true"),this._zoomOutButton=this._createButton("maplibregl-ctrl-zoom-
out",t=>this._map.zoomOut({},{originalEvent:t}))),i.create("span","maplibregl-
ctrl-icon",this._zoomOutButton).setAttribute("aria-
hidden","true")),this.options.showCompass&&
(this._compass=this._createButton("maplibregl-ctrl-compass",t=>
{this.options.visualizePitch?this._map.resetNorthPitch({},
{originalEvent:t}):this._map.resetNorth({},
{originalEvent:t});}),this._compassIcon=i.create("span","maplibregl-ctrl-
icon",this._compass),this._compassIcon.setAttribute("aria-
hidden","true"));}onAdd(t){return this._map=t,this.options.showZoom&&
(this._setButtonTitle(this._zoomInButton,"ZoomIn"),this._setButtonTitle(this._zo
omOutButton,"ZoomOut"),this._map.on("zoom",this._updateZoomButtons),this._update
ZoomButtons()),this.options.showCompass&&
(this._setButtonTitle(this._compass,"ResetBearing"),this.options.visualizePitch&
&this._map.on("pitch",this._rotateCompassArrow),this._map.on("rotate",this._rota
teCompassArrow),this._rotateCompassArrow(),this._handler=new

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ya(this._map,this._compass,this.options.visualizePitch)),this._container}onRemov
e()
{i.remove(this._container),this.options.showZoom&&this._map.off("zoom",this._upd
ateZoomButtons),this.options.showCompass&&
(this.options.visualizePitch&&this._map.off("pitch",this._rotateCompassArrow),th
is._map.off("rotate",this._rotateCompassArrow),this._handler.off(),delete
this._handler),delete this._map;}_createButton(t,e){const
s=i.create("button",t,this._container);return
s.type="button",s.addEventListener("click",e),s}},Ba.GeolocateControl=class
extends t.E{constructor(e){super(),this._onSuccess=e=>{if(this._map)
{if(this._isOutOfMapMaxBounds(e))return this._setErrorState(),this.fire(new
t.k("outofmaxbounds",e)),this._updateMarker(),void
this._finish();if(this.options.trackUserLocation)switch(this._lastKnownPosition=
e,this._watchState)
{case"WAITING_ACTIVE":case"ACTIVE_LOCK":case"ACTIVE_ERROR":this._watchState="ACT
IVE_LOCK",this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-
waiting"),this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-
active-error"),this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-
active");break;case"BACKGROUND":case"BACKGROUND_ERROR":this._watchState="BACKGRO
UND",this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-
waiting"),this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-
background-error"),this._geolocateButton.classList.add("maplibregl-ctrl-
geolocate-background");break;default:throw new Error(`Unexpected watchState
${this._watchState}`)}this.options.showUserLocation&&"OFF"!==this._watchState&&t
his._updateMarker(e),this.options.trackUserLocation&&"ACTIVE_LOCK"!==this._watch
State||this._updateCamera(e),this.options.showUserLocation&&this._dotElement.cla
ssList.remove("maplibregl-user-location-dot-stale"),this.fire(new
t.k("geolocate",e)),this._finish();}},this._updateCamera=e=>{const i=new
t.L(e.coords.longitude,e.coords.latitude),s=e.coords.accuracy,a=this._map.getBea
ring(),o=t.e({bearing:a},this.options.fitBoundsOptions),r=L.fromLngLat(i,s);this
._map.fitBounds(r,o,{geolocateSource:!0});},this._updateMarker=e=>{if(e){const
i=new
t.L(e.coords.longitude,e.coords.latitude);this._accuracyCircleMarker.setLngLat(i
).addTo(this._map),this._userLocationDotMarker.setLngLat(i).addTo(this._map),thi
s._accuracy=e.coords.accuracy,this.options.showUserLocation&&this.options.showAcc
uracyCircle&&this._updateCircleRadius();}else
this._userLocationDotMarker.remove(),this._accuracyCircleMarker.remove();},this
._onZoom=()=>
{this.options.showUserLocation&&this.options.showAccuracyCircle&&this._updateCir
cleRadius();},this._onError=e=>{if(this._map)
{if(this.options.trackUserLocation)if(1===e.code)
{this._watchState="OFF",this._geolocateButton.classList.remove("maplibregl-ctrl-
geolocate-waiting"),this._geolocateButton.classList.remove("maplibregl-ctrl-
geolocate-active"),this._geolocateButton.classList.remove("maplibregl-ctrl-
geolocate-active-error"),this._geolocateButton.classList.remove("maplibregl-
ctrl-geolocate-background"),this._geolocateButton.classList.remove("maplibregl-
ctrl-geolocate-background-error"),this._geolocateButton.disabled=!0;const
t=this._map._getUIString("GeolocateControl.LocationNotAvailable");this._geolocat
eButton.title=t,this._geolocateButton.setAttribute("aria-label",t),void
0!==this._geolocationWatchID&&this._clearWatch();}else
{if(3===e.code&&Pa)return;this._setErrorState();}"OFF"!==this._watchState&&this
.options.showUserLocation&&this._dotElement.classList.add("maplibregl-user-
location-dot-stale"),this.fire(new
t.k("error",e)),this._finish();}},this._finish=()=>
{this._timeoutId&&clearTimeout(this._timeoutId),this._timeoutId=void
0;},this._setupUI=e=>{if(this._map)
{if(this._container.addEventListener("contextmenu",
(t=>t.preventDefault()),this._geolocateButton=i.create("button","maplibregl-
ctrl-geolocate",this._container),i.create("span","maplibregl-ctrl-
icon",this._geolocateButton).setAttribute("aria-
hidden","true"),this._geolocateButton.type="button",!1===e){t.w("Geolocation
support is not available so the GeolocateControl will be disabled.");const

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e=this._map._getUIString("GeolocateControl.LocationNotAvailable");this._geolocateButton.disabled=!0,this._geolocateButton.title=e,this._geolocateButton.setAttribute("aria-label",e);}else {const t=this._map._getUIString("GeolocateControl.FindMyLocation");this._geolocateButton.title=t,this._geolocateButton.setAttribute("aria-label",t);}this.options.trackUserLocation&&(this._geolocateButton.setAttribute("aria-pressed","false"),this._watchState="OFF"),this.options.showUserLocation&&(this._dotElement=i.create("div","maplibregl-user-location-dot"),this._userLocationDotMarker=new Ea({element:this._dotElement}),this._circleElement=i.create("div","maplibregl-user-location-accuracy-circle"),this._accuracyCircleMarker=new Ea({element:this._circleElement,pitchAlignment:"map"}),this.options.trackUserLocation&&(this._watchState="OFF"),this._map.on("zoom",this._onZoom)),this._geolocateButton.addEventListener("click",this.trigger.bind(this)),this._setup=!0,this.options.trackUserLocation&&this._map.on("movestart",(e=>{e.geolocateSource||"ACTIVE_LOCK"!==this._watchState||e.originalEvent&&"resize"===e.originalEvent.type||(this._watchState="BACKGROUND",this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-background"),this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-active"),this.fire(new t.k("trackuserlocationend")))}));}},this.options=t.e({},Sa,e);}onAdd(t){return this._map=t,this._container=i.create("div","maplibregl-ctrl maplibregl-ctrl-group"),function(t,e=!1){void 0===ba||e?void 0!==window.navigator.permissions?window.navigator.permissions.query({name:"geolocation"}).then((e=>{ba="denied"!==e.state,t(ba);})).catch((()=>{ba=!window.navigator.geolocation,t(ba);})):ba=!window.navigator.geolocation,t(ba):t(ba);}(this._setupUI),this._container).onRemove(){void 0!==this._geolocationWatchID&&(window.navigator.geolocation.clearWatch(this._geolocationWatchID),this._geolocationWatchID=void 0),this.options.showUserLocation&&this._userLocationDotMarker&&this._userLocationDotMarker.remove(),this.options.showAccuracyCircle&&this._accuracyCircleMarker&&this._accuracyCircleMarker.remove(),i.remove(this._container),this._map.off("zoom",this._onZoom),this._map=void 0,Ca=0,Pa=!1;}_isOutOfMapMaxBounds(t){const e=this._map.getMaxBounds(),i=t.coords;return e&&(i.longitude<e.getWest()||i.longitude>e.getEast()||i.latitude<e.getSouth()||i.latitude>e.getNorth())}_setErrorState(){switch(this._watchState){case"WAITING_ACTIVE":this._watchState="ACTIVE_ERROR",this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-active"),this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-active-error");break;case"ACTIVE_LOCK":this._watchState="ACTIVE_ERROR",this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-active"),this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-active-error"),this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-waiting");break;case"BACKGROUND":this._watchState="BACKGROUND_ERROR",this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-background"),this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-background-error"),this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-waiting");break;case"ACTIVE_ERROR":break;default:throw new Error(`Unexpected watchState ${this._watchState}`)}}_updateCircleRadius(){const t=this._map.getBounds(),e=t.getSouthEast(),i=t.getNorthEast(),s=e.distanceTo(i),a=Math.ceil(this._accuracy/(s/this._map._container.clientHeight)*2);this._circleElement.style.width=`${a}px`,this._circleElement.style.height=`${a}px`;}trigger(){if(!this._setup)return t.w("Geolocate control triggered before added to a map"),!1;if(this.options.trackUserLocation){switch(this._watchState){case"OFF":this._watchState="WAITING_ACTIVE",this.fire(new t.k("trackuserlocationstart"));break;case"WAITING_ACTIVE":case"ACTIVE_LOCK":case"ACTIVE_ERROR":case"BACKGROUND_ERROR":Ca--,Pa=!1,this._watchState="OFF",this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-waiting"),this._geolocateButton.classList.remove("maplibregl-

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ctrl-geolocate-active"),this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-active-error"),this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-background"),this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-background-error"),this.fire(new
t.k("trackuserlocationend"));break;case"BACKGROUND":this._watchState="ACTIVE_LOCK",this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-background"),this._lastKnownPosition&&this._updateCamera(this._lastKnownPosition),this.fire(new t.k("trackuserlocationstart"));break;default:throw new
Error(`Unexpected watchState ${this._watchState}`)}switch(this._watchState){case"WAITING_ACTIVE":this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-waiting"),this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-active");break;case"ACTIVE_LOCK":this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-active");break;case"OFF":break;default:throw new
Error(`Unexpected watchState ${this._watchState}`)}if("OFF"===this._watchState&&void 0!==this._geolocationWatchID)this._clearWatch();else if(void 0===this._geolocationWatchID){let
t;this._geolocateButton.classList.add("maplibregl-ctrl-geolocate-waiting"),this._geolocateButton.setAttribute("aria-pressed","true"),Ca++,Ca>1?(t={maximumAge:6e5,timeout:0},Pa=!0):(t=this.options.positionOptions,Pa=!1),this._geolocationWatchID=window.navigator.geolocation.watchPosition(this._onSuccess,this._onError,t);}}else
window.navigator.geolocation.getCurrentPosition(this._onSuccess,this._onError,this.options.positionOptions),this._timeoutId=setTimeout(this._finish,1e4);return
!0}_clearWatch(){window.navigator.geolocation.clearWatch(this._geolocationWatchID),this._geolocationWatchID=void 0,this._geolocateButton.classList.remove("maplibregl-ctrl-geolocate-waiting"),this._geolocateButton.setAttribute("aria-pressed","false"),this.options.showUserLocation&&this._updateMarker(null);}},Ba.AttributionControl=ra,Ba.LogoControl=na,Ba.ScaleControl=class{constructor(e){this._onMove=()=>{Ma(this._map,this._container,this.options);},this.setUnit=t=>{this.options.unit=t,Ma(this._map,this._container,this.options);},this.options=t.e({},Da,e);}getDefaultPosition(){return "bottom-left"}onAdd(t){return
this._map=t,this._container=i.create("div","maplibregl-ctrl maplibregl-ctrl-scale",t.getContainer()),this._map.on("move",this._onMove),this._onMove(),this._container}onRemove(){i.remove(this._container),this._map.off("move",this._onMove),this._map=void 0;}},Ba.FullscreenControl=class extends t.E{constructor(e={}){super(),this._onFullscreenChange=()=>>{(window.document.fullscreenElement||window.document.mozFullScreenElement||window.document.webkitFullscreenElement||window.document.msFullscreenElement)===this._container!&&this._fullscreen&&this._handleFullscreenChange();},this._onClickFullscreen=()=>>{this._isFullscreen()?this._exitFullscreen():this._requestFullscreen();},this._fullscreen=!1,e&&e.container&&(e.container instanceof HTMLElement?this._container=e.container:t.w("Fullscreen control 'container' must be a DOM element."),"onfullscreenchange" in document?this._fullscreenchange="fullscreenchange":"onmozfullscreenchange" in document?this._fullscreenchange="mozfullscreenchange":"onwebkitfullscreenchange" in document?this._fullscreenchange="webkitfullscreenchange":"onmsfullscreenchange" in document&&(this._fullscreenchange="MSFullscreenChange");}onAdd(t){return
this._map=t,this._container||this._container=this._map.getContainer()),this._controlContainer=i.create("div","maplibregl-ctrl maplibregl-ctrl-group"),this._setupUI(),this._controlContainer}onRemove(){i.remove(this._controlContainer),this._map=null>window.document.removeEventListener(this._fullscreenchange,this._onFullscreenChange);}_setupUI(){const
t=this._fullscreenButton=i.create("button","maplibregl-ctrl-fullscreen",this._controlContainer);i.create("span","maplibregl-ctrl-icon",t).setAttribute("aria-

```



```

hidden","true"),t.type="button",this._updateTitle(),this._fullscreenButton.addEvent
entListener("click",this._onClickFullscreen),window.document.addEventListener(th
is._fullscreenchange,this._onFullscreenChange);}_updateTitle(){const
t=this._getTitle();this._fullscreenButton.setAttribute("aria-
label",t),this._fullscreenButton.title=t;}_getTitle(){return
this._map._getUIString(this._isFullscreen()?"FullscreenControl.Exit":"Fullscreen
Control.Enter");}_isFullscreen(){return
this._fullscreen}_handleFullscreenChange()
{this._fullscreen=!this._fullscreen,this._fullscreenButton.classList.toggle("map
libregl-ctrl-shrink"),this._fullscreenButton.classList.toggle("maplibregl-ctrl-
fullscreen"),this._updateTitle(),this._fullscreen?(this.fire(new
t.k("fullscreenstart")),this._map._cooperativeGestures&&
(this._prevCooperativeGestures=this._map._cooperativeGestures,this._map.setCoope
rativeGestures())):(this.fire(new
t.k("fullscreenend")),this._prevCooperativeGestures&&
(this._map.setCooperativeGestures(this._prevCooperativeGestures),delete
this._prevCooperativeGestures));}_exitFullscreen()
{window.document.exitFullscreen?
window.document.exitFullscreen():window.document.mozCancelFullScreen?
window.document.mozCancelFullScreen():window.document.msExitFullscreen?
window.document.msExitFullscreen():window.document.webkitCancelFullScreen?
window.document.webkitCancelFullScreen():this._togglePseudoFullScreen();}_reques
tFullscreen(){this._container.requestFullscreen?
this._container.requestFullscreen():this._container.mozRequestFullScreen?
this._container.mozRequestFullScreen():this._container.msRequestFullscreen?
this._container.msRequestFullscreen():this._container.webkitRequestFullscreen?
this._container.webkitRequestFullscreen():this._togglePseudoFullScreen();}_toggl
ePseudoFullScreen(){this._container.classList.toggle("maplibregl-pseudo-
fullscreen"),this._handleFullscreenChange(),this._map.resize();},Ba.TerrainCont
rol=class{constructor(t){this._toggleTerrain=()=>{this._map.getTerrain()?
this._map.setTerrain(null):this._map.setTerrain(this.options),this._updateTerrai
nIcon();},this._updateTerrainIcon=()=>
{this._terrainButton.classList.remove("maplibregl-ctrl-
terrain"),this._terrainButton.classList.remove("maplibregl-ctrl-terrain-
enabled"),this._map.terrain?(this._terrainButton.classList.add("maplibregl-ctrl-
terrain-
enabled"),this._terrainButton.title=this._map._getUIString("TerrainControl.disab
leTerrain")):(this._terrainButton.classList.add("maplibregl-ctrl-
terrain"),this._terrainButton.title=this._map._getUIString("TerrainControl.enabl
eTerrain"));},this.options=t;}_onAdd(t){return
this._map=t,this._container=i.create("div","maplibregl-ctrl maplibregl-ctrl-
group"),this._terrainButton=i.create("button","maplibregl-ctrl-
terrain",this._container),i.create("span","maplibregl-ctrl-
icon",this._terrainButton).setAttribute("aria-
hidden","true"),this._terrainButton.type="button",this._terrainButton.addEventLi
stener("click",this._toggleTerrain),this._updateTerrainIcon(),this._map.on("terr
ain",this._updateTerrainIcon),this._container}_onRemove()
{i.remove(this._container),this._map.off("terrain",this._updateTerrainIcon),this
._map=void 0;},Ba.Popup=class extends t.E{constructor(e){super(),this.remove=
()=>(this._content&&i.remove(this._content),this._container&&
(i.remove(this._container),delete this._container),this._map&&
(this._map.off("move",this._update),this._map.off("move",this._onClose),this._ma
p.off("click",this._onClose),this._map.off("remove",this.remove),this._map.off("
mousemove",this._onMouseMove),this._map.off("mouseup",this._onMouseUp),this._map
.off("drag",this._onDrag),delete this._map),this.fire(new
t.k("close")),this),this._onMouseUp=t=>
{this._update(t.point);},this._onMouseMove=t=>
{this._update(t.point);},this._onDrag=t=>
{this._update(t.point);},this._update=t=>
{if(!this._map||!this._lngLat&&!this._trackPointer||!this._content)return;if(!th
is._container){if(this._container=i.create("div","maplibregl-
popup",this._map.getContainer()),this._tip=i.create("div","maplibregl-popup-

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tip",this._container),this._container.appendChild(this._content),this.options.className)for(const t of this.options.className.split("
"))this._container.classList.add(t);this._trackPointer&&this._container.classList.add("maplibregl-popup-track-
pointer");}if(this.options.maxWidth&&this._container.style.maxWidth!==this.options.maxWidth&&
(this._container.style.maxWidth=this.options.maxWidth),this._map.transform.renderWorldCopies&&!this._trackPointer&&
(this._lngLat=wa(this._lngLat,this._pos,this._map.transform)),this._trackPointer&&!t)return;const e=this._pos=this._trackPointer&&t?
t:this._map.project(this._lngLat);let s=this.options.anchor;const
a=Ra(this.options.offset);if(!s){const
o=this._container.offsetWidth,i=this._container.offsetHeight;let
o;o=e.y+a.bottom.y<i?["top"]:e.y>this._map.transform.height-i?["bottom"]:
[],e.x<t/2?o.push("left"):e.x>this._map.transform.width-
t/2&&o.push("right"),s=0===o.length?"bottom":o.join("-");}const
o=e.add(a[s]).round();i.setTransform(this._container,`${Ta[s]}
translate(${o.x}px,${o.y}px)`),la(this._container,s,"popup");},this._onClose=
()=>{this.remove();},this.options=t.e(Object.create(La),e);}addTo(e){return
this._map&&this.remove(),this._map=e,this.options.closeOnClick&&this._map.on("cl
ick",this._onClose),this.options.closeOnMove&&this._map.on("move",this._onClose)
,this._map.on("remove",this.remove),this._update(),this._focusFirstElement(),thi
s._trackPointer?
(this._map.on("mousemove",this._onMouseMove),this._map.on("mouseup",this._onMous
eUp),this._container&&this._container.classList.add("maplibregl-popup-track-
pointer"),this._map._canvasContainer.classList.add("maplibregl-track-
pointer")):this._map.on("move",this._update),this.fire(new
t.k("open")),this.isOpen(){return !!this._map.getLngLat(){return
this._lngLat}setLngLat(e){return
this._lngLat=t.L.convert(e),this._pos=null,this._trackPointer=!1,this._update(),
this._map&&
(this._map.on("move",this._update),this._map.off("mousemove",this._onMouseMove),
this._container&&this._container.classList.remove("maplibregl-popup-track-
pointer"),this._map._canvasContainer.classList.remove("maplibregl-track-
pointer")),this.trackPointer(){return
this._trackPointer=!0,this._pos=null,this._update(),this._map&&
(this._map.off("move",this._update),this._map.on("mousemove",this._onMouseMove),
this._map.on("drag",this._onDrag),this._container&&this._container.classList.add
("maplibregl-popup-track-
pointer"),this._map._canvasContainer.classList.add("maplibregl-track-
pointer")),this.getElement(){return this._container}setText(t){return
this.setDOMContent(document.createTextNode(t))}setHTML(t){const
e=document.createElementFragment(),i=document.createElement("body");let
s;for(i.innerHTML=t;s=i.firstChild,s)e.appendChild(s);return
this.setDOMContent(e)}getMaxWidth(){var t;return null===
(t=this._container)||void 0===t?void 0:t.style.maxWidth}setMaxWidth(t){return
this.options.maxWidth=t,this._update(),this.setDOMContent(t)
{if(this._content)for(;this._content.hasChildNodes();)this._content.firstChild&&
this._content.removeChild(this._content.firstChild);else
this._content=i.create("div","maplibregl-popup-content",this._container);return
this._content.appendChild(t),this._createCloseButton(),this._update(),this._focu
sFirstElement(),this.addClass(t)
{this._container&&this._container.classList.add(t);}removeClassName(t)
{this._container&&this._container.classList.remove(t);}setOffset(t){return
this.options.offset=t,this._update(),this.toggleClassName(t)
{if(this._container)return
this._container.classList.toggle(t)}_createCloseButton()
{this.options.closeButton&&(this._closeButton=i.create("button","maplibregl-
popup-close-
button",this._content),this._closeButton.type="button",this._closeButton.setAttr
ibute("aria-label","Close
popup"),this._closeButton.innerHTML="#215;",this._closeButton.addEventListener(

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"click",this._onClose));}_focusFirstElement()
{if(!this.options.focusAfterOpen||!this._container)return;const
t=this._container.querySelector(Aa);t&&t.focus();}},Ba.Marker=Ea,Ba.Style=se,Ba.
LngLat=t.L,Ba.LngLatBounds=L,Ba.Point=t.P,Ba.MercatorCoordinate=t.U,Ba.Evented=t
.E,Ba.AJAXError=t.bi,Ba.config=t.c,Ba.CanvasSource=Z,Ba.GeoJSONSource=B,Ba.Image
Source=N,Ba.RasterDEMTileSource=F,Ba.RasterTileSource=k,Ba.VectorTileSource=R,Ba
.VideoSource=U,Ba.setRTLTextPlugin=t.bj,Ba.getRTLTextPluginStatus=t.bk,Ba.prewar
m=function(){st().acquire(J);},Ba.clearPrewarmedResources=function(){const
t=it;t&&(t.isPreloaded())&&l===t.numActive()?
(t.release(J),it=null):console.warn("Could not clear WebWorkers since there are
active Map instances that still reference it. The pre-warmed WebWorker pool can
only be cleared when all map instances have been removed with
map.remove()"));},ka.extend(Ba,
{isSafari:t.ac,getPerformanceMetrics:t.bg.getPerformanceMetrics}),Ba}));

//
// Our custom intro provides a specialized "define()" function, called by the
// AMD modules below, that sets up the worker blob URL and then executes the
// main module, storing its exported value as 'maplibregl'

var maplibregl$1 = maplibregl;

return maplibregl$1;

});
//# sourceMappingURL=maplibre-gl.js.map

```

```

=====
FILE: https://cdn.jsdelivr.net/npm/maplibre-gl@3.6.2/dist/maplibre-gl.css
63722 chars · code · shown in full · found via <link stylesheet>, loaded by the
browser (link)
=====
.maplibregl-map{-webkit-tap-highlight-color:rgb(0 0 0/0);font:12px/20px
Helvetica Neue,Arial,Helvetica,sans-
serif;overflow:hidden;position:relative}.maplibregl-
canvas{left:0;position:absolute;top:0}.maplibregl-
map:fullscreen{height:100%;width:100%}.maplibregl-ctrl-group button.maplibregl-
ctrl-compass{touch-action:none}.maplibregl-canvas-container.maplibregl-
interactive,.maplibregl-ctrl-group button.maplibregl-ctrl-compass{cursor:grab;-
moz-user-select:none;-webkit-user-select:none;-ms-user-select:none;user-
select:none}.maplibregl-canvas-container.maplibregl-interactive.maplibregl-
track-pointer{cursor:pointer}.maplibregl-canvas-container.maplibregl-
interactive:active,.maplibregl-ctrl-group button.maplibregl-ctrl-
compass:active{cursor:grabbing}.maplibregl-canvas-container.maplibregl-touch-
zoom-rotate,.maplibregl-canvas-container.maplibregl-touch-zoom-rotate
.maplibregl-canvas{touch-action:pan-x pan-y}.maplibregl-canvas-
container.maplibregl-touch-drag-pan,.maplibregl-canvas-container.maplibregl-
touch-drag-pan .maplibregl-canvas{touch-action:pinch-zoom}.maplibregl-canvas-
container.maplibregl-touch-zoom-rotate.maplibregl-touch-drag-pan,.maplibregl-
canvas-container.maplibregl-touch-zoom-rotate.maplibregl-touch-drag-pan
.maplibregl-canvas{touch-action:none}.maplibregl-canvas-container.maplibregl-
touch-drag-pan.maplibregl-cooperative-gestures,.maplibregl-canvas-
container.maplibregl-touch-drag-pan.maplibregl-cooperative-gestures .maplibregl-
canvas{touch-action:pan-x pan-y}.maplibregl-ctrl-bottom-left,.maplibregl-ctrl-
bottom-right,.maplibregl-ctrl-top-left,.maplibregl-ctrl-top-right{pointer-
events:none;position:absolute;z-index:2}.maplibregl-ctrl-top-
left{left:0;top:0}.maplibregl-ctrl-top-right{right:0;top:0}.maplibregl-ctrl-
bottom-left{bottom:0;left:0}.maplibregl-ctrl-bottom-
right{bottom:0;right:0}.maplibregl-ctrl{clear:both;pointer-
events:auto;transform:translate(0)}.maplibregl-ctrl-top-left .maplibregl-

```

```

ctrl{float:left;margin:10px 0 0 10px}.maplibregl-ctrl-top-right .maplibregl-
ctrl{float:right;margin:10px 10px 0 0}.maplibregl-ctrl-bottom-left .maplibregl-
ctrl{float:left;margin:0 0 10px 10px}.maplibregl-ctrl-bottom-right .maplibregl-
ctrl{float:right;margin:0 10px 10px 0}.maplibregl-ctrl-
group{background:#fff;border-radius:4px}.maplibregl-ctrl-group:not(:empty){box-
shadow:0 0 0 2px rgba(0,0,0,.1)}@media (-ms-high-contrast:active){.maplibregl-
ctrl-group:not(:empty){box-shadow:0 0 0 2px ButtonText}}.maplibregl-ctrl-group
button{background-color:transparent;border:0;box-sizing:border-
box;cursor:pointer;display:block;height:29px;outline:none;padding:0;width:29px}.
maplibregl-ctrl-group button+button{border-top:1px solid #ddd}.maplibregl-ctrl
button .maplibregl-ctrl-icon{background-position:50%;background-repeat:no-
repeat;display:block;height:100%;width:100%}@media (-ms-high-contrast:active)
{.maplibregl-ctrl-icon{background-color:transparent}.maplibregl-ctrl-group
button+button{border-top:1px solid ButtonText}}.maplibregl-ctrl button::-moz-
focus-inner{border:0;padding:0}.maplibregl-ctrl-attrib-button:focus,.maplibregl-
ctrl-group button:focus{box-shadow:0 0 2px 2px #0096ff}.maplibregl-ctrl
button:disabled{cursor:not-allowed}.maplibregl-ctrl button:disabled .maplibregl-
ctrl-icon{opacity:.25}.maplibregl-ctrl button:not(:disabled):hover{background-
color:rgb(0 0 0/5%)}.maplibregl-ctrl-group button:focus:focus-visible{box-
shadow:0 0 2px 2px #0096ff}.maplibregl-ctrl-group button:focus:not(:focus-
visible){box-shadow:none}.maplibregl-ctrl-group button:focus:first-child{border-
radius:4px 4px 0 0}.maplibregl-ctrl-group button:focus:last-child{border-
radius:0 0 4px 4px}.maplibregl-ctrl-group button:focus:only-child{border-
radius:inherit}.maplibregl-ctrl button.maplibregl-ctrl-zoom-out .maplibregl-
ctrl-icon{background-image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' fill='%233333'
viewBox='0 0 29 29'%3E%3Cpath d='M10 13c-.75 0-1.5.75-1.5 1.5S9.25 16 10
16h9c.75 0 1.5-.75 1.5-1.5S19.75 13 19 13h-9z'/%3E%3C/svg%3E")}).maplibregl-ctrl
button.maplibregl-ctrl-zoom-in .maplibregl-ctrl-icon{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' fill='%233333'
viewBox='0 0 29 29'%3E%3Cpath d='M14.5 8.5c-.75 0-1.5.75-1.5 1.5v3h-3c-.75 0-
1.5.75-1.5 1.5S9.25 16 10 16h3v3c0 .75.75 1.5 1.5 1.5S16 19.75 16 19v-3h3c.75 0
1.5-.75 1.5-1.5S19.75 13 19 13h-3v-3c0-.75-.75-1.5-1.5-
1.5z'/%3E%3C/svg%3E")}}@media (-ms-high-contrast:active){.maplibregl-ctrl
button.maplibregl-ctrl-zoom-out .maplibregl-ctrl-icon{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' fill='%23fff'
viewBox='0 0 29 29'%3E%3Cpath d='M10 13c-.75 0-1.5.75-1.5 1.5S9.25 16 10
16h9c.75 0 1.5-.75 1.5-1.5S19.75 13 19 13h-9z'/%3E%3C/svg%3E")}).maplibregl-ctrl
button.maplibregl-ctrl-zoom-in .maplibregl-ctrl-icon{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' fill='%23fff'
viewBox='0 0 29 29'%3E%3Cpath d='M14.5 8.5c-.75 0-1.5.75-1.5 1.5v3h-3c-.75 0-
1.5.75-1.5 1.5S9.25 16 10 16h3v3c0 .75.75 1.5 1.5 1.5S16 19.75 16 19v-3h3c.75 0
1.5-.75 1.5-1.5S19.75 13 19 13h-3v-3c0-.75-.75-1.5-1.5-
1.5z'/%3E%3C/svg%3E")}}@media (-ms-high-contrast:black-on-white){.maplibregl-
ctrl button.maplibregl-ctrl-zoom-out .maplibregl-ctrl-icon{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' viewBox='0 0 29
29'%3E%3Cpath d='M10 13c-.75 0-1.5.75-1.5 1.5S9.25 16 10 16h9c.75 0 1.5-.75 1.5-
1.5S19.75 13 19 13h-9z'/%3E%3C/svg%3E")}).maplibregl-ctrl button.maplibregl-ctrl-
zoom-in .maplibregl-ctrl-icon{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' viewBox='0 0 29
29'%3E%3Cpath d='M14.5 8.5c-.75 0-1.5.75-1.5 1.5v3h-3c-.75 0-1.5.75-1.5 1.5S9.25
16 10 16h3v3c0 .75.75 1.5 1.5 1.5S16 19.75 16 19v-3h3c.75 0 1.5-.75 1.5-
1.5S19.75 13 19 13h-3v-3c0-.75-.75-1.5-1.5-1.5z'/%3E%3C/svg%3E")}}}.maplibregl-
ctrl button.maplibregl-ctrl-fullscreen .maplibregl-ctrl-icon{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' fill='%233333'
viewBox='0 0 29 29'%3E%3Cpath d='M24 16v5.5c0 1.75-.75 2.5-2.5 2.5H16v-11.3-1.5-

```

```

4-5.5 1-1 5.5 4 1.5-3h1zm6 16l1.5 3 5.5-4 1 1-4 5.5 3 1.5v1h7.5C5.75 24 5 23.25
5 21.5V16h1zm7-11v1l-3 1.5 4 5.5-1 1-5.5-4L6 13H5V7.5C5 5.75 5.75 5 7.5 5H13zm11
2.5c0-1.75-.75-2.5-2.5-2.5H16v1l3 1.5-4 5.5 1 1 5.5-4 1.5
3h1V7.5z'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-shrink
.maplibregl-ctrl-icon{background-image:url("data:image/svg+xml;charset=utf-
8,%3Csvg xmlns='http://www.w3.org/2000/svg' width='29' height='29' viewBox='0 0
29 29'%3E%3Cpath d='M18.5 16c-1.75 0-2.5.75-2.5 2.5V24h11.5-3 5.5 4 1-1-4-5.5
3-1.5v-1h-5.5zM13 18.5c0-1.75-.75-2.5-2.5-2.5H5v1l3 1.5L4 24l1 1 5.5-4 1.5 3h1v-
5.5zm3-8c0 1.75.75 2.5 2.5 2.5H24v-1l-3-1.5L25 5l-1-1-5.5 4L17 5h-1v5.5zM10.5
13c1.75 0 2.5-.75 2.5-2.5V5h-1l-1.5 3L5 4 4 5l4 5.5L5
12v1h5.5z'/%3E%3C/svg%3E"))}@media (-ms-high-contrast:active){.maplibregl-ctrl
button.maplibregl-ctrl-fullscreen .maplibregl-ctrl-icon{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' fill='%23fff'
viewBox='0 0 29 29'%3E%3Cpath d='M24 16v5.5c0 1.75-.75 2.5-2.5 2.5H16v-1l3-1.5-
4-5.5 1-1 5.5 4 1.5-3h1zm6 16l1.5 3 5.5-4 1 1-4 5.5 3 1.5v1h7.5C5.75 24 5 23.25
5 21.5V16h1zm7-11v1l-3 1.5 4 5.5-1 1-5.5-4L6 13H5V7.5C5 5.75 5.75 5 7.5 5H13zm11
2.5c0-1.75-.75-2.5-2.5-2.5H16v1l3 1.5-4 5.5 1 1 5.5-4 1.5
3h1V7.5z'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-shrink
.maplibregl-ctrl-icon{background-image:url("data:image/svg+xml;charset=utf-
8,%3Csvg xmlns='http://www.w3.org/2000/svg' width='29' height='29' fill='%23fff'
viewBox='0 0 29 29'%3E%3Cpath d='M18.5 16c-1.75 0-2.5.75-2.5 2.5V24h11.5-3 5.5
4 1-1-4-5.5 3-1.5v-1h-5.5zM13 18.5c0-1.75-.75-2.5-2.5-2.5H5v1l3 1.5L4 24l1 1
5.5-4 1.5 3h1v-5.5zm3-8c0 1.75.75 2.5 2.5 2.5H24v-1l-3-1.5L25 5l-1-1-5.5 4L17
5h-1v5.5zM10.5 13c1.75 0 2.5-.75 2.5-2.5V5h-1l-1.5 3L5 4 4 5l4 5.5L5
12v1h5.5z'/%3E%3C/svg%3E"))}@media (-ms-high-contrast:black-on-white)
{.maplibregl-ctrl button.maplibregl-ctrl-fullscreen .maplibregl-ctrl-
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1.5-3h1zm6 16l1.5 3 5.5-4 1 1-4 5.5 3 1.5v1h7.5C5.75 24 5 23.25 5 21.5V16h1zm7-
11v1l-3 1.5 4 5.5-1 1-5.5-4L6 13H5V7.5C5 5.75 5.75 5 7.5 5H13zm11 2.5c0-
1.75-.75-2.5-2.5-2.5H16v1l3 1.5-4 5.5 1 1 5.5-4 1.5
3h1V7.5z'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-shrink
.maplibregl-ctrl-icon{background-image:url("data:image/svg+xml;charset=utf-
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3-1.5v-1h-5.5zM13 18.5c0-1.75-.75-2.5-2.5-2.5H5v1l3 1.5L4 24l1 1 5.5-4 1.5 3h1v-
5.5zm3-8c0 1.75.75 2.5 2.5 2.5H24v-1l-3-1.5L25 5l-1-1-5.5 4L17 5h-1v5.5zM10.5
13c1.75 0 2.5-.75 2.5-2.5V5h-1l-1.5 3L5 4 4 5l4 5.5L5
12v1h5.5z'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-compass
.maplibregl-ctrl-icon{background-image:url("data:image/svg+xml;charset=utf-
8,%3Csvg xmlns='http://www.w3.org/2000/svg' width='29' height='29' fill='%23333'
viewBox='0 0 29 29'%3E%3Cpath d='m10.5 14 4-8 4 8h-8z'/%3E%3Cpath fill='%23ccc'
d='m10.5 16 4 8 4-8h-8z'/%3E%3C/svg%3E"))}@media (-ms-high-contrast:active)
{.maplibregl-ctrl button.maplibregl-ctrl-compass .maplibregl-ctrl-
icon{background-image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' fill='%23fff'
viewBox='0 0 29 29'%3E%3Cpath d='m10.5 14 4-8 4 8h-8z'/%3E%3Cpath fill='%23999'
d='m10.5 16 4 8 4-8h-8z'/%3E%3C/svg%3E"))}@media (-ms-high-contrast:black-on-
white){.maplibregl-ctrl button.maplibregl-ctrl-compass .maplibregl-ctrl-
icon{background-image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' viewBox='0 0 29
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4-8h-8z'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-terrain
.maplibregl-ctrl-icon{background-image:url("data:image/svg+xml;charset=utf-
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viewBox='0 0 22 22'%3E%3Cpath d='m1.754 13.406 4.453-4.851 3.09 3.09 3.281
3.277.969-.969-3.309-3.312 3.844-4.121 6.148 6.886h1.082v-.855l-7.207-8.07-4.84
5.187L6.169 6.571-5.48 5.965v.871zM.688 16.844h20.625v1.375H.688Zm0
0'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-terrain-enabled
.maplibregl-ctrl-icon{background-image:url("data:image/svg+xml;charset=utf-

```

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3.09 3.281 3.277.969-.969-3.309-3.312 3.844-4.121 6.148 6.886h1.082v-.855l-
7.207-8.07-4.84 5.187L6.169 6.571-5.48 5.965v.871zM.688
16.844h20.625v1.375H.688Zm0 0'/%3E%3C/svg%3E"))}.maplibregl-ctrl
button.maplibregl-ctrl-geolocate .maplibregl-ctrl-icon{background-
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1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-1zm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3Ccircle cx='10' cy='10' r='2'/%3E%3C/svg%3E"))}.maplibregl-ctrl
button.maplibregl-ctrl-geolocate:disabled .maplibregl-ctrl-icon{background-
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viewBox='0 0 20 20'%3E%3Cpath d='M10 4C9 4 9 5 9 5v.1A5 5 0 0 0 5.1 9H5s-1 0-1 1
1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-1zm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3Ccircle cx='10' cy='10' r='2'/%3E%3Cpath fill='red' d='m14 5 1 1-9 9-1-
1 9-9z'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-
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viewBox='0 0 20 20'%3E%3Cpath d='M10 4C9 4 9 5 9 5v.1A5 5 0 0 0 5.1 9H5s-1 0-1 1
1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-1zm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3Ccircle cx='10' cy='10' r='2'/%3E%3C/svg%3E"))}.maplibregl-ctrl
button.maplibregl-ctrl-geolocate.maplibregl-ctrl-geolocate-active-error
.maplibregl-ctrl-icon{background-image:url("data:image/svg+xml;charset=utf-
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3.9-3.9h.1s1 0 1-1-1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-1zm0 2.5a3.5 3.5 0 1 1 0
7 3.5 3.5 0 1 1 0-7z'/%3E%3Ccircle cx='10' cy='10'
r='2'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-
geolocate.maplibregl-ctrl-geolocate-background .maplibregl-ctrl-icon{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
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viewBox='0 0 20 20'%3E%3Cpath d='M10 4C9 4 9 5 9 5v.1A5 5 0 0 0 5.1 9H5s-1 0-1 1
1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-1zm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-
geolocate.maplibregl-ctrl-geolocate-background-error .maplibregl-ctrl-
icon{background-image:url("data:image/svg+xml;charset=utf-8,%3Csvg
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viewBox='0 0 20 20'%3E%3Cpath d='M10 4C9 4 9 5 9 5v.1A5 5 0 0 0 5.1 9H5s-1 0-1 1
1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-1zm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-
geolocate.maplibregl-ctrl-geolocate-waiting .maplibregl-ctrl-
icon{animation:maplibregl-spin 2s linear infinite}@media (-ms-high-
contrast:active){.maplibregl-ctrl button.maplibregl-ctrl-geolocate .maplibregl-
ctrl-icon{background-image:url("data:image/svg+xml;charset=utf-8,%3Csvg
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viewBox='0 0 20 20'%3E%3Cpath d='M10 4C9 4 9 5 9 5v.1A5 5 0 0 0 5.1 9H5s-1 0-1 1
1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-1zm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3Ccircle cx='10' cy='10' r='2'/%3E%3C/svg%3E"))}.maplibregl-ctrl
button.maplibregl-ctrl-geolocate:disabled .maplibregl-ctrl-icon{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' fill='%23999'
viewBox='0 0 20 20'%3E%3Cpath d='M10 4C9 4 9 5 9 5v.1A5 5 0 0 0 5.1 9H5s-1 0-1 1

```

```
1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-lzm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3Ccircle cx='10' cy='10' r='2'/%3E%3Cpath fill='red' d='m14 5 1 1-9 9-1-
1 9-9z'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-
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viewBox='0 0 20 20'/%3E%3Cpath d='M10 4C9 4 9 5 9 5v.1A5 5 0 0 0 5.1 9H5s-1 0-1 1
1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-lzm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3Ccircle cx='10' cy='10' r='2'/%3E%3C/svg%3E"))}.maplibregl-ctrl
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5.1 9H5s-1 0-1 1 1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0
3.9-3.9h.1s1 0 1-1-1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-lzm0 2.5a3.5 3.5 0 1 1 0
7 3.5 3.5 0 1 1 0-7z'/%3E%3Ccircle cx='10' cy='10'
r='2'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-
geolocate.maplibregl-ctrl-geolocate-background .maplibregl-ctrl-icon{background-
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viewBox='0 0 20 20'/%3E%3Cpath d='M10 4C9 4 9 5 9 5v.1A5 5 0 0 0 5.1 9H5s-1 0-1 1
1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-lzm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-
geolocate.maplibregl-ctrl-geolocate-background-error .maplibregl-ctrl-
icon{background-image:url("data:image/svg+xml;charset=utf-8,%3Csvg
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1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-lzm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3C/svg%3E"))}@media (-ms-high-contrast:black-on-white){.maplibregl-ctrl
button.maplibregl-ctrl-geolocate .maplibregl-ctrl-icon{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='29' height='29' viewBox='0 0 20
20'/%3E%3Cpath d='M10 4C9 4 9 5 9 5v.1A5 5 0 0 0 5.1 9H5s-1 0-1 1 1 1 1 1h.1A5 5
0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-1-1-1-1h-.1A5 5
0 0 0 11 5.1V5s0-1-1-lzm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-7z'/%3E%3Ccircle
cx='10' cy='10' r='2'/%3E%3C/svg%3E"))}.maplibregl-ctrl button.maplibregl-ctrl-
geolocate:disabled .maplibregl-ctrl-icon{background-
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1 1 1 1h.1A5 5 0 0 0 9 14.9v.1s0 1 1 1 1-1 1-1v-.1a5 5 0 0 0 3.9-3.9h.1s1 0 1-1-
1-1-1-1h-.1A5 5 0 0 0 11 5.1V5s0-1-1-lzm0 2.5a3.5 3.5 0 1 1 0 7 3.5 3.5 0 1 1 0-
7z'/%3E%3Ccircle cx='10' cy='10' r='2'/%3E%3Cpath fill='red' d='m14 5 1 1-9 9-1-
1 9-9z'/%3E%3C/svg%3E"))}}@keyframes maplibregl-spin{0%
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5.644-.757-1.71-.345 1.71zm1.958-3.42-.726 3.663a1.255 1.255 0 0 1-1.232 1.011h-
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1.001h.197a1.255 1.255 0 0 1 1.12.685l3.19 6.273 3.125-6.263a1.255 1.255 0 0 1
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.314-.787l.009-.016a4.623 4.623 0 0 1 1.777-1.887c.782-.46 1.668-.667
2.611-.667a4.548 4.548 0 0 1
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1.255 1.255 0 0 1 1.212-.925h1.711c.284 0 .545.094.755.252.613-.3 1.312-.45
```

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.37.043.532.118v-4.77c0-.356.147-.678.385-.906a2.416 2.416 0 0 1-.682-
1.71c0-.665.267-1.253.735-1.7a2.448 2.448 0 0 1 1.722-.674 2.43 2.43 0 0 1
1.705.675c.211.2.381.43.504.683V4.7a1.255 1.255 0 0 1 1.255-1.255h1.744A1.255
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0 .451.064.639.174a3.407 3.407 0 0 1 1.567-.372c.346 0 .861.02 1.285.232a1.255
1.255 0 0 1 .689 1.004 4.73 4.73 0 0 1 .853-.588c.795-.44 1.675-.647 2.61-.647
1.385 0 2.65.39 3.525 1.396.836.938 1.168 2.173 1.168 3.528 0 .343-.02.694-.056
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1.9631-.004-.008a5.815 5.815 0 0 1-.548-2.512c0-.286.017-.567.053-.843a1.255
1.255 0 0 1-.333-.0861-.166-.004c-.223
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1-.764.258h-1.71a1.255 1.255 0 0 1-1.256-1.255V7.687a2.402 2.402 0 0
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2.495-.6321-.017-.01a4.983 4.983 0 0 1-1.081-.836 1.255 1.255 0 0 1-1.254
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1.332h1.678V9.242h-1.6941-.033
1.267c-.088-.22-.264-.438-.526-.6581-.032-.028a3.16 3.16 0 0
0-.668-.4281-.27-.12a3.293 3.293 0 0 0-1.235-.23c-.757 0-1.415.163-
1.974.493a3.36 3.36 0 0 0-1.3 1.382c-.297.593-.444 1.284-.444 2.074 0 .8.17
1.503.51 2.107a3.795 3.795 0 0 0 1.382 1.381 3.883 3.883 0 0 0 1.893.477c.53 0
1.015-.11 1.455-.33zm-2.789-5.38c-.384.45-.575 1.038-.575 1.762 0 .735.186
1.332.559 1.794.384.45.933.675 1.645.675a2.25 2.25 0 0 0 .934-.19 2.17 2.17 0 0
0 .468-.291.178-.161a2.163 2.163 0 0 0 .397-.561c.163-.333.244-.717.244-
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2.706c.163-.17.28-.407.28-.83v-.114c0-.292-.06-.508-.15-.68a.958.958 0 0
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0-.56.114-.664.2391-.01.012c-.148.174-.275.45-.275.945 0
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1.705a2.626 2.626 0 0 0 .331.423c.213.22.464.402.755.5481.173.074c.433.17.93.255
1.49.255.68 0 1.295-.165 1.844-.493a3.447 3.447 0 0 0 1.316-1.4c.329-.603.493-
1.299.493-2.089 0-1.273-.33-2.243-.988-2.913-.658-.68-1.52-1.02-2.584-1.02-.598
0-1.124.115-1.575.347a2.807 2.807 0 0 0-.415.2621-.199.166a3.35 3.35 0 0
0-.64.82V9.242h-1.712v11.553h1.7291-.017-5.134zm.53-
1.138c.137.193.297.36.48.51.155.11.053.034c.34.197.713.297 1.119.297.714 0
1.262-.225 1.645-.675.385-.46.576-1.048.576-1.762 0-.746-.192-1.338-.576-
1.777-.372-.45-.92-.675-1.645-.675-.29 0-.569.053-.835.16a2.366 2.366 0 0
0-.284.136 1.99 1.99 0 0 0-.363.254 2.237 2.237 0 0 0-.46.5691-.082.162a2.56
2.56 0 0 0-.213 1.072v.115c0 .471.098.894.296 1.2671.135.211zm.964-.818a1.11
1.11 0 0 0 .367.385.937.937 0 0 0 .476.118c.423 0
.59-.117.687-.23.159-.194.28-.478.28-.95
0-.53-.133-.8-.266-.9521-.021-.025c-.078-.094-.231-.221-.68-.221a.995.995 0 0
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6.222c.232-.23.346-.516.346-.856a1.053 1.053 0 0 0-.345-.79 1.175 1.175 0 0
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9.314.033-1.332c.128.269.324.518.59.7471.098.081a3.727 3.727 0 0 0
.316.2241.223.122a3.21 3.21 0 0 0 1.44.322 3.785 3.785 0 0 0 1.875-.477 3.52
3.52 0 0 0 1.382-1.366c.352-.593.526-1.29.526-2.09 0-.79-.147-1.48-.444-
2.073a3.235 3.235 0 0 0-1.283-1.399c-.549-.34-1.195-.51-1.942-.51a3.476 3.476 0
0 0-1.527.3441-.086.043-.165.09a3.412 3.412 0 0
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1.262-.225 1.645-.675.384-.46.576-1.053.576-1.778 0-.734-.192-1.327-.576-
1.777-.373-.46-.921-.692-1.645-.692a2.18 2.18 0 0 0-
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1.49 0 0 1 .466-.636 2.52 2.52 0 0 1 .399-.253 2.19 2.19 0 0 1 .224-.099zm9.784
2.656.05-.922c0-1.162-.285-2.062-.856-2.698-.559-.647-1.42-.97-2.584-.97-.746 0-
1.415.163-2.007.493a3.462 3.462 0 0 0-1.4 1.382c-.329.604-.493 1.306-.493 2.106
0 .714.143 1.371.428 1.975.285.593.73 1.07 1.332 1.432.604.351 1.355.526
2.255.526.649 0 1.204-.062
1.668-.1851.044-.012.135-.04c.409-.122.736-.263.984-.4211-.542-
1.267c-.2.108-.415.199-.642.2741-.297.087c-.34.088-.773.131-1.3.131-.636 0-
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0-.156-.176c-.307-.285-.746-.428-1.316-.428-.657 0-1.155.202-
1.494.604-.253.3-.417.712-.494 1.234zm-27.053 2.77V4.7h-
1.86v12.095h5.333V15.15zm7.103-5.908v7.553h-1.843V9.242h1.843z'/%3E%3Cpath
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0-.79.147-1.482.444-2.074a3.364 3.364 0 0 1 1.3-1.382c.559-.33 1.217-.494
1.974-.494a3.293 3.293 0 0 1 1.234.231 3.341 3.341 0 0 1
.97.575c.264.22.44.439.527.6591.033-1.267h1.694v7.553H37.181-.033-
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0 1.152-.214 1.58-.642.428-.44.642-1.01.642-1.711v-.115c0-.472-.098-.894-.296-
1.267a2.211 2.211 0 0 0-.807-.872 2.098 2.098 0 0 0-1.119-.313c-.702 0-
1.245.231-1.629.692-.384.45-.575 1.037-.575 1.76 0 .736.186 1.333.559
1.795.384.45.933.675 1.645.675zm6.521-6.237h1.711v1.4c.604-1.065 1.547-1.597
2.83-1.597 1.064 0 1.926.34 2.584 1.02.659.67.988 1.641.988 2.914 0 .79-.164
1.487-.493 2.09a3.456 3.456 0 0 1-1.316 1.399 3.51 3.51 0 0 1-1.844.493c-.636 0-
1.19-.11-1.662-.329a2.665 2.665 0 0 1-1.086-.971.017 5.134h-1.728V9.242zm4.048
6.22c.714 0 1.262-.224 1.645-.674.385-.46.576-1.048.576-1.762 0-.746-.192-
1.338-.576-1.777-.372-.45-.92-.675-1.645-.675-.395 0-.768.098-
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1.165 1.165 0 0 1-.346-.856 1.053 1.053 0 0 1
.346-.79c.23-.219.516-.329.856-.329.329 0 .609.11.839.33a1.053 1.053 0 0 1
.345.79 1.159 1.159 0 0 1-.345.855c-.22.23-.5.346-.84.346zm7.875 9.133a3.167
3.167 0 0 1-1.662-.444c-.482-.307-.817-.658-1.004-1.0531-.033 1.332h-
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0 1 1.778-.477c.746 0 1.393.17 1.942.51a3.235 3.235 0 0 1 1.283 1.4c.297.592.444
1.282.444 2.072 0 .8-.175 1.498-.526 2.09a3.52 3.52 0 0 1-1.382 1.366 3.785
3.785 0 0 1-1.876.477zm-.296-1.481c.713 0 1.26-.225 1.645-.675.384-.46.577-

1.053.577-1.778 0-.734-.193-1.327-.577-1.776-.373-.46-.921-.692-1.645-.692a2.115
2.115 0 0 0-1.58.659c-.428.428-.642.992-.642 1.694v.115c0 .473.098.895.296
1.267a2.385 2.385 0 0 0 .807.872 2.1 2.1 0 0 0 1.119.313zm5.927-
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1.651-.175-2.255-.526-.603-.362-1.047-.84-1.332-1.432a4.567 4.567 0 0 1-.428-
1.975c0-.8.164-1.502.493-2.106a3.462 3.462 0 0 1 1.4-1.382c.592-.33 1.262-.494
2.007-.494 1.163 0 2.024.324 2.584.97.57.637.856 1.537.856 2.7 0
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1.497.443.526 0 .96-.044 1.3-.131a4.123 4.123 0 0 0 .938-.362l.542
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6.192-5.88 3.338 0 6.193 2.561 6.193 5.881 0 2.34-1.22 4.387-2.376 5.822a16.898
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1.185-1.222c.638 0 1.186.529 1.186 1.222zm-1.186 2.477c1.348 0 2.442-1.11 2.442-
2.478S8.507 5.28 7.159 5.28 4.72 6.39 4.72 7.758s1.092 2.477 2.44 2.477zm2.048
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2.44 2.477z'/%3E%3Cpath fill='%23fff' d='M4.377 15.948a.657.657 0 0
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logo{background-color:transparent;background-
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xmlns='http://www.w3.org/2000/svg' width='88' height='23' fill='none'%3E%3Cpath
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12.095h-1.81l-1.218-5.644-.362-1.71-.658 1.71-2.929 5.644h-.098l-2.914-
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1.255 1.255 0 0 1 1.212-.925h1.711c.284 0 .545.094.755.252.613-.3 1.312-.45
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1.71c0-.665.267-1.253.735-1.7a2.448 2.448 0 0 1 1.722-.674 2.43 2.43 0 0 1
1.705.675c.211.2.381.43.504.683V4.7a1.255 1.255 0 0 1 1.255-1.255h1.744A1.255
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0 .451.064.639.174a3.407 3.407 0 0 1 1.567-.372c.346 0 .861.02 1.285.232a1.255
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0-.426.062-.643.228-.03.024-.142.139-.142.59v3.883a1.255 1.255 0 0 1-1.256
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1.86 1.833 5.04 5.04 0 0 1-2.484.634 4.47 4.47 0 0 1-1.935-.424 1.252 1.252 0 0
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1.332h1.678v9.242h-1.6941-.033
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0-.668-.4281-.27-.12a3.293 3.293 0 0 0-1.235-.23c-.757 0-1.415.163-
1.974.493a3.36 3.36 0 0 0-1.3 1.382c-.297.593-.444 1.284-.444 2.074 0 .8.17
1.503.51 2.107a3.795 3.795 0 0 0 1.382 1.381 3.883 3.883 0 0 0 1.893.477c.53 0
1.015-.11 1.455-.33zm-2.789-5.38c-.384.45-.575 1.038-.575 1.762 0 .735.186
1.332.559 1.794.384.45.933.675 1.645.675a2.25 2.25 0 0 0 .934-.19 2.17 2.17 0 0
0 .468-.291.178-.161a2.163 2.163 0 0 0 .397-.561c.163-.333.244-.717.244-
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0-.633-.7091-.13-.086-.047-.028a2.099 2.099 0 0 0-1.073-.285c-.702 0-1.244.231-
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2.706c.163-.17.28-.407.28-.83v-.114c0-.292-.06-.508-.15-.68a.958.958 0 0
0-.353-.389.851.851 0 0 0-.464-.127c-.4
0-.56.114-.664.2391-.01.012c-.148.174-.275.45-.275.945 0
.506.122.801.27.99.097.11.266.224.68.224.303 0 .504-.09.687-.269zm7.545
1.705a2.626 2.626 0 0 0 .331.423c.213.22.464.402.755.5481.173.074c.433.17.93.255
1.49.255.68 0 1.295-.165 1.844-.493a3.447 3.447 0 0 0 1.316-1.4c.329-.603.493-
1.299.493-2.089 0-1.273-.33-2.243-.988-2.913-.658-.68-1.52-1.02-2.584-1.02-.598
0-1.124.115-1.575.347a2.807 2.807 0 0 0-.415.2621-.199.166a3.35 3.35 0 0
0-.64.82v9.242h-1.712v11.553h1.7291-.017-5.134zm.53-
1.138c.137.193.297.36.48.51.155.11.053.034c.34.197.713.297 1.119.297.714 0
1.262-.225 1.645-.675.385-.46.576-1.048.576-1.762 0-.746-.192-1.338-.576-
1.777-.372-.45-.92-.675-1.645-.675-.29 0-.569.053-.835.16a2.366 2.366 0 0
0-.284.136 1.99 1.99 0 0 0-.363.254 2.237 2.237 0 0 0-.46.5691-.082.162a2.56
2.56 0 0 0-.213 1.072v.115c0 .471.098.894.296 1.2671.135.211zm.964-.818a1.11
1.11 0 0 0 .367.385.937.937 0 0 0 .476.118c.423 0
.59-.117.687-.23.159-.194.28-.478.28-.95
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0-.335.343c-.073.133-.132.324-.132.614v.115a1.43 1.43 0 0 0 .14.66zm15.7-
6.222c.232-.23.346-.516.346-.856a1.053 1.053 0 0 0-.345-.79 1.175 1.175 0 0
0-.84-.329c-.34 0-.625.11-.855.33a1.053 1.053 0 0 0-.346.79c0
.34.115.625.346.855.23.23.516.346.856.346.34 0 .62-.115.839-.346zm4.337
9.314.033-1.332c.128.269.324.518.59.7471.098.081a3.727 3.727 0 0 0
.316.2241.223.122a3.21 3.21 0 0 0 1.44.322 3.785 3.785 0 0 0 1.875-.477 3.52
3.52 0 0 0 1.382-1.366c.352-.593.526-1.29.526-2.09 0-.79-.147-1.48-.444-
2.073a3.235 3.235 0 0 0-1.283-1.399c-.549-.34-1.195-.51-1.942-.51a3.476 3.476 0
0 0-1.527.3441-.086.043-.165.09a3.412 3.412 0 0
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1.262-.225 1.645-.675.384-.46.576-1.053.576-1.778 0-.734-.192-1.327-.576-
1.777-.373-.46-.921-.692-1.645-.692a2.18 2.18 0 0 0-
1.015.235c-.147.075-.285.17-.415.282l-.15.142a2.086 2.086 0 0
0-.42.594c-.149.32-.223.685-.223 1.1v.115c0 .47.097.89.293
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0-.411.148 2.18 2.18 0 0 0-.4.249 2.482 2.482 0 0 0-.485.499 2.659 2.659 0 0
0-.32.581l-.05.137v-1.48h-1.778v7.553h1.777v-3.884c0-.364.053-.678.159-.943a1.49
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2.656.05-.922c0-1.162-.285-2.062-.856-2.698-.559-.647-1.42-.97-2.584-.97-.746 0-
1.415.163-2.007.493a3.462 3.462 0 0 0-1.4 1.382c-.329.604-.493 1.306-.493 2.106
0 .714.143 1.371.428 1.975.285.593.73 1.07 1.332 1.432.604.351 1.355.526
2.255.526.649 0 1.204-.062
1.668-.185l.044-.012.135-.04c.409-.122.736-.263.984-.421l-.542-
1.267c-.2.108-.415.199-.642.274l-.297.087c-.34.088-.773.131-1.3.131-.636 0-
1.135-.147-1.497-.444a1.573 1.573 0 0 1-.192-.193c-.244-.294-.415-.705-.512-
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1.86v12.095h5.333v15.15zm7.103-5.908v7.553h-1.843v9.242h1.843z'/%3E%3Cpath
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5.644-.362-1.71-.658 1.71-2.93 5.644h-.098l-2.913-5.644zm14.836 5.81c-.68 0-
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0-.79.147-1.482.444-2.074a3.364 3.364 0 0 1 1.3-1.382c.559-.33 1.217-.494
1.974-.494a3.293 3.293 0 0 1 1.234.231 3.341 3.341 0 0 1
.97.575c.264.22.44.439.527.659l.033-1.267h1.694v7.553H37.181-.033-
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0 1.152-.214 1.58-.642.428-.44.642-1.01.642-1.711v-.115c0-.472-.098-.894-.296-
1.267a2.211 2.211 0 0 0-.807-.872 2.098 2.098 0 0 0-1.119-.313c-.702 0-
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1.795.384.45.933.675 1.645.675zm6.521-6.237h1.711v1.4c.604-1.065 1.547-1.597
2.83-1.597 1.064 0 1.926.34 2.584 1.02.659.67.988 1.641.988 2.914 0 .79-.164
1.487-.493 2.09a3.456 3.456 0 0 1-1.316 1.399 3.51 3.51 0 0 1-1.844.493c-.636 0-
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6.22c.714 0 1.262-.224 1.645-.674.385-.46.576-1.048.576-1.762 0-.746-.192-
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3.167 0 0 1-1.662-.444c-.482-.307-.817-.658-1.004-1.053l-.033 1.332h-
1.71v4.701h1.743v4.657l-.082 1.283c.186-.438.548-.812 1.086-1.119a3.486 3.486 0
0 1 1.778-.477c.746 0 1.393.17 1.942.51a3.235 3.235 0 0 1 1.283 1.4c.297.592.444
1.282.444 2.072 0 .8-.175 1.498-.526 2.09a3.52 3.52 0 0 1-1.382 1.366 3.785
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1.053.577-1.778 0-.734-.193-1.327-.577-1.776-.373-.46-.921-.692-1.645-.692a2.115
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2.007-.494 1.163 0 2.024.324 2.584.97.57.637.856 1.537.856 2.7 0
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1.497.443.526 0 .96-.044 1.3-.131a4.123 4.123 0 0 0 .938-.3621.542
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1.185-1.222c.638 0 1.186.529 1.186 1.222zm-1.186 2.477c1.348 0 2.442-1.11 2.442-
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2.44 2.477z'/%3E%3Cpath fill='%23fff' d='M4.377 15.948a.657.657 0 0
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1.262-.225 1.645-.675.385-.46.576-1.048.576-1.762 0-.746-.192-1.338-.576-
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0-.335.343c-.073.133-.132.324-.132.614v.115a1.43 1.43 0 0 0 .14.66zm15.7-
6.222c.232-.23.346-.516.346-.856a1.053 1.053 0 0 0-.345-.79 1.175 1.175 0 0
0-.84-.329c-.34 0-.625.11-.855.33a1.053 1.053 0 0 0-.346.79c0
.34.115.625.346.855.23.23.516.346.856.346.34 0 .62-.115.839-.346zm4.337
9.314.033-1.332c.128.269.324.518.59.7471.098.081a3.727 3.727 0 0 0
.316.2241.223.122a3.21 3.21 0 0 0 1.44.322 3.785 3.785 0 0 0 1.875-.477 3.52
3.52 0 0 0 1.382-1.366c.352-.593.526-1.29.526-2.09 0-.79-.147-1.48-.444-
2.073a3.235 3.235 0 0 0-1.283-1.399c-.549-.34-1.195-.51-1.942-.51a3.476 3.476 0
0 0-1.527.3441-.086.043-.165.09a3.412 3.412 0 0
0-.33.214c-.288.21-.507.446-.656.707a1.893 1.893 0 0 0-.099.1981.082-
1.283V4.701h-1.744v12.095zm.473-2.509a2.482 2.482 0 0 0
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1.262-.225 1.645-.675.384-.46.576-1.053.576-1.778 0-.734-.192-1.327-.576-
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0-.411.148 2.18 2.18 0 0 0-.4.249 2.482 2.482 0 0 0-.485.499 2.659 2.659 0 0
0-.32.5811-.05.137v-1.48h-1.778v7.553h1.777v-3.884c0-.364.053-.678.159-.943a1.49
1.49 0 0 1 .466-.636 2.52 2.52 0 0 1 .399-.253 2.19 2.19 0 0 1 .224-.099zm9.784
2.656.05-.922c0-1.162-.285-2.062-.856-2.698-.559-.647-1.42-.97-2.584-.97-.746 0-
1.415.163-2.007.493a3.462 3.462 0 0 0-1.4 1.382c-.329.604-.493 1.306-.493 2.106
0 .714.143 1.371.428 1.975.285.593.73 1.07 1.332 1.432.604.351 1.355.526
2.255.526.649 0 1.204-.062
1.668-.1851.044-.012.135-.04c.409-.122.736-.263.984-.4211-.542-
1.267c-.2.108-.415.199-.642.2741-.297.087c-.34.088-.773.131-1.3.131-.636 0-
1.135-.147-1.497-.444a1.573 1.573 0 0 1-.192-.193c-.244-.294-.415-.705-.512-
1.2341-.004-.021h5.43zm-5.427-
1.256-.003.022h3.752v-.138c-.007-.485-.104-.857-.288-1.118a1.056 1.056 0 0
0-.156-.176c-.307-.285-.746-.428-1.316-.428-.657 0-1.155.202-
1.494.604-.253.3-.417.712-.494 1.234zm-27.053 2.77V4.7h-
1.86v12.095h5.333V15.15zm7.103-5.908v7.553h-1.843V9.242h1.843z'/%3E%3Cpath
fill='%23fff' d='m19.63 11.151-.757-1.71-.345 1.71-1.12 5.644h-1.827L18.083
4.7h.19713.325 6.533.988 2.19.988-2.19L26.839 4.7h.18112.6 12.095h-1.811-1.218-
5.644-.362-1.71-.658 1.71-2.93 5.644h-.0981-2.913-5.644zm14.836 5.81c-.68 0-
1.311-.16-1.893-.478a3.795 3.795 0 0 1-1.381-1.382c-.34-.604-.51-1.306-.51-2.106
0-.79.147-1.482.444-2.074a3.364 3.364 0 0 1 1.3-1.382c.559-.33 1.217-.494
1.974-.494a3.293 3.293 0 0 1 1.234.231 3.341 3.341 0 0 1
.97.575c.264.22.44.439.527.6591.033-1.267h1.694v7.553H37.181-.033-
1.332c-.186.395-.526.746-1.02 1.053a3.167 3.167 0 0 1-1.662.444zm.296-1.482c.626
0 1.152-.214 1.58-.642.428-.44.642-1.01.642-1.711v-.115c0-.472-.098-.894-.296-
1.267a2.211 2.211 0 0 0-.807-.872 2.098 2.098 0 0 0-1.119-.313c-.702 0-
1.245.231-1.629.692-.384.45-.575 1.037-.575 1.76 0 .736.186 1.333.559
1.795.384.45.933.675 1.645.675zm6.521-6.237h1.711v1.4c.604-1.065 1.547-1.597
2.83-1.597 1.064 0 1.926.34 2.584 1.02.659.67.988 1.641.988 2.914 0 .79-.164
1.487-.493 2.09a3.456 3.456 0 0 1-1.316 1.399 3.51 3.51 0 0 1-1.844.493c-.636 0-
1.19-.11-1.662-.329a2.665 2.665 0 0 1-1.086-.971.017 5.134h-1.728V9.242zm4.048
6.22c.714 0 1.262-.224 1.645-.674.385-.46.576-1.048.576-1.762 0-.746-.192-
1.338-.576-1.777-.372-.45-.92-.675-1.645-.675-.395 0-.768.098-
1.12.296-.34.187-.613.46-.822.823-.197.351-.296.763-.296 1.234v.115c0
.472.098.894.296 1.267.209.362.483.647.823.855.34.197.713.297
1.119.297z'/%3E%3Cpath fill='%23e1e3e9' d='M51.325 4.7h1.86v10.45h3.473v1.646h-
5.333zm7.12 4.542h1.843v7.553h-1.843zm.905-1.415a1.159 1.159 0 0 1-.856-.346
1.165 1.165 0 0 1-.346-.856 1.053 1.053 0 0 1
.346-.79c.23-.219.516-.329.856-.329.329 0 .609.11.839.33a1.053 1.053 0 0 1
.345.79 1.159 1.159 0 0 1-.345.855c-.22.23-.5.346-.84.346zm7.875 9.133a3.167
3.167 0 0 1-1.662-.444c-.482-.307-.817-.658-1.004-1.0531-.033 1.332h-
1.71V4.701h1.743v4.6571-.082 1.283c.186-.438.548-.812 1.086-1.119a3.486 3.486 0
0 1 1.778-.477c.746 0 1.393.17 1.942.51a3.235 3.235 0 0 1 1.283 1.4c.297.592.444
1.282.444 2.072 0 .8-.175 1.498-.526 2.09a3.52 3.52 0 0 1-1.382 1.366 3.785
3.785 0 0 1-1.876.477zm-.296-1.481c.713 0 1.26-.225 1.645-.675.384-.46.577-
1.053.577-1.778 0-.734-.193-1.327-.577-1.776-.373-.46-.921-.692-1.645-.692a2.115
2.115 0 0 0-1.58.659c-.428.428-.642.992-.642 1.694v.115c0 .473.098.895.296
1.267a2.385 2.385 0 0 0 .807.872 2.1 2.1 0 0 0 1.119.313zm5.927-
6.237h1.777v1.481c.176-.505.46-.91.856-1.217a2.14 2.14 0 0 1 1.349-.46c.351 0
.593.032.724.0981-.247 1.794c-.099-.066-.313-.099-.642-.099-.516 0-.988.164-
1.416.494-.417.329-.626.855-.626 1.58v3.883h-1.777V9.242zm9.534 7.718c-.9 0-
1.651-.175-2.255-.526-.603-.362-1.047-.84-1.332-1.432a4.567 4.567 0 0 1-.428-
1.975c0-.8.164-1.502.493-2.106a3.462 3.462 0 0 1 1.4-1.382c.592-.33 1.262-.494
2.007-.494 1.163 0 2.024.324 2.584.97.57.637.856 1.537.856 2.7 0
.296-.017.603-.05.92h-5.43c.12.67.356 1.153.708 1.45.362.296.86.443
1.497.443.526 0 .96-.044 1.3-.131a4.123 4.123 0 0 0 .938-.3621.542
1.267c-.274.175-.647.329-1.119.46-.472.132-1.042.197-1.711.197zm1.596-
4.558c.01-.68-.137-1.158-.444-1.432-.307-.285-.746-.428-1.316-.428-1.152 0-
1.815.62-1.991 1.86h3.752z'/%3E%3Cg fill-rule='evenodd' stroke-
width='1.036'%3E%3Cpath fill='%23000' fill-opacity='.4' d='m8.166
16.146-.002.002a1.54 1.54 0 0 1-2.009
01-.002-.002-.043-.034-.002-.002-.199-.162H4.377a.657.657 0 0

```

0-.659.659v1.84a.657.657 0 0 0 .659.659h5.565a.657.657 0 0 0 .659-.659v-
1.84a.657.657 0 0 0-.659-.659H8.411l-.202.164zm-1.121-.905a.29.29 0 0 0
.113.023.286.286 0 0 0 .189-.071.077-.063c.634-.508 4.672-3.743 4.672-7.575 0-
2.55-2.215-4.625-4.938-4.625S2.221 5.006 2.221 7.556c0 3.225 2.86 6.027 4.144
7.137h.004l.04.038.484.4.077.063a.628.628 0 0 0 .074.047zm-2.52-.548a16.898
16.898 0 0 1-1.183-1.315C2.187 11.942.967 9.897.967 7.555c0-3.319 2.855-5.88
6.192-5.88 3.338 0 6.193 2.561 6.193 5.881 0 2.34-1.22 4.387-2.376 5.822a16.898
16.898 0 0 1-1.182 1.315h.15a1.912 1.912 0 0 1 1.914 1.914v1.84a1.912 1.912 0 0
1-1.914 1.914H4.377a1.912 1.912 0 0 1-1.914-1.914v-1.84a1.912 1.912 0 0 1 1.914-
1.914zm3.82-6.935c0 .692-.55 1.222-1.187 1.222s-1.185-.529-1.185-1.222.548-1.222
1.185-1.222c.638 0 1.186.529 1.186 1.222zm-1.186 2.477c1.348 0 2.442-1.11 2.442-
2.478S8.507 5.28 7.159 5.28 4.72 6.39 4.72 7.758s1.092 2.477 2.44 2.477zm2.048
7.71H5.114v-.838h4.093z'/%3E%3Cpath fill='%23e1e3e9' d='M2.222 7.555c0-2.55
2.214-4.625 4.937-4.625 2.723 0 4.938 2.075 4.938 4.625 0 3.832-4.038 7.068-
4.672 7.575l-.077.063a.286.286 0 0 1-.189.07.286.286 0 0
1-.188-.071l-.077-.063c-.634-.507-4.672-3.743-4.672-7.575zm4.937 2.68c1.348 0
2.442-1.11 2.442-2.478S8.507 5.28 7.159 5.28 4.72 6.39 4.72 7.758s1.092 2.477
2.44 2.477z'/%3E%3Cpath fill='%23fff' d='M4.377 15.948a.657.657 0 0
0-.659.659v1.84a.657.657 0 0 0 .659.659h5.565a.657.657 0 0 0 .659-.659v-
1.84a.657.657 0 0 0-.659-.659zm4.83
1.16H5.114v.838h4.093z'/%3E%3C/g%3E%3C/svg%3E"))}.maplibregl-ctrl.maplibregl-
ctrl-attrib{background-color:hsla(0,0%,100%,.5);margin:0;padding:0 5px}@media
screen{.maplibregl-ctrl-attrib.maplibregl-compact{background-color:#fff;border-
radius:12px;box-sizing:content-box;margin:10px;min-height:20px;padding:2px 24px
2px 0;position:relative}.maplibregl-ctrl-attrib.maplibregl-compact-
show{padding:2px 28px 2px 8px;visibility:visible}.maplibregl-ctrl-bottom-
left>.maplibregl-ctrl-attrib.maplibregl-compact-show,.maplibregl-ctrl-top-
left>.maplibregl-ctrl-attrib.maplibregl-compact-show{border-
radius:12px;padding:2px 8px 2px 28px}.maplibregl-ctrl-attrib.maplibregl-compact
.maplibregl-ctrl-attrib-inner{display:none}.maplibregl-ctrl-attrib-
button{background-color:hsla(0,0%,100%,.5);background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='24' height='24' fill-rule='evenodd'
viewBox='0 0 20 20'%3E%3Cpath d='M4 10a6 6 0 1 0 12 0 6 6 0 1 0-12 0m5-3a1 1 0 1
0 2 0 1 1 0 1 0-2 0m0 3a1 1 0 1 1 2 0v3a1 1 0 1 1-2
0'/%3E%3C/svg%3E"));border:0;border-radius:12px;box-sizing:border-
box;cursor:pointer;display:none;height:24px;outline:none;position:absolute;right
:0;top:0;width:24px}.maplibregl-ctrl-attrib summary.maplibregl-ctrl-attrib-
button{appearance:none;list-style:none}.maplibregl-ctrl-attrib
summary.maplibregl-ctrl-attrib-button::-webkit-details-
marker{display:none}.maplibregl-ctrl-bottom-left .maplibregl-ctrl-attrib-
button,.maplibregl-ctrl-top-left .maplibregl-ctrl-attrib-
button{left:0}.maplibregl-ctrl-attrib.maplibregl-compact .maplibregl-ctrl-
attrib-button,.maplibregl-ctrl-attrib.maplibregl-compact-show .maplibregl-ctrl-
attrib-inner{display:block}.maplibregl-ctrl-attrib.maplibregl-compact-show
.maplibregl-ctrl-attrib-button{background-color:rgb(0 0 0/5%)}.maplibregl-ctrl-
bottom-right>.maplibregl-ctrl-attrib.maplibregl-
compact:after{bottom:0;right:0}.maplibregl-ctrl-top-right>.maplibregl-ctrl-
attrib.maplibregl-compact:after{right:0;top:0}.maplibregl-ctrl-top-
left>.maplibregl-ctrl-attrib.maplibregl-compact:after{left:0;top:0}.maplibregl-
ctrl-bottom-left>.maplibregl-ctrl-attrib.maplibregl-
compact:after{bottom:0;left:0}}@media screen and (-ms-high-contrast:active)
{.maplibregl-ctrl-attrib.maplibregl-compact:after{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='24' height='24' fill='%23fff' fill-
rule='evenodd' viewBox='0 0 20 20'%3E%3Cpath d='M4 10a6 6 0 1 0 12 0 6 6 0 1 0-
12 0m5-3a1 1 0 1 0 2 0 1 1 0 1 0-2 0m0 3a1 1 0 1 1 2 0v3a1 1 0 1 1-2
0'/%3E%3C/svg%3E"))}@media screen and (-ms-high-contrast:black-on-white)
{.maplibregl-ctrl-attrib.maplibregl-compact:after{background-
image:url("data:image/svg+xml;charset=utf-8,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='24' height='24' fill-rule='evenodd'
viewBox='0 0 20 20'%3E%3Cpath d='M4 10a6 6 0 1 0 12 0 6 6 0 1 0-12 0m5-3a1 1 0 1

```



```

0 2 0 1 1 0 1 0-2 0m0 3a1 1 0 1 1 2 0v3a1 1 0 1 1-2
0'/%3E%3C/svg%3E"))}.maplibregl-ctrl-attrb a{color:rgba(0,0,0,.75);text-
decoration:none}.maplibregl-ctrl-attrb a:hover{color:inherit;text-
decoration:underline}.maplibregl-attrb-empty{display:none}.maplibregl-ctrl-
scale{background-color:hsla(0,0%,100%,.75);border:2px solid #333;border-
top:#333;box-sizing:border-box;color:#333;font-size:10px;padding:0
5px}.maplibregl-popup{display:flex;left:0;pointer-
events:none;position:absolute;top:0;will-change:transform}.maplibregl-popup-
anchor-top,.maplibregl-popup-anchor-top-left,.maplibregl-popup-anchor-top-
right{flex-direction:column}.maplibregl-popup-anchor-bottom,.maplibregl-popup-
anchor-bottom-left,.maplibregl-popup-anchor-bottom-right{flex-direction:column-
reverse}.maplibregl-popup-anchor-left{flex-direction:row}.maplibregl-popup-
anchor-right{flex-direction:row-reverse}.maplibregl-popup-tip{border:10px solid
transparent;height:0;width:0;z-index:1}.maplibregl-popup-anchor-top .maplibregl-
popup-tip{align-self:center;border-bottom-color:#fff;border-
top:none}.maplibregl-popup-anchor-top-left .maplibregl-popup-tip{align-
self:flex-start;border-bottom-color:#fff;border-left:none;border-
top:none}.maplibregl-popup-anchor-top-right .maplibregl-popup-tip{align-
self:flex-end;border-bottom-color:#fff;border-right:none;border-
top:none}.maplibregl-popup-anchor-bottom .maplibregl-popup-tip{align-
self:center;border-bottom:none;border-top-color:#fff}.maplibregl-popup-anchor-
bottom-left .maplibregl-popup-tip{align-self:flex-start;border-
bottom:none;border-left:none;border-top-color:#fff}.maplibregl-popup-anchor-
bottom-right .maplibregl-popup-tip{align-self:flex-end;border-
bottom:none;border-right:none;border-top-color:#fff}.maplibregl-popup-anchor-
left .maplibregl-popup-tip{align-self:center;border-left:none;border-right-
color:#fff}.maplibregl-popup-anchor-right .maplibregl-popup-tip{align-
self:center;border-left-color:#fff;border-right:none}.maplibregl-popup-close-
button{background-color:transparent;border:0;border-radius:0 3px 0
0;cursor:pointer;position:absolute;right:0;top:0}.maplibregl-popup-close-
button:hover{background-color:rgb(0 0 0/5%)}.maplibregl-popup-
content{background:#fff;border-radius:3px;box-shadow:0 1px 2px
rgba(0,0,0,.1);padding:15px 10px;pointer-
events:auto;position:relative}.maplibregl-popup-anchor-top-left .maplibregl-
popup-content{border-top-left-radius:0}.maplibregl-popup-anchor-top-right
.maplibregl-popup-content{border-top-right-radius:0}.maplibregl-popup-anchor-
bottom-left .maplibregl-popup-content{border-bottom-left-radius:0}.maplibregl-
popup-anchor-bottom-right .maplibregl-popup-content{border-bottom-right-
radius:0}.maplibregl-popup-track-pointer{display:none}.maplibregl-popup-track-
pointer *{pointer-events:none;user-select:none}.maplibregl-map:hover
.maplibregl-popup-track-pointer{display:flex}.maplibregl-map:active .maplibregl-
popup-track-pointer{display:none}.maplibregl-
marker{left:0;position:absolute;top:0;will-change:transform}.maplibregl-user-
location-dot,.maplibregl-user-location-dot:before{background-
color:#1da1f2;border-radius:50%;height:15px;width:15px}.maplibregl-user-
location-dot:before{animation:maplibregl-user-location-dot-pulse 2s
infinite;content:"";position:absolute}.maplibregl-user-location-
dot:after{border:2px solid #fff;border-radius:50%;box-shadow:0 0 3px
rgba(0,0,0,.35);box-sizing:border-
box;content:"";height:19px;left:-2px;position:absolute;top:-2px;width:19px}@keyf
rames maplibregl-user-location-dot-pulse{0%{opacity:1;transform:scale(1)}70%
{opacity:0;transform:scale(3)}to{opacity:0;transform:scale(1)}}.maplibregl-user-
location-dot-stale{background-color:#aaa}.maplibregl-user-location-dot-
stale:after{display:none}.maplibregl-user-location-accuracy-circle{background-
color:#1da1f233;border-radius:100%;height:1px;width:1px}.maplibregl-
crosshair,.maplibregl-crosshair .maplibregl-interactive,.maplibregl-crosshair
.maplibregl-interactive:active{cursor:crosshair}.maplibregl-
boxzoom{background:#fff;border:2px dotted
#202020;height:0;left:0;opacity:.5;position:absolute;top:0;width:0}.maplibregl-
cooperative-gesture-screen{align-
items:center;background:rgba(0,0,0,.4);color:#fff;display:flex;font-
size:1.4em;inset:0;justify-content:center;line-

```

```
height:1.2;opacity:0;padding:1rem;pointer-
events:none;position:absolute;transition:opacity 1s ease 1s;z-
index:99999}.maplibregl-cooperative-gesture-screen.maplibregl-
show{opacity:1;transition:opacity .05s}.maplibregl-cooperative-gesture-screen
.maplibregl-mobile-message{display:none}@media (hover:none),(width <= 480px)
{.maplibregl-cooperative-gesture-screen .maplibregl-desktop-
message{display:none}.maplibregl-cooperative-gesture-screen .maplibregl-mobile-
message{display:block}}.maplibregl-pseudo-
fullscreen{height:100%!important;left:0!important;position:fixed!important;top:0
!important;width:100%!important;z-index:99999}
```

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FILE: <https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/ventusv8.css>

14877 chars · code · shown in full · found via <link stylesheet>, loaded by the browser (link)

=====

```
html, body { margin: 0; padding: 0; width: 100%; height: 100%; background: #000;
font-family: 'Courier New', monospace; color: white; overflow: hidden; }
.dashboard { display: flex; flex-direction: column; height: 100vh; height:
100dvh; width: 100vw; padding: 4px; gap: 4px; box-sizing: border-box; }
#fatal-banner { display: none; position: fixed; top: 0; left: 0; width: 100%;
background: #ff0000; color: #fff; text-align: center; padding: 10px; font-
weight: bold; z-index: 9999; text-transform: uppercase; font-size: 12px; letter-
spacing: 1px; }
.hud-header { background: #0a0a0a; border: 1px solid #333; border-radius: 6px;
padding: 6px 12px; display: flex; justify-content: space-between; align-items:
center; flex-shrink: 0; }
.hud-val { font-size: 16px; font-weight: bold; color: #00ffff; text-shadow: 0 0
5px #00ffff; }
.ventus-brand { text-align: center; display: flex; flex-direction: column;
align-items: center; justify-content: center; line-height: 1.2; }
.ventus-main { font-family: -apple-system, sans-serif; font-size: 17px; font-
weight: 800; color: #fff; letter-spacing: 5px; text-transform: uppercase;
margin-bottom: 2px; }
.ventus-sub { font-family: -apple-system, sans-serif; font-size: 6.5px; color:
#888; letter-spacing: 2px; text-transform: uppercase; }
.map-container { position: relative; flex-grow: 1; min-height: 0; border: 1px
solid #222; border-radius: 6px; overflow: hidden; background: #0b0e14; }
#map { width: 100%; height: 100%; }
.podcast-shoutout { position: absolute; bottom: 6px; right: 8px; font-family: -
apple-system, sans-serif; font-size: 8px; color: #fff; opacity: 0.5; text-align:
right; text-transform: uppercase; letter-spacing: 1px; pointer-events: none; z-
index: 10; text-shadow: 1px 1px 2px #000; }
.scada-wrapper { background: #050505; border: 1px solid #444; border-radius:
6px; padding: 12px; display: flex; flex-direction: column; flex-shrink: 0; max-
height: 38vh; }
.scada-brand { display: flex; align-items: center; gap: 8px; padding-bottom:
8px; margin-bottom: 8px; border-bottom: 1px solid #222; flex-shrink: 0; }
.scada-brand-main { font-family: -apple-system, sans-serif; font-size: 11px;
font-weight: 800; color: #fff; letter-spacing: 4px; text-transform: uppercase; }
.scada-brand-sub { font-family: -apple-system, sans-serif; font-size: 6px;
color: #555; letter-spacing: 2px; text-transform: uppercase; }
.scada-brand-tag { margin-left: auto; font-family: 'Courier New', monospace;
font-size: 7px; color: #333; text-transform: uppercase; letter-spacing: 1px; }
.scada-keys { display: grid; grid-template-columns: 1fr 1fr; gap: 8px 12px;
overflow-y: auto; padding-bottom: 5px; }
.key-group { border-left: 2px solid #333; padding-left: 10px; margin-bottom:
4px; }
.key-title { font-size: 10px; color: #66ccff; text-transform: uppercase; margin-
bottom: 6px; font-weight: bold; }
.key-item { display: flex; align-items: center; gap: 8px; font-size: 13px;
```

```

margin-bottom: 8px; cursor: pointer; }
input[type="checkbox"], input[type="radio"] { transform: scale(1.3); margin-
right: 4px; accent-color: #00ffff; }
.maplibregl-popup-content { background: #000; color: #00ffff; border: 1px solid
#444; font-family: monospace; font-size: 12px; padding: 8px; }

/* LCP fix: disclaimer box is the largest paint element – push it off the
critical path */
.disclaimer-box { font-family: -apple-system, BlinkMacSystemFont, "Segoe UI",
Roboto, Helvetica, Arial, sans-serif; text-align: justify; font-size: 9px; line-
height: 1.5; color: #666; margin-top: 10px; padding-top: 10px; border-top: 1px
solid #222; user-select: none; pointer-events: none; flex-shrink: 0; content-
visibility: auto; }

.custom-map-attrib { position: absolute; top: 10px; left: 10px; background:
rgba(5, 5, 5, 0.7); color: #888; font-family: 'Courier New', monospace; font-
size: 9px; border-radius: 4px; padding: 4px 8px; z-index: 10; border: 1px solid
#333; pointer-events: auto; max-width: calc(100% - 60px); line-height: 1.6; }
.custom-map-attrib a { color: #00ffff; text-decoration: none; }
.custom-map-attrib a:hover { text-decoration: underline; }
body.fs-active .custom-map-attrib { top: 44px; }
.popup-search-btns { display: flex; gap: 6px; margin-top: 8px; }
.popup-btn { flex: 1; padding: 5px 0; font-family: monospace; font-size: 11px;
font-weight: bold; border: 1px solid #444; border-radius: 3px; cursor: pointer;
text-align: center; text-decoration: none; display: block; transition: border-
color 0.15s, color 0.15s; }
.popup-btn-news { background: #0a0a0a; color: #ffae00; border-color: #ffae00; }
.popup-btn-news:hover { background: #ffae00; color: #000; }
.popup-btn-images { background: #0a0a0a; color: #00ffff; border-color: #00ffff;
}
.popup-btn-images:hover { background: #00ffff; color: #000; }
.search-bar-wrapper { position: absolute; top: 72px; right: 10px; z-index: 20;
display: flex; gap: 6px; }
body.fs-active .search-bar-wrapper { top: 100px; }
.search-input { background: rgba(5,5,5,0.85); border: 1px solid #444; border-
radius: 4px; color: #00ffff; font-family: 'Courier New', monospace; font-size:
11px; padding: 5px 10px; width: 220px; outline: none; }
.search-input::placeholder { color: #444; }
.search-input:focus { border-color: #00ffff; }
.search-btn { background: #0a0a0a; border: 1px solid #00ffff; border-radius:
4px; color: #00ffff; font-family: 'Courier New', monospace; font-size: 11px;
padding: 5px 10px; cursor: pointer; }
.search-btn:hover { background: #00ffff; color: #000; }
.search-results { position: absolute; top: 36px; right: 0; background: #050505;
border: 1px solid #333; border-radius: 4px; width: 300px; max-height: 220px;
overflow-y: auto; z-index: 21; display: none; }
.search-result-item { padding: 7px 10px; font-family: 'Courier New', monospace;
font-size: 10px; color: #888; cursor: pointer; border-bottom: 1px solid #111;
line-height: 1.5; }
.search-result-item:hover { background: #111; color: #00ffff; }
.search-result-item b { color: #ffcc00; }
.search-no-results { padding: 8px 10px; font-family: 'Courier New', monospace;
font-size: 10px; color: #444; }
.map-controls { position: absolute; bottom: 30px; left: 10px; z-index: 20;
display: flex; flex-direction: column; gap: 4px; }
.map-ctrl-btn { background: rgba(5,5,5,0.85); border: 1px solid #333; border-
radius: 4px; color: #888; font-family: 'Courier New', monospace; font-size: 9px;
padding: 5px 8px; cursor: pointer; text-transform: uppercase; letter-spacing:
1px; white-space: nowrap; }
.map-ctrl-btn:hover { border-color: #00ffff; color: #00ffff; }
.map-ctrl-btn.active { border-color: #00ffff; color: #00ffff; background:
rgba(0,255,255,0.08); }

```

```

/* Radius area popup — stacked above radius search popup */
#radius-area-popup { bottom: 60px; }

/* Poly Zone panel — stacked above radius area popup, orange accent */
#polyzone-panel { bottom: 90px; border-color: #ff6600; }
.polyzone-label { color: #ff6600 !important; }

/* Radius search popup base */
.radius-popup { position: absolute; bottom: 30px; left: 120px; z-index: 20;
background: #050505; border: 1px solid #444; border-radius: 4px; padding: 8px
12px; display: none; }
.radius-popup label { font-size: 9px; color: #888; text-transform: uppercase;
letter-spacing: 1px; display: block; margin-bottom: 4px; }
.radius-input-row { display: flex; align-items: center; gap: 4px; }
#radius-input { background: #0a0a0a; border: 1px solid #333; color: #00ffff;
font-family: 'Courier New', monospace; font-size: 12px; font-weight: bold;
padding: 3px 6px; border-radius: 3px; width: 56px; outline: none; text-align:
right; -moz-appearance: textfield; }
#radius-input::-webkit-outer-spin-button, #radius-input::-webkit-inner-spin-
button { -webkit-appearance: none; margin: 0; }
#radius-input:focus { border-color: #00ffff; }
#radius-input.invalid { border-color: #ff4444; color: #ff4444; }

/* Radius area input — matches radius-input style, magenta accent */
#radius-area-input { background: #0a0a0a; border: 1px solid #333; color:
#ff00ff; font-family: 'Courier New', monospace; font-size: 12px; font-weight:
bold; padding: 3px 6px; border-radius: 3px; width: 72px; outline: none; text-
align: right; -moz-appearance: textfield; }
#radius-area-input::-webkit-outer-spin-button, #radius-area-input::-webkit-
inner-spin-button { -webkit-appearance: none; margin: 0; }
#radius-area-input:focus { border-color: #ff00ff; }
#radius-area-input.invalid { border-color: #ff4444; color: #ff4444; }

.radius-unit { font-size: 10px; color: #555; font-family: 'Courier New',
monospace; white-space: nowrap; }
.radius-hint { font-size: 8px; color: #444; margin-top: 4px; letter-spacing:
0.5px; }

/* Measure display panel */
#measure-display { position: absolute; bottom: 30px; left: 50%; transform:
translateX(-50%); z-index: 20; background: rgba(5,5,5,0.92); border: 1px solid
#00ffff; border-radius: 4px; padding: 8px 14px; display: none; font-family:
'Courier New', monospace; font-size: 10px; color: #00ffff; text-align: center;
white-space: nowrap; pointer-events: none; max-width: calc(100vw - 40px); box-
sizing: border-box; }
@media (max-width: 480px) {
    #measure-display { white-space: normal; font-size: 9px; }
}
#measure-display .m-row { margin-bottom: 2px; color: #888; }
#measure-display .m-row span { color: #00ffff; }
#measure-display .m-hint { font-size: 8px; color: #444; margin-top: 4px; }
#btn-measure-undo { pointer-events: auto; background: #0a0a0a; border: 1px solid
#555; border-radius: 3px; color: #888; font-family: 'Courier New', monospace;
font-size: 8px; padding: 2px 6px; cursor: pointer; margin-top: 5px; text-
transform: uppercase; letter-spacing: 1px; display: none; }
#btn-measure-undo:hover { border-color: #ffff00; color: #ffff00; }

/* Poly Zone display panel — orange accent, mirrors measure-display */
#polyzone-display { position: absolute; bottom: 30px; left: 50%; transform:
translateX(-50%); z-index: 20; background: rgba(5,5,5,0.92); border: 1px solid
#ff6600; border-radius: 4px; padding: 8px 14px; display: none; font-family:

```

```

'Courier New', monospace; font-size: 10px; color: #ff6600; text-align: center;
white-space: nowrap; pointer-events: none; max-width: calc(100vw - 40px); box-
sizing: border-box; }
#polyzone-display .m-row { margin-bottom: 2px; color: #888; }
#polyzone-display .m-row span { color: #ff6600; }
#polyzone-display .m-hint { font-size: 8px; color: #444; margin-top: 4px; }
#btn-polyzone-undo { pointer-events: auto; background: #0a0a0a; border: 1px
solid #555; border-radius: 3px; color: #888; font-family: 'Courier New',
monospace; font-size: 8px; padding: 2px 6px; cursor: pointer; margin-top: 5px;
text-transform: uppercase; letter-spacing: 1px; display: none; }
#btn-polyzone-undo:hover { border-color: #ff6600; color: #ff6600; }
#btn-polyzone-undo:focus-visible { outline: 1px solid #ff6600; outline-offset:
2px; }
@media (max-width: 480px) {
    #polyzone-display { white-space: normal; font-size: 9px; }
}

/* Fullscreen letterhead */
#fs-letterhead { display: none; position: fixed; top: 8px; right: 12px; z-index:
512; text-align: right; pointer-events: none; }
#fs-letterhead .ventus-main { font-family: -apple-system, sans-serif; font-size:
15px; font-weight: 800; color: #fff; letter-spacing: 5px; text-transform:
uppercase; margin-bottom: 1px; }
#fs-letterhead .ventus-sub { font-family: -apple-system, sans-serif; font-size:
6px; color: #888; letter-spacing: 2px; text-transform: uppercase; }
body.fs-active #fs-letterhead { display: block; }

/* Status legend */
.status-legend { display: flex; gap: 8px; flex-wrap: wrap; padding: 4px 0 6px 0;
border-bottom: 1px solid #222; margin-bottom: 6px; flex-shrink: 0; }
.status-dot { display: inline-flex; align-items: center; gap: 4px; font-size:
8px; color: #555; }
.status-dot span { width: 7px; height: 7px; border-radius: 50%; display: inline-
block; }

/* Fullscreen map */
.map-container.is-fullscreen { position: fixed; top: 0; left: 0; width: 100vw;
height: 100dvh; border-radius: 0; border: none; z-index: 500; }
#btn-fullscreen { position: absolute; top: 10px; right: 10px; z-index: 25;
background: rgba(5,5,5,0.85); border: 1px solid #333; border-radius: 4px; color:
#888; font-family: 'Courier New', monospace; font-size: 11px; padding: 5px 8px;
cursor: pointer; letter-spacing: 1px; line-height: 1; }
#btn-fullscreen:hover { border-color: #00ffff; color: #00ffff; }
#btn-fullscreen-exit { display: none; position: fixed; top: 8px; left: 12px; z-
index: 510; background: rgba(5,5,5,0.9); border: 1px solid #00ffff; border-
radius: 4px; color: #00ffff; font-family: 'Courier New', monospace; font-size:
10px; padding: 5px 10px; cursor: pointer; letter-spacing: 1px; }
body.fs-active #btn-fullscreen-exit { display: block; }

/* Fullscreen curtain (layer panel slide-down) */
#fs-curtain { display: none; position: fixed; top: 0; left: 0; width: 100vw; z-
index: 505; background: rgba(5, 5, 5, 0.96); border-bottom: 1px solid #333;
transform: translateY(-100%); transition: transform 0.3s ease; max-height:
70dvh; overflow-y: auto; padding: 48px 16px 16px 16px; box-sizing: border-box; }
#fs-curtain.curtain-open { transform: translateY(0); }
body.fs-active #fs-curtain { display: block; }
#fs-curtain-tab { display: none; position: fixed; top: 0; left: 50%; transform:
translateX(-50%); z-index: 510; background: rgba(5,5,5,0.9); border: 1px solid
#333; border-top: none; border-radius: 0 0 6px 6px; color: #555; font-family:
'Courier New', monospace; font-size: 9px; padding: 4px 16px 5px 16px; cursor:
pointer; letter-spacing: 1px; text-transform: uppercase; white-space: nowrap;
transition: color 0.15s, border-color 0.15s; }

```

```

#fs-curtain-tab:hover { color: #00ffff; border-color: #00ffff; }
body.fs-active #fs-curtain-tab { display: block; }
#fs-curtain-keys { display: grid; grid-template-columns: 1fr 1fr; gap: 8px 16px;
}

/* Responsive: single column on very narrow screens (iPhone SE etc.) */
@media (max-width: 480px) {
    .scada-keys, #fs-curtain-keys { grid-template-columns: 1fr; }
}

/* Custom scrollbars – prevents jarring default grey on dark SCADA theme */
::-webkit-scrollbar { width: 6px; }
::-webkit-scrollbar-track { background: #050505; }
::-webkit-scrollbar-thumb { background: #333; border-radius: 3px; }
::-webkit-scrollbar-thumb:hover { background: #00ffff; }

/* Firefox scrollbar */
* { scrollbar-width: thin; scrollbar-color: #333 #050505; }

/* Focus-visible states – keyboard accessibility without affecting mouse users */
.map-ctrl-btn:focus-visible,
.search-btn:focus-visible,
#btn-fullscreen:focus-visible,
#btn-fullscreen-exit:focus-visible,
#btn-measure-undo:focus-visible { outline: 1px solid #00ffff; outline-offset:
2px; }

/* Short viewport – landscape mobile or compact laptop screens */
@media (max-height: 600px) {
    .scada-wrapper { max-height: 28vh; }
    .hud-header { padding: 3px 10px; }
    .hud-val { font-size: 13px; }
    .ventus-main { font-size: 13px; }
    .key-item { margin-bottom: 4px; font-size: 11px; }
}

```

```

=====
FILE: https://ventusltd.github.io/gridatlas/atlas/current.json
46738 chars · data · TRUNCATED: the first 4000 characters of 46738 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (fetch)
=====

```

```

{
  "schema": "gridatlas.current.v2",
  "generation": "202609051510",
  "previous_generation": "202609051503",
  "architecture": "IMMUTABLE_SHELL_PLUS_HASHED_CARTRIDGES",
  "live_route": "/gridatlas/atlas/",
  "release_id": "202608300453-atlas-v9",
  "release_route": "/gridatlas/atlas/releases/202608300453-atlas-v9/",
  "shell": {
    "release_id": "202608300453-atlas-v9",
    "index": "../releases/202608300453-atlas-v9/index.html",
    "base": "../releases/202608300453-atlas-v9/"
  },
  "cartridge_order": [
    "streaming-parquet-bridge",
    "uk-gazetteer-flyto",
    "substation-intelligence",
    "sld-sandbox"
  ]
}

```

```

],
"cartridges": [
  {
    "id": "streaming-parquet-bridge",
    "generation": "202609041945",
    "version": "v9.112",
    "type": "script",
    "slot": "replace-script",
    "replace_script": "202608292311-maplibre-worker-bridge.js",
    "path": "./cartridges/202609041945-streaming-parquet-bridge-v9-5.js",
    "sha256": "4a6b594f7705bc4d7d62cbe69d2e40726d12f9a759896430cfe5eb00d830d23e",
    "contract": "../ui/cartridges/202608301825-streaming-parquet-bridge-v9-
5.mjs",
    "capabilities": [
      "response-before-body-reconstruction",
      "duckdb-runtime-prewarm-after-400kv",
      "metro-partition-alias",
      "parquet-on-demand-preserved",
      "payload-cache-release-after-serialisation"
    ],
    "immutable_shell_modified": false
  },
  {
    "id": "uk-gazetteer-flyto",
    "generation": "202609041945",
    "version": "v9.112",
    "type": "script",
    "slot": "replace-script",
    "replace_script": "202608291818-place-postcode-search.js",
    "path": "./cartridges/202609041945-place-global-search-v9-5.js",
    "sha256": "21ab1507e98ee29961d03236e8b7853298104a33ecb1986d4d10ffc420a99c1c",
    "contract": "../ui/cartridges/202609040337-global-gazetteer-flyto-v9-
106.mjs",
    "capabilities": [
      "exact-repd-first",
      "uk-postcode",
      "uk-place",
      "global-address",
      "global-place",
      "deep-link-publishes-resolved-technology-and-capacity"
    ],
    "result_class": "LOCATION_ONLY",
    "sets_deep_link": false,
    "assembled_from": "./manifests/202609041945-place-global-search-v9-5-
parts.json"
  },
  {
    "id": "sld-sandbox",
    "generation": "202609050354",
    "version": "v9.121",
    "type": "script",
    "slot": "replace-script",
    "replace_script": "202608292126-pre-snapped-config-adapter.js",
    "path": "./cartridges/202609050354-sld-sandbox-v9-8.js",
    "sha256": "7c2df5dabb0082d4cef941b98bf31e92161d1e79a7d22bd6c84e81b171507022",
    "capabilities": [
      "pre-snapped-config-adapter-carried-forward",
      "project-or-substation-selection",
      "nearest-substations-33kv-and-above",
      "nearest-projects-from-a-substation",
      "distances-on-the-project-card",
      "beta-scope-and-limits-stated-on-card",

```

```

"animated-scada-link-layer",
"reduced-motion-honoured",
"sld-layout-ported-from-gis-sld-financial-sandbox",
"string-and-central-sizing",
"bess-compound",
"33kv-collector-trunk-and-branches",
"export-cable-with-editable-route",
"drag-array-and-rotate-handle",
"electron-flow-on-collectors-and-cable",
"single-earth-radius-throughout",
"panel-clear-of-the-atlas-toolbar",
"deep-link-runs-the-same-measurement-as-a-click",
"substation-layer-auto-enabled-on-arrival",
"layout-opens-from-the-project-card",
"denser-electron-flow-two-layers",
"fit-array-to-the-registers-stated-capacity",
"ac-dc-basis-declared-not-assumed",
"ratio-mismatch-descriptive-not-graded",
"own-ui-clicks-do-not-reach-the-map",
"unloaded-layer-reported-as-unloaded",
"cards-drag-minimise-and-close",
"minimised-cards-stay-identifiable",
"card-bounded-to-the-map",
"layers-survive-fullscreen",
"project-pin-drawn-by-the-cartridge-not-a-layer",
"project-layer-enabled-on-arrival",
"pin-toggles-on-and-off-from-the-card",
"central-ac-is-the-limiting-nameplate-not-a-product",
"inverter-and-transformer-nameplates-reported-separately",
"mv-block-overload-detected-against-its-skid",
"proj

```

```

=====
FILE: https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-
v9/index.html

```

```

21193 chars · code · shown in full · found via loaded by the browser (fetch)
=====

```

```

<!DOCTYPE html>

```

```

<html lang="en">

```

```

<head>

```

```

  <meta charset="utf-8" />

```

```

  <meta name="viewport" content="width=device-width, initial-scale=1.0,
maximum-scale=1.0, user-scalable=no" />

```

```

  <title>GlobalGrid2050 | UK Region v8 (Experimental)</title>

```

```

  <link href="https://cdn.jsdelivr.net/npm/maplibre-gl@3.6.2/dist/maplibre-
gl.css" rel="stylesheet" />

```

```

  <link href="ventusv8.css" rel="stylesheet" />

```

```

</head>

```

```

<body>

```

```

<div id="fatal-banner">CRITICAL ERROR: SYSTEM DEPENDENCIES FAILED TO LOAD</div>

```

```

<button id="btn-fullscreen-exit" onclick="exitFullscreen()">× Exit</button>

```

```

<div id="fs-curtain-tab">⬇ Layers</div>

```

```

<div id="fs-curtain"><div id="fs-curtain-keys"></div></div>

```

```

<div id="fs-letterhead">

```

```

  <div class="ventus-main">Ventus</div>

```

```

  <div class="ventus-sub">Cables & Connectivity&reg;</div>

```

```

</div>

```

```

<div class="dashboard">

```

```

  <div class="hud-header">

```



```

<div>
  <small style="color:#888">SYSTEM TIME</small><br>
  <span class="hud-val" id="clock">--:--:--</span><br>
  <span id="date" style="font-size: 11px; color: #888;">--/--/----
</span>
</div>
<div class="ventus-brand">
  <div class="ventus-main">Ventus</div>
  <div class="ventus-sub">Cables & Connectivity&reg;</div>
</div>
<div style="text-align:right"><small style="color:#888">2050
TARGET</small><br><span class="hud-val" id="days" style="color:#ff9d00">--
DAYS</span></div>
</div>

<div class="map-container" id="map-container">
  <div id="map"></div>
  <div class="custom-map-attrib">
    Data © <a href="https://www.openstreetmap.org/copyright"
target="_blank" rel="noopener noreferrer">OpenStreetMap contributors</a> | ©
CARTO | EV data © <a href="https://openchargemap.org" target="_blank"
rel="noopener noreferrer">Open Charge Map</a>
  </div>
  <button id="btn-fullscreen" onclick="enterFullscreen()"
title="Fullscreen">[]</button>
  <div class="search-bar-wrapper">
    <div style="position:relative">
      <input class="search-input" id="search-input" type="text"
placeholder="Search project name..." autocomplete="off" />
      <div class="search-results" id="search-results"></div>
    </div>
    <button class="search-btn" id="search-btn">GO</button>
  </div>

  <div class="map-controls">
    <button class="map-ctrl-btn" id="btn-export">📄 Export CSV</button>
    <button class="map-ctrl-btn" id="btn-radius">⊙ Radius
Search</button>
    <button class="map-ctrl-btn" id="btn-radius-area">⊙ Radius
Area</button>
    <button class="map-ctrl-btn" id="btn-zonedraw">○ Poly Zone</button>
    <button class="map-ctrl-btn" id="btn-status">● Status
Colours</button>
    <button class="map-ctrl-btn" id="btn-measure">📏 Measure</button>
  </div>

  <div class="radius-popup" id="radius-popup">
    <label>Radius from click</label>
    <div class="radius-input-row">
      <input type="number" id="radius-input" value="10" min="1"
max="160" step="1" />
      <span class="radius-unit">km</span>
    </div>
    <div class="radius-hint">1 – 160 km (100 mi) · press Enter to
apply</div>
  </div>

  <div class="radius-popup" id="radius-area-popup">
    <label>Measure Area from click</label>
    <div class="radius-input-row">

```

```

        <input type="number" id="radius-area-input" value="1"
min="0.001" max="160" step="0.001" />
        <span class="radius-unit">km</span>
    </div>
    <div class="radius-hint">Input radius: 0.001 (1m) to 160 km</div>
</div>

<div class="radius-popup" id="zonedraw-display">
    <label class="polyzone-label">🔹 Poly Zone Tool</label>
    <div class="radius-input-row">
        <input type="number" id="zonedraw-radius-input" value="2"
min="0.1" max="160" step="0.1" />
        <span class="radius-unit">km</span>
    </div>
    <div class="radius-hint">Set starting radius, then click map to
create zone</div>
    <button id="btn-zonedraw-undo" title="Undo last vertex"
style="margin-top:8px;background:#fff660;color:#000;border:none;padding:6px
12px;border-radius:4px;cursor:pointer;font-weight:bold;font-size:12px;">↶ Undo
Point</button>
</div>

<div id="measure-display">
    <div class="m-row" id="m-line" style="display:none">— <span id="m-
km"></span> km | <span id="m-m"></span> m | <span id="m-mi"></span> mi</div>
    <div class="m-row" id="m-perim" style="display:none">Perimeter:
<span id="m-pkm"></span> km | <span id="m-pm"></span> m</div>
    <div class="m-row" id="m-area" style="display:none">Area: <span
id="m-km2"></span> km² | <span id="m-ha"></span> ha | <span id="m-ac"></span>
ac</div>
    <div class="m-hint" id="m-hint">Click to add points • Double-click
to close polygon</div>
    <button id="btn-measure-undo" title="Undo last point">↶ Undo
Point</button>
</div>

<div id="polyzone-display">
    <div class="m-row" id="pz-area" style="display:none">
        <span id="pz-km2"></span> km² • <span id="pz-ha"></span> ha<br>
        <span id="pz-ac"></span> ac • <span id="pz-mi2"></span> mi²<br>
        <span id="pz-m2"></span> m²<br>
        <span style="color:#ffae00">⚽ <span id="pz-pitch"></span>
pitches</span>
    </div>
    <div class="m-row" id="pz-perim" style="display:none">
        <span style="color:#555;font-size:9px">perimeter <span id="pz-
pkm"></span> km</span>
    </div>
    <div class="m-hint" id="pz-hint">Click map to drop circle</div>
    <button id="btn-polyzone-undo" title="Undo last vertex">↶ Undo
Vertex</button>
</div>

<div class="podcast-shoutout">In support of The Future of Solar
Photovoltaics podcast<br>& all participants to date</div>
</div>

<div class="scada-wrapper">
    <div class="scada-brand">
        <div>

```

```

        <div class="scada-brand-main">Ventus</div>
        <div class="scada-brand-sub">Cables & Connectivity&reg;
</div>
    </div>
    <div class="scada-brand-tag">globalgrid2050.com | open
infrastructure intelligence</div>
</div>
    <div class="status-legend">
        <div class="status-dot"><span style="background:#00ff88">
</span>Operational</div>
        <div class="status-dot"><span style="background:#ffcc00">
</span>Under Construction</div>
        <div class="status-dot"><span style="background:#ff8800">
</span>Consented</div>
        <div class="status-dot"><span style="background:#8888ff">
</span>Applied</div>
    </div>
    <div class="scada-keys" id="scada-ui-container"></div>
    <div class="disclaimer-box">
        This interface is based on publicly available and open source
infrastructure data, including public energy datasets such as REPD where
applicable. It is presented for analytical and visualisation purposes only and
does not include live operational or other non public operational data.
    </div>
</div>
</div>

<script src="https://cdn.jsdelivr.net/npm/maplibre-gl@3.6.2/dist/maplibre-
gl.js"></script>

<script src="202608292311-maplibre-worker-bridge.js"></script>
<script src="202608291818-place-postcode-search.js"></script>

<script src="ventus-corev8engine.js"></script>
<script src="202608292126-pre-snapped-config-adapter.js"></script>

<script>
    const ukConfig = [
        {
            group: "Topology (GeoJSON)",
            layers: [
                { id: "400", label: "400kV", color: "#0054ff", type: "line",
width: 2.5, url: "data/grid_400kv.geojson", snap: true, preload: true },
                { id: "275", label: "275kV", color: "#ff0000", type: "line",
width: 2.0, url: "data/grid_275kv.geojson", snap: true, preload: true },
                { id: "220", label: "220kV", color: "#ff9900", type: "line",
width: 1.8, url: "data/grid_220kv.geojson", snap: true, preload: true },
                { id: "132", label: "132kV", color: "#00cc00", type: "line",
width: 1.5, url: "data/grid_132kv.geojson", snap: true, preload: true },
                { id: "66", label: "66kV", color: "#b200ff", type: "line",
width: 1.2, url: "data/grid_66kv.geojson", snap: true, preload: true },
                { id: "11kv", label: "11kV (UKPN)", color: "#ff00ff", type:
"point", radius: ['interpolate', ['linear'], ['zoom'], 13.5, 4, 15, 8, 18, 18],
url: "data/grid_11kv_ukpn.geojson", minzoom: 13.5, preload: false },
                { id: "subs", label: "Subs", color: "#ffffff", type: "point",
radius: ['interpolate', ['linear'], ['zoom'], 5, 3, 10, 5, 14, 10, 18, 22], url:
"data/grid_substations.geojson", isSubs: true, preload: true }
            ]
        },
        {
            group: "Assets (GeoJSON)",
            layers: [

```

```

        { id: "nuc", label: "Nuclear", color: "#39ff14", type:
"point", radius: 4.5, url: "data/power_plants.geojson", filter: ['==', ['get',
'source']], 'nuclear'], preload: true },
        { id: "gas", label: "Gas", color: "#ff4500", type:
"point", radius: 4.5, url: "data/power_plants.geojson", filter: ['!=', ['get',
'source']], 'nuclear'], preload: true },
        { id: "ind", label: "Industry", color: "#ff6600", type:
"point", radius: 4.5, url: "data/industrial_offtakers.geojson", preload: true },
        { id: "dc", label: "Data Ctrs", color: "#00ffff", type:
"point", radius: 4.5, url: "data/datacentres.geojson", preload: true },
        { id: "air", label: "Airports", color: "#ff00ff", type:
"point", radius: 4.5, url: "data/airports.geojson", preload: true },
        { id: "rail", label: "Railways", color: "#ffd700", type:
"point", radius: 4.5, url: "data/railways.geojson", preload: true }
    ],
    {
        group: "Supermarkets (GeoJSON)",
        layers: [
            { id: "tesco", label: "Tesco", color: "#eelc2e",
type: "point", radius: 4.5, url: "data/supermarkets_tesco.geojson", preload:
false },
            { id: "sainsburys", label: "Sainsbury's", color: "#ff8200",
type: "point", radius: 4.5, url: "data/supermarkets_sainsburys.geojson",
preload: false },
            { id: "asda", label: "Asda", color: "#78be20",
type: "point", radius: 4.5, url: "data/supermarkets_asda.geojson", preload:
false },
            { id: "morrison's", label: "Morrison's", color: "#ffd700",
type: "point", radius: 4.5, url: "data/supermarkets_morrison's.geojson", preload:
false },
            { id: "aldi", label: "Aldi", color: "#003087",
type: "point", radius: 4.5, url: "data/supermarkets_aldi.geojson", preload:
false },
            { id: "lidl", label: "Lidl", color: "#0050aa",
type: "point", radius: 4.5, url: "data/supermarkets_lidl.geojson", preload:
false },
            { id: "waitrose", label: "Waitrose", color: "#7ab800",
type: "point", radius: 4.5, url: "data/supermarkets_waitrose.geojson", preload:
false },
            { id: "ms", label: "M&S Food", color: "#009b77",
type: "point", radius: 4.5, url: "data/supermarkets_ms.geojson", preload: false
},
            { id: "coop", label: "Co-op", color: "#00b1a9",
type: "point", radius: 4.5, url: "data/supermarkets_coop.geojson", preload:
false },
            { id: "costco", label: "Costco", color: "#005daa",
type: "point", radius: 4.5, url: "data/supermarkets_costco.geojson", preload:
false },
            { id: "booths", label: "Booths", color: "#6d2077",
type: "point", radius: 4.5, url: "data/supermarkets_booths.geojson", preload:
false }
        ],
    },
    {
        group: "Transit (GeoJSON)",
        layers: [
            { id: "elizabeth", label: "Elizabeth Line", color: "#60399E",
type: "point", radius: 5, url: "data/elizabeth_line.geojson", preload: false },
            { id: "lu", label: "London Underground", color: "#e32017",
type: "point", radius: 5, url: "data/london_underground.geojson", preload: false
},
        ],
    },

```

```

        { id: "dlr", label: "DLR", color: "#00afad", type: "point",
radius: 5, url: "/uk_metros_trams.geojson", preload: false,
        filter: ['all', ['in', 'Docklands', ['get', 'operator']],
['!', ['in', 'London Underground', ['get', 'operator']]]] },
        { id: "metro", label: "UK Metro", color: "#f3a712", type:
"point", radius: 5, url: "/uk_metros_trams.geojson", preload: false,
        filter: ['any', ['in', 'Tyne', ['get', 'operator']], ['in',
'Metrolink', ['get', 'operator']], ['in', 'Supertram', ['get', 'operator']],
['in', 'Nottingham', ['get', 'operator']], ['in', 'Edinburgh', ['get',
'operator']], ['in', 'Midland Metro', ['get', 'operator']]] },
        { id: "tram", label: "Trams & Light Rail", color: "#84329b",
type: "point", radius: 5, url: "/uk_metros_trams.geojson", preload: false,
        filter: ['all', ['==', ['get', 'type'], 'Tram / Light Rail'],
['!', ['in', 'London Underground', ['get', 'operator']]]] },
        { id: "mainline_rail", label: "UK Mainline Railways", color:
"#ff8800", type: "line", width: 1.5, url: "/uk_mainline_railways.geojson",
preload: false },
        { id: "hs2", label: "HS2", color: "#2c6fad", type: "point",
radius: 4, url: "data/hs2.geojson", preload: false },
        { id: "eurostar_route", label: "Eurostar (HS1) UK & Europe",
color: "#001489", type: "line", width: 2.5, url: "data/eurostar.geojson",
filter: ['==', ['get', 'type'], 'route'], preload: false },
        { id: "eurostar_station", label: "Eurostar Stations UK &
Europe", color: "#ffd700", type: "point", radius: 5.5, url:
"data/eurostar.geojson", filter: ['==', ['get', 'type'], 'eurostar_station'],
preload: false }
    ],
    {
        group: "Leisure & Sports (GeoJSON)",
        layers: [
            { id: "stadiums", label: "Major Stadiums", color: "#e5ff00",
type: "point", radius: 5.5, url: "data/stadiums.geojson", preload: false }
        ]
    },
    {
        group: "Global Data Backbone",
        layers: [
            { id: "subsea_data_cables", label: "Subsea Data Cables (Fibre)
(not all connections available yet)", color: "#00e5ff", type: "line", width:
1.5, url: "data/subsea_data_cables.geojson", preload: false }
        ]
    },
    {
        group: "Global Supply Chain & Chokepoints",
        layers: [
            { id: "global_ports_major", label: "Major Ports (Deep
Water/Ferry)", color: "#00bfff", type: "point", radius: ['interpolate',
['linear'], ['get', 'area_ha'], 0, 8, 80, 10, 500, 16], url:
"data/global_ports.geojson", filter: ['==', ['get', 'type'], 'Major
Cargo/Container Port'], preload: false },
            { id: "global_ports_minor", label: "Inland / Minor Harbours",
color: "#0055aa", type: "point", radius: ['interpolate', ['linear'], ['get',
'area_ha'], 0, 2.5, 50, 4.5, 200, 6], url: "data/global_ports.geojson", filter:
['==', ['get', 'type'], 'Minor/Local Harbour'], preload: false },
            { id: "global_hydrocarbons", label: "Global Hydrocarbons (Oil,
Gas, LNG)", color: "#ff3300", type: "point", radius: ['interpolate', ['linear'],
['get', 'area_ha'], 0, 4, 100, 8, 1000, 15], url:
"data/global_hydrocarbons.geojson", preload: false }
        ]
    },
    {

```

```

    group: "Highway Infrastructure",
    layers: [
      { id: "motorways", label: "Motorways", color: "#0054ff", type:
"line", width: 2.5, url: "/uk_motorways.geojson", preload: false },
      { id: "trunk_roads", label: "Major A-Roads (Trunk)", color:
"#00cc00", type: "line", width: 1.8, url: "/uk_trunk_roads.geojson", preload:
false },
      { id: "primary_roads", label: "A-Roads (Primary)", color:
"#ff4400", type: "line", width: 1.2, url: "/uk_primary_roads.geojson", preload:
false },
      { id: "motorway_services", label: "Motorway Services", color:
"#ff00ff", type: "point", radius: 5.5, url: "data/motorway_services.geojson",
preload: false }
    ]
  },
  {
    group: "EV (GeoJSON)",
    layers: [
      { id: "ev", label: "EV Rapid (100kW+)", color: "#00ff88", type:
"point", radius: 5, url: "data/ev_chargers.geojson", preload: false }
    ]
  },
  {
    group: "Heavy Industry",
    layers: [
      { id: "naei_co2", label: "Major Industrial Sites", color:
"#ff4400", type: "point", url: "/heavy_emitters_uk.json", preload: false }
    ]
  },
  {
    group: "REPD (GeoJSON)",
    layers: [
      { id: "solar", label: "Solar PV", color:
"#ffff00", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'solar'], preload: false },
      { id: "solar_operational", label: "Solar PV (Operational Only)",
color: "#00ff88", type: "point", radius: ['interpolate', ['linear'],
['coalesce', ['get', 'capacity'], 0], 0, 10, 10, 14, 50, 18, 100, 22, 200, 28,
350, 35, 500, 42], url: "/dist/repd_master.json", filter: ['all', ['==', ['get',
'tech'], 'solar'], ['==', ['get', 'status'], 'operational']], preload: false },
      { id: "solar_roof", label: "Solar Roof", color:
"#ffcc00", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'solar_roof'], preload: false },
      { id: "wind", label: "Wind", color:
"#00ffff", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'wind'], preload: false },
      { id: "wind_onshore_operational", label: "Onshore Wind
(Operational Only)", color: "#00ffcc", type: "point", radius: ['interpolate',
['linear'], ['coalesce', ['get', 'capacity'], 0], 0, 8, 10, 12, 50, 16, 100, 20,
200, 26, 350, 32, 500, 38], url: "/dist/repd_master.json", filter: ['all',
['==', ['get', 'raw_tech'], 'Wind Onshore'], ['==', ['get', 'status'],
'operational']], preload: false },
      { id: "wind_offshore_operational", label: "Offshore Wind
(Operational Only)", color: "#0066ff", type: "point", radius: ['interpolate',
['linear'], ['coalesce', ['get', 'capacity'], 0], 0, 8, 10, 12, 50, 16, 100, 20,
200, 26, 350, 32, 500, 38], url: "/dist/repd_master.json", filter: ['all',
['==', ['get', 'raw_tech'], 'Wind Offshore'], ['==', ['get', 'status'],
'operational']], preload: false },
      { id: "bess", label: "Battery Storage", color:
"#ffae00", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'bess'], preload: false },
      { id: "bess_operational", label: "Battery Storage (Operational

```

```

Only)", color: "#ff69b4", type: "point", radius: ['interpolate', ['linear'],
['coalesce', ['get', 'capacity'], 0], 0, 8, 10, 12, 50, 16, 100, 20, 200, 26,
350, 32, 500, 38], url: "/dist/repd_master.json", filter: ['all', ['==', ['get',
'tech'], 'bess'], ['==', ['get', 'status'], 'operational']], preload: false },
    { id: "biomass", label: "Biomass & EfW", color:
"#39ff14", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'biomass'], preload: false },
    { id: "tidal", label: "Tidal & Wave", color:
"#00bfff", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'tidal'], preload: false },
    { id: "hydrogen", label: "Hydrogen", color:
"#ffffff", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'hydrogen'], preload: false },
    { id: "hydro", label: "Hydro & Pumped Storage", color:
"#00aaff", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'hydro'], preload: false },
    { id: "flywheel", label: "Flywheel", color:
"#ff69b4", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'flywheel'], preload: false },
    { id: "act", label: "Advanced Conversion", color:
"#ff6600", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'act'], preload: false },
    { id: "geothermal", label: "Geothermal", color:
"#ff3300", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'geothermal'], preload: false },
    { id: "caes", label: "Compressed Air Storage", color:
"#88aaff", type: "point", radius: 8, url: "/dist/repd_master.json", filter:
['==', ['get', 'tech'], 'caes'], preload: false }
  ]
};

window.initVentusMap({
  config: ukConfig,
  center: [-3.5, 54.0],
  zoom: 4.2
});
</script>

</body>
</html>

```

```

=====
FILE: https://ventusltd.github.io/gridatlas/atlas/cartridges/202609041945-
streaming-parquet-bridge-v9-5.js
13316 chars · code · shown in full · found via loaded by the browser (fetch)
=====
(() => {
  'use strict';

  const GENERATION = '202609041945';
  const DATA_BASE = 'https://ventusltd.github.io/data-gridatlas/202608291237-
data-gridatlas/';
  const MANIFEST_URL = `${DATA_BASE}data/manifest.json`;
  const MANIFEST_SHA256 =
'3246dbdaa042ae8352ec9b7128cb6c2fe65e4flaba0534302510661828df2526';
  const DUCKDB_MODULE = 'https://cdn.jsdelivr.net/npm/@duckdb/duckdb-
wasm@1.29.0/+esm';
  const CRITICAL_RELEASE_PATH = 'data/grid_400kv.geojson';
  const CRITICAL_RUNTIME_URL =
'../cartridges/5f5fbec83f9ce307b47ddc6e7277743f0bb1a2445b0f3ca50a9a1806146e993/

```

```

grid_400kv.geojson';
const MAP_READY_PATHS = new Set([
  'data/grid_400kv.geojson',
  'data/grid_275kv.geojson',
  'data/grid_220kv.geojson',
  'data/grid_132kv.geojson',
  'data/grid_66kv.geojson',
  'data/grid_substations.geojson',
  'data/power_plants.geojson',
  'data/industrial_offtakers.geojson',
  'data/datacentres.geojson',
  'data/airports.geojson',
  'data/railways.geojson'
]);

const nativeFetch = window.fetch.bind(window);
const sourceCache = new Map();
let manifestPromise = null;
let runtimePromise = null;

const state = {
  schema: 'gridatlas.maplibre-worker-fetch-bridge.v1',
  generation: GENERATION,
  data_release: '202608291237-data-gridatlas',
  architecture: {
    map_ready_same_origin_geojson: true,
    critical_400kv_maplibre_worker_source: true,
    critical_400kv_window_prefetch: false,
    critical_400kv_main_thread_json_parse: false,
    preload_browser_duckdb: 'AFTER_CRITICAL_SOURCE',
    serialized_preload_queue: false,
    topology_pre_snapped: true,
    analytical_search_duckdb_retained: true
  },
  critical_source: {
    release_copy_path: CRITICAL_RELEASE_PATH,
    runtime_url: CRITICAL_RUNTIME_URL,
    cache_identity: 'CONTENT_ADDRESSED_STABLE_URL',
    delivery: 'MAPLIBRE_WORKER_DIRECT_URL',
    eager_window_prefetch: false,
    window_fetch_hits: 0
  },
  map_ready_requests: 0,
  map_ready_paths: [...MAP_READY_PATHS],
  parquet_requests: 0,
  duckdb_runtime_started: false,
  duckdb_runtime_started_at_ms: null,
  duckdb_runtime_shared: false,
  intercepted_on_demand: 0,
  loaded_on_demand: {},
  failures: [],
  streamed_responses: 0,
  released_payloads: 0,
  stream_failures: [],
  runtime_prewarm: { requested: false, started: false, completed: false,
failed: null }
};
window.__GRIDATLAS_MAP_READY__ = state;

function invariant(condition, message) {
  if (!condition) throw new Error(message);
}

```



```

function requestPath(input) {
  try {
    const raw = typeof input === 'string' ? input : input?.url;
    return new URL(raw, window.location.href).pathname;
  } catch {
    return '';
  }
}

function mapReadyKey(pathname) {
  const marker = '/data/';
  const index = pathname.toLowerCase().lastIndexOf(marker);
  if (index < 0) return '';
  return `data/${decodeURIComponent(pathname.slice(index +
marker.length)).toLowerCase()}`;
}

function legacyStem(pathname) {
  const name = decodeURIComponent(pathname.split('/').pop() ||
'').toLowerCase();
  if (name === 'repd_master.json') return 'repd_master_v8_oracle';
  if (name === 'heavy_emitters_uk.json') return 'heavy_emitters_uk';
  if (name.endsWith('.geojson')) return name.slice(0, -8);
  return '';
}

async function sha256Hex(bytes) {
  const digest = await crypto.subtle.digest('SHA-256', bytes);
  return [...new Uint8Array(digest)]
    .map(value => value.toString(16).padStart(2, '0'))
    .join('');
}

async function getManifest() {
  manifestPromise ||= (async () => {
    const response = await nativeFetch(MANIFEST_URL, { cache: 'no-store' });
    invariant(response.ok, `manifest HTTP ${response.status}`);
    const bytes = await response.arrayBuffer();
    invariant(await sha256Hex(bytes) === MANIFEST_SHA256, 'data manifest SHA-
256 mismatch');
    const manifest = JSON.parse(new TextDecoder().decode(bytes));
    invariant(manifest?.schema === 'data-gridatlas.v8-transplant-manifest.v1',
'data manifest schema mismatch');
    invariant(manifest?.closure?.sources === 56, 'V8 source closure
mismatch');
    invariant(manifest?.closure?.layers === 60, 'V8 layer closure mismatch');
    invariant(manifest?.closure?.features === 541282, 'V8 feature closure
mismatch');
    return manifest;
  })();
  return manifestPromise;
}

/* ONE DuckDB runtime for the whole page.
-----
This cartridge and the uk-gazetteer-flyto search lane each built their
own. Both were correctly memoised internally, so neither was a fault by
itself; nothing coordinated between them because they are separate
cartridges by design. Measured on the live page at an iPhone 13 profile,
202609041500: duckdb-eh.wasm fetched TWICE, 5.92 MB each, 11.84 MB of a
12.81 MB arrival - and, which matters more on a phone, two WebAssembly

```

heaps and two workers alive at once. iOS Safari caps per-tab memory hard where a desktop does not.

The broker below is deliberately duplicated verbatim into the search lane rather than imported. Cartridges are composed as independent scripts with no shared module loader between them, so there is nothing to import from; the CONTRACT is the window key and the shape it holds, not this function. Whichever cartridge asks first builds the runtime and the other awaits the same promise, so it is order-independent - which matters because composition order is a manifest decision, not a guarantee this code should rely on.

A rejected runtime is dropped rather than cached, so a retry rebuilds instead of inheriting a dead promise. And note what is NOT here: nobody terminates a shared database. The search lane's retry closes only its own connection - see the note there. */

```
function sharedDuckDBRuntime(moduleUrl) {
  const KEY = '__GRIDATLAS_DUCKDB_RUNTIME__';
  const held = window[KEY];
  if (held && held.module_url === moduleUrl) return held.promise;
  const promise = (async () => {
    const duckdb = await import(moduleUrl);
    const bundle = await duckdb.selectBundle(duckdb.getJsDelivrBundles());
    invariant(bundle?.mainModule && bundle?.mainWorker, 'DuckDB-WASM bundle
unavailable');
    const workerUrl = URL.createObjectURL(new Blob([
      `importScripts(${JSON.stringify(bundle.mainWorker)});`
    ], { type: 'text/javascript' }));
    const worker = new Worker(workerUrl);
    const logger = new duckdb.ConsoleLogger(duckdb.LogLevel.WARNING);
    const database = new duckdb.AsyncDuckDB(logger, worker);
    try {
      await database.instantiate(bundle.mainModule, bundle.pthreadWorker);
    } finally {
      URL.revokeObjectURL(workerUrl);
    }
    return { duckdb, database, worker };
  })();
  /* Assigned synchronously, before the first await inside the IIFE can
  yield, so two callers in the same task cannot both build one. */
  window[KEY] = { module_url: moduleUrl, promise, claimed_at_ms:
performance.now() };
  promise.catch(() => {
    if (window[KEY] && window[KEY].promise === promise) delete window[KEY];
  });
  return promise;
}

async function getRuntime() {
  runtimePromise ||= (async () => {
    state.duckdb_runtime_started = true;
    state.duckdb_runtime_started_at_ms = performance.now();
    const shared = await sharedDuckDBRuntime(DUCKDB_MODULE);
    state.duckdb_runtime_shared = true;
    return { database: shared.database, worker: shared.worker };
  })();
  return runtimePromise;
}

function scheduleRuntimePrewarm() {
  state.runtime_prewarm.requested = true;
  let checks = 0;
```

```

const poll = setInterval(() => {
  checks += 1;
  let criticalReady = false;
  try {
    const map = window.__GRIDATLAS_V9_MAP__;
    criticalReady = Boolean(map && map.getSource('src-400') &&
map.isSourceLoaded('src-400'));
  } catch {}
  if (!criticalReady && checks < 240) return;
  clearInterval(poll);
  const start = () => {
    state.runtime_prewarm.started = true;
    getRuntime().then(() => {
      state.runtime_prewarm.completed = true;
    }).catch(error => {
      state.runtime_prewarm.failed = String(error?.message || error);
      state.failures.push({ pathname: 'duckdb-runtime-prewarm', message:
state.runtime_prewarm.failed });
    });
  };
  if (typeof requestIdleCallback === 'function') requestIdleCallback(start,
{ timeout: 2000 });
  else setTimeout(start, 0);
}, 250);
}

queueMicrotask(scheduleRuntimePrewarm);

async function resolvePartition(pathname) {
  const stem = legacyStem(pathname);
  invariant(stem, `unsupported V8 data path: ${pathname}`);
  const alias = stem === 'uk_metros_trams' ? 'uk_metros_trams_root' : stem;
  const expected = `partitions/${alias}.parquet`.toLowerCase();
  const manifest = await getManifest();
  const artifact = (manifest.artifacts || []).find(
    item => String(item.path || '').toLowerCase() === expected
  );
  invariant(artifact, `no V9 Parquet partition for V8 source ${pathname}`);
  invariant(/^([a-f0-9]{64})$/.test(artifact.sha256 || ''), `bad partition
digest for ${artifact.path}`);
  return artifact;
}

function rowObject(row) {
  return row && typeof row.toJSON === 'function' ? row.toJSON() : row;
}

async function queryOnDemand(pathname) {
  if (sourceCache.has(pathname)) return sourceCache.get(pathname);
  const task = (async () => {
    const artifact = await resolvePartition(pathname);
    const parquetUrl = `${DATA_BASE}data/${artifact.path}`;
    const { database } = await getRuntime();
    const connection = await database.connect();
    state.parquet_requests += 1;
    try {
      const escaped = parquetUrl.replaceAll('"', '""');
      const table = await connection.query(`
SELECT source_id, feature_index, feature_id, geometry_json,
properties_json
FROM read_parquet('${escaped}')
ORDER BY feature_index

```

```

`);
const features = table.toArray().map(raw => {
  const row = rowObject(raw);
  return {
    type: 'Feature',
    id: row.feature_id || `${row.source_id}:${row.feature_index}`,
    geometry: JSON.parse(String(row.geometry_json)),
    properties: JSON.parse(String(row.properties_json || '{}'))
  };
});
state.loaded_on_demand[pathname] = {
  parquet: artifact.path,
  rows: features.length,
  sha256: artifact.sha256
};
return { type: 'FeatureCollection', features };
} finally {
  await connection.close();
}
})();
sourceCache.set(pathname, task);
try {
  return await task;
} catch (error) {
  sourceCache.delete(pathname);
  state.failures.push({ pathname, message: String(error?.message || error)
});
  throw error;
}
}

window.fetch = async function gridAtlasMaplibreWorkerFetch(input, init =
undefined) {
  const pathname = requestPath(input);
  const readyKey = mapReadyKey(pathname);

  if (MAP_READY_PATHS.has(readyKey)) {
    state.map_ready_requests += 1;
    if (readyKey === CRITICAL_RELEASE_PATH)
state.critical_source.window_fetch_hits += 1;
    return nativeFetch(input, { ...(init || {}), cache: 'force-cache' });
  }

  if (!legacyStem(pathname)) return nativeFetch(input, init);

  state.intercepted_on_demand += 1;
  if (init?.signal?.aborted) throw new DOMException('Aborted', 'AbortError');

  // Return response headers before DuckDB imports, downloads, queries and
serialises the body.
  // This preserves the native fetch contract expected by the V8 engine: its
15 s timer protects
  // response establishment, while response.json() may continue consuming a
streamed body.
  state.streamed_responses += 1;
  const encoder = new TextEncoder();
  let bodyController = null;
  let aborted = false;
  const abort = () => {
    aborted = true;
    try { bodyController?.error(new DOMException('Aborted', 'AbortError')); }
catch {}

```

```

};
init?.signal?.addEventListener('abort', abort, { once: true });
const body = new ReadableStream({
  start(controller) {
    bodyController = controller;
    queryOnDemand(pathname).then(payload => {
      if (aborted || init?.signal?.aborted) return abort();
      const serialised = JSON.stringify(payload);
      sourceCache.delete(pathname);
      state.released_payloads += 1;
      controller.enqueue(encoder.encode(serialised));
      controller.close();
    }).catch(error => {
      sourceCache.delete(pathname);
      state.stream_failures.push({ pathname, message: String(error?.message
|| error) });
      try { controller.error(error); } catch {}
    });
  },
  cancel() { aborted = true; }
});
return new Response(body, {
  status: 200,
  headers: {
    'Content-Type': 'application/geo+json; charset=utf-8',
    'Cache-Control': 'private, max-age=3600',
    'X-GridAtlas-Data-Plane': 'V9-PARQUET-DUCKDB-STREAMED-RESPONSE'
  }
});
};
})();

```

```

=====
FILE: https://ventusltd.github.io/gridatlas/atlas/cartridges/202609041945-place-
global-search-v9-5.js
34719 chars · code · shown in full · found via loaded by the browser (fetch)
=====

```

```

/**
 * place-global-search-v9-5, generation 202609041945 (UTC).
 *
 * ASSEMBLED by tools/build-cartridge.mjs from the parts below. Do not edit
 * this file: edit a part and rebuild under a new generation. Each part is
 * hashed in manifests/202609041945-place-global-search-v9-5-parts.json.
 *
 * part atlas/parts/202609040229-place-global-search-
arrival-identity.js
 */

(() => {
  'use strict';

  const PARQUET_URL =
'https://ventusltd.github.io/gridatlas/data/repd_projects_202608290716.parquet';
  const PARQUET_SHA256 =
'174040c37f3d63742d6fdd7af722a8cfd3fb53de3ff85ff1142d22fdac4866b';
  const MANIFEST_URL =
'https://ventusltd.github.io/gridatlas/data/repd_v9_manifest_202608290716.json';
  const MANIFEST_SHA256 =
'8850567ff9f1d2b6996b4e0d9707320030f3466a0b821cdcf5325322b8be8c8';
  const GEOCODER_BASE = 'https://api.postcodes.io';
  const GLOBAL_GEOCODER_URL = 'https://nominatim.openstreetmap.org/search';

```

```

const DUCKDB_MODULE = 'https://cdn.jsdelivr.net/npm/@duckdb/duckdb-wasm@1.29.0/+esm';
const FALSE_ORIGIN = Object.freeze({ latitude: 49.766807, longitude: -7.55716 });
const SOURCE_GENERATION = '202609040229';
const RUNTIME_GENERATION = String(
  document.documentElement?.dataset?.gridatlasGeneration || SOURCE_GENERATION
);

const state = {
  schema: 'gridatlas.v9-place-global-search.v5',
  generation: RUNTIME_GENERATION,
  source_generation: SOURCE_GENERATION,
  version: 'v9.106',
  geocoder: GEOCODER_BASE,
  global_geocoder: GLOBAL_GEOCODER_URL,
  geocoder_providers: ['postcodes.io', 'Nominatim / OpenStreetMap'],
  geocoder_requests: 0,
  global_geocoder_requests: 0,
  geocoder_failures: [],
  last_location_selection: null,
  parquet_url: PARQUET_URL,
  parquet_sha256: PARQUET_SHA256,
  ready: false,
  map_captured: false,
  query_count: 0,
  identity_retry_count: 0,
  last_query: '',
  last_results: [],
  last_selection: null,
  failures: [],
  deep_link: { status: 'IDLE', repd_ref: null, resolved: false, mapped: false }
};

window.__GRIDATLAS_PLACE_SEARCH__ = state;

let runtimePromise = null;
let manifestPromise = null;
let debounceTimer = null;
let activeQuerySerial = 0;

function invariant(condition, message) {
  if (!condition) throw new Error(message);
}

function arrivalCoordinator() {
  const measure = window.__GRIDATLAS_NEON_LINKS__?.measure;
  if (!measure?.arrivalGate || typeof measure.claimPendingArrival !==
'function'
    || typeof measure.invalidatePendingArrival !== 'function') return null;
  return measure;
}

function markDeepLinkCancelled(reason, epoch = null) {
  const prior = state.deep_link;
  if (!prior?.repd_ref || prior.status === 'ABSENT'
    || prior.status === 'IDLE' || prior.status === 'CANCELLED') return;
  state.deep_link = {
    ...prior,
    status: 'CANCELLED',
    resolved: false,
    mapped: false,
  }
}

```

```

        cancelled_by: String(reason || 'user-navigation'),
        cancelled_epoch: Number.isInteger(epoch) ? epoch : null
    };
    document.body.dataset.gridatlasRepdDeepLink = 'cancelled';
}

function invalidatePendingDeepLink(reason) {
    const coordinator = arrivalCoordinator();
    if (!coordinator) {
        markDeepLinkCancelled(reason);
        return null;
    }
    const epoch = coordinator.invalidatePendingArrival(reason);
    /* The shared owner dispatches this synchronously. Keep the direct call as
       a fail-soft fallback for a DOM shim without CustomEvent. */
    markDeepLinkCancelled(reason, epoch);
    return epoch;
}

window.addEventListener('gridatlas:arrival-invalidated', (event) => {
    markDeepLinkCancelled(event?.detail?.reason, event?.detail?.epoch);
});

function escapeHtml(value) {
    return String(value ?? '')
        .replace(/&/g, '&amp;')
        .replace(/</g, '&lt;')
        .replace(/>/g, '&gt;')
        .replace(/"/g, '&quot;')
        .replace(/'/g, '&#39;');
}

function sqlString(value) {
    return `${String(value).replaceAll('"', "'')}`;
}

function normaliseCompact(value) {
    return String(value ?? '').toUpperCase().replace(/[^A-Z0-9]/g, '');
}

function parseGroups(query) {
    return String(query ?? '')
        .trim()
        .toLowerCase()
        .split(/\s+/)
        .map(group => group.split('/').map(term => term.replace(/^[a-z0-9]/g, '')))
        .filter(Boolean)
        .filter(group => group.length);
}

async function sha256Hex(bytes) {
    const digest = await crypto.subtle.digest('SHA-256', bytes);
    return [...new Uint8Array(digest)].map(value => value.toString(16).padStart(2, '0')).join('');
}

async function verifyManifest() {
    manifestPromise ||= (async () => {
        const response = await fetch(MANIFEST_URL, { cache: 'no-store' });
        invariant(response.ok, `REPD manifest HTTP ${response.status}`);
        const bytes = await response.arrayBuffer();
        invariant(await sha256Hex(bytes) === MANIFEST_SHA256, 'REPD manifest SHA-256 mismatch');
        const manifest = JSON.parse(new TextDecoder().decode(bytes));
        invariant(manifest?.schema === 'gridatlas.build-manifest.v1', 'REPD

```

```

manifest schema mismatch');
    invariant(manifest?.generation === '202608290716', 'REPD generation
mismatch');
    invariant(manifest?.closure?.rows === 11069, 'REPD row closure mismatch');
    invariant(manifest?.closure?.postcodes === 9505, 'REPD postcode closure
mismatch');
    invariant(manifest?.closure?.addresses === 11059, 'REPD address closure
mismatch');
    invariant(manifest?.parquet?.sha256 === PARQUET_SHA256, 'REPD Parquet
identity mismatch');
    return manifest;
  })();
  return manifestPromise;
}

```

/* ONE DuckDB runtime for the whole page. Duplicated verbatim from 202608301825-streaming-parquet-bridge-v9-5.js, deliberately: cartridges are composed as independent scripts with no shared module loader, so there is nothing to import from. The CONTRACT is the window key and the shape it holds, not this function, and whichever cartridge asks first builds it - so neither depends on composition order.

Why it exists: this lane and the bridge each built their own runtime. Measured live at an iPhone 13 profile, 202609041500: duckdb-eh.wasm fetched twice at 5.92 MB, 11.84 MB of a 12.81 MB arrival, and two WebAssembly heaps alive at once on a device that caps per-tab memory. */

```

function sharedDuckDBRuntime(moduleUrl) {
  const KEY = '__GRIDATLAS_DUCKDB_RUNTIME__';
  const held = window[KEY];
  if (held && held.module_url === moduleUrl) return held.promise;
  const promise = (async () => {
    const duckdb = await import(moduleUrl);
    const bundle = await duckdb.selectBundle(duckdb.getJsDelivrBundles());
    invariant(bundle?.mainModule && bundle?.mainWorker, 'DuckDB-WASM bundle
unavailable');
    const workerUrl = URL.createObjectURL(new Blob([
      `importScripts(${JSON.stringify(bundle.mainWorker)});`
    ], { type: 'text/javascript' }));
    const worker = new Worker(workerUrl);
    const logger = new duckdb.ConsoleLogger(duckdb.LogLevel.WARNING);
    const database = new duckdb.AsyncDuckDB(logger, worker);
    try {
      await database.instantiate(bundle.mainModule, bundle.pthreadWorker);
    } finally {
      URL.revokeObjectURL(workerUrl);
    }
    return { duckdb, database, worker };
  })();
  window[KEY] = { module_url: moduleUrl, promise, claimed_at_ms:
performance.now() };
  promise.catch(() => {
    if (window[KEY] && window[KEY].promise === promise) delete window[KEY];
  });
  return promise;
}

```

```

async function runtime() {
  runtimePromise ||= (async () => {
    await verifyManifest();
    const shared = await sharedDuckDBRuntime(DUCKDB_MODULE);
    /* A connection each. DuckDB supports many connections on one database,
       which is what makes the sharing safe: the lanes stay independent at

```



```

        the level they actually need to be. */
        const connection = await shared.database.connect();
        return { connection, database: shared.database, worker: shared.worker,
shared: true };
    })();
    return runtimePromise;
}

async function resetOfficialRuntime() {
    const previous = runtimePromise;
    runtimePromise = null;
    manifestPromise = null;
    if (!previous) return;
    try {
        const active = await previous;
        try { await active.connection?.close?.(); } catch (_) { /* best effort */
}
        /* A SHARED database is never terminated here. Retry owns this lane's
        connection, not the page's runtime: the streaming-parquet bridge is
        serving every V8 data layer from the same database, and tearing it
        down to retry a search would take the map's data plane with it.
        Closing the connection is the whole of what this lane owns.
        The `shared` flag is set by runtime() above; the terminate path is
        kept for a runtime this lane built alone, so behaviour is unchanged
        if the broker is ever absent. */
        if (!active.shared) {
            try { await active.database?.terminate?.(); } catch (_) { /* best effort
*/ }
            try { active.worker?.terminate?.(); } catch (_) { /* best effort */ }
        }
    } catch (_) {
        // A rejected cached promise is exactly what retry is replacing.
    }
}

function installMapCapture() {
    if (!window.maplibregl?.Map ||
window.__GRIDATLAS_V9_MAP_CAPTURE_INSTALLED__) return;
    const NativeMap = window.maplibregl.Map;
    const ProxyMap = new Proxy(NativeMap, {
        construct(target, args, newTarget) {
            const instance = Reflect.construct(target, args, newTarget);
            window.__GRIDATLAS_V9_MAP__ = instance;
            state.map_captured = true;
            return instance;
        }
    });
    window.maplibregl.Map = ProxyMap;
    window.__GRIDATLAS_V9_MAP_CAPTURE_INSTALLED__ = true;
}

installMapCapture();

function buildWhere(query) {
    const groups = parseGroups(query);
    invariant(groups.length > 0, 'empty search query');
    const searchable = `lower(concat_ws(' ', coalesce(name, ''),
coalesce(repd_address_display, ''), coalesce(repd_postcode, ''),
coalesce(county, ''), coalesce(planning_authority, ''), coalesce(repd_ref, '')))`;
    const compactPostcode = `regexp_replace(upper(coalesce(repd_postcode, '')),
'[^A-Z0-9]', '', 'g')`;
    const compactRef = `regexp_replace(upper(coalesce(repd_ref, '')), '[^A-Z0-

```

```

9]', '', 'g')`;
    return groups.map(group => {
      const alternatives = [];
      for (const term of group) {
        const compact = normaliseCompact(term);
        alternatives.push(`${searchable} LIKE ${sqlString(`%${term}%`)} `);
        if (compact) {
          alternatives.push(`${compactPostcode} LIKE
${sqlString(`%${compact}%`)} `);
          alternatives.push(`${compactRef} = ${sqlString(compact)} `);
        }
      }
      return `(${alternatives.join(' OR ')} )`;
    }).join(' AND ');
  }

function buildScore(query) {
  const compact = normaliseCompact(query);
  const firstTerm = parseGroups(query).flat()[0] || '';
  const clauses = [];
  if (compact) {
    clauses.push(`CASE WHEN regexp_replace(upper(coalesce(repd_postcode, '')),
'^A-Z0-9]', '', 'g') = ${sqlString(compact)} THEN 10000 ELSE 0 END`);
    clauses.push(`CASE WHEN upper(coalesce(repd_ref, '')) =
${sqlString(compact)} THEN 9000 ELSE 0 END`);
  }
  if (firstTerm) {
    clauses.push(`CASE WHEN lower(coalesce(name, '')) = ${sqlString(firstTerm)}
THEN 2000 ELSE 0 END`);
    clauses.push(`CASE WHEN lower(coalesce(name, '')) LIKE
${sqlString(`${firstTerm}%`)} THEN 500 ELSE 0 END`);
    clauses.push(`CASE WHEN lower(coalesce(repd_address_display, '')) LIKE
${sqlString(`${firstTerm}%`)} THEN 200 ELSE 0 END`);
  }
  return clauses.length ? clauses.join(' + ') : '0';
}

function rowObject(row) {
  return row && typeof row.toJSON === 'function' ? row.toJSON() : row;
}

async function queryOfficialRepd(query, serial = null, stillOwned = null) {
  const trimmed = String(query ?? '').trim();
  if (trimmed.length < 2) return [];
  const { connection } = await runtime();
  const sql = `
SELECT
  repd_ref, name, repd_address_display, repd_postcode, county,
  planning_authority, technology, status, capacity_mw, longitude,
latitude,
  (${buildScore(trimmed)}) AS search_score
FROM read_parquet(${sqlString(PARQUET_URL)})
WHERE ${buildWhere(trimmed)}
ORDER BY search_score DESC, TRY_CAST(repd_ref AS BIGINT) ASC NULLS LAST,
capacity_mw DESC NULLS LAST, name ASC
LIMIT 25
`;
  const table = await connection.query(sql);
  if (serial !== null && serial !== activeQuerySerial) return [];
  if (stillOwned && !stillOwned()) return [];
  const results = table.toArray().map(rowObject).map(row => ({
    repd_ref: String(row.repd_ref ?? ''),

```

```

        name: String(row.name ?? ''),
        address: String(row.repd_address_display ?? ''),
        postcode: String(row.repd_postcode ?? ''),
        county: String(row.county ?? ''),
        planning_authority: String(row.planning_authority ?? ''),
        technology: String(row.technology ?? ''),
        status: String(row.status ?? ''),
        capacity_mw: row.capacity_mw == null ? null : Number(row.capacity_mw),
        longitude: row.longitude == null ? null : Number(row.longitude),
        latitude: row.latitude == null ? null : Number(row.latitude),
        score: Number(row.search_score || 0)
    ));
    state.query_count += 1;
    state.last_query = trimmed;
    state.last_results = results.slice(0, 25);
    return results;
}

function hasSafeMapPoint(result) {
    if (!Number.isFinite(result.longitude) || !Number.isFinite(result.latitude))
return false;
    if (Math.abs(result.longitude) < 1e-12 && Math.abs(result.latitude) < 1e-12)
return false;
    if (Math.abs(result.latitude - FALSE_ORIGIN.latitude) < 1e-9 &&
Math.abs(result.longitude - FALSE_ORIGIN.longitude) < 1e-9) return false;
    return result.longitude >= -180 && result.longitude <= 180 &&
result.latitude >= -90 && result.latitude <= 90;
}

function setDeepLink(result) {
    const url = new URL(window.location.href);
    url.searchParams.set('repd_ref', result.repd_ref);
    history.replaceState(history.state, '', url);
}

function selectResult(result, options = {}) {
    const deepLinkEpoch = Number.isInteger(options.deepLinkEpoch)
    ? options.deepLinkEpoch : null;
    if (deepLinkEpoch === null) {
        invalidatePendingDeepLink('project-search-selection');
    } else if (!arrivalCoordinator()?.arrivalGate.isCurrent(deepLinkEpoch)) {
        return false;
    }
    setDeepLink(result);
    const map = window.__GRIDATLAS_V9_MAP__;
    const canMap = hasSafeMapPoint(result) && map && typeof map.flyTo ===
'function';
    state.last_selection = {
        repd_ref: result.repd_ref,
        name: result.name,
        postcode: result.postcode,
        mapped: Boolean(canMap),
        longitude: result.longitude,
        latitude: result.latitude,
        // Published since 202609011141: the measurement cartridge consumes a
        // resolved identity rather than requiring the URL to restate what the
        // register already knows. This lane resolved them; it publishes them.
        technology: result.technology,
        capacity_mw: result.capacity_mw
    };
    if (!canMap) return true;
    map.flyTo({ center: [result.longitude, result.latitude], zoom: 12, duration:

```

```

1200, essential: true });
    const cap = Number.isFinite(result.capacity_mw) ?
` ${result.capacity_mw.toLocaleString('en-GB')} MW` : '';
    const location = [result.address, result.postcode,
result.county].filter(Boolean).join(' · ');
    new window.maplibregl.Popup({ maxWidth: '340px' })
        .setLngLat([result.longitude, result.latitude])
        .setHTML(`<div style="font-family:monospace;background:#000;padding:6px">
<b style="color:#00ffff;font-size:13px">${escapeHtml(result.name)}</b><br><span
style="color:#888">${escapeHtml(result.technology)}</span>${cap ? `<br><span
style="color:#ffae00">${escapeHtml(cap)}</span>` : ''}<br><span
style="color:#aaa;font-size:10px">${escapeHtml(location)}</span><br><span
style="color:#555;font-size:9px">REPD ${escapeHtml(result.repd_ref)} ·
${escapeHtml(result.status)}</span></div>`);
        .addTo(map);
    return true;
}

// ---- 202608301136 UK gazetteer lane (LOCATION_ONLY, never claims REPD
identity) ----
const FULL_POSTCODE = /^[A-Z]{1,2}[0-9][A-Z0-9]?[0-9][A-Z]{2}$/;
const OUTCODE = /^[A-Z]{1,2}[0-9][A-Z0-9]?$/;
let locationPopup = null;

async function geocoderGet(path, query) {
    state.geocoder_requests += 1;
    try {
        const response = await fetch(`${GEOCODER_BASE}${path}`, { cache: 'default'
});
        if (response.status === 404) return null;
        if (!response.ok) throw new Error(`postcodes.io ${response.status} for
${path}`);
        const body = await response.json();
        return body?.result ?? null;
    } catch (error) {
        state.geocoder_failures.push({
            query: String(query ?? ''),
            path,
            message: String(error?.message || error)
        });
        throw error;
    }
}

function joinParts(...parts) {
    return parts.filter(Boolean).join(' · ');
}

async function queryUkGazetteer(query) {
    const raw = String(query ?? '').trim();
    if (raw.length < 2) return [];
    const compact = normaliseCompact(raw);
    const out = [];
    try {
        if (FULL_POSTCODE.test(compact)) {
            const result = await
geocoderGet(`/postcodes/${encodeURIComponent(compact)}`, raw);
            if (result && Number.isFinite(result.longitude) &&
Number.isFinite(result.latitude)) {
                out.push({
                    kind: 'postcode',
                    provider: 'postcodes.io',

```

```

        label: result.postcode,
        sublabel: joinParts(result.admin_district, result.admin_county,
result.region),
        longitude: result.longitude,
        latitude: result.latitude
    });
    return out;
}
} else if (OUTCODE.test(compact)) {
    const result = await
geocoderGet(`/outcodes/${encodeURIComponent(compact)}`, raw);
    if (result && Number.isFinite(result.longitude) &&
Number.isFinite(result.latitude)) {
        out.push({
            kind: 'postcode_district',
            provider: 'postcodes.io',
            label: result.outcode,
            sublabel: joinParts(
                Array.isArray(result.admin_district) ?
result.admin_district.join(', ') : result.admin_district,
                Array.isArray(result.region) ? result.region.join(', ') :
result.region
            ),
            longitude: result.longitude,
            latitude: result.latitude
        });
        return out;
    }
}

const places = await geocoderGet(`/places?
q=${encodeURIComponent(raw)}&limit=8`, raw);
for (const place of Array.isArray(places) ? places : []) {
    if (!Number.isFinite(place.longitude) ||
!Number.isFinite(place.latitude)) continue;
    out.push({
        kind: 'place',
        provider: 'postcodes.io',
        label: place.name_1,
        sublabel: joinParts(place.local_type, place.county_unitary,
place.region),
        longitude: place.longitude,
        latitude: place.latitude
    });
}
} catch (error) {
    console.warn('[V9 UK GAZETTEER]', error);
}
return out;
}

async function queryGlobalGazetteer(query) {
    const raw = String(query ?? '').trim();
    if (raw.length < 2) return [];
    const compact = normaliseCompact(raw);
    if (FULL_POSTCODE.test(compact) || OUTCODE.test(compact)) return [];
    state.global_geocoder_requests += 1;
    try {
        const url = new URL(GLOBAL_GEOCODER_URL);
        url.searchParams.set('format', 'jsonv2');
        url.searchParams.set('limit', '8');
        url.searchParams.set('addressdetails', '1');
    }

```

```

url.searchParams.set('accept-language', 'en');
url.searchParams.set('q', raw);
const response = await fetch(url, {
  cache: 'default',
  headers: { Accept: 'application/json' }
});
if (!response.ok) throw new Error(`Nominatim ${response.status}`);
const body = await response.json();
const out = [];
for (const row of Array.isArray(body) ? body : []) {
  const longitude = Number(row.lon);
  const latitude = Number(row.lat);
  if (!Number.isFinite(longitude) || !Number.isFinite(latitude)) continue;
  const display = String(row.display_name || '').trim();
  const label = String(row.name || row.address?.city || row.address?.town
|| row.address?.village || display.split(',')[0] || row).trim();
  out.push({
    kind: 'global_place',
    provider: 'Nominatim / OpenStreetMap',
    label,
    sublabel: display || row,
    longitude,
    latitude
  });
}
return out;
} catch (error) {
  state.geocoder_failures.push({
    query: raw,
    path: GLOBAL_GEOCODER_URL,
    provider: 'Nominatim / OpenStreetMap',
    message: String(error?.message || error)
  });
  console.warn('[V9.5 GLOBAL GAZETTEER]', error);
  return [];
}
}

function dedupeGlobalLocations(ukResults, globalResults) {
  return globalResults.filter(globalResult => !ukResults.some(ukResult => {
    const sameLabel = String(globalResult.label).trim().toLowerCase() ===
String(ukResult.label).trim().toLowerCase();
    const close = Math.abs(globalResult.longitude - ukResult.longitude) < 0.03
&& Math.abs(globalResult.latitude - ukResult.latitude) < 0.03;
    return sameLabel && close;
  }));
}

function selectLocation(result) {
  invalidatePendingDeepLink('location-search-selection');
  const url = new URL(window.location.href);
  url.searchParams.delete('repd_ref');
  history.replaceState(history.state, '', url);
  const map = window.__GRIDATLAS_V9_MAP__;
  const canMap = hasSafeMapPoint(result) && map && typeof map.flyTo ===
'function';
  state.last_location_selection = { ...result, mapped: Boolean(canMap) };
  if (!canMap) return;
  const zoom = result.kind === 'postcode' ? 13 : result.kind ===
'postcode_district' ? 11 : result.kind === 'global_place' ? 12 : 12;
  const provider = result.provider || 'postcodes.io';
  map.flyTo({ center: [result.longitude, result.latitude], zoom, duration:

```

```

1200, essential: true });
    if (locationPopup) locationPopup.remove();
    locationPopup = new window.maplibregl.Popup({ maxWidth: '300px' })
        .setLngLat([result.longitude, result.latitude])
        .setHTML(`<div style="font-family:monospace;background:#000;padding:6px">
<b style="color:#00ffff;font-size:13px">${escapeHtml(result.label)}</b><br><span
style="color:#aaa;font-size:10px">${escapeHtml(result.sublabel)}</span><br><span
style="color:#555;font-size:9px">Location only · ${escapeHtml(provider)} · no
project identity claimed</span></div>`)
        .addTo(map);
    }

function renderResults(repdResults, resultsEl, ukResults = [], globalResults =
[]) {
    resultsEl.innerHTML = '';
    if (!repdResults.length && !ukResults.length && !globalResults.length) {
        const empty = document.createElement('div');
        empty.className = 'search-no-results';
        empty.textContent = 'No REPD project, postcode, address or place match';
        resultsEl.appendChild(empty);
        resultsEl.style.display = 'block';
        return;
    }

    for (const result of repdResults) {
        const item = document.createElement('div');
        item.className = 'search-result-item';
        item.dataset.repdRef = result.repd_ref;
        const location = [result.address, result.postcode,
result.county].filter(Boolean).join(' · ');
        const capacity = Number.isFinite(result.capacity_mw) ?
`${result.capacity_mw.toLocaleString('en-GB')} MW` : '';
        item.innerHTML = `<b>${escapeHtml(result.name)}</b><br>
<span>${escapeHtml(location)}</span>${capacity ? `<br><span
style="color:#ffae00">${escapeHtml(capacity)}</span>` : ''}<span
style="color:#555"> · REPD ${escapeHtml(result.repd_ref)}</span>`;
        item.addEventListener('click', event => {
            event.preventDefault();
            event.stopPropagation();
            resultsEl.style.display = 'none';
            selectResult(result);
        });
        resultsEl.appendChild(item);
    }

function renderLocationLane(title, locations) {
    if (!locations.length) return;
    const divider = document.createElement('div');
    divider.className = 'search-no-results';
    divider.textContent = title;
    resultsEl.appendChild(divider);
    for (const result of locations) {
        const item = document.createElement('div');
        item.className = 'search-result-item';
        item.dataset.locationKind = result.kind;
        item.dataset.locationProvider = result.provider || '';
        item.innerHTML = `<b>${escapeHtml(result.label)}</b><br>
<span>${escapeHtml(result.sublabel)}</span><span style="color:#555"> · fly to
only, not a REPD project</span>`;
        item.addEventListener('click', event => {
            event.preventDefault();
            event.stopPropagation();

```

```

        resultsEl.style.display = 'none';
        selectLocation(result);
    });
    resultsEl.appendChild(item);
}
}

renderLocationLane('UK location', ukResults);
renderLocationLane('Global location', globalResults);
resultsEl.style.display = 'block';
}

async function executeSearch(input, resultsEl, includeGlobal = false) {
    const serial = ++activeQuerySerial;
    const query = input.value.trim();
    if (query.length < 2) {
        resultsEl.innerHTML = '';
        resultsEl.style.display = 'none';
        return;
    }
    resultsEl.style.display = 'block';
    resultsEl.innerHTML = '<div class="search-no-results">Searching REPD
projects, postcodes, addresses and places...</div>';
    try {
        const [results, ukLocations, rawGlobalLocations] = await Promise.all([
            queryOfficialRepd(query, serial),
            queryUkGazetteer(query),
            includeGlobal ? queryGlobalGazetteer(query) : Promise.resolve([])
        ]);
        if (serial !== activeQuerySerial) return;
        const globalLocations = dedupeGlobalLocations(ukLocations,
rawGlobalLocations);
        renderResults(results, resultsEl, ukLocations, globalLocations);
    } catch (error) {
        if (serial !== activeQuerySerial) return;
        state.failures.push({ query, message: String(error?.message || error) });
        resultsEl.innerHTML = '<div class="search-no-results">Search unavailable –
V8 map remains usable</div>';
        resultsEl.style.display = 'block';
        console.error('[V9.5 PLACE SEARCH]', error);
    }
}

async function waitForCapturedMap(timeoutMs = 60000) {
    const started = performance.now();
    while (performance.now() - started < timeoutMs) {
        const map = window.__GRIDATLAS_V9_MAP__;
        if (map && typeof map.flyTo === 'function') return map;
        await new Promise(resolve => setTimeout(resolve, 50));
    }
    throw new Error('GridAtlas map was not captured for exact REPD deep link');
}

function suppliedArrivalFields(params, repdRef) {
    const numberOrNull = (name) => {
        const raw = params.get(name);
        if (raw === null || String(raw).trim() === '') return null;
        const value = Number(raw);
        return Number.isFinite(value) ? value : null;
    };
    const textOrNull = (name) => {
        const value = String(params.get(name) || '').trim();

```



```

    return value || null;
  };
  return Object.freeze({
    repd_ref: repdRef,
    name: textOrNull('project'),
    technology: textOrNull('technology'),
    capacity_mw: numberOrNull('capacity_mw'),
    longitude: numberOrNull('longitude'),
    latitude: numberOrNull('latitude'),
    // Pipeline News 0144 does not send status. Retain it only when another
    // authorised producer explicitly supplies it; never present it as an
    // official active-register value.
    supplied_status: textOrNull('status')
  });
}

async function receiveExactRepdDeepLink(input, resultsEl,
  expectedOwnerEpoch = null) {
  const params = new URLSearchParams(window.location.search);
  const repdRef = String(params.get('repd_ref') || '').trim();
  if (!repdRef) {
    state.deep_link = { status: 'ABSENT', repd_ref: null, resolved: false,
mapped: false };
    return;
  }

  const arrival = suppliedArrivalFields(params, repdRef);
  let coordinator = null;
  let ownerEpoch = null;
  try {
    invariant(/^([A-Za-z0-9-]{1,40})$/.test(repdRef), 'invalid exact REPD deep-
link identity');
    coordinator = arrivalCoordinator();
    invariant(coordinator, 'shared arrival coordinator unavailable');
    if (Number.isInteger(expectedOwnerEpoch)) {
      if (!coordinator.arrivalGate.isCurrent(expectedOwnerEpoch)) return;
      ownerEpoch = expectedOwnerEpoch;
    } else {
      ownerEpoch = coordinator.claimPendingArrival(window.location.search);
    }
    state.deep_link = {
      ...arrival, owner_epoch: ownerEpoch, status: 'RECEIVING',
      resolved: false, mapped: false,
      identity_source: 'ACTIVE_REGISTER_PENDING'
    };
    const stillOwned = () => coordinator.arrivalGate.isCurrent(ownerEpoch);
    const querySerial = ++activeQuerySerial;
    input.value = repdRef;
    const results = await queryOfficialRepd(repdRef, querySerial, stillOwned);
    if (!stillOwned() || querySerial !== activeQuerySerial) return;
    const exact = results.find(result => String(result.repd_ref) === repdRef);
    if (!exact) {
      // A successful query with no exact row is evidence about this active
      // snapshot, not a network failure and not evidence that the supplied
      // project never existed. Keep the link's point and identity separate.
      renderResults(results, resultsEl);
      document.body.dataset.gridatlasRepdRef = repdRef;
      document.body.dataset.gridatlasRepdDeepLink = 'not-in-active-register';
      state.deep_link = {
        ...arrival,
        owner_epoch: ownerEpoch,
        status: 'NOT_IN_ACTIVE_REGISTER',

```

```

        resolved: false,
        mapped: false,
        supplied_point_usable: hasSafeMapPoint(arrival),
        identity_source: 'ARRIVAL_LINK',
        official_active_register_match: false,
        statement: 'No exact identity in the active-register snapshot;
supplied arrival fields retained.'
    };
    return;
}
renderResults(results, resultsEl);
await waitForCapturedMap();
if (!stillOwned()) return;
if (!selectResult(exact, { deepLinkEpoch: ownerEpoch })) return;
if (!stillOwned()) return;
invariant(state.last_selection?.repd_ref === repdRef, 'exact REPD
selection was not retained');
invariant(state.last_selection?.mapped === true, 'exact REPD identity did
not fly to a safe map point');
document.body.dataset.gridatlasRepdRef = repdRef;
document.body.dataset.gridatlasRepdDeepLink = 'resolved';
state.deep_link = {
    status: 'RESOLVED',
    owner_epoch: ownerEpoch,
    repd_ref: repdRef,
    resolved: true,
    mapped: true,
    name: exact.name,
    postcode: exact.postcode,
    longitude: exact.longitude,
    latitude: exact.latitude,
    technology: exact.technology,
    capacity_mw: exact.capacity_mw,
    status_value: exact.status,
    identity_source: 'OFFICIAL_ACTIVE_REGISTER',
    official_active_register_match: true
};
} catch (error) {
    if (coordinator && Number.isInteger(ownerEpoch)
        && !coordinator.arrivalGate.isCurrent(ownerEpoch)) return;
    const message = String(error?.message || error);
    state.failures.push({ phase: 'exact_repd_deep_link', repd_ref: repdRef,
message });
    state.deep_link = {
        ...arrival, owner_epoch: ownerEpoch, status: 'FAILED',
        resolved: false, mapped: false, message,
        identity_source: 'ACTIVE_REGISTER_CHECK_FAILED'
    };
    document.body.dataset.gridatlasRepdDeepLink = 'failed';
    console.error('[V9 EXACT REPD DEEP LINK]', error);
}
}

async function retryExactRepdDeepLink(input, resultsEl, ownerEpoch) {
    const coordinator = arrivalCoordinator();
    invariant(coordinator, 'shared arrival coordinator unavailable for retry');
    invariant(Number.isInteger(ownerEpoch), 'identity retry requires one shared
owner epoch');
    if (!coordinator.arrivalGate.isCurrent(ownerEpoch)) return state.deep_link;
    state.identity_retry_count += 1;
    await resetOfficialRuntime();
    if (!coordinator.arrivalGate.isCurrent(ownerEpoch)) return state.deep_link;

```

```

    return receiveExactRepdDeepLink(input, resultsEl, ownerEpoch);
}

function bindSearch() {
  const input = document.getElementById('search-input');
  const button = document.getElementById('search-btn');
  const resultsEl = document.getElementById('search-results');
  invariant(input && button && resultsEl, 'V8 search controls missing');
  input.setAttribute('placeholder', 'Search project, address, postcode or
place...');
  input.setAttribute('aria-label', 'Search project, address, postcode or
place');

  input.addEventListener('input', event => {
    event.stopImmediatePropagation();
    invalidatePendingDeepLink('user-search-input');
    clearTimeout(debounceTimer);
    debounceTimer = setTimeout(() => executeSearch(input, resultsEl, false),
180);
  }, true);

  input.addEventListener('keydown', event => {
    if (event.key === 'Enter') {
      event.preventDefault();
      event.stopImmediatePropagation();
      invalidatePendingDeepLink('user-search-submit');
      clearTimeout(debounceTimer);
      executeSearch(input, resultsEl, true);
    } else if (event.key === 'Escape') {
      event.stopImmediatePropagation();
      invalidatePendingDeepLink('user-search-dismiss');
      resultsEl.style.display = 'none';
    }
  }, true);

  button.addEventListener('click', event => {
    event.preventDefault();
    event.stopImmediatePropagation();
    invalidatePendingDeepLink('user-search-submit');
    clearTimeout(debounceTimer);
    executeSearch(input, resultsEl, true);
  }, true);

  state.retry_exact_deep_link = (ownerEpoch) =>
    retryExactRepdDeepLink(input, resultsEl, ownerEpoch);
  state.ready = true;
}

window.addEventListener('DOMContentLoaded', () => {
  try {
    bindSearch();
    const input = document.getElementById('search-input');
    const resultsEl = document.getElementById('search-results');
    void receiveExactRepdDeepLink(input, resultsEl);
  } catch (error) {
    state.failures.push({ phase: 'bind', message: String(error?.message ||
error) });
    console.error('[V9 PLACE SEARCH INIT]', error);
  }
}, { once: true });
})();

```

```

=====
FILE: https://ventusltd.github.io/gridatlas/atlas/cartridges/202609051510-
substation-intelligence-v9-63.js
422014 chars · code · shown in full · found via loaded by the browser (fetch)
=====
/**
 * substation-intelligence-v9-63, generation 202609051510 (UTC).
 *
 * ASSEMBLED by tools/build-cartridge.mjs from the parts below. Do not edit
 * this file: edit a part and rebuild under a new generation. Each part is
 * hashed in manifests/202609051510-substation-intelligence-v9-63-parts.json.
 *
 *   carried_shell_script    atlas/parts/202609040229-ventus-corev8engine-exact-
repd-delegation.js
 *   module                  atlas/modules/202609011950-geodesy.js
 *   module                  atlas/modules/202609012245-network-topology.js
 *   module                  atlas/modules/202609012245-electrical-distance.js
 *   module                  atlas/modules/202609012250-rating-envelope.js
 *   module                  atlas/modules/202609020015-injection-response.js
 *   module                  atlas/modules/202609012345-planned-change.js
 *   module                  atlas/modules/202609012350-owner-boundary.js
 *   module                  atlas/modules/202609030137-pinned-products.js
 *   module                  atlas/modules/202609030157-version-ledger.js
 *   module                  atlas/modules/202609030205-corridor-estimate.js
 *   module                  atlas/modules/202609031958-menu-bar.js
 *   module                  atlas/modules/202609032220-arrival-tidy.js
 *   module                  atlas/modules/202609030048-pipeline-news-layers.js
 *   module                  atlas/modules/202609040400-sld-styles.js
 *   module                  atlas/modules/202609051510-teleprint-controls.js
 *   part                    atlas/parts/202609012350-substation-intelligence-
body.js
 */

'use strict';

window.initVentusMap = function({ config, center, zoom }) {
  if (typeof maplibregl === 'undefined') {
    document.getElementById('fatal-banner').style.display = 'block';
    throw new Error('CRITICAL: MapLibre failed to load.');
```

```

  }

  // — Utilities

  function deepFreeze(obj) {
    Object.keys(obj).forEach(prop => {
      if (typeof obj[prop] === 'object' && obj[prop] !== null)
        deepFreeze(obj[prop]);
    });
    return Object.freeze(obj);
  }

  function escapeHTML(value) {
    return String(value ?? '')
      .replace(/&/g, '&amp;')
      .replace(/</g, '&lt;')
      .replace(/>/g, '&gt;')
      .replace(/"/g, '&quot;')
      .replace(/'/g, '&#39;');
  }

  function normalizeStatus(status) {
    return String(status ?? '').trim().toLowerCase();
  }

```

```

function fmt(n, decimals) {
  return n.toLocaleString('en-GB', { maximumFractionDigits: decimals,
minimumFractionDigits: decimals });
}

// — Earth model

```

```

const EARTH_RADIUS_KM = 6378.137;
const MAX_RADIUS_KM = Math.PI * EARTH_RADIUS_KM; // 20037.508 km

// — V5.1: Named constants

```

```

const DEG_TO_RAD = Math.PI / 180;
const HIT_RADIUS_VERTEX_PX = 18;
const HIT_RADIUS_EDGE_PX = 22;
const CLICK_DEBOUNCE_MS = 220;
const HOVER_THROTTLE_MS = 100;
const POPUP_MAX_WIDTH = '300px';
const ZONE_DRAW_VERTICES = 24;
const ZONE_DRAW_DEFAULT_KM = 0.337;

function haversine(lon1, lat1, lon2, lat2) {
  const R = EARTH_RADIUS_KM, r = Math.PI / 180;
  const dLat = (lat2 - lat1) * r, dLon = (lon2 - lon1) * r;
  const a = Math.sin(dLat / 2) ** 2 + Math.cos(lat1 * r) * Math.cos(lat2 *
r) * Math.sin(dLon / 2) ** 2;
  return R * 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
}

// — Config Loading

```

```

const GRID_CONFIG = deepFreeze(config);
const RUNTIME_STATE = {};
GRID_CONFIG.forEach(group => {
  group.layers.forEach(layer => {
    RUNTIME_STATE[layer.id] = { status: 'WAIT', loading: false, loaded:
false };
  });
});

const layerConfigById = new Map(
  GRID_CONFIG.flatMap(g => g.layers).map(l => [l.id, l])
);

// Removed naei_co2 from here so it gets its own dedicated source
const REPD_IDS =
['solar', 'solar_operational', 'solar_roof', 'wind', 'wind_onshore_operational', 'win
d_offshore_operational', 'bess', 'bess_operational', 'biomass', 'tidal', 'hydrogen', '
hydro', 'flywheel', 'act', 'geothermal', 'caes'];
const TRANSIT_IDS = ['elizabeth', 'lu', 'dlr', 'metro', 'tram', 'hs2'];
const TRANSIT_SOURCE_MAP = { 'elizabeth': 'src-elizabeth', 'lu': 'src-
lu', 'dlr': 'src-metros', 'metro': 'src-metros', 'tram': 'src-metros', 'hs2': 'src-hs2'
};
const TRANSIT_URLS = { 'src-elizabeth': '/elizabeth_line.geojson', 'src-
lu': '/london_underground.geojson', 'src-metros': '/uk_metros_trams.geojson', 'src-
hs2': '/hs2.geojson' };

const SEARCH_THRESHOLD = {
  'solar': 50, 'solar_roof': 0.5, 'wind': 50, 'bess': 50, 'biomass': 50,
  'tidal': 10, 'hydrogen': 10, 'hydro': 10, 'flywheel': 1, 'act': 10, 'geothermal': 1, 'caes':

```

```

1
    };

    const TECH_TERMS = new Map([
        ['solar', 'solar farm'], ['solar_roof', 'rooftop solar'], ['wind', 'wind
farm'],
        ['bess', 'battery storage'], ['biomass', 'biomass plant'], ['tidal', 'tidal
energy'],
        ['hydrogen', 'hydrogen plant'], ['hydro', 'hydro power'],
        ['flywheel', 'flywheel storage'],
        ['act', 'advanced conversion energy'], ['geothermal', 'geothermal energy'],
        ['caes', 'compressed air energy storage']
    ]);

    const TECH_COLOURS = new Map([
        ['solar', '#ffff00'], ['solar_roof', '#ffcc00'], ['wind', '#00ffff'],
        ['bess', '#ffae00'],
        ['biomass', '#39ff14'], ['tidal', '#00bfff'], ['hydrogen', '#ffffff'],
        ['hydro', '#00aaff'],
        ['flywheel', '#ff69b4'], ['act', '#ff6600'], ['geothermal', '#ff3300'],
        ['caes', '#88aaff']
    ]);

    const STATUS_COLOURS = {
        'operational': '#00ff88', 'under construction': '#ffcc00', 'awaiting
construction': '#ffaa00',
        'consented': '#ff8800', 'planning permission granted': '#ff8800', 'planning
approved': '#ff8800',
        'application submitted': '#8888ff', 'pre-construction': '#aaaaff'
    };

    let statusMode = false;
    let radiusMode = false;
    let radiusMarker = null;
    let radiusCenter = null;

    let radiusAreaMode = false;
    let radiusAreaMarker = null;
    let radiusAreaCenter = null;

    // — ZONE DRAW STATE

    const ZONE_DRAW_MAX_KM = MAX_RADIUS_KM;
    let zoneDrawMode = false;
    let zoneDrawPoints = [];
    let zoneDrawDragging = false;
    let zoneDrawDragIdx = -1;
    let zoneDrawJustDragged = false;
    let _zoneDrawCollapsed = false;

    function _zoneDrawGetRadius() {
        const input = document.getElementById('zonedraw-radius-input');
        if (!input) return ZONE_DRAW_DEFAULT_KM;
        const v = parseFloat(input.value);
        if (isNaN(v) || v <= 0) return ZONE_DRAW_DEFAULT_KM;
        if (v > ZONE_DRAW_MAX_KM) return ZONE_DRAW_MAX_KM;
        return v;
    }

    function _zoneDrawCirclePoints(lon, lat, radiusKm, n) {
        const R = EARTH_RADIUS_KM, DEG = Math.PI / 180;
        const ad = radiusKm / R;

```

```

    const lat1 = lat * DEG;
    return Array.from({ length: n }, (_, i) => {
      const b = (i / n) * 2 * Math.PI;
      const lat2 = Math.asin(Math.sin(lat1) * Math.cos(ad) +
Math.cos(lat1) * Math.sin(ad) * Math.cos(b));
      const lon2 = lon * DEG + Math.atan2(Math.sin(b) * Math.sin(ad) *
Math.cos(lat1), Math.cos(ad) - Math.sin(lat1) * Math.sin(lat2));
      return [lon2 / DEG, lat2 / DEG];
    });
  }

  function _zoneDrawCalcArea(pts) {
    if (pts.length < 3) return { areaKm2: 0, areaHa: 0, areaAc: 0, areaMi2:
0, areaM2: 0, perimKm: 0, pitches: 0 };
    let area = 0;
    const R = EARTH_RADIUS_KM;
    for (let i = 0; i < pts.length; i++) {
      const j = (i + 1) % pts.length;
      const xi = pts[i][0] * Math.PI / 180, yi = pts[i][1] * Math.PI /
180;
      const xj = pts[j][0] * Math.PI / 180, yj = pts[j][1] * Math.PI /
180;
      area += (xj - xi) * (2 + Math.sin(yi) + Math.sin(yj));
    }
    const areaKm2 = Math.abs(area) * R * R / 2;
    let perimKm = 0;
    for (let i = 0; i < pts.length; i++) perimKm += haversine(pts[i][0],
pts[i][1], pts[(i+1)%pts.length][0], pts[(i+1)%pts.length][1]);
    const areaM2 = areaKm2 * 1e6;
    return { areaKm2, areaHa: areaM2 / 10000, areaAc: areaM2 / 4046.85642,
areaMi2: areaKm2 * 0.386102, areaM2, perimKm, pitches: areaM2 / 7140 };
  }

  function _zoneDrawUpdateLayers(dragOnly) {
    if (!map.getSource('src-zonedraw-fill')) return;
    const n = zoneDrawPoints.length;
    if (n < 3) {
      ['fill', 'line', 'points'].forEach(s => map.getSource(`src-zonedraw-
${s}`).setData({ type: 'FeatureCollection', features: [] }));
      return;
    }
    const ring = [...zoneDrawPoints, zoneDrawPoints[0]];
    map.getSource('src-zonedraw-fill').setData({ type: 'FeatureCollection',
features: [{ type: 'Feature', geometry: { type: 'Polygon', coordinates: [ring] }
}] });
    map.getSource('src-zonedraw-line').setData({ type: 'FeatureCollection',
features: [{ type: 'Feature', geometry: { type: 'LineString', coordinates: ring
} }] });
    if (dragOnly) {
      map.getSource('src-zonedraw-points').setData({ type:
'FeatureCollection', features:
        zoneDrawPoints.map((c, i) => ({ type: 'Feature', properties: {
kind: 'vertex', idx: i }, geometry: { type: 'Point', coordinates: c } }));
    });
    } else {
      const vFeatures = zoneDrawPoints.map((c, i) => ({ type: 'Feature',
properties: { kind: 'vertex', idx: i }, geometry: { type: 'Point', coordinates:
c } }));
      const mFeatures = [];
      zoneDrawPoints.forEach((c, i) => {
        const b = zoneDrawPoints[(i + 1) % n];
        [0.33, 0.5, 0.66].forEach(t => {

```

```

        mFeatures.push({ type: 'Feature', properties: { kind: 'mid',
edgeIdx: i, t }, geometry: { type: 'Point', coordinates: [c[0]+(b[0]-c[0])*t,
c[1]+(b[1]-c[1])*t] } });
    });
    });
    map.getSource('src-zonedraw-points').setData({ type:
'FeatureCollection', features: [...vFeatures, ...mFeatures] });
    }

    let _zoneDrawPopupRaf = null;

    function _zoneDrawShowPopup() {
        if (zoneDrawPoints.length < 3) return;
        const { areaKm2, areaHa, areaAc, areaMi2, areaM2, perimKm, pitches } =
_zoneDrawCalcArea(zoneDrawPoints);
        const centLon = zoneDrawPoints.reduce((s, p) => s + p[0], 0) /
zoneDrawPoints.length;
        const centLat = zoneDrawPoints.reduce((s, p) => s + p[1], 0) /
zoneDrawPoints.length;
        if (_zoneDrawCollapsed) {
            openPopup([centLon, centLat], `
                <div onclick="window._zdExpand&&window._zdExpand()" style="font-
family:monospace;background:#000;padding:5px 10px;border:1px solid
#fff6600;border-radius:4px;cursor:pointer;color:#fff6600;font-size:11px;white-
space:nowrap;">
                    ● ${fmt(areaKm2,3)} km² · 🏈 ${fmt(pitches,0)} pitches
&nbsp;▼
                </div>`);
            window._zdExpand = () => { _zoneDrawCollapsed = false;
_zoneDrawShowPopup(); };
        } else {
            openPopup([centLon, centLat], `
                <div style="font-family:monospace;background:#000;padding:10px
12px;border:1px solid #fff6600;border-radius:4px;min-width:230px;">
                    <div style="display:flex;justify-content:space-
between;align-items:center;margin-bottom:8px;">
                        <b style="color:#fff6600;font-size:13px;">● Zone Draw</b>
                        <span onclick="window._zdCollapse&&window._zdCollapse()"
style="color:#555;font-size:12px;cursor:pointer;padding:0 4px;user-select:none;"
title="Collapse">▲ hide</span>
                    </div>
                    <div style="color:#ffae00;font-size:13px;margin-
bottom:10px;">🏈 ${fmt(pitches,1)} football pitches</div>
                    <div style="display:grid;grid-template-columns:auto
1fr;gap:4px 14px;font-size:12px;">
                        <span style="color:#888;">Square Metres</span><span
style="color:#fff;">${fmt(areaM2,0)}</span>
                        <span style="color:#888;">Hectares</span><span
style="color:#fff;">${fmt(areaHa,2)}</span>
                        <span style="color:#888;">Acres</span><span
style="color:#fff;">${fmt(areaAc,2)}</span>
                        <span style="color:#888;">Square Kilometres</span><span
style="color:#fff;">${fmt(areaKm2,4)}</span>
                        <span style="color:#888;">Square Miles</span><span
style="color:#fff;">${fmt(areaMi2,3)}</span>
                        <span style="color:#888;">Perimeter</span><span
style="color:#fff;">${fmt(perimKm,2)} km</span>
                    </div>
                    <div style="color:#555;font-size:10px;margin-top:8px;line-
height:1.4;">

```



```

        <b style="color:#ff6600;">HOW TO USE:</b><br>
        • <b>Drag orange dots</b> to reshape polygon<br>
        • <b>Click light dots</b> on edges to add points<br>
        • <b>Undo button</b> in top-left removes last point<br>
        • <b>Click elsewhere</b> to start new zone
    </div>
</div>`);
    window._zdCollapse = () => { _zoneDrawCollapsed = true;
_zoneDrawShowPopup(); };
    }

function _zoneDrawShowPopupDebounce() {
    if (_zoneDrawPopupRaf) return;
    _zoneDrawPopupRaf = requestAnimationFrame(() => { _zoneDrawPopupRaf =
null; _zoneDrawShowPopup(); });
}

function _zoneDrawClear() {
    zoneDrawPoints = [];
    zoneDrawDragging = false;
    zoneDrawDragIdx = -1;
    zoneDrawJustDragged = false;
    _zoneDrawCollapsed = false;
    window._zdExpand = null;
    window._zdCollapse = null;
    closeActivePopup();
    _zoneDrawUpdateLayers(false);
    const el = document.getElementById('zonedraw-display');
    if (el) el.style.display = 'none';
}

function zoneDrawUndo() {
    if (zoneDrawPoints.length <= 3) { _zoneDrawClear(); return; }
    zoneDrawPoints.pop();
    _zoneDrawUpdateLayers(false);
    _zoneDrawShowPopup();
}

function toggleZoneDrawMode() {
    zoneDrawMode = !zoneDrawMode;
    const btn = document.getElementById('btn-zonedraw');
    if (btn) { btn.classList.toggle('active', zoneDrawMode);
btn.setAttribute('aria-pressed', zoneDrawMode); }
    map.getCanvas().style.cursor = zoneDrawMode ? 'crosshair' : '';
    if (zoneDrawMode) {
        if (radiusMode) toggleRadiusMode();
        if (radiusAreaMode) toggleRadiusAreaMode();
        if (measureMode) toggleMeasureMode();
        const el = document.getElementById('zonedraw-display');
        if (el) el.style.display = 'block';
    } else {
        _zoneDrawClear();
    }
}

function _zoneDrawNearVertex(px) {
    for (let i = 0; i < zoneDrawPoints.length; i++) {
        const vpx = map.project(zoneDrawPoints[i]);
        const dx = px.x - vpx.x, dy = px.y - vpx.y;
        if (Math.sqrt(dx*dx + dy*dy) < HIT_RADIUS_VERTEX_PX) return i;
    }
}

```

```

    return -1;
}

function _zoneDrawNearEdgeDot(px) {
    for (let i = 0; i < zoneDrawPoints.length; i++) {
        const j = (i + 1) % zoneDrawPoints.length;
        const a = zoneDrawPoints[i], b = zoneDrawPoints[j];
        for (const t of [0.33, 0.5, 0.66]) {
            const dot = [a[0]+(b[0]-a[0])*t, a[1]+(b[1]-a[1])*t];
            const dpx = map.project(dot);
            const dx = px.x - dpx.x, dy = px.y - dpx.y;
            if (Math.sqrt(dx*dx + dy*dy) < HIT_RADIUS_EDGE_PX) return {
insertIdx: j, dot };
        }
    }
    return null;
}

function _zoneDrawOnClick(e) {
    if (zoneDrawDragging) return;
    if (zoneDrawJustDragged) { zoneDrawJustDragged = false; return; }
    const lon = e.lngLat.lng, lat = e.lngLat.lat;

    if (zoneDrawPoints.length === 0) {
        const km = _zoneDrawGetRadius();
        zoneDrawPoints = _zoneDrawCirclePoints(lon, lat, km,
ZONE_DRAW_VERTICES);
        _zoneDrawCollapsed = false;

        const mpp = (km * 2000) / (window.innerWidth * 0.6);
        const lat1 = lat * Math.PI / 180;
        const targetZoom = Math.log2(156543 * Math.cos(lat1) / mpp);
        const clampedZoom = Math.max(8, Math.min(19, targetZoom));
        map.easeTo({ center: [lon, lat], zoom: clampedZoom, duration: 600
});

        _zoneDrawUpdateLayers(false); _zoneDrawShowPopup();
        return;
    }

    const px = map.project([lon, lat]);
    if (_zoneDrawNearVertex(px) >= 0) return;

    const edgeHit = _zoneDrawNearEdgeDot(px);
    if (edgeHit) {
        zoneDrawPoints.splice(edgeHit.insertIdx, 0, [edgeHit.dot[0],
edgeHit.dot[1]]);
        _zoneDrawUpdateLayers(false); _zoneDrawShowPopup();
        return;
    }

    const km = _zoneDrawGetRadius();
    zoneDrawPoints = _zoneDrawCirclePoints(lon, lat, km,
ZONE_DRAW_VERTICES);
    _zoneDrawCollapsed = false;
    const mpp = (km * 2000) / (window.innerWidth * 0.6);
    const lat1 = lat * Math.PI / 180;
    const targetZoom = Math.log2(156543 * Math.cos(lat1) / mpp);
    map.easeTo({ center: [lon, lat], zoom: Math.max(8, Math.min(19,
targetZoom)), duration: 600 });
    _zoneDrawUpdateLayers(false); _zoneDrawShowPopup();
}

```

```

function _zoneDrawOnMouseDown(e) {
  if (!zoneDrawMode || zoneDrawPoints.length < 3) return;
  const px = map.project(e.lngLat);
  if (_zoneDrawNearEdgeDot(px)) return;
  const vi = _zoneDrawNearVertex(px);
  if (vi >= 0) {
    zoneDrawDragging = true; zoneDrawDragIdx = vi;
    map.dragPan.disable();
    map.getCanvas().style.cursor = 'grabbing';
    e.preventDefault();
  }
}

function _zoneDrawOnMouseMove(e) {
  if (!zoneDrawMode || zoneDrawPoints.length < 3) return;
  if (zoneDrawDragging && zoneDrawDragIdx >= 0) {
    zoneDrawPoints[zoneDrawDragIdx] = [e.lngLat.lng, e.lngLat.lat];
    _zoneDrawUpdateLayers(true);
    _zoneDrawShowPopupDebounce();
    return;
  }
  const px = map.project(e.lngLat);
  const vi = _zoneDrawNearVertex(px);
  const edgeHit = vi < 0 ? _zoneDrawNearEdgeDot(px) : null;
  map.getCanvas().style.cursor = vi >= 0 ? 'grab' : (edgeHit ? 'copy' :
'crosshair');
}

function _zoneDrawOnMouseUp() {
  if (!zoneDrawDragging) return;
  zoneDrawDragging = false;
  zoneDrawDragIdx = -1;
  zoneDrawJustDragged = true;
  map.dragPan.enable();
  map.getCanvas().style.cursor = 'crosshair';
  _zoneDrawUpdateLayers(false);
  _zoneDrawShowPopup();
  setTimeout(() => { zoneDrawJustDragged = false; }, 50);
}

const urlCache = {};
let globalSubsData = null;
let allREPDFeatures = [];
let searchIndex = [];

// — Single popup instance — prevents accumulation

let activePopup = null;
function openPopup(lngLat, html, maxWidth) {
  if (activePopup) { activePopup.remove(); activePopup = null; }
  activePopup = new maplibregl.Popup({ maxWidth: maxWidth ||
POPUP_MAX_WIDTH })
    .setLngLat(lngLat)
    .setHTML(html)
    .addTo(map);
  activePopup.on('close', () => { activePopup = null; });
  return activePopup;
}
function closeActivePopup() {
  if (activePopup) { activePopup.remove(); activePopup = null; }
}

```

```

}
window._closePopupKeepShape = () => closeActivePopup();

// — Fullscreen

```

```

let fsActive = false;
let curtainOpen = false;

window.enterFullscreen = function() {
  fsActive = true;
  document.body.classList.add('fs-active');
  document.documentElement.classList.add('fs-active');
  document.getElementById('map-container').classList.add('is-fullscreen');
  document.getElementById('btn-fullscreen').style.display = 'none';
  const el = document.getElementById('map-container');
  if (el.requestFullscreen) { el.requestFullscreen().catch(() => {}); }
  else if (el.webkitRequestFullscreen) { el.webkitRequestFullscreen(); }
  setTimeout(() => map.resize(), 50);
};

window.exitFullscreen = function() {
  fsActive = false;
  curtainOpen = false;
  document.body.classList.remove('fs-active');
  document.documentElement.classList.remove('fs-active');
  document.getElementById('map-container').classList.remove('is-
fullscreen');
  document.getElementById('btn-fullscreen').style.display = '';
  document.getElementById('fs-curtain').classList.remove('curtain-open');
  if (document.fullscreenElement || document.webkitFullscreenElement) {
    if (document.exitFullscreen) document.exitFullscreen().catch(() =>
{});
    else if (document.webkitExitFullscreen)
document.webkitExitFullscreen();
  }
  setTimeout(() => map.resize(), 50);
};

function toggleCurtain() {
  curtainOpen = !curtainOpen;
  const curtain = document.getElementById('fs-curtain');
  const tab = document.getElementById('fs-curtain-tab');
  curtain.classList.toggle('curtain-open', curtainOpen);
  tab.innerText = curtainOpen ? '⬆️ Close' : '⬇️ Layers';
}

document.addEventListener('fullscreenchange', () => { if
(!document.fullscreenElement && fsActive) exitFullscreen(); });
document.addEventListener('webkitfullscreenchange', () => { if
(!document.webkitFullscreenElement && fsActive) exitFullscreen(); });

// — Radius Tool

```

```

const RADIUS_MIN = 1;
const RADIUS_MAX = MAX_RADIUS_KM;

function getRadiusValue() {
  const raw = parseFloat(document.getElementById('radius-input').value);
  if (isNaN(raw) || raw < RADIUS_MIN) return RADIUS_MIN;
  if (raw > RADIUS_MAX) return RADIUS_MAX;
  return raw;
}

```

```

}

function validateRadiusInput() {
  const input = document.getElementById('radius-input');
  const raw = parseFloat(input.value);
  const invalid = isNaN(raw) || raw < RADIUS_MIN || raw > RADIUS_MAX;
  input.classList.toggle('invalid', invalid);
  return !invalid;
}

// — Measure Tool

```

```

let measureMode = false;
let measurePoints = [];
let measureClosed = false;
let _lastMouseMoveRaf = null;

function updateMeasureDisplay() {
  const lineEl = document.getElementById('m-line');
  const perimEl = document.getElementById('m-perim');
  const areaEl = document.getElementById('m-area');
  const hint = document.getElementById('m-hint');
  const undoBtn = document.getElementById('btn-measure-undo');

  undoBtn.style.display = (measurePoints.length > 0 && !measureClosed) ?
'inline-block' : 'none';

  if (measurePoints.length < 2) {
    lineEl.style.display = 'none'; perimEl.style.display = 'none';
areaEl.style.display = 'none';
    hint.innerText = 'Click to add points · Double-click to close
polygon';
    return;
  }

  let totalKm = 0;
  for (let i = 1; i < measurePoints.length; i++) {
    totalKm += haversine(measurePoints[i-1][0], measurePoints[i-1][1],
measurePoints[i][0], measurePoints[i][1]);
  }

  if (!measureClosed) {
    lineEl.style.display = 'block'; perimEl.style.display = 'none';
areaEl.style.display = 'none';
    document.getElementById('m-km').innerText = fmt(totalKm, 2);
    document.getElementById('m-m').innerText = fmt(totalKm * 1000, 0);
    document.getElementById('m-mi').innerText = fmt(totalKm * 0.621371,
2);
    hint.innerText = 'Double-click last point to close polygon';
  } else {
    const closingKm = haversine(measurePoints[measurePoints.length-1]
[0], measurePoints[measurePoints.length-1][1], measurePoints[0][0],
measurePoints[0][1]);
    const perimKm = totalKm + closingKm;
    let area = 0;
    const R = EARTH_RADIUS_KM;
    for (let i = 0; i < measurePoints.length; i++) {
      const j = (i + 1) % measurePoints.length;
      const xi = measurePoints[i][0] * Math.PI / 180; const yi =
measurePoints[i][1] * Math.PI / 180;
      const xj = measurePoints[j][0] * Math.PI / 180; const yj =
measurePoints[j][1] * Math.PI / 180;

```

```

        area += (xj - xi) * (2 + Math.sin(yi) + Math.sin(yj));
    }
    const areaKm2 = Math.abs(area) * R * R / 2;
    const areaHa = areaKm2 * 100; const areaAc = areaKm2 * 247.105;

    lineEl.style.display = 'none'; perimEl.style.display = 'block';
    areaEl.style.display = 'block';
    document.getElementById('m-pkm').innerText = fmt(perimKm, 2);
    document.getElementById('m-pm').innerText = fmt(perimKm * 1000, 0);
    document.getElementById('m-km2').innerText = fmt(areaKm2, 3);
    document.getElementById('m-ha').innerText = fmt(areaHa, 1);
    document.getElementById('m-ac').innerText = fmt(areaAc, 1);
    hint.innerText = 'Click  Measure again to reset';
}
}

function updateMeasureLayers() {
    if (!map.getSource('src-measure-line')) return;
    const lineCoords = [...measurePoints];
    if (measureClosed && measurePoints.length > 2)
lineCoords.push(measurePoints[0]);
    map.getSource('src-measure-line').setData({ type: 'FeatureCollection',
features: lineCoords.length > 1 ? [{ type: 'Feature', geometry: { type:
'LineString', coordinates: lineCoords } }] : [] });
    map.getSource('src-measure-fill').setData({ type: 'FeatureCollection',
features: measureClosed && measurePoints.length > 2 ? [{ type: 'Feature',
geometry: { type: 'Polygon', coordinates: [...measurePoints, measurePoints[0]]
} }] : [] });
    map.getSource('src-measure-points').setData({ type: 'FeatureCollection',
features: measurePoints.map(c => ({ type: 'Feature', geometry: { type: 'Point',
coordinates: c } }));
}

function clearMeasure() {
    measurePoints = []; measureClosed = false; updateMeasureLayers();
updateMeasureDisplay();
    document.getElementById('measure-display').style.display = 'none';
}

function undoLastMeasurePoint() {
    if (measurePoints.length === 0 || measureClosed) return;
    measurePoints.pop(); updateMeasureLayers(); updateMeasureDisplay();
}

function toggleMeasureMode() {
    measureMode = !measureMode;
    const btn = document.getElementById('btn-measure');
    btn.classList.toggle('active', measureMode); btn.setAttribute('aria-
pressed', measureMode);
    map.getCanvas().style.cursor = measureMode ? 'crosshair' : '';
    if (!measureMode) { clearMeasure(); } else {
        if (radiusMode) toggleRadiusMode();
        if (radiusAreaMode) toggleRadiusAreaMode();
        if (zoneDrawMode) toggleZoneDrawMode();
        document.getElementById('measure-display').style.display = 'block';
updateMeasureDisplay();
    }
}

// — Radius Area Tool

```

```

function toggleRadiusAreaMode() {
    radiusAreaMode = !radiusAreaMode;
    const btn = document.getElementById('btn-radius-area');
    if(btn) {
        btn.classList.toggle('active', radiusAreaMode);
        btn.setAttribute('aria-pressed', radiusAreaMode);
    }
    const popupEl = document.getElementById('radius-area-popup');
    if(popupEl) popupEl.style.display = radiusAreaMode ? 'block' : 'none';

    map.getCanvas().style.cursor = radiusAreaMode ? 'crosshair' : '';

    if (radiusAreaMode && radiusMode) toggleRadiusMode();
    if (radiusAreaMode && measureMode) toggleMeasureMode();
    if (radiusAreaMode && zoneDrawMode) toggleZoneDrawMode();

    if (!radiusAreaMode) {
        if(map.getSource('src-radius-area')) {
            map.getSource('src-radius-area').setData({ type:
'FeatureCollection', features: [] });
        }
        radiusAreaCenter = null;
        if (radiusAreaMarker) { radiusAreaMarker.remove(); radiusAreaMarker
= null; }
        closeActivePopup();
    }
}

function doRadiusAreaMeasure(lon, lat) {
    const input = document.getElementById('radius-area-input');
    if(!input) return;
    const km = parseFloat(input.value);
    if (isNaN(km) || km <= 0 || km > MAX_RADIUS_KM) {
        input.classList.add('invalid');
        return;
    }
    input.classList.remove('invalid');
    radiusAreaCenter = { lon, lat };

    if(map.getSource('src-radius-area')) {
        map.getSource('src-radius-area').setData(createGeoJSONCircle(lon,
lat, km));
    }
    if (radiusAreaMarker) radiusAreaMarker.remove(); radiusAreaMarker =
null;

    // Calculate Geodesic Spherical Cap Area
    const R = EARTH_RADIUS_KM;
    const areaKm2 = 2 * Math.PI * R * R * (1 - Math.cos(km / R));
    const areaM2 = areaKm2 * 1000000;
    const areaHa = areaM2 / 10000;
    const areaAc = areaM2 / 4046.85642;
    const areaMi2 = areaKm2 * 0.386102;
    const pitches = areaM2 / 7140;

    openPopup([lon, lat], `
        <div style="font-family:monospace;background:#000;padding:10px
12px;border:1px solid #ff00ff;border-radius:4px;min-
width:220px;position:relative;">
            <div style="display:flex;justify-content:space-between;align-
items:center;margin-bottom:8px;">
                <b style="color:#ff00ff;font-size:13px;">⊙ ${km}km

```

```

radius</b>
        <span onclick="window._closePopupKeepShape()"
style="color:#555;font-size:14px;cursor:pointer;line-height:1;padding:0
2px;user-select:none;" title="Close popup, keep circle">x</span>
        </div>
        <div style="color:#ffae00;font-size:13px;margin-bottom:10px;">⚽
${fmt(pitches, 1)} football pitches</div>
        <div style="display:grid;grid-template-columns:auto 1fr;gap:4px
14px;font-size:12px;">
            <span style="color:#888;">Square Metres</span><span
style="color:#fff;">${fmt(areaM2, 0)}</span>
            <span style="color:#888;">Hectares</span><span
style="color:#fff;">${fmt(areaHa, 2)}</span>
            <span style="color:#888;">Acres</span><span
style="color:#fff;">${fmt(areaAc, 2)}</span>
            <span style="color:#888;">Square Kilometres</span><span
style="color:#fff;">${fmt(areaKm2, 3)}</span>
            <span style="color:#888;">Square Miles</span><span
style="color:#fff;">${fmt(areaMi2, 3)}</span>
        </div>`);
    }

    // — Clock

    setInterval(() => {
        const now = new Date();
        const target = new Date(Date.UTC(2050, 0, 1, 0, 0, 0));
        document.getElementById('clock').innerText = now.toLocaleTimeString('en-
GB');
        document.getElementById('date').innerText = now.toLocaleDateString('en-
GB');
        document.getElementById('days').innerText = Math.floor((target - now) /
86400000) + ' DAYS';
    }, 1000);

    // — Map Init

    const map = new maplibregl.Map({
        container: 'map',
        style: 'https://basemaps.cartocdn.com/gl/dark-matter-gl-
style/style.json',
        center: center,
        zoom: zoom,
        attributionControl: false
    });

    new ResizeObserver(() =>
map.resize()).observe(document.getElementById('map'));

    // — UI State

    function updateUIState(id, state, stats) {
        RUNTIME_STATE[id].status = state;
        ['lbl-', 'fs-lbl-'].forEach(prefix => {
            const span = document.getElementById(`_${prefix}${id}`);
            if (span) {
                const baseText = span.getAttribute('data-base-label');
                if (stats && stats.count > 0) {
                    let unitStr = '';

```



```

        if (id === 'naei_co2') {
            unitStr = `${fmt(stats.mw, 0)} tCO2e`;
        } else {
            unitStr = stats.mw >= 1000 ? `${(stats.mw /
1000).toFixed(1)}GW` : `${Math.round(stats.mw)}MW`;
        }
        span.innerText = `${baseText} [${stats.count} |
`${unitStr}]`;
    } else {
        span.innerText = `${baseText} [${state}]`;
    }
    span.style.opacity = state === 'FAIL' ? '0.5' : '1';
    }
    });
}

```

// — Fetch Queue

```

class FetchQueue {
    constructor(concurrency) { this.concurrency = concurrency; this.active =
0; this.queue = []; }
    async add(task) {
        if (this.active >= this.concurrency) await new Promise(resolve =>
this.queue.push(resolve));
        this.active++;
        try { return await task(); }
        finally { this.active--; if (this.queue.length > 0)
this.queue.shift(); }
    }
}

const networkQueue = new FetchQueue(4);

async function fetchWithTimeout(url, ms = 15000) {
    const controller = new AbortController();
    const id = setTimeout(() => controller.abort(), ms);
    try {
        const response = await fetch(url, { signal: controller.signal,
cache: 'no-cache' });
        clearTimeout(id);
        if (!response.ok) throw new Error(`HTTP ${response.status} for
`${url}``);
        return response;
    } catch (err) { clearTimeout(id); throw err; }
}

async function fetchAndParseGeoJSON(url) {
    if (urlCache[url]) return await urlCache[url];
    const promise = fetchWithTimeout(url)
        .then(res => res.json())
        .then(data => {
            if (!data || !Array.isArray(data.features)) {
console.error(`[INVALID GEOJSON] ${url}`, data); return []; }
            console.log(`[DATA LOADED] ${url}: ${data.features.length}
features`); return data.features;
        })
        .catch(err => { delete urlCache[url]; console.error(`[FETCH ERROR]
`${url}`, err); throw err; });
    urlCache[url] = promise;
    return promise;
}

```

// — Geometry

```

function snapLines(features, subs) {
  if (!subs || !subs.length) return features;

  const TOLERANCE_DEG_SQ = 0.001 * 0.001;
  const RAD = Math.PI / 180;

  const snapCoordinate = (coord) => {
    let best = coord, min = Infinity;
    const latCos = Math.cos(coord[1] * RAD);
    subs.forEach(s => {
      const sc = s.geometry && s.geometry.coordinates;
      if (!sc) return;
      const dx = (coord[0] - sc[0]) * latCos;
      const dy = (coord[1] - sc[1]);
      const d = dx * dx + dy * dy;
      if (d < min && d <= TOLERANCE_DEG_SQ) { min = d; best = sc; }
    });
    return best;
  };

  return features.map(f => {
    const geom = f.geometry;
    if (!geom || !geom.coordinates) return f;
    if (geom.type === 'LineString') {
      const c = [...geom.coordinates];
      if (c.length > 0) {
        c[0] = snapCoordinate(c[0]);
        c[c.length - 1] = snapCoordinate(c[c.length - 1]);
      }
      return { ...f, geometry: { ...geom, coordinates: c } };
    }
    if (geom.type === 'MultiLineString') {
      const coords = geom.coordinates.map(line => {
        const l = [...line];
        if (l.length > 0) {
          l[0] = snapCoordinate(l[0]);
          l[l.length - 1] = snapCoordinate(l[l.length - 1]);
        }
        return l;
      });
      return { ...f, geometry: { ...geom, coordinates: coords } };
    }
    return f;
  });
}

function createGeoJSONCircle(lon, lat, radiusKm) {
  const points = radiusKm > 5000 ? 128 : radiusKm > 500 ? 96 : 64;
  const R = EARTH_RADIUS_KM, DEG = Math.PI / 180;
  const ad = radiusKm / R;
  const lat1 = lat * DEG;
  const coords = Array.from({ length: points }, (_, i) => {
    const b = (i / points) * 2 * Math.PI;
    const lat2 = Math.asin(Math.sin(lat1) * Math.cos(ad) +
Math.cos(lat1) * Math.sin(ad) * Math.cos(b));
    const lon2 = lon * DEG + Math.atan2(Math.sin(b) * Math.sin(ad) *
Math.cos(lat1), Math.cos(ad) - Math.sin(lat1) * Math.sin(lat2));
    return [lon2 / DEG, lat2 / DEG];
  });
  coords.push(coords[0]);
  return { type: 'FeatureCollection', features: [{ type: 'Feature',

```

```

geometry: { type: 'Polygon', coordinates: [coords] } }];
}

function drawRadiusCircle(lon, lat, radiusKm) { map.getSource('src-radius-circle').setData(createGeoJSONCircle(lon, lat, radiusKm)); }
function clearRadiusCircle() { map.getSource('src-radius-circle').setData({ type: 'FeatureCollection', features: [] }); }

let _visibleInteractiveIds = [];
let _visibleHoverIds = [];

function _rebuildVisibleCache(allLayerIds) {
  _visibleInteractiveIds = allLayerIds.filter(id => {
    try { return map.getLayoutProperty(id, 'visibility') === 'visible';
  }
    catch(e) { return false; }
  });
  _visibleHoverIds = [..._visibleInteractiveIds];
}

let _lastHoverMs = 0;

// — Popup / Search

function buildSearchButtons(name, capacity, tech) {
  const threshold = SEARCH_THRESHOLD[tech] !== undefined ?
SEARCH_THRESHOLD[tech] : 50;
  if (capacity < threshold) return '';
  const term = TECH_TERMS.get(tech) || 'energy project';
  const q = encodeURIComponent(`${name} ${term} UK`);
  const newsUrl = `https://news.google.com/search?q=${q}`;
  const imageUrl = `https://www.google.com/search?q=${q}&tbm=isch`;
  return ` NEWS</a>
    <a class="popup-btn popup-btn-images" href="${imageUrl}"
target="_blank" rel="noopener noreferrer"> IMAGES</a>
  </div>`;
}

function buildSearchIndex() {
  searchIndex = allREPDFeatures
    .filter(f => f && f.properties && f.properties.name)
    .map(f => ({ feature: f, nameLower:
String(f.properties.name).toLowerCase(), capacity: Number(f.properties.capacity)
|| 0 }));
}

function flyToProject(feature) {
  const [lon, lat] = feature.geometry.coordinates;
  const p = feature.properties;
  const cap = p.capacity ? `${p.capacity} MW` : '';
  const mounting = (p.mounting && p.mounting !== 'nan') ? ` |
${escapeHTML(p.mounting)}` : '';
  map.flyTo({ center: [lon, lat], zoom: 12, duration: 1800, essential:
true });
  setTimeout(() => {
    openPopup([lon, lat], `

blob:https://ventusltd.github.io/184a2ca1-a0ae-42ea-b3ba-72ceefc889c4



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```

```

                <span style="color:#888">${escapeHTML(p.raw_tech ||
p.tech)}${mounting}</span><br>
                <span style="color:#ffae00">${escapeHTML(cap)}</span>
                <span style="color:#666"> | ${escapeHTML(p.status)}</span>
<br>
                <span style="color:#555;font-
size:10px">${escapeHTML(p.operator)}</span>
                ${REPD_IDS.includes(p.tech) ? buildSearchButtons(p.name,
parseFloat(p.capacity) || 0, p.tech) : ''}
                </div>`);
        }, 1900);
    }

    // V9 canonical project deep links. Identity is resolved only by official
    REPD Ref;
    // URL names and coordinates are never used to manufacture a match.
    async function focusCanonicalProjectDeepLink() {
        const params = new URLSearchParams(window.location.search);
        const repdRef = String(params.get('repd_ref') || '').trim();
        if (!/^[A-Za-z0-9-]{1,40}$/.test(repdRef)) return;

        try {
            const requestedTechnology = String(params.get('technology') ||
            '').trim();
            const allowedTechnologies = new Set([
                'solar', 'solar_operational', 'solar_roof',
                'bess', 'bess_operational',
                'wind', 'wind_onshore', 'wind_onshore_operational',
                'wind_offshore', 'wind_offshore_operational',
                'biomass', 'tidal', 'hydrogen', 'hydro', 'flywheel',
                'act', 'geothermal', 'caes', 'other'
            ]);
            /* One exact identity owner. The search cartridge queries the pinned
            active-register product and publishes RESOLVED,
            NOT_IN_ACTIVE_REGISTER or FAILED. This carried V8 receiver used
            a site-relative /uk_renewables_pipeline path owned by another
            domain, so on GridAtlas Pages it issued a redundant 404 before
            falling back to the same supplied point. Delegate every REPD
            identity before any legacy fetch, including a ref-only arrival
            that legitimately supplies no technology. The measurement
            cartridge already uses valid supplied coordinates immediately
            and reconciles against the exact owner concurrently. */
            window.__GRIDATLAS_V8_DEEP_LINK__ = {
                status: 'DEFERRED_TO_EXACT_REPD_RECEIVER',
                repd_ref: repdRef,
                technology: requestedTechnology || null,
                technology_recognised: requestedTechnology
                    ? allowedTechnologies.has(requestedTechnology) : null,
                legacy_fetches: 0
            };
            return;
        } catch (error) {
            console.error('[V9 DEEP LINK FAILED]', error);
            const lon = Number(params.get('longitude'));
            const lat = Number(params.get('latitude'));
            if (Number.isFinite(lon) && Number.isFinite(lat) && Math.abs(lon) <=
180 && Math.abs(lat) <= 90) {
                map.flyTo({ center: [lon, lat], zoom: 12, duration: 1800,
essential: true });
            }
        }
    }
}

```

```

function searchProjects(query) {
  const resultsEl = document.getElementById('search-results');
  if (!query || query.length < 2) { resultsEl.style.display = 'none';
return; }
  if (!allREPDFeatures.length) {
    resultsEl.innerHTML = '<div class="search-no-results">Load a REPD
layer first to enable search</div>';
    resultsEl.style.display = 'block'; return;
  }
  const q = query.toLowerCase();
  const matches = searchIndex.filter(item =>
item.nameLower.includes(q)).sort((a, b) => b.capacity - a.capacity).slice(0,
12).map(item => item.feature);
  if (!matches.length) { resultsEl.innerHTML = '<div class="search-no-
results">No projects found</div>'; resultsEl.style.display = 'block'; return; }
  resultsEl.innerHTML = matches.map((f, i) => {
    const p = f.properties;
    const cap = p.capacity ? ` - ${p.capacity} MW` : '';
    const col = TECH_COLOURS.get(p.tech) || '#888';
    return `<div class="search-result-item" data-idx="${i}">
<b>${escapeHTML(p.name)}</b><span style="color:#555">${escapeHTML(cap)}</span>
<br>
    <span style="color:${col};font-size:9px">${escapeHTML(p.raw_tech
|| p.tech)}</span>
    <span style="color:#444;font-size:9px"> | ${escapeHTML(p.status
|| '')}</span></div>`;
  }).join('');
  resultsEl.querySelectorAll('.search-result-item').forEach((el, i) => {
    el.addEventListener('click', () => { flyToProject(matches[i]);
resultsEl.style.display = 'none'; document.getElementById('search-input').value
= matches[i].properties.name; });
  });
  resultsEl.style.display = 'block';
}

// — Export

```

```

function exportCSV() {
  if (!allREPDFeatures.length) { alert('Load a REPD layer first'); return;
}
  const visibleTechs = REPD_IDS.filter(id => { const cb =
document.querySelector(`input[data-layer-id="${id}"]`); return cb && cb.checked;
});
  const rows = allREPDFeatures.filter(f =>
visibleTechs.includes(f.properties.tech));
  if (!rows.length) { alert('No visible REPD layers to export – tick some
layers first'); return; }
  const headers =
['name', 'tech', 'raw_tech', 'capacity_mw', 'status', 'operator', 'mounting', 'longitud
e', 'latitude'];
  const csv = [headers.join(','), ...rows.map(f => {
    const p = f.properties; const [lon, lat] = f.geometry.coordinates;
    return [`${(p.name||'').replace(/"/g,
'""')}``, `${(p.tech||'').replace(/"/g,
'""')}``, `${(p.raw_tech||'').replace(/"/g,
'""')}``, p.capacity, `${(p.status||'').replace(/"/g,
'""')}``, `${(p.operator||'').replace(/"/g,
'""')}``, `${(p.mounting||'').replace(/"/g, '""')}``, lon, lat].join(',');
  })].join('\n');
  const blob = new Blob([csv], { type: 'text/csv' });
  const objectUrl = URL.createObjectURL(blob);
}

```

```

        const a = document.createElement('a'); a.href = objectUrl;
a.download = `globalgrid2050_export_${new Date().toISOString().slice(0,
10)}.csv`;
        document.body.appendChild(a);
        try { a.click(); } finally { a.remove(); setTimeout(() =>
URL.revokeObjectURL(objectUrl), 1000); }
    }

    // — Status / Radius

function toggleStatusMode() {
    statusMode = !statusMode;
    const btn = document.getElementById('btn-status');
    btn.classList.toggle('active', statusMode); btn.setAttribute('aria-
pressed', statusMode);

    if (map.getLayer('l-naei_co2-glow')) {
        const isBaseVisible = document.querySelector('input[data-layer-
id="naei_co2"]')?.checked;
        map.setLayoutProperty('l-naei_co2-glow', 'visibility', statusMode ?
'none' : (isBaseVisible ? 'visible' : 'none'));
    }

    REPD_IDS.forEach(id => {
        if (!map.getLayer(`l-${id}`)) return;
        if (id === 'solar' || id === 'solar_roof') {
            if (map.getLayer(`l-${id}-glow`)) {
                const isBaseVisible = document.querySelector(`input[data-
layer-id="${id}"]`).checked;
                map.setLayoutProperty(`l-${id}-glow`, 'visibility',
statusMode ? 'none' : (isBaseVisible ? 'visible' : 'none'));
            }
        }
        if (statusMode) {
            map.setPaintProperty(`l-${id}`, 'circle-color', ['match',
['downcase', ['coalesce', ['get', 'status'], '']],
            'operational', '#00ff88', 'under
construction', '#ffcc00', 'awaiting construction', '#ffaa00',
            'consented', '#ff8800', 'planning permission
granted', '#ff8800', 'planning approved', '#ff8800',
            'application submitted', '#8888ff', 'pre-
construction', '#aaaaff', '#444']);
        } else {
            const layer = layerConfigById.get(id);
            if (id === 'solar_roof') {
                map.setPaintProperty(`l-${id}`, 'circle-color',
['interpolate', ['linear'], ['coalesce',
['get', 'capacity'], 0], 0, '#ffcc00', 0.99, '#ffcc00', 1.0, '#ff8c00', 5.0, '#ff6600', 10.
0, '#ff4400']);
            } else if (id === 'solar') {
                map.setPaintProperty(`l-${id}`, 'circle-color',
['interpolate', ['linear'], ['coalesce',
['get', 'capacity'], 0], 0, '#ffff00', 20.0, '#ffcc00', 50.0, '#ffaa00', 200.0, '#ff6600',
500.0, '#ff2200']);
            } else {
                map.setPaintProperty(`l-${id}`, 'circle-color',
layer.color);
            }
        }
    });
}

```

```

function toggleRadiusMode() {
    radiusMode = !radiusMode;
    const btn = document.getElementById('btn-radius');
    btn.classList.toggle('active', radiusMode); btn.setAttribute('aria-
pressed', radiusMode);
    document.getElementById('radius-popup').style.display = radiusMode ?
'block' : 'none';
    map.getCanvas().style.cursor = radiusMode ? 'crosshair' : '';

    if (radiusMode && measureMode) toggleMeasureMode();
    if (radiusMode && radiusAreaMode) toggleRadiusAreaMode();
    if (radiusMode && zoneDrawMode) toggleZoneDrawMode();

    if (!radiusMode) { clearRadiusCircle(); radiusCenter = null; if
(radiusMarker) { radiusMarker.remove(); radiusMarker = null; } }
}

function doRadiusSearch(lon, lat) {
    if (!validateRadiusInput()) return;
    const km = getRadiusValue(); radiusCenter = { lon, lat };
drawRadiusCircle(lon, lat, km);
    if (radiusMarker) radiusMarker.remove(); radiusMarker = null;
    const nearby = allREPDFeatures.filter(f => { const [flon, flat] =
f.geometry.coordinates; return haversine(lon, lat, flon, flat) <= km;
}).sort((a, b) => (b.properties.capacity || 0) - (a.properties.capacity || 0));
    if (!nearby.length) {
        openPopup([lon, lat], `
            <div style="font-family:monospace;background:#000;padding:8px">
                <b style="color:#00ffff">⊙ ${km}km radius active</b><br><br>
                <span style="color:#888;font-size:10px">No REPD assets found
in this area.</span><br>
                <span style="color:#555;font-size:9px;line-height:1.6">Tick
layers in the panel below<br>to explore assets within this circle.</span>
            </div>`);
        return;
    }
    const totalMW = nearby.reduce((s, f) => s +
(parseFloat(f.properties.capacity) || 0), 0);
    const byTech = {};
    nearby.forEach(f => { const t = f.properties.tech; byTech[t] =
(byTech[t] || 0) + 1; });
    const techSummary = Object.entries(byTech).sort((a, b) => b[1] -
a[1]).map(([t, n]) => `<span style="color:#888">${escapeHTML(t)}: ${n}
</span>` ).join('<br>');
    const topAssets = nearby.slice(0, 5).map(f => {
        const p = f.properties;
        return `<div style="border-top:1px solid #222;padding-
top:4px;margin-top:4px">
            <b style="color:#ffcc00;font-size:11px">${escapeHTML(p.name)}
</b><br>
            <span style="color:#888;font-
size:10px">${escapeHTML(p.raw_tech)}</span>
            <span style="color:#ffae00;font-size:10px"> ${p.capacity || '?'}
MW</span></div>`;
    }).join('');
    openPopup([lon, lat], `
        <div style="font-family:monospace;background:#000;padding:6px">
            <b style="color:#00ffff">⊙ ${km}km - ${nearby.length} assets |
${totalMW.toFixed(1)} MW</b><br>
            <span style="color:#555;font-size:9px;line-height:1.8">Tick
layers in the panel to explore this area</span><br><br>
            ${techSummary}${topAssets}
        </div>`);
}

```

```

        </div>`);
    }

    // — DOM Builder

    function buildLayerRow(layer, idPrefix) {
        const label = document.createElement('label'); label.className = 'key-
item';
        const input = document.createElement('input'); input.type = 'checkbox';
input.dataset.layerId = layer.id; input.setAttribute('data-layer-id', layer.id);
        const span = document.createElement('span'); span.id =
`${idPrefix}${layer.id}`; span.setAttribute('data-base-label', layer.label);
span.style.color = layer.color; span.style.fontSize = '11px';
        const existing = document.getElementById(`lbl-${layer.id}`);
span.innerText = existing ? existing.innerText : `${layer.label} [WAIT]`;
        const mainCb = document.querySelector(`input[data-layer-
id="${layer.id}"]`); if (mainCb) input.checked = mainCb.checked;
        label.appendChild(input); label.appendChild(document.createTextNode('
')); label.appendChild(span);
        return label;
    }

    function buildDOM() {
        const container = document.getElementById('scada-ui-container');
        const fsContainer = document.getElementById('fs-curtain-keys');
        container.innerHTML = ''; fsContainer.innerHTML = '';
        const fragment = document.createDocumentFragment();
        const fsFragment = document.createDocumentFragment();

        GRID_CONFIG.forEach(group => {
            const groupDiv = document.createElement('div'); groupDiv.className
= 'key-group';
            const fsGroupDiv = document.createElement('div');
fsGroupDiv.className = 'key-group';
            groupDiv.innerHTML = fsGroupDiv.innerHTML = `<div class="key-
title">${group.group}</div>`;
            group.layers.forEach(layer => {
                const label = document.createElement('label'); label.className =
'key-item';
                const input = document.createElement('input'); input.type =
'checkbox'; input.dataset.layerId = layer.id; input.setAttribute('data-layer-
id', layer.id);
                const span = document.createElement('span'); span.id = `lbl-
${layer.id}`; span.setAttribute('data-base-label', layer.label);
span.style.color = layer.color; span.innerText = `${layer.label} [WAIT]`;
                label.appendChild(input);
                label.appendChild(document.createTextNode(' ')); label.appendChild(span);
                groupDiv.appendChild(label);
                fsGroupDiv.appendChild(buildLayerRow(layer, 'fs-lbl-'));
            });
            fragment.appendChild(groupDiv); fsFragment.appendChild(fsGroupDiv);
        });

        const bmHTML = `<div class="key-title">Basemap</div><label class="key-
item"><input type="radio" name="bm" value="dark" checked> Dark</label><label
class="key-item"><input type="radio" name="bm" value="sat"> Satellite</label>`;
        const bmGroup = document.createElement('div'); bmGroup.className = 'key-
group'; bmGroup.innerHTML = bmHTML; fragment.appendChild(bmGroup);
        const fsBmGroup = document.createElement('div'); fsBmGroup.className =
'key-group'; fsBmGroup.innerHTML = bmHTML.replace(/name="bm"/g, 'name="bm-fs"');
fsFragment.appendChild(fsBmGroup);

```



```

        container.appendChild(fragment); fsContainer.appendChild(fsFragment);

        container.addEventListener('change', e => {
            if (e.target.type === 'checkbox' && e.target.dataset.layerId) {
                const layerId = e.target.dataset.layerId; const isVisible =
e.target.checked;
                const fsCb = document.querySelector(`#fs-curtain-keys
input[data-layer-id="${layerId}"]`); if (fsCb) fsCb.checked = isVisible;
                handleLayerToggle(layerId, isVisible);
            } else if (e.target.name === 'bm') {
                map.setLayoutProperty('l-sat', 'visibility', e.target.value ===
'sat' ? 'visible' : 'none');
                const fsBm = document.querySelector(`input[name="bm-fs"]
[value="${e.target.value}"]`); if (fsBm) fsBm.checked = true;
            }
        });

        fsContainer.addEventListener('change', e => {
            if (e.target.type === 'checkbox' && e.target.dataset.layerId) {
                const layerId = e.target.dataset.layerId; const isVisible =
e.target.checked;
                const mainCb = document.querySelector(`#scada-ui-container
input[data-layer-id="${layerId}"]`); if (mainCb) mainCb.checked = isVisible;
                handleLayerToggle(layerId, isVisible);
            } else if (e.target.name === 'bm-fs') {
                map.setLayoutProperty('l-sat', 'visibility', e.target.value ===
'sat' ? 'visible' : 'none');
                const mainBm = document.querySelector(`input[name="bm"]
[value="${e.target.value}"]`); if (mainBm) mainBm.checked = true;
            }
        });

        document.getElementById('fs-curtain-tab').addEventListener('click',
toggleCurtain);

        const input = document.getElementById('search-input'); const btn =
document.getElementById('search-btn'); const resultsEl =
document.getElementById('search-results');
        input.addEventListener('input', () => searchProjects(input.value));
        input.addEventListener('keydown', e => { if (e.key === 'Enter')
searchProjects(input.value); if (e.key === 'Escape') resultsEl.style.display =
'none'; });
        btn.addEventListener('click', () => searchProjects(input.value));
        document.getElementById('map').addEventListener('click', () => {
resultsEl.style.display = 'none'; });

        document.getElementById('btn-export').addEventListener('click',
exportCSV); document.getElementById('btn-status').addEventListener('click',
toggleStatusMode);
        document.getElementById('btn-radius').addEventListener('click',
toggleRadiusMode); document.getElementById('btn-
measure').addEventListener('click', toggleMeasureMode);
        document.getElementById('btn-measure-undo').addEventListener('click',
undoLastMeasurePoint);

        const radiusInput = document.getElementById('radius-input');
        if(radiusInput) {
            radiusInput.addEventListener('input', () => validateRadiusInput());
            radiusInput.addEventListener('keydown', e => { if (e.key ===
'Enter') { e.preventDefault(); if (validateRadiusInput() && radiusCenter)
doRadiusSearch(radiusCenter.lon, radiusCenter.lat); } e.stopPropagation(); });
            radiusInput.addEventListener('blur', () => {

```

```

        const raw = parseFloat(radiusInput.value);
        if (isNaN(raw) || raw < RADIUS_MIN) radiusInput.value =
RADIUS_MIN; else if (raw > RADIUS_MAX) radiusInput.value = RADIUS_MAX;
        radiusInput.classList.remove('invalid'); if (radiusCenter)
doRadiusSearch(radiusCenter.lon, radiusCenter.lat);
    });
}

    const btnRadiusArea = document.getElementById('btn-radius-area');
    if (btnRadiusArea) btnRadiusArea.addEventListener('click',
toggleRadiusAreaMode);

    const btnZoneDraw = document.getElementById('btn-zonedraw');
    if (btnZoneDraw) btnZoneDraw.addEventListener('click',
toggleZoneDrawMode);

    const btnZoneDrawUndo = document.getElementById('btn-zonedraw-undo');
    if (btnZoneDrawUndo) btnZoneDrawUndo.addEventListener('click',
zoneDrawUndo);

    const zdRadiusInput = document.getElementById('zonedraw-radius-input');
    if (zdRadiusInput) {
        zdRadiusInput.addEventListener('keydown', e => {
e.stopPropagation(); });
        zdRadiusInput.addEventListener('blur', () => {
            const raw = parseFloat(zdRadiusInput.value);
            if (isNaN(raw) || raw <= 0) zdRadiusInput.value =
String(ZONE_DRAW_DEFAULT_KM);
            else if (raw > ZONE_DRAW_MAX_KM) zdRadiusInput.value =
String(ZONE_DRAW_MAX_KM);
        });
    }

    const rAreaInput = document.getElementById('radius-area-input');
    if (rAreaInput) {
        rAreaInput.addEventListener('keydown', e => {
            if (e.key === 'Enter') {
                e.preventDefault();
                if (radiusAreaCenter)
doRadiusAreaMeasure(radiusAreaCenter.lon, radiusAreaCenter.lat);
            }
            e.stopPropagation();
        });
        rAreaInput.addEventListener('blur', () => {
            const raw = parseFloat(rAreaInput.value);
            if (isNaN(raw) || raw <= 0) rAreaInput.value = 1; else if (raw >
MAX_RADIUS_KM) rAreaInput.value = MAX_RADIUS_KM;
            rAreaInput.classList.remove('invalid');
            if (radiusAreaCenter) doRadiusAreaMeasure(radiusAreaCenter.lon,
radiusAreaCenter.lat);
        });
    }
}

```

// — Layer Hydration

```

function handleLayerToggle(layerId, isVisible) {
    if (map.getLayer(`l-${layerId}`)) map.setLayoutProperty(`l-${layerId}`,
'visibility', isVisible ? 'visible' : 'none');
    if (map.getLayer(`l-${layerId}-glow`)) map.setLayoutProperty(`l-
${layerId}-glow`, 'visibility', (isVisible && !statusMode) ? 'visible' :
'none');
}

```

```

    const mapId = `l-${layerId}`;
    if (isVisible) {
        if (!_visibleInteractiveIds.includes(mapId))
            _visibleInteractiveIds.push(mapId);
        if (!_visibleHoverIds.includes(mapId)) _visibleHoverIds.push(mapId);
    } else {
        _visibleInteractiveIds = _visibleInteractiveIds.filter(id => id !==
mapId);
        _visibleHoverIds = _visibleHoverIds.filter(id => id !== mapId);
    }
    if (isVisible && layerId !== '400') hydrateLayer(layerId);
}

function getLayerConfig(layerId) { return layerConfigById.get(layerId); }

function getSourceIdForLayer(layerId) {
    if (REPD_IDS.includes(layerId)) return 'src-repd';
    if (TRANSIT_IDS.includes(layerId)) return TRANSIT_SOURCE_MAP[layerId];
    if (layerId === 'naei_co2') return 'src-naei_co2';
    return `src-${layerId}`;
}

/* A successful fetch is not proof that a layer can draw a feature.
-----
The deployed metro/tram partition is made entirely of LineStrings,
while the inherited V8 transit layers are MapLibre circle layers. The
old path called every sibling [OK] as soon as their shared source had
bytes, leaving checked switches which could never paint anything.

Evaluate only the small expression vocabulary used by these transit
filters. An unknown expression deliberately returns true: it must not
be used as evidence for disabling a control. Geometry compatibility is
checked against the layer MapLibre actually created, not the friendly
`point`/`line` name in the configuration.
*/
function transitExpressionValue(expression, feature) {
    if (!Array.isArray(expression)) return expression;
    if (expression[0] === 'get') return feature && feature.properties
        ? feature.properties[expression[1]] : undefined;
    if (expression[0] === 'literal') return expression[1];
    return undefined;
}

function transitFilterMatches(filter, feature) {
    if (!Array.isArray(filter) || filter.length === 0) return true;
    const operator = filter[0];
    if (operator === 'all') return filter.slice(1).every(item =>
transitFilterMatches(item, feature));
    if (operator === 'any') return filter.slice(1).some(item =>
transitFilterMatches(item, feature));
    if (operator === '!') return !transitFilterMatches(filter[1], feature);
    if (operator === '=') return transitExpressionValue(filter[1], feature)
=== transitExpressionValue(filter[2], feature);
    if (operator === '!=') return transitExpressionValue(filter[1], feature)
!== transitExpressionValue(filter[2], feature);
    if (operator === 'in' || operator === '!in') {
        const needle = transitExpressionValue(filter[1], feature);
        const haystack = transitExpressionValue(filter[2], feature);
        const contains = typeof haystack === 'string'
            ? haystack.includes(String(needle))
            : Array.isArray(haystack) && haystack.includes(needle);
        return operator === 'in' ? contains : !contains;
    }
}

```

```

    }
    return true;
}

function geometryFitsMapLayer(layerType, geometryType) {
    if (layerType === 'circle' || layerType === 'heatmap') return
    geometryType === 'Point' || geometryType === 'MultiPoint';
    if (layerType === 'line') return geometryType === 'LineString' ||
    geometryType === 'MultiLineString';
    if (layerType === 'fill' || layerType === 'fill-extrusion') return
    geometryType === 'Polygon' || geometryType === 'MultiPolygon';
    return true;
}

function countTransitFeaturesLayerCanDraw(layerId, features) {
    const layerConfig = getLayerConfig(layerId);
    const mapLayer = map.getLayer(`l-${layerId}`);
    if (!layerConfig || !mapLayer) return 0;
    return features.reduce((count, feature) => {
        const geometryType = feature && feature.geometry &&
        feature.geometry.type;
        if (!geometryFitsMapLayer(mapLayer.type, geometryType)) return
        count;
        return transitFilterMatches(layerConfig.filter, feature) ? count + 1
        : count;
    }, 0);
}

function setLayerControlAvailability(layerId, available) {
    document.querySelectorAll(`input[data-layer-id="${layerId}"]`).forEach(input => {
        input.disabled = !available;
        if (!available) input.checked = false;
    });
    if (available) return;
    [`l-${layerId}`, `l-${layerId}-glow`].forEach(mapLayerId => {
        if (map.getLayer(mapLayerId)) map.setLayoutProperty(mapLayerId,
        'visibility', 'none');
    });
    const mapLayerId = `l-${layerId}`;
    _visibleInteractiveIds = _visibleInteractiveIds.filter(id => id !==
    mapLayerId);
    _visibleHoverIds = _visibleHoverIds.filter(id => id !== mapLayerId);
}

function updateTransitSourceStates(sourceId, features) {
    TRANSIT_IDS.forEach(transitId => {
        if (TRANSIT_SOURCE_MAP[transitId] !== sourceId ||
        !RUNTIME_STATE[transitId]) return;
        const available = countTransitFeaturesLayerCanDraw(transitId,
        features) > 0;
        RUNTIME_STATE[transitId].loaded = true;
        RUNTIME_STATE[transitId].loading = false;
        setLayerControlAvailability(transitId, available);
        updateUIState(transitId, available ? 'OK' : 'EMPTY');
    });
}

async function hydrateLayer(layerId) {
    const state = RUNTIME_STATE[layerId];
    if (!state || state.loaded || state.loading) return;
    state.loading = true; updateUIState(layerId, 'LOAD');
}

```

```

    const layerConfig = getLayerConfig(layerId);
    if (!layerConfig) { updateUIState(layerId, 'FAIL'); state.loading =
false; return; }

    await networkQueue.add(async () => {
      try {
        let features = await fetchAndParseGeoJSON(layerConfig.url);
        if (features.length === 0) {
          if (TRANSIT_IDS.includes(layerId))
updateTransitSourceStates(TRANSIT_SOURCE_MAP[layerId], features);
          else { state.loaded = true; state.loading = false;
updateUIState(layerId, 'EMPTY'); }
          return;
        }
        if (layerConfig.isSubs) globalSubsData = features;
        if (layerConfig.snap) {
          if (!globalSubsData) { const subsLayer =
getLayerConfig('subs'); globalSubsData = await
fetchAndParseGeoJSON(subsLayer.url); }
          console.warn(`[SNAP] Runtime snapping active for
"${layerId}" - ${features.length} features. Move to build pipeline when
possible.`);
          features = snapLines(features, globalSubsData);
        }
        const sourceId = getSourceIdForLayer(layerId);
        const source = map.getSource(sourceId);
        if (!source) { console.error(`[SOURCE MISSING] ${sourceId}`);
updateUIState(layerId, 'FAIL'); state.loading = false; return; }
        source.setData({ type: 'FeatureCollection', features });
        state.loaded = true; state.loading = false;

        if (REPD_IDS.includes(layerId)) {
          allREPDFeatures = features; buildSearchIndex();
          function evalFilter(filter, props) {
            if (!filter) return true;
            const op = filter[0];
            if (op === '==') { const v = filter[1][0] === 'get' ?
props[filter[1][1]] : null; return String(v).toLowerCase() ===
String(filter[2]).toLowerCase(); }
            if (op === 'all') { return filter.slice(1).every(f =>
evalFilter(f, props)); }
            if (op === '>=') { const v = filter[1][0] === 'coalesce'
? (parseFloat(props[filter[1][1][1]]) || 0) : 0; return v >= filter[2]; }
            return true;
          }
          REPD_IDS.forEach(id => {
            if (!RUNTIME_STATE[id]) return;
            RUNTIME_STATE[id].loaded = true;
            RUNTIME_STATE[id].loading = false;
            const lcfg = getLayerConfig(id);
            const filtered = lcfg && lcfg.filter ? features.filter(f
=> evalFilter(lcfg.filter, f.properties)) : features.filter(f =>
f.properties.tech === id);
            const idStats = filtered.reduce((acc, f) => {
              acc.count++;
              acc.mw += parseFloat(f.properties.capacity) || 0;
              return acc;
            }, { count: 0, mw: 0 });
            updateUIState(id, idStats.count > 0 ? 'OK' : 'EMPTY',
idStats.count > 0 ? idStats : null);
          });
          if (statusMode) { toggleStatusMode(); toggleStatusMode(); }
        }
      }
    });
  }

```

```

    } else if (layerId === 'naei_co2') {
      const stats = features.reduce((acc, f) => {
        acc.count++;
        acc.mw += parseFloat(f.properties.emission_tco2e) || 0;
        return acc;
      }, { count: 0, mw: 0 });
      updateUIState(layerId, stats.count > 0 ? 'OK' : 'EMPTY',
stats.count > 0 ? stats : null);
    } else if (TRANSIT_IDS.includes(layerId)) {
      updateTransitSourceStates(TRANSIT_SOURCE_MAP[layerId],
features);
    } else {
      updateUIState(layerId, 'OK');
    }
  } catch (err) { console.error(`[LAYER FAILED] ${layerId}:`, err);
state.loading = false; updateUIState(layerId, 'FAIL'); }
  });
}

```

// — Map Load

```

map.on('load', () => {
  buildDOM();
  map.addSource('sat-s', { type: 'raster', tiles:
['https://server.arcgisonline.com/ArcGIS/rest/services/World_Imagery/MapServer/t
ile/{z}/{y}/{x}'], tileSize: 256 });
  map.addLayer({ id: 'l-sat', type: 'raster', source: 'sat-s', layout: {
visibility: 'none' } });

  map.addSource('src-radius-circle', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
  map.addLayer({ id: 'l-radius-circle-fill', type: 'fill', source: 'src-
radius-circle', paint: { 'fill-color': '#00ffff', 'fill-opacity': 0.04 } });
  map.addLayer({ id: 'l-radius-circle-stroke', type: 'line', source: 'src-
radius-circle', paint: { 'line-color': '#00ffff', 'line-width': 1.5, 'line-
opacity': 0.7, 'line-dasharray': [4, 3] } });

  map.addSource('src-radius-area', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
  map.addLayer({ id: 'l-radius-area-fill', type: 'fill', source: 'src-
radius-area', paint: { 'fill-color': '#ff00ff', 'fill-opacity': 0.08 } });
  map.addLayer({ id: 'l-radius-area-stroke', type: 'line', source: 'src-
radius-area', paint: { 'line-color': '#ff00ff', 'line-width': 1.5, 'line-
opacity': 0.8, 'line-dasharray': [2, 2] } });

  map.addSource('src-measure-line', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
  map.addSource('src-measure-fill', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
  map.addSource('src-measure-points', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
  map.addLayer({ id: 'l-measure-fill', type: 'fill', source: 'src-
measure-fill', paint: { 'fill-color': '#ffff00', 'fill-opacity': 0.08 } });
  map.addLayer({ id: 'l-measure-line', type: 'line', source: 'src-
measure-line', paint: { 'line-color': '#ffff00', 'line-width': 2, 'line-
dasharray': [3, 2] } });
  map.addLayer({ id: 'l-measure-points', type: 'circle', source: 'src-
measure-points', paint: { 'circle-color': '#ffff00', 'circle-radius': 5,
'circle-stroke-width': 1.5, 'circle-stroke-color': '#000' } });

```

// — Zone Draw layers (orange accent)

```

        map.addSource('src-zonedraw-fill', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
        map.addSource('src-zonedraw-line', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
        map.addSource('src-zonedraw-points', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
        map.addLayer({ id: 'l-zonedraw-fill', type: 'fill', source: 'src-
zonedraw-fill', paint: { 'fill-color': '#ff6600', 'fill-opacity': 0.18 } });
        map.addLayer({ id: 'l-zonedraw-line', type: 'line', source: 'src-
zonedraw-line', paint: { 'line-color': '#ff6600', 'line-width': 3, 'line-
dasharray': [4, 2] } });
        map.addLayer({ id: 'l-zonedraw-points', type: 'circle', source: 'src-
zonedraw-points', paint: {
        'circle-color': ['case', ['==', ['get', 'kind'], 'vertex'],
'#ff6600', '#ffaa44'],
        'circle-radius': ['case', ['==', ['get', 'kind'], 'vertex'], 9, 6],
        'circle-stroke-width': 2, 'circle-stroke-color': '#000',
        'circle-opacity': ['case', ['==', ['get', 'kind'], 'vertex'], 1,
0.85]
        } });

const allLayerIds = [];

GRID_CONFIG.forEach(group => {
    group.layers.forEach(layer => {
        if (REPD_IDS.includes(layer.id) ||
TRANSIT_IDS.includes(layer.id) || layer.id === 'ev' || layer.id === 'naei_co2')
return;

        if (layer.id === '400') {
            map.addSource('src-400', {
                type: 'geojson',
                data:
'../cartridges/5f5fbec83f9ce307b47ddc6e7277743f0bbala2445b0f3ca50a9a1806146e993/
grid_400kv.geojson'
            });
        } else {
            map.addSource(`src-${layer.id}`, { type: 'geojson', data: {
type: 'FeatureCollection', features: [] } });
            const layerObject = {
                id: `l-${layer.id}`, type: layer.type === 'line' ? 'line' :
'circle', source: `src-${layer.id}`, layout: { visibility: 'none' },
                paint: layer.type === 'line' ? { 'line-color': layer.color,
'line-width': layer.width } : { 'circle-color': layer.color, 'circle-radius':
layer.radius, 'circle-stroke-width': 1, 'circle-stroke-color': '#000' }
            };
            if (layer.filter) layerObject.filter = layer.filter; if
(layer.minzoom) layerObject.minzoom = layer.minzoom;
            map.addLayer(layerObject); allLayerIds.push(`l-${layer.id}`);
        }
    });
});

// — Heavy Industry (Custom VIP styling) —
map.addSource('src-naei_co2', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
map.addLayer({
    id: `l-naei_co2-glow`,
    type: 'circle',
    source: 'src-naei_co2',
    filter: ['>=', ['coalesce', ['get', 'emission_tco2e'], 0], 50000],
    layout: { visibility: 'none' },
    paint: {

```

```

        'circle-color': ['interpolate', ['linear'], ['coalesce',
['get', 'emission_tco2e'], 0], 50000, '#ffaa00', 200000, '#ff6600', 1000000, '#ff0000'],
        'circle-radius': ['interpolate', ['linear'], ['coalesce',
['get', 'emission_tco2e'], 0], 50000, 20, 200000, 40, 1000000, 60, 5000000, 90],
        'circle-opacity': ['interpolate', ['linear'], ['coalesce',
['get', 'emission_tco2e'], 0], 50000, 0.15, 200000, 0.25, 1000000, 0.35],
        'circle-blur': 1.0,
        'circle-stroke-width': 0
    }
});
map.addLayer({
    id: 'l-naei_co2',
    type: 'circle',
    source: 'src-naei_co2',
    layout: { visibility: 'none' },
    paint: {
        'circle-color': ['interpolate', ['linear'], ['coalesce',
['get', 'emission_tco2e'], 0], 0, '#ffcc00', 50000, '#ffaa00', 200000, '#ff6600', 1000000,
        '#ff0000'],
        'circle-radius': ['interpolate', ['linear'], ['coalesce',
['get', 'emission_tco2e'], 0], 0, 6, 50000, 10, 200000, 14, 1000000, 20, 5000000, 28],
        'circle-stroke-width': 1.5,
        'circle-stroke-color': '#000',
        'circle-opacity': 0.85
    }
});
allLayerIds.push('l-naei_co2-glow', 'l-naei_co2');

Object.keys(TRANSIT_URLS).forEach(sourceId => { map.addSource(sourceId,
{ type: 'geojson', data: { type: 'FeatureCollection', features: [] } }); });
TRANSIT_IDS.forEach(id => {
    const layer = getLayerConfig(id);
    const layerObject = {
        id: `l-${id}`, type: 'circle', source: TRANSIT_SOURCE_MAP[id],
        layout: { visibility: 'none' },
        paint: { 'circle-color': layer.color, 'circle-radius':
layer.radius, 'circle-stroke-width': 1, 'circle-stroke-color': '#000', 'circle-
opacity': 0.9 }
    };
    if (layer.filter) layerObject.filter = layer.filter; if
(layer.minzoom) layerObject.minzoom = layer.minzoom;
    map.addLayer(layerObject); allLayerIds.push(`l-${id}`);
});

map.addSource('src-ev', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
map.addLayer({ id: 'l-ev', type: 'circle', source: 'src-ev', layout: {
visibility: 'none' }, paint: { 'circle-color': '#00ff88', 'circle-radius': 5,
'circle-stroke-width': 1, 'circle-stroke-color': '#000', 'circle-opacity': 0.9 }
});
allLayerIds.push('l-ev');

map.addSource('src-repd', { type: 'geojson', data: { type:
'FeatureCollection', features: [] } });
REPD_IDS.forEach(id => {
    const layer = getLayerConfig(id);
    if (id === 'solar_roof') {
        map.addLayer({ id: `l-${id}-glow`, type: 'circle', source: 'src-
repd', filter: ['all', layer.filter, ['>=', ['coalesce', ['get', 'capacity'],
0], 1.0]], layout: { visibility: 'none' }, paint: { 'circle-color':
['interpolate', ['linear'], ['coalesce',
['get', 'capacity'], 0], 1.0, '#ff8c00', 5.0, '#ff6600', 10.0, '#ff4400'], 'circle-

```



```

radius': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],1.0,26,2.0,30,5.0,36,10.0,44], 'circle-opacity': 0.15,
'circle-blur': 1.0, 'circle-stroke-width': 0 } }));
    }
    if (id === 'solar') {
        map.addLayer({ id: `l-${id}-glow`, type: 'circle', source: 'src-
repd', filter: ['all', layer.filter, ['>=', ['coalesce', ['get', 'capacity'],
0], 4.0]], layout: { visibility: 'none' }, paint: { 'circle-color':
['interpolate',['linear'],['coalesce',
['get','capacity'],0],4.0,'#ffff00',20.0,'#ffaa00',50.0,'#ff4400',200.0,'#ff0000
'], 'circle-radius': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],4.0,22,20.0,32,50.0,44,200.0,60,500.0,80], 'circle-
opacity': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],4.0,0.12,20.0,0.18,50.0,0.25,200.0,0.35], 'circle-blur':
1.0, 'circle-stroke-width': 0 } }));
    }
    if (id === 'solar_operational') {
        map.addLayer({ id: `l-${id}-glow`, type: 'circle', source: 'src-
repd', filter: ['all', layer.filter, ['>=', ['coalesce', ['get', 'capacity'],
0], 10.0]], layout: { visibility: 'none' }, paint: { 'circle-color':
['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,'#00ff88',50.0,'#00cc66',200.0,'#009944',350.0,'#0066
22'], 'circle-radius': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,28,50.0,36,200.0,56,350.0,70,500.0,88], 'circle-
opacity': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,0.15,50.0,0.22,200.0,0.30,350.0,0.38], 'circle-blur':
1.0, 'circle-stroke-width': 0 } }));
    }
    if (id === 'bess_operational') {
        map.addLayer({ id: `l-${id}-glow`, type: 'circle', source: 'src-
repd', filter: ['all', layer.filter, ['>=', ['coalesce', ['get', 'capacity'],
0], 10.0]], layout: { visibility: 'none' }, paint: { 'circle-color':
['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,'#ffb3d9',50.0,'#ff69b4',200.0,'#ff1493',350.0,'#cc00
66'], 'circle-radius': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,24,50.0,32,200.0,50,350.0,62,500.0,78], 'circle-
opacity': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,0.15,50.0,0.22,200.0,0.30,350.0,0.38], 'circle-blur':
1.0, 'circle-stroke-width': 0 } }));
    }
    if (id === 'wind_onshore_operational') {
        map.addLayer({ id: `l-${id}-glow`, type: 'circle', source: 'src-
repd', filter: ['all', layer.filter, ['>=', ['coalesce', ['get', 'capacity'],
0], 10.0]], layout: { visibility: 'none' }, paint: { 'circle-color':
['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,'#99ffee',50.0,'#00ffcc',200.0,'#00ccaa',350.0,'#0088
77'], 'circle-radius': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,24,50.0,32,200.0,50,350.0,62,500.0,78], 'circle-
opacity': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,0.15,50.0,0.22,200.0,0.30,350.0,0.38], 'circle-blur':
1.0, 'circle-stroke-width': 0 } }));
    }
    if (id === 'wind_offshore_operational') {
        map.addLayer({ id: `l-${id}-glow`, type: 'circle', source: 'src-
repd', filter: ['all', layer.filter, ['>=', ['coalesce', ['get', 'capacity'],
0], 10.0]], layout: { visibility: 'none' }, paint: { 'circle-color':
['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,'#99ccff',50.0,'#3399ff',200.0,'#0055dd',350.0,'#0033
99'], 'circle-radius': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,24,50.0,32,200.0,50,350.0,62,500.0,78], 'circle-
opacity': ['interpolate',['linear'],['coalesce',
['get','capacity'],0],10.0,0.15,50.0,0.22,200.0,0.30,350.0,0.38], 'circle-blur':

```

```

1.0, 'circle-stroke-width': 0 } }));
    }
    const circlePaint = id === 'solar_roof'
      ? { 'circle-color': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, '#ffcc00', 0.99, '#ffcc00', 1.0, '#ff8c00', 5.0, '#ff6600', 10.0, '#ff4400'], 'circle-radius': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, 7, 0.5, 7, 0.99, 8, 1.0, 16, 2.0, 18, 5.0, 22, 10.0, 28], 'circle-stroke-width': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, 1, 0.99, 1, 1.0, 2], 'circle-stroke-color': '#000', 'circle-opacity': 0.9 }
      : id === 'solar'
      ? { 'circle-color': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, '#ffff00', 20.0, '#ffcc00', 50.0, '#ffaa00', 200.0, '#ff6600', 500.0, '#ff2200'], 'circle-radius': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, 8, 10, 10, 50, 13, 200, 17, 500, 22, 1000, 28], 'circle-stroke-width': 1.5, 'circle-stroke-color': '#000', 'circle-opacity': 0.85 }
      : id === 'solar_operational'
      ? { 'circle-color': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, '#66ff99', 10, '#33ff77', 50, '#00dd55', 100, '#00bb44', 200, '#008833', 350, '#006622', 500, '#004411'], 'circle-radius': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, 10, 10, 14, 50, 18, 100, 22, 200, 28, 350, 35, 500, 42], 'circle-stroke-width': 2, 'circle-stroke-color': '#000', 'circle-opacity': 0.90 }
      : id === 'bess_operational'
      ? { 'circle-color': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, '#ffcc00', 10, '#ff8c00', 50, '#ff6600', 100, '#ff4400', 200, '#ff2200', 350, '#ff0000', 500, '#ff0000'], 'circle-radius': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, 8, 10, 12, 50, 16, 100, 20, 200, 26, 350, 32, 500, 38], 'circle-stroke-width': 2, 'circle-stroke-color': '#000', 'circle-opacity': 0.90 }
      : id === 'wind_onshore_operational'
      ? { 'circle-color': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, '#ccff00', 10, '#99ff00', 50, '#00ff00', 100, '#00dd00', 200, '#00aa00', 350, '#007700', 500, '#004400'], 'circle-radius': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, 8, 10, 12, 50, 16, 100, 20, 200, 26, 350, 32, 500, 38], 'circle-stroke-width': 2, 'circle-stroke-color': '#000', 'circle-opacity': 0.90 }
      : id === 'wind_offshore_operational'
      ? { 'circle-color': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, '#cce5ff', 10, '#99ccff', 50, '#3399ff', 100, '#0066ee', 200, '#0044bb', 350, '#003399', 500, '#001166'], 'circle-radius': ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, 8, 10, 12, 50, 16, 100, 20, 200, 26, 350, 32, 500, 38], 'circle-stroke-width': 2, 'circle-stroke-color': '#000', 'circle-opacity': 0.90 }
      : { 'circle-color': layer.color, 'circle-radius':
        ['interpolate', ['linear'], ['coalesce',
        ['get', 'capacity'], 0], 0, 8, 10, 10, 50, 13, 200, 17, 500, 22, 1000, 28], 'circle-stroke-width': 1.5, 'circle-stroke-color': '#000', 'circle-opacity': 0.85 };

    map.addLayer({ id: `l-${id}`, type: 'circle', source: 'src-repd',
    filter: layer.filter, layout: { visibility: 'none' }, paint: circlePaint });
    allLayerIds.push(`l-${id}`);
  });

  // — PERF: seed the visible layer cache from actual map state after all
  layers are added
  _rebuildVisibleCache(allLayerIds);

  // — Map Events

```

```

let _pendingToolClick = null;

```

```

        map.getCanvas().addEventListener('mousedown', e => {
            if (!zoneDrawMode) return;
            const lngLat = map.unproject([e.offsetX, e.offsetY]);
            _zoneDrawOnMouseDown({ lngLat, preventDefault: () =>
e.preventDefault() });
        });

        map.on('click', e => {
            if (measureMode) {
                _pendingToolClick = setTimeout(() => {
                    _pendingToolClick = null;
                    if (!measureClosed) {
                        measurePoints.push([e.lngLat.lng, e.lngLat.lat]);
                        updateMeasureLayers();
                        updateMeasureDisplay();
                    }
                }, CLICK_DEBOUNCE_MS);
                return;
            }
            if (zoneDrawMode) { _zoneDrawOnClick(e); return; }
            if (radiusMode) { doRadiusSearch(e.lngLat.lng, e.lngLat.lat);
return; }
            if (radiusAreaMode) { doRadiusAreaMeasure(e.lngLat.lng,
e.lngLat.lat); return; }

            if (!_visibleInteractiveIds.length) return;
            const features = map.queryRenderedFeatures(e.point, { layers:
_visibleInteractiveIds });

            if (!features.length) return;
            const p = features[0].properties || {}; const name = p.name ||
p.SiteName || p['Site Name'] || 'Unnamed Asset';

            if (p.type === 'supermarket') {
                const address = [p.street, p.city,
p.postcode].filter(Boolean).join(', '); const area = p.area_m2 ?
`${p.area_m2.toLocaleString()} m²` : '';
                openPopup(e.lngLat, `

blob:https://ventusltd.github.io/184a2ca1-a0ae-42ea-b3ba-72ceefc889c4



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```

```

style="color:#888">${escapeHTML(p.sport)}</span><br><span
style="color:#ffae00">${escapeHTML(cap)}</span></div>`); return;
}

    if (p.type === 'naei_emitter') {
        const tonnes = p.emission_tco2e ?
Number(p.emission_tco2e).toLocaleString('en-GB', { maximumFractionDigits: 0 }) :
'Unknown';
        const dataLabel = p.datatype === 'O' ? 'Self-reported by the
company' : p.datatype === 'M' ? 'Estimated by the government' : 'Official
figures';
        openPopup(e.lngLat, `

2 equivalent tonnes</span><br><br><span style="color:#444;font-
size:9px">${escapeHTML(dataLabel)} · UK Government emissions database</span>
</div>`); return;
    }

    const tech = p.tech || ''; const rawTech = p.raw_tech || p.type ||
tech; const voltage = p.voltage || ''; const capacity = parseFloat(p.capacity)
|| 0; const powerKw = p.power_kw || null; const connectors = p.connectors || '';
const status = p.status || ''; const operator = p.operator || ''; const mounting
= (p.mounting && p.mounting !== 'nan') ? ` | ${escapeHTML(p.mounting)} ` : '';
const capStr = capacity ? `${capacity} MW` : ''; const statusCol =
STATUS_COLOURS[normalizeStatus(status)] || '#888'; const searchBtns =
REPD_IDS.includes(tech) ? buildSearchButtons(name, capacity, tech) : ''; const
evFields = powerKw ? `<span style="color:#00ff88;font-size:10px">${powerKw}
kW</span>${connectors ? `<span style="color:#555;font-size:10px"> |
${escapeHTML(connectors)}</span>` : ''}<br>` : '';
    openPopup(e.lngLat, `

blob:https://ventusltd.github.io/184a2ca1-a0ae-42ea-b3ba-72ceefc889c4



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```

```

        map.on('mousemove', e => {
            if (zoneDrawMode) { _zoneDrawOnMouseMove(e); return; }

            if (measureMode || radiusMode || radiusAreaMode) {
map.getCanvas().style.cursor = 'crosshair'; return; }

            if (!_visibleHoverIds.length) { map.getCanvas().style.cursor = '';
return; }

            const now = Date.now();
            if (now - _lastHoverMs < HOVER_THROTTLE_MS) return;
            _lastHoverMs = now;

            if (_lastMouseMoveRaf) return;
            _lastMouseMoveRaf = requestAnimationFrame(() => {
                _lastMouseMoveRaf = null;
                const features = map.queryRenderedFeatures(e.point, { layers:
_visibleHoverIds });
                map.getCanvas().style.cursor = features.length ? 'pointer' : '';
            });
        });

        GRID_CONFIG.forEach(group => { group.layers.forEach(layer => { if
(layer.preload && layer.id !== '400') hydrateLayer(layer.id); }); });
        const state400 = RUNTIME_STATE['400'];
        if (state400) { state400.loaded = true; state400.loading = false;
updateUIState('400', 'OK'); }
        focusCanonicalProjectDeepLink();
    });
};

/**
 * Module: geodesy
 *
 * One Earth radius for the whole estate, and the three operations every
 * measurement here is built from. This existed three times tonight - in
 * the sandbox, in the substation cartridge and in the data repository -
 * which is exactly how two of them end up on different radii without
 * anyone noticing.
 *
 * Radius 6378.137 km, matching Ventusltd/grid-distance-maths. Haversine.
 * No projection, no turf, no second radius for geometry.
 *
 * Pure functions. No DOM, no network, no state.
 */
(() => {
    'use strict';

    const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||
{});
    if (NS.geodesy) return;

    const EARTH_RADIUS_KM = 6378.137;
    const DEG = Math.PI / 180;

    function distanceKm(lon1, lat1, lon2, lat2) {
        const dLat = (lat2 - lat1) * DEG;
        const dLon = (lon2 - lon1) * DEG;
        const a = Math.sin(dLat / 2) ** 2
            + Math.cos(lat1 * DEG) * Math.cos(lat2 * DEG) * Math.sin(dLon / 2) ** 2;
        /* atan2, in this operand order, because that is the form every version

```

of this estate has shipped - ventus-corev8engine.js haversine() and every cartridge carried from it.

The extraction wrote $2 * R * \text{asin}(\sqrt{a})$ instead. Algebraically the same; numerically one unit in the last place apart, which the all-versions proof caught on West Burton Solar to Cottam: 7.050150827184836 shipped, 7.050150827184837 from the module. It is 1e-15 km and changes no figure any reader will ever see - and it is still wrong, because the claim being made is PARITY. A module that is nearly the incumbent is a module that has to be argued about every time a digit differs. */

```
return EARTH_RADIUS_KM * 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
}
```

/* A polygon reduces to the mean of its outer ring, not its first corner. A substation drawn as a compound outline would otherwise be measured from whichever vertex the mapper happened to start at.

Point, Polygon and MultiPolygon, and NOTHING ELSE. The first draft of this module accepted any nested coordinate array and so returned a mean for a LineString where the incumbent returns null; the parity proof caught it against the live cartridge. Extraction is not the moment to change behaviour, so the behaviour is pinned here and any widening becomes its own version with its own reasoning.

One deliberate difference, on malformed input only: this returns null where the incumbent would throw on a Point with no coordinates. No real geometry reaches that path, and a proof asserts it. */

```
function representativePoint(geometry) {
  if (!geometry) return null;
  const { type, coordinates } = geometry;
  if (type === 'Point') {
    return Array.isArray(coordinates) && coordinates.length >= 2
      ? [coordinates[0], coordinates[1]] : null;
  }
  const ring = type === 'Polygon' ? coordinates && coordinates[0]
    : type === 'MultiPolygon' ? coordinates && coordinates[0] &&
coordinates[0][0]
    : null;
  if (!Array.isArray(ring) || !ring.length) return null;
  let sumLon = 0;
  let sumLat = 0;
  for (const point of ring) {
    sumLon += point[0];
    sumLat += point[1];
  }
  return [sumLon / ring.length, sumLat / ring.length];
}
```

/* OpenStreetMap's `voltage` is VOLTS at every magnitude, and a feature may carry several separated by a semicolon. Magnitude is not the unit: 750 is a DC traction supply at a railway depot, not 750 kV. An audit of the served payload found 229 features (3.95%) carrying a token below 1,000, every one of which had been misread. An explicit `kv` property is already kilovolts and is trusted as such. */

```
function voltagesKv(properties) {
  if (!properties) return [];
  const out = [];
  const explicit = properties.kv ?? properties.KV;
  if (explicit != null && String(explicit).trim() !== '') {
    for (const token of String(explicit).match(/\d+(?:\.\d+)?/g) || []) {
      const value = Number(token);
```

```

        if (Number.isFinite(value) && value > 0) out.push(value);
    }
}
const volts = properties.voltage ?? properties.VOLTAGE;
if (volts != null) {
    for (const token of String(volts).match(/\d+(?:\.\d+)?/g) || []) {
        const value = Number(token);
        if (Number.isFinite(value) && value > 0) out.push(value / 1000);
    }
}
return [...new Set(out)].sort((a, b) => b - a);
}

/* Projection and bearing, carried in from the sandbox verbatim.
-----
The deep scan found the body carrying a SECOND geodesy section - "the
geodesy the layout needs, all on R_ATLAS" - four hundred lines away
from the first. Two geodesies in one file, on a constant that must
never differ, is the configuration that produced the divergence the
all-versions proof caught. Both belong here, on the one radius, and
the body now delegates rather than defining.

The bodies below are the incumbent's, character for character apart
from the radius identifier, so parity is a property of the move rather
than something to argue about afterwards. */
function destinationPoint(lon, lat, km, bearingDeg) {
    const ad = km / EARTH_RADIUS_KM;
    const brg = bearingDeg * DEG;
    const p1 = lat * DEG;
    const p2 = Math.asin(Math.sin(p1) * Math.cos(ad)
        + Math.cos(p1) * Math.sin(ad) * Math.cos(brg));
    const l2 = lon * DEG + Math.atan2(
        Math.sin(brg) * Math.sin(ad) * Math.cos(p1),
        Math.cos(ad) - Math.sin(p1) * Math.sin(p2));
    return [l2 / DEG, p2 / DEG];
}

function initialBearingDeg(lon1, lat1, lon2, lat2) {
    const p1 = lat1 * DEG; const p2 = lat2 * DEG;
    const dl = (lon2 - lon1) * DEG;
    const y = Math.sin(dl) * Math.cos(p2);
    const x = Math.cos(p1) * Math.sin(p2) - Math.sin(p1) * Math.cos(p2) *
Math.cos(dl);
    return (Math.atan2(y, x) / DEG + 360) % 360;
}

NS.geodesy = Object.freeze({
    schema: 'gridatlas.module.geodesy.v1',
    EARTH_RADIUS_KM,
    distanceKm,
    destinationPoint,
    initialBearingDeg,
    representativePoint,
    voltagesKv
});
})();

/**
 * Module: network-topology
 *
 * What the network operator publishes about ONE site: its nodes, the
 * circuits that land on them, the transformers between them, the changes

```

```

* it has published for future years, and which other sites those circuits
* reach. Appendix B of the Electricity Ten Year Statement, read as a
* node/branch model and reported as facts.
*
* Successor at generation 202609012245: the node-level adjacency this module
* already builds is now handed OUT, so that the electrical-distance module
* can traverse it instead of building a second one. A second implementation
* of voltageOf would be a second opinion about which voltages are real, and
* this estate has already shipped one cartridge carrying two geodesies that
* disagreed in the last place. Nothing else changed: at() is byte-for-byte
* the incumbent's, and the parity proof holds the two to identical answers
* on the published payload.
*
* It answers "what is here, and what is it connected to". It does not
* answer "can this project connect", and it cannot: that depends on queue
* position, committed connections, consent and commercial terms which no
* published appendix contains. The product says so itself and the refusal
* travels inside every result, in the same object as the numbers, because
* a caveat in a different place from the figure is a caveat nobody reads.
*
* Three disciplines, each of them a defect this estate has already shipped:
*
* VOLTAGE IS NEVER MIXED. A card printed "5.1-49.6 kA" across a 132 kV
* and a 400 kV busbar and an engineer would have read it as one number
* for one point. So every answer here is grouped by the voltage of the
* node the circuit lands on, and a caller asking for one voltage gets
* only that voltage. There is no site-wide range in this module at all.
*
* VOLTAGE IS NEVER DECODED. The node-code convention (digit 1->132,
* 2->275, 4->400) is derived, not documented, and the product reports
* 726 of 2,679 nodes whose voltage their site does not declare. This
* reads `voltage_kv` and honours `voltage_consistent_with_site`; where
* that is false the voltage is `null` and the node is grouped under
* `undeclared`, never guessed from its name.
*
* R, X AND B ARE NOT A LOAD FLOW. They are published percentages on a
* 100 MVA base. Carrying them is publishing; solving with them would
* need a declared model, generation and load assumptions, tap positions
* and contingencies, and validation against a trusted solver. This
* module carries them and says what base they are on. It computes
* nothing from them.
*
* Fail closed: an unrecognised schema yields no index and therefore no
* answers, rather than plausible ones from a shape that has moved.
*
* Depends on: nothing. Topology is not geometry - this module never
* measures a distance and never touches a coordinate.
*/
((() => {
  'use strict';

  const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||
{}));
  if (NS.networkTopology) return;

  const ACCEPTS = 'data-grid-gb.transmission-network.v1';

  const NOT_AN_ASSESSMENT =
    'Counts, lengths, ratings and impedances are what the network operator '
    + 'publishes about this site. None of them states whether any project can '
    + 'connect here, which depends on queue position, committed connections, '
    + 'consent and commercial terms that no published appendix contains.';

```



```

const IMPEDANCE_BASIS =
  'R, X and B are percentages on a 100 MVA base, as published. They are '
  + 'network parameters, not a solved power flow.';

const UNDECLARED = 'undeclared';

/* A node's voltage is trusted only where the product says the site
  declares it. Everything else is undeclared - never inferred from the
  digit in the node code, which is a derived convention the product
  itself marks as undocumented. */
function voltageOf(node) {
  if (!node) return null;
  if (node.voltage_consistent_with_site !== true) return null;
  return Number.isFinite(node.voltage_kv) ? node.voltage_kv : null;
}

const bandKey = (kv) => (kv == null ? UNDECLARED : String(kv));

function ratingsOf(row) {
  const seasons = { winter: row.winter_mva, spring: row.spring_mva,
    summer: row.summer_mva, autumn: row.autumn_mva };
  const published = {};
  for (const [season, value] of Object.entries(seasons)) {
    if (Number.isFinite(value)) published[season] = value;
  }
  return Object.keys(published).length ? published : null;
}

```

/* A published branch is seen once from EACH of its ends, so a site that owns both ends of a branch publishes it twice.

 A transformer's two windings are at the same site by construction, so almost every transformer lands twice: 1,394 of the 1,472 published transformers have both ends at one site. Counting landings therefore reported 2,944 machines where 1,550 site-held machines exist, and Cowley - five machines, COWL41 to COWL11 and COWL12, 269 to 278 MVA - said ten.

It is NOT only transformers. Measured against gb-transmission-network.v1 on 2026-09-03:

transformers	2,944 landings -> 1,550 units, 484 of 525 sites differ
circuits	2,784 landings -> 2,638 units, 78 of 636 sites differ
planned changes	4,460 landings -> 3,696 units, 282 of 645 sites differ

so the same correction is applied to all three site-wide aggregates. The PER-VOLTAGE lists are untouched and must stay as they are: "at 400 kV, 5 transformers" and "at 132 kV, 5 transformers" are the same five machines seen from each winding, which is what a reader standing at a busbar is asking for.

Halving was rejected: it is wrong at 57 of the 525 sites that hold a transformer, and 24 of them publish an odd number of landings, so halving would invent a fractional machine. The pair is keyed instead, and a pair seen from BOTH directions was published twice while a pair seen from one - which is what a voltage-filtered query sees of an internal machine - was published once. */

```

function physicalUnits(records) {
  const pairs = new Map();
  for (const record of records) {
    const near = String(record.from_node);

```

```

    const far = String(record.to_node);
    const forward = near < far;
    const key = forward ? near + '\u0000' + far : far + '\u0000' + near;
    if (!pairs.has(key)) pairs.set(key, { forward: 0, reverse: 0 });
    const seen = pairs.get(key);
    if (forward) seen.forward += 1; else seen.reverse += 1;
  }
  let units = 0;
  for (const seen of pairs.values()) {
    units += (seen.forward && seen.reverse)
      ? Math.max(seen.forward, seen.reverse)
      : seen.forward + seen.reverse;
  }
  return units;
}

function parametersOf(row) {
  const published = {};
  for (const [key, field] of [['r_pct', 'r_pct_100mva'], ['x_pct', 'x_pct_100mva'], ['b_pct', 'b_pct_100mva']]) {
    if (Number.isFinite(row[field])) published[key] = row[field];
  }
  return Object.keys(published).length ? published : null;
}

/**
 * @param product the parsed data-grid-gb transmission-network payload
 * @returns an index, or null if the schema is not the one this reads
 */
function index(product) {
  if (!product || product.schema !== ACCEPTS) return null;

  const nodes = new Map();
  for (const node of product.nodes || []) {
    if (node && node.node) nodes.set(node.node, node);
  }

  const sitesByCode = new Map();
  const sitesByName = new Map();
  for (const site of product.sites || []) {
    if (!site || !site.code) continue;
    sitesByCode.set(String(site.code).toUpperCase(), site);
    if (site.name) sitesByName.set(String(site.name).toUpperCase().trim(), site);
  }

  /* Branches are indexed by the node they land on, both ends, because a
     circuit is a fact about both of its sites. */
  const byNode = new Map();
  function land(nodeName, entry) {
    if (!nodeName) return;
    if (!byNode.has(nodeName)) byNode.set(nodeName, []);
    byNode.get(nodeName).push(entry);
  }
  for (const [kind, rows] of [['circuit', product.circuits], ['transformer', product.transformers], ['planned_change', product.planned_changes]]) {
    for (const row of rows || []) {
      if (!row) continue;
      land(row.node_1, { kind, row, near: 'node_1', far: 'node_2' });
      land(row.node_2, { kind, row, near: 'node_2', far: 'node_1' });
    }
  }
}

```

```

    }
}

function siteOf(nodeName) {
    const node = nodes.get(nodeName);
    return node ? node.site_code : null;
}

/* The adjacency, handed out rather than rebuilt.
-----
Every accessor here is a READ of the structures at() already uses,
so a traversal cannot disagree with a one-hop view about which
nodes exist, which site a node belongs to, or whether a node's
voltage is trustworthy. planned_change rows are excluded from
edges: a change published for 2029 is not a path a current can
take today, and treating it as one would be the headroom lie in a
new costume. They remain available through at(). */
function graph() {
    return {
        schema: 'gridatlas.module.network-topology.graph.v1',
        has: (name) => nodes.has(name),
        nodeVoltageKv: (name) => voltageOf(nodes.get(name)),
        nodeSiteCode: (name) => {
            const node = nodes.get(name);
            return node ? node.site_code : null;
        },
        /* circuits and transformers only - see above */
        edgesAt: (name) => (byNode.get(name) || [])
            .filter((entry) => entry.kind !== 'planned_change'),
        nodesOfSite: (code) => {
            const wanted = String(code || '').toUpperCase();
            const out = [];
            for (const node of nodes.values()) {
                if (String(node.site_code || '').toUpperCase() === wanted)
                    out.push(node.node);
            }
            return out.sort();
        },
        siteByCode: (code) => sitesByCode.get(String(code || '').toUpperCase())
    || null,
        ratingsOf,
        parametersOf
    };
}

function resolve(key) {
    if (!key) return null;
    const wanted = String(key).toUpperCase().trim();
    return sitesByCode.get(wanted) || sitesByName.get(wanted) || null;
}

/**
 * Everything published about one site, grouped by the voltage of the
 * node each branch lands on. Never a site-wide range.
 *
 * @param key          site code or exact site name
 * @param options      { voltageKv } to restrict to one voltage
 */
function at(key, options) {
    const site = resolve(key);
    if (!site) return null;
    const wantedKv = options && Number.isFinite(options.voltageKv)

```

```

    ? options.voltageKv : null;

const siteNodes = [];
for (const node of nodes.values()) {
    if (node.site_code !== site.code) continue;
    const kv = voltageOf(node);
    if (wantedKv !== null && kv !== wantedKv) continue;
    siteNodes.push({ node: node.node, voltage_kv: kv });
}
siteNodes.sort((a, b) => a.node.localeCompare(b.node));

const byVoltage = new Map();
const neighbours = new Map();

for (const entry of siteNodes) {
    for (const landing of byNode.get(entry.node) || []) {
        const farNode = landing.row[landing.far];
        const farSiteCode = siteOf(farNode);
        const farSite = farSiteCode ? sitesByCode.get(farSiteCode) : null;
        const internal = farSiteCode === site.code;

        const key2 = bandKey(entry.voltage_kv);
        if (!byVoltage.has(key2)) {
            byVoltage.set(key2, { voltage_kv: entry.voltage_kv,
                circuits: [], transformers: [], planned_changes: [] });
        }
        const band = byVoltage.get(key2);

        const published = {
            from_node: entry.node,
            to_node: farNode,
            to_site_code: farSiteCode,
            to_site_name: farSite ? farSite.name : null,
            within_this_site: internal,
            transmission_owner: landing.row.transmission_owner || null,
            parameters_pct_100mva: parametersOf(landing.row),
            ratings_mva: ratingsOf(landing.row)
        };

        if (landing.kind === 'circuit') {
            published.circuit_type = landing.row.circuit_type || null;
            if (Number.isFinite(landing.row.ohl_km)) published.ohl_km =
landing.row.ohl_km;
            if (Number.isFinite(landing.row.cable_km)) published.cable_km =
landing.row.cable_km;
            band.circuits.push(published);
        } else if (landing.kind === 'transformer') {
            if (Number.isFinite(landing.row.rating_mva)) published.rating_mva =
landing.row.rating_mva;
            delete published.ratings_mva;
            band.transformers.push(published);
        } else {
            published.year = landing.row.year || null;
            published.status = landing.row.status || null;
            published.asset = landing.row.asset || null;
            band.planned_changes.push(published);
        }

        /* A neighbour is another SITE this site's circuits reach. An
            internal branch is not a neighbour, and a planned change is
            not a neighbour either - it has not been built. */
        if (landing.kind === 'circuit' && !internal && farSiteCode) {

```

```

        if (!neighbours.has(farSiteCode)) {
            neighbours.set(farSiteCode, {
                site_code: farSiteCode,
                site_name: farSite ? farSite.name : null,
                circuits: 0
            });
        }
        neighbours.get(farSiteCode).circuits += 1;
    }
}

const voltages = [...byVoltage.entries()]
    .sort((a, b) => {
        if (a[0] === UNDECLARED) return 1;
        if (b[0] === UNDECLARED) return -1;
        return Number(b[0]) - Number(a[0]);
    })
    .map(([ , band]) => band);

return {
    schema: 'gridatlas.module.network-topology.v1',
    source: ACCEPTS,
    site: {
        code: site.code,
        name: site.name,
        transmission_owner: site.transmission_owner || null,
        voltages_kv: Array.isArray(site.voltages_kv) ?
site.voltages_kv.slice() : []
    },
    requested_voltage_kv: wantedKv,
    nodes: siteNodes,
    by_voltage: voltages,
    neighbours: [...neighbours.values()].sort((a, b) => b.circuits -
a.circuits),
    counts: {
        nodes: siteNodes.length,
        /* Physical units, deduplicated across the two ends a site may
        hold of the same branch. The landing tallies are published
        beside them so a reader can see the difference rather than
        wonder which number the per-voltage lists add up to. */
        circuits: physicalUnits(voltages.flatMap(band => band.circuits)),
        transformers: physicalUnits(voltages.flatMap(band =>
band.transformers)),
        planned_changes: physicalUnits(voltages.flatMap(band =>
band.planned_changes)),
        circuit_landings: voltages.reduce((sum, band) => sum +
band.circuits.length, 0),
        transformer_landings: voltages.reduce((sum, band) => sum +
band.transformers.length, 0),
        planned_change_landings: voltages.reduce((sum, band) => sum +
band.planned_changes.length, 0),
        neighbour_sites: neighbours.size
    },
    counts_are_units: 'A site holds both ends of a transformer and of any '
        + 'internal circuit, so the same branch lands twice. The counts above
        + 'are physical units; the landing tallies beside them are what the '
        + 'per-voltage lists contain.',
    impedance_basis: IMPEDANCE_BASIS,
    not_an_assessment: NOT_AN_ASSESSMENT
};

```

```

    }

    return {
      schema: 'gridatlas.module.network-topology.v1',
      source: ACCEPTS,
      counts: {
        sites: sitesByCode.size,
        nodes: nodes.size,
        branch_landings: byNode.size
      },
      site: resolve,
      at,
      graph
    };
  }

  NS.networkTopology = Object.freeze({
    schema: 'gridatlas.module.network-topology.v1',
    accepts: ACCEPTS,
    not_an_assessment: NOT_AN_ASSESSMENT,
    impedance_basis: IMPEDANCE_BASIS,
    index
  });
})();

/**
 * Module: electrical-distance
 *
 * How far away a substation is, measured in the network operator's own
 * published circuits rather than in kilometres.
 *
 * WHY THIS EXISTS
 * -----
 * Everything this estate has measured until now has been geometry. The
 * geodesy module answers "how many kilometres from this project to that
 * substation", and it answers it correctly, to the last place. But a
 * kilometre is not a connection. Two substations 7 km apart can be on
 * opposite sides of a network boundary with no circuit between them; two
 * substations 90 km apart can be the two ends of a single published
 * circuit. A map that shows only the first number invites the reader to
 * infer the second, and that inference is wrong often enough to be
 * dangerous in a document someone spends money on.
 *
 * So this module answers a different question, from a different source:
 * on the network Appendix B actually publishes, how many circuits lie
 * between these two sites, and which ones? Every hop is a published row
 * with an identity. The answer is a citation, not an estimate.
 *
 * WHAT IT IS NOT
 * -----
 * A hop count is not a distance. Two hops is not "twice as far" as one,
 * and a site one hop away is not thereby available to connect to. It is
 * not an impedance either: R, X and B are carried on every hop exactly as
 * published, and this module contains no arithmetic over them at all -
 * summing impedance along a path is the first step of a load flow, and a
 * load flow needs a declared model, base values, taps, generation and load
 * assumptions, contingencies and validation against a trusted solver. None
 * of those are in this file, so neither is the sum.
 *
 * THE ONE RULE THAT SHAPES THE TRAVERSAL
 * -----
 * A voltage may only change across a transformer, and when it does the

```

```

* transformer is named in the path. A circuit whose two ends carry
* different declared voltages is not a voltage change - it is a
* contradiction in the data, and the traversal refuses it and says so
* rather than quietly walking through. Undeclared voltages are carried as
* undeclared and never guessed from a node code.
*
* node tools/proofs/modules/202609012245-electrical-distance.proof.mjs
*/
(() => {
  const NS = window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ || {};
  if (NS.electricalDistance) return;

  const SCHEMA = 'gridatlas.module.electrical-distance.v1';
  const REQUIRES = 'gridatlas.module.network-topology.graph.v1';

  const NOT_A_DISTANCE =
    'Hops are published circuits between two sites, not a distance. A site '
    + 'one hop away may be a hundred kilometres away, and a site ten '
    + 'kilometres away may be on no shared circuit at all.';

  const NOT_A_CAPACITY =
    'A path existing on the published network says nothing about whether '
    + 'anything can flow along it for a new project. Ratings are the '
    + 'circuit\'s, not a spare allowance, and queue position, committed '
    + 'connections, consent and commercial terms appear in no appendix.';

  const IMPEDANCE_CARRIED =
    'R, X and B are reproduced on each hop exactly as published, on a '
    + '100 MVA base. They are not added, scaled or combined anywhere in '
    + 'this module. A sum of them would be the beginning of a load flow, '
    + 'which needs a declared model this data does not contain.';

  const UNDECLARED = 'undeclared';

  /* A traversal must not invent a voltage, so the two ends of an edge are
     compared only when BOTH are declared. */
  function crossing(graph, nearNode, farNode) {
    const near = graph.nodeVoltageKv(nearNode);
    const far = graph.nodeVoltageKv(farNode);
    return {
      near_kv: near,
      far_kv: far,
      both_declared: near != null && far != null,
      changes: near != null && far != null && near !== far
    };
  }

  function describe(graph, entry, nearNode) {
    const farNode = entry.row[entry.far];
    const cross = crossing(graph, nearNode, farNode);
    return {
      kind: entry.kind,
      from_node: nearNode,
      to_node: farNode,
      from_site_code: graph.nodeSiteCode(nearNode) || null,
      to_site_code: graph.nodeSiteCode(farNode) || null,
      from_voltage_kv: cross.near_kv,
      to_voltage_kv: cross.far_kv,
      voltage_changed: cross.changes,
      /* The transformer's own published ratio, where it has one. Only 140
         of 1,472 transformers carry it, so it is reported when present and
         never reconstructed from the two node voltages when absent - a

```

```

        reconstructed ratio would be this module's opinion wearing the
        product's authority. */
    voltage_ratio_kv: entry.kind === 'transformer'
        && typeof entry.row.voltage_ratio_kv === 'string'
        ? entry.row.voltage_ratio_kv : null,
    /* carried, never computed with */
    ratings_mva: graph.ratingsOf(entry.row),
    transformer_rating_mva: entry.kind === 'transformer'
        && Number.isFinite(entry.row.rating_mva) ? entry.row.rating_mva : null,
    parameters_pct_100mva: graph.parametersOf(entry.row)
    };
}

/**
 * Is this edge legal to walk?
 *
 * A transformer is the only thing that may change voltage. A circuit
 * that appears to change voltage is a contradiction between two
 * published node records, and it is refused and reported rather than
 * traversed - a silent walk through it would mix voltages, which is the
 * exact failure this estate holds itself to never repeating.
 */
function legality(kind, cross) {
    if (!cross.changes) return { legal: true, refusal: null };
    if (kind === 'transformer') return { legal: true, refusal: null };
    return {
        legal: false,
        refusal: 'a ' + kind + ' whose two ends carry different declared '
            + 'voltages (' + cross.near_kv + ' kV and ' + cross.far_kv + ' kV); '
            + 'only a transformer may change voltage, so this edge is not walked'
    };
}

function startNodes(graph, site, voltageKv) {
    const nodes = graph.nodesOfSite(site.code);
    if (voltageKv == null) return nodes;
    return nodes.filter((name) => graph.nodeVoltageKv(name) === voltageKv);
}

/**
 * The shortest published path between two sites, in circuits.
 *
 * Breadth-first, so the first arrival is a fewest-hop path. Where
 * several paths tie, the one found first by sorted node order is
 * returned and `ties` says how many others arrived at the same depth,
 * because "the" path implies a uniqueness the network does not have.
 *
 * @param index      a network-topology index (must expose graph())
 * @param fromKey    site code or exact site name
 * @param toKey      site code or exact site name
 * @param options    { voltageKv, maxHops }
 * @returns a result object, or null if either site is unknown
 */
function between(index, fromKey, toKey, options) {
    if (!index || typeof index.graph !== 'function') return null;
    const graph = index.graph();
    if (!graph || graph.schema !== REQUIRES) return null;

    const from = index.site(fromKey);
    const to = index.site(toKey);
    if (!from || !to) return null;

```



```

const opts = options || {};
const voltageKv = Number.isFinite(opts.voltageKv) ? opts.voltageKv : null;
const maxHops = Number.isFinite(opts.maxHops) ? opts.maxHops : 6;

const targets = new Set(graph.nodesOfSite(to.code));
const origins = startNodes(graph, from, voltageKv);

const base = {
  schema: SCHEMA,
  from: { code: from.code, name: from.name },
  to: { code: to.code, name: to.name },
  requested_voltage_kv: voltageKv,
  max_hops: maxHops,
  not_a_distance: NOT_A_DISTANCE,
  not_a_capacity: NOT_A_CAPACITY,
  impedance_basis: IMPEDANCE_CARRIED
};

if (!origins.length) {
  return Object.assign({}, base, {
    reached: false,
    reason: voltageKv == null
      ? 'the origin site publishes no nodes in this product'
      : 'the origin site publishes no node at ' + voltageKv + ' kv',
    hops: null, path: [], refusals: [], ties: 0, explored_nodes: 0
  });
}

if (from.code === to.code) {
  return Object.assign({}, base, {
    reached: true, hops: 0, path: [], refusals: [], ties: 0,
    explored_nodes: origins.length,
    reason: 'the same site'
  });
}

const seen = new Map();
const refusals = [];
let frontier = [];
for (const name of origins.slice().sort()) {
  if (targets.has(name)) {
    return Object.assign({}, base, {
      reached: true, hops: 0, path: [], refusals: [], ties: 0,
      explored_nodes: 1,
      reason: 'both site codes resolve to the same node'
    });
  }
  seen.set(name, null);
  frontier.push(name);
}

for (let depth = 1; depth <= maxHops; depth += 1) {
  const next = [];
  const arrivals = [];
  for (const nearNode of frontier) {
    for (const entry of graph.edgesAt(nearNode)) {
      const farNode = entry.row[entry.far];
      if (!farNode || !graph.has(farNode)) continue;
      const cross = crossing(graph, nearNode, farNode);
      const verdict = legality(entry.kind, cross);
      if (!verdict.legal) {
        refusals.push({

```

```

        at_node: nearNode, to_node: farNode,
        kind: entry.kind, reason: verdict.refusal
    });
    continue;
}
if (seen.has(farNode)) continue;
seen.set(farNode, { via: entry, from: nearNode });
if (targets.has(farNode)) arrivals.push(farNode);
else next.push(farNode);
}
}

if (arrivals.length) {
    arrivals.sort();
    const path = [];
    let cursor = arrivals[0];
    while (cursor) {
        const step = seen.get(cursor);
        if (!step) break;
        path.unshift(describe(graph, step.via, step.from));
        cursor = step.from;
    }
    return Object.assign({}, base, {
        reached: true,
        hops: path.length,
        path,
        transformers_crossed: path.filter((h) => h.kind ===
'transformer').length,
        voltage_changes: path.filter((h) => h.voltage_changed).length,
        ties: arrivals.length - 1,
        refusals,
        explored_nodes: seen.size,
        arrival_node: arrivals[0]
    });
}

if (!next.length) break;
frontier = next.sort();
}

return Object.assign({}, base, {
    reached: false,
    reason: 'no published path within ' + maxHops + ' hops'
    + (voltageKv == null ? '' : ' from a ' + voltageKv + ' kV node')
    + '; this is a statement about the published network, not about '
    + 'whether the two sites are connected in reality',
    hops: null, path: [], refusals, ties: 0, explored_nodes: seen.size
});
}

/**
 * Every site reachable within N hops, with the hop count at which it was
 * first reached. The neighbourhood a click is actually in, electrically.
 *
 * @param index    a network-topology index
 * @param key      site code or exact site name
 * @param options  { hops, voltageKv }
 */
function within(index, key, options) {
    if (!index || typeof index.graph !== 'function') return null;
    const graph = index.graph();
    if (!graph || graph.schema !== REQUIRES) return null;

```

```

const site = index.site(key);
if (!site) return null;

const opts = options || {};
const voltageKv = Number.isFinite(opts.voltageKv) ? opts.voltageKv : null;
const limit = Number.isFinite(opts.hops) ? opts.hops : 2;

const origins = startNodes(graph, site, voltageKv);
const seen = new Set(origins);
const bySite = new Map();
const refusals = [];
let frontier = origins.slice().sort();

for (let depth = 1; depth <= limit; depth += 1) {
  const next = [];
  for (const nearNode of frontier) {
    for (const entry of graph.edgesAt(nearNode)) {
      const farNode = entry.row[entry.far];
      if (!farNode || !graph.has(farNode) || seen.has(farNode)) continue;
      const cross = crossing(graph, nearNode, farNode);
      const verdict = legality(entry.kind, cross);
      if (!verdict.legal) {
        refusals.push({ at_node: nearNode, to_node: farNode,
          kind: entry.kind, reason: verdict.refusal });
        continue;
      }
      seen.add(farNode);
      next.push(farNode);
      const code = graph.nodeSiteCode(farNode);
      if (!code || String(code).toUpperCase() ===
String(site.code).toUpperCase()) continue;
      if (bySite.has(code)) continue;
      const far = graph.siteByCode(code);
      bySite.set(code, {
        code,
        name: far ? far.name : null,
        hops: depth,
        first_node: farNode,
        voltage_kv: cross.far_kv,
        via: entry.kind
      });
    }
  }
  if (!next.length) break;
  frontier = next.sort();
}

const sites = [...bySite.values()].sort((a, b) =>
  a.hops - b.hops || String(a.code).localeCompare(String(b.code)));

return {
  schema: SCHEMA,
  site: { code: site.code, name: site.name },
  requested_voltage_kv: voltageKv,
  hop_limit: limit,
  origin_nodes: origins.length,
  sites,
  counts: {
    sites: sites.length,
    by_hop: sites.reduce((acc, s) => {
      acc[s.hops] = (acc[s.hops] || 0) + 1;
    }, {});
  }
};

```

```

        return acc;
    }, {}))
    },
    refusals,
    not_a_distance: NOT_A_DISTANCE,
    not_a_capacity: NOT_A_CAPACITY
  };
}

NS.electricalDistance = Object.freeze({
  schema: SCHEMA,
  requires: REQUIRES,
  not_a_distance: NOT_A_DISTANCE,
  not_a_capacity: NOT_A_CAPACITY,
  impedance_basis: IMPEDANCE_CARRIED,
  undeclared: UNDECLARED,
  between,
  within
});
})();

/**
 * Module: rating-envelope
 *
 * What the operator publishes a circuit can carry, season by season -
 * and a structural refusal to add those numbers up.
 *
 * WHY THE REFUSAL IS THE FEATURE
 * -----
 * A substation with eight 400 kV circuits rated 3,000 MVA each does not
 * have 24,000 MVA of anything. The ratings are thermal limits on
 * individual branches under stated seasonal conditions; they are not
 * additive, they are not simultaneous, and the sum is not a quantity that
 * exists in the network. Yet a sum is the single easiest number to
 * produce from this data and the single most persuasive to a reader, and
 * once printed it is indistinguishable from a capacity figure. That is
 * how a published rating becomes an invented headroom number.
 *
 * So this module reports each circuit's ratings individually, names the
 * season each one belongs to, and contains no code path that produces a
 * site total. The proof asserts the absence, not merely the intent.
 *
 * WHAT IT ADDS BEYOND "THE MINIMUM AND THE MAXIMUM"
 * -----
 * The owner product already publishes a site-wide winter envelope
 * (`circuit_winter_rating_mva.min/max`). Three things it does not do,
 * which are done here:
 *
 * 1. All four seasons, not winter alone. Summer ratings are the binding
 *    ones for a thermally limited circuit, and the product carries them
 *    on 1,276 of 1,392 circuits - the 116 without are OFTO-labelled and
 *    are reported as not published rather than filled in.
 *
 * 2. Scoped to a voltage. A site-wide range across a 132 kV and a
 *    400 kV busbar is a number about no busbar at all.
 *
 * 3. Implausible values named rather than averaged away. Four circuits
 *    publish winter_mva of exactly 9999, on spans of a kilometre or
 *    less with zero impedance; planned changes reach 69,275. These have
 *    the shape of placeholders, not ratings. A module that quietly
 *    includes them in a maximum reports a lie with a citation attached,
 *    so they are carried, flagged, and excluded from the envelope with

```

```

*      the exclusion stated.
*
*      node tools/proofs/modules/202609012250-rating-envelope.proof.mjs
*/
((() => {
  const NS = window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ || {};
  if (NS.ratingEnvelope) return;

  const SCHEMA = 'gridatlas.module.rating-envelope.v1';
  const REQUIRES = 'gridatlas.module.network-topology.graph.v1';

  const SEASONS = Object.freeze(['winter', 'spring', 'summer', 'autumn']);
  const FIELD = Object.freeze({
    winter: 'winter_mva', spring: 'spring_mva',
    summer: 'summer_mva', autumn: 'autumn_mva'
  });

  const NEVER_SUMMED =
    'These are per-circuit thermal ratings under stated seasonal '
    + 'conditions. They are not additive and they are not simultaneous: '
    + 'the sum of the circuits at a site is not a quantity that exists in '
    + 'the network, and this module contains no code that produces one.';

  const NOT_A_CAPACITY =
    'A rating is what a circuit is rated to carry, not what is free on '
    + 'it. Existing flows, committed connections, queue position, outage '
    + 'conditions and commercial terms decide what a project could use, '
    + 'and no published appendix contains any of them.';

  /* A rating that is obviously not a rating.
  -----
  9999 on a one-kilometre span with zero impedance is a placeholder,
  not a thermal limit; so is 69,275 on a hundred-metre cable. The test
  is deliberately narrow - a value at or above this threshold is
  flagged, nothing else is second-guessed - because a module that
  starts judging which published numbers it believes has stopped
  reporting the published record. */
  const IMPLAUSIBLE_MVA = 9999;

  function seasonsOf(row) {
    const published = {};
    const absent = [];
    for (const season of SEASONS) {
      const value = row[FIELD[season]];
      if (Number.isFinite(value)) published[season] = value;
      else absent.push(season);
    }
    return { published, absent };
  }

  function flagsFor(published) {
    const flags = [];
    for (const [season, value] of Object.entries(published)) {
      if (value >= IMPLAUSIBLE_MVA) {
        flags.push({
          season,
          value,
          reason: 'at or above ' + IMPLAUSIBLE_MVA + ' MVA, which has the '
            + 'shape of a placeholder rather than a thermal rating; it is '
            + 'reported and excluded from the range below'
        });
      }
    }
  }
})

```

```

    }
    return flags;
}

/**
 * Every circuit landing at a site, at one voltage, with its own
 * seasonal ratings. No total anywhere.
 *
 * @param index      a network-topology index exposing graph()
 * @param key        site code or exact site name
 * @param options    { voltageKv }
 */
function at(index, key, options) {
    if (!index || typeof index.graph !== 'function') return null;
    const graph = index.graph();
    if (!graph || graph.schema !== REQUIRES) return null;

    const site = index.site(key);
    if (!site) return null;

    const opts = options || {};
    const voltageKv = Number.isFinite(opts.voltageKv) ? opts.voltageKv : null;

    const nodes = graph.nodesOfSite(site.code)
        .filter((name) => voltageKv == null || graph.nodeVoltageKv(name) ===
voltageKv);

    const circuits = [];
    const seen = new Set();
    for (const nodeName of nodes) {
        for (const entry of graph.edgesAt(nodeName)) {
            if (entry.kind !== 'circuit') continue;
            const far = entry.row[entry.far];
            const id = [nodeName, far].sort().join('|');
            if (seen.has(id)) continue;
            seen.add(id);
            const { published, absent } = seasonsOf(entry.row);
            if (!Object.keys(published).length) continue;
            circuits.push({
                from_node: nodeName,
                to_node: far,
                to_site_code: graph.nodeSiteCode(far) || null,
                voltage_kv: graph.nodeVoltageKv(nodeName),
                circuit_type: typeof entry.row.circuit_type === 'string' ?
entry.row.circuit_type : null,
                ohl_km: Number.isFinite(entry.row.ohl_km) ? entry.row.ohl_km : null,
                cable_km: Number.isFinite(entry.row.cable_km) ? entry.row.cable_km :
null,
                ratings_mva: published,
                seasons_not_published: absent,
                flags: flagsFor(published),
                parameters_pct_100mva: graph.parametersOf(entry.row)
            });
        }
    }

    circuits.sort((a, b) => String(a.to_node).localeCompare(String(b.to_node)));

    /* The per-season RANGE across circuits - a lowest and a highest
       rating, which are two real published values - never a sum, and
       never a mean, which would be a number no circuit is rated at. */
    const by_season = {};

```

```

for (const season of SEASONS) {
  const values = circuits
    .filter((c) => Number.isFinite(c.ratings_mva[season])
      && c.ratings_mva[season] < IMPLAUSIBLE_MVA)
    .map((c) => c.ratings_mva[season]);
  const excluded = circuits
    .filter((c) => Number.isFinite(c.ratings_mva[season])
      && c.ratings_mva[season] >= IMPLAUSIBLE_MVA).length;
  by_season[season] = values.length
    ? {
      lowest_circuit_mva: Math.min.apply(null, values),
      highest_circuit_mva: Math.max.apply(null, values),
      circuits: values.length,
      excluded_as_implausible: excluded
    }
    : { circuits: 0, excluded_as_implausible: excluded, published: false };
}

const flagged = circuits.filter((c) => c.flags.length);
const missingSeasons = circuits.filter((c) =>
c.seasons_not_published.length);

return {
  schema: SCHEMA,
  site: { code: site.code, name: site.name },
  requested_voltage_kv: voltageKv,
  scope: voltageKv == null
    ? 'every voltage at this site; a range across two busbar voltages '
      + 'is a number about neither of them'
    : voltageKv + ' kV nodes at this site only',
  circuits,
  by_season,
  counts: {
    circuits: circuits.length,
    with_a_flagged_value: flagged.length,
    with_a_season_not_published: missingSeasons.length
  },
  never_summed: NEVER_SUMMED,
  not_a_capacity: NOT_A_CAPACITY
};
}

NS.ratingEnvelope = Object.freeze({
  schema: SCHEMA,
  requires: REQUIRES,
  seasons: SEASONS,
  implausible_mva: IMPLAUSIBLE_MVA,
  never_summed: NEVER_SUMMED,
  not_a_capacity: NOT_A_CAPACITY,
  at
});
})();

/**
 * Module: injection-response
 *
 * A DECLARED DC power-flow model of the published GB transmission network,
 * used to answer one question: if power is injected here, which circuits
 * carry it, and what fraction of it does each one carry?
 *
 * _____
 * WHY THIS IS A POWER FLOW AND NOT A PRETENDED ONE

```



```

*
* ZERO-REACTANCE BRANCHES
*
*
* The product publishes circuits with x_pct_100mva of exactly 0 - zero
* length spans, busbar couplers, some series devices. 1/x is undefined for
* these, and substituting a small number would silently invent a
* reactance. They are instead treated as what they physically are: a
* short, meaning the two nodes are electrically the same bus. The nodes
* are merged before the matrix is built, the merge is counted, and the
* count is reported in the answer.
*
*
* WHAT THE FIRST VERSION GOT WRONG
*
*
* Codex reviewed this adversarially on request and found two defects that
* the proof could not see, both of them the same shape: the proof tested a
* case the production caller never takes.
*
* 1. NO ARBITRARY SLACK. The caller chose the first bus that was not the
* injection bus. The 400 kV induced graph is not connected - 573 buses,
* 437 modelled edges, 238 components - so that bus is usually in a
* different component from the injection, and a transfer between two
* components does not exist. West Burton was paired with ABHA41, which
* is disconnected from 572 of the 573 buses; the solve ran 2,292
* iterations to a residual of about 1.7e13.
*
* Worse, acceptance checked Kirchhoff at the INJECTION BUS ONLY, which
* a disconnected pair can satisfy while the solve has not converged at
* all. A page gated on that check could print an impossible transfer.
*
* Now: the withdrawal bus must be in the same component as the
* injection, the component is computed and named, and where no sink is
* declared the rule that picks one is stated in the answer rather than
* left implicit. Acceptance requires convergence AND a global residual
* AND Kirchhoff at EVERY bus. Anything else is an explicit refusal.
*
* 2. PARALLEL CIRCUITS ARE NOT ONE CIRCUIT. Edges were de-duplicated by
* endpoints, kind and reactance, so two genuine parallel circuits that
* publish the same reactance collapsed into one - 22 groups covering 45
* published rows at 400 kV. Two parallel circuits carry twice what one
* carries at the same angle, so the collapse understated the network.
* Every landing is still visited twice, once from each end; the row
* ITSELF is now the identity, which distinguishes "seen twice" from
* "two rows that look alike".
*
* node tools/proofs/modules/202609020015-injection-response.proof.mjs
*/
(( ) => {
  const NS = window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ || {};
  if (NS.injectionResponse) return;

  const SCHEMA = 'gridatlas.module.injection-response.v2';
  const REQUIRES = 'gridatlas.module.network-topology.graph.v1';

  const BASE_MVA = 100;

  const DECLARED_MODEL = Object.freeze({
    method: 'linear DC power flow (injection response / power transfer
distribution factor)',

```

```

equations: 'P = B' ·  $\theta$  ; branch flow  $f_{ij} = (\theta_i - \theta_j) / x_{ij}$ ',
base_mva: BASE_MVA,
reactance: 'x = x_pct_100mva / 100, per unit, as published',
resistance: 'not used; the DC approximation neglects series resistance',
shunt_susceptance: 'not used; line charging does not appear in a DC model',
voltages: 'assumed flat at 1.0 per unit; not published and not solved',
angles: 'assumed small, so  $\sin \theta \approx \theta$ ',
losses: 'zero by construction; real losses are of order 1-2% and are not
represented',
transformer_taps: 'not published, therefore not modelled; transformers are
their series reactance only',
contingencies: 'none; this is the intact network',
slack: 'declared explicitly on every answer, never inferred silently'
});

const NOT_A_LOADING =
  'This is the response to a NEW injection, not a loading. What is '
+ 'already flowing on these circuits is published nowhere in this '
+ 'product, so the total flow - which is what decides whether a '
+ 'circuit is full - cannot be computed here by anyone.';

const NOT_A_CONNECTION_OFFER =
  'A fraction of an injection appearing on a circuit is not permission '
+ 'to use that circuit. Queue position, committed connections, outage '
+ 'conditions, consent and commercial terms decide what a project may '
+ 'connect, and no published appendix contains any of them.';

/* — union-find, for shorting zero-reactance branches ————— */
function makeUnionFind() {
  const parent = new Map();
  function find(x) {
    if (!parent.has(x)) { parent.set(x, x); return x; }
    let root = x;
    while (parent.get(root) !== root) root = parent.get(root);
    let cursor = x;
    while (parent.get(cursor) !== cursor) {
      const next = parent.get(cursor);
      parent.set(cursor, root);
      cursor = next;
    }
    return root;
  }
  return {
    find,
    union(a, b) {
      const ra = find(a);
      const rb = find(b);
      if (ra === rb) return false;
      parent.set(ra, rb);
      return true;
    }
  };
}

/* The matrix assembly, taking an explicit node list so the caller
   decides the scope - one site, one voltage, or the whole product -
   without a second copy of this code existing for each case. */
function assemble(graph, nodeNames, { voltageKv, includeTransformers }) {
  const inScope = new Set(nodeNames);
  const uf = makeUnionFind();
  for (const n of nodeNames) uf.find(n);

```

```

const branches = [];
const seen = new Set();
let shorted = 0;
let skippedNoReactance = 0;

for (const name of nodeNames) {
  for (const entry of graph.edgesAt(name)) {
    if (entry.kind === 'transformer' && !includeTransformers) continue;
    const far = entry.row[entry.far];
    if (!inScope.has(far)) continue;
    /* The published ROW is the identity, not its values.
    -----
    Keying on endpoints + kind + reactance collapsed genuine
    parallel circuits that happen to publish the same reactance:
    22 groups covering 45 rows at 400 kV. Two parallel circuits
    carry twice what one carries at the same angle, so collapsing
    them understates the network - and it understates it silently,
    which is worse. Every landing is visited twice, once from each
    end, and the row object distinguishes that from two rows that
    merely look alike. (Codex 202609020030, finding 2.) */
    if (seen.has(entry.row)) continue;
    seen.add(entry.row);

    const xPct = entry.row.x_pct_100mva;
    if (!Number.isFinite(xPct)) { skippedNoReactance += 1; continue; }
    if (xPct === 0) {
      /* physically a short: the two nodes are the same bus */
      if (uf.union(name, far)) shorted += 1;
      continue;
    }
    branches.push({
      from: name, to: far, kind: entry.kind,
      x_pu: xPct / 100,
      row: entry.row
    });
  }
}

/* After shorting, work in terms of bus representatives. */
const busOf = (name) => uf.find(name);
const buses = [...new Set(nodeNames.map(busOf))].sort();
const busIndex = new Map(buses.map((b, i) => [b, i]));

const edges = [];
for (const b of branches) {
  const i = busIndex.get(busOf(b.from));
  const j = busIndex.get(busOf(b.to));
  if (i === undefined || j === undefined || i === j) continue;
  edges.push({ i, j, b: 1 / b.x_pu, meta: b });
}

/* Which buses can actually reach each other.
-----
A DC transfer between two components is not a transfer; it is a
singular system, and asking a solver for it produces a large number
rather than an error. So the components are computed here, once,
and every answer names the one it was solved in. */
const comp = makeUnionFind();
for (const b of buses) comp.find(b);
for (const e of edges) comp.union(buses[e.i], buses[e.j]);
const componentOf = (bus) => comp.find(bus);
const componentSizes = new Map();

```

```

for (const b of buses) {
  const root = componentOf(b);
  componentSizes.set(root, (componentSizes.get(root) || 0) + 1);
}
/* the degree of each bus, for the declared sink rule below */
const degree = new Map();
for (const e of edges) {
  degree.set(buses[e.i], (degree.get(buses[e.i]) || 0) + 1);
  degree.set(buses[e.j], (degree.get(buses[e.j]) || 0) + 1);
}

return {
  schema: SCHEMA,
  declared_model: DECLARED_MODEL,
  componentOf,
  componentSize: (bus) => componentSizes.get(componentOf(bus)) || 0,
  degreeOf: (bus) => degree.get(bus) || 0,
  voltage_kv: voltageKv,
  includes_transformers: includeTransformers,
  buses, busIndex, busOf, edges,
  counts: {
    nodes: nodeNames.length,
    buses: buses.length,
    branches: edges.length,
    shorted_zero_reactance: shorted,
    skipped_no_published_reactance: skippedNoReactance,
    components: componentSizes.size,
    largest_component: Math.max(0, ...componentSizes.values())
  }
};
}

/* — sparse conjugate gradient on the reduced B' matrix ————— */
function multiply(model, x, slackIndex) {
  const y = new Float64Array(x.length);
  for (const e of model.edges) {
    if (e.i === slackIndex || e.j === slackIndex) {
      /* the slack angle is pinned at zero, so its column contributes
         nothing and its row is not solved */
      if (e.i !== slackIndex) y[e.i] += e.b * x[e.i];
      if (e.j !== slackIndex) y[e.j] += e.b * x[e.j];
      continue;
    }
    const d = x[e.i] - x[e.j];
    y[e.i] += e.b * d;
    y[e.j] -= e.b * d;
  }
  return y;
}

function solve(model, injection, slackIndex, tolerance, maxIterations) {
  const n = model.buses.length;
  const x = new Float64Array(n);
  let r = new Float64Array(injection);
  r[slackIndex] = 0;
  let p = new Float64Array(r);
  let rr = 0;
  for (let k = 0; k < n; k += 1) rr += r[k] * r[k];
  const target = tolerance * tolerance * Math.max(rr, 1e-30);
  let iterations = 0;
  for (; iterations < maxIterations && rr > target; iterations += 1) {
    const ap = multiply(model, p, slackIndex);

```

```

    let pap = 0;
    for (let k = 0; k < n; k += 1) pap += p[k] * ap[k];
    if (!(Math.abs(pap) > 1e-30)) break;
    const alpha = rr / pap;
    let rrNext = 0;
    for (let k = 0; k < n; k += 1) {
        x[k] += alpha * p[k];
        r[k] -= alpha * ap[k];
        rrNext += r[k] * r[k];
    }
    const beta = rrNext / rr;
    for (let k = 0; k < n; k += 1) p[k] = r[k] + beta * p[k];
    rr = rrNext;
}
x[slackIndex] = 0;
return { theta: x, iterations, residual: Math.sqrt(rr) };
}

const SINK_RULE =
    'Where no withdrawal bus is declared, the sink is the most connected '
    + 'bus in the SAME component as the injection - the bus with the most '
    + 'published branches landing on it. It is a stated rule, not a '
    + 'convenience: a transfer has two ends and the answer is meaningless '
    + 'without naming both. Declare a sink to override it.';

/**
 * The withdrawal bus this model would choose, and why.
 *
 * Returns null when the injection bus has no one to transfer to, which
 * is a real answer about an isolated bus and not a failure.
 */
function sinkFor(model, atNode) {
    if (!model || typeof model.componentOf !== 'function') return null;
    const atBus = model.busOf(atNode);
    const component = model.componentOf(atBus);
    let best = null;
    let bestDegree = -1;
    for (const bus of model.buses) {
        if (bus === atBus) continue;
        if (model.componentOf(bus) !== component) continue;
        const d = model.degreeOf(bus);
        if (d > bestDegree || (d === bestDegree && best !== null && bus < best)) {
            best = bus;
            bestDegree = d;
        }
    }
    return best;
}

/**
 * Inject `mw` at one node and withdraw it at the declared slack; report
 * the flow this puts on every branch that carries a meaningful share.
 *
 * @param model      from modelFor()
 * @param options    { atNode, slackNode, mw, minimumShare }
 */
function respond(model, options) {
    const opts = options || {};
    const mw = Number.isFinite(opts.mw) ? opts.mw : 100;
    const atBus = model.busOf(opts.atNode);
    const slackBus = model.busOf(opts.slackNode);
    const i = model.busIndex.get(atBus);

```

```

const s = model.busIndex.get(slackBus);
if (i === undefined || s === undefined) return null;
if (i === s) {
  return {
    schema: SCHEMA,
    declared_model: DECLARED_MODEL,
    injected_mw: mw,
    at_node: opts.atNode,
    slack_node: opts.slackNode,
    same_bus: true,
    reason: 'the injection point and the slack are the same electrical '
      + 'bus once zero-reactance branches are shorted, so there is no '
      + 'transfer to distribute',
    branches: [],
    not_a_loading: NOT_A_LOADING,
    not_a_connection_offer: NOT_A_CONNECTION_OFFER
  };
}

/* A transfer between two components does not exist. The solver would
answer anyway - with a large number - so the refusal is here, before
it is asked. (Codex 202609020030, finding 1.) */
if (typeof model.componentOf === 'function'
&& model.componentOf(atBus) !== model.componentOf(slackBus)) {
  return {
    schema: SCHEMA,
    declared_model: DECLARED_MODEL,
    injected_mw: mw,
    at_node: opts.atNode,
    slack_node: opts.slackNode,
    same_bus: false,
    publishable: false,
    reason: 'the injection bus and the withdrawal bus are in different '
      + 'connected components of the published network at this voltage, '
      + 'so there is no transfer between them to distribute. The model '
      + 'has ' + (model.counts ? model.counts.components : 'several')
      + ' components at this voltage; a transfer must be solved within
one.',
    branches: [],
    component: {
      injection: model.componentOf(atBus),
      slack: model.componentOf(slackBus),
      injection_component_buses: model.componentSize(atBus)
    },
    sink_rule: SINK_RULE,
    not_a_loading: NOT_A_LOADING,
    not_a_connection_offer: NOT_A_CONNECTION_OFFER
  };
}

const n = model.buses.length;
const p = new Float64Array(n);
p[i] = mw / BASE_MVA; /* per unit on the declared base */
p[s] = -mw / BASE_MVA;

const solved = solve(model, p, s, 1e-10, Math.min(4 * n, 20000));

const minimumShare = Number.isFinite(opts.minimumShare) ? opts.minimumShare
: 0.01;
const flows = [];
for (const e of model.edges) {
  const flowPu = (solved.theta[e.i] - solved.theta[e.j]) * e.b;

```

```

const flowMw = flowPu * BASE_MVA;
const share = mw === 0 ? 0 : flowMw / mw;
if (Math.abs(share) < minimumShare) continue;
const row = e.meta.row;
const ratings = {};
for (const [season, field] of [['winter', 'winter_mva'], ['spring',
'spring_mva'],
['summer', 'summer_mva'], ['autumn', 'autumn_mva']]) {
  if (Number.isFinite(row[field])) ratings[season] = row[field];
}
flows.push({
  from_node: e.meta.from,
  to_node: e.meta.to,
  kind: e.meta.kind,
  x_pct_100mva: e.meta.x_pu * 100,
  flow_mw: flowMw,
  share_of_injection: share,
  published_ratings_mva: Object.keys(ratings).length ? ratings : null,
  transformer_rating_mva: e.meta.kind === 'transformer'
    && Number.isFinite(row.rating_mva) ? row.rating_mva : null
});
}
flows.sort((a, b) => Math.abs(b.share_of_injection) -
Math.abs(a.share_of_injection));

/* Validation carried in the answer, and GLOBAL.
-----
This checked Kirchhoff at the injection bus alone, which a
disconnected pair can satisfy while the solve has not converged at
all: a minimal A-B / C-D fixture injecting at A and withdrawing at D
returned validation.passes true with converged false. A page gated on
that could print an impossible transfer.

Now every bus is checked. Net flow must be the injection at the
injection bus, minus it at the slack, and zero everywhere else. */
const net = new Float64Array(model.buses.length);
for (const e of model.edges) {
  const flowPu = (solved.theta[e.i] - solved.theta[e.j]) * e.b;
  net[e.i] += flowPu;
  net[e.j] -= flowPu;
}
const kirchhoff = net[i] * BASE_MVA / (mw || 1);
let worstBusError = 0;
let worstBus = null;
for (let k = 0; k < net.length; k += 1) {
  const expected = k === i ? mw / BASE_MVA : (k === s ? -mw / BASE_MVA : 0);
  const error = Math.abs(net[k] - expected);
  if (error > worstBusError) { worstBusError = error; worstBus =
model.buses[k]; }
}
const worstBusMw = worstBusError * BASE_MVA;

return {
  schema: SCHEMA,
  declared_model: DECLARED_MODEL,
  injected_mw: mw,
  at_node: opts.atNode,
  slack_node: opts.slackNode,
  same_bus: false,
  branches: flows,
  counts: {
    branches_in_model: model.edges.length,

```

```

    branches_carrying_at_least: minimumShare,
    branches_reported: flows.length
  },
  convergence: {
    iterations: solved.iterations,
    residual: solved.residual,
    converged: solved.residual < 1e-6
  },
  publishable: solved.residual < 1e-6 && worstBusMw < 1e-6 * Math.max(1, mw)
    && Math.abs(kirchhoff - 1) < 1e-6,
  sink_rule: SINK_RULE,
  component: {
    solved_in: typeof model.componentOf === 'function' ?
model.componentOf(atBus) : null,
    buses_in_component: typeof model.componentSize === 'function' ?
model.componentSize(atBus) : null
  },
  validation: {
    kirchhoff_at_injection: kirchhoff,
    kirchhoff_error: Math.abs(kirchhoff - 1),
    worst_bus_error_mw: worstBusMw,
    worst_bus: worstBus,
    /* ALL THREE, not one. Convergence alone is not conservation, and
    conservation at one bus is not conservation. */
    passes: Math.abs(kirchhoff - 1) < 1e-6
      && worstBusMw < 1e-6 * Math.max(1, mw)
      && solved.residual < 1e-6,
    /* Exact by Kirchhoff's current law under the DC model: everything
    injected at a bus must leave it along the branches. It is
    checked at runtime and carried in the answer rather than
    asserted in a comment, because a solve that has not converged
    produces a plausible-looking set of flows that are wrong. */
    what_it_checks: 'the shares leaving the injection bus must sum to 1.0, '
      + 'AND net flow must be zero at every other bus, AND the solve '
      + 'must have converged. Any one of the three alone can hold while '
      + 'the answer is wrong.'
  },
  not_a_loading: NOT_A_LOADING,
  not_a_connection_offer: NOT_A_CONNECTION_OFFER
};
}

/**
 * Convenience: build a model over every node at one voltage.
 */
function modelFor(index, options) {
  if (!index || typeof index.graph !== 'function') return null;
  const graph = index.graph();
  if (!graph || graph.schema !== REQUIRES) return null;
  const opts = options || {};
  const voltageKv = Number.isFinite(opts.voltageKv) ? opts.voltageKv : null;
  const includeTransformers = opts.includeTransformers === true;

  const names = [];
  for (const name of (opts.nodeNames || [])) {
    if (voltageKv == null || graph.nodeVoltageKv(name) === voltageKv)
names.push(name);
  }
  if (!names.length) return null;
  return assemble(graph, names, { voltageKv, includeTransformers });
}

```



```

NS.injectionResponse = Object.freeze({
  schema: SCHEMA,
  requires: REQUIRES,
  base_mva: BASE_MVA,
  declared_model: DECLARED_MODEL,
  not_a_loading: NOT_A_LOADING,
  not_a_connection_offer: NOT_A_CONNECTION_OFFER,
  sink_rule: SINK_RULE,
  modelFor,
  assemble,
  sinkFor,
  respond
});
})();
})();

/**
 * Module: planned-change
 *
 * What the network operator has PUBLISHED as planned for the circuits and
 * transformers that touch one site: additions, changes and removals, by
 * the year they are published against.
 *
 * WHY THIS EXISTS
 * -----
 * Appendix B carries 2,230 rows that are not the network. They are the
 * operator's statement of what the network is planned to look like in
 * 2026, 2028, 2030 and 2033: a circuit to be added, a circuit whose
 * parameters are to change, a transformer to be removed. The topology
 * module carries them and the graph it hands out deliberately refuses to
 * walk them, which is right - a circuit published for 2030 is not a path
 * a current can take today. But refusing to walk them is not the same as
 * reporting them, and a reader looking at a site with four published
 * additions in 2028 is entitled to be told so, with the year and the
 * status and the published parameters, in the operator's own words.
 *
 * So this module reports the planned rows that land at a site, grouped by
 * year and then by status, with real counts. It reads the same product the
 * topology module reads and resolves sites, nodes and voltages through the
 * topology index rather than through a second opinion of its own.
 *
 * WHAT IT IS NOT
 * -----
 * A published plan is not infrastructure. Every entry this module returns
 * is marked as a publication about a future year, it is never mixed into
 * a list of circuits that exist, and nothing here can be traversed: the
 * module contains no path, no hop and no neighbour, and the graph it
 * borrows excludes these rows from its edges by construction.
 *
 * A published plan is not a commitment either. The operator publishes
 * planned changes as its current view of network development; the view
 * moves between editions, an addition can be deferred or dropped, and a
 * year against a row is the year the row is published for, not a
 * consent, not a delivery date and not a date on which anything could
 * connect. A "Removed" row says a circuit is planned to be taken out; it
 * does not say why, and it does not say what replaces it.
 *
 * And, as everywhere in this estate: nothing here states whether a project
 * can connect. R, X and B on a planned row are carried as published on a
 * 100 MVA base and never computed with; ratings on a planned row are the
 * planned circuit's, not a spare allowance; voltages are trusted only where
 * the product says the site declares them and are never decoded from a
 * node code.

```

```

*
* ONE PUBLISHED FACT THAT IS WORTH CARRYING
* -----
* 552 of the 2,230 planned rows sit on a node pair that already has a
* circuit or transformer published for today, and 16 of those are marked
* "Addition" - a second circuit on an existing pair, on the face of it.
* Whether a pair is published today is a fact from the same product, so
* each entry carries it. It is a cross-reference, not a judgement about
* what the addition means.
*
* node tools/proofs/modules/202609012345-planned-change.proof.mjs
*/
(() => {
  'use strict';

  const NS = window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ || {};
  if (NS.plannedChange) return;

  const SCHEMA = 'gridatlas.module.planned-change.v1';
  const ACCEPTS = 'data-grid-gb.transmission-network.v1';
  const REQUIRES = 'gridatlas.module.network-topology.graph.v1';

  const NOT_EXISTING =
    'Every entry here is a change the network operator has published for '
    + 'a future year. None of it is a circuit or a transformer that exists '
    + 'today, none of it is a path, and none of it is counted among the '
    + 'site\'s circuits anywhere in this estate.';

  const NOT_A_COMMITMENT =
    'A published plan is the operator\'s current view of network '
    + 'development, and the view moves between editions. It is not a '
    + 'commitment to build, not a consent, and the year on a row is the '
    + 'year it is published for - not a delivery date and not a date on '
    + 'which anything could connect.';

  const NOT_AN_ASSESSMENT =
    'Nothing here states whether any project can connect at this site, '
    + 'before or after a planned change. That depends on queue position, '
    + 'committed connections, consent and commercial terms which no '
    + 'published appendix contains. A rating on a planned row is the '
    + 'planned asset\'s rating, not a spare allowance.';

  const IMPEDANCE_BASIS =
    'R, X and B on a planned row are percentages on a 100 MVA base, as '
    + 'published for the planned asset. They are carried and not computed '
    + 'with.';

  /* The order the statuses are presented in. Anything the product
     publishes that is not one of these three is kept and sorted after
     them by name, never dropped. */
  const STATUS_ORDER = Object.freeze(['Addition', 'Change', 'Removed']);
  const ASSETS = Object.freeze(['circuit', 'transformer']);

  const asString = (v) => (typeof v === 'string' && v.length ? v : null);
  const asNumber = (v) => (Number.isFinite(v) ? v : null);

  function statusRank(status) {
    const i = STATUS_ORDER.indexOf(status);
    return i === -1 ? STATUS_ORDER.length : i;
  }

  /* Years are published as strings ("2026"). They are sorted numerically

```

```

    where they parse and left in their published form on the entry. */
function yearRank(year) {
    const n = Number(year);
    return Number.isFinite(n) ? n : Number.POSITIVE_INFINITY;
}

/**
 * Is there a circuit or a transformer published for TODAY between
 * these two nodes? Read off the graph's edges, which are circuits and
 * transformers only, so a planned row can never vouch for itself.
 */
function publishedToday(graph, nearNode, farNode) {
    const today = { circuit: false, transformer: false };
    for (const entry of graph.edgesAt(nearNode)) {
        if (entry.row[entry.far] !== farNode) continue;
        if (entry.kind === 'circuit') today.circuit = true;
        if (entry.kind === 'transformer') today.transformer = true;
    }
    return today;
}

/**
 * @param product the parsed data-grid-gb transmission-network payload
 * @returns an index, or null if the schema is not the one this reads
 */
function index(product) {
    if (!product || product.schema !== ACCEPTS) return null;
    const topology = NS.networkTopology;
    if (!topology || typeof topology.index !== 'function') return null;
    const base = topology.index(product);
    if (!base || typeof base.graph !== 'function') return null;
    const graph = base.graph();
    if (!graph || graph.schema !== REQUIRES) return null;

    const rows = Array.isArray(product.planned_changes) ?
product.planned_changes : [];

    /* Planned rows land on their two nodes exactly as branches do in the
    topology module, so that a site query is a lookup and not a scan. */
    const byNode = new Map();
    for (const row of rows) {
        if (!row) continue;
        for (const [near, far] of [['node_1', 'node_2'], ['node_2', 'node_1']]) {
            const name = row[near];
            if (!name) continue;
            if (!byNode.has(name)) byNode.set(name, []);
            byNode.get(name).push({ row, near, far });
        }
    }

    /* Product-wide tallies. These are counts of published rows, each
    counted once, and they are the only place in this module where a
    row is counted without reference to a site. */
    const tally = { by_year: {}, by_status: {}, by_asset: {} };
    for (const row of rows) {
        if (!row) continue;
        const y = asString(row.year) || 'unstated';
        const s = asString(row.status) || 'unstated';
        const a = asString(row.asset) || 'unstated';
        tally.by_year[y] = (tally.by_year[y] || 0) + 1;
        tally.by_status[s] = (tally.by_status[s] || 0) + 1;
        tally.by_asset[a] = (tally.by_asset[a] || 0) + 1;
    }
}

```

```

}

function describe(landing, nearNode) {
  const row = landing.row;
  const farNode = row[landing.far];
  const farSiteCode = graph.nodeSiteCode(farNode) || null;
  const farSite = farSiteCode ? graph.siteByCode(farSiteCode) : null;
  const nearSiteCode = graph.nodeSiteCode(nearNode) || null;
  const asset = asString(row.asset);
  const entry = {
    publication: 'planned',
    year: asString(row.year),
    status: asString(row.status),
    asset,
    from_node: nearNode,
    to_node: farNode,
    from_site_code: nearSiteCode,
    to_site_code: farSiteCode,
    to_site_name: farSite ? farSite.name : null,
    within_this_site: !!farSiteCode && farSiteCode === nearSiteCode,
    /* trusted only where the site declares it; null otherwise */
    from_voltage_kv: graph.nodeVoltageKv(nearNode),
    to_voltage_kv: graph.has(farNode) ? graph.nodeVoltageKv(farNode) : null,
    transmission_owner: asString(row.transmission_owner),
    labels: Array.isArray(row.labels) ? row.labels.slice() : [],
    /* carried, never computed with */
    parameters_pct_100mva: graph.parametersOf(row),
    pair_published_today: graph.has(farNode)
      ? publishedToday(graph, nearNode, farNode)
      : { circuit: false, transformer: false }
  };
  if (asset === 'transformer') {
    entry.rating_mva = asNumber(row.rating_mva);
    entry.voltage_ratio_kv = asString(row.voltage_ratio_kv);
  } else {
    entry.circuit_type = asString(row.circuit_type);
    entry.ohl_km = asNumber(row.ohl_km);
    entry.cable_km = asNumber(row.cable_km);
    entry.ratings_mva = graph.ratingsOf(row);
  }
  return entry;
}

/**
 * Every planned change landing at one site, grouped by year and then
 * by status. A row landing on two nodes of the same site is reported
 * once, from the first node it is met at in sorted node order.
 *
 * @param key      site code or exact site name
 * @param options  { voltageKv } to restrict to rows landing on a node
 *                  the site declares at that voltage
 */
function at(key, options) {
  const site = base.site(key);
  if (!site) return null;
  const opts = options || {};
  const voltageKv = Number.isFinite(opts.voltageKv) ? opts.voltageKv : null;

  const nodes = graph.nodesOfSite(site.code)
    .filter((name) => voltageKv == null || graph.nodeVoltageKv(name) ===
voltageKv);

```

```

const seen = new Set();
const entries = [];
for (const nodeName of nodes) {
  for (const landing of byNode.get(nodeName) || []) {
    if (seen.has(landing.row)) continue;
    seen.add(landing.row);
    entries.push(describe(landing, nodeName));
  }
}

/* year -> status -> entries, in a stable presentation order */
const years = new Map();
for (const entry of entries) {
  const y = entry.year || 'unstated';
  if (!years.has(y)) years.set(y, new Map());
  const statuses = years.get(y);
  const s = entry.status || 'unstated';
  if (!statuses.has(s)) statuses.set(s, []);
  statuses.get(s).push(entry);
}

const by_year = [...years.entries()]
  .sort((a, b) => yearRank(a[0]) - yearRank(b[0]) ||
a[0].localeCompare(b[0]))
  .map(([year, statuses]) => {
    const by_status = [...statuses.entries()]
      .sort((a, b) => statusRank(a[0]) - statusRank(b[0]) ||
a[0].localeCompare(b[0]))
      .map(([status, list]) => {
        list.sort((a, b) =>
String(a.to_node).localeCompare(String(b.to_node)));
        const by_asset = {};
        for (const a of ASSETS) by_asset[a] = list.filter((e) => e.asset
=== a).length;
        return { status, entries: list, counts: { entries: list.length,
by_asset } };
      });
    const counts = { entries: 0, by_status: {} };
    for (const group of by_status) {
      counts.entries += group.counts.entries;
      counts.by_status[group.status] = group.counts.entries;
    }
    return { year, by_status, counts };
  });

const counts = { planned_changes: entries.length, by_year: {}, by_status:
{}, by_asset: {} };
for (const y of by_year) counts.by_year[y.year] = y.counts.entries;
for (const e of entries) {
  const s = e.status || 'unstated';
  const a = e.asset || 'unstated';
  counts.by_status[s] = (counts.by_status[s] || 0) + 1;
  counts.by_asset[a] = (counts.by_asset[a] || 0) + 1;
}
counts.on_a_pair_published_today = entries
  .filter((e) => e.pair_published_today.circuit ||
e.pair_published_today.transformer).length;

return {
  schema: SCHEMA,
  source: ACCEPTS,
  site: { code: site.code, name: site.name },

```

```

    requested_voltage_kv: voltageKv,
    scope: voltageKv == null
      ? 'rows landing on any node of this site; each entry carries the '
        + 'declared voltage of the node it lands on, and undeclared is '
        + 'undeclared'
      : 'rows landing on a node this site declares at ' + voltageKv + ' kV
only',
    nodes_considered: nodes.length,
    by_year,
    counts,
    not_existing: NOT_EXISTING,
    not_a_commitment: NOT_A_COMMITMENT,
    not_an_assessment: NOT_AN_ASSESSMENT,
    impedance_basis: IMPEDANCE_BASIS
  };
}

return {
  schema: SCHEMA,
  source: ACCEPTS,
  counts: Object.assign({ planned_changes: rows.length }, tally),
  site: base.site,
  at
};
}

NS.plannedChange = Object.freeze({
  schema: SCHEMA,
  accepts: ACCEPTS,
  requires: REQUIRES,
  status_order: STATUS_ORDER,
  not_existing: NOT_EXISTING,
  not_a_commitment: NOT_A_COMMITMENT,
  not_an_assessment: NOT_AN_ASSESSMENT,
  impedance_basis: IMPEDANCE_BASIS,
  index
});
})();

/**
 * Module: owner-boundary
 *
 * Which transmission owner the published assets at a site belong to, and
 * which circuits cross from one owner's network into another's.
 *
 * WHY THIS EXISTS
 * -----
 * Great Britain's transmission network is not one network. Appendix B
 * publishes a `transmission_owner` on every site, node, circuit and
 * transformer, and four values occur: NGET in England and Wales, SPT in
 * southern Scotland, SHET in the north of Scotland, and OFTO for the
 * offshore assets. Most circuits sit wholly inside one owner's network.
 * Sixty-two do not: their two ends are nodes that different owners
 * publish, and a circuit like that is the seam between two networks.
 *
 * That seam is worth naming because a connection near it involves more
 * than one party. It is a fact about who publishes what, read straight
 * off the product, and it is reported here with both owners named on
 * every boundary circuit so that nobody has to infer it from a map colour.
 *
 * WHAT IT IS NOT
 * -----

```

```

* Ownership is not a statement about who a project would contract with.
* Connection agreements in Great Britain are made with the system operator
* and the relevant owner under a framework this data does not describe,
* and a site being NGET's says nothing about the counterparty, the process
* or the terms of any connection at it. This module reports the published
* owner of the published assets and stops there.
*
* Nor is an owner ever inferred. Forty-nine nodes publish no owner - all
* of them on placeholder site codes such as OFFS and ONSH that the product
* does not list as sites - and where a node's owner is not published it is
* reported as unknown. A circuit with an unknown end is reported as
* undetermined, not as a boundary and not as internal. Nothing is read
* from a site name, a node code or a neighbour.
*
* TWO DIFFERENT FACTS, KEPT APART
* -----
* A circuit carries its own `transmission_owner`, and so do the nodes at
* its two ends. A BOUNDARY circuit is one whose two END nodes belong to
* different owners. Separately, seven circuits in the product carry an
* owner that matches neither end - SPT and OFTO circuits between SHET
* nodes at Hunterston, Inverness and Nedd. That is not a boundary by the
* definition above; it is the asset's own published owner differing from
* the owner of the nodes it lands on, and it is reported as exactly that.
*
* Voltages are trusted only where the site declares them and never decoded
* from a node code; assets are counted per voltage and never across
* voltages. R, X and B are carried and never computed with. No rating here
* is headroom.
*
* node tools/proofs/modules/202609012350-owner-boundary.proof.mjs
*/
((() => {
  'use strict';

  const NS = window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ || {};
  if (NS.ownerBoundary) return;

  const SCHEMA = 'gridatlas.module.owner-boundary.v1';
  const ACCEPTS = 'data-grid-gb.transmission-network.v1';
  const REQUIRES = 'gridatlas.module.network-topology.graph.v1';

  const NOT_A_COUNTERPARTY =
    'The transmission owner is the party the network operator publishes '
    + 'as owning an asset. It is not a statement about who a project would '
    + 'contract with, under what process, or on what terms; none of that '
    + 'is in any published appendix.';

  const NOT_AN_ASSESSMENT =
    'An ownership boundary is a fact about who publishes which asset. It '
    + 'says nothing about whether any project can connect on either side '
    + 'of it, and a rating on a boundary circuit is that circuit\'s rating, '
    + 'not a spare allowance across the boundary.';

  const NEVER_INFERRED =
    'An owner is reported only where the product publishes one on the '
    + 'record in question. A node with no published owner is unknown, a '
    + 'circuit with an unknown end is undetermined, and nothing is read '
    + 'from a site name, a node code or a neighbour.';

  const UNKNOWN = 'unknown';
  const UNDECLARED = 'undeclared';

```

```

const asString = (v) => (typeof v === 'string' && v.length ? v : null);
const asNumber = (v) => (Number.isFinite(v) ? v : null);
const bandKey = (kv) => (kv == null ? UNDECLARED : String(kv));

/**
 * @param product the parsed data-grid-gb transmission-network payload
 * @returns an index, or null if the schema is not the one this reads
 */
function index(product) {
  if (!product || product.schema !== ACCEPTS) return null;
  const topology = NS.networkTopology;
  if (!topology || typeof topology.index !== 'function') return null;
  const base = topology.index(product);
  if (!base || typeof base.graph !== 'function') return null;
  const graph = base.graph();
  if (!graph || graph.schema !== REQUIRES) return null;

  /* The owner of a NODE, from the node record and nowhere else. The
   graph does not hand this out, so it is read from the product here;
   the graph is still the only authority on which nodes exist, which
   site they belong to and what voltage may be trusted. */
  const nodeOwner = new Map();
  for (const node of product.nodes || []) {
    if (node && node.node) nodeOwner.set(node.node,
asString(node.transmission_owner));
  }
  const ownerOfNode = (name) => (nodeOwner.has(name) ? nodeOwner.get(name) :
null);

  /**
   * The relation between the two ends of a branch, by published owner.
   * 'boundary' both ends published, and they differ
   * 'internal' both ends published, and they agree
   * 'undetermined' at least one end publishes no owner
   */
  function relation(nearOwner, farOwner) {
    if (nearOwner == null || farOwner == null) return 'undetermined';
    return nearOwner === farOwner ? 'internal' : 'boundary';
  }

  function describe(entry, nearNode) {
    const row = entry.row;
    const farNode = row[entry.far];
    const nearOwner = ownerOfNode(nearNode);
    const farOwner = graph.has(farNode) ? ownerOfNode(farNode) : null;
    const assetOwner = asString(row.transmission_owner);
    const nearSiteCode = graph.nodeSiteCode(nearNode) || null;
    const farSiteCode = graph.has(farNode) ? graph.nodeSiteCode(farNode) ||
null : null;
    const farSite = farSiteCode ? graph.siteByCode(farSiteCode) : null;
    const out = {
      kind: entry.kind,
      from_node: nearNode,
      to_node: farNode,
      from_site_code: nearSiteCode,
      to_site_code: farSiteCode,
      to_site_name: farSite ? farSite.name : null,
      within_this_site: !!farSiteCode && farSiteCode === nearSiteCode,
      from_voltage_kv: graph.nodeVoltageKv(nearNode),
      to_voltage_kv: graph.has(farNode) ? graph.nodeVoltageKv(farNode) : null,
      /* the three published owners, each named for what it is */
      from_owner: nearOwner || UNKNOWN,

```



```

    to_owner: farOwner || UNKNOWN,
    asset_owner: assetOwner || UNKNOWN,
    ends: relation(nearOwner, farOwner),
    /* the asset's own owner set against the ends it lands on; null
       where either end is unknown, because "matches neither" cannot
       be said of an end that has not been published */
    asset_owner_matches_an_end: assetOwner && nearOwner && farOwner
      ? (assetOwner === nearOwner || assetOwner === farOwner)
      : null,
    /* carried, never computed with */
    parameters_pct_100mva: graph.parametersOf(row)
  };
  if (entry.kind === 'circuit') {
    out.circuit_type = asString(row.circuit_type);
    out.ohl_km = asNumber(row.ohl_km);
    out.cable_km = asNumber(row.cable_km);
    out.ratings_mva = graph.ratingsOf(row);
  } else {
    out.rating_mva = asNumber(row.rating_mva);
    out.voltage_ratio_kv = asString(row.voltage_ratio_kv);
  }
  return out;
}

/**
 * Ownership at one site: the site's own published owner, the owner of
 * each of its nodes, the assets landing on those nodes counted per
 * owner within each voltage, and every boundary branch named with both
 * owners.
 *
 * @param key      site code or exact site name
 * @param options  { voltageKv } to restrict to nodes the site declares
 *                  at that voltage
 */
function at(key, options) {
  const site = base.site(key);
  if (!site) return null;
  const opts = options || {};
  const voltageKv = Number.isFinite(opts.voltageKv) ? opts.voltageKv : null;

  const nodeNames = graph.nodesOfSite(site.code)
    .filter((name) => voltageKv == null || graph.nodeVoltageKv(name) ===
voltageKv);

  const nodes = nodeNames.map((name) => ({
    node: name,
    voltage_kv: graph.nodeVoltageKv(name),
    transmission_owner: ownerOfNode(name) || UNKNOWN
  }));

  /* Assets are counted once per site even when both ends are here,
     and grouped under the declared voltage of the node they were
     first met at in sorted node order. */
  const seen = new Set();
  const bands = new Map();
  const boundary_circuits = [];
  const boundary_transformers = [];
  const undetermined = [];
  const asset_owner_differs = [];

  function band(kv) {
    const k = bandKey(kv);

```

```

        if (!bands.has(k)) {
            bands.set(k, { voltage_kv: kv, by_owner: {}, circuits: 0,
transformers: 0, nodes: 0 });
        }
        return bands.get(k);
    }
    function count(b, owner, what) {
        const o = owner || UNKNOWN;
        if (!b.by_owner[o]) b.by_owner[o] = { nodes: 0, circuits: 0,
transformers: 0 };
        b.by_owner[o][what] += 1;
        b[what] += 1;
    }

    for (const n of nodes) count(band(n.voltage_kv), n.transmission_owner,
'nodes');

    for (const nodeName of nodeNames) {
        for (const entry of graph.edgesAt(nodeName)) {
            if (seen.has(entry.row)) continue;
            seen.add(entry.row);
            const d = describe(entry, nodeName);
            const b = band(d.from_voltage_kv);
            count(b, d.asset_owner === UNKNOWN ? null : d.asset_owner,
                entry.kind === 'circuit' ? 'circuits' : 'transformers');
            if (d.ends === 'boundary') {
                (entry.kind === 'circuit' ? boundary_circuits :
boundary_transformers).push(d);
            } else if (d.ends === 'undetermined') {
                undetermined.push(d);
            }
            if (d.asset_owner_matches_an_end === false)
asset_owner_differs.push(d);
        }
    }

    const by_voltage = [...bands.entries()]
        .sort((a, b) => {
            if (a[0] === UNDECLARED) return 1;
            if (b[0] === UNDECLARED) return -1;
            return Number(b[0]) - Number(a[0]);
        })
        .map(([, b]) => b);

    const owners = new Set();
    for (const b of by_voltage) for (const o of Object.keys(b.by_owner))
owners.add(o);

    const byPair = (list) => list.sort((a, b) =>
        String(a.from_node).localeCompare(String(b.from_node))
        || String(a.to_node).localeCompare(String(b.to_node)));

    return {
        schema: SCHEMA,
        source: ACCEPTS,
        site: {
            code: site.code,
            name: site.name,
            transmission_owner: asString(site.transmission_owner) || UNKNOWN
        },
        requested_voltage_kv: voltageKv,
        scope: voltageKv == null
    }

```

```

        ? 'every node of this site, counted within its own declared voltage; '
        + 'no count here spans two voltages'
        : 'nodes this site declares at ' + voltageKv + ' kV only',
nodes,
by_voltage,
owners_present: [...owners].sort(),
boundary_circuits: byPair(boundary_circuits),
boundary_transformers: byPair(boundary_transformers),
undetermined: byPair(undetermined),
asset_owner_differs_from_both_ends: byPair(asset_owner_differs),
counts: {
    nodes: nodes.length,
    nodes_with_unknown_owner: nodes.filter((n) => n.transmission_owner ===
UNKNOWN).length,
    owners_present: owners.size,
    circuits: by_voltage.reduce((s, b) => s + b.circuits, 0),
    transformers: by_voltage.reduce((s, b) => s + b.transformers, 0),
    boundary_circuits: boundary_circuits.length,
    boundary_transformers: boundary_transformers.length,
    undetermined: undetermined.length,
    asset_owner_differs_from_both_ends: asset_owner_differs.length
},
not_a_counterparty: NOT_A_COUNTERPARTY,
never_inferred: NEVER_INFERRED,
not_an_assessment: NOT_AN_ASSESSMENT
};
}

/**
 * Every boundary branch in the product, each reported once, with both
 * owners named. The seam between the networks as a list.
 */
function boundaries() {
    const out = [];
    const seen = new Set();
    const pairs = {};
    for (const [kind, rows] of [['circuit', product.circuits], ['transformer',
product.transformers]]) {
        for (const row of rows || []) {
            if (!row || seen.has(row)) continue;
            seen.add(row);
            const d = describe({ kind, row, near: 'node_1', far: 'node_2' },
row.node_1);
            if (d.ends !== 'boundary') continue;
            out.push(d);
            const pair = [d.from_owner, d.to_owner].sort().join('/');
            pairs[pair] = (pairs[pair] || 0) + 1;
        }
    }
    return {
        schema: SCHEMA,
        source: ACCEPTS,
        branches: out.sort((a, b) =>
            String(a.from_node).localeCompare(String(b.from_node))
            || String(a.to_node).localeCompare(String(b.to_node))),
        counts: {
            boundary_circuits: out.filter((d) => d.kind === 'circuit').length,
            boundary_transformers: out.filter((d) => d.kind ===
'transformer').length,
            by_owner_pair: pairs
        },
        not_a_counterparty: NOT_A_COUNTERPARTY,
    }
}

```

```

        never_inferred: NEVER_INFERRED,
        not_an_assessment: NOT_AN_ASSESSMENT
    };
}

const ownerTally = {};
for (const node of product.nodes || []) {
    const o = (node && asString(node.transmission_owner)) || UNKNOWN;
    ownerTally[o] = (ownerTally[o] || 0) + 1;
}

return {
    schema: SCHEMA,
    source: ACCEPTS,
    counts: {
        nodes: nodeOwner.size,
        nodes_by_owner: ownerTally
    },
    site: base.site,
    at,
    boundaries
};
}

NS.ownerBoundary = Object.freeze({
    schema: SCHEMA,
    accepts: ACCEPTS,
    requires: REQUIRES,
    unknown: UNKNOWN,
    not_a_counterparty: NOT_A_COUNTERPARTY,
    never_inferred: NEVER_INFERRED,
    not_an_assessment: NOT_AN_ASSESSMENT,
    index
});
})();

/* =====
pinned-products - the runtime data this Atlas reads, by commit and by digest
===== */

```

A published Atlas release is immutable. Three of its runtime fetches were not: they named a BRANCH.

```

data-grid-gb          main/derived/connection-points.v3.json
data-grid-gb          main/derived/gb-transmission-network.v1.json
data-gb-electricity main/derived/price-decade-rollup.json

```

The only defence was a schema string, and a schema string defends SHAPE and is blind to VALUES. On 2026-09-03 that stopped being theoretical. A correction on data-grid-gb - branch codex/20260903-phase0-integrity, commit b91e45b - publishes deduplicated transformer counts under the IDENTICAL schema `data-grid-gb.connection-points.v3`:

```

COWLEY   transformers  10 -> 5
ABHAM    transformers   4 -> 2
located                502 -> 489      (886 points either way)

```

Every record in the file differs. A factor of two on a number the card prints, invisible to every check the consumer had. An immutable release would have changed what it said with none of its own bytes changing.

So the ref is a commit and the bytes are hashed. The NESO inputs upstream

are pinned by SHA-256 with exactly this rationale - document ids are stable and "latest" links are not - and the discipline simply had not reached the estate's own last hop.

MOVING A PIN IS A CUT, AND THAT IS THE POINT. A data correction and a map release become one event a reader can see. It is also a real cost: the correction above does NOT reach a reader until the pin moves here. That is the right trade, because the alternative is not knowing which of the two numbers is on the card - and this file is where a human decides.

This module lives in the substation-intelligence cartridge, which the shell evaluates before the sandbox, so both consumers read one table rather than each carrying its own copy of the constants and the digest arithmetic.

```

===== */
((() => {
  'use strict';

  const SCHEMA = 'gridatlas.module.pinned-products.v1';
  const RAW = 'https://raw.githubusercontent.com/VentusLtd/';

  /* Each entry: the repository, the 40-character commit the Atlas was built
     against, the path, and the SHA-256 of the bytes served at that commit.
     `bytes` is recorded beside the digest because a truncated response is
     the failure mode a digest catches last and a length catches first. */
  const PINS = {
    'connection-points.v3': {
      repository: 'data-grid-gb',
      ref: '1c9909d1138704b29235c27fd769436dda8a0b18',
      path: 'derived/connection-points.v3.json',
      sha256:
'11e28859a6d17cc8ee4047c2032d55d043be98f7123743f3b2b03225e07a4c0c',
      bytes: 2896561,
      schema: 'data-grid-gb.connection-points.v3'
    },
    'gb-transmission-network.v1': {
      repository: 'data-grid-gb',
      ref: '1c9909d1138704b29235c27fd769436dda8a0b18',
      path: 'derived/gb-transmission-network.v1.json',
      sha256:
'fc331cc20b061f85adf18d890762a164328a1c5e84acef6a23d35d36f849fc8a',
      bytes: 10069966,
      schema: 'data-grid-gb.transmission-network.v1'
    },
    'price-decade-rollup': {
      repository: 'data-gb-electricity',
      ref: 'd310e3cec8cd14bc7cd3eef1e37037197bcb0798',
      path: 'derived/price-decade-rollup.json',
      sha256:
'18da5059c93cf09f6036bfcaabf56afaedf16d5f03e664c3cf0b0cff1dca970d',
      bytes: 6873,
      schema: 'data-gb-electricity.price-decade-rollup.v2'
    }
  };

  function pin(id) {
    return Object.prototype.hasOwnProperty.call(PINS, id) ? PINS[id] : null;
  }

  function url(id) {
    const entry = pin(id);
    return entry ? RAW + entry.repository + '/' + entry.ref + '/' + entry.path :

```

```

null;
}

/* Absent crypto is NOT a mismatch.
-----
`crypto.subtle` exists only in a secure context, so a page served over
plain http - a local check-out, a preview server - has no digest to
offer. That is reported as unverified and the product is still read.
Only a real disagreement between the bytes and the recorded digest
refuses, because refusing on absence would make the Atlas unusable
anywhere but production while proving nothing about the bytes. */
function encode(text) {
  try {
    return typeof TextEncoder === 'function'
      ? new TextEncoder().encode(text) : null;
  } catch (_) { return null; }
}

async function digestBytes(bytes) {
  try {
    const subtle = (window.crypto || {}).subtle;
    if (!subtle || !bytes) return null;
    const digest = await subtle.digest('SHA-256', bytes);
    return Array.from(new Uint8Array(digest))
      .map(byte => byte.toString(16).padStart(2, '0')).join('');
  } catch (_) {
    /* A thrown digest is an unavailable digest, not a wrong one. */
    return null;
  }
}

async function digestHex(text) {
  return digestBytes(encode(text));
}

/**
 * @returns { state, sha256, expected, ref, bytes_seen, bytes_expected }
 *   state is 'verified', 'MISMATCH', or a stated reason it is unverified.
 *   Only 'MISMATCH' means the caller must refuse.
 *
 * `bytes_seen` is BYTES. The first cut of this module reported
 * `text.length`, which is UTF-16 code units: the node/branch product is
 * 10,069,964 characters and 10,069,966 bytes, so the field disagreed with
 * the `bytes` it was being compared against by two, on a file that was
 * entirely correct. A length is checked as well as a digest because
 * truncation is the failure a length names immediately and a digest only
 * says "different" about.
 */
async function verify(id, text) {
  const entry = pin(id);
  const bytes = encode(text);
  const seen = bytes ? bytes.length : null;
  if (!entry) {
    return { state: 'unverified: no pin for ' + String(id), sha256: null,
      expected: null, ref: null, bytes_seen: seen, bytes_expected: null };
  }
  const digest = await digestBytes(bytes);
  const answer = { sha256: digest, expected: entry.sha256, ref: entry.ref,
    bytes_seen: seen, bytes_expected: entry.bytes };
  if (seen !== null && seen !== entry.bytes) {
    answer.state = 'MISMATCH';
    answer.detail = 'the response at ' + entry.ref + ' is ' + seen
  }
}

```

```

    + ' bytes, not the recorded ' + entry.bytes;
  } else if (digest === null) {
    answer.state = 'unverified: no subtle crypto in this context';
  } else if (digest === entry.sha256) {
    answer.state = 'verified';
  } else {
    answer.state = 'MISMATCH';
    answer.detail = 'bytes at ' + entry.ref + ' hash to ' + digest
      + ', not the recorded ' + entry.sha256;
  }
  return answer;
}

const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||
{}));
NS.pinnedProducts = Object.freeze({
  schema: SCHEMA,
  ids: Object.freeze(Object.keys(PINS)),
  pin,
  url,
  digestHex,
  verify,
  why: 'A branch ref lets an immutable release change what it says without '
    + 'any of its own bytes changing. The schema string defends shape and is '
    + 'blind to values: a correction to data-grid-gb halves published '
    + 'transformer counts under the same schema. The ref is a commit and the '
    + 'bytes are hashed, so a data correction and a map release are one
event.',
  not_an_assessment: 'A pin says which bytes were read. It says nothing '
    + 'about whether those bytes are right.'
});
})();

```

```

/* =====
version-ledger - what every generation of this Atlas changed
===== */

```

The page shows its own version history (202609010204, v9.40), and the cut writes the newest row rather than a human appending it (202609012130, v9.65, after v9.64 shipped a ledger whose newest entry named v9.63 and so told a reader it was running the generation before the one it was running).

It lived in the sandbox body. That is 13,655 characters of pure data in the cartridge with the least room: the sandbox stood at 339,367 characters against the 340,000 guard its own proof asserts, and every cut adds another row. Three card-facing changes could not be made without either raising that guard - which v9.76 considered and rejected, and so did every cut tonight - or moving something out.

So it moves to the cartridge that has room, exactly as the five network modules did at v9.76. The shell evaluates substation-intelligence before the sandbox, so the sandbox reads it the same way it reads geodesy.

The array literal below keeps its exact original spelling, because that spelling is what tools/recompose.mjs matches when it appends a row at the cut. Naming it here in full would be matched too - and was: the first version of this comment quoted the declaration, recompose found the comment before it found the data, and tried to JSON.parse this sentence. A comment that a tool reads is code. The appending happens for the parts of every RESTAMPED cartridge, and every cut in this repository restamps both - the sandbox proof derives the generation from its own filename and holds the manifest to it, so a cut that skipped

the sandbox would fail three identity checks. If one ever did skip it, the ledger would stop growing silently; the check that its newest row is the composed version is what would catch that, and it is asserted against atlas/current.json rather than against a string in the same file.

```

===== */
(( ) => {
  'use strict';

  const VERSION_LEDGER = [{"g":"202608312121","v":"v9.16","s":"the project
arriving from Pipeline News is visible: its own technology layer is enabled and
a pin owned by this cartridge is dropped on it, with a toggle on the card"},
{"g":"202608312133","v":"v9.17","s":"central AC sizing: the limiting nameplate,
not a squared product"}, {"g":"202608312140","v":"v9.18","s":"the project marker
is a ring, found by looking at it in Chrome"},
{"g":"202608312154","v":"v9.19","s":"the grid maths installs even when the
basemap never paints"}, {"g":"202608312157","v":"v9.20","s":"the Atlas says what
it is waiting for, sized for a phone"}, {"g":"202608312205","v":"v9.21","s":"the
MapLibre exception storm: symbol layers with no glyph atlas"},
{"g":"202608312208","v":"v9.22","s":"a symbol layer is added only once its text
can be drawn"}, {"g":"202608312222","v":"v9.23","s":"card geometry resets on
every selection"}, {"g":"202608312227","v":"v9.24","s":"the GB electricity
tracker is connected to the map"}, {"g":"202608312238","v":"v9.25","s":"one
source of truth for GB prices: the data repository"},
{"g":"202608312244","v":"v9.26","s":"late layer controls are used, and the
repository is LF everywhere"}, {"g":"202608312257","v":"v9.27","s":"the MAP
button works for every technology in the register"},
{"g":"202608312300","v":"v9.28","s":"voltage classes are explained, and the
whole dashboard is accepted"}, {"g":"202608312306","v":"v9.29","s":"the headline
capacity actually moves the layout"}, {"g":"202608312313","v":"v9.30","s":"the
neon flow no longer exhausts the renderer"},
{"g":"202608312315","v":"v9.31","s":"Codex's LineAtlas cardinality gate
passes"}, {"g":"202608312317","v":"v9.32","s":"no substation can display an
impossible voltage"}, {"g":"202608312321","v":"v9.33","s":"nothing can rewrite
the reference design, not even later"}, {"g":"202608312324","v":"v9.34","s":"a
missing source costs a drawing, never the session"},
{"g":"202609010021","v":"v9.35","s":"phone pointer operation, viewport
containment and named electrical ratios"},
{"g":"202609010040","v":"v9.36","s":"original financial-model parity with
explicit correction of the known central AC double-count"},
{"g":"202609010053","v":"v9.37","s":"complete the original finance interaction
contract by linking development stage, cost and success"},
{"g":"202609010058","v":"v9.38","s":"restore topology-isolated physical inputs
and the original mounting-to-bifacial linkage"},
{"g":"202609010106","v":"v9.39","s":"remove duplicate BESS truth, restore
original central defaults and reject fractional topology counts"},
{"g":"202609010204","v":"v9.40","s":"the version ledger itself, on the page"},
{"g":"202609010722","v":"v9.41","s":"exact GB price evidence, beside the
ledger"}, {"g":"202609010726","v":"v9.42","s":"the price panel revalidates
instead of pinning its first sight"},
{"g":"202609010902","v":"v9.43","s":"mobile: tools collapse behind one chip;
grid and subs are one tap"}, {"g":"202609011141","v":"v9.44","s":"a repd_ref-only
link computes the links: identity resolved by the search lane is consumed, not
re-required from the URL"}, {"g":"202609011205","v":"v9.45","s":"arrival:
fullscreen on touch, the identity wait runs to its end, and every stage says
what it is doing"}, {"g":"202609011215","v":"v9.46","s":"the distances survive
the card: a keeper re-attaches the measurement block when a late popup replaces
the one it decorated"}, {"g":"202609011242","v":"v9.47","s":"the arrival owns its
card: 2,421 register-absent projects (873 solar) get a card from the link's own
fields, yielded if the register's card lands"},
{"g":"202609011243","v":"v9.48","s":"supersedes v9.47's boundary: the
composition was sound, its proof shipped one stale check; re-sealed coherent"},
{"g":"202609011244","v":"v9.49","s":"supersedes v9.48, whose boundary shipped
```


without its proof file; same composition, re-sealed whole"}, {"g": "202609011251", "v": "v9.50", "s": "the card precedes the lines: a register-absent arrival opened its card after drawing, and the lines-belong-to-the-card watcher rightly wiped them"}, {"g": "202609011433", "v": "v9.51", "s": "the 400 kV public record: declared DCO connections drawn and carded, new customer substations named, nearest 400 kV measured for every project"}, {"g": "202609011434", "v": "v9.52", "s": "a recovered failure is not a failure: late-arrival entries move to their own ledger when the controls arrive, per the Codex supervision finding"}, {"g": "202609011435", "v": "v9.53", "s": "the consented works on the card: customer-substation and PoC interface quotes from the made Orders; the nearest-400 row gains its named companion"}, {"g": "202609011612", "v": "v9.54", "s": "the measurement no longer waits for the engine's layer controls: links draw at once, layers follow when they arrive"}, {"g": "202609011615", "v": "v9.55", "s": "the sales surface answers at once: a declared 400 kV connection and its citation are on the card before the payload arrives, the distance following it"}, {"g": "202609011718", "v": "v9.56", "s": "pink for a point of connection not yet built, and a declared connection that is a circuit rather than a substation draws no line at all"}, {"g": "202609011805", "v": "v9.58", "s": "service restored: v9.57 claimed a shell script the shell never loads, so the composer refused it and the map went dark"}, {"g": "202609011820", "v": "v9.59", "s": "the substation cartridge returns through the engine slot, and the card gains what NESO publishes: circuits, ratings, fault current and planned changes"}, {"g": "202609011845", "v": "v9.60", "s": "the manifest states its own identity, and the Subs control is found by its attribute rather than by label text that changes"}, {"g": "202609011915", "v": "v9.61", "s": "the network sentence names its own scope: a site-wide envelope across the voltages present, never a value for the declared bus"}, {"g": "202609012020", "v": "v9.62", "s": "the Grid Finding Scope: a click on blank space says what grid is mapped there, in bands, and what that cannot tell you"}, {"g": "202609012045", "v": "v9.63", "s": "fault current is quoted at the voltage the connection is made at, not across every busbar at the site"}, {"g": "202609012110", "v": "v9.64", "s": "the Grid Finding Scope stops censusing a voltage that did not parse, and every assembled cartridge is checked against the parts it was built from"}, {"g": "202609012130", "v": "v9.65", "s": "the page's version ledger is written by the cut, so it cannot tell a reader it is running the generation before the one it is running"}, {"g": "202609012155", "v": "v9.66", "s": "the geodesy module measures on the estate's canonical haversine, so every version ever shipped returns the same distance to the last digit"}, {"g": "202609012250", "v": "v9.67", "s": "one geodesy for the whole cartridge, a click reports which cartridges answered it, and the zoom the deep link has always sent is finally read"}, {"g": "202609012141", "v": "v9.68", "s": "sld-sandbox: the declared-connections table moves out of the body into a module proven value-for-value against the last inline copy; the network-topology module (proven 47/47 at 202609012145, composed into nothing until now) is wired on demand - the 10 MB ETYS node/branch product is fetched on first click, never at load, and project and scope cards fill an ETYS topology block in place; the source registry reports the loader's true state instead of ready for a module on a shelf; and this stamp is read from the clock, which is why it sorts before the typed one it succeeds."}, {"g": "202609012211", "v": "v9.69", "s": "sld-sandbox: the sizing arithmetic (physical inputs, the three named ratios, string and central nameplates, the finance port and the two-variable capacity fit) leaves the body for the sizing-arithmetic module, lifted mechanically expression for expression and proven value-for-value against the last inline copy; the body keeps one-line delegations so no caller changes"}, {"g": "202609012234", "v": "v9.70", "s": "the composition manifest is proven against the bytes: substation-intelligence declares connection-points.v3 as it has required since v9.65 (the entry said v2 from v9.63 until this cut), sld-sandbox declares the transmission network and the price rollup it fetches, a data-contract parity proof holds every entry to its bytes in both directions, the source registry states what each fetching source requires in every state and registers the GB conditions loader, whose state the sandbox now publishes"}, {"g": "202609012243", "v": "v9.71", "s": "the Atlas measures in the operator's own circuits as well as in kilometres: an electrical-

distance module traverses the published node/branch model, a voltage changes only across a named transformer and a circuit that appears to change voltage is refused and recorded, planned changes are never walked as if they existed today, every hop carries its published rating and its R/X/B untouched, and the card names what lies two hops away"}, {"g": "202609012249", "v": "v9.72", "s": "the card reports every season the operator publishes rather than winter alone, each circuit keeping its own rating and its own season, scoped to the connection voltage; four circuits published at 9,999 MVA on spans of a kilometre or less are named as placeholders and excluded from the range while still being reported; and the module that produces all of this contains no code path that adds two ratings together, which its proof asserts structurally"}, {"g": "202609012308", "v": "v9.73", "s": "the Atlas solves a declared DC injection response on the published node/branch model and reports which circuits would carry a project's stated capacity and what fraction each takes, with the equations, the 100 MVA base, the named slack and every assumption carried in the answer; validated to 1e-9 against networks whose solutions are exact by hand and checked for power conservation at every intermediate bus of the real 400 kV network; it never states a loading, because existing flows are published nowhere"}, {"g": "202609012317", "v": "v9.74", "s": "the grid computation is reachable from the map itself: a tool arms a point query, and a click on open map resolves the nearest published connection points and renders the published circuits, the seasonal ratings, the electrical distance in hops and the declared powerflow against the nearest one, saying every time that 384 of the 886 published points have no coordinates so the nearest MAPPED point may not be the nearest point; and the layers dash collapses and restores on any device without entering fullscreen, the choice remembered per browser"}, {"g": "202609012345", "v": "v9.75", "s": "the card reports what the operator has published as planned in its own sentence - by year, by status and by asset, with the published parameters carried - kept structurally apart from what exists today, because a row published for a future year is not a circuit now, not a commitment, not a consent and not a connection date"}, {"g": "202609020006", "v": "v9.76", "s": "the computation moves to the cartridge that owns it: the five modules that read the operator's published network leave the sandbox for substation-intelligence, which is split into the two halves it has always been - the V8 engine carried verbatim and the intelligence itself - and gains the parts manifest it should always have had; the sandbox drops from 95% of its 400 kB boundary to about 79%, and the new owner-boundary module lands beside its siblings, naming which transmission owners the assets at a site belong to and where two of them meet on one circuit"}, {"g": "202609020018", "v": "v9.77", "s": "the powerflow stops choosing an arbitrary withdrawal bus: the published 400 kV network has 238 connected components, a transfer across two of them does not exist, and the card was asking for one - it now uses a declared sink rule, refuses a cross-component transfer before the solver is asked, accepts an answer only on convergence AND a global residual AND Kirchhoff at every bus rather than at the injection bus alone, says plainly when no answer is available, and counts parallel circuits as the separate published rows they are rather than collapsing those that share a reactance"}, {"g": "202609030059", "v": "v9.78", "s": "a PIPELINE NEWS (REPD) section in the layer dashboard that summons the rest of the pipeline around the selected project"}, {"g": "202609030109", "v": "v9.79", "s": "the site card counts transformers as machines rather than as winding connections: a site owns both ends of its own transformers, so every internal machine was published twice and Cowley's five read as ten"}, {"g": "202609030116", "v": "v9.80", "s": "HIDE LAYERS collapsed the element that contains the map, so on a phone the one control that gets past the layer panel blanked the application and the remembered choice blanked it again on reload"}, {"g": "202609030119", "v": "v9.81", "s": "a deep link carrying coordinates but no register identity moved no camera at all: both lanes that own the arrival stand down at the same identity test, and the zoom was then eased without a centre ever being set"}, {"g": "202609030128", "v": "v9.82", "s": "an unrecognised project technology abandoned the whole arrival - the card, the ring, the measurement and the substation layer - when the only thing that needs to know what a project generates is the one technology layer"},

```

{"g":"202609030137","v":"v9.83","s":"the three products this Atlas reads at
runtime are pinned to a commit and checked by their SHA-256, because a schema
string defends shape and is blind to values"},
{"g":"202609030151","v":"v9.84","s":"the proof reads the published products
through the pin the composition declares, by commit and by digest, and a product
it cannot get fails those checks loudly instead of skipping them"},
{"g":"202609030156","v":"v9.85","s":"the version ledger leaves the sandbox body
for the cartridge that has room for it, because the sandbox stood 600 characters
short of the guard its own proof asserts and every cut adds another row"},
{"g":"202609030200","v":"v9.86","s":"the nearest 400 kV superlative carries the
sample it was drawn from, counted at render time from the payload actually
fetched rather than written into the sentence"},
{"g":"202609030233","v":"v9.87","s":"a straight line is not a route: beside
every measured distance the card now prints an indicative highway-corridor
estimate, with the calibration it came from and the questions it is not an
answer to"}, {"g":"202609030234","v":"v9.88","s":"a 44 pixel action opens the
corridor figures in full, reachable by button, by right-click and by a long
press that stands down for a pan, a pinch and any drag already under way"},
{"g":"202609031316","v":"v9.89","s":"the measurement stops reading technology:
the one branch that gated it is removed, so offshore measures to the nearest
mapped substation instead of withholding, and a module owns which sentences go
under the distances rather than which technologies get one"},
{"g":"202609031751","v":"v9.90","s":"the measurement arrives before the
intelligence that qualifies it, and on a phone the card docks to the bottom edge
instead of hanging off a marker: nothing is printed over it and nothing has to
be dragged to reach it"}, {"g":"202609031809","v":"v9.91","s":"the register owns
the coordinates on every repd_ref arrival, not only when the link is malformed,
and which source won is published with the distance between them"},
{"g":"202609032001","v":"v9.92","s":"collapse the chrome into File, Edit, View
and About at the top, closed at rest"}, {"g":"202609032005","v":"v9.93","s":"the
menu bar fails soft where there is no timer, as its own comment claimed"},
{"g":"202609032012","v":"v9.94","s":"restamp both cartridges so every manifest
names the generation it is part of"},
{"g":"202609032041","v":"v9.95","s":"withdraw the menu bar from the live Atlas
on two testers' evidence"}, {"g":"202609032213","v":"v9.96","s":"the layer panel
opens closed, so the map is the first impression"},
{"g":"202609032222","v":"v9.97","s":"a deep link is not a search: the results
list and the box it filled get out of the way"},
{"g":"202609032246","v":"v9.98","s":"the arrival frames the project by the
viewport, so a shared link is not wide and empty on a desktop"},
{"g":"202609032315","v":"v9.99","s":"the ceiling warning light reports the limit
that can actually fail the build"},
{"g":"202609040021","v":"v9.100","s":"Consolidate the existing interface into
exactly File, Edit, View, Scope, Grid, and About menus; expose all 60 engine and
3 Pipeline News layer controls through their original handlers; preserve nested
Scope and Clear actions; remain closed at rest and fail closed unless all 63
unique layer controls exist."}, {"g":"202609040046","v":"v9.101","s":"Valid
Pipeline coordinates measure before concurrent REPD verification; every
canonical technology reaches the exact receiver; the mobile menu remains
hittable above the docked project card; transit bridge hydration executes to a
populated source; pipeline layers move to the network cartridge to preserve SLD
headroom."}, {"g":"202609040047","v":"v9.102","s":"The production receiver uses
its exported measure-first plan for all valid Pipeline coordinates, records link
provenance, reconciles resolved identity atomically, accepts every canonical
technology, and preserves mobile layer hit targets plus executable transit
hydration."}, {"g":"202609040058","v":"v9.103","s":"A late REPD identity can no
longer resurrect an abandoned deep-link arrival: every user selection and clear
invalidates the pending token, while intended first and corrected selections
carry the current token through asynchronous measurement."},
{"g":"202609040134","v":"v9.104","s":"mobile fullscreen no longer creates an
ancestor cycle, and Grid keeps each layer tick plus its live V8 status
visible"}, {"g":"202609040219","v":"v9.105","s":"A successfully fetched transit

```

source is reported available only when its configured MapLibre layer can draw at least one matching feature; zero eligible features become visibly EMPTY, unticked, disabled and non-interactive while quarantined data and historical generations remain unchanged."},{ "g": "202609040337", "v": "v9.106", "s": "Pipeline 0144 arrivals absent from the active register retain their supplied identity and measurement with explicit provenance; real resolver failures remain failures with a working single-epoch owner-to-measurement retry, stale arrival ownership cannot reclaim the map, and the legacy receiver issues no cross-domain Pipeline request."},{ "g": "202609040403", "v": "v9.107", "s": "Move seven byte-identical SLD stylesheet templates into the earlier substation cartridge, fail closed if the module is absent, and restore bounded headroom without changing UI behaviour."},{ "g": "202609041221", "v": "v9.108", "s": "the OSM/CARTO/Open Charge Map credit clears the menu bar unconditionally, measured from the bar's own rendered height, at every width -- not only while body.fs-active happens to be set"}, {"g": "202609041244", "v": "v9.109", "s": "Pipeline News' three broken technology buckets (wind_onshore, wind_offshore, other) resolve to the engine's real layer id through one table instead of a set-membership test that read enabled while the layer sat off, on a third of the register; substation-intelligence carried forward unchanged to keep the on-page version ledger current"}, {"g": "202609041250", "v": "v9.110", "s": "the v8 VENTUS masthead is fused into the six-menu bar's own centre so it is present at every width and cannot be torn into a closed panel and vanish after arrival; the SCADA layer panel is restored with the real, moved-not-cloned .scada-brand and .status-legend nodes and every layer control on a full-size >=44px hit target; every one of the six panels is anchored to its own side group and clamped into the viewport by measurement; and the OSM/CARTO/Open Charge Map credit now outranks every open panel by z-index, not only clearing the bar's own height"}, {"g": "202609041330", "v": "v9.111", "s": "Hide the v8 fullscreen letterhead whenever the menu bar hosts the fused VENTUS masthead, so a phone arrival shows one wordmark in the bar instead of two and the SCOPE, GRID and ABOUT titles are unobscured."},{ "g": "202609041945", "v": "v9.112", "s": "Compose the written-but-never-composed iOS Safari visibility fix so a deep link opened in a background tab is not spent before anyone can see it; share one DuckDB runtime across the two cartridges that each built their own; and keep the GRID and SUBS chips on the map at phone widths instead of inside a menu."},{ "g": "202609041956", "v": "v9.114", "s": "Bring the on-page version ledger up to the composition a reader is actually looking at, so its newest entry names this generation rather than an earlier one."},{ "g": "202609041957", "v": "v9.115", "s": "Bring the arrival cartridge and the cartridge carrying the version ledger onto one generation, so every proof, the ledger's newest entry and the composition a reader is looking at all name the same thing."},{ "g": "202609042123", "v": "v9.116", "s": "Bring the v8 layers panel back beneath the menu bar, open on a desktop and collapsed with its toggle visible on a phone, while keeping every dropdown - FILE, EDIT, VIEW, SCOPE, GRID and ABOUT - exactly as it is."},{ "g": "202609050238", "v": "v9.117", "s": "About: the map attribution moves in, last and in small print, and an Estate group of three links sits above it. File: the engine's canonical modules are listed, fetched once from the engine's own published graph, each linking into it focused on itself. No other panel, control or layer changed."},{ "g": "202609050244", "v": "v9.118", "s": "File: list every node the engine graph publishes, grouped canonical / extract / reference / fragment, not only the canonical eleven; add one copyable command that clones the engine and runs its own gate. No other panel, control or layer changed."},{ "g": "202609050249", "v": "v9.119", "s": "The copyable engine command drops its install step, because the engine has no dependencies and its gate opens no socket. No other change."},{ "g": "202609050301", "v": "v9.120", "s": "View gains a Studies group carrying the GB electricity price and grid constraint series 2016-2026. No other panel, control or layer changed."},{ "g": "202609050354", "v": "v9.121", "s": "File gains an Export group: print the view as a slide, or save it as an image. The print sizes to whatever page the reader chose, portrait or landscape, on a phone or a desktop. The image capture happens inside a render frame and is sampled for non-transparent pixels before it is

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offered. Both paths carry the attribution."},
{"g":"202609051340","v":"v9.127","s":"Print fills the sheet edge to edge with
nothing on it but the map and its credit, and a new File control writes the PDF
in the page so no print dialog or printer driver can lose it."},
{"g":"202609051455","v":"v9.128","s":"File > Print emits a screen grab of the
reader's view: nothing hidden, layout pinned to the viewport, the WebGL canvas
substituted by its own captured pixels in its own box, provenance strip moved
below the view."},{ "g":"202609051503","v":"v9.129","s":"Two separate Teleprinter
commands in the File menu: Print PDF (a picture of this screen, 1:1, our own
writer) and Print source code (the entire source and its browser dependencies,
for AI review)."}, {"g":"202609051510","v":"v9.130","s":"Print source code keeps
CODE in full and marks DATA truncated, so the teleprint is small enough to
attach to a chat on a phone."}]];

```

```

const REJECTED_PRE_PROMOTION = Object.freeze({
  '202609040021': 'never live: the mobile project card could cover the engine
layer panel',
  '202609040046': 'never live: its generation proof reported 748/755',
  '202609040047': 'never live: a late identity could restore an arrival after
a user selection or clear'
});
for (const entry of VERSION_LEDGER) {
  if (!REJECTED_PRE_PROMOTION[entry.g]) continue;
  entry.status = 'REJECTED_PRE_PROMOTION';
  entry.reason = REJECTED_PRE_PROMOTION[entry.g];
}

```

```

const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||
{});
NS.versionLedger = Object.freeze({
  schema: 'gridatlas.module.version-ledger.v1',
  entries: VERSION_LEDGER,
  written_by: 'tools/recompose.mjs, at the cut, never by hand',
  not_an_assessment: 'A ledger row says what a generation changed. It does '
    + 'not say the change was right, and nothing here grades one.'
});
})();

```

```

/* =====
corridor-estimate - a straight line is not a route, and this says how far off
=====

```

Every distance this Atlas prints is a great-circle distance between two points. No cable is ever laid that way. The question a reader actually has is how much longer the built thing would be, and until now the card said nothing at all about it.

THE SCALAR, AND WHAT IT WAS MEASURED ON.

Route factor 1.245, calibrated against the published built lengths of GB transmission CABLE circuits in the operator's own node/branch model:

median absolute error	8.45%
within 15%	73% of circuits
sample	95 circuits spanning 59 DISTINCT SITE PAIRS

The sample is 59, not 95. Parallel circuits between the same two sites duplicate the geometry exactly, so quoting 95 would be counting the same measurement up to four times and claiming a precision the data has not got.

WHAT IT IS NOT FOR.

Not overhead line. The measured OHL factor is 1.13, and it is a different number for a physical reason rather than a statistical one: a tower line crosses open country in long straight spans, while a cable follows the highway network, its bends and its wayleaves. Applying a cable factor to an overhead question would overstate the route by about ten per cent and would be the wrong model regardless of the error.

Not below about a kilometre. Where the two ends are under 1 km apart the site-centroid resolution dominates the geometry: in that band the median published length is 0.59 km against a median error of 52.5%, which is not the route factor being wrong, it is the straight line not being a measurement of anything at that scale. Those separations are refused rather than scaled.

It is a screening estimate for a corridor. It is not a connection offer, a constructability assessment or a consenting design, and the caveat below travels with every number this module produces.

```
(() => {
  'use strict';

  const SCHEMA = 'gridatlas.module.corridor-estimate.v1';

  const CABLE_FACTOR = 1.245;
  const OHL_FACTOR = 1.13;
  const MINIMUM_KM = 1;

  const BASIS = Object.freeze({
    factor: CABLE_FACTOR,
    median_absolute_error_pct: 8.45,
    within_15_pct: 73,
    circuits: 95,
    distinct_site_pairs: 59,
    source: 'published built lengths of GB transmission cable circuits',
    sample_note: 'parallel circuits between the same two sites duplicate the '
      + 'geometry, so the sample is 59 distinct site pairs and not 95 circuits',
    minimum_separation_km: MINIMUM_KM,
    below_minimum: 'under about a kilometre the site-centroid resolution '
      + 'dominates: median published length 0.59 km against a median error of '
      + '52.5%, so a straight line between centroids is not measuring route '
      + 'factor and no estimate is offered'
  });

  const CAVEAT = 'Indicative highway-corridor screening only. Not a connection '
    + 'offer, not a constructability assessment and not a consenting design.';

  const NOT_FOR_OVERHEAD = 'Calibrated on cable circuits, which follow the '
    + 'highway network. Overhead line crosses open country and measures 1.13; '
    + 'this factor is not applied to an overhead-line question.';

  /**
   * The corridor estimate for a CABLE route of `km` straight-line distance.
   * @returns null when there is nothing honest to say - no distance, or a
   *   separation short enough that the straight line is not measuring
   *   route factor. Null is the answer, not zero.
   */
  function forCable(km) {
    const straight = Number(km);
    if (!Number.isFinite(straight) || straight <= 0) return null;
    if (straight < MINIMUM_KM) {
      return { km: null, factor: CABLE_FACTOR, straight_km: straight,
        withheld: BASIS.below_minimum };
    }
  }
}
```

```

    }
    return {
      km: straight * CABLE_FACTOR,
      factor: CABLE_FACTOR,
      straight_km: straight,
      withheld: null
    };
  }

  /* Deliberately no forOverhead(). A module that offered one would be used,
    and the 1.13 above is published here so a reader can see WHY the cable
    factor is not the answer to that question - not so that this cartridge
    can start answering it. */

  const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||
  {});
  NS.corridorEstimate = Object.freeze({
    schema: SCHEMA,
    factor: CABLE_FACTOR,
    overhead_factor: OHL_FACTOR,
    minimum_km: MINIMUM_KM,
    basis: BASIS,
    caveat: CAVEAT,
    not_for_overhead: NOT_FOR_OVERHEAD,
    forCable,
    not_an_assessment: 'An estimated corridor length says nothing about '
      + 'whether a connection is available, consentable or affordable.'
  });
  })();
  })();

  /* GridAtlas menu bar.

```

v9.94 proved that the conventional menu is the right shape and the wrong implementation can still strand the product. That version enumerated only direct children of .map-controls and then hid the whole owner container. Scope, Clear and their result surfaces were nested, so they disappeared.

This successor has a stricter admission rule:

- it installs nothing until the engine's 60 layer controls and the three Pipeline News controls are all present and uniquely identified;
- the Grid menu proxies those 63 ORIGINAL inputs, so their delegated engine handlers remain the only implementation of behaviour;
- action buttons are moved as the same DOM nodes, preserving listeners and state, while result panels stay with the map that owns them;
- legacy containers collapse only after the complete inventory is built. A missing control leaves the old interface reachable and is published as a failure instead of being silently skipped.

```

  The six names are the architect's current vocabulary. In particular Grid
  is not exposed through the abandoned "Select layers" alias. */
(function gridAtlasMenuBar() {
  'use strict';

  if (typeof window === 'undefined' || typeof document === 'undefined') return;

  var SCHEMA = 'gridatlas.menu-bar.v2';
  var BAR_ID = 'gridatlas-menu-bar';
  var STYLE_ID = BAR_ID + '-css';
  var FAILURE_ID = BAR_ID + '-failure';
  var MENUS = ['File', 'Edit', 'View', 'Scope', 'Grid', 'About'];
  var EXPECTED_ENGINE_LAYERS = 60;

```

```

var EXPECTED_PIPELINE_LAYERS = 3;
var EXPECTED_LAYER_CONTROLS = 63;
var MAX_TRIES = 160; // 40 s: the register UI is built after map
load

var NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ || {});
if (NS.menuBar && NS.menuBar.schema === SCHEMA) return;

var state = {
  schema: SCHEMA,
  menus: MENUS.slice(),
  bar_id: BAR_ID,
  installed: false,
  controls_moved: 0,
  layer_controls: 0,
  engine_layer_controls: 0,
  pipeline_layer_controls: 0,
  expected_layer_controls: EXPECTED_LAYER_CONTROLS,
  panel_counts: {},
  failure: null,
  tries: 0,
  listeners: 0,
  closed_at_rest: true,
  one_identity_surface: false
};
NS.menuBar = state;

var bar = null;
var panels = {};
var titles = [];
var layerTargets = Object.create(null);
var layerProxies = Object.create(null);
var forwardingLayerChoice = false;
var observer = null;
var timer = null;
var brandSlot = null; // holds the v8 .hud-header (VENTUS wordmark), fused
into the bar itself
var gridHead = null; // holds the v8 .scada-brand + .status-legend,
restored at the top of Grid
var gridBody = null; // holds the layer groups, so gridHead never enters
the 2-column flow

function array(value) {
  return Array.prototype.slice.call(value || []);
}

function cleanText(value) {
  return String(value == null ? '' : value).replace(/\s+/g, ' ').trim();
}

function layerKey(input) {
  if (!input || !input.getAttribute) return '';
  var engine = input.getAttribute('data-layer-id');
  if (engine) return 'engine:' + engine;
  var pipeline = input.getAttribute('data-pn-layer');
  return pipeline ? 'pipeline:' + pipeline : '';
}

function layerLabel(input) {
  var label = input && input.closest ? input.closest('label') : null;
  var span = label && label.querySelector
    ? label.querySelector('[data-base-label], [data-pn-label], span') : null;

```



```

    var base = span && span.getAttribute ? span.getAttribute('data-base-label')
: '';
    /* The V8 panel exposes WAIT/LOAD/OK/FAIL beside every layer. The first
    menu implementation preferred data-base-label, which deliberately
    strips that live suffix. That made a successful load and a failed
    load indistinguishable in the only layer surface left on a phone. */
    var text = cleanText((span && span.textContent) || base
    || (label && label.textContent) ||
layerKey(input).split(':').slice(1).join(':'));
    return text || layerKey(input);
}

function layerGroup(input) {
    var group = input && input.closest ? input.closest('.key-group') : null;
    var title = group && group.querySelector ? group.querySelector('.key-title')
: null;
    return cleanText(title && title.textContent) || 'Other layers';
}

function inventory(doc) {
    var host = doc.getElementById('scada-ui-container');
    var engine = host ? array(host.querySelectorAll(
        'input[type="checkbox"][data-layer-id]')) : [];
    var pipeline = host ? array(host.querySelectorAll(
        'input[type="checkbox"][data-pn-layer]')) : [];
    var controls = engine.concat(pipeline);
    var keys = controls.map(layerKey);
    var unique = new Set(keys);
    return {
        host: host,
        engine: engine,
        pipeline: pipeline,
        controls: controls,
        keys: keys,
        complete: engine.length === EXPECTED_ENGINE_LAYERS
            && pipeline.length === EXPECTED_PIPELINE_LAYERS
            && controls.length === EXPECTED_LAYER_CONTROLS
            && unique.size === EXPECTED_LAYER_CONTROLS
            && !keys.includes('')
    };
}

state.inspect = function () {
    var found = inventory(document);
    return {
        engine: found.engine.length,
        pipeline: found.pipeline.length,
        total: found.controls.length,
        unique: new Set(found.keys).size,
        complete: found.complete
    };
};

function required(doc) {
    var found = inventory(doc);
    var nodes = {
        host: doc.querySelector('.map-container'),
        stack: doc.querySelector('.map-controls'),
        search: doc.querySelector('.search-bar-wrapper'),
        header: doc.querySelector('.hud-header'),
        exportButton: doc.getElementById('btn-export'),
        statusButton: doc.getElementById('btn-status'),
    };
}

```

```

    fullscreenButton: doc.getElementById('btn-fullscreen'),
    radiusButton: doc.getElementById('btn-radius'),
    radiusAreaButton: doc.getElementById('btn-radius-area'),
    zoneButton: doc.getElementById('btn-zonedraw'),
    measureButton: doc.getElementById('btn-measure')
  };
  var missing = Object.keys(nodes).filter(function (key) { return !nodes[key];
});
  if (!found.complete) missing.push('63 unique layer controls');
  return { found: found, nodes: nodes, missing: missing };
}

function installStyle(doc) {
  if (doc.getElementById(STYLE_ID)) return;
  var style = doc.createElement('style');
  style.id = STYLE_ID;
  style.textContent = [
    '#' + BAR_ID + '{position:absolute;top:0;left:0;right:0;height:36px;z-
index:10020;',
    'display:flex;align-items:stretch;gap:0;padding-left:env(safe-area-inset-
left);',
    'padding-right:env(safe-area-inset-right);box-sizing:border-box;',
    'background:rgba(4,10,13,.95);border-bottom:1px solid
rgba(80,220,240,.3);',
    'font:11px/1 ui-monospace,SFMono-Regular,Menlo,monospace;',
    '-webkit-backdrop-filter:blur(7px);backdrop-filter:blur(7px);',
    'isolation:isolate;pointer-events:auto}',
    '#' + BAR_ID + '.gm-menu{position:relative;min-width:0}',
    '#' + BAR_ID + '.gm-
title{appearance:none;border:0;background:transparent;color:#cfeef6;',
    'min-height:36px;padding:0 11px;cursor:pointer;font:inherit;letter-
spacing:.05em;',
    'text-transform:uppercase;white-space:nowrap}',
    '#' + BAR_ID + '.gm-title:hover,#' + BAR_ID + '.gm-title:focus-
visible,',
    '#' + BAR_ID + '.gm-menu.gm-open>.gm-
title{background:rgba(80,220,240,.16);color:#fff}',
    '#' + BAR_ID + '.gm-title:focus-visible,#' + BAR_ID + '.gm-panel :focus-
visible{',
    'outline:2px solid #6bebff;outline-offset:-2px}',
    '#' + BAR_ID + '.gm-side{display:flex;align-items:stretch;flex:1 1 0;min-
width:0}',
    '#' + BAR_ID + '.gm-side-left{justify-content:flex-start}',
    '#' + BAR_ID + '.gm-side-right{justify-content:flex-end}',
    /* The VENTUS masthead, fused into the centre of this same 36px strip
    (see buildBar) rather than a second row, so it costs no map height
    on a phone and can never be torn out into a closed panel again. */
    /* "The VENTUS logo is the best part" -- the architect's own words.
    It is the hero of this strip: sized and weighted to outrank the
    six menu titles either side of it, not a corner credit shrunk to
    fit. Same face, tracking and two-line lockup as the v8 masthead
    and the fullscreen letterhead it is carried from verbatim. */
    '#' + BAR_ID + '.gm-brand-slot{flex:0 1 auto;min-width:0;max-width:64%;',
    'display:flex;align-items:center;justify-content:center;overflow:hidden;',
    'padding:0 6px;text-align:center}',
    '#' + BAR_ID + '.gm-brand-slot .hud-header{display:flex!important;',
    'position:static!important;width:auto!important;align-items:center;',
    'justify-content:center;gap:11px;margin:0!important;padding:0!important;',
    'background:none!important;border:0!important}',
    '#' + BAR_ID + '.gm-brand-slot .hud-header>div{flex:0 0 auto;line-
height:1.05}',
    '#' + BAR_ID + '.gm-brand-slot .hud-header small{font-size:6.5px;white-

```

```

space:nowrap}',
    '#' + BAR_ID + ' .gm-brand-slot .hud-header .hud-val{font-size:10.5px;',
    'text-shadow:none}',
    '#' + BAR_ID + ' .gm-brand-slot .ventus-main{font-size:14px;font-
weight:800;',
    'letter-spacing:.2em;margin:0;color:#fff}',
    '#' + BAR_ID + ' .gm-brand-slot .ventus-sub{font-size:5.5px;letter-
spacing:.14em}',
    '#' + BAR_ID + ' .gm-panel{position:absolute;top:100%;left:0;min-
width:240px;',
    'max-width:min(92vw,420px);max-height:min(72dvh,620px);overflow:auto;',
    'overscroll-behavior:contain;padding:6px;background:rgba(4,10,13,.98);',
    'border:1px solid rgba(80,220,240,.32);border-top:0;',
    'box-shadow:0 12px 34px rgba(0,0,0,.68);box-sizing:border-box}',
    '#' + BAR_ID + ' .gm-panel[hidden]{display:none!important}',
    /* Right-align every panel whose title lives in the right-hand group,
not "the last two of six flat siblings" -- that positional rule is
what let the About panel resolve to a negative x once the six
titles stopped being one undifferentiated row (measured live:
x=-95 at 1568px, a quarter of its own Versions control
unreachable). clampPanel() below is the second, JS-measured
guarantee: this CSS is the common case, not the only defence. */
    '#' + BAR_ID + ' .gm-side-right .gm-panel{left:auto;right:0}',
    '#' + BAR_ID + ' .gm-panel button,#' + BAR_ID + ' .gm-panel
[role="button"]{',
    'display:flex;align-items:center;width:100%;min-height:44px;box-
sizing:border-box;',

    'position:static!important;inset:auto!important;transform:none!important;margin:
0 0 3px;',
    'padding:7px 10px;border:0;border-
radius:2px;background:transparent;color:#cfeef6;',
    'font:11px/1.35 ui-monospace,SFMono-Regular,Menlo,monospace;text-
align:left;',
    'letter-spacing:.03em;text-transform:none;cursor:pointer}',
    '#' + BAR_ID + ' .gm-panel button:hover,#' + BAR_ID + ' .gm-panel
[role="button"]:hover{',
    'background:rgba(80,220,240,.14);color:#fff}',
    /* The estate links are anchors so they are real links -- middle-click,
copy, open in a new tab all work -- and they take the panel's own
button look rather than a second one. Only the underline has to go. */
    '#' + BAR_ID + ' .gm-panel a[data-gm-estate],#' + BAR_ID + ' .gm-panel
a[data-gm-engine]',
    '#' + BAR_ID + ' .gm-panel a[data-gm-study]',
    '{text-decoration:none}',
    /* Module paths are long. They stay on one row and lose their middle
rather than wrapping a 44px control into three lines on a phone. */
    '#' + BAR_ID + ' .gm-panel a[data-gm-engine]{white-
space:nowrap;overflow:hidden;',
    'text-overflow:ellipsis;display:block;line-height:30px;min-height:44px}',
    /* The attribution, once moved into About, is prose in a panel of
controls: it keeps its own small type and wraps rather than being
clipped to one 44px row. */
    '#' + BAR_ID + ' .gm-panel .custom-map-attrib{position:static!important;',
    'inset:auto!important;margin:4px 0 2px;padding:6px 8px;max-width:none;',
    'background:transparent;border:0;white-space:normal;line-height:1.45;',
    'font:10px/1.45 ui-monospace,SFMono-
Regular,Menlo,monospace;color:#8fb6c0}',
    '#' + BAR_ID + ' .gm-layer-group{margin:5px 0 2px;padding:6px 8px 3px;',
    'border-top:1px solid #19343b;color:#6fa2ae;font-size:10px;letter-
spacing:.08em;',
    'text-transform:uppercase}',

```

```

/* Every layer control's OWN <input> covers its whole label (see
   layerCheckbox / buildLayerControls): a measured audit found the raw
   v8 checkboxes at 17x17px, the input element itself and not just a
   padded label, so a re-measurement of the input's own rect is the
   bar this has to clear, not only a centre-point hit test. */
'#' + BAR_ID + ' .gm-layer{position:relative;display:flex;align-
items:center;gap:10px;',
  'min-height:44px;box-sizing:border-box;padding:7px
9px;color:#cfeef6;cursor:pointer;',
  'line-height:1.35}',
'#' + BAR_ID + ' .gm-layer:hover{background:rgba(80,220,240,.12)}',
'#' + BAR_ID + ' .gm-layer
input{position:absolute;inset:0;width:100%;height:100%;',
  'margin:0;opacity:0;cursor:pointer;z-index:1}',
'#' + BAR_ID + ' .gm-layer-box{width:20px;height:20px;flex:0 0 auto;',
  'border:1.5px solid #4a8b96;border-radius:4px;position:relative;',
  'background:rgba(255,255,255,.04)}',
'#' + BAR_ID + ' .gm-layer input:checked~.gm-layer-
box{background:#4fd7ee;',
  'border-color:#4fd7ee}',
'#' + BAR_ID + ' .gm-layer input:checked~.gm-layer-
box::after{content:"";',
  'position:absolute;left:6px;top:2px;width:5px;height:10px;',
  'border:solid #04141a;border-width:0 2px 2px 0;transform:rotate(38deg)}',
'#' + BAR_ID + ' .gm-layer input:focus-visible~.gm-layer-box{outline:2px
solid #6bebff;',
  'outline-offset:2px}',
'#' + BAR_ID + ' .gm-layer-name{overflow-wrap:anywhere}',
/* The restored SCADA panel: a branded head (the real .scada-brand and
   .status-legend nodes, moved in once -- see install()) above a
   scrollable body that never mixes with the head's own layout. */
'#' + BAR_ID + ' .gm-panel-grid{padding:0;overflow:hidden;display:flex;',
  'flex-direction:column;min-width:min(94vw,360px);max-
width:min(96vw,900px)}',
'#' + BAR_ID + ' .gm-panel-head{flex:0 0 auto;padding:8px 8px 0}',
'#' + BAR_ID + ' .gm-panel-head .scada-brand{padding:0 0 6px;margin:0 0
6px}',
'#' + BAR_ID + ' .gm-panel-head .status-legend{padding:0 0
8px;margin:0;border:0}',
'#' + BAR_ID + ' .gm-panel-body{flex:1 1 auto;overflow:auto;',
  'overscroll-behavior:contain;padding:6px;min-height:0}',
'@media(min-width:560px){#' + BAR_ID + ' .gm-panel-body{column-count:2;',
  'column-gap:14px}',
'#' + BAR_ID + ' .gm-panel-body .gm-layer-group{break-inside:avoid}',
'#' + BAR_ID + ' .gm-panel-body .gm-layer{break-inside:avoid}}',
'@media(min-width:900px){#' + BAR_ID + ' .gm-panel-body{column-count:3}}',
'#' + BAR_ID + ' .search-bar-
wrapper{position:static!important;display:grid!important;',
  'grid-template-columns:minmax(150px,1fr)
auto;width:min(82vw,390px);gap:5px;margin:2px 0 6px}',
'#' + BAR_ID + ' .search-bar-wrapper>div{position:relative}',
'#' + BAR_ID + ' .search-input{width:100%!important;min-height:44px;box-
sizing:border-box}',
'#' + BAR_ID + ' .search-results{position:static!important;max-
height:42vh;overflow:auto}',
'#' + BAR_ID + ' .disclaimer-box,#' + BAR_ID + ' .podcast-shoutout{',
  'display:block!important;position:static!important;max-
width:380px;padding:8px;',
  'box-sizing:border-box;text-align:left:pointer-events:auto}',
'.gridatlas-menu-hosted .map-controls[data-gridatlas-menu-emptied="1"]
{display:none!important}',
/* The v8 SCADA layers panel STAYS. It was hidden here unconditionally,

```

and the reasoning that justified it was circular: this rule set `display:none!important`, which is why "the container height never changes", which was then cited as evidence the panel's own toggle was inert, which justified the rule. Measured on the live page at 202609041957, with the toggle un-hidden and clicked: the label does flip (LAYERS -> HIDE LAYERS) and data-gridatlas-collapsed does clear. The toggle was never inert. Only its effect was invisible.

The cost of the rule was the whole product surface: all 60 engine layer switches sat in a container measured at 0x0, on desktop AND phone, with the page unable to scroll to it -- zero of 120 layer controls reachable without opening a menu. The register, the voltages, the supermarkets, the transit and the EV layers were all still in the DOM and none of them could be touched.

"One identity surface" is still honoured, and it was always about the VENTUS wordmark rather than the switches: the real .scada-brand node is MOVED into the Grid panel head by `install()`, not cloned, so the restored panel has no second wordmark to show. The Grid dropdown and this panel drive the SAME 63 original inputs -- the dropdown proxies them -- so the two cannot disagree about what is on.

Requested directly by the architect, whose product this is, on 2026-09-04: "restore v8 panels but keep dropdowns file, edit, scope, grid, about". Both, not either. */

```
'#gridatlas-menu-hosted .scada-wrapper{display:flex!important}',
/* At phone widths the panel starts collapsed and the toggle opens it:
   measured, an expanded panel held 31.6% of a 393x852 screen against
   the map's 29.3%, which is the wrong trade on the surface most
   readers arrive on. Desktop has the room and gets the panel open, as
   v8 always did. Either way the toggle is now visible, so the reader
   decides rather than the stylesheet. */
'#gridatlas-dash-toggle{display:inline-flex!important}',
/* v9.90 made the mobile project card a fixed, full-width bottom sheet.
   The old SCADA layer panel remained underneath it, so a visible layer
   checkbox could lose the hit test to text in the project card. Keep the
   conventional menu and its fixed phone panel in the higher, interactive
   stacking context whenever that sheet is open. */
'html.gridatlas-sheet-open #' + BAR_ID + '{z-
index:10020!important;pointer-events:auto!important}',
'html.gridatlas-sheet-open #' + BAR_ID + ' .gm-panel{pointer-
events:auto!important}',
'body:not(.fs-active) #' + BAR_ID + ' #btn-fullscreen-
exit{display:none!important}',
'body.fs-active #' + BAR_ID + ' #btn-fullscreen-
exit{display:flex!important}',
/* The shell's own .custom-map-attrb (OpenStreetMap / CARTO / Open
   Charge Map credit) only ever cleared this bar while body.fs-active
   was set. At rest -- and on every Pipeline News deep-link arrival,
   which does not always reach fs-active -- the credit painted at its
   default top:10px and sat directly under the bar, invisible under
   the ABOUT title. A licence credit that is painted but covered is
   not attribution. Clear it whenever this bar is hosted, not only in
   fullscreen; --gridatlas-menu-bar-clear is kept in step with the
   bar's own rendered height (see syncAttribClearance) rather than a
   second hard-coded constant, because the bar itself drops from 36px
   to 34px under the @media rule below and a fixed number sized for
   one breakpoint would leave the credit covered, or needlessly far
   down, at the other. 44px is only the pre-JS fallback. Z-INDEX, not
   only top: measured live, an open dropdown panel painted over the
   credit's right two-thirds (elementFromPoint at 50/70/90% of its
   width resolved to the panel's own button) even though the credit's
```

```

    TOP already cleared the bar -- the two are siblings in the same
    stacking context and the panel simply painted after it. The credit
    must outrank every panel this bar can ever open, present or future,
    so its z-index is set once here rather than chased per panel. */
    '.gridatlas-menu-hosted .custom-map-attrib{' ,
    'top:var(--gridatlas-menu-bar-clear,44px)!important;z-
index:10025!important}',
    /* The v8 fullscreen letterhead stands down once this bar hosts the
    brand. #fs-letterhead is painted only under body.fs-active, and
    fs-active is set by exactly one caller: the deep-link arrival's
    enterFullscreen(), which runs when trayTarget() is true. That is
    every phone and no desktop -- so the duplicate was invisible to
    every desktop check and present on every phone arrival. Measured
    on an iPhone 13 viewport at 202609041250: the fused masthead sat
    correctly at x=165 (11px) while this one painted at x=254 (15px),
    over the SCOPE, GRID and ABOUT titles in the right side group.
    The brand is not lost by hiding it -- it is the same wordmark,
    still on screen, now in the bar at every width and in every
    fullscreen state, which is what fusing it there was for. */
    '.gridatlas-menu-hosted #fs-letterhead{display:none!important}',
    '@media(max-width:700px){#' + BAR_ID + '{height:34px}',
    '#' + BAR_ID + ' .gm-title{min-height:34px;padding:0 6px;font-
size:9px;letter-spacing:.025em}',
    '#' + BAR_ID + ' .gm-brand-slot{max-width:48%;padding:0 2px}',
    '#' + BAR_ID + ' .gm-brand-slot .hud-header>div:first-child,',
    '#' + BAR_ID + ' .gm-brand-slot .hud-header>div:last-child{display:none}',
    '#' + BAR_ID + ' .gm-brand-slot .ventus-main{font-size:11px;letter-
spacing:.14em}',
    '#' + BAR_ID + ' .gm-brand-slot .ventus-sub{font-size:4.5px}',
    '#' + BAR_ID + ' .gm-
panel{position:fixed;top:34px;left:4px!important;right:4px!important;',
    'width:auto;max-width:none;max-height:calc(100dvh - 40px);padding-
bottom:',
    'calc(6px + env(safe-area-inset-bottom))}',
    '#' + BAR_ID + ' .gm-panel-grid{max-width:none}}'
  ].join('');
  (doc.head || doc.documentElement).appendChild(style);
}

function syncAttribClearance(doc) {
  /* Measured, not asserted: the bar is 36px at rest and 34px under the
  @media(max-width:700px) rule in installStyle, and either number could
  change again. Reading the live box keeps the credit clear of the bar
  at whatever height it actually rendered, on the phone width the
  fixture failed on as much as on desktop. Guarded so the DOM-fixture
  proof, which stubs neither getBoundingClientRect nor a CSSOM style
  object, runs through this as a no-op. */
  if (!bar || typeof bar.getBoundingClientRect !== 'function') return;
  var root = doc.documentElement;
  if (!root || !root.style || typeof root.style.setProperty !== 'function')
  return;
  var rect = bar.getBoundingClientRect();
  var height = Math.ceil(rect.height) || 36;
  var clearance = height + 8; // clear of the bar's own border-bottom, not
  flush against it
  root.style.setProperty('--gridatlas-menu-bar-clear', clearance + 'px');
  state.attrib_clearance_px = clearance;
}

function closeAll(focusTitle) {
  if (!bar) return;
  array(bar.querySelectorAll('.gm-menu.gm-open')).forEach(function (menu) {

```

```

        menu.classList.remove('gm-open');
        var title = menu.querySelector('.gm-title');
        var panel = menu.querySelector('.gm-panel');
        if (title) title.setAttribute('aria-expanded', 'false');
        if (panel) panel.hidden = true;
    });
    state.closed_at_rest = true;
    openPanelRefs = null;
    if (focusTitle && typeof focusTitle.focus === 'function')
focusTitle.focus();
    }

function syncLayer(key) {
    var original = layerTargets[key];
    var proxy = layerProxies[key];
    if (!original || !proxy) return;
    proxy.checked = !!original.checked;
    proxy.disabled = !!original.disabled;
    proxy.setAttribute('aria-label', layerLabel(original));
    var name = proxy.parentNode && proxy.parentNode.querySelector
        ? proxy.parentNode.querySelector('.gm-layer-name') : null;
    if (name) {
        var nextLabel = layerLabel(original);
        if (name.textContent !== nextLabel) name.textContent = nextLabel;
    }
}

function syncAll() {
    Object.keys(layerTargets).forEach(syncLayer);
}

/* Measured live: the About panel resolved to x=-95 at 1568px width, a
quarter of its own control off the left edge of the window -- the CSS
right:0 anchor (now scoped to the right-hand group, see installStyle)
covers the common case, but this is the second, JS-measured guarantee
that no panel this bar ever opens can resolve outside the viewport,
regardless of how its title happens to be positioned. Runs after the
panel is laid out (post layout, not pre-measured), clears any earlier
override before measuring so a panel that no longer overflows is not
left pinned from a previous, narrower viewport. */
function clampPanel(doc, menu, panel) {
    if (!panel || typeof panel.getBoundingClientRect !== 'function') return;
    if (!menu || typeof menu.getBoundingClientRect !== 'function') return;
    panel.style.left = '';
    panel.style.right = '';
    var view = doc.defaultView || (typeof window !== 'undefined' ? window :
null);
    var vw = (view && view.innerWidth) || doc.documentElement.clientWidth;
    if (!vw) return;
    var margin = 4;
    var panelRect = panel.getBoundingClientRect();
    var desiredLeft = panelRect.left;
    if (panelRect.left < margin) desiredLeft = margin;
    else if (panelRect.right > vw - margin) desiredLeft = Math.max(margin, vw -
margin - panelRect.width);
    if (Math.round(desiredLeft) === Math.round(panelRect.left)) return;
    var menuRect = menu.getBoundingClientRect();
    panel.style.left = (desiredLeft - menuRect.left) + 'px';
    panel.style.right = 'auto';
}

var openPanelRefs = null;    // {menu, panel} while a panel is open, so a

```

resize can re-clamp it

```
function openMenu(menu, title, panel) {
  var wasOpen = menu.classList.contains('gm-open');
  closeAll();
  if (wasOpen) { openPanelRefs = null; return; }
  if (title.textContent === 'Grid') syncAll();
  menu.classList.add('gm-open');
  title.setAttribute('aria-expanded', 'true');
  panel.hidden = false;
  state.closed_at_rest = false;
  openPanelRefs = { menu: menu, panel: panel };
  clampPanel(title.ownerDocument || document, menu, panel);
}

function buildBar(doc) {
  var nav = doc.createElement('nav');
  nav.id = BAR_ID;
  nav.setAttribute('aria-label', 'Atlas menu');

  /* Three zones, not six flat siblings. The architect's complaint was
  that the VENTUS identity had been torn out of view -- moved into a
  closed About panel, so the reader saw the v8 masthead for the first
  ~1.5s of every arrival and then watched it vanish. Fusing the brand
  into the CENTRE of the same 36px strip the menu titles already live
  in restores it permanently, at every width, with no extra row and
  therefore no map height stolen on a phone. Two flex:1 side groups
  keep the brand visually centred regardless of the (unequal) width
  of "File Edit View" versus "Scope Grid About", and give every panel
  a real left- or right-hand anchor to resolve against -- the flat
  row's "last two of six" rule is what let the About panel resolve to
  a negative x once the titles no longer filled the bar edge to edge. */
  var left = doc.createElement('div');
  left.className = 'gm-side gm-side-left';
  var right = doc.createElement('div');
  right.className = 'gm-side gm-side-right';
  var brand = doc.createElement('div');
  brand.className = 'gm-brand-slot';
  brandSlot = brand;

  MENUS.forEach(function (name, index) {
    var menu = doc.createElement('div');
    menu.className = 'gm-menu';
    var title = doc.createElement('button');
    title.type = 'button';
    title.className = 'gm-title';
    title.textContent = name;
    title.id = BAR_ID + '-title-' + index;
    title.setAttribute('aria-haspopup', 'menu');
    title.setAttribute('aria-expanded', 'false');
    title.setAttribute('aria-controls', BAR_ID + '-panel-' + index);
    var panel = doc.createElement('div');
    panel.className = 'gm-panel';
    panel.id = BAR_ID + '-panel-' + index;
    panel.hidden = true;
    panel.setAttribute('role', 'group');
    panel.setAttribute('aria-labelledby', title.id);
    if (name === 'Grid') {
      /* The real v8 SCADA panel, restored: a branded head (Ventus /
      Cables & Connectivity(r) / the status legend -- the exact shell
      nodes, moved in rather than cloned) above a scrollable body that
      carries the layer groups in the two-column shape v8 used. The
```



```

        head never enters that column flow. */
        var head = doc.createElement('div');
        head.className = 'gm-panel-head';
        var body = doc.createElement('div');
        body.className = 'gm-panel-body';
        panel.appendChild(head);
        panel.appendChild(body);
        panel.classList.add('gm-panel-grid');
        gridHead = head;
        gridBody = body;
    }
    title.addEventListener('click', function (event) {
        event.stopPropagation();
        openMenu(menu, title, panel);
    });
    menu.appendChild(title);
    menu.appendChild(panel);
    (index < 3 ? left : right).appendChild(menu);
    panels[name] = panel;
    titles.push(title);
});

nav.appendChild(left);
nav.appendChild(brand);
nav.appendChild(right);

nav.addEventListener('keydown', function (event) {
    var active = doc.activeElement;
    var index = titles.indexOf(active);
    if (event.key === 'Escape') {
        var owner = active && active.closest ? active.closest('.gm-menu') :
null;
        var ownerTitle = owner && owner.querySelector ?
owner.querySelector('.gm-title') : null;
        closeAll(ownerTitle);
        event.preventDefault();
        return;
    }
    if (index < 0) return;
    if (event.key === 'ArrowRight' || event.key === 'ArrowLeft') {
        var delta = event.key === 'ArrowRight' ? 1 : -1;
        titles[(index + delta + titles.length) % titles.length].focus();
        event.preventDefault();
    } else if (event.key === 'Home' || event.key === 'End') {
        titles[event.key === 'Home' ? 0 : titles.length - 1].focus();
        event.preventDefault();
    } else if (event.key === 'ArrowDown') {
        var ownerMenu = active.closest('.gm-menu');
        var ownerPanel = ownerMenu.querySelector('.gm-panel');
        openMenu(ownerMenu, active, ownerPanel);
        var first = ownerPanel.querySelector('button,input,[role="button"]');
        if (first && first.focus) first.focus();
        event.preventDefault();
    }
});
return nav;
}

function appendGroup(panel, text) {
    var heading = document.createElement('div');
    heading.className = 'gm-layer-group';
    heading.textContent = text;

```

```

    panel.appendChild(heading);
}

/* The estate's other published surfaces, reachable from About.
-----
These are method, not client material, and the publication boundary is
explicit that method is never withheld: "the mathematics, the method and
the derivations; the schemas, the object models and the contracts; the
code, the solvers and the validation suites" are published openly in all
cases (seed-data/07_CRITICALITY_AND_PUBLICATION_BOUNDARY.md, section 6).
The engine graph is the first of them: it is where a reader, human or
machine, sees which engine owns a calculation and which copies of it
exist elsewhere in the estate.

They are anchors carrying role="button" so they inherit the panel's own
button styling rather than introducing a second look, and they open in a
new tab so a reader never loses the map they were reading. */
var ESTATE_LINKS = [
  { href: 'https://ventusltd.github.io/ventus-grid-engine/?graph=engine-
graph',
    text: 'Grid engine · the maths' },
  { href: 'https://ventusltd.github.io/data-federation-map-for-globalgrid2050-
all-repos/dashboard/sandbox/spider_full_po_test.html',
    text: 'Federation map' },
  { href: 'https://ventusltd.github.io/spiders/spider_printer_v1/',
    text: 'Spider printer' }
];

/* The engine's own modules, listed in File, each linking into the graph.
-----
"we are heading towards a grid OS in our website the menus must be neat,
it should allow AI and humans to develop and use" -- the architect,
2026-09-05. A File menu that lists the mathematics an application runs on,
and opens each one, is the first honest step towards that: it is what an
IDE's File menu is for, and the publication boundary already says the
method is published in all cases.

The list is FETCHED from the engine's own published graph, never restated
here. Both surfaces are served from ventusltd.github.io, so this is a
same-origin request. If it fails -- offline, or the engine moved -- no
group is added and the menu is exactly what it was; a File panel that
silently lacks one group is a far better failure than a menu bar that
throws during install. */
var ENGINE_GRAPH_URL =
  'https://ventusltd.github.io/ventus-grid-engine/genome/engine-graph.json';
var ENGINE_VIEW_URL =
  'https://ventusltd.github.io/ventus-grid-engine/?graph=engine-graph&focus=';

/* Set SYNCHRONOUSLY, before the fetch is issued. The DOM marker below cannot
do this job on its own: adoptLate runs from a MutationObserver, so between
the request going out and the rows arriving there are hundreds of further
calls, and a guard that only checks the DOM starts a request on every one
of them. */
var engineFetchStarted = false;

function appendEngineModules(panel) {
  if (!panel || !window.fetch || engineFetchStarted) return;
  if (panel.querySelector('[data-gm-engine]')) return;
  engineFetchStarted = true;
  fetch(ENGINE_GRAPH_URL, { cache: 'no-cache' }).then(function (response) {
    if (!response.ok) throw new Error('HTTP ' + response.status);
    return response.json();
  });
}

```

```

}).then(function (graph) {
  var nodes = (graph && graph.nodes) || [];
  if (!nodes.length || panel.querySelector('[data-gm-engine]')) return;

  /* EVERY node, not only the canonical eleven.
  -----
  The first version of this listed `type === 'canonical'` and the
  architect caught it in one line: "Why are the mjs files not there?"
  The graph publishes 44 nodes -- 11 canonical, 1 extract, 1 reference
  and 31 fragments -- and every .mjs in the estate is in the three
  groups the filter had thrown away: sizing-arithmetic.mjs is an
  extract, grid-distance-maths/src/geodesy.mjs a reference,
  atlas-pointer-deep-link.mjs and wider-fleet.mjs are fragments.

  The fragments are the most useful rows here, not the least: they are
  where a calculation has been copied and left to drift, which is the
  defect class this estate keeps paying for. A File menu that hides
  them shows the tidy half of the truth. Groups are ordered canonical
  first, then the rest, and the graph's own kind_labels are used for
  the headings where it publishes them, so this menu does not invent a
  vocabulary the graph does not use. */
  var ORDER = ['canonical', 'extract', 'reference', 'fragment'];
  var FALLBACK_LABEL = {
    canonical: 'Engine · the maths this runs on',
    extract: 'Extracts',
    reference: 'References',
    fragment: 'Copies elsewhere in the estate'
  };
  var kindLabels = (graph && graph.kind_labels) || {};
  var byKind = {};
  nodes.forEach(function (node) {
    if (!node || !node.label) return;
    var kind = node.type || 'other';
    if (!byKind[kind]) byKind[kind] = [];
    byKind[kind].push(node);
  });
  var kinds = ORDER.filter(function (k) { return byKind[k]; })
    .concat(Object.keys(byKind).filter(function (k) { return
ORDER.indexOf(k) < 0; })).sort());

  /* Hand over something that RUNS, not only something that reads.
  -----
  "The MJS apps that AI and humans can deploy as an IDE via menu ...
  or text they could execute with AI in approved terminals or
  powershells or IDE lets hand power to the people" -- the architect,
  2026-09-05.

  This is one command, and it is one that actually works today: it
  clones the engine, installs it and runs its own fail-closed gate,
  which currently reports 8 proofs and 133 checks. It is copied to the
  clipboard rather than executed -- nothing here runs anything on
  anyone's machine, and the person or agent that pastes it into a
  terminal is the one who decides to. That is the whole of the
  approval step, and it belongs to them. */
  /* No `npm install`. Measured 202609050250: the engine declares no
  dependencies and no proof in it opens a socket, so `node verify.mjs`
  runs its whole fail-closed gate -- 8 proofs, 133 checks -- from a
  clone, offline, on any machine with node. The earlier version of this
  command included an install step, which implied a network dependency
  that does not exist and would have made the offline claim false. */
  var RUN_COMMAND =
    'git clone https://github.com/VentusLtd/ventus-grid-engine'

```

```

+ ' && cd ventus-grid-engine && node verify.mjs';

appendGroup(panel, 'Run the engine yourself · offline, no dependencies');
var run = document.createElement('button');
run.setAttribute('data-gm-engine', '1');
run.setAttribute('type', 'button');
run.title = RUN_COMMAND;
run.textContent = '📄 Copy: clone the engine and run its 133 checks';
run.addEventListener('click', function () {
  var done = function (ok) {
    run.textContent = ok
      ? '✓ Copied — paste it into a terminal'
      : '📄 ' + RUN_COMMAND;
  };
  try {
    if (navigator.clipboard && navigator.clipboard.writeText) {
      navigator.clipboard.writeText(RUN_COMMAND).then(function () {
done(true); },
      function () { done(false); });
    } else {
      /* No clipboard permission: show the command in full so it can
         still be selected by hand. A control that silently does
         nothing is worse than one that shows its own payload. */
      done(false);
    }
  } catch (_) { done(false); }
});
panel.appendChild(run);

var total = 0;
kinds.forEach(function (kind) {
  var group = byKind[kind];
  /* Alphabetical, as every non-version group in this estate is. */
  group.sort(function (a, b) { return
String(a.label).localeCompare(String(b.label), 'en-GB'); });
  appendGroup(panel, (FALLBACK_LABEL[kind] || kindLabels[kind] || kind) +
' · ' + group.length);
  group.forEach(function (node) {
    var a = document.createElement('a');
    a.setAttribute('data-gm-engine', '1');
    a.setAttribute('role', 'button');
    a.href = ENGINE_VIEW_URL + encodeURIComponent(node.label);
    a.target = '_blank';
    a.rel = 'noopener';
    a.textContent = node.label;
    /* The node's own one-line reason, as the graph publishes it. */
    if (node.reason) a.title = node.reason;
    panel.appendChild(a);
    total += 1;
  });
});
state.engine_modules = total;
}).catch(function () {
  /* Deliberately silent. See the note above: the menu must survive the
     engine being unreachable, and a reader who cannot reach it is not
     helped by an error row in a File menu. */
  state.engine_modules = 0;
});
}

/* Published studies, in View, beside the price control the reader is already
   using. A study is a reading of the network over time, and View is where

```

```

    this application keeps readings -- GB prices · historic is already there.
    Putting it in About would file it as provenance, which it is not: it is
    something to look at. */
var STUDY_LINKS = [
  { href: 'https://globalgrid2050.com/data/grid_studies_public/'
    +
    'great_britain_electricity_price_grid_constraint_trends_2016_2026.html',
    text: 'GB electricity price & grid constraint trends · 2016–2026' }
];

function appendStudies(panel) {
  if (!panel || panel.querySelector('[data-gm-study]')) return 0;
  appendGroup(panel, 'Studies');
  var added = 0;
  STUDY_LINKS.forEach(function (item) {
    var a = document.createElement('a');
    a.setAttribute('data-gm-study', '1');
    a.setAttribute('role', 'button');
    a.href = item.href;
    a.target = '_blank';
    a.rel = 'noopener';
    a.textContent = item.text;
    panel.appendChild(a);
    added += 1;
  });
  return added;
}

/* Export: print a slide, or save an image of exactly what is on screen.
-----
"allow the user to export screenshot prints via file save or file print
and they should produce beautiful slides of what's on display be it mobile
or desktop" -- the architect, 2026-09-05.

Two things this must get right, and both are failure modes this estate has
already paid for:

1. THE IMAGE MUST NOT BE BLANK. The map is a WebGL canvas created without
preserveDrawingBuffer, so reading it after the frame is composited
returns an empty image. The capture therefore happens INSIDE a render
frame, and the result is then CHECKED -- decoded and sampled for
non-transparent pixels -- before it is offered. If it comes back blank
the reader is told and sent to print instead. Handing someone an empty
PNG that looks like a successful save is worse than refusing.

2. THE CREDIT MUST TRAVEL WITH THE ARTEFACT. This generation moved the
attribution into About, which is right for the screen and wrong for an
export: OpenStreetMap, CARTO and Open Charge Map require attribution on
the thing that leaves the building. Both paths re-place it, visibly, on
the exported artefact. */
function exportStamp() {
  var now = new Date();
  return now.toISOString().replace('T', ' ').slice(0, 16) + ' UTC';
}

function attributionText(doc) {
  var node = doc.querySelector('.custom-map-attrib');
  return cleanText(node && node.textContent)
    || 'Data © OpenStreetMap contributors | © CARTO | EV data © Open Charge
Map';
}

```

```

/* THE GENERATION IS READ FROM WHAT THE COMPOSER ACTUALLY PUBLISHES.
v9.121 read `window.__GRIDATLAS_CURRENT__`, which exists nowhere in this
estate: the loader in atlas/index.html writes
`window.__GRIDATLAS_ATLAS__` and stamps
`document.documentElement.dataset.gridatlasGeneration`. So the stamp on
every printed sheet silently lost the build it came from -- measured at
202609050354, the furniture read "2026-09-05 10:35 UTC" and nothing else.
A slide that cannot be traced back to a composition is not evidence.
Both published sources are read, because the dataset attribute survives a
later document.write while a global would not. */
function generationText() {
  var atlas = window.__GRIDATLAS_ATLAS__;
  var generation = (atlas && atlas.generation)
    || (document.documentElement && document.documentElement.dataset
      && document.documentElement.dataset.gridatlasGeneration);
  return generation ? 'generation ' + generation : '';
}

/* The slide furniture: a title, the identity of whatever is on screen, the
credit, and the stamp. Created only while printing and removed after, so
nothing about the live page changes. */
/* Printing twice must not leave two of anything. Both export overlays are
addressed by id, and a second Print before the first one's cleanup timer
has run would otherwise append a duplicate id and stack two stamps on the
sheet. */
function dropById(doc, id) {
  var existing = doc.getElementById(id);
  if (existing && existing.parentNode)
    existing.parentNode.removeChild(existing);
}

function buildPrintFurniture(doc) {
  dropById(doc, 'gridatlas-print-furniture');
  var box = doc.createElement('div');
  box.id = 'gridatlas-print-furniture';
  var selected = doc.querySelector('.project-popup .name, .gm-panel .project-
name');
  var title = cleanText(selected && selected.textContent) || 'GlobalGrid2050 ·
Grid Atlas';
  box.innerHTML =
    '<div class="gpf-head"><span class="gpf-brand">VENTUS</span>'
    + '<span class="gpf-sub">GLOBAL GRID 2050 · GRID ATLAS</span></div>'
    + '<div class="gpf-title"></div>'
    + '<div class="gpf-foot"><span class="gpf-attrib"></span>'
    + '<span class="gpf-stamp"></span></div>';
  box.querySelector('.gpf-title').textContent = title;
  box.querySelector('.gpf-attrib').textContent = attributionText(doc);
  box.querySelector('.gpf-stamp').textContent =
    [generationText(), exportStamp()].filter(Boolean).join(' · ');
  doc.body.appendChild(box);
  return box;
}

function installPrintStyle(doc) {
  if (doc.getElementById('gridatlas-print-css')) return;
  var style = doc.createElement('style');
  style.id = 'gridatlas-print-css';
  style.textContent = [
    '#gridatlas-print-furniture{display:none}',
    '#gridatlas-print-map{display:none}',
    /* A SCREEN GRAB, NOT A SLIDE.
    -----

```

-

This block used to hide the entire page -- `body > \*{display:none}` -- and print the map raster edge to edge, as a presentation slide. The architect printed one and the layer panel was gone: "The layers are vital otherwise the reader doesn't know what is being shown on the map", and then the contract itself: "Keep EVERYTHING IN THE PRINT, DON'T TRY

TO

BE CLEVER, JUST A SCREEN GRAP OF WHAT THE USER SEES ... just print what is already being rendered on the display".

So nothing is hidden any more. The one substitution that remains is not a choice: a WebGL canvas built without preserveDrawingBuffer is transparent to every rasteriser outside the frame that drew it, so its OWN captured pixels are laid in its OWN box. Everything else on the sheet is the live DOM at the size it had on screen. */

```

'@media print{',
'  @page{size:auto;margin:0}',
/* The reader's paper is not a layout instruction. The view keeps the
width and height it had on screen -- published by printView() as
--gpf-vw / --gpf-vh -- so nothing reflows to paper width, which is
what turned a 1390px-wide desktop view into a phone-shaped column. */
'  html{background:#fff!important;height:auto!important;',
'    overflow:visible!important}',
'
body{background:#0b1416!important;margin:0!important;padding:0!important;',
'  width:var(--gpf-vw,100%)!important;height:auto!important;',
'    min-
height:0!important;overflow:visible!important;display:block!important}',
'  body>.dashboard{width:var(--gpf-vw,100%)!important;',
'    height:var(--gpf-vh,100vh)!important;max-height:none!important;',
'    min-height:0!important;overflow:hidden!important}',
/* The captured raster sits in the canvas's own box, at its own size.
Only ever shown when a capture actually succeeded: the body class is
added by printView() after the PNG decodes, so a failed capture prints
the live canvas exactly as before rather than a hole. */
'  #gridatlas-print-
map{display:block!important;position:absolute!important;',
'
left:0!important;top:0!important;width:100%!important;height:100%!important;',
'  max-width:none!important;max-height:none!important;object-fit:fill;',
'  z-index:1;pointer-events:none}',
'  body.gridatlas-print-raster .maplibregl-
canvas{visibility:hidden!important}',
/* PROVENANCE BELOW THE RECORD, NEVER ON TOP OF IT.
The furniture used to be position:fixed;inset:0 with scrim bands at
both ends, painted over the map -- which was fine when the map WAS the
sheet, and is not fine now that the sheet is the reader's screen: a
band over the top covers the menu bar and a band over the bottom covers
the legend. "I like the headers and footers that's nice ... KEEP THAT",
so it is kept, as a strip AFTER the view. Nothing the reader was
looking at is obscured, and the stamp is still on the sheet. */
'  #gridatlas-print-
furniture{display:block!important;position:static!important;',
'    width:var(--gpf-vw,100%)!important;box-sizing:border-box!important;',
'    inset:auto!important;padding:5mm 7mm
6mm!important;background:#040a0c!important;',
'    color:#eaf4f6!important;z-index:auto!important;',
'    font:11px/1.45 ui-monospace,SFMono-
Regular,Menlo,monospace!important}',
'  #gridatlas-print-furniture::before,#gridatlas-print-furniture::after{',
'    content:none!important}',
'  #gridatlas-print-furniture .gpf-head{letter-spacing:.28em;font-

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size:11px}',
' #gridatlas-print-furniture .gpf-brand{font-weight:700;margin-
right:10px}',
' #gridatlas-print-furniture .gpf-sub{opacity:.65;letter-spacing:.16em}',
' #gridatlas-print-furniture .gpf-title{margin-top:2mm;font-size:15px;',
' letter-spacing:.02em}',
' #gridatlas-print-furniture .gpf-foot{position:static;margin-top:2mm;',
' display:flex;justify-content:space-between;gap:6mm;',
' font-size:8.5px;opacity:.75}',
/* The map chain is given an explicit print height rather than `auto`.
Measured at 202609050354: with `height:auto` the canvas was 385x0 on a
393x852 phone and 1392x0 on a 1400x900 desktop, and Page.printToPDF
produced ZERO image XObjects -- a sheet with no map on it. The height
has to be stated. */
' .map-container{position:relative!important;width:100%!important;',
' height:100%!important;max-height:100%!important;min-
height:0!important}',
' #map,.maplibregl-map,.maplibregl-canvas-container{',
' height:100%!important;width:100%!important;max-
height:100%!important}',
' .maplibregl-canvas{width:100%!important;height:100%!important}',
/* One sheet. The view plus its strip is the record; nothing may spill. */
' body>{*{break-inside:avoid;page-break-inside:avoid}',
' body{page-break-after:avoid}',
' }'
].join('');
doc.head.appendChild(style);
}

```

/* THE SHEET NEEDS A RASTER, NOT A LIVE CANVAS.

Giving the map chain a print height was necessary and not sufficient. With the height restored, `Page.printToPDF` on the pinned candidate emitted an image XObject of exactly the map's size -- 383x838 -- and every one of its 962,862 inflated bytes was (0,0,0), behind an /SMask whose 320,954 bytes were all zero. A fully transparent rectangle the shape of the map. The reader still gets a blank sheet; the only difference is that the blankness now has dimensions.

The cause is the one `saveImage()` already knew about. The map is a WebGL canvas created without `preserveDrawingBuffer`, so its drawing buffer is gone by the time anything outside the frame that drew it goes looking -- and the print rasteriser is outside that frame, exactly as `toDataURL` was. No stylesheet can fix that, because it is not a layout problem.

So printing captures the map the same way saving does: inside a render frame, into a PNG, which is placed over the map for print media only and removed afterwards. A raster survives rasterisation. If the capture fails -- a tainted canvas, no handle, a blank read -- the sheet is printed anyway with the live canvas, which is no worse than before and still carries the furniture, title and credit. */

/* The raster goes in the CANVAS'S OWN BOX, not over the page.

It used to be appended to `<body>` and stretched ``position:fixed;inset:0``, because everything else was hidden and the map WAS the sheet. Now that the sheet is the reader's screen, a full-viewport image would cover the menu bar, the layer panel and the legend -- the very things the architect said were missing. So it is inserted as a sibling of the canvas, inside the positioned canvas container, and fills exactly that box.

Falls back to `<body>` only if the container cannot be found, which keeps a sheet with a map on it rather than none. */


```

function buildPrintMap(doc, dataUrl) {
  dropById(doc, 'gridatlas-print-map');
  var image = doc.createElement('img');
  image.id = 'gridatlas-print-map';
  image.alt = '';
  image.src = dataUrl;
  var canvas = doc.querySelector('.maplibregl-canvas');
  var host = canvas && canvas.parentNode;
  if (host && host.appendChild) host.appendChild(image);
  else doc.body.appendChild(image);
  return image;
}

/* PUBLISH THE SCREEN'S OWN SIZE TO THE STYLESHEET.
-----
A print stylesheet cannot read the viewport: @media print measures the
PAPER, so `100vw` inside it is the sheet, not the screen. Without these
two custom properties the layout reflows to paper width, which is what
turned a 1390px-wide desktop view into a phone-shaped column on the sheet.
Read once, at the moment Print is pressed, from the box the reader is
actually looking at. */
function pinViewportSize(doc) {
  var root = doc.documentElement;
  if (!root || !root.style || !root.style.setProperty) return;
  var width = Number(window.innerWidth) || (root.clientWidth || 0);
  var height = Number(window.innerHeight) || (root.clientHeight || 0);
  if (width > 0) root.style.setProperty('--gpf-vw', width + 'px');
  if (height > 0) root.style.setProperty('--gpf-vh', height + 'px');
}

function unpinViewportSize(doc) {
  var root = doc.documentElement;
  if (!root || !root.style || !root.style.removeProperty) return;
  root.style.removeProperty('--gpf-vw');
  root.style.removeProperty('--gpf-vh');
}

function printView(doc) {
  installPrintStyle(doc);
  pinViewportSize(doc);
  var furniture = buildPrintFurniture(doc);
  var shot = null;
  var clean = function () {
    if (furniture && furniture.parentNode)
furniture.parentNode.removeChild(furniture);
    if (shot && shot.parentNode) shot.parentNode.removeChild(shot);
    /* The canvas is only hidden while a raster stands in for it. Removing
       the class with the image keeps the two facts in one place: a failed
       capture never leaves an invisible map behind. */
    if (doc.body && doc.body.classList) {
      doc.body.classList.remove('gridatlas-print-raster');
    }
    unpinViewportSize(doc);
    window.removeEventListener('afterprint', clean);
  };
  window.addEventListener('afterprint', clean);
  var go = function () {
    /* Give the browser a frame to apply the print stylesheet before the
       dialog measures the page; without it the map is measured at its
       screen size. */
    window.setTimeout(function () { window.print(); }, 60);
    /* afterprint is not fired by every browser, notably some mobile ones,

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        so the furniture is removed on a timer as well. It is display:none
        off print anyway, so a late removal changes nothing a reader sees. */
        window.setTimeout(clean, 20000);
    };
    captureMap(doc, function (dataUrl) {
        if (dataUrl) {
            shot = buildPrintMap(doc, dataUrl);
            if (doc.body && doc.body.classList) {
                doc.body.classList.add('gridatlas-print-raster');
            }
            /* Wait for the browser to decode it: printing a raster it has not
               finished reading is the same blank sheet by another route. */
            if (shot.decode) { shot.decode().then(go, go); return; }
            shot.onload = go;
            shot.onerror = go;
            return;
        }
        go();
    });
}

/* Was anything actually drawn? A canvas read outside a render frame returns
   a fully transparent image, which encodes to a small PNG and downloads
   perfectly happily. Sample it rather than trust it. */
function looksBlank(canvas) {
    try {
        var probe = document.createElement('canvas');
        probe.width = 40; probe.height = 40;
        var context = probe.getContext('2d');
        context.drawImage(canvas, 0, 0, 40, 40);
        var data = context.getImageData(0, 0, 40, 40).data;
        for (var i = 3; i < data.length; i += 4) if (data[i] !== 0) return false;
        return true;
    } catch (_) {
        /* A tainted canvas throws here. That is not blank, and treating it as
           blank would send the reader to print for no reason. */
        return false;
    }
}

/* THE MAP HANDLE IS THE ONE THE ESTATE ACTUALLY PUBLISHES.
   v9.121 read `window.__GRIDATLAS_MAP__` || (window.map && ...). Neither
   resolves: `__GRIDATLAS_MAP__` is assigned nowhere in this estate, and
   `window.map` is the DIV `

blob:https://ventusltd.github.io/184a2ca1-a0ae-42ea-b3ba-72ceefc889c4



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```

function captureMap(doc, then) {
  var map = mapHandle();
  var canvas = doc.querySelector('.maplibregl-canvas')
    || (map && map.getCanvas ? map.getCanvas() : null);
  if (!canvas) { then(null, null); return; }
  var grab = function () {
    var url = null;
    try { url = canvas.toDataURL('image/png'); } catch (_) { url = null; }
    if (!url || looksBlank(canvas)) { then(null, canvas); return; }
    then(url, canvas);
  };
  if (map && map.once && map.triggerRepaint) {
    map.once('render', grab);
    map.triggerRepaint();
  } else {
    grab();
  }
}

function saveImage(doc, button) {
  var map = mapHandle();
  var canvas = doc.querySelector('.maplibregl-canvas')
    || (map && map.getCanvas ? map.getCanvas() : null);
  var say = function (text) { button.textContent = text; };

  if (!canvas) { say('⊘ No map canvas to save – use Print'); return; }

  var grab = function () {
    var url;
    try { url = canvas.toDataURL('image/png'); } catch (_) { url = null; }
    if (!url || looksBlank(canvas)) {
      /* Refused, and the reason is said out loud. The alternative is a file
         that opens as an empty rectangle an hour later, in front of someone
         else. */
      say('⊘ The map could not be captured – use Print instead');
      return;
    }
    var link = doc.createElement('a');
    link.href = url;
    link.download = 'gridatlas-' + exportStamp().replace(/^[^0-9]/g,
''.slice(0, 12) + '.png';
    doc.body.appendChild(link);
    link.click();
    doc.body.removeChild(link);
    say('✓ Image saved');
    window.setTimeout(function () { say('↓ Save an image of this view'); },
4000);
  };

  /* Inside a render frame, which is what makes the read non-blank on a
     canvas created without preserveDrawingBuffer. */
  if (map && map.once && map.triggerRepaint) {
    map.once('render', grab);
    map.triggerRepaint();
  } else {
    grab();
  }
}

/* A PDF THE PAGE WRITES ITSELF.
-----
"get the PDF PRINTER UP AND RUNNING so the WORLD CAN TAKE THE DATA" --

```

the architect, after Firefox rendered the print preview correctly and then produced NO FILE AT ALL when Print was pressed with a physical printer selected.

Every route to a PDF before this one went through the browser print pipeline: window.print(), a dialog, a destination, a driver. That pipeline differs in every browser, it is the part this code does not own, and on his machine it is the part that failed. So this does not use it at all. It writes the PDF bytes here and hands the reader a file.

ONE PAGE, SIZED TO THE IMAGE. The MediaBox takes the captured raster's own aspect ratio, so the map covers the sheet corner to corner with no margin AND nothing is cropped -- the brochure that object-fit:cover can only approximate on paper, done exactly.

The image is embedded as JPEG with /DCTDecode, which is why there is no compressor in this file: the canvas already produces JPEG bytes and PDF reads that stream natively. */

```
function pdfEscape(text) {
  return String(text == null ? '' : text)
    .replace(/\\/g, '\\\\')
    .replace(/\(/g, '\\(')
    .replace(/\)/g, '\\)')
    .replace(/\[\^x20-\x7e\]/g, '');
}

function buildMapPdf(jpegBinary, pixelWidth, pixelHeight, heading, leftFoot,
rightFoot) {
  /* ONE PAGE UNIT PER CAPTURED PIXEL. NO PAPER, NO REDUCTION.
  -----
  This scaled the long edge to 1190pt, "A3-ish". That is a paper
  assumption, and on a 1390x518 desktop capture it emitted a 1190x443
  page -- a 14% REDUCTION of the record. "THE PRINT MUST BE HIGH RES OF
  WHAT THE USER SEES NOT A REDUCED CRAP VERSION", "WE ARE NOT USING
  PAPER", "THIS IS A CALL FOR A 2026 era TELEPRINTER".

  A teleprinter emits the record as it was. So the page is exactly the
  captured raster: one PDF unit per pixel, no scaling in either
  direction, and the canvas is captured at devicePixelRatio -- 1149x2514
  on a phone at dpr 3, native on a desktop at dpr 1. Nothing is resampled
  on the way out, and a viewer showing it at 100% shows the reader's own
  pixels. */
  var pageW = pixelWidth;
  var pageH = pixelHeight;
  /* Furniture scaled to the record rather than to an assumed sheet, so a
  2514px-tall phone capture and a 518px-tall desktop one both carry a
  legible credit rather than one sized for A3. */
  var unit = Math.max(1, Math.min(pageW, pageH) / 520);
  var band = Math.round(Math.min(pageH * 0.14, 46 * unit));
  var headSize = Math.round(13 * unit);
  var footSize = Math.round(8 * unit);
  var pad = Math.round(14 * unit);
  var rightX = Math.max(pad, pageW - pad - String(rightFoot).length * footSize
* 0.56);

  var content = [
    'q', pageW + ' 0 0 ' + pageH + ' 0 0 cm', '/Im0 Do', 'Q',
    /* Scrim bands at 55% alpha so the text reads over a dark basemap
    without dimming the map itself. */
    'q', '/GsA gs', '0.02 0.06 0.07 rg',
    '0 ' + (pageH - band) + ' ' + pageW + ' ' + band + ' re f',
    '0 0 ' + pageW + ' ' + band + ' re f', 'Q',
```

```

    'BT /F1 ' + headSize + ' Tf 1 1 1 rg ' + pad + ' ' + (pageH - pad -
headSize)
    + ' Td ( ' + pdfEscape(heading) + ' ) Tj ET',
    'BT /F1 ' + footSize + ' Tf 0.86 0.93 0.94 rg ' + pad + ' ' +
Math.round(pad * 0.7)
    + ' Td ( ' + pdfEscape(leftFoot) + ' ) Tj ET',
    'BT /F1 ' + footSize + ' Tf 0.86 0.93 0.94 rg ' + rightX + ' ' +
Math.round(pad * 0.7)
    + ' Td ( ' + pdfEscape(rightFoot) + ' ) Tj ET'
  ].join('\n');

  var objects = [
    '<< /Type /Catalog /Pages 2 0 R >>',
    '<< /Type /Pages /Kids [3 0 R] /Count 1 >>',
    '<< /Type /Page /Parent 2 0 R /MediaBox [0 0 ' + pageW + ' ' + pageH + ']'
    + ' /Resources << /XObject << /Im0 5 0 R >> /Font << /F1 6 0 R >>'
    + ' /ExtGState << /GsA 7 0 R >> >> /Contents 4 0 R >>',
    '<< /Length ' + content.length + ' >>\nstream\n' + content +
'\nendstream',
    '<< /Type /XObject /Subtype /Image /Width ' + pixelWidth + ' /Height ' +
pixelHeight
    + ' /ColorSpace /DeviceRGB /BitsPerComponent 8 /Filter /DCTDecode
/Length '
    + jpegBinary.length + ' >>\nstream\n' + jpegBinary + '\nendstream',
    '<< /Type /Font /Subtype /Type1 /BaseFont /Helvetica /Encoding
/WinAnsiEncoding >>',
    '<< /Type /ExtGState /ca 0.55 >>'
  ];

  var out = '%PDF-1.4\n%\u00e2\u00e3\u00cf\u00d3\n';
  var offsets = [];
  var i;
  for (i = 0; i < objects.length; i += 1) {
    offsets.push(out.length);
    out += (i + 1) + ' 0 obj\n' + objects[i] + '\nendobj\n';
  }
  var xref = out.length;
  out += 'xref\n0 ' + (objects.length + 1) + '\n0000000000 65535 f \n';
  for (i = 0; i < offsets.length; i += 1) {
    out += ('0000000000' + offsets[i]).slice(-10) + ' 00000 n \n';
  }
  out += 'trailer\n<< /Size ' + (objects.length + 1) + ' /Root 1 0 R >>\n'
    + 'startxref\n' + xref + '\n%%EOF\n';

  var bytes = new Uint8Array(out.length);
  for (i = 0; i < out.length; i += 1) bytes[i] = out.charCodeAt(i) & 0xff;
  return { bytes: bytes, pageW: pageW, pageH: pageH };
}

```

/* JPEG rather than PNG: /DCTDecode embeds the bytes verbatim, so no compressor is needed here, and a basemap raster is what JPEG is for. Same render-frame discipline as captureMap -- a canvas without preserveDrawingBuffer is transparent to any reader outside the frame that drew it. */

```

function captureMapJpeg(doc, then) {
  var map = mapHandle();
  var canvas = doc.querySelector('.maplibregl-canvas')
  || (map && map.getCanvas ? map.getCanvas() : null);
  if (!canvas) { then(null, null); return; }
  var grab = function () {
    var url = null;
    try { url = canvas.toDataURL('image/jpeg', 0.92); } catch (_) { url =

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null; }
    if (!url || url.indexOf('data:image/jpeg') !== 0 || looksBlank(canvas)) {
        then(null, canvas);
        return;
    }
    then(url, canvas);
};
if (map && map.once && map.triggerRepaint) {
    map.once('render', grab);
    map.triggerRepaint();
} else {
    grab();
}
}

function pdfFileStamp() {
    var d = new Date();
    var pad = function (n) { return (n < 10 ? '0' : '') + n; };
    return d.getUTCFullYear() + pad(d.getUTCMonth() + 1) + pad(d.getUTCDate())
        + pad(d.getUTCHours()) + pad(d.getUTCMinutes());
}

function savePdf(doc, button) {
    var say = function (text) { button.textContent = text; };
    say('... building PDF');
    captureMapJpeg(doc, function (jpegDataUrl, canvas) {
        if (!jpegDataUrl) {
            say('\u2298 The map could not be captured \u2014 try again once it has
drawn');
            return;
        }
        var binary;
        try {
            binary = atob(jpegDataUrl.slice(jpegDataUrl.indexOf(',') + 1));
        } catch (_) {
            say('\u2298 The capture could not be decoded');
            return;
        }
        var built = buildMapPdf(binary, canvas.width, canvas.height,
            'GlobalGrid2050 \u00b7 Grid Atlas', attributionText(doc),
            (generationText() || 'generation unknown')
                + ' \u00b7 ' + new Date().toISOString().slice(0, 16).replace('T', ' ') + '
UTC');
        var blob = new Blob([built.bytes], { type: 'application/pdf' });
        var url = URL.createObjectURL(blob);
        var link = doc.createElement('a');
        link.id = 'gridatlas-pdf-download';
        link.href = url;
        link.download = 'globalgrid2050-grid-atlas-' + pdfFileStamp() + '.pdf';
        doc.body.appendChild(link);
        link.click();
        setTimeout(function () {
            URL.revokeObjectURL(url);
            if (link.parentNode) link.parentNode.removeChild(link);
        }, 30000);
        say('\u2713 PDF saved \u00b7 ' + built.pageW + ' \u00d7 ' + built.pageH + '
px, 1:1');
    });
}

function appendExport(panel, doc) {
    if (!panel || panel.querySelector('[data-gm-export]')) return 0;

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appendGroup(panel, 'Export this view');

/* EACH CONTROL IS NAMED, BECAUSE ITS LABEL IS NOT A NAME.
Both controls carried `data-gm-export="1"` and nothing else, so the only
way anything outside this function could tell them apart was by reading
their text. saveImage() REWRITES that text on click -- to "Image saved",
or to the refusal -- so a caller that found the button by its words lost
it the moment it used it. The first outcome proof written for this
feature did exactly that, and its failures were unattributable as a
result. An id and a typed data attribute are stable across the click,
across a relabelling, and across translation. The attribute keeps the
name `data-gm-export` so the "already built" guard above and every
existing check that looks for it still see it. */
var print = doc.createElement('button');
print.id = 'gridatlas-export-print';
print.setAttribute('data-gm-export', 'print');
print.setAttribute('type', 'button');
print.textContent = '🖨 Print · or save as PDF';
print.addEventListener('click', function () { printView(doc); });
panel.appendChild(print);

var pdf = doc.createElement('button');
pdf.id = 'gridatlas-export-pdf';
pdf.setAttribute('data-gm-export', 'pdf');
pdf.setAttribute('type', 'button');
pdf.textContent = '📄 Save this view as a PDF';
pdf.addEventListener('click', function () { savePdf(doc, pdf); });
panel.appendChild(pdf);

var image = doc.createElement('button');
image.id = 'gridatlas-export-image';
image.setAttribute('data-gm-export', 'image');
image.setAttribute('type', 'button');
image.textContent = '📷 Save an image of this view';
image.addEventListener('click', function () { saveImage(doc, image); });
panel.appendChild(image);

return 3;
}

function appendEstateLinks(panel) {
  if (!panel || panel.querySelector('[data-gm-estate]')) return 0;
  appendGroup(panel, 'Estate');
  var added = 0;
  ESTATE_LINKS.forEach(function (item) {
    var a = document.createElement('a');
    a.setAttribute('data-gm-estate', '1');
    a.setAttribute('role', 'button');
    a.href = item.href;
    a.target = '_blank';
    a.rel = 'noopener';
    a.textContent = item.text;
    panel.appendChild(a);
    added += 1;
  });
  return added;
}

/* A measured audit found the v8 panel's own checkboxes at 17x17 px --
the input element itself, not just its label. A label with a tall
min-height passes a hit-test at its centre but still measures 17x17
if something re-measures the <input> node's own rect, the way the

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live audit did. So the proxy <input> here is stretched, invisible,
over the FULL label (position:absolute;inset:0) -- its own
getBoundingClientRect() is therefore the whole >=44px control, under
any measurement method -- and a separate, normally-sized box (built
from CSS alone, no image) carries the visible tick. */
function layerCheckbox(kind) {
  var proxy = document.createElement('input');
  proxy.type = kind;
  var box = document.createElement('span');
  box.className = 'gm-layer-box';
  box.setAttribute('aria-hidden', 'true');
  return { proxy: proxy, box: box };
}

function buildLayerControls(found) {
  var lastGroup = '';
  found.controls.forEach(function (original) {
    var key = layerKey(original);
    var group = layerGroup(original);
    if (group !== lastGroup) {
      appendGroup(gridBody, group);
      lastGroup = group;
    }
    var label = document.createElement('label');
    label.className = 'gm-layer';
    label.setAttribute('data-gridatlas-layer-key', key);
    var built = layerCheckbox('checkbox');
    var proxy = built.proxy;
    proxy.setAttribute('data-gridatlas-layer-proxy', key);
    var name = document.createElement('span');
    name.className = 'gm-layer-name';
    label.appendChild(proxy);
    label.appendChild(built.box);
    label.appendChild(name);
    gridBody.appendChild(label);
    layerTargets[key] = original;
    layerProxies[key] = proxy;
    proxy.addEventListener('change', function () {
      if (!!original.checked !== !!proxy.checked && typeof original.click ===
'function') {
        /* original.click() must remain the implementation: its delegated
engine listener owns hydration. Suppress only the document-level
outside-click closer while that synchronous forwarding runs, so
the reader can see the tick and its live load state. */
        forwardingLayerChoice = true;
        try { original.click(); }
        finally { forwardingLayerChoice = false; }
      }
      syncLayer(key);
    });
    syncLayer(key);
  });

  var basemaps = array(found.host.querySelectorAll('input[type="radio"]
[name="bm"]'));
  if (basemaps.length) appendGroup(gridBody, 'Basemap');
  basemaps.forEach(function (original) {
    var label = document.createElement('label');
    label.className = 'gm-layer';
    var built = layerCheckbox('radio');
    var proxy = built.proxy;
    proxy.name = 'gridatlas-menu-basemap';

```



```

        proxy.value = original.value;
        proxy.checked = !!original.checked;
        var name = document.createElement('span');
        name.className = 'gm-layer-name';
        name.textContent = cleanText(original.closest('label').textContent) ||
original.value;
        proxy.addEventListener('change', function () {
            if (proxy.checked && !original.checked && typeof original.click ===
'function') original.click();
            closeAll();
        });
        label.appendChild(proxy);
        label.appendChild(built.box);
        label.appendChild(name);
        gridBody.appendChild(label);
    });
}

```

```

function move(panel, node, label) {
    if (!node || !panel || (bar && bar.contains(node))) return false;
    if (label && node.setAttribute) node.setAttribute('aria-label', label);
    panel.appendChild(node); // same node: its original listener
survives
    if (node.removeAttribute) node.removeAttribute('hidden');
    state.controls_moved += 1;
    return true;
}

```

```

function trayRoute(node) {
    var text = cleanText(node && node.textContent).toLowerCase();
    if (/\\bclear\\b|\\bscope\\b/.test(text)) return 'Scope';
    if (/\\bgrid\\b|\\bsub\\b/.test(text)) return 'Grid';
    return '';
}

```

/* The two chips that must NOT be swallowed by a menu on a phone.

GRID and SUBS were put on the map deliberately, and the reason is on the record: the grid-line and substation switches live in the SCADA panel below the map, "which a phone never scrolls to; activation looked broken" (composition manifest, mobile_tray, from phone acceptance on 2026-09-01). Moving every tray button into a dropdown re-created a milder form of exactly that fault - measured at an iPhone 13 viewport on 202609041330, zero layer controls were reachable without first opening a menu.

So on a touch screen or a narrow window these two stay where they were designed to be. Everything else still routes into the menus, and desktop is unchanged: there the chips are redundant with a menu that is already one click away and always visible. */

```

function chipStaysOnMap(node) {
    var text = cleanText(node && node.textContent).toLowerCase();
    if (!/\\bgrid\\b|\\bsub\\b/.test(text)) return false;
    /* An UNKNOWN width is not a phone. Reading `(window.innerWidth || 0) <=
700`
    makes a missing or zero width report narrow, which is the wrong way for
    a default to fail: it would strand these chips on the map in any host
    that does not publish a width, including a headless proof fixture. The
    width has to be a real positive number before it argues for a phone. */
    var coarse = false;
    try {
        coarse = !(window.matchMedia && window.matchMedia('(pointer:

```

```

coarse)').matches);
    } catch (error) {
        coarse = false;
    }
    var width = Number(window.innerWidth);
    var narrow = isFinite(width) && width > 0 && width <= 700;
    return coarse || narrow;
}

function adoptLate(doc) {
    if (!bar) return;
    move(panels.View, doc.getElementById('gridatlas-gb-conditions'));
    move(panels.About, doc.getElementById('gridatlas-version-ledger'));
    move(panels.View, doc.getElementById('btn-fullscreen-exit'), 'Exit full
screen');

    var curtain = doc.getElementById('fs-curtain-tab');
    if (curtain && !bar.contains(curtain)) {
        curtain.setAttribute('role', 'button');
        curtain.setAttribute('tabindex', '0');
        curtain.setAttribute('aria-label', 'Open the fullscreen layer curtain');
        move(panels.Grid, curtain);
    }

    var tray = doc.getElementById('gridatlas-mobile-tray');
    if (tray) {
        array(tray.querySelectorAll('button')).forEach(function (button) {
            if (chipStaysOnMap(button)) return; // one tap on a phone, not two
            var route = trayRoute(button);
            if (route) move(panels[route], button);
            else button.hidden = true; // Tools only revealed the controls now
in menus
        });
    }

    move(panels.Scope, doc.getElementById('btn-gridpoint'));

    var disclaimer = doc.querySelector('.disclaimer-box');
    var shoutout = doc.querySelector('.podcast-shoutout');
    move(panels.About, disclaimer);
    move(panels.About, shoutout);

    state.estate_links = appendEstateLinks(panels.About);
    appendEngineModules(panels.File);
    state.export_controls = appendExport(panels.File, doc);
    state.studies = appendStudies(panels.View);

    /* The map attribution moves off the map and into About, LAST, in small
    print.
    -----
    Measured live at generation 202609042123, 2026-09-05: .custom-map-attrib
    rendered at x=15 y=47, 401x24 px -- a boxed band immediately under the
    menu bar, over the top-left of the map, which is where a reader arriving
    on a deep link looks first. On a 393x852 phone it wrapped to two lines
    and the architect photographed the consequence: opening EDIT drew
    "Status Colours" UNDERNEATH the attribution box, the two overlapping in
    the same space. His instruction, verbatim: "Attribution bar clashes move
    that to about and in small print at the bottom".

    The credit is owed and is not dropped: it is the same node, moved once,
    text intact, so OpenStreetMap, CARTO and Open Charge Map are still named
    on the page and one tap away. It is appended AFTER the estate links, and

```

```

        re-appended on every later adoption pass, so a late DOM rebuild cannot
        leave it above the controls again. appendChild on a node already in the
        panel moves it to the end rather than duplicating it. */
var attrib = doc.querySelector('.custom-map-attrib');
if (attrib) {
    if (!bar || !bar.contains(attrib)) move(panels.About, attrib);
    else if (panels.About.lastElementChild !== attrib)
panels.About.appendChild(attrib);
    /* The `lastElementChild` test is not tidiness, it is the difference
    between this working and crashing the tab. adoptLate runs from a
    MutationObserver, so an unconditional appendChild here IS a mutation,
    which re-enters adoptLate, which appends again: a feedback loop that
    crashed the renderer outright under the 393x852 arrival gate, while
    the previous generation passed the same gate in the same harness.
    Once the node is already last, this is a no-op and the loop closes. */
}

/* The real .scada-brand (VENTUS, again) and .status-legend move once,
into the restored SCADA panel's head, during install() below -- not
hidden as a "duplicate" and not here, so a late DOM rebuild cannot
repeatedly fight over one node's location. */

/* The panel's own show/hide control stays REACHABLE. It was hidden here as
"superseded; measured inert", and it is neither: the stylesheet rule
above was hiding the panel it moved, so its effect could not be seen.
With the panel restored this is the only control that opens and closes
it, and hiding it would leave a phone with a collapsed panel and nothing
to open it with -- which is the fault this generation exists to end. */
var dashToggle = doc.getElementById('gridatlas-dash-toggle');
if (dashToggle) dashToggle.hidden = false;

var stack = doc.querySelector('.map-controls');
if (stack) {
    var leftovers =
array(stack.querySelectorAll('button,input,select,textarea,a'))
    .filter(function (node) { return !node.hidden; });
    if (leftovers.length === 0) stack.setAttribute('data-gridatlas-menu-
emptied', '1');
}

MENUS.forEach(function (name) {
    state.panel_counts[name] = panels[name] ? panels[name].children.length :
0;
});
}

function install(doc) {
    if (state.installed || doc.getElementById(BAR_ID)) return true;
    var ready = required(doc);
    state.engine_layer_controls = ready.found.engine.length;
    state.pipeline_layer_controls = ready.found.pipeline.length;
    state.layer_controls = ready.found.controls.length;
    if (ready.missing.length) {
        state.waiting_for = ready.missing.slice();
        return false;
    }

    installStyle(doc);
    bar = buildBar(doc);
    buildLayerControls(ready.found);

    move(panels.File, ready.nodes.search);

```

```

move(panels.File, ready.nodes.exportButton);
move(panels.Edit, ready.nodes.statusButton);
move(panels.View, ready.nodes.fullscreenButton);
move(panels.Scope, ready.nodes.radiusButton);
move(panels.Scope, ready.nodes.radiusAreaButton);
move(panels.Scope, ready.nodes.zoneButton);
move(panels.Scope, ready.nodes.measureButton);

/* The VENTUS masthead: fused into the bar's own centre (see buildBar),
   never a closed panel -- the architect's "VENTUS branding has been
   lost" was this node being moved into a collapsed About panel, and
   the measured "flash then vanish" (present for ~1.5s, then torn out)
   was that same move happening after the raw v8 page had already
   painted it once. Moving it here, into brandSlot, keeps it visible
   through the whole transition: raw markup, then fused into the bar,
   never hidden in between. */
move(brandSlot, ready.nodes.header);

/* The restored SCADA panel's head: the real .scada-brand (VENTUS,
   again, exactly as v8 rendered it) and .status-legend, moved once --
   not cloned, not hidden as a duplicate. gridHead sits above gridBody
   (built by buildLayerControls) and never enters its column flow. */
move(gridHead, doc.querySelector('.scada-brand'));
move(gridHead, doc.querySelector('.status-legend'));

ready.nodes.host.insertBefore(bar, ready.nodes.host.firstChild);
doc.documentElement.classList.add('gridatlas-menu-hosted');
syncAttribClearance(doc);
if (typeof ResizeObserver === 'function') {
    var barResize = new ResizeObserver(function () {
        syncAttribClearance(doc);
        if (openPanelRefs) clampPanel(doc, openPanelRefs.menu,
openPanelRefs.panel);
    });
    barResize.observe(bar);
    state.attrib_clearance_source = 'ResizeObserver';
} else if (doc.defaultView && typeof doc.defaultView.addEventListener ===
'function') {
    /* No ResizeObserver: a viewport resize is the only other way the
       bar's own height changes (the @media breakpoint), so fall back to
       watching that. */
    doc.defaultView.addEventListener('resize', function () {
        syncAttribClearance(doc);
        if (openPanelRefs) clampPanel(doc, openPanelRefs.menu,
openPanelRefs.panel);
    });
    state.attrib_clearance_source = 'resize-listener';
}

/* One document click listener and one change listener, installed once.
   The retry path cannot multiply effects. */
doc.addEventListener('click', function (event) {
    if (!bar.contains(event.target)) {
        if (!forwardingLayerChoice) closeAll();
    }
    else if (event.target && /^(BUTTON|INPUT)$/.test(event.target.tagName ||
'')) {
        if (event.target.type !== 'text'
            && !event.target.classList.contains('gm-title')
            && !event.target.hasAttribute('data-gridatlas-layer-proxy')) {
            window.setTimeout ? window.setTimeout(closeAll, 0) : closeAll();
        }
    }
}

```

```

    }
  });
  doc.addEventListener('change', function (event) {
    var key = layerKey(event.target);
    if (key && layerProxies[key]) syncLayer(key);
  });
  state.listeners = 2;

  adoptLate(doc);
  closeAll();
  state.installed = true;
  state.waiting_for = [];
  state.failure = null;
  state.one_identity_surface = true;
  state.mobile_sheet_hit_target_guard = true;
  state.layer_status_mirrored = true;
  state.layer_menu_stays_open = true;

  if (typeof MutationObserver === 'function') {
    observer = new MutationObserver(function () {
      adoptLate(doc);
      syncAll();
    });
    observer.observe(doc.body, { childList: true, subtree: true,
characterData: true });
  }
  return true;
}

state.install = function () { return install(document); };
state.closeAll = closeAll;

function loudFailure(doc) {
  if (state.installed || doc.getElementById(FAILURE_ID)) return;
  var found = inventory(doc);
  state.failure = 'menu not installed: expected 60 engine + 3 Pipeline News
layer controls; found '
    + found.engine.length + ' + ' + found.pipeline.length;
  if (window.console && typeof window.console.error === 'function') {
    window.console.error('[GRIDATLAS MENU] ' + state.failure);
  }
  var alert = doc.createElement('div');
  alert.id = FAILURE_ID;
  alert.setAttribute('role', 'alert');
  alert.textContent = state.failure + '. Original controls remain available.';
  alert.style.cssText = 'position:fixed;left:8px;right:8px;top:8px;z-
index:10030;'
    + 'padding:8px;background:#280b0b;color:#ffd0d0;border:1px solid #b44;'
    + 'font:11px/1.4 monospace';
  (doc.body || doc.documentElement).appendChild(alert);
}

function start() {
  /* Cartridge proofs and prerenderers can provide a deliberately partial
  document. Treat that exactly like any other missing dependency: publish
  the refusal and leave the owner's interface untouched. In particular,
  do not start the 40-second browser retry loop against a non-DOM stub. */
  if (!document.documentElement || typeof document.createElement !==
'function') {
    state.failure = 'menu not installed: full document unavailable';
    return;
  }
}

```

```

var probe = document.createElement('div');
if (!probe || typeof probe.setAttribute !== 'function' || !probe.classList)
{
    state.failure = 'menu not installed: full DOM element API unavailable';
    return;
}
if (install(document)) return;
if (typeof window.setInterval !== 'function'
    || typeof window.clearInterval !== 'function') return;
timer = window.setInterval(function () {
    state.tries += 1;
    if (install(document)) {
        window.clearInterval(timer);
        timer = null;
    } else if (state.tries >= MAX_TRIES) {
        window.clearInterval(timer);
        timer = null;
        loudFailure(document);
    }
}, 250);
}

if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', start, { once: true });
} else {
    start();
}
})();

```

/* Arrival tidy - the search machinery gets out of the way once it has answered.

MEASURED, on v9.96 at 393x852, arriving at ?repd_ref=12588 the way a shared link arrives: the map canvas was topmost at 13 per cent of 3,200 sampled viewport points and the app's own controls at 87. Three of those controls were the SEARCH, not the answer - the results list still open at [40,136,302,92] holding the project the reader had already been taken to, the box holding "12588", a reference the reader never typed, and the identity repeated in the results row, the search bar and the card header.

A deep link is not a search. The reader asked for one project by name and got it; the list of how it was found is scaffolding, and scaffolding left up is clutter. So on a RESOLVED arrival the list is dismissed and the box is emptied back to its placeholder.

WHAT THIS DELIBERATELY DOES NOT DO

- It does not touch a FAILED arrival. When the identity could not be resolved the results are the only thing on screen that explains why, and hiding them would leave a reader with a map and no account of it.
- It does not clear a box the reader has touched. Focus or a keystroke retires this permanently, so a reader who starts typing while the arrival is still resolving never has their text taken away.
- It does not clear a box holding something other than the ref, which is the same rule stated for the case where another lane wrote there first.
- It hides with `style.display = 'none'`, which is what the shell's own Escape key and map click do. This is the product's mechanism, not a new one, so nothing has to be taught how to bring it back.

It runs only where there is a MutationObserver. The cartridge proofs execute composed cartridges under node:vm against a window stub that has neither observers nor timers, and a module that throws there takes the whole cartridge down with it - which is exactly how v9.92 shipped a menu

```

    bar that broke the sandbox proof. Where there is no observer there is no
    arrival either, so doing nothing is the correct answer rather than a
    degraded one. */
(function arrivalTidy() {
    if (typeof document === 'undefined' || typeof window === 'undefined') return;
    if (typeof MutationObserver !== 'function') return;

    var ref = '';
    try {
        ref = String(new URLSearchParams(window.location.search).get('repd_ref') ||
        '').trim();
    } catch (_) { return; }
    if (!ref) return;

    var typed = false;
    var done = false;

    function tidy() {
        if (done) return true;
        var state = document.body && document.body.dataset
            ? document.body.dataset.gridatlasRepdDeepLink : '';
        if (state !== 'resolved') return state === 'failed'; /* failed: stop
watching, change nothing */
        var results = document.getElementById('search-results');
        var input = document.getElementById('search-input');
        if (!results || !input) return false;
        if (!typed) {
            results.style.display = 'none';
            if (input.value === ref) input.value = '';
        }
        done = true;
        return true;
    }

    function retire() { typed = true; }

    function watch() {
        var input = document.getElementById('search-input');
        if (input) {
            input.addEventListener('focus', retire, { once: true });
            input.addEventListener('input', retire, { once: true });
        }
        if (tidy()) return;
        var observer = new MutationObserver(function () {
            if (tidy()) observer.disconnect();
        });
        observer.observe(document.body, {
            attributes: true, attributeFilter: ['data-gridatlas-repd-deep-link']
        });
        window.__GRIDATLAS_ARRIVAL_TIDY__ = {
            installed: true,
            repd_ref: ref,
            get dismissed() { return done; },
            get retired_by_reader() { return typed; }
        };
    }

    if (document.readyState === 'loading') {
        document.addEventListener('DOMContentLoaded', watch, { once: true });
    } else {
        watch();
    }
}

```

```

})();

/**
 * Module: pipeline-news-layers
 *
 * A PIPELINE NEWS (REPD) section in the layer dashboard, beside TOPOLOGY,
 * ASSETS and TRANSIT, that summons the rest of the pipeline around whatever
 * project is currently selected.
 *
 * Vikram: "summon other pipeline items within the atlas after clicking the
 * map ... under REPD pipelinenews under its own section like topology, assets
 * etc". Arriving from Pipeline News you land on one project with five links to
 * substations and nothing else of the pipeline in view. These three controls
 * put the neighbours back: what else is being built within reach, of the same
 * technology, of the twenty technologies Pipeline News' own spine does not
 * carry, or of anything at all.
 *
 * WHY IT DOES NOT USE data-layer-id
 * -----
 * The engine delegates a `change` listener on #scada-ui-container and on
 * #fs-curtain-keys, and any checkbox carrying `data-layer-id` is routed to its
 * own handleLayerToggle -- which would be handed an id it has no config for.
 * These controls carry `data-pn-layer` instead and are handled here. Same
 * lesson as the wider-fleet tabs in Pipeline News: borrow the styling, never
 * the attribute that another owner dispatches on.
 *
 * WHAT IT IS NOT
 * -----
 * It draws register points near a selection. It does not measure them, rank
 * them, bind them to the selected project, or imply any relationship between
 * them. Two projects near each other share a map square and nothing else --
 * not a circuit, not a connection, not a queue position. The labels say
 * "within 25 km" and stop there.
 *
 * Depends on: geodesy.
 */
(() => {
  'use strict';

  const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||
  {});
  if (NS.pipelineNewsLayers) return;

  const geodesy = NS.geodesy;
  if (!geodesy) {
    throw new Error('pipeline-news-layers requires the geodesy module');
  }

  const GENERATION = '202609030048';
  const RADIUS_KM = 25;
  const GROUP_TITLE = 'PIPELINE NEWS (REPD)';

  /* The register comes from the engine, not from a URL.
  -----
  dist/repd_master.json is NOT a served file. Fetching it 404s on the live
  host and in a local checkout alike -- measured both ways -- because the
  streaming bridge reconstructs the register from parquet and hands it
  straight to MapLibre. Every REPD layer the engine draws (l-solar, l-wind,
  l-bess, l-biomass and the rest) is a filter over ONE shared source, and
  that source holds all 10,784 rows once any one of those layers has been
  switched on.
  So this reads src-repd. It is the engine's register, hydrated by the

```



```

    engine, and there is no second copy and no second fetch. If nothing has
    hydrated it yet, ticking the engine's own control is what fills it --
    the same move enableTechnologyLayer makes, for the same reason. */
const REGISTER_SOURCE = 'src-repd';
const REGISTER_PRIMER = 'biomass'; // any REPD control hydrates the shared
source

/* Pipeline News' spine carries four REPD technology types. Everything else
in the register is the wider fleet -- the 1,104 projects its own product
could not admit without changing what it is. Named by the register's own
`tech` classification, not by a nickname. */
const SPINE_TECHS = new Set(['solar', 'solar_roof', 'bess', 'wind']);

/* The engine's own technology colours, so a point reads the same here as it
does on the layer it belongs to. */
const TECH_COLOUR = {
  solar: '#ffff00', solar_roof: '#ffcc00', bess: '#ffae00', wind: '#00ffff',
  biomass: '#39ff14', hydro: '#00aaff', hydrogen: '#ffffff', tidal: '#00bfff',
  act: '#ff6600', geothermal: '#ff3300', flywheel: '#ff69b4', caes: '#88aaff',
  other: '#888888'
};

const CONTROLS = [
  {
    id: 'same',
    label: 'Same technology',
    colour: '#5fbdc2',
    keep: (row, selection) => row.tech === selection.tech
  },
  {
    id: 'wider',
    label: 'Wider fleet',
    colour: '#39ff14',
    keep: (row) => !SPINE_TECHS.has(row.tech)
  },
  {
    id: 'all',
    label: 'All pipeline',
    colour: '#d8b64a',
    keep: () => true
  }
];

const state = {
  schema: 'gridatlas.pipeline-news-layers.v1',
  generation: GENERATION,
  installed: false,
  register_rows: 0,
  register_url: null,
  radius_km: RADIUS_KM,
  selection: null,
  counts: {},
  active: [],
  failures: []
};

window.__GRIDATLAS_PIPELINE_LAYERS__ = state;

function note(message) {
  const text = String(message && message.message ? message.message : message);
  if (!state.failures.includes(text)) state.failures.push(text);
}

```

```

let register = null;          // the engine's rows, read once and kept

function readRegisterSource(map) {
  try {
    const source = map.getSource(REGISTER_SOURCE);
    const features = source && source._data && source._data.features;
    if (!Array.isArray(features) || !features.length) return null;
    return features.map((feature) => {
      const properties = feature.properties || {};
      const coordinates = (feature.geometry || {}).coordinates || [];
      return {
        name: properties.name || '',
        operator: properties.operator || '',
        tech: properties.tech || 'other',
        raw: properties.raw_tech || '',
        status: properties.status || '',
        mw: Number(properties.capacity) || 0,
        lon: Number(coordinates[0]),
        lat: Number(coordinates[1])
      };
    }).filter((row) => Number.isFinite(row.lon) && Number.isFinite(row.lat));
  } catch (error) {
    note('register: ' + String(error && error.message || error));
    return null;
  }
}

/* Ask the engine to hydrate its own register, by ticking the control it
   owns rather than reaching past it into the map. The panel then tells the
   truth about what is on, which it would not if this added a source itself.
*/
function primeRegister() {
  const box = document.querySelector(
    '#scada-ui-container input[type=checkbox][data-layer-id="' +
REGISTER_PRIMER + '"]');
  if (!box) { note('register: no ' + REGISTER_PRIMER + ' control to prime
with'); return false; }
  if (!box.checked) box.click();
  state.primed_with = REGISTER_PRIMER;
  return true;
}

async function loadRegister(map) {
  if (register) return register;
  register = readRegisterSource(map);
  if (register) { state.register_rows = register.length; return register; }

  if (!primeRegister()) throw new Error('register unavailable');
  for (let attempt = 0; attempt < 40; attempt += 1) {
    await new Promise((resolve) => setTimeout(resolve, 250));
    register = readRegisterSource(map);
    if (register) { state.register_rows = register.length; return register; }
  }
  note('register: ' + REGISTER_SOURCE + ' did not hydrate within 10 s');
  throw new Error('register unavailable');
}

/* The selected project, read from the pin the sld-sandbox cartridge draws.
   There is no public selection surface carrying coordinates -- last_selection
   has the name, the technology and the nearest distance, but not the origin
   -- so this reads the pin source and corroborates it against the public
   project_pin.name before trusting it. If the cartridge ever publishes the

```

```

    origin properly, delete this and read that. */
function readSelection(map) {
  try {
    const source = map.getSource('gridatlas-project-pin');
    const features = source && source._data && source._data.features;
    if (!features || !features.length) return null;
    const [lon, lat] = features[0].geometry.coordinates || [];
    if (!Number.isFinite(lon) || !Number.isFinite(lat)) return null;
    const links = window.__GRIDATLAS_NEON_LINKS__ || {};
    const name = (features[0].properties || {}).name
      || (links.project_pin || {}).name || '';
    return { lon, lat, name, tech: (links.last_selection || {}).tech || '' };
  } catch (error) {
    note('selection: ' + String(error && error.message || error));
    return null;
  }
}

function near(rows, selection) {
  const found = [];
  for (const row of rows) {
    const km = geodesy.distanceKm(selection.lon, selection.lat, row.lon,
row.lat);
    if (km > RADIUS_KM) continue;
    // The selected project is not one of its own neighbours.
    if (km < 0.0005 && row.name === selection.name) continue;
    found.push({ ...row, km });
  }
  found.sort((a, b) => a.km - b.km);
  return found;
}

function collection(rows) {
  return {
    type: 'FeatureCollection',
    features: rows.map((row) => ({
      type: 'Feature',
      geometry: { type: 'Point', coordinates: [row.lon, row.lat] },
      properties: {
        name: row.name, operator: row.operator, tech: row.tech,
        raw_tech: row.raw, status: row.status, mw: row.mw,
        km: Number(row.km.toFixed(3)),
        colour: TECH_COLOUR[row.tech] || TECH_COLOUR.other
      }
    }
  )))
};
}

/* addSource throws if the style is not loaded, and a source that failed to
add reads back as null. The sld-sandbox body learned this the night the
basemap CDN served style.json and then no tiles at all, and its proof now
refuses any unguarded setData call site anywhere in the served
cartridge -- including, as it turns out, one written inside a comment.
This section is drawing, not plumbing: a missing source costs the drawing,
not the session. */
function setSourceData(map, id, data) {
  try {
    const source = map.getSource(id);
    if (!source || typeof source.setData !== 'function') {
      note('source missing, nothing drawn: ' + id);
      return false;
    }
  }
}

```

```

    source.setData(data);
    return true;
  } catch (error) {
    note('source ' + id + ': ' + String(error && error.message || error));
    return false;
  }
}

function ensureLayers(map, control) {
  const sourceId = 'pn-src-' + control.id;
  const ringId = 'l-pn-' + control.id + '-ring';
  const dotId = 'l-pn-' + control.id;
  if (map.getSource(sourceId)) return { sourceId, ringId, dotId };
  try {
    map.addSource(sourceId, { type: 'geojson', data: collection([]) });
  } catch (error) {
    note('addSource ' + sourceId + ': ' + String(error && error.message ||
error));
    return { sourceId, ringId, dotId };
  }
  // A ring in the control's colour, a dot in the technology's own. The ring
  // says which control summoned it; the dot says what it is.
  map.addLayer({
    id: ringId, type: 'circle', source: sourceId,
    paint: {
      'circle-radius': ['interpolate', ['linear'], ['zoom'], 6, 4, 12, 8, 16,
13],
      'circle-color': 'rgba(0,0,0,0)',
      'circle-stroke-color': control.colour,
      'circle-stroke-width': 1.4,
      'circle-stroke-opacity': 0.9
    }
  });
  map.addLayer({
    id: dotId, type: 'circle', source: sourceId,
    paint: {
      'circle-radius': ['interpolate', ['linear'], ['zoom'], 6, 1.8, 12, 3.4,
16, 5.5],
      'circle-color': ['coalesce', ['get', 'colour'], '#888888'],
      'circle-opacity': 0.95
    }
  });
  map.on('click', dotId, (event) => {
    const properties = (event.features && event.features[0] || {}).properties
|| {};
    try {
      new window.maplibregl.Popup({ closeButton: true })
        .setLngLat(event.lngLat)
        .setHTML(
          '<div style="font-family:monospace;background:#000;padding:6px;max-
width:260px">'
          + '<b style="color:#5fbdc2;font-size:12px">' +
escapeHtml(properties.name || 'Project') + '</b><br>'
          + '<span style="color:#888">' + escapeHtml(properties.raw_tech ||
properties.tech || '') + '</span><br>'
          + '<span style="color:#ffae00">' + escapeHtml(String(properties.mw
|| 0)) + ' MW</span> . '
          + '<span style="color:#aaa">' + escapeHtml(properties.status || '')
+ '</span><br>'
          + '<span style="color:#555;font-size:10px">' +
escapeHtml(String(properties.km)) + ' km from the selected project. '
          + 'Proximity only – not a connection, a circuit or a queue position.

```

```

</span></div>')
    .addTo(map);
  } catch (error) {
    note('popup: ' + String(error && error.message || error));
  }
});
map.on('mouseenter', dotId, () => { map.getCanvas().style.cursor =
'pointer'; });
map.on('mouseleave', dotId, () => { map.getCanvas().style.cursor = ''; });
return { sourceId, ringId, dotId };
}

function escapeHtml(value) {
  return String(value == null ? '' : value)
    .replace(/[\&<>"]/g, (character) => ({
      '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&quot;';
    }[character]));
}

function setVisibility(map, control, visible) {
  for (const id of ['l-pn-' + control.id + '-ring', 'l-pn-' + control.id]) {
    if (map.getLayer(id)) map.setLayoutProperty(id, 'visibility', visible ?
'visible' : 'none');
  }
}

function labelFor(control, count, selection) {
  if (!selection) return control.label + ' [SELECT A PROJECT]';
  if (count === null || count === undefined) return control.label + ' [WAIT]';
  return control.label + ' [' + count.toLocaleString('en-GB') + ' within ' +
RADIUS_KM + ' km]';
}

function paintLabels(selection) {
  for (const control of CONTROLS) {
    const text = labelFor(control, state.counts[control.id], selection);
    for (const span of document.querySelectorAll('[data-pn-label="' +
control.id + '"]')) {
      span.textContent = text;
    }
  }
}

async function refresh(map, control) {
  const selection = state.selection;
  if (!selection) return;
  const rows = await loadRegister(map);
  const found = near(rows.filter((row) => control.keep(row, selection)),
selection);
  state.counts[control.id] = found.length;
  const { sourceId } = ensureLayers(map, control);
  setSourceData(map, sourceId, collection(found));
  paintLabels(selection);
}

function buildGroup(container, isFullscreen) {
  if (container.querySelector('[data-pn-group]')) return;
  const group = document.createElement('div');
  group.className = 'key-group';
  group.setAttribute('data-pn-group', '1');
  const title = document.createElement('div');
  title.className = 'key-title';

```

```

title.textContent = GROUP_TITLE;
group.appendChild(title);

for (const control of CONTROLS) {
  const label = document.createElement('label');
  label.className = 'key-item';
  const input = document.createElement('input');
  input.type = 'checkbox';
  // NOT data-layer-id: the engine dispatches on that attribute.
  input.setAttribute('data-pn-layer', control.id);
  input.dataset.pnLayer = control.id;
  const span = document.createElement('span');
  span.setAttribute('data-pn-label', control.id);
  span.style.color = control.colour;
  span.textContent = labelFor(control, state.counts[control.id],
state.selection);
  label.appendChild(input);
  label.appendChild(document.createTextNode(' '));
  label.appendChild(span);
  group.appendChild(label);
}
container.appendChild(group);
state[isFullscreen ? 'installed_fullscreen' : 'installed_main'] = true;
}

function bind(map, container) {
  container.addEventListener('change', (event) => {
    const target = event.target;
    if (!target || target.type !== 'checkbox' || !target.dataset.pnLayer)
return;
    const control = CONTROLS.find((candidate) => candidate.id ===
target.dataset.pnLayer);
    if (!control) return;

    // Keep the two dashboards agreeing, as the engine does for its own.
    for (const twin of document.querySelectorAll(
      'input[data-pn-layer="' + control.id + '"]')) {
      twin.checked = target.checked;
    }

    state.active = CONTROLS
      .filter((candidate) => document.querySelector(
        'input[data-pn-layer="' + candidate.id + '"]:checked'))
      .map((candidate) => candidate.id);

    if (!target.checked) {
      setVisibility(map, control, false);
      return;
    }
    if (!state.selection) {
      // Nothing is selected, so there is no "near" to be near to. Say so on
      // the label rather than switching on an empty layer and looking broken.
      paintLabels(null);
      target.checked = false;
      return;
    }
    ensureLayers(map, control);
    setVisibility(map, control, true);
    refresh(map, control).catch((error) => {
      note('refresh: ' + String(error && error.message || error));
      paintLabels(state.selection);
    });
  });
}

```

```

    });
}

function install() {
    const map = window.__GRIDATLAS_V9_MAP__;
    const container = document.getElementById('scada-ui-container');
    if (!map || typeof map.addSource !== 'function') return false;
    if (!container || !container.querySelector('.key-group')) return false;

    buildGroup(container, false);
    bind(map, container);

    const curtain = document.getElementById('fs-curtain-keys');
    if (curtain) { buildGroup(curtain, true); bind(map, curtain); }

    /* Watch the pin rather than the cartridge. There is no selection event to
       subscribe to, and wrapping the cartridge's selectAt would make this a
       second owner of its behaviour. A one-second poll of a source it already
       maintains is the smaller coupling, and costs nothing measurable. */
    let lastKey = '';
    if (typeof setInterval !== 'function') return true;
    setInterval(() => {
        const selection = readSelection(map);
        const key = selection ? [selection.lon, selection.lat,
selection.tech].join('|') : '';
        if (key === lastKey) return;
        lastKey = key;
        state.selection = selection;
        state.counts = {};
        if (!selection) {
            for (const control of CONTROLS) setVisibility(map, control, false);
            paintLabels(null);
            return;
        }
        paintLabels(selection);
        for (const control of CONTROLS) {
            if (!document.querySelector('input[data-pn-layer="' + control.id +
'" ]:checked')) continue;
            refresh(map, control).catch((error) => note('refresh: '
+ String(error && error.message || error)));
        }
    }, 1000);

    state.installed = true;
    return true;
}

/* The engine builds its dashboard inside map.on('load'), so nothing here can
   assume a panel at module time. Poll until it exists, then stop.

   Guarded on the timer existing at all. The cartridge proof runs this file
   in a bare vm context with no DOM and no timers, to check the served bytes
   without a browser; an unguarded setInterval threw there and took the whole
   proof down. A context with no timers also has no map and no dashboard, so
   there is nothing for this to install and returning is the correct answer
   rather than a concession to the harness. */
if (typeof setInterval === 'function') {
    const started = Date.now();
    const boot = setInterval(() => {
        let done = false;
        try { done = install(); } catch (error) { note('install: ' + String(error
&& error.message || error)); }
    }, 1000);
}

```

```

    if (done || Date.now() - started > 120000) clearInterval(boot);
  }, 400);
}

NS.pipelineNewsLayers = Object.freeze({
  schema: 'gridatlas.module.pipeline-news-layers.v1',
  generation: GENERATION,
  RADIUS_KM,
  CONTROLS: CONTROLS.map((control) => control.id),
  install,
  state
});
})();

/**
 * SLD runtime styles, mechanically lifted from the v9.106 sandbox body.
 *
 * This module owns only the seven template-literal CSS values. The sandbox
 * still owns when each surface is installed and the DOM element receiving it.
 * Keeping those lifecycles in place makes this a byte-for-byte style move,
 * not a UI redesign. The sibling substation cartridge carries this module
 * because it executes before sld-sandbox and has the required headroom.
 */
(() => {
  'use strict';

  const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||
  {}));
  if (NS.sldStyles) throw new Error('sld-styles module registered twice');

  NS.sldStyles = Object.freeze({
    schema: 'gridatlas.module.sld-styles.v1',
    neonBlock(BLOCK_CLASS) {
      return `
.${BLOCK_CLASS}{margin-top:7px;padding-top:6px;border-top:1px solid #123;font-
family:monospace}
.${BLOCK_CLASS} .neon-hd{display:flex;align-items:center;gap:6px;font-
size:10px;letter-spacing:.06em;
color:#5fbdc2;font-weight:bold;text-transform:uppercase}
.${BLOCK_CLASS} .neon-beta{font-size:8px;letter-spacing:.06em;padding:1px
4px;border-radius:2px;
background:#3a2f12;color:#e0b050;border:1px solid #6a5320;text-
transform:uppercase}
.${BLOCK_CLASS} ol{list-style:none;margin:5px 0 0;padding:0}
.${BLOCK_CLASS} li{display:flex;align-items:baseline;gap:6px;padding:2px 0}
.${BLOCK_CLASS} .neon-km{color:#5fbdc2;font-weight:bold;font-variant-
numeric:tabular-nums;
min-width:54px;text-shadow:0 0 6px rgba(95,189,194,.35)}
.${BLOCK_CLASS} .neon-name{color:#9fb3ba;overflow:hidden;text-overflow:ellipsis;
white-space:nowrap;flex:1;max-width:150px}
.${BLOCK_CLASS} .neon-kv{color:#ffae00;font-size:9px;white-
space:nowrap;cursor:help}
.${BLOCK_CLASS} .neon-kvnote{margin-top:6px;font-size:10px;line-height:1.45}
.${BLOCK_CLASS} .neon-kvnote b{color:#ffae00;font-weight:normal}
.${BLOCK_CLASS} .neon-pin{display:block;width:100%;margin-top:7px;padding:5px
6px;
background:#0a1a1d;border:1px solid #2f6f75;border-radius:3px;color:#8b9aa1;
font:inherit;font-size:10px;letter-spacing:.05em;cursor:pointer;text-
transform:uppercase}
.${BLOCK_CLASS} .neon-pin:hover{border-color:#5fbdc2;color:#bfe9ee}
.${BLOCK_CLASS} .neon-pin[aria-pressed="false"]{color:#5f7a80;border-
color:#1d3238}

```



```

.${BLOCK_CLASS} .neon-layout{display:block;width:100%;margin-top:7px;padding:5px
6px;
  background:#0a1ald;border:1px solid #2f6f75;border-radius:3px;color:#5fbdc2;
  font:inherit;font-size:10px;letter-spacing:.05em;cursor:pointer;text-
transform:uppercase}
.${BLOCK_CLASS} .neon-layout:hover{border-
color:#5fbdc2;color:#bfe9ee;background:#0d2429}
/* The measurement and its qualifiers, as one element. A hairline rule and
  six pixels of padding: nothing here colours or grades the number. */
.${BLOCK_CLASS} .neon-answer{margin:0 0 8px;padding:6px 0 8px;
  border-bottom:1px solid #123}
.${BLOCK_CLASS} .neon-answer > .neon-caveat:first-child{margin-top:0}
/* The card sits over the map and used to be immovable, with only MapLibre's
  own hairline close cross. It gets a bar: grab it to move the card out of the
  way, and two controls big enough to hit without aiming. */
/* Measured on the live map: the card was 563px tall inside a 319px map and
  hung 403px below it, so the caveat and the layout button could not be
  reached at all. The content is now bounded to the map and scrolls, and the
  bar stays put at the top of that scroll so the controls never leave. */
.maplibregl-popup-content{max-height:var(--gridatlas-card-max, 60vh) !important;
  overflow-y:auto !important;overflow-x:hidden;overscroll-behavior:contain}
.gridatlas-card-bar{position:sticky;top:-6px;z-index:2;flex:0 0 auto;
  display:flex;align-items:center;gap:6px;margin:-6px -6px 6px;
  padding:5px 6px;background:#0a1ald;border-bottom:1px solid #1d3238;
  border-radius:3px 3px 0 0;cursor:grab;user-select:none;touch-action:none;font-
family:monospace}
.gridatlas-card-bar:active{cursor:grabbing}
.gridatlas-card-bar .grip{color:#3f6f75;letter-spacing:2px;font-size:11px}
.gridatlas-card-bar .label{color:#8b9aa1;font-size:10px;max-
width:190px;overflow:hidden;
  text-overflow:ellipsis;white-space:nowrap}
.maplibregl-popup.gridatlas-min .gridatlas-card-bar .label{color:#5fbdc2;font-
weight:bold;max-width:230px}
.gridatlas-card-bar .spacer{flex:1}
.gridatlas-card-bar button{background:#050a0d;border:1px solid
#2f6f75;color:#5fbdc2;
  font:inherit;font-size:14px;line-height:1;min-width:44px;height:44px;border-
radius:3px;
  cursor:pointer;padding:0 6px}
.gridatlas-card-bar button:hover{color:#bfe9ee;border-
color:#5fbdc2;background:#0d2429}
.gridatlas-card-bar button.close:hover{color:#ff8f8f;border-color:#ff5d5d}
.maplibregl-popup.gridatlas-free{position:fixed !important;transform:none
!important;
  left:var(--gx) !important;top:var(--gy) !important;z-index:12}
.maplibregl-popup.gridatlas-free .maplibregl-popup-tip{display:none !important}
.maplibregl-popup.gridatlas-min .maplibregl-popup-content > *:not(.gridatlas-
card-bar){display:none !important}
.maplibregl-popup.gridatlas-min .maplibregl-popup-content{padding:6px
!important;
  border:1px solid #2f6f75;border-radius:4px;box-shadow:0 0 14px
rgba(95,189,194,.25)}
.maplibregl-popup.gridatlas-min .gridatlas-card-bar{margin:0;border-bottom:0;
  border-radius:3px;background:#08171a}
.maplibregl-popup.gridatlas-min .gridatlas-card-bar button.min{border-
color:#5fbdc2;color:#bfe9ee}
.${BLOCK_CLASS} .neon-caveat{margin-top:6px;color:#68797f;font-size:9px;line-
height:1.5}
.${BLOCK_CLASS} .neon-caveat b{color:#8b9aa1;font-weight:bold}
/* The immutable shell predates phone-landscape use. These are composition
  overrides, not a mutation of the attested shell: the left control stack can
  scroll inside a short map, and search results cannot extend below it. */

```

```

@media (max-height:600px){
  .map-controls{max-height:min(70%,calc(100dvh - 100px));overflow-y:auto;
    overscroll-behavior:contain;scrollbar-width:thin}
  .search-results{max-height:calc(100dvh - 140px) !important}
}
@media (pointer:coarse){
  .map-ctrl-btn,.search-btn{min-height:44px}
  .search-input{min-height:44px;box-sizing:border-box}
  .${BLOCK_CLASS} .neon-pin,.${BLOCK_CLASS} .neon-layout{min-height:44px}
}

/* — THE CARD IS A DOCKED SHEET ON A PHONE —————
Measured on a verified iPhone-class device (393x852 at dpr 3,
pointer:coarse, hover:none, 5 touch points, document.hidden false): the
anchored card opened at y=426, 819px tall, so its bottom edge was 393px
BELOW the screen and the end of it was unreachable at every scroll
position. Its left edge sat at x=-89 at 390px wide. Anchoring is the wrong
idea on a phone - a 340px box hung off a marker in a 393px viewport has
nowhere to go - so on a coarse pointer or a narrow window the card docks
to the bottom edge, full width, and nothing has to be dragged.
No rule below reads technology: three buckets light no layer at all. */
html.gridatlas-sheet-open .maplibregl-popup.gridatlas-sheet{
  position:fixed !important;left:0 !important;right:0 !important;
  top:auto !important;bottom:0 !important;transform:none !important;
  width:100vw !important;max-width:100vw !important;
  margin:0 !important;padding:0 !important;z-index:400 !important}
.maplibregl-popup.gridatlas-sheet .maplibregl-popup-tip{display:none !important}
.maplibregl-popup.gridatlas-sheet .maplibregl-popup-content{
  max-height:var(--gridatlas-sheet-h,56dvh) !important;
  width:100% !important;max-width:100% !important;
  border-radius:12px 12px 0 0;box-sizing:border-box;
  border-top:1px solid #2f6f75;box-shadow:0 -8px 24px rgba(0,0,0,.55);
  padding-bottom:calc(8px + env(safe-area-inset-bottom,0px)) !important}
.maplibregl-popup.gridatlas-sheet.gridatlas-min .maplibregl-popup-content{
  max-height:none !important}
/* MapLibre's own hairline cross is 20x18 and now sits inside a full-width
sheet whose bar already carries a 44px close. Two closes, one of them
unhittable, is worse than one. */
.maplibregl-popup.gridatlas-sheet .maplibregl-popup-close-button{display:none
!important}
/* WHEN A CONTROL AND THE ANSWER WANT THE SAME PIXELS, THE ANSWER WINS.
Four bars printed over the card's text on every load: the tray with the
GB PRICES and VERSIONS bars (all inside .map-controls at y 698-822), HIDE
LAYERS at y 796 on z-index 9999, and the credit strip at y 827. None is
deleted and none goes UNDER the sheet, which would make it unreachable:
they are lifted clear, still on the map and still 44px. The offset reads
the same var the sheet is sized from, so the two cannot disagree. */
html.gridatlas-sheet-open .map-controls{
  bottom:calc(var(--gridatlas-sheet-h,56dvh) + 12px) !important;
  max-height:calc(100dvh - var(--gridatlas-sheet-h,56dvh) - 120px) !important;
  overflow-y:auto;overscroll-behavior:contain}
html.gridatlas-sheet-open #gridatlas-dash-toggle{
  bottom:calc(var(--gridatlas-sheet-h,56dvh) + 12px) !important}
/* Not an attribution: the OSM and CARTO credit is .custom-map-attrib at the
top of the map and does not move. This is a shout-out that was printing
across the card's sentences. Hidden only while a sheet is open. */
html.gridatlas-sheet-open .podcast-shoutout{display:none !important}`;
},
bootStatus(STATUS_ID) {
  return `
#{$STATUS_ID}{position:absolute;left:50%;top:14px;transform:translateX(-50%);
z-index:5;max-width:min(92vw,420px);padding:7px 11px;border-radius:4px;
background:rgba(6,18,21,.93);border:1px solid #21454b;color:#9fb3ba;

```

```

    font:11px/1.5 ui-monospace,SFMono-Regular,Menlo,monospace;letter-
spacing:.04em;
    text-align:center;pointer-events:auto}
#{STATUS_ID}[data-kind="failed"]{border-color:#7a4a4a;color:#d0a9a9}
#{STATUS_ID} button{display:block;margin:7px auto 0;padding:4px 12px;
    background:#0a1a1d;border:1px solid #2f6f75;border-radius:3px;color:#bfe9ee;
    font:inherit;text-transform:uppercase;letter-spacing:.06em;cursor:pointer}
#{STATUS_ID} button:hover{border-color:#5fbdc2}
@media (prefers-reduced-motion:no-preference){
    #{STATUS_ID}[data-kind="waiting"]{animation:ga-status-pulse 2.4s ease-in-out
infinite}
}
@keyframes ga-status-pulse{0%,100%{opacity:.72}50%{opacity:1}}`;
    },
    versionLedger(LEDGER_ID) {
        return `
#{LEDGER_ID}{margin-top:6px;font:11px/1.5 ui-monospace,SFMono-
Regular,Menlo,monospace}
#{LEDGER_ID} > button{display:block;width:100%;padding:6px
8px;background:#0a1a1d;
    border:1px solid #2f6f75;border-radius:3px;color:#8fb3b8;font:inherit;
    letter-spacing:.06em;text-transform:uppercase;cursor:pointer;text-align:left}
#{LEDGER_ID} > button:hover{border-color:#5fbdc2;color:#bfe9ee}
#{LEDGER_ID} .vl-body{display:none;margin-top:5px;padding:8px;border:1px solid
#1d3238;
    border-radius:3px;background:rgba(6,18,21,.94);max-width:min(88vw,300px);
    max-height:min(56vh,380px);overflow:auto;overscroll-behavior:contain}
#{LEDGER_ID}[data-open="1"] .vl-body{display:block}
#{LEDGER_ID} .vl-row{padding:4px 0;border-bottom:1px solid #142226}
#{LEDGER_ID} .vl-head{display:flex;justify-content:space-between;gap:8px}
#{LEDGER_ID} .vl-ver{color:#bfe9ee;font-weight:bold}
#{LEDGER_ID} .vl-status{margin-left:5px;color:#ff9b73;font-size:9px;font-
weight:bold}
#{LEDGER_ID} .vl-when{color:#5f7a80;font-size:10px}
#{LEDGER_ID} .vl-scope{color:#9fb3ba;font-size:10px;line-height:1.4;margin-
top:1px}
#{LEDGER_ID} .vl-reason{color:#df9b83;font-size:9px;line-height:1.35;margin-
top:2px}
#{LEDGER_ID} .vl-note{margin:7px 0 0;color:#6f8288;font-size:10px;line-
height:1.45}`;
    },
    mobileTray(TRAY_ID) {
        return `
#{TRAY_ID}{display:flex;gap:4px;font:11px/1.2 ui-monospace,SFMono-
Regular,Menlo,monospace}
#{TRAY_ID} button{min-height:44px;padding:6px 10px;background:#0a1a1d;
    border:1px solid #2f6f75;border-radius:3px;color:#8fb3b8;font:inherit;
    letter-spacing:.06em;text-transform:uppercase;cursor:pointer}
#{TRAY_ID} button[aria-pressed="true"]{border-color:#5fbdc2;color:#bfe9ee;
    background:rgba(0,255,255,0.08)}
#{TRAY_ID} button[disabled]{opacity:.45;cursor:default}
.map-controls.gm-tools-collapsed > .map-ctrl-btn{display:none}`;
    },
    gbConditions(GB_ID) {
        return `
#{GB_ID}{margin-top:6px;font:11px/1.5 ui-monospace,SFMono-
Regular,Menlo,monospace}
#{GB_ID} > button{display:block;width:100%;padding:6px 8px;background:#0a1a1d;
    border:1px solid #2f6f75;border-radius:3px;color:#8fb3b8;font:inherit;
    letter-spacing:.06em;text-transform:uppercase;cursor:pointer;text-align:left}
#{GB_ID} > button:hover{border-color:#5fbdc2;color:#bfe9ee}
#{GB_ID} .gb-body{display:none;margin-top:5px;padding:8px;border:1px solid

```

```

#1d3238;
  border-radius:3px;background:rgba(6,18,21,.94);max-width:min(88vw,260px);
  max-height:min(52vh,340px);overflow:auto;overscroll-behavior:contain}
#{GB_ID}[data-open="1"] .gb-body{display:block}
#{GB_ID} .gb-row{display:flex;justify-content:space-between;gap:8px;
  padding:2px 0;border-bottom:1px solid #142226}
#{GB_ID} .gb-k{color:#7d8f95}
#{GB_ID} .gb-v{color:#bfe9ee;font-weight:bold}
#{GB_ID} .gb-v em{color:#5f7a80;font-style:normal;font-weight:normal;font-
size:10px}
#{GB_ID} .gb-note{margin:7px 0 0;color:#6f8288;font-size:10px;line-height:1.45}
#{GB_ID} .gb-note.gb-point{color:#9fb3ba;border-top:1px solid #142226;padding-
top:6px}
#{GB_ID} .gb-note.gb-point b{color:#d8a76a}
#{GB_ID} .gb-more{display:block;margin-top:7px;color:#5fbdc2;font-size:10px;
  text-decoration:none;letter-spacing:.04em}
#{GB_ID} .gb-more:hover{text-decoration:underline}`;
},
sldPanel(PANEL_ID) {
  return `
/* Top RIGHT, below the search box. The Atlas keeps its own tool buttons down
the left edge -- EXPORT CSV, RADIUS SEARCH, ZONE DRAW, MEASURE -- and a
panel on that side covers them, and the search bar occupies 72-96px inside
the map container on the right, so the panel clears it at 112px. Both offsets
were measured on the live map: no headless test catches a collision with a
component the panel knows nothing about. */
#{PANEL_ID}{position:absolute;right:14px;top:112px;bottom:14px;z-
index:11;width:310px;
  max-width:calc(100% - 28px);box-sizing:border-box;overflow:auto;font:11px/1.5
'Courier New',monospace;
  color:#cfe9ee;background:rgba(2,8,11,.93);border:1px solid #0b5f63;border-
radius:5px;
  padding:11px 12px;box-shadow:0 0 22px rgba(0,255,255,.14);backdrop-
filter:blur(3px);display:none}
#{PANEL_ID}[data-open="true"]{display:block}
#{PANEL_ID} h4{margin:0 0 2px;font-size:10px;letter-
spacing:.09em;color:#5fbdc2;text-transform:uppercase;
  display:flex;align-items:center;gap:7px}
#{PANEL_ID} .sld-beta{font-size:8px;padding:1px 4px;border-
radius:2px;background:#3a2f12;
  color:#e0b050;border:1px solid #6a5320}
#{PANEL_ID} .sld-site{color:#fff;font-size:12px;font-weight:bold;margin:2px 0
8px;
  overflow:hidden;text-overflow:ellipsis;white-space:nowrap}
#{PANEL_ID} h4.sld-drag{cursor:grab;user-select:none;touch-action:none}
#{PANEL_ID} h4.sld-drag:active{cursor:grabbing}
#{PANEL_ID} .sld-min{margin-left:auto}
#{PANEL_ID} .sld-min,#{PANEL_ID} .sld-close{cursor:pointer;background:#050a0d;
  border:1px solid #2f6f75;color:#5fbdc2;font:inherit;font-size:12px;line-
height:1;
  min-width:44px;height:44px;border-radius:3px;padding:0 5px}
#{PANEL_ID} .sld-min:hover{color:#bfe9ee;border-color:#5fbdc2}
#{PANEL_ID} .sld-close:hover{color:#ff8f8f;border-color:#ff5d5d}
#{PANEL_ID}[data-min="true"] > *:not(h4){display:none}
#{PANEL_ID}[data-min="true"]{width:auto;padding:7px 9px;
  box-shadow:0 0 14px rgba(95,189,194,.25)}
#{PANEL_ID}[data-min="true"] h4{margin:0}
#{PANEL_ID}[data-min="true"] .sld-min{border-color:#5fbdc2;color:#bfe9ee}
#{PANEL_ID} .sld-to{color:#8b9aa1;font-size:9.5px;margin:-6px 0 8px}
#{PANEL_ID} .sld-target{margin:0 0 9px;padding:7px 8px;border:1px solid
#1d3238;
  border-radius:3px;background:#050a0d}

```

```

#{$PANEL_ID} .sld-target-row{display:flex;justify-content:space-between;align-
items:baseline}
#{$PANEL_ID} .sld-target-row b{color:#e0b050;font-variant-numeric:tabular-nums}
#{$PANEL_ID} .sld-basis{display:flex;align-items:center;gap:6px;margin-top:5px}
#{$PANEL_ID} .sld-basis span{color:#8b9aa1;font-size:10px;white-space:nowrap}
#{$PANEL_ID} .sld-basis select{flex:1}
#{$PANEL_ID} .sld-danger{margin-top:6px;color:#ff5d5d;font-size:9px;line-
height:1.5;
border-left:2px solid #ff5d5d;padding-left:6px}
#{$PANEL_ID} .sld-fitted{margin-top:6px;color:#8b9aa1;font-size:9px;line-
height:1.5}
#{$PANEL_ID} .sld-fitted b{color:#6fb582}
#{$PANEL_ID} .sld-fitted b.sld-off{color:#ff5d5d}
#{$PANEL_ID} .sld-ratio-note{margin-top:6px;color:#d9b45f;font-size:9px;line-
height:1.5;
border-left:2px solid #8b6c28;padding-left:6px}
#{$PANEL_ID} .sld-tabs{display:flex;gap:5px;margin-bottom:8px}
#{$PANEL_ID} .sld-tabs button{flex:1;background:#050a0d;border:1px solid
#1d3238;color:#7f939a;
font:inherit;font-size:9px;padding:4px;cursor:pointer;border-radius:3px;text-
transform:uppercase}
#{$PANEL_ID} .sld-tabs button[data-on="true"]{color:#5fbdc2;border-
color:#5fbdc2}
#{$PANEL_ID} .sld-grid{display:grid;grid-template-columns:1fr 62px;gap:3px
7px;align-items:center}
#{$PANEL_ID} label{color:#8b9aa1;font-size:10px}
#{$PANEL_ID} input,#{$PANEL_ID} select{width:100%;background:#050a0d;border:1px
solid #1d3238;
color:#d8dee6;font:inherit;font-size:10px;padding:2px 4px;border-radius:2px}
#{$PANEL_ID} input:focus,#{$PANEL_ID} select:focus{outline:1px solid #5fbdc2}
#{$PANEL_ID} .sld-out{margin-top:9px;padding-top:8px;border-top:1px solid
#10262b;
display:grid;grid-template-columns:1fr auto;gap:2px 8px}
#{$PANEL_ID} .sld-out b{color:#e0b050;font-variant-numeric:tabular-nums}
#{$PANEL_ID} .sld-out .lit{color:#5fbdc2}
#{$PANEL_ID} .sld-warn{margin-top:7px;color:#d9963c;font-size:9px;line-
height:1.45}
#{$PANEL_ID} .sld-caveat{margin-top:7px;padding-top:7px;border-top:1px solid
#10262b;
color:#68797f;font-size:9px;line-height:1.5}
#{$PANEL_ID} .sld-caveat b{color:#8b9aa1}
#{$PANEL_ID} .sld-hint{margin-top:6px;color:#5f7a80;font-size:9px;line-
height:1.45}
#{$PANEL_ID} .sld-finance{margin-top:9px;border-top:1px solid #214047;padding-
top:7px}
#{$PANEL_ID} .sld-finance summary{min-height:32px;display:flex;align-
items:center;cursor:pointer;
color:#d9b45f;font-weight:bold;letter-spacing:.05em;user-select:none}
#{$PANEL_ID} .sld-fin-grid{display:grid;grid-template-columns:1fr 76px;gap:3px
7px;align-items:center}
#{$PANEL_ID} .sld-fin-section{grid-column:1/-1;margin-top:7px;padding-top:5px;
border-top:1px solid #10262b;color:#5fbdc2;font-size:9px;text-
transform:uppercase}
#{$PANEL_ID} .sld-fin-grid input[type="checkbox"]{width:24px;justify-self:end}
#{$PANEL_ID} .sld-fin-out{margin:8px 0;padding:7px;background:#050a0d;border:1px
solid #1d3238;
display:grid;grid-template-columns:1fr auto;gap:2px 8px}
#{$PANEL_ID} .sld-fin-out b{color:#d9b45f;font-variant-numeric:tabular-
nums;text-align:right}
#{$PANEL_ID} .sld-fin-note{margin:6px 0;color:#8b9aa1;font-size:9px;line-
height:1.5}
@media (max-width:700px){#{$PANEL_ID}

```

```

{width:auto;left:8px;right:8px;top:96px;bottom:8px}}
@media (pointer:coarse){
  #${PANEL_ID} .sld-tabs button,#${PANEL_ID} input,#${PANEL_ID} select,
  #${PANEL_ID} .sld-finance summary{min-height:44px}
}`;
},
fullscreenLayers() {
  return `.gridatlas-fs-layers{position:absolute
!important;left:0;right:0;bottom:0;
  max-height:42vh;overflow:auto;z-index:9;background:rgba(2,8,11,.94);
  border-top:1px solid #0b5f63;backdrop-filter:blur(3px)}`;
}
});
})();
})();

/**
 * TELEPRINT CONTROLS -- GENERATED. DO NOT EDIT THIS FILE.
 *
 * Built by teleprinter/drivers/gridatlas/build-part.mjs from the drivers in
 * that directory, which are where any change belongs. Editing this copy makes
 * the served bytes disagree with the source they claim to come from, and
 * --verify will say so.
 *
 * Two SEPARATE functions, on the architect's instruction:
 *   Print PDF          a picture of the reader's screen, for a person
 *   Print source code  the entire source and its dependencies, for a machine
 *
 * Sources:
 *   screen-frame.js  sha256
e48826f121cc9d2f011351094dc40f14e68b7c4f96d2f95dcc9acabbe8a6ae96
 *   print-pdf.js    sha256
8eac02d8ea3bad87ae8f9609444e75a828617d1e26a36dc42delf58bcd40ee2e
 *   print-source-code.js  sha256
edc6029f40fa7722c3cb523c418b0ca32758a9belf13ac2a5837c198b27893a4
 *   gridatlas-wiring.js  sha256
7fb9d074cb85bee9df57928835b5aeda8dc0d514df676130f58522a9822ac5e8
 */
(function () {
  'use strict';

  /* ---- screen-frame.js (sha256 e48826f121cc9d2f) ---- */
  /**
   * GET THE FRAME THE READER IS LOOKING AT.
   * -----
   * One job, and one refusal.
   *
   * The job: return the pixels currently on the reader's screen, with their
real
   * dimensions and a note saying HOW they were obtained.
   *
   * The refusal: this never reconstructs the page. There is a tempting shortcut
   * -- serialise the DOM into an <svg><foreignObject>, draw that to a canvas,
and
   * call the result a screenshot. It is not one. It drops cross-origin images,
   * loses most pseudo-elements, cannot see into shadow roots, and silently
   * substitutes fallback fonts. A reader handed that would be looking at a
   * drawing of their screen while being told it was a record of it, which is
the
   * one thing a teleprinter must never do. If no honest frame can be had, this
   * throws and the caller says so.
   *
   * TWO SOURCES, IN ORDER, AND THE RECORD ALWAYS NAMES THE ONE USED

```

```

*
*   'host'      a capture function supplied by the embedder. This exists so a
*               test harness can hand in the browser's own screenshot and
*               exercise every downstream byte without a permission chooser --
*               and so a future native shell can supply a real device grab.
*
*   'display'
navigator.mediaDevices.getDisplayMedia({preferCurrentTab:true}).
*       The compositor's own output: every layer, WebGL and DOM
*       together, at the resolution the reader is actually looking at.
*       Costs one permission click. That click is the price of an
*       honest record and it is worth paying.
*
* WHY NOT canvas.toDataURL ALONE. GridAtlas draws its map into a WebGL canvas
* created without preserveDrawingBuffer, so the drawing buffer is gone by the
* time anything outside the frame that drew it goes looking. Reading it gives
* a fully transparent image that encodes to a perfectly valid, perfectly
blank
* PNG. That is the failure that produced white sheets, and it is why a canvas
* read is a MAP source, never a SCREEN source.
*/

/** Milliseconds to wait for the capture track to actually produce a frame. */
const FRAME_TIMEOUT_MS = 10000;

/* A display track that never paints must not hang the reader's browser
   forever. Chrome will happily keep an unstarted <video> pending, so the wait
   is bounded and the tracks are stopped whatever happens -- an abandoned
   capture leaves the tab's sharing indicator lit, which reads to the reader
as
   "this page is still watching my screen". */
function stopTracks(stream) {
  if (!stream || typeof stream.getTracks !== 'function') return;
  for (const track of stream.getTracks()) {
    try { track.stop(); } catch (_) { /* already stopped */ }
  }
}

function pixelsFromSource(source, width, height) {
  if (!width || !height) throw new Error('The captured frame has no size.');
```

/* 40 megapixels is about a 8K screen at dpr 2. Past that the RGBA buffer
alone is 160 MB and the tab is more likely to die than to print. */

```

  if (width * height > 40000000) {
    throw new Error('The screen is too large to print in one page.');
```

}

```

  const canvas = document.createElement('canvas');
  canvas.width = width;
  canvas.height = height;
  const context = canvas.getContext('2d', { willReadFrequently: true });
  if (!context) throw new Error('This browser refused a 2D context.');
```

context.drawImage(source, 0, 0, width, height);

```

  return { width, height, rgba: context.getImageData(0, 0, width, height).data
};
}

async function fromDisplay() {
  const media = navigator.mediaDevices;
  if (!media || typeof media.getDisplayMedia !== 'function') {
    throw new Error('This browser cannot capture the screen. '
      + 'On an iPhone, take a screenshot and use "Print a screenshot"
instead.');
```

}

```

/* preferCurrentTab is Chromium-only and is a HINT: it puts this tab at the
   top of the chooser. Elsewhere the reader picks, which is correct -- their
   screen, their choice. */
const stream = await media.getDisplayMedia({
  video: { frameRate: 1 },
  audio: false,
  preferCurrentTab: true,
  selfBrowserSurface: 'include'
});
try {
  const video = document.createElement('video');
  video.muted = true;
  video.playsInline = true;
  video.srcObject = stream;
  await video.play();
  await new Promise((resolve, reject) => {
    const timer = setTimeout(
      () => reject(new Error('No screen frame arrived within 10 seconds.')),
      FRAME_TIMEOUT_MS
    );
    /* requestVideoFrameCallback fires on a frame that has actually been
       composited. readyState alone can be satisfied by a track that has
       produced metadata and no picture. */
    if (typeof video.requestVideoFrameCallback === 'function') {
      video.requestVideoFrameCallback(() => { clearTimeout(timer);
resolve(); });
    } else {
      const poll = setInterval(() => {
        if (video.videoWidth > 0 && video.readyState >= 2) {
          clearInterval(poll);
          clearTimeout(timer);
          resolve();
        }
      }, 60);
    }
  });
  const frame = pixelsFromSource(video, video.videoWidth,
video.videoHeight);
  video.pause();
  video.srcObject = null;
  return { ...frame, method: 'display' };
} finally {
  stopTracks(stream);
}
}

/**
 * @param {object} [options]
 * @param {function} [options.capture] host-supplied capture returning a Blob,
 *   an ImageBitmap, or {width,height,rgba}. Used by the proof harness.
 * @returns
 {Promise<{width:number,height:number,rgba:Uint8ClampedArray,method:string}>}
 */
async function screenFrame({ capture } = {}) {
  if (typeof capture === 'function') {
    const supplied = await capture();
    if (supplied && supplied.rgba && supplied.width && supplied.height) {
      return { ...supplied, method: supplied.method || 'host' };
    }
  }
  if (supplied && typeof createImageBitmap === 'function') {
    /* colorSpaceConversion:'none' matters: the browser's image decoder will
       otherwise shift sample values by a level or two on a tagged image,

```


and

```
    a print engine that claims to preserve pixels must not quietly alter
    them. Codex measured exactly this in WebKit on 2026-09-05. */
    const bitmap = await createImageBitmap(supplied, {
      colorSpaceConversion: 'none',
      premultiplyAlpha: 'none'
    });
    try {
      return { ...pixelsFromSource(bitmap, bitmap.width, bitmap.height),
method: 'host' };
    } finally {
      bitmap.close();
    }
  }
  throw new Error('The supplied capture produced nothing usable.');
```

it

```
  }
  return fromDisplay();
}
```

```
/* ---- print-pdf.js (sha256 8eac02d8ea3bad87) ---- */
/**
 * PRINT PDF -- one PDF unit per captured pixel, and a strip that says where
 * came from.
 * -----
 * This is the FIRST of the two Teleprinter functions and it is entirely
 * separate from the second. It answers "give me what is on my screen as a
 * file". It knows nothing about source code.
 *
 * NO PAPER. The page box is exactly the captured raster's width, and its
 * height is the raster plus one provenance strip. Not A4, not A3, not "the
 * long edge scaled to 1190pt" -- that last one was real, it shipped, and on a
 * 1390x518 desktop capture it emitted a 1190x443 page: a 14% REDUCTION of the
 * record presented as the record. "THE PRINT MUST BE HIGH RES OF WHAT THE
USER
 * SEES NOT A REDUCED CRAP VERSION". "WE ARE NOT USING PAPER".
 *
 * WHY FLATE AND RAW RGB RATHER THAN JPEG. JPEG is lossy, and a lossy record
is
 * an edited one: text in the layer panel and the thin 400 kV lines are
exactly
 * the content its ringing artefacts damage most. /FlateDecode over raw RGB
 * samples is bit-exact, so the bytes in the file ARE the bytes that were on
 * the screen, and a reviewer can say so rather than hope so.
 *
 * WHY THE STRIP IS OUTSIDE THE IMAGE. The furniture used to be painted over
 * the map. It covered the menu bar at the top and the legend at the bottom,
 * and on a real sheet it truncated the generation stamp to "generation
 * 202609051211 - 2026-09-". The record must not be written on. So the page is
 * made taller than the capture and the strip lives in the space that adds:
is
 * every pixel of the reader's screen survives untouched, and the provenance
 * still on the sheet. "I like the headers and footers thats nice ... KEEP
 * THAT".
 */

const encoder = new TextEncoder();
const bytes = (text) => encoder.encode(text);

function join(parts) {
  let length = 0;
```

```

    for (const part of parts) length += part.length;
    const out = new Uint8Array(length);
    let at = 0;
    for (const part of parts) { out.set(part, at); at += part.length; }
    return out;
}

/* PDF strings are parenthesised, so a literal parenthesis or backslash in a
   project name would end the string early and corrupt every object offset
   after it. Non-ASCII is dropped rather than guessed at: WinAnsi is not UTF-8
   and a mojibake stamp is worse than a plain one. */
function pdfString(text) {
    return String(text == null ? '' : text)
        .replace(/[\(\)]/g, '\\$&')
        .replace(/[\x20-\x7e]/g, '');
}

async function deflate(data) {
    if (typeof CompressionStream !== 'function') {
        throw new Error('This browser cannot compress the page.');
```

}

const stream = new Blob([data]).stream().pipeThrough(new
CompressionStream('deflate'));

return new Uint8Array(await new Response(stream).arrayBuffer());

}

/**

* Build a one-page PDF whose page is the capture plus a provenance strip.

*

* @param {{width:number,height:number,rgba:Uint8ClampedArray}} frame

* @param

{{brand?:string,title?:string,url?:string,stamp?:string,credit?:string,method?:s
tring}} [note]

*/

async function screenPdf(frame, note = {}) {

const { width, height, rgba } = frame || {};

if (!Number.isSafeInteger(width) || !Number.isSafeInteger(height)
 || width < 1 || height < 1) {

throw new Error('The captured frame has invalid dimensions.');

}

if (!rgba || rgba.length !== width * height * 4) {

throw new Error('The captured frame has the wrong number of samples.');

}

/* The strip scales with the record, not with an assumed sheet: a 2514px-
tall

phone capture and a 518px-tall desktop one both need a legible credit,
and

a fixed 9pt is illegible on one and enormous on the other. */

const unit = Math.max(1, Math.min(width, height) / 900);

const headSize = Math.max(9, Math.round(13 * unit));

const footSize = Math.max(7, Math.round(9 * unit));

const pad = Math.max(10, Math.round(14 * unit));

const strip = headSize + footSize + pad * 2 + Math.round(pad * 0.6);

const pageH = height + strip;

const rgb = new Uint8Array(width * height * 3);

for (let i = 0, o = 0; i < rgba.length; i += 4, o += 3) {

rgb[o] = rgba[i]; rgb[o + 1] = rgba[i + 1]; rgb[o + 2] = rgba[i + 2];

}

const image = await deflate(rgb);

```

    const heading = pdfString(note.title || 'GlobalGrid2050 · Grid Atlas');
    const brand = pdfString(note.brand || 'VENTUS · GLOBALGRID2050 · GRID
ATLAS');
    const left = pdfString(note.credit || '');
    const right = pdfString([note.stamp, note.method && ('capture: ' +
note.method)]
    .filter(Boolean).join(' · '));
    const link = pdfString(note.url || '');

    /* The image is placed at y=strip so the strip occupies the BOTTOM of the
page, below the record, in PDF's origin-at-bottom-left space. */
    const content = [
        'q', `${width} 0 0 ${height} 0 ${strip} cm`, '/Screen Do', 'Q',
        'q', '0.016 0.039 0.047 rg', `0 0 ${width} ${strip} re f`, 'Q',
        `BT /F1 ${headSize} Tf 1 1 1 rg ${pad} ${strip - pad - headSize +
Math.round(headSize * 0.25)} Td (${brand}) Tj ET`,
        `BT /F1 ${footSize} Tf 0.86 0.93 0.94 rg ${pad} ${Math.round(pad * 0.9) +
footSize} Td (${heading}) Tj ET`,
        `BT /F1 ${footSize} Tf 0.66 0.78 0.80 rg ${pad} ${Math.round(pad * 0.55)}
Td (${link}) Tj ET`,
        `BT /F1 ${footSize} Tf 0.66 0.78 0.80 rg ${pad} ${Math.round(pad * 0.55) +
footSize + 2} Td (${left}) Tj ET`,
        `BT /F1 ${footSize} Tf 0.86 0.93 0.94 rg ${Math.max(pad, width - pad -
right.length * footSize * 0.52)} ${Math.round(pad * 0.9) + footSize} Td
(${right}) Tj ET`
    ].join('\n');
    const contentBytes = await deflate(bytes(content));

    const objects = [
        bytes('<< /Type /Catalog /Pages 2 0 R >>'),
        bytes('<< /Type /Pages /Kids [3 0 R] /Count 1 >>'),
        bytes('<< /Type /Page /Parent 2 0 R /MediaBox [0 0 ${width} ${pageH}] `
+ /Resources << /XObject << /Screen 5 0 R >> /Font << /F1 6 0 R >> >> `
+ /Contents 4 0 R >>'),
        join([bytes('<< /Length ${contentBytes.length} /Filter /FlateDecode
>>\nstream\n`'),
            contentBytes, bytes('\nendstream')]),
        join([bytes('<< /Type /XObject /Subtype /Image `
+ /Width ${width} /Height ${height} /ColorSpace /DeviceRGB `
+ /BitsPerComponent 8 /Interpolate false /Filter /FlateDecode `
+ /Length ${image.length} >>\nstream\n`'), image,
            bytes('\nendstream')]),
        bytes('<< /Type /Font /Subtype /Type1 /BaseFont /Helvetica /Encoding
/WinAnsiEncoding >>')
    ];

    const chunks = [bytes('%PDF-1.4\n')];
    const offsets = [];
    let at = chunks[0].length;
    objects.forEach((body, index) => {
        offsets.push(at);
        const piece = join([bytes(`${index + 1} 0 obj\n`), body,
            bytes('\nendobj\n')]);
        chunks.push(piece);
        at += piece.length;
    });
    let xref = `xref\n0 ${objects.length + 1}\n0000000000 65535 f \n`;
    for (const offset of offsets) xref += String(offset).padStart(10, '0') + '
00000 n \n';
    chunks.push(bytes(xref + `trailer\n<< /Size ${objects.length + 1} /Root 1 0
R >>\n`
+ `startxref\n${at}\n%%EOF\n`));

```

```

    return { bytes: join(chunks), pageWidth: width, pageHeight: pageH, strip };
}

/**
 * Capture the screen and put a PDF of it on the reader's disk.
 *
 * Separate from printSourceCode() on purpose: one produces a picture of the
 * screen, the other produces text for a machine to read, and conflating them
 * gives a reader a file that is bad at both.
 */
async function printPdf({ capture, note = {}, filename } = {}) {
    const frame = await screenFrame({ capture });
    const built = await screenPdf(frame, { ...note, method: frame.method });
    const blob = new Blob([built.bytes], { type: 'application/pdf' });
    const name = filename
        || `gridatlas-screen-${new Date().toISOString().replace(/[:.]/g, '-')}
    ).pdf`;
    const url = URL.createObjectURL(blob);
    const link = document.createElement('a');
    link.href = url;
    link.download = name;
    link.rel = 'noopener';
    document.body.appendChild(link);
    link.click();
    /* Revoking immediately races the browser's own fetch of the blob on some
    builds; 30 s is long enough for any of them and short enough that a
reader
    printing repeatedly does not accumulate them. */
    setTimeout(() => {
        URL.revokeObjectURL(url);
        if (link.parentNode) link.parentNode.removeChild(link);
    }, 30000);
    return {
        method: frame.method,
        width: frame.width,
        height: frame.height,
        pageWidth: built.pageWidth,
        pageHeight: built.pageHeight,
        bytes: built.bytes.length,
        filename: name
    };
}

/* ---- print-source-code.js (sha256 edc6029f40fa7722) ---- */
/**
 * PRINT SOURCE CODE -- the entire source and its dependencies, as the browser
 * actually has them, in one plain text file an AI can read.
 * -----
 * This is the SECOND Teleprinter function and it shares nothing with the
 * first. Print PDF makes a picture of the screen for a person. This makes
text
 * for a machine: "the ENTIRE SOURCE CODE AND DEPENDENCIES IN THE BROWSER FOR
 * AI REVIEW", so a reader on a phone can attach one file in ChatGPT and get a
 * real answer about the thing they are looking at.
 *
 * The reader is not assumed to know what GitHub is, what a commit is, or
where
 * any of these files live. They press one button and get one .txt.
 *
 * WHY .txt AND NOT .zip OR .pdf. A .zip does not open on an iPhone and cannot

```

```

* be attached to a chat as readable text. A PDF of code is worse to read than
* code -- it reflows, it loses indentation, and line numbers stop meaning
* anything. Plain UTF-8 text attaches and pastes everywhere.
*
* HOW DEPENDENCIES ARE FOUND, AND WHY THIS WAY. The spine is
* performance.getEntriesByType('resource') -- what the browser ACTUALLY
* fetched to build this page -- rather than a scan of the source for things
* that look like imports. The two disagree in both directions and the
* disagreement matters:
*
*   - a literal scan finds files that were never loaded (a dead import behind
*     a feature flag, a string that happens to end in .js), and printing them
*     tells a reviewer the page runs code that it does not;
*   - the browser's list finds files a scan cannot see: anything assembled at
*     runtime, a worker started from a blob, a cartridge whose URL is built
by concatenation -- which is exactly how GridAtlas loads its cartridges.
*
* The Codex driver in ../codex takes the literal-scan approach deliberately.
* Where the two disagree, that disagreement is the finding, not a bug.
*
* WHAT IT CANNOT DO, STATED IN THE FILE ITSELF. A cross-origin response
* without CORS is opaque: the browser has the bytes and will not let the page
* read them. Basemap tiles, some CDN fonts and any third-party script are in
* that category. Those are LISTED WITH THEIR URL AND THE REASON, never
* silently dropped and never replaced with a guess -- a reviewer who cannot
* see a gap will reason as though it is not there.
*/

const MAX_RESOURCES = 400;
const MAX_TOTAL_BYTES = 48 * 1024 * 1024;
const FETCH_TIMEOUT_MS = 20000;

/* CODE IN FULL, DATA IN SUMMARY -- and the difference stated in the file.
-----
The first working version of this printed everything it could read and
produced a 13,237,685-byte file: 33 resources, of which one was the 10 MB
decoded REPD dataset. That file is useless for the job it exists to do. The
architect's purpose is "so we can debug in chatgpt chat on mobile", and no
phone is attaching 13 MB of mostly numbers to a chat.

The distinction that fixes it is not "big vs small", it is CODE vs DATA. An
AI reviewing why a screen is wrong needs every line of the code that drew
it. It does not need all 7,680 REPD rows to answer that; it needs to know
the dataset is there, how large it is, and what its first rows look like.

So code is never truncated -- truncating code is how a reviewer is led to a
wrong conclusion about a branch they cannot see -- and data is headed,
measured and marked TRUNCATED in the file itself. */
const CODE = /\.(m?js|cjs|css|html?|svg)(\?|#|$)/i;
const DATA_HEAD_CHARS = 4000;

const TEXTUAL = /\.(m?js|cjs|css|json|html?|txt|svg|map|geojson|csv)(\?
|#|$)/i;

function textual(url, type) {
  if (TEXTUAL.test(url)) return true;
  return type === 'script' || type === 'link' || type === 'css' || type ===
'fetch'
  || type === 'xmlhttprequest';
}

```

```

async function readText(url) {
  const controller = new AbortController();
  const timer = setTimeout(() => controller.abort(), FETCH_TIMEOUT_MS);
  try {
    /* force-cache so printing does not re-download a 10 MB dataset the page
       already has, and so the text printed is the text the page is RUNNING
       rather than whatever the server would serve now. */
    const response = await fetch(url, { cache: 'force-cache', signal:
controller.signal });
    if (!response.ok) {
      return { ok: false, reason: `HTTP ${response.status}` };
    }
    if (response.type === 'opaque') {
      return { ok: false, reason: 'opaque cross-origin response; the browser
will not let this page read it' };
    }
    return { ok: true, text: await response.text() };
  } catch (error) {
    return { ok: false, reason: String((error && error.message) || error) };
  } finally {
    clearTimeout(timer);
  }
}

/* What is on the screen right now, in words a reviewer can act on. Without
   this the source is a pile of files with no indication of which branch the
   reader was standing in when it went wrong. */
function screenState() {
  const layers = Array.from(document.querySelectorAll('input[type=checkbox]'))
    .filter(node => {
      const rect = node.getBoundingClientRect();
      return rect.width > 0 && rect.height > 0;
    })
    .map(node => {
      const label = node.closest('label') || node.parentElement;
      const text = ((label && label.textContent) || node.name || node.id ||
'').trim();
      return { control: text.replace(/\s+/g, ' ').slice(0, 80), checked:
!!node.checked };
    });
  const selected = document.querySelector('.project-popup .name, .gm-panel
.project-name');
  return {
    url: location.href,
    title: document.title,
    generation: (document.documentElement.dataset || {}).gridatlasGeneration
|| null,
    viewport: {
      width: window.innerWidth,
      height: window.innerHeight,
      devicePixelRatio: window.devicePixelRatio,
      orientation: window.innerWidth >= window.innerHeight ? 'landscape' :
'portrait'
    },
    userAgent: navigator.userAgent,
    capturedAt: new Date().toISOString(),
    selectedProject: selected ? selected.textContent.trim().slice(0, 160) :
null,
    layersOn: layers.filter(l => l.checked).map(l => l.control),
    layersOff: layers.filter(l => !l.checked).map(l => l.control),
    attribution: Array.from(document.querySelectorAll('.maplibregl-ctrl-
attrib-inner'))

```

```

        .map(n => n.textContent.trim()).filter(Boolean).join(' | ')
    };
}

function discover() {
    const seen = new Map();
    const add = (url, how) => {
        if (!url) return;
        let absolute;
        try { absolute = new URL(url, location.href).href; } catch (_) { return; }
        if (absolute.startsWith('blob:') || absolute.startsWith('data:')) return;
        if (seen.has(absolute)) { seen.get(absolute).how.add(how); return; }
        seen.set(absolute, { url: absolute, how: new Set([how]) });
    };

    add(location.href, 'the page itself');
    for (const node of document.querySelectorAll('script[src]')) add(node.src,
'<script src>');
    for (const node of document.querySelectorAll('link[rel~="stylesheet"
[href]')) {
        add(node.href, '<link stylesheet>');
    }
    if (typeof performance !== 'undefined' && performance.getEntriesByType) {
        for (const entry of performance.getEntriesByType('resource')) {
            if (textual(entry.name, entry.initiatorType)) {
                add(entry.name, `loaded by the browser (${entry.initiatorType})`);
            }
        }
    }
    return Array.from(seen.values()).slice(0, MAX_RESOURCES);
}

/**
 * Collect everything and render one text file.
 * @returns
 {Promise<{text:string,filename:string,included:number,missing:Array}>}
 */
async function collectSourceCode({ appName = 'GridAtlas', inlineDom = true } =
{})) {
    const state = screenState();
    const targets = discover();
    const included = [];
    const missing = [];
    let total = 0;

    /* Four at a time. Serial is needlessly slow on a page with fifty
    dependencies; unbounded parallelism on a phone on mobile data drops
    requests and produces gaps that look like defects in the app. */
    for (let i = 0; i < targets.length; i += 4) {
        const batch = targets.slice(i, i + 4);
        const results = await Promise.all(batch.map(async target => {
            const read = await readText(target.url);
            return { target, read };
        }));
        for (const { target, read } of results) {
            if (!read.ok) {
                missing.push({ url: target.url, reason: read.reason });
                continue;
            }
            if (total > MAX_TOTAL_BYTES) {
                missing.push({ url: target.url, reason: 'size budget reached before
this file' });
            }
        }
    }
}

```

```

        continue;
    }
    const isCode = CODE.test(target.url) || target.url === location.href;
    const full = read.text;
    const truncated = !isCode && full.length > DATA_HEAD_CHARS;
    const body = truncated ? full.slice(0, DATA_HEAD_CHARS) : full;
    total += body.length;
    included.push({
        url: target.url,
        how: Array.from(target.how).join(', '),
        chars: full.length,
        kind: isCode ? 'code' : 'data',
        truncated,
        text: body
    });
}
}

const rule = '='.repeat(78);
const lines = [];
lines.push(rule);
lines.push(`TELEPRINT OF THE SOURCE CODE -- ${appName}`);
lines.push(rule);
lines.push('');
lines.push('WHAT THIS FILE IS');
lines.push(' Everything the browser loaded to build the screen this was
printed');
lines.push(' from, in full, plus a description of what was on that screen.
It is');
lines.push(' meant to be attached to an AI chat and asked about
directly. ');
lines.push(' ');
lines.push('HOW TO USE IT');
lines.push(' Attach this file in ChatGPT, Claude or Gemini and describe
what you');
lines.push(' saw. You do not need to know how to code, and you do not need
GitHub. ');
lines.push(' ');
lines.push('WHAT IS NOT HERE');
lines.push(' Map tiles and any other cross-origin response the browser will
not let');
lines.push(' this page read. Every one of those is listed by URL under NOT
READ, ');
lines.push(' with the reason. Nothing has been guessed at or
substituted. ');
lines.push(' ');
lines.push(' CODE is here in full and is never shortened. DATA files
(datasets, ');
lines.push(' GeoJSON, CSV, JSON) are shown as their first ' +
DATA_HEAD_CHARS + ' characters and');
lines.push(' marked TRUNCATED, with their true size given, so this file
stays small');
lines.push(' enough to attach to a chat on a phone. ');
lines.push(' ');
lines.push(rule);
lines.push('THE SCREEN THIS CAME FROM');
lines.push(rule);
lines.push(JSON.stringify(state, null, 2));
lines.push(' ');
lines.push(rule);
lines.push(`CONTENTS -- ${included.length} file(s), ${total} characters`);
lines.push(rule);

```



```

    included.forEach((item, index) => {
      lines.push(`${String(index + 1).padStart(3, ' ')} ${item.url}`);
      lines.push(`${item.chars} chars · ${item.kind}`
        + (item.truncated ? ` · TRUNCATED to first ${DATA_HEAD_CHARS}` : ` · in
full`)
        + ` · found via ${item.how}`);
    });
    lines.push('');
    if (missing.length) {
      lines.push(rule);
      lines.push(`NOT READ -- ${missing.length} resource(s)`);
      lines.push(rule);
      for (const item of missing) lines.push(`- ${item.url}\n
${item.reason}`);
      lines.push('');
    }
    if (inlineDom) {
      lines.push(rule);
      lines.push('THE LIVE PAGE AS IT STOOD
(document.documentElement.outerHTML)');
      lines.push(rule);
      lines.push(document.documentElement.outerHTML);
      lines.push('');
    }
    for (const item of included) {
      lines.push(rule);
      lines.push(`FILE: ${item.url}`);
      lines.push(`${item.chars} chars · ${item.kind}`
        + (item.truncated
          ? ` · TRUNCATED: the first ${DATA_HEAD_CHARS} characters of
${item.chars} are shown,`
          + ` because this is data rather than code. Nothing has been
summarised or`
          + ` rewritten; the rest is simply not here.`
          : ` · shown in full`)
          + ` · found via ${item.how}`);
      lines.push(rule);
      lines.push(item.text);
      lines.push('');
    }
    lines.push(rule);
    lines.push('END OF TELEPRINT');
    lines.push(rule);

    return {
      text: lines.join('\n'),
      filename: `${appName}-source-code-${new
Date().toISOString().replace(/[:.]/g, '-')}.txt`,
      included: included.length,
      missing,
      state
    };
  }
}

```

/* OUR OWN WAY OUT, ON EVERY PLATFORM.

 "WE ARE NOT DEPENDENT ON APPLE WE ARE BUILDING OUR OWN NATIVE SOFTWARE OPEN
 SOURCE ON THE WEB." So the primary route is the one WE own: a panel drawn
 by
 this code, holding the whole teleprint, with its own Copy and Download
 buttons. It cannot be withdrawn by a vendor, it needs no permission, and it
 behaves the same on a phone, a laptop and a tablet.

The platform's own conveniences are OFFERED, never depended on. A download is started because on a desktop that is what a reader expects; a share sheet is exposed only where the browser admits it can take a file. If both are absent or refused the reader still has the text in front of them, selected, with a Copy button under their thumb -- which is the whole point.

GETTING THE FILE OFF THE PHONE, WHICH IS THE WHOLE POINT.

The architect's use for this is: print the source on an iPhone, attach it in ChatGPT, and show an agent what that phone is actually being served. So the delivery path has to work on iOS Safari, where the desktop assumption -- `a[download]` on a blob URL -- is the least reliable of the four options, not the most.

Four ways out, tried in order, and the returned record names the one that worked so a failure on a real device is attributable:

share	navigator.share({files}) -- the native share sheet, which is how a file actually reaches another app on iOS.
download	a[download] -- correct everywhere else.
clipboard	navigator.clipboard.writeText -- pasting works even when no file ever lands.
shown	a selectable panel of the text on the page itself. Never pretty, always available, and it is the difference between a reader who can get their answer and one who cannot.

THE USER GESTURE IS THE CONSTRAINT. iOS grants share and clipboard only inside a real gesture, and this function has to fetch fifty resources first, which ends it. So the bytes are prepared when the FILE MENU OPENS and the button click only DELIVERS them. That is why prepareSourceCode() and deliverSourceCode() are separate exports. */

```
function downloadText(text, filename) {
  const blob = new Blob([text], { type: 'text/plain;charset=utf-8' });
  const url = URL.createObjectURL(blob);
  const link = document.createElement('a');
  link.href = url;
  link.download = filename;
  link.rel = 'noopener';
  document.body.appendChild(link);
  link.click();
  setTimeout(() => {
    URL.revokeObjectURL(url);
    if (link.parentNode) link.parentNode.removeChild(link);
  }, 30000);
}

function showTeleprintPanel(text, filename) {
  const existing = document.getElementById('gridatlas-teleprint-fallback');
  if (existing && existing.parentNode)
    existing.parentNode.removeChild(existing);
  const box = document.createElement('div');
  box.id = 'gridatlas-teleprint-fallback';
  box.style.cssText = 'position:fixed;inset:5% 4%;z-index:100000;display:flex;'
    + 'flex-direction:column;gap:8px;background:#04141a;color:#eaf4f6;'
    + 'border:1px solid rgba(80,220,240,.4);border-radius:6px;padding:12px;'
    + 'font:12px/1.4 ui-monospace,SFMono-Regular,Menlo,monospace';
```

```

    const head = document.createElement('div');
    head.textContent = filename + ' - select all, copy, and paste into your AI
chat.';
    const area = document.createElement('textarea');
    area.readOnly = true;
    area.value = text;
    area.style.cssText = 'flex:1 1 auto;width:100%;box-sizing:border-box;'
    + 'background:#02090c;color:#cfeef6;border:1px solid
rgba(80,220,240,.25);'
    + 'font:11px/1.35 ui-monospace,SFMono-
Regular,Menlo,monospace;padding:8px';
    const row = document.createElement('div');
    row.style.cssText = 'display:flex;gap:8px;flex-wrap:wrap;justify-
content:flex-end';
    const button = (label) => {
        const node = document.createElement('button');
        node.type = 'button';
        node.textContent = label;
        /* 44px because a control smaller than that is not reliably hittable with
a
        thumb, and this panel exists to be used on a phone. */
        node.style.cssText = 'min-height:44px;padding:0 16px;background:#0b2b33;'
        + 'color:#eaf4f6;border:1px solid rgba(80,220,240,.4);border-
radius:4px;'
        + 'font:inherit;cursor:pointer';
        row.appendChild(node);
        return node;
    };
    const copy = button('Copy all');
    copy.setAttribute('data-teleprint', 'copy');
    copy.addEventListener('click', async () => {
        try {
            area.focus();
            area.setSelectionRange(0, area.value.length);
            if (navigator.clipboard && navigator.clipboard.writeText) {
                await navigator.clipboard.writeText(text);
            } else {
                document.execCommand('copy');
            }
            copy.textContent = 'Copied';
        } catch (_) {
            /* Selection is already made, so the reader can still copy by hand. Say
            that rather than claim a success that did not happen. */
            copy.textContent = 'Selected - press copy';
        }
    });
    const save = button('Download .txt');
    save.setAttribute('data-teleprint', 'download');
    save.addEventListener('click', () => downloadText(text, filename));
    const close = button('Close');
    close.addEventListener('click', () => {
        if (box.parentNode) box.parentNode.removeChild(box);
    });
    box.appendChild(head);
    box.appendChild(area);
    box.appendChild(row);
    document.body.appendChild(box);
    /* Pre-selecting means one tap to "Copy" on a phone rather than a drag
    across half a megabyte of text. */
    try { area.focus(); area.setSelectionRange(0, area.value.length); } catch
(_) { /* ignore */ }
    return box;

```

```

}

/**
 * Prepare the bytes. Call this when the menu OPENS, not when the button is
 * pressed, so the press is still a user gesture.
 */
function prepareSourceCode(options = {}) {
  return collectSourceCode(options);
}

/**
 * Deliver already-prepared bytes. Tries share, download, clipboard, then
shows
 * the text. Returns which path was used.
 */
async function deliverSourceCode(collected, { prefer, panel = true } = {}) {
  const filename = collected.filename;
  const text = collected.text;
  const record = {
    filename,
    bytes: new Blob([text]).size,
    included: collected.included,
    missing: collected.missing.length,
    state: collected.state,
    via: [],
    offered: []
  };

  /* OURS FIRST. Whatever any platform does or refuses to do below, the reader
   is now looking at the whole teleprint with a Copy button under it. */
  if (panel) {
    showTeleprintPanel(text, filename);
    record.via.push('panel');
  }

  /* A download is what a desktop reader expects, so it is started -- but
   `download` is advisory, there is no event that says a file was written,
   and a browser that ignores it navigates instead. Recorded as "requested",
   never as "saved". */
  try {
    downloadText(text, filename);
    record.via.push('download-requested');
  } catch (error) {
    record.offered.push('download: ' + String((error && error.message) ||
error));
  }

  /* Offered, not depended on. canShare({files}) is the only honest test: a
   browser can expose share() and still refuse files. */
  if (prefer === 'share' && typeof navigator !== 'undefined' &&
navigator.share
    && typeof File === 'function') {
    try {
      const file = new File([text], filename, { type: 'text/plain' });
      if (!navigator.canShare || navigator.canShare({ files: [file] })) {
        await navigator.share({ files: [file], title: filename });
        record.via.push('share');
      } else {
        record.offered.push('share: this browser will not share files');
      }
    } catch (error) {
      record.offered.push(error && error.name === 'AbortError'

```

```

        ? 'share: the reader cancelled'
        : 'share: ' + String((error && error.message) || error));
    }
}

return record;
}

/** Collect and deliver in one call, for callers outside a gesture. */
async function printSourceCode(options = {}) {
    const collected = await collectSourceCode(options);
    return deliverSourceCode(collected, options);
}

/* WIRING: put the two functions in the File menu, and nowhere else.
-----
The reader already knows where printing lives -- they found it once and
printed a sheet from it. A floating button somewhere else would be a second
place to learn, so both controls go into the same EXPORT THIS VIEW group,
next to the print command that is already there.

The two are SEPARATE COMMANDS and are never merged into one "export":
    Print PDF          a picture of the screen, for a person to look at
    Print source code  the whole source and its dependencies, for a machine
                      to read

NOTHING EXISTING IS REPLACED. The older map-only PDF control stays exactly
as it is. Two implementations of the same idea sitting side by side is the
point -- where they disagree, the disagreement is the finding. */

var PANEL_ANCHOR = '#gridatlas-export-print';
var STATUS_ID = 'gridatlas-teleprint-status';
var MAX_TRIES = 80;

function statusLine(anchor) {
    var node = document.getElementById(STATUS_ID);
    if (node) return node;
    node = document.createElement('div');
    node.id = STATUS_ID;
    node.setAttribute('role', 'status');
    node.setAttribute('aria-live', 'polite');
    node.style.cssText = 'padding:6px 11px;font:10px/1.4 ui-monospace,'
        + 'SFMono-Regular, Menlo, monospace;color:#9fd6e4;opacity:.9;'
        + 'white-space:normal;max-width:34ch';
    if (anchor && anchor.parentNode) anchor.parentNode.appendChild(node);
    return node;
}

function say(text) {
    var node = document.getElementById(STATUS_ID);
    if (node) node.textContent = text;
}

/* The teleprint of the source is prepared when the FILE MENU OPENS, not when
the button is pressed.

Collecting it means reading every resource the browser loaded, which takes
seconds and ends the user gesture. A share sheet or a clipboard write
requested after that gesture has ended is refused, and the reader is told
nothing happened for no reason they can see. Preparing early costs one
speculative pass and makes the press instant. */

```

```

var prepared = null;
var preparing = null;

function beginPreparing() {
  if (prepared || preparing) return preparing;
  preparing = collectSourceCode({ appName: 'GridAtlas' })
    .then(function (result) {
      prepared = result;
      preparing = null;
      say('Source ready · ' + result.included + ' files'
        + (result.missing.length ? ' · ' + result.missing.length + ' not
readable' : ''));
      return result;
    })
    .catch(function (error) {
      preparing = null;
      say('Source could not be prepared: ' + (error && error.message));
      throw error;
    });
  return preparing;
}

function button(label, id, onClick) {
  var node = document.createElement('button');
  node.id = id;
  node.setAttribute('type', 'button');
  node.setAttribute('data-gm-export', id.replace('gridatlas-', ''));
  node.setAttribute('data-teleprint', '1');
  node.textContent = label;
  node.addEventListener('click', function () { onClick(node); });
  return node;
}

function install() {
  var anchor = document.querySelector(PANEL_ANCHOR);
  if (!anchor || !anchor.parentNode) return false;
  if (document.getElementById('gridatlas-teleprint-source')) return true;

  var pdf = button('🖨 Print PDF · exactly this screen',
    'gridatlas-teleprint-pdf', function (node) {
    var was = node.textContent;
    say('Capturing the screen...');
    printPdf({
      note: {
        brand: 'VENTUS · GLOBALGRID2050 · GRID ATLAS',
        title: document.title,
        url: location.href,
        stamp: new Date().toISOString().replace('T', ' ').slice(0, 16) + '
UTC',
        credit: Array.prototype.map.call(
          document.querySelectorAll('.maplibregl-ctrl-attrb-inner'),
          function (n) { return n.textContent.trim(); }
        ).filter(Boolean).join(' | ')
      }
    }).then(function (receipt) {
      say('PDF · ' + receipt.width + 'x' + receipt.height
        + ' px, 1:1, page ' + receipt.pageWidth + 'x' + receipt.pageHeight
        + ' · via ' + receipt.method);
      node.textContent = was;
    }, function (error) {
      /* The reader cancelling the screen-share chooser is a decision, not a
      fault, and must not be reported as one. */
    });
  });
}

```

```

        say(error && /denied|Permission|abort/i.test(String(error.message ||
error)))
        ? 'Screen capture was not allowed, so nothing was printed.'
        : 'Could not print: ' + (error && error.message));
        node.textContent = was;
    });
});

var source = button('🖨 Print source code · for AI review',
'gridatlas-teleprint-source', function () {
    say('Preparing the source...');
    Promise.resolve(prepared || beginPreparing()).then(function (collected)
{
        return deliverSourceCode(collected, { panel: true });
    }).then(function (receipt) {
        say('Source code · ' + receipt.included + ' files, '
+ receipt.bytes + ' bytes · ' + receipt.via.join(' + ')
+ (receipt.missing ? ' · ' + receipt.missing + ' not readable' :
''));
    }, function (error) {
        say('Could not print the source: ' + (error && error.message));
    });
});

anchor.parentNode.insertBefore(pdf, anchor.nextSibling);
pdf.parentNode.insertBefore(source, pdf.nextSibling);
statusLine(source);
say('Print PDF saves this screen. Print source code saves the code behind
it.');
```

```

/* Opening File is the signal to start work. Delegated from the document so
it survives the menu being rebuilt. */
document.addEventListener('click', function (event) {
    var title = event.target && event.target.closest
    ? event.target.closest('.gm-title') : null;
    if (title && /file/i.test(title.textContent || '')) beginPreparing();
}, true);
return true;
}

/* The export group is built by the menu bar, which itself installs only once
63 layer controls exist. Polling rather than assuming: a fixed delay is a
guess that is wrong on a slow phone and wasteful on a fast desktop. */
if (!install()) {
    var tries = 0;
    var timer = setInterval(function () {
        tries += 1;
        if (install() || tries >= MAX_TRIES) clearInterval(timer);
    }, 250);
}

})();

/* =====
PART 2 - the network, as its operator publishes it
===== */
((() => {
    'use strict';

    const GENERATION = '202609012045';
    /* Pinned to a commit and hashed. The reasoning, the table and the
    digest arithmetic are in the pinned-products module beside this file;
```

```

    what matters here is that `main` was the ref and the schema string was
    the whole defence, and a schema string is blind to values. */
const PINS = (window.__GRIDATLAS_MODULES__ || {}).pinnedProducts || null;
const PRODUCT_ID = 'connection-points.v3';
const PRODUCT = PINS ? PINS.url(PRODUCT_ID) : null;
const REQUIRED_SCHEMA = 'data-grid-gb.connection-points.v3';
/* Appendix D publishes eight current metrics and they are NOT
   interchangeable, so one is quoted and named rather than any of them
   being called "the fault level".
   Codex, 202609011852: an earlier version of this comment and of the
   card said the RMS break current is "the one switchgear is rated
   against". That overclaims. Switchgear carries several relevant
   ratings - making capacity, short-time withstand, peak withstand -
   and this is ONE published breaker-duty metric among the eight. */
const QUOTED_METRIC = 'three_phase_rms_break_current_ka';
const QUOTED_METRIC_LABEL = 'three-phase RMS break current';
const DEG = Math.PI / 180;

const state = {
  schema: 'gridatlas.substation-intelligence.v2',
  generation: GENERATION,
  product: PRODUCT,
  loaded: false,
  points: 0,
  located: 0,
  product_schema: null,
  quoted_metric: QUOTED_METRIC,
  failures: []
};
window.__GRIDATLAS_NETWORK__ = state;

/* ONE geodesy, and it is the module's.
   -----
   This carried its own haversine using 2*R*asin(sqrt(a)) while the
   estate canonical form is R*2*atan2(sqrt(a), sqrt(1-a)). They
   agree algebraically and differ in the last place, and the
   difference was invisible for as long as this half of the
   cartridge was a monolith the all-versions scan could not read.
   202609012350 extracted it, the scan found it immediately, and
   the answer is not to retype the right form here but to stop
   having a second implementation at all. */
const GEODESY = (window.__GRIDATLAS_MODULES__ || {}).geodesy;
if (!GEODESY) throw new Error("substation-intelligence requires the geodesy
module");
const distanceKm = GEODESY.distanceKm;

const NOISE = /\b(SUBSTATION|SUB
STATION|SUBSTN|GRID|SUPPLY|POINT|GSP|NATIONAL|POWER|STATION|WIND|FARM|WINDFARM|O
FFSHORE|ONSHORE|EXTENSION|400KV|275KV|132KV|66KV|33KV|11KV|NGET|SSE|SP|SHE)\b/g;
function normalise(name) {
  return String(name || '').toUpperCase()
    .replace(/[^\A-Z0-9 ]/g, ' ').replace(NOISE, ' ')
    .split(/\s+/).filter(Boolean).join(' ');
}

const byName = new Map();
const located = [];
/* Every point, not only the ones that survived the name join. `byName`
   keeps the first of a colliding key, so counting from it would report a
   smaller network than the operator publishes. */
const published = [];

```



```

const ready = (async () => {
  try {
    if (!PRODUCT) throw new Error('the pinned-products module is not composed,

    + 'so this cartridge has no pinned ref to read and will not guess one');
    const response = await fetch(PRODUCT, { cache: 'no-cache' });
    if (!response.ok) throw new Error('HTTP ' + response.status);
    const text = await response.text();
    const seal = await PINS.verify(PRODUCT_ID, text);
    state.product_pin = seal;
    if (seal.state === 'MISMATCH') {
      state.failures.push(seal.detail
        + '; refusing to answer from bytes this composition has not seen');
      return false;
    }
    const product = JSON.parse(text);
    state.product_schema = product?.schema || null;
    if (product?.schema !== REQUIRED_SCHEMA) {
      state.failures.push('schema is ' + String(product?.schema)
        + ', this cartridge answers only ' + REQUIRED_SCHEMA);
      return false;
    }
    for (const point of product.connection_points || []) {
      const key = normalise(point.name);
      if (key && !byName.has(key)) byName.set(key, point);
      published.push(point);
      if (point.location) located.push(point);
    }
    state.points = (product.connection_points || []).length;
    state.located = located.length;
    state.counts = product.counts || null;
    state.join = product.join || null;
    state.source = product.source || null;
    state.loaded = true;
    return true;
  } catch (error) {
    state.failures.push('network product: ' + String(error?.message ||
error));
    return false;
  }
})();
state.ready = ready;

state.byName = (name) => state.loaded
  ? (byName.get(normalise(name)) || null) : null;

/* The owner product's coordinates are NOT used for anything a reader
sees. Codex, 202609011852: WBUR's exact-name join binds a different
West Burton 96.42 km from the project, and exact text equality is not
exact identity. The Atlas measures on its own substation payload and
always has; this stays available for callers that want it, marked,
and the card is proven never to print a distance from here. */
state.location_join_is_unverified = true;
state.nearest = (lon, lat, options) => {
  if (!state.loaded) return null;
  const minimumKv = options?.minimumKv ?? 0;
  const limit = options?.limit ?? 1;
  const found = [];
  for (const point of located) {
    if (Math.max(...point.voltages_kv) < minimumKv) continue;
    found.push({ point, km: distanceKm(lon, lat, point.location.lon,
point.location.lat) });
  }

```

```

    }
    found.sort((a, b) => a.km - b.km);
    return limit === 1 ? (found[0] || null) : found.slice(0, limit);
  };

  /* One line a card can print, built only from what is published, or null
  when nothing is. An empty sentence about a substation is worse than
  silence. */
  /* connectionKv is the voltage the connection is actually made at: the
  declared point of connection's class, or the class of the substation
  being measured to. Given one, the fault current is quoted at THAT
  busbar group rather than across the site.

  An outside review put the reason plainly: fault duty at a 400 kV
  busbar and at a 132 kV busbar are different physical quantities
  governing different switchgear, so a range spanning both is
  meaningless to the engineer reading it - and the more correctly the
  metric is named, the more readily the eye trusts it. */
  /* WHAT A DISTANCE SEARCH COULD SEE, at the voltage it was asked for.
  -----
  ETYS names substations and does not locate them; the only geometry
  this estate holds comes from OpenStreetMap through a GridAtlas
  release. So a proportion of what the operator publishes is invisible
  to any search by distance, and a card that says "nearest" without
  saying that is asserting something it cannot know.

  COMPUTED, never written down. The numbers move when the product moves
  - Codex's join correction takes located from 502 to 489 - and a
  sentence with a literal in it would go quietly false the day the pin
  is bumped. This counts the payload that was actually fetched, using
  exactly the predicate state.nearest uses to decide what is eligible,
  so the denominator on the card is the denominator of the search. */
  state.coverage = (minimumKv) => {
    if (!state.loaded) return null;
    const floor = Number(minimumKv) || 0;
    const eligible = (point) => Array.isArray(point.voltages_kv)
      && point.voltages_kv.length
      && Math.max(...point.voltages_kv) >= floor;
    const inBand = published.filter(eligible);
    const seen = located.filter(eligible);
    return {
      minimum_kv: floor,
      published: inBand.length,
      located: seen.length,
      unlocated: inBand.length - seen.length,
      /* Named rather than computed by the reader, and never graded. */
      basis: 'counted from the connection-points payload this session fetched'
    };
  };

  state.summarise = (name, options) => {
    const point = state.byName(name);
    if (!point) return null;
    const connectionKv = options && Number(options.connectionKv);
    /* connection-points.v3 publishes `circuits` and `transformers` as
    LANDINGS - one per node-end at the site - not as machines. A
    transformer's two windings are both at the site by construction, so
    almost every transformer is counted twice: Cowley publishes 10 for
    its five machines, and 484 of the 525 sites that hold a transformer
    are overstated, 2,944 landings against 1,550 units (1.90x).
  */

```

This cartridge holds no node pairs, so it cannot deduplicate on its

```

    own; the caller may pass `units` derived from the node/branch model
    by the network-topology module, which does hold them. Where it does,
    the machine count is stated. Where it does not, the published figure
    is still shown but is named for what it is, because a landing tally
    presented as a machine count is the defect, not the tally. */
/* Number(null) is 0, not NaN, so `Number(units && units.x)` reports a
    finite ZERO the moment no units are passed - which is every call
    that does not supply them. The proof caught it saying "0 circuits"
    for a site publishing eight. Read the field only when there is one. */
const units = (options && options.units) || null;
const unitCount = (field) => {
    if (!units) return null;
    const value = Number(units[field]);
    return Number.isFinite(value) ? value : null;
};
const transformerUnits = unitCount('transformers');
const circuitUnits = unitCount('circuits');
const parts = [];
if (circuitUnits !== null) {
    parts.push(circuitUnits + (circuitUnits === 1 ? ' circuit' : '
circuits'));
} else if (point.circuits) {
    parts.push(point.circuits + (point.circuits === 1 ? ' circuit' : '
circuits'));
}
if (transformerUnits !== null) {
    parts.push(transformerUnits
        + (transformerUnits === 1 ? ' transformer' : ' transformers'));
} else if (point.transformers) {
    parts.push(point.transformers + ' transformer winding connections at the
site '
        + '(a transformer whose windings are both here is published at each of
them, '
        + 'so this is not a count of machines)');
}
const rating = point.circuit_winter_rating_mva;
if (rating) {
    /* The product does not split ratings by voltage, and a site with
        several voltages will show a range no single circuit could span -
        Blackhilllock publishes 23 to 1,995 MVA. So it is marked site-wide
        wherever it appears, rather than sitting beside a bus-specific
        fault figure as though it shared its scope. */
    parts.push('circuit winter ratings across the site '
        + rating.min.toLocaleString('en-GB')
        + '\u2013' + rating.max.toLocaleString('en-GB') + ' MVA');
}
/* Prefer the busbar group the connection is made at. Fall back to the
    site-wide envelope only when the voltage is unknown or the product
    does not publish that group, and say which was used either way. */
const byVoltage = point.fault_current_by_voltage || null;
let peak = point.fault_current?.peak || null;
let faultScope = 'site';
let faultKv = null;
if (Number.isFinite(connectionKv) && byVoltage) {
    const key = Object.keys(byVoltage)
        .find(k => Math.abs(Number(k) - connectionKv) < 0.5);
    if (key && byVoltage[key]?.peak) {
        peak = byVoltage[key].peak;
        faultScope = 'bus';
        faultKv = Number(key);
    }
}
}

```

```

const metric = peak?.metrics?.[QUOTED_METRIC];
if (metric) {
  parts.push(QUOTED_METRIC_LABEL + ' ' + metric.min.toFixed(1) + '\u2013'
    + metric.max.toFixed(1) + ' ' + metric.unit
    + (faultScope === 'bus'
      ? ' at the ' + faultKv + ' kV busbars'
      : ' across every busbar at this site')
    + ' over ' + peak.scenarios + ' peak-demand rows'
    + (peak.locations?.length ? ' at ' + peak.locations.length
    + (peak.locations.length === 1 ? ' bus' : ' buses') : '')
    + (peak.winters?.length
      ? ' (' + peak.winters[0] + ' to ' + peak.winters[peak.winters.length -
1] + ') '
      : ''));
}
if (point.reactive_compensation?.units) {
  parts.push(point.reactive_compensation.units + ' reactive compensation
units');
}
if (point.planned_changes) {
  const years = point.planned_change_years || [];
  parts.push(point.planned_changes + ' changes published for '
    + (years.length ? years[0] + '\u2013' + years[years.length - 1] : 'later
years'));
}
if (!parts.length) return null;
/* Everything above is aggregated at SITE CODE, not selected for a
bus. Where a site carries more than one voltage the numbers span
them, so the reader is told that before reading any of them -
otherwise a sentence under a 400 kV point of connection reads as a
400 kV result. West Burton is exactly this case: WBUR1 is 132 kV
and WBUR4 is 400 kV, and its published fault range spans both. */
const voltages = point.voltages_kv || [];
/* Site-wide is now about what remains site-wide. Once the fault
current is quoted at a busbar group, the label must not claim the
whole sentence is site-wide - only the parts that still are. */
const siteWide = voltages.length > 1;
const busLocations = point.fault_current?.peak?.locations || [];
return {
  site_code: point.site_code,
  transmission_owner: point.transmission_owner,
  voltages_kv: voltages,
  site_wide: siteWide,
  bus_locations: busLocations,
  fault_scope: faultScope,
  fault_kv: faultKv,
  scope_label: faultScope === 'bus'
    ? ('Fault current is quoted at the ' + faultKv + ' kV busbars, the '
      + 'voltage this connection is made at. Circuit counts, ratings, '
      + 'transformers and planned changes remain site-wide across the '
      + voltages.slice().sort((a, b) => b - a).join('/') + ' kV buses
here')
    : (siteWide
      ? ('Site-wide published envelope across the '
        + voltages.slice().sort((a, b) => b - a).join('/') + ' kV buses at
this site, '
        + 'not a value for any one bus')
      : ('Published for this site, which carries one voltage: '
        + (voltages[0] || '?') + ' kV')),
  sentence: parts.join(' \u00b7 '),
  metric_named: QUOTED_METRIC_LABEL,
  metrics_not_interchangeable: 'Appendix D publishes eight current '

```

```

    + 'metrics and they are not interchangeable; this is one published '
    + 'breaker-duty metric, and switchgear carries several relevant '
    + 'ratings besides it.',
    attribution: 'NESO Electricity Ten Year Statement 2025, appendices B and
D, '
    + 'via VentusLtd/data-grid-gb',
    not_an_assessment: 'Published parameters. Not a statement about whether '
    + 'any project can connect here.'
  };
};
})();

```

```

=====
FILE: https://ventusltd.github.io/gridatlas/atlas/cartridges/202609050354-sld-
sandbox-v9-8.js
359603 chars · code · shown in full · found via loaded by the browser (fetch)
=====

```

```

/**
 * sld-sandbox-v9-8, generation 202609050354 (UTC).
 *
 * ASSEMBLED by tools/build-cartridge.mjs from the parts below. Do not edit
 * this file: edit a part and rebuild under a new generation. Each part is
 * hashed in manifests/202609050354-sld-sandbox-v9-8-parts.json.
 *
 *   module      atlas/modules/202609012040-grid-scope.js
 *   module      atlas/modules/202609012217-source-registry.js
 *   module      atlas/modules/202609012128-declared-connections.js
 *   module      atlas/modules/202609012205-sizing-arithmetic.js
 *   module      atlas/modules/202609031310-technology-coverage.js
 *   part        atlas/parts/202609041234-sld-sandbox-technology-
buckets.js
 */

/**
 * Module: grid-scope
 *
 * "When you click on a blank space, the user should be able to see grid in
 * the vicinity. Call it the GRID FINDING SCOPE – analysis of what is
 * there, NOT indicative of capacity." – Vikram, 2026-09-01.
 *
 * So this answers exactly one question: WHAT IS MAPPED HERE. It counts
 * what the served payload contains around a point, by voltage class and
 * by distance band, and names the nearest few. It is a census of the map,
 * not a study of the network.
 *
 * WHAT IT WILL NOT DO, EVER
 * It does not say whether a connection is available, likely, cheap or
 * possible. Nothing in a payload of substation positions can support any
 * of that: capacity depends on queue position, committed connections,
 * thermal and fault headroom, consent and commercial terms, and none of
 * those is a distance. A scope that counted substations and implied
 * opportunity would be the most dangerous thing this estate could ship,
 * because it would look like analysis.
 *
 * Pure. No DOM, no network, no state. Depends on: geodesy.
 */
(() => {
  'use strict';

  const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||
{}));

```

```

if (NS.gridScope) return;

const geodesy = NS.geodesy;
if (!geodesy) throw new Error('grid-scope requires the geodesy module');

/* Bands, not a single radius. A reader asking "what is around here"
   wants the shape of the answer - is the nearest thing on top of me or
   twenty kilometres away - and one number hides that. */
const DEFAULT_BANDS_KM = [2, 5, 10, 25];
const CLASSES_KV = [400, 275, 220, 132, 66, 33];

/* A voltage is classified ONLY as a class it actually is.
   -----
   The first version walked the classes and returned the first one the
   value exceeded, so it labelled 750 kV as 400, 110 kV as 66 and 50 kV
   as 33 - a false label on anything the list does not contain, which is
   exactly the sort of quiet relabelling this estate exists to avoid.
   Codex caught it on the committed module before it reached a card
   (stop-ship 202609012025).

   Now: membership, within a tolerance for the fractions OSM carries.
   Anything else is UNCLASSIFIED and counted as such, because a voltage
   the standard classes do not contain is a fact about the data, not a
   value to be rounded into the nearest familiar number. */
const CLASS_TOLERANCE_KV = 0.5;

function classOf(kv) {
  if (!Number.isFinite(kv)) return null;
  for (const known of CLASSES_KV) {
    if (Math.abs(kv - known) <= CLASS_TOLERANCE_KV) return known;
  }
  return null;
}

/**
 * @param origin [lon, lat]
 * @param substations [{ at:[lon,lat], kv:[numbers], name, operator }]
 * @param options { bandsKm, minimumKv, nearestCount }
 */
function scope(origin, substations, options) {
  const bandsKm = (options && options.bandsKm) || DEFAULT_BANDS_KM;
  const minimumKv = (options && options.minimumKv) || 0;
  const nearestCount = (options && options.nearestCount) || 5;
  const maximumKm = bandsKm[bandsKm.length - 1];

  const within = [];
  for (const substation of substations || []) {
    if (!substation || !Array.isArray(substation.at)) continue;
    const voltages = (Array.isArray(substation.kv) ? substation.kv : [])
      .filter(Number.isFinite);
    /* Non-finite voltages are dropped BEFORE the maximum.
       -----
       Codex, 202609012055: Math.max over a NaN gives NaN, and NaN < floor
       is false, so a substation whose voltage did not parse survived a
       132 kV floor and was censused as though it qualified. A voltage
       that is not a number is not a voltage above the floor. */
    const top = voltages.length ? Math.max(...voltages) : 0;
    if (top < minimumKv) continue;
    const km = geodesy.distanceKm(origin[0], origin[1],
      substation.at[0], substation.at[1]);
    if (km > maximumKm) continue;
    within.push({

```

```

        name: substation.name || '',
        operator: substation.operator || '',
        kv: top,
        class_kv: classOf(top),
        km,
        at: substation.at
    });
}
within.sort((a, b) => a.km - b.km);

const bands = bandsKm.map((band) => {
    const inBand = within.filter(entry => entry.km <= band);
    const counts = {};
    let unclassified = 0;
    const unclassifiedKv = [];
    for (const entry of inBand) {
        if (entry.class_kv == null) {
            // Counted, never folded into a class it is not.
            unclassified += 1;
            if (Number.isFinite(entry.kv) && !unclassifiedKv.includes(entry.kv)) {
                unclassifiedKv.push(entry.kv);
            }
            continue;
        }
        counts[entry.class_kv] = (counts[entry.class_kv] || 0) + 1;
    }
    const highest = inBand.reduce(
        (best, entry) => (entry.class_kv != null && (best == null ||
entry.class_kv > best)
        ? entry.class_kv : best), null);
    return {
        within_km: band,
        substations: inBand.length,
        by_class_kv: counts,
        highest_class_kv: highest,
        unclassified_voltage: unclassified,
        unclassified_kv: unclassifiedKv.sort((a, b) => b - a)
    };
});

/* Named first, because an unnamed OSM node is a fact about the map
   rather than a place anyone can look up. Both are reported: the
   nearest thing, and the nearest thing with an identity. */
const named = within.filter(entry => entry.name);
return {
    schema: 'gridatlas.grid-scope.v1',
    origin: [origin[0], origin[1]],
    radius_km: maximumKm,
    minimum_kv: minimumKv,
    counted: within.length,
    bands,
    nearest: within.slice(0, nearestCount),
    nearest_named: named.slice(0, nearestCount),
    nearest_transmission: within.find(entry => entry.kv >= 275 - 0.5) || null,
    /* Carried in the result itself so it cannot be separated from the
       numbers by a renderer, a screenshot or a quote. */
    what_this_is: 'A census of the substations in the served map payload '
        + 'around this point, by voltage class and distance band.',
    what_this_is_not: 'Not a statement about capacity, headroom, '
        + 'availability or the cost of connecting here. Distance is not '
        + 'capacity: queue position, committed connections, thermal and '
        + 'fault headroom, consent and commercial terms decide that, and '

```

```

    + 'none of them is in this payload.',
    method: 'haversine on a single Earth radius of '
    + geodesy.EARTH_RADIUS_KM + ' km, straight line to mapped geometry'
  };
}

NS.gridScope = Object.freeze({
  schema: 'gridatlas.module.grid-scope.v2',
  CLASS_TOLERANCE_KV,
  DEFAULT_BANDS_KM,
  CLASSES_KV,
  classOf,
  scope
});
})();

/**
 * Module: source-registry
 *
 * "Click anywhere on a map and the neons that already work via Pipeline News
 * look for cartridges and code." – Vikram, 2026-09-01.
 *
 * The looking is this module. The Atlas is a composition of cartridges that
 * find each other through `window.__GRIDATLAS_*` globals, and the deep scan
 * of 1 Sep 2026 found fifteen such surfaces ever registered, thirteen live,
 * and nothing anywhere that documents them. Every consumer therefore does
 * its own `window.__GRIDATLAS_NETWORK__?.something` and quietly does less
 * when the answer is undefined. That is how a click on blank space came to
 * report only what OpenStreetMap has mapped, while the cartridge holding
 * NESO's 886 published connection points sat loaded in the same page.
 *
 * So: one registry, declared once, that answers three questions.
 *
 * WHAT COULD ANSWER      the sources this estate knows about, each with
 *                          what it contributes and whether it is required.
 * WHAT IS ANSWERING NOW  probed live, by looking for the surface AND the
 *                          specific capability, because a cartridge that
 *                          has loaded but not yet fetched is present and
 *                          not yet useful, and those are different states.
 * WHAT DID NOT           named, with the reason, in the result itself.
 *
 * The third is the point. A reader who is told "3 of 4 sources answered;
 * NESO's published network did not, because its payload had not loaded" can
 * judge the answer. A reader shown a shorter answer cannot, and will
 * reasonably assume the map has told them everything it knows.
 *
 * It reads. It never fetches, never renders, and never decides what a
 * finding means.
 *
 * Successor to 202609012135 at generation 202609012217: every source that
 * fetches a product declares what it REQUIRES (repository, product, schema)
 * and the survey carries that in every state, because a contract stated
 * only once the load has succeeded is no help to the reader of a failure.
 * The GB price rollup, fetched since v9.41 without a row here, is
 * registered with the loader state the sandbox now publishes.
 *
 * Depends on: nothing.
 */
(() => {
  'use strict';

  const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||

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{}));
if (NS.sourceRegistry) return;

/* The registry is DECLARED, not discovered by scanning window.
-----
Enumerating every __GRIDATLAS_* global would report whatever happens to
be there, including surfaces this estate has never agreed to consume,
and would silently start using a new one the day someone adds it. A
declared list is a contract: adding a source is an edit here, with a
reason, and a proof that the probe actually works. */
const SOURCES = [
  {
    id: 'map',
    surface: '__GRIDATLAS_V9_MAP__',
    contributes: 'the map itself: where the click happened, and what is
drawn',
    probe: (w) => (w.__GRIDATLAS_V9_MAP__ ? 'ready' : 'absent')
  },
  {
    id: 'mapped-substations',
    surface: '__GRIDATLAS_NEON_LINKS__',
    contributes: 'substations as OpenStreetMap has them mapped, and the '
      + 'measurement the neon links already use',
    probe: (w) => {
      const links = w.__GRIDATLAS_NEON_LINKS__;
      if (!links) return 'absent';
      if (typeof links.measure?.distanceKm !== 'function') return 'loaded,
cannot measure';
      if (!links.substations_loaded) return 'loaded, no substations yet';
      return 'ready';
    },
    detail: (w) => ({ substations:
w.__GRIDATLAS_NEON_LINKS__?.substations_loaded || 0 })
  },
  {
    id: 'neso-connection-points',
    surface: '__GRIDATLAS_NETWORK__',
    requires: { repository: 'Ventusltd/data-grid-gb',
      product: 'derived/connection-points.v3.json',
      schema: 'data-grid-gb.connection-points.v3' },
    contributes: "NESO's published connection points: circuits, transformers,
"
      + 'per-voltage fault current and planned changes',
    probe: (w) => {
      const network = w.__GRIDATLAS_NETWORK__;
      if (!network) return 'absent';
      if (network.failed) return 'failed to load';
      if (!network.loaded) return 'loading';
      return 'ready';
    },
    detail: (w) => ({ connection_points: w.__GRIDATLAS_NETWORK__?.count ||
null,
      schema: w.__GRIDATLAS_NETWORK__?.schema || null })
  },
  {
    id: 'grid-scope',
    surface: '__GRIDATLAS_MODULES__.gridScope',
    contributes: 'the census of what is mapped around a point, in distance
bands',
    probe: (w) => (w.__GRIDATLAS_MODULES__?.gridScope ? 'ready' : 'absent')
  },
  {

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id: 'network-topology',
surface: '__GRIDATLAS_MODULES__.networkTopology + __GRIDATLAS_TOPOLOGY__',
requires: { repository: 'Ventusltd/data-grid-gb',
  product: 'derived/gb-transmission-network.v1.json',
  schema: 'data-grid-gb.transmission-network.v1' },
contributes: 'circuits, transformers, planned changes and neighbouring '
+ 'sites at a named substation, per voltage',
/* Generation 202609012135: the module alone is not the source. At v9.67
  this probe said "ready" because the module object existed, while
  the ten-megabyte product it indexes had never been fetched by any
  cartridge - the module was on disk and answered nothing. Ready now
  means the product is indexed; idle means it will load on the first
  click that asks; the other states are what the loader says. */
probe: (w) => {
  if (!w.__GRIDATLAS_MODULES__.networkTopology) return 'absent';
  const loader = w.__GRIDATLAS_TOPOLOGY__;
  if (!loader) return 'module present, no loader in this composition';
  if (loader.state === 'ready') return 'ready';
  if (loader.state === 'loading') return 'loading';
  if (loader.state === 'failed') return 'failed to load';
  return 'idle, loads on first use';
},
detail: (w) => ({ sites: w.__GRIDATLAS_TOPOLOGY__.sites || null,
  bytes: w.__GRIDATLAS_TOPOLOGY__.bytes || null,
  schema: w.__GRIDATLAS_TOPOLOGY__.schema || null })
},
{
  id: 'declared-connections',
  surface: '__GRIDATLAS_MODULES__.declaredConnections',
  contributes: 'points of connection bound to a made Order or a published '
+ 'planning document',
  probe: (w) => (w.__GRIDATLAS_MODULES__.declaredConnections?.count > 0 ?
'ready' : 'absent'),
  detail: (w) => ({ records:
w.__GRIDATLAS_MODULES__.declaredConnections?.count || null })
},
{
  id: 'gb-electricity-conditions',
  surface: '__GRIDATLAS_GB_CONDITIONS__',
  contributes: 'the GB wholesale price context a project card carries: '
+ 'negative-price days and the record daily mean, from the owner
rollup',
  requires: { repository: 'Ventusltd/data-gb-electricity',
    product: 'derived/price-decade-rollup.json',
    schema: 'data-gb-electricity.price-decade-rollup.v2' },
  /* Withheld is its own state: the product was reached and was not the
    schema this consumer answers, so the panel shows nothing and says
    why. That is neither a failure of the network nor a source ready. */
  probe: (w) => {
    const loader = w.__GRIDATLAS_GB_CONDITIONS__;
    if (!loader) return 'absent';
    if (loader.state === 'ready') return 'ready';
    if (loader.state === 'loading') return 'loading';
    if (loader.state === 'failed') return 'failed to load';
    if (loader.state === 'withheld') return 'withheld: ' +
String(loader.reason || 'schema not supported');
    return 'idle, loads on first use';
  },
  detail: (w) => ({ schema: w.__GRIDATLAS_GB_CONDITIONS__.schema || null,
    renders: w.__GRIDATLAS_GB_CONDITIONS__.renders || 0 })
}
}
];

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const READY = 'ready';

/**
 * Probe every declared source against a window.
 * @param scope the window to read; defaults to this one. Passing it in is
 * what lets a proof drive the probe without a browser.
 */
function survey(scope) {
  const w = scope || window;
  const sources = SOURCES.map((source) => {
    let state = 'absent';
    let detail = null;
    try { state = source.probe(w) || 'absent'; }
    catch (error) { state = `probe threw: ${error} && error.message`; }
    if (state === READY && typeof source.detail === 'function') {
      try { detail = source.detail(w); } catch (_) { detail = null; }
    }
    return { id: source.id, surface: source.surface,
      contributes: source.contributes, requires: source.requires || null,
      state, ready: state === READY, detail };
  });

  const ready = sources.filter(s => s.ready);
  const missing = sources.filter(s => !s.ready);

  return {
    schema: 'gridatlas.module.source-registry.v1',
    sources,
    ready: ready.map(s => s.id),
    missing: missing.map(s => ({ id: s.id, state: s.state })),
    counts: { declared: sources.length, ready: ready.length, missing:
missing.length },
    /* Written as a sentence here so a card cannot compose its own and get
       it wrong, and so an absence is never presented as an absence in the
       world rather than in this page. */
    sentence: missing.length === 0
      ? `All ${sources.length} sources answered.`
      : `${ready.length} of ${sources.length} sources answered. Not answering:

      + missing.map(s => `${s.id} (${s.state})`).join(', ')
      + '. What they would have added is missing from this answer, not '
      + 'absent from the world.'

    };
  }

  /** Is one source usable right now. */
  function ready(id, scope) {
    const source = SOURCES.find(s => s.id === id);
    if (!source) return false;
    try { return source.probe(scope || window) === READY; }
    catch (_) { return false; }
  }

  NS.sourceRegistry = Object.freeze({
    schema: 'gridatlas.module.source-registry.v1',
    declared: SOURCES.map(s => s.id),
    survey,
    ready
  });
})();

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/**
 * Module: declared-connections
 *
 * The 400 kV public record: what each DCO-scale scheme has DECLARED as its
 * point of connection, taken from Development Consent Orders, Planning
 * Inspectorate documents and public project statements. The table binds a
 * register identity (REPD ref) to a NAMED substation, and the functions
 * here bind that name to the served payload and measure the distance -
 * measured, never asserted.
 *
 * The rule this exists to keep: bind to the public record or say nothing.
 * A nearest-substations list is a measurement; it was listing closer 33 and
 * 132 kV points under schemes whose Order names a 400 kV connection, which
 * read as connecting them to the wrong network. This table is the answer,
 * and it is data with three small functions, so it lives in a module where
 * a proof can read every record and a cut can hash it on its own.
 *
 * WHAT IT WILL NOT DO
 * It does not say whether a connection is available, likely or adequate. A
 * declared point of connection is a fact about a consent, not a judgement
 * about the network. `poc_status` distinguishes a far end that exists from
 * one not yet built or under construction, because drawing both the same
 * would say something untrue.
 *
 * Extracted from the sld-sandbox body at generation 202609012128 (UTC),
 * record for record; the parity proof reads the previously served bytes
 * and asserts the table is unchanged.
 *
 * Pure. No DOM, no network, no state. Depends on: geodesy.
 */
((() => {
  'use strict';

  const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||
  {}));
  if (NS.declaredConnections) return;

  const geodesy = NS.geodesy;
  if (!geodesy) throw new Error('declared-connections requires the geodesy
module');
  const distanceKm = geodesy.distanceKm;

  const RECORDS = Object.freeze({
    '10914': { works: "an up to 400 kV substation collating the satellite sites
at 132 kV and site generation at 33 kV (Work No. 4A)",
      poc_works: "reuse of an ex-generation bay: busbars, a 400 kV 3-phase 4000
A breaker, metering and protection (Work No. 5)",
      substation: 'Cottam Substation',
      via: 'a new 400 kV scheme substation consented within the DCO',
      source: 'Cottam Solar Project Order 2024, granted 5 Sep 2024 (EN010133)'
    },
    '10915': { works: "an up to 400 kV substation collating the satellite sites
at 132 kV and site generation at 33 kV (Work No. 4A)",
      poc_works: "reuse of an ex-generation bay: busbars, a 400 kV 3-phase 4000
A breaker, metering and protection (Work No. 5)",
      substation: 'Cottam Substation',
      via: 'a new 400 kV scheme substation consented within the DCO',
      source: 'Cottam Solar Project Order 2024, granted 5 Sep 2024 (EN010133)'
    },
    '10916': { works: "an up to 400 kV customer substation at West Burton 3 with
reactive power units; up to 132 kV site substations at WB1 and WB2 (Works 3A-
3C)",

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    poc_works: "a new GIS bay by extension of main busbar 4 and reserve busbar
3/4 gas zones (Work No. 4)",
    substation: 'West Burton Substation',
    via: 'a new 400 kV customer substation at West Burton 3 and a 400 kV cable
to the former generator bay',
    source: 'West Burton Solar Project Order, granted 24 Jan 2025 (EN010132)'
},
    '10917': { works: "an up to 400 kV customer substation at West Burton 3 with
reactive power units; up to 132 kV site substations at WB1 and WB2 (Works 3A-
3C)",
    poc_works: "a new GIS bay by extension of main busbar 4 and reserve busbar
3/4 gas zones (Work No. 4)",
    substation: 'West Burton Substation',
    via: 'a new 400 kV customer substation at West Burton 3 and a 400 kV cable
to the former generator bay',
    source: 'West Burton Solar Project Order, granted 24 Jan 2025 (EN010132)'
},
    '9809': { works: "a scheme substation with reactive power units and a 400 kV
harmonic filter compound (Work No. 3)",
    poc_works: "one new 400 kV generation bay at Cottam (Work No. 4C)",
    substation: 'Cottam Substation',
    via: 'a new 400 kV scheme substation and a 7.5 km 400 kV underground
cable',
    source: 'Gate Burton Energy Park Order, granted 2024 (EN010131)' },
    '9810': { works: "a scheme substation with reactive power units and a 400 kV
harmonic filter compound (Work No. 3)",
    poc_works: "one new 400 kV generation bay at Cottam (Work No. 4C)",
    substation: 'Cottam Substation',
    via: 'a new 400 kV scheme substation and a 7.5 km 400 kV underground
cable',
    source: 'Gate Burton Energy Park Order, granted 2024 (EN010131)' },
    '12281': { works: "two scheme substations, each 2 x 400/33 kV 150/75/75 MVA
transformers with 400 kV GIS (Works 3A-3B)",
    poc_works: "the standard 400 kV bay kit at a free bay at Cottam (Work No.
5)",
    substation: 'Cottam Substation',
    via: 'an 18.5 km 400 kV underground cable to a free bay',
    source: 'Tillbridge Solar Order 2025 (EN010142)' },
    '12282': { works: "two scheme substations, each 2 x 400/33 kV 150/75/75 MVA
transformers with 400 kV GIS (Works 3A-3B)",
    poc_works: "the standard 400 kV bay kit at a free bay at Cottam (Work No.
5)",
    substation: 'Cottam Substation',
    via: 'an 18.5 km 400 kV underground cable to a free bay',
    source: 'Tillbridge Solar Order 2025 (EN010142)' },
    '14806': { poc_status: 'not_built',
    poc_status_note: 'the point of connection is NGET\u2019s new substation
beside the existing High Marnham, built as Great Grid Upgrade works; the line is
drawn to the existing site',
    substation: 'High Marnham Substation',
    via: "NGET's new substation adjacent to the existing High Marnham (Great
Grid Upgrade)",
    source: 'One Earth Solar Farm DCO, consented (EN010159)' },
    '14807': { poc_status: 'not_built',
    poc_status_note: 'the point of connection is NGET\u2019s new substation
beside the existing High Marnham, built as Great Grid Upgrade works; the line is
drawn to the existing site',
    substation: 'High Marnham Substation',
    via: "NGET's new substation adjacent to the existing High Marnham (Great
Grid Upgrade)",
    source: 'One Earth Solar Farm DCO, consented (EN010159)' },
    '13599': { works: "up to four 33-400 kV transformers (160 t, up to 15 x 9.5

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x 10.5 m each) in a compound of up to 40,000 m2 (ES Ch.2 s2.8)",
  poc_works: "a National Grid-delivered extension of Bicker Fen, AIS or GIS,
sited for multiple customers (s2.13)",
  substation: 'Bicker Fen Substation',
  via: 'a 400 kV cable and a consented extension of Bicker Fen shared with
Heckington Fen',
  source: 'Beacon Fen Energy Park DCO, granted Aug 2026 (EN010151)' },
'13600': { works: "up to four 33-400 kV transformers (160 t, up to 15 x 9.5
x 10.5 m each) in a compound of up to 40,000 m2 (ES Ch.2 s2.8)",
  poc_works: "a National Grid-delivered extension of Bicker Fen, AIS or GIS,
sited for multiple customers (s2.13)",
  substation: 'Bicker Fen Substation',
  via: 'a 400 kV cable and a consented extension of Bicker Fen shared with
Heckington Fen',
  source: 'Beacon Fen Energy Park DCO, granted Aug 2026 (EN010151)' },
'9806': { works: "transformers with bunding and blast walls, switchgear, and
harmonic filtering reactive power compensation (Work No. 4)",
  poc_works: "a new generation bay plus an AIS-or-GIS extension and a cable
sealing end compound at Bicker Fen (Works 6A-6C)",
  substation: 'Bicker Fen Substation',
  via: 'the consented Bicker Fen extension shared with Beacon Fen',
  source: 'Heckington Fen Solar Park DCO, granted (EN010123)' },
'9807': { works: "transformers with bunding and blast walls, switchgear, and
harmonic filtering reactive power compensation (Work No. 4)",
  poc_works: "a new generation bay plus an AIS-or-GIS extension and a cable
sealing end compound at Bicker Fen (Works 6A-6C)",
  substation: 'Bicker Fen Substation',
  via: 'the consented Bicker Fen extension shared with Beacon Fen',
  source: 'Heckington Fen Solar Park DCO, granted (EN010123)' },
'13644': { poc_status: 'under_construction',
  poc_status_note: 'a new 400 kV four-bay substation is under construction
at Thorpe Marsh',
  substation: 'Thorpe Marsh Substation',
  via: 'a new 400 kV four-bay substation under construction at Thorpe
Marsh',
  source: 'public planning and contractor records; construction under way'
},
'19801': { poc_status: 'under_construction',
  poc_status_note: 'a new 400 kV four-bay substation is under construction
at Thorpe Marsh',
  substation: 'Thorpe Marsh Substation',
  via: 'a new 400 kV four-bay substation under construction at Thorpe
Marsh',
  source: 'public planning and contractor records; construction under way'
},
/* Little Crow is the counter-archetype and belongs here precisely
because it is NOT a 400 kV story: no customer transmission
substation, no long cable, and a point of connection that is a
circuit crossing the site rather than a substation to draw a line
to. Stating that plainly is worth more than drawing nothing. */
'6557': { poc_kind: 'circuit', poc_status: 'existing',
  circuit: 'the Keadby \u2013 Broughton \u2013 Teed \u2013 Scawby Brook
overhead 132 kV line circuit (Northern Powergrid)',
  via: 'a looped connection into an existing 132 kV circuit within the site,
with 99.9 MW of export capacity secured',
  kv: 132,
  source: 'Little Crow Solar Park Grid Network Constraints Report, EN010101,
November 2020' },
'7175': { poc_kind: 'circuit', poc_status: 'existing',
  circuit: 'the Keadby \u2013 Broughton \u2013 Teed \u2013 Scawby Brook
overhead 132 kV line circuit (Northern Powergrid)',
  via: 'a looped connection into an existing 132 kV circuit within the site,

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with 99.9 MW of export capacity secured',
    kv: 132,
    source: 'Little Crow Solar Park Grid Network Constraints Report, EN010101,
November 2020' },
    '11928': { substation: 'West Burton Substation',
    via: 'a 400 kV grid connection at the former power station site (West
Burton C); financial close July 2026',
    source: 'public project records' }
});

/* Public works at named substations, shown wherever the name is - the
"customer and NG substations that do not exist yet" half of the logic.
Descriptions of the network, never advice about a scheme. */

const SUBSTATION_WORKS = Object.freeze({
    'thorpe marsh substation':
        'A new 400 kV four-bay substation is under construction here (public
record).',
    'high marnham substation':
        'NGET is building a new substation adjacent to the existing one (Great
Grid Upgrade, public record).',
    'bicker fen substation':
        'A consented extension here will connect Beacon Fen and Heckington Fen
(public record).',
});

const worksAt = (name) => SUBSTATION_WORKS[String(name || '').toLowerCase()]
|| null;

/* What the Order says is known the moment the identity is known: the
substation, the voltage class, the route, the consented works and the
citation need no payload, no fetch and no map. The distance is the one
part that must be measured, so it is the one part marked pending. */
function provisional(repdRef) {
    const declared = RECORDS[String(repdRef || '')];
    if (!declared) return null;
    if (declared.poc_kind === 'circuit') {
        // Nothing to measure to and nothing to draw: say what is declared.
        return { poc: declared.circuit, kv: declared.kv || null, at: null,
            km: null, pending: false, kind: 'circuit',
            poc_status: declared.poc_status || 'existing',
            via: declared.via, source: declared.source, works: null,
            customer_works: declared.works || null, poc_works: declared.poc_works ||
null };
    }
    return {
        poc: declared.substation, kv: 400, at: null, km: null, pending: true,
        kind: 'substation', poc_status: declared.poc_status || 'existing',
        poc_status_note: declared.poc_status_note || null,
        via: declared.via, source: declared.source,
        works: worksAt(declared.substation),
        customer_works: declared.works || null,
        poc_works: declared.poc_works || null
    };
}

/* Bind the declared name to the served payload. Only a substation of the
declared class (>= 400 kV) with exactly that name counts; a 132 kV site
that happens to share the name is not the point of connection. */
function resolve(repdRef, origin, subs) {
    const declared = RECORDS[String(repdRef || '')];
    if (!declared) return null;

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if (declared.poc_kind === 'circuit') return provisional(repdRef);
const wanted = declared.substation.toLowerCase();
const works = SUBSTATION_WORKS[wanted] || null;
const match = (Array.isArray(subs) ? subs : [])
  .filter(s => String(s.name).toLowerCase() === wanted
    && Array.isArray(s.kv) && s.kv[0] >= 400)
  .sort((a, b) => b.kv[0] - a.kv[0])[0] || null;
if (!match) {
  return { poc: declared.substation, kv: 400, at: null, km: null,
    kind: 'substation', poc_status: declared.poc_status || 'existing',
    poc_status_note: declared.poc_status_note || null,
    via: declared.via, source: declared.source, works,
    customer_works: declared.works || null,
    poc_works: declared.poc_works || null };
}
return { poc: match.name, kv: Math.round(match.kv[0]), at: match.at,
  km: distanceKm(origin[0], origin[1], match.at[0], match.at[1]),
  kind: 'substation', poc_status: declared.poc_status || 'existing',
  poc_status_note: declared.poc_status_note || null,
  via: declared.via, source: declared.source, works,
  customer_works: declared.works || null,
  poc_works: declared.poc_works || null };
}

/* The nearest transmission (>= 400 kV) substation in the payload, and
separately the nearest one WITH A NAME: an unnamed OSM node can win on
raw distance and the reader still wants an identity. Two measurements,
no judgement about either. */
function nearestTransmission(origin, subs) {
  let best = null;
  let bestNamed = null;
  /* The size of the sample this superlative is drawn from. Counted here
rather than recounted by a caller, because the predicate below is
what decides eligibility and a second implementation of it would
drift from this one. */
  let considered = 0;
  for (const s of (Array.isArray(subs) ? subs : [])) {
    if (!(Array.isArray(s.kv) && s.kv[0] >= 400)) continue;
    considered += 1;
    const km = distanceKm(origin[0], origin[1], s.at[0], s.at[1]);
    if (!best || km < best.km) {
      best = { name: s.name || 'Unnamed substation', km, at: s.at };
    }
    if (s.name && (!bestNamed || km < bestNamed.km)) {
      bestNamed = { name: s.name, km, at: s.at };
    }
  }
  if (best) {
    best.considered = considered;
    best.works = worksAt(best.name);
    if (bestNamed && bestNamed.name !== best.name) {
      best.named = bestNamed;
      best.named.works = worksAt(bestNamed.name);
    }
  }
  return best;
}

NS.declaredConnections = Object.freeze({
  schema: 'gridatlas.module.declared-connections.v1',
  records: RECORDS,
  substationWorks: SUBSTATION_WORKS,

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    count: Object.keys(RECORDS).length,
    isDeclared: (repdRef) => Object.prototype.hasOwnProperty.call(RECORDS,
String(repdRef || '')),
    worksAt,
    provisional,
    resolve,
    nearestTransmission
  });
})();

/**
 * Module: sizing-arithmetic
 *
 * The screening arithmetic of the SLD sandbox: physical inputs to array
 * statistics, the three named ratios (design, export, headroom), the
 * string and central topologies with their corrected nameplates, the
 * finance port of gis-sld-v5-finance.js, and the two-variable fit that
 * lands a layout on the capacity the register states.
 *
 * Lifted out of the sld-sandbox body at generation 202609012205 (UTC),
 * expression for expression. The body closed over its state object and
 * its finance defaults; here both are parameters. Nothing else changed,
 * and the parity proof evaluates the last inline copy beside this module
 * on the same inputs and asserts identical values.
 *
 * WHAT IT WILL NOT DO
 * It grades nothing. A ratio below one is stated with its meaning; an
 * export set by the transformers is stated as the design fact it is. The
 * finance figures are a screening model with the reference's own inputs
 * and are labelled as such by the panel that shows them.
 *
 * Pure. No DOM, no network, no state of its own: fitToStatedCapacity
 * mutates the state object it is handed, as the body's did, and says so.
 */
((() => {
  const NS = window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ || {};
  if (NS.sizingArithmetic) return;

  function physicalInputs(inputs) {
    const i = inputs;
    if (i.mode === 'central') {
      return {
        mod_wp: i.mod_wp_c, mod_l: i.mod_l_c, mod_w: i.mod_w_c,
        gcr: i.gcr_c, gross_factor: i.gross_factor_c,
      };
    }
    return {
      mod_wp: i.mod_wp, mod_l: i.mod_l, mod_w: i.mod_w,
      gcr: i.gcr, gross_factor: i.gross_factor,
    };
  }

  function buildStats(inputs, o) {
    const p = physicalInputs(inputs);
    const dcMwp = (o.module_count * p.mod_wp) / 1e6;
    const acMw = o.ac_mw_direct !== null ? o.ac_mw_direct
      : (o.dc_ac_ratio > 0 ? dcMwp / o.dc_ac_ratio : 0);
    const netModArea = o.module_count * p.mod_l * p.mod_w;
    const netArrayArea = p.gcr > 0 ? netModArea / p.gcr : 0;
    return {
      total_blocks: o.total_blocks,

```

```

module_count: o.module_count,
dc_mwp: dcMwp,
ac_mw: acMw,
dc_ac_ratio: acMw > 0 ? dcMwp / acMw : o.dc_ac_ratio,
net_array_area_m2: netArrayArea,
gross_site_area_m2: netArrayArea * p.gross_factor,
block_ground_area_m2: o.total_blocks > 0 ? netArrayArea / o.total_blocks :
0,
production_substation_ac_mva: o.production_substation_ac_mva || 0,
ring_main_ac_mva: o.ring_main_ac_mva || 0,
warning: o.warning || 'Check skid rating, transformer rating, cable
ratings, protection, losses and grid compliance.'
};
}

```

/* Three numbers that must agree, and did not.

Measured on the shipped defaults, the panel produced three different values for one quantity:

string	stated DC/AC input	1.200
	reported DC/AC	1.040
	implied by the hardware	0.945

A DC/AC ratio below one is not a design choice, it is a contradiction: it says the array is smaller than the inverters it feeds, which nobody builds. And in central mode the reported ratio was 2.402 against an inverter ratio of 1.200 – exactly double, because AC had correctly become the LIMITING nameplate (the transformers) while the ratio was still being read as though it were the inverter nameplate. Both numbers were right about different things and both were called DC/AC.

There are three distinct quantities here and the panel now keeps them apart by name:

DC	the array, MWp
inverter AC	the inverters can convert, MW
export	the smaller of the inverters and the transformers, MVA

The DESIGN ratio is DC over inverter AC, which is the number the industry means by DC/AC and the one a stated 1.2 refers to. The EXPORT ratio is DC over the export limit, which is what determines clipping and curtailment. Reporting one of them under the other's name is how a plant ends up described as 2.4 when it was specified as 1.2.

Nothing here changes a layout. It changes what the numbers are called, and says so out loud when they disagree with each other. */

/* There was an auto-reconciler here. It is deleted, not disabled.

It computed a "consistent" strings-per-inverter count from the stated DC/AC ratio and assigned it to sld.inputs.z_strings, on the reasoning that the original's 18 gives a block DC/AC of 0.945 and that nobody builds an array smaller than its own inverters. That reasoning was wrong: the reference documents 28 string inverters at 352 kVA making 9,856 kVA ahead of an 8.96 MVA skid, and the oversizing is the design.

The default was reverted, and the reconciler was left behind uncalled. Flagged by the Codex source gate as a stop-ship, and it was right. Dead code that ASSIGNS to a reference input is not inert: it is one future handler away from silently rewriting the design this cartridge exists to reproduce, and it would do so quietly, in a place nobody would look.

This is the same lesson as the dead .grid-cell grading CSS removed from Pipeline News earlier tonight – a rule with no caller is one edit from having one – and I repeated the mistake within hours of writing it down. Deleted rather than commented out, for the same reason. */

```
function consistency(inputs, stats) {
  const i = inputs;
  const string = i.mode === 'string';

  const inverterAcMw = string
    ? (stats.total_blocks * i.y_invs * i.string_inv_kva) / 1000
    : stats.total_blocks * i.inv_ac_mw_c;
  const skidAcMva = string
    ? stats.total_blocks * i.string_skid_mva
    : (i.mv_per_ring_c * i.rings_c) * i.central_skid_mva_c;
  const exportMva = Math.min(inverterAcMw, skidAcMva);

  /* Three ratios, three names. They describe different pairs of things and
     collapsing them is how a plant specified at 1.2 gets reported as 2.4.

     design      array DC MWp / inverter AC MW      what "DC/AC" means
     export      array DC MWp / export MVA          what drives clipping
     headroom    inverter AC MW / export MVA        how hard the inverters are
                                                    pushed against their skids

     The third is the one that says whether the inverters are oversized
     against the transformers, and in this design they deliberately are. */
  const designRatio = inverterAcMw > 0 ? stats.dc_mwp / inverterAcMw : null;
  const exportRatio = exportMva > 0 ? stats.dc_mwp / exportMva : null;
  const headroomRatio = exportMva > 0 ? inverterAcMw / exportMva : null;
  const statedRatio = string ? Number(i.dc_ac_ratio) : (
    i.inv_ac_mw_c > 0 ? i.inv_dc_mw_c / i.inv_ac_mw_c : null);

  const notes = [];
  /* Descriptive, not a verdict.
     An earlier version of this called a design ratio below one a
     contradiction that "nobody builds". That was wrong about this design:
     the reference sandbox documents 28 string inverters at 352 kVA making
     9,856 kVA ahead of an 8.96 MVA skid, and oversizing inverters against
     the transformer is a deliberate choice, not an arithmetic fault. The
     panel states the number and what it means; it does not grade it. */
  if (Number.isFinite(designRatio) && designRatio < 1) {
    notes.push('Array DC divided by inverter AC is ' + designRatio.toFixed(2)
      + ' from the module, string and inverter counts shown.');
```

```

        + 'lower nameplate in this screening model. The connection agreement '
        + 'and electrical design determine the applicable export constraint.');
```

```

    }
    return {
      dc_mwp: stats.dc_mwp,
      inverter_ac_mw: inverterAcMw,
      skid_ac_mva: skidAcMva,
      export_mva: exportMva,
      design_dc_ac: designRatio,
      export_dc_ac: exportRatio,
      inverter_to_export: headroomRatio,
      stated_dc_ac: Number.isFinite(statedRatio) ? statedRatio : null,
      notes,
    };
  }
}
```

```

function stringStats(inputs) {
  const i = inputs;
  if (i.mod_wp <= 0 || i.mod_l <= 0 || i.mod_w <= 0 || i.x_mods <= 0) {
    return buildStats(i, { total_blocks: 0, module_count: 0, dc_ac_ratio:
i.dc_ac_ratio });
  }
  const total_blocks = i.b_cols * i.s_subs;
  const module_count = total_blocks * i.y_invs * i.z_strings * i.x_mods;
  const inverterAcMaxMva = (i.y_invs * i.string_inv_kva) / 1000;
  const production = i.string_skid_mva;
  let warning;
  if (inverterAcMaxMva > production) {
    warning = 'Inverter ACmax exceeds the skid transformer rating. Verify
temperature rating, overload strategy and clipping assumptions.';
  } else if (i.string_inv_kva > 500) {
    warning = 'Large string inverter rating selected. Verify LV switchgear,
transformer, cable loading and protection.';
  }
  return buildStats(i, {
    total_blocks, module_count, dc_ac_ratio: i.dc_ac_ratio,
    ac_mw_direct: total_blocks * production,
    production_substation_ac_mva: production,
    ring_main_ac_mva: production * i.s_subs,
    warning
  });
}
}
```

```

function centralStats(inputs) {
  const i = inputs;
  if (i.mod_wp_c <= 0 || i.mod_l_c <= 0 || i.mod_w_c <= 0 || i.x_mods_c <= 0)
{
    return buildStats(i, { total_blocks: 0, module_count: 0, dc_ac_ratio: 1.2
});
  }
  const strDcKwp = (i.x_mods_c * i.mod_wp_c) / 1000;
  const reqStrings = strDcKwp > 0 ? Math.ceil((i.inv_dc_mw_c * 1000) /
strDcKwp) : 0;
  // total_blocks counts INVERTERS: inverters per MV skid, times skids per
  // ring, times rings. The skids are the level above it.
  const total_blocks = i.inv_per_mv_c * i.mv_per_ring_c * i.rings_c;
  const skid_count = i.mv_per_ring_c * i.rings_c;
  const module_count = reqStrings * i.x_mods_c * total_blocks;

  /* Two nameplates, and they are not the same number.
```

```

  -----
  The inverters and the MV skid transformers they share are rated
```

separately, and the plant can export no more than the smaller of the two. On the shipped defaults they are a factor of two apart: 24 inverters at 4.4 MW is 105.6 MW of inverter, sitting on 12 skids at 4.4 MVA, which is 52.8 MVA of transformer.

The figure shown was 211.2 MW -- neither of those, and larger than both. `total\_blocks` already contains `inv\_per\_mv\_c`, and the AC line multiplied by it a second time, so the count of inverters sharing a skid entered the answer squared. It also multiplied a count of inverters by a TRANSFORMER rating, which is not a quantity that exists.

This is a deliberate divergence from the sandbox this was ported from. gis-sld-v5-calculations.js line 147 computes the same expression, so the fault is in the original and was carried across faithfully by a port whose whole contract was to carry the arithmetic unchanged.

Reported by the Codex session auditing this estate in parallel; confirmed here dimensionally and against those defaults. */

```
const inverter_ac_total = total_blocks * i.inv_ac_mw_c;
const skid_ac_total = skid_count * i.central_skid_mva_c;
const ac_mw_direct = Math.min(inverter_ac_total, skid_ac_total);

// A skid carries every inverter fed into it, so the comparison that
// matters is the whole MV block against its transformer, not one
// inverter against it. One-to-one it never fires; on the defaults the
// block is 8.8 MW on a 4.4 MVA skid and it should.
const block_ac_mw = i.inv_ac_mw_c * i.inv_per_mv_c;
let warning;
if (block_ac_mw > i.central_skid_mva_c) {
  warning = `The ${i.inv_per_mv_c} inverters on each MV skid total `
    + `${block_ac_mw.toFixed(2)} MW against a skid rated `
    + `${i.central_skid_mva_c} MVA. Export is limited by the transformer, `
    + `not the inverters. Verify thermal rating, overload strategy and `
    + `the export limit in the connection agreement.`;
} else if (i.inv_ac_mw_c > 10) {
  warning = 'Large central inverter or power block selected. Verify
transformer, MV switchgear, harmonics, thermal loading, protection and grid code
compliance.';
}
return buildStats(i, {
  total_blocks, module_count,
  dc_ac_ratio: i.inv_ac_mw_c > 0 ? i.inv_dc_mw_c / i.inv_ac_mw_c : 1.2,
  ac_mw_direct,
  // One skid's rating. The label on the control is "Skid MVA", so it is
  // the skid, and multiplying it by the inverters on that skid described
  // no piece of equipment.
  production_substation_ac_mva: i.central_skid_mva_c,
  ring_main_ac_mva: i.central_skid_mva_c * i.mv_per_ring_c,
  central_inverter_ac_total: inverter_ac_total,
  central_skid_ac_total: skid_ac_total,
  warning
});
}

const DEVELOPMENT_STAGES = Object.freeze({
  '0.003': 'Land Option Signed',
  '0.015': 'Grid Connection Application Accepted',
  '0.035': 'Planning Application Submitted',
  '0.055': 'Planning Permission Granted',
  '0.070': 'Grid Connection Terms Reviewed and Agreed',
  '0.080': 'Buyer or Revenue Agreement Reviewed (Power Purchase Agreement
(PPA) / Offtaker)',
```

```

    '0.100': 'Construction Contract Signed and Finance Committed (Financial
Close)',
  });

  const financeNumber = value => {
    const number = Number(value);
    return Number.isFinite(number) ? number : 0;
  };

  const DEVELOPMENT_SUCCESS = Object.freeze({
    '0.003': 10,
    '0.015': 15,
    '0.035': 30,
    '0.055': 55,
    '0.070': 70,
    '0.080': 80,
    '0.100': 95,
  });

  const BIFACIAL_BY_GCR = Object.freeze({
    '0.35': 8,
    '0.45': 5,
    '0.75': 2,
  });

  /* The original stage selector is not only a label: its change handler sets
  development cost to the selected GBP/Wp value and success probability to
  a stage-specific percentage. Keep that linked behavior explicit so a
  stage change cannot leave the old stage's costs behind. */
  function applyDevelopmentStageDefaults(financeInputs, stageValue) {
    const stage = String(stageValue);
    if (!Object.prototype.hasOwnProperty.call(DEVELOPMENT_STAGES, stage)) return
false;
    financeInputs.dev_stage = stage;
    financeInputs.dev_cost_mw = financeNumber(stage);
    financeInputs.dev_success = DEVELOPMENT_SUCCESS[stage];
    return true;
  }

  /* Original Mounting & GCR presets also set the financial bifacial gain.
  Apply the exact three preset mappings to the active topology only. A
  free-form GCR value does not invent a gain. */
  function applyMountingBifacial(financeByMode, mode, gcrValue) {
    const values = (financeByMode || {})[mode];
    if (!values) return false;
    const key = String(Number(gcrValue));
    if (!Object.prototype.hasOwnProperty.call(BIFACIAL_BY_GCR, key)) return
false;
    values.bifacial = BIFACIAL_BY_GCR[key];
    return true;
  }

  /* Direct port of gis-sld-v5-finance.js computeFinance(). The original
  executable fixture is the authority, not this comment. The one deliberate
  divergence is inherited from the corrected electrical port: annual OPEX
  uses the corrected central inverter nameplate, so the inv_per_mv > 1 case
  must match the fixture's explicit corrected surplus rather than repeat the
  original AC double-count. Every unaffected output remains exact. */
  function screeningFinance(financeInputs, stats, context) {
    const f = financeInputs || (context && context.defaults) || {};
    const dcMwp = financeNumber(stats?.dc_mwp);
    // The reference's OPEX input is GBP/MWac/year. In string mode its AC

```

```

// quantity is skid-limited export. In central mode, once the known square
// is removed, it is inverter count x inverter MWac. Do not silently swap
// that to transformer-limited export: those are separately named values.
const centralInverterAc = (stats?.mode || (context && context.fallbackMode))
=== 'central'
  ? financeNumber(stats?.consistency?.inverter_ac_mw) : 0;
const acMw = centralInverterAc > 0 ? centralInverterAc :
financeNumber(stats?.ac_mw);
const price = financeNumber(f.price);
const other = financeNumber(f.other);
const yieldVal = financeNumber(f.yield);
const bifacial = financeNumber(f.bifacial);
const baseLoss = financeNumber(f.losses);
const deg = financeNumber(f.deg);
const opexRate = financeNumber(f.opex);
const epcEx = financeNumber(f.epc_ex);
const floodRate = financeNumber(f.flood_rate);
const floodAdder = f.flood ? floodRate : 0;
const modules = financeNumber(f.modules);
const otherCapex = financeNumber(f.other_capex);
const fixedCapex = financeNumber(f.fixed_capex);
const cont = financeNumber(f.cont);
const lossExtras = financeNumber(f.loss_dc_string) +
financeNumber(f.loss_lv_dc)
  + financeNumber(f.loss_lv_ac) + financeNumber(f.loss_tx) +
financeNumber(f.loss_other);
const totalLoss = baseLoss + lossExtras;
const bessMw = financeNumber(f.bess_mw);
const bessMwh = financeNumber(f.bess_mwh);
const bessCapexRate = financeNumber(f.bess_capex);
const bessCycles = financeNumber(f.bess_cycles);
const bessRevenuePerMwh = financeNumber(f.bess_spread);
const bessEffPercent = financeNumber(f.bess_eff);
const safeLoss = Math.min(Math.max(totalLoss, 0), 100);
const safeBessEff = Math.min(Math.max(bessEffPercent / 100, 0), 1);
const effectiveYield = yieldVal * (1 + bifacial / 100);
const year1Gen = dcMwp * effectiveYield * (1 - safeLoss / 100);
let gen25 = 0;
let gen35 = 0;
for (let year = 1; year <= 35; year += 1) {
  const generation = year1Gen * Math.pow(1 - deg / 100, year - 1);
  if (year <= 25) gen25 += generation;
  gen35 += generation;
}
const annualSolarRevenue = year1Gen * (price + other);
const bessAnnualValue = bessMwh * bessCycles * bessRevenuePerMwh *
safeBessEff;
const annualRevenue = annualSolarRevenue + bessAnnualValue;
const revenue25 = gen25 * (price + other) + bessAnnualValue * 25;
const revenue35 = gen35 * (price + other) + bessAnnualValue * 35;
const annualOpex = acMw * opexRate;
const baseCapexWp = epcEx + modules + otherCapex + floodAdder;
const baseCapex = dcMwp * 1_000_000 * baseCapexWp;
const contingency = baseCapex * (cont / 100);
const bessCapex = bessMwh * bessCapexRate;
const totalCapex = baseCapex + contingency + fixedCapex + bessCapex;
const capexPerWp = dcMwp > 0 ? totalCapex / (dcMwp * 1_000_000) : 0;
const surplus25 = revenue25 - annualOpex * 25 - totalCapex;
const surplus35 = revenue35 - annualOpex * 35 - totalCapex;
const devCostPerMw = financeNumber(f.dev_cost_mw);
const devModulePerMwp = financeNumber(f.dev_module_mwp);
const devEpcPerMw = financeNumber(f.dev_epc_mw);

```

```

const devOwnerPerMw = financeNumber(f.dev_owner_mw);
const devGridPerMw = financeNumber(f.dev_grid_mw);
const devExitPerMwp = financeNumber(f.dev_exit_mwp);
const devNpvPerMwp = financeNumber(f.dev_npv_mwp);
const devSuccessPct = financeNumber(f.dev_success);
const devYears = financeNumber(f.dev_years);
const devStage = DEVELOPMENT_STAGES[String(f.dev_stage)] || 'Manual';
const wpCapacity = dcMwp * 1_000_000;
const devCapitalAtRisk = wpCapacity * devCostPerMw;
const devModuleCost = wpCapacity * devModulePerMwp;
const devEpcCost = wpCapacity * devEpcPerMwp;
const devOwnerCost = wpCapacity * devOwnerPerMw;
const devGridCost = wpCapacity * devGridPerMw;
const devTotalBuildCost = devCapitalAtRisk + devModuleCost + devEpcCost
  + devOwnerCost + devGridCost;
const devExitValue = wpCapacity * devExitPerMwp;
const devOperatingNpv = wpCapacity * devNpvPerMwp;
const devGrossMargin = devExitValue - devTotalBuildCost;
const devRiskAdjustedValue = devGrossMargin * (devSuccessPct / 100);
const devReturnMultiple = devCapitalAtRisk > 0 ? devGrossMargin /
devCapitalAtRisk : 0;
return {
  annualRevenue, revenue25, revenue35, totalCapex, capexPerWp, surplus25,
surplus35,
  devStage, devCostPerMw, devModulePerMwp, devEpcPerMw, devOwnerPerMw,
  devGridPerMw, devExitPerMwp, devNpvPerMwp, devSuccessPct, devYears,
  devCapitalAtRisk, devModuleCost, devEpcCost, devOwnerCost, devGridCost,
  devTotalBuildCost, devExitValue, devOperatingNpv, devGrossMargin,
  devRiskAdjustedValue, devReturnMultiple, price, other, yieldVal, bifacial,
  baseLoss, deg, opexRate, epcEx, floodActive: Boolean(f.flood), floodRate,
  modules, otherCapex, fixedCapex, cont, totalLoss, bessMw, bessMwh,
  bessCapexRate, bessCycles, bessSpread: bessRevenuePerMwh,
  bessEff: bessEffPercent, epcIncModules: epcEx + modules,
};
}

function computeStats(inputs, financeByMode, defaults) {
  const stats = inputs.mode === 'string'
    ? stringStats(inputs) : centralStats(inputs);
  // Same object, so nothing can read a capacity without the check that says
  // whether the capacities agree with each other.
  stats.mode = inputs.mode;
  stats.consistency = consistency(inputs, stats);
  stats.finance = screeningFinance((financeByMode || {})[inputs.mode], stats,
    { fallbackMode: inputs.mode, defaults });
  return stats;
}

/**
 * Size the array so its capacity lands on the figure the register states.
 *
 * WHAT IS ADJUSTED, AND WHAT IS NOT
 * Two integer topology counts move -- circuits and skids per circuit in
 * string mode, rings and MV skids per ring in central mode. Everything a
 * supplier fixes stays where the user put it:
 * module rating, string length, inverter and skid ratings. That keeps the
 * result buildable rather than a number reverse-engineered into nonsense.
 *
 * Blocks are integers, so an exact hit is usually impossible. The residual
 * is reported rather than hidden, because a layout that quietly lands 7%
 * off the stated capacity is worse than one that says so.
 *

```



```

* WHICH CAPACITY IS BEING MATCHED
* That is the caller's declared basis, never a guess. REPD's figure is
* nominally MWelec, but it is reported inconsistently: some schemes state
* DC, some AC, and the register does not carry the distinction reliably.
* Matching AC when the figure was DC oversizes the connection by the DC/AC
* ratio, which is exactly the error that matters for export limitation.
*/
/* Fit on two variables, because one cannot reach a small project.
-----
Reported: the numbers do not change when the headline capacity changes.
Measured, and they do not:

    string    5, 10, 20, 30, 40, 49.9 and 50 MW all produced 44.80 MW
    central   5, 10 and 20 MW all produced 17.60 MW

The fit moved ONE variable. In string mode that is b_cols, and because
total_blocks is b_cols x s_subs with s_subs pinned at five, one step of
b_cols is five blocks – 44.8 MW at the default skid rating. Nothing below
that is reachable, so a 30 MW solar farm was drawn as a 44.8 MW one, an
overstatement of half as much again, and every target under 50 MW
collapsed onto the same layout. The register starts at 1 MW.

A block is 8.96 MW in string mode and a skid is 4.4 MVA in central. Those
are the real quanta, and they are reachable as soon as the inner variable
is allowed to move too. So the search is over both, and it prefers the
candidate that stays closest to the shape the user already had – a fit
that reaches the right capacity by rearranging the whole plant is a worse
answer than one that reaches it by adding a column.

Bounds are physical rather than generous: a ring main carries a handful of
skids, not four hundred, so the inner variable stops at twelve. */
const FIT_OUTER_MAX = 120;
const FIT_INNER_MAX = 12;

function fitToStatedCapacity(sld, computeSldStats) {
    sld.fitResidualPct = null;
    sld.fitQuantumMw = null;
    const target = Number(sld.targetMw);
    if (!Number.isFinite(target) || target <= 0) return;
    if (sld.targetBasis !== 'ac' && sld.targetBasis !== 'dc') return;

    const string = sld.inputs.mode === 'string';
    const outerKey = string ? 'b_cols' : 'rings_c';
    const innerKey = string ? 's_subs' : 'mv_per_ring_c';
    const outer0 = sld.inputs[outerKey];
    const inner0 = sld.inputs[innerKey];

    let best = null;
    for (let inner = 1; inner <= FIT_INNER_MAX; inner += 1) {
        sld.inputs[innerKey] = inner;
        for (let outer = 1; outer <= FIT_OUTER_MAX; outer += 1) {
            sld.inputs[outerKey] = outer;
            const s = computeSldStats();
            const got = sld.targetBasis === 'ac' ? s.ac_mw : s.dc_mwp;
            if (!Number.isFinite(got) || got <= 0) continue;
            const error = Math.abs(got - target);
            // Ties, and near-ties, go to the layout closest to the one already on
            // screen. Without this the fit rearranges the plant for a rounding
            // difference and the drawing jumps for no reason the user can see.
            const drift = Math.abs(inner - inner0) + Math.abs(outer - outer0) / 100;
            if (!best
                || error < best.error - 1e-9

```

```

        || (Math.abs(error - best.error) <= 1e-9 && drift < best.drift)) {
      best = { outer, inner, error, got, drift };
    }
  }
}
if (!best) {
  sld.inputs[outerKey] = outer0;
  sld.inputs[innerKey] = inner0;
  return;
}
sld.inputs[outerKey] = best.outer;
sld.inputs[innerKey] = best.inner;
sld.fitResidualPct = ((best.got - target) / target) * 100;

// What one more block would have added. A residual means nothing without
// it: 10% off a plant whose smallest step is 9 MW is exact, and 10% off
// one whose step is 0.5 MW is a miss.
const oneMore = (() => {
  sld.inputs[outerKey] = best.outer + 1;
  const s = computeSldStats();
  sld.inputs[outerKey] = best.outer;
  const got = sld.targetBasis === 'ac' ? s.ac_mw : s.dc_mwp;
  return Number.isFinite(got) ? Math.abs(got - best.got) : null;
})();
sld.fitQuantumMw = oneMore;
}

NS.sizingArithmetic = Object.freeze({
  generation: '202609012205',
  DEVELOPMENT_STAGES,
  DEVELOPMENT_SUCCESS,
  BIFACIAL_BY_GCR,
  FIT_OUTER_MAX,
  FIT_INNER_MAX,
  financeNumber,
  physicalInputs,
  buildStats,
  consistency,
  stringStats,
  centralStats,
  applyDevelopmentStageDefaults,
  applyMountingBifacial,
  screeningFinance,
  computeStats,
  fitToStatedCapacity
});
})();
})();

```

```

/*
technology-coverage - which technologies the measurement runs for, and
what each measurement is a measurement OF

```

WHAT WAS MEASURED BEFORE THIS MODULE WAS WRITTEN.

The request was to extend the nearest-substation computation to the twenty wider-fleet REPD technologies, on the report that it ran only for the spine. Driven in Chrome against the live composition (v9.88, generation 202609030234) on the wider fleet's own MAP link:

```

?repd_ref=8795 ... &technology=biomass    Caledon Green, Landfill Gas
?repd_ref=626  ... &technology=biomass    Pitsea Tipp,   Landfill Gas

```

Both arrived, flew, opened the card, ticked the Subs control and DREW THE LINKS. Caledon Green measured 1.74 km at 132 kV, 2.98 km at 275 kV and 5.70 km at 33 kV; Pitsea Tipp named Coryton South Substation at 7.70 km and printed its 400 kV scope sentence with the denominator in it. So the wider fleet was ALREADY computing, and a change that claimed to enable it would have been a change that did nothing while saying it did something.

There IS a four-member technology allow-list in this estate:

```
const allowedTechnologies = new
Set(['solar', 'bess', 'wind_onshore', 'wind_offshore']);
```

It lives at line 805 of the IMMUTABLE SHELL, atlas/releases/202608300453-atlas-v9/ventus-corev8engine.js, inside the shell's own focusCanonicalProjectDeepLink(). It rejects every value the wider fleet can send, because those four are exactly the four types the wider fleet is DEFINED as excluding. Its rejection is caught, and its only effects are a console line and a flyTo the arrival lane in this cartridge has already performed - which is why the measurement runs anyway. The shell is carried forward verbatim by contract and this module does not reach into it.

WHAT THIS MODULE ACTUALLY CHANGES.

One thing: OFFSHORE WIND NOW MEASURES. It used to open a card and withhold the distance, on the reasoning that a turbine in the North Sea does not reach the nearest onshore substation by a straight line. The reasoning about routes was right and is kept in full below; the conclusion was over-cautious. An offshore project's export cable does land at an onshore substation, so the distance to the nearest mapped substation is a real measurement of a real thing, provided the card says what it measured.

WHAT "NEAREST" CANNOT BE MADE TO MEAN HERE, MEASURED RATHER THAN ASSUMED.

The coordinator asked for an onshore-only filter, so that "nearest onshore substation" would mean onshore. The pinned substation product cannot carry one. Counted over all 5,800 features of

```
atlas/releases/202608300453-atlas-v9/data/grid_substations.geojson
```

the only properties present are voltage (5,800), name (4,460), operator (3,264), brand (1,310), source (684), type (72), capacity (15) and colour (3). The OSM `location` tag - the field that would say offshore, platform or underwater - is present on ZERO of them.

That leaves the name, and the name does not separate them either. Fourteen features carry "offshore" in their name; read against their coordinates, at least four are ONSHORE substations serving an offshore wind farm, which is precisely what an offshore project should be measured to - Hornsea (-0.2598, 53.6582) and Hornsea Two (-0.2604, 53.6568) at 400/220 kV, Thanet's explicitly-named onshore substation (1.3459, 51.3089), and the European Offshore Wind Deployment Centre (-2.0650, 57.2158). The rest are genuinely platforms at sea: Neart na Gaoithe North and South, Sheringham Shoal 1 and 2, Humber Gateway, Westernmost Rough, Rampion, Burbo Bank 2.

So a name filter would drop Hornsea - a landfall connection - from the very search it was supposed to sharpen. No onshore filter is applied, and this module says so on the card instead of pretending to one. An offshore project is measured against the SAME 5,800 features, at the SAME ≥ 33 kV floor, by the SAME straight line as every other technology, and the card carries two extra sentences: that the line crosses water and is not the export cable, and that the set searched contains substations that are

themselves offshore - named where one is returned, so the reader can see which. A stated limit, rather than a hidden one. A filter whose predicate is wrong four times in fourteen is worse than no filter, because it looks like precision.

WHAT DOES NOT CHANGE, AND MUST NOT.

- The straight line stays. It is the measurement, it is first, and the corridor-estimate module still sits beside it saying how far off a built route typically is. Nothing here replaces it.
- Every superlative keeps carrying its sample. This module supplies the sample LABEL for each policy so the card cannot print "nearest" without printing what it searched.
- The coordinate denominator stays. The operator publishes connection points and only a fraction carry coordinates; that count is computed at render time by nearestScope() and is not restated here, because a literal would go quietly false the day the pinned product moves.
- Nothing here grades anything. No verdict word appears anywhere in this module, not even to disown one: the sandbox proof greps the served bytes for them and cannot tell a comment from a card, which is the right way round. A distance and a voltage, stated, and the things a distance cannot answer named rather than implied.

THE ROSTER IS BY NAME ON PURPOSE.

The Atlas never receives a raw REPD technology. Pipeline News' MAP link sends `technology=<t>`, the COLOUR BUCKET - so twenty raw types arrive as nine bucket values, and "Landfill Gas" reaches this cartridge as "biomass". Every one of the twenty is listed below against its bucket, so a proof can assert coverage by REPD name rather than by bucket, and so anyone reading this can see that the twenty are accounted for rather than assumed.

*/

```
(function installTechnologyCoverage() {
  'use strict';

  const SCHEMA = 'gridatlas.technology-coverage.v1';

  /* The twenty wider-fleet REPD technologies, each against the bucket the
     MAP link actually sends. Counts are the live wider-fleet payload,
     202609030009, 1,104 rows - carried so a drift in either side is visible
     rather than silent. Every one of the 1,104 rows carries a usable
     coordinate pair; none is withheld for want of a location. */
  const WIDER_FLEET = Object.freeze([
    Object.freeze({ repd: 'Landfill Gas',          bucket:
'biomass',      rows: 275 }),
    Object.freeze({ repd: 'Anaerobic Digestion',    bucket:
'biomass',      rows: 253 }),
    Object.freeze({ repd: 'Biomass (dedicated)',     bucket:
'biomass',      rows: 159 }),
    Object.freeze({ repd: 'EfW Incineration',        bucket:
'biomass',      rows: 122 }),
    Object.freeze({ repd: 'Small Hydro',             bucket: 'hydro',
rows: 108 }),
    Object.freeze({ repd: 'Hydrogen',               bucket:
'hydrogen',     rows: 60 }),
    Object.freeze({ repd: 'Advanced Conversion Technologies', bucket: 'act',
rows: 37 }),
    Object.freeze({ repd: 'Large Hydro',            bucket: 'hydro',
rows: 28 }),
    Object.freeze({ repd: 'Pumped Storage Hydroelectricity', bucket: 'hydro',
```

```

rows: 15 }},
    Object.freeze({ repd: 'Tidal Stream',                bucket: 'tidal',
rows: 14 }},
    Object.freeze({ repd: 'Sewage Sludge Digestion',      bucket:
'biomass',      rows: 12 }},
    Object.freeze({ repd: 'Geothermal',                  bucket:
'geothermal', rows: 5 }},
    Object.freeze({ repd: 'Shoreline Wave',              bucket: 'tidal',
rows: 4 }},
    Object.freeze({ repd: 'Liquid Air Energy Storage',    bucket: 'caes',
rows: 2 }},
    Object.freeze({ repd: 'Biomass (co-firing)',          bucket:
'biomass',      rows: 2 }},
    Object.freeze({ repd: 'Hot Dry Rocks (HDR)',          bucket:
'geothermal', rows: 2 }},
    Object.freeze({ repd: 'Compressed Air Energy Storage', bucket: 'caes',
rows: 2 }},
    Object.freeze({ repd: 'Fuel Cell (Hydrogen)',         bucket:
'hydrogen',     rows: 2 }},
    Object.freeze({ repd: 'Flywheels',                   bucket:
'flywheel',     rows: 1 }},
    Object.freeze({ repd: 'Unknown',                     bucket: 'other',
rows: 1 })
  });

  /* The spine, for completeness: the four the wider fleet is defined as
     excluding, and the four the shell's allow-list accepts. */
  const SPINE = Object.freeze(['solar', 'bess', 'wind_onshore',
'wind_offshore']);

  /* Offshore wind, in every spelling the register and the engine use. This
     is no longer a withholding set - it selects a DIFFERENT NOTE, not a
     different answer. */
  const OFFSHORE_TECHS = Object.freeze([
    'wind_offshore', 'wind_offshore_operational'
  ]);
  const OFFSHORE = new Set(OFFSHORE_TECHS);

  /* Named as offshore in the substation product. Used only to LABEL a
     returned row, never to remove one - see the header for the four onshore
     substations this pattern also matches, which is exactly why it does not
     filter. */
  const OFFSHORE_NAMED = /\boffshore\b/i;
  const ONSHORE_NAMED = /\bonshore\b/i;

  const PRODUCT = Object.freeze({
    features: 5800,
    with_location_tag: 0,
    offshore_in_name: 14,
    of_those_onshore: 4,
    source: 'atlas/releases/202608300453-atlas-v9/data/grid_substations.geojson'
  });

  /* The sentence the card prints under an offshore project's distances. It
     keeps every word of the old withholding note that was ABOUT ROUTES,
     because none of that reasoning was wrong; it drops only the conclusion
     that therefore nothing should be measured. */
  const OFFSHORE_NOTE =
    'This is a straight line from the project to the nearest mapped '
    + 'substation, and for an offshore project that line crosses water. It is '
    + 'not the export cable and not its length. An offshore project reaches an '
    + 'offshore substation, an export cable and a landfall before anything '

```

```

+ 'onshore, and the route inland is then chosen for consent and ground '
+ 'conditions rather than for distance, so the built length is longer than '
+ 'this by an amount no distance can tell you.';

const OFFSHORE_SET_NOTE =
  'The set searched is the same 5,800 mapped substations used for every '
+ 'other technology. It carries no field saying which of them are onshore: '
+ 'the OSM location tag is absent from all 5,800, and of the 14 whose name '
+ 'contains "offshore" at least 4 are onshore substations serving an '
+ 'offshore wind farm. No onshore filter is applied, because one built on '
+ 'the name would drop those 4 - including Hornsea at 400/220 kV, which is '
+ 'a landfall connection. Where a result is itself named as an offshore '
+ 'substation it is marked below.';

/* The label under which a measurement is made. The card must never print
   the word "nearest" without one of these beside it. */
const SAMPLE = Object.freeze({
  mapped_substations: 'nearest of the mapped substations at or above the '
+ 'voltage floor that this search could see'
});

function bucketOf(tech) {
  return String(tech == null ? '' : tech).trim();
}

/**
 * What the measurement is, for one technology id.
 *
 * There is no `measure: false` branch. Every technology the register or
 * the wider fleet can send is measured; what differs is the note that
 * goes with it. A technology this module has never heard of is measured
 * too - the arrival lane already continues past an unknown id, and
 * refusing arithmetic over two coordinates because a string was
 * unfamiliar is how 109 offshore projects got nothing at all.
 */
function policy(tech) {
  const id = bucketOf(tech);
  const offshore = OFFSHORE.has(id);
  return Object.freeze({
    technology: id || null,
    measure: true,
    offshore,
    sample: SAMPLE.mapped_substations,
    /* Both notes, in order, for offshore; nothing extra for the rest.
       The generic straight-line-is-not-a-route caveat is the card's own
       and is printed for every technology either way. */
    notes: Object.freeze(offshore ? [OFFSHORE_NOTE, OFFSHORE_SET_NOTE] : [])
  });
}

/**
 * Is this substation NAME one of the ones the product calls offshore?
 * Labelling only. A true here marks a row; it never removes one.
 */
function namedOffshore(name) {
  const text = String(name == null ? '' : name);
  return OFFSHORE_NAMED.test(text) && !ONSHORE_NAMED.test(text);
}

/** The roster, by REPD name, for a proof to assert against. */
function widerFleetNames() {
  return WIDER_FLEET.map(entry => entry.repd);
}

```

```

}

/** The bucket values the wider fleet's MAP link can actually send. */
function widerFleetBuckets() {
  return [...new Set(WIDER_FLEET.map(entry => entry.bucket))].sort();
}

/** Every wider-fleet technology measures. Stated as a function so a proof
  cannot pass by reading a literal that stopped being true. */
function measuredCount() {
  return WIDER_FLEET.filter(entry => policy(entry.bucket).measure).length;
}

const NS = (window.__GRIDATLAS_MODULES__ = window.__GRIDATLAS_MODULES__ ||
{});
NS.technologyCoverage = Object.freeze({
  schema: SCHEMA,
  wider_fleet: WIDER_FLEET,
  spine: SPINE,
  offshore_techs: OFFSHORE_TECHS,
  product: PRODUCT,
  offshore_note: OFFSHORE_NOTE,
  offshore_set_note: OFFSHORE_SET_NOTE,
  sample: SAMPLE,
  policy,
  namedOffshore,
  widerFleetNames,
  widerFleetBuckets,
  measuredCount,
  /* Said once, here, so no caller has to phrase it and none can soften it. */
  not_a_connection: 'A distance to a mapped substation is not a connection, '
    + 'a capacity, a queue position or an offer, for any technology on this '
    + 'list.',
  shell_allow_list_note: 'The four-member technology allow-list in the '
    + 'immutable shell rejects every wider-fleet value and is caught; the '
    + 'arrival lane in this cartridge has already flown and carded by then, '
    + 'so it costs a console line and nothing else. The shell is carried '
    + 'forward verbatim and is not edited here.'
});
})();

/**
 * GridAtlas cartridge – neon substation links and the SLD layout sandbox.
 *
 * Assembled under the generation named in the header above; this part
 * states no generation of its own, because a second identity inside the
 * same file is one that can disagree with the first. Slot: replace-script for
 * 202608292126-pre-snapped-config-adapter.js.
 *
 * WHAT IT DOES
 * -----
 * Select a solar, battery or onshore wind project and the map draws animated
 * neon lines from it to the nearest substations at 33 kV and above. The same
 * distances are written onto the project card the engine has just opened,
 * marked BETA, with the reasons a distance is not a connection stated on the
 * card itself rather than hidden in a tooltip.
 *
 * WHY IT REPLACES THE CONFIG ADAPTER RATHER THAN ADDING A SCRIPT
 * -----
 * The composer in atlas/index.html supports exactly one slot, `replace-script`.
 * There is no append slot, and inventing one would mean changing the composer,
 * which is a larger contract change than this feature is worth. So this

```

```

* cartridge carries the pre-snapped config adapter's behaviour VERBATIM -- same
* layer ids, same closure assertion, same failure mode, same public state
* object -- and adds the link layer beside it. The immutable shell is not
* touched, and `__GRIDATLAS_PRE_SNAPPED_CONFIG__` still reports exactly what it
* reported before, so anything asserting on it keeps working.
*
* HOW IT HOOKS IN WITHOUT SHELL MUTATION
* -----
* Two decorators, both of things the engine has already published by the time
* this script runs:
*
*   window.initVentusMap wrapped for the pre-snap config, as before.
*   maplibregl.Map       wrapped to capture the instance, because the engine
*                         keeps `map` in a closure and returns nothing. The
*                         engine constructs its map inside initVentusMap, which
*                         runs after this file, so the constructor is still
*                         ours to wrap.
*
* THE MEASUREMENT
* -----
* Haversine on R = 6378.137 km, the same constant as ventus-corev8engine.js,
* pipelinenews and Ventusltd/grid-distance-maths, so a distance read here
* equals the same distance read there. Substations are mapped as points AND as
* polygons; a polygon is reduced to its ring mean, because its first vertex is
* a corner rather than the site.
*
* Scope is 33 kV and above. 11 kV is rare for utility-scale export and where it
* occurs is often a private network behind the meter, so it is not a screening
* signal. `voltage` is written `33000`, `33000;11000` for two voltages, and
* `33000:11000` for a transformer ratio -- a 33/11 primary still carries 33 kV.
*
* WHAT A LINE IS NOT
* -----
* A straight line to mapped geometry. Not a cable route, not a connection
* length, no wayleave, crossing, terrain or consent content. A mapped
* substation does not confirm capacity, voltage suitability or connection
* rights, and fault level and thermal headroom cannot be inferred from distance
* at all -- they need DNO network data such as source impedance and a
* connection study, alongside right of way, wayleaves and easements, land
* control and consent. The card says all of that on screen.
*/
(() => {
  'use strict';

  const SLD_STYLES = (window.__GRIDATLAS_MODULES__ || {}).sldStyles;
  if (SLD_STYLES?.schema !== 'gridatlas.module.sld-styles.v1') {
    throw new Error('sld-sandbox requires the sld-styles module');
  }

  const GENERATION = '202609012045';

  /* =====
  PART 1 – the pre-snapped config adapter, carried forward unchanged.
  ===== */

  const PRE_SNAPPED_LAYER_IDS = new Set(['400', '275', '220', '132', '66']);
  const originalInit = window.initVentusMap;

  if (typeof originalInit !== 'function') {
    throw new Error('V8 engine init function is unavailable before map-ready
adapter');
  }

```



```

const state = {
  schema: 'gridatlas.pre-snapped-config-adapter.v1',
  generation: '202608292126',
  applied: false,
  changed_layer_ids: [],
  preserved_preload_flags: true,
  failures: []
};
window.__GRIDATLAS_PRE_SNAPPED_CONFIG__ = state;

window.initVentusMap = function gridAtlasMapReadyInit(options) {
  try {
    const changed = [];
    const config = options.config.map(group => ({
      ...group,
      layers: group.layers.map(layer => {
        if (!PRE_SNAPPED_LAYER_IDS.has(String(layer.id))) return layer;
        if (layer.snap !== true) {
          throw new Error(`expected V8 snap=true for topology layer
${layer.id}`);
        }
        changed.push(String(layer.id));
        return { ...layer, snap: false };
      })
    }));

    const expected = [...PRE_SNAPPED_LAYER_IDS].sort();
    if (JSON.stringify([...changed].sort()) !== JSON.stringify(expected)) {
      throw new Error(`pre-snapped layer closure mismatch:
${JSON.stringify(changed)}`);
    }

    state.applied = true;
    state.changed_layer_ids = changed;
    return originalInit({ ...options, config });
  } catch (error) {
    state.failures.push(String(error?.message || error));
    throw error;
  }
};

/* =====
PART 2 – neon substation links.
===== */

/* ONE geodesy, and it is the module's.
-----
This file used to declare R_ATLAS = 6378.137 and its own distanceKm,
four hundred lines from a second geodesy section, while the assembled
cartridge ALSO carried the geodesy module in front of it. Two radii and
two haversines in one served file, agreeing only because nobody had
changed one of them yet - and on 1 Sep 2026 the all-versions proof
found that they had in fact stopped agreeing in the last bit.

The Grid Finding Scope already did this correctly: it calls its module
and computes nothing itself. Geodesy was the exception. It is not now.
A missing module is a hard failure at load, not a fallback: a fallback
would silently restore exactly the duplication this removes. */
const GEODESY = (window.__GRIDATLAS_MODULES__ || {}).geodesy;
if (!GEODESY) throw new Error('sld-sandbox requires the geodesy module');
const R_ATLAS = GEODESY.EARTH_RADIUS_KM;

```

```

const DEG = Math.PI / 180;
const MIN_KV = 33;
const LINK_COUNT = 5;           // how many substations to reach for
const MAX_LINK_KM = 40;         // beyond this, silence is more honest
const SUBS_URL = 'data/grid_substations.geojson';
const SUBS_LAYER_ID = 'l-subs'; // engine convention: layer `l-<id>`,
source `src-<id>`

// Project technologies this fires for. Onshore only: an offshore turbine's
// export route is nothing like a straight line to the nearest onshore
// substation, so drawing one would be a picture of a lie.
/* Every technology the register actually uses, and then some.
-----
This set was solar, bess and two spellings of wind, and it silently
rejected the rest. Counted against the shipped register: 2,399 onshore
wind projects and 109 offshore, so 2,508 of 7,680 – a third of the
register – had a MAP button that did nothing at all. Not an error, not a
message, nothing: the deep link tested membership and returned.

The register writes `wind_onshore`. The engine has had a `wind_onshore`
layer the whole time. Only this list disagreed with both.

So it no longer decides alone. The list below is the fast path, and
anything the ENGINE has a layer control for is accepted too – the engine
owns the layers, so the engine's vocabulary is the authority and this
stops being a place a technology can be forgotten. */
const PROJECT_TECHS = new Set([
'solar', 'solar_operational', 'solar_roof',
'bess', 'bess_operational',
'wind', 'wind_onshore', 'wind_onshore_operational',
'wind_offshore', 'wind_offshore_operational',
// The rest of the engine's own generation and storage dashboard, read off
// the live page rather than guessed. Note wind_onshore is NOT among the
// engine's layer ids -- it has `wind` and `wind_onshore_operational` --
// yet the register writes wind_onshore for 2,399 projects. Asking the
// engine alone would still have missed every one of them, which is why the
// explicit entry above is not redundant with the lookup below.
'biomass', 'hydro', 'hydrogen', 'tidal', 'geothermal',
'flywheel', 'caes', 'act',
/* `other` is what the register writes for a category it has no
bucket for. Over the 11,069-row REPD product this Atlas's search
lane reads, 25 DESNZ categories normalise to 14 ids and `other` was
the only one missing here – 4 projects, 2 Unknown and 2 Air Source
Heat Pumps. Not many; not zero; and written again the next time the
register gains a category. */
'other'
]);

/* Which technologies measure, and what each measurement is OF, is owned by
the technology-coverage module rather than by this file. It is assembled
into this cartridge ahead of this part, so it is here by the time any of
this runs; the fallback below exists only so a proof can load this part
alone, and it reproduces the module's answer rather than a different one.

Offshore NO LONGER WITHHOLDS. It used to open a card and draw nothing, on
reasoning about export cables and landfalls that was right about routes
and wrong about whether to measure at all. An offshore project's export
cable does land at an onshore substation, so the distance is a
measurement of something real. What changed is the answer; what did not
change is a single word of the route reasoning, which the module now
prints beside the number instead of instead of it. */
const coverage = (() => {

```

```

try {
  const module = window.__GRIDATLAS_MODULES__?.technologyCoverage;
  if (module && typeof module.policy === 'function') return module;
} catch (_) { /* fall through to the local reproduction */ }
const OFF = new Set(['wind_offshore', 'wind_offshore_operational']);
return {
  policy: (tech) => ({ technology: tech || null, measure: true,
    offshore: OFF.has(String(tech || '')),
    sample: 'nearest of the mapped substations at or above the voltage '
      + 'floor that this search could see',
    notes: [] })),
  namedOffshore: () => false
};
})();
const OFFSHORE_TECHS = new Set(['wind_offshore',
'wind_offshore_operational']);

function isProjectTech(tech) {
  if (!tech) return false;
  if (PROJECT_TECHS.has(tech)) return true;
  // Ask the engine. If it has a control for this layer, it is a technology
  // this map knows about, whatever this cartridge was written knowing.
  try {
    return Boolean(document.querySelector(
      'input[type=checkbox][data-layer-id="' + String(tech).replace(/"/g, '')
+ '"]'));
  } catch (error) {
    return false;
  }
}

```

/* Pipeline News' MAP link sends a technology BUCKET, not a layer id, and the two are not the same vocabulary. There are exactly thirteen buckets it can send -- the four-member spine solar/bess/wind_onshore/wind_offshore, plus the nine wider-fleet buckets biomass/hydro/hydrogen/act/tidal/geothermal/caes/flywheel/other -- see atlas/modules/202609031310-technology-coverage.js SPINE and widerFleetBuckets(), which is the one place that list is owned.

isProjectTech() above tests membership of PROJECT_TECHS, which deliberately contains wind_onshore, wind_offshore and other so that an arrival for one of them is not abandoned. That membership test answers "is this a technology the map recognises", not "is there a layer control with this exact id" -- and those are different questions here:

- wind_onshore and wind_offshore are not layer ids. The engine publishes one combined `wind` layer, filtered on tech === 'wind', which is inclusive of both orientations and every status (see ukConfig's REPD layer group); wind_onshore_operational and wind_offshore_operational are narrower operational-only subsets, not the general layer. A deep link for either bucket wants the general layer switched on, so both resolve to 'wind'.
- other has never had a layer control at all. 25 DESNZ categories normalise to 14 register ids and `other` is the one bucket that genuinely has nothing to switch on -- not a bug to retry, a fact to state.

Measured live on v9.107: because isProjectTech('wind_onshore') is true, the arrival's own technology_layer.enabled read true while the DOM search for a control literally named "wind_onshore" failed every time -- 2,508 of 7,680 register rows, a third of it. One table here, consulted at the one place a control is actually looked up, so a

```

    bucket cannot go missing from it the way these three did while still
    passing the membership test that was supposed to catch that. */
const LAYER_ID_FOR_BUCKET = Object.freeze({
  wind_onshore: 'wind',
  wind_offshore: 'wind',
  other: null    // no layer exists; the caller must say so, not search for one
});

function layerIdForBucket(tech) {
  const id = String(tech == null ? '' : tech);
  return Object.prototype.hasOwnProperty.call(LAYER_ID_FOR_BUCKET, id)
    ? LAYER_ID_FOR_BUCKET[id] : id;
}

// SCADA on a dark map, not arcade neon. These are the muted siblings of the
// engine's own layer colours: enough saturation to read as live, low enough
// not to shout over the basemap or the grid layers underneath.
const TECH_COLOUR = {
  solar: '#d8c96a', solar_operational: '#d8c96a', solar_roof: '#d8c96a',
  bess: '#d9963c', bess_operational: '#d9963c',
  wind: '#6fb582', wind_onshore: '#6fb582', wind_onshore_operational:
'#6fb582',
  // Offshore reads cooler than onshore: it is the one technology here whose
  // links are deliberately not drawn, and it should not look like the others.
  wind_offshore: '#5f9fb5', wind_offshore_operational: '#5f9fb5',
  biomass: '#b58f6f', hydro: '#6f9fd8', hydrogen: '#a98fd8',
  tidal: '#5fb5a8', geothermal: '#b57f6f',
  flywheel: '#9f9fb5', caes: '#9f9fb5', act: '#9f9fb5'
};
const SUBSTATION_COLOUR = '#5fbdc2';    // teal, the substation end of a link

/* — the 400 kV public record —————
The nearest-substations list is a measurement and nothing more. But
the DCO-scale schemes each carry a PUBLIC declared point of connection
at 400 kV - usually through a new customer substation consented within
the scheme itself - and this card used to say nothing about it while
listing closer 33 and 132 kV points, which read as connecting them to
the wrong network. The rule holds: bind to the public record or say
nothing. Every entry below is sourced from Development Consent Orders,
Planning Inspectorate documents, or public project statements; the
table binds a register identity to a NAMED substation in the served
payload, and the distance shown is measured, never asserted. */
const DECLARED_COLOUR = '#d8b64a';    // gold: declared, and the far end exists
today
/* Pink: declared, and the thing at the far end has not been built. One
Earth's point of connection is a National Grid substation that does not
exist yet; Thorpe Marsh's is under construction. Drawing those in the
same gold as a connection into a live substation would say something
untrue about the network, so they get their own colour and the card
says which it is. The distinction is taken from the public record, not
from the map: an unbuilt substation can be absent from OSM, or present
because someone mapped the consented site. */
const DECLARED_UNBUILT_COLOUR = '#d87aa8';
/* The table itself, and the three functions that bind it to the payload,
moved to the declared-connections module at generation 202609012128.
The body keeps the colours (rendering) and the state; the record is
read from the module, which a proof can open on its own and a cut
hashes on its own. A missing module is a hard failure at load, for
the same reason as geodesy: a fallback would quietly restore a second
copy of the table. */
const DECLARED = (window.__GRIDATLAS_MODULES__ || {}).declaredConnections;
if (!DECLARED) throw new Error('sld-sandbox requires the declared-connections

```

```

module');

let currentRepdRef = null;
let currentDeclared = null;
let currentNearest400 = null;
/* The coverage policy for the CURRENT selection. It carries no arithmetic
   and changes no result - it decides only which sentences go under the
   distances. The measurement itself is coordinates, a candidate set and a
   distance, and reads no technology at all. */
let currentPolicy = null;
/* The capacity Pipeline News sent, kept where the network card can
   reach it. Without this the powerflow answer would have to invent a
   figure, and an invented megawatt is exactly the kind of number that
   gets quoted back as the project's own. */
let currentCapacityMw = null;

/* What the Order says is known the moment the identity is known: the
   substation, the voltage class, the route, the consented works and the
   citation need no payload, no fetch and no map. On a phone that is the
   difference between a card that answers and a card that waits. The
   distance is the one part that must be measured, so it is the one part
   marked pending until it has been. */
function provisionalDeclaredConnection(repdRef) {
  return DECLARED.provisional(repdRef);
}

function resolveDeclaredConnection(repdRef, origin, subs) {
  return DECLARED.resolve(repdRef, origin, subs);
}

function nearestTransmission(origin, subs) {
  return DECLARED.nearestTransmission(origin, subs);
}

/* EVERY SUPERLATIVE CARRIES ITS SAMPLE.
-----
"Nearest 400 kV substation" is nearest among what this search could
see, and two different things limit that. The search runs over the
substation features the map has loaded, so the first number is how
many of them were actually eligible - counted by the measurement
itself, not assumed. And the operator's own published list is only
partly located: ETYS names substations and does not place them, so
the geometry comes from OpenStreetMap through a GridAtlas release and
a fraction of the network is invisible to any search by distance.

Both numbers are COMPUTED at render time from what was fetched. A
literal would go quietly false the day the pinned product moves -
Codex's join correction alone takes located points from 502 to 489 -
and a stale denominator under the word "nearest" is worse than none.

It states the sample. It does not grade the result. */
/* A straight line is not a route, and the card said nothing about it.
-----
ADDITIVE. The straight-line distance is unchanged, still first, still
the measurement; the corridor figure sits beside it and is labelled an
estimate every time it appears.

Only for a CABLE question. The factor is calibrated on cable circuits,
which follow the highway network; overhead line crosses open country
and measures 1.13. The module publishes that number and deliberately
offers no forOverhead(), so this cannot quietly become the answer to a
question it was not measured on.

```

```

    Under about a kilometre the module withholds the estimate rather than
    scaling, and the card says why: at that separation the straight line
    between two site centroids is not measuring route factor at all. */
function corridorBeside(km) {
  const module = (() => {
    try { return window.__GRIDATLAS_MODULES__?.corridorEstimate || null; }
    catch (_) { return null; }
  })();
  if (!module) return '';
  const estimate = module.forCable(km);
  if (!estimate) return '';
  const basis = module.basis;
  if (estimate.withheld) {
    return ` <span class="neon-caveat">No corridor estimate at this `
      + `separation: ${escapeHtml(estimate.withheld)}</span>`;
  }
  return ` &middot; ~${estimate.km.toFixed(1)} km corridor estimate `
    + `(&times;${estimate.factor}, ${basis.within_15_pct}% of GB transmission

    + `cable circuits within 15% of published length, `
    + `${basis.distinct_site_pairs} distinct site pairs). `
    + `<span class="neon-caveat">${escapeHtml(module.caveat)} `
    + `${escapeHtml(module.not_for_overhead)}</span>`;
}

```

/* A 44 px action, and a sheet that is the corridor figures in full.

The card line beside each distance is one sentence. A reader who wants the basis - what the factor was calibrated on, how wrong it typically is, and what it is not for - should not have to take that from a sentence, and on a phone there is no room to print it inline.

44 px because that is the smallest target a thumb hits reliably; the estate's own layers toggle was widened to the same at 202609030116.

The sheet is opened three ways and all of them end in one function: the button, a right-click on the map (the engine binds no contextmenu - measured, zero occurrences in the shell, in index.html and in every composed part), and a long press.

THE LONG PRESS DEFERS TO EVERYTHING. It arms on touchstart, and it is cancelled by a movement of more than 10 px, by a second finger, by touchend, and by an SLD drag already being in progress - the sandbox binds its own touchstart for dragging the array, its handle and its route pins, so this must never fire during one. A pan is a movement; a pinch is a second finger; both cancel before the 500 ms is up. */

```

const CORRIDOR_SHEET = 'gridatlas-corridor-sheet';
let corridorSheetInstalled = false;

```

```

function corridorTargets() {
  const rows = [];
  if (currentDeclared && currentDeclared.kind !== 'circuit'
    && Number.isFinite(currentDeclared.km)) {
    rows.push({ name: currentDeclared.poc, km: currentDeclared.km,
      note: 'declared point of connection, from the public record' });
  }
  if (currentNearest400) {
    rows.push({ name: currentNearest400.name, km: currentNearest400.km,
      note: 'nearest mapped substation at 400 kV or above' });
    if (currentNearest400.named && currentNearest400.named.name !==
currentNearest400.name) {

```

```

        rows.push({ name: currentNearest400.named.name, km:
currentNearest400.named.km,
        note: 'nearest NAMED substation at 400 kV or above' });
    }
}
return rows;
}

function corridorAction() {
    if (!corridorTargets().length) return '';
    /* The card can render before the map is wired, and the button is
    useless without the delegated listener. Installing here as well is
    idempotent and removes the ordering dependency entirely. */
    installCorridorSheet();
    return `<button type="button" class="gridatlas-corridor-open" `
        + `data-gridatlas-corridor="1" `
        + `aria-haspopup="dialog">Explore route corridors \u203a</button>`;
}

function installCorridorSheet() {
    if (corridorSheetInstalled) return;
    corridorSheetInstalled = true;

    const style = document.createElement('style');
    style.textContent =
        '.gridatlas-corridor-open{display:block;width:100%;min-
height:44px;margin:8px 0 2px;\'
        + \'font:600 12px/1.2 ui-monospace,SFMono-Regular,Menlo,monospace;letter-
spacing:.06em;\'
        + \'text-align:left;padding:12px 14px;border-radius:6px;cursor:pointer;\'
        + \'background:#0d1117;color:#7fe3d0;border:1px solid #2b3a44;\'
        + \'.gridatlas-corridor-open:focus-visible{outline:2px solid
#7fe3d0;outline-offset:2px;\'
        + \'#\' + CORRIDOR_SHEET + '{position:fixed;left:0;right:0;bottom:0;z-
index:10000;\'
        + \'max-height:min(70vh,560px);overflow:auto;transform:translateY(101%);\'
        + \'transition:transform .18s ease-out;background:#070d11;color:#cfe6e8;\'
        + \'border-top:1px solid #0b5f63;padding:14px 16px calc(16px + env(safe-
area-inset-bottom));\'
        + \'font:12px/1.5 ui-monospace,SFMono-Regular,Menlo,monospace;\'
        + \'#\' + CORRIDOR_SHEET + '[data-open="1"]{transform:translateY(0);\'
        + \'#\' + CORRIDOR_SHEET + ' h4{margin:0 0 8px;font-
size:13px;color:#7fe3d0;\'
        + \'#\' + CORRIDOR_SHEET + ' .r{padding:8px 0;border-top:1px solid
#14252b;\'
        + \'#\' + CORRIDOR_SHEET + ' .c{color:#8fa6ab;\'
        + \'#\' + CORRIDOR_SHEET + ' button{min-height:44px;min-
width:44px;cursor:pointer;\'
        + \'background:#0d1117;color:#7fe3d0;border:1px solid #2b3a44;border-
radius:6px;\'
        + \'padding:10px 14px;font:600 12px/1 ui-monospace,monospace;\'
        + '@media (prefers-reduced-motion:reduce){#\' + CORRIDOR_SHEET +
'{transition:none;}}';
    document.head.appendChild(style);

    const sheet = document.createElement('div');
    sheet.id = CORRIDOR_SHEET;
    sheet.setAttribute('role', 'dialog');
    sheet.setAttribute('aria-label', 'Route corridor estimates');
    sheet.dataset.open = '0';
    sheet.hidden = true;
    document.body.appendChild(sheet);

```

```

const close = () => { sheet.dataset.open = '0'; sheet.hidden = true; };

window.__GRIDATLAS_CORRIDOR_SHEET__ = {
  get open() { return sheet.dataset.open === '1'; },
  close,
  open: () => openCorridorSheet(sheet)
};

document.addEventListener('click', (event) => {
  const opener = event.target?.closest?(['data-gridatlas-corridor']);
  if (opener) { event.preventDefault(); openCorridorSheet(sheet); return; }
  if (event.target?.closest?(['data-gridatlas-corridor-close'])) close();
});
document.addEventListener('keydown', (event) => {
  if (event.key === 'Escape' && sheet.dataset.open === '1') close();
});
}

function openCorridorSheet(sheet) {
  const module = (() => {
    try { return window.__GRIDATLAS_MODULES__?.corridorEstimate || null; }
    catch (_) { return null; }
  })();
  const rows = corridorTargets();
  if (!rows.length) return;
  const body = rows.map(row => {
    const estimate = module ? module.forCable(row.km) : null;
    const corridor = estimate && estimate.km !== null
      ? `~${estimate.km.toFixed(1)} km corridor estimate`
      : (estimate ? 'no corridor estimate at this separation' : '');
    return `

blob:https://ventusltd.github.io/184a2ca1-a0ae-42ea-b3ba-72ceefc889c4



05/09/2026, 16:22  
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```



```

installCorridorSheet();
try {
  map.on('contextmenu', (event) => {
    if (!corridorTargets().length) return;
    event.preventDefault?();
    openCorridorSheet(document.getElementById(CORRIDOR_SHEET));
  });
} catch (_) { /* a shimmed map in a proof */ }

const canvas = (() => {
  try { return map.getCanvas ? map.getCanvas() : null; } catch (_) { return
null; }
})();
if (!canvas) return;
let timer = null;
let from = null;
const cancel = () => { if (timer) clearTimeout(timer); timer = null; from =
null; };
canvas.addEventListener('touchstart', (event) => {
  cancel();
  /* Never during an SLD drag: this cartridge binds its own touchstart
  for the array, the rotate handle and the route pins. */
  if (sld && sld.dragging) return;
  if (event.touches.length !== 1) return;
  if (!corridorTargets().length) return;
  from = { x: event.touches[0].clientX, y: event.touches[0].clientY };
  timer = setTimeout(() => {
    timer = null;
    if (sld && sld.dragging) return;
    openCorridorSheet(document.getElementById(CORRIDOR_SHEET));
  }, 500);
}, { passive: true });
canvas.addEventListener('touchmove', (event) => {
  if (!timer || !from) return;
  const touch = event.touches[0];
  if (!touch) { cancel(); return; }
  /* A pan is a movement. Ten pixels is below the threshold a deliberate
  press produces and above the jitter a still thumb produces. */
  if (Math.abs(touch.clientX - from.x) > 10
    || Math.abs(touch.clientY - from.y) > 10) cancel();
}, { passive: true });
canvas.addEventListener('touchend', cancel, { passive: true });
canvas.addEventListener('touchcancel', cancel, { passive: true });
}

function nearestScope(n) {
  const considered = Number(n && n.considered);
  const network = (() => {
    try { return window.__GRIDATLAS_NETWORK__?.coverage?.(400) || null; }
    catch (_) { return null; }
  })();
  if (!Number.isFinite(considered) && !network) return '';
  const said = [];
  if (Number.isFinite(considered)) {
    said.push(`nearest of the ${considered.toLocaleString('en-GB')} mapped `
      + `substation${considered === 1 ? '' : 's'} at 400 kV or above that this
      + `search could see`);
  }
  if (network && network.published) {
    said.push(`the operator publishes ${network.published.toLocaleString('en-
GB')}`);
  }
}

```

```

    + `connection point${network.published === 1 ? '' : 's'} at that class
and `
    + `${network.located.toLocaleString('en-GB')} of them carry coordinates,
so `
    + `${network.unlocated.toLocaleString('en-GB')} cannot be measured to at
all`);
}
return `
```

```

const networkName = currentDeclared?.kind !== 'circuit'
  ? (currentDeclared?.poc || currentNearest400?.name) :
currentNearest400?.name;
/* Tell it the voltage the connection is actually made at: the declared
point of connection's class where there is one, otherwise the class
of the substation being measured to. Without this the answer can
only be a site-wide envelope. */
const connectionKv = currentDeclared?.kind !== 'circuit'
  ? (currentDeclared?.kv || currentNearest400?.kv || null)
  : null;
/* connection-points counts LANDINGS: a transformer's windings are both
at the site, so Cowley's five read as ten and 484 of 525 sites were
1.90x over. Only the node/branch model holds the pairs. */
const publishedUnits = (() => {
  if (topology.state !== 'ready' || !topology.index) return null;
  try {
    const point = window.__GRIDATLAS_NETWORK__?.byName?.(networkName);
    if (!point?.site_code) return null;
    const facts = topology.index.at(point.site_code);
    if (!facts?.counts) return null;
    return { circuits: facts.counts.circuits,
      transformers: facts.counts.transformers };
  } catch (_) { return null; }
})();
const published = (() => {
  try {
    return window.__GRIDATLAS_NETWORK__?.summarise?.(
      networkName, { connectionKv, units: publishedUnits }) || null;
  } catch (_) { return null; }
})();
/* The envelope is built HERE and appended BELOW the measurement.
-----
WHY THE ORDER CHANGED. Driven on a verified iPhone-class device
(393x852, pointer:coarse, hover:none, 5 touch points), 101 MAP taps
across all 25 technologies: the distance was on the page 99 times out
of 99 and on the first screen zero times. Measured here, it landed at
y=907 in an 852px viewport with about 270px of published-envelope
detail above it. The envelope is neither deleted nor shortened; it
moves below the answer it is context for. The measurement paragraph
names the substation, so nothing above it is needed to read it. It is
still built before the block is emitted, because link.network_published
and the topology block both depend on it having been asked for. */
const publishedHtml = (() => {
  if (!published) return '';
  link.network_published = { name: networkName, site: published.site_code };
  /* The scope label goes FIRST. A site-wide envelope printed under a
400 kV declared connection reads as a 400 kV result unless the
reader is told otherwise before the numbers, not after. */
  return `<div class="neon-hd">${escapeHtml(networkName)}`
    + (published.fault_scope === 'bus'
      ? `<span class="neon-beta" style="background:#12323a;color:#8fd8e0">`
      + `${published.fault_kv} kV bus</span>`
      : (published.site_wide
        ? `<span class="neon-beta"
style="background:#3a3a2a;color:#d8c96a">Site-wide</span>`
        : ''))
    + `<span class="neon-beta">NESO published</span></div>`
    + `<p class="neon-caveat"><b>${escapeHtml(published.scope_label)}</b>`
</p>`
    + `<p class="neon-caveat">${escapeHtml(published.sentence)}</p>`
    + `<p class="neon-
caveat">${escapeHtml(published.metrics_not_interchangeable)} `

```

```

    + `${escapeHtml(published.attribution)}. `
    + `${escapeHtml(published.not_an_assessment)}</p>`;
  })());
  if (currentNearest400) {
    const n = currentNearest400;
    /* The measurement and everything that makes it honest are ONE element,
       which is why nearestScope(n) is inside the wrapper rather than
       appended after it. Whatever a layout does to this block - move it,
       dock it, put it in a sheet - the number cannot arrive without the
       word "straight", without the corridor estimate's own caveat, and
       without the sentence naming the sample the superlative searched. */
    out += `

<b>Nearest 400 kV substation:</b> `
    + `${escapeHtml(n.name)} · ${n.km.toFixed(2)} km straight`
    + corridorBeside(n.km)
    + (n.works ? `· ${escapeHtml(n.works)}` : '')
    + (n.named
      ? ` (nearest named: ${escapeHtml(n.named.name)} ·
    ${n.named.km.toFixed(2)} km`
      + (n.named.works ? `· ${escapeHtml(n.named.works)}` : '') + `)`
      : '')
    + `</p>`
    + nearestScope(n)
    + `</div>`;
  }
  out += publishedHtml;
  if (networkName) {
    out += topologyBlockHtml([{ name: networkName, kv: connectionKv }]);
  }
  out += corridorAction();
  return out;
}


```

/* — the transmission network, on demand —————

Vikram: click anywhere and the neons should "look for cartridges and code". Until this generation the looking stopped at NESO's connection points - one row per site. The node/branch model behind them (ETYS Appendix B as VentusLtd/data-grid-gb's gb-transmission-network.v1) was indexed by the network-topology module, proven 47/47, and composed into nothing: the module was on disk and not in any served cartridge. The deep scan of 202609012230 listed it as alive; it was alive the way a book on a shelf is.

The product is ten megabytes. It is NOT fetched at load: a phone on a hillside should not pay for it until a click asks a question it answers. Fetched once, indexed once by the module, every state on `__GRIDATLAS_TOPOLOGY__` so the source registry can say whether this source answered, and the cards fill in when it arrives rather than waiting for it.

The join from a mapped substation to a published site is BY NAME, through the connection-points cartridge's own join, and the block says which site it joined to - Codex's WBUR finding stands: exact text equality is not exact identity, so the reader is shown the identity. */

/* Pinned; the table and the reasoning are in the pinned-products module. */

```

const PINS = (window.__GRIDATLAS_MODULES__ || {}).pinnedProducts || null;
const TOPOLOGY_ID = 'gb-transmission-network.v1';
const TOPOLOGY_PRODUCT = PINS ? PINS.url(TOPOLOGY_ID) : null;
const TOPOLOGY_BLOCK = 'gridatlas-topology';
const topology = { state: 'idle', product: TOPOLOGY_PRODUCT, schema: null,
  schema_required: topologyModule()?.accepts || null,
  bytes: null, sites: null, index: null, error: null,

```

```

    started_at: null, ready_at: null, blocks_filled: 0 };
window.__GRIDATLAS_TOPOLOGY__ = topology;

/* Published so a reviewer can ask the page how many cards the traversal
   actually answered, and how many published branches it refused to walk.
   A refusal is a finding about the data, not a failure of the page. */
const electrical = { answered: 0, refusals: 0 };
window.__GRIDATLAS_ELECTRICAL__ = electrical;

/* How many cards quoted a seasonal rating, and how many placeholder
   values the published record turned out to contain. */
const rating = { answered: 0, flagged: 0 };
window.__GRIDATLAS_RATINGS__ = rating;

/* Published so a reviewer can ask the page whether any answer it gave
   failed its own conservation check. A solve that has not converged
   produces plausible-looking flows that are wrong, so the error is
   surfaced rather than trusted. */
const powerflow = { answered: 0, refused: 0, worst_kirchhoff_error: 0 };
window.__GRIDATLAS_POWERFLOW__ = powerflow;

/* How many point queries the reader ran. A feature nobody can reach is
   indistinguishable from a feature that does not work, and this is the
   number that tells them apart. */
const pointQuery = { answered: 0 };
window.__GRIDATLAS_POINT_QUERY__ = pointQuery;

/* How many cards reported published plans, and how many rows they came
   from. A count of zero where the product has 2,230 rows would mean the
   wiring is broken, not that nothing is planned. */
const plannedState = { answered: 0, rows: 0 };
window.__GRIDATLAS_PLANNED__ = plannedState;

/* How many cards named an owner, and how many seams they found. */
const ownerState = { answered: 0, seams: 0 };
window.__GRIDATLAS_OWNERSHIP__ = ownerState;

/* The layers dash collapses without entering fullscreen.
   -----
   The dash is 816 px tall on a desktop and takes most of a phone, and
   the only way past it has been fullscreen - a different mode with a
   different layout, which is a large thing to ask of a reader who just
   wants to see the map. This collapses it in place and leaves a tab to
   bring it back.

   The choice is remembered per browser and every storage access is
   wrapped: a private window, cleared site data or a browser set to
   block storage all throw here, and a thrown error must not take the
   control with it. */
(function dashCollapse() {
  const KEY = 'gridatlas.dash.collapsed';
  /* HIDE LAYERS collapsed `.dashboard`, which is the whole app: in the
     shipped shell it opens at index.html:22 and holds BOTH
     `.map-container` (line 36, containing #map) and `.scada-wrapper`
     (line 112, the layer keys and legend), so max-height:0 took the WebGL
     canvas down with the checkboxes - and the choice is remembered, so
     the reload a reader reaches for blanked the page again.

     It targets `.scada-wrapper` now, with NO fallback: the last fallback
     is what blanked the map. It also hides itself in fullscreen, because
     the keeper below MOVES the panel into the fullscreen element. */
  const dash = document.querySelector('.scada-wrapper');

```

```

if (!dash) {
  /* `link` is declared further down this file and is in its temporal
     dead zone here, so the refusal is published on its own surface
     rather than through the failure ledger. */
  window.__GRIDATLAS_DASH__ = { installed: false, target: '.scada-wrapper',
    reason: 'no .scada-wrapper in this shell; the control was not installed
    + 'rather than pointed at .dashboard, which contains the map' };
  return;
}
if (document.getElementById('gridatlas-dash-toggle')) return;

const style = document.createElement('style');
style.textContent = '.scada-wrapper[data-gridatlas-collapsed="1"]{max-
height:0;
  + 'overflow:hidden;padding-top:0;padding-bottom:0;border:0;}'
  + '#gridatlas-dash-toggle{position:fixed;right:12px;bottom:12px;z-
index:9999;
  + 'min-height:44px;min-width:44px;'
  + 'font:600 12px/1 ui-monospace,SFMono-Regular,Menlo,monospace;'
  + 'letter-spacing:.08em;padding:8px 12px;border-
radius:6px;cursor:pointer;'
  + 'background:#0d1117;color:#7fe3d0;border:1px solid #2b3a44;}'
  + '#gridatlas-dash-toggle[hidden]{display:none !important;}'
  + '#gridatlas-dash-toggle:focus-visible{outline:2px solid #7fe3d0;outline-
offset:2px;}}';
document.head.appendChild(style);

const toggle = document.createElement('button');
toggle.id = 'gridatlas-dash-toggle';
toggle.type = 'button';

/* Closed on a FIRST arrival ON A PHONE only - measured, the panel held
   31.6% of a 393x852 screen against the map's 29.3%, which is the wrong
   trade on the surface most readers arrive on. A desktop has the room and
   opens with the panel showing, exactly as v8 always did: that is the
   product surface, and hiding it made the Atlas look less capable than the
   version it replaced. A choice once made still wins over both defaults.

   An UNKNOWN width is not a phone - the width has to be a real positive
   number before it argues for starting collapsed, so a host that publishes
   no width gets the desktop default rather than an empty-looking page. */
const width = Number(window.innerWidth);
const coarse = (() => {
  try { return !(window.matchMedia
    && window.matchMedia('(pointer: coarse)').matches); } catch (_) { return
false; }
})();
let collapsed = coarse || (isFinite(width) && width > 0 && width <= 700);
try {
  const v = window.localStorage.getItem(KEY);
  if (v !== null) collapsed = v === '1';
} catch (_) { /* the width-derived default above stands */ }

function reflect() {
  if (collapsed) dash.setAttribute('data-gridatlas-collapsed', '1');
  else dash.removeAttribute('data-gridatlas-collapsed');
  toggle.textContent = collapsed ? '\u25b4 LAYERS' : '\u25be HIDE LAYERS';
  toggle.setAttribute('aria-pressed', String(collapsed));
  toggle.setAttribute('aria-label', collapsed
    ? 'Show the layers panel' : 'Hide the layers panel');
  /* MapLibre sizes itself to its container and will not notice the

```

```

        page reflowing under it. */
        try { if (window.map && typeof window.map.resize === 'function')
window.map.resize(); }
        catch (_) { /* the control still works without the resize */ }
    }

    toggle.addEventListener('click', () => {
        collapsed = !collapsed;
        try { window.localStorage.setItem(KEY, collapsed ? '1' : '0'); } catch (_)
{ /* per-viewer nicety only */ }
        reflect();
    });

    document.body.appendChild(toggle);
    reflect();

    const reflectFullscreen = () => {
        const full = !(document.fullscreenElement ||
document.webkitFullscreenElement);
        toggle.hidden = full;
    };
    document.addEventListener('fullscreenchange', reflectFullscreen);
    document.addEventListener('webkitfullscreenchange', reflectFullscreen);
    reflectFullscreen();

    window.__GRIDATLAS_DASH__ = {
        /* Named so a reader - and a proof - can see WHAT collapses. */
        installed: true,
        target: '.scada-wrapper',
        get collapsed() { return collapsed; },
        get hidden_by_fullscreen() { return toggle.hidden; },
        toggle: () => { toggle.click(); return collapsed; }
    };
})();
let topologyPromise = null;

function topologyModule() {
    try { return window.__GRIDATLAS_MODULES__?.networkTopology || null; }
    catch (_) { return null; }
}

function ensureTopology() {
    if (topologyPromise) return topologyPromise;
    const module = topologyModule();
    if (!module) {
        topology.state = 'failed';
        topology.error = 'network-topology module absent from this composition';
        noteFailure('transmission network: ' + topology.error);
        return Promise.resolve(null);
    }
    topology.state = 'loading';
    topology.started_at = Date.now();
    if (!TOPOLOGY_PRODUCT) {
        topology.state = 'failed';
        topology.error = 'no pinned ref: pinned-products is not composed';
        noteFailure('transmission network: ' + topology.error);
        return Promise.resolve(null);
    }
    topologyPromise = fetch(TOPOLOGY_PRODUCT)
        .then(async response => {
            if (!response.ok) throw new Error('HTTP ' + response.status);
            const text = await response.text();

```

```

        topology.bytes = text.length;
        topology.pin = await PINS.verify(TOPOLOGY_ID, text);
        if (topology.pin.state === 'MISMATCH') throw new
Error(topology.pin.detail);
        return JSON.parse(text);
    })
    .then(product => {
        const index = module.index(product);
        if (!index) {
            throw new Error('schema ' + String(product?.schema || 'absent')
+ ' is not ' + module.accepts + '; this cartridge answers nothing
from it');
        }
        topology.index = index;
        /* the buses the declared DC model is built over. 400 kV only:
        a DC model that walks a transformer without its tap position
        is modelling something the product does not describe, and the
        taps are not published. */
        /* Kept because the planned-change and owner-boundary readers
        take the PRODUCT, not the graph - graph() withholds planned
        rows by design and does not carry transmission_owner. This is
        one reference to an object already in memory, not a copy. */
        topology.parsedProduct = product;
        topology.nodes400 = (product.nodes || [])
            .filter(n => n && n.voltage_consistent_with_site === true &&
n.voltage_kv === 400)
            .map(n => n.node);
        topology.schema = product.schema;
        topology.sites = index.counts.sites;
        topology.state = 'ready';
        topology.ready_at = Date.now();
        return index;
    })
    .catch(error => {
        topology.state = 'failed';
        topology.error = String(error?.message || error);
        noteFailure('transmission network: ' + topology.error);
        return null;
    });
    return topologyPromise;
}

/* Render for a list of { name, kv } queries: the mapped substation's
name and the voltage the question is about. Every card that names a
substation gets one of these; it is a placeholder until the product
is indexed and is filled in place, so no card holds a reference to
another card. */
/* The electrical-distance module, read the same way the topology
module is read: absent is absent, never an excuse to guess. */
function distanceModule() {
    try { return window.__GRIDATLAS_MODULES__?.electricalDistance || null; }
    catch (_) { return null; }
}

/* The planned-change module takes the PRODUCT, not the topology
index: graph() deliberately withholds planned rows, which is the
right default and the reason this needs its own reader. The parsed
product is not kept alive after indexing, so the index is built once
and cached here beside the others. */
let ownerIndex;
function ownerModule() {
    try { return window.__GRIDATLAS_MODULES__?.ownerBoundary || null; }

```



```

    catch (_) { return null; }
}

let plannedIndex;
function plannedModule() {
    try { return window.__GRIDATLAS_MODULES__?.plannedChange || null; }
    catch (_) { return null; }
}

function flowModule() {
    try { return window.__GRIDATLAS_MODULES__?.injectionResponse || null; }
    catch (_) { return null; }
}

/* The 400 kV model is built once per session and kept: assembling it
walks every published node, and a card that rebuilt it per click
would be doing that work again for an answer it already had.

The node list comes from topology.nodes400, which ensureTopology
records while it already has the parsed product in hand. Rebuilding
it here would mean either keeping a second copy of the 10 MB payload
alive or asking the graph for something it does not expose. */
let flowModel = null;
function flowModelFor(index) {
    if (flowModel) return flowModel;
    const mod = flowModule();
    if (!mod || !index || typeof index.graph !== 'function') return null;
    const names = topology.nodes400;
    if (!Array.isArray(names) || !names.length) return null;
    try {
        flowModel = mod.assemble(index.graph(), names,
            { voltageKv: 400, includeTransformers: false });
        return flowModel;
    } catch (_) { return null; }
}

function ratingModule() {
    try { return window.__GRIDATLAS_MODULES__?.ratingEnvelope || null; }
    catch (_) { return null; }
}

/* The published network at an arbitrary point.
-----
The connection-points cartridge already resolves a position to the
nearest published sites, so this does not measure anything itself -
a second distance implementation in this file is exactly the drift
that put two geodesies in one cartridge. It resolves, states the
limit of what "nearest" can mean here, and hands the nearest name to
the block that already renders circuits, ratings, hops and flow. */
async function runGridAtPoint(map, lon, lat) {
    const gl = window.maplibregl;
    if (!gl?.Popup) return;
    const network = window.__GRIDATLAS_NETWORK__ || null;
    const show = (html) => {
        try {
            return new gl.Popup({ maxWidth: '380px', closeOnClick: false })
                .setLngLat([lon, lat]).setHTML(html).addTo(map);
        } catch (error) {
            noteFailure('grid at point: ' + String(error?.message || error));
            return null;
        }
    };
};

```

```

if (!network || typeof network.nearest !== 'function' || !network.loaded) {
  show('<p class="neon-caveat"><b>Grid at point:</b> the connection-points '
    + 'cartridge has not loaded, so no published site can be named. '
    + 'Nothing is inferred from its absence.</p>');
  return;
}

/* nearest() returns { point, km } pairs, sorted. It owns the distance;
   measuring again here would be the second implementation that put
   two geodesies in one cartridge earlier tonight. */
let found = [];
try { found = network.nearest(lon, lat, { limit: 5 }) || []; }
catch (_) { found = []; }
if (!Array.isArray(found)) found = found ? [found] : [];

const points = Number(network.points || 0);
const located = Number(network.located || 0);
const unlocated = points - located;

if (!found.length) {
  show('<p class="neon-caveat"><b>Grid at point:</b> no published connection
    + 'point with coordinates resolved here. ' + unlocated + ' of the '
    + points + ' published points carry no coordinates at all, so this is '
    + 'a statement about the mapped set and not about the network.</p>');
  pointQuery.answered += 1;
  return;
}

const rows = found.map(entry => escapeHtml(String(entry.point?.name || '?'))
  + (Number.isFinite(entry.km) ? ' \u00b7 ' + entry.km.toFixed(1) + ' km' :
  ''));

const popup = show('<div class="neon-hd">Grid at this point'
  + '<span class="neon-beta">published network</span></div>'
  + '<p class="neon-caveat"><b>Nearest mapped connection points:</b> '
  + rows.join(', ') + '</p>'
  + '<p class="neon-caveat">Straight-line distance from where you clicked -
    + 'not a cable route, and not a statement that anything can connect at any
    + 'of them. ' + unlocated + ' of the ' + points + ' published connection '
    + 'points have no coordinates, so the nearest <i>mapped</i> point may not
    + 'be the nearest point.</p>'
  /* fillTopologyBlocks() selects by CLASS, not by a data attribute.
     Getting that wrong would have produced a block that renders its
     loading line forever and never fills. */
  + '<div class="' + TOPOLOGY_BLOCK + '" data-queries="'
  + escapeHtml(JSON.stringify([ { name: String(found[0].point?.name || ''),
kv: null } ])) + '">'
  + '<p class="neon-caveat"><b>Transmission network:</b> loading\u2026</p>
</div>');

  pointQuery.answered += 1;
  if (!popup) return;
  try { await ensureTopology(); } catch (_) { /* the block reports its own
state */ }
  fillTopologyBlocks();
}

```

```

function topologyBlockHtml(queries) {
  const wanted = (queries || []).filter(q => q && q.name);
  if (!wanted.length) return '';
  const attr = escapeHtml(JSON.stringify(wanted.map(q => ({ name: q.name, kv:
q.kv ?? null })))));
  if (topology.state === 'idle' || topology.state === 'loading') {
    ensureTopology().then(() => fillTopologyBlocks());
  }
  return `<div class="${TOPOLOGY_BLOCK}" data-queries="${attr}">`
    + topologyInnerHtml(wanted) + `</div>`;
}

function fillTopologyBlocks() {
  document.querySelectorAll('.' + TOPOLOGY_BLOCK).forEach(node => {
    try {
      node.innerHTML = topologyInnerHtml(JSON.parse(node.getAttribute('data-
queries') || '[]'));
      topology.blocks_filled += 1;
    } catch (error) {
      noteFailure('topology block: ' + String(error?.message || error));
    }
  });
}

function topologyInnerHtml(queries) {
  const caveat = (text) => `<p class="neon-caveat">${text}</p>`;
  if (topology.state === 'idle' || topology.state === 'loading') {
    return caveat(`<b>Transmission network:</b> loading the published
node/branch `
      + `model (about 10 MB, once per session). This card fills in when it
arrives.`);
  }
  if (topology.state === 'failed') {
    return caveat(`<b>Transmission network:</b> not available - `
      + `${escapeHtml(topology.error || 'unknown failure')}. Nothing here is
inferred from its absence.`);
  }
  const network = window.__GRIDATLAS_NETWORK__ || null;
  if (!network?.loaded) {
    return caveat(`<b>Transmission network:</b> indexed (${topology.sites
sites}), but the `
      + `connection-points cartridge that joins a mapped name to a published
site `
      + `${network?.failed ? 'failed to load' : 'has not loaded yet'}, so no
site is named.`);
  }
  let out = '';
  let shown = 0;
  const module = topologyModule();
  for (const q of queries) {
    const kv = Number.isFinite(q.kv) ? q.kv : null;
    let point = null;
    try { point = network.byName(q.name); } catch (_) { point = null; }
    if (!point?.site_code) continue;
    const facts = topology.index.at(point.site_code, kv != null ? { voltageKv:
kv } : undefined);
    if (!facts) continue;
    shown += 1;
    out += `<div class="neon-hd">${escapeHtml(facts.site.name)}`
      + `<span class="neon-beta">ETYS topology</span></div>`
      + caveat(`Joined by name from <i>${escapeHtml(q.name)}</i> to NESO site
`

```

```

+ `${escapeHtml(facts.site.code)}`
+ (facts.site.transmission_owner ? `
`${escapeHtml(facts.site.transmission_owner)}): `)
+ `; declared voltages ${facts.site.voltages_kv.length
? escapeHtml(facts.site.voltages_kv.join(', ')) + ' kV' : 'not
stated'}`);
if (!facts.by_voltage.length) {
  out += caveat(kv != null
    ? `No published branch lands at ${kv} kV at this site.`
    : `No published branch lands at this site.`);
} else {
  /* Site totals are UNITS; the lines below are landings and will
  not add up, because a transformer with both windings here
  appears under each of its voltages. */
  const c = facts.counts;
  out += caveat(`At the site: ${c.circuits} circuit`
    + `${c.circuits === 1 ? '' : 's'} and ${c.transformers} transformer`
    + `${c.transformers === 1 ? '' : 's'}, counted as units.`
    + `The per-voltage lines below count landings, one for each end held`
    + `here, so a transformer with both windings at this site appears`
    + `under both of its voltages.`);
}
for (const band of facts.by_voltage) {
  const label = band.voltage_kv == null ? 'an undeclared voltage' :
`${band.voltage_kv} kV`;
  const years = [...new Set(band.planned_changes.map(c =>
c.year).filter(Boolean))].sort();
  out += caveat(`At ${label}: ${band.circuits.length}
circuit${band.circuits.length === 1 ? '' : 's'},`
    + `${band.transformers.length} transformer${band.transformers.length
=== 1 ? '' : 's}`
    + (band.planned_changes.length
      ? `, ${band.planned_changes.length} published
change${band.planned_changes.length === 1 ? '' : 's}`
      + (years.length ? ` (${escapeHtml(years.join(', ')))` : ''
      : ''
    + `.`);
}
/* Seasonal ratings, per circuit, never added together.
-----
The lowest and the highest are two REAL published values, not a
range around a mean, and each is labelled with the season it
belongs to. Where the operator publishes a placeholder rather
than a rating, the card says so rather than quietly carrying the
larger number into the maximum. */
const ratings = (() => {
  const mod = ratingModule();
  if (!mod) return null;
  try { return mod.at(topology.index, point.site_code, kv != null ? {
voltageKv: kv } : undefined); }
  catch (_) { return null; }
})();
if (ratings && ratings.circuits.length) {
  const said = [];
  for (const season of ['winter', 'summer']) {
    const band = ratings.by_season[season];
    if (!band || !band.circuits) continue;
    said.push(`${season} ${band.lowest_circuit_mva ===
band.highest_circuit_mva
? band.lowest_circuit_mva
: `${band.lowest_circuit_mva}-${band.highest_circuit_mva}`} MVA`);
  }
}

```

```

        if (said.length) {
            const flagged = ratings.counts.with_a_flagged_value;
            out += caveat(`<b>Circuit ratings:</b> ${escapeHtml(said.join(', '))},`
                + `across ${ratings.counts.circuits}
circuit${ratings.counts.circuits === 1 ? '' : 's'}`. `
                + `Each figure is one circuit's rating in that season. They are not
added `
                + `together: the sum of the circuits at a site is not a quantity
that exists `
                + `in the network, and a rating is not what is free on the circuit.`
                + (flagged ? ` ${flagged} circuit${flagged === 1 ? ' publishes a
value' : 's publishes'}` `
                    + `at or above 9,999 MVA on spans of a kilometre or less; `
                    + `${flagged === 1 ? 'it reads' : 'they read'} as a placeholder
and `
                    + `${flagged === 1 ? 'is' : 'are'} excluded from the range above.`
: ''));
            rating.answered += 1;
            rating.flagged += flagged;
        }
    }

    /* Who owns what lands here, and whether two owners meet.
    -----
    Printed before the planned sentence because it is a fact about
    what is there now. A single owner is a small fact; two owners on
    one circuit is a seam, and a connection across a seam involves
    more than one party. */
    const ownership = (() => {
        const mod = ownerModule();
        if (!mod || !topology.parsedProduct) return null;
        try {
            if (ownerIndex === undefined) ownerIndex =
mod.index(topology.parsedProduct);
            if (!ownerIndex) return null;
            return ownerIndex.at(point.site_code, kv != null ? { voltageKv: kv } :
undefined);
        } catch (_) { return null; }
    })();
    if (ownership && Array.isArray(ownership.owners_present) &&
ownership.owners_present.length) {
        const owners = ownership.owners_present.map((o) =>
escapeHtml(String(o)));
        const seams = (ownership.boundary_circuits || []).length
            + (ownership.boundary_transformers || []).length;
        const counts = ownership.counts || {};
        out += caveat(`<b>Transmission owner${owners.length === 1 ? '' : 's'}:
</b> `
            + `${owners.join(', ')}.`
            + (seams
                ? ` ${seams} branch${seams === 1 ? '' : 'es'} here ${seams === 1 ?
'is' : 'are'}` `
                    + `a boundary: the two ends are published under different owners.`
                    : '')
            + (counts.nodes_with_unknown_owner
                ? ` ${counts.nodes_with_unknown_owner} node here publishes no owner
and is `
                    + `reported as unknown, never taken from the site.`
                    : '')
            + (counts.asset_owner_differs_from_both_ends
                ? ` ${counts.asset_owner_differs_from_both_ends} asset carries an

```

```

owner `
    + `matching neither of its ends; that is reported as itself, not
as a boundary.`
    : ``)
    + ` Ownership is a published fact about an asset. It is not a
statement `
    + `about who a project would contract with, which depends on
connection `
    + `agreements and commercial terms no appendix contains.`);
ownerState.answered += 1;
ownerState.seams += seams;
}

/* What is published as planned, in its own sentence.
-----
Never folded into the counts above. A row published for 2030 is
a statement about a future year; presenting it beside today's
circuits would let a reader take it for one. */
const planned = (() => {
    const mod = plannedModule();
    if (!mod || !topology.parsedProduct) return null;
    try {
        if (plannedIndex === undefined) plannedIndex =
mod.index(topology.parsedProduct);
        if (!plannedIndex) return null;
        return plannedIndex.at(point.site_code, kv != null ? { voltageKv: kv }
: undefined);
    } catch (_) { return null; }
})();
if (planned && planned.counts && planned.counts.planned_changes) {
    /* by_year is an ORDERED ARRAY of { year, by_status: [{ status,
    entries }] }, not a map - the module keeps publication order
    rather than letting object key order decide what the reader
    sees first. */
    const years = (planned.by_year || []).map((band) => {
        const parts = (band.by_status || []).map((s) =>
            `${(s.entries || []).length}`
        );
        return `${escapeHtml(String(s.status).toLowerCase())}
            <b>${escapeHtml(String(band.year))}</b> ${parts.join(', ')}`
        );
    });
    if (years.length) {
        out += caveat(`<b>Published as planned:</b> ${years.join('; ')}. `
            + `These are rows NESO publishes for a future year. None of them is
a `
            + `circuit today, a commitment, a consent, or a connection date, and
`
            + `none is counted among the circuits above.`);
        plannedState.answered += 1;
        plannedState.rows += planned.counts.planned_changes;
    }
}

/* Where the project's own power would go.
-----
Pipeline News sends capacity_mw on every deep link, so the
question "where would MY output flow" is answerable the moment a
project is selected. The slack is NAMED in the sentence: a
transfer has two ends and quoting one of them is meaningless.
What is deliberately absent is any statement about room. */
const injection = (() => {
    const mod = flowModule();
    if (!mod || kv !== 400) return null;

```

```

/* The project's own stated capacity where the deep link carried
   one, and a declared 100 MW probe otherwise - labelled as such in
   the sentence, never presented as the project's figure. */
const mw = Number.isFinite(currentCapacityMw) && currentCapacityMw > 0
  ? currentCapacityMw : 100;
const model = flowModelFor(topology.index);
if (!model) return null;
try {
  const graph = topology.index.graph();
  const here = graph.nodesOfSite(point.site_code)
    .filter(n => graph.nodeVoltageKv(n) === 400).sort()[0];
  if (!here) return null;
  /* The withdrawal bus is DECLARED, not the first one to hand.
  -----
  This took model.buses.find(b => b !== injection), which on a
  network with 238 components is almost always a bus the
  injection cannot reach. Codex found it at 202609020030. The
  module now publishes the rule it uses and the component it
  solved in, and refuses a cross-component transfer outright. */
  const slackNode = typeof mod.sinkFor === 'function'
    ? mod.sinkFor(model, here) : null;
  if (!slackNode) return null;
  const r = mod.respond(model, { atNode: here, slackNode, mw,
minimumShare: 0.05 });
  /* publishable, not validation.passes: a disconnected pair can
     balance at the injection bus while the solve has not converged
     at all, and the old gate would have let that print. */
  return r && r.publishable === true ? r : (r || null);
} catch (_) { return null; }
})();
if (injection && injection.publishable !== true) {
  /* Saying nothing looks identical to having nothing to say. When
     the model cannot solve this transfer the reader is told, with
     the reason, rather than left with a card that quietly lost a
     section. */
  out += caveat(`<b>Where the power would flow:</b> not available here. `
    + escapeHtml(String(injection.reason
      || 'the solve did not meet its acceptance conditions'))
    + ` No figure is shown rather than one that has not converged.`);
  powerflow.refused += 1;
}
if (injection && injection.publishable === true &&
injection.branches.length) {
  const top = injection.branches.slice(0, 3);
  out += caveat(`<b>Where ${injection.injected_mw} MW would flow</b> `
    + `(declared DC model, 100 MVA base, transfer to
${escapeHtml(injection.slack_node)}, `
    + `solved in a component of ${injection.component
  ? injection.component.buses_in_component : '?'} buses): `
    + top.map(b => `${escapeHtml(b.from_node)}-${escapeHtml(b.to_node)} `
    + `${Math.round(Math.abs(b.share_of_injection) * 100)}%`
    + `(b.published_ratings_mva && b.published_ratings_mva.summer
      ? `(summer rating ${b.published_ratings_mva.summer} MVA)` :
    + ` `)).join(', ')
    + ` . Flat 1.0 pu voltages, small angles, no losses, no taps, intact
network. `
    + `This is the response to a NEW injection, not a loading: what is
already `
    + `flowing on these circuits is published nowhere, so whether there is
room `
    + `for it cannot be computed here by anyone.`);
  powerflow.answered += 1;
}

```

```

    powerflow.worst_kirchhoff_error =
Math.max(powerflow.worst_kirchhoff_error,
    injection.validation.kirchhoff_error);
}

/* Electrical distance, beside the one-hop view.
-----
"Circuits reach" above is one hop. This is the second, and it is
reported as a COUNT of sites at each hop rather than as a claim
about any of them: naming a site two hops away and nothing else
would read as a recommendation, which no published appendix
supports. The hop count is never called a distance. */
const reach = (() => {
    const mod = distanceModule();
    if (!mod) return null;
    try { return mod.within(topology.index, point.site_code, { hops: 2,
voltageKv: kv }); }
    catch (_) { return null; }
})();
if (reach && reach.sites.length) {
    const atOne = reach.counts.by_hop[1] || 0;
    const atTwo = reach.counts.by_hop[2] || 0;
    out += caveat(`<b>On the published network:</b> ${atOne} site${atOne ===
1 ? '' : 's'} `
        + `one circuit away${atTwo ? `, ${atTwo} more at two` : ''}. `
        + `A hop is a published circuit, not a distance - a site one hop away
may be a `
        + `hundred kilometres away.${reach.refusals.length
? ` ${reach.refusals.length} branch${reach.refusals.length === 1 ? '
was' : 'es were'}`
        + `not walked because a circuit cannot change voltage; only a
transformer can.`
        : ''}`);
    electrical.answered += 1;
    electrical.refusals += reach.refusals.length;
}
if (facts.neighbours.length) {
    const shownNeighbours = facts.neighbours.slice(0, 6);
    out += caveat(`<b>Circuits reach:</b> `
        + shownNeighbours.map(n => `${escapeHtml(n.site_name) | n.site_code}}
(${n.circuits})`).join(', ')
        + (facts.neighbours.length > shownNeighbours.length
? ` and ${facts.neighbours.length - shownNeighbours.length} more` :
'')
        + `.`);
}
}
if (!shown) {
    return caveat(`<b>Transmission network:</b> none of the substations named
here joins by `
        + `name to a published NESO site, so nothing is stated about their
circuits.`);
}
link.topology = { shown, sites: topology.sites, bytes: topology.bytes };
out += caveat(`${escapeHtml(module ? module.not_an_assessment : '')}`
    + `Counts are branches landing on this site's nodes at the stated voltage,
from `
    + `ETYS Appendix B via Ventusltd/data-grid-gb.`);
return out;
}

const FLOW_COLOUR = '#bfe9ee'; // pale cyan travelling pulse, not

```


white

```
// The flow. MapLibre repeats a dash array along the line, so a short period
// puts several electrons on the wire at once instead of one dot going round.
// Two layers half a period apart double the density without doubling the
// speed, which would only look frantic.
const FLOW_PERIOD = 1.5;
const FLOW_SPEED = 0.055;
const FLOW_PULSE = 0.42;
```

```
/* A fixed set of dash patterns, cycled – not a new one every frame.
```

```
-----
MapLibre rasterises every distinct line-dasharray into a texture atlas
(its LineAtlas) and keeps it for the lifetime of the map. A continuously
varying dasharray therefore asks for a NEW entry sixty times a second,
and the atlas fills: it runs out of space in about twenty seconds, after
which lines stop drawing correctly and the renderer spends its time
managing a texture nobody will reuse.
```

Reported by the Codex session's LineAtlas cardinality gate, which counted five continuously varying writes and refused to call the storm fixed. It was right: the glyph fault in v9.21 and v9.22 was a different fault with a similar symptom, and fixing one did not fix the other.

The animation only needs to LOOK continuous. Twenty-four phases around the cycle is finer than the eye resolves on a moving dash at this speed, and it bounds the atlas at twenty-four entries forever. The patterns are built once, at module load, so the running loop only ever hands back an array it has already handed back before, and MapLibre reuses the raster.

Interpolating the phase against a frame index rather than a clock also makes the flow independent of frame rate, which it was not: a slow phone ran the electrons slower than a desktop. */

```
const FLOW_STEPS = 24;
```

```
const FLOW_PATTERNS = (() => {
  const patterns = [];
  for (let step = 0; step < FLOW_STEPS; step += 1) {
    const phase = (step / FLOW_STEPS) * FLOW_PERIOD;
    const lead = Math.max(0.001, phase);
    const tail = Math.max(0.001, FLOW_PERIOD - phase);
    // Frozen: a caller that mutated one of these would poison every frame
    // that reuses it, and the reuse is the whole point.
    patterns.push(Object.freeze([0.001, lead, FLOW_PULSE, tail]));
  }
  return Object.freeze(patterns);
})();
```

```
// Quantise to one of the prepared patterns. Same input band, same array
// identity, so the atlas never grows past FLOW_STEPS.
/* Write a dash only when it changes.
```

```
-----
Bounding the atlas to twenty-four patterns stopped it filling, but the
call sites still handed MapLibre a value sixty times a second, and
twenty-three of every twenty-four of those were the value it already had.
```

Codex's cardinality gate went on failing on exactly that, and it was asking the right question: a paint-property write per frame is a promise to the renderer that something changed, and it is cheaper not to make it when nothing has.

MEASURED, and smaller than it looks. At FLOW_SPEED 0.055 over a period of

1.5 the phase advances 3.7% of the cycle per frame while a step is 4.2%, so the pattern really does change on most frames: 3,168 writes in 3,600 frames, a reduction of 1.1x rather than the 3.5x this comment first claimed. The saving grows with frame rate, which is the case it is for -- at 120 Hz half the frames become redundant, and on a slow phone almost none do.

The bound on the atlas is the substantive fix. This is tidiness on top of it, and worth having because it is free.

The index is remembered per layer, because the two flow layers run half a period apart and would otherwise fight over one memo. */
const lastDashIndex = new Map();

```
function setFlowDash(map, layerId, phase) {
  const index = flowIndex(phase);
  if (lastDashIndex.get(layerId) === index) return false;
  lastDashIndex.set(layerId, index);
  map.setPaintProperty(layerId, 'line-dasharray', FLOW_PATTERNS[index]);
  return true;
}
```

// Forgotten when the layers go, or a rebuilt layer keeps a stale memo and
// misses its first write.

```
function forgetDashMemo() { lastDashIndex.clear(); }
```

```
function flowIndex(phase) {
  const wrapped = ((phase % FLOW_PERIOD) + FLOW_PERIOD) % FLOW_PERIOD;
  return Math.floor((wrapped / FLOW_PERIOD) * FLOW_STEPS) % FLOW_STEPS;
}
```

```
function flowDash(phase) {
  return FLOW_PATTERNS[flowIndex(phase)];
}
flowDash.patterns = FLOW_PATTERNS;
```

```
const SRC = 'gridatlas-neon-links';
const SRC_NODES = 'gridatlas-neon-nodes';
const L_GLOW = 'l-neon-glow';
const L_CORE = 'l-neon-core';
const L_FLOW = 'l-neon-flow';
const L_FLOW_B = 'l-neon-flow-b';
const L_NODE = 'l-neon-node';
const L_NODE_RING = 'l-neon-node-ring';
const L_LABEL = 'l-neon-label';
```

```
const link = {
  schema: 'gridatlas.neon-substation-links.v1',
  generation: GENERATION,
  minimum_kv: MIN_KV,
  map_captured: false,
  installed: false,
  substations_loaded: 0,
  substations_qualifying: 0,
  last_selection: null,
  links_drawn: 0,
  deep_linked: false,
  boot_trigger: null,
  layer_controls_ready_ms: null,
  layer_controls_arrived_late: false,
  status_message: null,
  labels_drawn: null,
```

```

    gb_panel_installed: false,
    version_ledger: null,
    gb_conditions: null,
    project_layer_enabled: null,
    // iOS Safari background-tab fix: the arrival never starts until the
    // document is visible, and is re-run on visibilitychange if it has not
    // yet produced a visible outcome (see attemptArrival below).
    arrival_deferred_for_visibility: false,
    arrival_attempts: 0,
    arrival_resumed_on_visibility: 0,
    project_pin: { shown: false, name: null },
    substation_layer_enabled: false,
    reduced_motion: false,
    failures: [],
    /* Codex supervision, 202609011446: entries that later recovered stayed
       in `failures`, making a recovered event indistinguishable from a
       terminal fault. Recovery moves an entry here, at the moment the late
       control actually arrives - the entry is preserved as history, and
       `failures` speaks only for what is still failing. */
    recovered: [],
    grid_scope: null,
    grid_scope_armed: false
  };
  window.__GRIDATLAS_NEON_LINKS__ = link;

  /* One entry per distinct active failure. A retry loop used to append the
     same sentence on every attempt, so three tries read as three faults
     and a reader could not tell repetition from spread. Deduplicated
     against what is CURRENTLY failing, never against what has already
     recovered - a fault that comes back deserves to be recorded again. */
  function noteFailure(message) {
    if (!link.failures.includes(message)) link.failures.push(message);
    return false;
  }

  function recoverFailures(pattern) {
    const kept = [];
    for (const entry of link.failures) {
      if (pattern.test(entry)) link.recovered.push(entry);
      else kept.push(entry);
    }
    link.failures = kept;
  }

  /* — geodesy ————— */

  /* Delegated, not reimplemented. Kept as a function declaration rather
     than a const binding so that hoisting behaves exactly as it did before
     - callers earlier in the file are unchanged. */
  function distanceKm(lon1, lat1, lon2, lat2) {
    return GEODESY.distanceKm(lon1, lat1, lon2, lat2);
  }

  /* — substation layer ————— */

  // `33000`, `33000;11000` (two voltages) and `33000:11000` (a transformer
  // ratio) all mean 33 kV is present. Splitting only on ';' drops the ratios.
  /* The unit comes from the property, never from the size of the number.
     -----
     This guessed: anything over 1,000 was volts, anything under was already
     kilovolts. That is not what the source says. OSM's `voltage` tag is in
     VOLTS throughout, including its small values, while an explicit `kv`

```

property is already kilovolts. Magnitude is not the unit.

Audited by the Codex session against the pinned 5,800-feature substation payload: all 5,800 use `voltage`, 229 of them (3.95%) carry a token below 1,000, and every one of those was misread. 204 then displayed a primary voltage ABOVE 400 kV, which does not exist anywhere on this network. The low tokens are 230 (10), 240 (1), 400 (14), 415 (2) and 750 (202) volts – the 750s are DC traction supplies at railway depots, so a depot's third rail was being shown as a 750 kV substation.

Measured project impact, from that audit:

19709	Selhurst Traincare Depot	33000;750	750 kV	->	33 kV
18128	Thames Way, Northfleet	33000;750	750 kV	->	33 kV
14596	Ford Halewood Transmissions	33000;11000;415	415 kV	->	33 kV

Each of those is a real project whose nearest displayed candidate carried an impossible voltage, on a card that also carries a distance. A wrong voltage beside a right distance is worse than either alone, because the distance lends it credibility.

A bare `33` under `voltage` is therefore 33 volts and correctly falls out of a 33 kV-and-above scope. That reads oddly until you remember it is the source's own unit; and Codex confirmed the pinned payload contains no such token, so nothing real is lost by obeying the contract rather than second-guessing it. */

```
function voltagesKv(properties) {
  const out = [];
  const push = (raw, divisor) => {
    for (const token of String(raw ?? '').split(/[;|:\s|+/)) {
      if (!token) continue;
      const value = Number(token);
      if (!Number.isFinite(value) || value <= 0) continue;
      out.push(value / divisor);
    }
  };
  // OSM `voltage` is volts. Always, at every magnitude.
  push(properties?.voltage, 1000);
  // An explicit `kv` is already kilovolts.
  if (!out.length) push(properties?.kv, 1);
  return out;
}

// A polygon's first ring vertex is a corner, not the site.
function representativePoint(geometry) {
  if (!geometry) return null;
  const { type, coordinates } = geometry;
  if (type === 'Point') return [coordinates[0], coordinates[1]];
  const ring = type === 'Polygon' ? coordinates[0]
    : type === 'MultiPolygon' ? coordinates[0]?.[0] : null;
  if (!Array.isArray(ring) || !ring.length) return null;
  let x = 0; let y = 0;
  for (const p of ring) { x += p[0]; y += p[1]; }
  return [x / ring.length, y / ring.length];
}

// Exposed so a proof can check this arithmetic against
// Ventusltd/grid-distance-maths rather than trusting the comment above it.
// Pure functions only; nothing here touches the map or the DOM.
link.measure = { distanceKm, voltagesKv, representativePoint };

/* The receiver decision is pure and exported because Pipeline News owns
```

```

    the complete link corpus. The product path below consumes this exact
    plan; the corpus proof can therefore pass all 8,756 derived source
    points (8,753 served coordinate rows and 8,743 clickable actions)
    through the same decision without booting a map or a 35.7 MB register. */
function deepLinkPlan(rawLon, rawLat, rawRepdRef) {
  const longitude = rawLon === null ? NaN : Number(rawLon);
  const latitude = rawLat === null ? NaN : Number(rawLat);
  const repdRef = String(rawRepdRef || '').trim();
  const coordinatesUsable = Number.isFinite(longitude) &&
Number.isFinite(latitude)
    && Math.abs(longitude) <= 180 && Math.abs(latitude) <= 90
    && !(Math.abs(longitude) < 1e-9 && Math.abs(latitude) < 1e-9);
  return Object.freeze({
    longitude, latitude, repd_ref: repdRef, coordinates_usable:
coordinatesUsable,
    route: coordinatesUsable ? 'MEASURE_LINK_FIRST'
      : (repdRef ? 'WAIT_FOR_REGISTER' : 'NO_USABLE_POINT')
  });
}
link.measure.deepLinkPlan = deepLinkPlan;

/* A resolved register identity is an asynchronous continuation of one
particular arrival, not permission to take the map back later. User
selection and clear paths invalidate the token. The same coordinator is
exported so the generation proof can resolve a deliberately late promise
through the production gate rather than restating this race in a mock. */
function createArrivalGate() {
  let epoch = 0;
  let activeKey = null;
  const invalidations = [];
  return Object.freeze({
    begin(key = null) {
      const candidate = key == null ? null : String(key);
      /* The search owner and the measurement owner claim the same URL
independently as their scripts boot. A keyed claim is idempotent,
so whichever starts first gives both halves one cancellation
epoch rather than making the second half stale immediately. */
      if (candidate && candidate === activeKey) return epoch;
      epoch += 1;
      activeKey = candidate;
      return epoch;
    },
    invalidate(reason) {
      epoch += 1;
      activeKey = null;
      invalidations.push({ epoch, reason: String(reason) || 'unspecified' });
      return epoch;
    },
    isCurrent(candidate) { return candidate === epoch; },
    snapshot() { return Object.freeze({ epoch, activeKey,
      invalidations: [...invalidations] }); }
  });
}
async function continueVerifiedArrival(gate, epoch, verification, apply) {
  const result = await verification;
  if (!gate.isCurrent(epoch)) return false;
  await apply(result);
  return gate.isCurrent(epoch);
}
const arrivalGate = createArrivalGate();
function arrivalKey(search = window.location.search) {
  return 'deep-link:' + String(search || '');
}

```

```

}
function claimPendingArrival(search = window.location.search) {
  return arrivalGate.begin(arrivalKey(search));
}
function invalidatePendingArrival(reason) {
  const epoch = arrivalGate.invalidate(reason);
  link.arrival_reconciliation = { status: 'INVALIDATED', reason, epoch };
  try {
    window.dispatchEvent(new CustomEvent('gridatlas:arrival-invalidated', {
      detail: { reason: String(reason) || 'unspecified', epoch }
    }));
  } catch (_) { /* cancellation still lives in the shared gate */ }
  return epoch;
}
link.measure.createArrivalGate = createArrivalGate;
link.measure.continueVerifiedArrival = continueVerifiedArrival;
link.measure.arrivalGate = arrivalGate;
link.measure.arrivalKey = arrivalKey;
link.measure.claimPendingArrival = claimPendingArrival;
link.measure.invalidatePendingArrival = invalidatePendingArrival;
link.enableSubstationLayer = () => enableSubstationLayer();
link.armGridScope = (on) => { scopeArmed = Boolean(on); return scopeArmed; };
link.clearGridScope = () => clearScope();
link.noteFailure = (message) => noteFailure(message);

/* A browser-history move is a user navigation, not permission for the
   asynchronous identity query from the URL being left to reclaim the map.
   replaceState used by the owner does not emit these events. */
window.addEventListener('popstate', () => invalidatePendingArrival('history-
navigation'));
window.addEventListener('hashchange', () => invalidatePendingArrival('history-
navigation'));

let substationsPromise = null;
function loadSubstations() {
  if (substationsPromise) return substationsPromise;
  substationsPromise = (async () => {
    // The engine may already hold the layer; prefer that over a second fetch.
    const response = await fetch(new URL(SUBS_URL, document.baseURI), { cache:
'force-cache' });
    if (!response.ok) throw new Error(`substations HTTP ${response.status}`);
    const collection = await response.json();
    const features = Array.isArray(collection?.features) ? collection.features
: [];
    link.substations_loaded = features.length;
    const out = [];
    for (const feature of features) {
      const kv = voltagesKv(feature.properties);
      if (!kv.length || Math.max(...kv) < MIN_KV - 0.5) continue;
      const at = representativePoint(feature.geometry);
      if (!at) continue;
      out.push({
        at,
        kv: kv.filter(v => v >= MIN_KV - 0.5).sort((a, b) => b - a),
        name: feature.properties?.name || '',
        operator: feature.properties?.operator
          || feature.properties?.['operator:short'] || ''
      });
    }
    link.substations_qualifying = out.length;
    return out;
  })().catch(error => {

```

```

    link.failures.push(String(error?.message || error));
    substationsPromise = null;
    return [];
  });
  return substationsPromise;
}

function nearestSubstations(lon, lat, subs) {
  const scored = [];
  for (const sub of subs) {
    const km = distanceKm(lon, lat, sub.at[0], sub.at[1]);
    if (km > MAX_LINK_KM) continue;
    scored.push({ ...sub, km });
  }
  scored.sort((a, b) => a.km - b.km);
  return scored.slice(0, LINK_COUNT);
}

// The mirror of nearestSubstations: given a substation, the projects around
// it. Read from the loaded source rather than the viewport, so panning the
// map does not change the answer -- querySourceFeatures returns what the
// GeoJSON source holds, queryRenderedFeatures returns only what is on screen.
function nearestProjects(map, lon, lat) {
  let features = [];
  try { features = map.querySourceFeatures('src-repd') || []; }
  catch (_) { return { loaded: false, links: [] }; }
  // querySourceFeatures reads loaded tiles. With every project layer switched
  // off there are none, and returning an empty list here made the card say
  // "no mapped project within 40 km" of a substation with a 840 MW scheme
  // beside it. Absence from a layer that is not loaded is not absence on the
  // ground, and this is exactly where that rule has to hold.
  if (!features.length) return { loaded: false, links: [] };
  const seen = new Set();
  const scored = [];
  for (const feature of features) {
    const properties = feature.properties || {};
    const tech = String(properties.tech || properties.type || '');
    if (!isProjectTech(tech)) continue;
    const at = representativePoint(feature.geometry);
    if (!at) continue;
    // One source, many tiles: the same project surfaces more than once.
    const key = properties.repd_ref || properties.repdRef
      || `${at[0].toFixed(5)},${at[1].toFixed(5)}`;
    if (seen.has(key)) continue;
    seen.add(key);
    const km = distanceKm(lon, lat, at[0], at[1]);
    if (km > MAX_LINK_KM) continue;
    const capacity = parseFloat(properties.capacity);
    scored.push({
      at, km, tech,
      kv: [],
      name: properties.name || properties.SiteName || properties['Site Name']
    || '',
      mw: Number.isFinite(capacity) ? capacity : null
    });
  }
  scored.sort((a, b) => a.km - b.km);
  return { loaded: true, links: scored.slice(0, LINK_COUNT) };
}

link.measure.nearestSubstations = nearestSubstations;
link.measure.MIN_KV = MIN_KV;

```

```

link.measure.MAX_LINK_KM = MAX_LINK_KM;
link.measure.LINK_COUNT = LINK_COUNT;
link.measure.PROJECT_TECHS = PROJECT_TECHS;
link.measure.flowDash = flowDash;
link.measure.flowIndex = flowIndex;
link.measure.OFFSHORE_TECHS = OFFSHORE_TECHS;
link.measure.isProjectTech = isProjectTech;
link.measure.LAYER_ID_FOR_BUCKET = LAYER_ID_FOR_BUCKET;
link.measure.layerIdForBucket = layerIdForBucket;
/* Exposed so a proof can hold the coordinates constant, vary the technology
   and assert the measurement does not move. nearestSubstations() takes a
   longitude, a latitude and a candidate set and reads no technology at all;
   coverage.policy() takes a technology and produces only sentences. That
   separation IS the invariant, and it is checkable from here. */
link.measure.coverage = coverage;

/* — the project card ————— */

const BLOCK_CLASS = 'gridatlas-neon-block';
const CSS_ID = 'gridatlas-neon-css';

// The distances belong ON the card the user just opened, not in a separate
// panel they have to notice. The engine builds that card with openPopup(),
// which is a closure, so this appends to the rendered popup instead --
// matching the engine's own idiom: monospace on black, cyan heading, amber
// for a figure, grey for provenance.
function installStyles() {
  if (document.getElementById(CSS_ID)) return;
  const style = document.createElement('style');
  style.id = CSS_ID;
  style.textContent = SLD_STYLES.neonBlock(BLOCK_CLASS);
  document.head.appendChild(style);
}

function escapeHtml(value) {
  return String(value).replace(/([<>"']/g, c => (
    { '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&quot;', "'": '&#39;' }
  [c]));
}

// BETA here marks SCOPE, not doubt about the arithmetic. The measurement is
// published and checked; what it does not cover is stated, and the things a
// distance cannot answer at any precision are named rather than implied.
function caveatHtml() {
  return `<div class="neon-caveat">`
    + `<b>Beta analytics, not an actual grid connection.</b> Straight-line
distance to mapped `
    + `geometry &mdash; not a cable route, not a connection length, and no
route has been `
    + `walked. A real connection depends on factors that must be studied:
network impedance `
    + `and fault level, thermal headroom, existing committed connections and
queue position, `
    + `right of way, wayleaves and easements, crossings, terrain, land control
and consent. `
    + `None of those can be inferred from a distance. A mapped substation does
not confirm `
    + `capacity, voltage suitability or acceptance by any network party, and
absence from a `
    + `mapped layer is not absence on the ground.`
    + `</div>`;
}

```



```
// Remembered so the LAYOUT button knows what it was opened from.
let lastSelection = null;

/* The offshore note used to live here and to be the reason there was no
number. It now lives in the technology-coverage module and is the caveat
BESIDE the number. Nothing was deleted: the route reasoning is quoted
there in full, and the module adds the second sentence this file could
not have written - what the candidate set does and does not say about
onshore, counted over the pinned product rather than assumed. */
const OFFSHORE_NOTE = null;

/* What a voltage class means, on the card, beside the number.
-----
A distance to a "132 kV" substation is not the same proposition as a
distance to a "66 kV" one, and the register's reader usually knows that
while the map does not say it. These are descriptions of the network as it
is, not advice about any scheme:

400 / 275 kV  transmission, bulk power
220 kV        transmission, and the class being built out for offshore
               wind landfalls
132 kV        distribution in England and Wales, transmission in
               Scotland - the same number meaning two different things
               depending on where you are standing
66 kV         largely legacy industrial distribution, much of it being
               reinforced to 132 kV and above as old heavy load is
               replaced and offshore wind arrives
33 kV         primary distribution, and the usual class for a
               utility-scale solar or storage connection

Deliberately descriptive. It says what a class generally is, never what a
particular project should do with it, because the estate's rule is that
the maths and the tools do the talking. */
const KV_CONTEXT = {
  400: 'transmission, bulk power',
  275: 'transmission, bulk power',
  220: 'transmission; the class being built out for offshore wind landfalls',
  132: 'distribution in England and Wales, transmission in Scotland',
  66:  'largely legacy industrial distribution, much of it being reinforced '
      + 'to 132 kV and above',
  33:  'primary distribution; the usual class for a utility-scale solar or '
      + 'storage connection',
};

function kvContext(kv) {
  const n = Number(kv);
  return Number.isFinite(n) ? (KV_CONTEXT[n] || null) : null;
}

/* The sentences a policy adds under the distances.
-----
ADDITIVE ONLY, and deliberately so. These change no number, no ordering
and no candidate set; they say what the number is a measurement OF. The
card printed nothing here for every technology until offshore stopped
withholding, and an offshore straight line is the one that most needs
saying out loud - it crosses water. */
function policyNotesHtml() {
  const notes = (currentPolicy && Array.isArray(currentPolicy.notes))
    ? currentPolicy.notes : [];
  if (!notes.length) return '';
  return notes.map(text => `
```

```

    .join('');
}

function cardBlockHtml(links, direction, layerLoaded = true) {
    installStyles();
    const toSubstations = direction !== 'from-substation';
    const title = toSubstations
        ? `Nearest substations &ge;${MIN_KV} kV`
        : 'Nearest projects';
    const fallbackName = toSubstations ? 'Unnamed substation' : 'Unnamed
project';
    const head = `

<span class="neon-km">${l.km.toFixed(2)} km</span>`
            + `



blob:https://ventusltd.github.io/184a2ca1-a0ae-42ea-b3ba-72ceefc889c4



05/09/2026, 16:22  
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```

```

/* One line naming the classes actually present, rather than a legend for
   classes that are not. If every substation found is 33 kV, a paragraph
   about 400 kV transmission is noise. */
const classes = [...new Set(links
  .map(l => (l.kv && l.kv.length ? Number(l.kv[0]) : null))
  .filter(kv => Number.isFinite(kv) && kvContext(kv)))]
  .sort((a, b) => b - a);
const classNote = classes.length
  ? `

${kv} kv</b>`
  ${escapeHtml(kvContext(kv))}`).join(' ')
    + `'. Descriptions of the network, not advice about this scheme.</p>`
    : '';
// The way into the layout. Without this there is no route from a project
// to the sandbox at all, which is exactly how it felt to use.
const kvNoteHtml = classNote;
const button = toSubstations
  ? `


```

```

    if (controls) {
        const rect = controls.getBoundingClientRect();
        if (rect.width > 0 && rect.right > x) x = rect.right + 12;
    }
} catch (_) { /* the default is still inside the map */ }
return { x, y: map.top + 12 };
}

/* The sheet height is not a taste. Measured live at 393x852 with the
answer at the top of the block: grab bar 55px, the identity the engine
renders 134px, the measurement paragraph with the corridor estimate and
its caveat 108px, the scope sentence with the coverage denominator 67px.
364px that must arrive together or not at all, because a measurement may
not appear without the sample its superlative searched or without the
word "straight". 58dvh is 494px at 852 and 464px at 800, so all of it
clears the fold; capped in pixels too, because 58dvh of a tall tablet is
most of the screen and the map is what the sheet stands on. */
const SHEET_MIN = 320;
function sheetTarget() {
    try { return trayTarget(); } catch (_) { return false; }
}
function dockAsSheet(popup, content) {
    const viewport = window.visualViewport?.height || window.innerHeight || 0;
    /* Minimised, the sheet is its bar and nothing else, so the stack on it
comes back down. Measured off the bar, not assumed: it is 55px with a
44px control in it, and a hard-coded 44 would overlap by eleven. */
    const height = popup.classList.contains('gridatlas-min')
        ? Math.max(44, Math.round(
            popup.querySelector('.gridatlas-card-
bar')?.getBoundingClientRect().height || 55) + 12)
        : Math.max(SHEET_MIN, Math.min(520, Math.round(viewport * 0.58)));
    popup.classList.add('gridatlas-sheet');
    popup.classList.remove('gridatlas-free');
    popup.style.removeProperty('--gx');
    popup.style.removeProperty('--gy');
    content.style.removeProperty('max-height');
    document.documentElement.classList.add('gridatlas-sheet-open');
    document.documentElement.style.setProperty('--gridatlas-sheet-h', height +
'px');
}
function undockSheet() {
    document.documentElement.classList.remove('gridatlas-sheet-open');
    document.documentElement.style.removeProperty('--gridatlas-sheet-h');
    document.querySelectorAll('.maplibregl-popup.gridatlas-sheet')
        .forEach(node => node.classList.remove('gridatlas-sheet'));
}

function boundCardToMap() {
    try {
        const container = capturedMap?.getContainer();
        if (!container) return;
        const map = container.getBoundingClientRect();
        const popup = document.querySelector('.maplibregl-popup');
        const content = popup?.querySelector('.maplibregl-popup-content');
        if (!popup || !content) {
            undockSheet();
            document.documentElement.style.setProperty(
                '--gridatlas-card-max', Math.max(160, map.height - 60) + 'px');
            return;
        }
        if (sheetTarget()) { dockAsSheet(popup, content); return; }
        undockSheet();
    }
}

```

```

    if (popup.classList.contains('gridatlas-free')) {
        // A freed card is wherever the user put it, and the same
        // anchor-blindness applies: dragging it low while minimised and then
        // restoring it made it 277px tall starting 88px above the bottom of
        // the map, so it hung 189px underneath. Cap to the room below where it
        // now sits, and if that is not enough, lift it rather than shrink it
        // into a slot.
        const rect = popup.getBoundingClientRect();
        let available = map.bottom - rect.top - 12;
        if (available < MIN_ANCHORED_CARD) {
            const lifted = Math.max(map.top + 12, map.bottom - MIN_ANCHORED_CARD -
12);
            popup.style.setProperty('--gy', lifted + 'px');
            available = map.bottom - lifted - 12;
        }
        content.style.maxHeight = Math.max(120, Math.min(available, map.height -
48)) + 'px';
        return;
    }
    const rect = popup.getBoundingClientRect();
    const available = map.bottom - rect.top - 12;
    if (available < MIN_ANCHORED_CARD) {
        popup.classList.add('gridatlas-free');
        const parked = parkingSpot(map);
        popup.style.setProperty('--gx', parked.x + 'px');
        popup.style.setProperty('--gy', parked.y + 'px');
        content.style.maxHeight = Math.max(160, map.height - 48) + 'px';
        return;
    }
    content.style.maxHeight = available + 'px';
} catch (_) { /* leave the CSS default */ }
}

function addCardBar(content) {
    if (!content) return;
    // The bar is built once and kept, but the fit is not: a card that already
    // has a bar is a card being reused for a different project, and its height
    // is the one thing that must be measured again.
    if (content.querySelector('.gridatlas-card-bar')) { boundCardToMap();
return; }
    const popup = content.closest('.maplibregl-popup');
    if (!popup) return;
    boundCardToMap();

    // Carry the card's own title into the bar. Minimised, the bar is all that
    // is left, and a nameless strip on a map is a puzzle rather than a card you
    // put down on purpose.
    const heading = content.querySelector('b, strong, h1, h2, h3');
    const title = (heading?.textContent || 'Card').replace(/\s+/g, ' ').trim();

    const bar = document.createElement('div');
    bar.className = 'gridatlas-card-bar';
    bar.innerHTML = '<span class="grip">&#8942;&#8942;</span>'
        + '<span class="label">${escapeHtml(title)}</span>'
        + '<span class="spacer"></span>'
        + '<button type="button" class="min" title="Minimise">&minus;</button>'
        + '<button type="button" class="close" title="Close">&times;</button>';
    content.insertBefore(bar, content.firstChild);

    bar.querySelector('.min').addEventListener('click', (event) => {
        event.stopPropagation();
        popup.classList.toggle('gridatlas-min');
    });
}

```

```

        bar.querySelector('.min').innerHTML = popup.classList.contains('gridatlas-
min')
        ? '&plus;' : '&minus;';
        // Restoring gives the card its height back, which is exactly when it can
        // fall off the bottom of the map again.
        requestAnimationFrame(boundCardToMap);
    });
    bar.querySelector('.close').addEventListener('click', (event) => {
        event.stopPropagation();
        clearLinks();
        popup.remove();
        // The sheet is what lifted the control stack and hid the credit strip.
        // Closing the card has to put the map back exactly as it was, or the
        // reader is left with a gap where a card used to be.
        undockSheet();
    });

    // Dragging frees the popup from its anchor. Fixed positioning with an
    // explicit left/top beats MapLibre's transform, which it rewrites on every
    // map move; without that the card would snap back the moment you panned.
    let dragging = null;
    bar.addEventListener('pointerdown', (event) => {
        if (event.target.closest('button')) return;
        /* A docked sheet is not dragged. Dragging the bar was the only route
        to the measurement, and it slid the bar - with the minus and the
        close on it - under the map's search box: tapping the close focused
        the search field and opened the keyboard, 3 times out of 3. On a
        sheet the answer is already on screen, so there is nothing to drag
        for, and the bar is pinned where nothing can be laid over it. */
        if (popup.classList.contains('gridatlas-sheet')) return;
        event.stopPropagation();
        event.preventDefault();
        const rect = popup.getBoundingClientRect();
        dragging = {
            pointerId: event.pointerId,
            dx: event.clientX - rect.left,
            dy: event.clientY - rect.top,
        };
        bar.setPointerCapture?.(event.pointerId);
        popup.classList.add('gridatlas-free');
        popup.style.setProperty('--gx', rect.left + 'px');
        popup.style.setProperty('--gy', rect.top + 'px');
    });
    const move = (event) => {
        if (!dragging || event.pointerId !== dragging.pointerId) return;
        event.preventDefault();
        const map = capturedMap?.getContainer()?.getBoundingClientRect()
            || { left: 0, top: 0, right: window.innerWidth, bottom:
window.innerHeight };
        const card = popup.getBoundingClientRect();
        const minX = map.left + 4;
        const maxX = Math.max(minX, map.right - card.width - 4);
        const minY = map.top + 4;
        // Keep the whole width and at least the 44px drag bar inside the map.
        const maxY = Math.max(minY, map.bottom - Math.min(44, Math.max(1,
card.height)) - 4);
        const x = Math.max(minX, Math.min(maxX, event.clientX - dragging.dx));
        const y = Math.max(minY, Math.min(maxY, event.clientY - dragging.dy));
        popup.style.setProperty('--gx', x + 'px');
        popup.style.setProperty('--gy', y + 'px');
    };
    const up = (event) => {

```

```

    if (!dragging || event.pointerId !== dragging.pointerId) return;
    try { bar.releasePointerCapture?.(event.pointerId); } catch (_) { /*
already released */ }
    if (dragging) requestAnimationFrame(boundCardToMap);
    dragging = null;
  };
  bar.addEventListener('pointermove', move);
  bar.addEventListener('pointerup', up);
  bar.addEventListener('pointercancel', up);
}

```

/* A second project gets a second card, not the first one's shape.

 Reported: arrive from Pipeline News, then click another solar pixel, and the card is the wrong size.

The popup element is reused between selections, and everything this cartridge does to a card was written onto it and never taken off -- the max-height computed for the previous card's contents, the gridatlas-free class if that one had been freed, the --gx/--gy it was parked at, and the minimised state. A card for Botley West would open at the height of the card before it, in the place the card before it had been dragged to.

Worse, addCardBar returns early once the bar exists, and the only call to boundCardToMap on that path was inside it. So on every selection after the first, nothing measured anything: the stale numbers were not merely inherited, they were never recomputed.

Geometry is per selection. The content is not: the bar, its listeners and the drag handlers are kept, because rebuilding them would drop the listeners and cost a card that could no longer be moved. */
 function resetCardGeometry(content) {
 const popup = content?.closest?('.maplibregl-popup');
 if (!popup) return;
 popup.classList.remove('gridatlas-free');
 popup.classList.remove('gridatlas-min');
 popup.style.removeProperty('--gx');
 popup.style.removeProperty('--gy');
 content.style.removeProperty('max-height');
 content.style.removeProperty('display');
 // The bar's own control has to agree with the class it toggles, or a card
 // restored by this reset still shows a plus that no longer minimises
 // anything. The control is .min and it carries an HTML entity, not text.
 const toggle = content.querySelector('.gridatlas-card-bar .min');
 if (toggle) toggle.innerHTML = '−';
 }

```

function injectIntoCard(links, direction, layerLoaded = true) {
  const content = document.querySelector('.maplibregl-popup-content');
  if (!content) return false;
  resetCardGeometry(content);
  addCardBar(content);
  content.querySelectorAll(`.${BLOCK_CLASS}`).forEach(node => node.remove());
  const holder = document.createElement('div');
  holder.innerHTML = cardBlockHtml(links, direction, layerLoaded);
  const block = holder.firstElementChild;
  if (!block) return false;
  // Straight onto the content, never onto firstElementChild. Once the grab
  // bar exists it IS the first element, and appending there put the whole
  // block inside the bar: measured live, a bar that should be 30px tall came
  // out at 401px with the card's contents crammed into a flex row.
  content.appendChild(block);
}

```

```

// The card only has its real height once the block is in it.
requestAnimationFrame(boundCardToMap);
block.querySelector?('.neon-pin')?.addEventListener('click', (event) => {
  event.stopPropagation();
  event.preventDefault();
  const shown = togglePin();
  const control = block.querySelector('.neon-pin');
  if (control) {
    control.textContent = `${shown ? 'Hide' : 'Show'} the project ring`;
    control.setAttribute('aria-pressed', String(shown));
  }
});
block.querySelector?('.neon-layout')?.addEventListener('click', (event) =>
{
  // The card sits inside the map container, so without this the click
  // carries on to the map, lands on the substation underneath and the
  // substation handler overwrites the layout that was just opened.
  event.stopPropagation();
  event.preventDefault();
  if (!lastSelection || !capturedMap) return;
  // The array goes at the project and the cable runs to the nearest
  // substation found for it, which is the direction a scheme is actually
  // built: generation first, then the route to the network.
  openSldFromProject(capturedMap, lastSelection);
});
return true;
}

```

/* — the card keeper —————

The measurement block lives inside the popup, and the popup is not ours: the search lane creates it when identity resolution completes, which on a phone can be seconds after the links were drawn. Watched live on the Pipeline News MAP journey: five links on the map, and a card with no distances, because the popup that had been decorated was replaced by the one that arrived late. The lines live in map layers and survived; the block died with the popup.

So the block is kept, not just written: while a selection is active, an observer re-attaches it whenever the current card lacks it. The payload is per selection - a new selection re-arms it, a cleared selection disarms it, so a radius or measure popup after deselection is never decorated with another project's distances. */

/* — the arrival card —————

The card on a deep-link arrival was the identity lane's popup, and the identity lane resolves against a register that deliberately excludes dead-pipeline statuses - Refused, Revised, Withdrawn, Expired, Abandoned. Pipeline News rightly reports on exactly those schemes. In the exact 0144 corpus, 2,430 of 8,743 unique clickable REPD refs are absent from the active snapshot while 6,313 match it; absent arrivals drew links and no card at all, which on a phone reads as nothing working.

The link itself carries the project - name, technology, capacity, coordinates - so when no card has appeared by the end of the arrival, this cartridge opens one from those fields. It says where it came from, and if the register's own card lands later, the fallback yields to it rather than standing beside it. */

```
let arrivalFallbackPopup = null;
```

```
function removeArrivalFallback() {
  if (!arrivalFallbackPopup) return;
  try { arrivalFallbackPopup.remove(); } catch (_) { /* already gone */ }
  arrivalFallbackPopup = null;
}
```



```

}

function markArrivalIdentityState(status, repdRef, message = '') {
  const node = document.querySelector('.gridatlas-arrival-identity');
  if (!node) return;
  const ref = String(repdRef || '').trim();
  node.dataset.state = String(status || 'UNKNOWN');
  if (status === 'NOT_IN_ACTIVE_REGISTER') {
    node.textContent = 'REPD ' + ref + ' • not in the active-register
snapshot; '
    + 'project details and point are from the arrival link.';
  } else if (status === 'FAILED') {
    node.textContent = 'REPD ' + ref + ' • active-register check failed'
    + (message ? ': ' + message : '') + '; supplied point retained.';
  } else if (status === 'RESOLVED' || status === 'VERIFIED') {
    node.textContent = 'REPD ' + ref + ' • verified in the active-register
snapshot.';
  }
}

function ensureArrivalCard(lon, lat, name, tech, statedMw, repdRef,
suppliedStatus) {
  if (document.querySelector('.maplibregl-popup-content')) return;
  const gl = window.maplibregl;
  if (!gl?.Popup || !capturedMap) return;
  try {
    const cap = Number.isFinite(statedMw) && statedMw > 0
    ? `${statedMw} MW` : '';
    const ref = String(repdRef || '').trim();
    const status = String(suppliedStatus || '').trim();
    arrivalFallbackPopup = new gl.Popup({ maxWidth: '340px', closeOnClick:
false })
    .setLngLat([lon, lat])
    .setHTML('<div style="font-
family:monospace;background:#000;padding:6px">'
    + '<b style="color:#00ffff;font-size:13px">${escapeHtml(name)}</b>
<br>'
    + '<span style="color:#888">${escapeHtml(tech)}</span>'
    + (cap ? '<br><span style="color:#ffae00">${escapeHtml(cap)}</span>' :
'')
    + (ref ? '<br><span class="gridatlas-arrival-identity" data-
state="PENDING" style="color:#8fb3b8;font-size:10px">REPD ${escapeHtml(ref)} •
checking the active-register snapshot</span>' : '')
    + (status ? '<br><span class="gridatlas-arrival-status"
style="color:#888;font-size:10px">Status supplied by arrival link:
${escapeHtml(status)}</span>' : '')
    + '<br><span style="color:#555;font-size:9px">Card and point built
from the arrival link.</span></div>')
    .addTo(capturedMap);
    link.arrival_card = 'from-link-fields';
    link.arrival_card_identity = {
      repd_ref: ref || null,
      name: String(name || ''),
      technology: String(tech || ''),
      capacity_mw: Number.isFinite(statedMw) && statedMw > 0 ? statedMw :
null,
      supplied_status: status || null,
      provenance: 'ARRIVAL_LINK'
    };
  } catch (error) {
    link.failures.push('arrival card: ' + String(error?.message || error));
  }
}

```

```

}

function injectDeclaredOnly() {
  const content = document.querySelector('.maplibregl-popup-content');
  if (!content || !currentDeclared) return false;
  if (content.querySelector(`.${BLOCK_CLASS}`)) return false; // never over a
measured block
  try {
    installStyles();
    resetCardGeometry(content);
    addCardBar(content);
    const holder = document.createElement('div');
    holder.innerHTML = `<div class="${BLOCK_CLASS}">`
      + declaredBlockHtml(true) + caveatHtml() + `</div>`;
    const block = holder.firstElementChild;
    if (!block) return false;
    content.appendChild(block);
    requestAnimationFrame(boundCardToMap);
    link.declared_shown_before_measurement = true;
    return true;
  } catch (error) {
    link.failures.push('provisional card: ' + String(error?.message ||
error));
    return false;
  }
}

let cardKeeper = null;
let cardKeeperPayload = null;

function armCardKeeper(links, direction, layerLoaded) {
  cardKeeperPayload = { links, direction, layerLoaded };
  if (cardKeeper || typeof MutationObserver !== 'function') return;
  try {
    cardKeeper = new MutationObserver(() => {
      const payload = cardKeeperPayload;
      if (!payload) return;
      // If the register's own card has landed beside the fallback, the
      // fallback yields: one card, and the resolved one wins.
      if (arrivalFallbackPopup
        && document.querySelectorAll('.maplibregl-popup-content').length >
1) {
        removeArrivalFallback();
      }
      const content = document.querySelector('.maplibregl-popup-content');
      if (!content || content.querySelector(`.${BLOCK_CLASS}`)) return;
      injectIntoCard(payload.links, payload.direction, payload.layerLoaded);
    });
    cardKeeper.observe(document.body, { childList: true, subtree: true });
  } catch (error) {
    link.failures.push('card keeper: ' + String(error?.message || error));
    cardKeeper = null;
  }
}

function disarmCardKeeper() {
  cardKeeperPayload = null;
}

function removeCardBlock() {
  document.querySelectorAll(`.${BLOCK_CLASS}`).forEach(node => node.remove());
}

```

```

/* — the map layers ————— */

let capturedMap = null;
let animationHandle = null;
let dashPhase = 0;

function emptyCollection() {
  return { type: 'FeatureCollection', features: [] };
}

function ensureLayers(map) {
  if (map.getSource(SRC)) return;

  map.addSource(SRC, { type: 'geojson', data: emptyCollection() });
  map.addSource(SRC_NODES, { type: 'geojson', data: emptyCollection() });

  // Three stacked strokes make the neon: a wide soft glow, a bright core, and
  // a dashed overlay whose offset is animated so the line reads as flowing
  // towards the substation.
  map.addLayer({
    id: L_GLOW, type: 'line', source: SRC,
    layout: { 'line-cap': 'round', 'line-join': 'round' },
    paint: {
      'line-color': ['get', 'colour'],
      'line-width': ['interpolate', ['linear'], ['zoom'], 6, 5, 12, 12],
      'line-opacity': 0.10,
      'line-blur': ['interpolate', ['linear'], ['zoom'], 6, 3, 12, 8]
    }
  });
  map.addLayer({
    id: L_CORE, type: 'line', source: SRC,
    layout: { 'line-cap': 'round', 'line-join': 'round' },
    paint: {
      'line-color': ['get', 'colour'],
      'line-width': ['interpolate', ['linear'], ['zoom'], 6, 0.9, 12, 1.8],
      'line-opacity': ['get', 'strength']
    }
  });
  // Two flow layers, half a period apart, so a link reads as a stream of
  // electrons rather than one dot going round.
  map.addLayer({
    id: L_FLOW, type: 'line', source: SRC,
    layout: { 'line-cap': 'round' },
    paint: {
      'line-color': FLOW_COLOUR,
      'line-width': ['interpolate', ['linear'], ['zoom'], 6, 1.2, 12, 2.4],
      'line-opacity': 0.8,
      'line-dasharray': [0.2, 3.2]
    }
  });
  map.addLayer({
    id: L_FLOW_B, type: 'line', source: SRC,
    layout: { 'line-cap': 'round' },
    paint: {
      'line-color': FLOW_COLOUR,
      'line-width': ['interpolate', ['linear'], ['zoom'], 6, 0.9, 12, 1.8],
      'line-opacity': 0.45,
      'line-dasharray': [0.2, 3.2]
    }
  });
}

```

```

map.addLayer({
  id: L_NODE_RING, type: 'circle', source: SRC_NODES,
  paint: {
    'circle-radius': ['interpolate', ['linear'], ['zoom'], 6, 6, 12, 13],
    'circle-color': 'rgba(0,0,0,0)',
    'circle-stroke-color': SUBSTATION_COLOUR,
    'circle-stroke-width': 1,
    'circle-stroke-opacity': 0.4
  }
});
map.addLayer({
  id: L_NODE, type: 'circle', source: SRC_NODES,
  paint: {
    'circle-radius': ['interpolate', ['linear'], ['zoom'], 6, 2.2, 12, 4],
    'circle-color': SUBSTATION_COLOUR,
    'circle-opacity': 0.8
  }
});
const neonFont = styleTextFont(map);
if (!neonFont) {
  link.labels_drawn = false;
  link.failures.push('the basemap serves no glyphs, so link labels are
omitted');
} else {
  // Defer until a glyph range actually comes back. Nothing waits on
  // labels: the links are already drawn and the distances are on the
  // card.
  addLabelLayerWhenDrawable(map, neonFont, {
    id: L_LABEL, type: 'symbol', source: SRC_NODES,
    layout: {
      'text-field': ['get', 'label'],
      'text-size': 10,
      'text-offset': [0, -1.5],
      'text-anchor': 'bottom',
      'text-allow-overlap': false,
      'text-font': neonFont
    },
    paint: {
      'text-color': '#a9c4c9',
      'text-halo-color': '#000c10',
      'text-halo-width': 1.5,
      'text-opacity': 0.9
    }
  }, 'link');
}
link.installed = true;
}

function stopAnimation() {
  forgetDashMemo();
  if (animationHandle !== null) {
    cancelAnimationFrame(animationHandle);
    animationHandle = null;
  }
}

function startAnimation(map) {
  stopAnimation();
  const reduced = window.matchMedia?.('(prefers-reduced-motion:
reduce)')?.matches === true;
  link.reduced_motion = reduced;
  if (reduced) {

```

```

    // Motion is a preference, not a requirement. The lines, the nodes and
    // every number stay; only the travelling dash stops.
    try {
        map.setPaintProperty(L_FLOW, 'line-opacity', 0);
        map.setPaintProperty(L_FLOW_B, 'line-opacity', 0);
    } catch (_) { /* layer gone */ }
    return;
}
const step = () => {
    dashPhase = (dashPhase + FLOW_SPEED) % FLOW_PERIOD;
    const half = (dashPhase + FLOW_PERIOD / 2) % FLOW_PERIOD;
    try {
        setFlowDash(map, L_FLOW, dashPhase);
        setFlowDash(map, L_FLOW_B, half);
    } catch (_) {
        stopAnimation();
        return;
    }
    animationHandle = requestAnimationFrame(step);
};
animationHandle = requestAnimationFrame(step);
}

function clearLinks() {
    invalidatePendingArrival('clear');
    stopAnimation();
    const map = capturedMap;
    if (map && map.getSource(SRC)) {
        setData(map, SRC, emptyCollection());
        setData(map, SRC_NODES, emptyCollection());
    }
    disarmCardKeeper();
    removeArrivalFallback();
    removeCardBlock();
    clearPin(capturedMap);
    link.links_drawn = 0;
    link.last_selection = null;
    currentRepdRef = null;
    currentDeclared = null;
    currentNearest400 = null;
}

function drawLinks(map, origin, name, tech, links, direction, statedMw,
layerLoaded = true) {
    ensureLayers(map);
    // A link takes the colour of the project end, whichever end was clicked.
    const colour = direction === 'from-substation'
        ? SUBSTATION_COLOUR
        : (TECH_COLOUR[tech] || SUBSTATION_COLOUR);

    const lines = links.map((l, index) => ({
        type: 'Feature',
        properties: {
            colour,
            // The nearest link burns brightest; the rest fade back in order, so
            // rank is legible without reading the numbers.
            strength: Math.max(0.20, 0.62 - index * 0.10),
            km: 1.km
        },
        geometry: { type: 'LineString', coordinates: [origin, l.at] }
    }));
}

```

```

const nodes = links.map(l => {
  const tail = l.kv && l.kv.length ? `${l.kv[0]} kV`
    : (l.mw != null ? `${l.mw} MW` : '');
  return {
    type: 'Feature',
    properties: {
      colour,
      label: tail ? `${l.km.toFixed(2)} km · ${tail}` : `${l.km.toFixed(2)}
km`
    },
    geometry: { type: 'Point', coordinates: l.at }
  };
});

if (direction === 'to-substation' && currentDeclared?.at) {
  // The declared link is public record, drawn in its own colour so it
  // never reads as one more nearest-neighbour measurement.
  const unbuilt = currentDeclared.poc_status === 'not_built'
    || currentDeclared.poc_status === 'under_construction';
  const declaredColour = unbuilt ? DECLARED_UNBUILT_COLOUR :
DECLARED_COLOUR;
  lines.push({ type: 'Feature',
    properties: { colour: declaredColour, strength: 0.85, km:
currentDeclared.km },
    geometry: { type: 'LineString', coordinates: [origin,
currentDeclared.at] } });
  nodes.push({ type: 'Feature',
    properties: { colour: declaredColour,
      label: `PoC \u00b7 ${currentDeclared.km.toFixed(2)} km \u00b7
${currentDeclared.kv} kV`,
    geometry: { type: 'Point', coordinates: currentDeclared.at } });
}
setSourceData(map, SRC, { type: 'FeatureCollection', features: lines });
setSourceData(map, SRC_NODES, { type: 'FeatureCollection', features: nodes
});

// The popup is built by the engine and rendered synchronously in its own
// click handler, but MapLibre attaches it on the next frame in some paths.
// One retry covers that without polling forever.
armCardKeeper(links, direction, layerLoaded);
if (!injectIntoCard(links, direction, layerLoaded)) {
  requestAnimationFrame(() => injectIntoCard(links, direction,
layerLoaded));
}
startAnimation(map);

link.links_drawn = links.length;
link.last_selection = { name, tech, direction, count: links.length,
  nearest_km: links.length ? Number(links[0].km.toFixed(3)) : null };
lastSelection = { origin, name, tech, direction, links, statedMw: statedMw
|| null };
if (direction !== 'from-substation') setPin(map, origin, name, tech);
}

/* — selection ————— */

// Tick the engine's own Subs control. Going through the checkbox means the
// engine hydrates the layer, updates its UI state and stays the owner of it;
// adding the source here instead would leave its panel lying about what is
on.
function enableSubstationLayer() {
  try {

```

```

/* The engine tags its own controls with data-layer-id, and that is
what enableTechnologyLayer and the mobile tray have both used for
generations. This searched LABEL TEXT for "subs " instead, which
is the one part of a control guaranteed to change: the labels
carry live counts and a [WAIT]/[OK]/[LOAD] state. Attribute
first; the label stays only as a fallback for a control that
somehow lacks it. Codex, 202609011823. */
const box = document.querySelector('input[type=checkbox][data-layer-
id="subs"]')
|| [...document.querySelectorAll('input[type=checkbox]')].find((input)
=> {
    const label = (input.closest('label') ||
input.parentElement)?.textContent || '';
    return label.replace(/\s+/g, '
').trim().toLowerCase().startsWith('subs ');
});
if (!box) { noteFailure('subs: control not found'); return false; }
if (!box.checked) box.click();
link.substation_layer_enabled = true;
recoverFailures(/^subs: control not found$/);
return true;
} catch (error) {
    link.failures.push('subs: ' + String(error?.message || error));
    return false;
}
}

// True when a click came from one of our own surfaces -- the card block or
// the layout panel -- rather than from the map itself. MapLibre delivers
// container clicks as map clicks, so without this every button we add fires
// whatever is under it.
function fromOwnUi(event) {
    const target = event?.originalEvent?.target;
    if (!target || typeof target.closest !== 'function') return false;
    return Boolean(target.closest('.maplibregl-popup')
|| target.closest('#gridatlas-sld-panel'));
}

// The engine's own layer control for a technology. Arriving from Pipeline
// News the project itself was invisible: the deep link switched the
// substations on and left the project's layer off, so the card described a
// scheme with no pixel under it and the links appeared to start from nowhere.
// The engine tags each layer control with the layer it drives:
// <input type=checkbox data-layer-id="solar">
// so the technology IS the hook, and no mapping table is needed. Matching on
// the label text worked, but the labels carry live counts -- "Solar PV [2819
// | 52.3GW]" -- so it was matching prose that changes with the data. The
// label match stays as a fallback for a control the engine has not tagged.
const TECH_LABEL_FALLBACK = {
    solar: "Solar PV [", solar_operational: "Solar PV (Operational",
    solar_roof: "Solar Roof [",
    bess: "Battery Storage [", bess_operational: "Battery Storage (Operational",
    wind: "Wind [", wind_onshore_operational: "Onshore Wind (Operational",
};

/* — say what is happening —————
Vikram, tonight: "the map feature from pipelinenews doesnt load on
iphone". Reproduced in kind on the desktop: a black rectangle, no
controls, and a deep link waiting for substations that could not arrive.
Nothing on screen said so. A black map is indistinguishable from a broken
one, and the reader is left to guess which they have.

```

The Atlas boots a 35.7 MB query engine before it can answer anything. On a phone over cellular that is a long wait and sometimes not a wait at all, and the honest thing is to say which. This chip says what is being waited for, and when the wait has failed it says that too, with a way to try again -- because a retry after the network recovers is usually all it needs, and a reload throws away the deep link.

It removes itself the moment the controls arrive, so a working Atlas is never decorated with news about itself. */

```
const STATUS_ID = 'gridatlas-boot-status';

function statusHost() {
  return document.querySelector('.maplibregl-map') || document.body;
}

function showStatus(message, kind) {
  try {
    let el = document.getElementById(STATUS_ID);
    if (!el) {
      el = document.createElement('div');
      el.id = STATUS_ID;
      el.setAttribute('role', 'status');
      el.setAttribute('aria-live', 'polite');
      statusHost().appendChild(el);
    }
    el.dataset.kind = kind || 'waiting';
    el.textContent = message;
    if (kind === 'failed') {
      const again = document.createElement('button');
      again.type = 'button';
      again.textContent = 'Try again';
      again.addEventListener('click', (event) => {
        event.stopPropagation();
        event.preventDefault();
        retryArrival();
      });
      el.appendChild(again);
    }
    link.status_message = message;
  } catch (error) {
    link.failures.push('status: ' + String(error?.message || error));
  }
}

function clearStatus() {
  document.getElementById(STATUS_ID)?.remove();
  link.status_message = null;
}

// Re-run the arrival rather than reloading: a reload on a phone repeats the
// whole 35.7 MB boot, and the deep link is in the URL either way.
let retryArrival = () => {};
let rerunDeepLink = null;

async function retryIdentityOwnerThenArrival() {
  clearStatus();
  const owner = window.__GRIDATLAS_PLACE_SEARCH__;
  if (typeof owner?.retry_exact_deep_link !== 'function') {
    showStatus('The active-register identity owner cannot retry in this
composition.',
      'failed');
  }
}
```



```

    return false;
}
const invalidatedEpoch = invalidatePendingArrival('identity-retry');
const retryEpoch = claimPendingArrival(window.location.search);
link.arrival_retry = {
  status: 'OWNER_PENDING', invalidated_epoch: invalidatedEpoch,
  owner_epoch: retryEpoch, measurement_epoch: retryEpoch
};
try {
  await owner.retry_exact_deep_link(retryEpoch);
} catch (error) {
  link.arrival_retry.status = 'FAILED';
  showStatus('The active-register identity retry failed: '
    + String(error?.message || error), 'failed');
  return false;
}
const terminal = window.__GRIDATLAS_PLACE_SEARCH__.deep_link;
if (!arrivalGate.isCurrent(retryEpoch)
  || terminal?.owner_epoch !== retryEpoch
  || terminal?.status === 'CANCELLED') {
  link.arrival_retry.status = 'CANCELLED';
  return false;
}
if (terminal?.status === 'FAILED') {
  link.arrival_retry.status = 'FAILED';
  showStatus('The active-register identity check failed: '
    + String(terminal.message || 'identity loader unavailable'), 'failed');
  return false;
}
if (typeof rerunDeepLink !== 'function') {
  link.arrival_retry.status = 'FAILED';
  showStatus('The grid measurement owner cannot retry in this composition.',
    'failed');
  return false;
}
link.arrival_retry.status = 'MEASUREMENT_PENDING';
const completed = await rerunDeepLink(retryEpoch);
if (!arrivalGate.isCurrent(retryEpoch)) {
  link.arrival_retry.status = 'CANCELLED';
  return false;
}
link.arrival_retry.status = terminal.status === 'NOT_IN_ACTIVE_REGISTER'
  ? 'NOT_IN_ACTIVE_REGISTER' : (completed === false ? 'FAILED' :
'RESOLVED');
return completed !== false;
}

```

```

function injectStatusStyle() {
  if (document.getElementById(STATUS_ID + '-style')) return;
  const style = document.createElement('style');
  style.id = STATUS_ID + '-style';
  style.textContent = SLD_STYLES.bootStatus(STATUS_ID);
  document.head.appendChild(style);
}

```

```

// Resolve when the engine has rendered its layer dashboard, or when the
// wait is up. Returning false is a fact worth having, not an error: it says
// the engine had not finished, which is a different problem from the layer
// being missing.
/* Watch for the controls; do not guess how long they will take.

```

 A fixed budget is always the wrong number. Twelve seconds was generous on

one load and hopeless on the next: the engine builds its layer dashboard from its own data, and that has been measured arriving in two seconds and not arriving at all in eighty-six.

Giving up after a budget also gave up permanently. If the dashboard appeared at thirteen seconds -- which it often does -- the layers the arrival depends on stayed off for the rest of the session, with a card on screen saying the grid data had not loaded while the controls sat there.

So: the wait still bounds how long the user is asked to look at a spinner, because that is a promise about the interface. But an observer keeps watching afterwards, and switches the layers on whenever they arrive, however late. The status line is cleared at the same moment, because a failure notice that outlives the failure is its own bug.

The observer disconnects the first time it fires. It is not a subscription to the page; it is one deferred question. */

```
const LAYER_CONTROL = 'input[type=checkbox][data-layer-id]';
let layerWatcher = null;

function watchForLayerControls(onReady) {
  if (layerWatcher || typeof MutationObserver !== 'function') return;
  try {
    layerWatcher = new MutationObserver(() => {
      if (!document.querySelector(LAYER_CONTROL)) return;
      layerWatcher.disconnect();
      layerWatcher = null;
      link.layer_controls_arrived_late = true;
      recoverFailures(/^the engine had not rendered its layer controls
within/);
      clearStatus();
      try { onReady(); } catch (error) {
        link.failures.push('late layers: ' + String(error?.message || error));
      }
    });
    layerWatcher.observe(document.body, { childList: true, subtree: true });
  } catch (error) {
    link.failures.push('layer watcher: ' + String(error?.message || error));
    layerWatcher = null;
  }
}

/* iOS Safari, reported live by the architect and reproduced independently
against a page opened hidden: this budget used to be charged in WALL
CLOCK time regardless of whether anyone could see the result of
spending it. Pipeline News' MAP control opens on touch devices with
target="_blank", and on iOS Safari a background tab is not guaranteed
to be composited while the reader is still looking at the page they
tapped from -- requestAnimationFrame does not tick there, so the
engine's own dashboard build (which the boot trigger gates behind
style.load/load, both paint-driven) can stall for the WHOLE budget
without ever having had a real chance. Elapsed time while hidden buys
nothing observable and is not charged against the budget; only time
the tab was actually visible counts down. */
async function waitForLayerControls(budgetMs) {
  let elapsed = 0;
  const started = Date.now();
  /* Visible time governs the BUDGET; wall clock still bounds the WAIT.
-----
  Charging the budget in visible time is what stops a background tab
spending it before anyone can see the result. But a tab that is never
```

```

    made visible would then poll every 200ms forever, which is a hang by any
    honest definition and costs battery on the device least able to afford
    it. So an absolute ceiling backstops the visible-time budget: whichever
    runs out first ends the wait. Ten minutes matches the last-resort stop
    the arrival lane already uses for a lane that died silently. */
const HARD_CEILING_MS = 600000;
if (!document.querySelector(LAYER_CONTROL)) {
    injectStatusStyle();
    showStatus('Switching the grid layers on as soon as the map\u2019s own '
        + 'controls appear. The distances do not wait for them.', 'waiting');
}
while (elapsed < budgetMs && Date.now() - started < HARD_CEILING_MS) {
    if (document.querySelector(LAYER_CONTROL)) {
        /* Both clocks are published, because they answer different questions:
           visible time is what the budget was actually spent from, wall time
           is what the reader sat through. Reporting only one of them makes a
           background arrival look instant or a foreground one look slow. */
        link.layer_controls_ready_ms = Date.now() - started;
        link.layer_controls_ready_visible_ms = elapsed;
        clearStatus();
        return true;
    }
    await new Promise(resolve => setTimeout(resolve, 200));
    if (document.visibilityState === 'visible') elapsed += 200;
}
link.layer_controls_ready_ms = null;
link.failures.push(
    'the engine had not rendered its layer controls within '
    + Math.round(budgetMs / 1000) + 's; still watching, and the layers will '
    + 'be switched on if they arrive');
if (link.links_drawn > 0) {
    // The answer is already on the map. Late layers are not a failure the
    // reader has to act on, and a red notice over a working map is noise.
    clearStatus();
} else {
    injectStatusStyle();
    showStatus('The grid data has not finished loading yet. The distances '
        + 'below are already measured; the layers will switch on by themselves '
        + 'if it arrives.', 'failed');
}
return false;
}

function enableTechnologyLayer(tech) {
    if (!tech) return false;
    // Resolved through the ONE table above. 'wind_onshore' and
    // 'wind_offshore' are Pipeline buckets, not layer ids -- searching the
    // DOM for a control literally named that always failed. 'other' has no
    // layer at all, and is said plainly rather than searched for.
    const layerId = layerIdForBucket(tech);
    if (layerId === null) {
        link.technology_layer = Object.assign({}, link.technology_layer, {
            requested: tech, layer_id: null, enabled: false,
            reason: 'GridAtlas has no map layer for the "' + tech + '" technology; '
                + 'nothing to switch on. The card and the distances above it are '
                + 'unaffected.'
        });
        return false;
    }
    try {
        const boxes = [...document.querySelectorAll('input[type=checkbox]')];
        let box = boxes.find((input) => input.dataset?.layerId === layerId);
    }
}

```

```

    if (!box) {
      const label = TECH_LABEL_FALLBACK[layerId];
      if (label) {
        box = boxes.find((input) => {
          const text = (input.closest('label') ||
            input.parentElement)?.textContent || "";
          return text.replace(/\s+/g, " ").trim().toLowerCase()
            .startsWith(label.toLowerCase());
        });
      }
    }
    if (!box) { noteFailure('layer control not found: ' + layerId); return
false; }
    if (!box.checked) box.click();
    link.project_layer_enabled = layerId;
    // The field a reader (and every prior proof) actually trusted must
    // say what happened, not what the request's bucket merely belonged
    // to. Set here, on the ONE path that turns a control on, rather than
    // synthesised from set membership before this ever ran.
    link.technology_layer = Object.assign({}, link.technology_layer, {
      requested: tech, layer_id: layerId, enabled: true, reason: null
    });
    recoverFailures(new RegExp('^layer control not found: '
      + String(layerId).replace(/[\.*+?^${}()|[\]\\"/g, '\\$&') + '$'));
    return true;
  } catch (error) {
    link.failures.push('layer: ' + String(error?.message || error));
    return false;
  }
}

```

/* — the project pin —————
 A marker for the selected project, drawn by this cartridge rather than
 borrowed from a layer.

The engine's technology layers are hydrated on demand and can be switched
 off by the user, so a project arriving by deep link may have no pixel at
 all. This one does not depend on any of that: it is the thing the card is
 about, and while a card is open its subject should be visible on the map.
 It toggles, because a pin over the site is exactly what you want out of the
 way when you are looking at the site. */

```

const SRC_PIN = 'gridatlas-project-pin';
const L_PIN_HALO = 'l-project-pin-halo';
const L_PIN = 'l-project-pin';
let pinVisible = true;

```

```

function ensurePinLayers(map) {
  // addSource throws if the style is not loaded, and a source that failed to
  // add reads back as null. The pin is a convenience: it may not be the
  // reason a card fails to open.
  if (!map || typeof map.addSource !== 'function') return false;
  if (map.getSource(SRC_PIN)) return true;
  try {
    map.addSource(SRC_PIN, { type: 'geojson', data: { type: 'FeatureCollection',
features: [] } });
  }
  /* A ring around the site, not a dot on it.

```

 A filled dot in the technology colour was invisible: it sat under the
 engine's own pixel for the same project, and the neon links converging
 on it are drawn in that same colour, so it disappeared into its own
 arrival point. Seen in Chrome at zoom 12 on Botley West -- position

exactly right, nothing to look at.

A ring solves all three. It does not duplicate the engine's pixel, because it surrounds it. It reads against the links, because it crosses them rather than joining them. And it answers the question the marker exists for -- which of these is the one the card is about -- which a second dot among dots cannot. */

```
map.addLayer({
  id: L_PIN_HALO, type: 'circle', source: SRC_PIN,
  paint: {
    'circle-radius': ['interpolate', ['linear'], ['zoom'], 6, 11, 14, 26],
    'circle-color': 'rgba(0,0,0,0)',
    'circle-stroke-color': ['get', 'colour'],
    'circle-stroke-width': 6,
    'circle-stroke-opacity': 0.13,
    'circle-blur': 0.4,
  },
});
map.addLayer({
  id: L_PIN, type: 'circle', source: SRC_PIN,
  paint: {
    'circle-radius': ['interpolate', ['linear'], ['zoom'], 6, 11, 14, 26],
    // Hollow: whatever the engine draws for this project stays readable
    // inside it.
    'circle-color': 'rgba(0,0,0,0)',
    // Pale, not white. It has to separate from the technology colour it
    // encircles without becoming the loudest thing on a dark map.
    'circle-stroke-color': '#cfe9ed',
    'circle-stroke-width': 1.6,
    'circle-stroke-opacity': 0.85,
  },
});
} catch (error) {
  link.failures.push('pin: ' + String(error?.message || error));
  return false;
}
return Boolean(map.getSource(SRC_PIN));
}

function setPin(map, origin, name, tech) {
  if (!ensurePinLayers(map)) return;
  const source = map.getSource(SRC_PIN);
  if (!source || typeof source.setData !== 'function') return;
  const colour = TECH_COLOUR[tech] || SUBSTATION_COLOUR;
  source.setData({
    type: 'FeatureCollection',
    features: origin && pinVisible ? [{
      type: 'Feature',
      properties: { colour, name: name || '' },
      geometry: { type: 'Point', coordinates: origin },
    }] : [],
  });
  link.project_pin = { shown: Boolean(origin && pinVisible), name: name ||
null };
}

function clearPin(map) {
  const source = map && map.getSource && map.getSource(SRC_PIN);
  if (source && typeof source.setData === 'function') {
    source.setData({ type: 'FeatureCollection', features: [] });
  }
  link.project_pin = { shown: false, name: null };
}
```

```

}

function togglePin() {
  pinVisible = !pinVisible;
  if (capturedMap && lastSelection) {
    setPin(capturedMap, lastSelection.origin, lastSelection.name,
lastSelection.tech);
  }
  return pinVisible;
}
link.togglePin = togglePin;

/* — labels need glyphs, and glyphs can be absent —————
A symbol layer cannot draw text without a glyph atlas, and maplibre does
not degrade when it cannot build one: it throws reading `width` off a null
atlas, and it does it again on the NEXT frame, and the next. Both of us
watching this estate tonight found the same storm from different ends --
Codex counted 50+ in about 20 seconds on mounting the layout, and a cold
load here produced 4,218. Same exception, and the two symbol layers in
this cartridge are the only text it draws.

Two ways to have no atlas: the style carries no `glyphs` endpoint at all,
or it has one and the named font is not served by it. The font name here
was assumed -- 'Open Sans Bold' -- rather than taken from the style that
has to serve it, so a basemap with a different font family produced text
that could never resolve.

So: ask the style. No glyphs endpoint means no labels, which is a quiet
map rather than a broken one. Otherwise use a font the style already uses
for its own labels, because that one is definitely served.

This matters most on a phone. An exception per frame is a main thread that
never idles, and on a phone that is heat, battery and a page that stops
answering touches. */
function styleTextFont(map) {
  try {
    const style = map.getStyle?();
    if (!style || !style.glyphs) return null;
    for (const layer of style.layers || []) {
      const font = layer?.layout?.['text-font'];
      if (Array.isArray(font) && font.length && typeof font[0] === 'string') {
        return font;
      }
    }
    // A glyph endpoint with no symbol layer to learn from. This is the
    // Mapbox/MapLibre default family and the one CARTO serves.
    return ['Open Sans Bold', 'Arial Unicode MS Bold'];
  } catch (error) {
    link.failures.push('glyphs: ' + String(error?.message || error));
    return null;
  }
}

/* Having a glyphs endpoint is not the same as being able to reach it.
-----
Naming a font the style serves fixed one half. The other half was watched
live and is worse: the style declared

  https://tiles.basemaps.cartocdn.com/fonts/{fontstack}/{range}.pbf

on the same CDN that had just returned 200 for style.json and then served
no vector tiles at all. A declared endpoint that cannot be reached gives

```

exactly the same null atlas as no endpoint, and 5,362 exceptions in thirty seconds. Checking that the property exists proves nothing; the only honest test is to ask for a range and see.

So fetch one - the first 256 codepoints, a few kilobytes, the same request the renderer would make. If it does not come back, there are no labels. That is a map without text, which is a great deal better than a map that throws on every frame it ever draws.

```
Deferring the labels costs nothing: they are decoration over links that
are already on screen, and the distances they annotate are on the card. */
async function glyphsReachable(map, font) {
  let template;
  try { template = map.getStyle?().?.glyphs; } catch (error) { template =
null; }
  if (!template || !font) return false;
  const url = String(template)
    .replace('{fontstack}', encodeURIComponent(font.join(',')))
    .replace('{range}', '0-255');
  try {
    const response = await fetch(url, { cache: 'force-cache' });
    if (!response.ok) {
      link.failures.push('glyph range ' + response.status + '; labels
omitted');
      return false;
    }
    return true;
  } catch (error) {
    link.failures.push('glyph range unreachable; labels omitted');
    return false;
  }
}

// Add a symbol layer only once its text can actually be drawn. Callers do
// not await this: the labels arrive when they arrive, or never, and nothing
// else waits on them.
function addLabelLayerWhenDrawable(map, font, spec, what) {
  glyphsReachable(map, font).then((ok) => {
    link.labels_drawn = ok;
    if (!ok) return;
    try {
      if (!map.getLayer(spec.id)) map.addLayer(spec);
    } catch (error) {
      link.failures.push(what + ' labels: ' + String(error?.message ||
error));
    }
  });
}
```

```
/* — GB grid conditions, from the tracker that already measures them —
The estate already has an application that tracks GB electricity:
globalgrid2050.com/uk_energy_tracking_v6, backed by Ventusltd/
data-gb-electricity. The Atlas had no idea it existed, so a map of where
the country is building generation could not tell you what the system was
doing.
```

This does not port that application. It is 49 MB, and the Atlas already boots a 35.7 MB query engine before it can answer anything -- adding a second one would be a way of making both worse. It reads the small published feeds the tracker writes, one to four kilobytes each, and links to the full application for everything else. The tracker stays the place the analysis lives.

HONESTY ABOUT AGE. Measured when this was written, those feeds were stamped 2026-06-18: about ten weeks old. A panel that prints a price with no date implies it is current, and a stale number presented as live is worse than no number. So the age is always shown, and past a day it is labelled as not current rather than merely dated. If the feeds start updating again the same panel gets better on its own.

Mobile first: it opens collapsed, sized against the viewport, and is a single column on a narrow screen. */

/* What a megawatt hour has been worth in the available historic record.

A map of where the country is building generation should be able to say what the system has been doing while it was built.

IT READS THE DATA REPOSITORY, NOT A COPY. The governing rule in the estate's migration scope is "data before charts": a consumer must read a data product that already sits clean, and must never own source data or become a second source of truth. So this reads Ventusltd/data-gb-electricity, which owns the Parquet and publishes a browser-sized rollup derived from it. An earlier version of this panel read

a

copy derived inside globalgrid2050; that copy was a second definition of the same numbers and has been retired in favour of this one.

A ROLLUP, NOT A HUNDRED MEGABYTES. Settlement-period history is the right size for a chart someone chose to open and the wrong size for a panel inside a map on a phone, which is where most readers arrive.

THE MEASUREMENT STOPS WHERE THE PRODUCT STOPS. A negative system price is an observed market value. It does not by itself establish a local network constraint, curtailment, connection capacity, a usable charging window or the economics of any project on this map.

SOLAR IS ABSENT, AND SAYS SO. PVLive has not been decided into the data repository, so the product declares solar absent rather than carrying a series from somewhere else. A panel that quietly filled that gap from a second source would be the exact thing the discipline forbids. */

```
const GB_PIN_ID = 'price-decade-rollup';
const GB_ROLLUP = PINS ? PINS.url(GB_PIN_ID) : null;
const GB_APP = 'https://globalgrid2050.com/uk_energy_tracking_v6/';
const GB_ID = 'gridatlas-gb-conditions';
const GB_SCHEMA = 'data-gb-electricity.price-decade-rollup.v2';
/* Loader state on the window, as the topology loader's is, so the source
   registry can say whether this product answered, was withheld (reached,
   not the schema this consumer answers) or failed - generation 202609012217.
```

*/

```
const gbLoader = { state: 'idle', product: GB_ROLLUP, schema_required:
GB_SCHEMA,
  schema: null, reason: null, error: null, renders: 0 };
window.__GRIDATLAS_GB_CONDITIONS__ = gbLoader;
```

```
const gbNumber = (value, digits) =>
  Number.isFinite(Number(value)) ? Number(value).toFixed(digits) : '--';
```

```
function gbRow(label, value, unit) {
  return `

<span class="gb-k">${label}</span>`
    + `



blob:https://ventusltd.github.io/184a2ca1-a0ae-42ea-b3ba-72ceefc889c4



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```



```

function gbProductError(product) {
  if (!product || product.schema !== GB_SCHEMA) return 'owner product v2 is
not available';
  const price = product.price || {};
  const derived = product.derived_from || {};
  const years = Array.isArray(price.by_year) ? price.by_year : [];
  if (!years.length) return 'year rows are absent';
  const included = years.reduce((sum, row) => sum + Number(row.days_included),
0);
  const negative = years.reduce((sum, row) =>
    sum + Number(row.days_with_a_negative_settlement_period), 0);
  if (included !== Number(derived.included_days)) return 'included-date total
disagrees';
  if (negative !== Number(price.days_with_a_negative_settlement_period)) {
    return 'negative-date total disagrees';
  }
  const validYears = years.every(row => {
    const days = Number(row.days_included);
    const calendarDays = Number(row.calendar_days);
    const status = days === calendarDays ? 'FULL_DATE_COVERAGE' :
'PARTIAL_DATE_COVERAGE';
    const coverage = 100 * days / calendarDays;
    const share = 100 * Number(row.days_with_a_negative_settlement_period) /
days;
    return days === Number(row.days)
      && calendarDays >= days
      && row.calendar_date_coverage === status
      && Math.abs(Number(row.calendar_date_coverage_pct) - coverage) < 0.011
      && Math.abs(Number(row.negative_period_day_share_pct) - share) < 0.011;
  });
  if (!validYears) return 'year coverage or share disagrees';
  if (Math.abs(Number(price.negative_period_day_share_pct)
    - (100 * negative / included)) >= 0.011) return 'record share disagrees';
  for (const extreme of [price.lowest_settlement_period,
price.highest_settlement_period]) {
    if (!Number.isInteger(extreme?.settlement_period)
      || !/^\\d{4}-\\d{2}-\\d{2}T\\d{2}:\\d{2}:\\d{2}Z$/u.test(extreme?.period_start_utc)) {
      return 'extreme-period identity is incomplete';
    }
  }
  return '';
}

async function renderGbConditions(body) {
  let product = null;
  gbLoader.state = 'loading';
  try {
    /* no-cache, not force-cache. Watched live on v9.41's first minutes:
    the owner had published v2, a fresh fetch returned v2, and the panel
    showed its withheld state - because force-cache handed back the v1
    this browser had fetched hours earlier and never revalidated. A
    consumer that gates on a product's version must not pin itself to
    whichever version it saw first. no-cache revalidates every open,
    which costs a 304 when nothing changed and correctness when it did. */
    const response = await fetch(GB_ROLLUP, { cache: 'no-cache' });
    if (response.ok) {
      const text = await response.text();
      gbLoader.pin = await PINS.verify(GB_PIN_ID, text);
      if (gbLoader.pin.state === 'MISMATCH') gbLoader.error =
gbLoader.pin.detail;
      else product = JSON.parse(text);
    }
  }
}

```

```

    }
  } catch (error) {
    product = null;
    gbLoader.error = String(error && error.message || error);
  }
  if (!product) {
    gbLoader.state = 'failed';
    body.innerHTML = '<p class="gb-note">The price rollup could not be '
      + 'reached. This says nothing about the grid, only about the network '
      + 'between here and the data repository.</p>';
    link.gb_conditions = { reached: false };
    return;
  }

  const productError = gbProductError(product);
  if (productError) {
    gbLoader.state = 'withheld';
    gbLoader.reason = productError;
    gbLoader.schema = product.schema || null;
    body.innerHTML = '<p class="gb-note">The owner price product did not pass
      + `the v2 evidence gate (${productError}), so no values are shown.</p>`;
    link.gb_conditions = { reached: true, schema_supported: false,
      source: 'data-gb-electricity', reason: productError, project_bindings: 0
    };
    return;
  }

  const price = product.price || {};
  const years = Array.isArray(price.by_year) ? price.by_year : [];
  const latest = years.length ? years[years.length - 1] : null;
  const low = price.lowest_settlement_period || null;
  const includedDays = Number((product.derived_from || {}).included_days);
  const negativeDays = Number(price.days_with_a_negative_settlement_period);
  const partialYears = years.filter(row =>
    row.calendar_date_coverage === 'PARTIAL_DATE_COVERAGE');

  const rows = [];
  rows.push(gbRow('Available-record daily mean',
    gbNumber(price.available_record_daily_mean, 2), ' &pound;/MWh'));
  rows.push(gbRow('Dates with a negative period',
    gbNumber(price.negative_period_day_share_pct, 2) + '%',
    `${negativeDays} of ${includedDays}`));
  if (latest) {
    const partial = latest.calendar_date_coverage === 'PARTIAL_DATE_COVERAGE';
    rows.push(gbRow(latest.year + (partial ? ' partial' : ''),
      gbNumber(latest.mean_gbp_per_mwh, 2), ' &pound;/MWh'));
  }

  const lowLine = low && Number.isFinite(Number(low.value))
    ? '<p class="gb-note gb-point">Lowest observed settlement price: '
      + '<b>${gbNumber(low.value, 2)} &pound;/MWh</b>, SP
    ${low.settlement_period}, `
      + `${low.period_start_utc}. This is a historic GB system-price `
      + 'observation, not evidence about a project on this map.</p>'
    : '';

  // Absent by decision, so it is stated rather than left as a silent gap.
  const solarLine = product.solar && product.solar.present === false
    ? '<p class="gb-note">Solar is not in this product yet: the data '
      + 'repository has not taken PVLive, and filling the gap from somewhere '
      + 'else would make a second source of truth.</p>'

```

```

: '';

const span = Array.isArray(price.span) ? price.span.join('-') : '';
body.innerHTML = rows.join('')
+ lowLine
+ solarLine
+ `
```

/* — the version ledger, on the page _____
The estate's whole method is sealed, timestamped compositions - build a new one, never edit the last - and in one overnight session that produced twenty-four of them, each a correction with a one-line scope. Pipeline News ledgers its releases on the homepage; the Atlas's generations were visible only in git, which a visitor does not have.

The ledger below is extracted from git at BUILD time: every composition manifest that ever existed, read at its last commit. It is pinned history carried by the page, not prose about it, and nothing is fetched at runtime. Rollback doctrine, stated where the versions are: a bad composition is never repaired in place - an earlier one is composed again under a new timestamp.

Mobile first, like everything since Vikram said the link travels by WhatsApp: a collapsed chip, viewport-sized body, newest first. */
/* The ledger itself is a module now, in the cartridge with room for it - see atlas/modules/202609030157-version-ledger.js. The name is kept so every reader below is unchanged, and an absent module gives an empty

```

    ledger the panel reports rather than a throw that costs the session. */
const VERSION_LEDGER =
  (window.__GRIDATLAS_MODULES__ || {}).versionLedger?.entries || [];
const PRE_SCOPE_COMPOSITIONS = 16;
const LEDGER_ID = 'gridatlas-version-ledger';
const LEDGER_DOCTRINE = 'Each row is immutable audit evidence, not a promise '
  + 'that it was live. REJECTED_PRE_PROMOTION entries were never live and are
  + 'not rollback targets. A deployed bad composition is never repaired in
place; '
  + 'an earlier deployed one is composed again under a new timestamp.';

function ledgerStamp(generation) {
  return generation.slice(0, 4) + '-' + generation.slice(4, 6) + '-'
    + generation.slice(6, 8) + ' ' + generation.slice(8, 10) + ':'
    + generation.slice(10, 12) + ' UTC';
}

function installVersionLedger() {
  if (document.getElementById(LEDGER_ID)) return;
  const stack = document.querySelector('.map-controls');
  if (!stack) { link.failures.push('no map-controls for the ledger'); return;
}

const style = document.createElement('style');
style.id = LEDGER_ID + '-style';
style.textContent = SLD_STYLES.versionLedger(LEDGER_ID);
document.head.appendChild(style);

const panel = document.createElement('div');
panel.id = LEDGER_ID;
panel.dataset.open = '0';
const button = document.createElement('button');
button.type = 'button';
const newest = VERSION_LEDGER[VERSION_LEDGER.length - 1];
button.textContent = 'Versions \u00b7' + newest.v + ' \u25b8';
button.setAttribute('aria-expanded', 'false');
const body = document.createElement('div');
body.className = 'vl-body';

const rows = [...VERSION_LEDGER].reverse().map(entry =>
  '<div class="vl-row"><div class="vl-head">'
  + '<span class="vl-ver">' + entry.v
  + (entry.status ? '<span class="vl-status">' + entry.status + '</span>' :
  + '</span>'
  + '<span class="vl-when">' + ledgerStamp(entry.g) + '</span></div>'
  + '<div class="vl-scope">' + entry.s + '</div>'
  + (entry.reason ? '<div class="vl-reason">' + entry.reason + '</div>' :
  + '</div>').join('');
body.innerHTML = rows
  + '<p class="vl-note">' + LEDGER_DOCTRINE + ' ' + PRE_SCOPE_COMPOSITIONS
  + ' earlier compositions predate '
  + 'the scope line. Extracted from the repository history at build time; '
  + 'nothing here is fetched or editable at runtime.</p>';

panel.appendChild(button);
panel.appendChild(body);
button.addEventListener('click', (event) => {
  event.stopPropagation();
  event.preventDefault();

```

```

    const open = panel.dataset.open === '1';
    panel.dataset.open = open ? '0' : '1';
    button.textContent = 'Versions \u00b7 ' + newest.v + (open ? ' \u25b8' : ' \u25be');
    button.setAttribute('aria-expanded', String(!open));
  });
  panel.addEventListener('click', (event) => event.stopPropagation());
  stack.appendChild(panel);
  link.version_ledger = { entries: VERSION_LEDGER.length, newest: newest.v,
    rejected_pre_promotion: VERSION_LEDGER
    .filter(entry => entry.status === 'REJECTED_PRE_PROMOTION').map(entry =>
entry.v) };
  }

```

/* — the Grid Finding Scope —

Vikram: clicking blank space should show what grid is in the vicinity. Until v9.62 a blank click cleared the map, which treated everywhere that is not a consented project as empty. It is not empty; it is unexamined.

The computation is the grid-scope module: this half only arms it, draws it and writes it down. The neon project path is untouched and remains the anchor - a scope never runs where a project or substation was hit. */

```

const SCOPE_SRC = 'gridatlas-scope-src';
const SCOPE_RING_LAYER = 'gridatlas-scope-rings';
const SCOPE_DOT_LAYER = 'gridatlas-scope-dots';
const SCOPE_COLOUR = '#7fb5d8'; // cool blue: neither project nor
declared
let scopeArmed = false;
/* Armed explicitly, like the scope. Published on the link object so a
  reviewer can ask the page which modes are live. */
let pointArmed = false;
let scopeResult = null;

function scopeModule() {
  try { return window.__GRIDATLAS_MODULES__.gridScope || null; }
  catch (_) { return null; }
}

function ensureScopeLayers(map) {
  if (map.getSource(SCOPE_SRC)) return true;
  if (!guardedAddSource(map, SCOPE_SRC, { type: 'geojson', data:
emptyCollection() }))) {
    return false;
  }
  map.addLayer({ id: SCOPE_RING_LAYER, type: 'circle', source: SCOPE_SRC,
    filter: ['==', ['get', 'kind'], 'band'],
    paint: { 'circle-radius': ['get', 'pixels'], 'circle-color':
'transparent',
    'circle-stroke-color': SCOPE_COLOUR, 'circle-stroke-width': 1,
    'circle-stroke-opacity': 0.5 } });
  map.addLayer({ id: SCOPE_DOT_LAYER, type: 'circle', source: SCOPE_SRC,
    filter: ['==', ['get', 'kind'], 'site'],
    paint: { 'circle-radius': 4, 'circle-color': SCOPE_COLOUR,
    'circle-stroke-color': '#04141c', 'circle-stroke-width': 1 } });
  return true;
}

function clearScope() {

```

```

    scopeResult = null;
    const map = capturedMap;
    if (map && map.getSource(SCOPE_SRC)) setData(map, SCOPE_SRC,
emptyCollection());
    document.querySelectorAll('.' + SCOPE_BLOCK).forEach(node => node.remove());
    link.grid_scope = null;
}

const SCOPE_BLOCK = 'gridatlas-scope-block';

function scopeHtml(result) {
    const bands = result.bands.map(band => {
        const classes = Object.keys(band.by_class_kv)
            .map(Number).sort((a, b) => b - a)
            .map(kv => `${band.by_class_kv[kv]} \u00d7 ${kv} km`).join(', ');
        return `- ${band.within_km} km
+ ${band.substations
? escapeHtml(classes) : 'nothing mapped'}
`;
    }).join('');
    const nearest = result.nearest_named.slice(0, 3).map(entry =>
`- ${entry.km.toFixed(2)} km
+ ${escapeHtml(entry.name)}
+ ${entry.kv} km
`).join('');
    /* Which cartridges answered, on the card, not in a console.
    -----
    Vikram: a click anywhere should make the neons "look for cartridges
    and code". The looking is the source-registry module; this prints
    what it found. A reader told that five of six sources answered and
    which one did not can judge the answer in front of them. A reader
    shown a quietly shorter answer cannot, and will reasonably assume
    the map has told them everything it knows. */
    const survey = result.sources || null;
    const sourcesLine = survey
        ? `

Sources: ${escapeHtml(survey.sentence)}

`
        : '';
    return `

Grid finding scopeBeta


+ 

${escapeHtml(result.what_this_is)}


+ 

${bands}

+ (nearest ? 

Nearest named



${nearest}

: '')
+ topologyBlockHtml(result.nearest_named.slice(0, 3).map(entry => ({ name:
entry.name, kv: entry.kv })))
+ 

${escapeHtml(result.what_this_is_not)}


+ sourcesLine
+ 

${escapeHtml(result.method)}. `
+ `Substations as mapped in this release's payload; an absence here is `
+ `an absence from the map, not from the ground.


```

```

    injectStatusStyle();
    showStatus('The substation data did not load, so nothing can be counted '
        + 'here. This is usually the network rather than the place.', 'failed');
    return;
}
clearStatus();
const result = scope.scope(origin, subs, { nearestCount: 5 });

/* Ask what could have answered, before saying what did.
-----
The deep scan of 202609012230 found this path reporting only what
OpenStreetMap has mapped while the cartridge holding NESO's 886
published connection points sat loaded in the same page. The scope
still counts only what it can count - widening the computation is a
separate, provable change - but it no longer stays silent about the
sources it did not use. */
const registry = window.__GRIDATLAS_MODULES__?.sourceRegistry || null;
if (registry) {
    try { result.sources = registry.survey(window); }
    catch (error) { noteFailure('source registry: ' + String(error?.message ||
error)); }
}

scopeResult = result;
link.grid_scope = { counted: result.counted, radius_km: result.radius_km,
    nearest_km: result.nearest[0]?.km ?? null,
    sources_ready: result.sources?.ready || null,
    sources_missing: result.sources?.missing || null };

if (ensureScopeLayers(map)) {
    /* The bands are drawn in metres-per-pixel at the current zoom, so a
        ring means the distance it says at the zoom it was drawn. */
    const metresPerPixel = 156543.03392
        * Math.cos(origin[1] * Math.PI / 180) / Math.pow(2, map.getZoom());
    const features = result.bands.map(band => ({
        type: 'Feature',
        properties: { kind: 'band', pixels: (band.within_km * 1000) /
metresPerPixel },
        geometry: { type: 'Point', coordinates: origin }
    })).concat(result.nearest.map(entry => ({
        type: 'Feature', properties: { kind: 'site' },
        geometry: { type: 'Point', coordinates: entry.at }
    })));
    setSourceData(map, SCOPE_SRC, { type: 'FeatureCollection', features });
}

const gl = window.maplibregl;
if (gl?.Popup) {
    try {
        new gl.Popup({ maxWidth: '360px', closeOnClick: false })
            .setLngLat(origin)
            .setHTML(scopeHtml(result))
            .addTo(map);
    } catch (error) {
        noteFailure('grid scope card: ' + String(error?.message || error));
    }
}
}

/* — the mobile tray —
Vikram's phone acceptance, 2026-09-01: the six shell tool buttons at
44px touch height are the right size to hit and the wrong size to keep

```

on screen - a third of a portrait map behind buttons - and the switches that turn the grid lines and substations on live below the map, where a phone never looks. Both are composition faults, not engine faults: the 44px rule is this cartridge's own coarse-pointer override, and the engine's switches work - toggled live in Chrome, the lines drew at once.

So on a touch screen or a narrow window the six tool buttons collapse behind one chip, and two first-class chips - GRID and SUBS - stand on the map itself. They drive the engine's own checkboxes with real clicks, the same path enableTechnologyLayer has used all along, so the scada panel, the fullscreen curtain and these chips cannot disagree. */

```
const TRAY_ID = 'gridatlas-mobile-tray';
const GRID_LINE_LAYERS = ['400', '275', '220', '132', '66'];

function trayTarget() {
  try {
    return matchMedia('(pointer: coarse)').matches || window.innerWidth <=
700;
  } catch (_) { return false; }
}

function engineLayerBox(id) {
  return document.querySelector(
    '#scada-ui-container input[type=checkbox][data-layer-id="' + id + '"]');
}

function installMobileTray() {
  if (document.getElementById(TRAY_ID)) return;
  if (!trayTarget()) {
    link.mobile_tray = { installed: false, reason: 'fine pointer, wide window'
};
    return;
  }
  const stack = document.querySelector('.map-controls');
  if (!stack) { link.failures.push('no map-controls for the tray'); return; }

  const style = document.createElement('style');
  style.id = TRAY_ID + '-style';
  style.textContent = SLD_STYLES.mobileTray(TRAY_ID);
  document.head.appendChild(style);

  stack.classList.add('gm-tools-collapsed');

  const tray = document.createElement('div');
  tray.id = TRAY_ID;

  const tools = document.createElement('button');
  tools.type = 'button';
  tools.textContent = 'Tools \u25b8';
  tools.setAttribute('aria-expanded', 'false');
  tools.addEventListener('click', (event) => {
    event.stopPropagation();
    const collapsed = stack.classList.toggle('gm-tools-collapsed');
    tools.textContent = collapsed ? 'Tools \u25b8' : 'Tools \u25be';
    tools.setAttribute('aria-expanded', String(!collapsed));
  });

  function quickChip(labelText, ids) {
    const chip = document.createElement('button');
    chip.type = 'button';
    chip.textContent = labelText;
    chip.disabled = true;
  }
}
```



```

chip.setAttribute('aria-pressed', 'false');
function reflect() {
  const boxes = ids.map(engineLayerBox).filter(Boolean);
  chip.disabled = boxes.length === 0;
  chip.setAttribute('aria-pressed',
    String(boxes.length > 0 && boxes.every((box) => box.checked)));
}
chip.addEventListener('click', (event) => {
  event.stopPropagation();
  const boxes = ids.map(engineLayerBox).filter(Boolean);
  if (!boxes.length) return;
  const turnOn = boxes.some((box) => !box.checked);
  boxes.forEach((box) => { if (box.checked !== turnOn) box.click(); });
  reflect();
});
chip.gmReflect = reflect;
return chip;
}

/* The tray tool. The shell owns .map-controls and the shell is
immutable, so the button is added by this cartridge at runtime and
removed again if the cartridge is not composed - there is no orphan
control left behind claiming a feature that is not present. */
(function addGridPointTool() {
  const tray = document.querySelector('.map-controls');
  if (!tray || document.getElementById('btn-gridpoint')) return;
  const button = document.createElement('button');
  button.className = 'map-ctrl-btn';
  button.id = 'btn-gridpoint';
  button.type = 'button';
  button.textContent = '\u25c8 Grid At Point';
  button.setAttribute('aria-pressed', 'false');
  button.title = 'Click anywhere on the map for the published network at the
nearest mapped connection point';
  button.addEventListener('click', (event) => {
    event.stopPropagation();
    pointArmed = !pointArmed;
    button.setAttribute('aria-pressed', String(pointArmed));
    button.style.outline = pointArmed ? '1px solid currentColor' : '';
    link.grid_point_armed = pointArmed;
    if (!pointArmed) clearScope();
  });
  tray.appendChild(button);
})();

const grid = quickChip('\u26a1 Grid', GRID_LINE_LAYERS);
const subs = quickChip('\u25c9 Subs', ['subs']);

/* Arming is explicit. A map that analysed every stray tap would put a
card over the thing the reader was trying to look at. */
const scopeChip = document.createElement('button');
scopeChip.type = 'button';
scopeChip.textContent = '\u25ce Scope';
scopeChip.setAttribute('aria-pressed', 'false');
scopeChip.addEventListener('click', (event) => {
  event.stopPropagation();
  scopeArmed = !scopeArmed;
  scopeChip.setAttribute('aria-pressed', String(scopeArmed));
  if (!scopeArmed) clearScope();
  link.grid_scope_armed = scopeArmed;
});

```

```

const clearChip = document.createElement('button');
clearChip.type = 'button';
clearChip.textContent = '\u2715 Clear';
clearChip.addEventListener('click', (event) => {
    event.stopPropagation();
    clearScope();
    clearLinks();
});

tray.appendChild(tools);
tray.appendChild(grid);
tray.appendChild(subs);
tray.appendChild(scopeChip);
tray.appendChild(clearChip);
tray.addEventListener('click', (event) => event.stopPropagation());
stack.insertBefore(tray, stack.firstChild);

// The engine builds its switches only after the map loads; the chips
// wake when the switches exist and follow them wherever they are
// toggled from - scada panel, fullscreen curtain, or a deep link.
document.addEventListener('change', (event) => {
    if (event.target?.dataset?.layerId) { grid.gmReflect(); subs.gmReflect();
}
});
let polls = 0;
const poll = setInterval(() => {
    grid.gmReflect(); subs.gmReflect(); polls += 1;
    if (!grid.disabled || polls > 200) clearInterval(poll);
}, 300);

link.mobile_tray = {
    installed: true,
    tools_collapsed: true,
    grid_quick_layers: GRID_LINE_LAYERS.length,
    subs_quick: true
};

function installGbConditions() {
    if (document.getElementById(GB_ID)) return;
    const stack = document.querySelector('.map-controls');
    if (!stack) { link.failures.push('no map-controls for the GB panel');
return; }

    const style = document.createElement('style');
    style.id = GB_ID + '-style';
    style.textContent = SLD_STYLES.gbConditions(GB_ID);
    document.head.appendChild(style);

    const panel = document.createElement('div');
    panel.id = GB_ID;
    panel.dataset.open = '0';
    const button = document.createElement('button');
    button.type = 'button';
    button.textContent = 'GB prices · historic ▶';
    button.setAttribute('aria-expanded', 'false');
    const body = document.createElement('div');
    body.className = 'gb-body';
    panel.appendChild(button);
    panel.appendChild(body);

    let loaded = false;

```

```

button.addEventListener('click', (event) => {
  // The panel lives inside the map container, so without this the click
  // carries on to the map underneath and selects whatever is there.
  event.stopPropagation();
  event.preventDefault();
  const open = panel.dataset.open === '1';
  panel.dataset.open = open ? '0' : '1';
  button.textContent = open ? 'GB prices · historic ►' : 'GB prices ·
historic ▼';
  button.setAttribute('aria-expanded', String(!open));
  // Fetched on first open, never at boot: nothing about the map should
  // wait on a third party, and most sessions never open this.
  if (!open && !loaded) {
    loaded = true;
    body.innerHTML = '<p class="gb-note">Reading the decade...</p>';
    renderGbConditions(body);
  }
});
panel.addEventListener('click', (event) => event.stopPropagation());
stack.appendChild(panel);
link.gb_panel_installed = true;
}

/* Never dereference a source without checking it is there.
-----
addSource throws if the style is not loaded, and a source that failed to
add reads back as null. Both happen: the basemap CDN served style.json and
then no tiles at all on this estate tonight, and the cartridge now boots on
the style rather than a painted frame precisely so it can work in that
condition.

The pin was guarded when that was found. Five call sites were not, and
they are the ones that draw the links, the nodes and the whole layout – so
the guarded convenience survived while the substance would have thrown.
Codex's gate has been finding this class all night; this is the last of it
in this file.

Returning false rather than throwing means a missing source costs the
drawing, not the card, the distances or the session. */
function setSourceData(map, id, data) {
  try {
    const source = map?.getSource?.(id);
    if (!source || typeof source.setData !== 'function') {
      link.failures.push('source missing, nothing drawn: ' + id);
      return false;
    }
    source.setData(data);
    return true;
  } catch (error) {
    link.failures.push('source ' + id + ': ' + String(error?.message ||
error));
    return false;
  }
}

function interactiveLayerIds(map) {
  // Whatever the engine has made visible and interactive. Reading the style
  // rather than hard-coding ids keeps this working as layers come and go.
  try {
    return map.getStyle().layers
      .filter(layer => /^1-/.test(layer.id) && layer.type !== 'background')
      .map(layer => layer.id)

```

```

        .filter(id => {
            try { return map.getLayoutProperty(id, 'visibility') !== 'none'; }
            catch (_) { return false; }
        });
    } catch (_) {
        return [];
    }
}

function install(map) {
    installStyles();
    ensureLayers(map);

    /* Warm the substation payload immediately. It is 1.2 MB and every
       measurement needs it; until v9.54 it was first requested only after
       the arrival had finished waiting for the engine's layer controls,
       so a phone paid for the wait AND the fetch in series. The promise is
       cached, so the arrival reuses whatever this started. */
    try { loadSubstations(); } catch (_) { /* the arrival will retry */ }

    // The lines belong to the card. When the card closes, they go with it --
    // leaving neon on the map with nothing explaining it is how a screenshot
    // ends up quoted without its caveat.
    const popupWatcher = new MutationObserver(() => {
        if (link.links_drawn > 0 && !document.querySelector('.maplibregl-popup'))
clearLinks();
    });
    try {
        popupWatcher.observe(map.getContainer(), { childList: true, subtree: true
});
    } catch (error) {
        link.failures.push(String(error?.message || error));
    }

    // Registered after the engine's own click handler, so the engine's popup
    // opens first and this decorates it rather than racing it.
    // Measure and draw for one selection. Split out of the click handler so a
    // deep link, which opens a card without anybody clicking, goes through
    // exactly the same path.
    async function selectAt(origin, name, tech, fromSubstation, statedMw,
        expectedArrivalEpoch = null) {
        const fromCurrentArrival = Number.isInteger(expectedArrivalEpoch);
        if (fromCurrentArrival) {
            if (!arrivalGate.isCurrent(expectedArrivalEpoch)) return false;
        } else {
            invalidatePendingArrival('new-selection');
        }
        clearScope();
        // Keep a provisional declared block standing until the measured one
        // replaces it; clearing here would blank the card mid-arrival.
        if (!currentDeclared?.pending) currentDeclared = null;
        currentNearest400 = null;
        currentPolicy = null;
        if (fromSubstation) {
            // No fetch needed: the projects are already in the engine's own
            // source, and reading them there keeps one set of coordinates.
            const found = nearestProjects(map, origin[0], origin[1]);
            drawLinks(map, origin, name, tech, found.links, 'from-substation',
                null, found.loaded);
            return;
        }
    }
    /* OFFSHORE MEASURES NOW.

```

This used to return here with an empty link list and a card that explained why nothing was measured. It measures instead. An offshore project reaches an offshore substation, an export cable and a landfall before anything onshore - every word of that is still true and still printed - but the export cable does land at an onshore substation, so the distance to the nearest mapped one is a measurement of a real thing rather than a number with nothing behind it. Withholding it was over-caution.

The route reasoning is not deleted; it moves from being the reason there is no number to being the caveat beside the number, which is where the estate puts every other thing a distance cannot answer.

```
No onshore-only filter is applied, and the card says so. The pinned
substation product carries the OSM `location` tag on zero of its
5,800 features, and of the 14 whose NAME contains "offshore" at least
4 are onshore substations serving an offshore farm - Hornsea at
400/220 kV among them, which is exactly a landfall connection. A
name-based filter would drop those four from the search it was meant
to sharpen. Measured, not assumed; the module carries the counts. */
const policy = coverage.policy(tech);
currentPolicy = policy;
if (!link.substations_qualifying) {
  injectStatusStyle();
  showStatus('Loading the substation data \u2014 the links need it.',
    'waiting');
}
const subs = await loadSubstations();
if (fromCurrentArrival && !arrivalGate.isCurrent(expectedArrivalEpoch))
return false;
if (!subs.length) {
  /* The register carries 5,800 substations; zero qualifying means the
  payload did not arrive, not that the map is empty here. Drawing
  nothing silently is how this looked broken on a phone. */
  injectStatusStyle();
  showStatus('The substation data did not load, so no links can be '
    + 'drawn. This is usually the network rather than the project.',
    'failed');
  return;
}
clearStatus();
currentDeclared = resolveDeclaredConnection(currentRepdRef, origin, subs);
currentNearest400 = nearestTransmission(origin, subs);
drawLinks(map, origin, name, tech,
  nearestSubstations(origin[0], origin[1], subs), 'to-substation',
statedMw);
  return true;
}
link.selectAt = selectAt;

map.on('click', async (event) => {
  try {
    if (fromOwnUi(event)) return;
    const ids = interactiveLayerIds(map);
    if (!ids.length) return;
    let features = [];
    try { features = map.queryRenderedFeatures(event.point, { layers: ids
  }); }
  catch (_) { return; }
  if (!features.length) { clearLinks(); return; }
```

```

// Either end of a link is a valid place to start. Whichever pixel was
// clicked, the card that came up is the one the distances are written
// onto, and the lines run to the other end.
const hit = features.find(feature => {
  const properties = feature.properties || {};
  const tech = String(properties.tech || properties.type || '');
  return isProjectTech(tech) || feature.layer?.id === SUBS_LAYER_ID;
});
if (!hit) {
  /* The anchor is untouched: a project or substation click still
  draws the neons. Only a click that hit NEITHER reaches here,
  and only then if the reader armed the scope. */
  clearLinks();
  if (pointArmed) await runGridAtPoint(map, event.lngLat.lng,
event.lngLat.lat);
  if (scopeArmed) await runGridScope(map, [event.lngLat.lng,
event.lngLat.lat]);
  return;
}

const properties = hit.properties || {};
const fromSubstation = hit.layer?.id === SUBS_LAYER_ID;
const tech = String(properties.tech || properties.type || '');
const origin = representativePoint(hit.geometry)
  || [event.lngLat.lng, event.lngLat.lat];
const name = properties.name || properties.SiteName || properties['Site
Name']
  || (fromSubstation ? 'Unnamed substation' : 'Unnamed project');
const stated = Number(properties.capacity);
currentRepdRef = fromSubstation
  ? null : String(properties.repd_ref || properties.repdRef || '');
await selectAt(origin, name, tech, fromSubstation,
  Number.isFinite(stated) && stated > 0 ? stated : null);
} catch (error) {
  link.failures.push(String(error?.message || error));
}
});

/* — arrival by identity —————
The search cartridge resolves repd_ref against the register - DuckDB
over the pinned parquet - flies there and opens the card. When a link
carries only that identity, this lane used to stop at its URL guards
while the card opened anyway, which reads as a broken map. The register
knows the coordinates and technology better than any URL restatement,
so when the identity lane has resolved, its published result is the
arrival. One resolver per composition; this lane only consumes.

The budget is generous because the identity lane boots a 35.7 MB query
engine first on a cold phone. Terminal failure or absence returns null
and is the caller's decision to record. */
async function waitForResolvedIdentity(options = {}) {
  const announce = options.announce !== false;
  /* v9.44 gave this a fixed 120s budget and Vikram's phone exceeded it:
  the identity lane boots a 35.7 MB query engine first, the budget
  expired, this lane gave up permanently, and the card then opened
  anyway - a resolved identity with nothing computed for it. The
  estate's own late-layers lesson, relearned: a fixed budget is always
  the wrong number. RESOLVED, FAILED and ABSENT are terminal; the only
  unbounded case is "still working", and waiting through it is the
  correct behaviour. The user is told what is being waited for; ten
  minutes is kept as a last-resort stop for a lane that died silently. */
  const started = Date.now();

```

```

let told = false;
for (;;) {
  const dl = window.__GRIDATLAS_PLACE_SEARCH__.deep_link;
  if (dl) {
    if (dl.status === 'RESOLVED') { if (told) clearStatus(); return dl; }
    if (dl.status === 'FAILED' || dl.status === 'ABSENT'
        || dl.status === 'NOT_IN_ACTIVE_REGISTER'
        || dl.status === 'CANCELLED') {
      if (told) clearStatus();
      return dl;
    }
  }
  const waited = Date.now() - started;
  if (announce && !told && waited > 6000) {
    told = true;
    injectStatusStyle();
    showStatus('Resolving the project against the register \u2014 a cold '
      + 'phone boots the query engine first, and this can take a minute.',
      'waiting');
  }
  if (waited > 600000) {
    link.failures.push('identity lane still not terminal after 10 minutes');
    if (told) clearStatus();
    return null;
  }
  await new Promise(resolve => setTimeout(resolve, 250));
}
}

// A deep link opens the project card on its own, with no click anywhere.
// Arriving that way is how most people reach the Atlas -- the MAP button in
// Pipeline News sends them here -- so the measurement has to run for it
// too, or the card that brought them arrives with nothing on it.
async function runDeepLink(expectedArrivalEpoch = null) {
  try {
    const q = new URLSearchParams(window.location.search);
    const epoch = Number.isInteger(expectedArrivalEpoch)
      ? expectedArrivalEpoch : claimPendingArrival(window.location.search);
    if (!arrivalGate.isCurrent(epoch)) return false;
    link.arrival_reconciliation = {
      status: 'MEASUREMENT_CLAIMED', epoch,
      owner_epoch: window.__GRIDATLAS_PLACE_SEARCH__.deep_link?.owner_epoch
    };
  } ?? null;
  /* Number(null) is 0, not NaN, so a link with no coordinates used to
   pass the finite guard as Null Island and only the technology
   guard stopped it. Absent now means absent. */
  const rawLon = q.get('longitude');
  const rawLat = q.get('latitude');
  const receiverPlan = deepLinkPlan(rawLon, rawLat, q.get('repd_ref'));
  let lon = receiverPlan.longitude;
  let lat = receiverPlan.latitude;
  const repdRef = receiverPlan.repd_ref;
  let tech = String(q.get('technology') || '');
  let name = q.get('project') || 'Deep-linked project';
  let stated = Number(q.get('capacity_mw'));
  const suppliedStatus = String(q.get('status') || '').trim();
  currentCapacityMw = Number.isFinite(stated) && stated > 0 ? stated :
  null;

  /* zoom: set on every deep link, and until now read by nobody.
  -----

```

The deep scan of 202609012230 compared both sides of the contract: Pipeline News sets seven parameters and GridAtlas read six. There is no get('zoom') anywhere in this repository. Arrival zoom came from `map.flyTo({ zoom: 12 })` hard-coded in the immutable shell, and Pipeline News happens to send 12 - so the two agreed by coincidence, and the day somebody tuned the sending side nothing would have moved.

The shell cannot be edited, so the cartridge honours the parameter after the shell has finished its own move. Bounded to what MapLibre and the payload can actually render, and a value outside that range is recorded rather than clamped silently. */

```
const rawZoom = q.get('zoom');
const requestedZoom = rawZoom === null ? null : Number(rawZoom);
const zoomUsable = requestedZoom !== null &&
Number.isFinite(requestedZoom)
  && requestedZoom >= 3 && requestedZoom <= 18;
if (rawZoom !== null && !zoomUsable) {
  link.failures.push('deep link: unusable zoom "' + rawZoom + '"');
}
link.requested_zoom = zoomUsable ? requestedZoom : null;

/* 12 shows 3.6x more ground at 1400 px than at 393, so a shared
link opens wide on a desktop. Frame by viewport, not by number. */
const framed = () => Math.min(18, Math.max(3, requestedZoom
+ Math.log2(Math.max(innerWidth, 320) / 393)));
function honourRequestedZoom(map) {
  if (!zoomUsable) return;
  /* One shot, after the shell's own flyTo has settled. Racing it
  would be a fight the shell wins, and re-applying on every idle
  would take the map away from a user who has since zoomed. */
  let done = false;
  const apply = () => {
    if (done) return;
    done = true;
    try {
      map.off('idle', apply);
      if (Math.abs(map.getZoom() - framed()) < 0.01) {
        link.zoom_applied = 'already there';
        return;
      }
      map.easeTo({ zoom: framed(), duration: 400 });
      link.zoom_applied = framed();
    } catch (error) {
      noteFailure('deep link zoom: ' + String(error?.message || error));
    }
  };
  try { map.once('idle', apply); } catch (_) { /* shimmed map in a proof
*/ }

  // A map that never goes idle must not swallow the request.
  setTimeout(apply, 2600);
}

const coordsUsable = () => Number.isFinite(lon) && Number.isFinite(lat)
  && Math.abs(lon) <= 180 && Math.abs(lat) <= 90
  && !(Math.abs(lon) < 1e-9 && Math.abs(lat) < 1e-9);

/* Vikram, phone acceptance 13:01: "arrive in full screen mode from
pipeline news with all the clutter minimised". On a touch screen
the normal page is a small map fighting a popup, chips, a HUD and
a panel below; fullscreen is the only honest arrival surface. The
shell's own control does it - CSS classes, and on an iPhone the
element fullscreen API simply does not exist, so nothing here is
gesture-gated - and the tray keeps the tool buttons collapsed. */
```



```

    if ((q.get('repd_ref') !== null || coordsUsable()) && trayTarget()) {
      try {
        window.enterFullscreen?();
        link.arrival_fullscreen = true;
        setTimeout(() => { try { map.resize(); } catch (_) { /* cosmetic */
} }, 120);
      } catch (error) {
        link.failures.push('arrival fullscreen: ' + String(error?.message ||
error));
      }
    }

    /* A VALID LINK POINT ANSWERS FIRST; THE REGISTER VERIFIES IT.
    -----
    v9.91 put `await waitForResolvedIdentity()` before selectAt for every
    repd_ref. Across Pipeline News that serialized a 35.7 MB query
    engine ahead of 8,743 links that already carried a usable point;
    2,430 then waited only to fall back to that same point. Coordinates
    are enough for a geometric measurement. They are therefore used at
    once, explicitly as link-supplied, while the one identity owner
    verifies them concurrently. A different resolved point replaces the
    selection and is measured again; FAILED/ABSENT never erases a valid
    supplied point. A ref-only link still waits because it has no point
    from which an honest measurement can be made. */
    let identityVerification = null;
    if (receiverPlan.route === 'MEASURE_LINK_FIRST' && repdRef) {
      link.origin_source = 'link-supplied';
      link.deep_link_identity = 'verifying-concurrently';
      link.identity_verification = {
        status: 'PENDING', repd_ref: repdRef, supplied_coordinates_used:
true
      };
      identityVerification = waitForResolvedIdentity({ announce: false })
        .then((owner) => ({
          resolved: owner?.status === 'RESOLVED' ? owner : null,
          terminal: owner?.status || 'UNKNOWN'
        }))
        .catch((error) => ({ resolved: null, terminal: 'FAILED', error }));
    } else if (receiverPlan.route === 'WAIT_FOR_REGISTER') {
      /* A ref-only link has no safe provisional geometry. This is the one
      case that must await the identity owner before measuring. */
      const owner = await waitForResolvedIdentity();
      if (!arrivalGate.isCurrent(epoch) || owner?.status === 'CANCELLED')
return;
      if (owner?.status === 'RESOLVED') {
        const resolved = owner;
        const rLon = Number(resolved.longitude);
        const rLat = Number(resolved.latitude);
        if (Number.isFinite(rLon) && Number.isFinite(rLat)
          && Math.abs(rLon) <= 180 && Math.abs(rLat) <= 90
          && !(Math.abs(rLon) < 1e-9 && Math.abs(rLat) < 1e-9)) {
          lon = rLon;
          lat = rLat;
          link.origin_source = 'register';
        }
        if (typeof resolved.technology === 'string' && resolved.technology)
{
          tech = resolved.technology;
        }
        if (resolved.name) name = String(resolved.name);
        const cap = Number(resolved.capacity_mw);
        if (Number.isFinite(cap) && cap > 0) stated = cap;
      }
    }
  }

```

```

    currentCapacityMw = Number.isFinite(stated) && stated > 0 ? stated :
null;
    link.deep_link_identity = 'resolved-by-search-lane';
  } else if (owner?.status === 'NOT_IN_ACTIVE_REGISTER') {
    link.origin_source = 'not-in-active-register-no-supplied-point';
    link.deep_link_identity = 'terminal-not-in-active-register';
    link.identity_verification = {
      status: 'NOT_IN_ACTIVE_REGISTER', repd_ref: repdRef,
      supplied_coordinates_kept: false,
      official_active_register_match: false
    };
    injectStatusStyle();
    showStatus('REPD ' + repdRef + ' is not in the active-register '
      + 'snapshot, and this link supplies no coordinates from which '
      + 'to measure. No official status or location is inferred.',
      'unavailable');
    return;
  } else {
    injectStatusStyle();
    const message = String(owner?.message || 'identity loader
unavailable');
    showStatus('The active-register identity check failed: ' + message
      + '. No location was supplied, so the grid measurement cannot '
      + 'start until the check succeeds.', 'failed');
    retryArrival = retryIdentityOwnerThenArrival;
    return;
  }
} else {
  link.origin_source = 'link-supplied';
}

if (!coordsUsable()) return;

/* Put the supplied point on screen with the supplied-point answer.
The old rule flew only links without repd_ref and therefore left a
valid coordinate link waiting for the register just to move the
camera. If verification later finds a different point, the identity
lane and the reconciliation below replace it together. */
try {
  const arrivalZoom = zoomUsable ? requestedZoom : 12;
  map.flyTo({ center: [lon, lat], zoom: arrivalZoom,
    duration: 1200, essential: true });
  link.camera_from_link = { longitude: lon, latitude: lat,
    zoom: arrivalZoom, reason: identityVerification
      ? 'supplied coordinates while register verification runs'
      : (repdRef ? 'resolved repd_ref coordinates'
        : 'no repd_ref, so no other lane flies') };
} catch (error) {
  noteFailure('deep link camera: ' + String(error?.message || error));
}
honourRequestedZoom(map);
/* An unrecognised technology used to abandon the whole arrival.
`return` cost the card, the ring, the nearest-substation
measurement, the declared connection and the substation layer -
all arithmetic over two coordinates and a register row. Only the
one technology layer needs the id, so that is all it costs now.
PROJECT_TECHS accepts 11,065 of the 11,069 ids the register
writes. What the guard really catches is a link that omits or
garbles the parameter, or carries an id from a newer register,
and for all three the answer is the map, not a blank. Recorded on
its own surface, not in `link.failures`, which since 202609011434
means the arrival lost something. This one did not. */

```

```

let technologyKnown = isProjectTech(tech);
/* enabled starts false and STAYS false until enableTechnologyLayer()
   actually turns a control on -- that is the one place the truth
   lives. It used to read `enabled: technologyKnown`, which is
   membership of PROJECT_TECHS, not the state of any control: for
   wind_onshore, wind_offshore and other, that set says true while
   no such data-layer-id has ever existed, so the field read green
   on 2,508 of 7,680 register rows while the layer sat off. A field
   nothing else corrects is a field that lies for as long as the
   page is open, and this was read by every prior proof. */
link.technology_layer = {
  requested: tech || null,
  layer_id: technologyKnown ? layerIdForBucket(tech) : null,
  enabled: false,
  reason: technologyKnown ? null
    : 'deep link: unknown technology "' + tech + '" - the arrival '
      + 'continues and this layer alone is not switched on'
};
// Turn the substations on. Arriving from the MAP button in Pipeline
// News, the whole point is to see the project against the network, and
// a user who has to find a checkbox first has been handed a puzzle
// rather than an answer. The engine owns the layer, so this ticks its
// own control rather than reaching past it into the map.
// The dashboard is built from the engine's own data and does not
// exist yet on a cold load -- measured at zero checkboxes twenty
// seconds in. Ticking a control that has not been rendered silently
// did nothing, and the layers the arrival depends on stayed off.
// Named, so Try again re-runs exactly the arrival rather than
// reloading and paying for the whole engine a second time.
let currentArrival = Object.freeze({ lon, lat, name, tech, stated,
repdRef, suppliedStatus });
const enableBoth = () => {
  if (!arrivalGate.isCurrent(epoch)) return false;
  enableSubstationLayer();
  if (technologyKnown) enableTechnologyLayer(currentArrival.tech);
  return true;
};
const arrive = async () => {
  clearStatus();
  const ready = await waitForLayerControls(12000);
  enableBoth();
  // Late is not never. If the dashboard turns up after the budget, the
  // layers still go on, without the user having to do anything.
  if (!ready) watchForLayerControls(enableBoth);
  return ready;
};
retryArrival = () => { runDeepLink(); };
/* Measure first. The distances are arithmetic over substation
   coordinates and need no layer control, no dashboard and no
   painted basemap; only the layers need the engine's controls.
   Until v9.54 this awaited arrive() - up to twelve seconds - before
   the measurement was even attempted, and Vikram's West Burton
   journey on a phone showed exactly what that buys: a card, and
   nothing beside it, for long enough to conclude the map is
   broken. The layer switch-on runs alongside and finishes whenever
   the engine is ready. */
const layersReady = arrive();
async function runArrivalSelection(arrival, waitForOwnerCard = false,
  expectedArrivalEpoch = epoch) {
  if (!arrivalGate.isCurrent(expectedArrivalEpoch)) return false;
  /* A ref-only arrival already paid for canonical identity, so it can
     briefly yield to that owner's richer card. A supplied coordinate

```

```

        arrival must not wait for a card before it can measure: it creates
        the explicit link-provenance card below on the same turn. */
    if (waitForOwnerCard) {
        for (let i = 0; i < 40; i += 1) {
            if (document.querySelector('.maplibregl-popup-content')) break;
            const idStatus =
window.__GRIDATLAS_PLACE_SEARCH__?.deep_link?.status;
            if (idStatus === 'FAILED' || idStatus === 'ABSENT') break;
            await new Promise(resolve => setTimeout(resolve, 250));
            if (!arrivalGate.isCurrent(expectedArrivalEpoch)) return false;
        }
    }
    /* The card must exist BEFORE the lines. The popup watcher enforces
    "the lines belong to the card" and clears any drawing standing
    with no card on screen - watched live: a register-absent arrival
    drew five links and the watcher wiped them in the same breath,
    because the fallback card was opened after the measurement.
    ensureArrivalCard is a no-op when a card is already up, so the
    resolved-register path is unchanged. */
    currentRepdRef = arrival.repdRef;
    ensureArrivalCard(arrival.lon, arrival.lat, arrival.name,
        arrival.tech, arrival.stated, arrival.repdRef,
arrival.suppliedStatus);
    /* Answer now, measure next. Everything in this block came from
    the made Order and the link; nothing here waits on a network. */
    currentDeclared = provisionalDeclaredConnection(currentRepdRef);
    if (currentDeclared) injectDeclaredOnly();
    try {
        if (capturedMap) setPin(capturedMap,
            [arrival.lon, arrival.lat], arrival.name, arrival.tech);
    }
    catch (_) { /* the measurement will draw it */ }
    link.deep_linked = true;
    const selected = await selectAt([arrival.lon, arrival.lat],
arrival.name,
        arrival.tech, false,
        Number.isFinite(arrival.stated) && arrival.stated > 0
            ? arrival.stated : null, expectedArrivalEpoch);
    return selected !== false &&
arrivalGate.isCurrent(expectedArrivalEpoch);
}
const firstStarted = performance.now();
const firstSelectionCurrent = await runArrivalSelection(currentArrival,
    Boolean(repdRef && !identityVerification), epoch);
if (!firstSelectionCurrent) return;
link.first_coordinate_answer_ms = Math.round((performance.now() -
firstStarted) * 10) / 10;
link.first_coordinate_origin = link.origin_source;

if (identityVerification) {
    /* Attach reconciliation only after the supplied-point selection has
    completed. A warm identity can resolve on the first microtask;
    sequencing it here prevents two selections racing each other. */
    continueVerifiedArrival(arrivalGate, epoch, identityVerification,
        async ({ resolved, terminal, error }) => {
            if (!resolved) {
                const ownerState = window.__GRIDATLAS_PLACE_SEARCH__?.deep_link ||
null;
                const message = error ? String(error?.message || error)
                    : String(ownerState?.message || '');
                link.origin_source = terminal === 'NOT_IN_ACTIVE_REGISTER'
                    ? 'link-supplied-not-in-active-register'

```

```

        : 'link-supplied-register-' + String(terminal).toLowerCase();
link.deep_link_identity = 'terminal-' +
String(terminal).toLowerCase();
link.identity_verification = {
  status: terminal, repd_ref: repdRef,
  supplied_coordinates_kept: true,
  arrival_fields: {
    name: currentArrival.name,
    technology: currentArrival.tech,
    capacity_mw: Number.isFinite(currentArrival.stated)
      ? currentArrival.stated : null,
    supplied_status: currentArrival.suppliedStatus || null
  },
  official_active_register_match: false,
  message: message || null
};
markArrivalIdentityState(terminal, repdRef, message);
if (terminal === 'FAILED') {
  injectStatusStyle();
  retryArrival = retryIdentityOwnerThenArrival;
  showStatus('The active-register identity check failed'
    + (message ? ': ' + message : '.'))
    + ' The supplied point and measurement remain on the map.',
    'failed');
}
return;
}

const rLon = Number(resolved.longitude);
const rLat = Number(resolved.latitude);
const resolvedPointUsable = Number.isFinite(rLon) &&
Number.isFinite(rLat)
  && Math.abs(rLon) <= 180 && Math.abs(rLat) <= 90
  && !(Math.abs(rLon) < 1e-9 && Math.abs(rLat) < 1e-9);
if (!resolvedPointUsable) {
  link.origin_source = 'link-supplied-register-without-point';
  link.deep_link_identity = 'resolved-without-usable-point';
  link.identity_verification = {
    status: 'RESOLVED_WITHOUT_USABLE_POINT', repd_ref: repdRef,
    supplied_coordinates_kept: true
  };
  return;
}

const discrepancyKm = Math.round(
  distanceKm(currentArrival.lon, currentArrival.lat, rLon, rLat) *
1000
) / 1000;
link.origin_discrepancy_km = discrepancyKm;
link.deep_link_identity = 'resolved-by-search-lane';
if (discrepancyKm <= 0.001) {
  link.origin_source = 'link-supplied-register-verified';
  markArrivalIdentityState('VERIFIED', repdRef);
  link.identity_verification = {
    status: 'VERIFIED', repd_ref: repdRef,
    discrepancy_km: discrepancyKm, recomputed: false
  };
  return;
}

const rTech = typeof resolved.technology === 'string' &&
resolved.technology

```

```

        ? resolved.technology : currentArrival.tech;
    const rName = resolved.name ? String(resolved.name) :
currentArrival.name;
    const rCap = Number(resolved.capacity_mw);
    const rStated = Number.isFinite(rCap) && rCap > 0 ? rCap :
currentArrival.stated;
    const verifiedArrival = Object.freeze({
        lon: rLon, lat: rLat, name: rName, tech: rTech,
        stated: rStated, repdRef, suppliedStatus:
currentArrival.suppliedStatus
    });
    currentArrival = verifiedArrival;
    technologyKnown = isProjectTech(verifiedArrival.tech);
    currentCapacityMw = Number.isFinite(verifiedArrival.stated)
        && verifiedArrival.stated > 0 ? verifiedArrival.stated : null;
    link.origin_source = 'register-corrected-after-link';
    link.identity_verification = {
        status: 'RECOMPUTING', repd_ref: repdRef,
        discrepancy_km: discrepancyKm, recomputed: false
    };
    try {
        map.flyTo({ center: [verifiedArrival.lon, verifiedArrival.lat],
            zoom: zoomUsable ? requestedZoom : 12,
            duration: 800, essential: true });
        await runArrivalSelection(verifiedArrival, false, epoch);
        enableBoth();
        link.identity_verification.status = 'RECOMPUTED';
        link.identity_verification.recomputed = true;
    } catch (reconcileError) {
        link.identity_verification.status = 'RECOMPUTE_FAILED';
        link.identity_verification.message =
            String(reconcileError?.message || reconcileError);
        noteFailure('deep link identity reconciliation: '
            + link.identity_verification.message);
    }
    }).catch((reconcileError) => {
        if (!arrivalGate.isCurrent(epoch)) return;
        link.identity_verification = {
            status: 'RECONCILIATION_FAILED', repd_ref: repdRef,
            supplied_coordinates_kept: true,
            message: String(reconcileError?.message || reconcileError)
        };
        noteFailure('deep link identity reconciliation: '
            + link.identity_verification.message);
    });
    }
    await layersReady;
    return arrivalGate.isCurrent(epoch);
} catch (error) {
    link.failures.push('deep link: ' + String(error?.message || error));
    return false;
}
}
rerunDeepLink = runDeepLink;

```

```

/* THE ARRIVAL MUST NEVER RUN WHILE NOBODY CAN SEE IT, AND MUST NEVER
BE ONE-SHOT.

```

Reported live by the architect on his own iPhone, twice, on two different projects: menu bar fine, attribution fine, basemap fully painted -- and the camera sat at the default UK-wide view, no card, no links. Both arrivals were opened from Pipeline News' MAP control,

which carries target="_blank" on touch devices. An independent audit today reproduced the mechanism exactly: a deep link loaded with document.hidden === true drew zero layer controls at 40s, the camera never left its default position, and it recovered 2.5s after being made visible -- because MapLibre's flyTo() and the engine's own paint-driven boot both depend on requestAnimationFrame, which iOS Safari does not tick in a tab that is not composited. Calling map.flyTo() there does not throw and is not a failure this cartridge can see: the animation is simply never given a frame to advance, so the camera stays exactly where it started, forever, even once the tab is later brought to the front -- because this function had already run to its own conclusion and nothing called it again.

So: never START the arrival until the document is actually visible, and never leave an arrival that has not produced a visible outcome stranded -- run it again the first time the tab is genuinely seen. */

```

let arrivalAttempts = 0;
const MAX_AUTO_ARRIVAL_ATTEMPTS = 5; // a real, non-visibility failure
must still stop retrying
function arrivalHasVisibleOutcome() {
  // Both already-published, already-relied-upon fields: links_drawn is
  // read the same way by the electron-flow visibility listener just
  // below, and the not-in-active-register message is this cartridge's
  // own genuine "nothing more to show" terminal state.
  return link.links_drawn > 0
    || link.origin_source === 'not-in-active-register-no-supplied-point';
}
function attemptArrival() {
  if (document.visibilityState !== 'visible') return;
  arrivalAttempts += 1;
  link.arrival_attempts = arrivalAttempts;
  void runDeepLink();
}
if (document.visibilityState === 'visible') {
  attemptArrival();
} else {
  link.arrival_deferred_for_visibility = true;
}
document.addEventListener('visibilitychange', () => {
  if (document.visibilityState !== 'visible') return;
  if (arrivalHasVisibleOutcome()) return;
  if (arrivalAttempts >= MAX_AUTO_ARRIVAL_ATTEMPTS) return;
  link.arrival_resumed_on_visibility = (link.arrival_resumed_on_visibility
|| 0) + 1;
  attemptArrival();
});

// Escape clears, the way a game HUD does.
document.addEventListener('keydown', (event) => {
  if (event.key === 'Escape') clearLinks();
});

installGbConditions();
installVersionLedger();
installMobileTray();

window.addEventListener('resize', boundCardToMap);
map.on('resize', boundCardToMap);
boundCardToMap();

// A backgrounded tab should not keep an animation frame loop alive.
document.addEventListener('visibilitychange', () => {

```

```

    if (document.hidden) stopAnimation();
    else if (link.links_drawn > 0 && capturedMap) startAnimation(capturedMap);
  });
}

/* — capture the map ————— */

function attach(map) {
  if (capturedMap) return;
  capturedMap = map;
  link.map_captured = true;
  const boot = () => {
    try { install(map); }
    catch (error) { link.failures.push(String(error?.message || error)); }
    try { installSld(map); }
    catch (error) { link.failures.push('sld: ' + String(error?.message ||
error)); }
  };
  /* Boot when the style is ready, not when a frame has painted.
  -----
  This waited on map.once('load'), which maplibre fires only after the
  first frame is on screen -- and that needs basemap tiles. Watched live:
  the CARTO style.json, tiles.json and sprite all returned 200 and then
  not one vector tile was fetched, so the map stayed black, 'load' never
  came, and the whole grid-maths layer never installed. The bare shell
  failed identically, which is how the cartridge was ruled out.

  Nothing here needs a painted frame. Sources and layers need a parsed
  STYLE, and the distances need no map at all: they are arithmetic over
  substation coordinates. Tying them to the basemap made an unrelated CDN
  a single point of failure for the measurement.

  So: whichever of style.load or load arrives first, and failing both, a
  timer. A basemap that never paints is a bad map, not a reason to have
  no maths. */
  if (map.isStyleLoaded?.()) { link.boot_trigger = 'already-loaded'; boot(); }
  else {
    let booted = false;
    const bootOnce = (trigger) => {
      if (booted) return;
      booted = true;
      link.boot_trigger = trigger;
      boot();
    };
    map.once('style.load', () => bootOnce('style.load'));
    map.once('load', () => bootOnce('load'));
    setTimeout(() => {
      // Only if a style is actually there to hang layers on. Booting without
      // one would fail on the first addSource and lose the real reason.
      if (booted) return;
      let hasStyle = false;
      try { hasStyle = Boolean(map.getStyle?.()); } catch (error) { hasStyle =
false; }
      if (hasStyle) {
        link.failures.push('basemap never finished painting; booted on the
style alone');
        bootOnce('timeout');
      } else {
        link.failures.push('no style after 8s; the grid maths cannot
install');
      }
    }, 8000);
  }
}

```



```

    }
  }

  // The engine keeps its map in a closure and returns nothing, so the only
  // clean handle is the constructor -- and the engine builds its map inside
  // initVentusMap, which runs after this file, so it is still ours to wrap.
  try {
    const gl = window.maplibregl;
    if (gl && typeof gl.Map === 'function' && !gl.Map.__gridatlasNeonWrapped) {
      const OriginalMap = gl.Map;
      function PatchedMap(...args) {
        const instance = new OriginalMap(...args);
        try { attach(instance); }
        catch (error) { link.failures.push(String(error?.message || error)); }
        return instance;
      }
      PatchedMap.prototype = OriginalMap.prototype;
      PatchedMap.__gridatlasNeonWrapped = true;
      Object.setPrototypeOf(PatchedMap, OriginalMap);
      gl.Map = PatchedMap;
    } else if (!gl) {
      link.failures.push('maplibregl unavailable when the neon cartridge
loaded');
    }
  } catch (error) {
    link.failures.push(String(error?.message || error));
  }
}

```

```

/*
PART 3 – the SLD sandbox, ported from
globalgrid2050/solar-bess-topology-v7/gis-sld-financial-sandbox.

```

WHAT CHANGED IN THE PORT, AND WHY

The sandbox is a working engine and the arithmetic below is its arithmetic, carried across unchanged. Three things are deliberately different.

1. ONE EARTH RADIUS. The sandbox measures cable length with atlasHaversineKm on $R = 6378.137$ but builds every rectangle, offset and projection with turf.destination, whose default is 6371.0088. It therefore mixes two radii inside one drawing: a 0.112% disagreement between where a thing IS and how far away it is said to be. That is the exact defect Ventusltd/grid-distance-maths was created to end. The Atlas ships no turf, so every geometric operation here is the canonical one on R_{ATLAS} and the mixture cannot recur.
2. GRAB, DO NOT MODE-SWITCH. The sandbox moves the array by arming a mode and clicking a destination, and edits a route by dropping pins and committing them. Here the array is dragged by grabbing it, the rotation has a handle on its boundary, and route vertices are dragged, inserted on a segment and removed with a double click. Everything recomputes live while the pointer is down.
3. THE ELECTRON FLOW CARRIES THROUGH. The travelling pulse used for the substation links runs along the 33 kV collectors and the export cable too, in the direction power actually flows: block, to customer substation, to grid node.

WHAT IT STILL IS NOT

A layout, not a design. Straight-line geometry with no wayleave,

crossing, terrain, ground condition or consent content, and no confirmation that any of it can connect. The caveat block travels with it.

*/

```
const SLD = {
  M2_PER_ACRE: 4046.86,
  BESS_M2_PER_MWH: 85,
  BESS_ASPECT: 2.5,
  BLOCK_SPACING_KM: 0.01,
  BOUNDARY_BUFFER_KM: 0.02,
  ARRAY_OFFSET_KM: 0.2
};

const SRC_SLD = 'gridatlas-sld';
const SLD_LAYERS = {
  boundary: 'l-sld-boundary',
  boundaryLine: 'l-sld-boundary-line',
  block: 'l-sld-block',
  bess: 'l-sld-bess',
  radial: 'l-sld-radial',
  radialFlow: 'l-sld-radial-flow',
  cable: 'l-sld-cable',
  cableGlow: 'l-sld-cable-glow',
  cableFlow: 'l-sld-cable-flow',
  cableFlowB: 'l-sld-cable-flow-b',
  node: 'l-sld-node',
  pin: 'l-sld-pin',
  handle: 'l-sld-handle',
  label: 'l-sld-label'
};

// Muted SCADA, same family as the substation links.
const SLD_COLOUR = {
  boundary: '#3f7fbf',
  block: '#5fbdc2',
  bess: '#b06ac0',
  radial: '#6fb582',
  cable: '#d9963c',
  node: '#e0b050',
  pin: '#bfe9ee',
  handle: '#d8c96a'
};

/* Financial inputs are kept per topology because that is how the original
   sandbox works. These are its post-migration defaults: the old HTML stores
   several development values as GBP/MW and migrateFinanceUnitsToWp converts
   them to GBP/Wp before the first calculation. Keeping the converted values
   here makes the units visible and avoids a hidden one-million multiplier. */
const FINANCE_DEFAULTS = Object.freeze({
  price: 65, other: 0, yield: 1000, bifacial: 5, losses: 2, deg: 0.4,
  opex: 25000, epc_ex: 0.30, flood: false, flood_rate: 0.03,
  modules: 0.15, other_capex: 0.20, fixed_capex: 1500000, cont: 7,
  loss_dc_string: 0, loss_lv_dc: 0, loss_lv_ac: 0, loss_tx: 0,
  loss_other: 0, bess_mw: 0, bess_mwh: 0, bess_capex: 0,
  bess_cycles: 0, bess_spread: 0, bess_eff: 88,
  dev_stage: '0.100', dev_cost_mw: 0.1, dev_module_mwp: 0.15,
  dev_epc_mw: 0.5, dev_owner_mw: 0.1, dev_grid_mw: 0.1,
  dev_exit_mwp: 1.35, dev_npv_mwp: 1.2, dev_success: 15, dev_years: 4,
});
const freshFinanceInputs = () => ({ ...FINANCE_DEFAULTS });
```

```

const sld = {
  active: false,
  gridNode: null,           // the substation the scheme connects to
  gridNodeName: '',
  gridNodeVoltage: '',
  arrayCentre: null,       // null = derived from the grid node and array size
  rotationDeg: 0,
  routePins: [],          // user vertices between customer substation and
grid node
  stats: null,
  projectName: null,
  // The capacity the register states for this project, and what that figure
  // is taken to mean. REPD does not reliably distinguish, which is why the
  // basis is a user choice and not an assumption.
  targetMw: null,
  targetBasis: 'unstated',
  fitResidualPct: null,
  financeOpen: false,
  cableKm: 0,
  straightKm: 0,
  dragging: null,
  inputs: {
    mode: 'string',
    mod_wp: 660, mod_l: 2.38, mod_w: 1.30, gcr: 0.45, gross_factor: 1.35,
    // 18 strings, which is the ORIGINAL sandbox's own default. It was briefly
    // changed to 23 here on the reasoning that 18 gives a block DC/AC of
    // 0.945 - an array smaller than its inverters. That reasoning was applied
    // without checking the reference, and the reference is explicit: "28
    // string inverters rated at 352 kVA create a skid block of approximately
    // 9,856 kVA BEFORE TRANSFORMER AND GRID LIMITATIONS". Oversizing the
    // inverters against an 8.96 MVA skid is the design, not an error in it.
    // A port does not get to improve its reference by guessing.
    x_mods: 28, z_strings: 18, y_invs: 28, s_subs: 5, b_cols: 6,
    dc_ac_ratio: 1.20, string_inv_kva: 352, string_skid_mva: 8.96,
    // The original has a complete second physical-input set for central
    // topology. Editing one tab must not rewrite the other tab's module,
    // mounting, land or BESS case.
    mod_wp_c: 660, mod_l_c: 2.38, mod_w_c: 1.30, gcr_c: 0.45,
    gross_factor_c: 1.35,
    inv_ac_mw_c: 4.4, inv_dc_mw_c: 5.28, central_skid_mva_c: 4.4,
    x_mods_c: 28, str_per_cb_c: 24, inv_per_mv_c: 1, mv_per_ring_c: 4,
rings_c: 4
  },
  finance: {
    string: freshFinanceInputs(),
    central: freshFinanceInputs(),
  }
};
window.__GRIDATLAS_SLD__ = sld;

/* — geodesy the layout needs, all on R_ATLAS ————— */

/* Both moved into the geodesy module, where the radius lives. Delegated
   here so every existing caller in the layout is untouched. */
function destinationPoint(lon, lat, km, bearingDeg) {
  return GEODESY.destinationPoint(lon, lat, km, bearingDeg);
}

function initialBearingDeg(lon1, lat1, lon2, lat2) {
  return GEODESY.initialBearingDeg(lon1, lat1, lon2, lat2);
}

```

```

function pathLengthKm(coords) {
  let total = 0;
  for (let i = 1; i < coords.length; i += 1) {
    total += distanceKm(coords[i - 1][0], coords[i - 1][1], coords[i][0],
coords[i][1]);
  }
  return total;
}

function normBearing(deg) { return ((deg % 360) + 360) % 360; }

// Scale factors from the WGS84 radii of curvature at a latitude, so a local
// tangent plane is correct rather than merely convenient.
function localScaleKm(latDeg) {
  /* The semi-major axis, taken from the module rather than written again.
  It is the same number as the haversine radius for a good reason - the
  estate measures on the WGS-84 equatorial axis - and writing it twice
  is how the two stop being the same number. */
  const a = R_ATLAS;
  const e2 = (1 / 298.257223563) * (2 - 1 / 298.257223563);
  const s = Math.sin(latDeg * DEG);
  const t = 1 - e2 * s * s;
  return {
    kx: (a / Math.sqrt(t)) * Math.cos(latDeg * DEG) * DEG,
    ky: ((a * (1 - e2)) / t ** 1.5) * DEG
  };
}

// Perpendicular distance to a SEGMENT, and the foot of that perpendicular.
// Measuring to an endpoint instead can only overstate; this is the function
// whose absence caused the original circuit_km defect, and it is what
// replaces turf.nearestPointOnLine in the ported layout.
function distanceToSegmentKm(lon, lat, aLon, aLat, bLon, bLat) {
  const { kx, ky } = localScaleKm(lat);
  const ax = (aLon - lon) * kx; const ay = (aLat - lat) * ky;
  const bx = (bLon - lon) * kx; const by = (bLat - lat) * ky;
  const dx = bx - ax; const dy = by - ay;
  const len2 = dx * dx + dy * dy;
  let t = 0;
  if (len2 > 0) {
    t = -(ax * dx + ay * dy) / len2;
    t = t < 0 ? 0 : t > 1 ? 1 : t;
  }
  const foot = [aLon + (bLon - aLon) * t, aLat + (bLat - aLat) * t];
  return { km: distanceKm(lon, lat, foot[0], foot[1]), foot, t };
}

// The sandbox's getRectPolygon, on one radius.
function rectPolygon(centre, widthKm, lengthKm, rotationDeg) {
  const axis = normBearing(rotationDeg);
  const n = destinationPoint(centre[0], centre[1], lengthKm / 2, axis);
  const s = destinationPoint(centre[0], centre[1], lengthKm / 2, axis + 180);
  const nw = destinationPoint(n[0], n[1], widthKm / 2, axis - 90);
  const ne = destinationPoint(n[0], n[1], widthKm / 2, axis + 90);
  const se = destinationPoint(s[0], s[1], widthKm / 2, axis + 90);
  const sw = destinationPoint(s[0], s[1], widthKm / 2, axis - 90);
  return [[nw, ne, se, sw, nw]];
}

// The sandbox uses turf.nearestPointOnLine to drop each block onto the
// collector trunk. distanceToSegmentKm already returns that foot.
function footOnSegment(lon, lat, a, b) {

```

```

    return distanceToSegmentKm(lon, lat, a[0], a[1], b[0], b[1]).foot;
}

/* — the sizing arithmetic, in its module —————
    Lifted out at 202609012205 into atlas/modules/202609012205-sizing-
    arithmetic.js, proven value-for-value against the last inline copy.
    These delegations keep every caller's name; the module is handed the
    state and the defaults the body used to close over. Absent module:
    fail by name, never quietly compute nothing. */
const SIZING = (window.__GRIDATLAS_MODULES__ || {}).sizingArithmetic;
if (!SIZING) throw new Error('sld-sandbox: the sizing-arithmetic module is not
composed');
const { DEVELOPMENT_STAGES, financeNumber } = SIZING;

function activePhysicalInputs() { return SIZING.physicalInputs(sld.inputs); }
function applyDevelopmentStageDefaults(financeInputs, stageValue) {
    return SIZING.applyDevelopmentStageDefaults(financeInputs, stageValue);
}
function applyMountingBifacial(mode, gcrValue) {
    return SIZING.applyMountingBifacial(sld.finance, mode, gcrValue);
}
function computeScreeningFinance(financeInputs, stats) {
    return SIZING.screeningFinance(financeInputs, stats,
    { fallbackMode: sld.inputs.mode, defaults: FINANCE_DEFAULTS });
}
const computeSldStats = () => SIZING.computeStats(sld.inputs, sld.finance,
FINANCE_DEFAULTS);
sld.computeFinance = computeScreeningFinance;
sld.applyDevelopmentStage = applyDevelopmentStageDefaults;
sld.applyMountingBifacial = applyMountingBifacial;

function fitToStatedCapacity() { return SIZING.fitToStatedCapacity(sld,
computeSldStats); }
sld.fitToStatedCapacity = fitToStatedCapacity;

/* — the layout ————— */

function buildLayout() {
    const stats = computeSldStats();
    sld.stats = stats;
    if (!sld.gridNode || stats.total_blocks === 0) {
        return { type: 'FeatureCollection', features: [] };
    }

    const axis = normBearing(sld.rotationDeg);
    const N = stats.total_blocks;
    const cols = Math.ceil(Math.sqrt(N));
    const rows = Math.ceil(N / cols);
    const blockAreaKm2 = stats.block_ground_area_m2 / 1e6;
    const physical = activePhysicalInputs();
    const aspect = physical.gcr === 0.45 ? 1 / 1.4 : physical.gcr === 0.75 ? 1.0
: 1.4;
    const blockW = Math.sqrt(blockAreaKm2 / aspect);
    const blockL = blockW * aspect;
    const gap = SLD.BLOCK_SPACING_KM;
    const gridW = cols * blockW + (cols - 1) * gap;
    const gridL = rows * blockL + (rows - 1) * gap;

    const gridNode = sld.gridNode;
    const offset = gridL / 2 + SLD.ARRAY_OFFSET_KM;
    const centre = sld.arrayCentre
    || destinationPoint(gridNode[0], gridNode[1], offset, axis);

```

```

// The customer substation sits on the array edge nearest the grid node.
const customerSub = destinationPoint(centre[0], centre[1], gridL / 2, axis +
180);

const features = [];
const push = (geometry, properties) =>
  features.push({ type: 'Feature', geometry, properties });

// Site boundary, and the grab surface for dragging.
push({ type: 'Polygon', coordinates: rectPolygon(centre, gridW +
SLD.BOUNDARY_BUFFER_KM, gridL + SLD.BOUNDARY_BUFFER_KM, axis) },
  { kind: 'boundary', colour: SLD_COLOUR.boundary });

// Blocks, laid out from the north-west corner along the axis.
const ptN = destinationPoint(centre[0], centre[1], gridL / 2, axis);
const ptNW = destinationPoint(ptN[0], ptN[1], gridW / 2, axis - 90);
const blocks = [];
let placed = 0;
for (let r = 0; r < rows && placed < N; r += 1) {
  for (let c = 0; c < cols && placed < N; c += 1) {
    const across = destinationPoint(ptNW[0], ptNW[1],
      c * blockW + c * gap + blockW / 2, axis + 90);
    const at = destinationPoint(across[0], across[1],
      r * blockL + r * gap + blockL / 2, axis + 180);
    push({ type: 'Polygon', coordinates: rectPolygon(at, blockW, blockL,
axis) },
      { kind: 'block', colour: SLD_COLOUR.block });
    blocks.push(at);
    placed += 1;
  }
}

// 33 kV collectors: each block drops onto a trunk running up the axis
// from the customer substation, and the trunk is clipped to the furthest
// block rather than drawn to the far edge of nothing.
if (blocks.length) {
  const trunkEnd = destinationPoint(customerSub[0], customerSub[1], gridL,
axis);
  let furthest = 0;
  const branches = [];
  for (const at of blocks) {
    const foot = footOnSegment(at[0], at[1], customerSub, trunkEnd);
    furthest = Math.max(furthest,
      distanceKm(customerSub[0], customerSub[1], foot[0], foot[1]));
    branches.push([at, foot]);
  }
  if (furthest > 0) {
    const clipped = destinationPoint(customerSub[0], customerSub[1],
furthest, axis);
    push({ type: 'LineString', coordinates: [customerSub, clipped] },
      { kind: 'radial', role: 'collector_trunk', colour: SLD_COLOUR.radial
});
  }
  for (const [at, foot] of branches) {
    push({ type: 'LineString', coordinates: [at, foot] },
      { kind: 'radial', role: 'block_branch', colour: SLD_COLOUR.radial });
  }
}

// BESS compound alongside the customer substation.
// The original drawing reads the same topology-local financial BESS MWh
// that drives CAPEX and revenue. There is no second layout-BESS input.

```

```

const bessMwh = financeNumber(sld.finance[sld.inputs.mode]?.bess_mwh);
if (bessMwh > 0) {
  const areaKm2 = (bessMwh * SLD.BESS_M2_PER_MWH) / 1e6;
  const w = Math.sqrt(areaKm2 * SLD.BESS_ASPECT);
  const l = areaKm2 / w;
  const at = destinationPoint(customerSub[0], customerSub[1], w / 2 + 0.05,
axis - 90);
  push({ type: 'Polygon', coordinates: rectPolygon(at, w, l, axis) },
    { kind: 'bess', colour: SLD_COLOUR.bess });
  push({ type: 'LineString', coordinates: [at, customerSub] },
    { kind: 'radial', role: 'bess_tie', colour: SLD_COLOUR.radial });
}

// Export cable: customer substation, through the user's vertices, to the
// grid node. Measured along its own path, and against the straight line
// so the detour is visible rather than implied.
const route = [customerSub, ...sld.routePins, gridNode];
sld.cableKm = pathLengthKm(route);
sld.straightKm = distanceKm(customerSub[0], customerSub[1], gridNode[0],
gridNode[1]);
push({ type: 'LineString', coordinates: route },
  { kind: 'cable', colour: SLD_COLOUR.cable, km: sld.cableKm });

sld.routePins.forEach((at, index) => {
  push({ type: 'Point', coordinates: at },
    { kind: 'pin', index, colour: SLD_COLOUR.pin });
});

push({ type: 'Point', coordinates: customerSub },
  { kind: 'node', role: 'customer_substation', colour: SLD_COLOUR.node,
    label: `CUSTOMER SUB · ${stats.production_substation_ac_mva.toFixed(2)}
MVA` });
push({ type: 'Point', coordinates: gridNode },
  { kind: 'node', role: 'grid_node', colour: SLD_COLOUR.node,
    label: `${sld.gridNodeName || 'GRID NODE'}${sld.gridNodeVoltage ? ` ·
${sld.gridNodeVoltage}` : ''}` });

// Rotation handle, off the far edge of the array.
const handle = destinationPoint(centre[0], centre[1], gridL / 2 + 0.12,
axis);
push({ type: 'Point', coordinates: handle },
  { kind: 'handle', colour: SLD_COLOUR.handle });
push({ type: 'LineString', coordinates: [centre, handle] },
  { kind: 'radial', role: 'handle_stem', colour: SLD_COLOUR.handle });

sld.geometry = { centre, customerSub, gridW, gridL, axis, handle, blocks:
blocks.length };
return { type: 'FeatureCollection', features };
}

function ensureSldLayers(map) {
  if (map.getSource(SRC_SLD)) return;
  map.addSource(SRC_SLD, { type: 'geojson', data: { type: 'FeatureCollection',
features: [] } });

  map.addLayer({ id: SLD_LAYERS.boundary, type: 'fill', source: SRC_SLD,
    filter: ['==', ['get', 'kind'], 'boundary'],
    paint: { 'fill-color': SLD_COLOUR.boundary, 'fill-opacity': 0.07 } });
  map.addLayer({ id: SLD_LAYERS.boundaryLine, type: 'line', source: SRC_SLD,
    filter: ['==', ['get', 'kind'], 'boundary'],
    paint: { 'line-color': SLD_COLOUR.boundary, 'line-width': 1.2, 'line-
opacity': 0.65 } });

```

```

map.addLayer({ id: SLD_LAYERS.block, type: 'fill', source: SRC_SLD,
  filter: ['==', ['get', 'kind'], 'block'],
  paint: { 'fill-color': SLD_COLOUR.block, 'fill-opacity': 0.16,
    'fill-outline-color': SLD_COLOUR.block } });
map.addLayer({ id: SLD_LAYERS.bess, type: 'fill', source: SRC_SLD,
  filter: ['==', ['get', 'kind'], 'bess'],
  paint: { 'fill-color': SLD_COLOUR.bess, 'fill-opacity': 0.22,
    'fill-outline-color': SLD_COLOUR.bess } });

map.addLayer({ id: SLD_LAYERS.radial, type: 'line', source: SRC_SLD,
  filter: ['==', ['get', 'kind'], 'radial'],
  layout: { 'line-cap': 'round' },
  paint: { 'line-color': ['get', 'colour'], 'line-width': 0.9, 'line-
opacity': 0.5 } });
// The electron flow, on the collectors.
map.addLayer({ id: SLD_LAYERS.radialFlow, type: 'line', source: SRC_SLD,
  filter: ['all', ['==', ['get', 'kind'], 'radial'], ['!=', ['get', 'role'],
'handle_stem']],
  layout: { 'line-cap': 'round' },
  paint: { 'line-color': FLOW_COLOUR, 'line-width': 1.3, 'line-opacity':
0.65,
    'line-dasharray': [0.2, 3.2] } });

map.addLayer({ id: SLD_LAYERS.cableGlow, type: 'line', source: SRC_SLD,
  filter: ['==', ['get', 'kind'], 'cable'],
  layout: { 'line-cap': 'round', 'line-join': 'round' },
  paint: { 'line-color': SLD_COLOUR.cable, 'line-width': 8, 'line-opacity':
0.12,
    'line-blur': 5 } });
map.addLayer({ id: SLD_LAYERS.cable, type: 'line', source: SRC_SLD,
  filter: ['==', ['get', 'kind'], 'cable'],
  layout: { 'line-cap': 'round', 'line-join': 'round' },
  paint: { 'line-color': SLD_COLOUR.cable, 'line-width': 1.8, 'line-
opacity': 0.85 } });
map.addLayer({ id: SLD_LAYERS.cableFlow, type: 'line', source: SRC_SLD,
  filter: ['==', ['get', 'kind'], 'cable'],
  layout: { 'line-cap': 'round' },
  paint: { 'line-color': FLOW_COLOUR, 'line-width': 2.4, 'line-opacity':
0.9,
    'line-dasharray': [0.2, 3.2] } });
map.addLayer({ id: SLD_LAYERS.cableFlowB, type: 'line', source: SRC_SLD,
  filter: ['==', ['get', 'kind'], 'cable'],
  layout: { 'line-cap': 'round' },
  paint: { 'line-color': FLOW_COLOUR, 'line-width': 1.6, 'line-opacity':
0.55,
    'line-dasharray': [0.2, 3.2] } });

map.addLayer({ id: SLD_LAYERS.node, type: 'circle', source: SRC_SLD,
  filter: ['==', ['get', 'kind'], 'node'],
  paint: { 'circle-radius': 5, 'circle-color': SLD_COLOUR.node,
    'circle-stroke-color': '#000c10', 'circle-stroke-width': 1.5 } });
map.addLayer({ id: SLD_LAYERS.pin, type: 'circle', source: SRC_SLD,
  filter: ['==', ['get', 'kind'], 'pin'],
  paint: { 'circle-radius': 5, 'circle-color': SLD_COLOUR.pin, 'circle-
opacity': 0.9,
    'circle-stroke-color': '#04343a', 'circle-stroke-width': 1.5 } });
map.addLayer({ id: SLD_LAYERS.handle, type: 'circle', source: SRC_SLD,
  filter: ['==', ['get', 'kind'], 'handle'],
  paint: { 'circle-radius': 6, 'circle-color': 'rgba(0,0,0,0)',
    'circle-stroke-color': SLD_COLOUR.handle, 'circle-stroke-width': 1.8 }
});
const sldFont = styleTextFont(map);

```



```

    if (!sldFont) {
      link.failures.push('the basemap serves no glyphs, so layout labels are
omitted');
    } else {
      addLabelLayerWhenDrawable(map, sldFont, { id: SLD_LAYERS.label, type:
'symbol', source: SRC_SLD,
      filter: ['==', ['get', 'kind'], 'node'],
      layout: { 'text-field': ['get', 'label'], 'text-size': 9.5,
        'text-offset': [0, -1.4], 'text-anchor': 'bottom',
        'text-font': sldFont },
      paint: { 'text-color': '#a9c4c9', 'text-halo-color': '#000c10',
        'text-halo-width': 1.5 } }, 'layout');
    }
  }

let sldFlowHandle = null;
let sldPhase = 0;
function animateSld(map) {
  if (sldFlowHandle !== null) cancelAnimationFrame(sldFlowHandle);
  if (window.matchMedia?.('(prefers-reduced-motion: reduce)')?.matches) {
    try {
      map.setPaintProperty(SLD_LAYERS.cableFlow, 'line-opacity', 0);
      map.setPaintProperty(SLD_LAYERS.cableFlowB, 'line-opacity', 0);
      map.setPaintProperty(SLD_LAYERS.radialFlow, 'line-opacity', 0);
    } catch (_) { /* layer gone */ }
    return;
  }
  const step = () => {
    sldPhase = (sldPhase + FLOW_SPEED) % FLOW_PERIOD;
    const half = (sldPhase + FLOW_PERIOD / 2) % FLOW_PERIOD;
    try {
      setFlowDash(map, SLD_LAYERS.cableFlow, sldPhase);
      setFlowDash(map, SLD_LAYERS.cableFlowB, half);
      setFlowDash(map, SLD_LAYERS.radialFlow, sldPhase);
    } catch (_) { sldFlowHandle = null; return; }
    sldFlowHandle = requestAnimationFrame(step);
  };
  sldFlowHandle = requestAnimationFrame(step);
}

function redrawSld(map, { fit = false } = {}) {
  ensureSldLayers(map);
  const data = buildLayout();
  setSourceData(map, SRC_SLD, data);
  renderSldPanel();
  if (data.features.length) animateSld(map);
  if (fit && data.features.length && sld.geometry) {
    const lons = []; const lats = [];
    for (const f of data.features) {
      const walk = (c) => {
        if (typeof c[0] === 'number') { lons.push(c[0]); lats.push(c[1]);
return; }
        c.forEach(walk);
      };
      walk(f.geometry.coordinates);
    }
    map.fitBounds([[Math.min(...lons), Math.min(...lats)],
      [Math.max(...lons), Math.max(...lats)]], { padding: 70, duration: 700
});
  }
}

```

```

/* — dragging ————— */

function attachSldDragging(map) {
  const canvas = map.getCanvas();
  const grabbable = [SLD_LAYERS.handle, SLD_LAYERS.pin, SLD_LAYERS.boundary];

  map.on('mousemove', (event) => {
    if (!sld.active || sld.dragging) return;
    const hits = map.queryRenderedFeatures(event.point, { layers:
grabbable.filter(id => map.getLayer(id)) });
    canvas.style.cursor = hits.length ? 'grab' : '';
  });

  const beginDrag = (event) => {
    if (!sld.active || sld.dragging || fromOwnUi(event)) return;
    const layers = grabbable.filter(id => map.getLayer(id));
    if (!layers.length) return;
    const hits = map.queryRenderedFeatures(event.point, { layers });
    if (!hits.length) return;
    const kind = hits[0].properties?.kind;
    let nextDrag = null;
    if (kind === 'handle') nextDrag = { what: 'rotate' };
    else if (kind === 'pin') nextDrag = { what: 'pin', index:
Number(hits[0].properties.index) };
    else if (kind === 'boundary') nextDrag = { what: 'array' };
    else return;
    nextDrag.dragPanWasEnabled = map.dragPan.isEnabled?(). !== false;
    nextDrag.touchWasEnabled = map.touchZoomRotate?.isEnabled?(). !== false;
    sld.dragging = nextDrag;
    event.preventDefault();
    map.dragPan.disable();
    map.touchZoomRotate?.disable();
    canvas.style.cursor = 'grabbing';
  };
  map.on('mousedown', beginDrag);
  map.on('touchstart', beginDrag);

  const moveDrag = (event) => {
    if (!sld.dragging) return;
    event.preventDefault();
    const at = [event.lngLat.lng, event.lngLat.lat];
    if (sld.dragging.what === 'array') {
      sld.arrayCentre = at;
    } else if (sld.dragging.what === 'rotate') {
      const c = sld.geometry?.centre;
      if (c) sld.rotationDeg = initialBearingDeg(c[0], c[1], at[0], at[1]);
    } else if (sld.dragging.what === 'pin') {
      sld.routePins[sld.dragging.index] = at;
    }
    redrawSld(map);
  };
  map.on('mousemove', moveDrag);
  map.on('touchmove', moveDrag);

  const release = () => {
    if (!sld.dragging) return;
    const finished = sld.dragging;
    sld.dragging = null;
    if (finished.dragPanWasEnabled) map.dragPan.enable();
    if (finished.touchWasEnabled) map.touchZoomRotate?.enable();
    canvas.style.cursor = '';
  };
};

```

```

map.on('mouseup', release);
map.on('touchend', release);
map.on('mouseout', release);
canvas.addEventListener?>('pointercancel', release);

// Click the cable to insert a vertex where you clicked; double-click a
// vertex to remove it. No modes, no commit step.
map.on('click', (event) => {
  if (fromOwnUi(event)) return;
  if (!sld.active || !map.getLayer(SLD_LAYERS.cable)) return;
  const onPin = map.queryRenderedFeatures(event.point, { layers:
[SLD_LAYERS.pin] });
  if (onPin.length) return;
  const onCable = map.queryRenderedFeatures(event.point, { layers:
[SLD_LAYERS.cable] });
  if (!onCable.length) return;
  const at = [event.lngLat.lng, event.lngLat.lat];
  const route = [sld.geometry.customerSub, ...sld.routePins, sld.gridNode];
  let best = 0; let bestKm = Infinity;
  for (let i = 0; i < route.length - 1; i += 1) {
    const km = distanceToSegmentKm(at[0], at[1], route[i][0], route[i][1],
      route[i + 1][0], route[i + 1][1]).km;
    if (km < bestKm) { bestKm = km; best = i; }
  }
  sld.routePins.splice(best, 0, at);
  redrawSld(map);
});

map.on('dblclick', (event) => {
  if (!sld.active || !map.getLayer(SLD_LAYERS.pin)) return;
  const hits = map.queryRenderedFeatures(event.point, { layers:
[SLD_LAYERS.pin] });
  if (!hits.length) return;
  event.preventDefault();
  sld.routePins.splice(Number(hits[0].properties.index), 1);
  redrawSld(map);
});
}

/* — the panel ————— */

const PANEL_ID = 'gridatlas-sld-panel';

function installSldStyles() {
  if (document.getElementById('gridatlas-sld-css')) return;
  const style = document.createElement('style');
  style.id = 'gridatlas-sld-css';
  style.textContent = SLD_STYLES.sldPanel(PANEL_ID);
  document.head.appendChild(style);
}

function sldPanel() {
  let el = document.getElementById(PANEL_ID);
  if (el) return el;
  installSldStyles();
  el = document.createElement('div');
  el.id = PANEL_ID;
  (capturedMap?.getContainer() || document.body).appendChild(el);
  return el;
}

const ELECTRICAL_RULES = Object.freeze({

```

```

    mod_wp: { min: 1, step: 1, integer: true },
    mod_l: { min: 0.01, step: 0.01 },
    mod_w: { min: 0.01, step: 0.01 },
    gcr: { min: 0.01, max: 1, step: 0.01 },
    gross_factor: { min: 1, step: 0.05 },
    x_mods: { min: 1, step: 1, integer: true },
    z_strings: { min: 1, step: 1, integer: true },
    y_invs: { min: 1, step: 1, integer: true },
    s_subs: { min: 1, step: 1, integer: true },
    b_cols: { min: 1, step: 1, integer: true },
    string_inv_kva: { min: 1, step: 1 },
    string_skid_mva: { min: 0.1, step: 0.01 },
    dc_ac_ratio: { min: 0.01, step: 0.05 },
    mod_wp_c: { min: 1, step: 1, integer: true },
    mod_l_c: { min: 0.01, step: 0.01 },
    mod_w_c: { min: 0.01, step: 0.01 },
    gcr_c: { min: 0.01, max: 1, step: 0.01 },
    gross_factor_c: { min: 1, step: 0.05 },
    x_mods_c: { min: 1, step: 1, integer: true },
    str_per_cb_c: { min: 1, step: 1, integer: true },
    inv_ac_mw_c: { min: 0.1, max: 20, step: 0.01 },
    inv_dc_mw_c: { min: 0.1, max: 30, step: 0.01 },
    central_skid_mva_c: { min: 0.1, max: 25, step: 0.01 },
    inv_per_mv_c: { min: 1, step: 1, integer: true },
    mv_per_ring_c: { min: 1, step: 1, integer: true },
    rings_c: { min: 1, step: 1, integer: true },
  });

function normalizeElectricalInput(key, rawValue) {
  const rule = ELECTRICAL_RULES[key];
  const value = Number(rawValue);
  if (!rule || !Number.isFinite(value)) return null;
  if (value < rule.min || (rule.max != null && value > rule.max)) return null;
  if (rule.integer && !Number.isInteger(value)) return null;
  return value;
}

function electricalInputAttributes(key) {
  const rule = ELECTRICAL_RULES[key];
  return `min="${rule.min}"${rule.max == null ? '' : ` max="${rule.max}"`}
step="${rule.step}"`;
}

sld.normalizeElectricalInput = normalizeElectricalInput;

const FIELDS_STRING = [
  ['mod_wp', 'Module rating Wp'], ['mod_l', 'Module length m'], ['mod_w',
'Module width m'],
  ['gcr', 'Ground cover ratio'], ['gross_factor', 'Gross site factor'],
  ['x_mods', 'Modules / string'], ['z_strings', 'Strings / inverter'],
  ['y_invs', 'Inverters / skid'], ['s_subs', 'Skids / ring main'], ['b_cols',
'Ring main circuits'],
  ['string_inv_kva', 'String inverter kVA'], ['string_skid_mva', 'Skid
transformer MVA'],
  ['dc_ac_ratio', 'DC/AC ratio']
];

const FIELDS_CENTRAL = [
  ['mod_wp_c', 'Module rating Wp'], ['mod_l_c', 'Module length m'],
  ['mod_w_c', 'Module width m'],
  ['gcr_c', 'Ground cover ratio'], ['gross_factor_c', 'Gross site factor'],
  ['x_mods_c', 'Modules / string'], ['str_per_cb_c', 'Strings / combiner'],
  ['inv_ac_mw_c', 'Inverter AC MW'], ['inv_dc_mw_c', 'Inverter DC MWp'],
  ['central_skid_mva_c', 'Skid MVA'], ['inv_per_mv_c', 'Inverters / MV'],

```

```

    ['mv_per_ring_c', 'MV / ring'], ['rings_c', 'Rings']
];

const FINANCE_FIELDS = [
  ['@', 'Revenue and operating case'],
  ['price', 'Energy price GBP/MWh'], ['other', 'Other income GBP/MWh'],
  ['yield', 'Base yield kWh/kWp'], ['bifacial', 'Bifacial gain %'],
  ['losses', 'Base losses %'], ['deg', 'Degradation %'], ['opex', 'OPEX
GBP/MWac/yr'],
  ['@', 'CAPEX'],
  ['modules', 'Modules GBP/Wp'], ['epc_ex', 'EPC ex modules GBP/Wp'],
  ['flood', 'Flood resilience', 'checkbox'], ['flood_rate', 'Flood adder
GBP/Wp'],
  ['other_capex', 'Other CAPEX GBP/Wp'], ['fixed_capex', 'Fixed CAPEX GBP'],
  ['cont', 'Contingency %'],
  ['@', 'Electrical loss allowances'],
  ['loss_dc_string', 'DC string loss %'], ['loss_lv_dc', 'LV main DC loss %'],
  ['loss_lv_ac', 'LV AC loss %'], ['loss_tx', 'Transformer loss %'],
  ['loss_other', 'Other electrical loss %'],
  ['@', 'BESS finance'],
  ['bess_mw', 'BESS power MW'], ['bess_mwh', 'BESS energy MWh'],
  ['bess_capex', 'BESS CAPEX GBP/MWh'], ['bess_cycles', 'BESS cycles / year'],
  ['bess_spread', 'BESS revenue GBP/MWh'], ['bess_eff', 'BESS efficiency %'],
  ['@', 'Development case'],
  ['dev_stage', 'Development stage', 'stage'], ['dev_cost_mw', 'Development
cost GBP/Wp'],
  ['dev_module_mwp', 'Module supply GBP/Wp'], ['dev_epc_mw', 'EPC cost
GBP/Wp'],
  ['dev_owner_mw', 'Owner costs GBP/Wp'], ['dev_grid_mw', 'Grid connection
GBP/Wp'],
  ['dev_exit_mwp', 'Target exit value GBP/Wp'], ['dev_npv_mwp', 'Operating NPV
GBP/Wp'],
  ['dev_success', 'Success probability %'], ['dev_years', 'Development
years'],
];

const financeFieldHtml = (field, values) => {
  const [key, label, kind] = field;
  if (key === '@') return `<div class="sld-fin-section">${escapeHtml(label)}
</div>`;
  if (kind === 'checkbox') {
    return `<label for="sld_fin_${key}">${escapeHtml(label)}</label>`
      + `<input id="sld_fin_${key}" data-fin-key="${key}" type="checkbox"
${values[key] ? 'checked' : ''}>`;
  }
  if (kind === 'stage') {
    const options = Object.entries(DEVELOPMENT_STAGES).map(([value, text]) =>
      `<option value="${value}" ${String(values[key]) === value ? 'selected' :
''}>${escapeHtml(text)}</option>`
    ).join('');
    return `<label for="sld_fin_${key}">${escapeHtml(label)}</label>`
      + `<select id="sld_fin_${key}" data-fin-key="${key}">${options}
</select>`;
  }
  const maximum = key === 'bess_eff' || key === 'dev_success' ? ' max="100"' :
'';
  return `<label for="sld_fin_${key}">${escapeHtml(label)}</label>`
    + `<input id="sld_fin_${key}" data-fin-key="${key}" type="number"
min="0"${maximum} step="any" value="${values[key]}">`;
};

const moneyText = value => `GBP

```

```

${Math.round(financeNumber(value)).toLocaleString('en-GB')}}`;

function renderSldPanel() {
  const el = sldPanel();
  const s = sld.stats;
  const fields = sld.inputs.mode === 'string' ? FIELDS_STRING :
FIELDS_CENTRAL;
  const financeInputs = sld.finance[sld.inputs.mode];
  const finance = s?.finance;
  const detour = sld.straightKm > 0 ? sld.cableKm / sld.straightKm : 1;
  const acres = s ? s.gross_site_area_m2 / SLD.M2_PER_ACRE : 0;

  el.innerHTML = `
    <h4 class="sld-drag">Layout sandbox<span class="sld-beta">Beta</span>
      <button class="sld-min" title="Minimise">&minus;</button>
      <button class="sld-close" title="Close">&times;</button></h4>
      <div class="sld-site">${escapeHtml(sld.projectName || sld.gridNodeName ||
'Grid node')}</div>
      ${sld.projectName ? `<div class="sld-to">to ${escapeHtml(sld.gridNodeName
|| 'grid node')}`
+ `${sld.gridNodeVoltage ? ` &middot;`
${escapeHtml(sld.gridNodeVoltage)}` : ''}</div>` : ''}
      <div class="sld-tabs">
        <button data-mode="string" data-on="${sld.inputs.mode ===
'string'}">String</button>
        <button data-mode="central" data-on="${sld.inputs.mode ===
'central'}">Central</button>
      </div>
      ${sld.targetMw ? `
      <div class="sld-target">
        <div class="sld-target-row"><span>Register states</span>
<b>${sld.targetMw} MW</b></div>
        <div class="sld-basis">
          <span>That figure is</span>
          <select id="sld_basis">
            <option value="unstated" ${sld.targetBasis === 'unstated' ?
'selected' : ''}>not stated</option>
            <option value="ac" ${sld.targetBasis === 'ac' ? 'selected' : ''}>AC
export MW</option>
            <option value="dc" ${sld.targetBasis === 'dc' ? 'selected' : ''}>DC
MWp</option>
          </select>
        </div>
        ${sld.targetBasis === 'unstated'
? `<div class="sld-danger">REPD does not reliably distinguish AC from
DC.
        Its figure is nominally MWelec, but schemes report it both ways
and the
        register does not carry the distinction. Nothing is fitted until
you say
        which this is: matching AC when the figure was DC oversizes the
export
        limitation, curtailment and the size of the offer.</div>`
: `<div class="sld-fitted">Fitted to ${sld.targetBasis === 'ac' ? 'AC
export' : 'DC'} by
        ${sld.inputs.mode === 'string' ? 'ring main circuits' : 'rings'}
        ${sld.fitResidualPct != null
? ` &middot; <b class="${Math.abs(sld.fitResidualPct) > 5 ?
'sld-off' : ''}">${sld.fitResidualPct >= 0 ? '+' :
''}${sld.fitResidualPct.toFixed(1)}%</b> against the stated figure`
: ''}.
`

```

```

        Ratings, string length and module choice are untouched.</div>`
</div>` : ''}
<div class="sld-grid">
    ${fields.map([key, label]) =>
        `<label for="sld_${key}">${label}</label>`
        + `<input id="sld_${key}" data-key="${key}" type="number"
    ${electricalInputAttributes(key)} value="${sld.inputs[key]}">`
    ).join('')}
</div>
<div class="sld-out">
    <span>Array DC</span><b>${s ? s.dc_mwp.toFixed(1) : '0.0'} MWp</b>
    <span>Inverter AC</span><b>${s?.consistency?.inverter_ac_mw != null
        ? s.consistency.inverter_ac_mw.toFixed(1) : '0.0'} MW</b>
    <span>Export limit</span><b>${s?.consistency?.export_mva != null
        ? s.consistency.export_mva.toFixed(1) : '0.0'} MVA</b>
    <span>Design DC/AC</span><b>${s?.consistency?.design_dc_ac != null
        ? s.consistency.design_dc_ac.toFixed(2) : '0.00'}</b>
    <span>DC / export</span><b>${s?.consistency?.export_dc_ac != null
        ? s.consistency.export_dc_ac.toFixed(2) : '0.00'}</b>
    <span>Inverter / export</span><b>${s?.consistency?.inverter_to_export !=
null
        ? s.consistency.inverter_to_export.toFixed(2) : '0.00'}</b>
    <span>Modules</span><b>${s ? s.module_count.toLocaleString('en-GB') :
'0'}</b>
    <span>Blocks</span><b>${s ? s.total_blocks : 0}</b>
    <span>Gross site</span><b>${acres.toFixed(0)} acres</b>
    <span>Ring main</span><b>${s ? s.ring_main_ac_mva.toFixed(2) : '0.00'}
MVA</b>
    <span class="lit">Export cable</span><b
class="lit">${sld.cableKm.toFixed(3)} km</b>
    <span>Straight line</span><b>${sld.straightKm.toFixed(3)} km</b>
    <span>Detour factor</span><b>${detour.toFixed(2)}&times;</b>
    <span>Route vertices</span><b>${sld.routePins.length}</b>
    <span>Rotation</span><b>${normBearing(sld.rotationDeg).toFixed(0)}&deg;
</b>
</div>
${(() => {
    const c = s?.consistency;
    if (!(c?.stated_dc_ac > 0) || !(c?.design_dc_ac > 0)) return '';
    if (Math.abs(c.design_dc_ac - c.stated_dc_ac) / c.stated_dc_ac <= 0.05)
return '';
    return `<div class="sld-ratio-note">Entered DC/AC
    ${c.stated_dc_ac.toFixed(2)}; `
        + `the equipment counts and ratings shown give
    ${c.design_dc_ac.toFixed(2)}. `
        + `Both values remain visible and no input is changed automatically.
    </div>`;
    })()}
    ${s && s.warning ? `<div class="sld-warn">${escapeHtml(s.warning)}</div>`
: ''}
    <details class="sld-finance" ${sld.financeOpen ? 'open' : ''}>
        <summary>Financial screening inputs and outputs</summary>
        <div class="sld-fin-out">
            <span>Year 1 revenue</span><b>${moneyText(finance?.annualRevenue)}</b>
            <span>25-year revenue</span><b>${moneyText(finance?.revenue25)}</b>
            <span>35-year revenue</span><b>${moneyText(finance?.revenue35)}</b>
            <span>Total CAPEX</span><b>${moneyText(finance?.totalCapex)}</b>
            <span>CAPEX / Wp</span><b>GBP
    ${financeNumber(finance?.capexPerWp).toFixed(2)}</b>
            <span>25-year surplus</span><b>${moneyText(finance?.surplus25)}</b>
            <span>35-year surplus</span><b>${moneyText(finance?.surplus35)}</b>
            <span>Development capital at risk</span>

```

```

<b>${moneyText(finance?.devCapitalAtRisk)}</b>
    <span>Total build cost</span>
<b>${moneyText(finance?.devTotalBuildCost)}</b>
    <span>Target exit value</span><b>${moneyText(finance?.devExitValue)}
</b>
    <span>Operating NPV</span><b>${moneyText(finance?.devOperatingNpv)}
</b>
    <span>Gross development margin</span>
<b>${moneyText(finance?.devGrossMargin)}</b>
    <span>Risk-adjusted value</span>
<b>${moneyText(finance?.devRiskAdjustedValue)}</b>
    <span>Equity money multiple</span>
<b>${financeNumber(finance?.devReturnMultiple).toFixed(2)}x</b>
</div>
<div class="sld-fin-grid">${FINANCE_FIELDS.map(field =>
financeFieldHtml(field, financeInputs)).join('')}</div>
<div class="sld-fin-note"><b>Screening values only, not financial
advice.</b> Revenue, CAPEX,
    OPEX, development value and BESS outputs depend entirely on the
visible assumptions. They do
    not replace project-specific yield, degradation, route-to-market, tax,
debt, grid, EPC,
    insurance, degradation, augmentation or investment-committee models.
</div>
</details>
<div class="sld-hint">Drag the site to move it. Drag the handle to rotate.
Click the
    cable to add a vertex, drag a vertex to shape the route, double-click
one to remove it.</div>
<div class="sld-caveat"><b>Beta analytics, not an actual grid connection.
</b> A layout, not
    a design. Every length is straight-line between the points shown, with
no wayleave,
    easement, right of way, crossing, terrain, ground condition or consent
content, and no
    route has been walked. A real connection depends on factors that must be
studied:
    network impedance and fault level, thermal headroom, existing committed
connections and
    queue position, and land control. A mapped substation does not confirm
capacity, voltage
    suitability or acceptance by any network party.</div>`;

el.dataset.open = 'true';
// Optional throughout: a panel that cannot find its own controls must not
// take the layout down with it. The geometry is the product; the panel is
// how it is driven.
el.querySelector?('.sld-close')?.addEventListener('click', closeSld);
el.querySelector?('.sld-min')?.addEventListener('click', () => {
    const min = el.dataset.min === 'true';
    el.dataset.min = min ? 'false' : 'true';
    const button = el.querySelector('.sld-min');
    if (button) button.innerHTML = min ? '&minus;' : '&plus;';
});
// The panel is draggable by its heading for the same reason the card is:
// on a map, anything fixed in a corner is eventually in the way.
const heading = el.querySelector?('.h4.sld-drag');
if (heading && !heading.dataset.bound) {
    heading.dataset.bound = '1';
    let drag = null;
    heading.addEventListener('pointerdown', (event) => {
        if (event.target.closest('button')) return;

```



```

    event.preventDefault();
    const rect = el.getBoundingClientRect();
    drag = {
      pointerId: event.pointerId,
      dx: event.clientX - rect.left,
      dy: event.clientY - rect.top,
    };
    heading.setPointerCapture?.(event.pointerId);
    el.style.right = 'auto';
    el.style.bottom = 'auto';
    el.style.left = rect.left + 'px';
    el.style.top = rect.top + 'px';
    const map = capturedMap?.getContainer()?.getBoundingClientRect();
    if (map) el.style.maxHeight = Math.max(120, map.bottom - rect.top - 8) +
'px';
  });
  heading.addEventListener('pointermove', (event) => {
    if (!drag || event.pointerId !== drag.pointerId) return;
    event.preventDefault();
    const map = capturedMap?.getContainer()?.getBoundingClientRect()
      || { left: 0, top: 0, right: window.innerWidth, bottom:
window.innerHeight };
    const panel = el.getBoundingClientRect();
    const minX = map.left + 4;
    const maxX = Math.max(minX, map.right - panel.width - 4);
    const minY = map.top + 4;
    const maxY = Math.max(minY, map.bottom - 44 - 4);
    const left = Math.max(minX, Math.min(maxX, event.clientX - drag.dx));
    const top = Math.max(minY, Math.min(maxY, event.clientY - drag.dy));
    el.style.left = left + 'px';
    el.style.top = top + 'px';
    el.style.maxHeight = Math.max(120, map.bottom - top - 8) + 'px';
  });
  const finish = (event) => {
    if (!drag || event.pointerId !== drag.pointerId) return;
    try { heading.releasePointerCapture?.(event.pointerId); } catch (_) { /*
already released */ }
    drag = null;
  };
  heading.addEventListener('pointerup', finish);
  heading.addEventListener('pointercancel', finish);
}
(el.querySelectorAll?('.sld-tabs button') || []).forEach(button => {
  button.addEventListener('click', () => {
    sld.inputs.mode = button.dataset.mode;
    if (capturedMap) redrawSld(capturedMap);
  });
});
el.querySelector?('#sld_basis')?.addEventListener('change', (event) => {
  sld.targetBasis = event.target.value;
  fitToStatedCapacity();
  if (capturedMap) redrawSld(capturedMap, { fit: true });
});
el.querySelector?('.details.sld-finance')?.addEventListener('toggle',
(event) => {
  sld.financeOpen = Boolean(event.currentTarget.open);
});
(el.querySelectorAll?('[data-fin-key]') || []).forEach(input => {
  input.addEventListener('change', () => {
    const values = sld.finance[sld.inputs.mode];
    if (input.type === 'checkbox') values[input.dataset.finKey] =
Boolean(input.checked);

```

```

        else if (input.dataset.finKey === 'dev_stage') {
            applyDevelopmentStageDefaults(values, input.value);
        } else if (input.tagName === 'SELECT') values[input.dataset.finKey] =
input.value;
        else {
            const value = Number(input.value);
            if (Number.isFinite(value)) values[input.dataset.finKey] = value;
        }
        if (capturedMap) redrawSld(capturedMap);
    });
});
(el.querySelector?(['input[data-key]']) || []).forEach(input => {
    input.addEventListener('change', () => {
        const key = input.dataset.key;
        const value = normalizeElectricalInput(key, input.value);
        if (value == null) {
            input.value = String(sld.inputs[key]);
            return;
        }
        sld.inputs[key] = value;
        if (key === 'gcr' || key === 'gcr_c') {
            applyMountingBifacial(sld.inputs.mode, value);
        }
        // Editing by hand wins. Re-fitting here would silently undo the change
        // the user just made; the residual simply moves and says so.
        if (sld.targetBasis === 'ac' || sld.targetBasis === 'dc') {
            const s = computeSldStats();
            const got = sld.targetBasis === 'ac' ? s.ac_mw : s.dc_mwp;
            sld.fitResidualPct = sld.targetMw > 0
                ? ((got - sld.targetMw) / sld.targetMw) * 100 : null;
        }
        if (capturedMap) redrawSld(capturedMap);
    });
});
});
}

function closeSld() {
    sld.active = false;
    sld.projectName = null;
    sld.routePins = [];
    sld.arrayCentre = null;
    sld.rotationDeg = 0;
    if (sldFlowHandle !== null) { cancelAnimationFrame(sldFlowHandle);
sldFlowHandle = null; }
    const el = document.getElementById(PANEL_ID);
    if (el) el.dataset.open = 'false';
    if (capturedMap && capturedMap.getSource(SRC_SLD)) {
        setSourceData(capturedMap, SRC_SLD, { type: 'FeatureCollection', features:
[] });
    }
}

// Opened from the substation card the neon links already produce, so the
// sandbox is one click from the thing it connects to.
function openSldAt(map, gridNode, name, voltage) {
    sld.active = true;
    sld.projectName = null;
    sld.targetMw = null;
    sld.targetBasis = 'unstated';
    sld.fitResidualPct = null;
    sld.gridNode = gridNode;
    sld.gridNodeName = name;

```

```

    sld.gridNodeVoltage = voltage;
    sld.arrayCentre = null;
    sld.rotationDeg = 0;
    sld.routePins = [];
    redrawSld(map, { fit: true });
}
sld.openAt = openSldAt;

// Opened from a project card. The scheme sits at the project and the export
// cable runs to the nearest substation the links already found, which is the
// order a scheme is actually built: generation first, then the route to the
// network. Falls back to the project's own point if nothing was in range, so
// the button never does nothing.
function openSldFromProject(map, selection) {
    const nearest = selection.links && selection.links[0];
    if (!nearest) {
        sld.active = false;
        link.failures.push('layout: no substation within '
            + `${MAX_LINK_KM} km of ${selection.name}`);
        return;
    }
    sld.active = true;
    sld.gridNode = nearest.at;
    sld.gridNodeName = nearest.name || 'Grid node';
    sld.gridNodeVoltage = nearest.kv && nearest.kv.length ? `${nearest.kv[0]}
kv` : '';
    sld.projectName = selection.name;
    sld.targetMw = selection.statedMw || null;
    // Unstated until the user says. The register's figure is not self-
describing
    // and the layout must not pretend otherwise.
    sld.targetBasis = 'unstated';
    // The array starts on the project, not offset from the substation, because
    // the project is the thing that exists.
    sld.arrayCentre = selection.origin;
    sld.rotationDeg = initialBearingDeg(
        nearest.at[0], nearest.at[1], selection.origin[0], selection.origin[1]);
    sld.routePins = [];
    enableSubstationLayer();
    redrawSld(map, { fit: true });
}
sld.openFromProject = openSldFromProject;

/**
 * Keep the layer controls reachable in fullscreen.
 *
 * The shell fullscreens the map element alone, so on desktop every layer
 * checkbox -- the whole dashboard below the map -- vanishes the moment you
 * maximise, and there is no way to turn anything on until you come back out.
 * Mobile carries its own drop-down curtain inside #map-container. When that
 * container is already fullscreen, moving its dashboard ancestor into it
 * would create a DOM cycle and throw HierarchyRequestError.
 *
 * The dashboard node is MOVED into the fullscreen element and moved back on
 * exit, rather than cloned. A clone would look right and do nothing, because
 * every checkbox listener belongs to the original.
 */
function keepLayersInFullscreen() {
    const dashboard = document.getElementById('dashboard')
        || document.querySelector('.dashboard');
    if (!dashboard) { link.failures.push('fullscreen: dashboard not found');
return; }

```

```

let home = null;

const onChange = () => {
  const full = document.fullscreenElement;
  /* Relocate only between disjoint trees. If either element contains the
  other, the dashboard is already represented in the fullscreen tree;
  appending an ancestor to its descendant is invalid DOM. */
  if (full && !full.contains(dashboard) && !dashboard.contains(full)) {
    home = { parent: dashboard.parentNode, next: dashboard.nextSibling };
    dashboard.classList.add('gridatlas-fs-layers');
    full.appendChild(dashboard);
  } else if (!full && home) {
    dashboard.classList.remove('gridatlas-fs-layers');
    home.parent.insertBefore(dashboard, home.next);
    home = null;
  }
  boundCardToMap();
};
document.addEventListener('fullscreenchange', onChange);
document.addEventListener('webkitfullscreenchange', onChange);

const style = document.createElement('style');
style.textContent = SLD_STYLES.fullscreenLayers();
document.head.appendChild(style);
}

function installSld(map) {
  installSldStyles();
  ensureSldLayers(map);
  attachSldDragging(map);
  try { keepLayersInFullscreen(); }
  catch (error) { link.failures.push('fullscreen: ' + String(error?.message ||
error)); }
  try { armCorridorGestures(map); }
  catch (error) { link.failures.push('corridor sheet: ' +
String(error?.message || error)); }
  // A substation click offers the layout; the neon links still draw.
  map.on('click', (event) => {
    if (fromOwnUi(event)) return;
    if (!map.getLayer(SUBS_LAYER_ID)) return;
    const hits = map.queryRenderedFeatures(event.point, { layers:
[SUBS_LAYER_ID] });
    if (!hits.length) return;
    const properties = hits[0].properties || {};
    const at = representativePoint(hits[0].geometry);
    if (!at) return;
    openSldAt(map, at, properties.name || 'Grid node',
(voltagesKv(properties)[0] ? `${voltagesKv(properties)[0]} kV` : ''));
  });
}
})();

```

```

=====
FILE: https://raw.githubusercontent.com/VentusLtd/data-grid-
gb/1c9909d1138704b29235c27fd769436dda8a0b18/derived/connection-points.v3.json
2896561 chars · data · TRUNCATED: the first 4000 characters of 2896561 are
shown, because this is data rather than code. Nothing has been summarised or
rewritten; the rest is simply not here. · found via loaded by the browser
(fetch)
=====
{"schema":"data-grid-gb.connection-points.v3","what_this_is":"Every transmission

```

substation NESO names at 132 kV and above, with what the operator publishes about it: circuits and their seasonal ratings, transformers, reactive plant, eight separately named fault-current metrics, and changes planned to 2033/34. Coordinates are joined from the OpenStreetMap-derived substation payload where a join exists.", "not_a_connection_assessment": "Nothing here says a project can or cannot connect at a node. Queue position, committed connections, consent and commercial terms decide that, and no published appendix contains them.", "source": {"publisher": "NESO", "publication": "Electricity Ten Year Statement 2025", "appendices": ["B - system technical data", "D - fault levels (peak and minimum)"], "note": "parameters as published; no power flow is solved here"}, "minimum_kv": 132, "join": {"why": "ETYS names substations and does not locate them", "geometry_source": "OpenStreetMap contributors, via the GridAtlas release", "exact_name": 461, "distinctive_tokens": 41, "ambiguous_exact_name": 25, "ambiguous_distinctive_tokens": 47, "unlocated": 384, "unlocated_are_published": "a site nobody has mapped is published without coordinates rather than dropped"}, "counts": {"connection_points": 886, "with_location": 502, "with_fault_current": 605, "with_planned_changes": 629}, "connection_points": [{"site_code": "ABBA", "name": "ABERDEEN BAY WINDFARM", "transmission_owner": "SHET", "voltages_kv": [132], "circuits": 1, "transformers": 0, "circuit_winter_rating_mva": {"min": 173, "max": 173}, "reactive_compensation": null, "fault_current": null, "fault_current_by_voltage": null, "planned_changes": 0, "planned_change_years": []}, {"site_code": "ABEB", "name": "ABERTHAW (NGED)", "transmission_owner": "NGET", "voltages_kv": [132], "circuits": 0, "transformers": 0, "circuit_winter_rating_mva": null, "reactive_compensation": null, "fault_current": null, "fault_current_by_voltage": null, "planned_changes": 2, "planned_change_years": ["2030"]}, {"site_code": "ABED", "name": "ABERARDER ONSHORE WIND FARM", "transmission_owner": "SHET", "voltages_kv": [132, 33], "circuits": 0, "transformers": 0, "circuit_winter_rating_mva": null, "reactive_compensation": null, "fault_current": null, "fault_current_by_voltage": null, "planned_changes": 3, "planned_change_years": ["2028"]}, {"site_code": "ABHA", "name": "ABHAM", "transmission_owner": "NGET", "voltages_kv": [400, 132], "circuits": 4, "transformers": 4, "circuit_winter_rating_mva": {"min": 2078, "max": 2423}, "reactive_compensation": {"units": 2, "mvar_generation": 90, "mvar_absorption": 0}, "fault_current": {"peak": {"scenarios": 20, "winters": ["2025/26", "2026/27", "2028/29", "2030/31", "2033/34"], "locations": ["ABHA1 M2", "ABHA1 R1", "ABHA4A", "ABHA4B"], "voltages_kv": [132.0, 400.0], "metrics": {"three_phase_initial_peak_current_ka": {"min": 31.28, "max": 47.52, "unit": "kA"}, "three_phase_rms_break_current_ka": {"min": 12.78, "max": 16.93, "unit": "kA"}, "three_phase_dc_break_current_ka": {"min": 4.16, "max": 8.24, "unit": "kA"}, "three_phase_peak_break_current_ka": {"min": 25.2, "max": 28.7, "unit": "kA"}, "single_phase_initial_peak_current_ka": {"min": 33.46, "max": 38.58, "unit": "kA"}, "single_phase_rms_break_current_ka": {"min": 13.86, "max": 15.71, "unit": "kA"}, "single_phase_dc_break_current_ka": {"min": 2.65, "max": 8.25, "unit": "kA"}, "single_phase_peak_break_current_ka": {"min": 22.58, "max": 30.42, "unit": "kA"}}, "scope": "site-wide envelope; may combine voltage levels and buses", "aggregation": "envelope across the listed published rows; metrics, voltages and buses are not interchangeable"}, "minimum": {"scenarios": 4, "winters": ["2025/26"], "locations": ["ABHA1 M2", "ABHA1 R1", "ABHA4A", "ABHA4B"], "voltages_kv": [132.0, 400.0], "metrics": {"three_phase_initial_peak_current_ka": {"min": 26.35, "max": 29.13, "unit": "kA"}, "three_phase_rms_break_current_ka": {"min": 9.92, "max": 11.76, "unit": "kA"}, "three_phase_dc_break_current_ka": {"min": 1.57, "max": 6.61, "unit": "kA"}, "three_phase_peak_break_current_ka": {"min": 15.6, "max": 23.25, "unit": "kA"}, "single_phase_initial_peak_current_ka": {"min": 27.6, "max": 32.44, "unit": "kA"}, "single_phase_rms_break

FILE: <https://basemaps.cartocdn.com/gl/dark-matter-gl-style/style.json>
70431 chars · data · TRUNCATED: the first 4000 characters of 70431 are shown,

because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (fetch)

```
=====
{
  "version": 8,
  "name": "Dark Matter",
  "metadata": { "maputnik:renderer": "mbgljs" },
  "sources": {
    "carto": {
      "type": "vector",
      "url":
"https://tiles.basemaps.cartocdn.com/vector/carto.streets/v1/tiles.json"
    }
  },
  "sprite": "https://tiles.basemaps.cartocdn.com/gl/dark-matter-gl-
style/sprite",
  "glyphs": "https://tiles.basemaps.cartocdn.com/fonts/{fontstack}/{range}.pbf",
  "layers": [
    {
      "id": "background",
      "type": "background",
      "layout": { "visibility": "visible" },
      "paint": { "background-color": "#0e0e0e", "background-opacity": 1 }
    },
    {
      "id": "landcover",
      "type": "fill",
      "source": "carto",
      "source-layer": "landcover",
      "filter": [
        "any",
        ["==", "class", "wood"],
        ["==", "class", "grass"],
        ["==", "subclass", "recreation_ground"]
      ],
      "paint": {
        "fill-color": {
          "stops": [
            [8, "#0e0e0e"],
            [9, "#0e0e0e"],
            [11, "#0e0e0e"],
            [13, "#0e0e0e"],
            [15, "#0e0e0e"]
          ]
        },
        "fill-opacity": 1
      }
    },
    {
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      "type": "fill",
      "source": "carto",
      "source-layer": "park",
      "minzoom": 9,
      "filter": ["all", ["==", "class", "national_park"]],
      "layout": { "visibility": "visible" },
      "paint": {
        "fill-color": {
          "stops": [
            [8, "#0e0e0e"],
            [9, "#0e0e0e"],
            [11, "#0e0e0e"],

```

```

        [13, "#0e0e0e"],
        [15, "#0e0e0e"]
    ],
    },
    "fill-opacity": 1,
    "fill-translate-anchor": "map"
  }
},
{
  "id": "park_nature_reserve",
  "type": "fill",
  "source": "carto",
  "source-layer": "park",
  "minzoom": 0,
  "filter": ["all", ["==", "class", "nature_reserve"]],
  "layout": { "visibility": "visible" },
  "paint": {
    "fill-color": {
      "stops": [
        [8, "#0e0e0e"],
        [9, "#0e0e0e"],
        [11, "#0e0e0e"],
        [13, "#0e0e0e"],
        [15, "#0e0e0e"]
      ]
    },
    "fill-antialias": true,
    "fill-opacity": {
      "stops": [
        [6, 0.7],
        [9, 0.9]
      ]
    }
  }
},
{
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  "type": "fill",
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  "source-layer": "landuse",
  "minzoom": 6,
  "filter": ["any", ["==", "class", "residential"]],
  "paint": {
    "fill-color": {
      "stops": [
        [5, "rgba(0, 0, 0, 0.5)"],
        [8, "rgba(0, 0, 0, 0.45)"],
        [9, "rgba(0, 0, 0, 0.4)"],
        [11, "rgba(0, 0, 0, 0.35)"],
        [13, "rgba(0, 0, 0, 0.3)"],
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FILE: <https://tiles.basemaps.cartocdn.com/vector/carto.streets/v1/tiles.json>
44256 chars · data · TRUNCATED: the first 4000 characters of 44256 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (fetch)

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a.basemaps.cartocdn.com/vectortiles/carto.streets/v1/{z}/{x}/{y}.mvt",
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c.basemaps.cartocdn.com/vectortiles/carto.streets/v1/{z}/{x}/{y}.mvt",
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href=\"http://www.openstreetmap.org/about/\" target=\"_blank\">OpenStreetMap</a>
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FILE: https://tiles.basemaps.cartocdn.com/gl/dark-matter-gl-style/sprite@2x.json
103 chars · data · shown in full · found via loaded by the browser (fetch)
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FILE: https://tiles.basemaps.cartocdn.com/gl/dark-matter-gl-style/sprite@2x.png
378 chars · data · shown in full · found via loaded by the browser (fetch)
=====

```

🌀PNG

```
=====
FILE:
https://tiles.basemaps.cartocdn.com/fonts/Montserrat%20Regular%20Italic%20Open%20Sans%20Italic%20Noto%20Sans%20Regular%20ChanWangHeiLight%20Regular%20NanumBarungothic%20Regular/0-255.pbf
74101 chars · data · TRUNCATED: the first 4000 characters of 74101 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (fetch)
=====
```

05/09/2026, 16:22
Page 571 of 591

```
=====
FILE: https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-
v9/data/grid_substations.geojson
1192742 chars · data · TRUNCATED: the first 4000 characters of 1192742 are
shown, because this is data rather than code. Nothing has been summarised or
rewritten; the rest is simply not here. · found via loaded by the browser
(fetch)
```

lob:https://ventusltd.github.io/184a2ca1-a0ae-42ea-b3ba-72ceefc889c4 05/09/2026, 16:22
Page 572 of 591

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FILE:

<https://tiles.basemaps.cartocdn.com/fonts/Montserrat%20Medium,Open%20Sans%20Bold,Noto%20Sans%20Regular,HanWangHeiLight%20Regular,NanumBarunGothic%20Regular/0-255.pbf>

77323 chars · data · TRUNCATED: the first 4000 characters of 77323 are shown, because this is data rather than code. Nothing has been summarised or rewritten; the rest is simply not here. · found via loaded by the browser (fetch)

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Page 574 of 591

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FILE: https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-
v9/data/grid_275kv.geojson
1022277 chars · data · TRUNCATED: the first 4000 characters of 1022277 are
shown, because this is data rather than code. Nothing has been summarised or
rewritten; the rest is simply not here. · found via loaded by the browser
(fetch)
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FILE: https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/grid_66kv.geojson
597382 chars · data · TRUNCATED: the first 4000 characters of 597382 are shown, because this is data rather than code. Nothing has been summarised or rewritten; the rest is simply not here. · found via loaded by the browser (fetch)

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62037 chars · data · TRUNCATED: the first 4000 characters of 62037 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (fetch)
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shown, because this is data rather than code. Nothing has been summarised or
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FILE: https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-
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FILE: https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-
v9/data/datacentres.geojson
42847 chars · data · TRUNCATED: the first 4000 characters of 42847 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (fetch)
=====
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FILE: <https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-v9/data/airports.geojson>
 119564 chars · data · TRUNCATED: the first 4000 characters of 119564 are shown, because this is data rather than code. Nothing has been summarised or rewritten; the rest is simply not here. · found via loaded by the browser (fetch)

=====

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=====
FILE: https://ventusltd.github.io/gridatlas/atlas/releases/202608300453-atlas-
v9/data/railways.geojson
508859 chars · data · TRUNCATED: the first 4000 characters of 508859 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (fetch)
=====

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=====
FILE: https://cdn.jsdelivr.net/npm/@duckdb/duckdb-wasm@1.29.0/+esm
26009 chars · data · TRUNCATED: the first 4000 characters of 26009 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (script)
=====

```

```

/**

```

```

 * Bundled by jsDelivr using Rollup v4.62.2 and esbuild v0.28.1.
 * Original file: /npm/@duckdb/duckdb-wasm@1.29.0/dist/duckdb-browser.mjs
 *

```

```

 * Do NOT use SRI with dynamically generated files! More information:
https://www.jsdelivr.com/using-sri-with-dynamic-files
 */

```

```

import*as r from"/npm/apache-arrow@17.0.0/+esm";var
v=Object.create,N=Object.defineProperty,G=Object.getOwnPropertyDescriptor,q=Obje
ct.getOwnPropertyNames,B=Object.getPrototypeOf,Y=Object.prototype.hasOwnProperty
,Q=(e,s)=>()=>(s|e((s={exports:{}}).exports,s),s.exports),H=(e,s,t,a)=>
{if(s&&typeof s=="object"||typeof s=="function")for(let i of
q(s))!Y.call(e,i)&&i!==t&&N(e,i,{get:()=>s[i],enumerable:!
(a=G(s,i))||a.enumerable});return e},K=(e,s,t)=>(t=e!=null?v(B(e)):
{}),H(!e||!e.__esModule?N(t,"default",{value:e,enumerable:!0}):t,e),W=Q((e,s)=>
{s.exports=Worker}),g=(e=>
(e[e.UNDEFINED=0]="UNDEFINED",e[e.AUTOMATIC=1]="AUTOMATIC",e[e.READ_ONLY=2]="REA
D_ONLY",e[e.READ_WRITE=3]="READ_WRITE",e))(g|{}),O=(e=>
(e[e.IDENTIFIER=0]="IDENTIFIER",e[e.NUMERIC_CONSTANT=1]="NUMERIC_CONSTANT",e[e.S
TRING_CONSTANT=2]="STRING_CONSTANT",e[e.OPERATOR=3]="OPERATOR",e[e.KEYWORD=4]="K
EYWORD",e[e.COMMENT=5]="COMMENT",e))(O|{}),y=(e=>
(e[e.NONE=0]="NONE",e[e.DEBUG=1]="DEBUG",e[e.INFO=2]="INFO",e[e.WARNING=3]="WARN
ING",e[e.ERROR=4]="ERROR",e))(y|{}),m=(e=>
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{}),F=(e=>
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.RUN=4]="RUN",e[e.CAPTURE=5]="CAPTURE",e))(F|{}),A=(e=>
(e[e.NONE=0]="NONE",e[e.WEB_WORKER=1]="WEB_WORKER",e[e.NODE_WORKER=2]="NODE_WOR
KER",e[e.BINDINGS=3]="BINDINGS",e[e.ASYNC_DUCKDB=4]="ASYNC_DUCKDB",e))(A|
{}),x=class{log(e){}},V=class{constructor(e=2){this.level=e}log(e)

```

```

{e.level>=this.level&&console.log(e)}};function z(e){switch(e){case
0:return"NONE";case 1:return"DEBUG";case 2:return"INFO";case
3:return"WARNING";case 4:return"ERROR";default:return"?"}}function j(e)
{switch(e){case 0:return"NONE";case 1:return"OK";case 2:return"ERROR";case
3:return"START";case 4:return"RUN";case
5:return"CAPTURE";default:return"?"}}function J(e){switch(e){case
1:return"CONNECT";case 2:return"DISCONNECT";case 5:return"INstantiate";case
3:return"OPEN";case 4:return"QUERY";default:return"?"}}function Z(e){switch(e)
{case 0:return"NONE";case 1:return"WEB WORKER";case 2:return"NODE WORKER";case
3:return"DUCKDB BINDINGS";case 4:return"DUCKDB";default:return"?"}}var P=(e=>
(e[e.SUCCESS=0]="SUCCESS",e))(P||{}),L=class{constructor(e,s)
{this._bindings=e,this._conn=s}get bindings(){return this._bindings}async
close(){return this._bindings.disconnect(this._conn)}useUnsafe(e){return
e(this._bindings,this._conn)}async query(e)
{this._bindings.logger.log({timestamp:new
Date,level:2,origin:4,topic:4,event:4,value:e});let s=await
this._bindings.runQuery(this._conn,e),t=r.RecordBatchReader.from(s);return
console.assert(t.isSync(),"Reader is not
sync"),console.assert(t.isFile(),"Reader is not file"),new r.Table(t)}async
send(e){this._bindings.logger.log({timestamp:new
Date,level:2,origin:4,topic:4,event:4,value:e});let s=await
this._bindings.startPendingQuery(this._conn,e);for(;s==null;)s=await
this._bindings.pollPendingQuery(this._conn);let t=new
d(this._bindings,this._conn,s),a=await r.RecordBatchReader.from(t);return
console.assert(a.isAsync()),console.assert(a.isStream()),a}async cancelSent()
{return await this._bindings.cancelPendingQuery(this._conn)}async
getTableNames(e){return await this._bindings.getTableNames(this._conn,e)}async
prepare(e){let s=await this._bindings.createPrepared(this._conn,e);return new
w(this._bindings,this._conn,s)}async insertArrowTable(e,s){let
t=r.tableToIPC(e,"stream");await this.insertArrowFromIPCStream(t,s)}async
insertArrowFromIPCStream(e,s){await
this._bindings.insertArrowFromIPCStream(this._conn,e,s)}async
insertCSVFromPath(e,s){await
this._bindings.insertCSVFromPath(this._conn,e,s)}async

```

=====

FILE: <https://ventusltd.github.io/ventus-grid-engine/genome/engine-graph.json>
34334 chars · data · TRUNCATED: the first 4000 characters of 34334 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (fetch)

=====

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{
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engine's own maths",
  "generated_utc": "2026-09-05T13:37:14.883Z",
  "note": "Nodes and edges match the node shape used by the live Spider Sandbox
(data-federation-map-for-globalgrid2050-all-
repos/dashboard/sandbox/spider_full_po_test.html: SCOPES[key].nodes =
[{{label,type,rag,reason,gh,ext,child}}, edges = [from,to,type]]. Edges here are
objects, not bare [from,to,type] tuples, because every edge in this genome must
carry evidence; edge.type stays a plain string for compatibility with a receiver
expecting the tuple shape (read edges[i].type, .from, .to as the tuple).",
  "focus_default": 0,
  "kind_labels": {
    "duplicates": "duplicates",
    "supersedes": "supersedes",
    "imports": "imports",
    "should_import": "should import",
    "drifts_from": "drifts from"
  },
  "nodes": [

```

```

{
  "label": "engine/geo-core.js",
  "type": "canonical",
  "rag": "green",
  "reason": "the one haversine + R_ATLAS/R_UK/R_MEAN, (lon,lat) order",
  "gh": "https://github.com/Ventusltd/ventus-grid-
engine/blob/main/engine/geo-core.js"
},
{
  "label": "engine/geo-area.js",
  "type": "canonical",
  "rag": "green",
  "reason": "polygonAreaKm2, polylinePerimeterKm, circleCapAreaKm2",
  "gh": "https://github.com/Ventusltd/ventus-grid-
engine/blob/main/engine/geo-area.js"
},
{
  "label": "engine/geo-shapes.js",
  "type": "canonical",
  "rag": "green",
  "reason": "destinationCirclePoints, the deduplicated circle generator",
  "gh": "https://github.com/Ventusltd/ventus-grid-
engine/blob/main/engine/geo-shapes.js"
},
{
  "label": "engine/geo-geojson.js",
  "type": "canonical",
  "rag": "green",
  "reason": "circleFeatureCollection, GeoJSON shaping kept out of the
maths",
  "gh": "https://github.com/Ventusltd/ventus-grid-
engine/blob/main/engine/geo-geojson.js"
},
{
  "label": "engine/v9-geodesy.js",
  "type": "canonical",
  "rag": "green",
  "reason": "distanceKm, destinationPoint, initialBearingDeg, voltagesKv,
representativePoint",
  "gh": "https://github.com/Ventusltd/ventus-grid-
engine/blob/main/engine/v9-geodesy.js"
},
{
  "label": "engine/v9-nearest-search.js",
  "type": "canonical",
  "rag": "green",
  "reason": "normalise + index, exhaustive scan, proven free of the ring-
search bug",
  "gh": "https://github.com/Ventusltd/ventus-grid-
engine/blob/main/engine/v9-nearest-search.js"
},
{
  "label": "deeplink/contract.js",
  "type": "canonical",
  "rag": "green",
  "reason": "buildDeepLink/parseDeepLink, LAYER_ID_FOR_BUCKET, the MAP
button as one testable thing",
  "gh": "https://github.com/Ventusltd/ventus-grid-
engine/blob/main/deeplink/contract.js"
},
{
  "label": "engine/network-topology.js",

```

```

    "type": "canonical",
    "rag": "green",
    "reason": "PROMOTED this session: index/at/graph over one site's published
nodes/branches",
    "gh": "https://github.com/Ventusltd/ventus-grid-
engine/blob/main/engine/network-topology.js"
  },
  {
    "label": "engine/electrical-distance.js",
    "type": "canonical",
    "rag": "green",
    "reason": "PROMOTED this session: between/within, BFS hop-count over
network-topology.graph()",
    "gh": "https://github.com/Ventusltd/ventus-grid-
engine/blob/main/engine/electrical-distance.js"
  },
  {
    "label": "engine/rating-envelope.js",
    "type": "canonical",
    "rag": "green",
    "reason": "PROMOTED this session: at(), per-season lowest/highest range,
never summed",
    "gh": "https://github.com/Ventusltd/ventus-grid-
engine/blob/main/engine/rating-envelope.js"
  },
  {
    "label": "engine/corridor-estimate.js",
    "type": "canonical",
    "rag": "green",
    "reason": "PROMOTED this session: forCable(), calibrated straight-line-to-
corridor multi

```

```

=====
FILE: https://cdn.jsdelivr.net/npm/apache-arrow@17.0.0/+esm
200034 chars · data · TRUNCATED: the first 4000 characters of 200034 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (script)
=====

```

```

/**
 * Bundled by jsDelivr using Rollup v4.62.2 and esbuild v0.28.1.
 * Original file: /npm/apache-arrow@17.0.0/Arrow.dom.mjs
 *
 * Do NOT use SRI with dynamically generated files! More information:
https://www.jsdelivr.com/using-sri-with-dynamic-files
 */
import{__asyncGenerator as mt,__await as L,__asyncDelegator as Ti,__asyncValues
as Gt,__awaiter as p,__rest as Ka}from"/npm/tslib@2.6.3/+esm";import*as N
from"/npm/flatbuffers@24.3.25/+esm";import{ByteBuffer as
es}from"/npm/flatbuffers@24.3.25/+esm";const qa=new TextDecoder("utf-
8"),mn=i=>qa.decode(i),Qa=new TextEncoder,Ze=i=>Qa.encode(i),tc=i=>typeof
i=="number",is=i=>typeof i=="boolean",H=i=>typeof
i=="function",nt=i=>i!=null&&Object(i)===i,Zt=i=>nt(i)&&H(i.then),we=i=>nt(i)&&H
(i[Symbol.iterator]),ae=i=>nt(i)&&H(i[Symbol.asyncIterator]),bn=i=>nt(i)&&nt(i.s
chema),ns=i=>nt(i)&&"done"in i&&"value"in
i,rs=i=>nt(i)&&H(i.stat)&&tc(i.fd),ss=i=>nt(i)&&_n(i.body),Ni=i=>"_getDOMStream"
in i&&"_getNodeStream"in
i,ec=i=>nt(i)&&H(i.abort)&&H(i.getWriter)&&!Ni(i),_n=i=>nt(i)&&H(i.cancel)&&H(i.
getReader)&&!Ni(i),ic=i=>nt(i)&&H(i.end)&&H(i.write)&&is(i.writable)&&!Ni(i),os=
i=>nt(i)&&H(i.read)&&H(i.pipe)&&is(i.readable)&&!Ni(i),nc=i=>nt(i)&&H(i.clear)&&
H(i.bytes)&&H(i.position)&&H(i.setPosition)&&H(i.capacity)&&H(i.getBufferIdentif
ier)&&H(i.createLong),vn=typeof SharedArrayBuffer<"u"?
SharedArrayBuffer:ArrayBuffer;function rc(i){const t=i[0]?[i[0]]:[];let

```

```

e,n,r,s;for(let o,a,c=0,u=0,d=i.length;++c<d;){
  if(o=t[u],a=i[c],!o||!a||o.buffer!==a.buffer||a.byteOffset<o.byteOffset){a&&
    (t[++u]=a);continue}if({byteOffset:e,byteLength:r}=o,
    {byteOffset:n,byteLength:s}=a,e+r<n||n+s<e){a&&(t[++u]=a);continue}t[u]=new
    Uint8Array(o.buffer,e,n-e+s)}return t}function Ui(i,t,e=0,n=t.byteLength){const
    r=i.byteLength,s=new Uint8Array(i.buffer,i.byteOffset,r),o=new
    Uint8Array(t.buffer,t.byteOffset,Math.min(n,r));return s.set(o,e),i}function
    bt(i,t){const e=rc(i),n=e.reduce((d,h)=>d+h.byteLength,0);let
    r,s,o,a=0,c=-1;const u=Math.min(t||Number.POSITIVE_INFINITY,n);for(const
    d=e.length;++c<d;){if(r=e[c],s=r.subarray(0,Math.min(r.length,u-
    a)),u<=a+s.length){s.length<r.length?
    e[c]=r.subarray(s.length):s.length===r.length&&c++,o?Ui(o,s,a):o=s;break}Ui(o||
    (o=new Uint8Array(u)),s,a,a+=s.length)}return[o||new Uint8Array(0),e.slice(c),n-
    (o?o.byteLength:0)]}function T(i,t){let e=ns(t)?t.value:t;return e instanceof i?
    i===Uint8Array?new i(e.buffer,e.byteOffset,e.byteLength):e:e?(typeof
    e=="string"&&(e=Ze(e)),e instanceof ArrayBuffer?new i(e):e instanceof vn?new
    i(e):nc(e)?T(i,e.bytes()):ArrayBuffer.isView(e)?e.byteLength<=0?new i(0):new
    i(e.buffer,e.byteOffset,e.byteLength/i.BYTES_PER_ELEMENT):i.from(e)):new
    i(0)}const
    sc=i=>T(Int8Array,i),oc=i=>T(Int16Array,i),Ie=i=>T(Int32Array,i),wn=i=>T(BigInt6
    4Array,i),M=i=>T(Uint8Array,i),ac=i=>T(Uint16Array,i),cc=i=>T(Uint32Array,i),lc=
    i=>T(BigUint64Array,i),uc=i=>T(Float32Array,i),dc=i=>T(Float64Array,i),hc=i=>T(U
    int8ClampedArray,i),In=i=>(i.next(),i);function*At(i,t){const e=function*(r)
    {yield r},n=typeof t=="string"||ArrayBuffer.isView(t)||t instanceof
    ArrayBuffer||t instanceof vn?e(t):we(t)?t:e(t);return yield*In((function*(r){let
    s=null;do s=r.next(yield T(i,s));while(!s.done)))(n[Symbol.iterator]()),new
    i}const
    fc=i=>At(Int8Array,i),yc=i=>At(Int16Array,i),pc=i=>At(Int32Array,i),as=i=>At(Uin
    t8Array,i),gc=i=>At(Uint16Array,i),mc=i=>At(Uint32Array,i),bc=i=>At(Float32Array
    ,i),_c=i=>At(Float64Array,i),vc=i=>At(Uint8ClampedArray,i);function _t(i,t)
    {return mt(this,arguments,function*(){if(Zt(t))return yield L(yield
    L(yield*Ti(Gt(_t(i,yield L(t))))));const n=function(o){return
    mt(this,arguments,function*(){yield yield L(yield L(o))}),r=function(o){return
    mt(this,arguments,function*(){yield L(yield*Ti(Gt(In((function*(a){let c=null;do
    c=a.next(yield c?.value);while(!c.done)))(o[Symbol.iterator]()))))}),s=typeof
    t=="string"||ArrayBuffer.isView(t)||t instanceof ArrayBuffer||t instanceof vn?
    n(t):we(t)?r(t):ae(t)?t:n(t);return yield L(yield*Ti(Gt(In((function(o){return
    mt(this,arguments,f

```

```

=====
FILE: https://cdn.jsdelivr.net/npm/flatbuffers@24.3.25/+esm
9269 chars · data · TRUNCATED: the first 4000 characters of 9269 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (script)
=====

```

```

/**
 * Bundled by jsDelivr using Rollup v4.62.2 and esbuild v0.28.1.
 * Original file: /npm/flatbuffers@24.3.25/mjs/flatbuffers.js
 *
 * Do NOT use SRI with dynamically generated files! More information:
https://www.jsdelivr.com/using-sri-with-dynamic-files
 */
const p=2,F=4,w=4,E=4,n=new Int32Array(2),d=new Float32Array(n.buffer),I=new
Float64Array(n.buffer),a=new Uint16Array(new Uint8Array([1,0]).buffer)
[0]==1;var l;(function(c)
{c[c.UTF8_BYTES=1]="UTF8_BYTES",c[c.UTF16_STRING=2]="UTF16_STRING"})(1||(1=
{}));class o{constructor(t)
{this.bytes_=t,this.position_=0,this.text_decoder_=new TextDecoder}static
allocate(t){return new o(new Uint8Array(t))}clear(){this.position_=0}bytes()
{return this.bytes_.position()}setPosition(t)
{this.position_=t}capacity(){return this.bytes_.length}readInt8(t){return
this.readUint8(t)<<24>>24}readUint8(t){return this.bytes_[t]}readInt16(t){return

```

```

this.readUint16(t)<<16>>16}readUint16(t){return this.bytes_[t]|this.bytes_[t+1]
<<8}readInt32(t){return this.bytes_[t]|this.bytes_[t+1]<<8|this.bytes_[t+2]
<<16|this.bytes_[t+3]<<24}readUint32(t){return
this.readInt32(t)>>>0}readInt64(t){return
BigInt.asIntN(64,BigInt(this.readUint32(t))+(BigInt(this.readUint32(t+4))
<<BigInt(32)))}readUint64(t){return
BigInt.asUintN(64,BigInt(this.readUint32(t))+(BigInt(this.readUint32(t+4))
<<BigInt(32)))}readFloat32(t){return n[0]=this.readInt32(t),d[0]}readFloat64(t)
{return n[a?0:1]=this.readInt32(t),n[a?
1:0]=this.readInt32(t+4),I[0]}writeInt8(t,i){this.bytes_[t]=i}writeUint8(t,i)
{this.bytes_[t]=i}writeInt16(t,i)
{this.bytes_[t]=i,this.bytes_[t+1]=i>>8}writeUint16(t,i)
{this.bytes_[t]=i,this.bytes_[t+1]=i>>8}writeInt32(t,i)
{this.bytes_[t]=i,this.bytes_[t+1]=i>>8,this.bytes_[t+2]=i>>16,this.bytes_[t+3]=
i>>24}writeUint32(t,i)
{this.bytes_[t]=i,this.bytes_[t+1]=i>>8,this.bytes_[t+2]=i>>16,this.bytes_[t+3]=
i>>24}writeInt64(t,i)
{this.writeInt32(t,Number(BigInt.asIntN(32,i))),this.writeInt32(t+4,Number(BigIn
t.asIntN(32,i>>BigInt(32))))}writeUint64(t,i)
{this.writeUint32(t,Number(BigInt.asUintN(32,i))),this.writeUint32(t+4,Number(Bi
gInt.asUintN(32,i>>BigInt(32))))}writeFloat32(t,i)
{d[0]=i,this.writeInt32(t,n[0]}writeFloat64(t,i){I[0]=i,this.writeInt32(t,n[a?
0:1]),this.writeInt32(t+4,n[a?1:0])}getBufferIdentifier()
{if(this.bytes_.length<this.position_+4+4)throw new Error("FlatBuffers:
ByteBuffer is too short to contain an identifier.");let t="";for(let
i=0;i<4;i++)t+=String.fromCharCode(this.readInt8(this.position_+4+i));return
t}__offset(t,i){const e=t-this.readInt32(t);return i<this.readInt16(e)?
this.readInt16(e+i):0}__union(t,i){return
t.bb_pos+i+this.readInt32(i),t.bb=this,t}__string(t,i)
{t+=this.readInt32(t);const e=this.readInt32(t);t+=4;const
s=this.bytes_.subarray(t,t+e);return i===1.UTF8_BYTES?
s:this.text_decoder.decode(s)}__union_with_string(t,i){return typeof
t=="string"?this.__string(i):this.__union(t,i)}__indirect(t){return
t+this.readInt32(t)}__vector(t){return t+this.readInt32(t)+4}__vector_len(t)
{return this.readInt32(t+this.readInt32(t))}__has_identifier(t)
{if(t.length!=4)throw new Error("FlatBuffers: file identifier must be length
4");for(let
i=0;i<4;i++)if(t.charCodeAt(i)!=this.readInt8(this.position()+4+i))return!1;retu
rn!0}createScalarList(t,i){const e=[];for(let s=0;s<i;++s){const
r=t(s);r!=null&&e.push(r)}return e}createObjList(t,i){const e=[];for(let
s=0;s<i;++s){const r=t(s);r!=null&&e.push(r.unpack())}return e}}class
__{constructor(t)
{this.minalign=1,this.vtable=null,this.vtable_in_use=0,this.isNested=!1,this.obj
ect_start=0,this.vtables=
[],this.vector_num_elems=0,this.force_defaults=!1,this.string_maps=null,this.tex
t_encoder=new TextEncoder;let i;t?
i=t:i=1024,this.bb=o.allocate(i),this.space=i}clear()
{this.bb.clear(),this.space=this.bb.capacity(),this.minalign=1,this.vtable=null,
this.vtable_in_use=0,this.isNested=!1,this.object_start=0,this.vtables=
[],this.vector_num_elems=0,this.force_defaults=!1,this.string_maps=null}forceDef
aults(t){this.force_defaults=t}dataBuffer(){return this.bb}

```

```

=====
FILE: https://cdn.jsdelivr.net/npm/tslib@2.6.3/+esm
10708 chars · data · TRUNCATED: the first 4000 characters of 10708 are shown,
because this is data rather than code. Nothing has been summarised or rewritten;
the rest is simply not here. · found via loaded by the browser (script)
=====

```

```

/**
 * Bundled by jsDelivr using Rollup v4.62.2 and esbuild v0.28.1.
 * Original file: /npm/tslib@2.6.3/tslib.es6.mjs
 *

```

* Do NOT use SRI with dynamically generated files! More information:
<https://www.jsdelivr.com/using-sri-with-dynamic-files>

```
*/
var O=function(e,t){return O=Object.setPrototypeOf||{__proto__:[]}instanceof
Array&&function(n,r){n.__proto__=r}||function(n,r){for(var i in
r)Object.prototype.hasOwnProperty.call(r,i)&&(n[i]=r[i])},O(e,t)};function
S(e,t){if(typeof t!="function"&&t!=null)throw new TypeError("Class extends
value "+String(t)+" is not a constructor or null");O(e,t);function n()
{this.constructor=e}e.prototype=t===null?Object.create(t):
(n.prototype=t.prototype,new n)}var v=function(){return
v=Object.assign||function(t){for(var n,r=1,i=arguments.length;r<i;r++)
{n=arguments[r];for(var o in n)Object.prototype.hasOwnProperty.call(n,o)&&
(t[o]=n[o])}return t},v.apply(this,arguments)};function P(e,t){var n={};for(var
r in e)Object.prototype.hasOwnProperty.call(e,r)&&t.indexOf(r)<0&&
(n[r]=e[r]);if(e!=null&&typeof Object.getOwnPropertySymbols=="function")for(var
i=0,r=Object.getOwnPropertySymbols(e);i<r.length;i++)t.indexOf(r[i])
<0&&Object.prototype.propertyIsEnumerable.call(e,r[i])&&(n[r[i]]=e[r[i]]);return
n}function E(e,t,n,r){var i=arguments.length,o=i<3?t:r===null?
r=Object.getOwnPropertyDescriptor(t,n):r,a;if(typeof Reflect=="object"&&typeof
Reflect.decorate=="function")o=Reflect.decorate(e,t,n,r);else for(var
u=e.length-1;u>=0;u--)(a=e[u])&&(o=(i<3?a(o):i>3?a(t,n,o):a(t,n))||o);return
i>3&&o&&Object.defineProperty(t,n,o),o}function T(e,t){return function(n,r)
{t(n,r,e)}}function Q(e,t,n,r,i,o){function a(_){if(!_==void 0&&typeof
_!="function")throw new TypeError("Function expected");return _}for(var
u=r.kind,y=u=="getter"? "get":u=="setter"? "set": "value",f=!t&&e.r.static?
e.e.prototype:null,l=t|| (f?Object.getOwnPropertyDescriptor(f,r.name):
{}),s,w=!l,c=n.length-1;c>=0;c--){var p={};for(var h in r)p[h]=h=="access"?
{r[h]}:r[h];for(var h in
r.access)p.access[h]=r.access[h];p.addInitializer=function(_){if(w)throw new
TypeError("Cannot add initializers after decoration has
completed");o.push(a(_||null))};var d=(0,n[c])(u=="accessor"?
{get:l.get,set:l.set}:l[y],p);if(u=="accessor"){if(d===void
0)continue;if(d===null||typeof d!="object")throw new TypeError("Object
expected");(s=a(d.get))&&(l.get=s),(s=a(d.set))&&(l.set=s),
(s=a(d.init))&&i.unshift(s)}else(s=a(d))&&(u=="field"?
i.unshift(s):l[y]=s)}f&&Object.defineProperty(f,r.name,l),w=!0}function U(e,t,n)
{for(var r=arguments.length>2,i=0;i<t.length;i++)n=r?
t[i].call(e,n):t[i].call(e);return r?n:void 0}function W(e){return typeof
e=="symbol"?e:"".concat(e)}function X(e,t,n){return typeof t=="symbol"&&
(t=t.description?
["".concat(t.description,"")]:""),Object.defineProperty(e,"name",
{configurable:!0,value:n?"".concat(n,"",t):t})}function D(e,t){if(typeof
Reflect=="object"&&typeof Reflect.metadata=="function")return
Reflect.metadata(e,t)}function x(e,t,n,r){function i(o){return o instanceof n?
o:new n(function(a){a(o)})}return new(n||(n=Promise))(function(o,a){function
u(l){try{f(r.next(l))}catch(s){a(s)}}function y(l){try{f(r.throw(l))}catch(s)
{a(s)}}function f(l){l.done?o(l.value):i(l.value).then(u,y)}f((r=r.apply(e,t||
[])).next())})}function R(e,t){var n={label:0,sent:function(){if(o[0]&l)throw
o[1];return o[1]},trys:[],ops:[],r,i,o,a;return a=
{next:u(0),throw:u(1),return:u(2)},typeof Symbol=="function"&&
(a[Symbol.iterator]=function(){return this}),a;function u(f){return function(l)
{return y([f,l])}}function y(f){if(r)throw new TypeError("Generator is already
executing.");for(;a&&(a=0,f[0]&&(n=0)),n;)try{if(r=1,i&&(o=f[0]&2?i.return:f[0]?
i.throw|((o=i.return)&&o.call(i),0):i.next)&&!(o=o.call(i,f[1])).done)return
o;switch(i=0,o&&(f=[f[0]&2,o.value]),f[0]){case 0:case 1:o=f;break;case 4:return
n.label++,{value:f[1],done:!1};case 5:n.label++,i=f[1],f=[0];continue;case
7:f=n.ops.pop(),n.trys.pop();continue;default:if(o=n.trys,!
(o=o.length>0&&o[o.length-1])&&(f[0]===6
```

=====

END OF TELEPRINT

=====